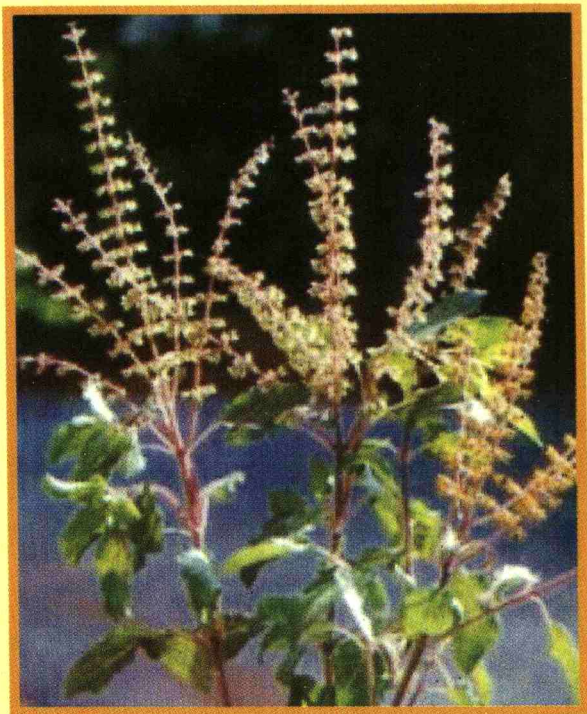


AYURVEDIC HOME REMEDIES



TULSI



CENTRAL COUNCIL FOR RESEARCH IN AYURVEDA AND SIDDHA

**Department of ISM&H, Ministry of Health &
Family Welfare, Govt of India**

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IN AYURVEDA AND SIDDHA**

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& Family Welfare, Govt of India

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FOREWORD

Most of the diseases arise due to improper food habits and faulty life style besides hereditary and other reasons. Mankind has constantly grappled with ill health and tried to achieve health & happiness. These efforts are on from time immemorial. These endeavors have thrown up various systems of medicine. Ayurveda is one such system evolved in India, which tries to bring harmony between man & nature by using holistic methods of diagnosis, prevention and treatment. Like all other Indian contributions it is intertwined with Indian philosophy, yoga and science.

The modern science is successful in inventing medicines for many diseases. But complete health is still a mirage for the mankind. Strong synthetic drugs are creating innumerable side effects known & unknown sometimes making the treatment more dangerous than the disease itself. As a contrast Ayurveda preaches gentle way of treatment mainly based on herbs and diet. Many of most effected drugs are simple, easily available and can be prepared at home.

This Booklet offers some simple recipes prepared from the materials available in and around the kitchen. For simple problems like cough, cold, indigestion etc, these preparations are very effective. For chronic problems like diabetes, joint pain, skin diseases these remedies may be used even along with other medication.

I hope that this booklet will be beneficial to the readers.



G. VELUCHAMY

Director

INSTRUCTIONS

- ◆ Measurements
5gm. = 1 Tea spoon full
5ml. = 1 Tea spoon full
- ◆ These preparations are for mild and chronic conditions. In case patient does not get relief within 2-3 days of taking medicine, he/she may consult the nearest doctor.
- ◆ The dose given in the booklet is for adult. For children $\frac{1}{2}$ or $\frac{1}{4}$ of this adult dose may be used.
- ◆ The dosage may be adjusted with little alterations according to the tolerance and desire.
- ◆ These preparations may be used regularly for many days. But in case of any discomfort drug should be stopped immediately.
- ◆ Decoction should be prepared by boiling crushed/ coarsely powdered drug in four parts of water and reducing to one fourth .
- ◆ Juice should be prepared by crushing/ grinding in mixi the fresh drug with little water if required and the juice should be expressed through a clean cloth.
- ◆ Paste should be prepared by crushing/ grinding the drug very finely with desired liquid if required.

- ◆ In general too spicy, salty, chilly, sour, preserved items fried food, heavy, indigestible, too cold & hot, stale food and the food that do not suit the health should be avoided.
- ◆ Irregular food habit, sleep and lack of physical exercise is main cause for many diseases.
- ◆ Too much tea, coffee, should be avoided. Tobacco, alcohol and drugs should not be taken.
- ◆ Mental stress should be tackled by recreation such as meditation, prayer, sports, exercises, yoga & other activities of individual's choice.

Adrak/ Sonth

(*Zingiber officinale* Rosc., Ardraka/Sunthi)

Indigestion	5 gm. crushed rhizome with salt or jaggery twice daily before meal.
Ear Pain	2-4 drops of fresh warm juice to be instilled in the ear (don't use when there is discharge)
Hoarseness of voice	1-3 gm. powder of dry rhizome with honey in three divided doses.
Aches & pains	10-20 ml. decoction prepared from 2 gm. of dry rhizome twice a day.
Cold/Cough	2-5 gm. powder of dry rhizome with jaggery thrice a day in divided doses. 10ml. decoction prepared from a piece of ginger every morning prevents recurrent attack of cold.
Headache	Warm paste should be applied over forehead upto 3-4 times a day.
Abdominal pain	5 ml. juice in a glass of butter milk with lemon & salt.

AJVAIN

(*Trachyspermum ammi* Spragne, Yavani)

Piles	1 gm. powder and 1 gm. black-salt with butter-milk twice daily.
Painful Menses	1-2 gm. powder of seeds with warm milk thrice a day for 2-3 days .
Urticaria (Skin allergy)	1-2 gm. powder of seeds with water twice daily.
Abdominal pain	1 gm. powder with luke warm water twice or thrice.
Flatulence (gas)	2 gm. ajvain powder with equal quantity of <i>Saunf</i> powder with warm water.
Sinusitis	warm paste should be applied over forehead & little below the eyes in the morning.
Nasal block	1-2 gm powder should be put in steaming water & vapour should be inhaled; 2-3 time a day.
Loss of appetite	1 gm. powder with warm water 1/2 hour before meals.

ANAR

(*Punica granatum* Linn., Dadima)

This is very nutritious fruit suitable for all & can be used in diet in any disease condition.

Indigestion	10 ml. juice of fruit, 1 gm. black-salt or fried Jira powder with honey or sugar to hold inside mouth for sometime before food.
Bleeding piles	10 ml. juice of fruit with sugar twice a day. or 10 gm. powder of dry fruit rind with equal quantity of sugar twice a day.
Diahhorria/Dysentery	10 ml. decoction of fruit skin thrice a day. Fruit may be eaten in plenty.
Hyper-acidity	10 ml. juice of fruit twice a day. Even the fruit may be eaten.
Bad breath	Gargling with the warm decoction prepared from fruit skin 3-4 time a day.
Acne	Paste of seeds should be applied to affected part twice daily.

AMLA

(*Emblica officinalis* Gaertn., Amalaki)

For general health	Regular use of Amla provides nutrition and boosts good health.
Hyper Acidity/ Peptic Ulcer/ Constipation	3-5 gm. powdered fruit rind twice a day with milk. or 10-20 ml. fruit juice twice a day. or 3-5 gm. powdered fruit rind twice a day with milk (Even raw <i>Amla</i> can be eaten).
Stress	External application of 25-50 gm. of fruit rind ground in butter milk on the forehead.
Diabetes	10-20 ml. Fruit juice with 10-20 ml. juice of fresh rhizome of <i>Haldi</i> twice a day.
Greying of Hair / Hair fall/Dandruff	Fruit rind soaked overnight in water for application on scalp two hours before bath. or paste prepared from fruit rind should be applied two hours before bath. 1-2 fresh fruits eaten every morning prevents hair fall & early greying.
Bleeding gums	Fine powder should be gently massaged to gums after brushing at least twice a day regularly.

DALCHINI

(*Cinnamomum zeylanicum* Blume ,Tvak)

This commonly used spice is a good digestive & its pleasant flavour has soothing effect on mind.

Indigestion	2 gm. powder of bark with water twice a day.
Loss of Appetite	2 gm. powder of equal parts of <i>Dalchini</i> and <i>Ajvain</i> for chewing in three divided doses before food.
Vomiting	1-2 gm. powder with honey thrice a day in divided doses.
Tension headache	To be rubbed with water on a rough surface & applied to forehead.
Mental tension	The aroma has soothing action on mind. Crushed pieces may be kept in hanky or near the pillow.
Dry cough	Chewing controls the throat irritation and helps in dry cough.

DHANIA

(*Coriandrum sativum* Linn. , Dhanyaka)

Cold/ Cough	20 ml. decoction prepared from 5 gm.coarse powder with sugar and turmeric powder thrice a day. or use of dhania powder as herbal tea every morning prevents cold, cough & problems related to digestion.
Intestinal worms	3-5 gm. powder with jaggery twice a day for 5 days.
Sun stroke/ Dehydration	20 ml. decoction prepared from coarse powder with sugar and a pinch of salt frequently.
Indigestion	20 ml. decoction prepared from 5 gm. coarse powder with a pinch of ginger powder thrice a day.
Fever	20 ml. decoction from 5 gm. powder with sugar 3-4 times a day.

ELAICHI

(*Elettaria cardamomum* Maton, Ela)

Hiccough	1-2 fruits to be chewed frequently (not more than 4/day).
Vomiting	250-500 gm. powder of seeds fried in ghee thrice daily with honey.
Bad breath	1-2 seeds to be chewed frequently (not more than 4/day).
Diarrhoea/Vomiting	Ash of the <i>Elaichi</i> skin 2 gm. with a little honey 4-5 times a day.
Cold	20 ml. decoction prepared from. 5 gm. <i>Dhania</i> , 1 gm <i>Methi</i> seeds, little turmeric powder should be taken 2-3 times/day.
Cough	Little elaichi powder with a tea spoon full of honey 3-4 times day. Even chewing elaichi (Not more than 3 per day) frequently is help full in dry as well as productive cough.

GHEE
(Clarified Butter)

Ghee is advocated as vehicle for number of medicines in Ayurvedic system. Judicious use of ghee is very good for physical as well as psychological health.

Ulcer/ Wounds/ Burns	Application over affected part frequently.
Loss of Appetite	With <i>Hing</i> and <i>Jeera</i> powder with food.
Memory	Use of ghee everyday in children improves memory.
Constipation	5 ml. ghee put in a cup of warm milk should be taken with sugar at bed time.

HALDI

(*Curcuma longa* Linn., Haridra)

Diabetes	10 ml. fresh juice with 10 ml. juice of <i>Amla</i> twice daily.
Acne	Application of paste to the effected part twice a day. Use of <i>Haldi</i> with water, milk or cream on face gives glow to the skin & removes unwanted hair if used regularly.
Cold	2 gm. powder with warm milk and sugar twice a day. Decoction of 1 gm. <i>Haldi</i> powder or use of <i>Haldi</i> in herbal tea prevents all allergic problems.
Wound/ulcer/ skin disease	To be washed with decoction of <i>Haldi</i> & paste of haldi should be applied. Paste made by mixing with <i>Ghee</i> /coconut oil should be applied over affected part.
Skin allergy	1-3 gm. powder should be taken with jaggery twice a day.

HING

(*Ferula narthex* Boiss., Hingu)

Use of hing in daily diet is good for degestion & disorders related to it. Best way is to fry hing in little ghee before use.

Abdominal Pain

Stir and dissolve in water and apply on & around umbilicus.

Specially in children or infants where there is abdominal distention this is very helpful.

1 gm. *Hing* fried in ghee with butter milk twice a day.

Toothache

Keep fried *Hing* in carious tooth.

Loss of appetite

A pinch of *Hing* fried in ghee & a piece of adrak crushed should be taken with butter milk before food.

JAYPHAL

(*Myristica fragrans* Houtt., Jatiphala)

- | | |
|-----------------------|--|
| Diarrhoea in children | A pinch of powder or fruit rubbed on a clean surface with milk or water should be given 3-4 times a day. |
| Irritability | If child is restless & irritable 1-2 pinch of powder given with milk acts as mild sedative. This may be used 3-4 times a day. |
| Black pigmentation | Black pigmentation over the face is common complaint in ladies especially around menopausal age. <i>Jayphal</i> rubbed in milk and applied on such areas is helpful. |
| Abdominal pain | If pain is due to diarrhoea 2 gm. powder with warm water should be taken 4-5 time a day. It reduces the motility of intestine their by reducing pain. |

JEERA

(*Cuminum cyminum* Linn., Jiraka)

Indigestion	3-6 gm. powder of fried <i>Jeera</i> and rock-salt with warm water thrice daily.
Diarrhoea/Dysentery	1-2 gm. powder of fried <i>Jeera</i> with 250 ml. butter milk four times daily.
Hyper-acidity	5-10 gm. ghee boiled with <i>Jeera</i> should be taken with rice during meals.
Skin disease	1-2 gm. powder of fried <i>Jeera</i> with milk twice daily.
Cold	Warm decoction of 2 gm. <i>Jeera</i> , 5 gm. <i>Dhania</i> , 1 gm. <i>Haldi</i> , 1 gm. <i>Methi</i> powder and little pepper should be taken with honey/sugar with lemon two to three times.
Cough	Either decoction as mentioned above or chewing few grains frequently helps in dry as well as productive cough.

KALIMIRACH

(*Piper nigrum* Linn., Maricha)

Cough	1 gm. powder of seeds with ghee and honey twice daily.
Skin diseases	Little powder with coconut oil for local application.
Hoarseness of voice	1-2 gm. powder of seed fried with ghee twice daily, to keep in mouth.
Hiccough	1-2 gm. powder of seeds with sugar twice daily.
Indigestion	A pinch of powder with a piece of crushed ginger & rock salt should be taken before meal.
Loss of appetite	A pinch of powder with 2 tea spoon full of lemon juice with water $\frac{1}{2}$ hour before food.
Bleeding gums/ tartar/bad breath	Pinch of powder mixed with honey should be applied to gums after gargling with warm salt water twice a day.

KARELA

(*Momordica charantia* Linn., Karavellaka)

Indigestion	5-10 ml. juice of fruit twice daily.
Diabetes	1-3 gm. powder of seeds with water twice a day.
Loss of appetite	5-10 ml. juice with butter milk 1-2 times.
Skin diseases	5-10 ml. juice in every morning in empty stomach.
Intestinal worms	10 ml juice with jaggery in the morning for 3 days.
Acne	5-10 ml. juice in empty stomach once a day.

LAHSUN

(*Allium sativum* Linn., Rasona)

Ear pain 2-4 drops of warm fresh juice for
instillation in ear twice a day
(Don't use when there is discharge).

or little paste boiled in coconut or
mustard oil should be filled in ear
2-3 time a day.

Flatulence 6 ml. juice with honey twice a day.

Cough/Cold Crushed bulb boiled in water is to be
taken with sugar.

Joint pain 5 gm. paste of bulb with honey or with
food twice a day.

Warm paste mixed in any oil
should be applied over the joint

LAUNG

(*Syzygium aromaticum* (L.) Merr & Perry, Lavanga)

Cough	To be chewed frequently or 1 gm. powder with honey 2-3 times in divided doses. 20 ml warm decoction prepared by putting 1 gm. clove 3-4 times daily. It is useful both in dry and productive cough.
Cold / Hiccough	1-2 gm. of powder with honey in three divided doses.
Indigestion	1-2 gm. powder with warm water.
Tooth ache	Crushed clove should be kept in carious tooth.
Bad breath	Small piece should be chewed frequently.
Ear pain	Warm coconut oil prepared by boiling the powder of <i>Laung</i> should be filled in ear twice daily (Do not use when there is discharge).

MADHU(Honey)

It is commonly prescribed by Ayurvedic doctors as vehicle with medicine especially in respiratory disorders.

Obesity	One teaspoon with glass of water in the morning.
Loss Appetite	With a pinch of pepper powder three to four times a day .
Cough	With a pinch of powder of <i>Laung</i> three to four times.
Wounds/Ulcer/Burn	Honey and ghee should be mixed and applied.
Pyorrhoea/ (Bleeding gums)	Apply mixture of ginger, pepper, rock salt, honey & <i>Ghee</i> to the gums twice a day.
Face pack	Application of honey gives glow to the skin.

Note : Honey should not be boiled.

METHI

(*Trigonella foenum-graecum* Linn., Methika)

Diabetes	2 gm. powder with milk twice daily.
Body ache	2 gm. powder mixed with 2 gm. Jeera powder with warm milk and sugar twice daily.
Lactation (To increase mother's milk)	5 gm. powder with milk and sugar in the morning.
Dandruff	Scalp should be massaged with paste mixed in the water $\frac{1}{2}$ hour before bath. Coconut oil boiled with methi powder should be used regularly. Use of methi paste acts as a good hair conditioner.

NARIYAL

(*Cocos nucifera* Linn., Narikela)

Hyper-acidity	Tender coconut water twice daily.
Kidney stone	Tender coconut water twice daily.
Wounds/Ulcer /Burn	Oil should be applied frequently.
Dehydration	Tender coconut water should be taken frequently.
Hair fall	Coconut oil processed with <i>Methi</i> powder and <i>Amla</i> powder should be used regularly.
Sun stroke	Tender coconut water should be taken frequently.

NEEM

(*Azadirachta indica* A. juss., Nimba)

Skin disease	10 ml. juice of leaves with honey twice a day.
Wounds/Ulcer	Warm application of paste prepared from leaves over affected part.
Intestinal worms	20 ml decoction made from hand full of leaves should be given in empty stomach for three days.
Loss of appetite	20 ml. decoction made from hand full of leaves should be given in empty stomach for three days.
Dandruff/ Ring worm	Decoction made from hand full of leaves should be applied to scalp one hour before bath.

NIMBU

(*Citrus limon* Linn., Nimbu)

Indigestion	5-10 ml. juice with a pinch of salt and pepper before food.
Dehydration/ sun stroke	One Nimbu squeezed into a glass of water mixed with salt and sugar.
Bleeding gums	The fruit skin should be crushed & rubbed gently twice a day regularly to the gums.
Loss of appetite	5 ml. juice with salt and a pinch of pepper before meals.
Hair fall	Coconut oil processed with the fruit skin dried & powdered.
Vomiting	5-10 ml juice mixed with water, sugar and a pinch of salt frequently in small quantities.

PYAJ

(*Allium cepa* Linn., Palandu)

White variety of onion is considered as best.

Sun stroke	Use plenty of onion for prevention of son stroke
Cold/ cough	In children the decoction made from a piece should be given with Jaggery thrice daily.
Dysentery	White onion chopped & fried in ghee should be eaten with rice.

PIPPALI

(*Piper longum* Linn., Pippali)

Indigestion	2gm. powder with guda twice daily before meal.
Cold/Cough	2 gm. powder with honey thrice a day in divided doses. 10-20 ml. decoction with 2-3 gm powder of <i>Pippali</i> , ginger, black pepper with sugar or honey 2-3 times a day.
Pyorrhoea	Decoction of Pippali with honey & ghee to keep inside mouth for some time in the morning & before bed time.
Diarrhoea	Mix 2-3 gm. powder in 1 litre butter milk and divide equally into 4 parts. Take 1 part 6 th hourly.
Cold with fever	2 gm. powder with honey twice daily.

SAUNF

(Foeniculum vulgare Mill., Shatapushpa)

Indigestion	3-5 gm. powder with powder of fried <i>Jeera</i> and rock-salt with warm water thrice daily.
Diarrhoea/Dysentery	3-5 gm. powder with butter milk 3-4 times daily.
Abdominal pain	3-5 gm. powder with butter milk 3-4 times daily.
Bad breath	Little quantity should be chewed after food.
Loss of appetite	3-5 gm. powder of roasted saunf with butter milk, little pepper powder & salt twice daily.

TULSI

(*Ocimum sanctum*, Tulasi)

Cough/ Cold	5-10 ml. juice twice or thrice a day with honey.
Fever	30 ml. decoction from handful of leaves & 5 gm. <i>Dhania</i> thrice a day.
Skin allergy	5-10 ml. juice twice or thrice daily.
Indigestion/ Loss of appetite	5-10 ml. juice twice or thrice daily.
Greying of hair	Coconut oil processed with <i>Tulsi</i> juice for regular use.
Wound/ulcer	Juice mixed with honey & <i>Haldi</i> powder for application.
Ear pain	2-3 luke warm drops 2 times daily (Do not use when there is discharge).

SCIENTIFIC BASIS FOR AYURVEDIC THERAPIES

SCIENTIFIC BASIS FOR AYURVEDIC THERAPIES

EDITED BY
LAKSHMI CHANDRA MISHRA



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Preface

There has been increasing public interest in complementary alternative medicine during the past decade in the U.S. that led to the establishment of the Center for Alternative and Complementary Medicine at the National Institutes of Health (NIH), Bethesda, MD, by the U.S. Congress to conduct scientific evaluation of these therapies. The Ayurvedic system of medicine (traditional medical system of India) is recognized by NIH as a complementary and alternative medicine (CAM). Ayurveda has been practiced in India for over 5000 years and is recognized as a complete medical system comparable with allopathic medicine by the government of India. In India, Ayurveda has a complete infrastructure, medical colleges, hospitals integrated with allopathic medicine, research institutes, and scientific journals devoted to Ayurveda. In addition, India's Ayurvedic pharmaceutical industry is governed by the same food and drug laws that regulate conventional drugs. Research in pharmacology, biochemistry, phytochemistry, and clinical trials of Ayurvedic therapies currently constitutes a substantial portion of the total research conducted in government institutes and medical colleges in India.

There has been considerable scientific research effort in Ayurvedic therapies during the past 50 years. This research has not been adequately disseminated to Ayurvedic students and physicians. In order to accomplish the assimilation of this research into practice, three major goals were set for this book: (1) to provide information on pharmacological, biochemical, and clinical investigations on Ayurvedic therapies; (2) to explore the scientific basis of Ayurvedic concepts of diseases, diagnosis, and treatments; and (3) to develop new interpretation of Ayurvedic concepts of therapies based on modern knowledge where possible.

As editor of the book, I used my combination of medical education in Ayurveda and research experience in conventional medicinal pharmacy and pharmacology in developing this text. Disease topics were carefully selected based on the scientific studies available and the prevalence of a disease. Knowledgeable physicians and scientists in these fields were selected based on their scientific contributions in the field to review and evaluate the worldwide literature. In addition, I, along with one Ayurvedic physician, one biochemist, and one overall expert with knowledge in conventional and CAM research methods, read each chapter.

General topics such as Ayurvedic disease management, *panchakarma*, Ayurvedic *bhasmas*, the current status of Ayurveda in India, and clinical research design and evaluation of typical clinical trials of certain diseases were written by experts in their fields.

It is my hope that the book will prove useful to all Ayurvedic and conventional medicine physicians, students and scientists, and the general public. In addition, the book is expected to bring awareness to the community of health-care providers that Ayurveda is significantly more than a few home remedies, yoga, and meditation. It is a fully established system of medicine, has a considerable scientific base, and has therapies that can be used alone or as an adjuvant with conventional health care as practiced in India.

Lakshmi Chandra Mishra
Editor

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I gratefully thank first and foremost Dr. Betsy B. Singh, professor and dean of the Research Division, Southern California University of Health Sciences (SCUHS), for the encouragement and sincere help given to me in preparing this book. Dr. Singh took great interest in reviewing all the chapters and provided valuable comments in the area of science and clinical research methods. I am thankful to Dr. Sivarama Prasad Vinjamury, associate professor; Dinesha Ninjegowda, Research Division, SCUHS; and Vijay Singh for help in reviewing the chapters. Finally, I thank Neil Shepard and Raheleh Khorsan of the Research Division and Dr. Tarek Adra and Dyba Kalyani, SCUHS students, for the technical support provided.

I sincerely acknowledge the effort of all contributors, particularly those who contributed more than one chapter and made all the required changes requested by reviewers and editors on time. These contributors worked hard to secure the information required for the chapters, to take a scientific approach to the presentation of both the Ayurvedic and conventional medicine data on a particular topic.

Finally, I thank my children, Aparna Rani, Gyan Sagar, Anita Rani, and Dhyan Sagar, for their encouragement during the preparation of this book.

Lakshmi Chandra Mishra
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Lakshmi Chandra Mishra, Ph.D., is a professor in the Research Division of Southern California University of Health Sciences and an Ayurvedic medicine practitioner at the University Health Center, Southern California University, Whittier. He has been conducting research on Ayurvedic therapies for musculoskeletal disorders at the university, and stroke in collaboration with faculty at the University of Southern California, School of Pharmacy.

Dr. Mishra received his Bachelor of Indian Medicine and Surgery degree from Bundelkand Ayurvedic College, Jhansi, UP, India, in 1954 and practiced Ayurvedic medicine for several years thereafter. He received his bachelor's and master's degrees in pharmacy from Banaras Hindu University, Varanasi, UP, India, in 1958 and 1959, respectively. He conducted phytochemical and

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After coming to the U.S. in 1963, Dr. Mishra received his Ph.D. in biochemical pharmacology from the State University of New York at Buffalo in 1967. Dr. Mishra worked on cancer chemotherapy research at Roswell Park Memorial Institute in Buffalo for 1 year before joining Microbiological Associates, Bethesda, MD, where he conducted research on anticancer agents that was sponsored by the National Cancer Institute, National Institutes of Health (NIH), Bethesda, MD. He later worked for several federal regulatory government agencies as a pharmacologist, toxicologist, and branch chief. Dr. Mishra has served on many scientific committees in government and industry. He was an ex-officio member of the National Cancer Advisory Board of the National Cancer Institute, NIH, for 18 years.

Dr. Mishra has extensive background and experience both as a clinician in the traditional and modern practice of Ayurveda and as a bench scientist. His research expertise includes pharmacology, toxicology, pharmaceutical chemistry, pharmacognosy, and phytochemistry. He has authored more than 100 scientific papers, documents, and reports in the fields of Ayurvedic medicine, pharmacology, toxicology, cancer chemotherapy, and health risk assessment, particularly cancer and neurotoxicity risk for exposure to environmental pollutants.

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Introduction

The objective of this book is to explore a scientific basis for Ayurvedic medicine and create a better understanding of Ayurveda among health-care providers and consumers. Books published in the West on Ayurveda generally discuss only a few aspects of Ayurveda such as dietary and lifestyle changes, yoga, breathing exercises, meditation practices, and aroma, gem-stones, color, music therapy, herbs, and other therapies. In fact, these areas cover only a small fraction of the body of classical Ayurveda. These areas offer little or no information about current biochemical, pharmacological, and clinical investigations. During the past 100 years, several hundred Ayurvedic herbs have been investigated with respect to plant chemistry, active chemical constituents, pharmacological effects, safety, and efficacy. However, the basic research on Ayurvedic therapeutic agents has not been adequately integrated into disease management protocols available to most consumers in the West.

The origin of Ayurveda is traced back to four books of knowledge called *vedas*: *Rigveda*, *Samveda*, *Yjurveda*, and *Atharveda* (4500 to 1600 B.C.). The information on health care was subsequently developed by many Ayurvedic practitioners and finally compiled into three important books known in Ayurveda as the senior triad (*vriddha traya*): *Charak Samhita*, *Sushrut Samhita*, and *Ashtang Hridaya Samhita*. The subsequent three books that are commentaries on the senior triads are known as the junior triad (*laghu traya*): *Madhava-nidana*, *Sarangdhar Samhita*, and *Bhavaprakash Nighantu*. These books contain basic concepts of health and disease, disease management, anatomy and physiology, hygiene, materia medica, pharmacology and therapeutics, herbal formulations, pharmacy, and synthesis of herbo-mineral formulas. Diseases are classified according to organ systems and functions. Specialties such as internal medicine, surgery, pediatrics, gynecology, obstetrics, eye, ear, nose, and throat diseases, geriatrics, eugenics and aphrodisiacs, psychiatry, pharmacology, toxicology, and pharmacy are clearly delineated and discussed in detail in Ayurveda.¹

Although there is no record of pharmacological testing during the time period when Ayurvedic texts were written, 50 distinct pharmacological categories of medicinal plants were described. The categories include anti-inflammatory, analgesic, antiallergic, antihistaminic, antidiuretic, diuretic, antiemetic, emetic, purgative, astringent, antiasthmatic, antipyretic, and others.² Similar to conventional medicine, Ayurvedic medicine has also benefited from advances in science and technology. These advances facilitated the understanding of diseases, the development of better pharmaceutical products, and the implementation of diagnostic techniques. Scientific studies in laboratory animals have now confirmed the pharmacological properties of many Ayurvedic herbs. The old concept of accepting the last word of an Ayurvedic teacher without questioning the scientific underpinning has begun to disappear. A large number of medical schools and medical research institutions, both private and governmental, are currently involved in Ayurvedic research. The government of India has established an agency, the Central Council for Research in Ayurveda and Siddha (CCRAS) under the Ministry of Health and Family Welfare, to sponsor and conduct research in Ayurveda. The agency has been conducting and sponsoring Ayurvedic research all over India for more than 30 years in 70 regional Ayurvedic research institutions. The agency is conducting clinical trials to investigate the effectiveness of Ayurvedic products as well as basic research. Some of the clinical trials conducted by

the agency are discussed in [Chapter 3](#), “Clinical Research Design: Limited Systematic Review on Five Diagnosis Categories.” The basic research includes pharmacology, biochemistry, plant chemistry, product development, the cultivation of medicinal plants, and manufacturing practices for safe and effective Ayurvedic products.

Basic Concept

Whereas conventional medicine is primarily oriented toward the treatment of disease, Ayurvedic medicine is oriented toward prevention, health maintenance, and treatment. In conventional medicine, drugs are developed based on the concept that the elimination of specific causes of a disease, such as microorganisms, will cure a disease. On the other hand, the belief in Ayurvedic medicine is that a disease is the product of an imbalance in the body and mental elements that reduce the body’s resistance to diseases. If the imbalance is corrected and the body’s defense mechanisms are strengthened by herbal formulas, lifestyle changes, and diet, then the body will resist a disease with a goal of eliminating it. Herbal and herbomineral products regularly used in Ayurveda are believed to strengthen the body’s defenses. Scientific evidence is gradually developing in support of the Ayurvedic concept.³

Ayurvedic Physician

Scientific, technical, and engineering advances in the 20th century have been extraordinary. They created an explosion of new information and understanding in molecular biology, biochemistry, physiology, antibiotics, vaccines, genetics, mapping of genome, and the identification of genes responsible for inherited diseases. Ayurvedic physicians have also benefited from these new medical advances. Scientifically based information is the core of the preclinical, clinical, residency, and continuing medical education phases of training.

An Ayurvedic physician must be able to understand and appreciate new scientific knowledge in order to maximize the benefits of Ayurvedic therapies. In order to do so, an Ayurvedic physician must be aware of the current information on the disease processes, inter- and intracellular message transmission, membrane transport, protein interactions, chemical and microbial toxins, autoimmunity, cellular and humoral immunity, and dynamics of neurotransmitters. A physician needs to know about genetics, congenital inborn metabolic problems, pharmacological and biochemical research, and the use of various clinical chemistry data in diagnosis and monitoring disease processes in order to interpret the clinical data. He must understand the basic principles of toxicology and how the knowledge can be used in developing protocols for clinical studies. He must also know how to apply basic science to the Ayurvedic concept of disease management and the use of modern diagnostic methods involving urine and blood chemistry, enzyme assays, histology, pathology, radiology, etc. Finally, the Ayurvedic physician must understand the interaction of spiritual and psychological elements of patients as described in Ayurveda and be able to earn the patient’s trust through compassionate and holistic care.

Educational Standards

An Internet search revealed a dozen Ayurvedic colleges in the U.S. offering weekend courses ranging from 1 to 3 years. The curriculums are designed to familiarize students with Ayurvedic therapies, but are inadequate for actual clinical practice. Because Ayurveda is currently unregulated in the U.S., there are no standards to meet or pass any state examination. Because herbs or herbominerals are included under the U.S. Dietary Supplement Health and Education Act (U.S. DSHEA, 1994), Ayurvedic practitioners in the U.S. recommend herbal formulas as dietary supplements along with dietary changes, yoga, and breathing exercises to help remedy certain health problems.

Ayurveda has been recognized as an independent medical system by the government of India for a long time. The Central Council of Indian Medicine (CCIM), Ministry of Health (Regulations 1986, Minimum Education in Indian Medicine) regulates Ayurvedic education and training in India.⁴ The curriculum includes 2820 hours of theory and 780 hours of practical and laboratory work over a period of 54 months; this is followed by 1 year of residency leading to a Bachelor of Ayurvedic Medicine and Surgery,⁵ formerly Bachelor of Indian Medicine and Surgery. This is an integrated program that teaches Ayurvedic and allopathic courses so that therapeutic options for patients may be maximized.

Reasons for the Current Interest in Ayurveda

The great therapeutic success of synthetic antibiotics, hormones, and vaccines has created an expectation that conventional medicine will be able to discover a cure for every ailment. This expectation has been only minimally met for many diseases (e.g., cancer, arthritis, autoimmune diseases, and AIDS) even after spending hundreds of billions of dollars in research worldwide over the past 30 years. In addition, the synthetic antibiotics and steroids sometimes result in serious adverse effects, such as immunosuppression, gastrointestinal bleeding, and ulcers, after prolonged administration. Ayurvedic therapies generally provide relief without such adverse effects even after prolonged administration. Some formulas known as *rasayanas* are believed to improve the body's defense mechanisms. For example, in one short study for 90 days with *chyanprash*, a *rasayana*, the following improvements were observed: increased stress tolerance; improved endocrine functions (adrenal and testicular); positive nitrogen balance as indicated by increased serum protein level; and a decrease in urinary levels of nitrogen, creatinine, mucopolysaccharide, and hydroxyproline. The general well-being of the volunteers improved and none of them complained of any physical disorder.³

Ayurvedic herbs and formulas often have a wide spectrum of therapeutic activity. For example, *guggul* is recommended in Ayurveda for 25 or more ailments (e.g., inflammatory diseases, a variety of infections, muscle spasm, cough, bronchitis, anemia, endometritis, neurological diseases, skin diseases, urinary system disorders, obesity, osteoarthritis, and rheumatoid arthritis). The reason for this wide spectrum of activity is that *guggul* has anti-inflammatory, anticoagulant, hypolipidemic, and antibacterial activity; it can be beneficial in many health ailments associated with inflammation, infection, obesity, or blood clotting. Additionally, pharmacological activities of herbs may not be confined to one specific chemical constituent. *Guggul* was found to have anti-inflammatory activity in both polar

and nonpolar solvents, indicating that several chemical constituents present in it may have anti-inflammatory activity.⁵ Similarly, herbs showing neuropharmacological activity *in vitro* did not have the activity concentrated in any one solvent extract. This indicates that several constituents with different chemical physical properties may have the same neuropharmacological activity.⁶

Ayurvedic therapies are known to be relatively economic. For example, a 1-month dose of an Ayurvedic formula for arthritis may be obtained at this time for about \$20, which is often the co-payment for brand name prescription drugs for many health insurance and health maintenance organizations. Other alternative nondrug complementary therapies may be even more expensive.

The relative safety of Ayurvedic medicine is another reason for its popularity. Ayurvedic formulas are time tested for safety. These formulas contain vitamins; minerals; biologically active steroids, alkaloids, glycosides, and tannins; and a variety of antioxidants in a natural state. A single herb extract or a pure active chemical constituent may cause some adverse effects under certain conditions and dose levels. For example, *guggul* extract has been shown to produce some anticoagulant effect under certain conditions.^{7,8} Ayurvedic text formulas containing *guggul* may be safer than the *guggul* extract; the formulas that have *guggul* in relatively small amounts, along with many other herbs, act as synergists and possibly counteract some of the side effects.

Quality of Ayurvedic Formulas

In order to assure quality of Ayurvedic formulas, the government of India, Ministry of Health, amended the Drugs and Cosmetic Act of 1940 in 1964 to include Ayurvedic drugs (Ayurvedic herbs and herbal formulas). The Act requires the raw materials to be genuine and adequately identified; the formulas must contain ingredients listed on the label and manufacturing must be conducted under prescribed good manufacturing practices conditions. CCRAS developed a formulary of Ayurvedic text formulas called *The Ayurvedic Formulary of India* published in 1978. CCRAS conducted research to establish enforceable standards for Ayurvedic formulas similar to allopathic drugs and promulgated the standards. The book *Pharmacopoeial Standards of Ayurvedic Formulations* was first published in 1976 and subsequently revised in 1987. This effort was followed by the development of *The Ayurvedic Pharmacopoeia of India* in 1989. The pharmacopoeia contains the popular names; macroscopic and microscopic description of herbs; and limits for foreign matter, total ash, acid-insoluble ash, alcohol-soluble extractive, water-soluble extractive, and heavy metals in the same way as prescribed in allopathic pharmacopoeias. It also gives the levels of known chemical constituents, therapeutic uses, and doses.

Organization of the Book

Only a few diseases are selected based on the scientific data available on Ayurvedic therapies. Specialists in treating each disease searched the worldwide literature, critically reviewed the information on the basis of their expertise, and summarized the information in an understandable and easily usable form. Ayurvedic description and therapy of a disease are first described followed by a discussion using current knowledge. Available

pharmacological, biochemical, and chemical studies on herbs used in the management of the disease are evaluated to determine if there is an adequate scientific basis for their use. Finally, the overall management strategy of the disease, commonly used therapies, and scientific data are presented.

An attempt is made to first use the English translation or meaning of the Ayurvedic word, followed by the Ayurvedic word in parentheses as often as possible to make the chapters easy to read and understand. In addition, a list of Ayurvedic words is provided in [Appendix 1](#). Other lists provided are botanical and Ayurvedic names of herbs, names and addresses of the manufacturers of Ayurvedic formulas, and a list of journals publishing articles on Ayurvedic therapies. The words *Ayurvedic texts* or *Ayurveda* refer to the senior and junior triads in all chapters.

The book is organized to give a global picture of Ayurveda and to show that these therapies are relatively safe, effective, and have supporting scientific data. It is intended to provide better understanding of Ayurvedic medicine so that wherever appropriate, the integration of these therapies by the health-care providers may become possible.

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1

Ayurveda — A Potential Global Medical System

P.N.V. Kurup

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1.1 Introduction

It is a universally accepted fact that good health plays an important role in human development. According to the concepts of Ayurveda, good health is based on the equilibrium of *dosha* (humor), *agni* (digestive fire), *dhatu* (seven body tissues: lymph, blood, muscle, adipose tissue, bone, bone marrow, semen), and *mala* (feces, urine, and other waste products). Furthermore, in Ayurveda there is clear-cut emphasis on maintaining physical, mental, and spiritual well-being as part of good health.¹ The World Health Organization (WHO) defines good health as a state of complete physical, mental, and social well-being and not merely an absence of disease or infirmity,² which is in close proximity to the definition of good health mentioned in Ayurvedic classics. Creation of the infrastructure, generation of requisite human resources, and framing of appropriate policies required to meet the health-care needs of its citizens are accepted as some of the main and fundamental responsibilities of a modern state. Every country in the global fraternity aspires to meet the health-care needs of its people through an appropriate and cost-effective approach.

The contributions of Traditional Systems of Medicine (TSM) for global health care in the past and their importance for the health-care needs of the present and the future are well recognized. The traditional systems of India, which are now called Indian Systems of Medicine (ISM), have a very strong conceptual base and have been practiced uninterruptedly for a very long time; hence they are considered as independent medical systems. Ayurveda, Siddha, and Unani are the three important traditional systems practiced in India. Ayurveda is the oldest and the most widely practiced system among the three. It takes into consideration all aspects of health including mental, physical, and social components.

Considering the comprehensive manner in which all matters related to health are addressed in Ayurveda, it has potential to become a global medical system. However, to achieve this status, some of the shortfalls that are perceived to hinder its progress must be rectified. In the following pages the present status in different subsectors of Ayurveda are reviewed and some of the steps required for further development in the future are suggested.

The matters that require consideration are national policy for the development of Ayurveda in India; role of Ayurveda in the country's health-care delivery; regulatory mechanism to control and regulate manufacturing and utilization of the drugs manufactured in this sector; and facilities available for the generation of trained manpower, including clinical and paramedical personnel, research and development aspects, and globalization of the system.

1.2 National Policy on ISM in India

One of the most important requirements for any system to play a meaningful role in the health care of a country is for that system to receive due recognition from the government. In India, the highest policymaking body regarding the matters concerned with health and family welfare is the Central Council for Health and Family Welfare, set up under Article 226 of the constitution of India. It consists of central health ministers, state health ministers, eminent health experts, nongovernmental organizations with an interest in the health sector, and officials of the central and state governments. It strongly advocates systematic use of the ISM in the primary and secondary infrastructure. In its last meeting, the Council recommended that at least one physician from the ISM and Homeopathy (ISM & H) should be available in every primary health center. Further, it was also resolved to introduce special treatment centers for ISM & H in rural hospitals and to create a wing for these systems in the existing state and district level government hospitals.

The government of India has taken up the matter quite seriously and drafted a national policy on traditional systems practiced in India (www.indianmedicine.nic.in).³ The policy provides comprehensive coverage of different sectors. Feedback and suggestions have been collected from concerned organizations. This will form the basic material to frame the National Policy after completing the process of consulting different state governments and different ministries of government of India. Some highlights of the draft policy are as follows:

- It addresses the system in the areas of current relevance while at the same time delineating the immense opportunities that lay ahead.

- It plans to build upon the positive features of ISM & H, which are their modest cost, low level of technological input, and growing popularity. There is a possibility of expansion of activity in a wide range of related fields. To maximize utilization of this opportunity a range of strategies has been recommended.
- It seeks to revamp the curriculum of the educational institutes to reorient the approach of the practitioners of these systems to increase their relevance, credibility, and professionalism.
- It strongly advocates enforcement of good manufacturing practices (GMP) by placing acceptable levels of regulation and enforcement covering manufacture and certification of drugs.
- Several measures have been mentioned for the scientific and sustainable utilization of the medicinal plant-based resources of the country. It advocates adoption of a collaborative approach. Emphasis has been placed on utilizing the experience and scientific base available in the research councils of the country; for example, the Central Council for Scientific and Industrial Research (CSIR), the Indian Council for Agriculture Research (ICAR), and institutes under the Department of Science and Technology.
- To protect the intellectual property rights (IPR) of the resources of this sector, creation of an extensive database and a traditional knowledge digital library (TKDL) has been advocated.
- It also highlights the importance of utilization of local health traditions in the national health-care programs. It seeks to provide support of the operational research studies and efficacy trials for this purpose.
- The policy encourages propagation of Ayurveda and other ISM throughout the world, especially in areas where there is special interest in these systems, through Indian missions abroad. This policy will promote creation of a larger constituency for the practitioners of these systems. Promoting ISM as part of health tourism is being planned. Other approaches include international collaboration and academic exchange between interested groups.
- It has been strongly recommended that there should be an increase in the share of ISM in the national health budget.
- The policy also seeks to involve and promote the importance of ISM physicians in various national health programs.
- It seeks to establish and build the credibility of the ISM sector by encouraging certification and establishment of quality marking of products to allay the concern expressed in some quarters about the quality, safety, and efficacy of the products used.
- It recommends effecting policy changes to cover nutraceuticals and food supplements. The Drugs and Cosmetic Act would be amended to cover intermediates and partially processed plant-based products. The enactment of the ISM Product Information Promotion and Regulation Act is under consideration.
- There will be policy support and taxation incentives to promote high standards of manufacture.
- It seeks to support evidence-based research to determine the efficacy of ISM drugs and therapies, generation of data on safety and efficacy, along with standardization.
- There will be strong encouragement for undertaking research on basic principles of Ayurveda and the medico-historical approach.

TABLE 1.1

Summary of Medical Care, Medical Manpower, and Facilities Available under ISM as of 1 April 1999

Facilities	Ayurveda	Unani	Siddha	Yoga	Naturopathy	Homeopathy	Total
Hospitals	2,258	196	224	8	21	297	3,004
Beds	40,313	4,872	1,811	101	733	12,836	60,666
Dispensaries	14,416	970	363	42	56	7,155	23,028
Registered practitioners	367,528	41,221	12,915	—	388	189,361	611,413

Source: www.indianmedicine.nac.in; Annual Report 2000–2001.

There is a feeling that formulation of a national policy has been delayed inordinately; however, the fact that a comprehensive draft has been prepared and is being circulated among all the agencies involved in its implementation indicates that the matter has been taken seriously and there will be a discernible change in the Ayurvedic sector for the better in the future. The fact that such a detailed draft policy has been prepared and the government of India has initiated steps to enforce GMP in the ISM drug-manufacturing sector clearly indicates that it is earnest in its desire to develop and ensure mainstreaming of the ISM sector and its practitioners. The expenditure in this sector has gradually and significantly increased from Rs. 21683.60 million (US \$433.66 million) in 1996–1997 to Rs. 40792.60 million (US \$815.95 million) in 2000–2001.

1.3 Role of Ayurveda in India's Health Care

Ayurveda is quite popular, being practiced throughout the country including tribal and remote areas where other modes of therapies are not readily available. Though it does not have as an elaborate organized structure as its modern counterpart, it plays a major role in meeting the health-care needs of a large section of India. There are separate directorates of the ISM & H in 18 states. Though Ayurveda is popular in all these states, this system is more prevalent in the states of Kerala, Himachal Pradesh, Gujarat, Karnataka, Madhya Pradesh, Rajasthan, Uttar Pradesh, and Orissa. In the state of Tamilnadu the Siddha system is prevalent.

At present there are 611,413 practitioners of TSM, 26,032 hospitals/dispensaries, and approximately 8,500 Ayurvedic drug manufacturers in the country. There are more than 190 Ayurvedic colleges turning out more than 7,000 Ayurvedic graduates and around 700 postgraduates every year (Tables 1.1 and 1.2). The government of India is well aware of the resources and is taking steps to integrate ISM in primary health care and national health programs. This growing attitude change is demonstrated by the drafted national policy mentioned above.

1.4 Academic Role of Ayurveda in Future Health Care

The concepts of proper lifestyles, dietary habits, and daily and seasonal routines followed in Ayurveda can be adopted with suitable modification to different countries in different

TABLE 1.2
Number of Undergraduate and Postgraduate Colleges and Institutions In India (1 April 1999)

Facilities	Ayurveda	Unani	Siddha	Total
Undergraduate colleges	196	40	2	238
Admission capacity	7070	1280	150	8500
Postgraduate colleges	49	3	2	54
Admission capacity	645	35	70	750

Source: www.indianmedicine.nac.in; Annual Report 2000–2001.

parts of the globe after giving due consideration to the cultural milieu existing in each country and also the constitutional profile of each population. Attempts can also be made to utilize the medicinal plant resources of these countries for meeting the health-care needs of their people after categorization of the plants according to Ayurvedic concepts. Drugs used in ISM can be used as adjuvant to the main drugs used in conventional medicine. Therapeutic approaches such as *Panchakarma*, *Ksarasutra*, etc. can certainly be integrated into other health systems, broadening the choices available to physicians and patients.

1.5 Education and Training

At present, more than 190 undergraduate Ayurvedic colleges in India offer a curriculum for a Bachelor of Ayurvedic Medicine and Surgery (BAMS) degree. This program takes 5½ years to complete and runs according to the standards of the Central Council of Indian Medicine (CCIM), which is a statutory body that regulates the ISM education in the country. During the 5½ years of education, the student must go through internship for a period of 1 year. The 10 + 2 (10 years of school education followed by 2 years of predegree study) students with Science Group are eligible to take admissions in the degree course. This is similar to the requirement for the admission to the MBBS (allopathy) degree program. In fact, in many states there is a common entrance test to admit candidates to these courses. BAMS contains many modern subjects in its course material; however, nothing is taught in MBBS colleges about Ayurveda or any other ISM. This is a paradox, as many modern medicine graduates prescribe Ayurvedic drugs — especially in difficult-to-cure diseases like hepatitis — without any training in Ayurveda. The Department of ISM, being perceptive of this situation, has prepared course material containing basic concepts and fundamentals of ISM & H for incorporation in MBBS curricula. This has been forwarded to the Medical Council of India for appropriate action. In most states, e.g., Uttar Pradesh, Rajasthan, and Gujarat, Ayurvedic colleges are state supported.

Because it is necessary that Ayurvedic graduates understand modern advances in medical diagnostic methods, medical technology, and drug treatment, the present curriculum contains about 50% conventional medicine, and clinical and preclinical subjects. The CCIM, which is the apex body in matters related to the education and practice of Ayurveda in the country, is initiating steps to revise the curriculum to suit the present-day requirement by placing emphasis on practical-oriented teaching. Furthermore, at present there is no facility to impart training in some of the important disciplines, like *Vriksha* Ayurveda (a subdivision of Ayurveda that deals with matters related to cultivation of plants) and *Pasu* Ayurveda (Ayurvedic veterinary science), at the undergraduate level. Steps have to be initiated to include them as subjects for study at the undergraduate level.

1.5.1 Postgraduate Education

Postgraduate education is available in over 30 research institutes and offers specialization in 16 clinical and preclinical Ayurvedic specialty areas such as medicine, surgery, pediatrics, pharmacology, pathology, pharmacy, and *Rasa Vigyan*. Besides the mainstream institutes, the National Academy of Ayurveda, run by the government of India, was established to impart intensive training in different specialties for graduates and postgraduates of Ayurveda under the guidance of eminent scholars (similar to the traditional custom of *Guru Shishya* Parampara; interested readers can obtain more information from the Web site of the Department of ISM as well as a CD released by the department³⁻⁵).

At the present time, Gujarat Ayurvedic University is the only university exclusively devoted to Ayurveda and allied sciences in India. Its constituent institutes include the (1) Institute for Postgraduate Training and Research in Ayurveda financed by the government of India, (2) Shri Gulabkunverba Mahavidyalaya funded by the Gujarat State, (3) Institute of Ayurvedic Pharmaceutical Sciences, (4) Institute of Ayurvedic Medicinal Plant Sciences, (5) International Center for Ayurvedic Studies, and (6) Mahrishi Pananjali Institute for Yoga and Naturopathy Education and Research. The last three institutes are self-financed. The Gujarat Ayurvedic University has signed the Memorandum of Understanding (MOU) with nine Ayurvedic institutions functioning in Japan, Australia, the Netherlands, Italy, Argentina, and Germany to coordinate and facilitate the globalization of Ayurveda through academic collaboration. Earlier, Medical (Ayu) Institute of Russia had signed the MOU with the government of India, in which the Gujarat Ayurvedic University is also one of the implementing authorities.

The Ayurveda Faculty of the Institute of Medical Sciences, Banaras Hindu University, Varanasi, Uttar Pradesh, which is under the Central Government administration, also has similar programs. It has excellent facilities for imparting postgraduate training in different disciplines of Ayurveda. It is collaborating in international research programs and attracting graduates for training and research in Ayurveda, conducting basic research and publishing in international journals. It has an advantage of sharing the research facilities and clinical research with the medical institute's research program. National Institute of Ayurveda, Jaipur, Rajasthan also has an excellent research facility and similar training programs and collaborations. The other postgraduate research centers also have very good facilities and are involved in various important research collaborations.

1.5.2 Para Ayurvedic Staff

At present there is only one institute attached to Gujarat Ayurved University that conducts courses in Ayurvedic pharmacy. There is an urgent need to establish many such institutes to generate requisite manpower for the Ayurvedic drug manufacturing and dispensing sector. This can be facilitated by formulating a national policy making it compulsory after an initial grace period to appoint only qualified persons in the Ayurvedic manufacturing, dispensing, and drug regulatory sectors. There is also a dearth of institutions to impart training in Ayurvedic nursing. There is a need to open many such institutes and the present practice of recruiting nurses trained in modern medicine has to be phased out. If such persons are recruited, they should be made to undergo orientation training in Ayurveda. Many special therapeutic approaches like Panchakarma, Ksarasutra, and Marmavidhya are of great importance in Ayurveda.

1.6 Research and Development

There are research councils and institutes functioning throughout the country on different aspects of Ayurveda. In 1971, the government of India established a research council, the Central Council for Research in Indian Medicine, Homoeopathy & Yoga (CCRIMH), which was subsequently developed into four independent councils in 1978. The Central Council for Research in Ayurveda and Siddha (CCRAS) is an apex body for the formation, coordination, development, and promotion of research on scientific lines in Ayurveda and the Siddha System of Medicine. The council has 89 field units under it, and they have been reorganized into 30 institutes and units including the Headquarters Office (List 2 in this chapter). The research activities in various fields can be broadly categorized as follows:

- *Clinical research*, encompassing clinical studies and programs in survey and surveillance, community health, and tribal health.
- *Drug research*, encompassing medico-botanical surveys, cultivation of medicinal plants, pharmacognostical studies, and phytochemical profiling of plants used in Ayurveda, plant tissue culture, pharmacological and toxicological studies, and drug standardization. It also has a breeding program for musk deer, which is the source of the well-known drug *Kasturi*.
- *Literary research*, encompassing publication of rare and classical manuscripts of Ayurveda and Siddha, monographs on the basis of the studies undertaken by the council, scientific journals and bulletins, newsletters featuring activities of the council, and pamphlets on research findings; and preparation of video films on various research achievements.
- *Family welfare research*, including studies on family welfare such as antifertility.

Complete details of the activities and research of CCRAS can be found on its Web site: www.ccras.com.

Besides CCRAS, research activities are carried out in other postgraduate institutes as part of an M.D. dissertation and Ph.D. thesis. Research studies on Ayurvedic drugs and therapies are carried out throughout the country in many conventional medical colleges and research institutes, giving MBBS degrees — though not from an Ayurvedic perspective but as part of conventional drug research. However, some institutes have done important research work on Ayurvedic herbs. The major institutes conducting research on medicinal plants are Seth GS Medical College (Mumbai), Central Drug Research Institute (CDRI, Lucknow), Regional Research Laboratory (Jammu), National Institute for Pharmaceutical Education and Research (NIPER, Mohali), Tropical Botanical Garden and Research Institute (TBGRI, Trivandrum), Central Institute of Medicinal and Aromatic Plants (CIMAP, Lucknow), National Botanical Research Institute (NBRI, Lucknow), pharmacology departments attached to the Institute of Medical Sciences, Banaras Hindu University (Varanasi), KG Medical College (Lucknow), SPARC (Mumbai), University Department of Pharmaceutical Sciences, Punjab University, etc. In-house research activity is undertaken by some of the large Ayurvedic drug manufacturers like Himalaya Drug Company, Dabur Research Center, Zandu Research Foundation, etc. (Lists 1 and 2 in this chapter).

The present research approach, especially in the area of drug development and therapeutics, needs to consider single plant as well as Ayurvedic concepts and text formulas. Research protocols are often prepared without giving due consideration to the Ayurvedic conceptual base that underlies employing a drug in a particular disease or clinical condi-

tion. The tendency is to treat medicinal plants used in Ayurvedic therapeutics as a source material for drug prospecting for a single herb or chemical constituent and not for a drug formulation. Ayurveda has a very well-developed discipline of Ayurvedic pharmacy and drug formulation called *Bhaisajya Kalpana*, which deals in great detail with different methods of drug preparations, use of adjuvant, maintaining ideal conditions, collecting and processing drugs in a particular season at a particular stage and site, and others. There is urgent need to study the impact of changes made in drug formulation and manufacturing processes on the expression of biological activity and therapeutic efficacy.

Existence of the “therapeutic gap” in modern medicine is well known.⁶ Though tremendous progress has been made in the treatment of many dreaded diseases, remedies are yet to be found for treating diseases like tuberculosis, cancer, rheumatoid arthritis, AIDS, etc. In these areas TSM drugs and procedures may have beneficial effects. Similarly, TSM drugs could be beneficial in the treatment of iatrogenic disorders like parkinsonism and to attenuate drug-induced toxicity when administered as adjuvant. Intensive research efforts are required to explore these possibilities.

1.7 Medicinal Plant Resources

The drugs used in the ISM are mainly plant based, in addition to a few materials of mineral and animal origin. Thus the therapeutic efficacy of the drugs used in these systems is greatly dependent on the use of pure and genuine raw materials in their preparation. According to the estimation of the Department of ISM,³ about 1100 medicinal plants are estimated to find use in ISM & H drug preparation, and 500 of these are more commonly used.

There are many problems in ensuring a constant supply of drugs; avoiding overexploitation of medicinal plant resources; and also ensuring the quality of raw drugs and the conservation, cultivation, and preservation of medicinal plants. To look into these matters, the government of India has set up a Medicinal Plant Board, which has been given the responsibility of coordinating all matters related to medicinal plants. The Planning Commission set up a task force under Dr. D.N. Tewari, to *inter alia* provide policy directives on (1) conservation and sustainable use of medicinal plants, (2) growth of domestic and foreign trade, (3) development of an equitable market system, (4) regulation of this sector to maintain quality control, and (5) protection of IPR of medicinal plants.

1.8 Ayurvedic Herbs and Herbal Formula Manufacturing

Ayurvedic medicines are marketed in various forms. They are available in the classical forms like *gutikas* (tablets), *churnas* (powder), *asavas* and *aristas* (fermented products), *ghritas* (medicated ghee), and *kashayams* (decoctions). For topical use, drops, creams, lotions, liniments, and ointments are available. Dried plant extracts in capsule form are also in use. In addition, many patent drugs are sold in other modern drug presentation forms like syrups, granules, creams, lotions, etc., which constitute around 65% of the market share. There are more than 8500 manufacturers of Ayurvedic drugs in the country, of which the annual return of 10 firms is more than Rs. 50 crores (approximately US \$5

million — \$1.00 is roughly equal to Rs. 48 crores), of 25 firms is between Rs. 5 and 50 crores, of 965 firms is between Rs. 1 and 5 crores, and the remaining is very small with an annual turnover of less than Rs. 1 crore. The market share of Ayurvedic drugs is around Rs. 3500 crores and that of Siddha drugs is around Rs. 5 crores.³

Ayurveda is covered by the Drugs and Cosmetic Act (1940) and Rules (1945) of the country, and the manufacturer is expected to comply with the rules delineated under this act. In 1960, Chapter IV and other related chapters were added. Chapter IV deals with the manufacture, sale, and distribution of drugs and cosmetics; Chapter IV-A, which was subsequently added, deals with provisions related to Ayurvedic, Siddha, and Unani drugs. In this chapter, 19 sections dealing with different aspects have been described. All the drug manufacturing and dispensing activity in the country is covered under this act. According to Clause 3(a) of this act, the definition of Ayurvedic drugs includes all medicines intended for internal and external use for or in the diagnosis, treatment, mitigation, or prevention of disease or disorder in human beings or animals and manufactured exclusively in accordance with the formulae prescribed in the authoritative texts of Ayurveda (which have been specified in the First Schedule of the act). Clause 3(h) describes patent and proprietary medicines. This refers to formulations prepared utilizing the drugs listed in the formulae mentioned in the authoritative texts and mentioned under Schedule I but excludes drugs administered by parenteral route and formulations mentioned in the books in the First Schedule.⁷ Schedule E₁ of the act contains a list of poisonous substances under the Ayurvedic and Unani system. In addition to the above, other acts that are relevant to the manufacturing of Ayurvedic drugs are the (1) Medicine and Toiletary Preparation Act, (2) Magic Remedies and Objectionable Advertisement Act, (3) Poisons Act, (4) Weights and Measurement Act, (5) Shops and Establishment Act, (6) Dangerous Drugs (Psychotropic and Narcotic Substances) Act, (7) Patents Act, and, in the future, the (8) Biodiversity Act (which is in the offing).

Three types of agencies are involved in the administration of the acts and rules mentioned above: the Drug Technical Advisory Board and Drug Consultative Committee, which act as advisory bodies;⁸ drug testing laboratories, which provide support by shouldering the analytical responsibilities; and licensing and controlling authorities, which function as the executive wing. The Technical Advisory Board and the Drug Consultative Committee have been set up to advise the government on matters relating to ISM drugs. The Drug Technical Advisory Board (Ayurveda, Siddha, and Unani Technical Advisory Board) has been functioning regularly. The Drug Controller General of India is in charge of licensing and enforcing the above acts and rules. At the state level are drug controllers or Food and Drug Administration Commissioners who shoulder this responsibility. The drugs manufactured in the Ayurvedic sector are also under their controlling authority. In some states, like Kerala, there are officers trained in Ayurveda who provide technical assistance to the enforcing authorities.

Manufacturing of Ayurvedic drugs, except in accordance with the prescribed standards, is prohibited. It is essential to obtain a license from the licensing authority to manufacture Ayurvedic drugs. Separate licensing is required for each of the manufacturing premises maintained by the manufacturer. Manufacturing activity can also be undertaken through a loan license. For drug manufacturing, it is necessary to maintain a certain level of hygiene and optimum manufacturing conditions. These conditions are specified in Schedule T of the act (mentioned under the schedule to the rules). On 23rd June 2000 an amendment was made in the above act to specify GMP for Ayurveda, Siddha, and Unani drugs. Notice was made to ensure that (1) raw materials used in the manufacture of drugs are authentic, of prescribed quality, and are free from contamination; (2) drugs are manufactured according to standard conditions; (3) adequate quality control measures are adopted; and (4) the manufactured preparations released to the marketplace are of acceptable quality.

The ISM drug-manufacturing industry is in the process of technical upgrading to comply with GMP norms prescribed by the government of India. To achieve the objectives listed above, each licensee is expected to evolve methodology and procedures for following the prescribed process of manufacture of drugs, which should be documented as a manual and kept for reference and inspection. However, teaching institutions and registered, qualified *Vaidyas*, *Siddhas*, and *Hakeems* who prepare medicines on their own to dispense to their patients and not to sell such drugs in the market are exempt from the purview of this practice. Part I of two parts contains specification regarding the maintenance of factory premises; Part II contains the list of machinery, equipment, and minimum manufacturing premises required.

For the implementation of drug testing provisions under the Drugs and Cosmetics Act (1940) and Rules (1945), it is necessary to evolve pharmacopoeial standards. At present, pharmacopoeial standards are available for 258 drugs, and 654 formulations have been published in the *Ayurvedic Formulary of India*. The Department of ISM is developing pharmacopoeial standards through pharmacopoeial committees. Three volumes of Part I of the pharmacopoeia, called *Ayurvedic Pharmacopoeia*, have been published containing 258 monographs. The target is to cover 600 single drugs.

At present, there are more than 8000 licensed pharmacies in the country manufacturing Ayurvedic drugs. There is the Mumbai-based Ayurvedic Drug Manufacturer's Association which can be contacted for further details about issues related to Ayurvedic drug manufacturing.

A drug-control cell in the Department of ISM deals with various issues pertaining to quality control, import, export, classification of drugs under the Drugs and Cosmetics Act, patents, and the establishment of a TKDL. It monitors and coordinates implementation of legislation relating to drugs of ISM & H. The agency may be a good source for information on matters related to regulations.

1.9 Globalization of Ayurveda

Globalization of Ayurveda has gained momentum. Many active groups have been formed in many parts of the world, including developed countries, to spread the concept and practice of Ayurveda. This is due primarily to the following three reasons: (1) the holistic approach advocated by Ayurveda in therapeutic practice, (2) it has one of the most extensive and profound conceptual bases among the TMSs of the world, and (3) its survival for more than 2 millennium as a vibrant medical system. It is believed that Ayurveda has the potential to develop into a global health-care system.

The first requirement is to undertake globalization of Ayurvedic education to generate high-quality, competent manpower with the requisite communication skills to teach the principles and practice of the system. There is a requirement to start introductory, short and long-term courses as per the local requirement and situation in different parts of the world. It is also necessary to start similar types of courses in the premier Ayurvedic institutes in India. Some universities have already taken a step in this direction by starting short- and long-term courses under its International Center for Ayurvedic Studies. Another important requirement is to translate important Ayurvedic literature to major international languages.

The second requirement is to globalize Ayurvedic practice and marketing of Ayurvedic drugs. There are many obstacles to achieve this, especially in the developed countries. The laws regulating these aspects are quite rigid and a lot of time, effort, and finance will

be required to comply with them. They do not take into consideration the conceptual uniqueness of Ayurveda. A two-pronged strategy is required to overcome this problem. The first one is to undertake multicenter collaborative studies on internationally acceptable guidelines to prove therapeutic utility and safety of Ayurvedic drugs and practices. The second one is to establish Ayurvedic clinics and hospitals in countries where there is no such barrier. Standardization of Ayurvedic drugs and formulations should be given top priority — without this it would not be possible to promote the utilization of Ayurvedic drugs at the global level. Another aspect needs to be taken into consideration is the possibility of existence of constitutional differences among different races and communities as per the concepts of Ayurveda. Also, it would be necessary to explore the possibility of utilization and integration of locally available flora into Ayurvedic practice of the particular country or region.

Facilities have been established in many countries to impart short- and long-term training in Ayurveda. Such facilities are available in the U.S., Argentina, Australia, Brazil, New Zealand, South Africa, Czech Republic, Greece, Italy, Hungary, the Netherlands, Russia, U.K., Israel, Japan, Nepal, and Sri Lanka.

1.10 Research at the International Level

At present, there are no international collaborative studies to validate the therapeutic claims of Ayurvedic preparations except for some studies carried out on medicinal plants used in Ayurveda. A recent review⁹ points out that more than 13,000 plants have been investigated during the past 5 years. A number of medicinal plants have been shown to possess important pharmacological activities in preclinical testing; however, the generated leads have not been adequately followed up with double-blind, placebo-controlled clinical trials. The following drugs are identified for such studies based on existing biological and clinical data: *Curcuma longa*, *Boswellia serrata*, *Picrorhiza kurroa*, *Terminalia chebula*, *Emblica officinalis*, *Bacopa monnieri*, *Boerhaavia diffusa*, *Phyllanthus niruri*, *Celastrus paniculatus*, *Ocimum sanctum*, *Gymnema sylvestre*, *Momordica charantia*, *Commiphora mukul*, *Withania somnifera*, *Pterocarpus marsupium*, *Tinospora cordifolia*, *Trichopus zeylanicum*, and *Terminalia arjuna*. One of the main lacunae that becomes apparent for any reviewer of this sector is that most of the studies are undertaken on an individual pharmacologic-effect basis. These studies need to be organized and reviewed. Such a review of these studies is likely to provide scientific basis to the traditional usage of Ayurvedic therapies. It is also of utmost importance that sufficient attention be paid to the conceptual basis that underlies selection of a drug or group of drugs to treat a particular disease condition. Furthermore, many more studies are required to be undertaken on multicomponent formulations and assessment of possible drug interactions when a person who is already on modern drugs takes herbal preparations. If the recent trends are any indications, interest in medicinal plant-based preparations and traditional medical systems is bound to increase.

List 1: Postgraduate Study Centers

1. Central Drug Research Institute, Chattar Manzil Palace, Lucknow 226 001

2. Dabur Research Foundation, 22, Site IV, Sahibabad, Ghaziabad 201 010
3. Himalaya Drug Company, Research and Development Centre, Makali, Bangalore 563 123
4. Institute of Medical Sciences, Faculty of Ayurveda, Banaras Hindu University, Varanasi 226 004
5. Institute of Postgraduate Teaching and Research in Ayurveda, Gujarat Ayurved University, Post Box no. 4, Jamnagar 361 008
6. National Institute of Ayurveda, Madhav Vilas Palace, Ajmer Road, Jaipur 302 002
7. National Institute of Pharmaceutical Education and Research (NIPER), Sector 67, SAS Nagar, Mohali, Punjab
8. National Botanical Research Institute (NBRI, Lucknow), Rana Pratap Marg Lucknow 226 001
9. R.A. Podar Medical College, Dr. Annie, Bessant Road, Worli, Mumbai 400 018
10. Regional Research Laboratory, Canal Road, Jammu, Jammu and Kashmir 181 001
11. State Ayurvedic College, Tulsidas Marg, Turia Ganj, Lucknow 226 004
12. Swami Prakashananda Ayurveda Research Centre (SPARC), 13th N A S Road, Mittal Nagar, JBPD-Scheme, Juhu, Mumbai 400 049
13. Tropical Botanical Garden and Research Institute (TBGRI), Pacha, Palode Trivandrum 695 562
14. Zandu Research Foundation attached to Zandu Pharmaceutical Works Ltd., 70, Gokhale Road South (Dadar), Mumbai 400 025

List 2: Regional Research Institutes (RRIs)

RRIs under Central Council for Research in Ayurveda and Siddha (CCRAS), 61-65, Institutional Area, Opp. D-Block, New Delhi 110058, Ph: 011-5528748/5536520/5624457 (O), 5614971 (R).

Addresses

1. Road No. 66, Punjabi Bagh, New Delhi 110026, Ph: 011-5919128/5411059, Fax: 5464546
2. R.A. Podar College, Worli, Mumbai 400 018, Ph: 022-4947822, Fax: 4947833 (O) 0250-462991 (R)
3. 4, CN Block, Sector V, Bidhan Nagar, Calcutta 700 091, Ph/Fax: 033-3673808
4. RCRI, AA Govt. Hospital Campus, Arumbakkam, Chennai 600 106, Ph: 044-6214809(O), 6214925 (G), 6265857 (R)
5. Cheruthuruthy, Via. Shoranur, Trichur, Kerala 679 531, Ph/Fax: 0488-462366, 462543 (O), 462544 (H), 622175 (R)
6. No. 1, Bhubaneswar 751 009, Ph/Fax: 0674-530125, 531941 (H), 570650/570705 (R)
7. Moti Bagh Road, Patiala 147 001, Ph/Fax: 0175-212393, 228361(H), 212348 (R)

8. Govt. Ayurvedic Hospital, Amhho, Gwalior 474 009, Ph/Fax: 0751-323307, 430317 (Fax), 327959 (R).
9. 474/6, Sitapur Road, Lucknow 206 020, Ph/Fax: 0522-369156, 362341/732238(R)
10. Indira Colony, Jhotwara, Bani Park, Jaipur 320 016, Ph/Fax: 0141-200812/206063, 392174/399646 (R)
11. 1044 Jagnada Chowk, KDK College Rd., Nandanwan, Nagpur 440 009, Ph/Fax: 0712-714230, 242310 (R)
12. Govt. Central Pharmacy Annexe, Ashoka Pillar, Jayanagar, Bangalore 560 011, Ph/Fax: 080-6562030, 2272208 (R)
13. Poojapura, Trivandrum 695 012, Ph/Fax: 0471-340628, 342070 (R)
14. Indira Gandhi Municipal Stadium Complex A, First Floor, North Wing, Vijayawada 520 010, Ph/Fax: 0866-472535, 481512 (R)
15. Govt. Pharmacy (Upstairs), Indira Nagar, Gorimedu, Pondicherry 605 5006, Ph: 0413-272420 (O), 202097 (R)
16. D Block, Rajindra Memorial Res. Institute Bldg., Agam Kuan, Patna – 800 007, Ph/fax: 0612-631678, 345775 (R).
17. Borsojai (Bhetapara), Beltola, Guwahati 781 028, Assam, Ph: 0361-303714
18. Tadung, Gangtok, Sikkim 737 102, Ph/Fax: 03592-31494, 81649/81662 (R)
19. New Itanagar 791 111 (Arunachal Pradesh), Ph/Fax: 0361-212284, 211498 (O), 212520 (R)
20. 20, Rewari Chowk, Jammu 180 005, Ph/Fax: 0191-546475
21. Gandhi Bhawan, Mandi 175 001 (HP), Ph/Fax: 01905-35236
22. Tarikhet 263 663 (Uttaranchal), Ph/Fax: 05966-64227, 64222(R)
23. Gwalior Road, Jhansi 284 003 (UP), Ph/Fax: 05174-442132
24. Tajmanzil, Sardar Bagh, Junagadh 362 001, Ph/Fax: 0285-631631
25. Nehru Garden, Kothrud, Pune 411 029, Ph: 020-5383138, Fax: 5386715, 5442338 (R)
26. Indian Institute of History of Medicine, Osmania Medical College Buildings, Putlibowli, Hyderabad 500 095, Ph/Fax: 040-4657388, 3511259 (R)
27. CSMDRIA, A.A. Govt. Hospital Campus, Arumbakkam, Chennai 600 106, Ph: 044-6214823/6207566 (O), 6282487 (R)
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2

Health Care and Disease Management

Lakshmi Chandra Mishra

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2.1 Introduction

In Ayurveda, health is defined as the state where physical body, senses, and psyche are in original or natural state with respect to body and function. Although the genetic makeup of an individual determines the basic body constitution with respect to *dosas* (biomaterials) and psychological factors, total health is determined by physical and psychological environment. It is currently believed that the expression of genes largely depends on environmental factors. Consistent with this belief, Ayurveda emphasizes the role of environmental factors, daily routine, seasonal changes, lifestyle, diet, regular exercise, and body tonics (*rasayana*) in maintaining health. It also emphasizes that all needs of the body and senses must be in balance in order to avoid illness and maintain good health — a scientifically valid concept.¹

The founding fathers of Ayurveda believed that the world is made up of five elements called *Panch-mahabhuta*. Their criterion was that an element must not be “divisible into” a new material. They determined that earth, water, fire, air, and space are indivisible and therefore designated them as the five basic elements (*Panchmahabhuta*). The human body in Ayurveda was also believed to be made up of these five elements. In order to understand the whole body physiology, Ayurveda postulates three main factors called *tridosas*, which are also called biomaterials, biofactors, or bioenergies. These *tridosas* or biomaterials are *vata*, *pitta*, and *kapha*. The primary dominant elements in *vata*, *pitta*, and *kapha* are air, fire, and earth, respectively. According to the *tridosas* theory the total human body consists of an intensive interplay of a solid material substrate referred to as *kapha*, chemical activity (biofire) referred to as *pitta*, and an energy pool of motion and movement referred to as *vata*. These three *dosas* coexist in a predetermined proportion and function in a complementary manner to overall function of the total organism in spite of their opposite properties and functions. The existence of the *dosas* can be understood at both the macromolecular and micromolecular levels. A balance in the activity of these *dosas* is necessary for health.

The human body according to Ayurveda is made up of somatic *dosas* (*vata*, *pitta*, and *kapha*) and psychic components (*dosas*), body tissues (*dhatus*), and waste products (*malas*). The three psychological components are *satogun*, *rajogun*, and *tamogun*. A close interdependence among the somatic and psychological components exists; if one component is out of balance, the others are also out of balance. The imbalance or vitiation of *vata*, *pitta*, or *kapha* is considered the major factor in the causation of a disease.¹

Ayurveda has a concept of *agni* (fire) for all digestive and metabolic activities. Use of the word “fire” can be rationalized based on the fact that fire generates carbon dioxide as a result of burning carbon-based materials, and metabolism of carbon-based nutrients in the body also generates carbon dioxide. Digestion and absorption processes are called digestive fire (*pakwagni*) and the enzymic action causing transformation of nutrients into various tissue materials is called tissue fire (*dhatwagni*). Food must not only be digested and absorbed from the intestine and circulate in the blood plasma, but also must be absorbed by the tissue cells in order to be assimilated by the body. There are seven primary supportive tissues (*dhatus*) in the body: (1) plasma, (2) blood, (3) muscular tissue, (4) osseous tissue, (5) adipose tissue, (6) marrow and myeloid tissue, and (7) reproductive tissue. All tissues are made up of cells with different structures and functions connected with other cells through minute channels or pores (*suksma srotas*) to receive and distribute nutrients and excrete metabolic waste products. This description of cellular communication in Ayurveda is consistent with the modern understanding of intercellular uptake and release of substrates and metabolites. All tissues have a material called activator, essence, or hormone (*ojas*) which has specific properties and functions that are perceived to produce a healthy essence. The properties and balance of excretory products (stool and urine) and breakdown products of tissues (nitrogenous products) are also important factors in determining health and disease.¹

2.2 *Tridosas*¹

2.2.1 *Vata*

Vata literally means “air.” The description of *vata* in Ayurveda includes the following: dry, cold, light, penetrating, mobile, transparent, and rough-like air and responsible for all

TABLE 2.1
Classification of *Vata*

<i>Vata</i>	Location	Function
<i>Prana</i>	Central nervous system and some vital centers of the brain	Controls the action of other <i>vatās</i>
<i>Apana</i>	Lumbo-sacral, gluteal, genito-urinary, inguinal, and visceral regions	Regulates discharge of seminal fluid, menstrual fluid, feces, urine, and aids in childbirth
<i>Vyana</i>	The heart (the fluid pumping system), nonconscious reflexes produced along the spinal cord (corresponds to autonomic nervous system)	Controls contraction and relaxation of muscles, blood circulation, lymph circulation, sweating, and other secretions of the body through the stimulation of nerves
<i>Udana</i>	The chest, umbilical area, nasal and pharyngeal passages	Controls speech, respiration, energy, body luster, and consciousness and also influences the respective centers in the brain
<i>Samana</i>	The entire abdomen	Controls digestion and assimilation of food, as well as excretion of waste products

Signs and symptoms of deranged vata are indicative of neurological problems

Causes of Deranged <i>Vata</i>	Symptoms of Deranged <i>Vata</i>
Overworking, grief, worry, lack of sleep, intentional retention of bodily waste, excess exercise or sexual indulgence, and ingestion of bitter, pungent, astringent, or dry foods	Irregular movement of the limbs, tearing pain, throbbing suppression of secretions, contraction of muscles (spasms), roughness of the skin, formation of bone cavities, suspension of bowel movements, and an astringent taste in the mouth

kinds of movements in the body, speed, impulse, and odor. *Vata* is a part of every cell; however, the major sites of *vata* are believed to be the colon, lower back, calf muscle, ear, bones, and joints. Some organs of the body also have a higher level of *vata* biofactor than other organs, such as the brain, spinal cord, autonomic nervous systems, and motor and sensory systems. *Vata* biofactor is responsible for functions of the central, autonomic, and peripheral nervous systems. *Vata* controls the respiratory, blood, lymphatic, excretory, and reproductive systems, as well as all types of movements. It is also responsible for the cognitive and neocognitive function of the brain, and secretion of various chemical neurotransmitters and hormones. This description of *vata* resembles the functions of central, peripheral, and autonomic nervous systems. The following six main *chakras* (extrasensory channels) of *vata* correspond to modern autonomic nervous system plexuses, further substantiating that *vata*, in fact, resembles the nervous system:

1. *Agnya chakra* – nasociliary extension of the cavernous plexus of the sympathetic nervous system
2. *Vishuddha chakra* – pharyngeal plexus of the sympathetic nervous system
3. *Anahat chakra* – cardiac plexus of the sympathetic nervous system
4. *Manipur chakra* – plexus of the cholinergic
5. *Swadhistan chakra* – hypogastric plexus
6. *Muladhar chakra* – pelvic plexus

Among *tridosas*, *vata* is more critical because it controls *pitta* and *kapha* in order to accomplish various physiological functions. *Vata* is classified into five types based on the organ site and function as shown in Table 2.1.

TABLE 2.2
Classification of *Pitta*

<i>Pitta</i>	Location	Function
<i>Pachak</i>	Stomach and small intestine	Controls bile and pancreatic juices
<i>Ranjaka</i>	Stomach, duodenum, liver, spleen, and pancreas	Enzymes responsible for digestion and metabolism; provide nutrients, blood constituents, and the formation of the blood cells in the spleen
<i>Sadhak</i>	Heart	Responsible for life span, complexion, energy, body temperature, body luster
<i>Bhrajak</i>	Eye	Responsible for vision
<i>Alochak</i>	Skin	Responsible for the heat, moisture, and health of the skin, as well as the vasomotor mechanism and adaptation to the surrounding environment

Signs and symptoms of deranged pitta are indicative of digestive, metabolic, and enzymatic problems

Causes of Deranged <i>Pitta</i>	Symptoms of Deranged <i>Pitta</i>
<i>Pitta</i> may be deranged by ingestion of salty, hot, irritating, or sharp-tasting foods, as well as by anger	Burning sensation in the stomach, redness, digestive system disturbances, excessive sweating, fainting, symptoms of intoxication, pungent and sour taste, and inability to see white and dark red color

2.2.2 *Pitta*

The second *dosa* is *pitta*, which indicates *agni* (fire, heat energy), and the description in Ayurvedic texts includes heat, sharpness, sourness, and moisture. It is responsible for appetite, thirst, digestion, metabolism, body heat, normal eyesight, softness of the body, luster, mental calmness, and intelligence. This description of *pitta* resembles that of digestion, metabolism, oxidation, conjugation, reduction, phosphorylation, enzymes, and hormones. Although *pitta* is present in every cell in the body, it is classified based on five major anatomical locations shown in Table 2.2. Although *pitta*, like *vata*, exists in every cell of the body, the major sites of *pitta* are believed to be the stomach, duodenum, liver, spleen, pancreas, heart, eyes, and skin.

2.2.3 *Kapha*

The third *dosa* is *kapha*, which is described in Ayurveda as moist, steady, cool, heavy, soft, and slimy materials. *Kapha* is responsible for the normal body moisture, stability of the joints, firmness of the body, a proportionate bulk, weight, strength, endurance, and courage. The description of *kapha* resembles those of the lymphatic system, immune system, body fat, and mucous and mucoid systems. *Kapha* balances *pitta* in functions by providing basic materials for conversion into body tissues by enzymes, a kind of biochemical feedback mechanism. *Pitta* generates heat from the enzymatic activities, while *kapha* provides ways to eliminate the heat through the skin, lungs, urine, and feces via fat and moisture.

Although *kapha* is basically the same in all cells of the body, it is classified into five types based on location and functions as shown in Table 2.3. Again, like *vata* and *pitta*, *kapha* is present in every cell of the body and its major sites are believed to be the chest, stomach, brain, tongue, and synovial membrane of bone joints.

TABLE 2.3
Classification of *Kapha*

<i>Kapha</i>	Location	Function
<i>Avalambak</i>	Chest	Enables other <i>dosas</i> to perform their function by supplying humidity and body water, blood, fluid
<i>Kledak</i>	Stomach	It is liquid, mucoid, sweet, cooling in nature, and moistens the bolus
<i>Bodhak</i>	Tongue	Controls saliva and other juices that moisten the mouth, helps the bolus descend, and creates taste
<i>Tarpak</i>	Brain	Controls vision, hearing, smell, taste, and touch
<i>Shleshaka</i>	Synovial areas of bone joints	Provides lubrication

Signs and symptoms of deranged kapha are indicative of lipid metabolism and mucus-secreting system

Causes of Deranged <i>Kapha</i>	Symptoms of Deranged <i>Kapha</i>
Overexposure to cold weather; excessive sleep; day sleeping; excessive, sweet, fatty, and oily food; low level of physical activity	Excess of moisture, hardness of tissues, itching, feeling cold or heavy, low secretory activity, mucous membranes heavily coated with mucous, reduced activity of limbs, swelling, weak digestive power, sleepiness, pale skin, salty taste, and slow recovery after an illness

2.3 Body Constitution (*Prakriti*)¹

Although all biomaterials are natural elements of the body, people may have the predominance of one or a combination of two of the biomaterials that determine general physical and psychological characteristics and the tendency (susceptibility) to develop certain diseases. Table 2.4 shows the characteristics of body constitutions according to predominance of biomaterials and tendency to develop certain diseases.

2.4 Mental Components (*Mano Guna*)

Charak describes attributes of mental components as desire for pleasure, aversion to pain or misery, intelligence, mind, will power, logical faculty (*vicarna*), memory, knowledge gained from experience (*viijnana*), ability, and perception. Based on these attributes, mental components are divided into three classes: godly (*satogun*), kingly (*rajogun*), and evil (*tamogun*). Manifestations of these classes of mental components and characteristic constitutions are shown in Tables 2.5 and 2.6, respectively.¹

2.5 Health Care

Health care in Ayurveda is essentially aimed at balancing the *dosas* and *gunas*, bringing them into equilibrium. With this goal in mind, management of a disease is designed to

TABLE 2.4
Characteristics of Dosha Constitutions

Vata Constitution	Pitta Constitution	Kapha Constitution
<i>Characteristics</i>		
Low body weight, light bone structure, fast to act, forgetful, artistic, loves esoteric material, is shy and sensitive, enthusiastic	Healthy body, well-developed muscles, well-functioning metabolism, an urge for physical activities, an interest in science and technical subjects, good speaking ability, aggressiveness, hostile tendency, impatience, tendency to baldness, good mental ability, excellent memory, loves intellectual activities	Heavy body weight, solid body structure, strength, endurance, secure feeling, self-confident, not easily rattled, somewhat lethargic, patient, polite, generous, materialistic tendency, greedy, passive, tendency to sleep too much
<i>Diseases Prone To</i>		
Rheumatism, nervous disorders, sciatica, insomnia, dry skin, constipation, receding gums, weak bones, infertility, impotence, colic, flatulence, stuttering, ringing in the ears, irregular menstruation with cramps, varicose veins, paralysis, blood clotting, anorexia, shivering fits, poor blood circulation	Stress-related diseases; high blood pressure; coronary diseases; thrombosis; ulcers; cancer of the stomach, intestine, and skin; psoriasis; inflammation of the lymph system; infectious diseases; inflammation of the spleen; hepatitis, urinary tract infection; heartburn	Nausea, colds, bronchitis, asthma, kidney stones, swollen lymph nodes, benign tumors, dropsy, goiter, lung and breast cancer, fungal infections, digestive system problems, obesity

TABLE 2.5
Characteristics of Mental Components

Godly (<i>Satogun</i>)	Kingly (<i>Rajogun</i>)	Evil (<i>Tamogun</i>)
Compassion, self-control, forbearance, devotion to truth, proper conduct, faith in god, knowledge gained from self-realization (<i>Jnana</i>), intelligence, resolution, comprehension, nonattachment	Sensitivity to pain, restlessness of mind, lack of true understanding, egoism, lack of honesty, lack of compassion, false pride, overconfidence, buoyancy of spirit, strong desires, anger	Despondency, agnosticism and atheism, tendency to evil conduct (<i>adharma</i>), dull intellect, lack of knowledge and insight, perverted mind, physical and mental inertia, somnolence

TABLE 2.6
Characteristics of Mental Constitutions (*Manasik Prakriti*)

Godly (<i>Satvik</i>)	Kingly (<i>Rajsik</i>)	Evil (<i>Tamsik</i>)
Pure-minded, compassionate, loving, and righteous person who is capable of enduring all degrees of hardship and pain through willpower	Egotistic, ambitious, proud, competitive, controlling, and power-hungry person who may be made to submit to pain and unpleasant medical treatment through persuasion and logic	Lazy, sleepy, depressed, greedy, irritable, irresponsible, and selfish person with a large appetite who is horrified at the prospect of bodily pain

TABLE 2.7

Eight-Point Diagnosis

Examination	Vata	Pitta	Kapha
1. Pulse	Thread-like, feeble, snake-like motion	Moderately heavy, and jumps like a frog	Heavy, slow, flows like swan
2. Urine Drop of sesame oil spreads on the surface of urine gives:	Black-brown color Wave-like movement	Dark brown color Multiple colors, like rainbow	Cloudy appearance Pearl-like droplets
3. Stool	Uniform, dark color	Yellowish color	Bulky foul smell
4. Tongue	Black to brown, dry, coarse, furred, pigmented	Red, yellow, or green, soft, sharp, moist	Whitish color, pale coated, big, soft, flabby
5. Eye	Small, conjunctiva, muddy, iris is dark gray or brown	Moderate size, sharp, more sensitive to light	Large, moist, oily, conjunctiva is white
6. Skin	Dry, coarse, wrinkled, dusky	Wheat color, copper-like color, shiny, moist	Soft, off-white, smooth, moist
7. Speech and voice	Coarse and dry	Sharp	Heavy
8. General appearance	Mental disorders, anxiety, distress, pain, weakness, and factors in diagnosis		strength are important

diagnose which biomaterials and components are in excess or not and how they can be brought into balance.²

2.5.1 Diagnosis

Diagnosis of a disease in Ayurveda is essentially done by inspection, palpation, and interrogation. The specific examination includes the standard eight-point examination: (1) pulse, (2) urine, (3) stool, (4) tongue, (5) eye, (6) skin, (7) speech and voice, and (8) general appearance (Table 2.7). Additional examinations are made to assess digestive capacity, personal habits, body appearance, and patient's resilience. The other elements of diagnosis are the most recent reason for the illness, warning symptoms, clear symptoms, various diagnostic tests, and pathogenesis.

Management of an illness considers elimination of causes of the imbalance of *dosas*, administration of various herbal formulas, dietary and lifestyle interventions to bring *dosas* back into balance, elimination of *chinta* (serious worry), and nurturing the soul to regain spiritual health (*Samana*).

2.5.2 Disease Management

Charak Samhita describes four components of a disease management scheme: (1) the physician, (2) the drug, (3) the patient, and (4) the attendant. A physician must have proper training, knowledge, and experience. A remedy must be abundantly available, effective, and relatively safe. A patient must provide all information to the physician about the disorder and be compliant. An attendant (a nurse) must have the knowledge of patient care, dexterity, loyalty, and cleanliness.

Management of illness primarily consists of four procedures: (1) cleansing (*samsodhan*), (2) palliation (*samsaman*), (3) rejuvenation (*kaya kalp*), and (4) mental and spiritual healing (*sattvavajaya*, or psychotherapy). The management of an illness starts with cleansing and

TABLE 2.8

Group of Herbs Used To Mitigate *Vata*³

Medicinal Plant	Botanical Name	Medicinal Plant	Botanical Name
<i>Bhadradaru</i>		<i>Vrsciva</i>	<i>Boerhavia diffusa</i>
<i>Nata</i>	<i>Caesalpinia bonduc</i> (L.)	<i>Devahvaya</i>	<i>Cedrus deodru</i>
<i>Kustha</i>	<i>Saussurea lappa</i> cob	<i>Two Salpaparni</i>	
<i>Dashmula</i>		<i>Kandukari</i>	<i>Mueuna pruriens</i>
<i>Two Bala</i>	<i>Sida cordifolia</i> Linn.	<i>Abhiru</i>	<i>Asparagus racemoses</i>
<i>Vellantara</i>	<i>Vativera zizanoides</i>	<i>Vira</i>	<i>Roscoea procera</i>
<i>Aranika</i>	<i>Premna integrefolia</i> Linn.	<i>Jivanti</i>	<i>Leptadenia reticulata</i>
<i>Buka</i>	<i>Osmanthus fragrans</i>	<i>Rsbhaka</i>	<i>Microstullus wallachi</i> Lindl
<i>Vrsa</i>	<i>Addhatoda vasica</i> Nees	<i>Brihatee</i>	<i>Solanum anguivi</i> Lam.
<i>Asmabheda</i>	<i>Bergenia ligulala</i> wall	<i>Kantakaree</i>	<i>Solanum surattense</i> Burm. F.
<i>Gokantaka</i>	<i>Tribulus teriatris</i>	<i>Saliparni</i>	<i>Desmodium gangeticum</i>
<i>Itkata</i>	<i>Sesbania bispinosa</i>	<i>Prasniparni</i>	<i>Uraria picta</i>
<i>Sahacara</i>	<i>Baeleria prionitis</i> Linn.	<i>Goksuraka</i>	<i>Tribulus terrestris</i>
<i>Bana</i>	<i>Bareleria stringosa</i>	<i>Hansapadi-Adiantum</i>	<i>Podophyllum hexandrum</i> royle
		<i>Lunilatum-Burm</i>	
<i>Gundra</i>	<i>Typha elephantia</i> Roxb	<i>Kasa</i>	<i>Saccharum spontaneum</i> Linn
<i>Bhalluka</i>	<i>Oroxylum indicum</i> Vent.	<i>Vrksadani</i>	<i>Loranthus longifolius</i>
<i>Karambha</i>	<i>Pergularia extensa</i>	<i>Nala</i>	<i>Arundo donax</i> Linn
<i>Partha</i>	<i>Gynandropsis pentaphylla</i>	<i>Two Kusa</i>	<i>Desmostachya bipinnata</i>
<i>Vrscikali</i>	<i>Pergularia extensa</i>	<i>Guntha</i>	<i>Typha angustata</i>

includes five procedures called *panchakarma*, all of which are not necessarily done at the same time or to all patients. This topic is described in detail in [Chapter 4](#).

Panchakarma is a major therapeutic procedure in Ayurveda known to be useful for all diseases of all the body organs and functions because the elimination of toxic products, endogenous or exogenous, can contribute to overall management of an ailment. It is understandable that if the body organs are not eliminating toxic products from the body, toxic symptoms will develop over a period of time. The major cause of body ailments is the toxic products produced by body metabolism, microorganisms, synthetic chemicals, xenobiotics, or drugs. *Panchakarma* is discussed in detail in Chapter 4.

Palliation essentially consists of compound preparations of herbs and minerals for diet and lifestyle interventions. Ayurvedic texts describe seven types of palliation: (1) digestive power enhancement, (2) toxic waste (*ama*) elimination, (3) fasting, (4) observing thirst, (5) yoga exercise, (6) lying in the sunlight or sunbathing, and (7) breathing exercise and meditation. A variety of formulas are used to improve digestion, eliminate *ama*, and balance *dosas*. Examples of herbs used to subdue deranged or vitiated *dosas* are shown in Tables 2.8 through 2.10. Fasting also helps in acute indigestion, dysentery, diarrhea, and acute fever.

2.5.2.1 An Example of Ayurvedic Case Management

To illustrate a typical Ayurvedic therapy, the management of *Ama-vata* (rheumatoid arthritis) is given below. It includes biopurification and detoxification by dietary interventions and necessary *panchakarma* procedures and administration of herbal formulas. The first step is elimination of *ama* (the incompletely digested food material in the intestine, collagen materials, allergens, mucus, mucoid materials, crystals like uric acid) by reducing the diet to near or total fasting if possible. The diet consists of very small quantities of double-boiled rice and watery *mung dal* (lentils) for the first 5 days, with the addition of small amounts of vegetables for the next 5 days, and then one piece of bread (*roti*) for an

TABLE 2.9

Herbs Used To Mitigate *Pitta*³

Herb	Botanical Name	Herb	Botanical Name
<i>Durva</i>	<i>Cynodon dactylon</i>	<i>Abhiru</i>	<i>Asparagus racemoses</i>
<i>Ananta</i>	<i>Alhagi camelorum</i>	<i>Sitapaki</i>	<i>Abrus precatorius</i>
<i>Nimba</i>	<i>Azadirachta indica</i> A.	<i>Priyangu</i>	<i>Setaria italica</i> Beauv
<i>Vasa</i>	<i>Litesa glutinosa</i>	<i>Two Sthira</i>	<i>Desmodium gangeticum</i>
<i>Atmagupta</i>	<i>Mueuna pruriens</i>	<i>Padmak</i>	<i>Prunus cerasoides</i>
<i>Gundra</i>	<i>Typha elephantine</i> Roxb	<i>Vanya</i>	<i>Cyperus rotundus</i> Linn.
<i>Nyagrodhadi</i> group			
<i>Nyagrodha</i>	<i>Ficus vengakensis</i> Linn.	<i>Two Meda (fat)</i>	<i>Litsea monpetala</i>
<i>Tuga</i>	<i>Bamboo mann</i>	<i>Mudgaparni</i>	<i>Vigna trilobata</i>
<i>Rddhi</i>	<i>Sphaeranthus indicus</i>	<i>Mashaparni</i>	<i>Teramnus labialis</i>
<i>Shringee</i>	<i>Pistacia chinensis</i> Bunge	<i>Rsabhaka</i>	<i>Microstullus wallachi</i>
<i>Amrita</i>	<i>Tinospora cordifolia</i>	<i>Jivaka</i>	<i>Microstylus wallachi</i> Lindl
<i>Jivanti</i>	<i>Leptadenia reticulata</i>	<i>Madhuka</i>	<i>Sapotaceae</i>
<i>Two Kakoli</i>	<i>Roscoeia procera</i>		
<i>Sarivadi</i> group (used to treat burning sensation, bleeding disease, thirst, and fever)			
<i>Pippala</i>	<i>Piper longum</i> Linn.	<i>Usira</i>	<i>Vativeria zizanoides</i>
<i>Sadaphala</i>	<i>Ficus glomerata</i>	<i>Kasmarda</i>	<i>Cassia sophera</i>
<i>Two Rodhra</i>	<i>Symplocos racemosa</i> Roxb	<i>Yashti-madhu</i>	<i>Glycyrrhiza glabra</i>
<i>Arjun</i>	<i>Terminalia arjuna</i>	<i>Kadamba</i>	<i>Anthocephalus chinensis</i>
<i>Kapitana</i>	<i>Thespesia populnea</i>	<i>Virala</i>	<i>Diospyros tomentosa</i>
<i>Somavalka</i>	<i>Acacia suma</i> Kurz	<i>Madhuka</i>	<i>Sapotaceae</i>
<i>Plaksha</i>	<i>Ficus lucescens</i> Blume		
<i>Amaara</i>	<i>Vitex nigundo</i> Linn.		
<i>Padmakadi</i> group (increase breast milk to subjugate <i>vata</i> and <i>pitta</i>)			
<i>Padmaka</i>	<i>Prunus cerasoides</i>		

additional 5 days. This treatment is generally given for a minimum of 1 month. During this period, hot fomentations are applied and no ghee (dehydrated milk fat), vegetable oil, meat, or animal fat is given. It is believed that this treatment increases the metabolic power of the body, resulting in breakdown and excretion of the disease-producing vitiated materials (*ama*), collagen, uric acid, sugar, urea, fat, etc. To augment the process of cleansing, ginger juice or dry ginger is given daily in the early morning on an empty stomach. *Chirata* or *Swertia chirayita*, a bitter herb, is also given to improve digestion.

After 1 month of this treatment, ghee (dehydrated butter) may be given to help in removing fat-soluble materials from the body. A medicated enema (*basti*) containing sesame (*til*) oil may be started and continued for 1 month for this purpose. At this point the treatment is stopped for 4 to 8 weeks and the reduced diet treatment is resumed. Once the acute phase has subsided, the patient is asked to exercise, first with passive movements and then with active movements (yoga exercises using various body positions). Yoga instructions are recommended to help the patients do these exercises correctly.

This treatment is often accompanied by an oral administration of Ayurvedic herbal formulas two to three times per day. The formulas often used are *Yograj-guggul* or *Mahayograj-guggul* (formulas from Ayurvedic text) with gold *bhasma* (gold formula made by oxidizing gold with heat using herbs). A physician must carefully examine the patient for other illnesses and organ disorders to determine a suitable antiarthritis formula, depending on accompanying health conditions.

TABLE 2.10

Group of Herbs To Mitigate *Kapha*³

Aragvadhadi Group (used to treat vomiting, leprosy, fevers, mitigate kapha, itching, diabetes, and clean bad wounds)

Medicinal Plant	Botanical Name
<i>Aragvadhha</i>	<i>Cassia fistula</i>
<i>Indrajaya</i>	<i>Holerrhina antidysenterica</i> wall
<i>Patali</i>	<i>Stereospermum colais</i>
<i>Kakatitka</i>	<i>Abrus precatorius</i> Linn.
<i>Nimba</i>	<i>Azadirachta indica</i> A.
<i>Amrata</i>	<i>Tinospora cordifolia</i> Miers
<i>Madhuras</i>	<i>Marsedenia tenacissima</i>
<i>Sruvavrksa</i>	<i>Flacourtia indica</i>
<i>Patha</i>	<i>Steohania japonica</i> Miers
<i>Bhunimba</i>	<i>Andrographis paniculata</i>
<i>Sairyaka</i>	<i>Barleria prionitis</i> Linn.
<i>Patala</i>	<i>Cydonia oblonga</i>
<i>Two Karanja</i>	<i>Pongamia pinnata</i>
<i>Saptacchada</i>	<i>Alstonia scholaris</i> R.
<i>Agni</i>	<i>Plymbago zeylanica</i> Linn.
<i>Susavi</i>	<i>Momordca chirantia</i> Linn.
<i>Phalsa</i>	<i>Grewia subinaequalis</i> DC
<i>Bana</i>	<i>Barleria stringosa</i>
<i>Ghonta</i>	<i>Acaia catechu</i> Willd <i>badari</i> (Hem), <i>Zyziphus jujuba</i> Lam.
<i>Arka</i>	<i>Calotropis gigantea</i> (L.)
<i>Alarka</i>	<i>Calotropis procera</i>
<i>Nagadanti</i>	<i>Croton oblongifolius</i>
<i>Visalya</i>	<i>Gloriosa superba</i> Linn.
<i>Bharangi</i>	<i>Clerodendrum indicum</i>
<i>Rasna</i>	<i>Acampe papikkosa</i>
<i>Vrscikali</i>	<i>Pergularia extensa</i>
<i>Prakirya</i>	<i>Cesalpinia bonducella</i> Fleming
<i>Pratyakpuspi</i>	<i>Achyranthes aspera</i> Linn.
<i>Pitataila</i>	<i>Abrus precatorius</i> Linn.
<i>Udakirya</i>	<i>Pongamia glabra</i> Vent.

2.5.3 Dietary Management and Lifestyle Changes

Dietary and lifestyle interventions are initiated according to the disturbed *dosas* and the physical and mental constitution of a person; these interventions are accompanied by spiritual nurturing, removing serious worry, exercise, and yoga practice. If *vata* were disturbed, the diet would include oils, butter, and sweet food. If *kapha* were disturbed, the diet would include bitter, sour, vinegary, spicy, dry food. If *pitta* were disturbed, the diet would include mild tasting food, grains, lentils, and moderate amounts of sweets and oils. Patients are given a restricted diet of small quantities of *mung dal* (a type of lentil) or yogurt and rice. Depriving the patient of drinking water helps in ascites, edema, and certain kidney diseases where large amounts of water are retained in the body.

TABLE 2.11

Ayurvedic Herbs Known To Have Confirmed Therapeutic Effects

Disease/Symptoms	Herb	Popular Name
Edema	<i>Achyranthes aspera</i> Linn.	Apamarga
Mental distress	<i>Acorus calamus</i> Linn.	Vacha
Bronchiolar constriction	<i>Adhatoda zeylanica</i> Medic.	Vasa
Intestinal worms	<i>Butea monosperma</i> Lam	Palasha
Arthritis, inflammation, and high cholesterol	<i>Commiphora mukul</i> Wightii	Guggul
Bone fracture	<i>Cissus quadrangularis</i> Linn.	Vajravalli
Malaria	<i>Alstonia scholaris</i> R. Br.	Saptaparna
Liver disorders	<i>Andrographis paniculata</i> Nees	Bhumimba
Cardiac disorders	<i>Artemisia vulgaris</i> Linn.	Nimba
	<i>Terminalia arjuna</i>	Arjuna
Bacterial or virus	<i>Azadirachta indica</i> A. Juss	Nimba
Memory problem	<i>Bacopa monnieri</i> (Linn.) Pennell	Brahmi
Inflammation and diuretic need	<i>Boerhaavia diffusa</i>	Punarnava
Severe constipation	<i>Cassia angustifolia</i> Vahl.	Rechani
Muscle spasm	<i>Cedrus deodara</i> (Roxb.)	Devadaaru
Memory problem	<i>Celastrus paniculatus</i> Willd.	Jyotishmati
Nerve weakness	<i>Centella asiatica</i> (Linn.) Urban.	Mandukaparni
Malaria and fevers	<i>Caesalpinia crista</i> Linn.	Natakaranja
Inflammation	<i>Curcuma longa</i> Linn.	Haridraa
Diabetes	<i>Eugenia jambolana</i> Lam.	Jamboo
Dysentery	<i>Holarrhena antidysenterica</i> Wall.	Kutaja
Convulsions	<i>Masilea minuta</i> Linn.	Sunisannak
Liver toxicity	<i>Picrorhiza kurroa</i> Royle.	Kutuka
Mild constipation	<i>Plantago ovata</i> Forsk	Ishadgola
Skin white spot	<i>Psoralea corylifolia</i> Linn.	Bakuchi
Low breast milk	<i>Pueraria tuberosa</i> DC.	Vidarikanda
Low muscle mass	<i>Sida rhombifolia</i> Linn.	Mahabala
Fever	<i>Swertia chirata</i> Buch.-Ham.	Kairata
Heart weakness	<i>Terminalia arjuna</i> W. and A.	Arjuna
Mental distress, physical exhaustion, and inflammation	<i>Withania somnifera</i> Dunal	Ashwagandha

Healthy lifestyle is very important to maximize the effect of palliative treatment. Ayurveda strongly recommends that the patient utilize a regular sleeping schedule, about 8 hours of sleep each night, rising early in the morning; do regular exercise; and eat breakfast and engage in other dietary recommendations according to seasons. In order to maximize the effectiveness of palliative treatment, *Vyayam* (exercise, yoga, stretching), breathing exercises, and sunbathing are necessary. It is important that some appropriate form of exercise be done at a maximum tolerated effect (MTE) level during the treatment period; this may be with the help of a physical fitness trainer. The concept of MTE is similar to a therapeutic dose of pharmacologic agents to get maximum beneficial effect. Exercise is likely to create a need for nutrients inside the cells. It is believed to open up the microchannels (*shrotas*) in the body cell for nutrients and for medicine to enter. It is particularly important in maximizing the effectiveness of therapies in management as well as prevention of chronic musculoskeletal disorders.

2.5.4 *Bhasma*

The oxidized form of metal and mineral preparations, called *bhasma*, is also extensively used in Ayurvedic medicine. *Bhasmas* are metal or mineral powder formulas made by specific Ayurvedic text procedures with several herbs and herbal extracts and subjected to very precise heat treatment. The common metals and minerals used in making *bhasmas* for therapeutic use are gold, silver, copper, mercury, iron, zinc, tin, arsenic, gypsum, lime, alum, borax, silica, diamond, ruby, emerald, sapphire, jade, moonstone, sunstone, turquoise, and mica.

Procedures to make *bhasmas* are very complicated and involve dozens of steps and many herbs and herbal extracts. These procedures detoxify toxic metals like mercury. If any of the steps in the procedure are either omitted or not properly carried out, then safety is compromised. The pure metal salts do not account for unique pharmacological properties of their *bhasmas*. For example, pure mercury salts do not account for the antiarthritic activity of mercury *bhasma*. There are several hundred metal *bhasmas* described in texts. The science and art of *bhasmas*, a separate branch of pharmacy known as *Rasa Shastra*, is described in [Chapter 6](#).

2.5.5 *Rasayana*

The word *rasayana* is made up of two Sanskrit words, *rasa* (nutrition) and *ayana* (transportation in the body). Thus, *rasayana* refers to compound preparations containing multiple herbs and minerals that improve transportation of nutritional materials to body tissues. The fundamental underlying theme of *rasayanas* is nutrition. They are part of the overall balanced diet. *Rasayanas* are claimed to improve vitality, rejuvenate body tissues, improve immunity, and prevent aging. *Rasayanas* may act in a variety of ways by improving the nutritional value of the food, digestion and absorption of nutrients, transportation of nutrients to tissues, bioavailability of nutrients, metabolism of nutrients in tissues, immunity, and by cleaning the *srotas* (microcirculatory channels or pores) which improve uptake of micronutrients. Basically, *rasayanas* are classified into three categories: (1) *Ajasrika rasayana* (nutrition, dietary) is taken regularly with food as nutrition; (2) *Kamyas rasayana* (normal health promoter) is indicated to improve vigor, vitality, and to make a healthy person feel better; and (3) *Naimittika rasayana* is indicated to promote vitality in a particular disease. *Kamyas rasayana* is further classified into three categories: (1) *Pranakamyas rasayana* (promoter of life vitality and longevity); (2) *Medhakamyas rasayana* (promoter of intellect); and (3) *Srikamyas rasayana* (promoter of skin complexion and luster). Ayurvedic physicians should be consulted to determine the appropriate *rasayana* required depending upon the health needs.

Rasayanas are the Ayurvedic equivalent to modern dietary supplements. It is understandable that vitamins, minerals, and biologically active steroids, alkaloids, glycosides, tannins, and a variety of antioxidants present in *rasayanas* would be a good source in providing necessary nutrients to the body. Sometimes an overdose of pure vitamins in modern dietary supplements can cause serious side effects. There is virtually no possibility of an overdose of any specific vitamin or mineral from a *rasayana*.

2.5.6 *Formulas*

The drug treatment in Ayurveda primarily consists of herbal formulas, *bhasmas*, *rasas* (mercury is always an ingredient of *rasas*), and medicated oils and ghee (for topical

application, nasal drops, and enemas). A single herb is rarely administered to a patient in Ayurveda; generally a formula made up of several herbs is used. They are always given with other foods or herbal items, e.g., honey, ghee, ginger, etc., which help mitigate toxicity and may increase absorption of certain ingredients, thus obtaining the desired therapeutic effect. Herbal formulas are favored in Ayurveda because the founders of Ayurveda recognized the possible synergistic and counterbalancing effects of herbs.

Sometimes none of the herbs in a formula exhibit therapeutic effects individually, but the formula could nevertheless be effective. Recently, a formula (*Trasina*) made up of *Withania somnifera*, *Tinospora cordifolia* (Wild) Miers, *Eclipt alba* Hassk, *Ocimum sanctum* Linn, *Picrorrhiza kurroa* Royal ex Benth, and *Shilajit* (rock sweat, collected from rocks) showed beneficial activity in diabetic rats, whereas no such effect was seen with individual herbs.⁴ This example illustrates the fact that a rapid screening program to assess biologic activity of single herbs or purified constituents of a herb may not necessarily confirm or disprove a health claim because of possible synergistic effects of the combination of drugs or adjuvant. Ayurvedic formulas are analogous to vegetarian food where an experienced cook makes the difference in taste and benefits. All the chemical constituents of milk consumed individually are not likely to give benefits equal to that of an equivalent amount of whole milk consumed. It is therefore important that Ayurvedic formulas as described in Ayurvedic texts be evaluated for therapeutic benefits along with the individual active herbs.

There are several thousand formulas described in Ayurvedic texts that are recommended for over 200 well-identified diseases. In the 1970s, a panel of Ayurvedic experts was appointed by the government of India to evaluate the formulas in Ayurvedic texts. The panel compiled an "Ayurvedic Formulary of India."⁵ It has over 560 evaluated formulas including 22 *bhasmas* and 55 *rasas* and *yogas* (preparations containing mineral drugs as main ingredients). Some formulas have dozens of herbs; for example, *Jirakadi Modaka* has 46 herbs and *Shiva Gutika* has 49 herbs.

Individual plants and plant constituents have been studied for therapeutic activity in recent scientific studies. Therapeutic effects have been confirmed by clinical and biological testing for many of them. A few examples are given in [Table 2.11](#).

2.5.7 Yoga

Yoga is used not only as a therapy, but also to improve general health and vitality. Yoga is defined as the inhibition of fluctuations of consciousness.⁶ In effect, it restrains mental activity. Mind in total is composed of three faculties: brain, intellect, and ego. Yoga teaches the means by which the mind is controlled and redirected into constructive channels.

Patanjali, a famous sage, advocated two methods to control the mind: constant practice and detachment. Although Yoga is conceived today as a combination of physical postures and breathing exercises, Ayurvedic texts actually describe several components of Yoga such as ethical practices, punctuality of daily routine, physical-posture exercises, breathing exercise, sensorial practices, and meditation practices. Today, breathing and physical-posture exercises are collectively known as Yoga. The practice of Yoga alone or in combination with other Ayurvedic therapies has been noted in Ayurvedic texts as beneficial in certain diseases, such as hypertension, bronchial asthma, anxiety, neurosis, gastrointestinal disorders, headache, insomnia, obesity, anxiety, and depression. It is suggested that the autonomic nervous system is improved by Yoga, resulting in a relatively hypometabolic state, thereby lessening the energy demands on the body. The efficiency of the cardiovascular and respiratory systems is also significantly enhanced by Yoga.

2.5.8 Massage

Vital points exist where muscle, cartilage, nerves, and bones join each other. The life energies are believed to be concentrated at these points. There are 22 vital points on the upper extremities, 22 on the lower extremities, 12 in the abdominal areas, 14 in the back, and 37 in the neck and head. Manipulation and massage of these points has been used in Ayurveda to treat diseases and strengthen the body. Ayurvedic massage is the combination of massage with medicinal oils and acupressure. Because lipophilic materials can be easily absorbed through skin, medicated oil massage can be a very effective therapy. Specific medicated oils and types of strokes are chosen based on the disturbed *dosas*, body constitution, injury, and disease condition.

The basic strokes of massage include (1) friction where pressure is applied by thumb and finger tips more on muscular part, and gentle pressure on bony parts; (2) kneading, deep or superficial; (3) rounding; (4) wringing or twisting; (5) chucking; (6) stroking; (7) percussion; (8) vibrations; (9) joint movement; and (10) soft, gentle massage with thumbs and palm. Ayurvedic massage is useful for general weakness, fatigue, arthritis, musculoskeletal disorders, tennis elbow, lumbago, frozen shoulder, backache sprains, and aches. Massage is also found useful for nervous system disorders, psychosis, drug addiction, gynecological disorders, menopausal syndrome, postdelivery recovery, myocardial infarction, ischemic heart disease, functional heart disease, cold hands and feet due to bad circulation, varicose veins, psychological problems, obesity, loss of weight, stamina, and disturbed digestive system.⁷ Contraindications for massage therapy are fever, indigestion, and patients undergoing cleansing processes.

Massage is also advised for healthy people to maintain their health and relieve muscular fatigue after a heavy physical activity or exercise.

2.6 Summary

Basically, Ayurvedic health care is based on the principle that body, mind, and soul must be in harmony for health and happiness. Each of these has to be nurtured for an individual to create health. Ayurveda recommends daily activities in detail, e.g., getting up before sunrise, routine physical exercise, meditation, and diet according to the body constitution with seasonal adjustments in lifestyle and sexual activities to maintain an optimum balance of *dosas* to prevent illness. Unhealthy lifestyles, exposure to various physical, chemical, and biological agents, extreme weather conditions, unhealthy diets, and overexertion from physical or mental activities are considered the major causes of imbalance of *dosas*. Cleansing of the body from waste products of the illness is necessary to eradicate an illness and prevent its recurrence. Finally, management of an illness essentially consists of elimination of causes of the imbalance of *dosas*, administration of various Ayurvedic formulas, dietary changes to bring *dosas* back into balance, elimination of serious worry, and nurturing the soul to regain spiritual health.

Research activity on Indian medicinal plants over the past 50 years has been extensive. Data on more than 5000 Indian medicinal plants have been compiled in the *Compendium of Indian Medicinal Plants* by Rastogi and Mehrotra.⁸ The data include research studies on pharmacological and therapeutic activity, pharmacognosy, plant chemistry, and established chemical structures of chemical constituents. Research on more than 500 popular Indian medicinal plants are similarly described in *The Treatise of Indian Medicinal Plants* by

Chatterjee and Pakarashi.⁹ Data on several hundred selected medicinal plants are also compiled in other books.

Ayurveda, the traditional health-care system of India, is a complete and holistic health-care system that contains both preventive and therapeutic aspects. It has a defined diagnostic system covering all physiological functions and organs, as well as specific treatments and management techniques for each disease. Although Ayurveda is thousands of years old, many of its core principles are consistent with modern medicine. Ayurvedic medicine offers a wealth of relatively effective, safe, and economic health-care therapies. Its therapies provide relief for many chronic illnesses such as musculoskeletal disorders, rheumatoid arthritis, osteoarthritis, diabetes, obesity, nervous disorders, etc. Ayurvedic medicine has been successfully integrated with allopathic medicine in India. Certainly, a great potential exists for its integration into the health-care system in the U.S.

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3

Clinical Research Design: Limited Systematic Review of Five Diagnostic Categories

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3.1 Introduction

Ayurveda, the Indian System of Medicine (ISM), may be the least well known of the complete systems of medicine in the West, despite being one of the oldest. According to the definition used by the National Center for Complementary and Alternative Medicine (NCCAM),¹ World Health Organization (WHO), and other organizations, Ayurveda qualifies as a complementary and alternative medicine (CAM).²

Certain techniques used as therapies in this system of medicine have been assimilated in the West but not practiced as a whole system. For example, yoga, oil massages, meditation practices, and body cleansing (both internal and external) are used frequently in the West as part of health-promoting behaviors; they are rarely perceived as part of a larger holistic traditional medical practice. Often the practitioners of these techniques have not completed full Ayurvedic medical training and have merely focused on learning

the technique via the apprenticeship method. The training is limited to studies with practitioners in India, whereas others have merely taken weekend seminars. Some training is now available in the U.S. for those who wish to get a certificate or some other training credential. These programs are not full time and last between 1 and 2 years.³ No equivalent program of the Ayurvedic medical schools in India, which require 5½ years of training before practice is allowed, exists.⁴ As a result, there are quite a variety of credentials held by the practitioners of Ayurvedic therapeutic techniques. In addition, because licensing is not required by most states in America, there is not a consistent and focused movement toward establishing the practice. It is also questionable as to whether those programs that teach both professionals and paraprofessionals certain Ayurvedic treatment techniques (e.g., yoga) without the whole context of Ayurvedic theoretical underpinning will do more than produce isolated therapies originally designed to be part of a whole.

This disassociation of the Ayurvedic treatment repertoire will minimize optimization of patient recovery potential as the synergistic advantages will be lost. Because these practices are rarely coordinated by a trained Ayurvedic physician in the West, research into the utilization of Ayurvedic bodywork practices, cleansing, and Ayurvedic herbals used independently or in an integrated medical scenario with allopathic treatment techniques is essential. Research in Ayurveda must be done with rigor and with an expectation of perusal by Western trained physicians who wish to integrate these techniques into their health-care practices. Without research data that will convince the Western trained physician that a patient can benefit or at least will not be harmed, Ayurveda will continue to remain an underutilized therapeutic option for patients.

Research must be done to identify which practices and herbals are both safe for specific conditions, with or without certain co-morbidities, and that are able to produce optimized patient recovery. Because individual practices have begun to be used primarily as a consumer-driven enterprise, research respecting the character and tradition of Ayurvedic medical parameters must be conducted within the context in which the practice is likely to occur.

3.2 Empiricism in Medical Knowledge Development

It is interesting to note that many Western-trained physicians question, for example, the scientific underpinning or rationale for the use of Ayurvedic medicine, homeopathy, and traditional Chinese medicine. However, empirical explanations for use of health treatments, much like the role of empiricism in a biological lab, are often ignored as a source of evidence. Empiricism draws upon close observation and experimentation.⁵ So, when past Ayurvedic physicians observed changes in a patient's condition after administering a treatment, it should not have been considered philosophical or traditional knowledge, *per se*. It is not less valid than observations made and recorded in a laboratory experiment.

When clinical case notes are passed on to colleagues and to the next generation of doctors being trained, it is no less valid than descriptive case studies published today in peer-reviewed medical journals. In balance, it might be helpful to note that if the highest scientific rigor were demanded of allopathic medicine as is being demanded of CAM practices, including Ayurveda, conventional medical practice would not meet the mark either. In 1983 the U.S. Office of Technology stated that only 10 to 20% of all procedures currently used in medical practice had been shown effective by controlled trial.⁶ Currently,

data indicate that only 20 to 37% of procedures in conventional medicine have been subjected to the same standards as demanded for CAM.⁷

3.3 Ayurveda: An Evidence-Based Medicine

An accurate understanding of what evidence-based medicine (EBM) really means might be helpful in understanding the relative role of science and clinical knowledge. EBM, which has become a buzz word concerning medical decision making, refers to a triangulated set of information that the physician should use to determine the best treatment for a particular patient.⁸ Here, Western medicine is coming closer to a holistic treatment approach, closer to the parameters of treatment in Ayurvedic practice. The three factors of EBM are the following:

1. Best available relevant scientific evidence concerning the effectiveness and efficacy of the proposed treatment
2. Physician knowledge based on practice experience
3. Patient's own preferences for treatment modalities if they do not contradict 1 or 2 above

In order to make the best utilization of this triangulation of information, physicians must be familiar with the treatment perspectives described in the scientific literature. If there is a dearth of information or the studies are conducted without scientific rigor, patients are put into jeopardy; either therapies that might help will not be considered, or physicians will be concerned about validity and reliability of data based on quality of data in studies reported.

3.3.1 Overcoming Barriers of Research

There have indeed been barriers to CAM research, in general, which also affected Ayurvedic research. They have included structural barriers in the West where conventional medical institutions received the largest share of funds to investigate therapies for which no one in the institution receiving the funds had any firsthand knowledge. Collaborations were often not equitable, even when CAM therapists were co-investigators. Publication bias has been a problem in the past. With specific journals dedicated to publishing CAM literature, the situation may improve. However, both CAM investigators and journal editors must produce and ensure the quality of data presented. In this way, those persons who are skeptical will learn that CAM researchers can produce good, relevant science.

Expectations for randomized controlled trials (RCTs) can be another barrier. It is appropriate for preclinical studies to be done along with quasi-experimental designs prior to conducting RCTs whenever possible.⁹ It is important for the investigators in RCTs to have established factors that will allow for accurate calculation of sample size, to determine possible sources of bias in executing such a trial, the usefulness of outcome measures chosen, and that the dosages (if applicable) and delivery schedule of the therapeutic product or intervention are optimized before moving to an RCT. If these factors, minimally, are not known, then the investigators must do Phase II clinical work to establish this

information. It behooves all of us to recognize research at the case study, case series, or outcome study level for the scientific building blocks that they are.¹⁰

3.4 Research Designs: A Look at a Hierarchy of Design

Several clinical study types can be implemented by those clinicians who do not have the funds or expertise to engage in larger research projects. These clinicians may consider adding to the body of knowledge through executing one of the following designs.

3.4.1 Case Studies

3.4.1.1 Retrospective Case Study

This case-study design can offer information about an unusual case that might help colleagues problem solve if they have a similar case. In this research design, there would be no prior planning to investigate the diagnosis or treatment. The term *retrospective* is used to refer to its structure. Generally, the authors of such a study give detailed information about the demographic, diagnosis profile, and general health parameters of the patient of focus. The therapy utilized and the patient progress are noted in detail. This is a particularly useful tool when the disease presents in an unusual manner, the patient's progress is unanticipated, or a unique therapy has been tried.

A prospective descriptive case study is structured identically to the retrospective model given above except that the clinician plans in advance to conduct the study. Like the retrospective case study described above, the information is for the most part qualitative. It is hoped that the prospective nature of the study will encourage the clinician to take even more detailed SOAP (clinical) notes than would have been the case normally; this creates a rich information source for those clinicians who might find the information useful in the future. Both retrospective case studies and prospective case studies can be case series. This change in name merely means that multiple (generally 5 to 10) patients with a similar diagnosis or similar treatment will be presented.

Probably the most useful study of this design type is the structured or experimental case study and case series. This design involves *a priori* design time. The specific patient diagnostic parameters, type and duration of treatment, outcome measures, number of patients in the sample ($n = 1$ to 10), and analyses would be planned in advance. This structure allows for quantitative information in addition to the standard qualitative information to be gleaned from the investigation. This type of design is particularly helpful when details of the therapy and consequences are required to determine optimum delivery of a new therapy or new combination of therapies or when there are questions about the effectiveness of a therapy for an esoteric group of sufferers from a particular diagnosis.

3.4.1.2 Outcome Studies

Another design that might be considered is an outcome study. This design is similar in structure to the experimental case study described above. However, it would best be built

upon a case study where notions of percentage of improvement are available so that power calculations could be generated. Whenever possible, knowing the number of participants needed to show differences, generally moderate differences, is important. Otherwise, investigators are left using cookbook guesses that may defeat the purpose of the study. Patients serve as their own control, and their improvement is noted through standardized measures from baseline (period just prior to treatment beginning) to various measurement points until the end of treatment. Often 25 to 30 persons are enrolled if prior data are not available to determine sample size needs. Because patients serve as their own control, it is a very useful design to refine the therapy, refine study protocol, and have larger sample size data to inform an RCT design.

Because there is no control group and patients serve as their own control (a better match would be hard to imagine), attention to possible sources of bias in the design is important. If covariants are known, efforts should be made to eliminate them in the design or to collect data on these factors and control them in the analysis. This design does not address issues of potential placebo effect; whenever possible, these studies are followed by homogeneous and heterogeneous RCTs. Many questions that can be answered by this design cannot be answered by RCTs. The scientific question to be asked coupled with the data that already exist in the literature may mean that an RCT is not the best design.

3.4.1.3 Randomized Controlled Trials

RCTs are best done when there is information in the existing literature at a quasi-experimental level (e.g., outcome study) to inform the RCT. It is important to minimally:

1. Have a specific research question that can be answered by a multigroup comparison.
2. Know the optimum effective dose or equivalent for the medical practice before starting.
3. Know the precise number of subjects needed to be able to detect differences among the groups.
4. Know that the placebo or sham arm of the study will not be obvious to the participants (blinding is maintained).
5. Know the approximate number of dropouts expected due to possible rigors of the protocol.
6. Take precautions to control potential biases and not merely rely on random assignment to condition or pure randomization to equally disburse individual differences among the groups in the study.
7. Have a precise analytical plan with the person manipulating the data (biostatistician or methodologist) blinded to group assignment as completely as the persons in contact with the patients.

It is obvious that preliminary data at quasi-experimental levels must be collected before the sophistication of an RCT design is cost effective. Guessing on information required to inform an RCT will result in poor data, which may be considered valid in today's medical culture merely because it uses the word *randomized* in the title. It is far better to contribute to the body of knowledge with a well-done case study or outcome study than to report a flawed RCT.

3.4.1.4 Systematic Reviews and Meta-Analyses

Two additional designs that contribute to a consensus building about a therapy or diagnosis are systematic reviews and meta-analyses. Both designs are secondary analyses of previously done work. Generally, these two designs have utilized RCT data and not included information from quasi-experimental designs with the assumption that RCTs constitute the highest level of data collection for individual trials. Both homogeneous and heterogeneous RCTs are used in both.

A systematic review is a qualitative method of evaluation of a group of RCTs on a specific therapy or diagnosis for the most part. The fewer the RCTs, the smaller the data pool from which conclusions are drawn. Currently, systematic reviews in CAM suffer from too few RCTs on the topic, design flaws of the RCTs (e.g., inadequate sample size), and changing expectations of expected research rigor in CAM during the most recent past. Other levels of data must continue to be relied upon for clues to prescribing and safety parameters.

A meta-analysis is difficult at this point, but some have been attempted. This research strategy is similar to systematic reviews except that the analysis is a quantitative one. The authors of such analyses generally attempt to get raw data from authors of RCTs, normalize the data across the trials they have collected, and use this pooled data attempt to determine the overall safety or efficacy of a particular therapy. This is even a more difficult task than a Systematic Review strategy. These designs are best left to biostatisticians or methodologists at this time. Even they have difficulty getting sufficient information in which to conduct a meta-analysis, as the total information required is rarely reported in full in a published article.

3.5 Southern California University Research Model

The authors of this chapter have pursued a stepwise approach to development of a research track in Ayurveda, just as in other medical treatment modalities. Even though effectiveness and efficacy of certain therapies were established via traditional knowledge and through clinical data used in conventional medicine, this research group has chosen to investigate Ayurvedic therapies holistically and to investigate specific herbals within a reductionistic science-based approach as well. Animal work concerning mechanism of action and safety and toxicity has been conducted with standard models. Additionally, each diagnosis and therapy pursued are first tested in a structured case study design, a case series, and outcome study before moving onward to an RCT. In this way, the team utilizes the traditional knowledge base to develop the research questions and design both basic science and clinical studies. Design structure and outcome measures are then used that will satisfy scientists regardless of the continent on which they live.

Unlike their knowledge of recently formulated conventional drugs, these authors have the advantage of knowing the long history of use of many of the herbal formulas or processes in Ayurveda. However, we still collect adverse event data and report them (lack of them) so that those physicians who read the studies will have as much detail as possible about the potential benefit to their patients. Until there are many fully trained Ayurvedic physicians in the West, access to knowledge about the appropriate use of therapies and formulas may have to come from doctors initially trained in the West. It is best that we

become “bilingual”: Ayurvedic medical benefits should be presented both in the context of Ayurvedic practice and in a framework that the conventional physician will understand and ultimately feel comfortable using in his or her practice with appropriate patients.

3.6 Review of Quasi-Experimental Data

The clinical trial data in five monographs on epilepsy,¹¹ asthma,¹² hyperlipidemia,¹³ urinary diseases,¹⁴ and liver diseases¹⁵ generated by Central Council for Research in Ayurveda and Siddha, Ministry of Health, India, were reviewed and analyzed in this chapter for effectiveness of the following herbs and herbal formulas:

1. AYUSH 65 in epilepsy
2. *Naradeeya lakshmi vilasa rasa* in bronchial asthma
3. *Shwasa kesari* tablets in bronchial asthma
4. *Guggulu* in hyperlipidemia (short-term study)
5. *Guggulu* in hyperlipidemia/hyperlipoproteinemia (long-term study)
6. *Varuna* in urinary tract infection, benign prostatic hyperplasia (BPH), urolithiasis, and miscellaneous disorders
7. *Kulattha* in urolithiasis
8. *Varuna* and *Kulattha* in urolithiasis
9. *Kulattha* and *Gokshuru* in urolithiasis
10. *Katuki* in hepatocellular jaundice, chronic hepatitis, and cirrhosis
11. *Katukyadi* yoga in hepatocellular jaundice, chronic hepatitis, postinfective hepatitis, obstructive jaundice, chronic cholecystitis, and postcholecystectomy syndrome
12. *Kumariasava* in hepatocellular jaundice, chronic hepatitis, and obstructive jaundice
13. *Kumariasava* in hepatocellular jaundice, chronic hepatitis, and obstructive jaundice
14. *Daruharidra* in hepatocellular jaundice, chronic hepatitis, postinfective hepatitis, and obstructive jaundice

3.6.1 Clinical Data

Table 3.1 shows the rating of the data included in the monographs along two parameters: (1) quality assessment of the data using the Singh Scale for quasi-experimental designs and (2) safety assessment using the Safety Assessment Score for Controlled Clinical Trials® (SAS-CT). The Singh Q-E Quality Assessment Scale® scores 10 factors with 15 possible points. The 10 areas are the following:

1. Background and significance of literature review
2. Adequacy of treatment
3. Power calculation

4. Description of product and procedure
5. Description of sample selected
6. Description of outcome measures
7. Data report
8. Attention to possible bias in experimental design
9. Bias present in design or operationalization of design
10. Comparison of dropouts and competitors

Of the factors listed, 1, 2, 3, 5, and 6 have two points possible due to the specific subfactors listed (see [Table 3.1](#) for more details). The safety score was derived by using the SAS-CT system which queries six global issues with subfactors being scored to produce a possible point score of 100. As the clinical trials reviewed here were not controlled, the scores were modified for the safety assessment. Full scores were given if the scoring category was satisfied for the single group receiving the therapy. Table 3.1 gives both summed scores for each data set reviewed within the monograph for both scales and gives partials across factors for all data sets reviewed.

All authors and three research assistants acknowledged here participated in the systematic review process using both scales. Each article was primarily reviewed by two people with a third person as an arbitrator for reconciliation purposes. This mechanism guaranteed that four to five people reviewed each article in order to produce the scoring offered in the table.

The studies reviewed here with the Singh Q-E Quality Assessment Scale received full points on the factors of the background and significance of study, sample description, clarity of outcome measures utilized, and clarity of tables and data reported. Twelve of thirteen data sets utilized outcome measures with established, known, or reported reliability and validity. Ten of the thirteen articles were believed to be without operationalization flaws and presented the protocol in sufficient detail to allow for replication. Nine of the data sets optimized treatment to patients and seven of the thirteen justified the dosage of herbal treatments or therapies offered. Overall, this is quite a good showing. These results should encourage others to read the literature that is originating in India and other non-Western venues with the attitude that the data presented are going to be consistent with the quality of trials available in the West.

The SAS-CT scores were a bit more mixed. This factor is important, because people in the West are particularly interested in safety information on herbals and procedures that are less well known to them. Of the 13 studies reviewed, the range of scores was from 0 to 100. Four studies got perfect scores for reporting safety issues and five got no points. Other scores were 17, 21, 44.1, 47, and 88. It must be noted that an investigator should address the issues around safety: the occurrence of adverse events, the severity of them should they occur, whether they are likely related to the trial or not, and what the dropout rate is and were these due to adverse events minimally. For example, it is impossible to tell whether five articles did not report adverse events because none occurred, or the information was merely not reported. A simple sentence saying (1) that there were no adverse events and (2) there were no dropouts would go a long way to assist those reading data to assess safety. A reader cannot assume that a product is safe if there is nothing about adverse events mentioned; he or she can only wonder.

TABLE 3.1
Evaluation of Clinical Trials of Typical Ayurvedic Therapies

Quasi-Experimental Designs: Singh's Scale	Factors and Study Codes*													
	EPI	ASTH 1	ASTH 2	LIPID 1	LIPID 2	VARUNA	KUL A	V+ K	K+ G	KATUK I	KYOGA	ASAV	DAR U	MEAN
Background and significance or literature review														
Adequate knowledge of disease	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Adequate knowledge of treatment	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Adequacy of treatment														
Optimum dose, duration	1	1	1	1	1	0	0	0	0	1	1	1	1	0.69
Dose justification	1	1	1	0	0	0	0	0	0	1	1	1	1	0.53
Power calculation														
Number needed reported	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Sufficient number recruited	0	0	0	1	1	0	0	0	0	1	1	0	0	0.3
Description of product and procedure														
Sufficient for replication	1	1	1	1	1	1	0	0	0	1	1	1	1	0.76
Description of sample selected														
Demographics	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Diagnosis-related information	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Outcome meaasures														
Clearly stated	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Validity and reliability established	0	1	1	1	1	1	1	1	1	1	1	1	1	0.92
Data reported														
Report consistent with data tables etc.	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Attention to possible biases	0	0	0	1	1	1	0	0	0	0	1	1	1	0.46
No operationalization flaws	1	1	1	1	1	1	1	1	1	0	1	0	0	0.76
Comparison of dropouts vs. completers	0	0	0	1	1	0	0	0	0	0	0	0	0	0.15
Total	10	11	11	13	13	10	8	8	8	11	13	11	11	10.46

TABLE 3.1 (continued)

Evaluation of Clinical Trials of Typical Ayurvedic Therapies

SAS-CT (Safety Assessment Scores for Controlled Clinical Trials)	Factors and Study Codes*													
	EPI	ASTH 1	ASTH 2	LIPID 1	LIPID 2	VARUNA	KUL A	V+ K	K+ G	KATUK I	KYOGA	ASAV	DAR U	MEAN
AEs NR (adverse events not related)	27.0	0.0	27.0	27.0	27.0	0.0	0.0	0.0	0.0	0.0	0.0	27.0	27.0	12.5
ADRs (adverse drug reactions)	27.0	0.0	0.0	27.0	27.0	0.0	0.0	0.0	0.0	17.0	21.0	27.0	27.0	13.3
SAEs NR (serious adverse events not related)	13.5	0.0	0.0	13.5	13.5	0.0	0.0	0.0	0.0	0.0	0.0	13.5	13.5	5.2
SADRs (serious adverse drug reactions)	13.5	0.0	13.0	13.5	13.5	0.0	0.0	0.0	0.0	0.0	0.0	13.5	13.5	6.2
DOs AEs/SAEs NR (dropouts due to AEs and SAEs NR)	3.5	0.0	3.5	9.5	9.5	0.0	0.0	0.0	0.0	0.0	0.0	9.5	9.5	3.5
DOs ADRs/SADRs (dropouts due to ADRs and SADRs)	3.5	0.0	3.5	9.5	9.5	0.0	0.0	0.0	0.0	0.0	0.0	9.5	9.5	3.5
Total	88.0	0.0	47.0	100.0	100.0	0.0	0.0	0.0	0.0	17.0	21.0	100.0	100.0	44.1

*Study codes:

EPI = Effect of AYUSH 65 in epilepsy.

ASTH 1 = Effectiveness of *Naradeeya Lakshmi vilasa rasa* in bronchial asthma.ASTH 2 = Effectiveness of *Shwasa kesari* tablets in bronchial asthma.LIPID 1 = Effectiveness of *Guggulu* in hyperlipidemia (short-term study).LIPID 2 = Effectiveness of *Guggulu* in hyperlipidemia/hyperlipoproteinemia (long-term study).VARUNA = Effectiveness of *Varuna* in urinary tract infection, BPH, urolithiasis, and miscellaneous disorders.KULA = Effectiveness of *Kulattha* in urolithiasis.V+K = Effectiveness of *Varuna* and *Kulattha* in urolithiasis.K+G = Effectiveness of *Kulattha* and *Gokshuru* in urolithiasis.KATUKI = Effectiveness of *Katuki* in hepatocellular jaundice, chronic hepatitis, and cirrhosis.KYOGA = Effectiveness of *Katukyadi* yoga in hepatocellular jaundice, chronic hepatitis, postinfective hepatitis, obstructive jaundice, chronic cholecystitis, and postcholecystectomy syndrome.ASAV = Effectiveness of *Kumariasava* in hepatocellular jaundice, chronic hepatitis, and obstructive jaundice.DARU = Effectiveness of *Daruharidra* in hepatocellular jaundice, chronic hepatitis, postinfective hepatitis, and obstructive jaundice.

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4

Panchakarma Therapy in Ayurvedic Medicine

Ajay Kumar Sharma

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4.1 Introduction

One of the fundamental concepts of Ayurvedic management of diseases is to eliminate toxic materials (vitiated *dosas*) from the body in order to cure a disease. *Panchakarma therapy* (PKT) is designed to eliminate the toxic materials. It is postulated that the toxic materials of the body need to be eliminated radically before a palliative therapy is given. The palliative therapy in the form of drugs and diets may not be effective unless the body channels are properly cleansed and toxic materials are eliminated.

PKT is believed to purify or cleanse all the body tissues and to bring about the harmony of neurohumours (*tridosas*) (i.e., *vata*, *pitta*, *kapha*, and *manasa dosas* [i.e., *satva*, *raja*, and *tama*]) and to obtain long-lasting beneficial effects. PKT is not merely a therapeutic regi-

men, but it may be considered a management tool when used at certain tissue and body parts. It promotes and preserves the individual's normal health.

PKT¹⁻³ is an important component of Ayurvedic management of diseases. It is the comprehensive method of internal purification of the body through emesis (*vaman karma*), purgation (*virechana karma*), enema (*vasti karma*), errhines (*nasya karma*), and bloodletting (*raktamokshana*). This chapter will review the ancient classical concepts, traditional practices, and recent advances made in this important field with proper evaluation and rational assessment. Evidence obtained at our hospitals and at other research institutes in treating a variety of diseases with PKT is discussed along with its possible use as an adjunct to allopathic therapies.

PKT is indicated in arthritis, paralysis, neuromuscular diseases and in respiratory, gastrointestinal, ENT, and several blood-related disorders with great benefits. PKT is contraindicated in certain conditions like acute fevers, in various debilitating diseases, and in certain tumors and cancers of different organs; it is also contraindicated in children, the elderly, and pregnant women. PKT is indicated for both the healthy and diseased. The five elimination procedures are usually advised in the sequence of emesis, purgation, enema, errhines, and bloodletting, although it is not mandatory. Either one or all five procedures are advised as per the need and condition of the person undergoing PKT. Based on the health of the individual and stage and type of the disease, only one of the five procedures may be done without following a sequence. However, proper preparation and follow-up treatment are implemented for even one cleansing procedure.

The classical PKT is done in three stages:

1. Preparatory procedures (PREP) (*purva karma*) — These procedures are done to prepare the body to undergo a proper and thorough cleansing. They involve applying as well as ingesting oils and fats, sweating, and also advising which herbs to use to improve the digestion and metabolism in tissues.
2. Main cleansing procedures (MCP) (*pradhana karma*) — These procedures consist of five purification procedures especially designed to eliminate toxic materials from the imbalanced dosas of the body. They are emesis, purgation, enema, errhines, and bloodletting.
3. Postprocedures (*pashchata karma*) — These procedures consist mainly of recuperative measures in the form of diet, lifestyle changes, and rejuvenating herbs.

4.2 Panchakarma Therapy Procedures

4.2.1 Preparatory Procedures^{4,5}

PREP are used to facilitate PKT effectiveness. They include (1) digestive juice stimulants (*dipana*), (2) digestant (*pachana*), (3) oleation (*snehana*), and (4) sudation (*swedana*). All procedures are discussed below.

4.2.1.1 Digestive Juice Stimulants (*Dipana*) and Digestants (*Pacana*)

Digestive juice stimulants are agents that directly stimulate biofire (*agni*) and allow undigested food to be processed without stimulating digestive enzymes. Administration of

TABLE 4.1

List of Commonly Used Digestants and Digestive Stimulant Drugs

No.	Formula Name	Dose	References
1	<i>Panchakoladi churna</i>	1–3 g	BR ^a 30/35
2	<i>Hingvasshtaka churna</i>	1–3 g	BR 10/59
3	<i>Lavana bhaskara churna</i>	1–3 g	BR 10/79–87
4	<i>Chitrakadi vati</i>	500 mg to 1 g	BR 8/26,27
5	<i>Arka vati</i>	500 mg to 1 g	BR 857
6	<i>Agnitundi vati</i>	250–500 mg	BR 10/93,94
7	<i>Shunthi ghrita</i>	20–50 g	BR 29/200, 201
8	<i>Pippalyadi ghrita</i>	20–50 g	BR 5/1319–1322
9	<i>Dashmoolarishta</i>	20–30 ml	BR 74/357–371
10	<i>Drakshasava</i>	20–30 ml	BR 8/170–174

^aBR = Bhaishajaya Ratnavali.

digestants and digestive juice stimulants is an essential prerequisite of PKT; the objective is to improve the digestion both at the cellular and gastrointestinal tract level.

Normal digestion is achieved with the administration of medicated dehydrated butter (*ghee*) mixed with digestants and digestive juice stimulants. Dehydrated butter is a potent biofire stimulant agent. Commonly used digestants and digestive juice stimulants are described in Table 4.1.

Any preparation is usually administered for 3 to 7 days, depending on the age, disease, and condition of the patient. Signs and symptoms of satisfactory stimulation of digestion are (1) feeling of lightness in the body, (2) improved appetite, (3) feeling of thirst, and (4) well-formed stool without any mucus.

4.2.1.2 Oleation Therapy (*Snehana Karma*)^{4,5}

Any procedure or substance that increases the availability of lubricants, which produce lubrication in the body externally or internally, is called oleation therapy (OT). It is often used as an independent therapeutic procedure for disorders of *vata* as well as PREP for PKT. It is essential to administer OT to an individual before subjecting him or her to MCP to mobilize the toxic materials from their respective sites.

OT may be given externally by applying the oily materials on the skin or internally via ingestion, enema, or nasal route. External application consists of massage, application as a thin layer on the skin, application on the scalp, as ear drops, holding the oily material in the mouth for a few minutes, applying on the feet, etc.

4.2.1.2.1 Classification of Oleating Drugs and Agents

OT materials may be of animal or vegetable origin. Examples of animal origin materials include dehydrated butter, animal fat, bone marrow, fish oil, and milk. Vegetable origin materials include sesame oil and mustard oil.

The oleation substance selected for administration to the patient on the basis of a underlying disease is shown in Table 4.2. Some standard preparations used for internal oleation are listed in Table 4.3. These are to be swallowed orally with lukewarm milk or water in the prescribed dose as indicated in Table 4.3.

OT is indicated prior to sudation as PREP, dry skin, *vata* dominance, excessive loss of blood, and eye disorders. It is contraindicated in patients with aggravated *kapha* and all conditions where PKT is contraindicated.

Internally medicated ghee is given for 3 to 7 days at the break of dawn (6 to 7 A.M.), based on the person's constitution and digestive power (*agni* and *kostha*). Usually this is

TABLE 4.2

Selection of Oleating Materials Based on Underlying Diseases

No.	Oleating Drug	Disease Indicated
1	<i>Tikta ghita, mahatikta ghita, or vasa ghita</i>	Skin and blood disorders
2	<i>Kalyana ghita, brahmi ghita, mahakalyana ghita</i>	Mental disorders, epilepsy, insanity, etc.
3	<i>Maha sneha, pancatikta ghita</i>	Paraplegia

Note: Adjuvants are recommended along with oleating materials.

No.	Adjuvant	Oleating Material
1	Lukewarm water	In all types until specified; otherwise, dehydrated butter
2	Vegetable soups	Oils
3	Gruel made of rice	Fat, bone marrow

TABLE 4.3

Some Standard Preparations of Internal Oleation

No.	Disease	Oil/Dehydrated Butter	Dose	Reference
1	Bronchial asthma	<i>Manhanshiladi ghrita</i>	20–30 ml	CC ^a 17/145
2	Skin disease	<i>Mahatikta ghrita</i>	20–30 ml	CC 7/144–150
3	Psychosis	<i>Mahakalyanaka ghrita</i>	20–30 ml	CC 9/42–44
4	Epilepsy	<i>Panchagavya ghrita</i>	20–30 ml	CC 10/17
5	Cardiovascular accident (stroke)	<i>Bala taila</i>	20–30 ml	CC 28/148–156
6	Osteoarthritis, arthritis	<i>Eranda taila</i>	20–30 ml	CD ^b 25/6
7	Hepatobiliary disorders	<i>Draksha ghrita</i>	20–30 ml	BR ^c 12/135
8	Irritable bowel syndrome	<i>Shunthi ghrita</i>	20–30 ml	BR 8/555
9	Gynecological disorders	<i>Phala ghrita</i>	20–30 ml	BR 67/74–77
10	Tuberculosis	<i>Pipali ghrita</i>	20–30 ml	BR 14/237
11	Hemorrhagic diathesis	<i>Vasa ghrita</i>	20–30 ml	CC 4/88
12	Hemorrhoids	<i>Vyoshadya ghrita</i>	20–30 ml	BR 8/187
13	Anemia	<i>Pathya ghrita</i>	20–30 ml	CC 16/50

^aCC = *Charak Chikitsa*.

^bCD = *Chakradatta*.

^cBR = *Bhaishajaya Ratnavali*.

preceded with easily digestible and compatible food with plenty of fluids a day earlier in order to obtain proper oleation.

4.2.1.2.2 Massage Therapy (External Oleation Therapy)

External OT is achieved through massage and is done all over the body. For massaging the head, cold or lukewarm oil should be used. The rest of the body should be massaged with lukewarm oils. Massaging the body is to be done from 15 to 35 min. The massage should be carried out in seven different postures:

1. Sitting posture with extended legs
2. Lying down in supine posture
3. Lying down in left lateral position
4. Lying down on the front side of the body (over the back region)

5. Lying down in right lateral position
6. Lying down in supine posture
7. Sitting postures with extended legs from 2 to 5 min in each posture, depending upon the underlying disease

The massage should be done in a downward direction (i.e., away from the heart) over larger areas or big organs and in circular form over joints or the lower back.

Proper and regular practice of massage therapy results in the following benefits:

1. Improves eyesight
2. Induces sound sleep
3. Improves the quality of skin by making it tender, delicate, and strong
4. Corrects stiffness, rigidity, and induces elasticity in the body by softening the muscles, ligaments, and tendons
5. Relieves exertion, tiredness, and weakness due to various physical activities
6. Relieves and controls vitiated *vata dosa* in the body
7. Delays the aging process and strengthens the body

Massage therapy produces all these effects by lubricating the microcirculatory channels, displacing the exudates which may relieve tension and pain, and preparing smooth passages (microchannels) for elimination of toxic materials during sudation therapy.

The standard preparations used for massage therapy are described below:

1. Oils used for head massage
 - a. *Candanadi taila* (Bhaishajaya Ratnavali [BR] 14/292–295)
 - b. *Brahmi taila*
 - c. *Jyotismati taila*
2. Oils used for body massage
 - a. *Vatika* disorders
 - i. *Narayana taila* (Chakradatta [CD] 22/120–130)
 - ii. *Bala taila* (BR 26/273–283)
 - iii. *Dasmula taila* (BR 65/89)
 - b. *Paittika* disorders
 - i. *Kshira Bala taila*
 - ii. *Candanadi taila*
 - iii. *Candan Balalaksadi taila*
 - c. *Kaphaja* disorders
 - i. *Sahacaradi taila* (BR 61/134–135)
 - ii. *Dasmula taila*

Oleating materials given alone without any other carrier substances are also effective in providing oleation. In certain cases, some other substances such as rice, fruit juices, soup, curd, buttermilk, and lentils are added.

The following instructions are given to patients to maintain equilibrium of physiological factors during and after OT:

1. Avoid day sleeping.
2. Avoid a coarse and constipating diet.
3. Drink warm water.
4. Abstain from sexual activities.
5. Suppress normal urges.
6. Avoid speaking loudly.
7. Avoid anger and anxiety.
8. Avoid exposure to heat and cold.
9. Avoid walking in open air.
10. Avoid exercise and hard work.

OT is done to obtain maximum benefit. Either excess or inadequate oleation may cause problems. Features of adequate oleation include improved digestion, lightness of the body, oiliness of the skin, neurasthenia, lethargy, semisolid stool, and visual fat in the stool. OT is considered inadequate if the patient passes dry and hard stools and presents with features like movement of flatus upward (bumping), weak digestive power, burning sensation in chest, and dryness of skin. Features of excessive oleation are heaviness and stiffness in the body, drowsiness, anorexia, nausea, dyspepsia, excessive salivation, and frequent evacuation of mucoid stools.

The possible complications of OT include indigestion, nausea, anorexia, gastrointestinal tract upsets, drowsiness, stiffness, skin diseases, anemia, edema, and piles. These complications can be easily treated by frequent use of lukewarm water externally and internally, with the administration of herbs like *Piper longum*, *Terminalia chebula*, *Terminalia belarica*, *Embllica officinalis* and *Comiphora mukul*, honey, and symptomatic treatment.

4.2.1.3 Sudation and Fomentation Therapy (Svedana Karma)^{4,5}

The therapeutic production and induction of sweat by a variety of methods is termed sudation therapy (ST). It relieves stiffness, heaviness, and coldness of the body and induces sweating. It is administered to liquefy the oleated toxic materials (brought about by OT), which are spread throughout the body, and direct them to the alimentary canal for elimination by any one of the four cleansing procedures.

ST is administered after OT and it precedes emesis therapy in the sequence of PKT. Besides being the principal PREP of PKT, ST may also be a specific treatment for a number of disorders, especially in *vata*-dominant diseases where it may be a main treatment.

4.2.1.3.1 Types of Sudation Therapy

ST may be applied with the use of direct application of heat (e.g., hot bed, affusion, steam kettle, sudatorium, etc.) or indirect application of heat (e.g., exercise, wearing heavy clothes, exposure to sun rays, etc.). It may be used on the basis of properties of drug used, like unctuous or wet ST and dry ST, which are indicated in *vata* and *kapha*-dominant diseases, respectively. ST may be applied either to one part of the body as a localized ST or to the whole body as a generalized ST.

ST is indicated in patients of various types of paralysis, musculoskeletal disorders, coryza, cough, stiffness, and need for undergoing PKT. ST is contraindicated in patients

with hemorrhages, diarrhea, eruptive skin diseases, alcoholism, pregnancy, and in patients of *pitta*-dominant constitution or who are emaciated.

4.2.1.3.2 Procedure of Sudation Therapy

ST must be performed after proper OT. In localized sudation, steam vapors of decoction of dashamula (or any other drug) are to be used on the affected parts. In generalized ST, medicated vapors are made to pass all over the oleated patient's body. ST should be done for a maximum of 30 to 45 min. Medicated vapors are generated by boiling the herbs in water.

The process of ST should be stopped when sweat appears over the forehead and there is relief from pain, stiffness, and heaviness. ST is considered inadequate if the patient does not produce any sweating and there is no relief from pain or coldness. A burning sensation of the body, fainting, vertigo, and appearance of blisters may occur if ST is done more than advised.

During and after ST, the blood pressure, pulse rate, respiratory rate, and temperature are monitored to make sure they are within normal limits. Direct exposure to cold water or cold air should be avoided at least for 1 h, and a bath in lukewarm water is indicated after ST. Generally, emesis should be given on the second day after ST and purgation after about a 3-day interval.

The standard preparations used for ST (for producing medicated steam) are:

1. *Dashamula kwatha* (decoction) — Sushruta Samhita Sutra Sthana 39/68–69
2. *Rasanasaptaka kwatha* (decoction) — CD Amavata– 8

4.2.1.3.2.1 *Physical Effects of Sudation Therapy* — Various procedures of ST therapeutically induce sweating, which cleanses microchannels, liquefies toxic materials, and expels toxic materials along with sweat. It recovers vascular insufficiency of the joints and muscles and produces relaxation. These factors might be responsible for improving blood circulation and local metabolic processes, causing relaxation of local structures and producing relief of local symptoms, functional recovery, and a slowdown of the disease process being treated. These therapeutic measures are classically *antivata* and relieve all *vata* manifestations and control the disease process. During ST, the body temperature rises to more than 2 to 3°C, which results in increased sweating.

4.2.1.3.3 Review of Scientific Data on Oleation, Massage, and Sudation Therapies^{6–11}

R.H. Singh and G.N. Chaturvedi at Banaras Hindu University (BHU), Varanasi, studied the effect of internal oleation clinically^{7,10} in patients with hyperchlorhydria and peptic ulcer after internal administration of *dadimadya ghrita* as a palliative therapy. They observed significant clinical improvement in these patients, including a decrease in gastric acid secretions and evidence of healing ulcers confirmed on radiological examinations. Other researchers also tried internal oleation with different *ghrita* preparations for peptic ulcers and reported similar observations.¹¹

Scientific studies conducted on the role of oleation and sudation on a series of patients of rheumatoid arthritis and other chronic arthropathies by R.H. Singh at BHU, Varanasi, are reproduced here.^{8,9} Similar studies have been conducted by this author and his team on rheumatoid arthritis at the National Institute of Ayurveda, Jaipur, and similar observations were recorded.⁶

Another clinical trial tested the role of *Pinda Sweda*, a type of preparatory procedure, as a main therapeutic procedure in rheumatoid arthritis, in which application of oil over the affected part is followed by fomentation with a hot rice paste cloth bundle. At the end of

TABLE 4.4

Herbs Used in PKT as Emetics

Medicinal Plant	Botanical Name
<i>Madanaphala</i>	<i>Randia dumatorium</i>
<i>Madhuka</i>	<i>Sapotaceae</i>
<i>Kututumbi</i>	<i>Lageneria vulgaris</i>
<i>Nimba</i>	<i>Azadirachta indica</i> A.
<i>Bimbi</i>	<i>Coccinia grandis</i> (L.) Voight
<i>Visala</i>	<i>Citrullus colocynthis</i> Schrad
<i>Trapusa</i>	<i>Cucumis sativus</i> Linn.
<i>Kutaja</i>	<i>Holarrhena antidysenterica</i> Linn.
<i>Murca</i>	<i>Celosia cristata</i> Linn.
<i>Devdali</i>	<i>Luffa echinata</i>
<i>Vidanga</i>	<i>Embelia ribes</i> Burm
<i>Viduli</i>	<i>Salix caprea</i> Linn.
<i>Dahana</i>	<i>Toddalia asiatica</i> Linn.
<i>Citra</i>	<i>Ipomea remiformis</i> Choisy
<i>Kosavati</i>	<i>Luffa aegyptica</i> Mill-Hock
<i>Karanja</i>	<i>Pongamia pinnata</i> Pierre

30 days the researchers observed significant changes in both clinical symptoms, such as stiffness and pain, as well as in laboratory tests such as erythrocyte sedimentation rate and urinary hydroxyproline. Improved vascularity at the joints was also observed when the pre- and postangiograms were compared.

4.2.2 Main Cleansing Procedures (*Pradhana Karma*)

4.2.2.1 *Biopurificatory Therapeutic Emesis — Emesis Therapy (Vamana Karma)*¹²

Emesis therapy (ET) is a process by which the contents of the stomach including *kapha* and *pitta dosas* are expelled out of body through the mouth. It is one of the MCPs that eliminates toxic materials from upper parts of the body.

Herbs used in PKT as emetics are detailed in Table 4.4.

ET is indicated for patients of asthma, respiratory disorders, sinusitis, rhinitis, anorexia, dyspepsia, peptic ulcers, and skin diseases and in healthy individuals in different states where *kapha dosa* is aggravated within normal limits (e.g., in the spring season for preserving normal health and preventing diseases). It is contraindicated in patients of hematemesis, cardiovascular diseases, and cachexia and in children and the elderly.

4.2.2.1.1 *Procedure of Emesis Therapy*

ET is usually the first main procedure of PKT done after a proper preparation of the body. After the PREP, OT and ST are done; the out-of-balance *kapha* is further increased in the body by giving oily food, fatty food, animal meat, milk, and curd for the next 24 h before the induction of emesis. On the morning of ET (between 7 and 8 A.M.), a gruel mixed with ghee in large quantities or 2.5 l of milk or decoction of *Glycyrrhiza glabra* or sugarcane juice is given to the patient. After 10 min, emesis is induced by administering a certain combination of emetic herbs made into a paste mixed with ghee. Milk or decoction of *Glycyrrhiza glabra* or sugarcane juice is used as a vehicle to push the paste down.

TABLE 4.5

Criterion of Assessment of Purification of the Body by Emesis Therapy

No.	Parameters	Active Purification	Moderate Purification	Mild Purification
1	No. of bouts	8	6	4
2	Total output and input	Output > input	Output > input	Output < Input
3	Contents vomited	Bile and <i>pitta</i>	Bile and <i>pitta</i>	Bile and <i>pitta</i>
4	Signs and symptoms	To be observed for adequate emesis	To be observed for adequate emesis	To be observed for adequate emesis

Items commonly used in standard preparation are:

1. Powder seeds of *Randia dumatorum* (6 to 10 g [4 parts])
2. Powder of *Acorus calamus* (3 to 5 g [2 parts])
3. Rock salt (3 to 5 g [1 part])
4. Honey (20 ml)

Thoroughly mix the preparations and make them into a paste. Mix in 100 ml of warm milk or lukewarm water and ask the patient to swallow. If the patient is getting an urge to vomit, instruct him or her to do so. Start counting the number of bouts of vomiting and observe its contents.

4.2.2.1.2 Criteria for Proper Emesis Therapy

A bout is the number of times the emetic drug is vomited out after its administration to the patient. Adequacy of emesis is assessed in terms of the number of bouts, ratio of total output/input, its contents, and signs and symptoms produced after ET. A proper emesis will have the sequence of expulsion of saliva, drug, *kapha*, *pitta* (yellowish bile), and *vata dosa* as the end point. The criteria of assessment for effective ET are described in Table 4.5.

After completing emesis, the patient is asked to inhale smoke from medicated cigars prepared with specific drugs that expel residual *dosas*, toxic materials, *kapha*, and unctuousness in the throat, mouth, nose, and microchannels. The patient is advised to avoid speaking loudly, overeating, physical exertion, anger, and anxiety during and after the therapy to maintain normal physiology of the body and avoid complications.

The persistent urge to vomit, heaviness in the body, itching, and abdominal pain may occur in the case of an incomplete emesis of gastric contents. Complications such as hematemesis, frothy vomiting, fainting, cardiac pain, clouding of vision, throat pain, and weakness may occur in uncontrolled excessive vomiting. The physician should monitor the subject constantly and stop the whole process at the right time.

4.2.2.1.3 Review of Scientific Data on Emesis Therapy

A clinical trial for assessing the role of ET in 30 patients with bronchial asthma was carried out by this author and his team on various scientific parameters. After the completion of the ET, it was observed that there was significant improvement in clinical symptoms, including marked improvement in breath-holding time, vital capacity, and respiratory rate. Similarly, statistically significant changes ($p > 0.05$) were observed in ESR, total serum proteins, serum electrolytes, lipid profile, and radiological changes.

TABLE 4.6

Herbs Used in PKT as Purgatives

Medicinal Plant	Botanical Name
<i>Kutaki</i>	<i>Picrorrhiza kurrora</i>
<i>Eranda</i>	<i>Ricinus communis</i>
<i>Triphala</i> (<i>haritaki</i> , <i>bibhitaka</i> , <i>amalaki</i>)	3 fruits = <i>Terminalia chebula</i> , <i>Termanilia belerica</i> , <i>Embelica officenalis</i>
<i>Gavaksi</i>	<i>Citrullus colocynthis</i> schrad
<i>Snuhi</i>	<i>Euphorbia nerifolia</i> Linn.
<i>Trivritta</i>	<i>Operculina turpenthum</i>
<i>Nilini</i>	<i>Indigofera tinctoria</i> Linn.
<i>Tilvaka</i>	<i>Locos racemosa</i> roxb.
<i>Aragvadha</i>	<i>Cassia fistula</i> Linn.
<i>Kampillaka</i>	<i>Mallotus philippinensis</i>

TABLE 4.7

Dose Schedule for Various Purgatives^a

No.	Drug of Choice	Soft Bowel	Medium Bowel	Costive Bowel
1	<i>Ricinus communis</i> oil	5–20 ml	20–50 ml	50–100 ml
2	<i>Operculina turpenthum</i> powder	1–3 g	3–6 g	5–10 g
3	<i>Vitis vinifera</i> , <i>cassia fistula</i> , <i>terminalia chebula</i> decoction	10–20 ml	25–50 ml	50–100 ml
4	<i>Croton tiglium</i> , <i>Euphorbia</i> <i>nerifolia</i>	60–125 mg	125–250 mg	500–1000 mg
5	<i>Pluntago ovata</i>	3 g	3–6 g	6–12 g

^aPurgatives are based on practical experiences.

All the patients reported relief in heaviness in the chest, expectoration, cough, breathlessness, and a feeling of well-being and lightness in the chest immediately after emesis. The success rate of ET was approximately 75% in cases of bronchial asthma.

4.2.2.2 Purificatory Purgation — Purgation Therapy (*Virecana Karma*)

Purgation therapy (PT) is a specific process for elimination of *pitta dosa*. PT procedure involves elimination of *pitta* dominating *dosas* and toxins of the body through the rectal route.

PT is indicated mainly in patients suffering with hemorrhage from the upper parts of the body, poisoning, chronic jaundice, various gastrointestinal tract disturbances, asthma, skin disorders, epilepsy, insanity, and other *pitta* disorders. PT is contraindicated in patients with anorectal injury, prolapse of the rectum, bleeding from lower parts of the body, diarrhea, emaciating chest diseases, excessive oleation, and after enema. It is also contraindicated in children, the elderly, and patients who are weak, tired, have fasted, pregnant women, or persons desirous of coitus and pregnancy.

4.2.2.2.1 Procedures of Purgation Therapy

Internal oleation is done prior to PT. A diet containing fatty materials, liquid, warm/hot liquid or solid food, and meat soups is given to the patient. General instructions to be followed in ET are also to be followed in PT.

The dose of the drug to be administered for purgation is decided according to both the nature of the purgative drug and the patient, as described in Table 4.7. The purgative recipe is administered at least 2 h after sunrise (between 7 and 8 A.M.). During PT, the

TABLE 4.8

Assessing Success of Therapeutic Purgation on the Basis of Clinical Manifestations

No.	Parameters	Good Purgation	Medium Purgation	Mild Purgation
1	No. of bouts	15–30	10–15	5–10
2	Quantity of feces in liters	1.5–2	1–1.5	0.05–1
3	Order of contents	Mucin/ <i>kapha</i>	Mucin	Mucin
4	Symptoms	Signs and symptoms as per textual description	Signs and symptoms as per textual description	Signs and symptoms as per textual description

stomach should preferably be empty, in contrast to emesis where the stomach must be full. Depending on their mode of action and degree of purgation produced, purgative drugs may be grouped as mild purgatives (e.g., *Operculina turpenthum* root), moderate purgatives (e.g., *Cassia fistula*), and drastic purgatives (e.g., *Euphorbia nerrifolia* milk). Various herbs and drugs used in PKT as purgatives are described in Table 4.6.

4.2.2.2.2 Criteria for Proper Purgation Therapy

The patient should be assessed for proper purgative effect as shown in Table 4.8. A proper purgation induces a feeling of lightness in the body and abdomen with improved appetite. The sequence of expulsion in proper PT is urine, stool, *pitta*, drug, *kapha*, and *vata*. If the patient has dyspepsia, heaviness in the abdomen or body, vomiting, and constipation, this suggests inadequate purgation. Signs and symptoms of excessive purgation include pain in the abdomen, blood-mixed serous discharge through the anus, syncope, weakness, and drowsiness. The physician should constantly monitor the subject and stop the whole process at the right time.

4.2.2.2.3 Review of Scientific Data on Purgation Therapy^{14–16}

A clinical trial for assessing the role of PT in 30 patients with hepatitis (jaundice) was carried out by this author and his team on various scientific parameters. After the completion of the PT, it was observed that there was significant improvement in clinical symptoms such as anorexia, pruritis, nausea, and abdominal discomforts. There were significant changes in laboratory tests such as ESR, 24-h stercobilinogen, and D-xylose excretion.

All the patients reported relief in the feeling of lightness in the body and abdomen and improved appetite. The success rate of PT was approximately 85% in cases of hepatitis (jaundice).

After purgation, these patients were put on a short course of posttherapy dietetic regimen for 2 to 3 days. Within 7 days after purgation, these patients showed notable symptomatic improvement ($p < 0.05$). The D-xylose excretion test applied on these patients within a week of completion of the total procedure showed a statistically significant ($p < 0.001$) increase in D-xylose excretion rate, indicating improved absorption power, which determines the rate of purification of microchannels in these patients.

4.2.2.3 Biopurificatory Therapeutic Enema — Enema Therapy (*Vasti Karma*)¹⁷

Enema therapy (ENT) is a procedure in which medicated oils, decoctions, and decoctions with pastes of herbs or oils are introduced into the large intestines through the rectum with the help of an enema apparatus.

TABLE 4.9
Herbs Used in PKT to Make Decoctions for Enema

Medicinal Plant	Botanical Name
Madanaphala	<i>Randia dumatorum</i>
Kutaja	<i>Holarrhena antidysenterica</i> Linn.
Kustha	<i>Saussurea costus</i>
Devadali	<i>Luffa echinata</i>
Vaca	<i>Acorus calamus</i>
Dasmula (group of ten drugs: bilva, agnimantha, syonaka, kasmarya, patala, saliparni, prsniparni, brhati, kantakari, goksura)	<i>Aegle marmelose</i> , <i>Premna integrifolia</i> , <i>Oroxylum indicum</i> , <i>Gmelina arboria</i> , <i>Stereospermum suaveolens</i> , <i>Desmodium gangeticum</i> , <i>Ureria picta</i> , <i>Solanum indicum</i> , <i>Solanum surrattense</i> , <i>Tribulus terrestris</i>
Devadaru	<i>Cedrus deodaru</i> (Roxb)
Rasna	<i>Acampe papillosa</i>
Yava	<i>Hordeum vulgare</i>
Methi	<i>Foeniculum vulgare</i> Mill
Krtavedhanam	<i>Luffa acutangula</i> Roxb
Kulattha	<i>Dolichos biflorus</i> Linn.
Saindhava lavan	Rock salt
Madhu (honey)	Honey

TABLE 4.10
Proposed Quantity of Decoction-Based Enema According to Various Age Groups

No.	Age	Maximum Quantity of Decoction-Based Enema
1	1 year	50 ml
2	10 years	400 ml
3	15 years	900 ml
4	18–70 years	1200 ml
5	Above 70 years	1000 ml

Note: The dose of oil-based enema according to age of the patient is recommend. Examples are *Sneha vasti* – 300 ml, *Anuvasana vasti* – 150 ml, and *Matra vasti* – 75 ml.

4.2.2.3.1 *Classification of Enema Therapy*

ENT can be classified in different groups on the basis of drugs that are used in an enema, such as (1) a decoction-based enema in which drugs used for the enema contain decoction in a larger quantity, or (2) an oil-based enema in which oil is the main ingredient. Usually, a combination of decoction and oil enema is given; decoction alone is not indicated at any time. According to therapeutic actions of the drugs, the enema can be an oleation enema that produces oleation of the body, a roborant enema that improves strength and general status of health, a purifying enema that removes toxic materials from the body, or a depleting enema that reduces fat content of the body tissue leading to sound health.

ENT may also be classified according to the number of enemas administered during the full course of therapy:

- 1. *Karma vasti* — Total of 30 enemas, 12 decoction-based and 18 oil-based enemas administered on alternate days
- 2. *Kala vasti* — Total of 16 enemas, 6 decoction-based and 10 oil-based

TABLE 4.11

Two Types of Topical Errhines (*Nasya*)

No.	<i>Pratimarsha Nasya</i> (Topical)	<i>Marsha Nasya</i> (Topical)
1	Never produces complications	May produce complications
2	Dose: 2 drops B.D.	Dose: Maximum – 10 drops Moderate – 8 drops Minimum – 4 drops
3	Indicated in all seasons and all age groups	Needs consideration of seasons and age
4	Least oleus material used	Excess oleus material used
5	Slow acting and less potent	Quick action and highly potent

3. *Yoga vasti* — Total of 8 enemas, 3 decoction-based and 5 oil-based

Various herbs used in PKT to make decoctions for enemas are described in [Table 4.10](#).

ENT is indicated in various gastrointestinal tract disorders, helminthiasis, urogenital disorders, lithiasis, neuromuscular disorders, articular diseases, venereal diseases, convulsions and paralytic disorders, and anorectal disorders. ENT is contraindicated in patients with intestinal obstruction, perforation, ascites, cholera, dysentery, anal inflammation, anemia, and anasarca.

4.2.2.3.2 Procedure of Enema Therapy

A decoction-based enema is given on an empty stomach between 5 and 7 P.M. after subjecting the patient to gentle massage and mild fomentation. An oil-based enema is administered 15 to 30 min after having a light diet in the evening. In both types of ENT, the quantity of enema is decided according to the schedule as described in Table 4.11.

Duration of elimination of the enema should be a maximum of 48 min. If expulsion does not occur within the stipulated time, the following complications may occur: tympanitis, distention of abdomen, painful abdomen, renal colic, reverse peristalsis, and pressing pain in the chest region; these complications should be managed accordingly.

Proper release of fecal matter and flatus, a feeling of lightness in the abdomen, suggests proper ENT. It is considered inadequate if the patient has few urges for evacuation, gripping pain, flatulence, and dyspnea.

After resting, the patient may take a bath with warm water. Lukewarm water, milk, gruel made of rice, meat soup, or another light diet is given. Excess of physical and mental exertion is avoided during ENT.

4.2.2.3.3 Uttara Vasti

Uttara vasti is the means by which the drugs of enema are made to pass through the penis or vagina or through extra genitalia into the urinary bladder or uterus. A dose of *uttara vasti* is 20 to 40 ml. The remaining description, indications, contraindications, and mode of administration of *uttara vasti* are similar to decoction-based or oil-based enemas.

4.2.2.3.4 Review of Scientific Data on Enema Therapy^{18,19}

A clinical trial for evaluating the efficacy of ENT was conducted on 30 patients with rheumatoid arthritis with scientific parameters. Patients were subjected to a course of *Vaitarana vasti* (a type of decoction-based enema, *Chakradatta*, Chapter No. 73/32) for 30 days.

TABLE 4.12

Standard Preparations Commonly Used for Errhine Therapy

No.	Drug	Dose (Purification)	Reference
1	<i>Anu taila</i>	6 drops in each nostril	CS ^a 5/63–70
2	<i>Rasnadi taila</i>	6 drops in each nostril	CC ^b 26/160
3	<i>Shadbindu taila</i>	6 drops in each nostril	BR ^c 65/83
4	<i>Vyagghri taila</i>	6 drops in each nostril	BR 63/30
5	<i>Pathadi taila</i>	6 drops in each nostril	BR 63/29
6	<i>Apamarga taila</i>	6 drops in each nostril	BR 65/113

^aCS = *Charak Sutra*.^bCC = *Charak Chiikitsa*.^cBR = *Bhaishajaya Ratnavali*.

Significant changes such as stiffness, pain, and swelling, as well as laboratory tests like ESR, were observed in both clinical symptoms. A decrease in the rate of urinary excretion of hydroxyproline and a trend of restoration of adrenocortical functions, indicating remission of the disease process with control of connective tissue breakdown, were also noted. After the course of the therapy, patients showed statistically significant ($p < 0.001$) increase in D-xylose excretion rate, indicating significant improvement in GIT absorption capacity, clearly indicating the cleansing effect of microchannels after the MCP (i.e., ENT). Scientific study on a current series of patients with rheumatoid arthritis showed relief in symptoms in 70% of the patients.

4.2.2.4 Errhine Therapy (*Nasya Karma*)

Errhine therapy (ErT) refers to administration of medicines in various forms through the nostrils (i.e., instilling medicated oil in the nose or administering paste, powder, or fumes of errhine drugs in the nostrils). There are two types of ErT: (1) *Pratimarsha nasya* and (2) *Marsha nasya* (Table 4.11). ErT should be administered to patients who are 7 to 80 years old. ErT is indicated in various diseases of the supraclavicular region such as stiff neck and jaw, headache, migraine, graying of hair, baldness, facial palsy, aphonia, stammering or alteration of voice, hoarseness of voice, corrhyza, tonsillitis, sinusitis, rhinitis, and earache. ErT should be avoided in patients who have acute fevers or acute corrhyza or have had purgation or oleus enema, had fasted, and had indigestion. There are five methods of ErT:

1. Inunction (*navana*)
2. Instillation of nasal drops (*avapeedana*)
3. Insufflation (*dhamapana*)
4. Inhalation (*dhuma nasya*)
5. Topical application (*marsha* and *pratimarsha*)

When the oil is being instilled, the head should not be shaken; the patient should not lose his or her temper, speak, sneeze, or laugh. The unctuous material should not be swallowed and should be expectorated properly so that no part of it remains inside. The patient should be instructed to avoid talking, becoming angry, laughing, and swallowing of errhine drugs during the ErT procedure.

Standard preparations commonly used for ErT are listed in Table 4.12, and herbs used in PKT for nasal medication (*nasya*) are described in Table 4.13. The dose of ErT should be determined according to Table 4.14.

TABLE 4.13

Herbs Used in PKT for Nasal Medication (*Nasya*)

Medicinal Plant	Botanical Name
<i>Vidanga</i>	<i>Embelia ribes</i> Burm
<i>Apamarga</i>	<i>Achyranthes aspera</i> Linn.
<i>Vyosa</i> (3 pungents: <i>sunthi</i> , <i>pippali</i> , <i>marica</i>)	<i>Zinziber officenalis</i> , <i>Piper longum</i> , <i>Piper nigrum</i>
<i>Darvi</i>	<i>Berberis aristata</i>
<i>Surala</i>	<i>Vateria indica</i> Linn.
<i>Bija of sirisa</i>	<i>Albizia lebbek</i>
<i>Brihati</i>	<i>Solanum anguivi</i> Lam.
<i>Shigru</i>	<i>Moringa oleifera</i>
<i>Madhusarkara</i>	<i>Dolichos biflorus</i> Linn.
<i>Saindhava lavana</i>	Rock salt

TABLE 4.14

Determination of Dosage in Errhine Therapy

No.	Type of Errhine Therapy	Mild Purification	Moderate Purification	Maximum Purification
1	Snuffing errhine therapy	8 drops in each nostril	16 drops in each nostril	32 drops in each nostril
2	Purificatory errhine therapy	4 drops in each nostril	6 drops in each nostril	8 drops in each nostril
3	Blowing errhine therapy	250 mg	375 mg	500 mg
4	Topical (<i>marsa nasya</i>)	6 drops in each nostril	8 drops in each nostril	10 drops in each nostril
5	Topical (<i>pratimarsh nasya</i>)	2 drops in each nostril	2 drops in each nostril	2 drops in each nostril
6	Paste (<i>kalka</i>)	4 drops in each nostril	6 drops in each nostril	8 drops in each nostril
7	Smoke (<i>dhumapana</i>)	1–2 min	2–3 min	Up to 5 min

Adequate ErT produces lightness in the body and head and sensorial happiness. ErT is inadequate if the patient has excessive secretions from the nose, eyes, and mouth and heaviness in the body. Complications such as headache, confusion, and salivation may occur in uncontrolled excessive ErT. The physician should constantly monitor the subject to avoid complications.

After ErT; the patient is given mild sudation over the throat, cheeks, face, and forehead and a soft massage on the feet, palms, and back of the neck; hot water gargling of the mouth and medicated smoking (*dhumapana*) is advised to the patient to clear the mouth and throat of residual *kapha dosa*.

4.2.2.4.1 Review of Scientific Data on Errhine Therapy²⁰

Clinical efficacy of ErT was evaluated by the author on 30 patients with chronic sinusitis. At the end of 21 days, after administration of ErT with *Anu Taila*, the researchers noted significant changes such as a growing feeling of well-being, sneezing, nasal secretions, and headache in both clinical symptoms, as well as in laboratory findings such as ESR and radiological findings.

There was a statistically significant ($p < 0.001$) reduction in TLC and ESR. On radiological examinations there was a marked reduction in the haziness of sinuses along with restoration of normal mucosa. Nasal passages and sinuses were thoroughly cleansed after a course of therapy, indicating suppression of the disease process.

In the author's clinic, the success rate of ErT in a current series of patients with chronic sinusitis is about 90%.

4.2.2.5 Bloodletting Therapy (Rakta Mokshana)

Bloodletting therapy (BLT) may be performed with or without the help of metallic instruments. In the latter type, BLT may be administered by application of leech, cow's horn, dried bitter gourd, or coupling glass.

BLT is practiced to remove toxic materials in blood in blood-borne diseases; in *pitta*-predominant diseases; and also in a few *vata* disorders like erysipalis, boils and carbuncles, abscesses, blue and black pigmentation on the face, moles, eczema, leucoderma and vitiligo, scabies, red patches, anorectal inflammation, splenomegaly, jaundice, dyspepsia, anorexia, stomatitis, halitosis (foul-smelling mouth), gingivitis, and gout. It is contraindicated in patients with bleeding disorders, general anasarca, cachexia, anemia, piles, and all conditions where venepuncture is contraindicated.

4.2.2.5.1 Bloodletting Therapy with the Application of Leeches

The patient should be properly screened before leech therapy can be given. Before the BLT is done with a leech, oleation and sudation should be performed the day before. It can be done localized or generalized, depending upon the condition, and the part that is to be leeches should be dried. Then the purified leech is applied to the diseased part. The leech's mouth is covered with a smooth, white, moistened cotton swab. When the leech starts opening its mouth, which is shaped like a horse's hoof, and raising the shoulders, the leech is sucking the blood. If the leech does not start sucking the blood, a drop of milk or blood is placed over the affected part and the leech will start sucking the blood. If at the biting site needling pain and itching occur, then the leech is sucking pure blood. The leech is then removed from the site by pouring rock salt at its mouth. With the removal of vitiated blood after leech therapy, the redness and pain subside immediately.

After the leech is detached, the blood from the sucked area should be allowed to flow for 1 to 3 min and then dressing either with turmeric powder and alum powder or antiseptic solution is applied. Dressing should be kept in place for 6 to 12 h.

4.2.2.5.2 Bloodletting Therapy by the Venesection

Venesection is the process of cutting open a vein under strict aseptic conditions. It is done with sharp instruments and is a more severe form of bloodletting than the application of leeches. When the venesection is done properly, the blood (which is impure) flows automatically. Adequate venesection is considered if the patient feels lightness in the body, a decrease in pain, and a remission of symptoms. Signs of inadequate venesection include swelling, burning sensation, inflammation, pain, itching, and redness at the site of venesection.

4.2.2.5.3 Bloodletting Therapy by Superficial Wounding

Superficial incisions are made over the skin with the help of a scalpel, fine needles, or instruments under strict aseptic conditions. BLT with this method is commonly used in localized blood disorders and in certain skin disorders. It must be done after taking aseptic measures. Antiseptic dressing should be applied over the wound.

BLT improves the collateral circulation of affected parts and cleanses the microcirculatory channels directly by removing toxic materials and exudating from the affected parts. Hirudin, the chemical substance present in the saliva of a leech, has a potent anticoagulant property. It induces bleeding and encourages free flow of blood, removing the virulent

substances present at the site of the lesion and helping to restore healthy blood supply. Hirudin checks the inflammatory process especially at specific points where bloodletting is done, which produces rapid symptomatic improvement.

4.3 Beneficial Effects of *Panchakarma* Therapy²¹

The beneficial contribution of PKT is that it removes the toxic materials from the body and provides purification of the body at two levels: (1) the gross level, where various organs and systems of the body are thoroughly cleansed (e.g., cardiovascular system, gastrointestinal tract, chest, etc.); and (2) the cellular level, where purification and cleansing of the body is produced at the level of cells, cell membranes, and molecules. PKT helps bring the whole body to normalcy; starts rejuvenation and revitalization of all body tissues; potentiates the pharmacological actions of various drugs and medicines administered after PKT; removes waste products, unwanted materials, various toxins, and stagnant *dosas*; and potentiates physiological functions of all the body systems (e.g., gut absorption improves considerably and metabolism is also corrected). The prognosis of various diseases that are difficult to treat with simple administration of medicines becomes significantly improved (good prognosis) after administration of PKT. PKT not only is a prerequisite for all the therapeutic procedures and medications but also has a full therapeutic role in promoting preventive, curative, and rehabilitative procedures.

If properly performed, PKT does not produce any serious complications. If any minor complications are produced, they are easily manageable. Specialized techniques of PKT are simple to perform and can be carried out at both the outdoor patient and indoor patient (OPD and IPD) levels. There is no need to carry out all PKT practices at one sitting; specific techniques of PKT (any of the MCP) can be recommended to the patient after proper preparation. The toxic materials are eliminated from the body mainly through the alimentary canal.

4.4 Conclusions

Any type of main cleansing procedure of PKT is believed to affect the cleansing of microcirculatory channels by eliminating the toxic metabolites from the body. This helps in the process of curing a disease. Scientific studies indicate that cleansing procedures appear to help in eradicating chronic diseases more effectively.

It was observed by the author in a case study that when PKT was administered as an adjuvant therapy along with the allopathic system of medicines, the results were better than the PKT or allopathic system of medicines given alone.

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5

Immunomodulation: Therapeutic Strategy through Ayurveda

Ashish A. Mungantiwar and Aashish S. Phadke

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5.1 Introduction

Why is it that there is always some lucky soul who does not get sick when the flu is going around the office? Or why is it when the chicken pox is sending all the kids home from school, there are a few who remain untouched? Why do only some in a family develop allergies or bacterial infection? Some attribute this phenomenon to simple luck, whereas others attribute it to strong genes. The answer is a bit more complex.

A large number of diverse disease states threatening mankind, such as chronic and recurrent viral and fungal infections, human immunodeficiency virus (HIV) infection, allergies, and cancer, are believed to have resulted from altered functioning of the immune system. The immune system helps an individual toward establishing an infection-free state. Disturbances in the internal state of order of the host immune system results either in the terms of hyperactivity, hypoactivity, or the inability to distinguish self from nonself that would lead to disease. Such multidimensional response would require proper control through immunomodulation.

It is true that our genes affect our immune system; we call that *influence of natural constitution*. We cannot change our genetic makeup, but we can definitely strengthen our immune system to fight against diseases.

The immune system can be stimulated and suppressed by various physical, chemical, and biological agents. These include steroids, cytotoxic drugs, and various natural products of microbial, animal, and plant origin.¹ Although an antigen-specific immunomodulator is desirable, reliable agents for sustained therapeutic effects are being investigated.

The suppression of the immune system associated with tuberculosis (TB), cancer,² surgery,³ or HIV infection is characterized by a reduction in the number and phagocytic function of neutrophils and macrophages⁴ as well as a reduction in the intracellular bactericidal capacity of these cells. This profound suppression of the individual elements of the system allows opportunistic pathogens to overwhelm the host so that secondary infection becomes the most common cause of the mortality in such individuals.⁵

An attempt to overcome this problem has been made by introducing the concept of pro-host therapy.⁶ This approach aims at administering drugs to bolster immune defense against infections.⁷ Several naturally occurring and synthetic substances like bacillus calmette and guerin (BCG),⁸ muramyl dipeptide,⁹ interleukin-1 (IL-1),¹⁰ and colony-stimulating factors¹¹ have been used to enhance resistance against infections. However, these substances share several limitations. One of their major drawbacks is the necessity to administer them parenterally, which often leads to local reactions such as granuloma formation. Another factor is the cost of therapy, which can be prohibitive. Against this background, the potential of Ayurvedic therapy appears to be encouraging.¹²

Ayurveda emphasizes the promotion of health with a concept of strengthening host defense against diseases. The concept of immunostimulation through Ayurveda has been used successfully in the treatment of immunocompromised conditions like acquired immune deficiency syndrome (AIDS), TB, cancer, and hepatic disease.

5.2 Etiology

The cellular components of the immune system, which are essential in controlling infections, include macrophage, T-cell and B-cell subset, natural killer cell, lymphokine

TABLE 5.1

Cytokines Released by CD4 T-cells

Cytokines	Functions	T _H ¹ -Cells	T _H ² -Cells
IL-2	T-cell proliferation	+	–
IL-3	T- and B-cell proliferation	++	++
IL-4	Activation of T-cell, B-cell, and macrophage	–	++
IL-5	Eosinophil production	–	++
IL-10	Inhibition of T-cell	–	++
IFN- γ	Activation of macrophage	++	–
TNF- α	Stimulation of inflammation	++	–
GM-CSF	Stimulation of macrophage	++	+

activated killer cell, and cytotoxic lymphocyte. Generation of an effective immune response involves two major groups of cells: lymphocytes and antigen-presenting cells. Lymphocytes are one of many types of white blood cells produced in bone marrow that circulate in the blood and lymph system and reside in various lymph organs. The major population of lymphocytes is B-lymphocytes (B-cell) and T-lymphocytes (T-cell). There are two well-defined subpopulations of T-cells: T helper cells (T_H) and T cytotoxic cells (T_C). T_H- and T_C-cells can be distinguished from one another by the presence of either membrane glycoprotein CD4 or CD8 on their surfaces. T-cells displaying CD4 generally function as T_H-cells, whereas those displaying CD8 generally function as T_C-cells. The T_H-cell recognizes and interacts with the specific antigen (microbe). Interaction takes place with the help of major histocompatibility complex (MHC) II molecule. Once the interaction takes place, T_H-cell activates and secretes various growth factors known as cytokines. The secreted cytokines play an important role in activating various cells of the immune system, like the B-cell, T_C-cell, and macrophage.¹³

Recently, it has become clear that two distinct subsets of CD4 helper T-cells (T_H¹- and T_H²-cells) secrete different pattern of cytokines as shown in the Table 5.1. T-cells respond to antigenic stimulation with a transient burst of cytokine production.¹⁴ The key differences are expressions of interferon-gamma (IFN- γ), tissue necrosis factor-alpha (TNF- α), interleukin-2 (IL-2), and granulocyte macrophage colony-stimulating factor (GM-CSF) by T_H¹ clones and interleukin-4 (IL-4) and interleukin-5 (IL-5) by T_H² clones. T_H²-cells stimulate the production of three of the hallmark features of the allergic disease: mast cells, eosinophils, and immunoglobulin E (IgE) antibodies. The cytokines responsible for these activities are IL-4 for IgE production¹⁵ and IL-5 for eosinophilia.¹⁶ Cytokines released by T_H¹ lead to a substantial increase in the ability of macrophages to kill a wide variety of intracellular pathogens and to enhance the development of long-term immunity to those organisms.¹⁷

The understanding of T-cell-mediated immunoregulation suggests a number of potential therapeutic applications in TB, hepatic disorders, HIV infections, etc.

5.2.1 Apoptosis

Apoptosis is a mode of cell death in which single cells are deleted in the living tissue.¹⁸ The term is derived from ancient Greece, meaning the falling of leaves from trees or petals from flowers. It is characterized by structural changes that appear with great fidelity in cells of widely different lineage.

5.2.2 Signal Pathway to Trigger Apoptosis and Its Intervention through Ayurveda

Several pieces of evidence implicate a rise in cellular-free calcium as a critical event for apoptosis. A moderate but sustained increase in intracellular calcium has been demonstrated to produce cell death. However, because calcium is implicated in signaling for a multitude of cellular activities, including proliferation and differentiation, it seems unlikely that a simple influx of calcium from outside the cell activates a specific pathway for apoptotic death. Apart from increased calcium levels, there are signals that contribute toward apoptosis. These are protein kinase A and protein kinase C. Protein kinase C has a negative effect on apoptosis. Therefore, factors that stimulate protein kinase C will inhibit apoptosis. For example, activated macrophage, IL-1, and GM-CSF both activate protein kinase C and inhibit apoptosis.

Inhibition of apoptosis is useful in the treatment in AIDS, making Ayurveda a potential treatment as well. Many ayurvedic medicinal plants have been found to modulate the process of apoptosis. For example, the plants *Boerhaavia diffusa*¹⁹ and *Tinospora cordifolia*²⁰ have been shown to stimulate macrophage and GM-CSF and inhibit apoptosis.

Other diseases in which apoptosis is implicated are cancer, autoimmune disease, and neurodegenerative disease. The rate at which cancer grows depends on the comparative rates of cell division and cell death. Apoptosis is an efficient way of preventing malignant transformation because it removes genetically malformed cells. Abnormal apoptosis can promote the development of cancer by either allowing the accumulation of normal cells or preventing their removal. AIDS is another condition where the balance between the rate of CD4 death and cell replacement becomes abnormal. It is believed that apoptosis plays an important role in lymphocyte depletion in AIDS.

Neurodegenerative diseases are also becoming the focus of much research as the census of the aging population increases. For example, it is hypothesized that apoptosis may play a role in Alzheimer's disease or motor neurone disease. In addition, anticancer effects produced by several anticancer drugs have been shown to induce apoptosis, thus aiding in removal of cancerous cells. The modulation of apoptosis may assist the manipulation of the aging process. Indian medicinal plants offer prospects for the modulation of the balance between proliferation and apoptosis. The potential of several medicinal plants described in Ayurveda needs to be tapped.

5.3 Immunomodulation and Concept *Rasayana* in Ayurveda

Out of the eight branches of Ayurveda, the *rasayana* and *vajeekarana tantra* branch is the one that is concerned with the immunomodulation. According to Ayurveda, the common cause of all diseases may be due to contact with environmental factors and aging. The latter is inevitable and the other cause can be avoided only by scrupulously puritan patterns of living. Ayurveda claims to be able to slow down the process of aging successfully.

Rasayana is a treatment in which the body constituents are prepared to adapt to a selective tissue endowment program. This concept in modern scientific understanding would mean the enhancement of immune responsiveness of an organism against pathogens by nonspecifically activating the immune system with immunomodulatory agents of plant origin. It is now recognized that immunomodulation could provide an alternative or complement to conventional chemotherapy for a variety of diseased conditions; this is especially the case when host defense mechanisms have to be activated under the conditions of impaired immune responsiveness.

Rasayan drugs are believed to slow down the aging process (*jara*) and provide a defense against diseases (*vyadhi*). *Rasayanas* improve the host resistance of an individual, helping to prevent aging and diseases. Specific diets and lifestyle changes are also advised in *rasayana* therapy.

Rasayana is made up of *rasa* and *ayana*. *Rasa* primarily means essential seven vital tissues: *rasa* (lymphatics), *rakta* (blood), *mamsa* (muscle), *meda* (adipose tissues), *asthi* (bones), *majja* (bone marrow and nervous tissue), and *shukra* (reproductive element). *Ayana* means the path or channel for the same. So, *rasayanas* are those that bring about proper uptake, growth, and improvement of essential seven vital tissues. *Rasayanas* provide long life, good intellect, the ability to remain young, and general well-being.

5.3.1 Mode of Action of *Rasayanas*

Rasayana drugs act on the immune system. On ingesting the appropriate formulation of the *rasayana* in the appropriate season by a qualified Ayurvedic physician, the beneficial effects of *rasayana* are seen. What, then, are these *rasayana* drugs that have so many vital life-enhancing properties? The ancient writers of Ayurveda have endowed the drugs with multiple properties of delaying aging, improving mental functions, and giving freedom from several diseases including those caused by infections. How can these *rasayanas*, which otherwise seem to be simple herbs having a different number of phytochemicals, influence such a wide variety of body functions? To the scientific mind, it appears impossible to credit one plant with such an apparently disparate variety of effects — almost as though it were a magic potion! It seemed possible only if the drugs were modulating an endogenous system of the body, setting into motion a whole cascade of events that will act on the different organs and produce its myriad actions.

The immune system has connections with a number of other organs and can directly or indirectly influence the actions of many other organs, including the brain. By acting primarily on the immune system like macrophages, the simple chemical of the herbs through the activating cytokine network could produce all the actions that have been attributed to them. This hypothesis was proved scientifically by Dhanukar et al.²⁴ In one set of experiments, *rasayanas* were given to rats whose macrophages had been overloaded with a fat lipofundin. In this setup, the *rasayanas* were unable to stimulate macrophages further and were thus unable to prevent stress-induced organ damage. *Rasayana* shows myriad actions on other organs by acting on the immune system. The classification of *rasayanas* is presented in Table 5.2.²⁵

5.4 Scientific Basis of Ayurvedic Immunomodulation Therapies

Commercially available *rasayana* herbal formulas and supporting clinical and biological studies are listed in Table 5.3. Studies on single *rasayana* herbs are presented here.

5.4.1 *Terminalia Chebula* (*Haritaki*)

Terminalia chebula is an antibacterial, antioxidizing, anti-inflammatory, and immunomodulatory agent. The topical administration of an alcoholic extract of the leaves of *T. chebula* was found to heal much faster as indicated by decreased period of epithelialization. The tensile strength of the tissue, which was treated with *T. chebula*, was increased by 40%.²⁶

TABLE 5.2

Classification of Rasayanas

No.	Classification	Remarks
1	<i>Kutipravesika</i>	Patient has to stay in a very specialized manner in the <i>kuti</i> (hut or home) specifically made as per the needs of <i>rasayana</i>
2	<i>Vatatapika rasayana</i>	Can be given on an out-patient department basis; person can consume it while doing his or her normal duties and staying at his or her own home
3	<i>Kamyā rasayana</i>	<i>Rasayana</i> that are specifically used for benefit of specific system (e.g., for bone marrow <i>Withania somnifera</i> and <i>Asparagus racemosus</i>)
4	<i>Naimittika rasayana</i>	<i>Rasayana</i> given only for partial period for certain number of days
5	<i>Agasrik rasayana</i>	<i>Rasayana</i> commonly used to keep the process of formation of vital tissue (e.g., milk)
6	<i>Sharira rasayana</i>	<i>Rasayana</i> working on physical levels (e.g., <i>Emblica officinalis</i>)
7	<i>Medhya rasayana</i>	<i>Rasayana</i> working on mind for the improvement of intellect (e.g., <i>Bacopa manorial</i> , <i>Centella asiatica</i>)
8	<i>Vardhanana rasayana</i>	Dosage of drug has to be increased slowly and kept at highest level and then once again tapered down (e.g., <i>Piper longum</i> , <i>Semecarpus anacardium</i>)
9	<i>Achara rasayana</i>	Based on a code of conduct, which keeps oneself to attain good mental and spiritual health; if one follows <i>Achara rasayana</i> , one can be kept away from anxiety, stress, and thereby from all diseases that are generated due to stress
10	<i>Ahar rasayana</i>	Food is considered as <i>rasayana</i> (e.g., milk)
11	<i>Dravya rasayana</i>	Drug therapy is considered under this; drug therapy includes single herbal, polyherbal, and herbo-mineral formulation; modern scientific medicine also states that good diet, nutrition, proper lifestyle management, supportive family, good counseling, and stress control help reverse the damage already done to the immune system

Biochemical studies revealed a significant increase in total protein, DNA, and collagen contents in the granulation tissue of the treated wound. Reduced lipid peroxide levels in the treated wounds suggest that *T. chebula* possesses antioxidant activity. The extracts of *T. chebula* were found to significantly suppress yields of cytomegalo virus (CMV) in lungs of mice. Thus, *T. chebula* may be beneficial for the prophylaxis of CMV disease in immunocompromised patients.²⁷ The extract of *T. chebula* was also found to inhibit human immunodeficiency virus-1 reverse transcriptase with $(IC_{50}) \leq 50 \mu\text{g/ml}$.²⁸

5.4.2 *Emblica officinalis* (Amlaki)

Emblica officinalis is a rich source of vitamin C. The fruits of *E. officinalis* have been used by Ayurveda as potent *rasayana* and also for the treatment of diverse etiology disease. Perhaps there is no other drug in any other system of natural medicine with such a vast range of attributed effects. In an experimental study, it was reported that *E. officinalis* could bring about a significant weight gain in the subjects together with an increase in serum total protein content.²⁹ It is used as an antioxidizing, antibacterial, and anti-inflammatory agent.³⁰ The antioxidant activity of *E. officinalis* resides in tannoids. Tannoids have been found to increase superoxide, catalase, and glutathione peroxidase.³¹

TABLE 5.3
Clinical Evidence of Some Compound Formulas

Compound Formula	Clinical and Biological Studies	Ref.
<i>Septilin tablet^a</i>		
Each tablet contains <i>Balsamodendron mukul</i> , <i>Exts. Maharsnadi, Phyllanthu emblica</i> , <i>Tinospora cordifolia</i> , <i>Rubia cordifolia</i> , <i>Moringa pterygosperma</i> , <i>Glycyrrhizasperma</i>	• Septilin possesses immunomodulatory and anti-inflammatory plant principles, which strengthen the nonspecific immune responses of the body; Septilin stimulates the phagocytosis by macrophage activation, increases the polymorphonuclear cells, and helps overcome the infections; Septilin builds up resistance to disease and helps prevent reinfection	
	• Septilin significantly enhanced gain in tensile strength in incision wounds and wound contraction and epithelialization in excision wounds	79
	• One tablet was given to younger patients twice a day, and one tablet was given to elder patients three times a day; doses have been found to be useful for upper respiratory tract infections; 50% of the patients showed excellent results; 30% reported good results with no recurrent attacks upon 6-month follow-up ^c	80
	• Septilin enhances both primary and secondary immune response in mice immunized with sheep red blood cells and counteracts IgG and IgM suppression induced by prednisolone when administered orally with immunosuppressive drug (prednisolone); the mechanism of immune protection by Septilin treatment could probably be due to increased number of activated macrophages by Septilin treatment ^d	81
<i>Mentat Tablet^a</i>		
Each tablet contains <i>Hydrocotyl asiatica</i> , <i>Withania somnifera</i> , <i>Acorus calamus</i> , <i>Asparagus racemosus</i> , <i>Emblica officinalis</i> , <i>Evolvulus alsinoides</i> , triphala	• Mentat tablets have been used as a general tonic and vitalizer and as a memory enhancer	
	• Clinical studies have shown that it improves memory quotient in normal subjects of different age groups, increases memory span and attenuates fluctuations of attention in normal subjects, and improves learning ability in children ^e	82
<i>Liv 52^a</i>		
Each tablet contains <i>Capparis spinosa</i> , <i>Cinchorium intybus</i> , <i>Solanum nigrum</i> , <i>Cassia occidentalis</i> , <i>Tamarix gallica</i> , <i>Terminalia arjuna</i> , <i>Achillca millefolium</i> , <i>Mandhura bhasma</i>	• In a clinical trial of Liv 52 vs. placebo in 60 malnourished children, a dose of one tablet twice a day for 6 months showed significant improvement in body weight, height, total protein, albumin, and globulin levels; appetite also improved remarkably ^f	83
	• Liv 52 significantly improves chronic hepatitis and cirrhosis ^g as observed from clinical trials on 300 patients with hepatitis.	84

TABLE 5.3 (continued)

Clinical Evidence of Some Compound Formulas

Compound Formula	Clinical and Biological Studies	Ref.
<i>Geriforte Tablets^a</i>		
Each tablet contains <i>Chyavanprash</i> concentrate, <i>Exts. Capparis spinosa</i> , <i>Solanum nigrum</i> , <i>Cassia occidentalis</i> , <i>Terminalia arjuna</i> , <i>Achillea millefolium</i> , <i>Tamarix gallica</i> , <i>Mandhur bhasma</i> , saffron, amber, makardhwaj, <i>Asparagus adscendens</i> , <i>Caesalpinia digyna</i> , <i>Asparagus racemosus</i> , <i>Withania somnifera</i> , <i>Glycyrrhiza glabra</i> , <i>Centella asiatica</i> , shilajeet, <i>Terminalia chebula</i> , <i>Mucuna pruriens</i> , <i>Myristica fragrans</i> , <i>Piper longum</i> , <i>Maceeugenia caryophyllata</i> , <i>Elettaria cardomomum</i> , <i>Carum copticum</i> , <i>Curcuma longa</i> , <i>Exts. Berberis aristata</i> , <i>Adhatoda vasica</i> , <i>Eclipta alba</i> , <i>Celastrus paniculatus</i> , <i>Argyreia aepcosia</i> , <i>Abhrak bhasma</i> , loh bhasma, jasa bhasma	• Geriforte, an indigenous herbo-mineral preparation, claimed to be a useful comprehensive general tonic and restorative	
	• Geriforte prevented stress-induced changes in adrenal, restraint ulcers, liver toxicity, and milk-induced leucocytosis; these properties indicate the presence of antistress (adpatogenic properties) of Geriforte^h	85
	• A placebo-controlled clinical trial with Geriforte was conducted in 280 postsurgical patients; it was observed that Geriforte therapy promotes cellular regeneration and helps in wound healing by increasing the availability of essential amino acids and elements for primary healing of the operative wounds; a substantial improvement in well-being was observed in 78.57% of patients, and 75% of patients gained weightⁱ	86
	• In an open trial in 50 patients, two tablets twice a day for 15 days up to 6 months showed a significant improvement in physical and mental activity and improvement in appetite; weight gain was observed in 80 to 85% of the patients^j	87
<i>Chyawanprash^b</i>		
Each supplement contains vitamin C derived from <i>Emblica officinalis</i> (<i>amla</i>) and several other herbal products	• Ten normal healthy volunteers were given <i>chyawanprash</i> for 16 weeks; they received 15 g/day; it was observed that <i>chyawanprash</i> reduces postprandial glycemia in the oral glucose tolerance test and reduces blood cholesterol level to a significantly greater extent^l	88
	• In a controlled clinical study using <i>chyawanprash awaleha</i> as an adjuvant in the treatment of pulmonary TB, it was observed that the preparation could augment the recovery process as well as an improved nutritional status of the subjects^m	58

^aHimalaya Drug House.

^bZandu Pharmaceuticals, Dabur Pharmaceuticals, Baidynath Pharmaceuticals.

5.4.3 *Piper longum* (Pippali)

Piper longum is indicated in treatment of chronic dysentery and worm infestations. *Pippali rasayana*, an Ayurvedic herbal medicine prepared from *P. longum* and *Butea monosperma*, is prescribed for the treatment of chronic dysentery and worm infestations. *P. longum* significantly increases macrophage migration inhibition and phagocytic activity.³² Enhancement of host resistance is the possible mechanism of recovery from giardial infection.

5.4.4 *Glycyrrhiza glabra* (Yashtimadhu [Parts Used: Roots])

Glycyrrhiza glabra is an immunomodulator and antioxidant. Glycyrrhizin, a triterpenoid glycoside obtained from *G. glabra*, is found to inhibit ribonucleic acid (RNA) viruses such as measles; polio vaccine viruses type 1, 2, and 3; and deoxyribonucleic acid (DNA) viruses such as herpes type 1 and 2.³³ Polysaccharide fractions obtained from the root of *G. glabra* induce nitric oxide production from macrophages.³⁴

5.4.5 *Allium sativum* (Lahsuna)

Allium sativum (garlic) is an antimicrobial, antitumor, hypolipemic, antiarthritic, and hypoglycemic agent. These characteristics have been linked to their influences on immune functions in various ways.³⁵ The herb has been found to enhance human immune functions by stimulating peripheral blood mononuclear cells. Diallyl sulfide in *A. sativum* is known to exert anticarcinogenic activity.³⁶ Allicin from garlic has been found to induce programmed cell death and the arrest of proliferation in cancer cells.

5.4.6 *Semecarpus anacardium* (Bhallatak [Parts Used: Nuts])

Extract of nut preparation of *Semecarpus anacardium* is effective against a variety of diseases like arthritis, tumors, and infections. An extract of *S. anacardium* at a dose of 150 mg/kg significantly reduced the lysosomal enzyme activity in arthritic animals³⁷ and displayed significant inhibition of tumor cells with IC₅₀ of 1.6 µg/ml.³⁸

5.4.7 *Commiphora mukul* (Guggul)

Commiphora mukul is used as an anti-inflammatory, antihyperlipidemic, and immunomodulatory agent. A placebo double-blind randomized controlled trial of *C. mukul*, at 50 mg twice daily, was conducted on patients with hypercholesterolemia.³⁹ *C. mukul* decreased total cholesterol by 11.7%, low-density lipoprotein (LDL) by 12.5%, triglyceride by 12%, and the ratio of total cholesterol/high density lipoprotein (HDL) by 11.1%.

5.4.8 *Tinospora cordifolia* (Guduchi [Parts Used: Roots])

Tinospora cordifolia (TC) is used for the treatment of jaundice, skin disease, diabetes, anemia, emaciation, and infections. Plant extracts are known to stimulate macrophages and enhance their phagocytic activity and intracellular killing activity. Immunosuppression associated with deranged hepatic function and sepsis results in poor surgical outcome in

obstructive jaundice. *T. cordifolia* was reported to improve surgical outcome by strengthening host defenses.³¹ Active principles of *T. cordifolia* were found to possess anticomplementary and immunomodulatory activities. Syringin (TC-4) and cordiol (TC-7) inhibited classical complement pathway. The compound also gave rise to significant increase in immunoglobulin G (IgG) antibodies in serum. Both humoral and cell-mediated immunity enhancement were dose-dependent enhanced. *T. cordifolia* has been found to activate the mononuclear cells to release cytokines like GM-CSF and IL-1. Whole aqueous extract of *T. cordifolia* has been evaluated as an adjuvant in clinical conditions like obstructive jaundice, TB, and cancer chemotherapy and has been found to increase the efficiency of conventional therapies. *T. cordifolia* is also shown to specifically stimulate macrophages and enhance their phagocytic activity. In a recent double-blind randomized controlled trial,⁴⁰ it also was shown to increase the production of nitric oxide in TB patients, causing increased macrophage stimulations. TB patients who took *T. cordifolia* extract along with conventional anti-TB treatment showed increased radiological recovery and sputum conversion as compared with conventional anti-TB treatment alone.

5.4.9 *Azardicta indica* (Neem [Parts Used: Leaves])

Azardicta indica is reported to have several therapeutic effects, including being anti-infective and anxiolytic and having general immunopotentiating ability.⁴¹ It is widely studied for various beneficial properties. The aqueous extracts of neem leaves enhance the phagocytic activity of macrophages. *Neem* is found to enhance the production of IL-2, IFN- γ , and TNF- α . In human volunteers, it stimulated humoral immunity by increasing antibody levels and cell-mediated immunity by increasing total lymphocyte and T-cell count in 21 days. The therapeutic potential of *neem* is mediated through its influence on the immune system, which is in consonance of the Ayurvedic concept of *rasayana*.

5.4.10 *Asparagus racemosus* (Shatavari [Parts Used: Whole Plant])

Asparagus racemosus is used as an immunomodulator and antioxidant. The aqueous extract of the whole plant of *A. racemosus* gives protection from biological, physical, and chemical stresses.⁴² Treatment with *A. racemosus* significantly inhibited suppression of chemotactic activity and production of IL-1 and TNF- α by murine macrophages induced by 17 weeks of treatment with ochratoxin A.⁴³ Studies on the mechanisms of action revealed that it produced immunostimulation. Aqueous extracts of *A. racemosus* are also found to suppress the myelosuppressive effects of single and double doses of cyclophosphamide.⁴⁴

5.4.11 *Withania somnifera* (Ashwagandha [Parts Used: Roots])

Withania somnifera is one highly acclaimed *rasayana*. *W. somnifera* is an immunomodulatory, anti-inflammatory, and antioxidizing agent. *W. somnifera* effectively inhibits the inflammatory process. It can also bring about a specific reduction in α -2 macroglobulin synthesis, unlike the conventionally used nonsteroidal anti-inflammatory drugs (NSAIDs), and has anti-oxidant activity.⁴⁵ The herb is described to act as *rasayana* and *medhya*. In a clinical study on patients of anxiety neurosis, *ashwagandha* was observed to reduce the symptoms of anxiety. One month of treatment with the drug has shown symptomatic relief in mental fatigue, and immediate memory span. In another study, the herb has been shown to be effective in cases of depression. In yet another double-blind placebo-controlled clinical study⁴⁶ on the extract of *ashwagandha* involving both normal and depressive volunteers,

it was observed that the test extract showed an excellent improvement in mental functions. Experiments with extracts of *Ashwagandha* protected animals against infections in normal and immunosuppressed states induced by hemisplenectomy or surgery. This plant also produced leucocytosis with predominant neutrophilia and prevented leucopenia induced by cyclophosphamide.⁴⁷

5.4.12 *Aloe vera* (*Ghrithkumari*)

Aloe vera is a plant exclusively used in skin-care preparation and other health-care products. It is also found to possess anticancer activity. Supplementation of aloe vera gel has been found to suppress the incidence of hepatocarcinogenesis induced by diethylnitrosamine and 2-acetylaminofluorene in male rats.⁴⁸ The extract from leaf gel contains glutathione peroxidase activity.⁴⁹ The low molecular weight constituent of this extract inhibits the release of reactive oxygen species by human polymorphonuclear leucocytes. The inhibitory effect of these compounds has been shown to be an indirect result of the diminished availability of intracellular free calcium.⁵⁰

5.4.13 *Solanum nigrum* (*Kakamachi*)

Solanum nigrum is an anti-inflammatory and hepatoprotective agent. Steroidal glycosides isolated from *S. nigrum* are found to possess antineoplastic activity against solid tumor cell lines.⁵¹ The hepatoprotective activity of *S. nigrum* may be due to their ability to suppress the oxidative degradation of DNA in the tissue debris.⁵²

5.4.14 *Bacopa monnieri* (*Brahmi*)

Bacopa monnieri is used as an antiallergic, antistress, and memory-enhancing agent. In a double-blind randomized controlled clinical trial in 76 subjects,⁵³ *B. monnieri* showed significant effect on the retention of new information. Follow-up tests showed that the rate of learning was unaffected, suggesting that *B. monnieri* decreases the rate of forgetting newly acquired information.

5.4.15 *Boerhaavia diffusa* (*Punarnava* [Parts Used: Roots])

The plant *Boerhaavia diffusa* (BD) is used as an antiarthritic, immunomodulatory, and antistress agent. Alkaloidal fraction of roots of BD was found to attenuate myelosuppressive effects of cyclophosphamide. BD is known to potentiate macrophage phagocytic activity. It is also shown to increase GM-CSF levels upon oral administration, inhibiting apoptosis.¹⁹

5.4.16 *Ocimum sanctum* (*Tulasi*)

Ocimum sanctum is used as an antistress, antioxidizing, and immunomodulatory agent. The ethanolic extract of *O. sanctum* reversed the changes in plasma levels of corticosterone induced by exposure to acute and chronic stress.⁵⁴ Banerjee et al.⁵⁵ have studied the modulatory influence of the alcoholic extract of leaves of *O. sanctum* on various enzyme levels in the liver, lung, and stomach of mice. Oral treatment with extract significantly elevated the activities of cytochrome P 450, cytochrome b5, and arylhydrocarbon hydroxylase, all

of which are important in the detoxification of carcinogens as well as mutagens. These observations suggest that leaf extract or its active principle may have a potential role in the chemoprevention of chemical carcinogenesis.

5.4.17 *Shilajit* (Asphalt)

Shilajit is an exudate of selected rocks occurring in the Himalaya region. It is a complex mixture of both organic and inorganic matter and is considered to be an effective remedy for many diseases. It was investigated for its effect on memory, learning, and anxiety. It was reported that *Shilajit* enhanced the acquisition of learning and memory in aged rats while exhibiting a marked reduction in anxiety levels. This neurochemical effect of *Shilajit* is attributed to decrease in 5-hydroxytryptamine turnover with an increase in dopaminergic activity.⁵⁶ To add the multifaceted activity of *Shilajit*, a study was designed by Oureshi et al.⁵⁷ to investigate the cytological and biological effects of *Shilajit* and its response to the changes induced by cyclophosphamide. The study showed that *Shilajit* exhibited cytotoxic effects and inhibited the carcinogenic potential of cyclophosphamide. In experimental studies, it has also been observed that *Shilajit* activates macrophage and enhances cytokine release.³¹

5.4.18 *Chyawanprash Awaleha*: A Polyherbal *Rasayan* Formulation

Chyawanprash awaleha (CA) is a polyherbal Ayurvedic *rasayan* preparation described in Charak Samhita and contains the pulp of *E. officinalis* as the prime ingredient. This preparation is popularly used in India and many other countries as a health supplement.

In a controlled clinical study using CA as an adjuvant in the treatment of pulmonary TB, it was observed that the preparation could augment the recovery process in addition to an improved nutritional status of the subjects.⁵⁸

Recently, the anticarcinogenic property of CA has also been studied. It has been shown to reduce tumor weight in mice. A clinical study was undertaken to estimate the radioprotective effect of CA. The study included patients with head and neck cancers undergoing radiotherapy. It was observed that CA conferred an excellent protection to the tissues from the burning effects of radiation, in comparison with controls.⁵⁹ The tumor regression rate was also noted to be higher in the CA-treated group. These results may be suggestive of possible antioxidant effects of *E. officinalis* contained in the preparation.

5.5 Immunomodulation Therapies

5.5.1 Tuberculosis

There is considerable evidence that failure to control or resolve infectious disease often results from an inappropriate rather than insufficient immune response. A good example is TB. In this condition, T-cell immunity is decreased; the T_H-cell is not able to produce sufficient IFN- γ to activate macrophage, which can kill mycobacterium TB. Intervention in this situation could be aimed at expanding protective T_H¹-cells or alternatively enhancing the key protective response macrophage activation by IFN- γ .

Current therapy consists of antibacterial drugs like Rifampicin, Isoniazid, Pyrazinamide, and Ethambutol for a period of 4 to 6 months. Significant success rate is achieved with this regimen. Treatment of TB with *rasayana* has shown significant improvement in treatment period.

Treatment of TB with Ayurveda includes various *rasayana* like *T. cordifolia* and CA. The latter is a wonderful remedy that provides strength and sustenance to the patient. It was mentioned earlier that using CA as an adjuvant in the treatment of pulmonary TB⁵⁸ could augment recovery process. It also increases the appetite. The dosage in the beginning should be two teaspoonfuls with milk on an empty stomach twice daily; as the patient gains strength, the dosage should be increased.

A double-blind randomized clinical trial⁴⁰ of *T. cordifolia* was conducted in TB patients along with a conventional anti-TB regimen for an initial 2-month intensive period. It was observed that when conventional anti-TB regimen was given along with 500 mg of *T. cordifolia* three times daily, the sputum conversion time was 2 weeks as compared with 4 weeks in the conventional group; there was also hastened radiological recovery as compared with the conventional regimen. This effect was observed due to increased nitric oxide production by *T. cordifolia*, which caused increased macrophage phagocytic activity.

5.5.2 Hepatic Disorder

In recent years, rapid progress has been made in understanding the relationship between various cells of the liver and pathobiology of cellular events responsible for an acute and chronic hepatic damage. Kupffer cells, which are present in the liver, are the most versatile cells. The activity of these cells is a major determinant of outcome of any liver injury.⁶⁰ It has been reported that excess activation of Kupffer cells damages the liver,^{61–63} whereas hepatic fibrosis is linked with suppressed or subnormal activity of these cells. Thus, modulation of Kupffer⁶⁴ cells activity through the immune system would be a weapon against liver disorders.

5.5.2.1 Obstructive Jaundice

In the management of obstructive jaundice, present preoperative therapy aims at decompressing the biliary tract and controlling infection with antibiotics and vitamin K if there is a disturbance in hemostasis. Despite improvement in preoperative care, the surgical outcome of these patients has remained poor.^{65,66} Endotoxemia resulting from depressed Kupffer cells is a well-known feature of obstructive jaundice. Based on Ayurvedic literature, it has been observed that *rasayana* like *T. cordifolia* could be a useful therapeutic armamentarium in the treatment of obstructive jaundice. In a clinical trial of *T. cordifolia* as an add-on regimen to conventional therapy, two groups of patients were well matched with respect to clinical features and hepatic and immune functions.⁶⁷ The neutrophil activity of these patients was significantly lower as compared with normal healthy donors. After biliary drainage, bactobilia was observed.⁶⁷ Therapy with *T. cordifolia* significantly decreased the mortality from 92 to 40%. This was also associated with a decrease in septicemia. *T. cordifolia* was found to potentiate the depressed neutrophil activity.

A randomized double-blind placebo-controlled clinical trial also showed a significant reduction in mortality rates.²⁰ Moreover, at the cellular level remarkable results were shown; patients suffering from obstructive jaundice taking *T. cordifolia* showed an increase in serum GM-CSF levels. The neutrophils of these patients also showed an increased phagocytic activity, making *T. cordifolia* an effective treatment for patients with obstructive jaundice.²⁰

5.5.2.2 Asymptomatic Carriers of Hepatitis B Antigen

In asymptomatic carriers of hepatitis B antigen, elimination of viral antigen is necessary to prevent future hepatic dysfunction and to reduce transmission. Although the role of nonspecific cellular immunity is not clearly defined, defects in monocyte have been reported⁶⁸ and immunomodulators like levamisole and interferons have been shown to eliminate antigen.⁶⁹ Therapy with *rasayana* such as *T. cordifolia* (16 mg/kg/day) for 2 months led to seroconversion in 37.5% of patients. This rate was three times higher than that observed in the intracellular killing capacity of monocytes.²⁰

5.5.2.3 HIV Infection

HIV is a disease starting from the stage of latent infection, progressing into the severe immunodeficiency state of AIDS, and finally ending in death. HIV is a smart bug. Unlike the other lente viruses, which possess only three genes, HIV has six more functional genes. These are responsible for producing numerous drug-resistant mutant variants.

Review of the scenario of the modern highly active antiretroviral therapy (HAART) is depressing. The scenario consists of reverse transcriptase inhibitors and a protease inhibitor. Failure to eliminate the virus from sanctuaries, an increase in the number of resistant variants of the virus, severe toxicity manifestations, a significant increase in risk of TB, very poor improvement in longevity, and prohibitive costs are some of the reasons why HAART cannot be considered as a choice therapy. The practical utility of a vaccine in preventing the disease also holds very little hope.

Under these circumstances, certain studies mentioned in Ayurvedic texts have been examined to find possible solutions. Ayurveda mentions sexually transmitted communicable progressive wasting disorder. Excessive indulgence in sex and abnormal sexual practices are suggested to be the cause of loss of vitality and immunity (*ojakashaya*) leading to a wasting disorder.

According to Ayurveda, the loss of *ojas* is responsible for a progressively wasting disorder. *Ojas* is the creamy by-product of *shukra dhatu*. It is the ultimate strength of the body. Debility, wastage, fear, depression, a feeling of wretchedness, and a loss of coordination in sense and motor organs are all signs of damage to *ojas*. Ayurvedic texts also reveal that *shukra*, the seventh tissue of the body, is of two types. One type is present all over the body and responsible for strength, vitality, and glow (*shukra* is probably related to the steroid hormones of the body). The other type, *beej shukra*, is responsible for gamete production. Ayurveda authoritatively states that the formation of *shukra* begins only after the complete development of bone marrow (*majja dhatu*). The complete formation of *shukra dhatu* occurs in females and males at 16 and 23 years of age, respectively. According to modern endocrinology, adolescent changes occur when red bone marrow in the long bones of a child is transformed into fatty yellow marrow. This produces extra gonadal steroids, which send a positive feedback to the hypothalamus and activate the hypothalamic-pituitary-gonadal axis. This is the beginning of functioning of the gonads. The Ayurvedic hypothesis of the origin of *shukra* from *majja dhatu* appears meaningful.⁷⁰

Cells need oxygen for survival, but the uncoupled single oxygen radical also causes oxidation damage to the cell. It is indeed an oxygen paradox of the cell. In HIV, reactive oxygen species activate nuclear transcription factor, an obligatory factor for HIV replication. It also aids apoptosis of CD4 cells and thus helps disease progression.

With this background of HIV disease pathology, the mind-body holistic concept of Ayurveda should be an interesting option for treating psychological factors and for treating HIV. *Rasayana* that are classified as *achar rasayana*, *ahar rasayana*, and *dravya rasayana* are helpful in reversing the damage already done to the immune system. *Ahaar rasayana* (food

as a drug) such as black gram, milk, ghee, and meat are advised. Black grams are well known to be a good source of alpha lipoic acid. Ghee has a special place in increasing the *ojas*.

Some of the plants that have shown anti-HIV activity include *Cichorium intybus*,⁷⁰ *Glycyrrhiza glabra*,^{71,72} *Grifola frondosa*,⁷³ *Punica granatum*,⁷⁰ curcumin,⁷⁰ and aloe vera.⁷⁴ In one study, aloe vera juice (20 oz/day) was used in conjunction with essential fatty acids and amino acid supplements to treat 29 patients. There were 15 AIDS, 12 AIDS-related complex (ARC), and two HIV seropositive patients that continued with regular medication, including Azidothymidine (AZT),⁷⁴ an antiviral drug used to combat AIDS. After 6 months, all patients showed clinical improvement; 25% of those positive for p24 core antigen converted to nonreactive, anemia induced by AZT showed improvement in all patients, and the patients gained an average of 7% in body weight.

Acemannan, a water-soluble polysaccharide obtained from aloe vera, is found to be responsible for this anti-HIV activity. The anti-HIV activity of acemannan probably results from inhibiting glycosylation of viral glycoproteins. However, the prominent effect is by enhancing the effect of AZT. Researchers believe that use of acemannan may reduce the amount of AZT by as much as 90%.⁷⁵ This is quite significant. AZT is extremely expensive and is also associated with side effects such as anemia and a decrease of white blood cell production. Acemannan is commercially available as CARRISYN by Carrington Laboratories in Irving, TX.

In a clinical study in patients in the early stages of AIDS, Acemannan showed some promise. In this study, 14 HIV patients who were prescribed oral Acemannan (800 mg/day) demonstrated significant increases in circulating monocytes and macrophages. In particular, the number of large circulating monocytes increased significantly, indicating improvement in phagocytizing, processing, and presenting cells in blood.⁷⁵

Although Acemannan may be helpful in the beginning stages of AIDS, a recent study⁷⁶ indicates that it may be of little value in patients with advanced HIV disease.

Glycyrrhizin, a major constituent of *G. glabra*, is also showing great promise in the treatment of HIV-related disease, including AIDS. In one study, 10 HIV patients without AIDS took 150 to 225 mg of glycyrrhizin daily.⁷⁷ After 1 to 2 years, none developed symptoms associated with AIDS or ARC; however, one of the 10 patients in a matched control group developed ARC and two progressed to AIDS and subsequently died. In yet another study, when glycyrrhizin was administered in daily doses of 200 to 800 mg intravenously, after 8 weeks the groups presented increased helper T-cell count, improved helper/suppressor T-cell ratio, and improved liver function.⁷⁸

5.6 Conclusions and Future Prospective

The immune system is a complex system, involving an interwoven network of biochemical mechanism. The concept of *rasayana* as mentioned in Ayurveda is a holistic approach and constitutes an important approach to handle with subjects of immunity. The objective of research on *rasayana* should focus not only on its immunomodulatory activity, but also on other effects such as antistress, antiaging, antioxidant, adaptogenic, and anti-HIV. The list of herbs in *rasayana* is exhaustive by itself. *Rasayana* formulas provide tremendous potential to be tapped for immunomodulatory activity. It is equally important to arrive at a consensus in regards to the utilization of herbs in a holistic manner in order to deliver their multidimensional benefits to mankind.

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6

Ayurvedic Bhasmas

Sheikh Raisuddin

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6.1 Introduction

Accumulated toxicity data on the hazardous effects of heavy metals have made health scientists afraid of heavy metals. As a result, renewed interest in the beneficial effects of metals and minerals is often viewed with skepticism. However, available literature from all the ancient civilizations indicates that man has used metals in disease treatment since time immemorial. Ayurvedic literature is full of the use of metals. Not only have Ayurveda and other Indian systems of medicine used metals, but their use is also amply described in Chinese and Egyptian civilizations in 2500 B.C.¹ Gold in medicine was also mentioned by Roman physician Pliny and Greek philosopher Dioscrides. Later, Arabic and Persian physicians used gold in various forms in a number of their prescriptions. Besides gold, other metals that are extensively described in Indian and other ancient systems of medicine include silver, arsenic, copper, iron, lead, mercury, and zinc. As far as Ayurveda is concerned, metals have been used mainly as *bhasma* (ash).

Bhasma literally means anything inorganic or organic burnt into its ash. The process of burning in Ayurvedic terminology is known as *marana* (calcination). The process of calcination is also employed for preparation of *bhasmas* of coral, pearl, and shell. The well-known Ayurvedic texts, *Charaka Samhita* and *Susruta Samhita*, which are regarded as the texts scripted by the forefathers of Ayurveda, include ample description of the use of the metals and minerals in the treatment of diseases. It is also reflected in later texts that were attempted to simplify the knowledge of *charaka* and *susruta* (e.g., *Astanga Hridaya*, *Vagbhata*, *Mdhahva Nidan*, *Sharangadhara Samhita*, and *Bhava Prakash*).^{2,3} In this chapter, an attempt has been made to describe various medicinal uses of *bhasmas* in general and *swrana bhasma* in particular.

The principles of Ayurvedic treatment are for the most part the same as those of allopathic treatment. They consist of removing the injurious agent, soothing the injured body and mind, and eradicating the cause. The difference lies in the methods adopted by the two systems. In Ayurveda, great importance is given to the study of the various stages of vitiation of the three *dosas* or humours of the body. When an imbalance occurs among three *dosas*, they defile the normal functioning of the body, leading to the manifestation of disease.⁴ Consciousness or intelligence (*sattva*), motion or action (*rajas*), and the inertia that resist them (*tamas*) are the three omnipresent nonmaterial qualities (*gunas*) that govern all material forms of basic matter. These material and nonmaterial attributes subsequently dictate the medicinal and healing properties of plants and other healing processes.⁵

Ayurveda uses the concept of purification as a means to eradicate disease rather than to cure as perceived by modern medicine.⁵ When treating acute and chronic infections,

Ayurveda does not aim to kill the microbes; restoration of *dosa* balance and host immunity (*rasayna*) ensures elimination of the infectious agent. Numerous Ayurvedic medicinal plants have shown strong chemotherapeutic and immunomodifying effect in experimentally induced infections.⁶

6.2 History of Alchemy in Ayurveda

Ayurvedic literature places great emphasis on the pharmaceutical process known as *samskara*. *Samskara* literally means a process. It is designed to enhance the desirable qualities of the drug being processed. The *samskara* can be classified into two stages: *shodhana* (purification or detoxification of toxic substances) and *bhaishajya kalpana* (formulation of a dosage form). A number of toxic herbal drugs like *Strychnos nuxvomica*, *Aconitum ferox*, *Semicarpus anacardium*, *Commiphora mukul*, and almost all metal and mineral drugs are purified and rendered safe for use, before their use as drugs.

In Ayurveda, *rasayana* is a well-developed concept. Literally, *rasayana* means the augmentation of *rasa*, the vital fluid produced by the digestion of food, which sustains the body through the strengthening of the *dhatu*s. It is the *rasa* flowing in the body that sustains life; when it stops flowing, life comes to an end. *Rasayana* is the medium through which the *rasa* is maintained, replenished, and augmented.

In modern terms, the study and practice of *rasayana* such as *rasavidya* is referred to as alchemy. It is generally believed that alchemy appeared in India in the fifth or sixth century A.D. and greatly prospered for the next seven or eight centuries under royal patronage.⁷ Hindu *tantriks* developed the mercury-based alchemy in India and related it to the male–female symbolism (*Shiva* and *Parvati*). Mercury was regarded as male principle (*Shiva*) and sulfur as the female principle (*Parvati*).

The laboratory of an Indian alchemist was known as *rasashala*, a place where the alchemists carried out their various operations under the benign influence of the *rasalinga*, a symbol of esoteric potentiality. *Rasalinga* was either a gold amalgam, prepared by triturating gold and mercury, or a compound of mercury and sulfur shaped into a *linga*. The text *Rasaratna Samuchchaya* describes how and where a *rasashala* can be established. It also describes important apparatuses that should be available in a *rasashala*. It is amply clear that knowledge of science of metals and its medicinal usage was well developed in India.

Alchemical or related texts can also be found in other Indian languages like Tamil, Telugu, Kannada, Malayalam, Bengali, Marathi, Oriya, and Gujarati. There are more than 200 works in Tamil on the Siddha system of medicine. Plants occupied an important place in the practices of Indian alchemists. More than 200 names of plants are mentioned in different texts of *Rasashastra*. The Indian alchemists (*rasavadins*) treated the minerals and metals with one medicinal plant or the other to render them the desirable chemical properties. Even mercury considered to be divine was subjected to this process.⁷ According to the *Matrakabhedatantram*, “mercury cannot be reduced to bhasma without the help of one or more of medicinal plants.”

The plants used were referred to as *divyausadhi* (divine medicinal plants). Mercury is considered to be the king of *rasas* and it is referred to by various names like *parada*, *rasa*, *suta*, *maharasa*, *rasendra*, *svarnakaraka*, *sarvadhathupati*, etc. There are detailed descriptions of a number of compositions with mercury as the chief ingredient. In Indian alchemical texts, the chemical substances have been divided into five main categories: *maharasa*, *uparasa*, *dhatu*, *ratna*, and poisons (*visha*). There are eight *maharasas*, eight *uparasas*, and seven *dhatu*s,

including three alloys: brass (*pittala*), bell metal (*kamsya*), and a mixture of five metals (*varṭka*). The precious gems were placed under the category of *ratnas*. Various plant products and minerals having toxic properties are included in the category of poisons.

6.3 Ayurvedic Pharmaceuticals

Ayurvedic medicines are categorized according to whether they promote general health and longevity, enhance sexual vigor, or fight disease. The first category is known as *rasayana*, the second is called *vajikarana*, and the third is *aushadhis*. These categories are not mutually exclusive because some of the *aushadhis* may act as *rasayanas* and vice versa. On the basis of their origin, Ayurvedic medicines are also classified into three groups: (1) *kastha ausadhis* (herbal preparations), (2) *rasa ausadhis* (metallic preparations [e.g., *bhasmas*, *sindoor*]), and (3) *jangama ausadhis* (animal preparation — prepared from animal products). Depending on their form, method of preparation, ingredients, and pharmacological properties, Ayurvedic medicines are grouped as *Swaras*, *Kalka*, *Hima*, *Phanita*, *Kashaya*, *Asavas*, *Aristas*, *Awalehas*, *Churnas*, *Vati*, *Gutika*, *Ghrita*, *Taila*, *Guggulu*, *Bhasmas*, *Pishti*, *Parpati*, *Rasayoga*, *Sindoor*, *Lepa*, and *Anjana*.^{4,8–11}

6.4 Metal and Mineral Drugs of Ayurveda

Metals like gold, silver, copper, lead, tin, and iron, sand (*balu* from river banks), lime and minerals like red arsenic (*manassila*), gems (*manayah*), salts (*lavana*), and red chalk (*gairika*) are indicated as drugs pertaining to earth (*bhauma*). In Indian metallurgy, the term *loha* is often used for metals like gold and silver and minerals containing metals (ores) are called *dhatu*s. There are seven *dhatu*s: *suvarna* (gold), *rajata* (silver), *tamra* (copper), *trapa* (tin), *tiksna* or *ayas* (iron), *sisra* or *naga* (lead), and *vaikranta*.¹¹ Salts or *lavan*as are mentioned under the *parthiva* substances. According to Charaka, there are five salts: *sauvarcala*, *saindhava*, *vida*, *audbhida*, and *samudra*. *Mani* and *ratna*, being synonyms for each other, stand for the modern term “jewel” or “gem.”

Mercury is considered eighth metal in *rasa shastra*. It earned the supreme position among the minerals and metals. The learned Acharyas also studied the relation and effects between these metals and planets over the human body and called them *grahanga navaloha*.¹¹ Metals are grouped as *shuddha*, *sishra*, and *pooti loha*.¹¹

The calcined forms of metals that are termed *bhasmas* in Ayurveda are referred to as *parpams* and *kushta* in Siddha and Unani-tibb, respectively. *Kushta* literally means to kill; in medical terms it is detoxifying the toxic properties of a toxic metal.¹²

Although *bhasmas* are regarded as chief metal-containing pharmaceuticals of Ayurveda, there are several other preparations prepared from metals. Some of these pharmaceuticals are described below.

6.4.1 *Bhasma*

Animal derivatives such as horns, shells, feathers, and metallic and nonmetallic minerals are normally administered as *bhasmas*. A *bhasma* means an ash obtained through incineration. The starter material undergoes an elaborate process of purification (*shodhana*). This

process is followed by the reaction phase, which involves incorporation of some other mineral and herbal extracts. Then the material in pellet form is incinerated in a furnace. The end product is expected to be a nontoxic material. Examples include *swarn bhasma*, *shankha bhasma*, and *tamra bhasma*.

6.4.2 Parpati

These are specialized mercury preparations. The name is derived from the method by which flakes of the compound are obtained. A black sulfide of mercury is obtained by mixing purified mercury and sulfur. Other drugs as per the formula are added to this and mixed well by triturating them in mortar and pestle. A shallow pit is made in fresh cow dung and a banana leaf is placed. The melted compound is poured onto the leaf and is covered with another leaf. Fresh dung is spread on it evenly. When it is cooled the flakes are removed and powdered.

6.4.3 Rasayoga

Rasayogas are compound formulations containing mercury and sulfur (in the form of *kajjali*) with other metals or minerals. Most of the ingredients contained in a *rasayoga* are added in the form of *bhasmas*. The final form may be either a pill or powder.

6.4.4 Sindoor

Sindoor are prepared by the elaborate process of sublimation. This procedure is termed *kupipakwa vidhi* and the sublimed mineral available on the neck of the sublimation glass flask is called *sindoor*. *Sindoor* preparations are considered to be more potent than *bhasma* preparations.

6.5 Types of Bhasma

Attempts have been made to classify various *bhasmas*. They have been classified on the basis of color and appearance. A more scientific way of classification is on the basis of dominant metal and mineral group.¹³ According to this classification, *bhasmas* have been grouped as *rajata* group (silver), *tamra* group (copper), *loha* group (iron), *pravala* group (shells), etc. Often two metals and a metal with mineral are the ingredients of *bhasmas*. For example, *Trivanga Bhasma* contains lead, tin, and zinc. The metals yield three different types of *bhasma* corresponding to the nature of the ingredient used. They appear as best, medium, and inferior quality. Mercury is always used as a basic substance in the process of *marana*.¹¹

6.6 Preparation of Bhasma: General Procedures

The name *bhasma* is generally applied to all metallic and nonmetallic substances that are subjected to the process of incineration and reduction to ash. Here it is applied to the

metals, minerals, and animal products that are, by special processes, calcinated in closed crucibles in pits with cow dung cakes (*puttam*). *Bhasmas* are generally white, pale, or red. The color of the preparation primarily depends on the parent material.¹⁴ The following pharmaceutical steps are used to prepare *bhasmas*.

6.6.1 *Shodhana*

In Ayurveda, purification is called *shodhana*. *Shodhana* is the process through which the external and internal impurities of metals and minerals are removed. Chemical purification is different from medicinal purification. In chemical purification it is only elimination of foreign matters, whereas in medicinal purification the objects are involved in the

1. Elimination of harmful matter from the drug
2. Modification of undesirable physical properties of the drug
3. Conversion of some of the characteristics of the drug to different stages
4. Enhancement of the therapeutic action

There are two kinds of *shodhana*. The first type, *samanya shodhana* (general purification), is applicable to the large number of metals or minerals as heating the thin sheets of metals and immersing them in oil (*taila*), extract (*takra*), cow urine (*gomutra*), and other materials. The second type, *Vishesha shodhana* (special purification), is applicable only to specific metals, minerals, and in certain preparations. *Vishesha shodhana* includes *bhavana*, *svedana*, *nirvapana*, and *mardana*.¹⁴

After *shodhana* *bhasmas* become soft and malleable for further processing and their metallic property is improved. The main apparatus required includes *dola yantra*, *khalva yantra*, and *musha yantra*.¹¹ Various procedures employed for *shodhana* are described below.

When mineral drugs are heated in a furnace in the presence of *dravaka*, substances (liqueficients) like alkali and acid release their *satva*. This is the purest form of any herbal or mineral drug.¹¹ All the metals except mercury are found in nature in solid state, and they all fuse under high temperature to attain a liquid state. When the temperature lowers they again return to their natural physical form (i.e., in the solid state). But these fused metals in the presence of some liqueficients do not return into their natural solid state even when the temperature lowers (i.e., the metals remain in liquid form). This method of obtaining metals in liquid form is called *dravana* and the obtained liquid metal is called *druti*. *Druti* holds superior character with respect to efficacy, toxicity, and increased shelf life than its native metals and retains its fluidity for a longer time with proper preservation.¹¹

Shuddhavarta is a particular stage of heating when the fire becomes strong enough to yield the pure substance (metal, *satva*). At this time the flame becomes golden yellow.¹¹

6.6.2 *Marana*

Marana is essentially the burning process or calcination. The purified metal is placed into a mortar and, with a pestle, ground with the juice of specified plants or *kashayas*, mercury (in metallic state), or a compound of mercury such as mercury perchloride (*sauviram*), mercuric subchloride (*ras karpur*), cinnabar (*ingalekam*), or an amalgam of sulfur and mercury (*kajjali*) for a specified period of time. The metal that is intended for *marana* is known as a primary metal (*pradhan dhatu*); the other metal, which is taken in small proportions for the *marana* of the primary metal, is known as secondary metal (*sahaya*

dhatu).¹⁴ Small cakes (*chakrikas*) are made with the ground paste of the minerals and dried under the sun. The size and thickness of the cakes depend on the heaviness of the drug and size. The heavier the drug, the thinner the cakes. These cakes are dried well under the shade and placed in one single layer in a mud tray (*sharava*) and closed with another such tray; the clay-smeared cloth keeps both the lid and the container in apposition. The clay-smeared cloth is applied seven times and dried to seal the crucibles properly. A pit is dug in an open space and half the pit is filled with dried cow dung cakes. The crucibles are placed in the half-filled pit and are covered with cow dung cakes up to the brim of the pit. Fire is then ignited on all four sides and in the middle of the pit. When the burning is over, the contents are allowed to cool completely on their own.

Marana differs with the nature of the substance to be calcinated. For example, organic substances such as herbs are burnt in open air, whereas inorganic substances such as metals like *rajata* (silver) are burnt in closed containers. In either case the end product is a *bhasma* of substance taken for *marana*. For example, the end product in the case of silver (*rajata*) is called as *rajata bhasma*. *Marana* of inorganic substances is called *puta* and the process of *marana* of herbs in closed freshly made containers is known as *puta paka*.¹³

Bhasmas obtained by *marana* from primary metals together with herbs (*mulika*) are called *mulika marita bhasma*; the ones where the second metal is taken for the *marana* of primary metal are called *parada* (mercury) *marita*, or *talaka* (arsenic trisulphide) *marita bhasma*, depending upon the second metal used for the purpose. During the process the second metal would finally volatilize itself at the temperature of *marana*, leaving behind the *bhasma* of primary metal.

Very few metals like copper or iron still bear some impurities after the *marana*. In such cases the whole process is repeated until a purified and therapeutically safer product for internal use is obtained. In addition, a process called *amritikarana* is done to make these metals safer.¹¹ The process consists of heating the product from the *marana* procedure in the presence of some herbal materials to improve safety and therapeutic effect. In this process the required amounts of *triphala* decoction, cow's *ghritika*, and *dhatu bhasma* are placed in an iron pot. Mild heat is applied until the medicinal fluids are completely evaporated. *Bhasma* that remains at the end of this process is safer and possesses higher therapeutic efficacy.

6.6.3 Quality Control of *Bhasma*

Traditionally, the end points of incineration of a metal and its conversion to a *bhasma* state are evaluated based on the following criteria:

1. There should be no *chandrika* or metallic lusture (*nischandrika*).
2. When a *bhasma* is spread between the index finger and thumb, it should be so fine as to get easily into the lines and crevices of the fingers (*rekhapurita*).
3. When a small quantity is spread on cold and still water, it should float on the surface (*varitara*).
4. The *bhasma* should not revert to the original state (*apurnabhava*).¹⁴

A technique known as the phased spot test has been developed by the investigators of Central Council for Research in Ayurveda and Siddha (CCRAS) of India to identify *bhasmas* and *sindooras*. This technique is very effective and accurate in identifying genuine quality of *bhasmas*. Nearly 30 *bhasmas* and *sinduroos* have been studied based on this technique, and suitable criteria have been established for their identification and quality assessment.¹³

TABLE 6.1

Important *Bhasmas* and Their Main Ingredients

No.	<i>Bhasma</i>	Main Ingredient
1	<i>Abhrak bhasma</i>	Mica
2	<i>Hathidanta bhasma</i>	Charcoal of elephant tusk
3	<i>Jasad bhasma</i>	Zinc oxide
4	<i>Loha bhasma</i>	Iron oxide
5	<i>Mandur bhasma</i>	Iron oxide
6	<i>Mayurapicha bhasma</i>	Ashes of peacock feather
7	<i>Mukta bhasma</i>	Oxide of pearl
8	<i>Naga bhasma</i>	Lead
9	<i>Parada bhasma</i>	Mercury compound
10	<i>Pravala bhasma</i>	Oxide of coral
11	<i>Rajata bhasma</i>	Silver oxide
12	<i>Sankha bhasma</i>	Oxide of conch shell
13	<i>Mukta sukti bhasma</i>	Oxide of pearl, oyster shell
14	<i>Talaka bhasma</i>	Arsenic sulphide
15	<i>Tamra bhasma</i>	Cupric oxide
16	<i>Vanga bhasma</i>	Tin compounds
17	<i>Varatika bhasma</i>	Oxide of cowrie shell

The identification of gold in *makardhwaja*, a powdered mercury preparation, by spot test is simple and spectacular. This effort is one big step toward standardizing Ayurvedic preparations.

6.6.4 Preparation of Gold and Iron *Bhasmas* and Their Chemical Characteristics

Some of the common *bhasmas* and their main ingredients used in Ayurveda are listed Table 6.1. Methods to prepare two important Ayurvedic *bhasmas* — gold and iron — and their characteristics are presented here as examples.

6.6.4.1 Gold (*Swarna*) *Bhasma*

The general preparation of *swarna bhasma* involves the three processes of *shodhana*, *dravana*, and *marana*. The leaves of gold are heated over fire and dipped in *sesa* (*Sesamum indicum*) oil when they are red hot, and the process is continued seven times separately. The soft leaves are processed in the same manner with buttermilk, cow's urine, and the decoction of *kulatha* (*Dolichos biflorus*), *kanji* (sour gruel processed from rice [*Oryza sativa*]), and radish (*Raphanus sativus*). Finally, the leaves are dried by heat. Care must be taken that the weight of the gold remains unchanged. A measurement of 15 g each of arsenic disulfide, realger (As_2S_2), and red lead (Pb_3O_4) is taken in an earthenware container and mixed thoroughly with 30 ml of latex of *Calotropis gigantea*. The mixture is triturated and the paste thus obtained is dried in sunlight. The process of triturating and drying in sunlight is repeated 7 to 14 times using fresh aliquots of latex, and the final product (~200 g) is obtained. An aliquot of the above product (~10 g) is poured into liquefied metallic gold (10 g) in a closed earthen pot and the mixture is heated above 1000°C. The content is gently stirred and the heating is continued until the mass becomes disintegrated and a homogenous red-brown powder is formed.¹⁵

All the major Ayurvedic pharmaceutical manufacturers, including Dabur and Zhandu, make *swarna bhasma*. The ingredients of *swarna bhasma* made by Kalptaru Ayurvedic Works, Kolkata, has been reported.¹⁶ The gold content was reported to be 96.76%. It also contained trace quantities of copper and iron. Recently, physicochemical characteristics which

resulted in the laboratory following the procedure as per *Rasaratna Samuchchya* of *swarna bhasma* have been reported.¹⁵ An atomic absorption spectrometer (ASS) was employed to study the contents of gold and other metals. The gold content was found to be 20.3%. Some other important metals that were present included iron, magnesium, strontium, copper, arsenic, lead, nickel, zinc, and cobalt.

The organoleptic characteristics of *swarna bhasma* are that it is dark brown, has a faint smell, fine touch, and is tasteless. The standard *swarna bhasma* should contain the following:

1. Free sulfur — no less than 1.43% w/w and no more than 6.39% w/w
2. Sulfur — no more than 3.33% w/w
3. Calcium as Ca — no more than 1.625% w/w
4. Sodium as Na — no more than 0.922% w/w
5. Potassium as K — no more than 0.370% w/w
6. Sulfate — no more than 3.00% w/w
7. Copper — no more than 17.2% w/w
8. Iron oxide (ferric) — no more than 85.0% w/w
9. Iron oxide (ferrous) — no more than 5.7% w/w
10. Iron — no less than 36.0% w/w and not more than 51.96% w/w
11. Phosphate as PO₄ — no more than 1.101% w/w
12. Silica — no more than 3.8% w/w
13. Acid insolubles — no more than 11.93% w/w

Swarna bhasma contains an ash value between 92.10 and no more than 98.20% w/w and an acid-insoluble ash value between 21.20 and 31.18% w/w. The recommended dose is 100 to 250 mg.¹⁴

6.6.4.2 Iron (Loha) Bhasma

Iron *bhasma* preparation uses three basic processes: *shodhana*, *dravana*, and *marana*. Iron is purified by sinking the red-hot leaflet into the fresh *trifala* decoction (*nishechan*) repeatedly nine times. A freshly prepared decoction is used every time. Coarse pieces of sulfur are taken in *khalva yantra* and some amount of *dewadali swaras* are added for *bhavana* (i.e., the sulfur pieces remain in good contact within this medicinal fluid). It is rubbed thoroughly and the process is repeated for at least 7 days. When sulfur powder obtained at the end is sprinkled (*pravap*) over the fused iron, it is kept in liquidity.

Iron *bhasma* should always be prepared with mercury; otherwise, it is not absorbed properly in the intestine. Additional processes are used to obtain the best quality iron *bhasma*. This includes *loha maraka gana*, *amritkarana*, and *nirutthikarana*. In the *loha maraka*, fresh lemon juice is prepared and a specific amount of *hingula* powder is added. Then these ingredients are mixed thoroughly. Afterward, the process of repeated dipping (*nirvapana*) red-hot iron leaflets in the medicinal fluid is applied to obtain *loha bhasma*. In the *amritkarana* process, equal amounts of *loha bhasma* and *ghrita* are placed in an iron pan and mixed properly under mild heat until the fat disappears. This compound now bears a *yagavahi* character. Finally, the properly prepared *bhasma* and *trifala* decoction (*bhavana drava*) is exposed to heat by the sun (*suryaputa*) or by the process of burning the herbs in closed freshly made mud containers (*putapaka*). At the end of this procedure, *loha bhasma* becomes the end product (*niruttha* and *varttara*) of filtration and separation procedures (*nirutthikarana*).¹¹

The organoleptic characteristics of *loha bhasma* are that it is dark brown, has a faint smell, fine touch, and no taste. Iron as Fe_2O_3 is no more than 96.575% w/w, and iron as Fe is no more than 75% w/w present in this *bhasma*. *Loha bhasma* contains an ash value not less than 96.8% w/w and not more than 99.7% w/w. Its acid-insoluble ash is between 0.101 and 2.803% w/w. Its dose is 100 to 250 mg.¹⁴

6.7 Therapeutic Indications and Scientific Investigations on *Bhasmas*

6.7.1 *Swarna Bhasma*

6.7.1.1 *Indications*

Gold has been used since the Vedic era in India to enhance strength and potency, promote longevity, and combat the aging process in humans. Around the eighth century A.D., gold was used in the form of *bhasma* after proper purification and incineration as described in the *Ayurvedic Pharmacopoeia*. Gold preparations are recommended to promote longevity, combat aging, and treat impotency. They are also used as a tonic, hepatotonic, cardiostimulant, nervine tonic, detoxifier, and an anti-infective drug. *Swarna bhasma* is also used in the treatment of diseases such as anemia, dyspepsia, epilepsy, neurasthenia, loss of memory, bronchitis, asthma, tuberculosis, leucoderma, and rheumatoid arthritis.^{9,15} The recommended dose is 100 to 250 mg.¹⁴ In modern medicine, gold is primarily used in rheumatoid arthritis. Some gold compounds used as medicine, in modern medicine are listed in [Table 6.2](#). Of all these gold compounds, auranofin and gold sodium thiomalate (GST) are the most extensively studied.

6.7.1.2 *Animal Studies*

6.7.1.2.1 *Analgesic Activity*

Analgesic activity of *swarna bhasma* was reported in one study involving mice.¹⁷ A Unani calcined gold preparation, *kushta tilan kalan* (KTK), was also used in this study along with the widely used gold drug auranofin. *Swarna bhasma* at 12 to 50 mg/kg body weight by oral route showed analgesic activity against chemical, thermal, electrical, and mechanical stimuli. Whereas the analgesic effect of *swarna bhasma* could be blocked by the treatment of nolaxone, such antagonism was not possible in the case of auranofin. The study suggests involvement of opiodergenic mechanism in the observed analgesic activity of *swarna bhasma*.

6.7.1.2.2 *Immune Response*

Specific and nonspecific immune responses were modified in a positive manner in *swarna bhasma*-treated mice.^{18,19} The doses were in the range of 12.5 to 50 mg/kg body weight. *Swarna bhasma* had a stimulatory effect on peritoneal macrophages, which may be helpful to fight against infections. It was suggested that macrophages achieved stimulation possibly due to presentation of the metal to cells in fine emulsified form.

6.7.1.2.3 *Antioxidant Activity*

The antioxidant and restorative effects of *swarna bhasma* in rats have recently been demonstrated.²⁰ The study was done under ischemia, which was induced by bilateral carotid occlusion under phenobarbitone anesthesia. A number of antioxidant enzymes and lipid

TABLE 6.2

Biologically Active Gold Compounds on the Market

No.	Generic Name	Trade Name	Gold Concentration (%)
1	Gold sodium thiomalate	Myochristin, Myocrisin, Tauredon	50.5
2	Gold thioglucose	Solganal	50.5
3	Gold thioglycoanilid	Lauron	54.2
4	Gold sodium thiosulphate	Sanochrysine, Aurothion, Thiochrysine	402
5	Calcium aurothiothioglycolate	Myoral	64.1
6	Sodium 2-aurothiobenzidazole-4-carboxylate	Triphal	47.8
7	Sodium auroallylthiourea- <i>m</i> -benzoate	Lapion	43.4
8	<i>S</i> -triethylphosphine gold 2,3,4,6-tetra- <i>O</i> -acetyl-1-thio- β -D-glycopyranoside	Auranofin	29.1
9	Chloro (triethylphosphine) gold	SK&F 36914	56.2

Based on information from Lewis, A.J. and Walz, D.T., *Progress in Medicinal Chemistry*, Vol. 19, Elsevier Biomedical Press, New York, 1982, 1–59, and Insel, P.A., *Goodman's and Gilman's: The Pharmacological Basis of Therapeutics*, 9th ed., McGraw-Hill, New York, 1996, 644.

peroxidation were studied. *Swarna bhasma* (25 mg/kg) had a restorative effect, significantly restoring antioxidant levels in ischemic animals. Biochemical findings supported the histological examination of the brain. The free radical scavenging and antioxidant effects of *swarna bhasma* have also been investigated recently by several groups. Two antioxidant enzymes — superoxide dismutase (SOD) and catalase — were measured after oxidative insult with acetic acid in both *swarna bhasma*-treated mice as well as control serum and liver homogenate of mice.¹⁵ *Swarna bhasma* induced activity of SOD and catalase. Chronic administration of *swarna bhasma* had no toxic effects as assessed by liver function test enzymes and histological investigations.

Biological investigations on *swarna bhasma* are summarized in Table 6.3.

6.7.1.3 Clinical Studies

Gold containing Ayurvedic preparation, *swarna vasanti malti*, was investigated for safety. The *bhasma* was given to 20 male persons at a dose of 100 mg twice a day for 40 days under supervision of Ayurvedic physicians. The total cumulative intake of 160 mg of gold at the rate of 4 mg/day in this form did not have any toxic effect on human body, as evidenced by clinical examination, unaltered body weight, absence of urinary pathology, and by 30 sensitive biochemical and enzymatic tests.²¹

TABLE 6.3

Summary of Scientific Investigations on Swarna Bhasma

No.	Investigation	Model	Ref.
1	Analgesic activity	Mouse, rat	17
2	Immunomodulatory activity: specific immunity	Mouse	18
3	Immunomodulatory activity: nonspecific immunity	Mouse	19
4	Evaluation of safety	Human	22
5	Antioxidant effect	Rat	20
6	Analysis of chemical constituents	—	16
7	Free radical scavenging activity	Mouse	16

6.7.2 Loha Bhasma

6.7.2.1 Indications

Loha bhasma is a powerful hematinic Ayurvedic drug indicated in anemia. It stimulates the appetite and has a general vitalizing effect. It is readily assimilated in the body.

6.7.2.2 Animal Studies

Various *loha bhasmas* made by different manufacturers were screened, and one representative preparation was evaluated for its effect in managing experimentally induced anemia in Charles Foster albino rats. Comparison was made with Fefol[®],²² a widely used standard drug. Anemia was induced by dietary means (agar gel treatment) and by bloodletting (phlebotomy) through the tail vein. Various parameters of hemoglobin estimation, serum ferritin estimation, and total iron and iron binding capacity of serum were studied. *Loha bhasma* was helpful not only in resorting hemoglobin level, but in significantly increasing body weight gain in *bhasma*-treated animals. In some parameters results were better than those observed with Fefol.

No clinical study on iron *bhasma* was found in our literature search.

6.7.3 Copper (Tamra) Bhasma

6.7.3.1 Indications

Tamra bhasma is used as a single drug and also in combination with many medicinal plant extracts. *Tamra bhasma* is used for the management of liver disorder, arthritis, old age disorders, leucoderma, etc.

6.7.3.2 Animal Studies

The hepatoprotective effect of *tamra bhasma* has been studied on cumene hydroperoxide-induced peroxidation, reduced glutathione content, and SOD in rat liver homogenate.²³ The drug was orally given for 8 days at different dose levels (0.5, 1.0, and 5.0 mg/100 g body weight). It showed significant reduction in the level of lipid peroxidation. The results suggested that *tamra bhasma* is a strong antioxidant drug and could be used in the management of lipid peroxidation. It showed no acute detectable adverse effects. Levels of SOD were also enhanced by *tamra bhasma*.

The hepatoprotective effects of four Ayurvedic drugs — *kumari asav*, *kumari kalp*, *arogyavardhini*, and *tamra bhasma* — against the toxicity of carbon tetrachloride (CCl₄) was established in a series of experiments in rats.²³ The hepatoprotective effect of these drugs was in the following order: *tamra bhasma* < *arogyavardhini* < *kumari kalp* < *kumari asav*. Significant reduction in hepatic necrosis was recorded in all the treated animals.^{23,24} Effects of these drugs on acid lipase, alkaline lipase, lipoprotein lipase of the liver and kidney, adipose tissue and hormone-sensitive lipase, acid phosphatase, and beta-glucuronidase were also investigated in the animals treated with the above-mentioned drugs.²⁴⁻²⁶ Histologically, all the above formulas showed hepatoprotective effects in CCl₄-exposed animals.

6.7.4 Abhrak Bhasma

6.7.4.1 Indications

Abhrak bhasma is prepared by treating biotite (mica) with the juices of a number of reconstituent plants that make it a powerful cellular regenerator. It is a commonly used Ayurvedic drug against many diseases, including hepatitis. It is also a nervine tonic and is widely used in respiratory tract infections and anemia. It contains iron, magnesium,

potassium, calcium, and aluminum in traces. *Abhrak bhasma* is an amorphous powdery drug. Mica mainly contains silicates of iron, magnesium, and aluminum.²⁷

6.7.4.2 Animal Studies

The hepatoprotective action of *abhrak bhasma* has been reported in albino rats using a model of hepatitis induced by a single dose of CCl_4 (3 ml/kg body weight). Different doses of *abhrak bhasma* (10, 20, 30, and 40 mg/kg body weight) were used.²⁷ The centrilobular necrosis induced by a single dose of CCl_4 was reduced significantly by *abhrak bhasma* (10 mg), and liver histology was also protected by a 20-mg dose. Liver acid lipase activity was lowered, whereas alkaline and lipoprotein lipase activity was elevated due to the treatment of CCl_4 . *Abhrak bhasma*-treated animals that were exposed to CCl_4 showed marked improvement in enzyme profile.

6.7.5 Mandur Bhasma

6.7.5.1 Indications

Mandur bhasma is prepared by purifying and calcinating iron rust. It is especially useful in anemia, amenorrhea, dysmenorrhea, menorrhagia, chlorosis, and hepatic and splenic disorders. It is also used in diarrhea, chronic bowel complaints, dyspepsia, intestinal worms, nervous system diseases, neuralgia, kidney diseases, and albuminuria. It is a powerful hematinic and tonic and is valuable in the treatment of hemolytic jaundice and microcytic anemia.

6.7.5.2 Animal Studies

The protective effect of *mandur bhasma* (10 mg/kg body weight) has been demonstrated in CCl_4 -treated rats.²⁸ *Bhasmas* were orally administered to animals. Hepatoprotection was established by various enzymatic parameters, lipid peroxidation, and a histological examination of tissues.

6.7.6 Muktaashukti Bhasma

Muktaashukti bhasma is a compound *bhasma* consisting of pearl (*moti*), *Aloe vera* Linn. (*guar patha*), and vinegar (*kanji*). The *bhasma* is prepared from the outer covering of the shell (pearl-oyster), and is ground and triturated with *A. vera* and vinegar in sufficient quantity to make a homogeneous paste. The recommended proportion of pearl-oyster and *A. vera* is in the ratio of 1:4.

6.7.6.1 Indications

Muktaashukti bhasma has been used in the treatment of tuberculosis, cough, chronic fever, conjunctivitis, abdominal discomfort, biliary disturbances, asthma, heart diseases, vomiting, acidity, dyspepsia, dysmenorrhea, general weakness, arthritis, rheumatism, musculoskeletal disorders, etc.

6.7.6.2 Animal Studies

An anti-inflammatory effect of *muktaashukti bhasma* has been shown in albino rats by Chauhan et al.²⁹ The *bhasma* inhibited acute and subacute inflammation in albino rats as induced by subplanter injection of carageenan, histamine, serotonin (5-HT), nystatin, and subcutaneous implants of cotton pellets. In all the test procedures, the anti-inflammatory

response of 1000 mg/kg of *muktashukti bhasma* was comparable with the response observed with 300 mg/kg of acetyl salicylic acid, which was used as a standard anti-inflammatory drug. Oral premedication with delayed castor oil-induced diarrhea in rats, indicating its prostaglandin inhibitory activity.²⁹

6.7.7 *Sankha Bhasma*

Sankha bhasma is a powder prepared from the calcinated conch shell. It consists mainly of calcium, iron, and magnesium.

6.7.7.1 *Indications*

Sankha bhasma is well known for its antacid and digestive properties. It is useful in hyperchlorhydria, sprue, colic, and hepatosplenomegaly.

6.7.7.2 *Biological Studies*

A mixture of some Ayurvedic medicines that contained *sankha bhasma* and the herbs *Glycyrrhiza glabra*, *Terminalia chebula*, and *Piper longum* showed protection against duodenal ulcer in rats.³⁰ These drugs act on Bruner's gland by improving its secretory status.

6.7.8 *Miscellaneous Studies on Bhasmas*

Yashada bhasma is a specially processed zinc. It is administered in sprue, diabetes, leucorrhea, and hyperhydrosis. The role of the *bhasma* in arresting myopia has been examined in one study.³¹

Contamination of *bhasmas* directly through the herbs used in the preparation and formation of polycyclic aromatic hydrocarbons (PAHs) is expected. Sometimes *bhasmas* may be contaminated with very harmful substances. *Bhasmas* were analyzed and found to contain PAH. The levels of total PAH varied widely (2.32 to 9.55 ppm) among the preparation tested. The benzo[a]pyrene level also varied, the highest concentration being 9.7 ppm.³²

The studies presented here suggest *bhasmas* may have a hepatoprotective effect. However, efforts should be made to study their beneficial effects on other systems. Especially, evaluation of their immunomodulatory and neuroprotective actions may prove to be rewarding.

A summary of scientific investigations on the above-mentioned *bhasmas* is provided in Table 6.4.

TABLE 6.4

Summary of Scientific Investigations on Other Bhasmas

Bhasma	Investigation	Animal Model	Ref.
<i>Abhrak bhasma</i>	Hepatoprotective activity	Rat	28
<i>Loha bhasma</i>	Antianemic	Rat	23
<i>Mandur bhasma</i>	Hepatoprotective activity	Rat	29
<i>Muktashukti bhasma</i>	Anti-inflammatory activity	Rat	30
<i>Sankha bhasma</i>	Antiulcerogenic activity	Rat	31
<i>Tamra bhasma</i>	Hepatoprotective activity	Rat	24–27

TABLE 6.5

Metal and Mineral Drugs Used in Unani System of Medicine

No.	Metal and Mineral (Unani Name)	Name of Unani Drug (% of Metal and Mineral)
1	Aluminium	<i>Kushta-e-Zumurrud</i> (16.55%)
2	Ammonium chloride (<i>naushadar</i>)	<i>Habbe Kabid Naushadri</i>
3	Antimony (<i>surma siyah</i>)	<i>Kehul-ul-Jawahar</i>
4	Arsenic (<i>sankhya</i>)	<i>Kushta Sam-ul-far</i> (18.89%)
5	Borax (<i>suhaqa</i>)	<i>Safoof-e-Chutki Atfal</i>
6	Bromide	<i>Kushta-e-Sadaf</i> (2.4458%)
7	Calcium	<i>Khameera-e-Zaharmohra</i> (27.16 mg/g)
8	Copper	<i>Jawarish-e-Jalinoos</i> (7.82 mg/g ash)
9	Coral (<i>marjan</i>)	<i>Kushta Marjan</i>
10	Diamond (<i>almas</i>)	<i>Kushta Almas</i>
11	Gold (<i>sona</i>)	<i>Kushta Tilan Kalan</i>
12	Iron (<i>loha</i>)	<i>Kushta-e Sadaf</i> (77 mg%), <i>Kushta-e-Faulad</i> (48.75%)
13	Iron rust (<i>khabsul-hadid</i>)	<i>Kushta Khabsul-Hadid</i>
14	Lead	<i>Sharbat-e-Khas Khaash</i> (8.333 mg/g)
15	Magnesium	<i>Kushta-e-Busud</i> (44 mg/g)
16	Mercury (<i>para</i>)	<i>Marham Gulabi</i>
17	Pearl (<i>moti</i>)	<i>Kushta Sadaf</i>
18	Potassium	<i>Safoof-e-namak-e-shaikh-ur-raees</i> (1190 mg%)
19	Shell (<i>sadaf</i>)	<i>Kushta Sankh</i>
20	Silver	<i>Kushta-e-Nuqra</i> (93%)
21	Strontium	<i>Kushta-e-Kharmohra</i> (1.32%)
22	Sulphur (<i>kibreet</i>)	<i>Habbe Kibreet</i>
23	Talc	<i>Kushta Abhrak</i>
24	White lead (<i>safeda kashgiri</i>)	<i>Marham Kafoori</i>

Based on information from Council for Research in Unani Medicine, *Physicochemical Standards of Unani Formulations*, Parts I and II, Central India, New Delhi, India, 1986, and Ali, S.S., *Unani Adviya Mafrada*, Turkey Urdu Beaurou, New Delhi, India, 1982, chaps. 3 and 4 (in Urdu).

6.8 Metal and Mineral Preparations of Unani-Tibb

Like Ayurveda, a large number of metal and mineral preparations are used in the Unani system of medicine (Unani-tibb). In Unani-tibb, the procedures of calcination, purification, and trituration are also undertaken to purify and enhance the therapeutic efficacy of drugs. All the important metals (i.e., copper, gold, silver, lead, arsenic, mercury, etc.) are used in this system of medicine. The calcinated drugs of Unani-tibb are termed *kushta*. Drugs derived from shell, oyster, and coral are also used in Unani-tibb. Although there is some similarity between the procedural details of calcination, purification, and trituration, Ayurvedic and Unani-tibb systems of medicine have a number of differences.^{5,12} Some of the most widely used drugs prepared from metals and minerals that are used in Unani-tibb are listed in Table 6.5.

6.9 Discussion

Various processes of marana and sodhana involve the incorporation of herbal juices that presumably makes the drugs more efficacious. The drug that is presented in the final form

seems to have an enhanced bioavailability, especially through the digestive system. It is also likely that most of the *bhasmas* act as adjuvants and thus may be potentiating the nonspecific immune functions (macrophage activation). They may also activate drug-metabolizing enzymes. Further scientific investigations are warranted. Studies should be undertaken aiming at use of radio-labeled metallic compounds in Ayurvedic medicine and subsequently their distribution and disposition should be tracked down.

Metal and mineral preparations used in Ayurveda, Siddha, and Unani-tibb are often subjected to scientific scrutiny. Analytical laboratories sometimes publish news articles about the existence of heavy metals like arsenic, lead, and mercury in Ayurvedic formulations. This makes the public uneasy to use Ayurvedic and other traditional drugs. However, it is necessary to note that many important metals are essential components of vital molecules of the body; every year the essential nature of some hitherto nonessential elements is established in biological systems.

In Ayurveda, a great emphasis is placed on *shodhana* (purification) and detoxification of metals and other minerals. The process of *shodhana* is followed by the incorporation of various herbal juices to get the final product. This alters the metallic salt forms and the bioavailability. These processes can also convert the metalloids in emulsified form, acting as an adjuvant and eliciting various responses, especially the immune responses. The unabsorbed parts are normally eliminated in the stool after localized activities in the receptor sites of the gut. The activity of any pharmacologically active substance is based on the dose of administration, duration of administration, frequency of administration, and the adjuvant, which may influence bioavailability. In Ayurveda, these principles are the essential parts of therapy. Many of the *bhasmas* are recommended at very low doses (often in divided doses) and for a specific period of time. Sometimes the dose levels of elements may even normally occur in water sources. Thus, dose is a very important factor in using *bhasmas*.

Metals and minerals in Ayurveda are not essentially the first line of treatment for common diseases. Introduction of metallic preparations is the last resort only when all other methods of treatment have not yielded desired results. One common criticism that is associated with Ayurvedic preparations, particularly those prepared from metals and minerals, is their nephrotoxicity. This may be due to self-medication, which is a common practice because patients do not bother to consult an Ayurvedic physician for proper prescription and, unlike modern medicine, no proper treatment regimen is followed. Another problem relates to drug standardization. The sporadic incidence of environmental contamination may ultimately find its way into plant parts. Due to the use of improper apparatus, contamination may also occur. With the implementation of good manufacturing practices (GMP) in Ayurvedic pharmaceutical industry, these problems will be overcome.

There is also a need to have preclinical and clinical investigations on some selected *bhasmas*. World Health Organization guidelines clearly direct that it is not necessary to carry out detailed toxicological evaluation of the herbs or their compounds originating from traditional systems of medicine. Length of continuity of use is to be taken as testimony of safety.

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Diabetes Mellitus (Madhumeha)

Lakshmi Chandra Mishra and Tarek Adra

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7.1 Introduction

Diabetes mellitus (DM) is described in Ayurveda as *madhumeha kshaudrameha*, which literally means “excessive urine with sweet taste like honey,” or *dhatupak janya vikriti*, which means a disease caused by a defective metabolism leading to derangement in body tissue (seven *dhatu*s) transformation process. Historically, Ayurvedic texts have described 20 types of urinary disorders (*pramehas*) based on the predominant dosas (10 *kaphaja*, 6 *pittaja*, and 4 *vataja* urinary disorders) and physical characteristics of the urine (e.g., volume, color, odor, taste, sediments, solid particles, presence of seminal fluid, and mucus). The urine is discharged in excessive quantities and is generally turbid. DM is one of these *pramehas* that may occur in any of the three (*vata*, *kapha*, or *pitta*) body constitutions.

DM is a very common health problem; almost 3% of the world population or 100 million people suffer from it.¹ It is the fourth most common cause for patient visits to a physician in the U.S. It accounts for nearly 15% of health-care costs in the U.S. DM may result in premature disability, mortality, blindness, end-stage renal disease (ESRD), and nontraumatic limb amputations. DM is also known to increase the risk of heart, cerebral, and vascular diseases by two- to sevenfold. Many of the complications of DM are preventable or can be delayed by appropriate treatment of hyperglycemia and other cardiovascular risk factors.² Synthetic drugs like sulphonyl ureas and biguanides may be effective in controlling the blood sugar level for some time, but they may cause side effects like hypoglycemia, nausea, vomiting, cholestatic jaundice, and other health problems. Most type 2 DM patients initially respond to lowering of blood glucose levels (BGLs), but after some time about 20% become resistant and do not benefit from these agents. Patients may also not respond fully to these agents due to the loss of interest in regular diet and exercise, the progression of beta cell failure, drug resistance, and other medical problems.² Many patients eventually require insulin treatment.³

The Ayurvedic approach to DM management includes life-style dietary interventions, exercise, and a variety of hypoglycemic herbs and herbal formulas depending upon the predominant *dosa*. Cleansing procedures are unique to the Ayurvedic approach to DM. However, the Ayurvedic clinical description of DM, etiology, diagnosis, prognosis, and recommended lifestyle changes are basically similar to those described in Western medicine.

The aim of this chapter is to explore the scientific basis for the Ayurvedic management of DM with single herbs, herbal formulas, and dietary interventions, in addition to understanding the Ayurvedic concept of DM and exploring the underlying science. Available pharmacological, biochemical, and chemical studies on herbs are used to understand the scientific basis of Ayurvedic therapies of DM.

7.2 Clinical Description (*Rog Viakhya*) and Etiology (*Vyadhi Haitu*)

The major signs and symptoms of DM described in classic Ayurvedic texts consist of honeylike sweetness of urine, thirst, polyphagia, lassitude, tiredness, obesity, dysgeusia, constipation, burning sensation in the skin, seizures, insomnia, and numbness of the body. Boils, wounds, and abscesses are often difficult to heal in a diabetic patient and are recognized in Ayurveda. All these symptoms are very similar to those currently described

in Western medicine. Ayurvedic physicians also use modern diagnostic chemical analysis of urine and blood for confirmation.

The etiology of DM in Ayurveda is multifactorial. DM may be a familial trait, and overweight (fat — *meda*) patients with this diagnosis may be engaged in a lethargic lifestyle and unhealthy diet (e.g., idle sitting, excessive sleep, overeating sweet and fatty food items, and lack of physical exercise). All these factors are understandable because they all can lead to type 2 DM.¹ Ayurveda divides DM into two categories: (1) genetic (*sahaja*), occurring in young age from the very beginning of life that has some similarities with the juvenile diabetes or insulin-dependent diabetes; and (2) acquired (*apathyaja*) due to an unhealthy lifestyle that occurs in old age and obese people and has similarities with type 2 DM. In addition, Charak Samhita (100 to 400 A.D.) describes two types of DM: one that occurs in very underweight people (*krsa prameha*) and one that occurs in obese people (*sthula*). The former DM requires restorative (*santarpan*) treatment along the line of insulin treatment and the latter requires fat-reducing (*apatarparna*) treatment.⁴

Type 1 DM is associated with the absence of insulin-secreting capacity of the patient, and this problem accounts for less than 10% of DM patients in the U.S. Type 2 DM patients have some degree of insulin-secreting capacity and account for more than 90% of DM patients in the U.S. Miscellaneous types of DM are very small in number and have various etiological factors (e.g., genetic defect of beta cells, defects in insulin action, diseases of pancreas, endocrinopathies [Cushing's syndrome, hyperthyroidism, and others], and drugs and chemicals [glucocorticosteroids and others]). Other etiological factors are infections (congenital rubella, cytomegalovirus), immune-mediated DM, other genetic syndromes (Down, Klinefelter, and others), and gestational disorders. Because all types of diabetes may require insulin treatment at some point, the terms "insulin-dependent" and "insulin-nondependent" DM are no longer used.²

Recent studies have shown that there is a genetic component in the etiology of type 1 DM. In fact, Kyvik et al.⁵ have shown that the cumulative concordance among twins (both twins affected) from birth to age 25 is 70%, which indicates that genetic factors cannot account for 100% of the incidence, but that environmental, immunity, and other factors are contributory. Several genes (HLA-DR3 or HLA-DR4) have been found in 95% of Caucasians with type 1 DM, whereas these genes are found in only 45% of the general DM population.⁶ Autoimmunity is suggested to be an important contributory factor because a lymphocyte-rich infiltrate, an indicator of immunological response, was found in pancreatic islets.⁷ In addition, 70 to 80% of the patients are found to have islet cell autoantibodies against intracellular islet cell antigens such as glutamic acid decarboxylase islet auto-antigen 2, insulin, and gangliosides.⁸ Support for the role of environmental factors came from observations that Finnish children had 60- to 70-fold greater risk of type 1 DM than did Korean children. The incidence of DM in children under 15 in the northeastern U.S. has tripled since the late 1960s, indicating the involvement of environmental factors.¹ Epidemiological studies have shown an association between Coxsackie viruses of group B and pancreatic diseases, including DM, that indicates the possibility of a contributory role of viruses in the etiology of DM.⁹ Chemicals such as streptozotocin, alloxan, and pentamidine have also been shown to produce DM by destroying the insulin-producing islet cells.¹

Type 2 DM has even greater involvement of genetic factors than type 1 DM, but no specific gene has been linked to DM to account for the role. There is no evidence available to show the role of autoimmunity in type 2 DM. The two major problems in type 2 DM are the reduced secretion of insulin from beta cells and development of resistance to insulin in peripheral tissues. Obesity is another major etiological factor, for 80% of type 2 DM patients are obese.¹

7.3 Pathogenesis (*Samprapti*)

According to classic Ayurvedic texts, DM and all *pramehas* (urinary disorders) start with the derangement of *kapha* that spreads throughout the body and mixes with fat (*meda*) that is similar in physical properties to *kapha* (mucus). *Kapha* mixed with fat passes into the urinary system, thereby interfering with normal urine excretion. Vitiating *pitta*, *vata*, and other body fluids (*malas*) may also be involved in this blockade. This blockade is believed to be the cause of frequent urination observed in DM. *Pramehas* left untreated may lead to deranged development of the bone marrow, body tissues, nutritional materials (fat, proteins, and carbohydrates), and hormones (*ojas*). The incurable stage of *pramehas* is *madhumeha*, which is insulin-dependent DM. *Madhumeha* may not be described precisely in Ayurveda, but it points in the direction of the current knowledge we have about the disease with respect to neurological damage and insulin (*ojas*) malfunctioning at the production (degeneration of islets of Langerhans in the pancreas) or at the utilization levels. The involvement of tissues (*dushyas*) leading to blood vessels, kidney, eye, and nerve damage is also described in Ayurveda as major complications. DM is described not only as a condition of *madhumeha* (sugar loss in urine), but also as a condition of *ojameha* (immunity and hormone loss) in Ayurveda for the purpose of treatment.

DM is primarily a dysfunction of insulin, a hormone produced by beta cells in the pancreas. Insulin is necessary for cellular uptake of glucose, glycogen formation in the liver and skeletal muscles, conversion of glucose into triglycerides, as well as nucleic acid and protein synthesis. The BGLs are maintained by three factors: (1) the production of glucose by the liver, (2) the uptake and utilization of glucose by the peripheral tissues, and (3) the production and release of insulin by the pancreas. The blood insulin levels are maintained by glucose, which stimulates the synthesis of insulin, and calcium-dependent secretion of insulin by other agents (e.g., amino acids and gastrointestinal hormones). The key feature of DM is impaired glucose tolerance. In a normal person, the BGL returns to normal within an hour after a glucose challenge dose, whereas the level remains high in a diabetic or prediabetic person for many hours.¹

7.4 Clinical Course and Prognosis (*Sadhyata*)

According to classical Ayurveda, all *pramehas* have the potential to become incurable (*madhumeha*) if left untreated. The *kaphaja* urinary disorders (*pramehas*) are curable because the causative *dosa* and the affected tissues (*dushya*) have the same properties, thus requiring the same type of therapy. Although the *pittaja* urinary disorders are controllable (palliative), the resulting disorder may persist for life because the causative *dosa* is *pitta*, but the tissues and waste products (*dushya*) are different, requiring a different type of therapy. *Vataja* urinary disorders are considered incurable because tissues (*dhatu*s) and hormones (*ojas*) undergo deterioration.

Recent studies have observed a relationship between the body constitution and relative amounts of hyperglycemia and insulinemia consistent with the Ayurvedic prognosis.¹⁰⁻¹² *Kapha* constitution patients showed the highest level of insulinemia and the lowest levels of FBS and PPBS. *Vata* patients showed the lowest level of insulinemia and the highest levels of FBS and PPBS. *Pitta* patients were in the middle. Further studies are necessary to confirm these findings.

In Ayurveda the major complications of *kaphaja* urinary disorders are believed to be poor digestion, anorexia, vomiting, drowsiness, coughing, and nasal catarrh. *Pittaja* urinary disorder patients tend to exhibit a pricking pain in the urinary bladder, penis, and scrotum, as well as fever, burning sensations, thirst, sourness of the throat, fainting, and loose bowel movements. *Vata* urinary disorders (diabetes) patients often experience tremors, pain in the cardiac region, abdominal tenderness, insomnia, and dryness of the mouth. The major complications of *vata* DM most commonly include ulcers (eruptions) over joints, muscles, skin, blood vessels, as well as damage to the kidney and the retina.

7.5 Clinical Examination and Diagnosis (*Rog Pariksha* and *Nidan*)

Historically, Ayurveda diagnosis of DM was primarily based on the sweetness of urine that was identified by a swarm of flies and ants over the urine. Ayurvedic physicians currently use urine, blood sugar, and glycohemoglobin (Hb A_{1c}) levels to confirm the diagnosis. The eight-point diagnosis discussed in [Chapter 2](#) is applied to identify the predominant *dosa*. Ancient Ayurvedic texts give the following signs and symptoms of *kaphaja*, *pittaja*, and *vataja pramehas* for diagnosis. Recently, however, Ayurvedic doctors' diagnostic tools evolved to more modern clinical and laboratory methods consistent with those of Western medicine:

1. *Kaphaja pramehas*

- a. *Udaka meha* — The urine is clear; is in large amounts; is white, cold, and odorless; resembles water, sometimes with slight turbidity, and slimy.
- b. *Iksu meha* — The urine is like sugarcane juice and is very sweet.
- c. *Sandra meha* — The urine becomes thick when kept overnight.
- d. *Sura meha* — The urine resembles beer (*sura*) with a clear top and a cloudy bottom portion.
- e. *Pista meha* — The urine is white and thick, similar to a solution of corn flour.
- f. *Sukra meha* — The urine is like semen or mixed with semen.
- g. *Sita meha* — The urine is sweet and very cold.
- h. *Sikata meha* — The urine contains sandlike particles.
- i. *Sanair meha* — The urine is passed very slowly.
- j. *Laala meha* — The urine is slimy and contains threads like that of saliva.

2. *Pittaja pramehas*

- a. *Ksara meha* — The urine is like a solution of alkali in smell, color, and taste.
- b. *Kala meha* — The urine is black.
- c. *Nila meha* — The urine is bluish.
- d. *Haridra meha* — The urine is yellowish, similar to tumeric.
- e. *Manjistha meha* — The urine is foul smelling resembling *manjistha* (*Rubia cordifolia*), a slightly red solution.
- f. *Rakta meha* — The urine is foul smelling, slightly salty, and blood red.

3. *Vataja pramehas*

- a. *Majja meha* — The urine looks like marrow or marrow mixed.

TABLE 7.1

Decoctions to Treat *Kaphaja Pramehas*

Medicinal Plant	Botanical Name	Medicinal Plant	Botanical Name
Decoction #1		Decoction #2	
<i>Katphala</i>	<i>Myrica esculenta</i>	<i>Patha</i>	<i>Stephania japonica</i> Miers
<i>Musta</i>	<i>Cyperus rotundus</i>	<i>Vidanga</i>	<i>Embelia ribes</i>
<i>Lodhra</i>	<i>Symplocos cochinchinensis</i>	<i>Arjuna</i>	<i>Terminalia arjuna</i>
<i>Haritaki</i>	<i>Terminalia chebula</i>		
Decoction #3		Decoction #4	
<i>Haridra</i>	<i>Curcuma longa</i>	<i>Kadamba</i>	<i>Anthocephalus chinensis</i>
<i>Daru haridra</i>	<i>Berbers aristata</i>	<i>Sala</i>	<i>Shorea robusta</i> Geartn.
<i>Tagarai</i>	<i>Cassia tora</i> Linn.	<i>Arjuna</i>	<i>Terminalia arjuna</i>
<i>Vidanga</i>	<i>Embelia ribes</i>	<i>Yavani</i>	<i>Carium copticum</i> Benth Hook
Decoction #5		Decoction #6	
<i>Daruharidra</i>	<i>Berbers aristata</i>	<i>Devodaru</i>	<i>Polyalthia longifolia</i>
<i>Vidanga</i>	<i>Embelia ribes</i>	<i>Kushta</i>	<i>Saussurea</i>
<i>Khadira</i>	<i>Acacia catechu</i> willd	<i>Aguru</i>	<i>Aquillaria agallocha</i> roxb.
<i>Dhava</i>	<i>Anogeissus latifolia</i>	<i>Candana</i>	<i>Santalum album</i> Linn.
<i>Candana</i>	<i>Santalum album</i> Linn.		
Decoction #7		Decoction #8	
<i>Daruharidra</i>	<i>Berbers aristata</i>	<i>Patha</i>	<i>Stephania japonica</i> Miers
<i>Agnimantha</i>	<i>Clerodendrum multiflorum</i>	<i>Murva</i>	<i>Celosia cristata</i> Linn.
<i>Triphala</i>	<i>Terminalia chebula</i> , <i>terminalia bellerica</i> , and <i>Emblica officinalis</i> (mixture)	<i>Gokshura</i>	<i>Tribulus lanuginosus</i>
<i>Patha</i>	<i>Stephania japonica</i> Miers		
Decoction #9		Decoction #10	
<i>Yavani</i>	<i>Carium copticum</i> Benth	<i>Cavya</i>	<i>Piper chaba</i> Hunter
<i>Usira</i>	<i>Vativeria zizanioides</i> Linn.	<i>Haritaki</i>	<i>Terminalia chebula</i>
<i>Haritaki</i>	<i>Terminalia chebula</i>	<i>Chitraka</i>	<i>Plumbago zeylanica</i>
<i>Guđuci</i>	<i>Tinospora cordifolia</i>	<i>Saptaparna</i>	<i>Alstonia scholaris</i>

- b. *Ojas meha* — The urine looks like honey.
- c. *Vasa meha* — The urine looks like liquid muscle fat and may be passed frequently.
- d. *Hasti meha* — The urine is like that of an elephant in rut, being discharged continuously without force, mixed with lymph and without obstruction.

These urine characteristics may be found in a wide range of pathologies covering all kinds of urinary infections, obstructive uropathies, renal failures, and other health conditions. The *kaphaja iksu meha* and *vataja ojas meha* are the correlate of the modern understanding of DM.

The results of diagnosis are correlated with body constitutions in order to design an individualized therapy.

TABLE 7.2

Decoctions to Treat *Pittaja Pramehas*

Medicinal Plant	Botanical Name	Medicinal Plant	Botanical Name
Decoction #1		Decoction #6	
<i>Usira</i>	<i>Vativeria zizanoides</i> Linn.	<i>Nimba</i>	<i>Azadirachta indica</i> A.
<i>Lodhra</i>	<i>Symplocos cochinchinensis</i>	<i>Arjuna</i>	<i>Terminalia arjuna</i>
<i>Arjuna</i>	<i>Terminalia arjuna</i>	<i>Guduchi</i>	<i>Tinospora cordifolia</i>
<i>Candana</i>	<i>Santalum album</i> Linn.	<i>Haridra</i>	<i>Curcuma longa</i>
		<i>Utpala</i>	<i>Nymphaea stellata</i>
Decoction #2		Decoction #7	
<i>Usira</i>	<i>Vativeria zizanoides</i> Linn.	<i>Sirisha</i>	<i>Albizia lebbek</i>
<i>Musta</i>	<i>Cyperus rotundus</i>	<i>Sarja</i>	<i>Vateria indica</i> Linn.
<i>Amalata</i>	<i>Cayratia trifolia</i>	<i>Arjuna</i>	<i>Terminalia arjuna</i>
<i>Abhaya</i>	<i>Terminalia chebula</i> retz	<i>Nagakesar</i>	<i>Mammea suriga</i>
Decoction #3		Decoction #8	
<i>Patala</i>	<i>Cydonia oblonga</i>	<i>Priyangu</i>	<i>Aglaiia elaeagnoidea</i>
<i>Nimba</i>	<i>Azadirachta indica</i> A.	<i>Kamala</i>	<i>Nelumbo nucifera</i>
<i>Amalaka</i>	<i>Embilca officinalis</i> Geartn.	<i>Utpala</i>	<i>Nymphaea stellata</i>
<i>Gudici</i>	<i>Tinospora cordifolia</i> Miers.	<i>Palasha flower</i>	<i>Butea monosperma</i>
Decoction #4		Decoction #9	
<i>Musta</i>	<i>Cyperus rotundus</i>	<i>Asvattha</i>	<i>Ficus religiosa</i>
<i>Haaritaki</i>	<i>Terminalia chebula</i>	<i>Patha</i>	<i>Stephania japonica</i> Miers
<i>Padmaka</i>	<i>Prunus cerasoides</i> D.	<i>Asana</i>	<i>Pterocarpus marsupium</i>
<i>Kutaja</i>	<i>Holarrhena antidysenterica</i> wall.	<i>Vetasa</i>	<i>Salix caprea</i> Linn.
Decoction #5		Decoction #10	
<i>Lodhra</i>	<i>Symplocos cochinchinensis</i>	<i>Daruharidra</i>	<i>Berbers aristata</i>
<i>Hribera</i>	<i>Coleus vettiveroides</i> Jacob	<i>Utpala</i>	<i>Nymphaea stellata</i>
<i>Kaliyaka</i>	<i>Coscinium fenestratum</i>	<i>Muskaka</i>	<i>Schrebera swietenoides</i>
<i>Dhataki</i>	<i>Woodfordia floribunda</i> Salish		

Note: The decoctions are useful in all types of pramehas. They can be used as impregnated on barley, food, or drinks.

7.6 Therapy (*Chikitsa*)

The principles of Ayurvedic therapy as discussed in [Chapter 2](#) are also applied to treat DM patients. Traditional daily management of DM is carried out with appropriate palliative herbal therapies. These herbs are selected based on their properties, such as *rasa* (taste), *guna* (physicochemical properties), *veerya* (potency), *vipaka* (postdigestive effect), and *prabhava* (unique action), that are necessary to bring about balance in *dosas*. On the basis of this approach, Charak Samhita has prescribed the following palliative treatments specific for *dosa* constitutions: 10 water decoctions for *kapha*, 10 decoctions for *pitta*, and 1 ghee for *vata* in which 17 herbs are collectively cooked.^{4,13} These herbs are listed in [Tables 7.1 to 7.4](#).

It is important to note that vigorous exercise is contraindicated in lean and weak patients with severe diabetes. They are advised to perform yoga and breathing exercises (*pranayama*). In fact, certain yoga practices and breathing exercises are believed to stimulate better utilization and production of insulin by stimulating both the pancreas and muscles. Other lifestyle changes recommended are regular walking and reducing the consumption of fat-producing foods such as lard, butter, and oils. The use of bitter gourd, pungent- and astringent-tasting food, asparagus, spinach, turmeric, fenugreek seeds, black pepper, and ginger is encouraged in the diet. Current experience suggests that it is unnecessary to exclude the use of sucrose, provided it is consumed in balance with a normal diet² that is consistent with Ayurvedic recommendations. Honey and jaggery are believed to pacify *kapha* and *pitta*, respectively, and may be useful in respective DMs. Patients are also asked to stop smoking tobacco, eliminate alcohol consumption, and restrict sleeping time to about 8 h/day.

Individual herbs are generally not used in Ayurvedic therapies. Because therapies are based on a predominant *dosa* and body constitution, they always include formulas containing many herbs and sometimes various minerals. In Ayurveda, every disease has one or two predominant *dosas* that need to be balanced according to the constitution of the patient; therefore, one therapy may not be applicable to all patients even though the patients share the same disease. Some of the formulations commonly used to treat type 2 DM are listed in Table 7.5. These formulas may also be used concurrently with insulin for the treatment of DM type 1 patients in order to help in decreasing their reliance on this hormone.

Ayurveda recommends that the patient's lifestyle, age, type of work, psychosocial needs, and will power are considered in a management plan. The latter includes the necessary *panchkarmas*, herbal formulas, a healthy meal plan, and blood-sugar monitoring. A physician needs to evaluate the plan at each visit and make necessary modifications.

Ayurvedic physicians currently monitor conventional blood sugar levels in the management of DM. The goal is to keep the premeal glucose level within 80 to 110 mg/day, bedtime glucose level within 100 to 140 mg/day, the low density lipid cholesterol (bad) at <100 mg/day, and the high density lipid cholesterol (good) above 35 mg/dL.²

The preventive measures for DM are primarily the adoption of life style and food habits that reduce fat accumulation in the body, because 80% of type 2 DM patients are known to be obese. Regular use of *rasayanas* and hypoglycemic herbal formulas may be useful as a preventive measure, particularly for those who have a family history of type 2 diabetes.

7.7 Scientific Basis

Buanides and sulfonylureas, discovered in the 1920s and 1940s, respectively, are still the most frequently prescribed hypoglycemic agents, although Ciglitazone, Pioglitazone, alpha-glucosidase inhibitors, and others are available.³ In an 8-year large multicenter clinical trial conducted by the University Group Diabetes Program (UDGP), patients treated with tolbutamide suffered a twofold increase in mortality as compared with insulin-treated patients.¹⁴ The study has not been completely refuted and physicians continue to treat patients with oral hypoglycemic agents, including tolbutamide, because there are few choices available to patients other than insulin where they are resistant to dietary and lifestyle changes.³ Ayurvedic therapies for DM are effective in controlling blood-sugar levels, either given alone or as an adjunct with other hypoglycemic drugs or insulin. Ayurveda offers alternative

TABLE 7.3
Group of Herbs To Treat *Vataja Pramehas*

Medicinal Plant	Botanical Name
<i>Goksura</i>	<i>Tribulus terrestris</i>
<i>Asmantaka</i>	<i>Mulaka parna</i> (ARU) <i>Ficus cordifolia</i> Roxb.
<i>Somavalka</i>	<i>Acacia suma</i> Kurz
<i>Bhallataka</i>	<i>Semecarpus anacardium</i>
<i>Ativisa</i>	<i>Aconitum heterophyllum</i>
<i>Lodhra</i>	<i>Symplocos cocchinchinensis</i>
<i>Vaca</i>	<i>Acorus calamus</i>
<i>Patala</i>	<i>Stereospermum suaveolens</i> DC
<i>Arjuna</i>	<i>Terminalia arjuna</i>
<i>Nimba</i>	<i>Azadirachta indica</i> A.
<i>Musta</i>	<i>Cyperus rotundus</i>
<i>Haridra</i>	<i>Curcuma longa</i>
<i>Padmak</i>	<i>Prunus cerasoides</i>
<i>Yavani</i>	<i>Carium copticum</i> Benth
<i>Manjistha</i>	<i>Rubia cordifolia</i> Linn.
<i>Aguru</i>	<i>Aquilaria agalloocha</i> Roxb.
<i>Candana</i>	<i>Santalum album</i> Linn.

Note: Herbs should be collectively cooked in oil for *kapha-vataja* or ghee for *pitta-vataja*.

TABLE 7.4
Herbs to Treat *Kapha-Vataja Pramehas*

Medicinal Plant	Botanical Name
<i>Kampillaka</i>	<i>Mallotus philippinensis</i>
<i>Saptacchada</i>	<i>Alstonia scholaris</i> R.
<i>Sala</i>	<i>Shorea robusta</i> Geratn
<i>Bibhitika</i>	<i>Terminalia bellirica</i>
<i>Rohitka</i>	<i>Aphanamixis polystachya</i>
<i>Kutaja</i>	<i>Holarrhena antidysenterica</i> Linn.
<i>Kapittha</i>	<i>Limona elephantum</i>

Note: Herbs can be taken as powder or paste in the dose of 10 g.

choices to control type 2 DM. Our literature search has revealed more than 52 herbs that have been shown to have hypoglycemic activity (Table 7.6). Some of these herbs have been frequently used in Ayurvedic formulas. Studies on six herbs and seven formulas are presented to demonstrate the scientific basis of the Ayurvedic therapies of DM.

7.7.1 *Gymnema sylvestere* R.Br. (Asclepiadacea)

Gymnema sylvestere (GS), a plant popularly known as *gurmara* (meaning sugar destroyer), is derived from a large woody climber that grows in the hills of Behar, Orissa Madhya Pradesh, and the Deccan Peninsula of India. It has been used in Ayurveda to treat DM for 2000 years. The whole plant, seeds, leaves, and roots are taken as a powder or as a decoction in combination with other herbs. The alcohol extract of GS (known as GS4) contains gymnemic acids (a chemically complex mixture of saponins and gurmardin, a polypeptide of 35 amino acids).¹⁵

TABLE 7.5

Ayurvedic Formulas

Formula	Dose (per day)	Ref.
<i>Kaisora guggul</i>	3 g	AF ^a Section 5:2
<i>Dhanvantra ghrat</i>	12 g	AF Section 6:22
<i>Navayasa churn</i>	1–3 g	AF Section 7:17
<i>Nyagrodhadhi churn</i>	1–3 g	AF Section 7:21
<i>Traikantaka ghrat</i>	12 g	AF Section 6:15
<i>Vastyamayantaka ghrat</i>	12 g	AF Section 6:40
<i>Shatavaryadi ghrat</i>	12 g	AP ^b Part II
<i>Agnitundi bati</i>	125 mg twice	Bhasahya Ratnavali
<i>Madhumehar yog</i>	1 tablet twice	Ayurveda Sar Sangraha
<i>Shilajjatwadi bati</i>	1 pill twice	Bhasajya Ratnavali
Ayush-82	5 g 3 times	104, 105
<i>Abraha chenodooram</i>	200 mg twice	108
D-400	2 tablets 3 times	109
MA 471	500 mg twice	106
<i>Sandana Podia (Siddha formula)</i>	300 mg twice	110
<i>Kadal Azhinjil choornam (Siddha formula)</i>	500 mg twice	111
M-93	1 g	112

^aAF = *Ayurvedic Formulary of India* (Government of India publication).

^bAP = *The Ayurvedic Pharmacopoeia of India* (Government of India publication).

TABLE 7.6

Scientific Studies on Hypoglycemic Herbs

Plant Name	Ref.	Plant Name	Ref.
<i>Acacia arabica</i>	113	<i>Momordica charantia</i>	See text
<i>Achyranthes aspera</i>	114	<i>Momordica cymbalaria</i>	162, 163
<i>Aegle marmelos</i>	115, 116	<i>Mucuna pruriens</i>	164, 150
<i>Allium cepa</i>	117–119	<i>Murraya koengii</i>	165
<i>Allium sativum</i>	120–125	<i>Musa sapientum</i>	166–168
<i>Aloe vera</i>	126–130	<i>Nelumbo nucifera</i>	169
<i>Andrographis paniculata</i>	131, 132	<i>Ocimum sanctum</i>	134, 170–172
<i>Azadirachta indica</i>	133–136	<i>Phaseolus mungo</i>	173, 174
<i>Bombax ceiba</i>	137	<i>Phyllanthus amarus</i>	175
<i>Caesalpinia bonducella</i>	138	<i>Phyllanthus urinaria</i>	176
<i>Calendula officinalis</i>	139	<i>Picrorrhiza kurroa</i>	177
<i>Capparis decidua</i>	140	<i>Polygonatum officinale</i>	178
<i>Capsicum frutescens</i>	141	<i>Psidium guajava</i>	179
<i>Cassia fistula</i>	142	<i>Pterocarpus marsipium</i>	See text
<i>Catharanthus roseus</i>	134, 143	<i>Pterocarpus santalinus</i>	180
<i>Coccina indica</i>	See text	<i>Punica granatum</i>	181
<i>Colocassia esculenta</i>	144	<i>Swertia chirayita</i>	182–187
<i>Cryptostegia grandiflora</i>	145	<i>Syzygium alternifolium</i>	188
<i>Cyanopsis tetragonoloba</i>	146, 147	<i>Tecoma stans</i>	189, 190
<i>Eugenia jambolana</i>	148–152	<i>Teramus labialis</i>	191
<i>Ficus bengalensis</i>	See text	<i>Tinospora cordofilia</i>	150, 152 192, 193
<i>Ficus carica</i>	153, 154	<i>Tinospora crispa</i>	194–196
<i>Gymnema sylvestre</i>	See text	<i>Trigonella foenumgraecum</i>	See text
<i>Hamiltonia suaveolens</i>	155	<i>Vinca rosea</i>	197
<i>Hibiscus rosa-sinensis</i>	156–158	<i>Withania somnifera</i>	198
<i>Inula racemosa</i>	159		
<i>Mangifera indica</i>	160, 161		

7.7.1.1 Animal Studies

Hypoglycemic activity and mechanism of action of GS have been examined in many animal studies. The following data offer possible mechanisms by which GS might exert its hypoglycemic effect:

1. Increase in the secretion or release of insulin — GS has been shown to stimulate secretion or release of the insulin *in vivo* and *in vitro*.^{16–21}
2. Protection of the pancreas from chemical toxins — GS has been shown to have partial protective effect on the pancreas against potent pancreatic toxins like beryllium.²² It also has been suggested that GS may be promoting the regeneration of islet cells destroyed by streptozotocin in rats.^{23,24} Al-DM rats treated with GS lived longer than the untreated diabetic rats, further supporting the regeneration hypothesis.²⁵
3. Increased utilization of glucose — GS has been shown to increase the activities of enzymes responsible for the utilization of glucose by insulin-dependent pathways, an increase in phosphorylase activity, and a decrease in gluconeogenic enzymes and sorbitol dehydrogenase.¹⁷
4. Inhibition of glucose absorption from intestine — GS has been shown to inhibit the absorption of glucose from the intestine.²⁶

7.7.1.2 Clinical Studies

Several clinical studies have been done to help authenticate the hypoglycemic effect of GS. These studies are evaluated based on the considerations outlined in [Chapter 3](#).

Our literature search has found two case control studies and two smaller studies. Major drawbacks in these clinical trials are small sample size and inadequate study duration.

1. One group (n = 10) of 4 normal healthy females and 6 males (ages 19 to 25) (nondiabetic) and another group (n = 6) of 2 females and 4 males (ages 35 to 50) (type 2 DM) were studied for the antidiabetic effect of GS.²⁷ Diabetic patients were given an aqueous decoction of shade-dried GS leaves in a dose of 2 g 3 times/day for 15 days and healthy subjects were given the same dose for 10 days. The fasting blood sugar (FBS) (mg/day) was measured at 30 min postprandial blood sugar (PPBS) and at 2 h PPBS on day 1 before the treatment and on day 10 after daily treatments. Healthy subjects showed mean blood sugar levels of 80, 156, and 76 on day 1 and 69, 132, and 66 on day 10, respectively. Those in the diabetic group showed blood sugar levels of 135, 220, and 152 on day 1 and 110, 180, and 121 on day 10, respectively. Authors concluded that GS treatment demonstrated a blood sugar lowering effect in both normal and diabetic subjects.²⁷
2. Eight type 2 DM patients were treated with 10 g of GS leaf powder orally daily for 21 days. The study showed significant reduction ($p < 0.05$) in blood sugar levels, mean fasting BGL, and mean 2-h BGL as compared with baseline.²⁸
3. GS4, a water-soluble acid fraction of an alcoholic extract of GS, was supplemented in diet to a group of 22 diabetic type 2 patients.²⁴ These patients were given oral hypoglycemic therapy plus GS4 (400 mg/day) for 1 to 12 years (mean = 4.6 years). A second group of 25 type 2 DM patients was given only hypoglycemic therapy alone for 1 to 5 years (mean = 2.7 years). Group 1 showed significant improvement ($p < 0.001$) in blood sugar cholesterol and triglycerides levels, whereas group 2 showed fewer effects compared with group 1 ($p < 0.001$). Patients in group 1

developed symptoms of hypoglycemia and the dose of their hypoglycemic agents had to be changed or stopped. The authors suggested that GS supplementation showed an obvious advantage over conventional hypoglycemic therapy alone.²⁴

4. Another study²³ done on GS4 reported a synergistic effect of this GS extract on BGLs in 23 type 1 diabetic patients (ages 10 to 31) who continued on insulin therapy plus GS4 as compared with 37 type 1 diabetic patients (ages 8 to 30) who were on insulin therapy alone. The dose of GS4 was 400 mg/day for 6 to 8 months. All patients in the combination therapy developed hypoglycemia and their insulin dose had to be reduced. Insulin doses, fasting glucose, hemoglobin A_{1c}, cholesterol, triglycerides, and free fatty acids were reported for the longest follow-up intervals. Although this study had many design problems, the data derived from it were somewhat encouraging. In fact, the authors concluded that GS4 improved blood glucose homeostasis and showed a better control of hyperlipidemia.²³

7.7.2 *Momordica charantia* (Cucurbitaceae)

Momordica charantia (MC), a climbing vine, has been widely used in Ayurveda as an antidiabetic, abortifacient (whole plant), antirheumatic, and carminative (fruits) agent. It is believed to cure deranged *kapha* and *pitta*.²⁹ The gourd produced by the plant, known as *karela* in India (bitter gourd), is traditionally eaten as a fried vegetable or as a fresh juice. Pharmacological and clinical studies demonstrating the hypoglycemic activity of MC are summarized here.

7.7.2.1 *Animal Studies*

The hypoglycemic activity, mechanism of action, and toxicity of MC have been extensively investigated in streptozotocin-induced or alloxan-induced DM (SZ-DM or AL-DM) rat models.³⁰⁻⁴⁴

Oral feeding of MC at 200 mg/kg/day for 40 days prevented renal hypertrophy as compared with SZ-DM in treated control rats, indicating that MC may have a protective effect against renal toxins.⁴⁵ Moreover, an active hypoglycemic agent, polypeptide-p, with a minimum molecular weight of approximately 11,000 (166 residues), was isolated from fruits, seeds, and tissues of MC. The agent was very effective when administered subcutaneously to gerbils, langurs, and humans in reducing blood sugar levels.³¹ MC extracts were also studied in relation to the course of the development of eye cataracts, one of the major side effects of DM. In fact, MC fruit extract given 4 g/kg/day for 20 days to AL-DM rats delayed the cataract formation by 2 months as compared with untreated diabetic rats. Blood sugar level in the diabetic control rats was 307 mg/day compared with 149 mg/day in the diabetic-treated rats. These findings suggest that the protective effect may be dependent upon the maintenance of lower BGLs.⁴⁶

The mechanism of the antidiabetic action of MC has been studied extensively. Fractionation extraction studies using alkaline and acid wash of chloroform extract of MC have shown that the alkaline wash residue produced a hypoglycemic effect 1 h after administration in SZ-DM rats. This same effect was much slower to occur with the residue from the acid wash.³⁴ On the basis of these results, authors stated that MC may be acting partly via pancreatic and partly via extrapancreatic mechanisms.

7.7.2.1.1 Increase in Secretion or Release of Insulin

The increased secretion or release of insulin as a mechanism of the hypoglycemic activity is supported by several studies.^{33,35,41} These studies indicate that MC increases (regenerates) or possibly facilitates the recovery of pancreatic beta cells and that viable beta cells in the pancreas that are capable of secreting insulin are still necessary for MC to exert its hypoglycemic effect. Other biochemical effects, such as the reduction in lipid peroxidation reported in SZ-DM rat pancreas and islet cells,⁴³ may be contributing to MCs hypoglycemic activity.

7.7.2.1.2 Decreased Synthesis and Increased Utilization of Glucose

Biochemical studies done on MC have shown that hepatic glucose-6-phosphatase and fructose-1,6-biphosphatase activities are increased and hepatic glucose-6-phosphate dehydrogenase activity is decreased by streptozotocin in SZ-DM rats.³⁶ MC treatment of SZ-DM rats depressed the phosphatase activities and enhanced the dehydrogenase activity, resulting in depressed synthesis and increased oxidation of glucose, respectively, leading to lowered blood sugar levels. In a comparative study with oral hypoglycemic drugs, MC treatment caused 10 to 15% reduction in BGL, whereas tolbutamide caused 40% reduction in BGL 1 h postadministration.³⁷ The effect of MC was not accompanied by increased insulin secretion. In SZ-DM rats, MC caused 26% reduction in BGL, whereas metformin, a biguinide, caused 40 to 50% reduction. MC increased the rate of glycogen synthesis in the liver by four- to fivefold as detected by the incorporation of ¹⁴C-glucose into glycogen; this suggests that increased glucose utilization in the liver may be a more important mechanism of action of MC.

7.7.2.1.3 Inhibition of Intestinal Glucose Absorption

It has been suggested that the inhibition of glucose uptake from the intestine may be an important factor in lowering BGL by MC.⁴⁰ Two hypoglycemic chemical constituents were isolated from the plant: (1) the oleanolic acid 3-O-monodesmoside, momordin Ic, and (2) oleanolic acid 3-O-glucuronide. These chemicals showed a dose-response effect in inhibiting the increase in serum glucose levels in oral glucose-loaded rats, but showed no significant effect on serum glucose levels in normal rats, intraperitoneal glucose-loaded rats, or AL-DM mice. Both chemicals were also found to suppress gastric emptying time in rats, in addition to inhibiting glucose uptake in rat's small intestine in a dose-dependent manner *in vitro*. These results indicate that these chemicals have neither insulin-like activity nor insulin-releasing activity; they apparently inhibit glucose absorption by suppressing the transfer of glucose from the stomach to the small intestine and by inhibiting the glucose transport system at the small intestine brush border. Inhibition of gastric emptying is, at least in part, mediated by capsaicin-sensitive sensory nerves and the central nervous system.⁴²

7.7.2.1.4 Decrease in Insulin Resistance

A decrease in insulin resistance is also a possible mechanism of action for MC. In fact, a significant increase in the muscle content of facilitative glucose transporter isoform 4 (GLUT4) protein content was found in the plasma membrane of KK-Ay mice, an animal model for type 2 diabetes, after 3 weeks of oral administration of MC. This increase in receptors induced by MC was postulated to be a contributing factor to the lowered insulin resistance in these mice.³⁹

7.7.2.1.5 Biochemical Effects of MC

Daily MC fruit extract feeding over a period of 10 weeks was found to reverse the increase of malonedialdehyde, a plasma lipid peroxidation (LPO) product, and LPO in kidney, nonesterified cholesterol, triglycerides, phospholipids, and high density lipoprotein (HDL)-cholesterol in the blood of SZ-DM rats.⁴⁴ These biochemical effects may account for the hypolipidemic action of MC.

Other biochemical studies demonstrated the effect of MC on hepatic drug-metabolizing enzymes. One example is the reversal of a 50 to 100% increase in hepatic anilin hydroxylase (AH) and ethoxyresorufin-*O*-deethylase (EROD) activities in SZ-DM rats after MC feeding. In addition, a decrease (17 to 20%) in aminopyrene *N*-demethylase (ADP) activity and cytosolic glutathione concentration in SZ-DM rats was brought to normal, and ethoxycoumarin-*O*-deethylase (ECOD) was even further decreased to 60% of control values by MC. This latter finding was not shown to consistently reverse the effects on drug-metabolizing enzymes in SZ-DM rats. This finding is probably due to the changes in hepatic phase 1 and phase 2 drug-metabolizing enzyme levels in SZ-DM animals and may be associated with the altered gene expression of different cytochrome P450 (CYP) and glutathione transferase isozymes.³⁸ Another theory suggested that the altered metabolism of endogenous substrates and hormonal status in DM may be responsible for changes in the metabolism and oxidative stress in various tissues.⁴⁷

7.7.2.2 Clinical Studies

An insulin-like compound isolated from MC was delivered by injection to 8 male and 1 female patients (ages 16 to 52) and to 5 healthy normal control subjects of the same age range for a period of 3 months to 5 years. The compound produced a consistent hypoglycemic effect in a small number of both type 1 and type 2 patients.⁴⁸ In another study, 50 mg/kg of MC powder was administered twice a day for 7 days to 8 type 2 DM patients (ages 38 to 50). A significant ($p < 0.001$) hypoglycemic effect was also noted in these patients.⁴⁹

Another MC investigation was performed on 9 type 2 DM patients who were allowed to continue on oral hypoglycemic drugs, but were asked to stop the drug 48 h before the glucose tolerance tests. The results showed that the water extract of MC caused a significant reduction in BGLs during the 50-g oral glucose tolerance tests when compared with controls. Insulin levels were statistically higher with MC juice, but no effect on insulin was seen in patients eating fried MC fruits with food.³⁰ A follow-up to this study by the same authors investigated MC juice and fried MC fruit in 9 type 2 diabetic patients. The patients were treated for 8 to 11 weeks. They took 50 ml of the juice or 230 g of the fried fruit, along their regular oral hypoglycemic agents, but were asked to stop their intake 48 h before the glucose tolerance tests. The authors reported that MC taken as a juice or as fried fruits showed significant hypoglycemic activity.³⁰

The fruit juice of MC significantly improved the glucose tolerance of 73% of the patients under investigation in one study;⁵⁰ a decoction of MC fruit in 200 ml of boiling water, given 100 ml/day for 3 weeks, produced a significant decrease in hemoglobin A_{1c} from 8.37 to 6.95% in another study.⁵⁰ A dose of dry MC fruit powder, 5 g/day for 3 weeks, did not lower blood sugar significantly. Drinking the aqueous suspension of MC fruit pulp also led to a significant ($p < 0.001$) reduction of both FBS and PPBS levels in 86% of the cases studied out of a 100 moderate type 2 DM patients.⁵² A significant hypoglycemic activity was observed in 7 type 2 DM patients treated with MC aqueous extract (100 g of the fruit boiled in 200 ml of water, dose of 100 ml) given orally daily for 3 weeks.⁵¹

As can be deduced from the above data, the results shown in clinical studies are consistent with those obtained from the preclinical studies. This evidence provides a relatively adequate scientific basis for the use of MC in Ayurvedic therapies of DM.

7.7.3 *Trigonella foenum-graecum* Linn. (Leguminose Family)

Trigonella foenum-graecum (TF), known as *methika* in Hindi and Sanskrit and *methi* in Tamil, is an erect aromatic herb, 30 to 50 cm tall.⁵³ It is widely cultivated in many parts of India. Its seeds are used in cooking as well as treating diabetes, whereas its leaves are eaten as a vegetable. The endosperm of the seed is rich in galactomannan (14 to 15%), young seeds contain carbohydrates and sugar, and mature seeds yield amino acids and fatty acids on hydrolysis. TF seeds also contain carotene, vitamins, and saponins.⁵³ It is used in Ayurveda as a diuretic, tonic, carminative, astringent, and emoliant. It is also used to treat diabetes, colic, dysentery, diarrhea, coughing, dropsy, rheumatism, rickets, and anemia and to subdue deranged *vata*.⁵³

The antidiabetic activity of TF has been confirmed in both animal models^{54–64} and type 2 DM patients.^{65,66} The hypoglycemic effect of TF has been shown to reside in the defatted seed material. A clear decrease in hyperglycemia and glycosurea and a decreased hyperglycemic effect in an oral glucose tolerance test in TF-treated AL-DM dogs have been shown.⁶⁷ The effect was found in the fraction “a” of the defatted portion of the TF seed that contains testa and endosperm and is rich in fibers (79.6%). The fraction “b” that contains the cotyledons and axes and is rich in proteins (52.8%) did not show the effect.

It was also found that saponins isolated from TF seeds enhance food consumption and the motivation to eat and reduce plasma cholesterol levels in rats.⁶⁸ It was also postulated that the hypoglycemic activity of TF is not concentrated in any one TF constituents based on the retention of this activity in all TF parts such as the seed powder, methanol extract, and the residue remaining after the methanol extraction.⁵⁸ Chemical analysis of the water extract of the methanol extractive-free residue of the seed powder showed that the major active constituent of the soluble dietary fiber is a galactomannan. Additional hypoglycemic compounds were also present in other fractions. Aqueous extracts of TF leaves given orally and intraperitoneally (i.p.) were also shown to produce a hypoglycemic effect in normoglycemic and AL-DM rats. The oral and i.p. LD₅₀ values were shown to be 9 and 1.9 g, respectively, and the main organ affected following i.p. administration was the liver. No major target organ was affected following oral administration.⁵⁹

Several mechanisms of action for TF's hypoglycemic action were proposed. It appears to take effect, at least in part, at the cellular level. In fact, TF antagonized the hyperglycemia caused by alloxan or cadmium in rats. Cadmium has been shown to cause hyperglycemia by increasing the release of epinephrine in intact rats and by decreasing the release of insulin in isolated perfused rat pancreas.⁵⁵ A higher level of antioxidants in animals on a TF-supplemented diet as compared with animals on a control diet lead to the assumption that TF seeds used as a supplement in the diet may normalize the disrupted free radical metabolism.⁶⁹ Another theory sprung from the fact that TF brought the high glucose-6-phosphatase and fructose 1,6-phosphatase activities in the kidney and liver back to normal levels in diabetic rats. These enzymes are involved in increased production of glucose and fructose.⁷⁰ TF powder supplementation also led to the normalization of creatinine kinase's activities in the diabetic rats and restored normoglycemia comparable with insulin and vanadate (an antidiabetic agent).⁶¹ It was also shown that TF seed powder increased the glycolysis and decreased the gluconeogenesis activities back to normal levels in the liver and kidney in diabetic rats.⁶²

7.7.3.1 Clinical Studies

Investigation on the hypoglycemic effect of whole TF seeds, extracted TF seed powder, gum isolate from TF seed, cooked TF seed, and cooked TF leaves led to several findings. Gum isolate significantly ($p < 0.05$) reduced BGL and increased insulin levels at 30- and 60-min intervals after administration. The dose was not given. The author repeated the study in 5 diabetic patients given a daily dose of 25 g of defatted TF seeds for 21 days with meals. At the end of the study, the glucose tolerance test showed significant hypoglycemic activity. The dose, number of patients, patients' ages and genders, and diagnostic criteria were not mentioned. The author did conclude that TF was effective following acute as well as chronic administration, its effect was not destroyed by cooking, and it would be a good supplement to a diabetic diet.⁷¹

A significant hypoglycemic effect of TF was also reported in 10 type 1 diabetic patients in a randomized trial. TF diet significantly ($p < 0.05$) reduced FBS and improved glucose tolerance tests. It also produced 54% reduction in 24-h urinary glucose excretion, a significant reduction ($p < 0.05$) in total cholesterol, and very low density cholesterol and triglycerides.⁷²

A crossover randomized trial⁶⁶ with a diet enriched with TF seeds was performed on two groups. The first group of 15 type 2 DM patients (ages 32 to 60; 33% females) was randomized and given a TF-enriched diet during the first 10 days or during the second 10 days. The second group of 5 type 2 DM patients (ages 35 to 58) was randomized and given a TF-enriched diet either the first 20 days or the second 20 days of the trial. The authors of this study reported significant hypoglycemic activity ($p < 0.05$) in both groups. However, the participant's gender, demographic data, and data on the use of other hypoglycemic agents were not mentioned.

Another randomized trial⁷³ was performed on 10 type 2 DM patients (ages 38 to 54) in which 2 groups (5 patients/group) were crossover between placebo and the TF-enriched diet group. Patients were given 25 g of TF seeds incorporated in bread for 15 days. Authors of this study reported significant ($p < 0.05$) hypoglycemic activity of TF and suggested that the use of seeds in the treatment of DM merits further investigation.

A case control study⁷⁴ investigated the hypoglycemic effect of TF in 60 type 2 diabetic patients. Patients were kept on a restricted carbohydrate diet and given TF seeds 12.5 g twice a day for 24 weeks. A significant reduction in glycosylated hemoglobin was noted. The authors concluded that TF should be further investigated in the management of DM.

TF (9 g/day in 3 doses for 90 days) in 15 type 2 diabetic patients was found to produce a significant ($p < 0.05$) hypoglycemic effect. It also produced a significant ($p < 0.05$) reduction in serum cholesterol and triglycerides.⁷⁵ It was also found to exert a hypoglycemic effect in 20 healthy male volunteers (ages 20 to 30).⁶⁵

The animal studies supported by the clinical experience indicate that TF has a significant hypoglycemic activity and may prove useful in the management of DM. However, studies using various biochemical parameters and endpoints should be carried out to determine efficacy of TF as an adjunct to other hypoglycemic drugs.

7.7.4 *Coccinia indica* Wight (*C. grandis*, *C. cordifolia* Cogn)

Coccinia indica (CI), known as *Bimbi* or *Kundura* in India, is a climber with branched leaves. CI is found wild in hedges and waste places in different parts of India. Fresh juice from leaves, stem, and roots is used to treat diabetes, glycosuria, enlarged glands, and skin diseases. The leaves and stem are also used as an antispasmodic and expectorant agent in bronchitis.⁷⁶

The hypoglycemic activity of CI has been reported in several animal studies.^{77,78} In fact, the oral administration of pectins isolated from CI produced a significant hypoglycemic effect in normal rats.⁷⁹ The ethanol extract in glucose-loaded normal rats showed similar results.⁸⁰ It was suggested that the hypoglycemic activity of CI is mediated through an insulin secretagogue effect or through an influence on enzymes involved in glucose metabolism.⁸¹ Also, it has been shown that powdered leaf suspension of CI produced a significant hypoglycemic activity in AL-DM dogs but not in normal animals. In a glucose tolerance test, however, the suspension did reduce the blood sugar levels.^{78,82} CI root extracts (water or ethanol) have also been shown to have hypoglycemic activity in healthy rabbits.^{83–85}

In a clinical double-blind controlled trial,⁸⁶ a significant improvement ($p < 0.05$) in the glucose tolerance was seen in 10 of 16 type 2 patients who received CI leaf preparation. In another clinical study,⁸⁷ 500 mg/kg dried extract of CI given orally to 30 type 2 diabetic patients for 6 weeks restored the enzymes involved in glucose metabolism back to normal levels.

7.7.5 *Pterocarpus marsupium* (Papilionoidae)

Pterocarpus marsupium (PM), also known as *pitsara*, *bijasal*, *pitasa*, or red sandalwood in India, is a moderate to large deciduous tree up to 30 m in height. Cold, aqueous extract of the wood is used to treat diabetes. The paste of the leaves is used to treat abscesses and skin diseases, and the extract of the bark is used as an astringent for gum and is also useful in diarrhea.⁸⁸

The hypoglycemic activity of PM has been reported in several studies.^{64,89–94} In fact, an active constituent of PM, epicatechin, was reported to show the hypoglycemic effect in AL-DM rats if given within 24 h after the administration of alloxan; no effect was seen if given after 92 h.⁹⁰ This indicates that epicatechin protected the pancreas from the direct toxic effect of alloxan. Three phenolic constituents from PM were isolated and studied for their hypoglycemic activity: marsupin, ptersupin, and pterostilbene. This hypoglycemic effect in diabetic rats was comparable with that of 1,1-dimethylbiguinide.⁹²

A significant decrease in BGL in normal rats ($p < 0.001$) 2 h after oral administration of 1 g/kg of aqueous extract of PM was evident in some studies. A significant decrease in BGL in AL-DM rats after the oral dose of PM extract for 21 days was also reported.⁶⁴

In one study,⁹⁵ the antidiabetic effect of aqueous, ethanol, and hexane extracts of PM bark in normal and diabetic rats was investigated. The ethanol extract at 0.25 g/kg oral dose showed a more hypoglycemic activity than the aqueous and hexane extracts. The same dose of ethanol extract was also more effective than glibenclamide, an oral hypoglycemic agent.⁹⁴ The ethanol extract of PM also exhibited a hypoglycemic effect in healthy normal rats.⁹⁵

These biological studies indicate that PM has substantial hypoglycemic activity and its activity is not concentrated in any one constituent. The studies also show that these active constituents are soluble in water, ethanol, and hexane.

7.7.5.1 *Clinical Studies*

The Indian Council of Medical Research (ICMR), Collaborating Centers, New Delhi,⁹³ evaluated the efficacy of PM in newly diagnosed type 2 DM patients. Of 97 patients, 93 completed the daily dose of 2, 3, or 4 g of PM extract treatment for 12 weeks. The fasting and postprandial BGLs in these patients fell significantly ($p < 0.001$) from the initial mean of 151 and 216 mg/day, respectively. No significant change was noted in the mean plasma lipid levels.

The above clinical studies coupled with the previously mentioned biological data provide adequate scientific support for the use of PM in Ayurvedic management of DM.

7.7.6 *Ficus bengalensis* L. (Moraceae)

Ficus bengalensis (FB), also known as *Vata*, *bor*, *bot*, or Banyan tree, is a large evergreen tree that sends down aerial roots for lateral growth. It is effective in deranged *kapha* and *pitta*. Infusion of the bark is used to treat diabetes. The white milky juice of the plant is helpful when applied on sores, ulcers, and cracked soles of the feet; it is also helpful for inflammation and rheumatism. FB leaves contain quercetin-3-galactoside, rutin, and beta-sitosterol, and its bark contains leucoanthocyanin and two flavonoids.⁹⁶

FB has been shown to have hypoglycemic activity in several animal models. The results of a comparative study of an ethanolic extract of FB bark, a glucoside isolated from the bark, and tolbutamide in normal and AL-DM rabbits showed that the glucoside was more active than the crude extract and half as potent as tolbutamide.⁹⁷ Another study also reported that a dimethoxy derivative of leucocyandin 3-*O*-beta-D-galactosyl cellobioside isolated from the bark of FB decreased blood sugar levels very significantly on oral administration in both normal and moderately diabetic rats. It also increased the serum insulin significantly in the diabetic rats at a dose of 250 mg/kg for a 2-h period. In addition, the oral dose of 100 mg/kg of the active principle given to diabetic rats for 1 month produced a significant decrease in blood and urine sugar, certain lipid components in serum and tissues, and glucose-6-phosphatase activity in liver. It further caused a significant increase in body weight and the activities of hexokinase and human menopausal gonadotropin coenzyme A (HMGCoA) reductase in tissues as compared with diabetic control. The authors suggested that the mechanism of action of the principle may be related to its protective and inhibitory action against the insulin-degradative processes.⁹⁸

The extracts of FB bark also showed hypoglycemic activity in SZ-DM rats. In fact, the oral dose caused an enhancement in serum insulin levels in normoglycemic and diabetic rats. Each of these extracts stimulated insulin secretion when incubated *in vitro* with isolated islets of Langerhans from normal as well as from diabetic animals. However, the insulin secretion by beta cells was more apparent in the presence of pelargonidin derivative than in the presence of another leucocyanidin derivative isolated from the bark.⁹⁹

Another study¹⁰⁰ compared the antidiabetic effect of a dimethoxy derivative of pelargonidin 3-*O*-alpha-L rhamnoside isolated from FB bark with glibenclamide (2 mg/kg and 0.5 mg/kg/day for a single dose and chronic dose, respectively). In moderately diabetic rats, 250 mg/kg/day was used in the single-dose study and 100 mg/kg/day of the extract in the chronic-dose study. Fasting blood glucose decreased by 19% and improved glucose tolerance by 29% in the single-dose group. The corresponding effects of glibenclamide were 25 and 66%, respectively, over the control values. In the 1-month treatment group, the FBS levels decreased to almost half of the pretreatment levels in both groups; glucose tolerance improved by 41% in the glibenclamide group and by 15% in the glycoside-treated group. Glycosurea was also decreased in both groups and they appeared healthy.¹⁰⁰

A glycoside of leucopelargonidin isolated from the bark of FB was compared with a minimal dose of glibenclamide. Significant hypoglycemic, hypolipidemic, and serum insulin-raising effects of the glycoside were demonstrated in moderately diabetic rats. The effect was similar to that of glibenclamide. The main difference was that the former significantly enhanced the fecal excretion of sterols and bile acids, whereas the latter had no such action even though both controlled high blood cholesterol.¹⁰¹ A very similar isolate,

dimethoxy ether of leucopelargonidin-3-O- α -L-rhamnoside, given orally at 100 mg/kg, produced significant hypoglycemic and serum insulin-raising effects in normal and moderately AL-DM dogs during a 2-h period. The mechanism of action of the glycoside compound seems to be similar to that of drugs that stimulate insulin secretion.

In a toxicity study,¹⁰² a single oral dose (0.2 to 1.8 g/kg) was given to mice. In a different study, daily oral doses (100, 250, and 500 mg/kg) given to rats for a period of 1 month did not show notable toxic effects. The compound was not lethal even at the high dose of 1.8 g/kg in both species. A leucodelphinidin derivative isolated from the bark of FB also showed hypoglycemic activity at a dose of 250 mg/kg in normal and AL-DM rats. The effect was similar to that produced by glibenclamide (2 mg/kg) under the same conditions. However, the plant product was less effective after a glucose load than glibenclamide.¹⁰³

7.8 Clinical Studies on Combination Formulas

7.8.1 Ayush-82

Ayush-82 is a mixture of four herbs: the seeds of *Mangifera indica*, *Syzygium cumini*, and *Momordica charantia*, and the leaves of *Gymnema sylvestre*. One hundred type 2 DM patients (ages 40 to 70; 52% male) were given 5 g of Ayush-82 3 times a day for 6 weeks in conjunction with *Shuddha Shilajit*.¹⁰⁴ *Shuddha shilajit* is a mineral preparation of black bitumen purified by *triphala* water. (*Triphala* is a mixture of *Terminalia chebula*, *terminalia bellerica*, and *Embolica officinalis*.) *Shilajit* was given 500 mg twice a day for 2 weeks. Oral hypoglycemic drugs were withdrawn after 2 weeks of treatment with Ayush-82. All patients were advised to consume a daily 1200-calorie diet. Although the average blood sugar, FBS, and PPBS levels reduced significantly ($p < 0.001$) in some patients, the control of diabetes was not good because 47% of the patients did not have their diabetes controlled by the end of the study. The study had no comparison arm and the duration of the study was too short.

The duration of treatment was extended to 24 weeks in another clinical trial on 350 type 2 DM patients, and all other parameters were kept the same as in the above study.¹⁰⁵ On the basis of a physician's rating scale, the authors reported that 61% of the patients had a good response, 12.9% had a fair response, and 25.9% had poor response. After the treatment, FBS and PPBS levels were found to be significantly reduced ($p < 0.001$).¹⁰⁵ It is important to note that this clinical trial had no control arm and was not clear about the time and method of withdrawal of the hypoglycemic agent. The results remain encouraging and do support the concept that Ayurvedic formulas, because of their slow duration of action, might yield better results if the treatment duration is extended.

7.8.2 MA-471

MA-471 is a mixture of the following herbs: *Enicostemma littorale*, *Phyllanthus niruri*, *Eugenia jambolana*, *Melia azadirachta* (*indica*), *Terminalia arjuna*, *Aegle marmelos*, and *shilajit*. This mixture was found to produce a significant hypoglycemic activity and hypolipidemic activity even in patients resistant to an oral hypoglycemic agent. This activity was demonstrated while studying 69 type 2 DM patients, 6 of whom dropped out due to noncompliance and

3 of whom dropped out due to other illnesses. This study had three arms: (1) patients who never had a hypoglycemic drug and DM was out of control despite dietary intervention and exercise ($n = 15$), (2) patients who had DM under control through hypoglycemic agents ($n = 30$), and (3) DM patients who did not respond to maximal dose of oral hypoglycemic agents ($n = 15$). All patients ingested 500 mg tablets of MA-471 twice a day for an average of 9 months. Arm 3 patients continued to take both M-471 and an oral hypoglycemic agent. The authors observed a significant improvement in other clinical symptoms (e.g., polyurea, fatigue, constipation, moderate improvement in weakness, polydipsia, giddiness, muscle pain, palpitation, and anorexia). The authors deduced from the data that M-471 was beneficial in all groups and that it can be used in conjunction with oral hypoglycemic agents.¹⁰⁶ The results are encouraging and warrant further investigation.

7.8.3 *Abraga Chendooram*

Abraga chendooram (AC) is a mixture of abragum (purified black mica, 80 g), *vengaram* (dehydrated borax, 0.5 g), and *Saranaiver charu* (juice of root of *Trainthema decandra* Linn.); *Adathodaielai charu* (juice from the leaves of *Adhatoda zeylanica* Linn); and *Alam Vizhuthu Kudineer* (root of *Ficus benghalensis* Linn).

Sixty type 2 DM patients (ages 21 to 70) who failed dietary control intervention and showed no evidence of serious complications (e.g., ketoacidosis, nephropathy, neuropathy, or retinopathy) were given 200 mg of AC in capsules twice daily for 45 days while their caloric intake was restricted to 25 calories/kg of ideal body weight. The authors concluded that the study demonstrated hypoglycemic effect and warrants further investigations. The use of other hypoglycemic agents and the actual FBS levels were not shown in this trial.¹⁰⁷

Another study also investigated AC in 130 type 2 DM patients (57% males). The patients were given 200 mg of the drug in gelatin capsules twice daily for 45 days. The caloric intake was restricted to 25 calories/kg body weight during the treatment. A significant reduction ($p < 0.005$) in FBS and PPBS levels was observed. The authors reported that 57.9% of all patients gave a good response that was defined as FBS and PPBS lower than normal accepted levels and absence of all clinical symptoms; the authors concluded that the treatment could reduce blood sugar levels was combined with dietary controls.¹⁰⁸ Major flaws in the design of this study included its short duration and the lack of defining the use of other hypoglycemic agents. Although these results are encouraging, a long-term study is still necessary to further substantiate the effect of AC.

7.8.4 *D-400*

D-400 is a mixture of *Eugenia jambolana*, *Pterocarpus marsupium*, *Ficus glomerulata*, *Gymnema sylvestre*, *Momordica charantia*, *Ocimum sanctum*, and *shilajit*. This mixture investigated D-400 in 38 diabetic patients (ages 35 to 76). The study had three arms: (1) patients who did not respond to oral hypoglycemic agents ($n = 19$), (2) patients who did not respond to diet and exercise therapy ($n = 8$), and (3) patients who were dependent on insulin therapy ($n = 6$). All patients in all arms were given 2 tablets (weight not known) of the drug 3 times a day for 6 months. The authors concluded that D-400 lowered the blood sugar and cholesterol levels in the groups that failed the oral hypoglycemic therapy and also in arm 2 but was less effective in arm 3 patients.¹⁰⁹ The relative proportion of each ingredient and method of preparation were not cited in the study. Results are encouraging and warrant further investigation.

7.8.5 Sandan Podia

Sandana is a mixture of six parts of *Tinospora cordifolia* sugar and one part of each of the following herbs: *Santalum album* (sandalwood) sawdust, *Andropogon citratus* (lemongrass) root, *Vitiveria zizanioides* (vetiver) root, *Syzygium aromaticum* (clove) flower bud, *Anacyclus pyrethrum* (pyrethrum) root, and purified *Shilajit*. The sugar is extracted from *tinospora* by suspending the crushed plant in water overnight, decanting and drying the extract in the sun, and resuspending the material in water overnight. The procedure is repeated three times and the extract is dried and ground into a fine powder. The fine powders are mixed thoroughly and put into 300-mg capsules, which were given to 20 type 2 DM patients (ages 20 to 70) twice a day for 45 days. The FBS and PPBS of the patients decreased significantly ($p < 0.01$) from 164.5 to 114.7 and 281.7 to 171.2 mg/day, respectively. Improvements were also observed in other clinical symptoms, such as polyurea, excessive appetite, weight gain, giddiness, polydipsia, indigestion, vomiting, and abdominal pain.¹¹⁰ Results are encouraging and warrant a long-term study.

7.8.6 Kadal Azhinjil Choornam and Triphala Tablets

Kadal, a preparation containing roots and bark of *Salacia chinensis*, and *triphala* were given to 25 type 2 DM patients. *Kadal* was given at 500 mg twice a day and *triphala* was given at 2.5 g three times a day with water for 120 days.¹¹¹ Authors reported a hypoglycemic effect, but the results need to be confirmed.

7.8.7 M-93

M-93 is a mixture of four herbs: *Aegle marmelos* (*bilva*), *Azadirachta indica* (neem, *nimba*), *Ocimum sanctum* (*tulsai*), and *Piper longum* (*kalimircha*). Thirty type 2 DM patients (ages 41 to 60) were treated with M-93, 1 g daily for 3 months.¹¹² Patients with FBS >300 mg/day, chronic renal failure, diabetic neuropathy, and diabetic ketoacidosis were excluded from the study. The authors stated that M-93 showed a positive response after 30 days of therapy and no adverse effect was observed.¹¹² The results need to be confirmed.

7.9 Summary and Discussion

The review of literature indicates that the Ayurvedic concept of DM, including signs and symptoms, etiology, and strategy of management, is not far from the current knowledge. Herbs used in Ayurvedic therapies and the multi-herbal combination formulas have been found to have a hypoglycemic effect in animal models of DM. Scientific studies done on 52 herbs (Table 7.6) and 7 multiherbal formulas have demonstrated hypoglycemic activity. Considerable information was available about some of the herbs, namely, *gymnema*, *momordica*, *trigonella*, *coccinia*, *pterocarpus*, and *ficus*, whereas other herbs have not been studied as extensively. This makes Ayurveda a considerable choice as an antidiabetic therapy that is safe, effective, and economic. The mechanism of antidiabetic action of these herbs appears to be multifactorial: increased secretion or release of insulin, efficient use of insulin, increased sensitivity of peripheral tissue to insulin, protective or regenerative effect on pancreas, decreased synthesis, and increased breakdown of glucose. Because these herbs have been used for thousands of years to manage DM patients, biological and

mechanistic studies confirming the hypoglycemic effect and clinical trials have not been conducted as rigorously as would have been done for a new synthetic chemical with no human experience.

7.9.1 Future Research Areas

7.9.1.1 Toxicity Studies

Although no serious adverse effects were reported in clinical trials and many of the herbs are traditionally used as vegetables in regular diet, providing some degree of confidence, the following toxicity studies should still be carried out at least on the most frequently used herbs:

1. Single-dose toxicity study to determine LD₅₀ dose
2. Subchronic toxicity study (120 days daily dose) in healthy animals and DM animal models to determine the maximum tolerated dose and identify the target toxicity organ
3. Lifetime feeding studies in normal and diabetic animals to determine long-term beneficial effects (increase in survival time), possible protection of pancreas, or toxic effects

7.9.1.2 Biochemical and Pharmacological Studies

There is a need for more biochemical and pharmacological studies. These studies will help determine the optimum therapeutic dose and the major mechanism of the action of herbs, giving the chance to maximize the efficacy by combining different mechanisms of action at their optimum doses.

7.9.1.3 Clinical Studies

Clinical studies necessary to establish management of DM by Ayurvedic therapies are as follows:

1. Definitive clinical studies with a rigorous protocol design; an optimum dose based on toxicological data; Hb A_{1c}, a good marker of DM control; and FBS and PPBS levels
2. Possible relationship between the *dosa* predominance and response to Ayurvedic treatments to validate the Ayurvedic concept of prognosis
3. Potential of Ayurvedic therapies to reduce or eliminate the use of synthetic oral hypoglycemic agents
4. Possible resistance of DM patients to herbs or herbal formulas over a period of time similar to oral hypoglycemic drugs
5. Ayurvedic management and use of herbs in high-risk population of type 2 DM (family history of DM, lifestyle, and obesity) to determine if it can prevent or reduce the incidence of DM
6. Combination of yoga and Ayurvedic therapies to identify any synergistic effects

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8

Ayurvedic Therapies for Thyroid Dysfunction

Anand Kar and Sunanda Panda

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8.1 Introduction

Thyroid problems are among the most common endocrine disorders presently seen worldwide. Patients with thyroid disorders suffer either from hypofunctioning or from hyperfunctioning of the gland. Whereas the former leads to a decrease in the concentrations of circulating thyroid hormones, the latter increases the same. These two dysfunctions are commonly referred as hypothyroidism and hyperthyroidism, respectively.

About 1 to 2% of the adult population is known to suffer from thyroid disorders.¹ According to the 1999 World Health Assembly report, about 1.5 billion persons in more than 110 countries are threatened with thyroid disorders. A World Health Organization (WHO) estimation also indicates that about 200 million people have goiter, although most of the goiters are small and subclinical.

Although small in size, the thyroid is considered to be one of the most important organs of the endocrine system, as it regulates nearly all body functions, including metabolic, respiratory, cardiovascular, digestive, nervous, and reproductive systems either directly or indirectly. Many abnormalities in a person's body can be the result of thyroid dysfunction. In fact, some typical problems directly related to thyroid abnormalities were known since the existence of Ayurveda, the ancient system of medicine in India.

In Ayurveda, the common thyroid-related problem that has been described from time to time is enlarged thyroid gland (*galaganda*),² which is now known as the simple goiter, a state of hypothyroidism. Some other diseases that are commonly mentioned in endocrine text,^{3,4} such as myxedema, cretinism, Graves' disease, and nodular goiter, have not been well described in Ayurveda symptom-based herbal treatments.

Despite several investigations on the herbal therapies, the use of Ayurvedic medicines in the regulation of thyroid problems continues to require more investigation. The primary aim of this chapter is to review the available literature including the experimental studies conducted to date and then to highlight the scientific basis of Ayurvedic therapies for thyroid dysfunctions. However, before we deal with the treatment of thyroid problems, it is necessary to have a basic understanding of the thyroid gland, its hormone, and functions.

8.2 Basic Understanding of Thyroid Gland: Its Hormones and Functions

The thyroid is a two-lobed gland that lies over the trachea (windpipe) just below the larynx. It synthesizes and secretes two major hormones, thyroxine (T₄) and triiodothyronine

(T₃). Both T₄ and T₃ are iodine-containing chemicals. Because they are the only iodine-containing hormones in the body, an adequate iodine intake is necessary for the optimum functioning of the thyroid gland. Although the entire amount of circulating T₄ is produced in the gland itself, T₃ is mostly generated in liver and kidney by peripheral monodeiodination of T₄ with the help of an enzyme, 5-monodeiodinase (5-D).

Although basically metabolic, thyroid hormones regulate almost all physiological processes directly or indirectly. The hormones are vital to the well-being of a person, and an imbalance in their physiological levels leads to health problems.

8.3 Thyroid Abnormalities: Hypothyroidism and Hyperthyroidism

Hypothyroidism occurs when the thyroid is hypoactive and does not produce enough thyroid hormones. The most common form of hypothyroidic diseases described in Ayurveda is the enlarged thyroid gland (*galaganda*), now known as goiter. In fact, goiter is produced by the inadequate secretion of thyroid hormones, resulting in a positive feedback of a pituitary hormone, thyrotropin (thyrotropin-stimulating hormone [TSH]), on the thyroid gland that ultimately enlarges. This disease process was recognized by early Ayurvedic practitioners.

As thyroid hormones regulate growth and development (both physical and mental) of a person, hypothyroidic children suffer from cretinism (mental retardation). In Ayurveda this is known as a condition of low intelligence (*manda buddhi*).

Hyperthyroidism is the result of excess production and delivery of the thyroid hormones to the peripheral tissues often referred as thyrotoxicosis. In Ayurveda this is believed to be an air or energy disorder (*vata*). Conventional medical text relates it to the production of TSH receptor-stimulating antibodies, leading to enhanced secretion of thyroid hormones, which are responsible for increased metabolism and energy wastage.

At present, diseases such as thyrotoxicosis, exophthalmic goiter or Graves' disease, and thyroid carcinoma and adenoma all come under the category of hyperthyroidism. These diseases can be serious, as the patients may end up with heart ailments and diabetes (*madhumeha*), the two most common modern health problems.⁵

Several symptoms are associated with hypothyroidism and hyperthyroidism. The common symptoms are listed in Table 8.1. A rare symptom, the thickening of the skin over the lower legs (thyroid dermopathy), is also observed in chronic hypothyroid individuals. At the beginning of the thyroid dysfunction, visible symptoms may not be observed.

8.4 Etiology

8.4.1 Hypothyroidism

According to Ayurvedic literature,² impairment of air (*vayu*), mucus or water (*kapha*), and the fat (*meda*) leads to the enlargement of the thyroid gland (*galaganda*). This is often known as bronchocele. According to common belief, the most common cause of hypothyroidism is iodine deficiency, and this disease is prevalent only in iodine-deficient areas

TABLE 8.1

Common Signs and Symptoms of Hypo- and Hyperthyroidism

Hypothyroidism	Hyperthyroidism
Weight gain	Weight loss
Husky, hoarse voice	Thinning of hair
Tiredness/lethargy	Sweating
Intolerance to cold	Nervousness
Cold hands and feet	Vomiting
Dry skin and hair, brittle nails	Tachycardia
Puffiness of the body	Diarrhea
Periorbital swelling	Tremor
Memory loss	Muscle weakness and fatigue
Constipation	Insomnia
Bradycardia	Angina
Muscle stiffness	Increased appetite
Impotence	Eyelid pain
Anemia	Loss of libido
Depression	Exophthalmos

such as hilly regions or areas with high lime content. In urban areas, iodine deficiency can also be considered a major factor of this disease, as people in these areas are following low sodium diet and using less iodized salt (as a preventive measure for heart problems and high blood pressure). Doctors recommend about 150 µg/day of iodine for normal thyroid function; less than 50 µg/day for a long period may cause goiter.³ Because of the inadequate iodine in the diet, the thyroid cannot synthesize sufficient amount of T₄ and T₃, resulting in the abnormal increase in circulating TSH, which causes abnormal increase in the size of the thyroid gland (simple goiter).

Other factors causing hypothyroidism are stress, hereditary defects in thyroid iodide transport, thyroglobulin and hormone synthesis, reduction in TSH or thyrotropin-releasing hormone (TRH) (secondary or tertiary hypothyroidism, respectively) and conversion of T₄ to T₃, peripheral resistance to the action of thyroid hormones, excess consumption of adulterated and goitrogenic foods (e.g., cabbage, cauliflower, cassava, peanuts, sweet potatoes, etc.),⁴ selenium deficiency, and consumption of heavy metals and pesticides.⁶⁻⁸ Very often it may be a result of the development of autoimmune antibodies, which destroy the thyroidal tissues by an immunological process.³

8.4.2 Hyperthyroidism

According to classic Ayurvedic text, hyperthyroidism is the result of imbalance in air (*vata*) and fire (*pitta*), which govern the neurohormonal system. In the conventional system of medicine, physicians relate constant psychological stress and excess secretion of cortisol with thyrotoxicosis. Apart from genetic susceptibility, an immunological increase in immunoglobulin G (IgG) antibodies (sometimes referred as thyroid stimulating antibodies), which act on TSH receptors on the gland to stimulate hormone production, are considered as major factors for hyperthyroidism.⁹

Prevalent data indicate that more young women (ages 20 to 40) suffer from thyroid dysfunctions than do males. This difference may exist because of high-circulating sex steroids, estrogen and progesterone, in females. Thyrotoxicosis may be aggravated during pregnancy, as human chorionic gonadotropin (hCG), a placental hormone, also possesses TSH-like activity.⁹ Formation of thyroid carcinoma also increases T₄ and T₃ production.

8.5 Pathogenesis (*Samprapti*)

Although in Ayurveda the pathogenesis of thyroid dysfunctions has not been well described, most available literature relate hypothyroidism with excess of mucus (*kapha*) and fats (*meda*). When the above factors get excited and become out of balance, they produce hormonal (*ojas*) imbalance and then enlargement (a slow process) of the gland, known as *Galaganda*.² It may start with loss of appetite and disgust for food or with loss of immunity.

There is limited Ayurvedic literature on hyperthyroidism. However, some physical and behavioral changes do indicate that *vata* and *pitta* lead to insomnia, which may manifest a condition of hyperthyroidism. Excessive sweating, intolerance to heat, and warm soles and palms also indicate the onset of hyperthyroidism.²

Conventional medicine describes autoimmunity development as the major pathogenesis process for thyroid dysfunctions. This could be due to genetic predisposition or it could be hormone or virus induced.¹⁰ A characteristic finding in the pathogenesis of hypothyroidism is that there is an accumulation of glycosaminoglycans (mostly hyaluronic acid) in interstitial tissues that generally accounts for the edema. In Graves' disease, sensitized T-lymphocytes stimulate B-lymphocytes to synthesize antibodies that enhance thyroid cell function. Some factors that may also stimulate the immune response of hyperthyroidism are postpartum period of pregnancy, excess iodide or lithium therapy, and glucocorticoid withdrawal.¹⁰

8.6 Prognosis (*Sadhyata*)

Hypothyroidism always results in a gradual decrease in metabolism with the slowing of mental and physical activity. Patients may experience one or more common symptoms, including sensitivity to cold, dryness of the skin and hair, constipation, anorexia, angina, anemia, and disordered menstrual function. With proper Ayurvedic therapy, these symptoms can be overcome. Goiter in which there is a considerable dyspnea along with emaciation of body, and if there is more than 1 year of standing, may be incurable.² According to conventional medical text, untreated myxedema may eventually lead to myxedema coma and death. However, the prognosis has been vastly improved with the use of L-thyroxine with frequent monitoring of the thyroidal status.

With respect to hyperthyroidism, Ayurvedic therapies have been found to provide about 60 to 80% symptomatic relief.¹⁴ With conventional medicine, when patients are treated for 12 to 24 months with antithyroid drugs, prolonged remissions of their illness are generally observed and many patients develop hypothyroidism. With radioactive iodide, 40 to 70% of patients develop hypothyroidism as a side effect within 10 years. Lifetime follow-up is therefore required for all hyperthyroid patients.

8.7 Diagnosis (*Nidan*)

Diagnosis of the thyroid dysfunctions is made on the basis of inspection of the thyroid gland both in Ayurveda and in Western medicine. The size and consistency of the swelling

TABLE 8.2

Commonly Used Ayurvedic Herbal Formulations and Food Supplements for the Treatment of Hypothyroidism

Formulation and Food Supplement	Dose	Major Herbs Used	Manufacturer
Thyro-L#455	2 capsules three times/day	<i>Laminaria sarassum</i>	Herbal Doctor Remedies, Monterey Park, CA
B 37 K Para-Thy-Mix	1 tablet/day	Natural iodine from dulse with raw glandular extract	Penn Herb Comp., Philadelphia, PA
<i>Suddha guggulu</i>	2.25 g twice/day	<i>Commiphora mukul</i>	Himalya Drug, New Delhi, India
<i>Kanchmar guggulu</i>	500 mg–1.0 g	<i>Bauhinia variegata</i> , <i>C. mukul</i>	Dabur Pharmaceuticals, New Delhi, India ^a
Gayatrin	1 to 2 tablets, two to three times/day	<i>B. variegata</i> , <i>C. mukul</i>	Arya Aushadhi Pharmaceutical Works, Indore, India
Kelp	1 tablet three times/day (660 mg kelp)	<i>Fucus vesiculosus</i>	Herbal Doctor Remedies, Monterey Park, CA

Note: These products are not intended to cure or prevent the disease.

^aAlso from Ayurvedic Formulary of India, Central Council for Research in Ayurveda and Siddha, Ministry of Health and Family Welfare, Government of India, New Delhi.

differ based on the disorder (*dosa*) of *vata*, *pitta*, or *kapha* involved in causing the dysfunction. Confirmation of diagnosis is done by the estimation of thyroid hormones and thyrotropin by radioimmunoassay (RIA), examination of gland activity by the use of radioiodine, basal metabolic rate (BMR)/oxygen consumption studies, and measurement of autoimmune antibodies.

8.8 Ayurvedic Therapies (*Chikitsa*)

According to the Ayurvedic management of thyroid diseases, the first line of treatment is to clear the blocked channels (*srotas*) in order to balance *vata*, *pitta*, and *kapha*. Several herbal preparations are also administered to increase the digestive fire at a cellular level and restore metabolism. For the treatment of goiter, some ayurvedic preparations are used to clean *kapha* internally.

Few herbal products are already available for the treatment of thyroid dysfunctions; the popular ones have been mentioned in Tables 8.2 and 8.3. Besides the mentioned drug and food supplements, *Vimbadi* oil (with sesame oil) can be applied four times/day on the goiter.² *Yograj guggul* (0.5 to 2.25 g/day) and *Ashwagandha* powder (*churn*) (1 to 6 g/day) are also used in the treatment of hypothyroidism. A study on Liv-52, an Ayurvedic liver tonic, has revealed that it stimulates thyroid hormone secretion, particularly the T₃.¹¹ In fact, action of the most commonly recommended Ayurvedic drug, *Kanchmar guggul* (prepared from *Bauhinia variegata*), has been found to be similar to that of the allopathic drug Eltroxin.¹²

For the treatment of hyperthyroidism, Charaka has mentioned *Shankapushpi* extract (*rosa*) as a brain tonic (Medhya rasayana, Charaka, Ch.1/30), although this is known to be a very effective supplement. Gupta et al.¹⁴ studied the comparative effects of an allopathic

TABLE 8.3

Common Ayurvedic Herbal Formulations and Food Supplements for the Treatment of Hyperthyroidism

Formulation and Food Supplement	Dose	Major Herbs Used	Manufacturer
Herbal-supra Power (product no. HZ1003)	1 capsule/day	Valerian root, passion flower, hops, skullcap, and wood betony	Viable Herbal Solutions Comp., Morrisville, PA
Biolite	1 capsule/day	Ephedra, kola seed, white willow, kelp, ginger, <i>Ginkgo biloba</i> , juniper berries, buchu leaves, <i>Cascara sagrada</i>	Bionomics International Inc., Delray Beach, FL
Thyro-H #450	1 capsule three times/day	Oyster shell, nourish yin, scrophularia, <i>Arnemarrhena fritillaria</i>	Herbal Doctor Remedies, Monterey Park, CA
Thyrovedic	1 tablet	Garcina, ginger, cambogia	Herbal Doctor Remedies, Monterey Park, CA
Lemon balm	1.5–4.5 g several times/day	<i>Melissa officinalis</i>	Herbal Doctor Remedies, Monterey Park, CA
<i>Astragalus</i> extract	1 vegicapsule/day	Raw astragalus powder (250 mg in each capsule)	Suzannes com, Joplin, MO
<i>Shankpuspi</i> syrup	125 mg twice/day	<i>C. pluricaulis</i>	Dabur India Ltd., New Delhi, India

TABLE 8.4

Promising Plant Extracts Suggested for the Treatment of Hypothyroidism

Plants	Effective Dose	Tested in	Days Treated	Ref.
<i>Bauhinia variegata</i>	2 g/kg	Animal models	20	12
<i>B. variegata</i> , <i>C. mukul</i> , <i>G. glabra</i> , and <i>C. pluricaulis</i>	100 mg from each plant (Thyrocap)	Humans	15	19
<i>Echhorina crassipes</i>	2 g/kg	Animal models	20	12
<i>Bauhinia purpurea</i>	2.5 mg/kg	Animal models	20	20
<i>Commiphora mukul</i>	0.2 g/kg	Animal models	15	21, 22
<i>Withania somnifera</i>	1.4 g/kg	Animal models	20	23
<i>W. somnifera</i> , <i>C. mukul</i> , <i>B. purpurea</i>	1.4 g, 0.2 g, 2.5 mg/kg	Animal models	30	24
<i>Achyranthes aspera</i>	200 mg/kg	Animal models	7	25
<i>Saussurea lappa</i>	400 mg/kg	Animal models	14	26
<i>Inula racemosa</i>	400 mg/kg	Animal models	21	27

antithyroidic drug, Neomercazole (15 mg three times/day), and the *Shankapushpi* extract (125 mg twice/day) for 9 months. They reported that the plant extract is more effective (as evident by 60 to 80% symptomatic recovery and significant changes in I^{131} uptake and concentration of serum cholesterol ($p < 0.05$ and $p < 0.01$, respectively), with no side effects as compared with the allopathic medicine. Another formulation, *Sutshekhar rasa sada* of Zandu Pharmaceuticals, is sometimes suggested for the amelioration of hyperthyroidism, and the recommended dose is 130 to 250 mg twice/day for 3 to 4 months (as recommended

TABLE 8.5
Promising Plant Extracts Suggested for the Treatment of Hyperthyroidism

Plants	Effective Dose	Tested in	Days Treated	Ref.
<i>Lycopus virginicus</i>	1–2 g	Humans	50	28
<i>Azadirachta indica</i>	100 mg/kg	Animal models	20	29
<i>Momordica charantia</i>	500 mg/kg	Animal models	15	30
<i>Convolvulus plauricaulis</i>	400 mg/kg	Animal models	30	31
<i>Embllica officinalis</i>	30 mg/kg	Animal models	20	32
<i>Ocimum sanctum</i>	0.5 g/kg	Animal models	15	33
<i>Piper betel</i>	0.1 g/kg	Animal models	15	34
<i>Rauwolfia serpentina</i>	2.5 mg/kg	Animal models	30	35
<i>Trigonella foenum-graecum</i>	0.11 g/kg	Animal models	15	36
<i>Moringa oleifera</i>	175 mg/kg	Animal models	10	37
<i>Lithospermum officinale</i>	400 mg/kg	Animal models	1	38
<i>Melissa officinalis</i> , <i>L. virginicus</i> , <i>L. officinale</i>	2 µg, 3 µg, 2 µg	Humans	1	39
<i>Lithospermum rudérale</i>	23 mg/kg	Animal models	5	40

by Zandu). According to some reports, *Potentilla alba* and *Kalanchoe brasiliensis* are also effective for the treatment of thyrotoxicosis.^{15,16}

8.9 Conventional Medicine Treatment

Several preparations of L-Thyroxine, including Eltroxin and Synthroid, and thyroid extracts are used for hypothyroidism. Neomarcazole, Methimazole (Tapazole), Carbimazole, and Propyl thiouracil are commonly prescribed for hyperthyroidism. Radioiodine (not in pregnancy) and surgery are suggested in acute cases, but these medicines are not free from side effects^{17,18} and are also expensive. A report revealed that 21 patients indicated abnormal T₄ suppression after receiving Carbimazole for 10 to 12 months and the disease relapsed after 1 year of stopping the therapy.¹⁶ With Thyroxine therapy, side effects including bone fracture and sudden increase in heart rate have been observed.⁵ As suggested by some other authors,¹⁴ Ayurvedic medicines based on scientific studies appear to be the best choice.

8.10 Scientific Basis

For many years people have been using seaweed, including kelp, as a supplement of iodine, a basic component of thyroid hormones. Several plants have also been screened from time to time and few appear highly promising for the treatment of thyroid dysfunctions. Some important ones have been mentioned in Table 8.4 (for hypothyroidism) and in Table 8.5 (for hyperthyroidism) along with their respective references.^{12,19–40} All these scientific studies are presented separately as clinical trials and pharmacological investigations.

8.10.1 Clinical Trials

8.10.1.1 Thyrocap

Thyrocap is a herbal preparation containing solid extracts of *Bauhinia variegata*, *Commiphora mukul*, *Glycyrrhiza glabra*, and *Convolvulus plauricaulis* (100 mg of each extract/capsule). This drug was prepared and tried in 50 patients of simple diffuse goiter at a dose of one capsule three times a day for 3 months.¹⁹ After 3 months treatment, a marked improvement was noticed in weakness, fatigability, dyspnea, and reduction in neck swelling. A significant increase in serum T₄ and T₃ concentrations ($p < 0.001$ and < 0.02 , respectively) and a decrease in serum cholesterol concentration ($p < 0.02$) confirmed its thyroid-stimulating property.

8.10.1.2 Kanchnara Guggul

Prepared from gum resin of *Commiphora mukul* and bark powder of *Bauhinia variegata*, *kanchnara guggul* (237 mg three times/day) was given to 899 tribal people of India bearing simple goiter. They were also asked to apply *Kanchnara* ointment simultaneously. Treatment was continued for 10 to 20 months. A 90 to 100% improvement was observed in the swelling of the gland in 163 patients, a 50 to 100% improvement was seen in 148 patients, and up to a 50% improvement was observed in 149 patients.¹³ It was also observed that patients over 50 years old exhibited only 40 to 50% improvement.

8.10.1.3 Shankhapushpi Syrup

One hundred sixty thyrotoxicosis patients were divided into three groups. Group 1 ($n = 100$) received a standard allopathic drug, Neomercazole (15 mg/day), along with Diazepam (tranquilizer; 5 mg/day) B.D.; group 2 ($n = 30$) received 125 mg of Shankapushpi (*C. pluricaulis*) syrup alone B.D.; and group 3 ($n = 30$) was treated with equivalent doses of *Shankhapushpi* syrup and Neomercazole. The treatment was continued for 9 months. Maximum improvement (73 to 93%) in clinical features such as tremors, weakness, palpitation, increased appetite, and nervousness was observed in the *Shankhapushpi*-treated group. Thyroidal I¹³¹ uptake was also minimum in this group. These observations and a significant increase in serum cholesterol ($p < 0.01$) suggested that *Shankhapushpi* is a better thyroid inhibitor as compared with the modern allopathic drug, Neomercazole.¹⁴

8.10.1.4 Miscellaneous Studies

Some minor investigations have also been made on few plant extracts in relation to the regulation of hyperthyroidism in human beings.

It was discovered that a whole plant extract of *Lycopus virginicus* (bugle weed) at a dose of 1 to 2 g/day for 50 days could decrease serum TSH and T₄ levels in hyperthyroid patients, suggesting that it can ameliorate secondary hyperthyroidism caused by increased TSH.²⁸ In an *in vitro* study, extracts of *Melissa officinalis* (2 μ g), *L. virginicus* (3 μ g), and *L. officinalis* (2 μ g) were found to decrease the TSH-binding activity in human thyroidal membrane when incubated for 24 h. These results indicate that these plant extracts are effective in secondary hyperthyroidism.³⁹

Some scientific investigations have been made on animal models considering different plant extracts to suggest their potential use in further investigation toward the treatment of thyroid dysfunctions. Whereas some have been found to stimulate the production or

release of thyroid hormones, others have been found to inhibit the same, indicating their possible use in hypothyroidism and hyperthyroidism, respectively.

8.10.2 Promising Plants for the Treatment of Hypothyroidism (Table 8.4)

8.10.2.1 *Bauhinia variegata*

This plant's water-soluble fraction of total alcoholic extract at a dose of 2 g/kg was fed to Neomercazole (150 mg/kg)-induced hypothyroidic rats ($n = 12$ in each group) for 20 days. The experiment resulted in enhanced thyroid function as evidenced by increased thyroidal weight ($p < 0.001$), I^{131} uptake and decreased serum cholesterol ($p < 0.05$ for both), and active thyroidal histology.¹² It was found that the action of this plant extract was comparable with Eltroxine.

8.10.2.2 *Echhornia crassipes*

In the same investigation, water-soluble fraction of *E. crassipes* ash at 2 g/kg/day for 20 days was also found to stimulate the thyroid function in the similar manner ($p < 0.001$ in all) as *B. variegata*.¹²

8.10.2.3 *Bauhinia purpurea*

Bark extract of *B. purpurea* at 2.5 mg/kg orally administered to female mice ($n = 7$ in each group) significantly increased serum T_3 and T_4 concentrations ($p < 0.001$ for both) after 20 days of treatment.²⁰

8.10.2.4 *Commiphora mukul*

Commiphora mukul extract at a dose of 200 mg/kg administered for 15 days in mice ($n = 10$) significantly increased the T_3 concentration ($p < 0.001$) and also the food consumption ($p < 0.001$).²¹ A dose of 200 mg of its petroleum ether extract was also earlier reported²² to enhance thyroidal weight and I^{131} uptake ($p < 0.001$) in melatonin-induced (250 mg/kg) hypothyroid rats ($n = 36$).

8.10.2.5 *Withania somnifera*

Withania somnifera root extract at 1.4 g/kg, orally administered in albino mice ($n = 10$ in each group) for 20 days, stimulated both thyroid hormones ($p < 0.001$) without any hepatotoxic effect.²³ In another study, all three extracts (*B. purpurea*, *C. mukul*, and *W. somnifera*) were administered simultaneously to mice ($n = 8$) for 30 days at the doses mentioned above. The results showed an increase in both T_3 and T_4 levels ($p < 0.01$ and $p < 0.001$, respectively), suggesting that a combination of the three plant extracts may prove to be an effective treatment for hypothyroidism.²⁴

8.10.2.6 *Achyranthes aspera*

Achyranthes aspera leaf extract administered in rats ($n = 7$) at a dose of 200 mg/kg for 7 days caused an increase of T_3 and T_4 ($p < 0.001$ for both). An increase in blood glucose in this group ($p < 0.05$) further supported the extract's thyroid-stimulating nature.²⁵

8.10.2.7 *Saussura lapa*

Thyroid function stimulation as evidenced by thyroidal histology was reported in rats ($n = 7$) treated either with 400 mg/kg of *S. lapa* root extract for 14 days²⁶ or with the equivalent dose of *Inula racemosa* root extract for 20 days.²⁷

8.10.3 Promising Plants for the Treatment of Hyperthyroidism (Table 8.5)

T_3 is metabolically the most potent thyroid hormone. Plants that have been found to decrease at least the serum level of this hormone have been suggested to act in the treatment of hyperthyroidism.

8.10.3.1 *Azadirachta indica*

Two doses (40 and 100 mg/kg) of the leaf extract of *A. indica* was administered in albino mice ($n = 7$ in each group) for 15 days; the higher dose could decrease the serum concentration of T_3 ($p < 0.05$). It was postulated that the inhibition of T_3 production was mediated through T_4 to T_3 conversion, the principal source of T_3 generation.²⁹

8.10.3.2 *Momordica charantia*

Fruit extract of *M. charantia* (500 mg/kg) orally administered for 15 days decreased serum T_3 and T_4 concentrations ($p < 0.001$ for both) in mice ($n = 8$ in each group).³⁰ However, this dose was found to enhance hepatic lipid peroxidation (LPO).

8.10.3.3 *Convolvulus pluricaulis*

The root extract of *C. pluricaulis* (0.4 mg/kg/day for 30 days) administered to L-thyroxine-induced hyperthyroid mice ($n = 7$ in each group) decreased serum concentration of T_3 and hepatic 5-D activity. These results indicate that the plant extract-induced inhibition in thyroid function is primarily mediated through T_4 to T_3 conversion.³¹

8.10.3.4 *Emblica officinalis*

The fruit extract of *E. officinalis* at 30 mg/kg for 20 days decreased both serum T_3 and T_4 concentrations ($p < 0.05$ and < 0.001 , respectively) in mice ($n = 8$ in each group). It was thought to decrease T_3 production particularly by inhibition of peripheral conversion of T_4 to T_3 in extra thyroidal tissues.³²

8.10.3.5 *Ocimum sanctum*

The leaf extract (0.5 g/kg) of *O. sanctum* administered to male mice for 15 days ($n = 10$ in each group) significantly inhibited only T_4 concentration ($p < 0.001$).³³

8.10.3.6 *Piper betel*

Administration of the *P. betel* leaf extract (0.1 g/kg) to male mice ($n = 7$) for 15 days significantly decreased both serum T_3 and T_4 concentrations ($p < 0.01$ for both).³⁴

8.10.3.7 *Rauwolfia serpentina*

The *R. serpentina* root extract (2.5 mg/kg) administered to T_4 -induced hyperthyroid mice ($n = 7$ in each group) for 30 days significantly decreased both the serum T_3 and T_4 concentrations ($p < 0.001$ for both).³⁵

8.10.3.8 *Trigonella foenum graecum*

T. foenum graecum seed extract treatment (0.11 g/kg for 15 days) given to both mice and rats ($n = 7$ for both) decreased serum T_3 ($p < 0.01$ in mice and $p < 0.001$ in rats) and increased T_4 concentration ($p < 0.01$ in both mice and rats). The seed extract induced reduction in T_3 level could be the result of inhibition in peripheral conversion of T_4 to T_3 in extra thyroidal tissues.³⁶

8.10.3.9 *Moringa oleifera*

M. oleifera leaf extract treatment (175 mg/kg) of female rats for 10 days ($n = 14$ for each group) decreased serum T_3 concentration ($p < 0.05$) and increased in serum T_4 concentration ($p < 0.05$). This observation suggests the inhibitory activity of the plant extract in the peripheral conversion of T_4 to T_3 .³⁷

8.10.3.10 *Lithospermum officinale*

An extract from the powdered leaf of *L. officinale*, administered at a dose of 400 mg/kg for 24 h to rats ($n = 42$), decreased serum T_3 concentration ($p < 0.01$) in TSH-induced hyperthyroidic animals.³⁸

8.10.3.11 *Lithospermum ruderales*

The root extract of *L. ruderales* (23 mg/kg/day) administered to TSH-treated guinea pigs ($n = 5$) for 5 days decreased thyroid weight ($p < 0.01$) and exhibited low active thyroidal histology. The results suggested its antithyroidal nature.⁴⁰

8.10.3.12 *Allium sativum*

A. sativum bulb extract (500 mg/kg) administered to hyperthyroid rats ($n = 8$) for 15 days showed a decrease in serum T_3 ($p < 0.05$) and cholesterol ($p < 0.05$) concentration.⁴¹

8.10.3.13 *Nelumbo nucifera*

Rhizome extract of *N. nucifera* (400 mg/kg) administered to rats ($n = 8$) for 15 days decreased both serum T_3 and T_4 concentrations ($p < 0.01$ and < 0.05 , respectively).⁴²

8.10.3.14 *Aloe vera*

Diluted parenchyma of *Aloe vera* leaf at 125 mg/kg/day administered for 15 days in mice ($n = 7$ in each group) has been reported to inhibit both T_4 and T_3 concentrations ($p < 0.05$ and < 0.01 , respectively).⁴³

8.10.3.15 *Aegle marmelos*

Dried leaf powder extract of *A. marmelos* at a dose of 1 g/kg/day for 15 days significantly ($p < 0.001$) decreased only serum T_3 concentration.⁴³

All these findings indicate the plant-specific role in the regulation of thyroid functions. Although some were found to be thyroid stimulatory as evidenced by the increase in serum T_3 and/or T_4 concentrations, the others were inhibitory to either one or both of the thyroid hormones. Although these findings do not claim to cure or treat the thyroid dysfunctions, certainly the plants can be further studied for their potential use in the treatment of hypo- and hyperthyroidism.

8.11 Prevention Strategies

Because thyroid-related problems affect nearly all systems of the body, it is advisable to take some preventive measures, particularly when one or more symptoms of thyroid abnormalities (see [Table 8.1](#)) are observed. For the prevention of hypothyroidism, it is important to supply the thyroid gland with adequate quantity of nutrients so that it can manufacture the required amount of thyroid hormones. If hypofunctioning of the gland is suspected, one has to avoid the goitrogenic food such as rapeseed, cabbage, cauliflower, kale, and turnip. One should also ensure the iodine intake of 150 $\mu\text{g}/\text{day}$ through food and water supplementation apart from using filtered water with less chloride and fluoride.

Left untreated, hyperthyroidism has a serious consequence and much precaution should be taken for its prevention. Scientists believe that hyperthyroidism has a genetic susceptibility. People who have a family history of thyrotoxicosis should avoid stress, trauma (both physical and mental), and excessive steroid therapy, which are very often considered as causative factors. From a safer point of view, serum T_4 , T_3 , and TSH should be estimated every year to ascertain the thyroid status and accordingly follow medical advice, if warranted.

8.12 Summary

Considering all the literature available on Ayurvedic therapies in the regulation of thyroid dysfunction, it can be said that the management of the thyroid problems in Ayurveda is nearly similar to the conventional allopathic therapy, because both primarily aim to augment thyroid function (for hypothyroidism) or to reduce it (for hyperthyroidism). However, herbal therapy appears to be safe when compared with the modern allopathic treatment.¹⁴ This is because Ayurvedic preparations include mainly herbs that make the body work more efficiently while supporting many complex functions. Moreover, herbal extracts possess natural antioxidants, which not only help in curing the diseases, but also improve the body's defense system. On the downside, allopathic drugs and chemicals interfere in the body's natural process, resulting in unwanted side effects. Because they offer fewer or no side effects, herbal therapies should be preferred in the treatment of thyroid dysfunctions.

Herbal therapies should be specific, considering the specific type of thyroid diseases and imbalance in the particular type of thyroid or pituitary hormone. In fact, recent researches have revealed a differential role of herbal extracts in relation to the regulation of thyroid function. Although some plant extracts are potent in altering both thyroid hormones, others have been found to be effective in correcting either T_4 or T_3 levels.

8.13 Suggested Future Research

Although some Ayurvedic preparations are already available for the regulation of thyroid dysfunctions, further research is required. Thyroid problems are indirectly related to many other diseases, including diabetes and gastrointestinal, neural, and heart problems. Further investigations on the different aspects of herbal regulation may provide a better understanding on the Ayurvedic management of thyroid dysfunctions. It is suggested that research activities should be undertaken in the following areas:

1. Screening of more plants to find out the highly effective plant extracts for the treatment of thyroid diseases
2. Concentration-dependent studies to identify safe and effective doses
3. Long-term effectiveness studies of the plant extracts to optimize the duration of treatment
4. Comparative studies on the active compounds and crude extracts
5. Target-specific studies on plants acting either at glandular level or at the level of peripheral tissues, including the liver and kidney, which produce maximum amount of T_3 in both human and animal models

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Obesity (Medoroga) in Ayurveda

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9.1 Introduction

Several centuries ago before foreign rule, India was one of the world's most affluent nations, and the affluence and sedentary habits at that time were known to be the leading cause of obesity.^{1,2} This was due to a deadly imbalance of high energy input through rich foods and low energy expenditure due to lack of physical exercise. Daily and seasonal health regimens

(*swasthavritta* and *ritucharya*) and other modalities, such as detailed instructions on a proper balanced diet and appropriate levels of exercise as per the constitution (*prakriti*) of the person, have been laid out clearly in Ayurvedic texts.³ It is interesting to note that the world is now again focusing on a healthy lifestyle as key to avoiding risk factors like obesity.⁴

The pioneers of Ayurveda, 2 millennia ago, have described the unhealthy consequences of obesity. Charaka, who may be called as the Indian Hippocrates, described obesity as a disease of the fat tissue (*medoroga*) leading to hugeness (*sthoulyam*).⁵ The Sanskrit word *sthoulya* is derived from *sthoor*, meaning gross or material.⁶ Nondrug and drug modalities to prevent and reduce obesity were elaborated and regularly utilized. Even now there are millions of Indians who do avail themselves of some of these interventions in their daily lives. The prevalence of obesity is higher in urban areas than in rural populations of India, due to a steady erosion of the Ayurveda way of life in the cities as well as the sedentary and overeating habit factor.

Being a holistic system of health care, Ayurveda has emphasized the psychospiritual dimensions in its philosophy; mere drugging without holistic healing is not advocated. The hierarchy of the spiritual, psychological, and physical levels of human health and disease is given due importance.

The disease management approaches in Ayurveda have two basic principles: repletion of the body tissues (*brimhan*) or the depletion (*karshan*) of them, as required. For this process, the groups of modalities used are spiritual and the karmic interventions (*daivavya-pashraya*), astral and psychological measures (*satwavyajaya*), and nondrug and drug modalities (*yuktivyapashraya*). The holistic base of Ayurveda provides a different kind of evidence than the current reliance on randomized controlled trials.⁷ The clinical effects are evident in the field.

This chapter may embolden the open-minded physicians of the world to learn from Ayurveda — a live and growing system of health that is more than 5000 years old.⁸ There are leads which can be followed-up proactively for research, too.⁹ The current understanding of adipose tissue as an endocrine organ coupled with the successful leads from Ayurveda may form a scientific basis for the management of obesity.¹⁰ Already, *Commiphora wightii* (*guggulu*) is emerging as a drug of global potential for obesity and hyperlipidemia.¹¹

9.2 Agni and Srotas with Reference to Obesity

As per Ayurveda, *agni* is the cosmic fire — the principle of transformations of materials. It is the energy or capacity of the body to convert complex food materials to their constituents and then to build the body tissues (*dhatu*).¹² The first and foremost level of *agni* is digestive fire (*jatharagni*).¹³ It digests all types of food in the stomach and the small intestine. The digested and absorbed essence of the food material is called *ahar rasa*, which circulates, providing the substrates for tissues. For each of the seven tissues, there is a special energy or digestive power (*dhatwagni*)¹⁴ assimilating the digested substrates. The lipid precursors are acted upon by fat-specific energy (*medodhatwagni*) for its conversion into adipose tissue (*medadhātu*). The channels and the loci where these conversions take place are called *srotas* or *dhatuvahasrotas*. The quality of all the specific energies depends on the quality of the digestive fire, which is protected and maintained carefully. The impairment of the digestive fire and the specific tissue energies lead to poor availability of the constituents and depletion of tissues (*dhatukshaya*).

With the substrates and energies in balance, all the metabolic activities occur properly in the channels. Defects in the sources with undigested matter block tissue channels. A healthy person is one in whom the activities of humor (*dosa*), tissue (*dhatu*), wastes (*mala*), fire, mind (*manas*), soul (*atma*), and senses are harmonious and in balance.¹⁵

9.3 Adipose Tissue (*Medodhatu*) and Dynamics

Any disequilibrium in *kapha* humor, fat-specific energy, and waste products of adipose tissues (*kleda*) leads to a dysfunction of adipose tissue — obesity. Adipose channels have two origins: the kidneys, adrenals, and fat around them (*vrukkās*) and the visceral and omental fat (*vapavahan*).¹⁶ These channels draw the nutritive parts (*medoposhakansh*), including lipid, from the antedecent flesh (*mansa dhatwagni*) and the transient lipids (*asthayi*) and then they are converted into a stored form (*sthayi*) of lipid.

Adipose-tissue dynamics is a crucial link for the tissue metabolism. Low adipose energy, despite a normal food intake, could lead to a steady accumulation of fat (*apachith*) and obesity. Excessive consumption of fat and oils, lack of exercise, and overindulgence of wine and sleeping during daytime can disturb the fat channels, as mentioned in Charaka.¹⁷

9.4 Measurement of Obesity

To determine the normal physique of a person, Charaka gave a standard termed as proportionate body (*samasanhanan nara*), with appropriate measurements and compactness.¹⁸ Gangadhar, a later author, has given the measurements of various parts of body for a normal physic. According to Charaka, fat gets deposited mainly in the abdomen (*udara*), so the measurement of the abdomen size was considered to determine obesity.¹⁹ The exact dimensions given in the text are not relevant today.

9.5 Obesity, Health Risks, and Life Span

Recent importance of obesity control and its adverse impact on health was presaged 1000 years ago by Bhavamishra and Madhavakar. Today, with a better understanding of the reference to the importance of the visceral and abdominal obesity as a cardiovascular risk factor, the reference in ancient texts seems a valuable insight of the seers.¹⁹ We have yet to understand what they have foreseen in the reference about fat accumulation around the kidneys. The management of morbid obesity was considered difficult and the prognosis grave if other risk factors were also present. Decreased life span (*ayu-kshaya*) is stated to be an important consequence of obesity. Even today, early diagnosis and timely management of obesity is mostly neglected because few people realize that obesity is associated with increased mortality.²⁰ Charaka emphasized that serious diseases (*darun vikar*) arise when fat blocks channels. Sushruta describes several complications of obesity such as tumors and space-occupying lesions like tumorous growth, diabetes mellitus (DM), and

TABLE 9.1

Features of Obesity

Symptom	Charak	Sushrut	Vagbhat	Bhavamishra	Madhavkar
Visceral obesity (<i>udararudhi</i>)	+	+	+	+	+
Flaccidity (<i>angashaithilya</i>)	+	—	—	+	+
Pendulous abdomen (<i>chalodara</i>)	+	—	+	+	+
Pendulous breasts (<i>chala-stana</i>)	+	—	+	—	+
Pendulous buttocks (<i>chala-sphik</i>)	+	—	+	+	+
Heaviness in body (<i>gourav</i>)	+	—	—	—	—
Weakness (<i>dourbalya</i>)	+	—	—	—	—
Poor self-care (<i>swakriya-asamarthata</i>)	+	+	—	—	—
Less coitus (<i>krichha-vyavaya</i>)	+	+	—	+	+
Oversleeping (<i>nidradhikya</i>)	+	+	+	+	+
Lethargy (<i>utsaha-nash</i>)	+	+	+	—	—
Breathlessness (<i>kshudra-shwasa</i>)	+	+	—	+	+
Oversweating (<i>swedati-pravrutti</i>)	+	+	+	+	+
Bad body odor (<i>dourgandhya</i>)	+	+	+	+	+
Excessive thirst (<i>trishnadhikya</i>)	+	+	+	+	+
Polyphagia (<i>kshudhadikya</i>)	+	+	+	+	+
Delicate and tender (<i>sukumarata</i>)	+	+	—	—	—
Short life (<i>ayushyarrhasa</i>)	+	+	—	+	+

excessive perspiration. Thirty to fifty percent postmenopausal women in the U.S. are said to be overweight.²¹

9.6 Symptoms and Signs

The three major and three minor Ayurvedic treatises (i.e., Charak, Sushruta, and Vagbhat and Madhavnidhan, Bhavaprakasha, and Sharangdhar, respectively) give detailed descriptions of obese individuals and their afflictions. Table 9.1 summarizes these descriptions as given by the first five treatises.^{15,19,22–24} Bhavamishra describes additional features of obesity such as difficulty in breathing and the emergence of a snoring sound from the throat while breathing. Snoring and daytime sleepiness are important markers of obesity that were well recognized in Ayurveda.¹⁹ In extreme forms of obesity, obesity-hypoventilation (OHV), or Pickwickian syndrome, and obstructive sleep apnea (OSA) are severe as reported in conventional medical literature.²⁵ They lead to pulmonary hypertension and often to a degree of cardiac failure. There is a spectrum of sleep-related disordered breathing in obesity that varies from simple snoring to OHV. Recognition of the association of OSA, obesity, and daytime sleepiness in conventional medicine is quite recent. Daytime sleeping contributes to obesity. OSA is associated with increased rates of hypertension, stroke, ischemic heart disease, and mortality.²⁶

9.7 Obesity and Affect Disorders

Obese people are depressed and often experience sadness as per one of the Ayurvedic texts (*Bhavamishra*).¹⁹ As physicians, we must recognize that obesity, a heterogenous disorder, does have psychopathology as a cause in an individual with a possible familial, environmental,

TABLE 9.2
Etiology of Obesity as per Ayurveda

Ayurvedic Factors	Allopathic Correlates
No exercise (<i>avyayama</i>)	Lack of exercise
Dense food (<i>sleshmal ahara</i>)	Fattening diet and foods
Daytime sleeping (<i>diva swapna</i>)	Obesity sleep apnea
No sexual intercourse (<i>avyavaya</i>)	Difficulty in intercourse
No anxiety (<i>achinta</i>)	Affect disorders
Genetic (<i>beeja-dosa</i>)	Genetic basis
Prodromal signs (<i>prameha-poorvarupa</i>)	Hyperinsulinemia
Loss of appetite (<i>agnimandya</i>)	Low energy expenditure
Lipotoxicity (<i>medavrittavayu</i>)	Defective satiety cascade
Lack of restraint (<i>ahara-asamyama</i>)	Environmental food clues
Tissue indigestion (<i>apathya-dhatwagni</i>)	Stress and hormones

or hereditary basis. According to modern medicine, specific organic causes may rarely be present. Metabolic and psychological (e.g., depression) pathologies often present together and are associated with dysregulation of the hypothalamo-pituitary adrenal axis.²⁷ Affect disorders are also reported among obese binge eaters (OBEs). Sixty percent of OBE had one or more psychiatric disorders. Some of the OBEs also had a lifetime prevalence of affective disorders as well as more frequent mood fluctuations and panic attacks.

9.8 Obesity and Diabetes Type 2-NIDDM (*Madhumeha*)

Sushruta, though classifying disorders according to their causes, has stated obesity and DM as the disorders of adipose tissue.²⁸ Charaka attributes the voracious appetite of obese people to increase in humor (*vata*) and digestive fire, which together rapidly consume ingested food. This leads to increased frequency and amount of food intake. This further increases the fat accumulation, particularly in the abdomen, breasts, and buttocks. The extent of fat accumulation, in these parts is so great that they become pendulous. Clogged fat channels in Ayurveda, as described earlier, then cause preclinical symptoms of urinary disorders (*prameha*) such as perspiration, bad body odor, flaccidity of the body, a desire sleep or sit, thickening of layers on teeth, accumulation of waste in eyes, nose, ears, mouth, etc., sweet taste in mouth, burning sensation of palms and soles, and dryness of the throat and palate.²⁹

9.9 Etiological Factors for Obesity (*Karana-Hetu*)

“The central causes of all disease are the vitiated *Malas*, induced by harmful diet and lifestyles.”³⁰ According to Ayurveda, the central cause of almost all diseases is the vitiation of wastes (*malas* — the end products of *dhatu*s) and vitiation of humor (*dosa* provocation). This is due to harmful regimens of foods and beverages, rest and exercise, restraint and indulgence, etc. (see Table. 9.2). Both Ayurveda and conventional medicine have considered obesity as multifactorial.

TABLE 9.3

Body Mass Index (BMI) and Obesity

Classification	BMI (kg/m ²)
Normal	18.5–24.9
Overweight or pre-obese	25.0–29.9
Obese Class 1	30.0–34.9
Obese Class 2	35.0–39.9
Obese Class 3	>40

9.10 Diagnosis and Markers of Overweight and Obesity

Differences in overweight (*pushta*), obesity, and morbid obesity (*ati sthoulya*) were recognized in Ayurveda. The measures of normality of the body structure have been described as follows. Charaka described the assessment of obesity by the patient's own fingers (*angulapramana*), breadth, and length (midcarpal–middle finger level) to be borders defined on the abdomen. This may be investigated through the waist–hip ratio (WHR), waist circumference (WC), and body–fat ratio analysis.

According to conventional medicine, the body mass index (BMI) is a simple method of estimating adiposity. It is calculated as weight in kilograms divided by the square of the height in meters (kg/m²). The World Health Organization (WHO) has classified individuals as per BMI (Table 9.3). Class 3 obese is considered morbidly obese.

It is now well recognized that central or visceral obesity is more important as a risk factor for cardiovascular diseases (CVDs) and noninsulin-dependent DM (NIDDM), independent of general obesity.³¹ Accurate assessment for the central and visceral obesity is paramount in evaluation of obesity. WHR and WC correlate better with degree of CVD risks. High accumulation of the upper body fat (abdominal) is associated with high plasma triglyceride levels. A high WHR is shown to be associated with dyslipidemia.

For clinical services and epidemiological studies, the measurements of height, weight, BMI, WHR, or WC are important and suffice. For research purposes, more sophisticated methods like electrical impedance, dual absorptiometry (DEXA), and other imaging techniques such as computered-aided tomography scan (CAT-scan) and magnetic resonance imaging (MRI) may be required. Other markers of obesity are hypertension, hyperpigmentation (*Acanthosis nigricans*), skin tags, hirsutism, menstrual dysfunctions, and female infertility. Hyperinsulinemia, hyperglycemia (type II DM), and hyperlipidemia are given associated biochemical features.

In a series of 90 cases of polycystic ovarian syndrome (PCOS) with hirsutism at our clinic, only 23% had normal weight. The rest were either overweight or obese. Sixty percent of these women were in their teens.³² Infertility and menstrual dysfunctions due to anovulation, abortions, and increased fetal wastage, and gestational diabetes are common among women with PCOS.

9.11 Clinical Course and Prognosis of Obesity

The long-term adverse impact on health has been well described in Ayurveda, including diminished life span (*vide supra*). Obesity-related increase in complications like NIDDM

and CVDs (including hypertension and arthritis) are well known. However, obesity-related cryptogenic cirrhosis of the liver and hepatocellular carcinoma has recently been recognized. In spite of the rising incidence of obesity and NIDDM, obese individuals are often not screened for nonalcoholic fatty infiltration of the liver and nonalcoholic steatohepatitis (NASH). Early diagnosis and treatment of NASH in overweight and obese patients may prevent hepatocellular carcinoma in cryptogenic cirrhosis of the liver. The incidence of the hepatocellular carcinoma among these patients is reported to be 27%. This rate is similar to that found among hepatitis C virus-related cirrhosis (21%). Early and persistent long-term interventions can certainly alter the progressive course and complications of obesity. Even a 5 to 10% loss in body weight can reduce the risk caused by obesity. Diet, exercise, and stress management can also reduce insulin resistance (IR) in a patient. Ayurvedic way of life is the need for altering the course of obesity.

Weight gains during adolescence in boys and girls, enormous weight gains during and after pregnancy, and peri-post menopausal obesity are noted frequently in clinics that are dedicated to management of obesity. These transitional physiological phases are known to be associated with changes in insulin resistance (IR) and physiological hyperinsulinemia. An antiobesity preventive measure for such physiological phases for smooth transitions may save many individuals, particularly women, a lifetime burden of obesity and its health consequences.

9.12 Prevention of Obesity

The knowledge of the etiological factors and their lifelong avoidance, besides the counter-measures, constitute the foundation of prevention of obesity in Ayurveda and other systems of medicine.³³

Ayurveda has much to offer for prevention in terms of daily and seasonal regimens, healthy foods, yoga exercise, *panchakarma*, and medicines. The daily routine in Ayurveda involves the following:

1. Going to bed early and waking up early
2. Proper eliminations, tongue cleaning, washing, etc.
3. Meditation
4. Massage
5. Sun-yoga-asana (*surya-namaskar*)
6. Proper clothing
7. Suitable, balanced, and measured dietary intake
8. Adequate fluid intake
9. Family, friends, and community relationships
10. Avoidance of undue stress or exertion
11. Pleasant, healthy, and socially permissible sex life
12. Continuing self-education etc.³

Similarly, as per the seasons and climate and age of the person, certain regimens are to be followed to maintain health and prevent diseases. *Chyavanprash* in the winter, *gulkand*

in the summer, and ginger candy in the monsoon are taken still by millions of Indians as seasonal foods. Bitter and astringent items are also included in the diet.

Education in daily healthy lifestyle should begin very early. Functional information on health, diet, proper exercise, and stress management, given in an incremental manner at the school level, would prove beneficial for prevention of adult obesity. The predictors of weight gain — genetic, metabolic, and demographic — can be identified early and appropriate steps can be taken. Overweight parents have overweight children. In twin studies, heritability estimates are 30 to 50%. Early genetic diagnosis would help identify need for prevention strategies with the advances in genomics. This would assist early Ayurvedic and modern preventive interventions in highly obesity-prone persons.

Breathing yoga practices (*pranayama*) under guidance of a proper expert are advisable for all persons prone to obesity, besides other measures. Meditation and yoga physical and breathing practices (*asanas* and *pranayama*) diminish the sympathetic overdrive, which has a role in inducing IR and obesity. Only massive public health education and values of restraint despite affluence can help prevent the rising prevalence of obesity globally. Ayurveda and modern medical measures have to be individualized for their synergy by practicing doctors and health educators.

9.13 Management of Obesity (*Chikitsa*)

9.13.1 Purification Modalities

The five modes of eliminating Nature's waste by *panchakarma* to reestablish the harmony of *dosas* is a unique gift of Ayurveda.³⁴ These modes are often preceded by the preparatory oil-therapy (*snehana*) and induced sweating (*swedana*) (see [Chapter 4](#)). Oil therapy is not indicated for those who do not exercise or are very obese or very thin.³⁵ It is also not given to those who have voracious appetites or anorexia. Before oil therapy is carried out in the obese, the vitiated digestive fire will have to undergo drying through controlled dieting and then careful oleation will produce benefit. The oil dose frequency can be individualized as necessary.

Induced sweating usually after oil therapy is indicated to transport the vitiated *dosas* to the trunk (*koshtha*). As per Ayurveda, dry heat, hot poultice, slow hot shower and sauna-bath, and hot-water bottles are used to induce sweating.³⁶ Sweating can be induced for patients with *kapha* diseases without preceding oil therapy. The parts of the body to be warmed have been described and are followed carefully by the experts. Very obese people are contraindicated for sweating therapy before reducing their weight to a reasonable level with other measures like diet, exercise, and medicine.

Induced therapeutic vomiting is indicated in *kapha*-dominant diseases. The procedures, contraindications, and cautions of this procedure are well described elsewhere in this textbook.³⁷ Besides preparations for *panchakarma*, follow-up (*paschatkarma*) is equally vital. *Only qualified and experienced vaidyas should engage in panchakarma therapy.*

Of the five purification procedures, the enema (*basti*) is the major component, as it controls the *vata*, which is responsible for the disease. In addition, enemas also cause cleaning of bowels (*malashoodhi*), which reestablish proper balanced metabolism within the tissue elements. Enema decoction of roots of ten medicinal plants (*dashasmool*) and fruits of three medicinal plants (*triphal*) are indicated for their *vataghna* and *tridoshar* properties frequently used in obesity.³⁹ To prevent aggravation of *vata*, an oil enema

TABLE 9.4

Preparations To Be Taken after Panchakarma Therapy

Preparation	Frequency of Intake after Elimination Therapy
Gruel (<i>peya</i>)	3 times/day
Rice gruel (<i>vilepi</i>)	3 times/day
Incomplete soup (<i>akruta yusha</i>)	3 times/day
Complete soup (<i>kruta yusha</i>)	3 times/day
Incomplete meat soup (<i>akruta mansarasa</i>)	3 times/day
Complete meat soup (<i>kruta mansarasa</i>)	3 times/day

(*anuwasan basti*) is then given in succession initially. The decoction and oil enemas are given alternately until 8 or 15 days, depending upon symptomatic response. *Shirobasti* of oil is also advisable to counter stress element.

9.13.2 Palliative Therapy

For obese patients, according to the symptoms of *ama*, digestive (*dhatwagni deepan*) and carminative (*pachan*) medications are given. Ayurvedic formulation of a *guggulu kalpa* decoction (e.g., *triphala guggulu*, *medohara guggulu*, *arogyavardhini*, *chandraprabhavati*, *shilajitvati*, *loharasayan vati*, *dashamoolarishta*, *manjisthadi*) are also advised with appropriate doses (as per age, sex, severity of obesity) with water or decoction of *triphala*.

Gentle oil massage is followed by massage with powders of medicinal plants (turmeric, sandalwood, rosepowder, *manjistha* powder, chickpea flour) along the antidirection of the lanugo hairs, reducing perspiration and its smell.

A diet known as *sansarjankrama* is advised as per the quality results obtained after elimination therapy and the state of digestive fire.⁴⁰ It includes rice preparations (*manda*, *peya*, *vilepi*) and soups of meat in successive days. Some of the preparations are mentioned in Table 9.4.

The current common treatment of obesity in Ayurveda involves the following:

1. Nondrug modalities of reasonable fasting with dieting, exercise, and yoga and lifestyle changes and counseling to stress, sleep, sex, etc.
2. Diverse formulations and sometimes single plants with appropriate individualized dosage needs

The weight loss is expected to be gradual, long term, and lasting due to integral care rather than drastic weight loss advocated by crash dieting.³⁸ Individual *prakruti*, aptitude, tastes, profession, peer support, motivation, relationships, and stage of obesity are carefully looked into for a successful outcome of the therapeutic program.

9.13.3 Baseline Acceptance of the Patient's State

Hidden within every obese person, a thin person with a proportionate body is struggling to emerge. Similarly, in every adult there is a hidden child who may regress to oral phase of development under stress.

A physician's nonjudgmental attitude about the patient's body image earns the patient's trust and builds the foundation for a long-term, warm, and effective patient-physician relationship; it is also essential for a sustained motivation, continued long-term compliance management. The repressed sexuality has to be rekindled by subtle and persistent professional counseling, as lack of sexual intercourse is one of the

causative factors in obesity in Ayurveda. The aforesaid have to be complemented by lifestyle changes in exercise, sleep, and stress management. The patient's friend, spouse, or family member, as well as the team, has to be firm and supportive for long-term care and compliance (*chatushpada*).

9.13.4 Dietary and Nutrient Diet

The diet for treating obesity has to be individualized as per the level of the caloric needs, energy expenditure, tastes, lifestyle, weight-loss goals, medicines, and other measures used. No formula diets or regimens are usually followed in Ayurveda. In reducing the bulk of a person, food that is heavy but falls in the category of a low-fat diet (*apatarpan*) is beneficial. Roasted pulses (*bharjit dhanya*) such as green beans (*mudga*), dolichos (*kulatha*), and barley (*yava*) are beneficial.

9.13.5 Energy Expenditure and Hormonal Stress

As per Charaka, occasional staying awake at night, sexual intercourse, physical exercise, and mental exertion should also be gradually indulged in to reduce fat. The patients are advised to walk at least 1 h in the morning or evening with regularity and consistency. Adolescent and young adults are advised to engage in vigorous sports and exercise. Movements of shoulder girdle and pelvic girdle are advised to prevent sarcopenia (*man-sadaurbalya*). Yoga, particularly physical asanas (*hathayoga*) with mind control (*rajayoga*), is to be integrated in daily routine. Yoga in everyday life (i.e., sun salutations [*suryanamaskar*]) is an excellent dawn practice that has been and continues to be an integral part of Indian life.

9.14 Scientific Basis: Clinical Studies

At hundreds of Ayurvedic hospitals and clinics in India, obesity is being treated with diverse modes. It is desirable to analyze carefully maintained records of many obese patients by pharmacoepidemiological methods. At our Bhavan's Swami Prakashananda and Ayurveda Research Centre (SPARC), we have ambulant care of patients with obesity by Ayurvedic and modern experts. Table 9.5 shows in brief the basal and final body weights and the drug treatment given in female patients with their usual unmodified diet and exercise.

In one study at our clinic, patients were given standard doses of the formulations. It is interesting to note that only two women showed no change, despite the fact that one was given the formulations for 4 months. A weight loss of 1 to 3 kg was observed in four patients within 1 to 2.5 months. There were seven patients who showed a 4- to 6.5-kg weight loss and received the formulations for 3.5 to 6 months. Significant weight loss (4 to 5 kg) occurred in two patients within a month and surprisingly within 1 week in a single subject. Besides *arogyawardhini* there is a widespread use of *medohar-guggulu* (MHG), a formulation based on *Commiphora wightii*, in India. The experiential data suggest the need to increase the dose, as per the individual needs of the patient, and concurrently

TABLE 9.5

Response in Ambulant Obese Patients to Ayurvedic Therapy

Initials	Sex	Age (Years)	Body Weight (kg)			Treatment Given ^a	Duration (Months)	Change (Δ kg)
			Initial	During	End			
AJ	F	40	79	76	75	MHG, AW, SV	6	-4
AS	F	34	64	63	59	AV, TG	1	-5
AG	F	52	75	72	70	MHG, AW, SV	3.5	-5
SS	F	38	95	93	90	MHG, AW	8	-5
MB	F	39	75	—	71	MHG, AW, SV	1	-4
AS	F	33	90	—	86	MHG, AW, SV	1	-4
JP	F	44	60	55.5	53.5	MHG, AW	4.5	-6.5
PM	F	45	98	—	95	MHG, AW, SV	2	-3
LD	F	43	64	63	62	MHG, AW, SV	2	-2
EJ	F	35	63	62	63	MHG, AW, SV	4	0
BS	F	32	64	64	64	MHG, AW, SV	1.5	0
SM	F	54	91	91	90	AW, SV	1.5	-1
MT	F	53	79	78	77	MHG, AW, TG	2.5	-2

Source: Bhavan's Swami Prakashananda Ayurveda Research Centre (SPARC).

^aMHG = Medoharaguggulu; AW = Arogyawardhini; SV = Shankhvati; TG = Triphalaguggulu.

advise essential diet and exercise measures. Then the magnitude and the rate of weight loss is expected to be more. Our studies with *guggulu* preparations in arthritis have highlighted the need to use larger doses than usually, and hesitantly, prescribed.⁴¹ *Panchakarma*, too, should be judiciously combined for integral therapy. However, some patients who are given higher doses of *guggulu* may complain of gastrointestinal irritation and burning.

At Bhavan's SPARC, the experiential data on the larger doses of different *guggulu* preparations and with a compliance for moderate diet and exercise, as well as steam and massage, showed interesting confirmation of the aforesaid statement of titrating the dose. Thirty-two of 36 patients with an average weight of 81 kg had lost an average of 3.6 kg in 4 weeks with a weight-loss range of 2 to 20 kg; 7 of them had lost more than 10 kg. *Guggulu* has been considered lipolytic (*medohara*) since ancient times. Hypolipidemic effects have been well demonstrated both experimentally and clinically.

In a placebo-controlled trial of 70 obese subjects, Patwardhan et al.⁴² showed significant weight loss, decrease in body measurements, and a decline in serum cholesterol and triglycerides with Ayurvedic formulations (i.e., *guggulu* preparations used were *triphalaguggul*, *gokshuradhi guggul*, and *sinhnad guggul*). The placebo group did not show any change in body weight or other variables mentioned above.

9.15 Ayurvedic Drugs and Medicinal Plants for Obesity

For obesity, many plants and formulations have been described in the Ayurvedic texts and are also currently used by Ayurvedic physicians. The herbs are chosen based on Ayurvedic pharmacology, which relies on taste and other physical-chemical properties for its action. Medicinal plants having bitter (*tikta*), pungent (*katu*), astringent (*kashay*) tastes,

and light (*laghu*), dry (*ruksha*), rough (*khara*), subtle (*sukshma*), sharp (*tikshna guna*), hot (*ushna veerya*), and pungent (*katu vipak*) properties are used to treat obesity. Substances having sweet (*madhur*), sour (*amla*), and salty (*lavan*) tastes increase fat, whereas bitter, pungent, and astringent tastes have a reverse effect.

Bitter tastes are dry, cold, and light (*ruksha*, *sheeta*, and *laghu*); they stabilize skin and flesh in the body and absorb subtle liquid waste products (*kleda*), excessive fat, excessive mucus, pus, sweat, urine, stool, and humor (*agni* and *apa*)⁴³ (e.g., neem [*Azadirachta indica* Linn.] and *kutaki* [*Picrorrhiza kurroa*]).

Pungent tastes have similar properties like bitter, light, and dry tastes but are unlike hot tastes. They act on the fat tissue through scraping (*lekhan*) and channels its excretion through sweat, urine, stool, etc. They also destroy coughing (e.g., ginger [*Zingiber officinalis* Roscoe] and *pipali* [*Piper longum*]). Astringent tastes are dry, cool, and heavy; they basically suppress *pitta* but also act on *kapha*.

The property of a substance, called *guna*, is equally important when selecting it as a medicine. In obesity, substances having light, dry, and rough properties are used. It also should have minute and sharp properties. Substances that are light and easy to digest relieve *kapha* and increase *vata* and digestive power (*agni*). They also cause depletion of tissues in the body, scrape out (*lekhan*) excessive fat, and clear the channels of the body for waste. Together, this brings lightness in the body.⁴⁴

Dry substances absorb water and create hardness and dryness. They aggravate *vata* and reduce *kapha*.⁴⁵ The rough property of substances increases *vata*, depletes the constituents of the body, and absorbs water. Substances having these three *gunas* will reduce *kapha* and increase *vatta* in obese patients; combined action of these properties on fat will be through scraping (*lekhan*), drying, and depleting through various channels in the form of waste. Sharp (*tikshna*) substances act as sharp-edged weapons and cause depletion of tissues and thereby emaciation of the body — this process is called *lekhana*.⁴⁶ The hot quality and bitter taste of a substance has light, dry, and sharp and subtle (*sukshma*) properties that act on obesity as described above.

9.16 Formulations Used and Cited (*Aushadhi-Yogas*)

Various formulations containing three to eight medicinal plants in the form of a decoction (*quath*), powder (*churna*), and tablet (*vati*) are widely used based on its description in the ancient text and are given in [Table 9.6](#).^{47–55}

9.16.1 Single Plants Used for Obesity

Although Ayurveda commonly uses multiple plants in the formulations, with varying ingredients and concentrations, single plants have also been mentioned in the text. [Table 9.7](#) lists many plants and their parts used either individually or in combination or with a follow-through vehicle (*anupan*). Many of these *tikta/katu* in *rasa* are light, dry, and rapidly acting (*tikshna*) in *guna*, hot in *veerya*, and usually pungent in *vipaka*. Most of these are antiobesity (*medohara*) and pacifying *vata* and *kapha* *dosas*. The groups of scraping, digestive (*pachaniya*), and appetite stimulating (*deepaniya*) dominate.

TABLE 9.6

Formulations Used for Obesity

Formulations	Plants	Botanical Name	Parts Used
<i>Triphaladiquath</i>	<i>Haritaki</i>	<i>Terminalia chebula</i> Retz.	Fruits
	<i>Bibhitaki</i>	<i>Terminalia belerica</i> Roxb.	Fruits
	<i>Amla</i>	<i>Emblica officinalis</i> Gaertn.	Fruits
<i>Bilvadhiquath</i>	<i>Bilva</i>	<i>Aegle marmelos</i> Corr.	Roots
	<i>Adulsa/Vasa</i>	<i>Adhatoda vasica</i> Nees.	Roots
	<i>Kashmiri</i>	<i>Gmelina arborea</i> Linn.	Roots
	<i>Patala</i>	<i>Stereospermum suaveolens</i>	Roots
<i>Trayushanadya-churna</i>	<i>Pippali</i>	<i>Piper longum</i> Linn.	Fruits
	<i>Shunthhi</i>	<i>Zingiber officinale</i> Roscoe	Rhizome
	<i>Marich</i>	<i>Piper nigrum</i> Linn.	Seeds
<i>Vidangadi-churna</i>	<i>Vidanga</i>	<i>Embelia ribes</i> Brum. F.	Seeds
	<i>Shunthhi</i>	<i>Zingiber officinale</i> Roscoe	Rhizome
	<i>Yavamal/Yavakshar</i>	<i>Alhaji mourorum</i>	Seed
	<i>Kantloh Bhasma</i>	Iron formulation	—
	<i>Amla</i>	<i>Emblica officinalis</i> Gaertn.	Fruits
<i>Amrutadiguggulu</i>	<i>Guduchi</i>	<i>Tinospora cordifolia</i>	Stem
	<i>Chota Eliachi</i>	<i>Elettaria cardamomum</i>	Seeds
	<i>Vidanga</i>	<i>Embelia ribes</i> Burm.	Seeds
	<i>Indrajav</i>	<i>Holarrhena antidysentrica</i>	Roots
	<i>Harali</i>		Fruits
	<i>Behada</i>	<i>Terminalia belerica</i>	Fruits
	<i>Amala</i>	<i>Emblica officinalis</i> Gaertn.	Fruits
	<i>Guggulu</i>	<i>Commiphora wightii</i>	Gum

9.17 The Future of Ayurvedic Modalities for Obesity

Certain Ayurvedic modalities bear close resemblance to several nondrug approaches of modern medicines. These modalities can be judiciously combined for individualizing prevention and therapy of obesity and are listed below:

1. Ayurvedic modes of fasting and dietary regulations over long-term period⁵⁶
2. Yoga, *suryanamaskara*, and exercise integrated in daily activities⁵⁷
3. *Shodhana* complementary to obesity programs by *vaidya* experts for induced vomiting, laxation, etc., as per Ayurvedic fundamental principles⁵⁸
4. Advocating frequent sexual intercourse, keeping awake, experiencing anxiety and concern for dear ones and social issues⁵⁹
5. Adopting an inner mind-set change for a new body image

Commonly used drugs and herbs can be adopted by non-Ayurvedic family doctors and obesity specialists who are willing to learn from transcultural healing traditions. These would include the following:

TABLE 9.7

Medicinal Plants in Ayurveda for Obesity

Name	Parts	Dosage per Day	Anupan (With)	Unique Activity	Major Formulation	Ref.
<i>Commiphora wightii</i> (guggulu)	Gum	2–4 g three times/day	Hot water	Rejuvenative	Medohar-guggulu	47, 48
<i>Terminalia chelula</i> (haritaki)	Fruits	1–2 g two times/day	Honey	Memory enhancer	Triphala	49, 50
<i>Tereminalia belerica</i> (bibhitaka)	Fruits	0.75–1.5 g two times/day	Warm water	Sedative	Triphala	51
<i>Emblica officinalis</i> (amalaki)	Fruits	3–6 g three times/day	Cow's milk	Antiaging	Chyavanprash	52
<i>Picrorrhiza kurroa</i> (kutki)	Roots	0.5–1 g two times/day	Honey	Hydrocholeretic	Arogyawardhini	53
<i>Curcuma longa</i> (haridra)	Rhizome	2–4 g three times/day	Warm water	Hypolipidemic	Churna	54
<i>Plantago ovata</i> (isabgol)	Husk	5–10 g one time/day	Cow's milk	Antithirst	Sat-Isabgol	55

1. *Trikatu* — *Piper nigrum*, *Piper longum*, and *Zingiber officinale* with meals⁶⁰
2. *Triphala* — Powder of the fruits of *Terminalia chibula*, *Terminlia berberica*, and *Emblica officinale*⁶¹
3. Common widely used Indian guggulu (*Commiphora wightii*) preparations in an initial large dose and as subsequent maintenance doses offer 10 to 20% deductions in body weight⁶²

In the near future, other single plants and formulations can be investigated by fast-track reverse pharmacology — experiential, exploratory, and experimental research by global collaborative herbal research efforts.

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10

Rheumatoid Arthritis, Osteoarthritis, and Gout

Lakshmi Chandra Mishra

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10.1 Introduction

There are three basic types of musculoskeletal joint diseases described in Ayurveda, namely, *amavata*, *sandhivata*, and *vatarakta*, that are characterized by pain and swelling of the joints. In modern medicine, the diseases closely resembling *amavata*, *sandhigat vata*, and *vata rakta* are rheumatoid arthritis (RA), osteoarthritis (OA), and gouty arthritis, respectively. RA is described first followed by brief descriptions of OA, gout, and other *vataja* musculoskeletal diseases.

10.2 Rheumatoid Arthritis

Amavata (RA) is described in Ayurveda as a constitutional disorder with clinical manifestations of joint swelling, pain, and stiffness in the ankle, knee, hip joints, wrist, elbow, and shoulder. The incidence of RA ranges from 0.3 to 1.5% in most populations of the world and the rate is two to three times higher in females than in males. The peak incidence of onset of RA is in persons 30 to 60 years old, but no age is immune. The severity of RA may range from mild oligoarticular illness of a brief duration and very little or no joint damage to polyarthritis with marked functional impairment.

RA is commonly treated with nonsteroidal anti-inflammatory drugs (NSAIDs). These drugs do not modify disease progression.¹ They have a tendency to cause adverse gastrointestinal effects that may range from mild dyspepsia and heartburn to ulceration of the stomach and duodenum, and many produce fatal consequences.^{2,3} Despite the popularity of NSAIDs, their use to treat OA has been controversial.⁴ In one survey, 27% of the patients suffering from arthritis in the U.S. had used complementary alternative medicine therapies (CAM).⁵ In a recent survey in India, 43% had used CAM therapies.⁶ Disease-modifying therapies, such as methotrexate, leflunomide, sulfasalazine, gold salts, penicillamine, azathioprine, cyclophosphamide, and chlorambucil, are used to modify the course of the disease but have serious toxic effects. These toxic effects include aplastic anemia (potentially life threatening), liver toxicity, gastrointestinal toxicity, leukopenia, and skin rashes.

The mechanism of action of NSAIDs is believed to be via inhibition of cyclooxygenase (COX) enzyme that is now known to exist in two separate isoforms. The COX-1 isoform is responsible for maintaining prostaglandin synthesis in the gastric mucosa, platelets, and kidney and its inhibition results in toxic effects in the organs. The COX-2 isoform is responsible for prostaglandin production in inflamed tissues, including RA synovium,

and its inhibition results in beneficial effect in the effected tissues. Recently developed NSAIDs seem to offer some selective inhibition of COX-2 isoform resulting in less toxicity.⁷ Corticosteroids are given to patients who are not responsive to NSAIDs, but corticosteroids may produce serious side effects, particularly after prolonged use. Dozens of therapies for RA have been used in Ayurvedic medicine for thousands of years. No serious side effect has been reported even after prolonged use of these therapies. For these reasons, interest in Ayurvedic management of RA is steadily growing around the world.

10.2.1 Pathogenesis and Etiology (*Samprapti* and *Vyadhi Haitu*)

Pathogenesis of RA in Ayurveda is considered to be due to the formation of *ama* (metabolic toxic waste materials) in the intestine caused by poor digestive power. *Ama* physically resembles *kapha*, thus it tends to deposit in *kapha* locations, primarily the joints. When *ama* is vitiated (function impaired) by *vata*, *amavata* disease is produced.

The etiological factors in the pathogenesis of *amavata* are incompatible diet, poor digestion, and sedentary habits.⁸ Poor digestion due to weak digestive power leads to the formation of *ama* in the intestine and is absorbed and distributed to all parts of the body. When *ama* is vitiated by the other three *dosas*, all types of diseases develop in the body.

An attempt has been made to determine if a relationship exists between the incidence of RA and other factors such as body constitution, lifestyle, and dietary habits of arthritis patients.⁸ Patients positive for RA factor were of *vata-pittaja* body constitution. They had poor digestive power, a tendency to eat an incompatible diet, a high level of erythrocyte sedimentation rate (ESR), and low hemoglobin levels. This group has more patients with sedentary habits than the patients with negative RA factor. Patients with a negative RA factor were found to be of *vata-kaphaja* constitution. They had better digestive power, eating habits, a more active lifestyle, lower levels of ESR, and higher levels of hemoglobin than did the RA factor positive patients. The authors suggested that Ayurvedic pathology of RA based on body constitution seems to be of two types: *vata-pittaja* (RA positive) and *vata-kaphaja* (RA negative). They also concluded that RA-negative patients may have a better prognosis than do RA-positive patients. This is because immunity may be better developed by *rasayanas* in the RA-negative patients.

It is currently believed in conventional medicine that genetic susceptibility, primary exogenous arthritogen, autoimmune reactions in joint components, and mediators of the joint damage are the factors involved in pathogenesis of RA.⁹ RA is believed to be an autoimmune disease. It is likely that *ama* contains some arthritogenic autoantigens that trigger the autoimmune reaction in a susceptible host, causing the development of RA. Although rheumatoid factors are found in 80% of the RA patients, these factors are not known to cause the disease.¹ Rheumatoid factors are also found to occur in other diseases that are characterized by chronic antigenic stimulation. These include bacterial endocarditis, tuberculosis, syphilis, *kala-azar*, viral infections, intravenous drug abuse, and liver cirrhosis.

10.2.2 Clinical Description (*Rog Vivaran*)

In Ayurvedic texts, the description of signs and symptoms of RA (*amavata*) includes pain (scorpionlike bite) and inflammation of one or more joints, particularly the hand, foot, thigh, sacrum, knees, and metatarsophalangeal (MTP). Joints develop swelling, tenderness, and reddish or bluish discoloration.

The characteristic feature of RA in conventional medicine is the persistent inflammation in the synovial membranes, usually of peripheral joints. Patients show loss of appetite, indigestion, constipation, occasionally low fever, malaise, fatigue, pain in different parts of the body, feeling of heaviness in the body, sleeplessness, and stiffness in the chest area. The seven criteria formulated by American Rheumatology Association¹⁰ and the American College of Rheumatology¹ to classify RA are summarized as follows:

1. Morning stiffness in and around the joints for an hour or more, continuous for 4 weeks or more; swelling of soft tissue of three or more joints, continuous for 4 weeks or more
2. Swelling of soft tissue of three or more joints
3. Swelling of soft tissue of hand joints (proximal interphalangeal [PIP], metacarpophalangeal or [MCP], or wrist)
4. Symmetrical swelling of soft tissue
5. Subcutaneous nodules
6. Serum rheumatoid factor
7. Erosions or periarticular osteopenia in hand or wrist joints seen on radiograph

Criteria 1 to 4 must continue for 4 weeks or more, and criteria 2 to 5 must be observed by a physician.

The prominent clinical features of RA include swelling of the PIP; MCP in the hands; loss of motion in the wrists; pain in the thumb; synovial proliferation and effusion in the knees; painful deformities of the toes in the feet; and ankle or tarsal collapse, resulting in painful valgus deformities.

10.2.3 Clinical Examination and Diagnosis (*Nidan*)

The standard eight-point diagnosis of Ayurveda is used to diagnose RA and determine which *dosas* are vitiated as discussed in [Chapter 2](#). Additionally, patients are examined for the classical signs of arthritis as discussed above. There is no specific laboratory diagnostic test for RA. Although only 80% of the patients are found to be positive for RA factor, the standard eight-point diagnosis is still one of the tests often used by physicians for diagnosis.

10.2.4 Clinical Course and Prognosis (*Sadhyata*)

RA is often considered a benign disease, but it may cause considerable disability, crippling, and death. Patients sometimes recover spontaneously and achieve complete remission. In most cases, the disease becomes chronic, resulting in functional deterioration of the joints and disability.

10.2.5 Strategy for Prevention

As previously mentioned, the major causative factors in the pathogenesis of RA are weak digestive power, leading to accumulation of *ama*, and a sedentary lifestyle. The strategy to prevent RA needs to include herbal formulas that improve digestion and clean the intestine from *ama* and a healthy active lifestyle as discussed in Ayurveda under daily

TABLE 10.1

Common Anti-Inflammatory Herbs Used in Ayurvedic Formulas

Sanskrit and Botanical Names
<i>Ashwagandha</i> (<i>Withania somnifera</i>)
<i>Bhallataka</i> (<i>Semecarpus anacardium</i>)
<i>Bhutala</i> (<i>Circuligo orchoides</i>)
<i>Chitraka</i> (<i>Plumbago zeylanica</i>)
<i>Eranda</i> (<i>Ricinus communis</i>)
<i>Goraksa</i> (<i>Vitex negundo</i>)
<i>Guggulu</i> (<i>Commiphora mukul</i>)
<i>Haritika</i> (<i>Terminalia chebula</i>)
<i>Harsinger</i> (<i>Nyctanthes arbor tristis</i>)
<i>Karanja</i> (<i>Pongamia glabra</i>)
<i>Kirayat</i> (<i>Andrographis paniculata</i>)
<i>Kupilu</i> (<i>Strychnos nux-vomica</i>)
<i>Madhuka</i> (<i>Basia latifolia</i>)
<i>Madhuyasti</i> (<i>Glycyrrhiza glabra</i>)
<i>Punarnava</i> (<i>Boerrhavia diffusa</i>)
<i>Rasna</i> (<i>Inula racemosa</i>)
<i>Rasona</i> (<i>Allium sativum</i>)
<i>Sunthi</i> (<i>Zinziber officinalis</i>)
<i>Vatsanabha</i> (<i>Aconitum chasmenthum</i>)
<i>Vidanga</i> (<i>Embelia ribes</i>)
<i>Yavasaka</i> (<i>Alhagi pseudalhagi</i>)

routine. One must eat easily digestible food according to the body constitution and make sure that bowel movement is regular. Regular exercise is very important to prevent RA and arthritis.

10.2.6 Therapy (*Chikitsa*)

The basic strategy of treating RA is to reduce the accumulated *ama* in the body and balance *vata*, encourage proper exercise, maintain a normal healthy lifestyle, and provide adequate rest. First, the food intake is reduced to a bare minimum, giving only boiled rice and well cooked *mung dal* (lentil) soup for 1 to 2 months to help clean the body of *ama*. This diet is believed to increase metabolism and elimination of disease-producing *ama*. Herbs or herbal formulas (e.g., *Panchkola*, 2 g/day; equal parts of *Piper longum* root and fruits, *Piper chava*, *Plumbago zeylanica*, and *Zinziber officinale* powders) are given during this period to clean the body, improve the digestion, and reduce the inflammation (*Commiphora mukul*-containing formulas) and pain.¹¹

Suitable herbal formulas are determined based on body constitution, disturbed *dosa*, and other disease conditions. For example, *Gokksharadi guggul* is recommended for patients who also suffer from kidney disorders, *Pushkarmoola guggul* is suggested for patients with heart problems, and *Kanchinar guggul* is recommended for patients with lymphadenopathy.¹¹ Some of the commercially available commonly used herbs and formulas are presented in Table 10.1 through Table 10.3. The patient is slowly brought back to a normal diet consisting of easily digestible light food, is given ghee (dehydrated butter), and may be asked to use a medicated sesame-oil enema to remove fat soluble toxic materials from the gastrointestinal tract. The patient is asked to do easy exercises and yoga. Body massage with oil, and hot fomentation may also be added to the treatment. Classic Panchakarma

TABLE 10.2

Formulas from Ayurvedic Formulary (AF) of India for Arthritis Treatment (1978)

Name	Dose	AF Section	Name	Dose	AF Section
<i>Kanchnar guggul</i>	3 g	5:1	<i>Rasa parpati</i>	250 mg	16:3
<i>Goksharidi guggul</i>	3 g	5:3	<i>Anand Bhaijarava rasa</i>	250 mg	20:3
<i>Trayodashang guggul</i>	3 g	5:4	<i>Mahalakshmi vilas rasa</i>	250 mg	20:27
<i>Yograj guggul</i>	3 g	5:7	<i>Sarvanabhupati rasa</i>	250 mg	20:51
<i>Vyousadi guggul</i>	3 g	5:9	<i>Rasnadi kwath curpa</i>	48 g	4:27
<i>Vatari guggul</i>	3 g	5:10	<i>Rasnadi kwath curpa maha</i>	48 g	4:28
<i>Simhanad guggul</i>	3 g	5:12	<i>Das-mula-haritiki</i>	12 g	3:14
<i>Ajamodadi curpa</i>	6 g	7:1	<i>Amrta ghrit</i>	12 g	6:1
<i>Nimbadi curpa</i>	3 g	7:20	<i>Brhat saindhavadya taila</i>	massage	8:40
<i>Pancasma curpa</i>	3 g	7:22	<i>Kotamuckkadi taila</i>	massage	8:10
<i>Vaisvanara curpa</i>	3 g	7:30	<i>Jiraka-modka</i>	3 g	3:12

TABLE 10.3

Miscellaneous Ayurvedic Formulas for the Treatment of RA

Name	Reference	Name	Reference
<i>Brhat vata cintamani</i>	BR ^a	<i>Maha rasnadi kwath</i>	SS ^f
<i>Chandrprabha vati</i>	SYS ^b	<i>Sunthi guduci kwath</i>	13 ^d
<i>Copper bhasma</i>	RS ^c	<i>Puskarmoola guggul</i>	Shukla 1996
<i>Gudici kwatha</i>	20 ^d	<i>Suvarna samirapannag rasa</i>	BR ^a
<i>Gold bhasma</i>	SYS ^b	<i>Rasnadi guggul</i>	YR ^g
<i>Mrtyunjaya rasa</i>	BR ^a	<i>Trailokya cintamani</i>	BR ^a
<i>Maha Narayan taila</i>	CD ^e	<i>Vatari guggul</i>	BR ^a
		<i>Vatgajankush rasa</i>	BR ^a

^aBR = Bhesajya Ratnavali.

^bSYS = Sidha Yog Sangraha.

^cRS = Rasayan Sar.

^dSee [References](#) at end of chapter.

^eCD = Chakardatta.

^fSS = Sarangdhar Samhita.

^gYR = Yog Ratnakar.

therapy described in [Chapter 5](#) was evaluated in RA patients and was found to give encouraging results.¹²

Clinical trials based on the above basic strategy of treatment using a variety of Ayurvedic herbal formulas are summarized below.

10.2.6.1 *Zingiber officinale*–*Tinospora cordifolia* Decoction, *Vatagajanankusa Rasa*, and *Maharasnadi Kwath*

Z. officinalis–*T. cordifolia* decoction was given (50 ml three times/day) to 40 RA patients (group B) and *Yograj guggul* (1 g/day), *Vatagajanankusa rasa* (250 mg three times/day), and *Maharasnadi kwath* (50 ml three times/day) were administered to 37 arthritis patients (group A) for 6 to 8 weeks. There were a total of 48 males and 29 females. The diagnosis was confirmed based on 1959 American Rheumatism Association (ARA) criteria. All patients received a limited vegetarian diet without fish or chicken. Both treatments were

effective in reducing pain and swelling, and the decoction treatment was found to be relatively more effective than the *Yograj guggul* combination.¹³

10.2.6.2 *Dasmularista, Pippalyasava, and Vettumaran Gutika*

Dasmularista, Pippalyasava, and Vettumaran Gutika were examined in 83 arthritis patients (1959 ARA criteria). Forty-three percent of the patients were unable to walk, and remaining cases showed varying degree of functional impairment of the joints. Measurements of swollen elbow, wrist, ankle, and knee joints were taken before and after treatment. An average of 90 days of treatment was found to be the minimum to obtain satisfactory improvements. Time taken for disappearance of RA ranged from 20 to 30 days. The results suggested that the combination of the above three formulas was effective in the acute stage of RA.¹⁴

10.2.6.3 *Rasonadi Kvatha*¹⁵

Rasonadi Kvatha was evaluated in 50 RA patients. The diagnosis was confirmed based on ARA 1959 criteria.¹⁶ *Kvatha* is a decoction of equal parts of *Z. officinalis*, *Allium sativum*, and *Vitex negundo*. A dose of 25 ml of the decoction representing 25 g of each ingredient was given three times/day for 6 weeks. During the acute inflammatory stage of the disease, patients were treated with external application of poultice and *Baluka* hot sand fomentation. The participants' diet consisted of rice, bread, lentil soup, cream of wheat, and milk during the 6-week trial period. *Pippalyadi churna* or *Visatundika vati* was also administered during the acute phase to relieve pain. The results were assessed based on the degree of pain, swelling, tenderness, and restriction of affected joints. The change in functional capacity was measured by functional tests. Authors observed significant improvement in all the parameters. Most of the patients showed complete relief from pain.¹⁶

10.2.6.4 *Withania somnifera, Boswellia serrata, Curcuma longa, and a Zinc Complex (Articulin-F)*

A herbomineral formulation containing roots of *W. somnifera*, the stem of *B. serrata*, rhizomes of *C. longa*, and a zinc complex (Articulin-F) was evaluated in a randomized, double-blind, placebo-controlled, crossover study in 42 patients with OA. Clinical efficacy was evaluated every fortnight on the basis of severity of pain, morning stiffness, Ritchie, articular index, joint score, disability score, and grip strength. Other parameters like erythrocyte sedimentation rate and radiological examination were carried out on a monthly basis. Treatment with the herbomineral formulation produced a significant drop in the severity of pain ($p < 0.001$) and disability score ($p < 0.05$). Radiological assessment did not show any significant changes and no serious adverse effect was observed.¹⁷

10.2.6.5 *Ginger (Zingiber officinale)*

Ginger has been reported to be beneficial in seven arthritic patients.¹⁸ In one study, ginger was given to 56 patients (28 with RA, 18 with OA, and 10 with muscular discomfort). More than 75% of the patients experienced varying degrees of relief from pain and swelling. All the patients with muscular discomfort experienced relief from pain. None of the patients reported adverse effects during the treatment period from 3 months to 2.5 years. The authors suggested that at least one of the mechanisms by which ginger shows its ameliorative effects could be related to inhibition of prostaglandin and leukotriene biosynthesis (dual inhibitor of eicosanoid biosynthesis).¹⁹

10.2.6.6 *Tinospora cordifolia*, *Balsamodendrom mukul*, and *Alpinia officinarum*

Tinospora cordifolia, *Balsamodendrom mukul*, and *Alpinia officinarum* were evaluated in 40 RA patients selected based on 1959 ARA criteria.²⁰ Forty patients (ages 10 to 65) were selected for the study on the basis of pain, swelling of multiple joints, and elevation of erythrocyte sedimentation rate. All patients were treated for the first 10 days as follows:

Day 1 — Fasting

Days 2–6 — *Panchkola* (2 g/day) to improve digestion and clean *ama* for days 2 to 6

Day 7 — *Ricinus communis* oil (30 ml) at 6 A.M. with hot water

Days 8–10 — *Panchkola* powder (2 g/day) with hot water

On day 11, the patients were separated randomly into 2 groups of 20 each. The first group was given 60 ml of *T. cordifolia* decoction, 1 ml of oil extract (*sneha*) of *T. cordifolia*, and 2 g of *B. mukul* three times/day. The second group received the same regimen except *B. mukul* was replaced by 2 g of *A. officinarum*.

The decoction consisted of one part of leaves and stem of the herb, boiled in four parts of water until one part of the water was left. The oil extract fortified three times consisted of 1 part herbal paste, 4 parts of sesame seed oil, and 16 parts of the decoction, boiled 6 to 8 h/day for 3 days to reduce the volume to one fourth of the total amount. It was filtered and the procedure was repeated three times.

A light diet was given in the first stage of 10 days and a normal hospital diet was given for rest of the treatment period. Criteria for assessment and classification of results followed were those adopted by American Medical Association (AMA). The authors stated that the improvement in signs and symptoms were highly encouraging and statistically significant at 0.1% level.

10.2.6.7 Fasting

Because *ama* is considered a major causal factor for RA, elimination of *ama* by fasting was studied in RA patients (1959 ARA criteria). The patients were subjected to total fasting for 2 days and partial fasting for the next 43 days. Notable clinical remission of signs and symptoms were observed, including a decrease in rheumatoid factor (RF) titer and chemically reactive proteins (CRP), which may be attributed to a decrease in *ama*.²¹

10.2.6.8 Elimination of Ama by a Purgative (*Gandharva hasta kashaya*)

The effect of eliminating *ama* by purgatives was evaluated in 60 arthritic patients. *Gandharva hastadi kvath* (50 ml), a purgative, was given three to four times/day for 21 days. The ingredients of the decoction were equal parts of *Gandharva hasta* (*Ricinus communis*), *Karavya* (*Pongamia glabra*), *Chitraka* (*Plumbago zeylanica*), *Shunti* (*Zinziber officinalis*), *Haritaka* (*Terminalia chebula*), *Punarnava* (*Boerrhavia diffusa*), *Yavasaka* (*Alhagi pseudalhagi*), and *Bhutala* (*Circuligo orchioides*). Of 60 patients, 46 were cured (complete disappearance of disease symptoms) and 14 showed significant relief in symptoms. Three patients had a relapse over a period of 1 year after observation. The study further confirms the hypothesis that the origin of arthritis is the formation of *ama* in the intestine.²²

10.2.6.9 *Vatari Guggul* (*Bhasajya Ratnavali*) and *Maharasnadi Kvatha*

A dose of 1 g of *Vatari guggul* was given three times/day and *Maharasnadi kvatha* (*Sarangdhar samhita*) was given 20 ml three times/day in 24 RA patients. *Vatari guggul*

consisted of castor oil, sulfur, guggul (*Commiphora mukul*), haritiki (*Terminalia chebula*), amalki (*Phyllanthus emblica*), and vibhitiki (*Terminalia belerica*). The major ingredient of Maharasnadi kvath was rasna (*Inula recemosa*). The results indicate a significant improvement in pain ($p < 0.001$) and a decrease in ESR levels ($p < 0.001$) in subjective and objective parameters ($p < 0.001$).²³

10.2.6.10 Ashwagandha (*Withania somnifera*)

Ashwagandha (*Withania somnifera*) has been shown to have antiarthritic activity in animal models of RA.^{24–28} In a recent clinical trial, 3 g of the drug was given three times/day with milk to 77 RA patients. The improvement in signs and symptoms of RA was good in 22.7%, moderate in 53.4%, and poor in 22.07%; there was no response in 2.59% of the patients.²⁹

10.2.6.11 Purified Guggul and Guduchyadi Kvath

Purified guggul and Guduchyadi kvath were given to 34 RA patients for 6 weeks. Authors observed marked improvement in signs and symptoms in 27 patients, moderate improvement in 4 patients, and no improvement in 3 patients.

10.2.7 Scientific Basis for the Use of Ayurvedic Herbs in Arthritis

Ayurvedic herbs have been investigated in animal models of arthritis for anti-inflammatory effect in the same way as synthetic drugs have been for the past 30 years. The three most common animal models currently used to investigate anti-inflammatory effect of drugs are (1) adjuvant-induced, (2) streptococcal cell-wall induced, and (3) collagen-induced models.³⁰ Agents currently in clinical use or trials that are active in these models include corticosteroids, methotrexate, nonsteroidal anti-inflammatory drugs, cyclosporin A, leflunomide, interleukin-1 receptor antagonist, and soluble tumor necrosis factor receptors. Animal models of OA include mouse and guinea pig spontaneous OA, meniscotomy and ligament transection in guinea pigs, meniscotomy in rabbits, and meniscotomy and cruciate transection in dogs. None of these models have a proven track record of predictability in human disease presentation because none of the agents have been proven to provide anything other than symptomatic relief.³¹

The studies to assess anti-inflammatory activity and analgesic activity are evaluated, and the results supporting the use of Ayurvedic therapies in arthritis are summarized below.

10.2.7.1 Anti-Inflammatory Activity

10.2.7.1.1 *Vanda roxburghii*

Vanda roxburghii has been shown to have anti-inflammatory activity in albino rats.³²

10.2.7.1.2 Gum-Guggul from *Commiphora mukul*

Anti-inflammatory activity of guggul has been reported by several investigators.^{33–35} In another study, phenylbutazone, ibuprofen, and fraction "A" of gum-guggul from *Commiphora mukul* were administered orally at a daily dose of 100, 100, and 500 mg/kg, respectively, for a period of 5 months. All three drugs decreased the thickness of joint swelling during the course of the drug treatment.³⁶

The anti-inflammatory effect of extracts of *guggul* resins of four species of the Burseraceae plant family, *Boswellia dalzielii*, *Boswellia carteri* (gum olibanum), *Commiphora mukul*, and *Commiphora incisa*, were screened in arthritis models in rat. The aqueous extracts of the resins of *B. dalzielii*, *C. incisa*, and *C. mukul* significantly inhibited both the maximal edema response and the total edema response during 6 h of carrageenan-induced rat paw edema. The octanordammarane triterpenes, mansumbinone and mansumbinoic acid, were isolated from the resin of *C. incisa*. Prophylactical administration of mansumbinone proved to be more than 20 times less potent than indomethacin and prednisolone in inhibiting carrageenan-induced rat paw edema. The acid was able to reverse an established carrageenan-induced inflammatory response when administered 2 h after induction. Daily administration of mansumbinoic acid at 1.5×10^{-4} mol kg⁻¹ dose level significantly reduced joint swelling in adjuvant-induced arthritic rats.³⁷

New triterpenes, Myrrhanol A and Myrrhanone A, isolated from *guggul*-gum resins (*C. mukul*), were found to have a potent anti-inflammatory effect on adjuvant-induced air-pouch granuloma of mice. The study indicated that the anti-inflammatory effect of Myrrhanol A was greater than that produced by hydrocortisone and the 50% extract of the crude resin. Authors suggested that Myrrhanol A may be a potent anti-inflammatory agent.³⁸

10.2.7.1.3 *Harsingar (Nyctanthes arbor tristis)*

The water-soluble portion of the alcoholic extract of the leaves of *Nyctanthes arbor tristis* (NAT) was screened for the presence of anti-inflammatory activity. NAT inhibited acute inflammatory edema produced by different phlogistic agents, such as carrageenin, formalin, histamine, 5-hydroxytryptamine, turpentine oil, and hyaluronidase, in the hindpaw of rats. It also inhibited the inflammation in adjuvant-induced arthritic models. In subacute models, NAT was found to check granulation tissue formation significantly in the granuloma pouch and cotton-pellet test. Acute and chronic phases of formaldehyde-induced arthritis were significantly inhibited. Anti-inflammatory activity in leaves of NAT supports its use in various inflammatory conditions by the followers of the Ayurvedic system of medicine.³⁹

10.2.7.1.4 *Semecarpus anacardium nut*

A chloroform extract of *Semecarpus anacardium* nut was found to significantly reduce acute inflammation and was also active against the secondary lesions of adjuvant-induced arthritis in rats.⁴⁰

A milk extract of *S. anacardium* nut was found to have anti-inflammatory effect in adjuvant-induced arthritis in rats at the dose level of 150 mg/kg.⁴¹ The mechanistic studies indicated that the diseased state of adjuvant arthritis may be associated with augmented lipid peroxidation; the studies also concluded that the administration of the drug may exert its antiarthritic effect by retarding lipid peroxidation and causing a modulation in a cellular antioxidant (AO) defense system.⁴²

10.2.7.1.5 *Crataeva nurvala*

The effect of triterpenes from stem bark was studied for its effect on lipid peroxidation in adjuvant induced arthritis in rats. Lupeol, a pentacyclic triterpene of *C. nurvala* stem bark, and its ester, lupeol linoleate, were synthesized and tested for anti-inflammatory activity in complete Freund's adjuvant-induced arthritic rats. The arthritic rats showed a significant increase in lipid peroxide level in plasma; a decrease of lipid peroxide in the liver; an increase in the AO enzymes, SOD, GPX, and catalase in both the liver and hemolysate; and a decrease in blood glutathione. Lupeol and lupeol linoleate at 50 mg kg⁻¹ body weight daily for 8 days, from 11 to 18 days after the adjuvant injection brought back the alterations

to normal levels. The effect of lupeol linoleate was found to be better in this respect when compared with lupeol.⁴³

10.2.7.1.6 *Hemidesmus indicus* (HI)

The anti-inflammatory activity of the pure compound (2-hydroxy-4-methoxy benzoic acid) isolated and purified from anantamul root extract was investigated. 2-OH-4-MeO benzoic acid effectively neutralized inflammation induced by *Vipera russelli* venom in male albino mice and reduced cotton-pellet-induced granuloma in rats. The compound produced a significant fall in body temperature in yeast-induced pyrexia in rats but did not change the normothermic body temperature. The compound effectively neutralized viper-venom-induced changes in serum phosphatase and transaminase activity in male albino rats. It also neutralized free radical formation as estimated by thiobaraturic acid reactive substances (TBARS) and SOD activities.⁴⁴

10.2.7.1.7 *Andrographis paniculata*

Andrographolide, a diterpenoid lactone isolated from *Andrographis paniculata* (anti-inflammatory herb), was investigated for the ability to prevent phorbol-12-myristate-13-acetate (PMA)-induced reactive oxygen species (ROS) production, as well as *N*-formyl-methionyl-leucyl-phenylalanine (fMLP)-induced adhesion by rat neutrophils. The study showed that PMA (100 ng/ml) induced rapid accumulation of H₂O₂ and O₂ in neutrophils within 30 min. Andrographolide (0.1 to 10 μ M) pretreatment (10 min, 37°C) significantly reduced the accumulation of these two oxygen-radical metabolites. Administration of andrographolide also significantly prevented fMLP-induced neutrophils adhesion. These data suggest that the mechanism of anti-inflammatory action of andrographolide may be via preventing ROS production and neutrophils adhesion.⁴⁵

10.2.7.1.8 *Aglaia roxburghiana* (AG)

Alcoholic extracts of aerial portions and fruits of AR and the triterpines, roxburghiadiol A and B, were studied in carrageenin-induced rat-paw edema, cotton-pellet granuloma, and mast-cell degranulation induced by compound 48/48. The extract and the triterpines were found to be effective anti-inflammatory agents in this study.⁴⁶

10.2.7.2 Analgesic Activity

There are few data available on the analgesic activity of antiarthritic herbs. A p-quinone, embeline, derived from *Embelia ribes* was investigated for analgesic activity. The potassium salt of embeline was found to be effective by oral, i.m., and i.c. routes of administration and the results compared well with morphine. Although potassium embelate acts centrally to produce analgesia, its effect is not antagonized by naloxone indicating a different central site of action. It has no abstinence syndrome as observed with morphine and no demonstrable anti-inflammatory action. The analgesic effect of the alkaloid provides a scientific basis for its use in arthritis. In addition, lack of any adverse effects, high therapeutic index, and absence of abstinence syndrome provide basis for long-term safety of embelate as an analgesic.⁴⁷

10.2.7.3 Mechanism of Action

One of the mechanisms of anti-inflammatory activity of Ayurvedic herbs may be the AO property. Andrographolide isolated from leaves of *Andrographis paniculata* has been shown to inhibit nitric oxide generation in endotoxin-stimulated macrophages.⁴⁸

The AO activity of two polyherbal formulations, *Maharasnadhi quatha* (MRQ) and *Weldehi choornaya* (WC), used by Ayurvedic medical practitioners in Srilanka for the treatment of RA patients, was assessed on SOD, GPX and catalase, lipid peroxidation (as estimated by TBARS generation), concentrations of serum iron, hemoglobin (Hb), and the total iron-binding capacity (TIBC). The overall results of the study demonstrated that MRQ has greater AO potential than WC.⁴⁹

In another study,⁵⁰ the mechanism of suppression of inducible nitric oxide synthase (iNOS) and (COX-2) by ergolide, a sesquiterpene lactone from *Inula Britannica*, was investigated. Ergolide markedly attenuated the iNOS activity in a cell-free extract of LPS/IFN-gamma-stimulated RAW 264.7 macrophages and markedly decreased the production of prostaglandin E(2) (PGE(2)) in cell-free extract of the macrophages in a concentration-dependent manner, without alteration of the catalytic activity of COX-2. Ergolide decreased the level of iNOS and COX-2 protein, and iNOS RNA caused by stimulation of LPS/IFN-gamma in a concentration-dependent manner and inhibited nuclear factor-kappaB (NF-kappaB) activation, a transcription factor necessary for iNOS and COX-2 expression in response to LPS/IFN-gamma. This effect was accompanied by the parallel reduction of nuclear translocation of the subunit p65 of NF-kappaB as well as IkappaB-alpha degradation. These effects were completely blocked by treatment with cysteine. This finding suggests that the inhibitory effect of ergolide may be mediated by alkylation of NF-kappaB or an upstream molecule of NF-kappaB. Ergolide also directly inhibited the DNA-binding activity of active NF-kappaB in LPS/IFN-gamma-pretreated RAW 264.7 macrophages. The authors suggested that the overall effect of ergolide is the suppression of the expression of iNOS and COX-2, which play important roles in the inflammatory-signaling pathway.

10.3 Osteoarthritis (*Sandhigat Vata*)

10.3.1 Introduction

Sandhigat Vata (*Krostuka-sirsa*) is primarily a disease of vitiated *vata* settling into bone and bone marrow without any involvement of *ama*. The disease is characterized by acute pain, stiffness of the joints, impairment of the function of the joints, loss of muscle strength, and sleeplessness. This disorder resembles OA, which is also characterized by acute pain, involvement of diarthroidial joints, and functional limitations. Sedentary lifestyle as a causative factor of *amavata* is scientifically valid.⁵¹ Current experiments in rabbits have shown that short repetitive immobilization induces OA. Even a 4-day immobilization period had a cumulative effect in producing OA.⁵²

10.3.2 Clinical Description

OA starts out with pain in one joint but subsequently spreads to other joints. The joints commonly involved are the big toe, fingers, hip, knees, and both lumbar and cervical spine. Ankles are generally spared except in the secondary form of OA. These joints in the advanced stage of the disease exhibit deformity. Patients do not show the presence of the RA factor.

10.3.3 Treatment

Because *ama* is not involved, the treatment of OA is simpler than RA. On the basis of the constitution and involvement of other *dosas*, herbs and herbal preparations that alleviate vitiated *vata* are selected to treat OA. Because muscle atrophy commonly accompanies OA, the treatment also involves an adequate exercise program. Both muscle strength and range of motion can be improved with appropriate physical exercise. Regular bench-press and leg-press exercises with a certain amount of weight may slow down or even stop the progress of OA in early stages and may even help in later stages in some patients. Adequate amount of rest is also an important adjunct and should be included in the daily routine of OA patients.

Basically, the drug treatments for OA and RA are very similar because both have a common cause, vitiated *vata*. Single herbs and formulas listed in Table 10.1 to Table 10.3 that are commonly used to treat RA are also applicable to treat OA. In addition, *Kishore guggul* from the Ayurvedic Formulary of India is also used for OA. Singh⁵³ reported a case study showing usefulness of *guggul* (*C. mukul*) in treating OA of the knee. In a larger study in 30 OA patients, *guggul* extract (500 mg, three times/day for 2 months) provided significant relief in signs and symptoms of OA.^{54,11}

In conventional medicine, the treatment of OA includes a proper exercise program, rest, NSAIDs, and newer NSAIDs (COX-2 inhibitors). Intra-articular injections of steroids and hyaluronan and joint replacement surgery are also used to treat OA. It is interesting to note that the NSAIDs and newer NSAIDs are the drugs also used to relieve pain and inflammation in RA.

10.4 Gout (*Rakta Vata*)

10.4.1 General Description

In Ayurveda, the origin of gout is believed to be in blood. When functionally impaired (vitiated) *vata* invades the blood and bone joints, it produces gout. It is characterized by acute excruciating pain (worse at night) and inflammation of the joints, which generally begins from a toe or finger and then gradually spreads to the other joints of the body. The affected joints are warmer than the body, swollen, extremely tender with a burning sensation, and have shiny overlying skin and dilated veins. The other symptoms are excessive sweating, rigidity, numbness, and discoloration of the skin on the affected joints. The joints return to normal after a few days with desquamation of the overlying skin. An acute attack may cause fever, anorexia, and general malaise.

In conventional medicine, gout is known as an inflammatory arthritis induced by the pathogenic deposition of aggregated monosodium urate monohydrate crystals (*tophi*), which deposit in various tissues and joints. Urate crystals can also deposit in kidneys. Chronic high levels of uric acid in the blood are necessary for the development of gout, although some other factors may be involved. This etiology points in the direction of the Ayurvedic etiology that indicates that blood is involved in this disease. If untreated, it can cause painful, destructive arthropathy and urolithiasis resulting in renal failure.

10.4.2 Treatment

Ayurvedic treatments include the following:

1. *Guduchi* (*T. cordifolia*) decoction of 14 to 28 ml made from 6 to 8 g of powder of leaves and stem to be ingested with purified *guggul* (2 to 4 g, three times/day)
2. Decoction of equal parts of *vasa* (*Adhatoda vasica*) leaves, stem of *guduchi*, and fruit pulp of *Aaragvadha* (*Cassia*) of 14 to 18 ml taken with castor oil (7 to 14 ml two times/day)
3. *Kishora guggul* and *Punarnavadi guggul* (3 g/day)
4. Additional single herbs and formulas used for RA (see [Table 10.1](#) to [Table 10.3](#))

To apply the treatment on the affected joints, a poultice of black sesame seeds is prepared in milk.

Guda (*Jaggary*), rice, wheat, gram, pigeon peas, spinach, black coffee, ripe fruit of white gourd, *Guduchi*, *Deodar* (*Cidrus deodar*) are useful articles of diet. The food articles to avoid are seeds of *Masa* (black phaseolus bean), *Kulattha* (dolichos beans), pea, legumes, radish, sugarcane products, wine, and yogurt. Because uric acid is the end product of protein metabolism, it is important to avoid high-protein diet. Sleeping during the day, exposure to heat, and excessive exercising should be avoided.

10.5 Spasmodic Torticollis, Lockjaw (*Hanu-Stambha*), Lumbago (*Kati Shula*), and Fibromyalgia

10.5.1 General Description

Vitiated *vata* is the cause for torticollis (*Manya-stambha*), lockjaw (*Hanu-stambha*), and lumbago (*kati shula*). When the vitiated *vata* invades the neck muscles it causes muscle rigidity and immobility of the neck and pain in the neck muscles. Similarly, when it invades jaw muscles and joints, it causes lockjaw. When it invades the lower part of the back it causes lumbago, and when it invades all body muscles it causes fibromyalgia (*Sarvang shula*). Pain and stiffness of the muscles are the main features of these diseases.

Specific etiology of these diseases is not known at this time. Ayurvedic etiology of these diseases refers to food and lifestyle habits that can aggravate *vata*.

10.5.2 Treatment

Because the causative *dosa* is vitiated *vata*, single herbs and formulas for the treatment of RA listed in [Table 10.1](#) to [Table 10.3](#) for RA are applicable. Additional formulas not in these tables but also used for these diseases are *Kishora guggul* and *Triphala guggul* from the Ayurvedic Formulary of India. Both formulas are given at a dose of 3 g/day. It is interesting that there are so many choices of treatments in Ayurveda. A physician needs to assess the body constitution, diagnose accompanying diseases, and determine the predominance of other *dosas* in order to select suitable herbal formulas and other components of the management regimen.

10.6 Future Research on Ayurvedic Herbal Formulas for Musculoskeletal Disorders

Because the efficacy of many Ayurvedic herbs has been shown in animal models, the next step is to investigate the mechanism of action so that effective combination of herbs can be formulated. It is likely that these herbs may have selective inhibitory activity on COX-2 because they have not been known to cause any toxicity in the stomach or intestine. Ergolide, a sesquiterpene lactone from *Inula britannica*, has been shown to inhibit the COX-2 activity.⁴⁹ *Semicarpous anacardium* has been shown to bring back the levels of altered AO defense system in adjuvant-induced arthritic rats.⁴¹

These and other biochemical mechanisms of action of these herbs should be further investigated.

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11

Ayurvedic Therapies of Sciatica (Gridhrasi)

Subhash Singh

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11.1 Introduction

Neuropathic pain due to injury or dysfunction of the nervous system is a difficult therapeutic challenge in the treatment of chronic pain. The pain of sciatica is typically a radiating pain that usually is related to a specific nerve root. Nerve dysfunction may be present in both motor and sensory modalities. The response of neuropathic pain to conventional analgesics has been in many cases disappointing. Several studies have suggested that spinal clonidine may be effective in treating some forms of neuropathic pain; some of the authors do not advocate such procedures because of its major complications. Thousands of years ago, Charaka, a great physician and Susruta (800 B.C.), the father of the surgery, identified the disease and named it *gridhrasi*. This recent research was aimed to establish the ancient ayurvedic therapeutic values among the people of modern era, evaluating fundamental principles by scientific parameters. In this chapter, the current knowledge about pathophysiologic mechanism, diagnosis, and management of sciatica is reviewed and discussed in relation to its prognosis.

11.2 Epidemiology

Lower back pain (LBP) is the most common presentation of lumbar spine disorders. Reports from various research institutes (Central Council for Research in Ayurveda and Siddha [CCRAS]), different modern class-three level hospitals, and referral centers show that 3 to 5% of the patients visiting annually in India present with LBP. Of them, 1.5% have features of sciatica, in which 62 to 69% are males and 31 to 38% are females. Sciatica affects mostly adult males, particularly those 41 to 60 years old. The ratio of incidence is similar in the people of low income as well as in higher income groups. It is more prevalent in the month from May to July (rainy season) in which *vata dosa* naturally gets vitiated and people become sick. People of *vataj prakriti* are more prone to the disease. Annual incidence of LBP in the U.S. is 2 to 5%, and annual cost of direct medical care for LBP has been estimated at \$13 to \$16 billion. Total annual societal costs are estimated at \$20 to \$50 billion.¹ Incidents are the same among heavy, light, and sedentary workers, although a higher proportion of heavy workers are incapacitated with LBP. Sciatic pain is common in people who either sit or stand for prolonged periods of time.

11.3 Definition

When tendons of the toes and heel become paralyzed due to vitiated *vata* and confine the extension of the leg, the disease is known as sciatica.^{2,3} The pain arises in the gluteal region, gradually spreads and radiates down the back of the thigh and knee, and travels up to the leg and foot. Perception of numbness, excruciating pain, needling sensation, restrained movement, and muscle spasm repeatedly in the limb are the main characteristics of the disease. Patients sometime feel lassitude, heaviness in the body, and may also have lack of appetite.⁴

11.4 Clinical Description

Sciatica is a disease of the nerves. The sciatic nerve is the largest nerve in the body and has a long course. It is derived from fourth and fifth lumbar (L₄ and L₅) and first and second sacral (S₁ and S₂) roots. It provides motor innervation of the hamstring muscles and to all muscles below the knee. It carries sensory impulses from the posterior aspect of the thigh and the posterior and lateral aspect of the leg and entire sole. Any pathological process that impinges upon at this level may cause pain associated with sciatica. The mechanism involved in the disease process is distortion; stretching; and irritation or compression of the spinal root (most often central to the intervertebral foramen), which causes tingling, paresthesias and numbness or sensory impairment of the skin, soreness of the skin, and tenderness along the nerve. It usually accompanies radicular pain, loss of reflexes, weakness, atrophy, fascicular twitching, and occasionally stasis edema if motor fibers of the anterior root are involved.

Ayurveda divides this disease into two categories: *vataj* and *vat-kaphaj*. *Vataj* disease is caused by the vitiation of *vata dosa* alone and contains a needling sensation. The body becomes crooked and the joints of the thigh, knee, and hip are rendered stiff.⁵ *Vata-kaphaj* disease is caused mainly by *vata* in company with *kapha dosa*. Numbness, excruciating pain, paresthesias, restrained movement, and repeated muscle spasms are distinguishing features of *vataj* sciatica. It is characterized by lassitude, heaviness in the body, and anorexia, along with associated symptoms of *vataj* sciatica. Patients suffer from dyspepsia and experience lassitude, anorexia, and excessive salivation.

11.5 Etiology (Vyadhi Haitu)

The potential causes of sciatica are myriad, but these account for only 15% of the total cases. The mechanisms of underlying neuropathic pain in 85% of the cases are still not clear. Sciatica is one of the many *vataj* diseases (*nanatmaj vyadhis*). *Vata dosa* is the main culprit in the disease. The nucleus pulposus of intervertebral disc is probably not pain sensitive under normal circumstances but may induce some changes in the nerve root.⁶

The causes of sciatica may be classified in to two main categories: pathology based and injury based. Both categories are discussed below.

11.5.1 Pathology Based

The common causes of sciatica are *vata*-aggravating factors, such as grief, sorrowful thought and sickness, sitting or sleeping on uncomfortable beds, anger, day sleeping and night awakening, apprehension, suppression of natural urges, indigestion, injury, starvation, and systemic disorders (e.g., osteomalacia, hyperparathyroidism, hyperthyroidism, multiple myeloma, and glucocorticoid use). Osteoporosis may be associated with the causes of disk fracture. These degenerative changes may be considered under the cause described in Ayurveda in terms of *dhatu sanchhayat*, meaning the degeneration of essential constituents. Other causes of sciatica are spondylolisthesis (a degenerative spine disease), spinal stenosis resulting from bony encroachment by osteoarthritis, lumbar adhesive arachnoiditis (the result of a fibrotic process following an inflammatory response to local tissue injury within a subarachnoid space), systemic cancer with epidural metastases of tumor, diabetes mellitus (DM), and polyarteritis nodosa.

True sciatica or radicular pain related to nerve root compression occurs in a small percentage of patients with LBP, roughly 1%. Herniation of nucleus pulposus occurs in 95 to 98% of cases at the disc between the L₄ and L₅ vertebrae or between the L₅ and S₁; it sometimes plays an important role in developing sciatica. One third of the patients with undiagnosed LBP and known systemic cancer have epidural metastasis of tumor and one third have pain associated with vertebral metastasis alone.⁷ Increased levels of circulating antibodies against one or more glycosphingo lipids were detected in 72% of patients with acute sciatica in 61% of sciatica patients at a 4-year follow-up visit (eight antigens analyzed) and in 54% in patients undergoing discectomy.⁸ Nerve ischemia may constitute an important factor in the pathology of human peripheral nerve injury and neuropathy.^{9,10} Studies on rats indicated that the circulating monocyte and macrophages are well involved in the development of hyperalgesia and wallerian degeneration after nerve injury.¹¹ The idiopathic mononeuropathy in young adults seems to occur in 6 to 10% of cases of microneuropathies of sciatic nerve.¹²

11.5.2 Injury Based

Common injury-related causes of sciatica are the unusual and excessive activities of the lower body and legs, such as jumping, climbing, stretching and other movements, and spinal fracture or dislocation resulting from compression (in 80%). The other causes are flexion injuries presenting the features of sciatica, neurologic impairment associated with the injuries caused by falling from height or sudden deceleration in an automobile accident, disk herniation, or spondylolysis as a bony defect of the pars interarticularis of the lower lumbar vertebrae (probably caused by stress fracture in the congenitally abnormal segment). The causes of arachnoiditis are myelography, multiple lumbar operations, chronic spinal infections, spinal cord injury, intrathecal hemorrhage, intrathecal injection of steroids, and anesthetics, and sometimes play an important role in producing sciatica. Overweight individuals, especially females who weigh >85 kg, occasionally present the features of sciatica recurrently due to abdominal muscle pressure on lumbosacral spine.

11.6 Pathogenesis

According to the *Ayurvedic* concept, vitiated *vata dosa*, by its own aggravating factors, strongly fills up the hollowed streams that produce many kinds of diseases pertaining to either all or one part of the body.^{4,5,13} In this pathogenesis *vata dosa*, blood, muscle, bone, ligaments, and tendons are involved.

In conventional medicine, the lumbosacral nerve roots have been known for more than 6 decades to be intimately involved in sciatica. However, the basic pathophysiologic mechanisms are still not well understood. Intervertebral disk is a fibrocartilagenous structure that lies between two vertebral bodies. It consists of soft inner nucleus pulposus surrounded by thicker fibrous tissue wall, the annulus fibrosus. The nucleus pulposus is a gelatinous structure and acts as a shock absorber between adjacent vertebral bodies. As age advances, the entire disc structure becomes more susceptible to trauma and compression. Tears develop in the annulus fibrosus as a result of repeated minor trauma; eventually, if the tears become large enough, a portion of the soft nucleus pulposus herniates through the annulus. Asymptomatic herniation may occur into the center of the vertebral bodies bordering the disc. When disc material herniates into the vertebral canal, it can compress nerve endings and nerve roots, causing pain and other symptoms. Generally, the disc herniates lateral to the posterior longitudinal ligaments, compressing spinal roots as they enter the intervertebral foramen. Occasionally, the disc herniates more centrally, compressing either the spinal cord or the cauda equina.

An experimental study examined compression-induced impairment of vasculature. It was observed that sensory fibers are slightly more susceptible to compression than are the motor fibers and that nerve roots are more susceptible to compression injury than are the nerve fibers.¹⁴ The study further suggests that nucleus pulposus has the potential to injure the nerve root. Proteoglycans and tumor necrosis factors present in disc cells may also be involved in early pathogenic process of the nerve root.⁶

11.7 Clinical Course and Prognosis

The clinical course of sciatic pain associated with systemic illness is that of the underlying disease. Excluding patients with local infections, congenital anomalies, metastatic neoplastic tumors, and cauda equina syndrome, whose prognosis depends on the specific cause and available treatment, the disease is left with the score of the typical patient with sciatica. From an *Ayurvedic* perspective, the prognosis is good in patients with *vataj* sciatica; 70% of patients are relieved of pain, enabling them to return to full activity within a 3- to 4-week period, and complete remission is observed in more than 90% of cases within a 6- to 8-week period. Among the patients who have the symptoms of *vat-kaphaj gridhrasi*, recurrence has been noted in 8% of cases 6 months to 5 years, after the treatment. Recurrence is also observed in those suffering from DM and herniated disc.

11.8 Clinical Examination and Diagnosis

11.8.1 History

History may provide a definite clue to set up the diagnosis of the patient. A patient presenting history of low back trauma with inability to move the legs strongly suggests either acute lumbar disc herniation or fracture. In more severe trauma, the patient may sustain a fracture, dislocation, or a burst fracture that involves not only the vertebral body but the posterior elements as well. Trauma caused by falling from a height may be associated with the fracture of pars interarticularis of L₅ vertebra. Pain after a pure flexion injury indicates a pathologic condition of the disc end plates, whereas pain after torsional is more often associated with disruption of posterior structures (i.e., facet joint and ligament injury). The duration, quality, severity and location of pain all are important in a detailed history.

An acute sciatic pain can last up to 6 weeks in 90% of patients; if it persists beyond 2 months, specific diagnosis in most cases is needed. Patients presenting a history of bowel and bladder dysfunction (incontinence or retention), perianal paresthesia, or pain are usually associated with cauda equina syndrome. Patients suffering from sciatica symptoms should be examined for the following conditions: spondylosis, osteomyelitis, osteoporosis, diskitis, spinal epidural abscess, herniated lumbar disk, DM, metastatic tumors, hyperparathyroidism, and neurological diseases. Occupational, social, and psychological factors also must be taken into account in history taking. For example, job dissatisfaction can sometimes adversely affect the prognosis. All factors should be considered in a proper manner.

11.8.2 Physical Examination

11.8.2.1 Spine Examination

A normal spine shows a thoracic kyphosis, lumbar lordosis, and cervical lordosis in sagittal plain; exaggeration of these alignments may result in hyperkyphosis (*lameback*) of the thoracic spine or hyperlordosis (*swayback*) of the lumbar spine. A spasm of lumbar paravertebral muscles results in the flattening of the usual lumbar lordosis; examination may reveal lateral curvature of the spine (scoliosis) or an asymmetry in the appearance of paraspinal muscles, suggesting muscle spasm. Taut para spinal muscles limit the motion of the lumbar spine. Back pain of bony spine origin is often reproduced by palpation or percussion over the spinous process of the affected vertebrae. Nerve root compression is indicated by radicular pain, sensory disturbances, coarse twitching, fasciculation, muscle spasm, and the impairment of tendon reflexes. Motor abnormality may also occur but is usually less prominent than the pain and sensory disturbances.

Because the herniation of intervertebral lumbar disc most often occurs between L₄–L₅ and L₅–S₁ with irritation and compression of L₅ and S₁ root, respectively, it is important to recognize the clinical characteristics of lesions of these two roots. If a patient complains of pain in the hip region, groin, posterolateral thigh, lateral calf radiating to the external malleolus, or dorsal surface of the foot, the first or second and third toes indicate L₅ nerve root compression. Paresthesia may be in the entire territory or only in the lateral calf, foot, and toes. Tenderness is felt in lateral gluteal region and near the head of fibula. If there

is muscle weakness, difficult dorsiflexion of the great toe and, less often, of the foot is shown. Moderately depressed ankle jerk may be observed in some patients. Generally, knee and ankle reflexes are not impaired but may be observed in rare cases.

Walking on the heel may be more difficult for a patient because of weakness of dorsiflexion of the foot. In S_1 root compression, the patient feels pain in the mid-gluteal region, posterior part of the thigh, posterior of calf to the heel, and the plantar surface of the foot and fourth and fifth toes. Tenderness is most pronounced over the mid-gluteal region (sacroiliac joint), posterior thigh area, and calf. It may sometimes extend to the rectum, testicles, or labia. The patient's complaint of paresthesia and loss of sensation is more likely in the lower leg and outer toes. If nerve weakness is present, there is difficulty in plantar flexion of the foot and toes. In this nerve involvement majority of cases show diminished or absent ankle reflex (Achilles tendon reflex); walking on the toes is more difficult because of the weakness of plantar flexors.

Disc may herniate into the adjacent vertebral body during the process of degeneration and may give rise to a Schmorl's node. Such cases often show no signs of nerve root involvement, though the symptoms are same as in sciatica. The lumbar disc syndrome is usually unilateral. But sometimes massive derangements of disc or the extrusion of the large free fragment into the canal presents lateral symptoms and signs of sciatica that may be associated with the paralysis of the sphincters. The infrequent occurrence of L_3 and L_4 disk herniation with L_4 root compression is manifested as a diminished knee jerk (patellar tendon reflex), quadriceps weakness, and hyperesthesia over the lateral aspect of the thigh.

11.8.2.2 Postural Examination

The fully developed syndrome of ruptured lumbar disc consists of backache, abnormal posture, and limitation of motion. Lateral bending to the side opposite the involved spinal element may stretch the affected tissue, worsen pain, and limit motion. Reduction of the lateral flexion is more suggestive of spondyloarthropathy. Hyperextension of the spine (with patient prone or standing) is limited in nerve root compression, ligamentous or facet joint injury, and bony spine disease. Increased discomfort with spinal extension indicates spinal stenosis. Forward bending in the standing posture with legs straight stretches the sciatic nerve and its root presenting pain. Flexion of the hip is normal in patients with lumbar spine diseases, but flexion of the lumbar spine is limited and sometimes painful. The manual internal and external rotation at the hip with the knee and hip in flexion (Patrick sign) may produce pain.

The most useful, simple maneuver in assessing nerve root impingement is the straight leg raising (SLR) test. In supine position, a passive straight leg raising (possible up to 80 to 90° in a normal individual without pain) or passive dorsiflexion of foot during the maneuver adds to the stretch of L_5 and S_1 nerve root and sciatic nerve, resulting in pain that is readily identified by the patient. If this is the case, the test is positive. To exclude false positive results, the physician should attempt repeated trials. The angle associated with pain should be reproducible when the physician passively flexes the patient's hip with the knee in flexion, then slowly extends the foreleg. An important supplement to the direct SLR is the *crossed SLR*. The crossed SLR is positive when performance of the maneuver on one leg reproduces pain in the opposite leg or buttock. The nerve root lesion is always on the side of the pain. The reverse SLR is a similar but less quantified provocative test that is performed to detect L_2 and L_4 nerve root lesion. This sign is elicited by making the patient stand next to the examination table and passively extending each leg while the patient continues to stand. This maneuver stretches the L_2 and L_4 nerve roots and the femoral nerve because the nerve passes anterior to the hip.

11.9 Diagnosis

After taking a detailed history and performing the complete physical examination of the patient, diagnosis is made on the basis of findings, subjective, and objective parameters preceding laboratory studies for mere confirmation. Examples of subjective parameters include radiating pain starting from the gluteal region toward the leg and foot, tenderness along the course of nerve, severe pain when squatting, sensory and motor changes, sphincteric tone, and positive SLR signs. Examples of objective parameters include pricking and pulling pain, stiffness, ankle and knee jerk, plantar and dorsal reflexes, pressing power, muscle wasting, walking speed, posture, and sensory impairment.

11.9.1 Laboratory Investigations

Investigations required for the patients are always guided by the history taken, examination performed, progress observed, and risk factors to the underlying disease. A physician has to decide what type of critical investigation is required for his or her patient. Exhaustive, hazardous, more expensive, and complication-creating tests should always be avoided. A patient having duration of illness less than 2 weeks and presenting history of no trauma or any serious disease without risk factors for underlying disease needs no investigation. Only conservative treatments are required. If the patient does not respond by the end of the second week, routine laboratory studies such as routine and microscopic urine examination; complete blood count; percentage of hemoglobin; erythrocyte sedimentation rate (ESR); chemistry panel; human leukocyte antigen (HLA) B-27; rheumatologic tests; immunoglobulin; calcium phosphate; alkaline phosphatase; and a plain x-ray of the lumbosacral part of the spine in anteroposterior, lateral, and oblique planes are required. Physician should keep in mind that the lumbosacral spine radiographs are abnormal in most persons over 50 years old and are rarely helpful.

If the pain still persists for more than 4 weeks and the causative factor is not elicited, the next option of choice is magnetic resonance imaging (MRI) and computed tomography (CT) scan-myelography. MRI is an excellent noninvasive imaging method for the lumbar spine. MRI elicits well the soft tissue such as disc, nerve roots, thecal sac, and paraspinal soft tissue. CT-myelography provides the greatest information about osseous lesions in the region of the lateral recess and intervertebral foramen. CT-myelography is particularly helpful in evaluating spondylolysis, infection, or tumor associated with bony destruction. Electromyography (EMG) is also useful to assess the functional integrity of the peripheral nervous system. In sciatica, sensory nerve conduction studies are normal. Sometimes a radionuclide bone scan is used to elucidate sinister cause for sciatica, including occult spondylolysis, primary and metastatic tumors, diskitis and vertebral osteomyelitis, but the technique is inferior to MRI and CT-myelography.

11.10 Therapy (*Chikitsa*)

A comprehensive treatment plan should be made keeping in view the duration, onset, risk factors of the disease, and a goal of early return of the patient to maximum routine functions. The treatment of sciatica in Ayurveda is based on a set of principles involving palliative purification as well as enema therapy (see [Chapters 3](#) and [5](#)). Preparations of

Commiphora mukul (guggul), *Strychnos nuxvomica* (vistinduk), *Ricinus communis* (erand), *Allium sativum* (garlic), and *Vitex negundo* Linn. (*nirgundi*) are extensively recommended for the treatment of *vat vyadhis* in general and sciatica in particular.

11.10.1 Shaman Shodhan Therapy

Palliative treatment (Shaman therapy) includes *guggul* and *vistinduk* preparations orally (Table 11.1) along with local massage and fomentation. Complete bed rest for 4 to 5 days provides additional benefit to the patient. More than 65% of patients have been benefited within 2 to 3 weeks from this combination therapy. The response of steam fomentation with *rasnasaptak* decoction has been proven better than poultice or fomentation with hot sand in bags. If the patient's bowels are constipated, he or she may be administered castor oil. Garlic, castor, and vitex act against the factors responsible for this disease. *Trayodashang* and *yogaraj* are better formulations in *guggul* preparations. *Punarnavadi guggul* is commonly prescribed in the patients with stasis edema. Common formulations used in the treatment of sciatica are listed in Table 11.1.

Most of the Ayurvedic classics recommend enema therapy in the management of sciatica. The enema is an important part of purification therapy that eliminates the waste materials from channels. Palliative treatment given after purification is believed to act more effectively and disease is not relapsed once cured. If the patient does not respond within 2 weeks of palliative treatment alone, purification therapy should then be included in the treatment. Purification procedures, in which oleation, fomentation, purgation, and enemas are used in the treatment of sciatica, have been found most effective (see Chapter 4 on *panchakarma* therapy).

When severe pain and neuromotor deficit from sciatica persist despite 4 to 8 weeks of appropriate conservative therapy, bloodletting therapy is advised. Bloodletting provides quick relief in pain and other inflammatory conditions. Details of the procedure are discussed in Chapter 4.

11.10.2 Cautery Therapy

If the pain and neurologic findings do not disappear on these prolonged conservative managements, or if the patient suffers frequently recurring acute episodes, cautery may be indicated. Cauterization stimulates the nerve briskly like transcutaneous electrical nerve stimulation (TENS) or percutaneous electrical nerve stimulation percutaneous electrical nerve stimulation (PENS). It is performed either between the tendo calcaneus and malleolus (in L_5 impingement) or at the plantar surface of the little toe (in S_1 impingement) in a peculiar marklike dot by a copper wire of 4 mm diameter. The burnt place is smeared with micro fine powders of *Pterocarpus santalinus* Linn. (*rakt chandan*) and *Glycyrrhiza glabra* Linn. (*yashtimadhu*). Cauterization provides short-term relief to the patients. Occasionally, some patients may receive a permanent cure. In few studies, two to three burns on the limb along the course of nerve were found to be optimum. Cautery should be avoided in children, the elderly, apprehensive patients, patients with *pittaj prakriti*, pregnant women, hemoptytic or hematemetic, diarrheic, weakened, and leprotic patients.

A minor surgery for sciatica has been suggested by Datta (11th century A.D.),¹⁶ but the details are not clear. Patients impaired by chronic sciatica pain also pose a special challenge. Alcoholism, depression, headaches, disruptions in family dynamics, unnecessary regular medications and job dissatisfaction, psychosocial issues, and narcotic addiction may cause chronic pain condition in sciatica. Attention must be focused on these factors. Lifestyle changes, reassurance, and education are important. If there is a

TABLE 11.1

Formulations Used in Therapy of Sciatica, Monoplegia, and Paraplegia

No.	Name (Manufacturer*)	Doses, Vehicle, and Duration	Ref.
1	<i>Dwatrinshak guggul</i>	3 g in the morning with cow's <i>ghrit</i> or honey (12 g)	13
2	<i>Punarnavadya guggul</i> (a, b, c)	3 g/day with lukewarm water	31
3	<i>Pathyadi guggul</i>	3 g/day with lukewarm water	32
4	<i>Rasnadya guggul</i> (a)	1 g three times/day with lukewarm water or milk	13
5	<i>Singhnad guggul</i> (a, b, c)	3 g/day with lukewarm water	16
6	<i>Trayodashang guggul</i> (a, b, c, d, e)	1 g three times/day with milk or lukewarm water	17
7	<i>Yograj guggul</i> (a, b, c, d, e)	1 g three times/day with lukewarm water	18
8	<i>Vatari guggul</i>	3 g/day with lukewarm water	34
9	<i>Gorochanadi gutika</i> (d)	250 mg three times/day for 60 days with <i>Ashwagandha</i> decoction (60 ml)	21
10	<i>Vistinduk vati</i> (b, e)	250 mg three times/day with water	17
11	<i>Ajamodadi churna</i> (b)	3 g three times/day with lukewarm water	35
12	<i>Krishna churna</i> (b)	500 mg/day in addition to cow's urine (50 ml) and castor oil (15 ml)	32, 33
13	<i>Dashmool, khireti, rasna</i> leaves, <i>guduchi</i> , and <i>shunthi</i> (powdered and mixed in equal proportion)	3 g twice/day with castor oil (15 ml/day)	16
14	<i>Mahanimb</i> stem bark kalk	3 g three times/day with water	13, 32
15	Purified <i>kupilu</i> seed powder	100 mg twice/day with water for 15 days	18
16	<i>Arusha, danti</i> , and <i>chirayata</i> (equal parts) decoction	25 ml twice/day with castor oil (15 ml)	32
17	<i>Erand mool twak, bilwagiri, brihati</i> , and <i>chhoti kateri kwatha</i> (equal parts)	25 ml twice/day with black salt (<i>kala namak</i> as per requirement)	32, 33
18	<i>Gridhrasi</i> decoction	20 ml twice/day with <i>pushkar moola</i> powder (3 g) and pure <i>hingu</i> powder (500 mg)	35
19	<i>Maharasnadi kwatha</i> (a, b, c, d, e)	50 ml twice/day with 1.5 g purified <i>guggul</i> or adding <i>shunthi/pippali</i> powder (500 mg), or with <i>ajamodadi</i> powder (3 g), or castor oil (15 ml)	35
20	<i>Nirgundi</i> leaves decoction	20 ml twice/day with <i>dashmoola</i> powder (3 g) and pure <i>hingu</i> powder (500 mg)	16
21	<i>Panchmooli</i> decoction	50 ml/day with castor oil (15 ml) or <i>nishoth mool</i> (bark) powder (3 g)	13
22	<i>Rasnasaptak kwatha</i>	20 ml twice/day with <i>shunthi</i> powder (1 g)	32
23	<i>Shefalika</i> leaves decoction	50 ml/day in two divided doses with castor oil (12 ml)	33
24	<i>Lahshun kalk</i>	12 g with <i>erand mool twak kwatha</i> (25 ml/day) for 1 month	13, 16
25	<i>Bala ashwagandha lakshadi taila</i> (b, d)	Apply to the affected parts of the leg	21
26	<i>Hingu triguna taila</i> (d)	5 ml two to three times/day orally with water for 3 to 4 weeks	36
27	<i>Mahavishagarbh taila</i> (a, c, e)	Apply to the affected parts of the leg	13

TABLE 11.1 (continued)

Formulations Used in Therapy of Sciatica, Monoplegia, and Paraplegia

No.	Name (Manufacturer*)	Doses, Vehicle, and Duration	Ref.
28	<i>Mahamash taila</i> (a, b, c, d, e, f)	Apply to the affected parts of the leg	16
29	<i>Narayan taila</i> (c, d, e, f)	Apply to the affected parts of the leg	35
30	<i>Prasarini taila</i> (a, f)	Apply to the affected parts of the leg	34
31	<i>Prabhanjan vimardanam taila</i> (d)	Apply to the affected parts of the leg	20
32	<i>Saindhavadya taila</i> (a, b)	Apply to the affected parts of the leg	13
33	<i>Erand phal payas</i>	Oral, once/day (12 g/day) for 1 month (<i>Note:</i> <i>Payas</i> made of 30 to 50 uncoated cooked castor seeds in 350 ml cow's milk)	13, 33
34	<i>Vrihachhagaladya ghrut</i> (d)	Oral (12 g/day)	31
35	<i>Gunja phal/gunja pistadi lepa</i>	Apply once/day	13, 35
36	<i>Ekang vir ras</i> (a, c, e, f)	125 mg three times/day for 60 days with milk	37
37	<i>Khanjanikari ras</i> (a)	250 mg three times/day for 90 days	33
38	<i>Lashun ksheer pak</i> made of <i>Rason kalk</i> (15 g) in cow's milk (200 ml)	200 ml/day	4

- * (a) Baidyanath Pharmaceuticals;
 (b) Indian Medicines Pharmaceutical Corporation Ltd. (IMPCL);
 (c) Unjha Pharmaceuticals;
 (d) Indian Medical Practitioners Co-operative Pharmacy & Stores (IMPCOPS);
 (e) Zandu Pharmaceuticals;
 (f) Dabur Pharmaceuticals.

progressive neurologic deficit with increasing weakness after 12 weeks of conservative management with or without clinical evidences of epidural abscesses, malignancy, hematoma, nerve root compression with persistent pain, spondylolysthesis or spinal stenosis, surgery may be considered. Surgery should be considered only in patients who have clearly defined anatomic structural abnormality and have failed from other therapeutic treatments.

11.11 Strategy for Prevention of Sciatica

Changed dietary habits, rapid industrialization, stress, pollution, and altered lifestyle have made people prone to many serious diseases. Sciatica may be one of them. Preventive measures include avoiding fast food (little food), starvation, regular and unnecessary medication, frequent use of cold materials, sleeping during the day and staying awake at night, and strenuous stretching exercises. The recurrences of pain can be reduced and chronic disability can be prevented in patients with mechanical sciatica due to lumbosacral strain or sprain by using a properly supervised exercise program, avoiding prolonged sitting, prolonged standing, and improper lifting.

11.12 Scientific Basis for the Use of Ayurvedic Drugs in Sciatica

11.12.1 Clinical Studies

1. In one study, 52 sciatica patients were given *Trayodashang guggul* (2 g) and *vistinduk vati* one tab three times/day for 3 weeks. The patients were also treated with *Nirgundi taila* massage and poultice for 3 weeks.¹⁷ In another study, 60 sciatica patients were given *Yograj guggul* (1 g) three times/day plus purified *Nuxvomica* seed powder (100 mg) in one capsule twice/day for 15 days. The patients were also treated with *Mahavisgarbha taila* massage and steam fomentation of *Rasnasaptak* decoction.¹⁸ The criteria for assessment of the results were based on pricking and pulling pain, stiffness, tenderness over sciatic nerve, SLR, knee and ankle jerks, plantar reflexes, muscle wasting, pressing power, walking speed, sensory impairment, and posture, as per global score method. In both studies, results were significant ($p < 0.05$) in more than 70% of the patients in terms of relief from the symptoms.
2. *Nirgundi* oil and dehydrated butter prepared as per the *Ayurvedic Formulary of India* were administered classically under purification therapy on 66 sciatica patients along with purified *guggul* (1 g) and *Vitex negundo* decoction (60 ml) three times/day for 45 days. The results showed complete cure in 76% of sciatica patients.¹⁹
3. Sixty-eight sciatica patients were distributed equally in four groups at random: (1) palliation, (2) placebo, (3) purification, and (4) purification with palliation. They were given *Prabhanjan vimardanam taila* as per classical schedule. Although the results of treatment were highly encouraging in both purification and purification cum palliation groups, the results were better in the purification cum palliation than the purification group.²⁰
4. In a clinical trial conducted on 80 patients with paraplegia, a group of 40 patients received *panchakarma* therapy with *Moorchhit Sesamum indicum* (til) oil as per classical schedule, and the other group was treated with one tablet of *Gorochanadi gutika* (250 mg) after *Ashwagandha* decoction (60 ml) three times/day. Both groups were given massage with 50 ml of *bala ashwagandha lakshadi taila* for 60 days. Results of both the groups were quite significant ($p < 0.001$). The group with *panchakarma* therapy showed better results than did the *ashwagandha* group, even after a 6-month follow-up. The progress in relief was judged on the basis of pain, bladder control, rectum control, sensory changes, muscle powers, fasciculations, deep tendon reflexes, plantar reflexes, wasting, and walking speed, as per global score method.²¹

11.12.2 Animal Studies for Anti-Inflammatory Activity

11.12.2.1 Yograj Guggul and Rasnasaptak Decoction

The aqueous extract of *yograj guggul* and *rasnasaptak* decoction were administered intraperitoneally (i.p.) in albino rats 30 min prior to carrageenin challenge, in combination and

alone at the doses of 1 g/kg. Test drugs produced significant inhibition of paw edema. Phenylbutazone was taken for standard comparison in the study.²²

11.12.2.2 *Commiphora mukul* (Hook. ex stocks) (Burseraceae)

Phenylbutazone (100 mg/kg), hydrocortisone acetate (20 mg/kg i.p.), ibuprofen (50 mg/kg orally), acetylsalicylic acid (300 mg/kg orally), or steroidal compound from *C. mukul* (100 mg/kg i.p.) were given 1 h before the carrageenin challenge to groups of 6 rats. No treatment was given to a control group of six rats. The steroid isolated from *C. mukul* showed a pronounced anti-inflammatory ($p < 0.001$) in paw edema as compared to phenylbutazone ($p < 0.001$), hydrocortisone acetate ($p < 0.005$), ibuprofen ($p < 0.01$), and acetylsalicylic acid ($p < 0.001$).²³

11.12.2.3 *Vitex negundo* Linn. (Verbenaceae) and *Withania somnifera* Linn. (Solanaceae)

The aqueous extract of *Vitex negundo* Linn. at the dose of 1 g/kg i.p. produced anti-inflammatory activity by 45, 60, and 44% at 2, 3, and 4 h of treatment in the carrageenin-induced rat's paw edema model. *W. somnifera* also exhibited anti-inflammatory activity in carrageenin induced rat's paw edema. The inflammation was recorded as 61, 56, and 27% at the first, second, and third hour of treatment, respectively. The study further revealed that *W. somnifera* acts by blocking histamine h_1 and h_2 receptors in early phase followed by inhibition of prostaglandin in delayed phase of acute inflammatory reaction. However, anti-inflammatory activity of *V. negundo* is mediated through histamine and 5-HT, which is further maintained during delayed phase.²⁴

11.12.2.4 *Ricinus communis* Linn. (Euphorbiaceae)

Petroleum ether extract of *R. communis* exhibited significant anti-inflammatory activity against formaldehyde and adjuvant-induced rat's paw arthritis.²⁵

11.12.2.5 *Pluchea lanceolata* (DC) Clarke (Compositae) and *Allium sativum* Linn. (Liliaceae)

The water-soluble portion of the 90% alcoholic extract of *P. lanceolata* administered orally in the dose of 2000 mg/kg/day produced significant anti-inflammatory activity in formalin induced albino rat's hind paw arthritis and granuloma pouch in comparison with betamethasone. The pharmaceutical preparation allisatin, a concentrated formulation of fresh *Allium sativum*, was suspended in water and administered in the dose of 2000 mg/kg/day. The results showed slight anti-inflammatory activity against formalin-induced arthritis but not against granuloma pouch.²⁶

11.12.2.6 *Sida cordifolia* Linn. (Malvaceae)

An aqueous extract of *S. cordifolia* was administered orally at a dose of 400 mg/kg body weight. The results showed a significant inhibition of carrageenin induced rat's paw edema but did not block the edema induced by arachidonic acid.²⁷

11.12.2.7 *Tinospora cordifolia* (willd.) Miers. (Ex Hook. f.) and Thoms. (Menispermaceae)

Water extract of stem of *T. cordifolia* (neem giloe) at the dose of 500 mg/kg given orally and i.p. inhibited acute inflammatory response evoked by carrageenin, significantly ($p < 0.05$). Analgesic and antipyretic actions have also been observed in the study.²⁸

11.13 Discussion

Although sciatica is not a terminal disease, lack of proper treatment may lead to difficulty in walking and other movements. The most often prescribed drugs in the treatment of sciatica are nonsteroidal anti-inflammatory drugs (NSAIDs), acetaminophen, muscle relaxants, and opioids. Several randomized double-blind studies demonstrate that NSAIDs are superior to placebo, but they provide short-term relief to patients.¹ Allergic reactions have also been reported by the use of drugs. The use of opioids, benzodiazepines, and tricyclics may be helpful, but in many cases drug dependence, tolerance, and drowsiness have been observed. Muscle relaxants provide short-term relief, but drowsiness often limits their day-time use.⁷ Using chymopapain to dissolve herniated disc material proved most effective in relieving sciatica, but severe neurologic complications restrict its use. Traction, acupuncture, and exercise programs have failed to show benefits of these therapies in sciatica.

In spite of the fact that the spinal diseases are difficult to cure, the Ayurvedic concept of treatment for sciatica is more effective and suitable as compared with the modern mode of treatment. The therapies can be used for prolonged periods of time without any toxic side effects. Oleation is considered the most effective remedy in the treatment of *vata* diseases. Medicated oils either given orally or applied externally, or both, help in the restoration of muscle strength. Massage increases circulation, prevents nerve degeneration, and promotes tissue building. It renders the body soft and helps in the elimination of accumulated waste materials that obstruct the channels causing vitiation of *vata*. Oleation followed by fomentation liquefies *dosas* adhered in the channels and brings them to the gastrointestinal tract (*kostha*) where they are expelled by the downflow action of purgation. Hemoglobin levels increased in one study in both albino rats and humans after oleation.²⁹ Heat applied by steam, poultice, hot sandbags, or other means reduces the stiffness and muscle spasms by decreasing efferent activity, improving tissue flexibility, and hastening tissue healing by increasing the blood flow and nutrients to the injured area. Heat application has a vasodilatory effect that can lead to edema; therefore, it should be used cautiously.³⁰ The role of bloodletting in the management of sciatica has also proved to be significant.

11.14 Monoplegia and Paraplegia

11.14.1 Definition

When vitiated *vata* situated in the low back region hurts the tendons of single lower limb (*sakthi*), the person is considered affected by monoplegia. When both the limbs are paralyzed, the disease is called paraplegia.

11.14.2 Etiology

Monoplegia and paraplegia both are *vata*-predominant diseases. Degeneration of essential constituents, trauma (especially on the oscoccygis bone [*kukunder marma*]), and obstructions caused in normal passage are the basic factors causing vitiating *vata*. According to conventional medicine, monoplegia is usually caused by lower motor neuron diseases. A small thoracic cord lesion, tumor, or demyelinating plaque may result in upper motor neuron diseases. Paraplegia results from injuries at T₁ vertebra and below. Compression in the low thoracic and lumbar region causes a conus medularis or cauda equina syndrome. Upper motor neuron and occasionally lower motor neuron diseases, epidural metastases, and spinal cord ischemia may be associated with monoplegia. Acute onset of paraplegia results from diseases of cerebral hemisphere (i.e., anterior cerebral artery ischemia, superior sagittal sinus thrombosis, and cortical venous thrombosis). Gradual onset over weeks or months is almost always due to multiple sclerosis, intraparenchymal tumor, and chronic spinal cord compression from degenerative diseases of the spine.

11.14.3 Clinical Course and Prognosis (*Sadhyasadhyata*)

The overall prognosis of monoplegia and paraplegia can be improved by the use of *panchakarma* therapy. In acute onset these therapeutic modalities produced a remission in the number of patients. Prognosis is poorer in epidural metastases, intraparenchymal tumor, lumbar intraspinal metastases, and in patients ages 60 and over with acute onset. Over time, the disease fails to respond to Ayurvedic therapy. Disease usually requires prolonged course of treatment if the condition is more chronic and the prognosis is not good.

11.14.4 Diagnosis

Monoplegia with acute onset presenting sensory loss or pain is usually caused by lower motor neuron diseases. A localized lower motor neuron disease presents progressive weakness with atrophy in one limb that develops over weeks or months. Distribution of weakness is commonly localized to a single nerve root or peripheral nerve within one limb. EMG and nerve conduction studies confirm the diagnosis in such cases. Monoparesis of distal and nonantigravity muscle without sensory loss or pain indicate an upper motor neuron lesion. Weakness of one limb with numbness is generally due to an involved peripheral nerve, spinal nerve root, or lumbosacral plexus. If numbness is absent, segmental anterior horn cell disease is likely. Paraplegia with numbness, frequent micturition, and fecal incontinence results from acute spinal cord disease. Drowsiness, confusion, and seizures present the disease with cerebral origin. Gradual onset over weeks or months indicates spinal cord compression due to degenerative diseases of the spine, intraparenchymal tumor, or multiple sclerosis. MRI of the spinal cord or brain may be helpful in diagnosis.

11.14.5 Therapy (*Chikitsa*)

Treatment plan advised in Ayurveda for the management of monoplegia and paraplegia involves oleation, fomentation, purgation, and enema along with *guggul* orally. *Khanjanikari ras*, *Ekangvir ras*, and *Gorochanadi gutika* are the some Ayurvedic formulas that have shown better results in studies.²¹ These formulas are listed in [Table 11.1](#). Recent controlled

trials have documented the efficacy of *Sirovasti* and *nasya* (refer *panchakarma*) in monoplegia and paraplegia with cerebral or motor neuron disease origin.

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12

Allergic Reaction

Lakshmi Chandra Mishra

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12.1 Introduction

Allergic reaction illnesses (ARIs), drug allergies, or hypersensitivities are considered *kapha*-dominated diseases in Ayurveda. Examples are asthma and eczema. In Allopathic medicine, the treatment of these diseases clusters around the use of steroids, antihistamines, and bronchodilators (beta-adrenergic blockers). Ayurveda has approaches worth investigating for the management of ARI. These approaches and available studies on herbs are discussed.

12.2 Definition and Clinical Description

There is no specific definition of ARI given in Ayurveda, but these disorders fall under the broad category of *kaphaja* diseases. In the conventional system of medicine, ARIs are characterized as disorders of the immune system. Clinical signs and symptoms of ARI differ according to organs involved. For example, an allergic reaction occurring in the upper respiratory tract would show an inflammation of the tract and congestion (allergic rhinitis). An allergic reaction occurring in the lungs would show the symptoms of an asthma attack. If it occurs in the skin, it may show the signs and symptoms of urticaria, eczema, psoriasis, or contact dermatitis. One common ARI is allergic rhinitis, which

accounts for at least 2.5% of all physician's office visits and costs about \$2.4 billion on prescriptions and on counter medication and \$1.1 billion physician billings per year in the U.S.¹ Other common ARI are allergic asthma and skin reactions.

12.3 Etiology

No specific etiology of ARI is available in ancient Ayurvedic texts. ARI in Ayurveda is described as intolerance developed in the body due to a variety of unwanted toxic materials generated inside the body; this intolerance is either caused by poor digestion of food materials, inadequate metabolism of nutrients (*ama-visa*), or exposure to environmental pollutants (*dushi-visa*).

Ayurvedic etiology of ARI is related to a decrease in immunity or *vyadhiksamatva* (resistance to diseases) that protects a person from diseases. Immunity is recognized as natural as well as acquired. This resistance is associated with or due to a biological material present in humans called *ojas*. Ayurveda has a number of herbs and herbal formulas called *rasayanas* or dietary supplements that are used to strengthen immunity and improve resistance to diseases.

ARIs are currently known in conventional medicine to be mediated by the immune system. Allergic reactions are described as adverse reactions caused by a second exposure of a subject to the same chemical or a structurally similar chemical or an organic material after the subject has been sensitized earlier with the chemical or the organic material. The reactions are invoked not only by exposure to exogenous antigens but also by endogenous (intrinsic to the body) antigens. This etiology of ARI is similar to that stated in classic Ayurvedic texts. Intrinsic antigens are known to be responsible for many of the important immune diseases, such as autoimmune diseases (immune reaction against the individual's own tissue, e.g., rheumatoid arthritis, systemic lupus erythematosus), immunologic deficiency syndrome, and amyloidosis.

12.4 Pathogenesis

Ayurveda includes all body tissues as possible targets of ARI. The pathogenesis of ARI involves invasion of various organs by vitiated *kapha*. The pathology in general consists of inflammation and excessive secretion of fluid.

In conventional medicine, the pathology of allergic reactions in general is also similar to that described in Ayurveda with respect to inflammation of the tissue and excessive secretion of fluid. On the basis of the immunologic mechanisms mediating the disease, ARI has been categorized in four types.² Type I reactions, also called anaphylactic shock, are mediated by immunoglobulin E (IgE) antibodies. The examples are food allergies, skin allergies (urticaria and atopic dermatitis), allergic rhinitis, asthma, and anaphylactic shock. Type II or cytolytic reactions are mediated by immunoglobulins G and M (IgG and IgM), and the common examples of this reaction are penicillin-induced hemolytic anemia, quinidine-induced thrombocytopenic purpura, sulfonamide-induced granulocytopenia, and procainamide-induced systemic lupus erythematosus. Type III reactions, or serum sickness, are primarily mediated by IgG. It involves antigen-antibody complex formation that

deposits in the vascular epithelium, causing inflammation of the tissue, urticarial skin eruptions, arthritis, lymphadenopathy and fever. Type IV reactions (delayed sensitivity) are mediated by previously sensitized T-lymphocytes and macrophages. An example of this type of reaction is contact dermatitis caused by poison ivy.

12.5 Diagnosis and Prognosis

Diagnosis is made based on the history of occurrence of the reaction and possible association with certain chemicals or organic materials. Skin tests are used to determine specific allergens. An ARI may be discomforting and may take weeks to subside. The acute phase involving lungs may be fatal, although rare, if not treated appropriately.

12.6 Therapy

The treatment of allergic reactions in Ayurveda essentially consists of identifying the allergens, avoiding the exposure to them, using drugs to relieve acute symptoms, improving digestion, and cleaning the intestine of toxic materials in the gut (*ama*). Emesis is recommended to treat allergy and asthma because ARI are considered *kaphaja* disease. Dietary recommendations include diet that suppresses *kapha* as discussed in [Chapter 3](#). *Sitopaladi churn* is useful in upper respiratory tract ARI. The following single herbs are commonly used in ARI and scientific studies in support of the use are listed below:

1. *Clerodendrum seratum*³
2. *Benincasa hispida*⁴
4. *Aquilaria agallocha*⁵
4. *Albezia lebbeck*⁶
5. *Curcuma longa*⁷
6. *Inula racemosa*
7. *Galphimia glauca*
8. *Picrorhiza kurora*
9. *Adhatoda vasica*

Treatment specific to skin diseases and asthma are discussed in [Chapters 17](#) and [18](#).

12.7 Scientific Basis

The herbs listed in the previous section have been investigated primarily in type I hypersensitivity models. The studies summarized here appear to support the use of these herbs in Ayurvedic therapies of ARI.

Prolonged administration of a saponin from the *Clerodendron serratum* plant has been reported to exhibit antihistaminic and antiallergic activity.³ Alcoholic extract of the root of *Inula racemosa*, prepared by continuous heat extraction, was studied for its antiallergic effect in experimental models of type I hypersensitivity (egg-albumin-induced passive cutaneous anaphylaxis [PCA] and mast cell degranulation) in albino rats.⁸ LD₅₀ of this extract was found to be 2100 ± 60 mg/kg intraperitoneally (i.p.). The extract was given orally for 7 days or once only. As a positive control for mast cell degranulation, compound 48/80 was given with the same dosage schedule. Remarkable protection against egg-albumin-induced PCA was observed on the extract treated group after a single dose. The extract also protected against compound 48/80-induced mast cell degranulation, and the protection was similar to that of an antiallergic drug disodium cromoglycate. The 7-day drug treatment schedule showed greater protection than did disodium cromoglycate intraperitoneally. The results suggest that *I. racemosa* possesses potent antiallergic properties in rats.

The methanol extract of wax gourd, the fruits of *Benincasa hispida* (Kooshamanda),⁴ was found to show inhibitory activity on the histamine release from rat exudate cells induced by an antigen-antibody reaction. Four known triterpenes and two known sterols were isolated as active components based on the bioassay. The other constituents isolated were flavonoid C-glycoside, an acylated glucose, and a benzyl glycoside. Two triterpenes, alnusenol and multiflorenol, were found to be potent inhibitors of histamine release.

An aqueous extract of stems of *Aquilaria agallocha* (agaru) has been investigated on the immediate hypersensitivity reactions.⁵ The aqueous extract of stems showed inhibitory effects on passive cutaneous anaphylaxis, anaphylaxis induced by compound 48/80, and histamine release from rat peritoneal mast cells (RPMC). The extract also prevented the degranulation of RPMC in rats. The authors concluded that the aqueous extract inhibits the immediate hypersensitivity reaction by the inhibition of histamine release from mast cells.

Decoction of *Albizzia lebbek* bark has been investigated to determine its possible anti-allergic activity.⁶ The effect of the decoction on the degranulation rate of sensitized peritoneal mast cells of albino rats challenged with antigen (horse serum) was studied. Triple vaccine was used as an adjuvant and disodium cromoglycate (DCG) and prednisolone were used for comparison. Drugs were given during the first or second week of sensitization and the mast cells were studied at the end of the second or third week. Serum from these rats was used to passively sensitize recipient rats whose peritoneal mast cells were then studied. The effects of *A. lebbek* and DCG on the degranulation rate of the sensitized mast cells were also studied *in vitro*. The results show that *A. lebbek* had a significant cromoglycate-like action on the mast cells. In addition, it appears that it inhibited the early processes of sensitization, and synthesis of reaginic-type antibodies. If the decoction were given during the first week of sensitization, it markedly inhibited the early sensitizing processes; if given during the second week, it suppressed antibody production during the period of drug administration. The active ingredients of the bark appear to be heat stable and water soluble.

Several plants used in traditional medicine for the treatment of bronchial asthma have been investigated.⁹ Thiosulfinates and cepaenes have been identified as active antiallergic compounds in onion extracts. They exert a wide spectrum of pharmacologic activities, both *in vitro* and *in vivo*. Tetragalloyl quinic acid from *Galphimia glauca*, suppressed allergen- and platelet-activating factor (PAF)-induced bronchial obstruction, PAF-induced bronchial hyperreactivity (5 mg/kg orally) *in vivo*, and thromboxane biosynthesis *in vitro*. *Adhatoda vasica* alkaloids (not identified) showed pronounced protection against allergen-induced bronchial obstruction in guinea pigs (10 mg/ml aerosol). Androsin from *Picrorhiza kurroa* prevented allergen- and PAF-induced bronchial obstruction (10 mg/kg

orally; 0.5 mg inhalative). Histamine release *in vitro* was inhibited by other unidentified chemical constituents of the plant extract.

Compound 73/602 (AA) is a structural analogue of vasicinone, an alkaloid, present in the leaves and roots of *Adhatoda vasica* (acanthaceae); this compound was found to possess potent anti-allergic activity in mice, rats, and guinea pigs.¹⁰ The immunologic effect of curcumin, a natural product of plants obtained from *Curcuma longa* (turmeric), has been reported in animal models.^{7,11}

12.8 Summary

Ayurvedic management of ARI is basically similar to conventional medicine — identify the allergen and avoid the exposure to it. In addition, the Ayurvedic approach includes dietary *kapha*-suppressing foods, herbs, and herbal formulas for treatment.

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13

Bronchial Asthma

Rakesh Lodha and Sushil K. Kabra

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13.1 Introduction

Bronchial asthma is a common global health problem. Individuals of all ages are affected by this chronic airway disorder that can be severe and occasionally fatal. The prevalence of asthma is increasing worldwide, especially in children.¹

Over the past few decades, there have been significant scientific advances leading to improved understanding of the disease and better management. However, the current modes of therapy in conventional medicine do not cure the disorder but control the symptomatology. There is a need to explore safe alternative therapies, such as Ayurvedic medicine, so that they can be successfully integrated with conventional therapy to provide maximal benefits to patients.

13.2 Definition

Ayurveda: *Svasa* (increased or difficult breathing, dyspnea) refers to the disorders of the respiratory system. There are five kinds: *ksudraka*, *tamaka*, *chchinna*, *mahan*, and *urdhava*. Of these, *tamaka svasa* refers to asthma.²⁻⁵

Modern medicine: Asthma is a chronic inflammatory disorder of the airway in which many types of cells and cellular elements play a role. The chronic inflammation causes an associated increase in airway hyperresponsiveness that leads to recurrent episodes of wheezing, breathlessness, chest tightness, and coughing, especially at night or in the early morning. These episodes are usually associated with widespread but variable airflow obstruction that is often reversible either spontaneously or with treatment.

13.3 Epidemiology

Asthma is currently a worldwide problem, and there is evidence that the prevalence has been increasing in many countries.¹ There are no clear-cut data from the Ayurvedic texts about the extent of the problem in ancient times. It is likely that the prevalence of the disorder was much less in ancient India because of factors discussed later in the chapter.

13.3.1 Children

The prevalence of asthma symptoms in children may be as high as 30%. The differences among various populations may be consequences of responses to environment, industrialization, or different allergen loads.¹ Asthma may develop less frequently in children who are exposed to infections and parasitic infestations early in life.^{6,7}

13.3.2 Adults

The prevalence of asthma symptoms in adults varies from 1 to 25%.⁸⁻¹¹ Similar symptoms from cardiac failure and chronic obstructive pulmonary disease make accurate estimates of asthma in older individuals difficult.⁸⁻¹¹

Mortality data from developed countries show that the rates vary from 0.1 to 0.8 per 100,000 persons aged 5 to 34.¹²⁻¹⁴

13.4 Etiology

According to Ayurveda, the general etiology of *svasa roga* is that all things, materials, and conditions that could help increase *vata dosa* and *kapha dosa* are causally responsible for *tamaka svasa*. This develops from an increase in cough (*kasa*), undigested materials (*ama*), diarrhea, vomiting (*vamathu*), poison (*visa*), anemia (*pandu*), and fever (*jvara*); coming into contact with air containing dust, irritant gases, pollens, or smoke; injuring vital spots; using very cold water; and residing in cold and damp places.²⁻⁵ Excessive use of dry food and astringent food and irregular dietary habits may also trigger an attack. In addition, constipation, excessive fasting, excessive use of cold water, excessive sexual indulgence in adults, exposure to extremes of temperature, anxieties, grief, disturbance of peace of mind, and debility may all precipitate an attack. Cough and coryza have also been implicated as etiologic agents. The habitual use of lablab-bean, black gram, *til* preparations, irritant spicy food, and *kapha*-producing diet may also be involved in development of the disease.²⁻⁵

In modern medicine the risk factors for asthma are classified under two categories: host factors and environmental factors.

The host factors include genetic predisposition, atopy, airway hyperresponsiveness, gender, and ethnicity. Asthma can be considered to be a heritable disorder.^{15,16} More boys than girls develop asthma during childhood; this difference disappears by age 10.^{17,18} Subsequently, females are at greater risk for developing asthma. Environmental factors influence the susceptibility to developing asthma in predisposed individuals, precipitate asthma exacerbations, and cause persistence of asthma. The important allergens that play a role in asthma are domestic mites, cockroach allergens, animal allergens, various fungi, and pollen.¹⁹⁻²² Exposure to tobacco smoke and various pollutants may play a role in enhancing symptoms in susceptible individuals.²³ Infections do cause exacerbations in asthmatic individuals.^{24,25} Severe viral respiratory infections in early childhood have been shown to be associated with the development of asthma.^{26,27}

According to the hygiene hypothesis, improvement in hygiene and reduced incidence of common infections is associated with increased prevalence of asthma and atopy in developed countries.²⁸ This may explain the apparent low prevalence of the disorder in the ancient era.

Risk factors for acute exacerbations trigger inflammation or cause bronchoconstriction or both. There is a wide interindividual variation in triggers. The triggers include exposure to various allergens and irritant gases, exercise, cold air, various drugs, and emotional changes. Various indoor and outdoor allergens and air pollutants are implicated in precipitating acute exacerbations. There is strong evidence linking acute viral respiratory infections to acute exacerbations of asthma; the most common are rhinovirus and respiratory syncytial virus infections. They are particularly important in children. Emotional stress can trigger an asthma exacerbation. These possibly act via hyperventilation, which can cause airway narrowing.²⁹ Other conditions, like sinusitis, can be associated with exacerbations. In children, gastroesophageal reflux may lead to the exacerbation of asthma.

From the above it is clear that even in the ancient era there was an excellent understanding of the etiologic factors of bronchial asthma.

13.5 Pathology and Pathogenesis

The pathogenesis of asthma, according to Ayurvedic texts, appears to arise from an abnormal interaction between *vata* and *kapha*. The initial step is the increase in *vata*. Because of the obstruction to normal movements of *vata* by *kapha*, *vata* begins to move in all directions. This disturbs the channels of respiration (*prana*), food (*anna*), and water (*udaka*) located in the chest, and produces dyspnea (*svasa*) originating from the stomach (*amasaya*). This suggests that the root cause of asthma is related to the digestive tract.²⁻⁴

Modern medicine recognizes the inflammatory processes in the airway as the key pathogenetic mechanism for asthma. Current evidence shows that the T-lymphocytes have an important role in regulating the inflammatory response through the release of various cytokines. Eosinophils and mast cells are the key effector cells; these cells secrete a wide range of chemicals that act on airways both directly and indirectly through neural mechanisms.^{30,31} There is a continuum of disease: acute inflammation, persistent disease, and remodeling.³²

Asthma is frequently found in association with atopy (the production of abnormally high amounts of immunoglobulin E [IgE] directed against common environmental allergens such as animal proteins, dust mites, pollen and fungi).³³ Because of this, mast cells are sensitized and upon activation lead to inflammation. Atopy is one of the strongest risk factors for asthma. The process of sensitization may begin *in utero*. The potential for allergic sensitization and risk for wheezing appears to be influenced by many factors in early life, including viral respiratory infections, exposure to endotoxin, exposure to tobacco smoke, use of antibiotics, house dust-mite sensitization, and diet.³⁴

Airway hyperresponsiveness and acute airflow limitation are the two major manifestations of abnormalities of lung function in asthma. Hyperresponsiveness is the presence of an exaggerated bronchoconstrictor response to a wide variety of endogenous and exogenous stimuli. Airway inflammation is the key factor for hyperresponsiveness.

Airflow limitation is produced by acute bronchoconstriction, swelling of the airway wall, chronic mucus plug formation, and airway remodeling. Allergen-induced bronchoconstriction results from the IgE-dependent release of histamine, prostaglandins, and leukotrienes by the mast cells.³⁵

The lungs of individuals who died of asthma were hyperinflated; the small and large airways were filled with plugs formed by mucus, serum proteins, inflammatory cells, and debris. The microscopy (both autopsy material and now on endobronchial biopsies) reveals extensive infiltration by eosinophils and lymphocytes in the airway wall and lumen.^{36,37} In addition, there are features of microvascular leakage, epithelial disruption, and airway remodeling.

13.6 Clinical Features

The prodromal symptoms described in the Ayurvedic texts include pain in the region of the chest and flanks, movement of *prana* (*vata*) in an upward direction, distension of abdomen, and cutting pain in temples. Other premonitory symptoms are pain in the temporal region, restlessness, anorexia, abnormality in taste, flatulence, and pain.

There are several symptoms of *tamaka svasa*. *Vata*, undergoing an increase, travels in an upward direction in the passages, causes increase of *kapha*, and seizes the head and neck.

There is pain in the chest and flanks, a noisy cough (from the throat and the chest), an increase in breathing effort, and wheezing (*ghur ghur* sounds). An acute attack may lead to delusions, loss of appetite, nasal discharge, and thirst. With increasing intensity of bouts, the patient may become unconscious. The patient finds relief for a few moments after expectoration of sputum, breathes with great difficulty while lying down, and finds comfort while sitting. The patient's eyes are wide open, and he or she perspires from the forehead, suffers dryness of mouth, gets bouts of dyspnea (often with shivering), and desires additional warmth. The disease greatly increases on cloudy days and in rain, cold, or direct breeze and other factors that may cause an increase of *kapha*. In addition there is throat irritation, vomiting, and anorexia. This description is quite similar to what we understand today.

Tamaka svasa has been classified into two varieties: *Tamaka svasa vatapradhan* (*vata dosa* is predominant) and *Tamaka svasa shlesma pradhan* (*kapha dosa* is predominant). The indicators of the former are highly painful breathing with high frequency of noisy sounds, little expectoration, difficulty in expectoration, and insomnia. The latter is characterized by high vibrating noise in throat while breathing, coryza, easy and copious expectoration, and painful fast breathing.

The clinical manifestations of asthma as we see it today include episodes of shortness of breath, cough, wheezing, and chest tightness. Patients with more severe cases are likely to have persistent symptoms, worsening at night and early mornings. Often the chest congestion takes more than 10 days to clear up. The symptoms may be precipitated by exposure to the various stimuli discussed above. Very often, exacerbations occur in the absence of any well-defined exposure, particularly in patients with persistent disease. These symptoms usually respond to bronchodilators; this response supports the diagnosis of asthma.

The practitioners of Ayurveda confirm the diagnosis by a detailed case history of the patient and objective examination to determine the imbalances between various constituents. An evaluation is made about the patient's body constitution, state of digestion, and level of activity through interrogation. Interrogation also includes questions on precipitating and pacifying factors and lifestyle. A detailed physical examination of all systems, including checking the pulse and tongue, is done to identify the specific type of *dosa* involved in the pathology. Diagnosis is finally made based on the symptoms, which are dependent on the location of morbid *dosa* and nature of pathology.

Currently, many practitioners use the modern techniques such as auscultation of the chest with a stethoscope for confirming the diagnosis. As there is progress toward holistic medicine, the Ayurvedic practitioners will benefit from the advances made in the diagnosis of asthma.

In modern medicine, the diagnosis can often be made convincingly on the basis of clinical history and physical examination. However, the examination of respiratory system may be normal as the symptoms are variable. A physical examination is more likely to be fruitful if the individual is examined in the symptomatic period.

Measurement of pulmonary function has an important place in diagnosis especially if history and physical examination are inconclusive. In addition, there can be inaccuracies in assessment of severity of symptoms by both patient and physician.

Measurement of airflow limitation (such as peak expiratory flow rate [PEFR] and forced expiratory volume [FEV₁]) and its reversibility are important in establishing a clear diagnosis of asthma. Newer guidelines recommend serial measurements for monitoring the course of asthma. The tests can be performed with ease in individuals older than 5 years.

On the basis of the symptomatology and the result of spirometry, asthma can be classified into intermittent, mild persistent, moderate persistent, and severe persistent types (Table 13.1).

TABLE 13.1
Classification of Asthma According to Severity

Severity	Symptoms	Nighttime Symptoms	PEFR
Severe persistent	Continuous, limited physical activity	Frequent	≤60% predicted; variability >30%
Moderate persistent	Daily use of beta-2 agonist, daily attack affects activity	>One time/week	>60–80% predicted; variability >30%
Mild persistent	>One time/week but < one time/day	>Two times/month	≥80% predicted; variability 20–30%
Intermittent	<One time/week, asymptomatic and normal PEFR between attack	≤Two times/month	≥80% predicted; variability <20%

13.7 Clinical Course and Prognosis

According to Ayurveda, of all the significant *svasa* disorders, *tamaka svasa* (asthma) is *yapya* (i.e., controllable).¹ It is not curable and symptoms are likely to persist. The only exceptions to this may be disease of short duration in strong, healthy, young individuals and if the disease severity is mild.

In the past few decades, a lot of research has been conducted to determine the natural history of asthma. Asthma symptoms may disappear in one third to one half of children at puberty.^{38,39} Even though symptoms may be absent, abnormalities in lung function can be demonstrated. However, symptoms may appear later in adulthood. Nearly two thirds of children continue suffering from the disorder through adolescence and adulthood.

Asthma may have onset in adult life.⁴⁰ The exact proportion of such patients is not known. It is likely that exposure to allergens and development of atopy later in life are responsible for late onset of asthma. The role of viral infections in causation in adults is not clear; they may be a trigger for acute exacerbation.

The overall description of the prognosis of asthma in Ayurveda and the current practice appear to be similar.

13.8 Management

13.8.1 Ayurvedic Approach

Management of asthma in Ayurveda judiciously encompasses herbal and herbomineral drugs in addition to advising a healthy lifestyle and diet that are contrary to the cause of disease and the disease itself. As the pathogenesis of this disorder involves an imbalance between the *vata* and *kapha*, the therapy is directed at correcting this imbalance. In addition, there are a few therapies for controlling the acute symptoms.

The texts recommend that the patients should be given sudation and steaming therapy (*svedana*) after anointing their bodies with oils processed with salt.^{2–5} With this method, the solidified phlegm (*kapha*) adhering inside the channels gets liquefied and comes into the alimentary tract. The channels become soft and *vata* attains its normal downward

movement. After sweating, the patient is advised to eat food rich in fats and food that improves the secretion of mucus. In order to eliminate the pathological *kapha*, a therapeutic emesis is induced with herbs such as *Piper longum* (*pippali*). With this, channels are cleared and *vata* moves without any hindrances. Purgation therapy (*virechana*) may also help in a few patients. Because of the nature of these therapies, they should be avoided in young children and pregnant women.

The purification (*shodhan*) can also be achieved by a more elaborate process: *panchakarma*. *Panchakarma* involves therapeutic vomiting and emesis (*vamana*), purgation (*virechan*), enema (*basti*), elimination of toxins through the nose (*nasya*), and purification of the blood (*rakta mokshana*). *Rakta mokshana* may be useful in patients with allergies. This purification therapy should not be performed in debilitated individuals, the elderly, young children, pregnant women, and frail women.

Various medications are described in Ayurveda.²⁻⁵ These include single drugs and compound formulations (see [Tables 13.2](#) and [13.3](#)).^{5,41} Some of them are listed below the table.

1. *Krsnadi churna* — Powder of *Piper longum* (*pippali*), *Embolia officinalis* (*amlaka*), and *Zingiber officinalis* (*sunthi*) with honey and sugar added
2. *Bharangyadi churna* — *nagaradi churna* — Powder of *bharangi* (*Clerodendrum serratum*) and *sunthi* (*Zingiber officinalis*) with hot water or powder of *sunthi*, sugar, and *sauvarcala*
3. *Srangyadi churna* — *Pistacia chinensis* (*karkatasrngi*), *trikatu* (combination of ginger root, *pippali* berry, black pepper), *triphala* (combination of *Embolia officinalis*, *Terminalia chebula*, and *Terminalia bellirica*), *Solanum xanthocarpum* (*kantakari*), *Clerodendrum serratum* (*bharangi*), *Inula racemosa* (*puskarmula*), and five salts
4. *Parnasapancaka* — Decoction of *Tinospora cordifolia* (*guduci*), *Zingiber officinalis* (*sunthi*), *Rivea hypocrateriformis* (*phanji*), *Solanum xanthocarpum* (*kantakari*), and *Ocimum sanctum* (*parnasa*, *tulsi*) mixed with *Piper longum* (*pippali*) powder
5. Decoction of *dasamula* (a combination of ten roots: *shonyaka*, *patala*, *kasmari*, *agni-mantha*, *brhati*, *mahabalam*, *guduci*, *kakoli*, *ananta*) added with *Inula racemosa* (*puskaramula*)
6. Decoction of *Dolichos biflorus* (*kulattha*), *Zingiber officinalis* (*sunthi*), *Solanum xanthocarpum* (*kantakari*), and *Adhatoda vasica* (*vasa*) added with *Inula racemosa* (*puskuramala*)
7. Use of jaggery with equal mustard oil
8. Powder of *Pistacia chinensis* (*karkatasrngi*), *Zingiber officinalis* (*sunthi*), *Piper longum* (*pippali*), *Cyperus rotundus* (*musta*), *Inula racemosa* (*puskarmula*), *Hedychium spicatum* (*sati*), *Piper nigrum* (*marica*), and sugar taken with a decoction of *Tinospora cordifolia* (*guduci*), *Adhatoda vasica* (*vasa*), and *pancamulla*
9. Linctus made from *Curcuma longa* (*haridra*), *Piper nigrum* (*marica*), *Vitis vinifera* (*draksa*), jaggery, *Pluchea lanceolata* (*rasna*), *Piper longum* (*pippali*), and *Hedychium spicatum* (*sati*)
10. The powder of *Terminalia bellirica* (*bibhitaka*) fruit mixed with honey
11. Preparation made from ghee, milk, and the paste of *Capparis sepiaria* (*himsra*); *Embelia ribes* (*vidanga*), *Mentha spicata* (*putika*); *trikatu*; *triphala*; and *Plumbago zeylanica* (*citraka*)
12. Preparation made from ghee and paste of *tegovati*, *Terminalia chebula* (*haritaki*), *Piper longum* (*pippali*), *Strychnos potatorum* (*katuka*), *bhutika*, *Inula racemosa* (*puskara-*

TABLE 13.2**Single Drugs Used in Ayurvedic Anti-Asthmatic Preparations**

<i>Adhatoda vasica</i>
<i>Aegle marmelos</i>
<i>Alangium salviifolium</i>
<i>Aquilaria agallocha</i>
<i>Arsenic rubrum</i>
<i>Benincasa hispida</i>
<i>Bhasma</i> of horn of stag
<i>Boerhavia diffusa</i>
<i>Calotropis procera</i>
<i>Cedrus deodara</i>
<i>Garcinia pedunculata</i>
<i>Clerodendrum serratum</i>
<i>Curcuma longa</i>
<i>Curcuma zedaria</i>
<i>Datura stramonium</i>
<i>Dolichos biflorus</i>
<i>Elettaria cardamomum</i>
<i>Fagonia cretica</i>
<i>Ferula narthex</i>
<i>Glycyrrhiza glabra</i>
Gypsum
Honey
<i>Inula racemosa</i>
Jaggery
<i>Leptadenia reticulata</i>
Mica
<i>Mucuna pruriens</i>
<i>Ocimum sanctum</i>
<i>Phyllanthus embica</i>
<i>Phyllanthus urinaria</i>
<i>Piper longum</i>
<i>Piper nigrum</i>
<i>Rhus succedanea</i>
<i>Solanum xanthocarpum</i>
<i>Terminalia bellirica</i>
<i>Vitis vinifera</i>
White arsenic

Source: Modified from Goyal, H.R., *Tamaka Shwasa (Bronchial Asthma): A Clinical Study*, Central Council for Research in Ayurveda and Siddha, New Delhi, 1997.

mula), *Butea monosperma* (*palasa*), *Plumbago zeylanica* (*citraka*), *Hedychium spicatum* (*sati*), *sauvarcala*, *Cinnamomum tamala* (*tamalaki*), rock salt, *Aegle marmelos* (*bilva*), *Abies webbiana* (*talispaptra*), *Leptadenia reticulata* (*jivanti*), *Acorus calamus* (*vaca*), and *Ferula foetida* (*hinga*)

13. Preparation made from *Clerodendrum serratum* (*bharangi*) and *dasamula*, *Terminalia chebula* (*haritaki*), jaggery, honey, *trikatu*, and *trijata* (powdered) and *yavakasra*
14. Preparation made from *Dolichos biflorus* (*kulatha*), *dasamula*, *Clerodendrum serratum* (*bharangi*), jaggery, honey, *Bambusa arundinacea* (*vamsalocana*), *Piper longum* (*pippali*), and *trijata*

The above medications are likely to have both therapeutic and preventive effects. It is advised to continue any of these for a prolonged duration.

TABLE 13.3

Compound Preparations for the Management of Asthma

Name	Instructions
<i>Eladi churn</i>	1–3 with 4–6 g of honey twice/day
<i>Sitopalidi churn</i>	1–3 with 4–6 g of honey twice/day
<i>Srangyidi churn</i>	1–3 with 4–6 g of honey twice/day
<i>Talisadi churn</i>	1–3 with 4–6 g of honey twice/day
<i>Dashmula kvath</i>	14–28 ml twice/day
<i>Kantaryadi kvath</i>	14–28 ml twice/day
<i>Srmgyadi kvath</i>	14–28 ml twice/day
<i>Vasadi kvath</i>	14–28 ml twice/day
<i>Agastya haritiki avaleha</i>	12–24 g twice/day
<i>Chavanprash avaleha</i>	12–24 g twice/day
<i>Chitraka haritiki avaleha</i>	12–24 g twice/day
<i>Vasa avaleha</i>	12–24 g twice/day
<i>Vyaghriharitiki avaleha</i>	12–24 g twice/day
<i>Draksharishta</i>	14–28 ml with equal quantity of water twice/day after meals
<i>Vasarishta</i>	14–28 ml with equal quantity of water twice/day after meals
<i>Kanakasva</i>	5–10 ml with equal quantity of water twice/day after meals (due to its potency, caution is advised)
<i>Abhakra bhasma</i> (plain)	120–150 mg with honey twice/day
<i>Maygrapuccha bhasma</i>	1–2 g with honey twice/day
<i>Apamarga ksara</i>	1 g with warm water twice/day
<i>Arka lavana</i>	1 g with warm water twice/day
<i>Khadiradi vati</i>	2–4 vati three times/day
<i>Lavangadi vati</i>	1 vati six times/day

Source: Modified from *Swas roga*, in *Handbook of Domestic Medicine and Common Ayurvedic Remedies*, Central Council for Research in Ayurveda and Siddha, 1999. These preparations may be obtained from major Ayurvedic pharmaceutical companies in India.

For controlling acute symptomatology, inhalation of medicated smoke from the following preparations have been described in Ayurveda^{2–4}:

1. Inhalation of the smoke of the fruit, the stem, and leaves of *Dhatura fastuosa* from a *hooka*. A paper dipped in water in which saltpetre has been dissolved is dried in the sun and rolled up in the form of a cigar.
2. Paste of *Cedrus deodara*, *Sida cordifolia*, and *Nardostachys jatamansi* is dried in sun, laced in ghee. Patient inhales the smoke from a hollow stick made out of this.
3. Leaves of *haridra*, root of *Ricinus communis* (*eranda*), *Cocculus lack* (*laksa*), *Psidium guajava* (*manassial*), *Cedrus deodara* (*devadaru*), *Elettaria cardamomum* (*ela*), and *mamsi* are all macerated and made into cigarette, which is smeared with ghee and smoked.

After alleviating the intensity of asthmatic breathing, the following syrups may be used:

1. Peacock feathers are reduced to ashes in a slow fire. They are mixed with a quantity of fruit of *Piper longum* reduced to powder. A syrup is then made with the aid of honey. If licked occasionally, it alleviates the intensity of asthmatic breathing.
2. Turmeric (*Curcuma longa*), black pepper (*Piper nigrum*), *Uvae passae*, old molasses, *Vanda roxburghii*, *Piper longum*, and *Circuina zerumbet* are reduced to a powder and mixed with mustard oil.
3. A decoction of *Tinospora cordifolia*, dry ginger, *Siphonanthus indica*, *Solanum xanthocarpum*, and *Ocimum sanctum* mixed with powdered *Piper longum* is made.

4. *Parnasapancaka* — A decoction of *guduci*, *sunthi*, *phanji*, *kantakari*, and *parnasa* (*tulsi*) mixed with *pippali* powder is made.
5. A syrup made out of *Curcuma zedoria* (*kachur*), *Inula racemosa* (*pushkarmool*), *Citrus decumona* (*amla vetas*), *Elettaria cardamomum* (*choti Elaichi*), *Ferula narthex* (*hingu*), *Ocimum sanctum* (*tulsi*), *Aquilaria agallocha* (*agar*), *Phyllanthus urinaria* (*bhumya-malki*), *Leptadenia reticulata* (*jeewanti*), and *Santalum album* (*chanda*) is recommended as a bronchial antispasmodic.

Although inhaling the smoke of various herbs may cause relief through the bronchodilatory effect of the constituents such as *dhatura*, this practice has not been evaluated scientifically to recommend its routine usage. In addition, the smoke may actually worsen the bronchospasm. This form of therapy is neither feasible in children nor advisable.

13.8.1.1 Precautions

The drugs, diet, and practices that aggravate the disease should be avoided. These include dust, smoke, residing in cold places, excessive use of cold water, seasonal changes, excessive walking, excessive use of dry foods, astringent food, irregular dietary habits, indigestion, trauma to vital organs, and habitual use of *lablab*-bean, black gram, til paste, and other *kapha*-producing articles.

13.8.1.2 Diet

Foods and drinks that restore the normal course of *vata* are useful in treating asthma. If the *vata* is greatly excited, syrup made up from old tamarind pulp is helpful. Sugar candy with lemon (*Citrus medica*) juice is beneficial. Light foods should be eaten at night. Heavy and rich foods, which are difficult to digest, foods that are dry, curds, fish, and chillies should all be avoided.

13.8.1.3 Lifestyle

Staying awake at night, exercising, labor, exposing oneself to the heat of the sun or fire, and anxieties, grief, wrath, and everything that disturbs peace of mind should be avoided. A healthy lifestyle would have a preventive role.

13.8.1.4 Breathing Exercises

Breathing exercises, particularly *pranayam*, reduce the frequency and severity of symptoms,^{42–44} improve exercise tolerance, and enhance lung function. Two systematic reviews have highlighted the need for studying the beneficial aspects of various breathing exercises.^{45,46}

13.8.1.5 Meditation

Meditation helps in reducing the stress and may check recurrence. *Sahaja* yoga is an Indian system of meditation based on traditional yogic principles, which may be used for therapeutic purposes. Clinical trials of this therapy in patients with asthma have found evidence of improvement in lung function and reduced frequency of exacerbations.⁴⁷

Some commercial Ayurvedic formulas are also available that may be useful in asthma. Examples include Asmon (Herbochem Remedies, Kolkata) and Asmakure (Herbicure Pvt. Ltd., Bishanpur, West Bengal).

TABLE 13.4

Stepwise Treatment of Asthma

Steps	Long-Term Prevention
Step 4: Severe persistent	Inhaled short-acting beta-agonist as required; inhaled corticosteroids: Budesonide and Beclomethasone (400 µg twice/day) increase up to 2000 µg/day in selected cases; long-acting bronchodilator: long-acting inhaled β ₂ -agonist or sustained release theophylline; corticosteroids tablets: low dose on alternate days (if no relief with above treatment)
Step 3: Moderate persistent	Inhaled short-acting beta-agonist as required; inhaled corticosteroids: Budesonide and Beclomethasone (400–800 µg divided twice/day); long-acting bronchodilator (if needed): long-acting inhaled β ₂ -agonist salmeterol (50 µg once or twice/day or sustained release theophylline)
Step 2: Mild persistent	Inhaled short-acting β-agonist as required; inhaled corticosteroids: Budesonide and Beclomethasone (200–400 µg) or cromolyn or sustained release theophylline or leukotriene modifiers
Step 1: Intermittent	Inhaled short-acting β-agonist as required for symptoms relief; if they are needed more than three times/week, move to step 2

Note: If Fluticasone is used, the dose is half that of Budesonide/Beclomethasone.

13.8.2 Conventional Medicine Approach

There is no cure available for asthma in Western medicine. However, appropriate management leads to control of the disorder. The goals of management are to:

1. Control the symptoms.
2. Prevent acute exacerbations.
3. Maintain normal or near normal pulmonary functions.
4. Maintain normal levels of activity.
5. Avoid adverse effects from medications.
6. Prevent mortality due to asthma.

Currently, therapy of asthma is guided by the fact that it is a chronic inflammatory airway disorder; the control of airway inflammation is the key to effective control. Except for mild intermittent asthma, the therapy includes regular use of anti-inflammatory medication and bronchodilators (as required). In addition, environmental control to avoid exposure to certain risk factors should improve the control of symptoms.

The first step in management after the diagnosis is made is the correct assessment of severity of asthma (Table 13.1).

The pharmacological therapy of bronchial asthma involves the use of drugs that relax smooth muscle and dilate the airways and drugs that decrease inflammation and prevent exacerbations. The medications used for long-term treatment of asthma include bronchodilators, steroids, mast-cell stabilizers, leukotriene modifiers, and theophylline.^{48–75} Table 13.4 shows the treatment plan according to disease severity. The use of bronchodilators alone in persistent asthma is not recommended, as this does not control the airway inflammation and there is a false sense of security.

13.8.2.1 Immunotherapy

This therapeutic mode consists of gradually giving increasing quantities of an allergen extract to a clinically sensitive subject to ameliorate the symptoms associated with subse-

quent exposure to a causative allergen. This is considered only occasionally in highly selected children who are sensitive to a specific allergen such as grass pollen, mites, etc. It is done only under specialist supervision and must usually be given for 3 years.⁷⁶ Some studies suggest that specific immunotherapy may induce a diminution of nonspecific bronchial hyperresponsiveness and enable reduction of symptomatic treatment.^{77,78}

13.8.2.2 Management of Acute Exacerbation

Severe exacerbations of asthma are life-threatening medical emergencies requiring hospital-based care. The aims of treatment are to reverse airflow obstruction and hypoxemia as rapidly as possible. The severity of an acute exacerbation can be judged by clinical symptoms and signs, lung function tests, and arterial blood gas analysis. Rapid-acting inhaled bronchodilators, early introduction of systemic glucocorticoids, and supplemental oxygen are the mainstay of treatment of acute exacerbations. Close monitoring of the patient's condition and response to therapy are mandatory. In addition, a plan should be formulated to prevent future relapses.

13.8.2.3 Complications of Therapy

Use of systemic (oral, parenteral) steroids over a prolonged period is associated with significant side effects. However, the current day management relies more on inhalation drugs. The main concern with the use of inhalation steroids is the effect on growth. An approximate 20% reduction in the growth velocity during the first year of treatment with inhaled steroids is reported. Subsequently, the growth velocity recovers and children ultimately attain predicted adult height.^{56,57} Studies have also reported that children treated with inhaled steroids were more likely to reach predicted adult height than children whose asthma was not treated with preventive medication.^{57,58} These findings show that the concern about adverse effect of inhaled steroids on growth is inappropriate.

The complications associated with beta-2 agonists are cardiovascular stimulation, skeletal muscle tremor, hypokalemia, and irritability. These adverse effects are more commonly seen with oral drugs than with inhaled medications.

If asthma severity is assessed properly and inhaled medications are used judiciously, the benefits far outweigh the risks.

13.9 Scientific Basis

A large number of single drugs have been described for use in asthma (Table 13.2).⁴ In addition there are quite a few compound preparations available (Table 13.3).⁴¹ There is a great deal of scientific literature available that supports the use of Ayurvedic preparations in the management of asthma. Most of the data are experimental. These studies highlight the anti-inflammatory and bronchodilatory activity in various medications. The following list shows the evidence available for various herbs:

1. *Picrorrhiza kurroa* — *P. kurroa* is a widely used herb in the Ayurvedic system. It belongs to the Scrophulariaceae family, and the active constituents are obtained from the root and rhizomes.⁷⁹ It is traditionally used in treatment of respiratory conditions such as asthma and bronchitis. Studies on an alcoholic extract have shown antioxidant and anti-inflammatory effects.⁸⁰ In animal studies, antiallergic,

antianaphylactic, anti-inflammatory, and immunomodulatory activities have been demonstrated. The active ingredients include pikuroside II, picroliv (containing iridoid glycoside fraction), and androsin.⁸¹⁻⁸⁶

2. *Adhatoda vasica* — This is widely used in treatment of respiratory tract ailments. Alkaloids from this herb have been shown to possess anti-inflammatory and antiallergic properties.^{85,87,88} In addition, its extract has an antitussive effect.⁸⁹
3. *Albizia lebbek* — The decoction of the bark of *A. lebbek* is used in treatment of asthma and eczema. Studies on animals show that *A. lebbek* has a significant cromoglycate-like action on the mast cells. It also appears to inhibit the early processes of sensitization and synthesis of reaginic-type antibodies.^{90,91}
4. *Solanum* species — Powder of or a decoction made from the whole dry plant of *Solanum xanthocarpum* or *Solanum trilobatum* is used in the treatment of asthma and other respiratory disorders. The mechanism of action in asthma may involve bronchodilation, reduction of bronchial mucosal edema, and reduction of airway secretions.⁹²⁻⁹⁴
5. *Tylophora indica* — *T. indica* is widely used in Ayurvedic medications to provide relief to patients with bronchial asthma. Studies have shown that alkaloids of *Tylophora indica* suppress cellular immune response.⁹⁵
6. *Cedrus deodara* — The wood oil of *C. deodara* has been shown to have anti-inflammatory activity. This can be attributed to its mast-cell stabilizing activity and the inhibition of leukotriene synthesis.⁹⁶
7. *Boswellia serrata* — The gum resin of *B. serrata* contains boswellic acids which have been shown to inhibit biosynthesis of leukotrienes.⁹⁷
8. *Phyllanthus urinaria* — The extracts of the stems, leaves, and roots have been shown to have a relaxant effect on the respiratory tract smooth muscle; involvement of adenosine tri-phosphate- (ATP) sensitive potassium channels is postulated.⁹⁸
9. *Aquilaria agallocha* — Aqueous extracts of stems of *A. agallocha* have been shown to inhibit immediate hypersensitivity reaction by inhibition of histamine release from mast cells.⁹⁹
10. *Calotropis procera* — The latex of *Calotropis procera* has anti-inflammatory property demonstrated in a rat paw edema model.¹⁰⁰
11. *Elettaria cardamomum* oil — Anti-inflammatory and antispasmodic properties have been demonstrated in a study on rats.¹⁰¹
12. *Ocimum sanctum* — Extracts of this widely used plant have been shown to possess immunomodulatory potential and antioxidant and cyclooxygenase inhibitory properties.^{102,103} Its fixed oils can inhibit enhancement of the vascular and capillary permeability and leukocyte migration after inflammatory stimulus.¹⁰⁴
13. *Piper longum* — Piperine, isolated from this plant, has anti-inflammatory potential.¹⁰⁵

13.9.1 Review of Clinical Trials

13.9.1.1 *Picrorrhiza kurroa*

Doshi et al.¹⁰⁶ studied the efficacy of *P. kurroa* in a randomized, crossover, double-blind trial. They enrolled 72 patients (ages 14 to 60) for a 14-week study. The patients were given either *P. kurroa* root powder (300 mg three times/day) or an identical placebo. The study

had three arms: (1) group A had a long duration of 12 weeks and an active drug was given from week 3 to week 14, (2) group B had a short duration of 3 weeks and an active drug was given from week 3 to week 6 with placebo during the rest of the period, and (3) group C had an intermediate duration of 6 weeks and an active drug was given from week 3–6 and 9–12 with a placebo during intervening periods. The patients were asked to maintain a symptom diary. Weekly pulmonary function tests were performed. The authors did not observe any significant reduction in clinical exacerbation, need for bronchodilators, or improvement in pulmonary function. Fifty patients dropped out of study at different points during the trial. Significant side effects were seen in ten patients — four experienced vomiting, three had anorexia, two had diarrhea, two experienced itching, one had a skin rash, and one experienced giddiness. As this preparation is used frequently, there is need for further studies.

13.9.1.2 *Solanum spp.*

Govindan et al.¹⁰⁷ studied the efficacy of this herb in bronchial asthma. They enrolled 60 adults with bronchial asthma. Twenty patients each received 300 mg of dry powder of *S. xanthocarpum* or *S. trilobatum*, whereas 10 patients each received salbutamol 4 mg or deriphylline 200 mg. Pulmonary function tests were performed before and 2 h after drug administration. *S. xanthocarpum* and *S. trilobatum* increased FEV₁ by 65 and 67%, respectively, at 2 h. This effect was less than that with salbutamol or deriphylline. Subjective relief lasted 6 to 8 h.

Similar results have been reported in other studies as well; the response rates were poor.^{108,109} In these studies the duration for which the drug was given was small; these results may lead to an underestimation of the effects because most Ayurvedic herbal preparations have their best action after about 2 weeks.

13.9.1.3 *Tylophora indica*

Shivpuri et al.¹¹⁰ conducted a double-blind, crossover study in 110 patients over 10 years old. Fifty-three patients in the control group ate one spinach leaf daily. At the end of week 1, 62% in the *T. indica* group had moderate to complete relief of symptoms compared with 28% in the control group. At the end of the 12-week study period, improvements in the two groups were 16 and 0%, respectively.

In another double-blind, crossover study,¹¹¹ 195 asthmatic patients received either alcoholic tincture of *T. indica* or a placebo for 12 weeks. A daily diary of symptoms scores was maintained. After a week, 56% of the patients in *T. indica* group had moderate to complete improvement in symptoms compared with 31.6% in the placebo group. After the crossover, 34.2% had improved with *T. indica* and 13.5% with placebo. At the end of study period, 14.8% in the *T. indica* group and 7.2% in the placebo group improved.

The efficacy of alkaloids extracted from *T. indica* was studied in a double-blind trial of 123 patients.¹¹² The study group received alkaloid extract from *T. indica* in glucose, whereas the control group received glucose colored with spinach. Lung-function tests were evaluated along with symptom scores. The percentage of patients in whom FEV₁ improved by more than 15% was significantly greater in the study group than the control at 1, 2, 4, 8, and 12 weeks, peaking at 4 weeks. The symptom scores were significantly better in the study group, with the peak at 1 week.

Thiruvengadam et al.¹¹³ studied the efficacy of dried *T. indica* in 30 asthmatic patients. The patients were enrolled into a four-arm, double-blind randomized clinical trial for 16

days. Dried *T. indica* powder was compared with standard drugs and a placebo. Pulmonary function tests and symptom scores were evaluated. Among all the parameters, nocturnal dyspnea was the only one that showed significant improvement with *T. indica*.

A placebo-controlled double-blind study¹¹⁴ on 135 asthmatic patients was conducted to evaluate the efficacy of powdered *T. indica*. The drug or placebo was given for 1 week with another 2 weeks of follow-up. Pulmonary function tests and symptoms scores were evaluated. There were no differences between the two groups.

13.9.1.4 *Boswellia serrata*

The gum resins of *B. serrata* (*salai guggul*) have been used in asthma. Gupta et al.¹¹⁵ compared the effect of gum resins of *B. serrata* with placebo in a double-blind, randomized trial on 80 adult asthmatic patients. Pulmonary function tests were evaluated serially. A significant increase in FEV₁ was reported in the test group compared with the placebo group.

13.9.1.5 *Miscellaneous Herbs*

Iyenger et al.¹¹⁶ studied the effect of a combination of five plants — *Adhatoda vasica*, *Solanum xanthocarpum*, *Albizzia lebbek*, *Glycyrrhiza glabra*, and *Picrorrhiza kurroa* — in 14 adult patients with asthma. All the patients showed clinical improvement with prevention of recurrence and reduction in severity of symptoms. However, the medication was not effective during acute exacerbation.

Shanker et al.¹¹⁷ conducted a clinical trial on 15 patients with bronchial asthma using *Gardenia turgida* and *Gardenia latifolia*. The results were unimpressive.

Sharma et al.¹¹⁸ reported significant improvement in one third of the 15 patients they treated with *Euphorbia prostrata* for a period of 2 weeks.

Swamy et al.¹¹⁹ evaluated the efficacy of *sirisa twak kvatha* (*Albizzia lebbek*) in 19 patients with asthma. They reported significant improvement in symptoms in most patients.

Trivedi et al.¹²⁰ reported bronchodilatory, antispasmodic, and antiasthmatic effects of *vibhitakphal churna* (*Terminalia bellirica*) in a trial on 93 patients.

In another trial,⁵ 240 cases of bronchial asthma were administered *naradeeya lakshmi vilas rasa* and *godanti bhasma* in a dose of 0.5 and 1.0 g, respectively, three times/day with honey for a varying period of 4 to 12 weeks. Ninety-six patients received 100% relief, 27 patients received 75% relief, 50 patients received 50% relief, 18 cases received 25% relief, and 28 remained unchanged; 21 cases left against medical advice. In the second group of this study, 210 cases received *shvasa kesari* tablets (500 mg) made from *Solanum xanthocarpum* and *Godanti bhasma* three times/day for 2 to 12 weeks. Seventy-five cases showed complete relief, 49 cases showed 75% relief, 26 cases showed 50% relief, 25 cases showed 25% relief, and 24 did not respond to therapy; 10 cases left against medical advice.

As can be judged from these studies, most trials have demonstrated some benefit. When analyzing the available literature and Ayurvedic texts, we conclude that more clinical research is still required before these therapies can be routinely used. Although there are good experimental data to justify use of Ayurvedic preparations in asthma, issues about doses, combinations, and duration of therapy are still to be resolved. In addition, these preparations do not seem to offer effective medications for acute exacerbations of asthma, which can be life threatening. There are plenty of Ayurvedic preparations with anti-inflammatory and immunomodulatory effects (vide supra). Further research is required to determine the optimal combinations that will help in reducing the airway inflammation and therefore lead to better control of symptoms. It appears that a judicious mix of Ayurvedic

medications and modern medicine may be able to improve the control of this common respiratory disorder.

13.10 Areas of Research

Further research is needed to:

1. Establish efficacy of various Ayurvedic preparations used in asthma by good quality randomized controlled trials.
2. Determine the active principles in various Ayurvedic medications.
3. Evaluate the impact of Ayurvedic medications on the natural history.
4. Evaluate the role of immunomodulator medications available in Ayurveda.

13.11 Conclusions

The review of literature highlights need for more research in utility of Ayurvedic preparations in bronchial asthma. At present, the medications and the management are not well standardized. As a large number of experimental studies have documented the presence of anti-inflammatory properties in various Ayurvedic preparations, there is potential for discovering new compounds useful in the management of asthma.

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14

Hepatic Disorders

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14.1 Introduction

In Ayurvedic texts (or Ayurveda), the liver is called *yakrut* or *yakruta* or *yakrit*. The word *yakrut* has two verbs, “*ya*” and “*krut*.” *Ya* denotes activity and *krut* denotes several meanings such as “to breakdown” or “to be instrumental” in performing any activity. The liver is an organ that is continuously engaged in the activity of breaking down various stimuli (e.g., food, water, air, or other sensory stimuli) and is instrumental in sustaining the process of life. In some instances, the word *yakrut* is also used as a synonym to the word “restoration” because it helps bring back damaged body tissues to a normal physiological state.¹ Ayurveda describes the liver as a center of origin (*mula sthana*) of the circulatory system (*raktavaha srotas*).

In Ayurveda, the role of liver is explained in relation to pathogens (*dosa*), tissues (*dhatu*) and its development, muscle (*mamsa*), heart (*hridroga*), blood (*rakta*), respiration (*pranavaha srotas*), and excretion (*mala*). The liver may be affected in the diseases involving these systems.

In conventional medicine, the liver is known to play a central role in the maintenance of metabolic homeostasis by its involvement in carbohydrate, lipid, and protein metabolism. It converts sugar into glycogen, carbohydrates and proteins into fats, toxic ammonia into nontoxic urea, etc. It produces bile, blood coagulating and anti-coagulating factors, proteins, and enzymes. It stores critical trace elements and vitamins and is responsible for detoxification and elimination of various toxins, carcinogens, nitrogen-containing waste products, and alcohol.

The maintenance of a healthy liver is vital to overall health and well-being. Unfortunately, environmental toxins, poor eating habits, alcohol consumption, and therapeutic drug use often abuse this vital organ, and as a consequence there is an overall decline in metabolic functions of the liver. This hepatotoxicity eventually leads to serious diseases like hepatitis, cirrhosis, alcoholic liver disease, and ultimately results in hepatic cancers (liver tumors).

Hepatic diseases like acute or chronic hepatitis and cirrhosis are some of the major causes of significant human mortality, which may result from a wide variety of viral infections and a range of toxins including alcohol. Although the etiology and pathology of hepatic disorders are not clearly defined in detail in Ayurvedic texts, symptoms of hepatic diseases are well described. The symptoms include increased serum bilirubin (*pittavridhi*), decreased hepatic uptake (*yakruta dhatwagni manya*), decreased hepatic conjugation (*saman vikruti*), hepatocellular damage (*yakruta shotha*), biliary stasis (*sang*), steo-stasis (*medomay, yakrut medoj siragranthi*), and cirrhosis (*yakrutshosh*).² Modern diagnostic tools presented in [Table 14.1](#) are widely used by Ayurvedic practitioners currently for accurate and speedy diagnosis of hepatobiliary disorders and to understand the underlying mechanisms of the diseases.

The aim of this chapter is to provide information on the Ayurvedic treatment of hepatitis, hepatic coma, ascities, cirrhosis, and liver cancer. Scientific information available on Ayurvedic treatments is summarized.

14.2 Hepatitis (*Kamala*)

According to Ayurveda, hepatitis is a disease of the circulatory system and is categorized under biliary (*pitta*) diseases; *ranjaka pitta* is the type of biliary fluid involved in this

TABLE 14.1
Tests for Hepatobiliary Disorders

Test (Reference Range)	Diagnosis
<i>Biochemical Analysis</i>	
Alanine aminotransferase (up to 35 IU/l)	Elevation indicates degree of inflammation and possible causes for the disease
Aspartate aminotransferase (6–38 IU/l)	
Alkaline phosphatase (53–128 IU/l)	Elevation suggestive of obstructive liver diseases and liver cancer γ-GT is an indicator of alcohol usage
Gamma glutamyl transpeptidase (<50 IU/l)	
Total bilirubin (0.2–1.0 mg/dl)	Elevation indicates diseases of the liver and bile ducts
Direct bilirubin (0–0.2 mg/dl)	
Albumin (3.5–5.0 g/dl)	Decreased level suggests chronic liver disease and liver cancer, particularly worse cases, though a decrease is also noticed in generalized protein deficiency conditions (e.g., kwashiokar and marasmus)
Alphafetoprotein (up to 10 ng/ml)	
Abnormal clotting studies	Specifically rises to 500 ng/ml during hepatocarcinoma
Antibody test (blood sample)	Suggestive of worsening chronic liver disease
Hepatitis B DNA (blood sample)	Indicates hepatitis A, B, or C
Hepatitis C RNA (blood sample)	Indicates hepatitis B
Bile duct imaging	Indicates hepatitis C
HFE gene analysis for C282Y mutation	Indicates primary biliary cirrhosis
Liver biopsy	Indicates hemochromatosis
<i>Imaging Modalities^a</i>	
Plain abdomen x-ray	Indicates nonalcoholic steatohepatitis, chronic hepatitis B or C, primary biliary cirrhosis, autoimmune hepatitis, alcoholic liver disease
Oral cholecystography	Detects gallstones, calcification
IV cholangiography	Gives size, shape, position, and function of gall bladder
Percutaneous transhepatic cholangiography	Visualizes bile duct
Endoscopic retrograd cholangiography	Detects obstructive jaundice
Ultrasound scanning	Confirms obstructive jaundice
CT scanning	Diagnoses cysts, abscesses, neoplasm, degenerative changes, and metastatic deposits
Radionucleotide or isotope scanning	Diagnoses malignant tumors, focal nodular hyperplasia, hemangioma, cysts, and abscesses
MRI scanning	Detects and estimates metastatic lesions
	Produces images of gall bladder and biliary duct

^aDole, V.A., *Ayurved and Hepatic Disorders*, Sri Satguru Publications, Delhi, 2001, 71.

pathogenesis.³ The liver secretes *Pachaka pitta* (more than 500 cc in a day) and is stored in the gall bladder (*pittashaya*). The stored bile gets reabsorbed and leaves a fraction of original bile (*tyakta drava pitta*). The concentration of original bile in the circulation is critical and any derangement leads to diseases arising out of weak digestive and metabolic activity (*agni vaishamy*). The more dilution of bile in the gall bladder results predominantly in symptoms like nausea, vomiting, and fever. When the concentration is too high, it leads to symptoms like burning sensations, thirst, profused sweating, giddiness, and hemorrhagic conditions (Figure 14.1).¹

Ayurveda classifies two basic types of hepatitis: hemolytic jaundice (*kostashakhasrita kamala*, *paratantra kamala*, or *bahupittaja kamala*) and obstructive jaundice (*shakhasrita kamala*). Hepatitis could also be intrahepatic (*swatantra*) or extrahepatic jaundice (*paratantra*). The other minor types of hepatitis are chronic hepatic failure (*kumba kamala*), fulminant hepatic failure (*halimaka*), and hepatorenal syndrome (*panaki*).³ Hemolytic hepatitis is associated with moderate to severe anemia. Accelerated destruction of red blood cells and hemoglobin (increased *malarooapa ranjaka pitta*), due to the diet and lifestyle that increase *pitta* (*pittavardhaka ahara* and *vihara*), leads to enhanced bilirubin formation. The obstructive hepatitis is due to the obstruction in the biliary flow. Water (*kapha*) or fluid accumulation obstructing the passage channels of biliary system results in this manifestation. The stools will be clay colored (*tilapista nibha*) until bile comes to the intestine (*kosta*).³

Conventional medicine considers hepatitis as one of the most common liver diseases. Hepatitis is caused by agents like viruses (hepatitis A, B [HBV], C [HCV], D, E, G, cytomegalovirus [CM virus], Epstein-Barr virus [EB virus], yellow fever), parasites (*E. histolytica*), bacteria (leptospirosis, cholangitis, septicemia), drugs, and toxins. Although these agents can be distinguished by their antigenic properties, clinically they all are generally manifested as jaundice along with some associated constitutional symptoms. Hepatitis may be contacted through the intake of viral contaminated food stuff, virus-infected blood, serum, saliva, sexual contact, urine or feces, blood transfusion, or fecal-contaminated food stuff.

Viral infection in liver causes destruction of parenchymal cells at the deoxyribonucleic acid (DNA) level, which leads to hepatocellular failure, disturbances in nitrogen metabolism, inflammation, necrosis, and cirrhosis resulting in the obstruction of bile-canaliculi. This obstruction leads to resorption of bile into the peripheral bloodstream through the hepatic vein and lymphatics. The process then causes the serum bilirubin level to rise above 2 mg% and is sufficient to be termed clinically as jaundice.⁴

14.2.1 Treatment

Removing biliary tract blockade during hepatitis by purgatives forms the basis for its treatment. The elimination of vitiated pathogens (*shodana* approach of *panchakarma*), preceded by curative along with liver regenerative therapy (*snehana*), forms the primary treatment protocols. Supplementary treatment includes the administration of a daily dose of purgatives, chologogue drugs like *katuki*, and medicated ghees. *Samprapti vighatana* treatment strikes at the root of the disease process by destroying and neutralizing the inflammation, removing the obstruction, and bringing the pathogens to the intestine. Viral hepatitis patients were treated with *pancha gavyam ghritam* (5 g B.D.) and *Katuka rohini* (1 g) daily in the morning with milk and *bhumi amla swarasa* (50 ml/day). For nasal infiltration (*nasya karma*), *katu tumbi jal* was used. *Netranjan* has been used as *jyotishmati swarasa*. After 4 to 6 weeks of treatment, encouraging results were obtained and clinical symptoms of jaundice completely disappeared.⁴

This *panchakarma* treatment is more appropriate for hemolytic jaundice than hepatocellular jaundice. This is because a patient with a poor appetite and general disability would not be able to tolerate the above-mentioned exhaustive procedures.³ In such cases, hepatoprotective drugs would be beneficial.

14.2.2 Herbal Treatment of Hepatitis

Two reviews have been published so far on hepatoprotective Ayurvedic drugs. The first one gives extensive review of the experimental and clinical research on the

TABLE 14.2
Hepatoprotective Ayurvedic Herbal Therapies

Botanical Name	Liver Disorders	Type of Study	Ref.
<i>Acacia catechu</i>	Viral hepatitis	Clinical	47
<i>Adhatoda vasica</i>	Liver diseases	Clinical	47
<i>Aegle marmelos</i>	Hepatitis B	Clinical	7
<i>Aloe vera</i>	Hepatoma	Yoshida AH-130 ascite hepatoma cell line	48
<i>Anacardium occidentale</i>	Hepatoma	Hepatoma 129 cell line	49
<i>Azadirachta indica</i>	Hepatitis B	Clinical	47
<i>Bacopa monnieri</i>	Viral hepatitis	Clinical	7
<i>Chichorium intybus</i>	Chronic hepatitis	Clinical	50
<i>Citrullus lanthus</i>	Viral hepatitis	Clinical	7
<i>Embolia officinalis</i>	Hepatomegaly hepatitis C	Clinical	51
<i>Fummaria parviflora</i>	Hepatitis C	Clinical	52
<i>Glycyrrhiza glabra</i>	Hepatoma	<i>In vitro</i>	53
<i>Gynandropsis pentaphylla</i>	Hepatoma	Hepatoma 129 cell line	49
<i>Phyllanthus niruri</i>	Hepatitis B	<i>In vitro</i>	54
<i>Plumbago zeylanica</i>	Hepatoma	Animal studies	55
<i>Saussurea lappa</i>	Hepatoma	HepG2 cell line (human hepatoma cell line)	56
<i>Solanum nigrum</i>	Viral hepatitis	Clinical	47
<i>Sphaeramthus indicus</i>	Viral hepatitis	Clinical	7
<i>Tephrosia purpurea</i>	Viral hepatitis	Clinical	7
<i>Vitex nigundo</i>	Hepatitis B	Clinical	50
<i>Withania somnifera</i>	Hepatoma	<i>In vitro</i>	49

hepatoprotective effects of medicinal plants and their preparations.⁵ The second review gives a compilation of the information on various studies on these plant medicines.⁶ The herbs with their scientifically proven hepatoprotective property are discussed in the end of the chapter and also listed in Tables 14.2 and 14.3. These herbs also possess one or more properties, including antiviral, cholorectic, regeneration of hepatocytes, antifibrotic, etc.⁷ Individual treatment profiles for various types of hepatitis presented in Table 14.4 will help practitioners choose a better drug for these dreadful diseases. Classical Ayurvedic formulations for diseases related to circulatory system, liver ailments, and jaundice are listed in Table 14.5. The Ayurvedic formulas commercially available for the treatment of hepatic disorders are presented in Table 14.6.

14.3 Ascites (Udaram, Jalodhar, Asadhya)

According to Ayurveda, the accumulation of abnormal toxic fluids around the liver creates blockage (*srotorodh*) that inhibits the secretion of bile resulting in liver enlargement. Impaired digestive power (*agni*) has been considered the basic etiological factor for this disease and clinical features include abdominal distension, weakness, sluggish passage of feces, and flatus. The successful treatment depends on the treatment of underlying causes. *Phalatrikadi kasaya* formula has the potential to remove a toxic blockage of the liver. The herbs *triphal*, *katuki*, and *chira* have the properties to clear off the abnormal pathogens. Constipation associated with ascites may be relieved by *Katuki*. Necessary detoxification

TABLE 14.3

Ayurvedic Herbs Shown to Have Antitumor Activity

Research Protocol	Therapeutic Hepatoprotective Properties	Ref.
<i>Andrographis paniculata</i> (Kalmegh) Active constituent: Andrographolide		
Paracetamol-induced toxicity on isolated rat hepatocytes	Increases the viability percentage of the hepatocytes; antagonizes the toxic effects of paracetamol on liver function enzymes	57
Carbon tetrachloride (CCl ₄) and galactosamine toxicity in rats	Stimulates hepatic regeneration and increases resistance to damage by toxins; activates reticuloendothelial system	58
Animal tumors	Antihepatocarcinogenic and enhances carcinogen detoxification by the regulation of antioxidant defense system and microsomal drug metabolism	59
<i>Annona atemoya/muricata</i> (Sitaphala) Active constituent: Bullatacin		
<i>In vitro</i> studies in 2.2.15 cells, human hepatocarcinoma cell line	Induces cell death due to apoptosis, preceded by cell blebbing, chromatin margination, and condensation	60
Animal studies	Inhibits mitochondrial electron transport and NADH oxidase activity; arrests abnormal cell growth by inhibiting oxidative phosphorylation and lowering ATP levels	61
<i>Boerhavia diffusa</i> (Punarnava) Active constituent: Punarnavine alkaloid		
CCl ₄ -induced toxicity in rats	Shows strong choleretic activity and increases normal bile flow	62
Animal studies	Pharmacological potency depends on the morphological features and time of collection of herb; plant roots of 1–3 cm diameter, collected in the summer, exhibited very high hepatoprotective effect	63
<i>Eclipta alba</i> (Bhringaraj) Active components: Wedelolactone and demethyl-wedelolactone		
<i>In vitro</i> studies in rat hepatocytes	Anti hepatotoxic and stimulates liver cell regeneration	64
CCl ₄ -induced hepatotoxicity in rats	Regulates GSH levels, hepatic drug metabolism, and activities of lysosomal enzymes	54
Clinical trials — hepatitis patients	Used as a cholagogue and as deobstruent; down regulates HBAg	
<i>Piper longum</i> (Pippali) Active constituent: Piperine		
CCl ₄ -induced toxicity in animals and culture	Reduces lipid peroxidation by increasing glutathione levels	59
<i>Terminalia chebula</i> (Haritaki) Active constituent: Tannins		
Cholesterol-induced hypercholesterolemia in rabbits	Hypocholesterolemic effect	65
Clinical trials	Used in many anticancer formulations for the treatment of liver enlargement	

TABLE 14.3 (continued)
Ayurvedic Herbs Shown to Have Antitumor Activity

Research Protocol	Therapeutic Hepatoprotective Properties	Ref.
<i>Semecarpus anacardium</i> Linn. (Bhallataka)		
Animal studies — Aflatoxin-induced hepatocellular carcinoma in rats	Gradual increase in body weight; decreases bilirubin and abnormal nucleic acid content; normalizes cancer marker activities, lysosomal enzymes, glycoprotein content, immunosuppression, and hyperlipidemia; reduces alpha-fetoprotein to normal range and regulates abnormal mineral metabolism; cures hypoglycemia by activating gluconeogenic enzymes, thereby increasing the synthesis of glucose; controls abnormal lipid peroxidation and maintains antioxidant defense status of the host; as a bifunctional inducer induces both phase I and phase II biotransformation enzymes and causes carcinogen detoxification; prevents tumor initiation by the metabolic activation of carcinogens; replaces necrotic tissues by newly regenerated hepatocytes	66–73
Clinical trial — <i>Anacartin forte</i> an Ayurvedic preparation containing <i>Semecarpus anacardium</i> nuts and seeds in the proportion of 1:200 (90 g/day of divided doses for 3 months)	Gives subjective and objective improvement, alleviates troublesome symptoms, increases survival time and causes complete regression of hepatocarcinoma; patient recovers by gaining weight and appetite; normalizes liver abnormalities like tenderness and enlargement; blood count and liver function tests become normal	74
Ayurvedic formulation containing <i>Semecarpus anacardium</i> nuts, <i>Amura rohitaka</i> , <i>Glycyrrhiza glabra</i> Linn., and <i>Tamra bhasma</i>		75
Animal studies — C ₃ H jax strain breast tumor-bearing mice	Inhibits tumor development and increases survival period; tumor specific γ -amino butyric acid level decreases to normal	
Clinical studies — cancer patients (1.5 g of divided doses)	Response to this formulation was higher than conventional chemotherapy and combined chemo- and radiotherapy; reduces the toxic effects of other chemotherapeutic agents, prolongs the drug efficiency, and increases the body's resistance to cancer; normalizes γ -aminobutyric acid levels and glutamic acid decarboxylase activities	

and blood purification are carried out by *nimba* and *vasa* because they have the detoxification properties and often are used as blood purifiers. *Guduchi* and *triphala* are used to restore health.⁸

14.4 Hepatic Coma

Ayurveda does not have a clear description and diagnosis of hepatic coma except that it resembles an advanced stage of *kumbha kamala*. Viral-, toxic-, or drug-induced hepatocel-

TABLE 14.4

Therapeutic Modalities for Hepatitis

Disease	Comment and Treatment	Ref.
Hemolytic jaundice	Bilirubin seldom exceeds 6 mg% and more than 80% is indirect bilirubin Treatment modalities include antibiliary measures, internal oleation, <i>virechana</i> , hepatoprotective drugs, <i>ghritha</i> , and <i>rasayana</i> drugs (e.g., <i>Brahmarasayana</i> , <i>Indukantha ghritha</i> , <i>Panchathiktha ghritha</i> , <i>Rohitakarista</i> , and <i>Suvarnamalini vasantha</i>)	3, 50
Hepatitis B	Annual incidence in U.S.: 140,000–320,000 Mild cases: <i>Virechaka</i> like <i>Phalatrikadi kashaya</i> and viricidal agent, <i>Phyllanthus niruri</i> Moderate: <i>Arogyavardhini</i> with viricidal and <i>virechaka</i> drugs Severe cases (serum bilirubin exceeds 20 mg%): <i>Kamalihara rasa</i> (0.3 g), <i>Tapyadi lauha</i> (0.25 g), <i>Arogyavardhini</i> (0.5 g), <i>Nirgundi sallaki</i> (3 g), and <i>Guduchi</i> (3 g) are used in three divided doses/day for 3 to 4 weeks; minerals like <i>tapyadi lauha</i> and <i>srothoshodhaka</i> drugs like <i>shilajithu</i> , <i>nirgundi</i> , and <i>Semecarpus anacardium</i> are additionally used	10, 50
Acute viral hepatitis (HBAg negative)	<i>Arogyavardini</i> (500 mg), <i>hepax</i> (500 mg), <i>valiliv</i> (250 mg), <i>kamalahar forte</i> (250 mg) and <i>Liv.52</i> (275 mg); <i>Arogyavardini</i> and <i>hepax</i> are more beneficial Decrease in total serum bilirubin was evident from the 4th day of treatment; stimulates either hepatic or extra hepatic clearance of bilirubin; normalizes HDL cholesterol and other lipid levels	76
Acute viral hepatitis (HBAg positive)	<i>Swarasa</i> composed of <i>P. kurroa</i> , <i>Azadirachta indica</i> , <i>Amrita</i> , <i>Eclipta alba</i> , and <i>P. niruri</i> is beneficial in 17/20 patients in 30 days <i>Suvarnamalini vasantha</i> , <i>P. niruri</i> , and <i>R. emodi</i> (100 mg 2 times/day for 3 to 6 months)	50, 77
Hepatitis C	Complicates into hepatic cirrhosis followed by liver failure <i>N. stellata</i> seeds (250 mg), <i>abhrak bhasma</i> (250 mg), <i>P. longum</i> (100 mg), <i>Fummaria parviflora</i> (100 mg), <i>Tinospora cordifolia</i> (100 mg), <i>Boerhavia diffusa</i> (100 mg), <i>Emblica officinalis</i> (100 mg) (total 1 g three times/day dose)	52
Amebic hepatitis	Combination of <i>Shankhabhasma vati</i> , <i>Laghusutshekhara vati</i> , <i>Indrayav vati</i> (250 mg 3 times/day for 15 days) <i>Shankhabhasma vati</i> has the property of <i>ushna</i> , <i>laghu</i> , <i>ruksha</i> , and <i>teekshna</i> and described as <i>Grahanirognashan</i> <i>Indrayav vati</i> is a combination of <i>tridoshaghna</i> , <i>sangrahi</i> , and <i>katu</i> ; it acts as <i>deepan</i> , <i>shoolnashak</i> , <i>jwarnashak</i> , <i>atisarnashak</i> , and <i>krumighna</i>	78
Drug- or alcohol-induced hepatitis	<i>Eclipta alba</i> , <i>Andrographis paniculata</i> , <i>P. kurroa</i> , <i>Tephrosia purpurea</i> , and <i>trikatu</i> in combinations or <i>Liv.52</i> (275 mg twice daily)	
Chronic hepatitis	Hepatoprotective drugs (<i>Arogyavardini</i> , <i>E. alba</i> , <i>Solanum nigrum</i> , <i>Chichorium intybus</i> , <i>P. kurroa</i>), anti-inflammatory drugs (<i>Vitex nigundo</i> , <i>P. longum</i> , <i>sallaki</i> , <i>Tephrosia purpurea</i>), <i>rasayani</i> drugs (<i>Vardhamana</i> , <i>P. longum</i> , <i>rasayana</i>), and immunomodulators (<i>Amritha</i> , <i>Semecarpus anacardium</i> , <i>Punarnava mandoora</i>) are used in combinations	50

lular necrosis results in a hepatic coma — a clinical syndrome due to neuropsychiatric complications. This kind of hepatocellular failure and portal systemic shunts contributes to the genesis of hepatic encephalopathy. This process occurs by allowing nitrogenous products of bacterial fermentation in the colon (e.g., ammonia) to reach the brain, which is sensitive to their toxic actions. The aromatic amino acids formed from the conversion of branched chain amino acids act as weak neurotransmitters and may displace normal neurotransmitters (adrenaline and dopamine) in the brain, producing hepatic coma. It is accompanied by jaundice, gynecomastia, hepatomegaly, splenomegaly, and ascites. Because hepatic coma is not a single disease but a complication resulting from various

TABLE 14.5

Classic Ayurvedic Formulations for Diseases Related to *Raktavaha srotas* (e.g., Liver Ailments, Jaundice [Kamala])

Formulation, Manufacturer, and Ingredients	Indications and Dose	Ref.
<i>Abhrak bhasma</i> (UNJHA, UAP Pharma Pvt. Ltd.)	Viral hepatitis (250 mg)	52
<i>Abhra</i> , <i>Ravi ksira</i> , <i>Vata ksira</i> , <i>Snuhi ksira</i> , <i>Kakamaci</i> , <i>Goksura</i> , <i>Kharamanjari</i> , <i>Vata prarooha</i> , <i>Gomutra</i> , <i>Tulasi</i> , <i>Kadali sipha</i>		
<i>Bhallataka vati</i> (Imis Pharmaceuticals Pvt. Ltd.)	Cancer (1–3 pills after food)	
<i>Bhallataka</i> , <i>tila bija</i> , <i>Haritaki</i> , <i>Guda</i>		
<i>Brahmarasayana</i> (Nagarjuna Herbal Concentrates Ltd.)	Hemolytic jaundice (12 g)	50
<i>Pathya</i> , <i>Dhatri</i> , <i>Bilva</i> , <i>Ganikarika</i> , <i>Salaparni</i> , <i>Prasniparni</i> , <i>Brhati</i> , <i>Kantakari</i> , <i>Goksura</i> , <i>Bala</i> , <i>Punarnava</i> , <i>Eranda</i> , <i>Masaparni</i> , <i>Mudgaparni</i> , <i>Satavari</i> , <i>Meda</i> , <i>Jivati</i> , <i>Jivaka</i> , <i>Rsabhaka</i> , <i>Sali</i> , <i>Kasa</i> , <i>Sara</i> , <i>Darbha</i> , <i>Iksu</i> , <i>Tvak</i> , <i>Ela</i> , <i>Musta</i> , <i>Rajani</i> , <i>Pippali</i> , <i>Agaru</i> , <i>Candana</i> , <i>Mandukaparni</i> , <i>Kanaka</i> , <i>Sankhapuspi</i> , <i>Vaca</i> , <i>Plava</i> , <i>Yastyahvaya</i> , <i>Vidanga</i> , <i>Sitopala</i> , <i>Sarpi</i> , <i>Taila</i> , <i>Ksaudra</i>		
<i>Candanabalalaksadi taila</i> (Dabur Ayurvedic Specialties Ltd.)	Hepatitis (6 g)	
<i>Candana</i> , <i>Bala mula</i> , <i>Laksa</i> , <i>Lamajjaka</i> , <i>Taila</i> , <i>Usira</i> , <i>Nisa Madhuka</i> , <i>Satahva</i> , <i>Aguru</i> , <i>Katurohini</i> , <i>Devadaru</i> , <i>Bala</i> , <i>Kustha</i> , <i>Manjistha</i> , <i>Balaka</i> , <i>Asvagantha</i> , <i>Darvi</i> , <i>Murva</i> , <i>Musta</i> , <i>Mulaka</i> , <i>Ela</i> , <i>Tvak</i> , <i>Naga kusuma</i> , <i>Rasna</i> , <i>Laksa</i> , <i>Sugundhika</i> , <i>Campaka</i> , <i>Pitasara</i> , <i>Sariva</i> , <i>Sauvarcala</i> , <i>Saindhava</i> , <i>Ksira</i>		
<i>Draksadi arkom</i> (Nagarjuna Herbal Concentrates Ltd.)	Liver diseases (15–30 ml)	79
<i>Vitis vinifera</i> , <i>Madhuca latifolia</i> , <i>Glycyrrhiza glabra</i> , <i>Symplocos racemosa</i> , <i>Cyperus Rotundus</i>		
<i>Gudapippali</i> (SRI Dhanwantari Matam Ayurvedics Pvt. Ltd.)	Liver diseases	79
<i>Embelia ribes</i> , <i>Piper longum</i> , <i>Zingiber officinale</i> , <i>Piper nigrum</i> , <i>Plumbago rosea</i> , <i>Nigella sativa</i> , salt, and mineral additives		
<i>Indukantha ghritha</i> (Nagarjuna Herbal Concentrates Ltd.)	Hemolytic jaundice (12 g)	50
<i>Ptia</i> , <i>daru</i> , <i>Bilva</i> , <i>Agniatha</i> , <i>Syonaka</i> , <i>Gambhari</i> , <i>Patala</i> , <i>Salaparni</i> , <i>Prsniparni</i> , <i>Brhati</i> , <i>Kantakari</i> , <i>Goksura</i> , <i>Ksira</i> , <i>Ghrta</i> , <i>Pippali</i> , <i>Pippalimula</i> , <i>Cavya</i> , <i>Citraka</i> , <i>Sunthi</i> , <i>Saindhava</i>		
<i>Kutajavaleha</i> (Zandu Pharmaceutical Works Ltd.)	Hepatic diseases (6–12 g)	
<i>Kutaja tvak</i> , <i>Guda</i> , <i>Rasanjana</i> , <i>Mocarasa</i> , <i>Sunthi</i> , <i>Marica</i> , <i>Pippali</i> , <i>Haritaki</i> , <i>Bibhitaka</i> , <i>Amalaki</i> , <i>Lajjalu</i> , <i>Citraka</i> , <i>Patha</i> , <i>Bilva</i> , <i>Indrayava</i> , <i>Vaca</i> , <i>Bhallataka</i> , <i>Prativisa</i> , <i>vidanga</i> , <i>Balaka Ghrta</i> , <i>Madhu</i>		
<i>Laghusutshekhara vati</i> (Gary and Son, U.S.A.)	Amebic hepatitis (1 to 2 tablets)	78
<i>Svarna gairika</i> , <i>Sunthi</i> , <i>Nagavalli svaras</i>		
<i>Pancha Gavyam Ghritham</i> (Imis Pharmaceuticals Pvt. Ltd.)	Hepatitis (12 g)	4
<i>Gomaya svarasa</i> , <i>Dadhi</i> , <i>Gomtra</i> , <i>Goghita</i>		
<i>Panchathiktha ghritha</i> (Vyas Pharmaceuticals)	Hemolytic jaundice (6 g)	50
<i>Nimba</i> , <i>Patola</i> , <i>Vyaghri</i> , <i>Guduci</i> , <i>Vasaka</i> , <i>Haritaki</i> , <i>Bibhitaka</i> , <i>Amalaki</i> , <i>Ghrta</i>		
<i>Patolakaturohinyadi arkom</i> (Nagarjuna Herbal Concentrates Ltd.)	Liver diseases (15–30 ml)	79
<i>Trichosanthes cucumarina</i> , <i>Picrorrhiza kurroa</i> , <i>Santalum album</i> , <i>T. cordifolia</i> , <i>Chonemopha fragrans</i>		
<i>Phalatrikadi kasaya</i> (Rajashree Ayurvedic Pharmacy)	Cirrhosis, ascites (20 ml)	80
<i>Haritaki</i> , <i>Bibhitak</i> , <i>Amalaki</i> , <i>Amruta</i> , <i>Vasa</i> , <i>Katuki</i> , <i>Nimb</i> , <i>Kirattikta</i>		
<i>Punarnavasava</i> (Nagarjuna Herbal Concentrates Ltd.)	Hepatic diseases (12–24 ml)	
<i>Sunthi</i> , <i>Marica</i> , <i>Pippali</i> , <i>Haritaki</i> , <i>Bibhitaka</i> , <i>Amalaki</i> , <i>Darvi</i> , <i>Svadamstra</i> , <i>Brhati</i> , <i>Kantakari</i> , <i>Vasa</i> , <i>Eranda mula</i> , <i>Katuki</i> , <i>Gajapippali</i> , <i>Sothaghni</i> , <i>Picumarda</i> , <i>Guduci</i> , <i>Suska mulaka</i> , <i>Duralabha</i> , <i>Patola</i> , <i>Dhataki</i> , <i>Draksa</i> , <i>Sita</i> , <i>Maksika</i>		
<i>Rohitakarista</i> (UNJHA, UAP Pharma Pvt. Ltd.)	Hemolytic jaundice (12–24 ml)	50
<i>Rohitaka</i> , <i>Guda</i> , <i>Dhataki</i> , <i>Pippali mula</i> , <i>Cavya</i> , <i>Citraka</i> , <i>Sunthi</i> , <i>Tvak</i> , <i>Ela</i> , <i>Patra</i> , <i>Haritaki</i> , <i>Bibhitaka</i> , <i>Amalaki</i>		

TABLE 14.5 (continued)

Classic Ayurvedic Formulations for Diseases Related to *Raktavaha srotas* (e.g., Liver Ailments, Jaundice [Kamala])

Formulation, Manufacturer, and Ingredients	Indications and Dose	Ref.
<i>Shankhabhasma vati</i> (UNJHA, UAP Pharma Pvt. Ltd.)	Amebic hepatitis (250 mg)	78
<i>Sanka, Kanjika</i>		
<i>Suvarnamalini vasantha</i> (UNJHA, UAP Pharma Pvt. Ltd.)	Hemolytic jaundice (10 g)	50
<i>Mauktika bhasma, Hingula, Marica, Kaphari, Svarna bhasma</i>		
<i>Tapyadi lauha</i> (Vyas Pharmaceuticals)	Viral hepatitis (250–500 mg)	50
<i>Haritaki, Amalaki, Vibhitakaki, Sunthi, Marica, Pippali, Vidanga, Citraka mula, Musta, Pippalimula, Devadaru, Daruharidra, Dalcini, Cavya, Silajitu, Svarnamaksika bhasma, Raupya bhasma, Lauha bhasma, Mandura bhasma</i>		
<i>Vasagulu chyadi kvatha</i> (Vaidyaratnam Oushadhasala)	Liver diseases (15–30 ml)	79
<i>Adhatoda vasica, T. cordifolia, Glycyrrhiza glabra, Strychnos potatorum, Melia azadirachta</i>		

disorders, the choice of treatment depends on symptoms of individual patients. Treatment profile includes the combination of the following drugs and a vegetarian diet:

1. *Nasya of haridra* and *daru haridra quath* reduce ammonia saturation in encephalopathy.
2. *Basti* (a type of enema) with *dashmool quath* and *saindhava* can reduce toxicity in guts.
3. Sugarcane juice maintains electrolyte balance.
4. *Vasa* prevents gastrointestinal bleeding.
5. *Guduchi* is an antibiotic.
6. *Triphala* and *katuki* manage constipation.
7. *Bhunimb* and *nimb* help in regeneration of liver cells.⁹

14.5 Cirrhosis

Ayurveda does not have a clear description and diagnosis of cirrhosis. According to conventional medicine, the etiology of the liver cirrhosis is due to the diffuse degeneration and infiltration of parenchyma that results in the structural alteration of fat lobules, dense perilobular connective tissue formation, and development of regeneration areas. Cirrhosis is the seventh leading cause of death in the U.S., with 4000 annual deaths.¹⁰ Advanced cirrhosis leads to ammonia toxicity, hepatic coma, gastrointestinal hemorrhage, and kidney failure. It is not a primary disease but the end result of a diffuse liver injury secondary to poor nutrition, anorexia and action of toxins, alcoholism, hepatitis C, or biliary obstruction. As liver cells are destroyed, they are systematically replaced by scar tissue. Hepatic cirrhosis occurs following one or more clinical manifestations such as the loss of appetite, hepatomegaly, splenomegaly, ascites, edema of the legs, fever, diarrhea, hematemesis, jaundice, portal hypertension, and anemia.

TABLE 14.6

Patent Ayurvedic Formulas for Hepatobiliary Disorders

Formulation, Manufacturer, and Ingredients	Indications and Dose	Ref.
ARKA LIV Tablet (Ayurvedeeya Arkashala Ltd.) <i>Guduchi, Rohitaka, Daruharidra, Amalaki, Katuki, Shankha, Karpardika, Dagadiber bhasma</i>	Hepatitis, jaundice, hepatoprotective (2–3 tablets two times/day with honey)	80
Arogyavardhini (Zandu Pharmaceutical Works Ltd.) <i>Mercury, Sulfur, Lohabhasma, Tamrabhasma, Abhrakbhasma, Triphala churna, Shilajit-, Guggul, Chitrakamal Churna, Kutaki churna, decoction of A. indica</i>	Hepatitis (500 mg/day)	
Ayu Liv Tablet (AYUDRUGS) <i>Kalamegha, Sharpunkha, Chitraka, Punarnava, Triphala, Kakmachi, Senai leaves, Arjuna, Rohitakarishtha, Kutaja tvak, Mandoora bhasma, Ajamoda, Khus-Khus, bile salt</i>	Liver diseases, sluggish liver, jaundice (1–2 tablets/day)	81
Hepabex (Swastik Formulations Pvt. Ltd.) <i>Solanum nigrum, Tephrosia pupurea, Picrorrhiza kurroa, Ipomoea tupethum, Eclipta alba, Phyllanthus niruri, Tecomella undulata, Boerhaavia diffusa, Tinospora cordifolia, Glycyrrhiza glabra</i>	Hepatoprotective, hepatitis B, and jaundice (2 tsp two times/day)	
Hepa Cap (Capro Labs Exports India Ltd.) <i>Zizyphus jujuba, Terminalia chebula, Aegle marmelos, Calcium oximum, Berberis aristata, Vitis vinifera, Ipomoea digitata, Emblica officinalis</i>	Jaundice, cirrhosis, infective hepatitis (2 capsules two to three times/day)	76
Hepajun (Phyto Pharma Pvt. Ltd.) <i>Bhringaraja, Madayantika, Daruharidra, Kalamegha, Punarnava, Nimba, Chitraka, Mandura bhasma, Shankha bhasma, Kasamarda, Pathari, Nishothara, Katuki, Kharavath, Kapardika</i>	Impaired liver function, hepatitis, hepatobiliary disorders (2 capsules/day)	
Hepatovit (Millenium Herbal Care Ltd.) <i>Phyllanthus niruri, Terminalia chebula, Terminalia belerica, Embilica officinalis, Eclipta alba, Berberis aristata, Fumaria parviflora, Boerhaavia diffusa, Zingiber officinale, Plumbago zeylanica, Tinospora cordifolia, Andrographis paniculata, Picrorrhiza kurroa, Piper longum</i>	Hepatitis, hepatobiliary dysfunction, enlarged liver, cirrhosis (1–2 capsules/day)	82
Hepax (Anglo-French Drugs and Industries Ltd.) <i>Chitraka, Katuki, Maricha, Ardraka, Sarjikakshara, Amalaki, Yavakshara, Chuna, Haritaki</i>	Chronic hepatitis, cirrhosis, infective hepatitis (2 tablets three times/day)	
Hepin (Nupal Remedies Pvt. Ltd.) <i>Yashthimadhu, Daruharidra, Pippali, Bhumyamalaki, kakamachi, Gudardrakam, Musali, Ela, Tripadi, Kandasari</i>	Hepatitis A and B, non-A and non-B hepatitis (2 tablets three times/day)	76
Herboliv (Vyas Pharmaceuticals) <i>Arogyavardhini vati, Punarnava mandura, Bhumyamalaki, Bhringaraja, Punarnava, Vidanga, Haridra, Haritaki, Guduchi, Rohitaka, Nishothara, Mehandi, Varuna, Arjuna, Parpata Sugar</i>	Anemia, hepatitis, hepatomegaly, anorexia (1–2 tablets/day)	
Liv.52 (The Himalaya Drug Company) <i>Himsra, Kasani, Mandur bhasma, Kakamachi, Arjuna, kasamarda, Biranjasipha, Jhavuka; processed in Bhringaraja, Bhumyamalaki, Punarnava, Guduchi, Daruharidra, Mulaka, Amalaki, Chitraka, Vidanga, Haritaki, parpata</i>	Viral hepatitis, cirrhosis, hepatotoxicity (2–3 tablets/day)	76
Livotrit (Zandu Pharmaceutical Works Ltd.) <i>Arogyavardhini rasa, Eclipta alba, Mandura bhasma, Boerhaavia diffusa, Andrographis paniculata</i>	Infectious hepatitis, liver dysfunctions (2 tablets two times/day)	
Livshield syrup (Bajaj Consumer Care Ltd.) <i>Bhumyamalaki, Bhringaraja, Guduchi, Gokshura, Kasamarda, Chitraka, Kantakari, Krishna tulasi, Palasha pushpa, Punarva, Arjuna twak, Triphala, Karanaja, Maricha, Methika beeja, Nagavalli, Trivrut, Pippali, Ativisha, Ajamoda satva</i>	Liver dysfunction, jaundice (2 tsp three times/day)	

TABLE 14.6 (continued)

Patent Ayurvedic Formulas for Hepatobiliary Disorders

Formulation, Manufacturer, and Ingredients	Indications and Dose	Ref.
Stimuliv (Fraco-Indian Pharmaceuticals Ltd.) <i>Kalamegha, Bhringaraja, Parpata, Bhumyamalaki</i>	Viral hepatitis, hepatomegaly, loss of appetite, constipation (2 tablets/tsp two times/day)	83
Vasuliv (Vasu Pharmaceuticals Ltd.) <i>Bringaraj, Punarnava, Sharapunkha, Kasni, Bjumyamilaki, Kalamegh, Katuki</i>	Liver dysfunction, hepatomegaly, jaundice, viral hepatitis, alcoholic liver diseases (2 tablets three times/day)	84

In a case report of early hepatic cirrhosis due to biliary obstruction, patients with moderate jaundice and anorexia with a tender, enlarged liver were treated with *arogyavardhini* and *phalatrikadi kasaya* with milk on empty stomachs for 3 months. No other food was allowed. The main symptoms and signs were relieved completely in 3 months, and the patients' appetites increased. The liver regained normal morphology, jaundice disappeared, blood pressure became normal, and ascites were not detected. Hepatomegaly and splenomegaly were reduced. Although the cirrhotic changes in the liver are irreversible, the hepatic functions can be maintained with the rest of the healthy tissues by this treatment.¹¹

14.6 Liver Cancer

In Ayurveda, tumors are discussed under the names *granthi* and *arbuda* to designate basic common neoplasms, which can appear in any tissue or organ of the body. *Granthi*, or minor neoplasm, is a localized small swelling within the subcutaneous fat tissue, muscle, or blood veins; it is round, erect, and knotted. *Arbuda*, or major neoplasm, is a spherical, stable, massive, painless swelling occurring at one site; it expands slowly with deeper roots. Immobile recurrence of *arbuda* at the site of previous swelling is called *ashyarbuda*. When two *arbudas* appear simultaneously the condition is known as *dvirarbuda*, which indicates their malignant nature.^{12, 13}

The different abnormal growths of pre- and postcancerous states and malignant and nonmalignant stages arising from various organs are grouped and known as cystic growths (*gulma*), benign growths (*apaci*), lymphatic growths (*gandamala*), bone tumor (*asthila*), cystic tumors (*mutragranthi*), vaginal tumors (*yonikarnini*), and systemic tumors (*granthivisarpa* and *balmika*). These growths represent morbid anatomy as neoplasm.¹³ Both allopathic and Ayurvedic systems generally refer to neoplasms as the uncoordinated abnormal cell growth found within particular organs or body tissues.

The words *vataja*, *pittaja*, or *kaphaja* tumors (adenoma) or *medhaja* (lipoma) or a combination of any two of them is used to signify a benign neoplasm where one or two of the three major bodily systems (*tridosas*) are out of control. Benign tumors are not very harmful because there is still coordination among the systems, which to some extent controls the damage. Malignant abnormal growth has been indicated as *tridosaja* neoplasm (metastatic tumors), because the three major bodily systems have lost mutual coordination and cannot prevent damage to tissues, resulting in a deadly morbid condition.¹³

According to conventional medical pathology, primary malignant hepatic tumors may arise from any constituent cells of the liver. Cancer arising from two types of cells are very common: liver cell cancer (hepatocellular carcinoma) and carcinoma of the biliary epithelium (cholangiocarcinoma). Other rare tumors that arise from the liver tissue include fibrolamellar carcinoma, squamous cell carcinoma, epithelial hemangioendothelioma, angiosarcoma, Kaposi's sarcoma, hepatoblastoma, and hepatocellular adenoma. In addition to these malignant primary liver cancers, metastatic involvement of the liver is also very common because of the extreme chance of spreading cancer from several parts of the body to the liver.

14.6.1 Hepatocellular Carcinoma

Hepatocellular carcinoma (HCC), also referred to as a fatal malignancy, is regarded as one of the most aggressive tumors with poor prognosis. It has a worldwide incidence of 250,000 to 1,000,000 patients annually.¹⁴ HCC is prevalent in Asia and Africa where annual incidence of 500 cases occurs per 100,000 population.¹⁵ In China, HCC was the third cancer killer, and annual deaths from HCC have exceeded 110,000, accounting for approximately 45% of the total HCC deaths in the world.¹⁶ According to 1996 annual statistics, in the U.S. there have been more than an estimated 19,500 new cases of HCC and 15,200 deaths due to this cancer. This estimate is based on incidence rates of the Service, Epidemiology, and End Results (SEER) program from 1979 to 1992.¹⁷

14.6.1.1 Etiology of Liver Cancer

According to Ayurvedic texts, cancer is a disease resulting from the derangement of bodily systems due to dietary constituents (chemical carcinogenesis). It differs according to each person's pathogens (exogenous) and constitutions (genetic, *bijadosa*).¹³ Sushruta called the sixth layer of the skin *Rohini* (the skin layer with the nature of growth), which represents the Western medical word *epithelium*, the layer of cells forming the epidermis of the skin and the surface layer of the mucous membranes where most cancers originate. Pathogenic injuries to the inner layer of the dermis can be caused by the imbalances of three bodily systems or lifestyle errors, such as unhealthy foods, poor hygiene, or behavioral habits and physical trauma resulting in ulcerous conditions. Because of the fast-growing nature of these tissues, the process of healing often leaves tiny scars. These slow-healing ulcers may develop into early neoplasms, which are complicated and difficult to cure.¹³

HCC is a multifactorial and multistage process whose precise etiology is not known. Both chronic hepatitis and cirrhosis are considered major precipitating factors. HBV or HCV infection results in tumor initiation.¹⁴ Other factors include the ingestion of high amounts of aflatoxins, occupational exposure to vinyl chloride or similar toxic chemicals, and alcoholism. Long-term use of oral contraceptives and angiogenic therapy have also been found to cause liver cancer development in fewer cases.

14.6.1.2 Pathogenesis of Liver Cancer

In Ayurveda, *Sushruta* has proposed six stages of pathogenesis, which apply more to the neoplasms (especially liver tumors) because of their metastatic nature:

1. Localization of growth (*sanchaya*)
2. Transformation of growth into metastatic tumors (*prakopa*)

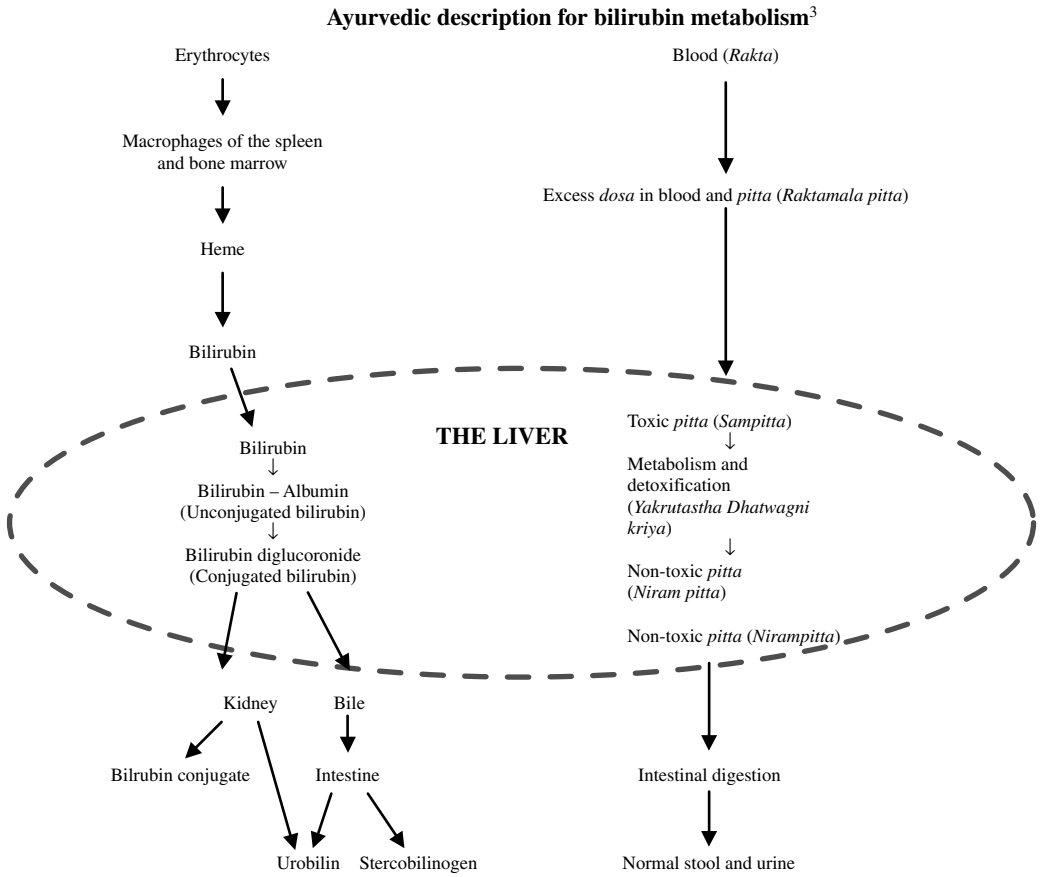


FIGURE 14.1
Ayurvedic description for bilirubin metabolism. (From Mali, M.D. and Kulkarni, P.H., *Ayurved and Hepatic Disorders*, Sri Satguru Publications, Delhi, 2001.)

3. Metastasis (*prasara*)
4. Secondary growth (*sthana samsraya*)
5. Expression of symptoms (*vyakti*)
6. Histopathological differentiation (*bheda*)

On the basis of *tridosas* concept, *pitta* present in each and every cell is responsible for digestion and metabolic function. In liver cancer, the decreased state of deranged metabolism (*dhatwagni* imbalance) results in the excessive growth of the liver tissue and creates metabolic crisis where anabolic phase exceeds the catabolic phase (aggravation of *vatha* forces and suppression of *kapha* forces) resulting in proliferation. It is clearly stated that liver enlargement (*ekadesavriddhi*) is accompanied by weight loss (*anyasthaniyakshaya*) in liver cancer.¹⁸

14.6.1.3 Diagnosis of Liver Cancer

In early stages, liver cancer usually has no noticeable symptoms. Hepatomegaly (*yakrutodara*), with moderate pain (*ruk*) or tenderness (*sparsahsnutwam*) that is localized to the

upper abdomen or the right upper quadrant, is often the major complaint from the HCC patients.¹⁹ Macroscopically, the tumor may be a solitary mass with multiple necrotic (*kotham*) nodules or may present diffusely, which infiltrates the liver and usually is cirrhotic. On microscopical examination, several abnormal architectural patterns of malignancy (e.g., trabecular, pseudoglandular, solid, etc.) exist, and these tumors are mitotically active. In certain cases, abdominal swelling (*sopham*) due to liver enlargement with or without ascites also occurs. Pyrexia, pain, or fractures due to bone metastasis are some of the rare presentations of preneoplastic syndromes. Anorexia (*aruchi*), weight loss, hypercalcemia, hypoglycemia, hyperlipidemia, hemotologic, and neurologic syndromes are also clinical manifestations that occur during HCC (see [Figure 14.1](#)).

Although serum alpha-fetoprotein (AFP) is a highly specific marker in HCC cases, ultrasonographic screening is also recommended because AFP levels are not always elevated in the early stages of HCC (in 30% of cases). Presence of alkaline phosphatase isoenzymes, carcinoembryonic antigen, and human chorionic gonadotropin are other associated complications in HCC patients. Bleeding of esophageal varices may present as a terminal sequele.

14.6.1.4 Treatment

HCC is an important cause of morbidity and mortality worldwide. The course of clinically apparent disease is rapid, and if untreated, most patients die within 3 to 6 months of diagnosis. Individual prognostic factors such as the extent of the disease, the site of metastasis, histological condition of the tumor, and the biochemical abnormalities have to be considered in selection of a therapy.

14.6.1.4.1 Ayurvedic Treatment Modalities

In Ayurvedic classical texts, specific treatment protocols for liver cancer were not mentioned. Ayurvedic practitioners are taking the advantage of available knowledge of treatment on cancer from the old classical texts and are applying those methods in liver cancer management. Empowerment, participation in the healing process, relief of cancer symptoms, time and personal attention, and reduced cost of care are essential elements in the Ayurvedic management of liver cancer.

The fundamental principles of Ayurveda are finding the cause of an illness and restoring the balance between the three major bodily systems by supplying deficient substances and by reducing the excessive ones (*tridosha siddhanta*). Anticancer therapeutic approach can be divided into four categories:

1. Restoration of balance (*prakritisthapani chikitsa*)
2. Curative therapy (*roghanashani chikitsa*)
3. Restoration of normal function (*rasayana chikitsa*) and spiritual therapy (*naishtiki chikitsa*)
4. Surgery (considered only as a last resort)

Unlike Western medicine, which mostly epitomizes the one symptom–one disease–one drug paradigm, Ayurvedic medicines are holistic and go a long way to substantiate the defects and deficiencies of medical practices.

14.6.1.4.2 General Line of Treatment

1. Purification process (*sodhana chikitsa* including *panchakarma*) using internal and external medications which can eliminate pathogens (*dosas*)

2. Curative therapy (*samana chikitsa*) pacifies pathogens (*dosas*)
3. Correction of metabolic defects (*dhatwagni chikitsa*)
4. Immunotherapy (*rasayana prayoga*)
5. Anticancerous drugs (*vyadhipratyanika chikitsa*)
6. Symptomatic treatment (*lakshanika chikitsa*)
7. Surgical treatment with herbal and mineral medicines (*Sastra chikitsa*)¹⁸

The herbal treasure chest of ancient Ayurveda offers a host of new phytochemicals that can be used both preventively and clinically to manage a spectrum of liver-related imbalances, including hepatocarcinoma. Tables 14.2 and 14.3 describes not only efficient and effective therapies for liver diseases, but also their underlying scientific principles based on an extensive literature search of Ayurvedic texts and research reports on hepatoprotective herbs with anticancerous property. Ayurvedic practitioners are using Ayurvedic text preparations that utilize the recent research outcomes on anticancer herbs (Table 14.5). Many Ayurvedic drug manufacturers are designing their own appropriate drug formulations (Table 14.6). These herbal formulas contain multiple chemical agents as active ingredients; these agents may operate synergistically, producing tremendous therapeutic benefits, lowering risks on adverse effects, and avoiding unnecessary supplemental therapy. The benefit of an herbal formula is that it can nourish the body as a whole by supporting various organ systems, yet its main focus will be on nourishing the liver and its functions. These formulations are reported to work on multiple biochemical pathways and are capable of influencing several organ systems simultaneously. This kind of orientation makes the Ayurvedic approach to liver cancer especially attractive.

14.6.1.4.3 Allopathic Treatment

Among allopathic treatment modalities, surgery is seldom of value for hepatic metastases nor is hormonal therapy. In radiotherapy using intraarterial ¹³¹iodine-lipiodol²⁰ or ⁹⁰yttrium, microspheres,²¹ the clinical benefits are of modest value because of side effects like fibrosis, pneumonitis, neurological abnormalities, colitis, enteritis, malabsorption, and sterility. Although chemotherapy is an essential component of an effective multidisciplinary cancer treatment program, current systematic therapies with 5-fluorouracil, floxuridine, doxorubicin, mitomycin-C, and cisplatin are less effective and responses are partial (5 to 25% range) and short lived.²²

Radiotherapy and chemotherapy also damage adjacent healthy cells and increase the possibility of developing a secondary cancer; many times, the mortality increases by the toxicity of the drug or radiation rather than the disease. Other approaches, namely, hepatic artery embolization with chemotherapy, alcohol ablation, ultrasound guided cryoablation, immunotherapy with monoclonal antibodies, and gene therapy with retroviral vectors, are also of only minimal benefit. Liver transplantation may be considered as another therapeutic option, but again the recurrence of tumor or metastases after transplantation has limited its usefulness. The major drawback with the treatment of HCC is that it has a significant propensity to develop metastases with major sites spreading to other parts of the liver, lymph nodes, lungs, bones, adrenal glands, and the peritoneal cavity.²³ Multifocal HCC is a very difficult neoplasm to treat due to the lack of efficacy of chemotherapeutic regimes and the inaccessibility of this tumor to surgical procedures. The World Health Organization (WHO) also categorized hepatic cancers under group 3 tumor, for which there are no effective drugs.²⁴

Given the lack of useful therapies for this disease, there is a great need to discover drugs that are selectively cytotoxic only to hepatic cancer cells without damaging the body's other normal cells. Because Ayurvedic herbal therapies have the potential to inhibit cancer cells without damaging normal cells, liver cancer patients who already are crippled with the disease and burdened by chemotherapy-induced toxic side effects are often turning to seek help from Ayurvedic physicians. Patients hope for a better cure without adverse side reactions from Ayurvedic therapies. The high cost of conventional medicine is also driving the search for alternatives in other medical systems such as Ayurveda.

14.7 Prevention of Hepatic Diseases

Prevention is a preferred strategy than cure, especially for chronic hepatic diseases such as liver cancer. Regular intake of hepatoprotective herbal tonics would be beneficial in this strategy. It may also be argued reasonably that prevention of viral infection by HBV vaccine may help prevent HCC. The antifibrotic agents prolyl 4-hydroxylase inhibitor (HOE 077), sho-saiko-to (TJ-9), and interferon can also reduce the risk of development of HCC.²⁵

14.8 Scientific Basis

14.8.1 *Phyllanthus niruri/amarus* (Bhumyamalaki)

14.8.1.1 Clinical Trials

In a clinical study²⁶ conducted on 55 patients with chronic viral hepatitis, all the 30 patients under *P. amarus* treatment were cured within 3 months with remarkable recovery of liver functions and inhibition of HBV replication. Another study²⁷ indicated that 59% of the *P. amarus*-treated patients became HBsAg negative.

14.8.1.2 In Vivo and In Vitro Studies

The aqueous extract of *P. amarus* not only increased the life span of the *N*-nitrosodiethylamine-induced HCC-bearing rats from 33 to 52 weeks, but also normalized gamma glutamyl transpeptidase and serum HCC marker activity.²⁸ The ethanolic extract exhibited a hepatoprotective effect on alcohol toxicity in rats and in carbon tetrachloride and galactosamine-induced cytotoxicity in primary cultured rat hepatocytes.²⁹ *P. amarus* extract inhibited HBsAg secretion in Alexander, a human hepatocellular carcinoma cell line, because of its ability to down-regulate HBV messenger ribonucleic acid (mRNA) transcription and up-regulate HBV enhancer I activity.³⁰ It also inhibited HBV polymerase activity and decreased episomal HBV DNA content.³¹

14.8.2 *Picrorrhiza kurroa* (Katuki)

14.8.2.1 Clinical Trials

The active principle of *katuki* is *kutkin*. The standardized preparation used in studies is Picroliv. In a placebo-controlled trial³² with acute viral hepatitis patients, *P. kurroa* (375 mg three times/day for 14 days) normalized liver function test (LFT) parameters with quicker clinical recovery and 100% clinical and biochemical clearance.

14.8.2.2 In Vivo and In Vitro Studies

P. kurroa acts as a direct viricidal agent and alters biotransformation of the toxins. In several studies^{33,34} it inhibited the uncoupling between respiration and oxidative phosphorylation and aflatoxin-induced lipid peroxidation in rats. In addition, it caused the reversal of low-density lipoprotein binding to paracetamol and lowered bilirubin level by biliary flow stimulation. It also showed a protective role against ethanol and carbon tetrachloride-induced toxicity in rat hepatocytes.³⁵ During *Plasmodium berghei* infection in *Mastomys caucha*, it showed hypolipidemic property by regulating lipolytic enzymes.³⁶ As an effective inhibitor of hepatocarcinogenesis, it normalized liver function enzymes and reduced liver enlargement and hepatic nodular formation in N-nitrosodiethylamine-induced hepatocarcinoma conditions.³⁷

14.8.3 *Podophyllum hexandrum* Linn. (Giriparpata, Vanatrapusi)

14.8.3.1 Clinical Studies

P. hexandrum (active constituent: Podophyllotoxin) is a hepatic stimulant, blood purifier, and *raktaarbudanasna* (anticancer drug). Etoposide (VP-16), a semisynthetic derivative of podophyllotoxin, has been used against HCC for several years. P-glycoprotein, an energy-dependent drug efflux pump that reduces intracellular concentration of drugs in tumor cells, seems to be less effective in reducing VP-16 concentration in hepatoma cell lines; this drug is found to be more efficient in hepatocarcinoma.³⁸ In a phase II study on anticancer effectiveness of VP-16 in combination with epirubicin, 3% of 36 HCC patients achieved complete response, 36% had partial response, and 31% exhibited stable disease; the disease progressed in another 31% of patients.³⁹

14.8.3.2 In Vivo and In Vitro Studies

P. hexandrum has the property of karyoplastic and causes mitotic arrest, nuclear fragmentation, and impaired spindle formation dispersing the chromosomes. Necrosis and rapid reduction of the cytochrome oxidase are suggested mechanisms for its antitumor property.⁴⁰

14.8.4 *Tinospora cordifolia* (Guduchi)

14.8.4.1 Clinical Trials

In a study⁴¹ with malignant obstructive jaundice patients, *T. cordifolia* decreased morbidity and mortality due to liver cell failure. In addition, a 92.4% survival rate after treatment without blood poisoning was noticed when compared with only a 40% survival rate on

conventional treatment. As an immunomodulator, it strengthened host defense and normalized phagocytic and killing capacities of neutrophils. In another study⁴² with hepatitis patients, the treatment showed symptomatic relief with improvement in yellow discoloration of urine and body and reduction of LFT parameters. Of a total of 20 cases, 75% were cured and 25% improved from their illness.

14.8.4.2 In Vivo and In Vitro Studies

T. cordifolia prevented fibrosis by hepatic tissue regeneration, membrane stabilization, and activation of kupffer cells in carbon tetrachloride-induced liver damage in rats.⁴³ In experimental animal tumors, it increased IgG antibodies and enhanced humoral- and cell-mediated immunity.⁴⁴ It not only stimulated the proliferation of stem cells and increased white blood and bone marrow cells, but also reduced tumor volume by 58.8%, which was comparable with cyclophosphamide.⁴⁵

Scientific basis for other hepatoprotective Ayurvedic herbs are continued in [Table 14.3](#).

14.9 Summary and Conclusion

The complete cure of liver diseases such as HCC is one of the biggest challenges in the medical world. Scientific researchers in medical sciences are conducting multidirectional research on various aspects of hepatic disorders in order to determine useful remedies for these dreadful diseases. New findings discussed are a result of the development of Ayurvedic science through the ages and their accomplishments in modern cancer therapeutics. The details of experimental and clinical studies conducted on single and compound Ayurvedic preparations for their efficacy against liver cancer and other hepatic ailments are also important; they strongly emphasize Ayurvedic therapy as a scientifically feasible medical practice and an unconventional entity. Research in this field should be sustained and strengthened in order to better understand its scientific merit and determine its utility for a heterogeneous population. Hepatic diseases are no longer inevitable and can be prevented and effectively treated. Ayurvedic medicine now has an opportunity to contribute to the end of liver cancer's long scourge of mankind.

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Antimutagenic Effect of Ayurvedic Therapies

Satwinderjeet Kaur

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15.1 Introduction

The purpose of this article is to explore the antimutagenic effect of Ayurvedic therapies, as this effect may be an underlying mechanism of their therapeutic uses. Also, these products may act to supplement antimutagens in our diets which have now become essential in order to combat the unavoidable environmental mutagens to which we are constantly exposed.

15.2 Background

A technologically advanced society creates an environment that warrants exposure to different types of genotoxic substances. The deoxyribonucleic acid (DNA) from living organisms is continuously exposed to exogenous or endogenous damaging agents as a result of which there is a substantial level of DNA damage. It has been hypothesized that DNA damage contributes to organismal senescence and the increased risk for specific age-related diseases.¹⁻⁴ DNA lesions can interfere with most DNA metabolisms, including replication, transcription, and recombination as well as other important cell functions (e.g., cell-cycle regulation or cell division). Cancer initiation is classically associated with the induction of mutations on specific oncogenes or tumor suppressor genes due to the presence of unrepaired DNA lesions produced by endogenous or exogenous genotoxic agents. The somatic mutation theory proposes that changes in DNA are responsible for changes that bring about senescence and death.

Exposure to mutagens and carcinogens either endogenously or exogenously is unavoidable. Cigarette smoking, diet, infection, and chronic inflammation account for a large proportion of cancers. Epidemiological studies have indicated that cigarette smoking significantly elevates the risk of tumor development in the upper-aerodigestive tract and in other organs.^{5,6} Polycyclic aromatic hydrocarbons, N-nitrosamines, and many mutagens and carcinogens and their precursors have been detected from cigarette smoke condensate.^{5,6} The total calorie intake and fat intake appear strongly associated with enhancement of human cancer development.⁷⁻¹⁰ Lipid peroxidation yields exocyclic propano-, etheno-, and malondialdehyde adducts in DNA to induce mutation.^{11,12} The mutagens and carcinogens may be present as microcomponents in various foodstuffs.^{13,14} The compounds include naturally occurring pyrrolizidine alkaloids, ptaquiloside, and cyasin. Mycotoxins like aflatoxin B₁, sterigmatocystin, and fumonisin exist as contaminants in food. Hepato-carcinomas developed in people who are living in areas where aflatoxin B₁ is highly contaminated in diet show a characteristic mutation in the p53 gene, namely, change of G to T at the third position of codon 249.^{15,16}

It is practically difficult to completely eliminate mycotoxin contamination of food materials. Heterocyclic amines (HCAs) are produced under normal cooking conditions, especially in the overcooked fish and meat. HCAs are metabolized by the cytochrome P450 family, especially by CYP1A2 to N-hydroxyamino derivatives, and are then esterified to

the ultimate forms that produce DNA base adducts.^{17,18} Infection and chronic inflammation may produce oxidative agents in addition to cytokines and chemokines that enhance cell proliferation, increasing the chance of replication error and genetic change.

15.3 Mechanism of Mutagenic Activity

The generation of reactive oxygen species (ROS) in biological systems, either by normal metabolic pathways or as a consequence of exposure to chemical carcinogens, has been extensively studied.¹⁹ Free radicals generate oxidative damage to main cellular components such as DNA, lipids, carbohydrates, and proteins; they have been implicated in the aging phenomena, ischemia,²⁰ cancer,²¹ autoimmune diseases,²² and neural cell death.²³ It is now universally agreed that the ROS generation contributes to the multistage process of carcinogenesis.^{19,24} Evidence has accumulated that lack of protection against free-radicals and lack of repair of oxidative damage in biological macromolecules have a significant role in mutagenesis and carcinogenesis. Kawanishi and co-workers²⁵ reported that sequence-specific oxidative damage to DNA may play an important role not only in carcinogenesis but also in aging.

15.4 Prevention

There are two complementary strategies for preventing chronic degenerative diseases. The first one is avoidance of exposure to recognized risk factors which evolves through risk assessment and risk management. The risk management is based either on risk communication and health education for controlling lifestyle risk factors or on suitable regulations for controlling environmental risk factors. The other strategy is to favor the intake of protective factors and to modulate the host defense mechanisms. This approach is more practical and is referred to as chemoprevention. It is based on dietary and pharmacological interventions, has already found extensive application in the field of cardiovascular disease, and is encountering a growing interest in cancer prevention.

Food is a complex mixture that may contain mutagenic or carcinogenic compounds such as polycyclic aromatic hydrocarbons and heterocyclic amines present in meat.^{26,27} On the other hand, food may contain protective antimutagenic or anticarcinogenic substances that are mostly present in plants. It is important to maintain a varied nutritionally balanced diet in which food antimutagens outweigh the mutagens. Inhibition of mutagenesis or carcinogenesis is generally not based on one mechanism. Protection against cancer can occur at different stages of the complicated processes of carcinogenesis. Compounds and complex mixtures with antimutagenic activity have different modes of action and act in parallel at different levels. As inhibitors, they may prevent the formation of mutagens, such as the endogenous formation of nitrosamines. As blocking agents, they can prevent the biotransformation of promutagens into reactive metabolites by inhibiting metabolic activation, stimulating detoxification enzymes, or scavenging reactive molecules. As suppressing agents, they may modulate intracellular processes, which are involved in DNA repair mechanisms, tumor promotion, and tumor progression.²⁸ Discovery and exploration

of compounds possessing antimutagenic and anticarcinogenic properties are of great importance.

15.5 Ayurvedic Herbs as Antimutagens

Phytochemicals are secondary metabolic products produced by plants in response to environmental stresses. Thousands of these phytochemicals have been identified and, when consumed in human diet, may affect chronic disease risk.²⁹ Of well over 2000 preparations known to the modern practitioner of Ayurveda, nearly 1500 are of plant origin. *Susruta Samhita* refers to 700 drugs including a small number which were not available in the country in that time. The list has grown substantially since then. Ancient Indian medical literature has references not only to plants that cure difficult and incurable diseases, but also to some endowed with many magical properties. Times have changed and people are looking at logical causes and effects. India has been exposed for well over a century to the application of allopathic medicines with their definite merits as well as failings.

There is a return to some kind of natural healing and to Ayurveda and Siddha medicines. The blind superstitious belief of the past has prompted extensive testing of several ancient medicines. Out of the known 1500, well over 100 medicines have qualified for entry in the Indian and British pharmaceutical Codex and the U.S. dispensatory. The investigations have involved clinical and pharmacological testing of principal components, such as phenolics and alkaloids, extracted from the herbs. Studies carried out on antimutagenic effects of the Ayurvedic therapies were critically reviewed, their protocol studied, and the significant results have been pointed out, as this effect may account for their therapeutic effect to great extent. The Ayurvedic herbs discussed here with respect to their antimutagenic and antiviral activity occupy an important place in the Ayurvedic system of medicine and are used in the treatment of various ailments either alone or from a part of various formulations.

15.6 Scientific Basis of Antimutagenic and Antiviral Activity of Ayurvedic Therapies

Antimutagenic and antiviral activities of the herbs used in Ayurveda are presented. The review indicates that the use of these herbs may be protective from exposure to environmental mutagens.

15.6.1 *Terminalia arjuna*

With a view to explore Ayurvedic plants for chemopreventive activities, *T. arjuna* was selected for testing of antimutagenic activity in our laboratory. *T. arjuna* is an important cardiogenic plant described in Ayurveda and is widely used in the preparation of important Ayurvedic formulations like *Arjunaristam*, *Cintamanirasam*, *Laksagulgulu*, *Liv.52*, etc.

15.6.1.1 Antimutagenic and Antitumor Activity

Ellagic acid was isolated from *T. arjuna* and its antimutagenic potential was evaluated in TA98 and TA100 strains of *Salmonella typhimurium* against direct- and indirect-acting mutagens. It was found to be quite effective against promutagen 2-aminofluorene.³⁰ Various other fractions were also isolated and were tested in Ames assay, comet, and micronucleus (MN) tests, which indicate that the bark harbors constituents with promising antimutagenic and anticarcinogenic potential.^{31–33} Gallic acid, ethyl gallate, and luteolin were reported to be active cancer-cell growth inhibitory constituents from *T. arjuna* by Pettit and co-workers³⁴ using bioassay-guided separation methods. Kandil and Nassar³⁵ have reported an ellagitannin from the leaves of *T. arjuna* to be an anticancer promoter. The effects of acetone and methanol extracts of *T. arjuna* were investigated on the growth of human normal fibroblasts (WI-38), osteosarcoma (U2OS), and glioblastoma (U251) cells *in vitro*.³⁶ It was found that both extracts inhibited the growth of human U251 and U2OS at 30 to 60 µg/ml. It was also discovered that the extracts contained components that can induce growth arrest of transformed cells by p53-dependent and independent pathways.

15.6.1.2 Antiviral Activity

Casuarinin, a hydrolysable tannin isolated from the bark, has been shown to possess anti-herpes virus activity.³⁷

15.6.2 *Ocimum sanctum*

Different parts of *O. sanctum* are traditionally used in Ayurveda and Siddha systems for treating diverse ailments. Examples include infections, skin diseases, hepatic disorders, common cold and cough, and malarial fever. *O. sanctum* is also used as an antidote for snake bite and scorpion sting.³⁸

15.6.2.1 Antimutagenic and Antitumor Activity

O. sanctum was one of the nine plant products that significantly decreased the incidence of both benzo[a]pyrene (B[a]P)-induced neoplasia in the stomachs of Swiss mice and 3'-methyl-4-dimethylaminoazobenzene (3'MeDAB)-induced hepatomas in Wistar rats.³⁹ The topical treatment of the ethanolic leaf extract of *O. sanctum* significantly reduced the values of tumor incidence, the average number of tumors per tumor bearing mice, and the cumulative number of papillomas in 7,12-dimethylbenz[a]anthracene (DMBA)-induced skin papillomagenesis in male Swiss albino mice at the periinitiation, postinitiation, or continuously at both the peri- and postinitiation stages of papillomagenesis as compared with the corresponding control group.⁴⁰ There was also significant twofold elevation of reduced glutathione content in the skin of mice ($p < 0.05$) and an elevation of glutathione S-transferase activity by 25% compared with the control group ($p < 0.05$) after 15 days of the treatment.

In a study carried out by Banerjee et al.,⁴¹ oral treatment with the alcoholic leaf extract at 400 and 800 mg/kg body weight given to mice for 15 days significantly elevated the activities of cytochrome P450 ($p < 0.05$), cytochrome b5 ($p < 0.01$, $p < 0.001$), aryl hydrocarbon hydroxylase ($p < 0.05$), and glutathione S-transferase ($p < 0.05$, $p < 0.01$), all of which are important in the detoxification of carcinogens as well as mutagens. The

reduced glutathione level in liver, lung, and stomach tissues was also elevated ($p < 0.01$, $p < 0.001$) by treatment with the leaf extract.

O. sanctum leaf extract was also reported by Prashar and coworkers⁴² to block or suppress the events associated with chemical carcinogenesis by inhibiting metabolic activation of the carcinogen. They treated the primary cultures of rat hepatocytes with up to 500 μg of *O. sanctum* extract for 24 h and then with DMBA (10 or 50 μg) for 18 h. A significant reduction in the levels of DMBA-DNA adducts was observed in all cultures pretreated with *O. sanctum* extract. *O. sanctum* also showed chemopreventive activity against DMBA-induced hamster buccal pouch carcinogenesis.⁴³ *O. sanctum* in the form of fresh leaf paste, aqueous extract, and ethanolic extract were topically applied and the extracts were orally administered to buccal pouch mucosa of animals exposed to 0.5% of DMBA. There was significant reduction in the incidence of papillomas and squamous cell carcinomas. In addition, there was an increase in the survival rate in the topically applied leaf paste and orally administered extracts to animals. From the observations, it was suggested that the orally administered extract of *O. sanctum* may have the ability to prevent the early events of carcinogenesis.

Antiproliferative activity of seed oil of *O. sanctum* against HeLa cells in culture has also been reported.⁴⁴ Prakash and Gupta⁴⁵ tested the chemopreventive activity of the seed oil of *O. sanctum* against subcutaneously injected 20-methylcholanthrene-induced fibrosarcoma tumors in the thigh region of Swiss albino mice. Supplementation of maximal tolerated dose of 100 $\mu\text{l/kg}$ body weight of the oil significantly reduced the 20-methylcholanthrene-induced tumor incidence and tumor volume. In liver enzymatic and non-enzymatic antioxidants and lipid peroxidation end product, malondialdehyde levels were significantly modulated with oil treatment as compared with untreated 20-methylcholanthrene-injected mice. The potential chemopreventive activity of the oil was partly attributed to its antioxidant properties.

15.6.2.2 Antiviral Activity

O. sanctum is an ingredient of *tefroli*, which is used in condition of viral hepatitis. Rajalakshmi and co-workers⁴⁶ have reported *O. sanctum* to show highly significant clinical and biochemical clearance of viral hepatitis (*manjal kamalai*) when used as a single drug within 2 weeks of treatment. Patients for the above study were screened and selected from the outpatient department of the Central Research Institute (Siddha) [C.R.I], Chennai, India, based on the symptoms such as fever, dull ache in the right costal margin, yellow tint of the sclera, anorexia, nausea and vomiting, dark-colored urine, clay-colored stools, and enlarged and tender liver. Clinical diagnosis was confirmed by laboratory parameters such as bile salts, bile pigment level in urine, and liver function test along with additional laboratory parameters. Serum alkaline phosphatase, serum amylase, cholecystogram, and stool tests were done to rule out the cases of surgical jaundice, neoplasm in the abdominal cavity, obstruction in the biliary tract due to stone, and infestation with worms, respectively. Twenty cases were screened and admitted in the inpatient department of CRI for trial.

All the cases were assessed once a week with both clinical and biochemical parameters. The leaves of *O. sanctum* were ground into paste and given in the dose of 10 g/day in two divided doses. A diet of fat, tamarind, and spices with plenty of glucose was given throughout the period of treatment. Within a week, anorexia, nausea, and vomiting disappeared and stool regained its normal color. After 2 weeks of treatment, 50% of the cases showed clearance of conjunctiva and dark-colored urine was seen only in 15% cases. The reduction in the icteric index level was observed to be slow when compared with reduction

in serum bilirubin level. After 3 weeks of treatment, all the symptoms had cleared except liver enlargement, which was persistent in 10% of the cases.

15.6.3 *Glycyrrhiza glabra*

The root of *G. glabra* is extensively used in traditional medicine and in food products as a sweetening and flavoring agent.

15.6.3.1 Antimutagenic and Antitumor Activity

Zani et al.⁴⁷ investigated the effects of *G. glabra* extract, glycyrrhizinic acid, and 18 α - and 18 β -glycyrrhetic acids on the mutagenicity of ethylmethanesulfonate (3 μ l/plate), *N*-methyl-*N'*-nitro-*N*-nitrosoguanidine (1 μ l/plate), and ribose-lysine (25 μ l/plate) Maillard model systems in TA100 tester strain of *S. typhimurium* with the *S. typhimurium* and microsome reversion assay. The compounds tested were also found to exhibit des-mutagenic activity, where the compound exerted its effect by direct interaction with the mutagen, and antimutagenic activity, where the compound acted at the cellular level suppressing the process of mutagenesis. All the compounds showed des-mutagenic activity only against ribose-lysine mutagenic browning mixture. *G. glabra* extract showed antimutagenic activity against ethylmethanesulfonate and ribose-lysine.

Glycyrrhizin, the main water-soluble constituent of licorice, was shown to possess considerable antitumorigenic activity in Sencar mice. In this study, Agarwal et al.⁴⁸ showed that oral feeding of glycyrrhizin to Sencar mice resulted in substantial protection against skin tumorigenesis caused by DMBA initiation and 12-*O*-tetradecanoylphorbol-13-acetate (TPA) promotion. The latent period prior to the onset of tumor development was considerably prolonged in glycyrrhizin-fed animals compared with animals not fed by glycyrrhizin. Results showed a significant decrease in the number of tumors per mouse both during and at the termination of the experiment. Oral feeding of glycyrrhizin in drinking water also resulted in inhibition in the binding of topically applied [³H]B[a]P and [³H]DMBA to epidermal DNA. The possible mechanism(s) of the antitumor-initiating activity was attributed to the possible involvement of glycyrrhizin as an inhibitor of the carcinogen metabolism followed by DNA adduct formation.

15.6.3.2 Antiviral Activity

Pompei et al.⁴⁹ studied the effect of glycyrrhizic acid on the growth of vaccinia, herpes simplex type 1 (HSV-1), Newcastle disease, vesicular stomatitis, and polio type 1 viruses in cultures of human aneuploid HEp2 cells. Twenty-four-hour-old cell monolayers (10⁷ cells per sample) were infected with five infectious units per cell of each virus at 20°C for 1 h, washed three times in Hank's balanced saline solution (BSS), and incubated at 37°C for 18 h in Eagle's essential medium supplemented with 2% calf serum (pH 7.4). Infectious virus yield was determined by the Dulbecco and Vogt technique and slightly modified for vaccinia and HSV-1 viruses. Cytopathic effects were evidenced by observing Giemsa-stained cells with a light microscope and by measuring spectrophotometrically at 530 nm the amount of neutral red incorporated by cell cultures (100 μ g/ml, 1-h pulses in drug-free medium) after solubilization in 1% sodium deoxycholate in BSS. It was reported that addition of 8mM glycyrrhizic acid after incubation completely inhibited both growth and cytopathic effects of viruses except poliovirus type 1. It produced irreversible inactivation of HSV-1 virus as the suspensions of this virus suffered a loss of infectivity of 10⁵ when incubated at 37°C with 8 mM glycyrrhizic acid for only 15 min. It was hypothesized that

glycyrrhizic acid interacted with sensitive virus proteins both at virionic stage and later, when these are synthesized in host cells.

Glycyrrhizin was reported to inhibit varicella-zoster virus (VZV) in human embryonic fibroblast (HEF) cells *in vitro*.⁵⁰ On treatment of HEF with glycyrrhizin after inoculation of virus (posttreatment), the average 50% inhibitory dose (ID_{50}) for five VZV strains was 0.71 mM, and the selectivity index (ratio of ID_{50} for host-cell DNA synthesis to ID_{50} for VZV replication) was 30. Glycyrrhizin was also effective against VZV replication when HEF cells were treated 24 h before the inoculation (pretreatment). Furthermore, at a concentration of 2.4 mM glycyrrhizin inactivated more than 99% of virus particles within 30 min at 37°C. Glycyrrhizin was reported to show a dose-dependent inhibition of the replication of human immunodeficiency virus type 1 in MOLT-4 (clone No. 8) cells within the concentration range of 0.075 to 0.6 mM.⁵¹ Within this concentration range, glycyrrhizin also effected a dose-dependent reduction in the protein kinase C (PKC) activity of MOLT-4 cells. PKC inhibition was considered as one of the mechanisms by which glycyrrhizin inhibited human immunodeficiency virus type-1 (HIV-1) replication as a PKC inhibitor, 1-(5-isoquinolinesulfonyl)-2-methylpiperazine dihydrochloride, also proved inhibitory to HIV-1 replication in MOLT-4 cells.

Badam⁵² reported that indigenously purified glycyrrhizin was a more potent antiviral agent than licorice from *G. glabra* and ammonium salt of glycyrrhizic acid (Sigma) in inhibiting Japanese encephalitis virus (JEV) *in vitro*. Glycyrrhizin was found to inhibit plaque formation in all the three strains of JEV — Nakayama, P-20778, and 821564 XY48 — at a concentration of 500 µg/ml at 96 h, whereas licorice and ammonium salt of glycyrrhizic acid inhibited at 1000 µg/ml concentration. A daily injection of glycyrrhizin (stronger neo-minophagen C [SNMC] containing 40 mg glycyrrhizin in a 20-ml ampoule) was reported to lower alanine aminotransferase (ALT) levels in patients with chronic viral hepatitis.⁵³ The therapeutic effects of intermittent administration of SNMC three times/week for 12 weeks were evaluated and compared between two doses (40 and 100 ml) in a randomized clinical trial. The therapeutic response was better in the 53 patients allocated 100 ml than in the 59 who were allocated to have 40 ml of SNMC. At the completion of SNMC treatment, ALT levels decreased more extensively in the patients on 100 ml than in those on 40 ml of SNMC. Minor side effects occurred in both the patients on 100 ml of (20%) and in those on 40 ml (12%) but did not require any therapies. It was suggested that intermittent SNMC would be efficient in suppressing ALT levels in patients with chronic viral hepatitis in a dose-dependent manner.

15.6.4 *Semecarpus anacardium*

S. anacardium Linn. of the family Anacardiaceae has many applications in the Ayurvedic and Siddha systems of medicine. It is popularly known as multipurpose medicine (*ardha vaidya*).

15.6.4.1 Antimutagenic and Antitumor Activity

Water, alcohol, and oil extracts of *S. anacardium* were found to be antimutagenic against B[a]P-induced mutagenicity in TA98 and TA100 tester strains of *S. typhimurium*.⁵⁴ Methanol extract, resinous fraction, and *Bhilawanol* isolated from *S. anacardium* showed antitumour activity against P388 lymphocytic leukemia in BDF1 mice as judged by their median survival time.⁵⁵ Smit and co-workers⁵⁶ studied 14 specimens from the list of Ayurvedic herbal drugs and collected from various parts of India and Nepal for cytostatic activity.

The ethanolic (70% v/v) extracts were tested for cytotoxicity on COLO 320 tumor cells, using the microculture tetrazolium assay. The nuts of *S. anacardium* displayed a cytotoxic effect with a IC_{50} -value of 1.6 μ g/ml.

Premalatha et al.⁵⁷ have reported the modulating effect of the extract of SA against aflatoxin B₁-induced experimental hepatocellular carcinoma. Anticancer property was attributed to its strong antioxidant capacity and its capability to induce *in vivo* antioxidant system.⁵⁸ In a study,⁵⁹ the beneficial effect of SA in the treatment of hepatocellular carcinoma was attributed to the stabilization of biomembranes by *S. anacardium* nut extract. *S. anacardium* nut milk extract was administered orally at a dose of 200 mg/kg/day for 14 days to normal rats as well to animals whose biomembranes were rendered fragile by the induction of hepatocellular carcinoma with aflatoxin B₁, where the discharge of lysosomal enzymes increased significantly with a subsequent increase in glycoprotein components. The nut extract administration reversed these adverse changes to near normal in treated animals. The administration of the same dose to male albino rats with aflatoxin B₁-induced hepatocellular carcinoma was found to be highly effective in inducing phase I and phase II biotransformation enzymes.⁶⁰

The administration of *S. anacardium* nut extract has also been reported to cause a significant decrease in the activity of glycolytic enzymes and an increase in gluconeogenic enzymes activities to near normal values in drug-treated animals.⁶¹ *Anacartin forte*, an Ayurvedic preparation, exhibited not only a broad spectrum of anticancer properties in clinical and animal studies, but also a wide margin of safety in therapeutic dosage even when used for long periods. It showed satisfactory results in cases of cancer of the esophagus, liver, urinary bladder, liver, and chronic leukemia by giving subjective and objective improvement, alleviation, or disappearance of troublesome symptoms and clinical benefit with extension of survival time. The preparation has selective action, attacking only the cancer cells without harming the normal cells.⁶²

15.6.5 *Terminalia chebula*

T. chebula is a *rasayana to vata*, increasing awareness, and has a tonic effect on the central nervous system. It improves digestion, promotes the absorption of nutrients, and regulates colon function. *T. chebula* is most useful in prolapsed organs, improving the strength and tone of the supporting musculature.

15.6.5.1 Antimutagenic and Antitumor Activity

Water extract of *T. chebula* was shown to significantly reduce 4-nitro-*o*-phenylenediamine (NPD) as well as 2-AF induced his⁺ revertants in Ames assay.⁶³ A tannin fraction (TC-E) obtained from the dried fruit pulp of *T. chebula* was subjected to column chromatography to yield four fractions — TC-EI, TC-EII, TC-EIII, and E-IV — which were evaluated for their antimutagenic potential. The fractions were quite effective in inhibiting the mutagenicity of 2-AF. The monomeric fraction (TC-EI) was the least effective in comparison with other oligomeric fractions.⁶⁴ A 70% methanol extract of *T. chebula* fruit was studied for its effects on growth in several malignant cell lines including a human (MCF-7) and mouse (S115) breast cancer cell line, a human osteosarcoma cell line (HOS-1), a human prostate cancer cell line (PC-3), and a nontumorigenic immortalized human prostate cell line (PNT1A) by using assays for proliferation, cell viability, and cell death. In all cell lines studied, the extract decreased cell viability, inhibited cell proliferation, and induced cell death in a dose-dependent manner.⁶⁵

15.6.5.2 Antiviral Activity

Badmaev and Nowakowski⁶⁶ tested a multicomponent herbal formula, ledretan-96, consisting of 23 components on an epithelial tissue culture cell line (MDCK) for its protective activity against cytopathic effects caused by influenza A virus. The whole formula and each of its 23 individual components were tested in the same system. The results indicated that the formula, when prepared according to established procedure, in the form of decoction was active in protecting epithelial cells against damage caused by influenza A virus used at different dosages. Of the 23 components tested, only *T. chebula* showed a significant protective effect when applied to the epithelial cells individually. The bioassay-directed isolation of *T. chebula* fruits afforded four HIV-1 integrase inhibitors, gallic acid, and three galloyl glucoses.⁶⁷ The galloyl moiety played a major role for inhibition against the 3'-processing of HIV-1 integrase.

T. chebula has also been reported to show a significant inhibitory activity on human immunodeficiency virus reverse transcriptase.⁶⁸ Anti-HSV-1 activity of *T. chebula* has been reported by Kurokawa et al.⁶⁹ *T. chebula* reduced virus yield in the brain and skin more strongly than acyclovir alone and exhibited stronger anti-HSV-1 activity in the brain than in the skin, in contrast to acyclovir treatment by itself. A group⁷⁰ from Japan showed *T. chebula* to possess anticytomegalovirus (CMV) activity. Shiraki et al.⁷¹ reported *T. chebula* as one of the medicinal plants to inhibit replication of human CMV and murine CMV *in vitro* and suggested it to be beneficial for the prophylaxis of CMV diseases in immunocompromized patients.

15.6.6 Terminalia bellerica

T. bellerica is also a strong rejuvenator of the body and is recommended as a daily supplement.

15.6.6.1 Antimutagenic Activity

Two polyphenolic fractions isolated from *T. bellerica* were significantly effective against mutagenic effects in *S. typhimurium*. Interaction of the polyphenols with S9 proteins may be the probable cause of the inhibitory effect.⁷²

15.6.6.2 Antiviral Activity

Suthienkul et al.⁷³ reported the extract of *T. bellerica* to show retroviral reverse transcriptase inhibitory activity. Hot water and methanol extracts of 57 Thai herbs and spices were examined for their retroviral reverse transcriptase inhibitory activity with Moloney murine leukemia virus reverse transcriptase. Hot water extract of *T. bellerica* at a concentration of 125 µg/ml showed relative inhibitory ratio of 75%, whereas its methanol extract exhibited an inhibitory ratio (IR) value of 83%. An extract of *T. bellerica* showed significant inhibitory activity on HIV-1 reverse transcriptase, with $IC_{50} \leq 50$ µg/ml.⁶⁸ Four lignans isolated possessed demonstrable anti-HIV-1 activity *in vitro*.⁷⁴

15.6.7 Emblica officinalis

E. officinalis is an important traditional medicine with broad prospects. The fruits of the plant have been used in Ayurveda as potent *rasayanas* and form the major constituent of *chyavanprash awaleha* and *triphala*.

15.6.7.1 Antimutagenic and Antitumor Activity

Water, acetone, and chloroform extracts of *E. officinalis* fruit were reported to significantly reduce the mutagenicity of sodium azide and NPD in TA100 and TA97a strains, respectively, of *S. typhimurium*.⁷⁵ *E. officinalis* extract was reported to show a pronounced protective effect in counteracting the genotoxicity induced by aluminium and lead.^{76,77} Oral administration of *E. officinalis* extract for 7 consecutive days before the exposure of mice to the metals by intraperitoneal injections reduced the frequencies of sister chromatid exchanges and micronuclei-induced in bone marrow cells by both metals. Aqueous extract of *E. officinalis* was reported to be quite effective in inhibiting mutagenicity of S9-dependent mutagens, aflatoxin B₁ (0.5 µg/plate), and B[a]P (1 µg/plate) in TA98 and TA100 tester strains of *S. typhimurium* in Ames assay.⁷⁸

Dietary supplementation with extract of fruit of *E. officinalis* to Swiss albino mice significantly reduced the cytotoxic effects of 3,4-B[a]P *in vivo*.⁷⁹ Age-matched Swiss albino mice were fed by gavaging the fruit extract daily for 28 days, and one dose of the carcinogen was given on alternate days up to a total of eight doses from day 9. On day 29, all mice were transferred to normal diet. Control sets received the extract alone, the carcinogen alone, and olive oil alone. All mice were sacrificed at 12 weeks and 14 weeks after the end of the experiment. Chromosome preparations were made from bone marrow after the usual colchicine-hypotonic-fixative-airdrying-Giemsa staining schedule. The end points screened were the frequencies of chromosomal aberrations and damaged cells that were induced, which clearly pointed toward the modulation of B[a]P-induced cytotoxic effects by the fruit extract.

Sharma and co-workers⁸⁰ studied the effect of *E. officinalis* extract administration on the *in vivo* genotoxicity of B[a]P and cyclophosphamide (CP) using bone marrow chromosomal aberration and micronucleus induction tests in mice. Three doses (50, 250, and 500 mg/kg body weight) of the plant extract were administered orally for 7 consecutive days prior to the administration of single dose of mutagens (B[a]P 125 mg/kg oral; CP 40 mg/kg intraperitoneal). It was found that administration of 250 and 500 mg/kg of *E. officinalis* extract significantly inhibited the genotoxicity of B[a]P as well as CP in both the assay systems. There was a significant induction in the levels of glutathione content (GSH) and of antioxidant and detoxification enzymes. Extract of *E. officinalis* significantly inhibited hepatocarcinogenesis induced by N-nitrosodiethylamine (NDEA) in a dose-dependent manner.⁸¹ Its anticarcinogenic activity was evaluated by its effect on tumour incidence, levels of carcinogen metabolizing enzymes, levels of liver cancer markers, and liver injury markers, which clearly indicated its protection against chemical carcinogenesis.

In a study,⁸² *in vitro* antiproliferative activity of extracts from medicinal plants were compared with human tumor cell lines, including human erythromyeloid K562, B-lymphoid Raji, T-lymphoid Jurkat, and erythroleukemic HeLa cell lines. Extracts from *E. officinalis* were the most active in inhibiting *in vitro* cell proliferation and contained pyrogallol as an active component. Aqueous extract of *E. officinalis* has also been reported by Jose et al.⁸³ to be cytotoxic to L929 cells in culture in a dose-dependent manner. The concentration needed for 50% inhibition was found to be 16.5 µg/ml. *E. officinalis* and chyavanaprash extracts were also found to reduce ascites and solid tumors in mice induced by Dalton's lymphoma ascites (DLA) cells. Animals treated with 1.25 g/kg body weight of *E. officinalis* extract increased the life span of tumor-bearing animals by 20%, whereas animals treated with 2.5 g/kg body weight of *chyavanaprash* produced a 60.9% increase in the life span. Antitumor activity of *E. officinalis* extract was attributed partially to its interaction with cell cycle regulation as was found to inhibit cell cycle-regulating enzymes.

15.6.7.2 Antiviral Activity

A bioassay-guided fractionation of a methanol extract of the fruit of *E. officinalis* yielded Putranjivain A (1) as a potent inhibitory substance on the effects of HIV-1 reverse transcriptase with $IC_{50} = 3.9 \mu M$, together with (2) 1,6-di-O-galloyl- β -D-glucose, (3) 1-O-galloyl- β -D-glucose, (4) kaempferol-3-O- β -D-glucoside, (5) quercetin-3-O- β -D-glucoside, and (6) digallic acid. The inhibitory mode of action by 1, 2, and 6 was noncompetitive with respect to the substrate but competitive with respect to a template-primer.⁶⁸

15.6.8 *Cinnamomum cassia*

C. cassia is known as *dalchini* in the Indian subcontinent. The bark of *C. cassia* is frequently used in Ayurveda and Unani preparations.

15.6.8.1 Antimutagenic Activity

Sharma et al.⁸⁴ evaluated the antimutagenic effect of *C. cassia* against two mutagens, B[a]P and CP, by *in vivo* chromosomal aberration and micronuclei tests after pretreatment with the extract orally for 7 consecutive days and with Ames assay. *C. cassia* pretreatment decreased cytochrome P450 content but increased GSH content and the activity of glutathione-dependent antioxidant enzymes. These results indicate its modulatory effect on the xenobiotic bioactivation and detoxification processes. 2'-Hydroxycinnamaldehyde isolated from *C. cassia* was found to strongly inhibit *in vitro* growth of 29 kinds of human cancer cells and *in vivo* growth of SW-620 human tumor xenograft without the loss of body weight in nude mice.⁸⁵ It prevented adherence of SW-620 cells to the culture surface but did not inhibit oncogenic K-Ras processing, implying its antitumor mechanisms at the cellular level. Kwon et al.⁸⁶ found the key functional group of the cinnamaldehyde-related compounds in the antitumor activity to be in the propenal group.

15.6.8.2 Antiviral Activity

C. cassia was one of the many medicinal plants possessing potent anti-HIV activity. Premanathan et al.⁸⁷ studied the inhibitory effect of plant extracts on HIV replication in terms of the inhibition of virus-induced cytopathogenicity in MT-4 cells. The MT-4 cells were infected with HIV. The HIV-infected MT-4 cells were incubated at 37°C in a CO₂ incubator in the presence of the plant extracts. After 5 days, cell viability was measured by a tetrazolium-based colorimetric assay.

15.6.9 *Withania somnifera*

W. somnifera is one of the Indian medicinal plants having a remarkable reputation, as a factor of health care, among the indigenous medical practitioners. Several studies over the past few years have indicated that *W. somnifera* has anti-inflammatory, antitumor, antistress, antioxidant, mind-boosting, and rejuvenating properties.

15.6.9.1 Antitumor Activity

The alcoholic extract of the dried roots of the plant, as well as the active component withaferin A isolated from the extract, showed significant antitumor and radiosensitizing effects in experimental tumors *in vivo*, without any noticeable systemic toxicity.⁸⁸ Russo

et al.⁸⁹ reported the effect of methanolic extract of *W. somnifera* in reducing the hydrogen peroxide-induced cytotoxicity and DNA damage in human nonimmortalized fibroblasts. The extract showed dose-dependent free radical scavenging capacity and a protective effect on DNA cleavage.

The chemopreventive activity of a hydroalcoholic extract of *W. somnifera* roots against 20-methylcholanthrene induced fibrosarcoma tumors in Swiss albino mice was reported by Prakash et al.⁹⁰ A single subcutaneous injection of 200 µg 20-methylcholanthrene in 0.1 ml of dimethyl-sulphoxide into the thigh region of mice produced a high incidence (96%) of tumors. Oral treatment of animals with 400 mg/kg body weight of extract (1 week before injecting 20-methylcholanthrene and continuing until 15 weeks thereafter) significantly reduced the tumor incidence and tumor volume and enhanced the survival of the mice, compared with 20-methylcholanthrene-injected mice. Liver biochemical parameters revealed a significant modulation of reduced glutathione, lipid peroxides, glutathione-S-transferase, catalase, and superoxide dismutase in extract-treated mice. The extract was also quite effective in preventing DMBA-induced squamous cell carcinoma of skin in Swiss albino mice.⁹¹ The skin lesions were induced by the twice-weekly topical application of DMBA (100 nmol/100 µl acetone) for 8 weeks on the shaved backs of mice. The extract was administered at the maximal tolerated dose of 400 mg/kg body weight three times/week on alternate days 1 week before DMBA and continued for 24 weeks thereafter. The chemopreventive activity was also linked to the antioxidant and free radical-scavenging constituents of the extract.

15.6.10 *Centella asiatica*

Centella asiatica (Umbelliferae) syn. *Hydrocotyl asiatica* has been used in various parts of India for different ailments. Examples include headaches, body aches, insanity, asthma, leprosy, ulcers, eczemas, and wound healing.

15.6.10.1 Antimutagenic and Antitumor Activity

The researchers at the Amala Cancer Research Centre in Kerala, India, tested both crude extract of *C. asiatica* and its partially purified fractions (AF) for their antitumor activity. AF inhibited the proliferation of the transformed cell lines significantly more than the crude extract and other solvent fractions in dose-dependent manner. Fifty percent effective doses of AF on its 3-h exposure for Ehrlich ascites tumor cells (EAC) and DLA were reported to be 17 and 22 µg/ml, respectively. AF also significantly suppressed the multiplication of mouse lung fibroblast (L-929) cells at a concentration of 8 µg/ml in long-term culture. Oral administration of the extracts retarded the development of solid and ascites tumors and increased the life span of these tumor-bearing mice. Tritiated thymidine, uridine, and leucine incorporation assay suggested that the fraction acted directly on DNA synthesis.⁹² Yen et al.⁹³ reported the inhibitory effect of *C. asiatica* against the mutagenicity of 2-amino-3-methyl-imidazole (4,5-f) quinoline.

15.6.10.2 Antiviral Activity

Aqueous extract of *C. asiatica* was one of the 500 herbs tested that showed significant anti-HSV-II action as determined by the virus inhibition logarithm.⁹⁴ Yoosook et al.⁹⁵ reported asiaticoside from *C. asiatica* to be an active constituent against antiherpes simplex virus. It showed both anti-HSV-1 and 2 activities in plaque inhibition assay.

15.7 Conclusion

A search of the literature on Ayurvedic herbs revealed ten herbs that have studies showing antimutagenic and antiviral activity: *T. arjuna*, *O. sanctum*, *G. glabra*, *S. anacardium*, *T. chebula*, *T. bellerica*, *E. officinalis*, *C. cassia*, *W. somnifera*, and *C. asiatica*. The data are indicative of their possible protective effect against the environmental mutagens. Future research is needed to further confirm their antimutagenic and antiviral effects in animals and humans.

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16

Benign Growths, Cysts, and Malignant Tumors

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16.1 Introduction

Approximately 1.2 million new cases of invasive cancer are confirmed each year in the U.S., and about 500,000 people die from the disease each year.¹ The second most deadly disease is cancer and is expected to be the first in the early 21st century. Over the past century there have been tremendous changes in the environment, both dietary and life-style, resulting in the increase of lung cancer in both men and women. Melanoma, non-Hodgkin's lymphoma, and brain tumors have also increased, particularly among the elderly. The overall mortality, especially with people under the age of 65, has declined mostly due to increasingly effective therapy for cancers of fetal and hematopoietic origin.¹

In spite of progress in the science of medicine, the understanding of the cancerous condition is still not complete. Although the knowledge has expanded considerably about the etiology and treatments, the exact cause of the disease and an effective management of cancer in general is still an unresolved mystery to the scientists. Cancer has been recognized and characterized by the science of Ayurveda. The descriptions regarding this

disease are available in a scattered form under the context of various diseases. Compilation of this information collectively, in a systemic manner, may help us in understanding the etiology and the pathology of the disease in a better way. Numerous drugs have been mentioned for the management of this condition; if evaluated methodically, they may generate some curative or supportive remedy for the sufferers of this disease. The present era of modern sciences has undergone many folds of technical advancement for the diagnosis and confirmation of cancer by histopathological studies under light and electron microscopes. Before this, doctors and researchers had to depend entirely on various clinical symptomatology with *dosa* theory.

The purpose of this chapter is to review and summarize the available information on Ayurvedic management of benign growths, cysts, and malignant tumors.

16.2 Historical Perspective

Cancer is derived from the Greek word meaning a crab, presumably because a cancer "adheres to any part that it seizes upon in an obstinate manner like the crab." The large veins surrounding a malignant growth, which to the ancients suggested the claws of a crab, medically inspired the use of this term.

The identification and description of malignant diseases are available in the literature of ancient India and Ayurveda. The earliest and foremost records are cited in *Atharva Veda* (2200 B.C.). During this period, the disease was probably described under the heading of *apachi* or *apachit*, which refers to the present knowledge of various types of lymph node swellings. In a later period, Sushruta (400 B.C.), in his classic *Sushruta Samhita*, described this *apachi* as multiple lymph node swellings that may arise at different places such as the neck, axilla, and groin.

The ancient Indian clinicians were aware of the malignant diseases and presented their views regarding neoplasia as follows:

1. As a swelling on the body surface or situated in deeper structure in relation to various systems and organs
2. As a chronic nonhealing ulcer

In classical texts of Ayurveda, the superficial swellings have been categorized under the heading of tumors (*arbuda*), whereas nonhealing ulcers listed as incurable ulcers (*asadhyavrana*).

The word *arbuda* has been derived from the root "Arb" with suffix "Ena" along with augmentation of "Nd," which means "to destroy" (particularly *mamsa dhatu*). Grammatically, it denotes the fleshy outgrowths. *Arbuda* has various meanings such as swelling, number of 10 billions, a mountain, a fleshy mass, a serpent, clouds, or a demon (i.e., a serpent). During the *Vedic* period, *arbuda* was considered as a serpentlike demon that was conquered by "Lord Indra," whereas the literary meaning of *arbuda* is a lump or a mass or a polyp. According to Sushruta, *arbuda* are gradually increasing, big, globular, slightly painful, fixed, deep-seated, fleshy masses that usually do not suppurate. They can arise from any part of the body surface. They are caused by the derangement of *mamsa* and *rakta* vitiated by *tridosha*.

The malignant lesions of deeper structures have been described in various contexts, such as intra-abdominal lesions having been described in the name of *gulma*. *Gulma* literally means a shrub (i.e., it can be understood as a lump or a mass in the abdomen and many of these lumps have been described to be incurable). The description of neoplastic growths is also available under the heading of *granthi*. The characteristic features of *arbuda* and *granthi* are similar. *Granthi* literally means a knotty growth. The etiology and clinical features of both *granthi* and *arbuda* are similar, with a difference that on breaking open, the *granthi* gives various discharges based on the involvement of the *dosas*. Thus, it can be understood that *granthi* may actually represent cystic growths.

The phenomenon of the spread of tumors or metastasis (*dwirarbuda*) was well known to the ancient Hindu physicians and surgeons. Several references are available regarding the local and distal spread of the tumor as well as its recurrence. Sushruta has described metastasis (i.e., distal spread of tumor under the heading of *dwirarbuda*), and the recurrence of tumor has been mentioned as *adhyarbuda*. Metastasis of tumors was described as an occurrence of a couple of similar types of tumors simultaneously serially. When a tumor arises on a preexisting site or near a primary tumor, it is called *adhyarbuda*. There are several other descriptions available in Ayurveda regarding distal spread and recurrence of tumor. While describing the treatment of tumors, Sushruta mentioned that all efforts should be made for the complete removal of tumors, as incomplete removal causes recurrence and ultimately destroys the person. To explain the graveness of recurrence, he gave an example that a small remnant tumor can destroy the body just as a small spark of fire can destroy a house. Distal spread of a tumor has also been described in connection with malignant ulcer (*asadhya vrana*).

16.3 Etiology

Sushruta has considered trauma and irritation precipitating or activating the formation of tumors as one of the etiological factors of soft-tissue tumor (*mamsarbuda*) and diseases of external genitalia (*sukaroga*) due to the use of irritants. Overeating of meat may lead to vitiation of muscle tissue and soft tissue (*mamsa*), considered an important factor for development of *mamsarbuda*. For the enlargement of external genitalia, local application of certain irritable medicines has been advised. Although this condition is rare, improper uses of these drugs used for the enlargement of external genitalia of male (*lingavidhikara yoga*) may lead to development of soft-tissue tumors (*mamsarbuda*) in external genitalia. The external application of irritable medicines and certain other internal medications have both been advised for enlarging the external genitalia. These medications may shed some light on the etiology of neoplasia, particularly the role of hormones in the development of tumors.

Vagbhata, another famous exponent of Ayurveda, also emphasizes that factors responsible for excessive formation of muscle and soft tissue (*mamsa dhatu*) may lead to the development of tumors and other pathological conditions. Examples include thyroid swelling (*galaganda*), cervical lymphadenopathy (*gandamala*), cysts (*granthi*), and fleshy growths (*adhimamsa*).

The genetic cause for the manifestation of cancer is also well documented in Ayurveda. In the etiology of familial polyposis coli (*sahaja arsha*), *Sushruta* has described that defects in ovum (*sonita*) and sperm (*shukra*) are responsible for the development of this disease.

From the above description, the factors responsible for the development of tumors may be categorized under the following headings:

1. Unwholesome diet (*mithya ahara*)
2. Unwholesome regimen (*mithya vihara*)
3. Trauma (*abhigata*)
4. Irritation
5. Genetic (*anuvanshaja*)

It can be concluded that unwholesome diet and lifestyle (*mithya ahara* and *vihara*) cause local and systemic biochemical changes. At the same time, these factors may alter the hemodynamics at the site of origin of tumors.

In conventional medicine, the common pathway in the induction (initiation) of cancer is a cellular genetic mutation that converts normal cells to cancer cells. The genes controlling normal cellular processes become mutated by a variety of chemical interactions with the dextroribonucleic acid (DNA) initiation process. These initiated genes promote cancer and are called promoter genes (oncogenes). The other normal genes suppress cancer and are called suppressor genes. Oncogenes produce cancer by encoding proteins that are further activated to induce cancer. The tumor suppressor genes cause cancer when their normal tumor suppressor function is blocked. Chemical agents, physical agents, chronic irritation, biologic agents, and even dietary factors contribute to carcinogenesis. Regulatory agencies in the U.S. and other countries inspect, monitor, and regulate environment and dietary sources of carcinogens.² The single most important agent known to contribute to increased incidence of cancer in the U.S. and Europe is tobacco, because it contributes to one third of all cancers, primarily cancers of the lung, esophageal, head, neck, and bladder.¹ Because genes define the cellular phenotype, this indicates that the mutational profile of a cancer cell may predict the clinical outcome, and this has become a reality.¹

16.4 Pathogenesis

In contemporary medicine, cancer is considered essentially as a disturbance of normal cell growth. Tumors of any cell type may show a wide range of behavior in their rate of growth, power and mode of spread, and degree of toxic effects on the host. There are tumors that grow slowly, remain quite localized, and do not invade the neighboring structures. They usually cause no harm and are known as benign tumors or harmless tumors. The other groups of tumors grow rapidly, penetrate the surrounding tissues, and spread to distant areas. They are invariably fatal and are termed cancerous tumors or malignant tumors.

The pathogenesis of tumors has been described in Ayurveda based on the theory of three *dosa* (*tridosha*). The vitiated *dosa* affects the local tissues (*dhatu* or *dushya*). *Vata*, *pitta*, and *kapha* are considered as *dosas*; muscle and soft tissue (*mamsa*), adipose tissue (*meda*), and blood (*rakta*) are termed *dushyas* (that which can be vitiated). Due to excess of *kapha* and adipose tissue, tumors usually do not suppurate. It is justified to suggest that excess of *kapha* might be responsible for the development of tumors. Moreover, as a tumor is a disorder of growth and as *kapha* is an essential factor for growth, it can also be postulated that preponderance of vitiated *kapha* is an essential factor for the manifestation of tumors.

TABLE 16.1

Clinical Features of Various Tumors in Ayurveda

Symptom	Vataja Arbuda	Pittaja Arbuda	Kaphaja Arbuda	Medaja Arbuda	Mamsaja Arbuda	Raktaja Arbuda
Pain	Variable	Burning	Mild pain with itching	Mild pain with itching	Painless or mild pain	—
Consistency	Hard or soft (variable)	—	Stony or hard	Soft	Stony	—
Mobility	Mobile	—	—	Freely mobile	Fixed	—
Skin over the tumor	Black	Red, yellowish	No change in color	Glossy	Normal or covered by prominent veins	—
Surface	Nodular	Smooth	Smooth	Smooth	Smooth	Covered with fleshy projections
Discharge	Nil	Nil	Nil	Nil	Nil	Blood
Shape and size	Raised above the surface like bladder	—	Large globular	Large	Large	Elevated fleshy growth
Growth rate	—	—	Slow	—	Slow	Rapid
Prognosis	Curable	Curable	Curable	Curable	Incurable	Incurable

In the etiopathogenesis of *raktaja tumors*, vitiated *dosa* cause the derangement of blood, and as a consequence the vessels (*siras*) of the local site become constricted. This results in a rapidly growing fleshy growth known as *raktarbuda* (ulcerated malignant tumor).

16.5 Clinical Manifestations of Tumors

Ayurveda has six major clinical manifestations of tumors. See Table 16.1 for an overview:

16.5.1 Benign Neoplasia of Fibrous Tissue and Hematoma (*Vataja Tumor*)

Various types of pain, such as pricking, stretching, and cutting, characterize *vataja* tumor. The growth assumes a black color, appears rough, and is elevated like a bladder. These tumors are mobile, soft in consistency, and change in size without reason.

16.5.2 Inflammatory State of Benign Neoplasia (*Pittaja Tumor*)

Pittaja arbuda are red or yellow and characterized by severe burning, fuming, and sucking or throbbing pain. They undergo early suppuration.

Like neoplastic growth, tumors usually do not suppurate due to excess *kapha* and *meda*. However, due to secondary infections, inflammatory changes may occur and cause sup-puration.

16.5.3 Benign Neoplasia of Osseous, Other Soft Tissues, and Adenomas (*Kaphaja* Tumor)

These swellings are cold to the touch, relatively painless, stony in consistency, and develop very slowly. The color of the tumor is similar to the color of the surrounding skin or is slightly glossy.

Based on the signs and symptoms, *vataja*, *pittaja*, and *kaphaja* tumors can be correlated to benign neoplastic conditions, irrespective of site. Examples include fibroma, hematoma, and neurofibroma (*vataja* tumors); inflammatory state of any benign neoplasia (*pittaja* tumor); and benign neoplasm of epithelial tissue, lymphoid tissue, and soft-tissue osteoma (*kaphaja* tumors).

16.5.4 Benign Conditions of Mesenchymal Tissues Like Lipoma (*Medaja* Tumor)

This tumor is slow growing, arises with slight pain, and is associated with itching. The tumor is freely mobile with a glossy appearance. The size varies according to the contents of adipose tissue (*medas*) in the body. If the content of adipose tissue in the body increases, the size of this tumor also increases and vice versa.

16.5.5 Ulcerated Malignant Lesions (*Raktaja* Tumor)

Raktaja tumors are very rapidly growing, slightly suppurated, exudating tumors covered with projectile small fleshy masses. The person affected by this tumor becomes pale due to continuous exudation of blood from this growth. This type of tumor is incurable.

16.5.6 Tumors of Muscle and Soft Tissues Like Breast Tissue (*Mamsaja* Tumor)

External trauma vitiates the soft tissues (*mamsa*) of any part of the body and gives rise to a smooth, large, painless, nonsuppurative, stony, skin-colored, fixed swelling called a *mams* tumor. Such swellings are usually covered with prominent veins. This is also an incurable tumor.

16.5.7 Tumors of Mixed Variety (*Tridoshaja* Tumor)

This type of tumor is incurable. It has mixed features of *vataja*, *pittaja*, and *kaphaja* tumor varieties.

16.6 Classification of Tumors

Based on the predominance of the vitiated *dosa* involved, tumors have been classified as *vataja arbuda*, *pittaja arbuda*, *kaphaja arbuda*, *tridoshaja arbuda*, *raktaja arbuda*, *mamsaja arbuda*, and *medaja* tumors. It is very difficult to give a specific modern nomenclature of each tumor mentioned in Ayurveda. In one study, 50 patients with various types of tumors and clinical features of different swellings mentioned in Ayurveda were compared with *vataja*, *pittaja*, and *kaphaja* benign neoplastic conditions irrespective of site. It was observed that

TABLE 16.2
Classification of Tumors According to Conventional Medicine

Tissue of Origin	Type of Tumor	
	Benign	Malignant
<i>Tumors of Connective Tissue</i>		
Fibrous tissue	Fibroma	Fibrosarcoma
Cartilage	Chondroma	Chondrosarcoma
Bone	Osteoma	Osteosarcoma
Adipose tissue	Lipoma	Liposarcoma
Blood vessels	Hemangioma	Hemangiosarcoma
Sinovial tissue	Sinovioma	Sinoviosarcoma
Smooth muscle	Leomyoma	Leomyosarcoma
Striated muscle	Rhabdomyoma	Rhabdomyosarcoma
<i>Reticulo-Endothelial Tissue</i>		
Lymphoid tissue		Hodgkin's lymphoma Non-Hodgkin's lymphoma
Granulocytes		Myeloid leukemia
Plasma cell		Myeloma
<i>Tumor of Epithelial Tissue</i>		
Squamous	Papilloma	Squamous cell carcinoma
Transitional	Papilloma	Transitional cell carcinoma
<i>Glandular</i>		
Papillary	Papilloma	Papillary carcinoma
Solid	Adenoma	Adenocarcinoma
Cystic	Cystadenoma	Cystadinocarcinoma
Mixed	Papillary cystadenoma	Papillary cystadinocarcinoma
<i>Neural Tissue</i>		
Glial tissue		Glioma
Nerve sheath	Neuroleimoma	Neurogenic sarcoma
Meningis	Meningioma	Meningeosarcoma
<i>Other Tissues</i>		
Placenta	Hydatidiform mole	Choriocarcinoma
Pigmented cells	Nevus	Malignant melanoma
Germ cell	Benign teratoma	Malignant teratoma

the *kaphaja* tumor is benign neoplasm of epithelial tissue, lymphoid tissue, and soft tissue; the *medaja* tumor is benign conditions of mesenchymal tissues like lipoma, fibrolipoma, and hemangioma. A *raktaja* tumor is an ulcerated malignant lesion and includes carcinoma of integument and other ulcerated malignant growth. The *mamsaja* tumor is a testicular tumor, ductal carcinoma of breast, lymphoid tumor, follicular carcinoma of thyroid, and soft-tissue sarcoma.³ Tumors classified according to conventional medicine are listed in Table 16.2.

Tumors may arise from any part of the body; no site is exempt. Tumors of the eye, ear, nose, oral cavity, and external genitalia have been dealt with separately (Table 16.3).

TABLE 16.3

Tumors of the Head and Neck in Ayurveda

Name of the Tumor	Site	Dosa	Clinical Presentation	Prognosis
Tumors of the eyelid (<i>vartmarbuda</i>)	Inner side of the eyelid	<i>Pitta</i> and <i>rakta</i>	Irregular, knotty, reddish mass with mild pain	May be curable
Tumors of the ear (<i>karnarbuda</i>)	External ear	Involvement of any one of the three <i>dosas</i> or in combination	Features based on the involved <i>dosa</i>	May be curable
Tumors of the nose (<i>nasa arbuda</i>)	Nasal cavity	Involvement of any one of the three <i>dosas</i> or in combination	Features based on the involved <i>dosa</i>	May be curable
<i>Tumors of Lips (oshtarbuda)</i>				
Cystic growths of lips (<i>jalarbuda</i>)	Lips	<i>Vata</i> and <i>kapha</i>	The growth appears like a water bubble	Fatal
Ulcerated malignant growths of lips (<i>raktarbuda</i>)	Lips	<i>Rakta</i>	Lips become red and swollen and discharge blood	Fatal
Tumors of the palate (<i>talv arbuda</i>)	Palate	<i>Rakta</i>	Swelling resembling the shape of lotus	Fatal
Tumors of the oral cavity (<i>mukharbuda</i>)	Inner surface of the mouth	<i>Kapha</i>	Pale, blackish mass that persists even after various surgical treatments	Fatal
Tumors of the throat (<i>galarbuda</i>)	Root of the tongue or in the middle of the throat		Nonuppurating, painless, hard, fixed and reddish tumor	Incurable
Tumors of the head (<i>shiro arbuda</i>)	<i>Shiras</i>		General features of <i>arbuda</i>	May be curable

16.6.1 Tumor of the Lip (*Mamsaja Oshta Roga*)

The lip becomes a heavy and thick projection of fleshy mass, subsequently leading to the formation of an ulcer, which may be infested with maggots.

16.6.2 Tumor of the Floor of the Mouth (*Alas*)

This tumor is a deep-seated mass on the under surface of the tongue due to vitiation of blood (*rakta*) and *kapha*. It is considered an incurable tumor, which increases gradually in size, fixes the tongue, and may suppurate. The tumor ultimately destroys the surrounding structures and exudates discharge with fishy odor.

Carcinoma of the floor of the mouth may give rise a similar picture like that of *alas*. Adenocystic and mucoepidermoid tumors of the salivary gland present themselves as an ulcerated growth that gradually holds the tongue and leads to progressive difficulties in

speech. These malignant lesions are usually complicated with secondary infection. Foul-smelling discharge and destruction of the surrounding tissue are very often found in these diseases.

16.6.3 Tumor of the Hard Palate (*Mamsa Kacchapa*)

The tumor develops on the palate because of vitiation of *kapha* and appears like a shell of a tortoise. It is a slow growing, painless, and considered an incurable tumor.

The tumors of the hard palate are characterized by slow growth and are usually not associated with pain. The tumors are smooth and hemispherical in shape, comparable with the convex shell of a tortoise. Similarly, carcinoma of the maxillary antrum may present as a convex mass in the hard palate.

16.6.4 Tumor of the Throat (*Galugha*)

This is considered an incurable tumor that is caused by vitiation of blood and *kapha*. The disease is characterized by a large swelling that develops on the inner and outer sides of the throat. It interferes with the functions of *udana vata*, resulting in the obstruction of food and air passages. The disease is also characterized by feeling of heaviness in the head, excessive salivation, and fever.

The oropharynx is the key structure for swallowing, breathing, and speaking. A malignant growth at the oropharynx may give rise to the obstructive features like that of throat tumor (*galugha*). Tumors at the oropharynx, palatine arch, and ulcerate become infected easily, causing pain in this nerve-rich area. Headache and otalgia are also common due to referred pain as the ninth cranial nerve supplies both the sites. Moreover, patients with carcinoma of the base of the tongue or primary malignant lesions of the oropharynx present with metastasis at the neck gland and may present only with neck gland swelling. The above features of oropharyngeal cancer can be correlated with the features of *galugha*.

16.6.5 Tumor of the Oropharynx (*Balaya*)

This is considered a malignant tumor of the throat that develops due to deranged *kapha*. The tumor is widespread and painless and ultimately obstructs the food passage.

These tumors could be malignant lesions of a deeper part of the oropharynx, which ulcerate later. Pain is less because of fewer nerve endings. The features of food-passage obstruction are found much earlier than the symptoms of airway obstruction due to their close vicinity to the esophagus.

16.6.6 Tumor of the Thyroid (*Asadhya Galaganda*)

Thyroid swellings (*galaganda*) of long duration (more than a year) lead to breathing difficulty, emaciation, anorexia, weakness, and hoarseness of the voice. This tumor is considered an incurable and may be due to malignant lesion of the gland.

The first symptom of thyroid carcinoma is the enlargement of the thyroid gland. It develops slowly, and choking attacks may occur due to pressure effect on the trachea. Hoarseness may be caused by the displacement of the larynx or involvement of the

recurrent laryngeal nerve, which is found in a good proportion of cases. Weight loss and anorexia are seldom present.

16.6.7 Intra-Abdominal Malignant Tumors (*Tridoshaja* and *Asadhya Gulma*)

Intra-abdominal malignant tumors (*tridoshaja gulma*) are characterized by an intra-abdominal stony mass with severe burning pain. Such patients lose mental peace, strength, and appetite.

Intra-abdominal malignant tumors (*asadhya gulma*) are gradually increasing, wide-spread, and fixed. They have a tortoise shell-like mass associated with weakness, anorexia, cough, vomiting, fever, and thirst. Multiple prominent veins usually cover the skin over the mass. Both of the above varieties are considered incurable.

In contemporary medicine, these conditions can be correlated with intra-abdominal malignant tumors, especially with tumors of liver. In hepatoma the lump is usually solitary, big, fixed, and convex; it may be comparable with the convex shell of the tortoise. The patient's general condition, physical and mental strength, and appetite decrease. Prominent abdominal veins may be due to portal hypertension or obstruction of the inferior vena cava. Cough and painful respiration are common symptoms that may result from pulmonary metastasis. Except primary hepatic carcinoma, a number of other abdominal conditions, such as retroperitoneal malignancies, ovarian and uterine carcinomas, lymphoma, and carcinoma of the stomach and bowels, may give rise to similar features.

16.6.8 Malignant Ascites (*Asadhya Udara Roga*)

This is considered an incurable ascites associated with flank pain, marked anorexia, generalized edema, and diarrhea. The recurrence of ascites is very fast even after removal of ascitic fluid by paracentesis. This very early recurrence of ascitic fluid in *asadhya udara roga* suggests the possibility of malignancy.

16.6.9 Familial Adenomatous Polyposis Coli (*Sahaja Arsha*)

The disease originates from defects in the genetic material (i.e., sperm and ovum). Such fleshy masses (*arsha*) are hardly visible (high enough), rough, painful, yellowish in color and directed inward.

The patients are emaciated and irritable, have a feeble voice and poor digestion, and tend to develop diseases of head and neck. Cracking sounds and shiny skin are the other symptoms of familial adenomatous polyposis. It is also considered an incurable disease.

Considering the etiology and clinical features of *sahaja arsha*, the disease can be compared with familial adenomatous polyposis coli. It has been described in the texts of contemporary medicine as a hereditary disease that is characterized by a large number of adenomatous polyps in the colon and rectum; if left untreated, it will almost certainly change into adenocarcinoma of the large bowel. Here again the disease originates from defects in the sperm and ovum. Bleeding, anemia, diarrhea, partial obstruction, pain in abdomen, and indigestion are the presenting symptoms. It is similar to the symptoms mentioned under *sahaja arsha*.

16.6.10 Tumor of the Lungs (*Kshayaja Kasa*)

This is considered an incurable tumor and may mimic other lung diseases (e.g., *kshayaja kasa*). It initially starts with a dry cough and becomes complicated with hemoptysis after a while. The patient feels a severe pricking pain in the chest and throat and suffers from malaise and fever. The patient also experiences difficulty breathing, choking, and wheezing during coughing. The disease is frequently associated with generalized body ache, fever, and a burning sensation. The patient gradually becomes emaciated and thin, devoid of flesh and vitality. The person spits mucus, blood, and pus.

The symptomatology and the sequence of symptoms of *kshayaja kasa* is similar to the symptomatology of carcinoma of the lungs. It has been described in the texts of conventional medicine that the cough initially is nonproductive but subsequently becomes productive with increasing amounts of mucoid expectoration, sometimes tinged with blood and a possible development of hemoptysis. Chest pain is severe and pleuritic, characterized by pinching pain. Paroxysmal hyperpnea may develop due to the presence of mucus buildup in the secondary and tertiary bronchi. Pleural effusion and empyema may cause dyspnea. Wheezing occurs when mucus partially obstructs the bronchus. Attacks of fever are common because of repeated obstructive pneumonia. Weight loss is a rather constant symptom of anorexia and metastasis.

16.6.11 Tumor of the External Genitalia of the Male (*Lingarsha*)

These lesions are ordinarily nonmalignant and do not grow like malignant tumors. Vitiating *dosas* affect the local flesh and blood of external genitalia, causing an itching sensation and ulceration. Such an ulcer usually grows over the foreskin or glans of the penis and is covered with papillomatous growth. The ulcer discharges slimy blood, ultimately destroys the penis, and halts the reproductive activity of the patients.

Penile neoplasms are very limited in number. Only two conditions, papilloma and carcinoma, produce new growth. The description of *lingarsha* is very close to the clinical features of carcinoma of the penis mentioned in conventional medicine. The patients with penile carcinoma usually present with a papillomatous growth in the glans with delayed features such as pain, hemorrhage, and discharge. The lesion initially is papillary, but gradually ulcerates and subsequently the ulcerative process destroys the entire penis, leaving a hornlike structure (penile horn) behind. Based on clinical presentation, course, treatment, and prognosis, the diseases listed in [Table 16.4](#) can be considered malignant.

16.7 Prognosis of Tumors and Ulcers

The prognosis of tumors has been explained based on the response to various treatment modalities. Depending upon the modality, the tumors that respond or are cured are considered *sadhya* or benign. Those that do not respond or are incurable are considered *asadhya* or malignant. *Vataja*, *pittaja*, and *kaphaja* tumors (benign neoplastic conditions) and benign conditions of mesenchymal tissue (*medaja* tumors) of any site are curable (*sadhya*). Tumors of muscle and soft tissue (*mamsarbuda*), ulcerated malignant growths (*raktaja* tumors), mixed type of tumors (*tridoshaja* tumors) of any site, cystic growths of lips, tumors of the palate and throat, and tumors of the oral cavity are considered incurable. Any tumor

TABLE 16.4

Neoplastic Diseases in Ayurveda

<i>Arbuda</i> (Neoplasia)	<i>Asadhya-Vrana</i> (Incurable Ulcers)	<i>Arbudavat Anya</i> <i>Vyadhi</i> (Diseases Comparable with Malignancy)
Tumors of muscle and soft tissue (<i>mamsarbuda</i>)		Head and neck Tumors of the lip (<i>mamsaja oshta roga</i>) Tumors of the floor of the mouth (<i>alas</i>) Tumors of the hard palate (<i>mamsakacchapa</i>)
Ulcerated malignant lesions (<i>raktarbuda</i>)		Tumors of the throat (<i>galugha</i>) Tumors of the oropharynx (<i>balaya</i>) Tumors of the thyroid (<i>asadhya galaganda</i>)
Mixed types of tumors (<i>tridosaja arbuda</i>)		Gastrointestinal tract Intra-abdominal malignant tumors (<i>tridoshaja gulma, asadhya gulma</i>) Malignant ascites (<i>asadhya udara roga</i>) Familial adenomatous polyposis coli (<i>sahaja arsha</i>)
Tumors of the palate (<i>talua arbuda</i>)		Respiratory system Tumors of the lungs (<i>kshayaja kasa</i>)
Tumors of the oral cavity (<i>mukharbuda</i>)		External genitalia of male Tumors of the penis (<i>lingarsha</i>)
Tumors of the throat (<i>galarbuda</i>)		

that reaches the stage of recurrence (*adhya* tumors) or metastasis (*dwiraar* tumor) is also considered incurable.

Almost all clinical presentations of incurable class of ulcers described by Sushruta can be included under malignant ulcers. Some of their specific features are:

1. Ulcers cropping up like a fleshy mass
2. Edges raised like those of external genitalia of a mare
3. Ulcers containing soft fleshy masses like the horn of a cow
4. Base of the ulcer being raised above the margins
5. Ulcers secreting various fluids, including vitiated blood and other substances similar to fatty (*vasa*) substance, adipose tissue (*meda*), bone marrow (*majja*), or brain matter
6. Ulcers manifested in an emaciated patient with discharge of blood and pus and the patient's condition being complicated by indigestion, cough, painful respiration, and anorexia
7. An ulcer that does not heal in spite of proper medication

In modern medicine, the description of malignant ulcers is similar. For example, the margins are raised above the base, rolled up, and hard in consistency; multiple fleshy masses develop (as seen in fungating basal cell carcinoma); and the base is raised above the margins many times. Cauliflowerlike ulcerative growths are an example of this type of malignant ulcer. Malignant ulcers bleed severely even with slight trauma. In addition, varieties of discharges are seen in malignant lesions. Painful respiration, cough, and

TABLE 16.5

Diagnosis

History/Interrogation	Signs and Symptoms ^a	Investigation
Family history	Change in wart or mole	Hemogram
Occupational history	Lump in the breast	TLC, DLC, Hb%, ESR, GBP, LFT, etc.
Dietary habits	Nonhealing sore	Radiogram
Geographical relation	Persistent cough	Plain x-ray
History of previous illness (may be predisposing factor etc.)	Unusual bleeding or discharge from any source	Barium contrast
	Indigestion or difficulty in swallowing	IVP, USG
	Change in bowel and urinary habits	CT scanning
		MRI etc.
		Histopathological and biochemical
		Aspiration
		Cytology
		Biopsy
		Enzyme
		Hormone
		Immunity factor
		Certain tumor markers
		Endoscopy
		Gastroscopy
		Colonoscopy
		Cystoscopy
		Bronchoscopy
		Laparoscopy
		Mediastinoscopy
		Histeroscopy

^a Seven danger signals by the American Cancer Society.

anorexia are additional symptoms of malignant ulcers that suggest metastasis of the tumor in the lungs.

16.8 Principles of Diagnosis

Ayurvedic diagnosis of tumors is carried out based on clinical manifestations, physical examination, and systemic examination. Apart from these methods, various diagnostic methods used by physicians of contemporary medicine are also adopted for a correct diagnosis.

The general condition of the patient (*rogi*) is examined after the standard eight-point examination procedures (i.e., pulse, feces, urine, various auscultatory findings, palpation, skin, eyes, and body strength).

Additional examinations includes history, type of pain, specific signs and symptoms, size and shape of the tumor (*akriti of arbuda*), consistency, color, tissue involved, site, and stage of the tumor (Tables 16.1 and 16.5).

In conventional medicine, the first principle of diagnosis is that adequate tissue for a biopsy be obtained from the tumor to establish the specific diagnosis and type of cancer. The second diagnostic principle is to establish the extent of the disease through physical examination, laboratory tests, and diagnostic imaging. In solid tumors, surgery is required to determine the extent of the disease or the stage of the tumor.¹

Early detection of cancer is very helpful in treating cancers. The following procedures and tumor markers have been proven useful as a secondary strategy in an attempt to reduce cancer mortality:¹

Procedures	Tumor markers
Sigmoidoscopy	Colorectal — CEA
Fecal occult blood test	Ovary — Ca-125
Digital rectal exam	Testicle — hCG, AFP
Prostate exam	Prostate — PSA
Papanicolaou test	Breast — CA15-2, CEA
Pelvic exam	Non-Hodgkin's lymphoma — LDH
Breast self-exam	Myeloma — Beta ₂ -M
Mammography	Hepatoma — AFP

Ayurvedic physicians also use these detection procedures and tumor markers for diagnosing and assessing the effect of therapeutic agents.

16.9 Management

16.9.1 Principle of Cancer Therapy

In Ayurveda, detoxification (*shodhana* therapy) is a major part of the treatment not only for *arbuda*, but also for all other diseases. The main principle of this procedure is to eliminate vitiated *dosas* to maintain their equilibrium, preserving the immunity (*rogibala*) of the patients. Detoxification is the pretherapy in the actual line of management, but not all patients are fit to undergo these procedures. The procedures are contraindicated in emaciated and severely ill patients who cannot tolerate therapeutic emesis or purgation.

The role of detoxification therapies on cancer patients as pretherapy to conventional line of treatment has been studied.^{3,4} The study showed that these procedures increased body weight, improved serum immunoglobulins, increased hemoglobin levels, and normalized liver functions. It was found helpful in minimizing the adverse effects of chemotherapeutic agents. The study concluded that the following procedures were helpful:

1. Oleation therapy (*snehana*) using medicated ghee prepared with *triphala* (*Terminalia chebula*, *Terminalia bellerica*, *Embolica officinalis*) in patients with breast cancer
2. Therapeutic purgation (*virechana*) in malignant conditions of hepatobiliary system
3. Medicated enema (*basti karma*) in malignancy of the genitor-urinary system

Sushruta has indicated that a patient's natural resistance or immunity to the disease (i.e., *rogibala*) is one the essential factors that should be preserved for the arrest of the progress of the disease. Fasting (*langhana*) is one of the principles of treatment for *kapha*-predominant diseases but has not been recommended because it may reduce the ability to defend

TABLE 16.6

Common Ayurvedic Formulations to Treat Tumors

Name of the Formulation	Main Constituents	Indication	Dose	Ref.
<i>Rudra rasa</i> (<i>arbudahara rasa</i>)	Mercury and sulphur ground with the decoctions of betel leaf, boerhavia, cow's urine, <i>Piper longum</i> , and amaranthus	Mentioned in all types of cancer	125–250 mg twice/day	58
<i>Lokanatha rasa</i> (<i>brihat</i>)	Mercury, sulphur, mica, aloe, iron oxide ground with <i>Solanum nigrum</i>	Mentioned in liver and spleen disorders	125–250mg twice/day	59
<i>Tamra basma</i>	Colloidal copper	Used in all types of intra-abdominal swellings (<i>gulma</i>)	15–125 mg ^a	60
<i>Abhraka basma</i>	Mica	Mentioned in all types of debilitating diseases	125–250 mg ^a	61
<i>Suvarna basma</i>	Gold dust	Used to improve immunity, strength, and body weight	2.5–6.0 mg ^a	62
<i>Manashila</i>	Arsenic disulphide	For external application on tumors	Quantity sufficient	63

^a As directed by the physician.

against the disease process (*rogibala*). Various drugs and formulations have been mentioned in Ayurveda for the management of tumors, but they have not been proven to have a direct cytotoxic effect on humans (Table 16.6).

16.9.2 General Treatment

The following treatments are used according to the involvement of *dosa* and the type of the tissues:

1. Detoxification therapies (*samshodhana chikitsa*) are advocated for the management of tumors based on the involvement of *dosa*. Oleation (*snehana*) is advised for *vata dosa*, purgation (*virechana*) is advised for *pitta dosa*, and emesis (*vamana*) is advised for *kapha dosa*.
2. After detoxification, administration of *dosa* subsiding drugs (*samshamana oushadha*) is advised as per involvement of *dosa*.
3. Local treatments like medicated poultice (*upanaha*), sudation (*swedana*), and local application of various drugs (*lepana*) are also indicated.
4. Parasurgical measures such as bloodletting (*raktamokshana*), application of caustics (*kshara karma*), and cauterization (*agnikarma*) are also mentioned.
5. Surgical excision (*shastrakarma*) of the tumor is performed.
6. Postoperative wound management by various Ayurvedic drugs is conducted.
7. Parasurgical approaches include bloodletting (*raktamokshana*), therapeutic cautery (*agnikarma*), application of caustics (*ksharakarma*), and maggotification.

16.9.3 Specific Treatments

In vivo and *in vitro* studies have been carried out on different Ayurvedic drugs in various institutions all over the world. The purpose of these studies is to evaluate drugs' cytotoxic action, their role in improving the hemopiesis, and level of immunity. Studies have also been used to show how drugs reduce the toxic effects of conventional therapies and improve the well-being of the patient, as an adjuvant to chemotherapy and radiotherapy.

1. A compound Ayurvedic drug containing *bhallataka* (*Semicarpus anacardium*), *rohitaka* (*Amoora rohitaka*), and *yastimadhu* (*Glyceriza glabra*) along with other conventional treatment modalities plays an important role in immunological alteration in cancer. This compound helps in terms of increases in mean lymphocyte count, improvements in cell-mediated immunity, increases in T-cell count, decreases in positivity to C-reactive protein, and increases in the levels of serum immunoglobulins (Ig) such as IgA, IgG, and IgM.⁵
2. *Rohitaka* and *bhallataka* reduce the hepatotoxicity induced by chemotherapy in hepatobiliary, gastrointestinal malignancy,⁶ and squamous cell carcinoma.⁷ These drugs normalize the liver functions after chemotherapy.
3. Various other clinical and experimental investigations have been done to evaluate the role of various Ayurvedic drugs as an adjuvant therapy. They were found effective in reducing the chemotherapy- and radiotherapy-induced toxicities in various types of cancers.⁸
4. Studies indicate that *Withania somnifera* (*ashwagandha*) could reduce the cyclophosphamide induced toxicity and hence be useful in cancer therapy.⁹ *W. somnifera* produces immunopotentiating and myeloprotective effects, as seen from the enhancement of cytokine production and stem-cell proliferation and its differentiation.¹⁰
5. The tablet Cystone, a polyherbal Ayurvedic preparation, protects against cisplatin-induced nephrotoxicity without interfering with its antitumor activity.¹¹
6. Certain vegetable oils rich in linoleic acid (e.g., sesame oil) that are recommended for topical use by Ayurveda may contain selective antineoplastic properties; these properties are similar to those demonstrated for essential polyunsaturated fatty acids and their metabolites. This suggests that vegetable oils may have potential clinical usefulness.¹²

16.9.3.1 Treatment of Vataja Tumors

Vataja tumors are treated locally by medicated poultice (*upanaha*), fomentation with steam (*nadi sweda*), and bloodletting (*raktamokshana*). They are treated systemically by medicated dehydrated butter and herbal formulations.

16.9.3.2 Medicated Poultice

Various medicated poultices are mentioned for the management of *vataja* tumors. *Kush-manda* (*Benincasa cerifera*), *ervaruka* (*Cucumis utilissimus*), *narikela* (*Cocos nucifera*), *priyala* (*Buchanania lanzan spreng*), and *eranda* (*Ricinus communis*) seeds that are boiled with milk, water, and ghee and mixed with oil should be applied. Another medicated poultice made up of boiled meat has also described as effective.¹³

TABLE 16.7Drugs Having a *Vata* Mitigating Effect (*Vatahara Dravya*)

Name of Plant	Botanical Name	Family Name	Parts Used
<i>Devadaru</i>	<i>Cedrus deodara</i> Roxb.	Pinaceae	Heart wood
<i>Kusta</i>	<i>Saussurea lappa</i> C.B Clarke.	Compositae	Root
<i>Haridra</i>	<i>Curcuma longa</i> Linn.	Zingiberaceae	Tuber
<i>Varuna</i>	<i>Crataeva nurvala</i> Buch-Ham.	Capparidaceae	Bark
<i>Meshashrungi</i>	<i>Gymnema sylvestre</i> R.Br.	Asclepiadaceae	Leaf, root
<i>Bala</i>	<i>Sida cordifolia</i> Linn.	Malvaceae	Root, seed
<i>Atibala</i>	<i>Abutilon indicum</i> Linn.	Malvaceae	Root, seed
<i>Kacchura</i>	<i>Curcuma zedoaria</i> Rose.	Zingiberaceae	Tuber
<i>Sallaki</i>	<i>Boswellia serrata</i> Roxb.	Burseraceae	Bark, latex
<i>Kuberaksha</i>	<i>Caesalpinia crista</i> Linn.	Caesalpinioideae	Seed
<i>Arjuna</i>	<i>Terminalia arjuna</i> Roxb.	Combretaceae	Bark
<i>Sahachara</i>	<i>Barleria prionitis</i> Linn.	Acanthaceae	Whole plant
<i>Kulattha</i>	<i>Dolichos biflorus</i> Linn.	Papilionatae	Seeds
<i>Kola</i>	<i>Zizyphus jujuba</i> Lam.	Rhamnaceae	Leaf, fruit
<i>Shatavari</i>	<i>Asparagus racemosus</i> Wild.	Liliaceae	Tuber
<i>Punarnava</i>	<i>Boerhavia diffusa</i> Linn	Nyctaginaceae	Root, seed, whole plant
<i>Eranda</i>	<i>Ricinus communis</i> Linn.	Euphorbiaceae	Root, leaf
<i>Arka</i>	<i>Calotropis procera</i> R.Br.	Asele piadaceae	Root bark, flower, leaf

Source: Ganekar, B.G. and Vaidya, L.C., *Samshodhana samshamaneeya Adhyaya in sutrasthana of Sushruta Samhita* by Acharya Sushruta, 5th ed., 1994, chap. 40, p. 142, verse 7. With permission.

16.9.3.3 Fomentation

Fomentation of local part in the form of *nadi sweda* is advised. Drugs such as moringa pterygosperma (*shigru*) and the juice of meat (*mamsa rasa*) are boiled and steam is to be passed through a tube over the site.¹⁴

16.9.3.4 Bloodletting (*Raktamokshana*)

Bloodletting with the help of horns (*shringa*) has been described to be effective in the treatment of *vataja* tumors.¹⁴

16.9.3.5 Systemic Treatment

Medicated ghee boiled with decoction of drugs that fix the aggravated *vata* (Table 16.7)⁶⁴ is given with milk or *kanji*.

16.9.3.6 Treatment of *Pittaja* Tumors

The local treatment of the tumor includes rubbing with the leaves of udumbara (*Ficus glomerata* Linn.) or other leaves having a rough surface. This is followed by sprinkling of fine dust of *sarjarasa* (*Viteria indica*), *priyangu* (*Callicarpa macrophylla*), *raktachandana* (*Pterocarpus santalinus*), *arjuna* (*Terminalia arjuna*), and *yastimadhu* (*Glycyrrhiza glabra*) mixed with honey.

A plaster composed of *aragvadha* (*Cassia fistula*), *soma* (*Sarcostemma acidum*), and *shyama* (*Callicarpa macrophylla*) is also prescribed. The above local application of medicaments is usually advised after bloodletting (*raktamokshana*).

16.9.3.7 Systemic Treatment

Medicated ghee prepared with *madhuyasthi* (*Glycyrrhiza glabra*), *draksha* (*Vitis venifera*), *shyama* (*Callicarpa macrophylla*), *girikwa* (*Clitoria ternatea*), *anjanaki* (*Strychnus colubrina*), and *yavatikta* (*Erythraea roxburgii*) is given internally.

16.9.3.8 Treatment of Kaphaja Tumors

Local application of various medicated pastes are used after detoxification (*samshodhana chikitsa*) especially after emesis (*vamana*). The paste of the drugs used for emesis (*vamana*) and purgation (*virechana*) may also be applied locally to arrest the disease.

Caustics (*kshara*) in a cow's urine are also prescribed as a local application for *kaphaja* tumors after the bloodletting procedure.

16.9.3.9 Treatment of Medaja Tumors

Surgical and parasurgical approaches for the management of *medaja* tumors are emphasized in Ayurveda. Nevertheless, medicated paste containing *haridra* (*Curcuma longa*), *lodhra* (*Symplocos recemosa*), *patanga* (*Caesalpinia sappan*), *manashila* (real gar), and *haritatala* (opiment) mixed with honey is also used as a local application.

16.9.3.10 Bloodletting (Raktamokshana)

Bloodletting is indicated after detoxification in the management of *vataja*, *pittaja*, *kaphaja*, and *medaja* tumors. The use of cow's horn, nonpoisonous leeches, and gourd (*Lagenaria vulgaris*) for bloodletting has been advised in *vataja*, *pittaja*, and *kaphaja* tumors, respectively. In *medaja* tumors, bloodletting has been advised after making an incision over the tumor.

16.9.3.11 Use of Cautery and Application of Caustics (Agnikarma and Ksharakarma)

Thermal cauterization (*agnikarma*) and application of caustics (*ksharakarma*) is used alone or in combination with surgery for the management of *kaphaja* tumors, *medaja* tumors, and tumors that do not respond to medical treatment.

The recurrence of a tumor after surgical excision was recognized by Sushruta. His idea was that even the last particle of *dosa* of a tumor left over would lead to a fresh growth and bring death, just like the last spark of an unextinguished fire. A radical excision was advised to avoid recurrence. Because it may not be possible to remove an entire tumor mass by surgery, therapeutic cautery and the application of caustics (*ksharakarma*) have been advised especially after surgery.

16.9.3.12 Maggotification

Maggotification (currently not practiced) is another important parasurgical treatment of tumors that was practiced by early Ayurvedic practitioners. In this unique technique, the gradual destruction of a tumor mass was achieved by maggotification of tumor. To attract the flies, *kulattha* (*Dolichos biflorus*), oil cakes of sesamum, and powder of dry meat are mixed with curd. This solution should be applied over the *arbuda* so that worms and parasites either grow locally or are attracted toward the tumor. They feed upon the tumor, and when only a small part of the tumor remains, the worms and the paste should be washed off the area and the remnant mass should be treated with cautery.

Local treatments mentioned in Ayurveda, like medicated poultice (*upanaha*), sudation (*swedana*), application of medicated pastes (*lepana*), and bloodletting (*raktamokshana*), are being practiced by Ayurvedic physicians in cancer as well as various other disorders. However, there is a lack of *in vitro* or *in vivo* trials and scientific studies to support their role in different types of cancer.

16.9.4 Surgical Management

16.9.4.1 Indications

If a tumor (*arbuda*) does not resolve or respond to proper medical treatment, it should be treated surgically. The main surgical treatments of tumors are excision (*chhedana*) and excision with scraping (*lekhana*).

Sushruta has described certain principles of the excision of tumors as follows:

1. Tumor mass is removed completely. The incomplete removal leads to recurrence and poor prognosis.
2. Efforts are made to prevent the intraoperative spread of the tumor to distal organs. Historically, application of metal tourniquets made up of iron, copper, zinc, and lead were used around the tumor to control the spread. The mass is then destroyed by cautery (*agni*), caustics (*kshara*), or surgically depending upon the depth and extent of the tumor.
3. Sushruta advocated a minimal invasive technique in inoperable tumors or tumors situated over vital areas (*marma*). In this case, a strong caustic thread (*ksharasutra*) is passed across the center of its base with the help of a needle. The ends of the thread should be tied firmly on either side, leading to avascular necrosis of the tumor mass.

These principles of treatment occupy an important place in surgical oncology even today. Application of tourniquets, the no-touch technique of excision, and the ligation of feeding vessels are some of the important operative techniques that are still in practice to prevent the spread of the tumor.

After surgical excision, the area is cauterized (*agni/kshara*) to achieve a complete cure. Cleansing (*shodhana karma*) of the wound should be undertaken after excision of the tumor. The wound has to be cleaned using the decoction of *aparajita* (*Clitorea teratea*), *jati* (*Jasminum grandiflorum*), and *karaveera* (*Neium odorum*). The oil prepared from *bharangi* (*Clerodendrum serratum*), *vidanga* (*Embelia ribes*), and the paste of *triphala* (*Terminalia chebula*, *Terminalia belerica*, *Embelia officinalis*) can enhance the healing of wounds. Suppurated wounds may be treated according to measures mentioned for the management of infected ulcers (*dushta vrana*).

The efficacy of the above-mentioned drugs in wound healing has been established by various experimental and clinical trials. Therefore, these drugs can be used safely in conjugation with conventional surgical treatment.

16.9.4.2 Conventional Medicine

There are three sets of goals in conventional management of cancer patients: therapeutic, human, and scientific.¹ Ayurvedic medicine has similar goals. The therapeutic goal is to

TABLE 16.8

Tumor Response to Chemotherapy in Conventional Medicine

Cure (>30%) of Advanced Disease	Significant Cures (5–30%)	Probably Increases Survival	Adjuvant Treatment Leading to Increased Cure
Choriocarcinoma, acute lymphocytic leukemia (childhood), malignant lymphoma (Hodgkin's disease)	Ovarian cancer Bladder cancer Small-cell lung cancer Gastric cancer	Breast cancer Multiple myeloma Head and neck cancer	Breast cancer Colon cancer Osteogenic sarcoma Early stage of large-cell lymphoma
Hairy-cell leukemia			
Testicular cancer			
Childhood tumors (e.g., embryonal rhabdomyosarcoma, Ewing's sarcoma, Wilm's tumor)			
Acute myelocytic leukemia			
Acute lymphocytic leukemia (adult)			
Prtomyelocytic leukemia			

Source: From Goldman, L. and Bennett, J.C., *Cecil Text Book of Medicine*, Part XIV, W.B. Saunders, Philadelphia, 2000. With permission.

cure patients and return them to normal life. If this is not attainable, then the next step in this goal is to help patients achieve long, qualitatively satisfactory remission, or on a tertiary level, a remission of any kind and finally terminal comfort care. The human goal in oncology is that the needs of the patient, family, and social environment be understood and met. The scientific goal requires special methods in oncology provided by specialists to deliver optimal clinical care.

In conventional medicine, effective anticancer therapy has integrated medical management with surgery and radiation therapy. The development of new cytotoxic and endocrine agents and the introduction of biologic therapy based on recombinant synthesis of interferon and cytokines have helped expand medical management. Not all patients are candidates for cancer therapy because of limitations in available drugs or comorbidity from other medical problems. In addition, not all tumors are responsive to chemotherapy (Table 16.8). Although many of the tumors are manageable by various chemotherapeutic agents, they are associated with the causation of various toxicities (Table 16.9).¹⁵

Antitumor drugs fail to cure cancer because they cannot kill 100% of the cancer cells. Even if one cell survives the chemotherapy, that cell can grow very quickly into millions of cells. The result is a relapse of cancer because the body immunity defense does not recognize the remaining cancer cells as foreign cells and thus does not kill them. Cancer cells can hide behind the blood barriers for drugs, such as for blood-brain barrier, and therefore do not get killed. The other mechanisms responsible for the failure of chemotherapy are the metabolism of drugs to inactive form, removal of drugs by binding to plasma proteins, rapid urinary excretion, and the development of resistance in the tumor cells due to an increase in certain critical enzymes that are inhibited by the drug. For example, tumors resistant to methotrexate were found to have high levels of dihydrofolate reductase, a key enzyme.^{16–22} Transport across the cell membrane is also an obstacle in reaching the agent to the malignant cell. This can sometimes present a formidable barrier

TABLE 16.9
Toxicity of Antitumor Drugs of Conventional Medicine

Type of Agent	Major Clinical Toxicities
Alylating agents	Cardiac, renal, hepatic, lungs, bone marrow, GI tract
Antifolates	Bone marrow depression, intestinal epithelium, interstitial pneumonitis, neurotoxicity
Pyrimidine analogs	GI toxicity, ulceration of GI mucosa, neurotoxicity, cardiotoxicity, myelosuppression, leucopenia
Purine analogs	Bone marrow suppression, GI tract toxicity, hepatic necrosis, immunosuppression
Vinca alkaloids	Leukopenia, neurotoxicity, GI tract toxicity
Paclitaxel	Neutropenia, bone marrow suppression, neurotoxicity, hypersensitivity reactions
Actinomycin D	GI tract toxicity, bone marrow suppression, hematopoietic suppression, ulceration of oral mucosa
Daunorubicin, doxorubicin, and idarubicin	Bone marrow suppression, GI tract toxicity, cardiac toxicity
Bleomycin	Myelosuppression, cutaneous toxicity, pulmonary toxicity, hyperthermia, hypotension, exacerbations of rheumatoid arthritis
Plicamycin	Bone marrow depression, liver and kidney toxicity, GI tract toxicity and neurotoxicity
Platinum compounds	Nephrotoxicity, hearing loss, peripheral neuropathy, myelosuppression, hemolytic anemia, cardiotoxicity

Source: From Calabresi, P. and Chabner, B.A., *Goodman & Gillman's The Pharmacological Basis of Therapeutics*, Sec. 10, McGraw-Hill, New York, 1996. With permission.

and may be one of the reasons why some tumors are insensitive to some of the chemotherapeutic agents.²³ Mechanisms of resistance to some common drugs are listed below.¹

Methotrexate	Impaired transport or amplification of dihydrofolate reductase
Cytarabine	Decrease in deoxycytidine kinase or increase in cytidine deaminase
5-Fluorouracil	Increase in thymidylate synthetase enzyme
Cisplatin	Decreased uptake, increase in repair enzymes
Taxol-Vinca alkaloids	Multidrug resistance (MDR) expression, mutations in tubulin (decreased binding)
Doxorubicin	MDR expression, decrease or alterations in topoisomerase II
Irinotecan, topotecan	Decrease in topoisomerase I

Cancer chemotherapy research in conventional medicine is primarily focused on developing analogs of purines, pyrimidines, and various vitamins like folic acid, plant products, and biological products. Various approaches have also been tried to overcome tumor resistance to various drugs. For example, an attempt has been made to take advantage of an increase in the enzyme responsible for the drug resistance. This approach is called enzyme dependent lethal synthesis. It was speculated that tetrahydrohomofolic acid, an analog of natural tetrahydrofolic acid (an inhibitor of thymidylate sythetase), may inhibit DNA synthesis and overcome the resistance. Studies have shown that dihydrohomofolic acid, a good substrate of dihydrofolate reductase, produced selective tumor inhibition only in methotrexate-resistant tumors containing high levels of dihydrofolate reductase enzyme.¹⁶⁻²² After administering dihydrohomofolic acid to tumor-bearing mice, several folds of higher levels of tetrahydrohomofolic acid were found only in resistant tumors containing high levels of dihydrofolate reductase enzyme than in the methotrexate sensitive tumors confirming the hypothesis. A similar approach can be used with other agents where an increase in a key enzyme is responsible for the drug resistance.

16.10 Scientific Basis

Giving due importance to the various drugs and formulations that have been mentioned for the management of various *arbuda*, the modern-day researchers have been trying these drugs in various animal models. The researchers' goal is to find the drug's mode of action, realizing the dream of generating a magical potion for the eradication of cancers.

Various herbs used in Ayurveda are being researched both experimentally and clinically in various institutions. Drugs and drug fractions with antitumor activity are summarized in Table 16.10. Clinical, animal, and *in vitro* studies on some of the Ayurvedic herbs are presented below.

16.10.1 Clinical Studies

There are only two clinical studies available on antitumor herbal formulations. In the first study,²⁴ an Ayurvedic formula containing *bhallataka* (*Semicarpus anacardium*), *rohitaka* (*Amoora rohitaka*), and *yastimadhu* (*Glyceriza glabra*) was studied on 100 patients with different types of malignancy, as an adjuvant therapy, in a dose of 1.5 g/day in two divided doses, given after food. The study concluded that this compound is effective in cancer patients as evidenced by increases in body weight, appetite status, and hemoglobin percentage. It also reduced the mortality rate in studied cases. It can be used along with conventional therapy to improve the general condition of the patient. This compound can be used in lymphoma patients along with chemotherapy and radiotherapy, as it reduces the lymphopenic effect of chemotherapy and improves the hemoglobin level of the patients.²⁵

In the second study,²⁶ the efficacy and side effects of an Ayurvedic arsenic-compound formula were evaluated. This is in light of the fact that just over 100 years ago arsenic was recognized in the West as useful in the control of blood counts in patients with chronic myeloid leukemia.

16.10.2 In Vitro studies

In one *in vitro* study,²⁷ extracts of the flowers of *Calotropis procera* (asclepiadaceae) and the nuts of *Semecarpus anacardium* (anacardiaceae) displayed the strongest cytotoxic effect with 50% inhibitory dose (ID₅₀) values of 1.4 and 1.6 µg/ml, respectively.

16.10.3 Animal Studies

16.10.3.1 Ashwagandha (*Withania somnifera*)

Withaferin A, isolated from the roots of *W. somnifera*, was found effective in Ehrlich ascitis carcinoma when given at a dose of 30 mg/kg along with radiation therapy.^{28,29} Antioxidant and detoxifying properties of *W. somnifera* may be responsible for chemopreventive activity.³⁰

Administration of 75% methanolic extract of *W. somnifera* was found to significantly increase the total white blood cell (WBC) count in normal Balb/c mice and reduce the leucopenia induced by a sublethal dose of gamma radiation. Treatment with *W. somnifera* was found to significantly increase the bone marrow cellularity in mice. *W. somnifera*

TABLE 16.10

Herbs and Herbal Fractions with Antitumor Activity

Plant	Botanical Name	Parts Used	Dose	Action as Mentioned in Classics	Proven Actions	Ref.
<i>Bhallataka</i>	<i>Semicarpus anacardium</i>	Fruit	3–6 g	<i>Lekhana</i> (excises unhealthy tissues), <i>rasayana</i> (rejuvenator)	Has antitumor activity against experimental mammary carcinoma in animals	65,66
<i>Ashwagandha</i>	<i>Withania somnifera</i>	Root	3–6 g	<i>Shothahara</i> (reduces swellings), <i>rasayana</i> (rejuvenator)	Alcoholic extract of the root in doses of 400 mg/kg and above produces complete regression of two-stage skin carcinogenesis induced by DMBA and croton oil	35,67,68
<i>Kumkuma</i>	<i>Crocus sativa</i>	Stamens	½–1 g	<i>Shothahara</i> (reduces swellings), <i>rasayana</i> (rejuvenator)	Has antitumor activity toward sarcoma-180 and Ehrlich ascites carcinoma (EAC) solid tumors in mice	69
<i>Bhunimba</i>	<i>Andrographis paniculata</i>	All parts	1–3 g	<i>Raktashodhaka</i> (purifies vitiated blood), <i>shothahara</i> (reduces swellings)	Prevents chemotoxicity including carcinogenicity	70
<i>Tulasi</i>	<i>Ocimum sanctum</i>	Leaf, seed, root	1–3 g	<i>Vishaghna</i> (detoxifies), <i>shothahara</i> (reduces swellings), <i>raktashodhaka</i> (purifies vitiated blood)	Reduces 20-methylcholanthrene-induced tumor incidence and tumor volume	71
<i>Manduka parni</i>	<i>Centella asiatica</i>	All parts	Juice 10–20 ml	<i>Kaphahara</i> (cleanses vitiated <i>kapha</i>), <i>shothahara</i> , (reduces swellings), <i>rasayana</i> (rejuvenator)	Retards the development of solid and ascites tumors and increases the life span of these tumor-bearing mice	72
<i>Arka</i>	<i>Calotropis procera</i>	Root, latex, flower	0.5–1 g	<i>Kaphashamaka</i> (pacifies vitiated <i>kapha</i>), <i>shothahara</i> (reduces swellings), <i>raktashodhaka</i> (purifies vitiated blood), <i>vishaghna</i> (detoxifier)	Displays the strongest cytotoxic effect with ID ₅₀ values of 1.4 µg/ml	27
<i>Chitraka</i>	<i>Plumbago rosea</i>	Root bark	1–2 g	<i>Lekhana</i> (excises unhealthy tissue), <i>shothahara</i> (reduces swellings), <i>rasayana</i> (rejuvenator)	Used with radiation to enhance the tumor-killing effect	73
<i>Bakuchi</i>	<i>Psoralea coryfolia</i>	Seed, seed oil	1–3 g	<i>Kaphaghna</i> (pacifies vitiated <i>kapha</i>), <i>shothahara</i> (reduces swellings), <i>vrana shodhana</i> (cleanses chronic wounds), <i>vrana ropana</i> (heals wounds)	100 mg/kg body weight (of the active fraction) inhibits the growth and delays the onset of papilloma formation in mice; the active fraction at the same dose, when administered orally inhibits the growth of subcutaneously injected 20-methylcholanthrene-induced soft tissue fibrosarcoma significantly	74

<i>Katuki</i>	<i>Picrorhiza kurroa</i>	Root	½–1 g	<i>Kaphagna</i> (pacifies vitiated <i>kapha</i>), <i>shothahara</i> (reduces swellings), <i>raktashodhaka</i> (purifies vitiated blood), <i>lekhana</i> (excises unhealthy tissue)	Picroliv (100 and 200 mg/kg) inhibits the sarcoma development by 47 and 53%; it also delays the onset of first skin tumor in the group of animals treated with Picroliv; Picroliv administration increases the life span of transplanted Dalton's lymphoma ascites (DLA) and Ehrlich ascites carcinoma (EAC)-harboring mice and reduces the volume of transplanted solid tumors	75
<i>Aragvadha</i>	<i>Cassia fistula</i>	Fruit pulp, root bark	5–10 g	<i>Kaphashodhaka</i> (pacifies vitiated <i>kapha</i>), <i>shothahara</i> (reduces swellings)	Increases life span by decreasing the tumor volume and viable tumor cell count in the EAC tumor hosts; improves the hematological factors after methanolic extract treatment, like hemoglobin content, red blood cell count and bone marrow cell count of the tumor-bearing mice	76
<i>Kumari</i>	<i>Aloe vera</i>	Leaf	10–20 ml	<i>Kaphahara</i> (pacifies vitiated <i>kapha</i>), <i>shothahara</i> (reduces swellings), <i>vrana ropana</i> (heals wounds), <i>raktashodhaka</i> (purifies vitiated blood)	Detoxifies reactive metabolites including chemical carcinogens and drugs	77
<i>Kalajaji</i>	<i>Nigella sativa</i>	Seed	1–3 g	<i>Kaphashamaka</i> (pacifies vitiated <i>kapha</i>), <i>lekhana</i> (excises unhealthy tissue), <i>shothahara</i> (reduces swellings)	<i>In vivo</i> EAC tumor development is completely inhibited by the active principle at the dose of 2 mg/mouse/day $\times 10^{16}$	78
<i>Vamsha</i>	<i>Bambusa arundinacea</i>	Root, leaf, Shoot	50–100 ml	<i>Kaphashamaka</i> (pacifies vitiated <i>kapha</i>), <i>lekhana</i> (excises unhealthy tissue), <i>vishaghna</i> (detoxifier)	Antitumor activity against benzopyrene and 4-nitroquinoline-1-oxide-induced tumors is highest with 1% bamboo leaf extracts (0.71 mg/ml) and a direct action of bamboo leaf extracts on tumor cells is indicated	79
<i>Hingu</i>	<i>Ferula narthex</i>	Latex	Up to 0.5 g	<i>Kaphahara</i> (normalizes vitiated <i>kapha</i>)	Asafoetida is a potent antioxidant and can afford protection against free radical mediated diseases such as carcinogenesis	80
<i>Methika</i>	<i>Trigonella foenum-Graecum</i>	Seed	1–3 g	<i>Shothahara</i> (reduces swellings)	Has an anti-inflammatory and antineoplastic effect	81
<i>Amala</i>	<i>Embilica officinalis Gaertn</i>	Fruit	4–10 g	<i>Shothahara</i> (reduces swellings)	Antioxidant, antitumor, chemopreventive, prostate cancer, immunomodulator, anticlastogenic radiation protection	82–96

treatment of irradiated mice normalized the ratio of normochromatic erythrocytes and polychromatic erythrocytes. Major activity of *W. somnifera* seems to be in the stimulation of stem cell proliferation.³¹

The alcoholic extract of dried roots of *W. somnifera* and Withaferin A showed significant antitumor and radio-sensitizing effects in experimental tumors *in vivo* without any noticeable systemic toxicity. The mechanism of action, however, is not clear. The studies indicate that *W. somnifera* could prove to be a good natural source of a potent and relatively safe radio-sensitizer and chemotherapeutic agent. Further studies are needed to explore the clinical potential of the plant for cancer therapy.³²

W. somnifera was injected at a dose of 500 mg/kg to tumor-bearing mice (tumor size: $50 \pm 5 \text{ mm}^3$) for 10 days with one local exposure of radiation therapy followed by hyperthermia. It significantly increased the tumor cure rate, produced a delay in the growth of partially responding tumors, and increased animal survival. Study concluded that *W. somnifera*, in addition to having a tumor-inhibitory effect, also acts as a radio sensitizer.³³ *W. somnifera* extract also inhibited 20-methylcholanthrene-induced sarcoma development in mice at a dose of 20 mg/day.³⁴ *W. somnifera* extract was also found to reduce two-stage skin carcinogenesis induced by DMBA and croton oil.³⁵

Withaferin A reduced survival of V79 cells in a dose-dependent manner. LD50 for survival was $16 \mu\text{M}$. One-hour treatment with a non-toxic dose of $2.1 \mu\text{M}$ before irradiation significantly enhanced cell killing, giving a sensitizer enhancement ratio (SER) of 1.5 for 37% survival and 1.4 for 10% survival.³⁶ *W. somnifera* was found to significantly reduce leucopenia induced by cyclophosphamide (CTX) treatment. The total WBC count on the 12th day of the CTX-treated group was 3720 cells/ mm^3 ; the WBC count in CTX-treated animals that were simultaneously treated with WS was 6120 cells/ mm^3 . Withania extract increased the number of alpha-esterase positive cells (1130/4000 cells) in the bone marrow of CTX plus *W. somnifera*-treated animals, compared with the CTX-treated group (687/4000 cells).³⁷

W. somnifera was also found to enhance the levels of interferon gamma (IFN-gamma) (75.87 pg/ml), Interleukin-2 (IL-2) (14.16 pg/ml), and granulocyte macrophage colony-stimulating factor (GM-CSF) (49.22 pg/ml) in normal Balb/c mice. The lowered levels of IFN-gamma (30 pg/ml), IL-2 (4.5 pg/ml), and GM-CSF (19.12 pg/ml) after treatment with CTX was reversed by the administration of *W. somnifera* extract (IFN gamma = 74 pg/ml; IL-2 $7.5 = \text{pg/ml}$; GM-CSF = 35.47 pg/ml). *W. somnifera* extract lowered the levels of tumor necrosis factor alpha (TNF alpha) production. Administration of bone marrow cells from donor mice treated with WS extract increased the spleen nodular colonies in irradiated mice (8.33) compared with those treated with normal bonemarrow cells (3.03). The number of nodular colonies increased significantly after continuous treatment with WS. These results indicate the immunopotentiating and myeloprotective effects of *W. somnifera*.³⁸

W. somnifera extract was also found effective as a chemopreventive agent. *W. somnifera* protected against 20-methylcholanthrene-induced fibrosarcoma tumors in Swiss albino mice. A single subcutaneous injection of 200 μg of 20-methylcholanthrene in 0.1 ml of dimethylsulphoxide into the thigh region of mice produced a high incidence (96%) of tumors. Oral treatment of animals with 400 mg/kg body weight of *W. somnifera* extract (1 week before injecting 20-methylcholanthrene and continuing until 15 weeks thereafter) significantly reduced the tumor incidence and tumor volume and enhanced the survival of the mice, compared with the untreated 20-methylcholanthrene-injected mice. The occurrence of tumors was also delayed in the treatment group. Liver biochemical parameters revealed a significant modulation of reduced glutathione, lipid peroxides, glutathione-S-transferase, catalase, and superoxide dismutase in extract-treated mice compared with 20-methylcholanthrene-injected mice. The authors suggested that the mechanism of chemopreventive activity of *Withania somnifera* extract may be due to its

antioxidant and detoxifying properties.³⁹ *W. somnifera* was also found to increase the neutrophil count in mice with paclitaxel-induced neutropenia.⁴⁰

16.10.3.2 *Aloe vera* Linn.

Di (2-ethylhexyl) phthalate (DEHP), an active principle isolated from *Aloe vera* Linn., has been shown to have antileukemic and antimutagenic effects *in vitro* in *Salmonella typhimurium* TA98 and TA 100 strains.⁴¹ Aloes potentiated the antitumor effect of 5-fluorouracil and cyclophosphamide as components of combination chemotherapy.⁴² Aloe-emodin, a hydroxyanthraquinone present in *Aloe vera* leaves, has been shown to have a specific *in vitro* and *in vivo* antineuroectodermal tumor activity.⁴³

Aloe vera has been claimed to contain several important therapeutic properties, including anticancerous effects. The effect of this drug was studied on a pleural tumor in rat (Yoshida AH-130, ascite hematoma cells) and proved its therapeutic use in cancer.⁴⁴

In one study, the antigenotoxic and chemopreventive effect of *Aloe barbadensis* Miller (polysaccharide fraction) on benzo[a]pyrene (B[a]P)-DNA adducts was investigated *in vitro* and *in vivo*. Aloe showed a time-course and dose-dependent inhibition of [3H]B[a]P-DNA adduct formation in primary rat hepatocytes (1×10^6 cells/ml) treated with [3H]B[a]P (4 nmol/ml).⁴⁵ The growth of human neuroectodermal tumors is inhibited in mice with severe combined immunodeficiency without any appreciable toxic effects on the animals. The compound does not inhibit the proliferation of normal fibroblasts nor that of hemopoietic progenitor cells. The cytotoxicity mechanism consists of the induction of apoptosis, whereas the selectivity against neuroectodermal tumor cells appear to depend upon a specific energy-dependent pathway of drug incorporation.⁴⁶

Crude modified aloe polysaccharide (MAP) activated macrophage cells and stimulated fibroblast growth. Under the same conditions, native *Aloe barbadensis* gel had no effect on macrophage activation. MAP prevented ultraviolet B (UVB) irradiation-induced immune suppression as determined by contact hypersensitivity (CHS) response in C3H/HeN mice.⁴⁷ Aloin- or sennoside-enriched diets (0.03%) did not promote incidence and growth of adenomas and carcinomas after 20 weeks in a model of dimethylhydrazine-induced colorectal tumors in male mice. In addition, no significant changes in serum electrolytes and parameters of hepato- and nephrotoxicity were observed in the aloe-fed mice.⁴⁷

A new immunostimulatory polysaccharide called Aloeride from commercial *Aloe vera* (*Aloe barbadensis*) juice has been isolated and characterized. It is between 4 and 7 million Da, and its glycosyl components include glucose (37.2%), galactose (23.9%), mannose (19.5%), and arabinose (10.3%). Although Aloeride comprises only 0.015% of the aloe juice dry weight, its potency for macrophage activation accounts fully for the activity of the crude juice.⁴⁸

16.10.3.3 *Coleus forskohlii*

Forskolin, a diterpene from *C. forskohlii*, is a potent platelet aggregation inhibitor. It has been found to strongly inhibit the melanoma cell-induced human platelet aggregation and tumor colonization. The study suggests that forskolin could prove to be valuable in the clinic for the prevention of cancer metastasis.⁴⁹

Roidex, a formulation of squalene, vitamin E, and *Aloe vera*, produced chemopreventive effect in chemically induced skin tumors in CD-1 mice. The tumors were induced by 7,12-dimethylbenz[a]-anthracene (DMBA) and promoted with 12-O-tetradecanoylphorbol-13-acetate (TPA). The mice were treated with either mineral oil, 5% squalene, or Roidex. There was a regression of 33.34% of the tumors in the Roidex-treated group (39 to 26 tumors) compared with the nontreated group, whose tumors regressed only 3.44% (29 to 28

tumors).⁵⁰ An immunomodulator fraction extracted from *Aloe vahombe* (Alva) was found to protect mice against bacterial, parasitic, and fungal infections. It also produced cures only in the case of the McC3-1 tumor; under different experimental conditions, the growth rate of tumors in animals treated was found to be slower than in those untreated.⁵¹

16.10.3.4 *Andrographis paniculata* Nees

The methanol extract of the aerial part of *Andrographis paniculata* Nees showed potent cell differentiation-inducing activity on mouse myeloid leukemia (M1) cells.⁵²

16.10.3.5 *Santalum album*

The essential oil, emulsion, or paste of *S. album* has been used in India as an Ayurvedic medicinal agent for the treatment of inflammatory and eruptive skin diseases. Sandlewood-oil treatment showed chemopreventive effect in DMBA-initiated and TPA-promoted skin papillomas tumor model and TPA-induced ornithine decarboxylase (ODC) activity in CD-1 mice. Sandalwood-oil treatment significantly decreased papilloma incidence by 67%, multiplicity by 96%, and TPA-induced ODC activity by 70%. The study suggests that the oil could be an effective chemopreventive agent against skin cancers.⁵³

16.10.3.6 *Picrorrhiza kurroa*

P. kurroa extract has been shown to have antitumor and anticarcinogenic activity in 20-methylcholanthrene (20 MC)-induced tumor model after oral administration. The extract also inhibited transplantable tumors.⁵⁴

16.10.3.7 *Cystone*

Cystone, a polyherbal ayurvedic preparation, was found to protect tumor-bearing mice from cisplatin-induced nephrotoxicity when given intraperitoneal 1 h before cisplatin. Pretreatment with cystone did not reduce the antitumor activity of cisplatin. The study shows potential for use in protecting patients from nephrotoxicity of cisplatin.⁵⁵

16.10.3.8 *Sesame Oil*

Sesame oil is extensively used as topical application in Ayurveda for skin health. In one study, it was found that sesame and safflower oils, both of which contain large amounts of linoleate in triglyceride form, selectively inhibited malignant melanoma growth over normal melanocytes; coconut, olive, and mineral oils, which contain little or no linoleate as triglyceride, did not. Further studies showed that only linoleic acid was selectively inhibitory, whereas palmitic and oleic acid were not. These fatty acids were tested in the range of 3 to 100 µg/ml. The study suggests that certain vegetable oils rich in linoleic acid, such as the sesame oil, recommended for topical use by Ayurveda, may actually contain selective antineoplastic properties.⁵⁶

16.10.3.9 *Terminalia arjuna*

Methanol extract of *T. arjuna* was found to inhibit the growth of human normal fibroblasts (WI-38) *in vitro* without any effect on normal cells. A cyclin-dependent kinase inhibitor, p21WAF1, was induced in the transformed cell by *T. arjuna*. It is likely that *T. arjuna* has components that can inhibit of transformed cell by p53-dependent and independent pathways.⁵⁷

16.11 Conclusions and Future Research

It is evident that early Ayurvedic physicians had a good understanding of etiology, clinical manifestations, symptoms, classification, malignant and benign nature of tumors, metastasis, recurrence diagnosis, prognosis, and treatment. It is remarkable that the basic information is fairly consistent with the current knowledge in these areas given the technology available 500 years ago. The physicians also recognized the fact that malignant tumors must be completely and extensively excised so that not a trace of tumor is left in the body for even a trace can grow back to a tumor. This is also consistent with the current knowledge about the nature of malignant tumors and their treatment. Various treatment methods, both local and systemic, and various herbal formulations found useful in many tumors are presented. The review has shown that Ayurvedic therapies are useful as an adjuvant to conventional chemotherapy. The following areas are identified for further research: (1) the use as an adjuvant to improve the well-being of the patient, (2) protection against the drug cytotoxicity (the major dose limiting side effect), (3) chemoprevention to reduce the cancer incidence, and (4) immunomodulation to help the body respond better to cancer chemotherapy.

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17

Indigestion (Ajirna)

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17.1 Introduction

Gastrointestinal (GI) symptoms are among the most common and widespread of health complaints among the general populace. An alternative term is dyspepsia. Dyspepsia is often used to refer to upper abdominal pain or discomfort but may also encompass symptoms of early satiety, postprandial abdominal bloating or distension, nausea, and vomiting. Dyspepsia can be episodic or persistent and is often exacerbated by eating. Indigestion is a general term used to describe discomfort or pain in the upper abdomen or chest, usually after meals.

Ayurvedic treatment is focused on the correction of gastrointestinal pathology for most of the endogenous disorders ranging from indigestion to arthritis. Ayurvedic texts describe the anatomy and physiology of the digestive system at great lengths. Here is a brief insight into the concepts of digestion and indigestion.

17.2 Ayurvedic Concept

17.2.1 The Stomach and Digestion

The stomach is one of the vital organs of the body (*marma*). The other important organs are the heart, lungs, kidneys, and the brain. The stomach not only stores food during the process of digestion, but also makes it more permeable through the juices it produces. The stomach has minute glands from which gastric juice (*pachaka pitta*) is produced. The gastric juice helps break down food into smaller molecules. The moment food enters the stomach, the wall of the organ starts a churning action to mix the food with the gastric juice so that it becomes absorbable. Foods that are easy to digest (*laghu ahara*), such as fat-free or low-fat milk and fat-free cereals or fruits, take far less time to digest than heavier foods (*guru ahara*), such as fried substances, which are rich in fat.

The partially digested food from the stomach enters the intestine and is then exposed to the action of bile, pancreatic and intestinal juices, and bacteria. The food is broken down into various easily absorbable substances, which are vital for growth and development, because these form the building blocks of the body. It is only when food has been eliminated from the intestine that the process of digestion is complete and the process of absorption starts.

17.2.2 Agni and Disease

Ayurvedic literature has recognized 13 types of enzymes (*agni*); the most important is the digestive enzyme (*jatharagni*), the primary factor of digestion. The others are the seven intracellular enzymes (*dhatu agni*) and the five organic catalysts (*bhuta agni*). The digestive enzyme (also called *koshthagni*) contained in the gastric juice is the source of all the enzymes of the body. In addition to the digestive enzyme, there are six other factors that help digestion process. They are gastric juice, gut motility factor (*samana vata*), moisture and water, mucin (*kledaka kapha*), time, and a proper combination of the first five. The gut

motility factor propels the food into the stomach, bringing it into contact with gastric juice, and supports the enzymes; moisture breaks down the compactness of the food and mucin softens the food. Time is required for completing the process of digestion, and a proper combination of all factors is vital for completion of the process. The food is digested with the help of the digestive enzyme. This process results in the separation of the nutrient factors from the food, which circulate in the body and help in tissue building through the help of intracellular enzymes.

As long as this enzyme is in homoeostasis, it results in a state of health. Disease is the result of a deviation from this state. The imbalance of the digestive enzyme (including the enzymes of the stomach and intestine) gives rise to most disorders of the stomach, particularly anorexia and dyspepsia. This enzyme occurs in four states: balanced (*samagni*), abnormal (*vishamagni*), increased (*tikshnagni*), and decreased (*mandagni*). Except the balanced state, others are pathological states. These pathological states of enzymes are due to the influence of three *dosas*: *vata*, *pitta*, and *kapha*, respectively.

Physiologically, food undergoes three stages of conversion in the GI tract. The first is a mucilaginous neutral stage (*madhurabhava*) under the influence of *kapha*. The second stage, due to the interference of *pitta*, is an acidic stage (*amlabhava*), and the final stage is a dry, pungent stage (*katubhava*) under the effect of *vata*. Due to etiological factors such as abnormal food habits, emotional disturbances, etc., a pathological state of enzyme results called *agnimandya*. *Agnimandya* is the principal cause for all metabolic disorders commencing with indigestion (*ajirna*).

17.2.3 Indigestion

Indigestion, a deviation in any of the steps described above, also known as upset stomach, dyspepsia, or gastric indigestion, is discomfort or a burning feeling in the upper abdomen. It is often accompanied by nausea, abdominal bloating, belching, and sometimes vomiting.¹ Dyspepsia, which means bad (dys) digestion (pepsia), is a term often used by doctors to describe a set of symptoms believed to have their cause somewhere in the upper part of the GI tract.

Ayurvedic physicians diagnose the disease with clinical interrogation about the state of digestive enzymes and digestion with certain clinical features such as absence of hunger, abdominal discomfort, and belching of undigested food. The physicians also examine the abdomen by palpation and percussion for the presence of flatulence.

To diagnose indigestion, the doctor of conventional medicine might perform tests for problems like ulcers. In the process of diagnosis, a person may have x-rays of the stomach and small intestine or undergo an endoscopy, in which the doctor uses an instrument to look at the inside of the upper GI tract.

17.3 Definition

According to classic Ayurveda, the word *ajirna* in Sanskrit means bad (a) digestion (*jirna*). It is defined as a pathological condition in which food is not digested easily and is the root cause for many internal diseases (metabolic). In conventional medicine, the term *dyspepsia*, derived from the Greek words *dys* (bad) and *pepsis* (digestion), refers to symptoms thought to originate in the upper GI tract.

17.4 Clinical Description

Indigestion, lassitude, heaviness in the body, the retention of flatus and constipation, or loose motion are the common signs and symptoms of *ajirna*, as described in classic texts. In conventional medicine, dyspepsia is characterized by upper abdominal pain or discomfort. It may also encompass symptoms of early satiety, postprandial abdominal bloating or distension, nausea, and vomiting.

17.5 History and Epidemiology

Dyspepsia is common; surveys in Western societies have recorded a prevalence of between 23 and 41%. For many people, dyspeptic symptoms are an acceptable part of living. Why some sufferers (about 25% of the population) seek help from doctors is not clear, but concern about symptoms seems to be as important as the symptoms themselves. The minority of sufferers (5% of the population) who do consult doctors are the major consumers of resources. In the U.K., in 1994, more than 400 million pounds was spent on ulcer-healing drug prescriptions issued by general practitioners. About 4% of general practice consultations are for dyspepsia and 2% of the entire populations receive either an endoscopy or barium swallow (upper GI) each year. Time lost from work and interference with quality of life is more difficult to measure but are likely to be considerable.²

Only 10% of patients attending their general practitioner with dyspepsia will be referred for hospital consultation or investigation.

The prevalence of dyspepsia ranges from 26% in the U.S. to 41% in the U.K. Although only 20 to 25% of persons with dyspepsia seek medical care; the problem is responsible for 2 to 5% of visits to primary care physicians. Nonulcer dyspepsia results in substantial health-care costs.²

17.6 Etiology

According to Ayurveda, the chief causes that affect digestion are the following:

1. Drinking too much liquid or water
2. Irregular eating habits, which include erratic schedules and quantities (i.e., eating at abnormal intervals [between meals, middle of the night, etc.] or too large or too small a serving)
3. Food allergies
4. Iatrogenic indigestion due to improper application of *panchakarma* (the detoxification procedures)
5. Suppression of nature's call or urges like hunger, defecation, etc.
6. Sleep disturbances
7. Emotional disturbances (e.g., envy, phobia, fury, depression, vengeance)

8. Seasonal changes
9. Debilitating chronic illnesses

In general, food selection, frequency of intake, seasonal changes, and emotional disturbances affect digestion.

According to conventional medicine, the main causes of dyspepsia are overeating, eating wrong food combinations, eating too rapidly, and neglecting proper mastication and salivation of food. Causes also include overeating and making the stomach, liver, kidneys, and bowels harder. When the food putrefies, its poisons are absorbed into the blood and consequently the whole system is poisoned. Certain foods, if not properly cooked, can cause dyspepsia. Other causes are intake of fried, rich, and spicy foods; excessive smoking; intake of alcohol; constipation; insomnia; lack of exercise; and emotions such as jealousy, fear, and anger.

17.7 Pathogenesis and Pathology

Ayurvedic texts state that diminished digestive enzymes lead to a state in which the patient is unable to digest even wholesome (*satmya*) and easily digestible food taken in an orderly time frame. This condition is known as *ajirna*.

Etiological factors disturb the homoeostasis of the *dosas*, which in turn influence the digestive enzyme and result in three states dependent on the three *dosas*. They are the *vata*-dependent abnormal state (*vishamagni*), *pitta*-dependent increased state (*tikshnagni*) and *kapha*-dependent decreased state (*mandagni*). These three pathological states of enzyme bring about *dosa*-specific indigestion. They are static indigestion (*vishtabdha ajirna*), acid indigestion (*vidagdha ajirna*), and endotoxic indigestion (*ama ajirna*), which are caused by *vata*, *pitta*, and *kapha dosas*, respectively. Section 17.9 details the types indigestion as described in Ayurvedic texts.

17.8 Clinical Diagnosis

The following clinical features as described in classic Ayurveda help to diagnose the three pathological states of enzyme:

1. Abnormal state of enzyme (*vishamagni*) — In this state, the enzyme is antagonistic due to involvement of vitiated *vata*. Symptoms include a very unpredictable digestive capacity. Sometimes the food is digested well, whereas at other times even light food in small quantities cannot be tolerated. Gaseous distension is common.
2. Increased state of enzyme (*tikshnagni*) — In this circumstance, the appetite is good, but the food that is eaten is not assimilated in the body. At such times, eating more food does not help at all. In fact, the enzyme will become more disturbed and so will the digestion process. A typical feature of this condition is that whatever the food that is eaten (and digested) seems to disappear down a slightly overburdened gut.

3. Decreased state of enzyme (*mandagni*) — In this condition, simple meals are difficult to digest and there is an uneasy feeling of indigestion. Other symptoms are malaise and vomiting.

17.9 Classification of Indigestion

Ayurvedic literatures describe a total of six types of indigestion, of which three are pathological types. They are static indigestion (*vishtabdha ajirna*), acid indigestion (*vidagdha ajirna*), and endotoxic indigestion (*ama ajirna*). The other three are transient indigestion (*rasasesha ajirna*), diurnal indigestion (*dinapaki ajirna*), and successive indigestion (*prativasara ajirna*). Although the latter three are not pathological types, they are transitional and occur physiologically in between any two meals every day. Dislike for food intake until the earlier food is digested is known as transient indigestion. The feeling of abdominal heaviness and lethargy soon after food ingestion is a natural process called diurnal indigestion and successive indigestion. The three pathological types are described below.

1. Endotoxic indigestion (*ama ajirna*) — The clinical features, such as abdominal heaviness, hypersalivation, puffiness of face, and belching out undigested food particles help diagnose this type of indigestion.
2. Acid indigestion (*vidagdha ajirna*) — This condition is presented with clinical features such as giddiness, intense thirst, daze, acid reflux, increased sweating, and a burning sensation all over the body.
3. Static indigestion (*vishtabdha ajirna*) — Clinical features, such as pain in the abdomen, abdominal distension, obstruction to bowel and flatus movements, bewilderment, and myalgia, are presented in this type of indigestion.

In conventional medicine, the clinical diagnosis is based on several signs and symptoms. As dyspepsia is a group of symptoms, it alerts doctors to suspect diseases of the upper GI tract. It includes symptoms of upper abdominal discomfort, retrosternal pain, anorexia, nausea, vomiting, bloating, fullness, early satiety, and heartburn, among others. A firm clinical diagnosis can be difficult based on these symptoms, as few symptoms are discriminatory. Many diseases cause dyspepsia; these include peptic ulcers, esophagitis, cancer of the stomach or pancreas, and gallstones. In a large proportion of cases, no clear pathological cause for a patient's symptoms can be determined. Other symptoms include a foul taste in the mouth, coated tongue, and foul breath. At times, a sensation of tightness in the throat is experienced. In most cases of indigestion, the patients suffer from constipation.

An organic cause is found in 40% of patients with dyspeptic symptoms. The most common organic disorders causing dyspepsia are gastroduodenal ulcer, gastroesophageal reflux disease, and gastric cancer. In 50% of patients, no cause is apparent and the dyspepsia is considered idiopathic, meaning the diagnosis is essential, functional, or nonulcer dyspepsia.² The history and physical examination do not reliably differentiate organic from nonulcer dyspepsia.

Many people do not require invasive investigation. However, as persistent indigestion may suggest a more serious underlying complaint, the doctor of conventional medicine may decide to arrange the following:

1. An endoscopy to enable the doctor to look into the stomach
2. A barium swallow test to enable the outline of the stomach to show up on an x-ray
3. An ultrasound to scans the abdominal organs
4. A blood test to detect anemia or any other abnormality

Ayurvedic physicians currently also use these diagnostic procedures.

17.10 Clinical Course and Prognosis

Indigestion not treated in time may lead to the development of sequelae such as stupor (*murcha*), incoherent talk (*pralapa*), vomiting (*vamathu*), hypersalivation (*praseka*), severe myalgia (*sadana*), vertigo (*bhrama*), and even death (*marana*).³ The GI disturbances that may be followed by metabolic imbalances, either due to toxic states (especially metabolic histotoxic anoxia) or malnutritional states (which are acute to begin with) may tend to become chronic.

The acute conditions are among others that are caused by the impairment of enzymes (*agni*) and the formation of endotoxin (*ama*). Among the subacute and chronic conditions, both GI and metabolic, that may occur in a kind of chain sequence, the following may be mentioned here: functional impairment of colon, liver damage, and hepatic diseases. In fact, according to Ayurveda, most of the diseases included under internal medicine may, from this point of view, be stated to be the outcome of this endotoxin and *dosa* combined with this, called *sama*.

There is increasing evidence that individuals with autoimmune diseases, like rheumatoid arthritis, have intestines that are more permeable to certain antigens, allowing these antigens to invade the body and stimulate the symptoms. These antigens may be akin with the Ayurvedic concept of *ama*.⁴

17.11 Management

17.11.1 Ayurvedic Therapy and Management of Indigestion

The management of indigestion begins with treating enzyme disturbance because it is considered the basic cause of this condition. *Agnimandya* is treated as follows⁵:

1. The abnormal state of enzyme is treated with food and drinks that have sour and salty tastes, along with some quantity of fat.
2. The increased state of enzyme is brought to equilibrium with dairy products such as curds, milk, and milk gruel (*payasam*).
3. The decreased state of enzyme is treated with food and drinks that have pungent, bitter, and astringent tastes, which stimulate enzymes.

Indigestion is generally treated with drugs that have digestive and carminative activities:

TABLE 17.1

Compound Formulas for Indigestion, Poor Digestion, and Dyspepsia

Name of Formulation	Activity	Dose	Adjuvant	Ref
<i>Bhaskaralavana curna</i>	Digestive, appetizer	1–3 g twice/day before meals	Warm water and lemon juice	10
<i>Hingvastaka curna</i>	Carminative, antiflatulent	1–2 g twice/day before or with meals	Warm water, buttermilk, and ghee	5
<i>Panchakola curna</i>	Digestive, carminative	1–2 g twice/day before or with meals	Warm water	10
<i>Agnitudi vati</i>	Carminative, antispasmodic	240–480 mg twice/ day before meals	Warm water	10
<i>Rasonadi vati</i>	Digestive, antiflatulent	240–480 mg twice/ day before meal	Warm water	10
<i>Sankha vati</i>	Digestive, antiflatulent	240–480 mg twice/ day before meals	Lime water	6

1. In endotoxic indigestion, digestive drugs have to be used. If there is excess *kapha*, then *vamana* (emesis) is done under a physician's supervision followed by digestive drugs such as *panchakola curna*, *agnitudi vati*, and *rasonadi vati* (Table 17.1).
2. Acid indigestion is treated with light and easily digestible food. If there is excess *pitta*, *virecana* (purgation) is followed by using antacids, such as *sankha vati* (Table 17.1).
3. To control *vata* in static indigestion, *virecana* with castor oil or *Terminalia chebula* is used followed by *basti* (enema) treatment done under medical supervision. To relieve constipation and indigestion, compound herbal powders such as *bhaskaralavana curna* and *hingvastaka curna* are used (Table 17.1).
4. In any stage of indigestion, ginger (*Zingiber officinale*), cumin (*Cuminum cyminum*), asafetida (*Ferula foetida*), and rock salt are used as digestives and carminatives.⁶

Therapy for functional gastrointestinal disorders comes under phytotherapeutic treatment. From ancient times, phytotherapeutics played a very important role in patients with dyspeptic symptoms. Bitter herbal drugs stimulate at even very small concentrations, sensorially, the secretions of the stomach as well as the digestive glands, and also strengthen the smooth musculature of the digestive tract (via the gustatory system, vagus, and the enteric nervous system). At higher dosages, herbs and food articles with bitter tastes probably directly affect the mucous membranes of the stomach and the bowels as stimulants. Bitters often are combined with essential oils (some volatile oils such as aromatic bitters, drug combinations of a volatile oil with a bitter taste). Essential oils act primarily as spasmolytic, carminative, and local anesthetic. In the past few years, several controlled studies were carried out with phytotherapeutic combinations (e.g., with *Iberis amara*, caraway oil, peppermint oil, curcuma, and ginger extracts) in which the herbal drugs proved to be superior compared with placebo and were as effective as prokinetics (studies according to evidence-based medicine).⁷

17.11.2 Clinical Experiences

1. In case of abdominal pain, *shankha vati* is used at 250 to 500 mg twice daily with warm water.

2. To control and relieve *borborygmi*, 1 to 3 g of *hingvashtaka curna* is used with $\frac{1}{2}$ tsp of cow's ghee along with the first morsel of cooked rice twice daily.
3. If there is constipation with indigestion, 1 to 3 g *bhaskaralavana curna* is used with warm water twice daily.

17.11.3 Lifestyle Changes

A person with an abnormal enzyme and indigestion should avoid stale, cold, and oily foodstuffs, cold drinks, and cold water. Onions, potatoes, sweet potatoes, eggplant, yogurts, bananas, pickles fried, hot and spicy foods, dairy products, liquor, and desserts should be avoided.

Items like ginger, garlic, cumin, coriander, coriander leaves, curry leaves, and lemon should be added to the diet. Vegetables such as radishes, yams, drumsticks, snake gourd, bitter gourd, and pumpkin or fruits (e.g., pomegranates) can be eaten freely. Whole wheat bread, whole cereals, and rice (harvested and stored for a long time) are also helpful.

To avoid indigestion, adhere to the following guidelines:

1. Eat meals at regular times.
2. Do not rush meals.
3. Enjoy eating and drinking, but do so in moderation.
4. Avoid those foods that are associated with symptoms.
5. Avoid nonsteroidal anti-inflammatory drugs (NSAIDs). If possible, consult a doctor before doing so.
6. Avoid stressful situations that cause emotional upset.
7. Stop smoking.⁸

17.11.4 Review of Ayurvedic Therapies

The provision of dosage information in this section does not constitute a recommendation or endorsement but rather indicates the range of doses commonly used in Ayurvedic practice. Doses are given for single herb use and must be adjusted when using herbs in combinations. Doses may also vary according to the type and severity of the condition treated and individual patient conditions.

17.11.4.1 Imbalanced States of Digestive Enzymes

1. 2 g of dried ginger (*Zingiber officinale*) powder taken, twice daily, with warm water stimulates enzymes and relieves indigestion.
2. 5 g of fresh ginger taken with salt or jaggery twice daily before meals.
3. 3 g of chebulic myrobalan (*Terminalia chebula*) powder taken twice daily before meals with salt or jaggery.
4. 7 to 14 ml of lemon juice taken three times daily after meals.
5. Heat 1 g each of long pepper (*Piper longum*) powder and salt on a half lemon fruit over blue flame. The juice of this cooked lemon is to be sucked two or three times daily with meals.

6. 1 g each of the powdered fruit of long pepper and salt to be taken with lemon juice twice daily with meals.
7. Take equal parts of fruit rind of chebulic myrobalan, dried ginger, and rock salt in powder form, and add to it jaggery (approximately one third of the total weight). This is to be taken in 1 to 3 g doses with warm water once daily before the first meal.⁹
8. Fennel is one of the ingredients in a compound pill called *rasonadi vati*. It is useful in low enzyme states.¹⁰
9. Distillate of peppermint leaves relieves nausea, stimulates enzyme secretion, and improves appetite and taste.¹¹

17.11.4.2 Indigestion

Various Ayurvedic formulas commonly used for indigestion are commercially available (Table 17.1). Other simple home remedies consisting of single herbs or multiple herbs are listed below.

1. Dried ginger tea should be taken several times a day.
2. Take 1 to 3 g of the powdered fruit rind of black chebulic myrobalan (*Terminalia chebula*) with an equal quantity of raw unrefined sugar twice daily before meals.
3. 1 to 3 g powdered dried ginger taken three times daily with jaggery.
4. 3 to 6 g of a paste prepared from equal parts of chebulic myrobalan fruit rind, raw unrefined sugar, and raisins is to be taken with honey twice daily before meals.
5. 1 g of the powder of equal parts of rock salt, fruit rind of chebulic myrobalan, fruit of long pepper, and root of leadwort (*Plumbago zeylanica*) should be taken with warm water twice daily after meals.
6. 14 to 28 ml of a decoction prepared from equal parts of clove (*Syzygium aromaticum*), fruit rind of chebulic myrobalan, and rock salt should be taken twice daily before meals.
7. 14 to 28 ml of decoction of equal parts of fruit of coriander (*Coriandrum sativum*) and dried ginger should be taken twice daily before meals.⁹
8. The powder of the turmeric rhizome (3 g) is useful in distaste and indigestion.¹¹

17.11.4.3 External Applications

1. A hot poultice prepared from equal parts of asafetida (*Ferula foetida*), rock salt, and hot water may be applied on the abdomen when it is warm to relieve abdominal discomfort.
2. A warm poultice prepared from equal parts of asafetida powder, black pepper (*Piper nigrum*), rock salt, and hot water may be applied on the abdomen to relieve abdominal discomfort.⁹

According to conventional medicine, the treatment of indigestion includes avoidance of the foods and situations that seem to cause indigestion; in some cases this is the most successful way to treat indigestion. Excess stomach acid does not usually cause or result from indigestion, so antacids are not an appropriate long-term treatment, although some people report that they do help. Smokers can help relieve their indigestion by quitting

smoking, or at least not smoking right before eating. Exercising with a full stomach may cause indigestion, so scheduling exercise before a meal or at least an hour after might help.

To treat indigestion caused by a functional problem in the digestive tract, the doctor may prescribe medicine that affects stomach motility. Because indigestion can be a sign of or mimic a more serious disease, patients should see a doctor if they have vomiting; weight or appetite loss; black tarry stools or blood in the vomit; severe pain in the upper right abdomen; discomfort unrelated to eating; and indigestion accompanied by shortness of breath, sweating, or pain radiating to the jaw, neck, or arm.

17.12 Scientific Basis

17.12.1 *Zingiber officinale* (Ginger)

17.12.1.1 *Antinausea and Antiemetic Effects*

17.12.1.1.1 *Animal Studies*

In mice, ginger's effect in enhancing intestinal motility was similar to metoclopramide's.¹² In shrews, dogs, and rats, ginger extracts effectively reduced chemotherapy-associated vomiting.^{13,14} Ginger also protected frogs against experimentally induced emesis.¹⁵ An herbal combination including ginger and ginkgo was as effective as metoclopramide in another animal study of experimentally induced nausea.¹⁶ Studies in rats and mice suggest that ginger produces its antiemetic effects by stimulating peripheral anticholinergic and antihistaminic receptors and by antagonizing 5-hydroxytryptamine (serotonin) receptors in the gut.^{17,18}

17.12.1.1.2 *Clinical Studies*

Both during fasting and after a standard test meal, ginger extracts significantly enhanced gastroduodenal motility in 12 normal volunteers.¹⁹ Several randomized controlled trials support ginger's use as an antiemetic for nausea secondary to several conditions: morning sickness, chemotherapy-associated nausea, postoperative nausea, and motion sickness. In a randomized, double-blind, placebo-controlled crossover trial of 30 women with hyperemesis gravidarum, ginger (250 mg four times daily) proved significantly more effective than placebo in preventing and reducing nausea.²⁰ Ginger also proved useful in treating chemotherapy-induced nausea in a small pilot study of 11 adult patients; their nausea scores fell from an average of 2 (out of maximum of 4) to 0.7 after taking 1.5 g of powdered ginger.²¹ Another case series also supported ginger's use as an antiemetic in patients undergoing chemotherapy.²²

Data on ginger's effectiveness in preventing postoperative nausea have been conflicting. In two randomized, double-blind studies of women undergoing gynecologic surgery, those treated with ginger had significantly less postoperative nausea and vomiting than those treated with placebo; ginger was as effective as metoclopramide in preventing postoperative GI symptoms.^{23,24} Two other randomized, controlled trials failed to document any statistically significant benefits of preoperative ginger (500 to 2000 mg) on postoperative nausea or vomiting.^{25,26} Several studies have evaluated ginger's effectiveness in preventing motion sickness or seasickness and the potential mechanisms for this effect. In an open study of 1741 tourists traveling by sea, ginger supplements (250 mg

every 2 h) were as effective as both nonprescription and prescription medications in preventing sea sickness.²⁷

In a randomized, crossover trial of eight healthy volunteers, ginger supplements were significantly more effective than placebo in alleviating vertigo associated with motion sickness.²⁸ In a randomized, controlled trial of naval cadets, ginger was significantly more effective than placebo in preventing seasickness, both vomiting and vertigo.²⁹ In an early trial involving 36 college students prone to motion sickness, ginger was as effective as a dimenhydrinate in preventing nausea.³⁰ In a randomized, controlled trial in healthy volunteers, ginger was an effective antiemetic, but its mechanism of action appeared not to rely on alterations in gastric emptying.²⁴ In a study evaluating potential mechanisms for ginger's ability to reduce motion sickness, ginger had no impact on experimentally induced nystagmus associated with motion sickness; the investigators concluded that ginger's primary effect was on the stomach rather than the central nervous system.³¹ In one National Aeronautics and Space Administration (NASA)-sponsored study in healthy volunteers, ginger (500 to 1000 mg) had no apparent effect on gastric emptying.³² Other studies have reported enhanced intestinal motility after oral administration of ginger.¹²

17.12.1.2 Carminative and Antiulcer Effect

17.12.1.2.1 Animal Studies

In mice, zingiberene and gingerol significantly reduced gastric ulceration experimentally induced by ethanol and hydrochloric acid.³³

These results were confirmed in several subsequent studies using several of ginger's constituents, including beta-sesquiphellandrene, beta-bisabolene, gingesulfonic acid, curcumen, and 6-shogaol. The results showed a demonstration of antiulcer effects and protection of gastric mucosa against alcohol, NSAIDs, and hydrochloric acid.^{34,35} Rats given ginger extracts (gingerols) had enhanced bile secretion.³⁶

17.12.1.2.2 Clinical Studies

A Chinese case series reported that an herbal mixture containing ginger was effective in halting upper gastrointestinal hemorrhage.³⁷ There are no randomized, controlled trials in humans evaluating ginger's effect as a carminative or ulcer remedy.

17.12.1.2.3 Dose

There is disagreement on the optimal form and dose of ginger. A pediatric dose is not established. Reputable physicians and herbalists recommend a range of doses. A dose of dried ginger at 250 mg, four times daily taken orally,³⁸ is commonly used. Some German herbalists recommend up to four times this amount.³⁹ Chinese herbalists may use up to ten times this amount. The following methods for preparing ginger can be used:

1. Tea — 1 tsp of fresh ginger root boiled in 1 to 2 cups of water for 10 to 20 min. Cool for 5 minutes and sweeten as desired. May be mixed with peppermint or chamomile.
2. Ginger tincture — 1.5 to 3.0 ml/dose.³⁸
3. Candied ginger — A 1-in. square piece is presumably equivalent to 500 to 1000 of dried ginger.^{40,41}

Ginger is used worldwide as a cooking spice, condiment, and herbal remedy. Rhizomes of ginger have long been used in traditional medicine for alleviating symptoms of GI illness.¹⁴ It is used as a carminative to enhance digestion and reduce intestinal gas and flatulence. In experimental studies, gingerol, an active constituent of ginger, had enhanced bile secretion.³⁶ Ginger in combination with long pepper and black pepper (the combination is called *trikatu* in Ayurvedic literature) promoted the secretion of digestive juices and an increased appetite; it is also reported useful in patients with gastric disorders accompanied with clinical symptoms of achlorhydria and hypochlorhydria.⁴²

17.12.2 *Foeniculum vulgare* (Fennel)

Fennel is a promoter of normal GI motility and is an antispasmodic.⁴³ In experimental studies, fennel was reported to stimulate bile flow from the liver, which could help in relieving GI discomfort.⁴⁴

17.12.3 *Curcuma longa* (Turmeric)

Experimental study with ethanolic extract of turmeric has proved its ability to inhibit gastric secretion and to protect gastroduodenal mucosa against the injuries caused by pyloric ligation; hypothermic-restraint stress; indomethacin; reserpine and cysteamine administration; and cytotoxic agents, including 80% ethanol, 0.6 M HCl, 0.2 M NaOH, and 25% NaCl. Turmeric extract has found to increase the gastric wall mucus and also restored the nonprotein sulfhydryl (NP-SH) content in the glandular stomachs of rats.⁴⁵

A number of clinical trials on turmeric have proved it beneficial in the treatment of peptic ulcer and nonulcer dyspepsia and right upper abdominal pain that may often be caused by biliary dyskinesia.^{46–50} Further acute and chronic oral toxicity studies on the turmeric rhizomes in mice, rats, guinea pigs, and monkeys found it to be safe.^{51,52}

17.12.4 *Mentha piperita* (Peppermint Oil) and *Carum carvi* (Caraway Oil)

A combination of peppermint and caraway oils in the treatment of functional dyspepsia has been extensively studied in experimental models and in clinical conditions of dyspepsia.^{53,54}

The peppermint and caraway oil combination was found to be a safe preparation that acted locally to cause GI smooth muscle relaxation.⁵⁵ The favorable risk-benefit ratio of a peppermint and caraway oil combination was demonstrated for the treatment of functional dyspepsia.⁵⁶ Peppermint oil was known to inhibit enterocyte glucose uptake via a direct action at the brush border membrane in the intestinal lumen.⁵⁷

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18

Constipation (Vibandha)

Shankar K. Mitra and Paramesh R. Rangesh

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18.1 Introduction

Constipation is the infrequent and difficult passage of stools. The frequency of bowel movements among healthy people varies greatly, ranging from three movements/day to three times/week. As a rule, if more than 3 days pass without a bowel movement, the intestinal contents may harden, and a person may have difficulty or even pain during elimination. Sometimes stools may harden and be painful to pass even after shorter intervals between bowel movements. Constipation is a common, often frustrating, and difficult to manage problem in the elderly. Approximately 25% of all Americans over 65 years old experience constipation.¹ Ayurvedic therapies are very effective treatments for constipation.

18.2 Ayurvedic Concept

Most people develop constipation after years of either an imbalanced lifestyle or diet. In general, there are some contributing factors. The most common is the suppression of natural urges resulting in the subsequent disappearance of the normal eliminative urges. For example, first thing in the morning (typically at or before dawn) there should be a natural urge to eliminate. We often find this natural urge inappropriate or inconvenient at certain times and force ourselves to suppress it. This habitual suppression can lead to a kind of psychosocial form of constipation whereby elimination may be regular but not complete. If elimination does not naturally conform to the Ayurvedic definition of normal (i.e., every morning), then the elimination cannot be complete and the colon is therefore constipated.

Normal elimination for one person may not be normal for someone else. In general, one is probably constipated if one passes hard stools fewer than three times/week.² In some cases, the person may also have a bloated feeling or abdominal cramping or pain.

A few commonsense lifestyle changes, including getting more exercise, eating high-fiber foods, and drinking plenty of fluids, especially water, can go a long way toward preventing and alleviating many cases of constipation.³

In Ayurveda, it is described that the normal stool has a definite consistency and hardness, which is called well-formed stool. During digestion, food undergoes an acidic state in which it is in semiliquid form in the small intestines. As it travels further in the large intestine, this digested food (called *ahara rasa*) is absorbed and the waste is left in the large intestine. The cecum, the first portion of this part of the gastrointestinal tract, has the dominance of *vata* and *pitta dosas* that helps in drying of waste and giving it a form. When these *dosas* are in a pathological state, it results in excess drying of the fecal matter. This stool becomes dry and hard and is expelled in small quantities with difficulty. In such situations, if an individual holds the urge to defecate for various reasons, there will be a possibility of collection of fecal matter that becomes very hard; this is a result of too much

absorption of water from the stool. In many diseases, constipation occurs as a symptom. In order to avoid this, one has to drink large amounts of water.

18.3 Definition

In Ayurveda, constipation is called *vibandha*. The word *vibandha* is derived from “vi” prefixed to “bandh,” meaning that which is especially bound (in the intestine) or obstructed. Its synonyms are *purishasanga*, *purishanaha* (both meaning the accumulation of feces), *vishtambha* (obstruction), *kricchravitka*, *alpavitka* (both meaning the passing of a small quantity of stool), and *anaha* (bloated abdomen due to accumulation of stools). Constipation is derived from Latin, “con” meaning together and “stipare,” meaning to cram or pack.

18.4 Clinical Description

The clinical feature of the disease, described in conventional medicine as having a bowel movement not every day, does not necessarily mean one is constipated. A person is most likely constipated if he or she experiences the following:

1. Passes a hard stool fewer than three times/week
2. Strains more than one of four times
3. Has abdominal bloating or discomfort³

Ayurvedic literature describes the clinical features of the disease as the condition that is manifested with the following signs and symptoms:

1. Pain in the abdomen (left hypochondriac, iliac, and umbilical regions)
 2. Reduced bowel movement
 3. Painful defecation
 4. Dry stools
 5. Indigestion
 6. Headache
 7. Churning pain in rectum
 8. Pain in the sacral region⁴
-

18.5 History and Epidemiology

This disease is referred to many times in *Caraka Samhita*.⁵ Constipation is the most common gastrointestinal complaint in the U.S., resulting in about 2 million annual visits to the

doctor.⁶ Most people treat themselves without seeking medical help, as is evident from the \$725 million Americans spend on laxatives each year.⁷

18.6 Etiology

According to Ayurveda, constipation is mainly caused by aggravation of *vata*, though it can sometimes be caused by aggravation of *pitta* and *kapha dosas*. Constipation is generally caused by wrong and untimely bowel habits, a controlled urge to defecate, improper eating habits and untimely food intake, eating food that is difficult to digest and dominated with astringent properties, and not eating enough vegetables or salads (dietary fiber). Frequent fasting is also considered as a cause of constipation. It is also caused by not sleeping well or sleeping very late at night, irritable colon, colitis, and negative emotions like stress, grief, fear, worry, etc.

The following are some of the most common causes of constipation, according to conventional medicine:

1. Improper diet — The most common cause of constipation may be a diet high in animal fats and refined sugar but low in fiber found in vegetables, fruits, and whole grains.
2. Not enough liquids — Liquids like water and juice add fluid to the colon and bulk to stools, making bowel movements softer and easier to pass. People who have problems with constipation should drink enough of these liquids every day, about eight 8-oz glasses. Other liquids that contain caffeine (e.g., coffee and cola) seem to have a dehydrating effect.
3. Lack of exercise — Lack of exercise can lead to constipation, although doctors do not know precisely why. For example, constipation often occurs after an accident or during an illness when one must take bed rest and cannot exercise.
4. Changes in life or routine — During pregnancy, women may be constipated because of hormonal changes or because the heavy uterus compresses the intestine. Aging may also affect bowel regularity, because a slower metabolism results in less intestinal activity and muscle tone.
5. Ignoring the urge to have a bowel movement — People who ignore the urge to have a bowel movement may eventually stop feeling the urge, which can lead to constipation.
6. Laxative abuse — People who habitually take laxatives become dependent upon them and may require increasing dosages until the intestine becomes insensitive and fails to work properly.
7. Travel — People often experience constipation when traveling long distances, which may relate to changes in lifestyle, schedule, diet, and drinking water.
8. Fissures and hemorrhoids — Painful conditions of the anus can produce a spasm of the anal sphincter muscle, which can delay a bowel movement.
9. Specific diseases — Diseases that cause constipation include neurological disorders, metabolic and endocrine disorders, and systemic conditions that affect organ systems. These disorders can slow the movement of stool through the colon, rectum, or anus.

10. Mechanical compression — Scarring, inflammation around diverticula, tumors, and cancer can produce mechanical compression of the intestine and result in constipation.
11. Irritable bowel syndrome (IBS) — Also known as spastic colon, IBS is one of the most common causes of constipation. Some people develop spasms of the colon that delay the speed with which the contents of the intestine move through the digestive tract, leading to constipation.
12. Nerve damage — Injuries to the spinal cord and tumors pressing on the spinal cord can produce constipation by affecting the nerves that lead to the intestine.
13. Medications — Many medications can cause constipation. These include pain medications (especially narcotics), antacids containing aluminum, antispasmodic drugs, antidepressant drugs, tranquilizers, iron supplements, anticonvulsants for epilepsy, antiparkinsonism drugs, and antihypertensive calcium channel blockers.
14. Problems with colon and rectum — The peristaltic activity of the intestine may be ineffective and result in colonic inertia or outlet obstruction. Intestinal obstruction, scar tissue (adhesions), diverticulosis, tumors, colorectal stricture, Hirschsprung's disease, or cancer can compress, squeeze, or narrow the intestine and rectum and cause constipation.

18.7 Pathogenesis and Pathology

Normally muscle contractions propel the waste products of digestion through the intestines. In the large intestine, reabsorption of up to 90% of the water and salt takes place because they are essential for many of our body's functions. If too much water is absorbed or if the waste moves too slowly, one may become constipated.³

According to Ayurveda, the *apana vata* affected (due to various etiological factors) dries up the stools, which obstructs the bowel movements and results in constipation. In short, the pathology of the disease is described as a result of the obstruction or reduced motility in the large intestine, the part of the excretory system, and due to the pathological changes in *apana vata* involving the fecal matter.

18.8 Clinical Features

The Ayurvedic literature describes two types of constipation:

1. Constipation due to *ama* (*amaja anaha*) — Constipation caused by *ama* presents with the following symptoms: thirst, burning sensation in the head, pain in the abdomen, and suppression of eructation and coryza.
2. Constipation due to feces (*purishaja anaha*) — In this type of constipation, retention of feces and urine, acute abdominal pain, and fainting are seen. Vomiting of undigested material and pedal edema may also occur in severe cases.

Current practitioners and information sources classify the types of constipation based on *dosa* dominance in a constitution.

18.8.1 *Vata* Constipation

In Ayurveda, excretory process is controlled by *vata*, the principle that governs all kinds of movement in the body. The particular *subdosa* of *vata* involved in constipation is called *apana vata*. *Apana vata* controls the movements in the pelvis and elimination and reproduction. Typically, when *apana vata* gets out of balance, it will first cause dryness in the colon where the stool can become hard and impacted.

18.8.2 *Pitta* Constipation

The dominance of *pitta*, whose property is heat, causes this form of constipation. An increased heat in the colon can also dry out the colon, aggravating *apana vata* and leading to constipation.

18.8.3 *Kapha* Constipation

When there is excess *vata* or dryness in the colon, the body will defend itself by producing more colonic mucus to combat dryness. When this happens in excess, the clogged colon with mucus causes a *kapha*-based constipation. This imbalance combined with a mucus-forming diet will result in a condition that could become chronic.⁸

18.9 Diagnosis

A diagnosis of constipation generally depends on the medical history and a physical examination. The doctor will first want to make sure there is no blockage (intestinal obstruction) in the small intestine or colon, an endocrine condition (e.g., hypothyroidism), or an electrolyte disturbance (e.g., excessive calcium in the blood [hypercalcemia]). The doctor may also check the medications in case they may be causing the constipation.

In some cases, the doctor may order a test for hidden (occult) blood in stools. Alternatively, one may have a barium enema. A sigmoidoscopy may help detect problems in the rectum and lower colon, in addition to routine blood, urine, and stool tests. One may also perform a proctosigmoidoscopy. In many cases, the doctor will be able to see the rectum and sigmoid colon more easily with this procedure than with a barium enema.⁹

18.9.1 Medical History

The doctor may ask a patient to describe the constipation, including duration of symptoms, frequency of bowel movements, consistency of stools, presence of blood in the stool, and toilet habits (frequency and place of bowel movements). Recording eating habits, medication, and level of physical activity or exercise under personal history also helps the doctor determine the cause of constipation.¹⁰

18.9.2 Physical Examination

A physical exam may include a digital rectal exam with a gloved, lubricated finger to evaluate the tone of the anal sphincter and to detect tenderness, obstruction, or blood.

18.9.3 Clinical Course and Prognosis

Constipation associated with intense thirst and weakness with severe abdominal pain is considered difficult to manage according to Ayurveda.

According to conventional medicine, medical intervention is necessary if one experiences a recent, unexplained onset of constipation or change in bowel habits or any of the following symptoms, which might indicate a more serious health condition:

1. Constipation that lasts longer than 7 days, despite changes in diet or exercise
2. Intense abdominal pain
3. Blood in the stool

Although constipation can be extremely bothersome, it usually is not serious. If it persists, and especially if straining results, one may develop complications such as hemorrhoids and cracks or tears in the anus called abrasions or fissures.

Very severe or chronic constipation can sometimes cause a fecal impaction, a mass of hardened stool not eliminated by a normal bowel movement. An impaction can be very dangerous, and one may need to have it manually removed by a nurse or doctor.

18.10 Therapy

Although treatment depends on the cause, severity, and duration, in most cases dietary and lifestyle changes will help relieve symptoms and help prevent constipation. As in all other gut problems, fasting is an important initial remedy. One should fast at least once a month so that the gut is clean and digestion is complete. This facilitates smooth passage of stools and a sense of complete evacuation. Along with oil, ghee (clarified butter), milk, and some roughage (i.e., green leafy vegetables) should be included in the diet.

If the staple diet consists of bread, boiled vegetables, vegetable or animal protein, salad or dry foodstuffs, the content of ghee in the diet should be increased. Paradoxical as this may sound, ghee in measured quantities does not increase bad cholesterol.¹¹ Ayurveda has instead attributed ghee to many useful effects, including the lubrication of blood vessels, which delays aging. One should pursue the habit of drinking milk before sleeping and add 1 tsp of ghee to it or use ghee with hot water; this is very beneficial in mild to moderate constipation. Administering a laxative will lubricate the walls and help elimination for the constipated patient and provides symptomatic relief at best. However, the goal of the Ayurvedic approach is to understand why and how the constipation manifested and to restore balance specifically while enlivening the body's natural ability to sustain normal elimination. With the proper diagnosis established, the treatment for constipation is relatively simple. The general line of treatment would include the following:

1. Application of oil massage (*abhyanga*) and sauna (*svedana*)

2. Intake of medicated fats (*snehapana*), purgation (*virecana*), and an enema (*anuvasana basti*) with honey, rock salt, and castor oil
3. Anal suppositories such as *phalavarti* and *snehavarti*

18.10.1 Purgation Therapy (*Virecana*)

Virecana, or purgation, is administered for cleansing of *pitta* and stools in the intestine. *Virecana* cleanses the small intestine and colon. Many herbs and other ingredients are used as laxatives or purgatives. These include senna, prunes, bran, flaxseed husk, psyllium husk, cow's milk, salt, castor oil, raisins, and mango juice. When taking these herbs, it is important to follow a restricted diet.

The dose and the selection of herbs also depend on the nature of one's bowel, called *koshta* in Ayurveda. The literature describes three types of bowels: extremely harsh, weak, and moderate. The extremely harsh bowel (*krura koshta*) is controlled by more *vata* and *kapha*, and the person with this type fails to purge and needs drastic purgatives in higher doses. The extremely weak bowel (*mrudu koshta*) is controlled by *pitta* and purges to the intake of milk and needs lower doses of purgatives. The moderate bowel (*madhya koshta*), which is under the influence of balanced *dosas*, requires moderate doses of purgatives.

Among the Ayurvedic medicines available for treatment of constipation, the most common one is triphala powder (compound powder of fruit rinds of Indian gooseberry [*Emblica officinalis*], chebulic myrobalan [*Terminalia chebula*], and Belliric myrobalan [*Terminalia bellirica*]) to be taken with ghee. Otherwise, the mucosal lining becomes dry and leads to further constipation, setting a vicious cycle into motion. In keeping with the emphasis laid on individualization of treatment by Ayurveda, the doses of these medicines vary in different individuals.

Some of the laxatives that can be effectively used in constipation are the following:

1. *Bhagottara curna* — The powders of the following ingredients are mixed in increasing ratio: resin of asfetida (*Ferula foetida*), rhizome of sweet flag (*Acorus calamus*), blacksalt, rhizome of dried ginger (*Zingiber officinale*), cumin seeds (*Cuminum cyminum*), fruit rind of chebulic myrobalan, and root of inula (*Inula racemosa*). The mixture is taken twice/day at the dose of 3 to 6 g.
2. An equal quantity of powders of rhizome of sweet flag, fruit rind of chebulic myrobalan, root of leadwort (*Plumbago zeylanica*), salt of potassium and sodium (*yavakshara*), long pepper (*Piper longum*) fruit, atis root (*Aconitum heterophyllum*), and rhizome of costus (*Saussurea lappa*) is taken twice/day with warm water at a dose of 3 to 6 g.
3. 5 g of fruit pulp of Indian laburnum (*Cassia fistula*) is taken with 50 ml of water and 5 to 10 g of unrefined sugar once/day.
4. 5 g of fruit rind of small Chebulic myrobalan is to be taken with 0.5 g of salt at bed time.
5. A powder of purified and fried resin of *asafetida*, garcinia fruit (*Garcinia indica*), fruit of bishop's weed (*Ptychotis ajowan*), rock salt, and seed of fennel (*Foeniculum vulgare*) is taken in equal parts. A dose of 2 to 6 g of the powder can be taken with 5 ml of fresh lemon juice twice/day.

Commercially available Ayurvedic formulas commonly used to treat constipation are listed in [Table 18.1](#).

TABLE 18.1

List of Ayurvedic Formulas Used for Constipation

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Triphala Curna</i>	Laxative, digestive	2–6 g two times/ day	50 ml warm water and ghee	5
<i>Hingvadi Curna</i>	Carminative, antiflatulent, laxative	1–2 g two times/ day before or with meals	Warm water, buttermilk and ghee	12
<i>Yashtyadi Curna syn. Madhukadi curna or Svadishta virecana</i>	Laxative	3–6 g at bedtime	Warm water	13
<i>Pancasakara curna</i>	Antiflatulent, laxative	3–6 g at bedtime	Warm water	13
<i>Trivritadi leha</i>	Laxative	6–12 g on an empty stomach in the morning	Warm water	5
<i>Ichhabhedhi Rasa</i>	Drastic purgative, contraindicated for regular use in constipation (for <i>virecana</i> therapy only)	120–240 mg on an empty stomach early morning	Cold water	12

Local applications:

1. Poultice of *asafetida* is applied around the umbilicus.
2. Poultice prepared from equal parts of salt petre, fruit rind of Indian gooseberry, ammonium chloride, and sesame seeds (*Sesamum indicum*) is applied around the umbilicus.

18.10.2 Enema (Basti)

The medication rectally administered to control *vata*, which is mainly located in the colon, is known as *basti* or medicated enema. In general, this treatment is used to flush out the loosened *dosas* (including stools) through the intestinal tract. This therapy is administered in chronic and extreme conditions of constipation. It is also administered to patients with habitual constipation and weak patients who cannot be subjected to purgation. This therapy is done under the supervision of a physician. There are over 100 specific enemas listed in Ayurveda to treat constipation and for other *panchakarma* procedures.

An enema involves introducing medicinal substances, such as sesame oil, cow's ghee, and other herbal decoctions, in a liquid medium into the rectum. This treatment is especially good for *vata* disorders and it alleviates constipation and abdominal distension.

The common type of enemas are the following:

1. Oil enema or *anuvasana basti* — $\frac{1}{2}$ to 1 cup of warm sesame oil or castor oil (for chronic constipation)
2. Decoction enema or *asthapana basti* (herbal enema) — $\frac{1}{2}$ cup of *gotukola* (*Centella asiatica*) decoction with $\frac{1}{2}$ cup of warm sesame oil or honey mixed with sesame oil and rock salt (for acute constipation)
3. Nutritional enema — 1 cup of warm milk, 1 cup of meat broth, or 1 cup of bone marrow soup⁴ (for chronic constipation where general weakness is a problem)

Contraindications for the enema are people suffering from chronic indigestion, bleeding from the rectum, cough, breathlessness, diarrhea, diabetes, and severe anemia; the elderly; or children under 7 years old. People suffering from acute fever, diarrhea, cold, paralysis, heart pain, or severe pain in the abdomen are not given decoction enemas.

18.10.3 Vata Constipation

Oleation (*snehana*) and purgation (*virecana*) is the general line of treatment for all *vata*, *pitta*, and *kapha* constipations. A monthly oleation with ghee followed by castor oil purgation provides eliminative support and cumulatively reinstates a more unctuous environment in the colon. To ensure such an effect, moistening and *vata*-balancing herbs are administered between oleations and purgations. This kind of therapy should not be continued beyond 3 months, and the following procedure is helpful:

1. While eating light food for a week, one should start each day with progressively increasing amounts of liquid ghee (2–4–6 tsp) taken orally. (*Note:* Avoid this procedure if fat intolerant.)
2. On the eve of the 4th day, take a warm bath before retiring and drink 6 tsp of castor oil as a purgative.
3. If there is any sign of weakness or fatigue, avoid the procedure above and simply take $\frac{1}{2}$ to 1 tsp of castor oil every night for 1 month. This should not produce a purgative effect. If it does, take less castor oil, because a continuous laxative effect can deplete body fluids and electrolytes.

For a proper diet, cold and dry foods should be avoided. Eat heavier warm foods with an emphasis on oily foods such as nuts, oils, and cooked grains.

18.10.4 Pitta Constipation

Purgation therapy provides a cooling, moistening, and eliminative effect, making it the treatment of choice for this type. Herb therapy includes the following:

1. Take 1 to 2 tsp of aloe gel (*Aloe vera*), three times/day.
2. Take 1 tsp of triphala with ghee to make a paste and take three times/day.
3. Take licorice (*Glycyrrhiza glabra*), *F. vulgare*, and coriander (*Coriandrum sativum*) tea three times/day.
4. Take 1 tsp of psyllium husk mixed well with 8 oz of warm water for 5 min before bedtime.
5. For severe cases, rhubarb root (*Rheum emodi*) and Indian senna (*Cassia angustifolia*) leaf can be taken individually or together as needed.

For a proper diet, one should favor foods that are slightly oily and cooked and avoid hot, spicy, and pungent foods.

18.10.5 Kapha Constipation

The aim of the treatment is to reduce *kapha* with dietary changes. Foods rich in hot and more pungent spices, such as ginger and black pepper, are recommended. Mucus-producing

foods, including cheese, sugar, yogurt, bread, and pastries, should be especially avoided at night. The following herbal remedies are useful:

1. 1 to 2 tsp of psyllium husk (*Plantago ovata*) taken with 8 oz of water three times/day.
2. 8 to 10 glasses of warm honey water daily.
3. $\frac{1}{2}$ to 1 tsp of triphala taken with honey three times/day.
4. Aloe, rhubarb, and Indian senna are bitter laxatives that will combat the intestinal *kapha* and provide an eliminative effect.

Note: While treating for *vata*, *pitta*, or *kapha* constipation there should not be an excess of bowel movements. If there is such an effect, reduce the dose of recommended therapy. If there is no improvement in 2 weeks of treatment, the dosages should be increased.⁸

18.10.6 General Dietary Recommendation

A diet with enough fiber (20 to 35 g/day) helps form soft, bulky stools. A doctor or dietitian can help plan an appropriate diet. High-fiber foods include beans, whole grains, bran cereals, fresh fruits, and vegetables such as asparagus, sprouts, cabbage, and carrots. For people prone to constipation, limiting foods that have little or no fiber such as ice cream, cheese, meat, pizza, and processed foods are also important.

18.10.7 General Lifestyle Changes

Other changes that can help treat and prevent constipation include drinking enough water and other liquids (e.g., fruit and vegetable juices and clear soup), engaging in daily exercise, walking a mile, and reserving enough time to have a bowel movement. In addition, the urge to have a bowel movement should not be ignored. People who are dependent on laxatives need to gradually stop using the medications with the help of a physician.

18.10.8 Prevention of Constipation

It is well known that prevention is the best approach to constipation. Although there is no way to ensure never experiencing constipation, the following guidelines should help. It is important to eat a well-balanced diet that includes unprocessed bran, whole-wheat grains, fresh fruits, and vegetables. Drink plenty of fluids, exercise regularly, have a regular time for breakfast, lunch, dinner, and have undisturbed visits to the toilet. It is very important not to ignore the urge to defecate and avoid a dependence on laxatives. Improve the digestion with the use of light spices such as cumin seeds, coriander, turmeric powder, fennel, and asafetida. Drinking a glass of warm milk at bedtime helps in evacuation the next morning. Regulate your sleeping hours. Ayurveda advises the common saying, “early to bed and early to rise.” In the morning after waking up, drink a glass or two of preferably warm water and then wait a few minutes before going for evacuation. Massaging the whole body with oil (*abhyanga*) once or twice a week and applying oil or ghee on the naval area daily helps in preventing constipation.

18.11 Scientific Basis

A review of pharmacological studies on various Ayurvedic plants used in the therapies of constipation are presented in this section.

18.11.1 *Aloe vera* (Aloe)

Barbaloin or aloin derived from the inner sheath cells of Aloe leaves is a laxative. *In vitro* studies have revealed the inhibitory effect on sodium and potassium pump and chloride channels at the colonic membrane.¹⁴ Aloe anthroquinones were reported to enhance large-intestinal propulsion and water secretion in rats and mice.^{15,16} Randomized controlled trials have documented its potency as a cathartic in chronically constipated adults.¹⁷

18.11.2 *Plantago ovata* (Psyllium)

The ground seeds or husks of psyllium are used in dietary supplements for increased fiber, cholesterol reduction, and laxative activity.¹⁸ In a randomized double-blind placebo study,¹⁹ psyllium was found to be effective for stool frequency and consistency in patients with chronic constipation. In an open study,^{20,21} conducted in patients having manifestations of irritable bowel syndrome with constipation, psyllium showed good results.

18.11.3 *Cassia senna* (Senna)

Senna leaf contains 1.5 to 3% hydroxyanthracene glycosides, mainly sennosides A and B, which are rheindianthrone, and smaller amounts of sennosides C and D, which are rhein-aloe-emodin-heterodianthrone. Modern human studies have investigated the use of senna for the following:

1. Treating severe constipation^{22,23}
2. Treating chronic constipation in long-stay elderly patients²⁴
3. Managing morphine-induced constipation²⁵
4. Improving colonoscopy preparation with lavage²⁶
5. Managing constipation in the immediate postpartum period²⁷
6. Managing postoperative constipation in anorectal surgery²⁸
7. Treating disorders characterized by slow intestinal transit time or constipation²⁹
8. Using as a laxative for terminal cancer patients treated with opiates³⁰

18.11.4 *Rheum officinale* (Rhubarb)

The active chemical constituents of rhubarb are anthraquinone glycosides, aloe-emodin, and physcion.^{31,32} Experimental studies³³ have revealed that the laxative effect is due to the inhibition of water and electrolyte reabsorption in the large colon and to a stimulant effect on intestinal motility. Clinically, it is used to soften stool in anal fissures and

hemorrhoids and is used postoperatively for anorectal surgeries. It is also effective as a cathartic and therefore used for colonoscopy preparations.³⁴

18.11.5 *Prunus persica* (Almond)

The leaf decoction of almond is used traditionally as anthelmintic and laxative. An experimental study³⁵ has revealed that the aqueous extract of leaves exhibits cholinomimetic activity, which may result in its laxative effect.

18.11.6 *Terminalia chebula* (Chebulic Myrobalan)

T. chebula is a commonly advocated agent in Ayurveda for improving gastrointestinal motility. Charles Foster rats were administered *T. chebula* (100 mg/kg/day for 15 days orally), metoclopramide, or atropine, which established prokinetic and antikinetic activities, respectively. *T. chebula* was found to increase the percent of gastric emptying. The enhancement of gastric emptying was comparable with that produced by metoclopramide. This indicates that *T. chebula* can be a useful alternative to the prokinetic drugs available today.^{36,37}

18.11.7 *Cassia fistula* (Indian Laburnum)

The seeds and dried pulp in the pod of this fruit act as a purgative. The flowers soothe the eyes and the pods suppress acidity, making it useful in treating constipation associated with burning pain in the stomach (*pitta* type). The water extract is also used for treating constipation in pregnant women, children, and elderly persons. The pulp, prepared from its fruits, is a laxative used in the treatment of constipation.³⁸

18.11.8 *Mallotus philippinensis* (Indian Kamala)

The *kampillaka* plant is excellent in treating constipation associated with worm infestation as it first kills worms and then, due to its purging effect, gets rid of them. A powder made of *kampillaka* is especially useful. Sometimes worm infestation causes itching of the skin. At such times, taking *kampillaka* powder at night in the dose of only 500 mg to 1 g is useful.³⁹

18.11.9 *Ricinus communis* (Castor)

Castor seed oil is a harmless laxative in small doses. In large doses it is and can be safely used throughout the year.⁴⁰

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Hyperacidity (Amlapitta)

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19.1 Introduction

Hyperacidity (*amlapitta*) is a disease caused by the excessive formation of acid in the stomach associated with the symptom of burning sensation in the throat and heart area. This condition appears in many diseases ranging from gastritis and peptic ulcer disease (PUD) to the currently named gastroesophageal reflux disease (GERD).

Gastritis includes myriad disorders that involve inflammatory changes in the gastric mucosa, including erosive gastritis caused by a noxious irritant, reflux gastritis from exposure to bile and pancreatic fluids, hemorrhagic gastritis, infectious gastritis, and gastric mucosal atrophy. PUD refers to a discrete mucosal defect in the portions of the gastrointestinal (GI) tract (gastric or duodenal) exposed to acid and pepsin secretion. Presentations of gastritis and PUD usually are indistinguishable, and the management is generally the same. GERD is a digestive disorder that affects the lower esophageal sphincter (LES), the muscle connecting the esophagus with the stomach. Many people, including pregnant women, suffer from heartburn or acid indigestion caused by GERD. Doctors believe that some people suffer from GERD due to a condition called hiatal hernia. In most cases, heartburn can be relieved through diet and lifestyle changes; some people may require medication or surgery.

Amlapitta is more generally classified as a GI disorder, which is due to imbalance in enzyme secretions. Caraka describes some of the common causes for such as deranged digestive enzymes. Fasting, overeating while having indigestion, practicing irregular eating habits, consuming unsuitable foods, eating in an unsupportive atmosphere or at the wrong time, or eating foods that have been improperly prepared will all produce acid indigestion.¹

19.2 Definition

Amlapitta is defined as the digestive secretions (*pitta*) that have undergone an acidic and acrid state.²

Gastroesophageal refers to the stomach and esophagus and reflux means to flow back or return. Therefore, gastroesophageal reflux is the return of the stomach's contents into the esophagus.

19.3 Clinical Description

Hyperacidity is characterized by reflux of acid juice from the stomach, vomiting, a burning pain in the chest region, and headache.²

As described in conventional medicine, the most prevalent symptom is a burning pain in the area of the lower chest or upper abdomen. The pain of an ulcer can last anywhere from 30 min to a few hours. Additional symptoms that may be experienced are weight loss, a decrease in appetite, anemia, nausea, and vomiting. There may be times when the pain of an ulcer seems to be gone and then it suddenly returns. Pain associated with ulcers affect different people in different ways. Some people experience pain immediately after eating, whereas others may not be bothered for several hours. It is beneficial to figure out what may be responsible for producing excess stomach acid so these symptoms can be avoided.

19.4 History and Epidemiology

Hyperacidity is not described in detail in most of the ancient Ayurvedic literatures, including those of Caraka, Susruta, and Vagbhata that belong to the period *circa* 1500 B.C. to 600 A.D. However, Caraka has mentioned, while explaining irritable colon (*grahani*) disease, that this condition is the outcome of endotoxin, and the horsegram (*Dolichos biflorus*) seeds are known to be aggravating the disease.¹ For the first time, Kashyapa (*circa* 800 A.D.) has described this disease separately. Madhava (1000 A.D.) has given details about this disease.³

Looking back at the history of conventional medicine, PUD was a rare and generally unrecognized cause of symptoms and complications or death until the early 19th century. Despite sporadic case reports beginning in late 18th century, PUD did not become widely appreciated until early 20th century. The first 6 decades saw the dominance of surgery in the treatment of PUD. With the introduction of acid-suppressive drugs like histamine-2 blockers in the 1970s, the treatment of PUD was revolutionized. By the 1980s, the advent of *Helicobacter pylori* brought about a dramatic twist and possibly the cure.

Reflux disease is common. Approximately 15 to 20% of adults experience heartburn at least once a week. Obesity (body mass index [BMI] >30 kg/m²), alcohol consumption (>7 standard drinks/week), and a first-degree relative with heartburn increase the risk of having reflux symptoms. Patients with connective tissue diseases, such as scleroderma, and chronic respiratory disease, such as asthma and cystic fibrosis, institutionalized and intellectually handicapped patients. Patients nursed in a supine position for prolonged periods are at increased risk of reflux disease.⁴

19.5 Etiology

According to Ayurvedic medicine, hyperacidity is caused by eating foods that are excessively sour, hot, and spicy; increasing the intake of alcohol; and eating oily, fried, or irritant foods. Violation of rules for eating and lifestyle changes, including overeating, having an incompatible diet, eating irritant and contaminated food, eating when suffering from indigestion, increasing the intake of horse gram, living in temperate climate, and experiencing factors that increase *pitta* (e.g., negative emotions and control of natural urges) are causes this disease.¹

Dietary and lifestyle choices may contribute to GERD. Certain foods and beverages, including chocolate, peppermint, fried or fatty foods, coffee, or alcoholic beverages, may

weaken the LES, causing reflux and heartburn. Studies show that cigarette smoking relaxes the LES. Obesity and pregnancy can also cause GERD.⁴

19.6 Pathogenesis and Pathology

In Ayurveda, hyperacidity may be an expression of excessive digestive enzymes (*pachaka pitta*) due to stimulation from *vata*. This excessive enzyme secretion will dry the mucous (*kaphadhara kala*) lining of the stomach, leading to irritation, hypersecretion, acidity, and eventually ulceration. There are two common causes of this type of *vata* stimulation:

1. *Samana vata* — This type of *vata* normally stimulates the digestive enzymes while eating. It is the controller of digestion and relates the state of mind. If one is stressed and hurried during a meal, skips a meal, or eats fast, nondigestible food, these stressors will affect the quality of digestion.
2. *Apana vata* — This type of *vata* is in the intestinal tract. It is often siphoned upward through the enteric cycle into the liver and digestive organs. This reversed flow of the *apana vata* can excessively stimulate the digestive enzyme in the stomach and irritate the stomach lining. Subsequently, *apana vata* can then be drawn upward from an already aggravated *pitta*. Also, *apana vata* may increase as a result of stress, wherein the nervous system is under such extreme influence that *apana vata* is drawn upward for nervous system support. Along the way, the *apana* would stimulate the digestive enzyme and leave the seat of *vata* in the intestines depleted.²

Pitta is at the root of such a disorder. This is typically seen in a *pitta* constitution with a *pitta* imbalance. *Pitta* constitution types are the ones who eat fast without awareness and have a very hectic lifestyle. *Pitta*, which increases by the etiological factors (especially with its acid state), results in hyperacidity. This state of *pitta* is burnt out to cause acid eructation.

Thus hyperacidity is a hypersecretory disorder of the stomach and duodenum. It involves the pathological state of *pitta* locally and blood systemically.

19.7 Diagnosis

The diagnosis of this condition is made by identifying a combination of signs and symptoms such as indigestion (*avipaka*), bitter and acid regurgitation (*tikta-amlodgara*), substernal burning (*hritkantha daha*), hypersalivation (*utklesha*), fatigue, and increased stress (*klama*).

19.7.1 Classification

Amlapitta is classified into two types: regurgitation through the mouth (*urdhvaga amlapitta*) and a similar type of movement in the lower GI tract (*adhoga amlapitta*). Based on the association of the *dosa* involvement, there are three types: *vata*, *kapha* and *vata kapha*.²

19.7.1.1 Hyperacidity Affecting the Upper GI Tract (Regurgitation through the Mouth)

Clinical features of this type are regurgitation of excessive sour, slimy, meat soup (with green, yellow, bluish, dark, blood-stained contents) and vomiting of bitter and sour contents despite having an empty or full stomach. Heartburn rising from the stomach or lower abdomen that continues to the chest and toward the neck, headache, a burning and hot feeling in the sole and palm, and distaste (usually associated with fever) are the other associated problems.

19.7.1.2 Hyperacidity Affecting the Lower GI Tract

This condition is characterized by the passing of loose stools with burning pain, excessive thirst, vertigo, stupor, hypersalivation, skin rashes, hypersweating, and horripilations.

19.7.1.3 Hyperacidity Due to Vata

This condition, in addition to the above symptoms, presents abdominal pain, tremors, horripilations, blackouts, delirious talk, stupor, and fatigue.

19.7.1.4 Hyperacidity Due to Kapha

This condition, in addition to the above symptoms, also has expectoration of sputum, heaviness of the body, loss of taste, vomiting, weakness, itching all over the body, and excessive sleep.²

19.7.2 Clinical Features of GERD**19.7.2.1 Heartburn**

This is the hallmark symptom of reflux disease. It is characterized by a burning feeling below the sternum, rising from the stomach or lower chest toward the neck. It is typically provoked by meals, especially those containing fatty and highly spicy food; bending forward; straining; or lying down. It is usually eased or relieved by antacids.

19.7.2.2 Regurgitation

This is the other characteristic symptom of reflux disease. Regurgitated material is usually tasted and reswallowed but is sometimes so voluminous that it may be mistaken for "vomiting." Some patients may experience regurgitation as their predominant symptom.

19.7.2.3 Water Brash

Esophageal acidification may cause such sudden and brisk stimulation of salivation that the patient's mouth fills with saliva.

19.7.2.4 Atypical Symptoms

Reflux may cause a number of other symptoms not easily identified as being due to reflux. These include angina-like chest pain, excessive belching, dyspepsia and nonspecific queasiness, uneasiness, or nausea.

19.7.3 Complications of Reflux Disease

19.7.3.1 Respiratory Symptoms

Reflux has been incriminated as a cause of respiratory and laryngeal diseases such as asthma, chronic cough, laryngitis, and sinusitis. Coughing, wheezing, hoarseness, or a sore throat may occur and are occasionally the presenting symptoms. It is uncertain how important reflux is in causing these nonspecific symptoms, but this applies probably only in a minority.

19.7.3.2 Dysphagia

Dysphagia is common but is usually variable and due either to defective esophageal peristalsis or heightened esophageal sensitivity. Dysphagia that is associated with symptoms of bolus impaction is highly suggestive of a stricture.

19.7.3.3 Odynophagia (Painful Swallowing)

This may be a prominent symptom due to excessive sensitivity of the esophageal mucosa. It is usually associated with severe esophagitis.

19.7.3.4 Bleeding from Esophagitis

Hematemesis may be a presenting symptom but is rarely severe. Occasionally, iron deficiency may result.⁵

19.8 Clinical Course and Prognosis

Hyperacidity is amenable to treatment if treated early. It becomes palliative if delayed and is difficult to cure in patients not observing preventive follow-up.²

19.9 Therapy

In Ayurveda, the following procedures are used in treating hyperacidity:

1. Emesis therapy (vamana) with the water extracts of snakeguard (*Trichosanthes dioica*) and neem (*Azadirachta indica*) followed by purgation (*virecana*) with powders of trivrit (*Operculina turpethum*) or any one of the following: *avipattikara curna*, *amalakyadi curna*, *abhayarishta*, or *triphala curna*.
2. If the burning pain is intensive, the patient is advised to apply whole body massage with *candana taila* or *lakshadi taila*.
3. In conditions of regurgitation through the mouth, purgation is advised. In case of lower GI symptoms due to hyperacidity, emesis is the choice.⁶

19.9.1 Lifestyle Changes

Three factors are essential for good and balanced digestion. First, food must be fresh and grown locally, or at least seasonally, and prepared with a caring and loving attitude. Second, food must be eaten with attention and awareness. Our sense of taste both prepares and ignites the digestive process before food ever reaches the stomach, as does our sense of composure and calm. Being distracted with books, driving, computers, and phones during a meal negates any hope of a balanced digestive process. Third, eating is best done during *pitta* period, between 10 A.M. and 2 P.M.; this is when the enzymes are the strongest and digestion will be the most effective. Eating at other times will eventually compromise the body's ability to digest and assimilate nutrition and give energy over a long period.

19.9.2 Home Remedies for Hyperacidity

1. Take tender coconut (*Cocos nucifera*) water. A dose of 100 to 500 ml should be taken twice/day.
2. Take the powdered fruit rind of emblic myrobalan (*Emblica officinalis*). A dose of 3 to 6 g should be taken with 100 to 250 ml milk twice/day.
3. Mix a gruel of rice corn, raw sugar, and honey in equal quantity. A dose of 100 to 200 g should be taken twice/day.
4. Take powdered fruit rind of chebulic myrobalan (*Terminalia chebula*) and whole plant of thistles (*Eclipta alba*) in equal quantity. A dose of 3 to 6 g should be taken with 12 g of jaggery and warm water twice/day.
5. Take dried rhizome of turmeric (*Curcuma longa*), leaf of snakeguard (*Trichosanthes dioica*), and fruit rind of emblic myrobalan (*Emblica officinalis*) in equal parts and make the powder. A dose of 3 to 6 g should be taken with the fresh juice of ginger and 30 drops of papaya latex twice/day.
6. Prepare a decoction from the equal parts of stem of tinospora (*Tinospora cordifolia*), fruit of neem, leaf of snakeguard, and *triphala*. A dose of 14 to 28 ml should be taken twice/day.
7. Prepare a decoction of equal parts of dried ginger, fruit of coriander, and leaf of snakeguard. A dose of 14 to 28 ml should be taken twice/day.
8. Prepare a decoction of equal parts of leaves of snakeguard and *vasaka*, dried ginger, stem of tinospora, and rhizome of katuki (*Picrorrhiza kurroa*). A dose of 14 to 28 ml should be taken with 4 to 6 g honey twice/day.
9. Prepare *mahasurdarshan churna*, containing triphala (*Emblica officinalis*, *Terminalia belerica*, *Terminalia chebula*), guduci (*Tinospora cordifolia*), katuka (*Picrorrhiza kurroa*), fumaria (*Fumaria officinalis*), lime juice, etc. A dose of 1/2 tsp three times/day after meals is recommended.
10. Prepare *avipattikara churna*, containing ginger, black pepper, triphala, nut grass (*Cyperus rotundus*), embelia fruits (*Embelia ribes*), cardamom (*Elettaria cardamomum*), clove (*Syzygium aromaticum*), cinnamon (*Cinnamomum cassia*), and sugar. This is the medicine of choice in treating hyperacidity. A dose of 1 tsp before meals three times/day with lemon juice is recommended.
11. Use aloe vera (*Aloe littoralis*). Prepare 1 tbsp of fresh gel from the plant and take three times/day.
12. Drink coriander (*Coriandrum sativum*) and cumin (*Cuminum cyminum*) tea.

TABLE 19.1

Common Ayurvedic Formulas Used To Treat Hyperacidity

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Avipattikara curna</i>	Digestive, pacifies <i>pitta</i>	3–6 g twice/day before meals	Warm water	6
<i>Dhatri loha</i>	Carminative, hemetinic	0.5–1 g twice/day	Ghee	8
<i>Kamadudha rasa</i>	Antacid	250–500 mg twice/day before meals	Honey and Water	8
<i>Sankha bhasma, Kapardika bhasma, or Pravala bhasma</i>	Antacid, antispasmodic	250–500 mg twice/day before meals	Honey and water	8
<i>Sutasekhara rasa</i>	Antacid, antispasmodic	250–500 mg twice/day before meals	Honey and water	8
<i>Narikela khanda</i>	Antacid, antifatulent	6–12 g twice/day before meals	Milk	8

13. Take 1 tsp of psyllium (*Plantago ovata*) 1 h after meals with 8 oz of warm water.
14. Prepare a milk decoction of 1 tsp emblica myrobalan (*Embolica officinalis*), 1 tsp raw sugar, 1 cup of milk, and 1 cup of water. Boil to 1 cup. Drink as a tea with meals or anytime three times/day.
15. Prepare a decoction of fenugreek (*Trigonella foenum-graecum*), honey, ½ tsp of powder with ½ tsp of turmeric on an empty stomach with warm water or warm milk.
16. Prepare a licorice (*Glycyrrhiza glabra*) ghee. Make the decoction with 1 tsp of licorice in 2 cups of water. Boil to 1 cup. Add 1 cup ghee and 1 cup water. Boil off water. Take ½ tsp three times/day on an empty stomach with hot water. This ghee can be used in cooking.⁷

Compound formulas commonly used to treat hyperacidity are listed in Table 19.1.

19.10 Scientific Basis

The scientific studies relating to the treatment of GERD through natural products are diverse, which include not only the studies on gastric acidity, but also those which are suggestive of antiulcer activity. Some of the current studies on therapeutic procedures as well as the plants and herbomineral preparations are listed below.

19.10.1 Fat Preparations (*Snehapana*)

Various dehydrated butter (ghee) or oil preparations have been prescribed in Ayurveda for the management of *parinama sula*. Ghee in the duodenum inhibits gastric acid secretion; gastric inhibitory peptide (GIP) has been proposed as a candidate for this enterogastrone action. Moreover, extracts of various indigenous herbs are mixed in Ayurvedic ghee and oil preparations, making these preparations more useful. Utility of ghee and oil preparations in PUD is discussed below.

Snehapana of *dadimadya ghrita* (2 oz/day for 2 to 6 weeks) exhibited symptomatic relief in peptic ulcer syndrome. A significant reduction in gastric acidity was observed particularly in cases of nonulcer dyspepsia (NUD).⁹

Dadimadya ghrta or *accha ghrta* produced excellent results in 60% of cases with peptic ulcer syndrome (15-patient case study) when given in dose of 50 g/day for 2 to 3 weeks. Experimental studies¹⁰ also confirmed the protective action of *ghrta* against histamine-induced ulcers.

In a control study, researchers have evaluated the efficacy of *panchakarma* (*snehana* and subsequent *virecana*) in 30 patients with *parinama sula*. A dose of 30 ml of *dadimadya ghrta* was given to the first group of 15 patients on the first day followed by a daily increase of 30 ml for 7 days. Thereafter, *virecana* with *avipattakara curna* was given. A control group of 15 patients was treated with *shankha vati* (6 tablets/day) for the same duration. Cases of both groups were followed for 3 months. Approximately 60% of the *snehana-virecana* group was completely cured from the disease for the whole period of follow-up along with radiological ulcer healing and reduction in gastric acidity. Other cases of this group had different degrees of relief. Patients of the control group were not completely cured. This cure probably works by virtue of elimination of *vata* and *pitta dosa* along with wound-healing effects.¹⁰

Tikta ghrta (*astangahrdaya*) was taken in doses of 40 ml/day. Results showed complete symptomatic relief in 82% of cases with *parinama sula* (38-patient case study). The study lacks laboratory or radiological evidence of improvement.¹⁰

Indukanta ghrta (*sahasrayoga*) was tested in 46 patients with *parinama sula* (28 with duodenal ulcer [DU], 10 with gastric ulcer, and 8 with hyperchlorhydria). *Snehapana* was given according to the *agnibala* of the patients for the first 7 days followed by *Svedana* (2 days). *Virecana* with castor oil (1 day) was followed by *samsarjana karma* for the next 3 days. During the *samana* period of 7 days, *indukanta ghrta* was given in doses of 30 ml/day. Complete symptomatic relief was observed in 84% of cases of this series. However, a gastric acid secretion augmented histamine test, barium meal, or endoscopic study has not been included in the evaluation of results.¹⁰

Kusadi ghrta has been studied in patients with peptic ulcers (18 with DU and 2 with gastric ulcer). A dose of 25 g of *ghrta* per day was given for 1 month. Apart from clinical improvement, *ghrta* reduced gastric acid secretion in the Fractional Test Meal test and caused complete ulcer healing confirmed radiologically in 65% of patients.¹⁰

Many single and combined herbs, herbominerals, and fat preparations have been found effective in treating PUD. *Amalaki*, *madhuyasthi*, *satavari*, and *bhringaraja* appear more useful among herbs. Real efficacy can be proved only by comparative and controlled study adopting endoscopic evaluation. These herbs have shown ulcer-healing property even on these parameters. Moreover, these herbs are cheap and nontoxic even with long-term use.¹⁰

19.10.2 Herbs, Herbal Preparations, Rasas, and Bhasmas

19.10.2.1 *Emblica officinalis* (Indian Gooseberry)

Dhatri lauha, frequently used in *amlapitta* and *parinama sula*, consists of two herbs (*amalaki* and *madhuyasthi*) and one mineral (iron). This preparation has been found to alleviate the common clinical complaints like epigastric pain, acid eructation, epigastric and retrosternal burning in the majority of patients with *amlapitta* along with an appreciable reduction in free and total acidity.¹² Fruits of *amalaki*, apart from having proven efficacy in experimental study, relieved the symptoms significantly in patients with *amlapitta* having hyperchlorhydria when given in doses of 3 g three times/day for 7 days. However, this study consisted of only a few patients.¹⁰

In a clinical trial on *amalaki* in a large series of patients with DU (27 cases) and NUD (12 cases), *amalaki* was given in a dose 9 to 13.5 g/day (in three divided doses) for an

initial 2 weeks. This was followed by a dose of 6 to 9 g/day (two divided doses) for the rest of total duration of 3 months to the patients of both groups. Gastric acid secretion was measured by an AHT. Pepsin and mucin were also estimated in gastric juice. A significant decrease in acid and pepsin secretion was observed. Complete radiological healing was reported in peptic-ulcer patients. Marked symptomatic relief was found in most of the patients (78-patient case study) with the use of *amalaki*. Moreover, the drug exhibited prophylactic effect against the histamine-induced ulcer in albino rats. Furthermore, *amalaki* exhibited some anabolic activity evidenced by weight gain and also an increase in hemoglobin concentration.¹³ Both the prophylactic and curative properties of *amalaki rasayana* have been observed in another study on experimental peptic ulcer.¹⁴ *Amalaki* powder produced 66% symptomatic improvement, significant reduction in gastric acidity response (AHT), and endoscopic healing in 57% of cases with DU. However, there were only seven trial cases.¹⁰

19.10.2.2 *Glycyrrhiza glabra* (Licorice)

The second and important ingredient of *dhatrilauha* is *madhuyasthi* or licorice. The antiulcerogenic effect of *madhuyasthi* was observed against both the pyloric ligation and histamine-induced ulcers. The drug powder in doses of 18 g/day in three divided doses exhibited marked response in 80% of patients with DU (10-case study) within a week. Further clinical and experimental study was done on *madhuyasthi*.¹⁵ The powder of the drug was given to 10 patients with NUD (6 g/day) and 15 patients with DU (6 g/day) for 1 to 3 months. Excellent response was reported in 30% of patients and good response was seen in 50% of patients with NUD. After 2 months of therapy in the duodenal ulcer group, pain was relieved completely in 56% of cases, whereas a burning sensation was reported in 78% of the cases. Complete radiological healing was observed in 50% of cases and partial in 40% of cases. No reduction in gastric acidity response was observed. Experimentally, *madhuyasthi* did not produce significant acid reduction. *Haridra* (*Curcuma longa*), *babula* (*Acacla arabica*), and *aparajita* (*Clitoria ternatia*) reduced acidity significantly in albino rats. Two herbs of *dhatrilauha*, *amalaki* and *madhuyasthi*, have been found quite effective for both types of patients, i.e., DU and NUD.¹⁰

19.10.2.3 *Asparagus racemosus* (Asparagus)

A clinical study has reported good response in 80% of patients with DU with the use of *satavari* (asparagus) powder in a dose of 20 g/day in three divided doses for 4 weeks. Free and total gastric acidity was reduced significantly in 15 patients with DU having hyperacidity. Workers also have observed a significant decline in free and total acidity of gastric juice of albino rats with *satavari*. In a double-blind, randomized drug trial, researchers have observed suppression of heartburn, nausea, or vomiting in 100% of cases and pain in 83% of cases with DU with the use of *satavari* powder (6 g/day for 6 weeks). Complete ulcer healing was observed in 50% of cases radiologically and in 62% of cases endoscopically. The drug also reduced hyperchlorhydria of these patients. Workers observed slightly better results with *satavari* in comparison with *amalaki*, but the study was conducted on only eight and seven patients, respectively.

A group of researchers have observed antiulcerogenic effect of *satavari* and *bhringaraja* in aspirin (200 mg/kg body weight for 4 days)-induced ulcers in albino rats. There was a decrease in ulcer score and severity in both the groups, along with reduction in gastric acidity and peptic activity and an increase in mucin activity. Continuity and thickness of gastric mucosa was maintained in comparison with aspirin-treated gastric mucosa. Histochémically, mucin content of gastric mucosa was increased in both the groups. Clinically,

both the drugs (*satavari* and *bhringaraja*) exhibited radiological (in 83 and 67% of cases, respectively) as well as endoscopical healing (in 55 and 50% of patients, respectively) along with significant reduction in hyperacidity pattern. However, again the number of patients was fewer: only six in *satavari* group and nine in *bhringaraja* group.

Another group has observed significant symptomatic improvement in 26 patients with proved DU disease with the use of *satavari* root powder (12 g/day for 6 weeks). *Satavari* inhibited basal acid output by 32%, histamine-induced maximum output by 38%, and alcohol-induced output by only 32%. Approximately 75% of patients had radiologic and endoscopic evidence of ulcer healing. The drug was not found to possess antacid property *in vitro*. Direct healing effect was proposed probably by enhancement of mucosal barrier, prolongation of life span of mucosal cells, or cytoprotection.¹⁰

The antiulcerogenic activity of juice of fresh roots of asparagus has been reported against cold-restraint stress and pylorus ligation-induced gastric ulcers. The activity was reported to be due to both a decrease in offensive acid-pepsin secretion and an increase in defensive mucin secretion. Mucin secretion was quantified in terms of TC:P ratio in the gastric juice. The strengthening of the mucin barrier led to a further decrease in the dextroribonucleic acid (DNA) content of the gastric juice, indicating a decrease in cell shedding.¹⁶ In continuation of earlier experimental work, a *satavari* containing Ayurvedic preparation, *satavari mandur*, has undergone clinical trails and shown promising results. It has been observed that when given *satavari mandur* at the dose of 1.5 g twice daily for a month, the preparation not only produced significant improvement in symptoms of peptic ulcer, but also healed endoscopically proved cases of ulcer dyspepsia by 75%. The biochemical estimations of offensive acid-pepsin and defensive mucin secretion and cell shedding before and after treatment with *satavari mandur* showed a tendency to decrease acid-pepsin secretion. The estimations also showed a significant decrease in cell shedding and an increase in mucin secretion indicating *satavari mandur's* predominant effect on mucosal defensive factors.¹⁰

19.10.2.4 *Eclipta alba* (Thistles)

Bhringaraja is used as a soaking agent for various Ayurvedic antiulcer preparations. As early as in 1968 researchers have reported the efficacy of *Eclipta alba* in treating hyperchlorhydria.¹⁷ Later on, in a clinical trial done on 22 patients with NUD and 8 patients with DU, researchers observed a significant reduction in gastric acidity when *bhringaraja* (whole plant) was given in syrup form in a dose of 20 ml/day (20 g crude drug) in three divided doses for 6 weeks. Approximately 90% of dyspeptic patients were cured, whereas about 87% of ulcer patients improved. Seeing its efficacy, a team of scientists further evaluated its effectiveness in NUD and DU in a large series of 60 cases (35 of NUD and 25 of DU). The powder of whole herb was given in a daily dose of 30 g (in three divided doses) for 1 month to the NUD group and for 3 months to the DU group. The drug exhibited marked symptomatic relief in the majority of the patients and reduced gastric acidity significantly in both groups. Approximately 80% of patients with NUD responded well to this therapy. Radiological improvement was observed in 75% of cases out of 12 patients with DU who came for follow-up. Approximately 48% of patients of this series gained excellent relief with *bhringaraja*. Later another group further confirmed the effectivity of *bhringaraja* in 60% of cases (935-patient case study) with DU when given in dose of 30 g/day for 3 to 6 months.¹⁰

19.10.2.5 *Trichosanthes dioica* (Snakeguard)

Patoladi kasaya (prepeptone), consisting of *patola*, *haritaki*, *bibhitaka*, *amalaki*, *kutaki*, *cirayata*, *amrta*, *pittapapada*, *sunthi*, and *bhringaraja*, exhibited complete improvement in 50% of cases and partial improvement in 40% of cases with PUD (10-patient case study). The decoction

was given in doses of 30 ml/day in three divided doses for 3 weeks. It decreased the acid response in hypersecretors and normalized the acid response in hyposecretors. Best results were observed in patients with *kaphaja parinama sula*.

Another *patoladi kasaya* consisting of only four herbs, *patola*, *sunthi*, *amrta*, and *kutaki*, was given in a follow-up study of 33 operated cases of DU. The drug kept the patients symptom- and complication-free when given in doses of 40 ml/day in two divided doses. It normalized both the hyper- and hypoacidity of these patients.

The efficacy of a single herb, *patola*, was studied in 20 patients with DU. Its decoction was given in doses of 150 ml/day (60 g crude drug) in three divided doses for three weeks. Drug exhibited marked clinical improvement and reduction in gastric hyperacidity. Excellent response with complete ulcer healing (radiological) was observed in 50% of patients, and good response in 35% of patients. Effectively of *patola* in DU (excellent response in 45% of 20 cases) has been further reported.¹⁰

19.10.2.6 *Adhatoda vasica* (Vasaka)

Vasaka in Ayurveda is indicated in bleeding disorders. PUD may cause hematemesis. Therefore, this drug was tried in 20 patients with DU. Approximately 60 ml of syrup (30 g crude drug) was given daily in four divided doses for 6 weeks. Clinical improvement along with reduction in gastric acidity was observed in 85% of patients in this series. Moreover, the drug also decreased gastric acid secretion in albino rats. The drug was not tried in patients with DU.^{18,19}

19.10.2.7 *Cocos nucifera* (Coconut)

Coconut water has been reported to possess protective action against aspirin- and histamine-induced gastric mucosal damage. Its kernal reduced the gastric acidity and improved the clinical features in patients with DU. *Narikela khanda*, a famous preparation for *parinama sula*, was given in a dose 60 g/day in five divided doses for three weeks to eight patients of PUD. Approximately 62% of patients got complete relief, whereas the remaining cases got partial improvement. Best results were obtained in the *paittika* type of *parinama sula*.

Two researchers evaluated the effectivity of an Ayurvedic compound preparation consisting of *narikela lavana*, *trisanakha bhasma*, and *avipittikara curna* in 60 cases of *amlapitta*. The compound clinically exhibited good response in 57% of cases and fair response in 26% of cases.¹⁰

19.10.2.8 *Tamra bhasma* (Copper Calx)

The use of herbomineral preparations in Ayurveda is well documented. *Tamra bhasma*, a traditional preparation of copper, has been suggested for its use in *amlapitta*. An earlier published study has given the composition of *tamra bhasma* as CuO < 44.45%, Fe₂O₃ < 6.03%, and S < 2.75%.²⁰

Extensive studies have been undertaken to unravel the antiulcerogenic activity of *tamrabhasma*. It has been shown to possess ulcer-protective activity against various gastric ulcers in 8-h immobilized, 4-h pylorus-ligated, and aspirin-induced rats and histamine-induced gastric and DUs in guinea pigs. The activity was due to both a decrease in offensive acid-pepsin and an increase in defensive mucin secretion.²¹ Further evaluation of *tamra bhasma* from different sources were compared with pure compound and mixture of major ingredients of *tamra bhasma* for antiulcerogenic activity. Results revealed that *tamra bhasma* preparation was better than pure copper compound or a mixture of known ingredients.

Quantitative differences in *tamra bhasma* preparation also showed the importance of pharmaceutical processing in the therapeutic activity of *tamra bhasma*.²² *Tamra bhasma* has an overall solubility in water of approximately 1% and 12 ng/ml of CuO forms a saturated solution at 50°C. This implies that when used orally it would have a local effect rather than being systemically absorbed, reducing the possibility of systemic toxicity with copper. Toxicity studies have shown that TMB was safe, and even 1000 times of the effective antiulcer dose was tolerable in rats.

Tamra bhasma was also found to have an antiulcerogenic effect against aspirin-induced ulcers without affecting the anti-inflammatory activity of aspirin. *Tamra bhasma* seemed to have prolonged effect, as the protective effect of *tamra bhasma* lasted up to 5 days after discontinuation of treatment. An aspirin-copper combination may reduce ulcerogenic activity of aspirin without affecting its anti-inflammatory effect. Researchers have reported the gastric protective effect of *tamra bhasma* against cold restraint stress-induced gastric ulcers in rats and guinea pigs and its healing effect against acetic acid-induced gastric ulcers in rats. *Tamra bhasma* was reported to increase prostaglandins (PGEs) and decrease leukotriene C4 (LTC4) in human gastric and colonic mucosal incubates. The mucosal protective effects of *tamra bhasma* and CuCl2 could be due to their effect on endogenous PGEs and LTs. *Tamra bhasma* showed better and potent effect when compared with CuCl2 on PGEs release both in gastric and colonic incubates, suggesting that other ingredients of TMB add to its effects. *Tamra bhasma* significantly protected rats against ethanol-induced ulcers, and the effect was ascribed to decrease in LTC4/D4 synthesis.²³

Tamra bhasma has also been reported to increase the defensive mucopolysaccharides, including sialomucin and fucose. It has also been reported to decrease DNA in gastric juice, suggesting decrease in cell shedding and increase in life span of mucosal cells, with no change in mucosal DNA and incorporation of [³H]-thymidine uptake in mucosal tissues, indicating absence of activity on cell proliferation. The effect of TMB on offensive acid-pepsin secretion and LTs and defensive mucus secretion, mucosal glycoproteins, prostaglandin, and cell shedding may contribute to its ulcer protective activity.²⁴

Tamra bhasma (1 mg/kg) has shown antiulcerogenic effect in 8-h immobilized, 4-h pylorous ligated, aspirin-induced and histamine-induced gastric ulcers in albino rats and guinea pigs. The drug decreases total acid and pepsin output and increases carbohydrate protein ratio, indicating an increase in mucosal barrier. Acute and subacute toxicity of the drug was not observed.²⁵ *Tamra bhasma* has better and prolonged antiulcer effect than pure copper preparation. Moreover, it does not antagonize the anti-inflammatory effect of aspirin. It has been suggested to take it on alternate days or even twice a week only with all beneficial effects in PUD.¹⁰

19.10.2.9 *Sutasekhara Rasa*

Sutasekhara rasa, an herbomineral mercury preparation, is widely used for the treatment of *amlapitta* and *parinama sula*. It contains copper, aconite, datura, and certain other herbs like *bhringaraja*. Two types of *sutasekhara rasa*, those with gold and those without gold, are popular in treating DUs.

Two researchers evaluated the efficacy of modified *sutasekhara rasa* (devoid of gold and copper) in seven patients with NUD and three with DU. The drug was given in doses of 600 mg/day for 6 weeks. Excellent response was reported in 85% of patients with NUD and in 67% of patients with DU.

A team of researchers, in a comparative clinical trial done on 63 patients with *parinama sula* (DU), did not observe any significant difference in the effectivity of *sutasekhara rasa* with gold and without gold. Gold preparation (375 mg/day in three divided doses)

exhibited relief in 70% of patients, whereas preparation without gold (750 mg/day in three divided doses) produced relief in 67% of patients in 45 days. Results were assessed only on symptomatic improvement and reduction in acid secretion (FTM) and not on radiological and endoscopical ulcer-healing criteria. They have recommended the use of *sutasekhara rasa* without gold because of the high cost and long-term toxicity of *sutasekhara rasa* with gold.²⁶ Another group has reported the relief in patients with DU after a course of *sutasekhara rasa* for 42 days. Workers believe that this preparation acts probably by modifying the homeostatic regulation of hormonal stimulation of H⁺ ion secretion.²⁷

Recently, *sutasekhara rasa* (without gold, 600 mg/day) along with *ashvangandha* powder (*Withania somnifera*, 6 g/day) exhibited symptomatic radiological and endoscopical improvement. It also exhibited along with a reduction in gastric acidity (FTM) in all the four treated patients with DU in 6 weeks; *sutasekhara rasa* alone improved three out of four patients. Similarly, *sutasekhara rasa* and *asvvangandha* separately exhibited good response in four out of five patients, whereas their combination showed response in all the five treated cases with NUD. The number of trial cases in each group was very low. A substantial inference cannot be drawn. Even then, this study highlights the utility of antianxiety indigenous herbs (medhya drugs) as an adjuvant in the management of PUD.¹⁰

There are many plants used in antiulcer herbal formulations that have not yet been individually investigated for their efficacy in this condition. A group of researchers has reported the antiulcer effect of an herbal formulation, UL-409, consisting of six medicinal plants: *Glycyrrhiza glabra* L. (root), *Saussurea lappa* C.B. Clarke (root), *Aegle marmelos* Corr. (fruit), *Foeniculum vulgare* Mill. (seed), *Rosa damascena* Mill. (flower petals), and *Santalum album* L. (stem). This herbal formulation was found to increase the stomach mucus and decrease the acid volume, free and total acid content, in rats. Moreover, this formulation significantly prevented the occurrence of stress-induced ulceration and significantly inhibited gastric ulceration induced by ethanol, aspirin, pylorus-ligation, histamine, and indomethacin in rats and guinea pigs. It was found to have antiulcer activity due to a modulation of defensive factors, thereby improving gastric cytoprotection.^{28,29}

A clinical trial was conducted in 56 patients with peptic ulcers, as confirmed by endoscopy, with UL-409. Results showed the formulation's beneficial effect in patients with PUD without any side effects in any of the patients.³⁰

In another clinical trial,³¹ patients with *H. pylori* infection (UL-409) showed clinical improvements without any side effects.

19.11 Future Ayurvedic Formulas

A variety of botanical products have been reported to possess antiulcer activity (especially from ethnopharmacological studies), but the documented literature has centered primarily on pharmacological action in experimental animals. Except for a few phytogenic compounds (i.e., liquorice and chilli), limited clinical data are available to support the use of herbs as gastroprotective agents and, thus, the data on efficacy and safety are limited. Despite this, there are several botanical products with potential therapeutic applications because of their high efficacy and low toxicity. Finally, it should be noted that substances, such as flavonoids, aescin, aloe gel, and many others, that possess both anti-inflammatory and antiulcer activity are of particular therapeutic importance. This is because most of the anti-inflammatory drugs used in modern medicine are ulcerogenic.

Several plants and herbs are used in folk medicine to treat gastrointestinal disorders, including peptic ulcers. Recently, there has been a growing interest in identifying new antiulcer agents from plant sources.

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Irritable Colon (Grahani)

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20.1 Introduction

Irritable colon is also known as Irritable Bowel Syndrome (IBS). It is a gastrointestinal (GI) motility disorder for which there is no organic or structural cause. Because the symptoms of IBS can mimic other disorders, such as hypothyroidism, it is diagnosed when all other local and systemic conditions have been ruled out.

In Ayurvedic literature, the symptom of a disease called *grahani* resembles most of IBS symptoms. According to Ayurveda, *grahani* can be due to a number of factors such as improper diet, heavy foods, overeating, constipation, hemorrhoids, drinking too much fluid (especially cold water when hungry), exercising too soon after eating, forcing or straining natural urges, and taking chemical drugs.

20.2 Definition

The disease condition is called *grahani*, which indicates the pathological state of function and integrity of the intestinal tract (mostly small intestine), a particular part of the GI system known as *grahani* in Sanskrit.¹ *Grahani* is the seat of enzymes (agni). Normally, it holds up the food (until it is digested) and releases it from the side after it is digested. But when it is deranged due to weak digestive enzymes, it releases the ingested material even in undigested conditions.¹

A disorder characterized by abnormally increased motility of the small and large intestines, producing abdominal pain, constipation, or diarrhea, is also known as irritable colon, spastic colon, or mucous colitis.

20.3 Clinical Description

Ayurvedic literature characterizes the disease by the passage of stools alternated with constipation or diarrhea and with undigested food particles. This disease is also associated with thirst, distaste, blackouts, pedal edema, pain in the bone, fever, and vomiting.² It is a motility disorder involving the entire GI tract, causing recurring upper and lower GI symptoms, including variable degrees of abdominal pain, constipation and diarrhea, and abdominal bloating.

20.4 History and Epidemiology

IBS has been a problem for people for thousands of years, as it is mentioned in the earlier literatures of Ayurveda. IBS, also known as spastic colon, is a common disorder affecting as many as 20% of the population, more so among women than men. Because it is a clinical

diagnosis, the prevalence figures vary depending upon the definition of IBS used. Although only about 10% of IBS patients actually seek medical attention, it accounts for about 3 million physician visits and over 2 million prescriptions in the U.S. annually. It is the seventh leading diagnosis among all physicians, and these patients utilize an additional \$300 per year of health-care costs than do their non-IBS counterparts.³

20.5 Etiology

Although no specific causes are mentioned in Ayurvedic literatures, most of them focus on the dietary injudiciousness as the foremost cause of IBS. Examples include the quality of food, such as extremely dry, astringent, spicy, and heavy to digest (protein rich) foods; the quantity of intake (e.g., overeating); and the improper frequency of intake (e.g., untimely eating habits). Certain lifestyles including excessive traveling and controlling various natural urges (e.g., defecation and urination) are described to be the foremost causes that affect the function and integrity of digestive system.⁴

In conventional medicine, the cause of IBS is unknown. No anatomical cause can be found. Emotional factors, diet, drugs, or hormones may precipitate or aggravate heightened GI motility. Some patients have anxiety disorders (particularly panic disorders), major depressive disorders, and somatization disorders. Stress and emotional conflict do not always coincide with symptom onset and recurrence. Some patients with IBS appear to have a learned aberrant illness behavior (i.e., they tend to express emotional conflict as a GI complaint, usually abdominal pain). The physician evaluating patients with IBS, particularly those with refractory symptoms, should investigate for unresolved psychological issues, including the possibility of sexual or physical abuse.³

20.5.1 Food Intolerance

True food allergy is mediated by the immune system and is associated with hives, asthma, eczema, nasal discharge, and positive skin-prick radioallergosorbent test (RAST) scores or other allergy tests.⁵ However, food intolerance, rather than true food allergy, is believed to be more significant in IBS.⁶ Between 33 and 66% of IBS patients report having one or more food intolerances.⁵ The most common culprits are dairy (40 to 44%) and grains (40 to 60%); the resulting GI bloating, flatulence, and pain caused by this reaction appears to be mediated by inflammatory prostaglandin synthesis.⁶

20.5.2 Neurochemical Imbalance

Interaction between the brain and the gut occurs via nerves that send neurotransmitter signals. An imbalance between two of these neurotransmitters, serotonin and norepinephrine, are implicated in IBS. Constipation may result when levels of norepinephrine increase, causing a reduction in serotonin levels and inhibition of another neurotransmitter called acetylcholine. Conversely, diarrhea can occur when increased serotonin inhibits norepinephrine and causes levels of acetylcholine to increase. For IBS patients, such an imbalance in the nervous system can lead to the fluctuating bowel symptoms of constipation and diarrhea.⁷

20.5.3 History of Analgesic Use

The use of acetaminophen, a common pain-relieving medication, is associated with diarrhea-predominant IBS. Its action may be due to an imbalance in the neurotransmitter serotonin. Because acetaminophen can cause elevated levels of the serotonin by-product 5-hydroxy-indole-acetic acid (HIAA) in the urine, it is possible that acetaminophen somehow interferes with serotonin metabolism. Plasma serotonin levels have indeed been shown to be elevated after eating by patients with diarrhea-predominant IBS. Clinically, a drug that blocks the 5-HT₃ serotonin receptor (5-HT₃ receptor antagonist) is effective for women with diarrhea-predominant IBS. It is interesting to note that asthma, another condition associated with disordered smooth muscle function, was recently found to be associated with acetaminophen.⁸

20.5.4 Reproductive Hormones

IBS occurs more than twice as frequently in women than in men and tends to follow a cyclic pattern, with aggravation during the postovulatory (progesterone predominant) and premenstrual phases of the menstrual cycle.⁹ Progesterone is known to delay gastric emptying and cause constipation; constipation with straining and the frequent passage of hard stools is a more prevalent IBS manifestation in women, especially during the postovulatory phase.¹⁰ At the end of the postovulatory phase, the sudden withdrawal of progesterone that occurs with the start of the premenstrual phase may trigger increased bowel activity. Women frequently report loose stools and diarrhea before or with the onset of menstruation. In contrast to progesterone, estrogen has not been associated with exacerbations of IBS symptoms.¹¹

Along with progesterone levels in women, prostaglandins E₂ and F₂ alpha also increase in the premenstrual phase. Because they are powerful stimulants of bowel contractions, it is possible that women with IBS may have an exaggerated response to these prostaglandins.⁹

20.5.5 Mood

Anxiety, hostile feelings, sadness, depression, and sleep disturbance are associated with IBS. Adverse life events such as family death, marital stress, financial difficulties, and especially physical and sexual abuse have also been reported more frequently in IBS patients than in the general population. However, it is possible that IBS patients with this social or psychological background may be more likely to seek medical treatment or participate in research studies.⁵

The impact of stress on bowel motility and pain were explored in one study⁵ by administering corticotrophin-releasing factor (CRF), a hormone released in the body during stress. CRF increases motility of the descending colon and can induce abdominal pain. The researchers found that IBS patients had greater colonic motility and more abdominal pain after receiving CRF.

20.5.6 Small Intestine Bacterial Overgrowth

Excess bacteria in the small intestine, an area that is normally relatively bacteria free, is being recognized as important in the development of IBS. When these bacteria are present in the small intestine, excessive gas, bloating, abdominal distension and pain, and altered gut motility can result.⁵

20.6 Pathogenesis and Pathology

According to Ayurveda, there is a certain pathogenesis of this disorder. Due to the above causative factors, the enzyme secretion gets disturbed by the dominant *dosa*, which results in the production of enterotoxin. This toxin causes the food to undergo acidification, which damages the mucous membrane of the intestines; the result is a decrease of intestinal transit time, causing early evacuation of bowel contents or alternated with a state of constipation, which is influenced by *apana vata*.²

In IBS, the circular and longitudinal muscles of the small bowel and sigmoid are particularly susceptible to motor abnormalities. The proximal small bowel appears to be hyperreactive to food or parasympathomimetic drugs. Small-bowel transit is variable in patients with IBS, and changes in bowel transit time often do not correlate with symptoms. Intraluminal pressure studies of the sigmoid show that functional constipation can occur when austral segmentation becomes hyperreactive (i.e., increased frequency and amplitude of contractions); in contrast, diarrhea is associated with diminished motor function.

Excess mucus production, which often occurs in IBS, is not related to mucosal injury. Its cause is unclear, but it may be related to cholinergic hyperactivity. Hypersensitivity to normal amounts of intraluminal distention exists, as does a heightened perception of pain in the presence of normal quantity and quality of intestinal gas. The pain of IBS seems to be caused by an abnormally strong contraction of the intestinal smooth muscle or by an increased sensitivity of the intestine to distention. Hypersensitivity to the hormones gastrin and cholecystokinin may also be present. However, hormonal fluctuations do not correlate with clinical symptoms. The caloric density of food intake may increase the magnitude and frequency of myoelectrical activity and gastric motility. Fat ingestion may cause a delayed peak of motor activity, which can be exaggerated in IBS. The first few days of menstruation can lead to transiently elevated prostaglandin E₂, resulting in increased pain and diarrhea. This is not caused by estrogen or progesterone but by the release of prostaglandins.³

20.7 Diagnosis

Ayurvedic literature indicates that there are four different types of IBS. They are constipation-predominant IBS (*vata grahani*), diarrhea-predominant IBS (*pitta grahani*), dysentery-predominant IBS (*kapha grahani*), and complex IBS (*tridosha grahani*). All three *dosas* are involved (*vata*, *pitta*, and *kapha*). Other literature also describes two chronic types of IBS: accrual IBS (*samgraha grahani*) and an incurable type called tympanitis predominant IBS (*ghatiyanthra grahani*).² The clinical features of all types of IBS are described below.

1. Constipation-predominant IBS — This type of IBS presents the clinical features of dryness in skin, mouth, or throat; more constipation or alternating constipation and diarrhea; thirst; bloating; flatulence; and a cold feeling. It is also associated with back or groin pain, weight loss, debility, anal fissures, insomnia, and anxiety.
2. Diarrhea-predominant IBS — This type of IBS clinically shows the presence of heartburn, thirst, feeling hot, irritable or angry, inflammation, sweating and fever, fluid, and foul smelling stools and eructation.

3. Dysentery-predominant IBS — This type of IBS exhibits the presence of nausea, indigestion, and excess sputum in the pharyngeal region, heaviness in the chest and abdomen, bad-smelling eructation, lethargy, sluggish bowels, and mucus in the stools.
4. Complex IBS — This type reveals the combined signs and symptoms of all the above types.
5. Accrual IBS — This type of IBS presents the combined features of all the *tridosas* and is more chronic. It is specially diagnosed with clinical features of *borborygmi*, diurnal changes that the bowel movements are increased in the daytime and stop at night. The nature of the stool will be pastier and slimy, have undigested food particles, and be eliminated with pain. The pattern of the bowel movement will be accumulation of stools for some days followed by passage of loose stools for several days.
6. Tympanites-predominant IBS — In this type of IBS, the clinical symptoms are rumbling sounds heard in abdomen and increased bowel movements with lots of undigested food particles.²

Conventional medicine states that IBS tends to begin in the second and third decades of life, causing bouts of symptoms that recur at irregular periods. Onset in late adult life is rare. Symptoms usually occur in the awake patient and rarely rouse the sleeping patient. Symptoms can be triggered by stress or the ingestion of food.

The features of IBS are pain relieved by defecation, an alternating pattern of bowel habits, abdominal distention, mucus in the stool, and sensation of incomplete evacuation after defecation. The more symptoms that are present, the likelier it is that the patient has IBS. In general, the character and location of pain, precipitating factors, and defecatory pattern are distinct for each patient. Variations or deviations from the usual symptoms may suggest intercurrent organic disease and should be thoroughly investigated. Patients with IBS may also have extra intestinal symptoms (e.g., fibromyalgia, headaches, dyspareunia, temporomandibular joint [TMJ] syndrome).

Two major clinical types of IBS have been described. In *constipation-predominant IBS*, constipation is common, but bowel habits vary. Most patients have pain over at least one area of the colon, associated with periodic constipation alternating with a more normal stool frequency. Stool often contains clear or white mucus. The pain is either colicky, comes in bouts, or has a continuous dull ache; it may be relieved by a bowel movement. Eating commonly triggers symptoms. Bloating, flatulence, nausea, dyspepsia, and pyrosis can also occur.

Diarrhea-predominant IBS is characterized by precipitous diarrhea that occurs immediately on rising or during or immediately after eating. Nocturnal diarrhea is unusual. Pain, bloating, and rectal urgency are common, and incontinence may occur. Painless diarrhea is not typical and should lead the physician to consider other diagnostic possibilities (e.g., malabsorption, osmotic diarrhea).³

Diagnosis of IBS is based on characteristic bowel patterns, time and character of pain, and exclusion of other disease processes through physical examination and routine diagnostic tests. Standardized criteria have been developed for IBS. The Rome criteria for IBS include abdominal pain relieved with defecation and a varying pattern of altered stool frequency or form, bloating, or mucus. The key to diagnosis is effective history taking. This requires attention to directed (but not controlled), elaboration of the presenting symptoms, history of present illness, past medical history, family history, familial inter-relationships, and drug and dietary histories. The patient's interpretation of personal

problems and overall emotional state are equally important. The quality of patient–physician interaction is key to diagnostic and therapeutic efficacy.

On *physical examination*, patients with IBS generally appear to be healthy. Palpation of the abdomen may reveal tenderness, particularly in the left lower quadrant, at times associated with a palpable, tender sigmoid. A routine digital rectal examination should be performed on all patients, and a pelvic examination should be performed on women. Stool examination for occult blood (preferably a 3-day series) should be performed. Routine testing for ova and parasites or a stool culture is rarely indicated without a supporting travel history or supporting symptoms (e.g., fever, bloody diarrhea, acute onset of severe diarrhea).

Proctosigmoidoscopy with a flexible fiber-optic instrument should be performed. Introduction of the sigmoidoscope and air insufflation frequently trigger bowel spasm and pain. The mucosal and vascular pattern in IBS usually appears normal. In patients with chronic diarrhea, particularly older women, mucosal biopsy can rule out possible microscopic colitis, which has two variants: collagenous colitis, seen on trichrome stain as increased submucosal collagen deposition, and lymphocytic colitis, characterized by increased numbers of mucosal lymphocytes. The mean age of presentation for these disorders is 60 to 65 years old, with a female predominance. Similar to IBS, presentation involves nonbloody, watery diarrhea. Diagnosis can be made via rectal mucosal biopsy.

Laboratory examination should include a complete blood count (CBC); erythrocyte sedimentation rate; 6- and 12-channel biochemical profile (sequential multiple analyses 6 and 12), including serum amylase; urinalysis; and thyroid-stimulating hormone. An abdominal sonogram, barium enema x-ray, and upper GI esophagogastroduodenoscopy or colonoscopy may be selectively used, based on the patient's history, physical examination, age, and follow-up evaluations. These studies should be undertaken only when less invasive and less expensive studies reveal objective abnormalities.

Diagnosis of IBS should never preclude suspicion of intercurrent disease. Changes in symptoms may signal another disease process. For example, a change in the location, type, or intensity of pain; a change in bowel habits; constipation and diarrhea; and new symptoms or complaints (e.g., nocturnal diarrhea) may be clinically significant. Other symptoms that require investigation include fresh blood in the stool, weight loss, very severe abdominal pain or unusual abdominal distention, steatorrhea or noticeably foul-smelling stools, fever or chills, persistent vomiting, hematemesis, symptoms that wake the patient from sleep (e.g., pain, the urge to defecate), or a steady progressive worsening of symptoms. Patients more than 40 years old are more likely than younger patients to have an intercurrent organic illness.³

Characteristic symptoms of IBS include recurrent abdominal pain; abdominal pain relieved by defecation; disordered bowel habit, including constipation, diarrhea, or an alternation between the two; and abdominal distension and bloating.¹²

IBS is also associated with nongastrointestinal conditions such as headache, low back pain, arthritis, noncardiac chest pain, difficult urination, and fibromyalgia.¹³

20.8 Clinical Course and Prognosis

According to Ayurvedic literature, IBS is curable in children, difficult to treat in middle age, and incurable in older patients. The chronic accrual IBS is difficult to cure whereas tympanites-predominant IBS is incurable.

20.9 Therapy

The Ayurvedic principle of the involvement of *dosas* is very helpful because it provides an understanding of the symptoms and how they vary from one type to another. It also provides a system of treatment specific to that particular *dosa* imbalance. Ayurveda utilizes not only diet and herbs, but also lifestyle advice so that treatment of IBS can be more specific and more successful.

Generally, Ayurveda suggests drinking warm water at first, when morbidity and enterotoxin is located in IBS and is flared up with the improperly digested food. Appetizers, a light diet (e.g., liquid gruel) followed by ghee mixed with appetizers, *vata*-alleviating drugs, and an enema are then advised as a line of treatment. Ayurvedic treatment is based on the nature of the stool with (*sama*) or without (*nirama*) enterotoxin stool. The therapy process is highlighted as follows:

1. According to Ayurveda, dyspepsia and enterotoxin in IBS are important pathological processes that have to be treated effectively along with consideration of bowel movements with respect of stools with or without mucus discharge. When enterotoxin dominates this stage, it is treated with fasting and administration of digestives and carminatives. Use of warm water is highly recommended.
2. In constipation-predominant IBS, the first line of treatment is administration of digestives like *chitrakadi vati* and *shankha vati* to eliminate enterotoxin. It is recommended to follow the digestive with an administration of medicinal ghee preparations such as *dashmuladi ghrita* and *thryushanadi ghrita*.
3. Emesis (*vamana*) is the first line of treatment in diarrhea-predominant IBS, followed by the herbal bitters.
4. The first line of treatment for dysentery-predominant IBS is laxatives (*virechana*) followed by the administration of herbs with spicy and sour tastes.
5. Complex IBS is treated with the *panchakarma* line of treatment and supported by digestives and carminative preparations.
6. Generally, in the treatment of IBS, the use of buttermilk (*takra*) is emphasized. The patient has to be kept on the diet of buttermilk, which is the treatment and nutrition in IBS. Buttermilk is digestive, astringent, and light to digest and helps in improving the consistency of the stool. Buttermilk is given along with asafetida, cumin, and rock-salt powder to control the bowel movements. It contains a good amount of lactobacillus bacteria, which helps restore the normal flora of the intestines.¹⁴

When treating IBS, a light diet is advised. Indigestible foods such as bread; cheese; red meat; and cold, hard, and raw foods should be avoided. A light fast can be helpful taking only vegetable soups and a little basmati rice and green gram beans. Herbs like ginger, fennel, and cumin that stimulate enzyme secretions will improve digestion, absorption, and clear *ama* from the digestive tract.

20.9.1 Common Herbs Used in IBS

20.9.1.1 Constipation-Predominant IBS

The following herbs are used in this type of IBS for their respective benefits:

1. Ginger, clove, fennel, cumin, cardamom, and a little cinnamon will all stimulate digestive enzymes.
2. Chebulic myrobalan (*Terminalia chebula*) clears excess *vata* from the bowel. *Triphala*, a mixture of chebulic myrobalan, Indian gooseberry (*Emblica Officinalis*), and bel-liric myrobalan (*Terminalia belerica*), is a gentle bowel tonic excellent for chronic constipation and IBS and for clearing toxins from the bowel. It can be taken in powder or capsule form at bedtime.
3. Asparagus (*Asparagus racemosus*), *ashwagandha* (*Withania somnifera*), and sesame oil all calm *vata*. This is often given as an enema but can also be massaged over the body, particularly over the abdomen 5 min before soaking in a warm bath. Sweet, sour, and salty foods are best.

20.9.1.2 Diarrhea-Predominant IBS

The following herbs are used in this type of IBS for their respective benefits:

1. Indian gooseberry (*Emblica officinalis*) is a rejuvenative tonic and renowned remedy for *pita* problems, which balances *dosas*.
2. Coriander water cools *pita*, as does sandalwood powder prepared in ghee, which is often mixed with fennel, long pepper, black pepper, and cloves.
3. Nutgrass (*Cyperus rotundus*) improves intestinal absorption and stops diarrhea.
4. Aloe vera juice, turmeric, Indian madder (*Rubia cordifolia*), guduchi (*Tinospora cordifolia*) and Indian asparagus (*Asparagus racemosus*) are all excellent for balancing *pitta*. Sweet, bitter, and astringent foods are best.

20.9.1.3 Dysentery-Predominant IBS

The following herbs are used in this type of IBS for their respective benefits:

1. Ginger, lime juice, and honey are excellent to control *kapha*.
2. *Trikatu*, a compound containing black pepper, long pepper, and ginger, is specific for low enzyme and high enterotoxin.
3. Turmeric, cardamom, cinnamon, cloves, sandalwood, cumin, nutmeg, rock salt, and long pepper raise digestive enzyme and clear *kapha*.
4. $\frac{1}{2}$ tsp of *Hingvastaka*, a mixture of asafetida, ginger, black pepper, rock salt, etc. taken in a little warm water 1 to 2 h before lunch and supper, increases enzymes and clears enterotoxin. Pungent, bitter, and astringent tasting foods are best.¹⁴

20.9.2 Lifestyle Changes

The following changes in lifestyle help to control IBS:

1. Identify and remove food intolerances — A trained practitioner can supervise an elimination diet. Many foods are removed from the diet for a brief period and then reintroduced sequentially to isolate the body's reaction to the offending foods. Because grains are a common culprit, it is important to remember that carbohydrate digestion begins in the mouth and that chewing grains thoroughly allows amylase, the digestive enzyme present in saliva, to digest the grains.
2. Improve gut motility — Soluble fiber increases bowel transit, stools, and relieves constipation. Wheat bran has been used in some research studies, but it is not recommended for people who may have intolerances to wheat. Psyllium is a good source of soluble fiber and is readily available. Sufficient water should be taken or fiber can have the opposite effect and result in greater constipation. Flaxseed (*Linum usitatissimum*) also acts as a gentle laxative. It is useful for chronic constipation, damage to the colon wall from laxative abuse, irritable colon, and soothing GI inflammation.
3. Restore a healthy balance of bacteria in the gut — *Lactobacillus acidophilus* and *Bifidobacterium bifidum* can help restore a healthy balance of bacteria in the gut. They can decrease the amount of bacteria with gas-producing abilities and relieve IBS symptoms such as abdominal distension and flatulence. Bifidobacterium acts as a barrier against colonization of the gastrointestinal tract by pathogenic bacteria, and lactobacillus inhibits the attachment of pathogens onto the intestinal mucous lining.⁵

Low fiber intake is associated with an overgrowth of toxin-producing bacteria and a lower percentage of lactobacillus bacteria. A diet high in dietary fiber increases the formation of short-chain fatty acids, such as butyrate, which is the preferred energy source of the cells that line the colon.¹⁵
4. Pancreatic enzymes — These enzymes help inhibit the growth of bacteria in the small intestine. They help improve protein digestion. Goldenseal (*Hydrastis canadensis*) also inhibits bacteria and prevents the conversion of proteins to vasoactive amines.⁶
5. Mind-body therapy — Brief psychotherapy was found to be helpful in improving IBS symptoms of pain and diarrhea. Relaxation training to induce whole-body relaxation and stress management, hypnosis, and biofeedback have all been helpful in treating IBS.¹⁶ Antidepressants have been shown to be very effective for treating bowel motility and visceral nerve responses, in addition to addressing the emotional component of IBS.

Therapy is supportive and palliative. A physician's sympathetic understanding and guidance are of overriding importance. The physician must explain the nature of the underlying condition and convincingly demonstrate to the patient that no organic disease is present. This requires time for listening and explaining normal bowel physiology and the bowel's hypersensitivity to stress, food, or drugs. These explanations form the foundation for attempting to reestablish a regular bowel routine and individualize therapy. The prevalence, chronicity, and need for continuing care of IBS should be emphasized. Psychological stress and anxiety or mood disorders should be sought, evaluated, and treated. Regular physical activity helps relieve stress and assists in bowel function, particularly in patients who present with constipation.

In general, a normal diet should be followed. Patients with abdominal distention and increased flatulence may benefit from dietary reduction or elimination of beans, cabbage, and other foods containing fermentable carbohydrates. Reduced intake of apple and grape juice, bananas, nuts, and raisins may also lessen the incidence of flatulence. Patients with evidence of lactose intolerance should reduce their intake of milk and dairy products. Bowel function may also be disturbed by the ingestion of sorbitol, mannitol, fructose, or combinations of sorbitol and fructose. Sorbitol and mannitol are artificial sweeteners used in dietetic foods and as drug vehicles, whereas fructose is a common constituent of fruits, berries, and plants. Patients with postprandial abdominal pain may try a low-fat diet supplemented with increased protein.

Increasing dietary fiber can help many patients with IBS, particularly those with constipation. A bland bulk-producing agent may be used (e.g., raw bran, starting with 15 ml [1 Tbsp] with each meal, supplemented with increased fluid intake). Alternatively, psyllium hydrophilic mucilloid with two glasses of water tends to stabilize the water content of the bowel and provide bulk. These agents help retain water in the bowel and prevent constipation. They also can reduce colonic transit time and act as a shock absorber to prevent spasm of the bowel walls against each other. Fiber added in small amounts may also help reduce IBS-induced diarrhea by absorbing water and solidifying stool. Excessive use of fiber can lead to bloating and diarrhea. Fiber doses must therefore be adjusted to individual patient needs.

Anticholinergic (antispasmodic) drugs (e.g., hyoscyamine [0.125 mg 30 to 60 min before meals]) may be used in combination with fiber agents. The use of narcotics, sedative hypnotics, and other drugs that produce dependency is discouraged. In patients with diarrhea, 2.5 to 5 mg of diphenoxylate (1 to 2 tablets) or 2 to 4 mg of loperamide (1 to 2 capsules) may be given before meals. Chronic use of antidiarrheals is discouraged because tolerance to the antidiarrheal effect may occur. Antidepressants (e.g., desipramine, imipramine, and amitriptyline [50 to 150 mg/day]) help many patients with both constipation- and diarrhea-predominant IBS. In addition to constipation and diarrhea, abdominal pain and bloating are relieved by antidepressants. These drugs can also reduce pain by down regulating the activity of spinal cord and cortical afferent pathways arriving from the intestine. Finally, certain aromatic oils (carminatives) can relax smooth muscles and relieve pain caused by cramps in some patients. Peppermint oil is the most commonly used agent in this class.³

Available information and indications on some of the treatments are given below.

1. Peppermint oil (*Mentha piperita*) — Abdominal pain, the most frequent and disabling symptom of IBS, improves when the intestinal smooth muscles are relaxed. Peppermint oil can reduce abdominal pain and distension of IBS, possibly by blocking the influx of calcium into muscle cells and inhibiting the excess contraction of intestinal smooth muscles.¹⁹ It is a carminative, which means it helps eliminate intestinal gas. Peppermint oil should be used only in enteric-coated capsules to ensure that it reaches the intestines intact; otherwise, the oil can relax the lower esophageal sphincter and cause heartburn.

In a randomized, double-blind controlled trial,²⁰ in children with irritable bowel syndrome (IBS), enteric-coated peppermint oil capsules proved to be effective by reducing the severity of pain associated with IBS.

TABLE 20.1

Ayurvedic Formulas Commonly Used in the Treatment of IBS

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Brhat gangadhara curna</i>	Astringent	1–3 g three times/ day before meals	Warm water	14
<i>Dadimastaka curna</i>	Carminative, hemetic	1–3 g three times/ day before meals	Warm water	17
<i>Jatiphaladi curna</i>	Antacid	1–3 g three times/ day before meals	Warm water	17
<i>Bilvadi lehyam</i>	Antacid, antispasmodic	3–6 g ½ h before meals	3–6 g butter three times/day	17
<i>Chitrakadi vati</i>	Relieves enterotoxin, carminative	250–500 mg two times/day	Warm water	18
<i>Shankha vati</i>	Digestive, antispasmodic	250–500 mg two times/day	Warm water	14
<i>Dashmuladi ghruta</i>	Carminative, nutritive	1–2 tsp two times/day	Warm water	1
<i>Pancamrta parpati</i>	Antacid, antispasmodic	120–360 mg three times/day	Honey and water	17
<i>Rasa-parpati</i>	Antacid, antifatulent	120–360 mg three times/day	Honey	17

2. Fennel seed (*Foeniculum vulgare*) — Fennel is another herb that is used to relieve spasm of the gastrointestinal tract, feelings of fullness, and flatulence.²¹
3. Psyllium (*Plantago ovata*) — An herbal product derived from seeds of the plantago plant, psyllium is a source of dietary fiber that may aid constipation and diarrhea by absorbing water and adding bulk to the stool. (Be sure to drink one to two glasses of water when you take psyllium, and drink plenty of extra water throughout the day.) If psyllium aggravates symptoms instead of relieving them, discontinue using it.
4. Acidophilus — This form of beneficial bacteria normally inhabits the digestive tract, helps digest food, and also fortifies the body against digestive disorders by stopping the harmful bacteria that cause disease from growing uncurbed. It is especially important to take after a round of antibiotics. (Obtain pills containing 1 to 2 billion "live" organisms per pill.) You can enhance the effect of the acidophilus by taking it in combination with fructo-oligosaccharides (FOS), a supplement containing indigestible carbohydrates that feed the friendly bacteria.
5. Gammaoryzanol — This is a natural substance isolated from rice-bran oil. Studies have shown that it protects the mucous lining of the GI tract by regulating nervous system control and exerting antioxidant activity. Clinically, gammaoryzanol has been found to be effective in a broad range of GI complaints, including IBS. Earlier studies suggested that gammaoryzanol may act on the hypothalamus and pituitary glands of the brain, resulting in an inhibition of leutinizing hormone; further research has not supported this finding.¹⁵

Compound formulations commonly used for the treatment of IBS are given in Table 20.1.

20.10 Scientific Basis

20.10.1 Review of Clinical Trials

20.10.1.1 *Holarrhena antidysenterica*, Beans of *Acacia arabica*, Fruit Pulp of *Aegle marmelos*, and Seeds of *Cuminum cyminum*

In a study on 60 cases of IBS diagnosed according to Ayurvedic guidelines, patients were separated into two equal groups. The first group received a herbal combination of equal quantities of the powders of the stem bark of *Holarrhena antidysenterica*, beans of *Acacia arabica*, fruit pulp of *Aegle marmelos*, and seeds of *Cuminum cyminum* in the dose of 1 g three times/day with water and showed better response in the character of the stools within a week of treatment. The second group received *kutaja ghanavati* (dry extract of the bark of *Holarrhena antidysenterica*) in the dose of 1 g three times/day with two drops *shankha drava* (dissolved in 1/2 oz of water). This group showed marked relief with more than 75% of the signs and symptoms of the disease being controlled along with significant improvement in body weight, hemoglobin and fat, and undigested food particles in the stool. The duration of the treatment was 3 weeks.²²

20.10.1.2 *Zingiber officinale*

Powder of the rhizome of *Zingiber officinale* at the dose of 3 g three times/day with warm water was given to 30 patients with *grahani* for 4 weeks. The results showed a definite effect on the bowel movement by way of controlling the number of motions and changes in the consistency of the stool within 7 days of the treatment. It also has shown improvement in hemoglobin and body weight. Observation of clearance of giardia and entamebic cyst in the stool was also made.²³ Further, it was reconfirmed on 111 cases of *grahani* that the administration of powder of the rhizome of *Zingiber officinale* is effective in controlling frequency of bowel movements, consistency, improvement in general health and anemia, and the relief of associated symptoms in most of the patients. The effect of this powder on the elimination of amoebiasis and giardiasis is also significantly observed, though it does not have any effect on ascaris.²⁴

20.10.1.3 *Acorus calamus* Rhizome, *Aegle marmelos* Fruit, *Withania somnifera* Roots, Sesame Oil, and Common Salt Powder

A randomized, placebo-controlled trial²⁵ studied the effect of *vasti* (medicated enema comprising of extracts of *Acorus calamus* rhizome, *Aegle marmelos* fruit, *Withania somnifera* roots, sesame oil, common salt powder) on 53 patients with IBS. Results showed a significant reduction in abdominal distention and pain and increased retention time and D-xylose excretion when compared with a control group. It is concluded that *vasti* alters the visceral pain perception, acting via the regulation of the enteric nervous system, and brings about anxiolytic and antidepressant effects.

20.10.1.4 *Aegle marmelos* and *Bacopa monniere*

Among 169 patients with IBS, standard therapy (with clidinium bromide, chlordiazepoxide, and isaphaghulla) and a compound Ayurvedic preparation (with *Aegle marmelos* correa and *Bacopa monniere* Linn.), along with a matching placebo were given in a double-blind, randomized trial for 6 weeks. The Ayurvedic preparation in 57 patients was found effective in 64.9%, whereas standard therapy (60 patients) was useful in 78.3%. Patients on placebo (52 patients) showed improvement in only 32.7%. Ayurvedic therapy was particularly beneficial in the diarrhea-predominant form as compared with the placebo. The standard therapy was more useful in the painful form of IBS as compared with the placebo and Ayurvedic preparation. In the gas-predominant form, the effects of standard and Ayurvedic therapy were similar to the placebo. Long-term follow-up (greater than 6 months) showed that both forms of therapy were no better than the placebo in limiting the relapse.²⁶

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21

Diarrhea (Atisara) and Dysentery (Pravahika)

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21.1 Diarrhea

21.1.1 Introduction

Diarrhea, a condition of loose, watery stools that occur more than three times in a day, is a common problem that usually lasts a day or two and goes away on its own without any special treatment. Prolonged diarrhea can be a sign of other problems. Diarrhea can cause dehydration, which is particularly dangerous in children and the elderly, and it must be treated promptly to avoid serious health problems. People of all ages can get diarrhea. The average adult has a bout of diarrhea about four times a year.

Diarrhea is called *atisara* in Ayurveda. Frequent passing of watery stools is the cardinal feature. According to Ayurveda, diarrhea results from the excess mass of stools and endotoxins (*ama*) in the gastrointestinal (GI) tract, and this excess is eliminated through the nearest outlet (rectum) as diarrhea. Digestive and excretory systems, the two parts of the GI tract called *mahasrotas*, influence each other. The common etiological factors in the diseases of these systems are dyspepsia and endotoxin.¹

21.1.2 Definition

Atisara in Sanskrit means frequent passing of excessively watery, unformed stool. Other descriptions include watery stools, frequent bowel movements, and loose bowel movements.

TABLE 21.1

Clinical Features of Different Types of Diarrhea

Nature	Vata	Pitta	Kapha	Tridosa	Emotional
Color of stool	Bluish	Yellow, reddish	Whitish	Fatty and meat-soup like	Blood red
Nature of stool	Frothy	Watery, foul smell	Mucus, foul smell	Watery	Watery, with or without smell
Frequency	Frequently with sound	Not frequently	Less, without sound, incompletely	Alternate	Frequently
Quantity	Less	More	Less	Sometimes more or less	More
Associated symptoms	Pain in abdomen and pelvis	Fever, thirst, sweating	Drowsy, nausea, horripulation	Combination	Headache, burning sensation
Complication	Rectum prolapse	Inflammation of rectum	Painful rectum	Combination of all	Rectum prolapse

21.1.3 Clinical Description

Clinical features of different types of diarrhea are listed in Table 21.1. The major clinical features of diarrhea are increased fluidity in stools, increased frequency of bowel movement, and increased stool volume.

Although diarrhea is not considered a disease or disorder on its own in conventional medicine, Ayurvedic literature records it as both a symptom and an independent disease.

The World Health Organization (WHO) defines diarrhea as the passage of loose or watery stools, at least thrice in a 24-h period. However, it is the consistency of the stools rather than the frequency that is most important. Frequent passing of formed stools is not diarrhea.²

21.1.4 History and Epidemiology

Diarrhea dates back to history of human life. According to Indian mythology, the disease was said to exist in the first age (*sathya yuga*) and was observed to be the outcome of abnormal food habits and emotional disturbance.³

More than 1 billion people worldwide suffer one or more episodes of acute diarrhea each year. Because of poor sanitation and more limited access to health care, acute infectious diarrhea remains one of the most common causes of mortality in developing countries, particularly among children younger than 3 years old. It accounts for 5 to 8 million deaths every year.

On average, children in developing countries experience three episodes of diarrhea each year. In 1993, an estimated 3.2 million children under 5 years old died from diarrheal disorders. Eight of 10 of these deaths occur in the first 2 years of life. In many countries, diarrheal disorders, including cholera, are also an important cause of morbidity among older children and adults.²

21.1.5 Etiology

According to Ayurveda, diarrhea is caused mainly by the consumption of excessive food that is hard to digest (e.g., protein-rich and fatty foods). In addition, eating foods that are too dry, too hot, or too cold; drinking contaminated water and alcoholic beverages; eating

foods with contradictory properties (*viruddhasana*); eating frequently (*adhyasana*); eating improperly cooked foods (*aparipakvasana*); and eating incompatible foods (*asathmya bhojana*) also result in diarrhea. Certain behaviors and lifestyles may also lead to diarrhea. Examples include emotional stress, fear, or grief and controlling natural urges such as urination and defecation (*vega dharana*). Disorders such as indigestion (*ajirna*), irritable colon (*grahani*), hemorrhoids (*arshas*), helminthiasis (including microbial infection [*krimi*]) also result in diarrhea.¹

Conventional medicine indicates that diarrhea may be caused by an infection or an intestinal disease. A few of the more common causes of diarrhea are bacterial or viral infections from consuming contaminated food or water. The common bacteria known to cause diarrhea are campylobacter, salmonella, shigella, *Escherichia coli*, and viral infections (e.g., rotavirus, Norwalk virus, cytomegalovirus, herpes simplex virus, viral hepatitis, etc.). Certain food intolerances such as lactose (sugar found in milk) and parasites that can enter the body through food or water (e.g., *Giardia lamblia*, *Entamoeba histolytica*, and *Cryptosporidium*) can cause diarrhea. Reaction to medicines, such as antibiotics, blood pressure medications, and antacids containing magnesium; intestinal diseases, such as inflammatory bowel disease and celiac disease; functional bowel disorders, such as irritable bowel syndrome (IBS); poststomach surgery; and gallbladder removal can result in diarrhea.

Certain people who travel to foreign countries are at risk for traveler's diarrhea due to consuming food or drinking water contaminated with bacteria, viruses, or parasites. Traveler's diarrhea is a particular problem for people visiting developing countries.² Children can have acute (short-term) or chronic (long-term) forms of diarrhea. Causes include bacteria, viruses, parasites, medications, functional disorders, and food sensitivities. Infection with the rotavirus is the most common cause of acute childhood diarrhea. Rotavirus diarrhea usually resolves in 5 to 8 days.

21.1.6 Pathogenesis and Pathology

Ayurveda describes the pathogenesis of diarrhea as caused by various etiological factors discussed above. The fluid content of the intestines increase, slowing down the activity of digestive enzymes. This fluid that mixes with the stool in the intestine increases the bulk and is propelled by the force of *vata*; as a result, the contents of the intestine are thrown out in the form of diarrhea.¹

In conventional medicine, the pathology of this disease is due to infections, generally short lived and self-limiting. Common forms of diarrhea are grouped under terms such as gastroenteritis. These conditions may include vomiting and often appear in mini-epidemics in schools, neighborhoods, or families. Most cases of diarrhea will stop without treatment in a few days.

Diarrhea broadly has infectious, drug-induced, food-related, postsurgical, inflammatory, transit-related, and psychological causes. These causes produce diarrhea by four distinct mechanisms: increased osmotic load, inflammation, fluid secretion, and decreased absorption time.

21.1.7 Diagnosis

21.1.7.1 Prodromal Features of Diarrhea

According to Ayurveda, the clinical manifestation of diarrhea is preceded with the following signs and symptoms: pricking pain in the epigastric, umbilical, anal, and general

abdominal regions; general body malaise; constipation and obstruction in flatus movement; abdominal bloating; and indigestion.⁴

21.1.7.2 Main Types of Diarrhea

There are broadly six types of diarrhea:

1. Diarrhea due to the imbalance of *vata* (*vata atisara*)
2. Diarrhea due to the imbalance of *pitta* (*pitta atisara*)
3. Diarrhea due to the imbalance of *kapha* (*kapha atisara*)
4. Diarrhea due to imbalance of *tridosha* (*sannipata atisara*)
5. Diarrhea due to emotional disturbances (*sokaja* including *bhayaja*)
6. Diarrhea due to the imbalance of and due to enterotoxin (*ama atisara*)

Table 21.1 shows the details of their presenting features.¹

21.1.7.2.1 Enterotoxic Diarrhea (Ama Atisara)

To aid treatment, enterotoxic diarrhea is classified into two kinds: with enterotoxin (*sama*) and without enterotoxin (*nirama* or *pakva*). The characteristic features of the stool with enterotoxin are described as improperly digested stool with a distinct foul smell that is slimy or incompacted and is associated with flatulence, abdominal pain, and salivation. The nonenterotoxic diarrhea is devoid of the above features and it also sinks in water.⁴ The diarrhea could be mild, moderate, or intense.

21.1.7.3 Other Types of Diarrhea

21.1.7.3.1 Hemorrhagic Diarrhea (Rakta Atisara)

Hemorrhagic diarrhea is an exacerbated variant or latter stage of *pitta-atisara*. If a patient is suffering from *pittaja* diarrhea, then ventures into excessive use of spicy foods, salt, or alcohol may result in hemorrhagic diarrhea, which is characterized by the passage of blood along with the stools.¹

21.1.7.3.2 Infective Diarrhea (Jvara Atisara)

Etiology of infective diarrhea is the result of etiological factors of fever and diarrhea, with the symptoms of diarrhea associated with fever.¹

21.1.7.3.3 Clinical Types of Diarrhea

In conventional medicine, four clinical types of diarrhea are recognized:

1. Acute bloody diarrhea (also known as dysentery) — The main dangers are intestinal damage, sepsis, and malnutrition. Other complications, including dehydration, may also occur.
2. Acute watery diarrhea (including cholera) — This type of diarrhea lasts several hours or days. The main danger is dehydration, and weight loss also occurs if feeding is not continued.
3. Persistent diarrhea — This type of diarrhea lasts 14 days or longer. The main danger is malnutrition and serious nonintestinal infection. Dehydration may also occur.

4. Diarrhea with severe malnutrition (marasmus or *kwashiorkor*) — The main dangers of this type of diarrhea are severe systemic infection, dehydration, heart failure, and vitamin and mineral deficiency.

The management of each type of diarrhea should prevent or treat the main cause that each one presents.²

Diagnostic tests to find the cause of diarrhea include the following:

1. Medical history and physical examination
2. Stool culture to determine the bacterial infection
3. Blood tests
4. Fasting tests
5. Sigmoidoscopy
6. Colonoscopy (similar to sigmoidoscopy, but the doctor looks at the entire colon)

21.1.8 Clinical Course and Prognosis

According to Ayurveda, diarrhea is difficult to cure if the stool is purple (ripe jamun fruit) or liver brown; amount of stool elimination is small; the consistency resembles dehydrated butter (*ghee*), oil, fat, or meat soup; embedded with shiny particles; and has a very fishy odor. Patients with severe thirst, generalized burning sensation, unconsciousness, dyspnea, relaxed rectal sphincters, rectal prolapse, loss of strength, muscular wasting, loss of blood, and severe arthritic pain along with diarrhea will be very difficult to cure.

The diarrhea due to the involvement of all three *dosas* and emotional types are also difficult to treat successfully.⁴

Diarrhea can be either acute or chronic. The acute form, which lasts less than 3 weeks, is usually related to a bacterial, viral, or parasitic infection. Chronic diarrhea lasts more than 3 weeks and is usually related to functional disorders like IBS or diseases like celiac disease or inflammatory bowel disease.

21.1.8.1 Complications

Common complications of diarrhea are pain in the abdomen, dysentery, tympanitis, rectal prolapse, anuria, edema, severe infection, and loss of consciousness.¹

According to conventional medicine, fluid loss with consequent dehydration, electrolyte loss (Na, K, Mg, Cl), and even vascular collapse may occur. Patients who are very young, old, or debilitated may collapse rapidly or have severe diarrhea (such as those with cholera). HCO₃ loss may cause metabolic acidosis. Serum sodium concentrations vary according to the composition of diarrheal losses relative to plasma. Hypokalemia may occur in severe or chronic diarrhea or if the stools contain excess mucus. Hypomagnesemia after prolonged diarrhea may cause tetany.

21.1.9 Therapy

The fundamental principle of treatment in diarrhea is to keep away from the causative factors as previously described. Treatment has to be undertaken based on the nature of the stool — with endotoxin (of recent origin) or without endotoxin (chronic).

21.1.9.1 Enterotoxic Diarrhea

In case of a mild degree of toxin, advice on diet and carminatives are used. While in moderate degree, digestives are administered. In case of an intense degree, the patient is subjected to detoxification (*shodhana-virecana, basti*), using herbs to eliminate toxins.²

The use of antimotility drugs is contraindicated in the beginning of enterotoxic diarrhea, in order to prevent the enterotoxin from remaining in the gut. The reduction of bowel movement results in convulsive disorders, abdominal distension, irritable colon, hemorrhoids, anal fissures, edema, anemia, and splenomegaly. Antimotility drugs are used in cases of infants, the elderly, dehydrated conditions, infective diarrhea, and excess diarrhea.⁵

The diet should be restricted to light food like soup of *mudga* (green gram) with ginger or cooked rice (100 to 250 g with 1 to 6 g of ghee). Depending upon the desire and taste, using buttermilk, rice gruel, or rice corn soaked in water is advised. These will be more nourishing and act as a digestive aid.

The following herbal mixtures are recommended (see [Section 21.1.10](#) for the names of the herbs):

1. Chebulic myrobalan (*Terminalia chebula*) fruit powder at a dose of 1 to 3 g with 250 mg of long pepper (*Piper longum*) fruit powder are advised along with warm water. Care should be taken, as large doses of this will result in more bowel movements. This formula helps eliminate toxins.
2. A tea made of rhizomes of calamus (*Acorus calamus*), aconitum species, nutgrass (*Cyperus rotundus*), aerial parts of fumaria (*Fumaria officinalis*), root of pavonia (*Pavonia odorata*), and rhizome of ginger taken at a dose of 50 ml two to three times/day is given to improve digestion.³
3. Powder of fried cumin (*Cuminum cyminum*), rhizome of nutgrass, long pepper, or ginger at a dose of 5 g is mixed with 1 l of buttermilk and divided into four parts. Each dose should be taken at 6-h intervals.
4. Ginger powder (1 to 3 g) taken with an equal quantity of sugar two times/day is useful in relieving the symptoms of enterotoxin.
5. Powder of kurchi seed (*Holarrhena antidysenterica*), bark of cinnamomum (*Cinnamomum zeylanicum*), root of vetiver (*Vetivera zizanioides*), seed of jambul (*Syzygium cumini*), and fruit pulp of bael (*Aegle marmelos*) are mixed in equal quantities. A dose of 3 to 6 g of this mixture is taken with buttermilk three times/day.
6. Equal quantities of the powder of pop of old paddy (*laja*), fruit of coriander (*Coriandrum sativum*), and resin of silk cotton tree (*Bombax ceiba*) are mixed with double-quantity fennel seeds (*Foeniculum vulgare*). A dose of 3 to 6 g of this mixture is taken with warm water two times/day.
7. Powder of nutmeg (*Myristica fragrans*), cumin seeds, and fruit pulp of bael are mixed in equal quantity and triturated with lime water to make pills each weighing 800 mg. One pill is taken in the morning with rice water in the case of *pitta* diarrhea and with 25 to 75 ml camphor water for *kapha* diarrhea.⁶

A few compound formulations used in the treatment of enterotoxic diarrhea are given below:

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Gangadhara curna</i>	Digestive, appetizer, antidiarrheal	1–3 g three times/day	Jaggery, buttermilk and honey	7
<i>Jatiphaladi curna</i>	Carminative, antidiarrheal	1–2 g three times/day	Honey, buttermilk and water	7
<i>Lai curna</i>	Antiflatulent, antidiarrheal	1–3 g three times/day	Buttermilk	5
<i>Jatiphaladi vati</i>	Antimotility, antispasmodic	60–120 mg three times/day	Buttermilk and water	8
<i>Karpura vati</i>	Antimotility, antispasmodic	120–240 mg two times/day	Honey	8
<i>Kutajarishtha</i> ⁸	Digestive, antidyenteric	15–30 ml two times/day	Equal quantity of water	8

21.1.9.2 Nonenterotoxic Diarrhea

This condition is treated with carminatives and antimotility herbs. The following herbal mixtures are recommended:

1. A dose of 3 to 6 g of a mixture of equal quantity of powders of the bark of kurchi and rhizome of aconitum species is taken three to four times/day with honey.⁵
2. A dose of 3 to 6 g of powder of fruit pulp of bael mixed with 100 to 250 ml buttermilk is taken three to four times/day.
3. A dose of 3 to 6 g of powder of bark or seeds of kurchi is taken with 100 to 250 ml buttermilk three to four times/day.
4. A decoction of equal parts of fruit pulp of bael, rhizomes of pavonia and aconite species, rhizomes of nutgrass, and sida root (*Sida cordifolia*) and bark of kurchi is taken at the dose of 14 to 28 ml three times/day.⁷

21.1.9.3 Vata Diarrhea

Vata diarrhea is treated with herbs that improves digestion, followed by antidiarrheal herbal formulas. A decoction of rhizomes of calamus, aconitum, and nutgrass, and seed of kurchi is given.⁹

21.1.9.4 Pitta Diarrhea

Pitta diarrhea is treated with a light diet, carminatives, fluids, and mineral supplements such as rice gruels and fruit juice. If this fails, resort to palliatives and antimotility agents and medicated enema. A mixture of equal quantity of powders of dry extract of berberis (*Berberis aristata*), rhizome of aconite, bark of kurchi, flowers of woodfordia (*Woodfordia fruticosa*), and ginger is given at a dose of 3 to 6 g with rice water two to three times/day. For a medicated decoction enema (asthapana basti), the paste of the fennel seeds (*Foeniculum vulgare*), roots of asparagus (*Asparagus racemosus*), cow's milk, roots of licorice (*Glycyrrhiza glabra*), sesame oil, cow's ghee, and fruit pulp of bael are used.³ This herbal

combination helps to decrease the motility of intestine. The diet includes goat milk and meat soup with rice.⁹

21.1.9.5 *Kapha Diarrhea*

Kapha diarrhea is treated best with fasting and by improving digestion because it occurs due to excessive eating. A decoction of chaba (*Piper chaba*), aconite, nutgrass, bael, ginger, kurchi seed and bark, and chebulic myrobalan in equal quantities are given at a dose of 15 to 30 ml.⁹

A few compound formulations used in the treatment of *vata* and *kapha* diarrheas are given here:

Name Formulation	Activity	Dose	Adjuvant	Ref.
<i>Kutajavaleha</i>	Antidiarrheal	5–10 g three times/day	Water	10
<i>Agasthisutaraja rasa</i>	Antidiarrheal	60–120 mg three times/day	Cumin, nutmeg, powders	5
<i>Kanakasundara rasa</i>	Antidiarrheal	60–120 mg three times/day	Buttermilk	8

21.1.9.6 *Hemorrhagic Diarrhea*

The bleeding in hemorrhagic diarrhea has to be stopped immediately. Analgesics (*vadanasthapana*) and styptics (*rakthasthambhaka*) as well as antidiarrheals have to be used. The following herbal mixtures are recommended:

1. A dose of 3 to 6 g of the powder of asparagus (*Asparagus racemosus*) root mixed with 120 ml of goat's milk, three to four times/day, is taken while following a strict rice-milk diet.³
2. A dose of 3 to 6 g of white sandalwood (*Santalum album*) powder is mixed with an equal quantity of honey and 5 to 10 g of sugar. It is taken two or three times/day along with rice water and relieves thirst, burning, and bleeding.
3. A dose of 5 to 10 g of cow's milk butter mixed with honey and candy sugar in equal quantities helps stop bleeding.⁹
4. A dose of 3 to 6 g of powder of the fruit pulp of bael is taken with an equal quantity of jaggery.
5. A dose of 3 to 6 g of banyan shoot (*Ficus bengalensis*) paste is taken with 100 to 250 ml of rice water three times/day.
6. A dose of 28 ml of the fresh juice of the leaves of jambul, Indian gooseberry, and mango in equal quantity is mixed with 14 g of honey and 100 ml of goat's milk. The solution is taken three times/day.
7. A dose of 2 to 6 g of red sandalwood (*Pterocarpus santalinus*) powder with 6 g of sugar and 6 g of honey, followed by a potion of 100 to 250 ml of rice water, is taken two times/day.⁵

Goat's milk is given with honey and sugar as a drink and even as a rectal douche.³

A few compound formulations used in the treatment of hemorrhagic diarrhea are given below:

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Dadimashtaka curna</i>	Antidysenteric	3–6 g three times/day	Water	5
<i>Dadimavaleha</i>	Antidysenteric	6–12 g three times/day	Honey and water	11
<i>Ushirasava</i>	Hemostatic	15–30 ml two times/day	Equal water	8
<i>Bola parpati</i>	Hemostatic	120–240 mg three times/day	Sugar, honey, buttermilk	5

21.1.9.7 Emotional Diarrhea

Treatment in emotional diarrhea is similar to *vata* diarrhea. In addition, entertainment and counseling is advised.

21.1.9.8 Infective Diarrhea

The line of treatment in this condition involves fasting or taking light and easily digestible food and digestive agents along with antipyretic drugs. A dose of 15 to 30 ml of a decoction of the equal parts of fruit pulp of bael, rhizomes of ginger and musk, fruit of coriander, and roots of sida should be taken once daily early in the morning. A 15- to 30-ml dose of the following decoction is given two times/day: equal quantities of aerial parts of velvet leaf (*Cissampelos pareira*), andrographis, fumaria, rhizomes of nutgrass, ginger, seeds of kurchi, and tinospora stem in water.⁷

The compound formulations used in the treatment of infective diarrhea are listed below:

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Karpura rasa</i>	Digestive, antipyretic, antidiarrheal	60–120 mg three times/day	Water	8
<i>Lakshminarayana rasa</i>	Antipyretic and antidiarrheal	120–240 mg three times/day	Ginger juice and honey	5
<i>Mritasanjivini rasa</i>	Antidiarrheal	360 mg two times/day	Honey	5

21.1.9.9 Rectal Prolapse

If the anal region is inflamed, a sitz bath with cold extract of snakeguard leaves (*Trichosanthes dioica*) and licorice is given. If it is associated with a burning sensation, goat's milk mixed with honey and sugar is douched or applied as a wash locally and is also taken internally. A poultice of Woodfordia flower, Lodh bark (*Symplocos racemosa*), and black-gram seeds helps reduce inflammation and bleeding and is applied on the end part. A cotton swab soaked in a medicated ghee (*shatadhoutha ghrita*) is also applied or plugged locally.^{9,12}

The compound formulations for all types of diarrheas are listed as follows:

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Rasa parpati</i>	Intestinal antiseptic	125 mg two times/day	Honey	13
<i>Panchamrita parpati</i>	Antidysenteric	125–250 mg two times/day	Honey, curd, and buttermilk	8
<i>Svarna parpati</i>	Antidysenteric, tonic	125 mg two times/day	Honey	5
<i>Vijaya parpati</i>	Antidysenteric, tonic	125–375 mg two times/day	Water and honey	8

21.1.9.10 Lifestyle Changes

Old rice, mung beans, asafetida, garlic, Indian gooseberry, a liquid diet, and raw bananas are all advised in treating diarrhea. Sleeping immediately after a meal also helps control diarrhea and helps in recovery. As one improves, soft and bland foods, raw bananas, guava fruit, plain rice, baked potatoes or dioscorea tubers, toast, crackers, cooked carrots, and baked chicken without the skin or fat are recommended. For children, the bananas, rice, apple sauce, and toast diet is recommended.¹⁴

Fatty foods (which are heavy for digestion); excess of meat; milk products; and foods that are greasy, high fiber, or very sweet are contraindicated. Negative emotions like fear, grief, and anger, and suppression of the urges of defecation and urination should also be avoided.

The treatment of any type of diarrhea is considered complete when urination and passage of flatus without the appearance of stool, increased appetite, and a feeling of lightness in the abdomen are observed.⁴

21.1.10 Scientific Basis

The inhibitory effect of ayurvedic plants on enteropathogenic bacterias is presented in [Table 21.2](#). Other studies on ayurvedic herbs are summarized below.

21.1.10.1 *Holarrhena antidysenterica* (Kurchi)

This herb has been indicated in many diseases apart from dysentery in all the forms described in Ayurveda.¹⁵ The total alkaloids of the bark has shown antibacterial activity on the intestinal bacteria, *E. coli*.¹⁶ Kurchamine, an alkaloidal fraction from the bark, effectively reduced the intestinal amoebic infection in rats and hepatic amoebiasis in hamsters. The activity was comparable with emetine hydrochloride.¹⁷ Clinical tests with conessine from the bark on patients with intestinal and hepatic amebiasis have been given results comparable with those obtained with emetine.¹⁸

21.1.10.2 *Myristica fragrans* (Nutmeg)

The seed or nutmeg of *Myristica fragrans* is considered a folk remedy for diarrhea. Experimental studies with petroleum ether, chloroform, and ethanol extract showed a reduction in fecal output, inhibition of castor oil-induced diarrhea, and enteropooling in rats. The protective effect of nutmeg in castor oil-induced diarrhea indicates the action could be due to the inhibition of prostaglandin biosynthesis.¹⁹

TABLE 21.2

Inhibitory Effect of Ayurvedic Plants on Enteropathogenic Bacterias

Name	Part	Solvent	A ^a	B ^a	C ^a	D ^a	MICD/IC ₅₀ (mg or mg/ml, or µg or µg/ml)	Ref.
<i>Ocimum basilicum</i>	Leaf	Ethanol	6	6	6	NT	<i>S. typhi</i> (>10 mg)	49
<i>Psidium guajava</i>	Leaf	Methanol	9	8	13	NT	<i>S. flexneri</i> (>10 mg)	49
<i>Punica granatum</i>	Fruit	Methanol	N/A	N/A	NT	NT	12 mg/ml	50
<i>Bergenia ciliata</i>	Rhizome	Methanol	15	NT	NT	12	1.0 mg	51
<i>Caesalpinia bonducella</i>	Seeds	Methanol	20	19	15	NT	<i>E. coli</i> (10 mg/ml), <i>S. typhi</i> (0.62 mg/ml), <i>S. flexneri</i> (2.5 mg/ml)	52
<i>Ficus racemosa</i>	Leaf	Petroleum ether	19	NT	NT	NT	25 mg/ml	53
<i>Capsicum annum</i>	Fruit	Ethanol	N/A	NT	NT	NT	100 mg/ml	54
<i>Holarrhena antidysenterica</i>	Stem bark	Aqueous	>25	>25	>20	>30	200 mg/ml	55
<i>Murraya koenigii</i>	Leaf	Chloroform	8	NT	10	NT	500 µg	56
<i>Syzygium cumini</i>	Stem bark	Ethanol	N/A	N/A	NT	NT	128 µg/ml	57
<i>Asterantha longifolia</i>	Whole plant	Chloroform	15	NT	NT	NT	100 mg/ml	58
<i>Leptadenia arborca</i>	Whole plant	Methanol	16	NT	NT	NT	100 mg/ml	58
<i>Tephrosia purpurea</i>	Whole plant	Methanol	17	NT	NT	NT	100 mg/ml	58
<i>Withania somnifera</i>	Whole plant	Methanol	15	NT	NT	NT	100 mg/ml	58
<i>Evolvulus alsinoides</i>	Leaf	Methanol	17	NT	NT	NT	100 mg/ml	59
		Chloroform	17	NT	NT	NT	100 mg/ml	59
<i>Abrus precatorius</i>	Stem	Methanol	18	NT	NT	NT	100 mg/ml	59
<i>Clitoria ternatea</i>	Whole plant	Methanol	15	NT	NT	NT	100 mg/ml	59
<i>Plumbago zeylanica</i>	Stem and leaf	Methanol	19	NT	NT	NT	100 mg/ml	59
<i>Calotropis procera</i>	Root	Ethanol	15	NT	NT	NT	10 mg/ml	60
<i>Trigonella foenum graecum</i>	Seeds	Ethanol	20	NT	NT	NT	100 mg/ml	61
<i>Allium sativum</i>	Cloves	Water	12	18	30	32	30 µg	62
<i>Cassia occidentalis</i>	Flower	Ethanol	8	NT	NT	NT	10 µg	63
	Callus	Ethanol	8	NT	NT	NT	10 µg	63
<i>Tachyspermum ammi</i>	Fruit	Ethanol	N/A	NT	NT	NT	6.25 mg/ml	64
<i>Hemidesmus indicus</i>	Root	Ethanol	N/A	NT	NT	NT	25 mg/ml	64
<i>Cassia fistula</i>	Seed	Ethanol	N/A	NT	NT	NT	12.5 mg/ml	64
<i>Terminalia bellerica</i>	Exocarp	Ethanol	N/A	NT	NT	NT	12.5 mg/ml	64
<i>Phyllanthus emblica</i>	Fruit	Ethanol	N/A	NT	NT	NT	12.5 mg/ml	64
<i>Oxalis corniculata</i>	Leaf	Ethanol	N/A	NT	NT	NT	12.5 mg/ml	64
<i>Plumbago indica</i>	Leaf	Ethanol	N/A	NT	NT	NT	12.5 mg/ml	64
<i>Aegle marmelos</i>	Leaf	Methanol	5	8	5	NT	50 µg/ml	65
<i>Terminalia chebula</i>	Fruit	Alcoholic	>20	>20	NT	NT	200 mg/ml	66
<i>Emblica officinalis</i>	Fruit	Alcoholic	>20	>20	NT	NT	200 mg/ml	66

^aA = *Escherichia coli*; B = *Salmonella typhi*; C = *Shigella flexneri*; D = *Vibrio cholerae*.

Note: Values indicate bacterial inhibition zone (mm); MICD = minimal inhibitory concentration in disk.

21.1.10.3 *Dioscorea oppositifolia* (Chinese Wild Yam)

The tuber extract is known to have antidiarrheal activity in infantile diarrhea.²⁰

21.1.10.4 *Psidium guajava* (Guava)

In fruits with therapeutic properties for antidiarrheal and laxative uses, the presence of lectins may be the bioactive properties that interfere with bacterial adhesion, which are thought to be competing for glycoside signal sites in the attachment. Guava has a galactose-specific lectin that prevents adhesion of *E. coli* O157:H7 to red cells; this lectin is mediated by galactose. Prevention could also be due to their capacity of agglutinating *E. coli* by guava.²¹ The antidiarrheal properties of the water and methanolic extracts of *Psidium guajava* leaves have been demonstrated with anteriority; their spasmolytic effect was attributed to quercetin, a flavonoid contained in this plant.²² The spasmolytic effects of *Psidium guajava* leaf methanol, hexane, and water extracts were demonstrated in the guinea pig isolated ileum-perfused model suggesting the existence of two different types of active components.²³

21.1.10.5 *Musa paradisiaca* (Banana)

Banana flakes can be used as a safe, cost-effective treatment for diarrhea in critically ill tube-fed patients. Banana flakes can be given concurrently with a workup for *Clostridium difficile* colitis, expediting the treatment of diarrhea.²⁴

21.1.10.6 *Nelumbo nucifera* (Sacred Lotus)

Extract of *Nelumbo nucifera* rhizome reduced not only the frequency of defecation, wetness of fecal dropping and prostaglandin-2 (PGE₂)-induced enteropooling, but also significantly reduced the propulsive movements of charcoal meal significantly in experimental rats.^{25,26}

21.1.10.7 *Ficus bengalensis* (Banyan Tree) and *Syzygium cumini* (Jamun)

Ethanol extracts of *Ficus bengalensis* Linn. (hanging roots) and *Syzygium cumini* Linn. (bark) showed significant inhibitory activity against castor oil-induced diarrhea and PGE₂-induced enteropooling in rats. These extracts also showed a significant reduction in GI motility in charcoal meal tests in rats. The results obtained establish the efficacy of all these plant materials as antidiarrheal agents.²⁷

21.1.10.8 *Acorus calamus* (Sweet Flag) and *Aegle marmelos* (Bael)

A study was conducted on the effect of aqueous and methanolic plant extracts of *Acorus calamus* rhizome, *Pongamia glabra* leaves, *Aegle marmelos* unripe fruit, and *Strychnos nux-vomica* root bark for their antidiarrheal potential against castor oil-induced diarrhea in mice. Results showed that methanolic extracts were more effective than were aqueous plant extracts. The methanolic extracts significantly reduced induction time of diarrhea and total weight of the feces. The result establishes the efficacy of these plant extracts as antidiarrheal agents.²⁸

21.1.10.9 *Clerodendrum phlomidis* (Clerodendrum)

The methanolic extract from the leaves of *Clerodendrum phlomidis* Linn. (*Agnimantha*) (MECP) was examined for its antidiarrheal potential against several experimental models

of diarrhea in Wistar albino rats. MECP showed significant inhibitory activity against castor oil-induced diarrhea and PGE₂-induced enteropooling in rats. The extract also showed a significant reduction in GI motility in charcoal meal test in rats. The results obtained establish the efficacy and substantiated the folklore claim as an antidiarrheal agent.²⁹

21.1.10.10 *Piper nigrum* (Black Pepper)

Antidiarrheal activity of piperine (*Piper nigrum*) against castor oil, magnesium sulphate, and arachidonic acid was studied in mice. It significantly inhibited diarrhea produced by these cathartics at an oral dose of 8 and 32 mg/kg. Inhibition of castor oil-induced enteropooling by piperine suggests its inhibitory effect on prostaglandins.³⁰

21.1.10.11 *Berberis aristata* (Indian Barberry)

Berberine was effective in reducing water and electrolyte secretions induced by *E. coli* heat-stable enterotoxin. These findings indicate that berberine may be an effective anti-diarrheal agent in *E. coli* heat-stable enterotoxin-mediated secretory diarrhea. The results also provided a basis for the frequent empirical use of berberine alkaloid and berberine-containing plants in gastroenteritis and infectious diarrhea in Asian and other countries.³¹

21.1.10.12 *Punica granatum* (Pomegranate)

Experimental studies³² on methanolic extract of *Punica granatum* fruit rind have reported antidiarrheal and antibacterial activity against *Staphylococcus aureus*, *E. coli*, *Klebsella pneumoniae*, *Proteus vulgaris*, and *Salmonella typhi*.

21.1.10.13 *Tinospora cordifolia* (*Tinospora gulancha*)

Tinospora cordifolia was reported to be active against *Entamoeba histolytica*.³³

21.1.10.14 Clinical Trials

21.1.10.14.1 Diarex

Diarex is a polyherbal formulation containing *Holarrhena antidysenterica*, *Tinospora cordifolia*, *Aegle marmelos*, *Punica granatum*, and *Cyperus rotundus* as its constituents. *Holarrhena antidysenterica* is reported to be more effective in treating amoebic dysentery.³⁴ *Aegle marmelos* is reported to be a very effective remedy in controlling acute diarrhea.³⁵ A combination of *Aegle marmelos*, *Punica granatum*, and *Tinospora cordifolia* along with other herbs is known to have potential antispasmodic activity.³⁶ Clinical trials of diarex have showed to be effective in acute chronic and infectious diarrhea.³⁷

21.1.10.14.2 *Calotropis procera* (Indian Madder)

A pilot study was conducted on the effect of the root bark of *Calotropis procera* (arka) in 73 patients of diarrhea. The powder was administered at 250 mg three times/day with buttermilk for 7 days. The consistency of the stools was achieved on the first day of the treatment. There were 49 patients (67.1%) who had complete relief and 18 patients (24.7%) who had marked relief.³⁸

21.1.10.14.3 *Compound herbal powder (Solanum torvum, Murraya koenigii, Mangifera indica, Carum roxburghianum, Emblica officinalis, Punica granatum, and Trigonella foenum-graecum)*

An open clinical trial³⁹ examined the role of a compound herbal powder (consisting of *Solanum torvum*, *Murraya koenigii*, *Mangifera indica*, *Carum roxburghianum*, *Emblica officinalis*, *Punica granatum*, and *Trigonella foenum-graecum*) in the management of nonspecific diarrhea in 25 subjects. Results showed marked relief in loose stools (76%), tenasmus (85%), acid belching (69.5%), borborygmus (66.6%), and bloating of abdomen (62.5%). During the study no adverse reactions were noted.

21.1.10.14.4 *Valeriana officinalis (Valerian)*

In a clinical observation and experimental study,⁴⁰ *Valeriana officinalis* was found to be an effective medicine in controlling infantile rota viral diarrhea.

21.1.10.14.5 *Green bananas*

In a double-blind trial,⁴¹ green bananas and pectin were shown to be useful in the dietary management of persistent diarrhea in hospitalized patients. The plantain flour-based solution proved effective for the treatment of dehydration due to acute diarrheal diseases and should be considered as an alternative when standard WHO oral rehydration solution (WHO-ORS) is not available.⁴²

21.1.10.14.6 *Soy fiber*

A randomized, blind clinical trial⁴³ was conducted on soy fiber and rice-based oral rehydration solutions. Results showed that these solutions reduced the duration of watery stools during acute diarrhea caused by bacterial and viral pathogens.

21.2 Dysentery

21.2.1 Introduction

Dysentery is a term used for diarrhea when there is evidence that organisms have invaded the intestinal wall and caused pus, mucus, and blood to appear in the stool. There is often fever and abdominal cramps as well. Although the term dysentery conjures up more emotion, there is no clear-cut line between diarrhea and dysentery, and treatments are often the same.

In its classic form, dysentery leads to symptoms of cramping abdominal pain, diarrhea, and blood and mucus in the feces. Initially, the diarrhea may be copious but soon becomes frequent and of small volume; the patient often complains of painful defecation. It is sometimes accompanied by other symptoms such as vomiting and fever. The *shigella* infection (Sd1) rarely involves other parts of the body. However, more severe forms of infection can occur, particularly in travellers abroad. In these cases, the patient may become very ill and dehydrated, and in the absence of medical treatment, dysentery can be fatal.⁴⁴

Some authors of Ayurvedic literature consider this disease as an advanced stage of *kapha* diarrhea as well as enterotoxic diarrhea.³ Others consider it an individual disease.⁴

21.2.2 Definition

In Ayurvedic medicine *pravahika* is the term used to indicate dysentery, which means moving with force. It is synonymous with *visramsā*, *antargranthi*, *annagranthi*, *niscaraka*, or *nissaraka*. Dysentery is an acute disease in which the stools and mucus secretion causes a delay in evacuation due to *kapha* and *vata dosas* resulting in straining down the stools.¹ The word *dysentery* is derived from the Greek *dusenteria*, meaning bad intestine.

21.2.3 Clinical Description

Passage of stool with mucus and tenesmus and a feeling of incomplete evacuation characterize the disease.¹ Dysentery is a disease in which the patient passes a small quantity of mucoid, slimy, or bloodstained stool with considerable tenesmus, gripping abdominal pain, and burning sensation.¹⁴ More or less fever, loss of appetite, sleeplessness, and restlessness at night also are noticed. Sometimes the abdomen is distended.

Severe symptoms include having a high fever, being very thirsty and having a red tongue; the abdomen may appear sunken in some cases, straining ceases, and the bowels become relaxed and may prolapse. The passage of urine is infrequent and is accompanied by a burning sensation. The pulse becomes slow, breathing becomes rapid, and generally, the patient looks pale and emaciated. This condition should not be allowed to continue.¹

21.2.4 History and Epidemiology

Man has known dysentery in its severe form since ancient times, but most cases acquired in the developed world today are mild. The earliest literature in Ayurveda describes this disease in the chapter on diarrhea, whereas the latest authors have recorded this disease as an individual entity.

Shigella infection has caused epidemics of dysentery throughout the world. It caused a 4-year epidemic in Central America beginning in 1968 that resulted in more than 500,000 cases and at least 20,000 deaths.⁴⁴

21.2.5 Etiology

According to Ayurveda, the main causes for this disease are similar to diarrhea. These causes especially include foods that are incompatible (*Ahita*), extremely spicy, and heavy to digest (e.g., protein-rich foods). Drinking contaminated water is also another important cause.¹

In conventional medicine, bacterial or viral infection, infestation of protozoa or parasitic worms, and chemical irritants are described as major causes of the disease. Some of these causes include the following:

1. Inflammation of the rectum and large intestine
2. Eating insufficient foods
3. Having an improper diet
4. Drinking too much liquid with meals

5. Overeating
6. Eating the wrong combinations of foods
7. Being in unhygienic surroundings
8. Eating fruits or vegetables that have started decomposing
9. Eating foods that have been in pantries that are not well ventilated
10. Eating improperly refrigerated, contaminated foods

Irritated bowels, habitual constipation, and taking certain types of medicine (e.g., laxatives) may also be the cause.

21.2.6 Pathogenesis

This disease can occur in two stages. The primary stage is due to food and other factors, whereas the secondary stage is due to *kapha* diarrhea.

21.2.6.1 Transmission

In conventional medicine, few studies have been done to determine how dysentery is spread. The most likely modes of transmission are person-to-person contact and contaminated water and food. Epidemics of Sd1 usually occur in impoverished areas. They affect people of all ages, with the highest age-specific incidence occurring among adults and the highest case-fatality rates occurring among children.⁴⁴

21.2.7 Diagnosis

According to Ayurveda, dysentery is diagnosed into four types. These categories are based on the nature of signs and symptoms and are diagnosed into the following four types:

1. *Vata* dysentery — Defecation with tenesmus and abdominal pain
2. *Pitta* dysentery — Defecation with a burning sensation
3. *Kapha* dysentery — Defecation with mucus
4. *Rakta* dysentery — Defecation with blood

The enterotoxic and nonenterotoxic stages of this disease are diagnosed based on the tests described in diarrhea.¹

The types of dysentery, such as amebic, bacillary, balantidial, malignant, and viral, are determined based on microscopic examination of the stool.⁴⁴

21.2.8 Clinical Course and Prognosis

Dysentery is a curable disease. If it is left untreated, it leads to irritable colon disease.¹

21.2.9 Therapy

The treatment of dysentery is aimed at eliminating the offending and toxic matter from the intestines, alleviating painful symptoms, stopping the virulence of the bacteria, and promoting the healing of any ulcers in the intestine. Antimotility agents should not be

used in the early stages of dysentery because of the presence of enterotoxin. In such conditions, mild laxatives are recommended. Using herbal formulas to improve digestion and produce a carminative effect is preferred to achieve a long-lasting effect.

The bark of kurchi (*Holarrhena antidysenterica*) and the raw, tender fruit pulp of bael (*Aegle marmelos*) are the most important herbal ingredients in dysentery. Bastard saffron (*Mesua ferrea*) flower bud, lodh bark (*Symplocos racemosa*), and silk cotton-tree resin (*Bombax malabarica*) are the styptic plants to be used. All prescriptions for diarrhea are also useful for dysentery.¹²

Among specific herbal remedies, bael fruit is, perhaps, the most efficacious in the treatment of dysentery of both the varieties. The pulp of this fruit is mixed with jaggery and should be given three times/day. To deal with a chronic case of dysentery, unripe bael fruit is roasted over a fire and the pulp is mixed with water. Large quantities of the infusion are administered with jaggery. The pulp of the unripe fruit mixed with an equal quantity of dried ginger can also be given with buttermilk.

The use of pomegranate rind is another effective remedy for dysentery. About 60 g of the rind is boiled in 250 g of milk until half of the milk has evaporated. It is administered to the patient in three equal doses at suitable intervals.⁴⁵

21.2.9.1 Food and Lifestyle

Adequate bedrest; a light diet; the use of potassium broth, soybean milk, or oatmeal milk; and drinking at least 1 pt/day of ginger or barley water are highly recommended. Chewing the food thoroughly before swallowing is most important.

Fasting is the best corrective remedy for dysentery. The patient should fast as long as acute symptoms are present. During the period of fasting, only fruit juice (orange) and water should be taken. In the alternative, the patient should subsist on buttermilk until the acute symptoms are over. Buttermilk combats offending bacteria and helps establish helpful microorganisms in the intestines.

The following foods are useful for patients with dysentery:

1. Old rice
2. Gruel prepared from pop of paddy
3. Soup prepared from phaseolus bean or lentil
4. Milk and butter from a goat or cow
5. Fresh or dried ginger
6. Leaves of wood sorrel
7. Fruits of jamun (*Eugenia jambulana*), pomegranate, and bael (*Aegle marmelos*)
8. Flower and fruit of banana

In addition, dysentery patients should avoid the following:

1. Uncherished food (*asatmya bhojana*)
2. Spicy and sour food
3. Incompatible articles of food (*viruddha ahara*)
4. Food that is difficult to digest
5. Products of sugarcane, wheat, black phaseolus bean, barley, and leaves of a variety of chenopodium

Excessive water intake and bathing, staying awake late in the night, and irregular hours of sleep should also be avoided.⁷

Compound formulations used in the treatment of dysentery are listed below:

Name of Formulation	Activity	Dose	Adjuvant	Ref.
<i>Gangadhara curna</i>	Digestive, appetiser, antidiarrheal	1–3 g three times/day	Jaggery, buttermilk and honey	6
<i>Hingvashtaka curna</i>	Carminative, digestive, antidyenteric	1–2 g two times/day	Warm water before meals	5
<i>Agnimukha curna</i>	Carminative, digestive, antidyenteric	1–3 g two times/day	Buttermilk	5
<i>Bilvadi curna</i>	Carminative, digestive, antidyenteric	2–6 g two times/day	Buttermilk (50–100 ml)	8
<i>Kutajaghana vati</i>	Antidyenteric	250–500 mg two times/day	Buttermilk (50–100 ml)	11
<i>Jatiphaladi vati</i>	Antimotility, Antispasmodic	60–120 mg three times/day	Buttermilk and water	8
<i>Karpura vati</i>	Antimotility, Antispasmodic	120–240 mg two times/day	Honey	8
<i>Kutajarishtha</i>	Digestive, antidyenteric	15–30 ml two times/day	Equal quantity of water	8
<i>Kutajavaleha</i>	Antidyenteric	5–10 g three times/day	Water	8
<i>Sankha bhasma, Kapardika bhasma, or Pravala bhasma</i>	Antiflatulent, digestive, and intestinal adsorbant	0.25–0.5 g two times/day	Lemon juice before meals	46
<i>Agasthisutaraja rasa</i>	Antidiarrheal	60–120 mg three times/day	Cumin, nutmeg powders	5
<i>Kanakasundara rasa</i>	Antidiarrheal	60–120 mg three times/day	Buttermilk	8

21.2.10 Scientific Basis

21.2.10.1 *Punica granatum* (Pomegranate)

A decoction or tea infusion of *Punica granatum* peel was used in the *in vitro* experiments against *Vibrio cholerae*. Results showed the best bactericidal effect, and it is suggested to use them to stop cholera spreading.¹⁴

21.2.10.2 *Berberis aristata* (Indian Barberry)

Berberine, an alkaloid from the plant *Berberis aristata*, inhibited by approximately 70% of the secretory responses of the heat-labile enterotoxins of *V. cholerae* and *E. coli* in the rabbit-ligated intestinal loop model. The drug was effective when given either before or after enterotoxin binding and when given either intraluminally or parenterally; it did not inhibit the stimulation of adenylate cyclase by cholera enterotoxin and caused no histological damage to intestinal mucosa. Berberine also markedly inhibited the secretory response of *E. coli* heat-stable enterotoxin in the infant mouse model. Although the mechanism of

action of the drug is not yet known, these data provide a rationale for its apparent clinical usefulness in treating acute diarrheal disease.⁴⁷

21.2.10.3 Medicinal Herbs (*Boerhaavia diffusa*, *Berberis aristata*, *Tinospora cordifolia*, *Terminalia chebula*, and *Zingiber officinale*)

The antiamebic effect of a crude drug formulation against *Entamoeba histolytica* was studied. The formula is composed of five medicinal herbs: *Boerhaavia diffusa*, *Berberis aristata*, *Tinospora cordifolia*, *Terminalia chebula*, and *Zingiber officinale*. The dried and pulverized plants were extracted in ethanol together and individually. *In vitro* amebicidal activity was studied to determine the minimal inhibitory concentration (MIC) values of all the constituent extracts as well as the whole formulation. The formulation had an MIC of 1000 µg/ml as compared with 10 µg/ml of metronidazole. In experimental cecal amebiasis in rats, the formulation had a curative rate of 89%; with the average degree of infection (ADI) reduced to 0.4 in a group dosed with 500 mg/kg/day as compared with an ADI of 3.8 for the sham-treated control group of rats. Metronidazole had a cure rate of 89% (ADI = 0.4) at a dose of 100 mg/kg/day and cured the infection completely (ADI = 0) when the dosage was doubled to 200 mg/kg/day. There were varying degrees of inhibition of the following enzyme activities of crude extracts of axenically cultured amoebae: deoxyribonuclease, ribonuclease, aldolase, alkaline phosphatase, acid phosphatase, alpha-amylase, and protease.⁴⁸

Ayurvedic plants having inhibitory activity on enteropathogenic bacteria are shown in [Table 21.2](#).

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22

Gastroduodenal Ulcers

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22.1 Introduction

In Ayurveda, peptic ulcers or gastroduodenal ulcers generally refer to *parinamasula*. It is the disease of the gastrointestinal tract, especially the stomach. Abdominal pain during digestion is the main symptom in these diseases. Earlier Ayurvedic classics like *Charaka Samhita* and *Sushrut Samhita* mentioned abdominal pain as relating to digestion but not as a separate clinical entity.^{1,2} For the first time, in 7th century A.D., Madhavakar described this disease separately with the name *parinamashula* in his famous treatise, *Madhava Nidana*, which mainly deals with the etiology and pathology of different diseases. He has given a precise description of *parinamasula*, but *Vijayarakshita*, the commentary of *Madhava Nidana*, later elaborated it. This disease has been mentioned in all the later books like *Bhavaprakash*, *Yogaratanakar*, *Sharangadhara Samhita*, and *Chakradatta*.³⁻⁵ Apart from the stress placed on food habits and personal hygiene, some herbal drugs have also been mentioned in the treatment of this disease. The gastric ulcer (GU) remains a major global health problem, eluding satisfactory therapeutic regimen. Modern medicine has not adequately evaluated the usefulness of natural drugs in ulcer therapy, although studies have been reported.

It is now assumed that the antiulcer drugs ultimately balance the aggressive factors (acid, pepsin, *H. pylori*, bile salts) and defensive factors (mucin secretion, cellular mucus, bicarbonate secretion, mucosal blood flow, cell turnover, etc.).⁶ New etiological factors, such as involvement of free radicals⁷ and *H. pylori*,^{8,9} have gained attention. Resistance of *H. pylori* to antimicrobials¹⁰ and poor patient compliance due to multiple doses¹¹ are reported to be major causes for the failure of antiulcer therapy. This necessitates using newer antibacterials and reducing doses during therapy. Antiulcer drugs with antioxidant and antibacterial effects would be superior to combined therapies.¹² These attributes are found with herbal drugs.

In this chapter, literature on Ayurvedic therapies of GUs and duodenal ulcers (DUs) is reviewed and compared with therapies in conventional medicine. Many similarities between the two systems are noted.

22.2 Clinical Description, History, and Epidemiology

According to Ayurveda, abdominal pain that is aggravated during digestion is known as *parinamashula*. Various synonyms of *parinamashula* mentioned in the texts (i.e., *annadravashula*, *paktishula*, *annavidahajashula*, etc.) are related to food and digestive factors. According to Ayurveda, the word *parinama* implies digestion. It actually starts after the

emptying of the stomach. The word *shula* refers to pain. Hence the pain that occurs during the digestion period is called *parinamashula*.

Similarly, in conventional medicine, peptic ulcer disease (PUD) is described to be associated with epigastric pain exacerbated by fasting and improvement with food. The patient has a history of alterations in bowel habits, especially diarrhea or constipation.¹³

As discussed earlier, the GU or DU has not been recognized as a separate clinical entity, making information on history and epidemiology limited.

22.3 Etiology

When we compare and contrast GUs and DUs, both seem to be very much similar. PUD, according to the modern concept, is a multifactorial disease.¹³ The causes include genetic, dietetic, psychological (stress and emotional factors), endocrine, and drug-induced infections. The common result of all these initiating factors is a breakage in mucous membrane of either the stomach or duodenum (common sites). There may be a difference in incidence, placement of ulcer, symptomatology, or prognosis of DUs and GUs, but the presence or absence of an ulcer is determined by a delicate interplay between the aggressive factors (acid and pepsin) and defensive factors (mucosal resistance).¹

The defense mechanism of the gastrointestinal mucosa against aggressive factors, such as hydrochloric acid, bile acid, free radicals, *H. pylori* colonization, nonsteroidal anti-inflammatory drugs (NSAIDs), etc., mainly consists of functional, humoral, and neuronal factors.¹⁴ These coincide with the Ayurvedic concept of PUDs, where it is believed to be an imbalance between *kapha* and *pitta*. The important factors responsible for ulcerogenesis are described below.

22.3.1 Offensive Factors

Pitta represents the aggressive factors such as acid and pepsin. *Vata* is the main factor in production of pain and represents the neuronal part of acid secretion. The symptomatology of *parinamashula* and PUD have similar symptoms such as pain in the epigastrium and other sites, nausea, vomiting, and pain relieving with intake of food (in DUs) or after digestion (i.e., in an empty stomach or with vomiting [in GUs], heartburn, abdominal pain, etc.)

22.3.1.1 Acid and Pepsin Secretion

The role of hydrochloric acid in the pathogenesis of gastric ulcer is well established.¹⁵ Many commonly used antiulcer agents heal ulcers by blocking acid secretion.

22.3.1.2 *H. pylori* colonization

H. pylori, a gram-negative bacterium, is found in more than 90% of DU cases and 75% of GU cases.¹⁶ Clinical outcomes associated with *H. pylori* infection include GU, DU, gastric adenocarcinoma, and gastric-mucosa-associated lymphoid tissue lymphoma. The pathogenicity of *H. pylori* depends on bacterial and host factors.¹⁶ Even though there have been advances in eradication of *H. pylori* in peptic ulcer patients by using multiple therapies,

there are several reports on the limitations of such treatments, such as resistance to *H. pylori*¹¹ and poor patient compliance due to multiple doses.¹² Further treatment can be accompanied by nausea, diarrhea, abdominal pain, and pseudomembranous colitis.¹⁷ These adverse effects contribute to patient noncompliance and reduce the efficacy of therapy.

22.3.1.3 Histamine

Histamine has been found in the gastric wall and is a powerful stimulant for gastric secretion. Histamine blockers such as ranitidine have been reported to prevent even psychological stress-induced gastric ulceration.¹⁸

22.3.1.4 Drugs (NSAIDs)

NSAIDs are widely prescribed for treating many conditions. The risk of ulcer complications, such as bleeding, perforation, and death, is increased approximately fourfold in NSAID users. NSAIDs disrupt the GI mucosal-protective and acid-limiting properties of prostaglandins (PGs) and also have a direct topical irritant effect. NSAIDs induce gastric damage through generation of reactive oxygen species.¹⁹ Adverse side effects may be significantly reduced through the use of cyclooxygenase-2 (COX-2)-specific inhibitors.²⁰

22.3.1.5 Free Radicals

Free radicals are defined as chemical species possessing unpaired electrons. The role of oxygen-derived free radicals has been demonstrated in acute and chronic ulceration.²¹ Several natural drugs have been reported to have antioxidant activity, which contributes to their activity.²²⁻²⁴

22.3.2 Gastric Mucosal Defense Mechanisms

The role of *kapha* is that of mucosal defensive mechanism and was clearly mentioned in the pathogenesis of *parinamashula* in 7th century A.D. According to modern medicine, the malfunctioning of certain defensive (protective) mechanisms of the GI tract mucosa are the important determinants in the development of ulcers. These mechanisms are presented here.

22.3.2.1 Mucus-Bicarbonate Barrier

The entire surface of the gastric mucosa is covered by a continuous layer of mucus gel, which has a variable thickness of less than 500 μ m.²⁵ Mucus is made up of glycoprotein and contains sulfhydryl groups (-SH), which protect the mucosa from free-radical-induced injury.²⁶ Mucus provides a mixing barrier at the mucosal surface and also prevents the activation of pepsinogen to pepsin apart from providing a microenvironment for repair and restitution.²⁷

22.3.2.2 Gastric Mucosal Renewal and Restitution

The rapid proliferation of gastric mucosa plays an important role in mucosal protection during normal state and after mucosal damage.²⁸

22.3.2.3 Gastric Motility

Gastric motility changes have been reported to cause both gastric and duodenal ulceration.²⁹ Incompetence of the pyloric sphincter and delayed gastric emptying may lead to

stasis and delayed clearing of refluxed duodenal contents; these actions can lead to increase gastric release and a subsequently higher rate of acid secretion.

22.3.2.4 Mucosal and Submucosal Blood Flow

Constituents of mucosal blood flow are important in mucosal defense. They play a vital role in protecting the mucosa by delivering oxygen, nutrients, and bicarbonate to the cells. They also remove hydrogen ions that have penetrated the mucus-bicarbonate and epithelial barrier.³⁰

22.3.2.5 Prostaglandins

PGs, namely, prostaglandin E and prostaglandin I (PGE and PGI), have been shown to protect against experimental necrotic damage. PGs increase gastric mucosal blood flow, mucus, and bicarbonate secretion and strengthen the mucus bicarbonate barrier. Thus, inhibition of the production predisposes the stomach to injury. The beneficial effects of COX-2 inhibitors are still controversial.³¹

22.3.2.6 Antioxidants

Antioxidants are stated to be any substance, even present at low concentrations compared with those of an oxidizable substrate, that significantly delays or prevents oxidation of that substrate. Uncontrolled oxidation in aerobic organisms produces oxidative stress, cell damage, and eventually cell death. An array of cellular defense systems exists to counterbalance reactive oxygen species (ROS).³² Antioxidants neutralize free radicals, thus protecting essential micro- and macromolecules in the body from the oxidative damage.

22.3.3 Other Causative Factors

Other causative factors can be broadly divided into three types: diet (*ahara*), lifestyle (*vihara*), and seasonal (*kala*) factors.

22.3.3.1 Dietetic Factors (Ahara)

Excessive use of sour, salty, pungent, spicy, astringent, and dry foods; alcohol intake; oil; mustard; pulses (e.g., peas, beans); and a nonvegetarian diet may all lead to ulcer development. Consuming food before digesting a previous meal, eating incompatible food, and being gluttonous may also cause peptic ulcers.

There are also recent reports from modern medicine on the influence of food on the genesis of GUs and DUs.³⁴⁻³⁶ Factors such as being a male, having a family history of ulcers, having an O-blood type, skipping breakfast or more than one meal, drinking excessive amount of coffee, and smoking cigarettes were reported to be associated with increased incidences of DUs.³⁷

22.3.3.2 Lifestyle Factors (Vihara)

Lifestyle factors include the following:

1. Physical factors — Examples include excessive exercise, hard work, excessive sexual indulgence, excessive exposure to sun or fire, waking up at night, and suppressing natural urges.

2. Mental factors — Examples include anger, grief, anxiety, and depression.
3. Psychological factors — Lifestyle factors can be equated with psychological factors. Although there has not been much information on patterns of life on ulcers in conventional medicine, there are reports on the psychological changes, which are an immediate fallout of these physical factors.
4. Stressful life events — According to modern medicine, these events have been associated with the onset or symptom exacerbation of some of the most common chronic disorders of the digestive system, including functional GI disorders, inflammatory bowel disease, gastroesophageal reflux disease, and PUD.³⁸ There is considerable evidence that supports the role of stressful life events in the etiology of PUD. These mechanisms involve both the cortical and subcortical levels.³⁹ Many investigators have suggested the role of ROS in stress-induced ulcers.⁴⁰
5. Cigarette smoking — This is a major lifestyle risk factor for the development and recurrence of peptic ulcers. The association between ulcerative disease and smoking cannot be explained on the basis of an effect on gastric secretion of acid or pepsin, blood flow, or pancreatic secretion.
6. Seasonal factors (*kala*) — Seasonal factors, including rainy season, autumn, spring, cold weather, evening time, midday, midnight, and sunrise, may trigger ulcer development.

22.4 Pathogenesis

According to Ayurveda, in his description of the gastroduodenal ulcers, Madhavakar has not mentioned the specific etiological factors. However, he did state that all the exogenous and endogenous factors that cause the vitiation of *vata* are responsible for the initiation of pathogenesis of ulcers; *kapha* and *pitta* are also involved.³³ Hence the provoking factors mentioned in reference to *shularoga* were found to be relevant for gastroduodenal ulcers. Because of the indulgence in specific etiological factors, the *dosas* may start accumulating at their own sites followed by vitiation. The vitiation of any one of these *dosas* may disturb other *dosas*. Even when one of the three *dosas* is in a state of vitiation, the biofire (i.e., *agni*) tends to be deranged, hindering the process of digestion and leading to the formation of defective juices from inadequate digestion of food (*ahara rasa*).

The stomach (*amashaya*) is the site of specific *kaphas* (*kledakakapha*, *pachakapitta*, and *samanavata*). According to Ayurveda, *kledakakapha* protects the stomach from the ill effects of *pachakapitta*. The role of *samana vata* is to stimulate the *pachakapitta*. So, the eroding effect of *pitta* is counteracted by *kledakakapha* and *vata* maintains the motility and movements of the stomach. Because of indulgence in etiological factors, the displaced *kapha* from its site admixtures with the *pitta* and is inactivated with the help of *vata*, which produces the pain during the period of digestion. *Parinamashula* is a *tridoshik* disease, but *pitta* is the predominant *dosa*, as it plays important role during the period of digestion (i.e., *parinamakala*).

22.5 Clinical Examination

The cardinal feature of this disease, according to Ayurveda, is pain in the abdomen during the process of food digestion. Important sites of the pain are the epigastrium (*kukshi*), flanks (*jathara parshwe*), umbilical region (*nabhau*), retrosternal region (*stanantare*), and back (*prish-tamoola pradesha*); the pain sometimes occurs in all the above mentioned sites simultaneously (*sarveshu*). There are seven clinical types mentioned, but basically they are derived from the permutations and combinations of three dosic states. Pain may be aggravated through the consumption of a specific variety rice (*raktashali*) and may be relieved by taking a meal and vomiting. As in modern medicine, the food habits and other personal clinical history features (e.g., previous drug intake) are taken into consideration, and modern techniques such as radiological and histological examinations are routinely done.

22.6 Clinical Course and Prognosis

The characters of pain are burning (*daha*), dull aching (*swalpa ruk*), flatulence (*borborygmi*), trembling (*vepana*), and acute continuous pain (*dirgha santata ruk*). *Borborygmi* is due to *vata* with systemic features of flatulence, constipation, and restlessness. These symptoms can be relieved by a hot and fatty diet. Burning sensation due to *pitta* with systemic features of thirst, restlessness, and excessive perspiration can be aggravated (*shula vridhhi*) by sour and salty food intake. Dull aching and continuous acute pain due to *kapha* with systemic features including nausea, drowsiness, and vomiting can be vitiated by pungent and bitter food.

The dual (*dvandaja*) type and triple (*tridoshaja*) type are clinical mixtures of above-mentioned three types of *parinamashula*. The *tridoshaja* variety is associated with excessive weakness and markedly diminished digestive capacity; it is considered incurable.

22.7 Therapy

In Ayurveda, the whole armamentarium of treatment of any disease has been broadly divided into two types: biopurificatory (*samshodhana*) and palliative (*samshamana*), along with avoidance of causative factors (*nidana parivarjana*):

1. *Samshodhan* therapy^{41,42} — This type of treatment is used in strong and suitable persons who can bear the impact of various procedures like *langhan*, *vaman*, *virechan*, *anuvasan*, and *niruha vasti*.
2. *Samshamana* — Limited drugs have been mentioned^{41,42} for this type of treatment (see [Tables 22.1](#) and [22.2](#)).
3. *Nidana parivarjana* — Also known as the withdrawal from the etiological factors, *nidana parivarjana* is considered to be crucial in the treatment of this disease.

TABLE 22.1
Single Herbal Drugs

Drug	Form of Drug	Dosage
<i>Patola</i>	Ghee prepared with fruits	25 ml twice daily with milk
<i>Shatavari</i>	Root powder	3–4 g three times with milk
	Ghee prepared with roots	25 ml twice daily with milk or hot water
<i>Yashtimadhu</i>	Root powder	4–5 g three times daily
<i>Aswagandha</i>	Root powder	4–5 g three times daily
<i>Bhringaraja</i>	Powder of whole plant	4–5 g four times daily
<i>Amalaki</i>	Powder of fruit pulp	1 g twice daily
<i>Aparajita</i>	Root paste	4–5 g twice daily

TABLE 22.2
Compound Ayurvedic Formulas Used in the
Treatment of Gastroduodenal Ulcers

Preparation Type	Name of Preparation
<i>Churna</i> [powder]	<i>Avipattikara churna</i> <i>Amalaki Rasayan</i>
<i>Avaleha</i>	<i>Kushmandavaleha</i> <i>Dadimavaleha</i>
<i>Gutika</i> [pill]	<i>Shankha vati</i> <i>Mahashankha vati</i>
<i>Bhasma</i> [ash]	<i>Shankha bhasma</i> <i>Varata bhasma</i> <i>Shambooka bhasma</i>
<i>Lauha</i> [iron preparations]	<i>Saptamrita lauha</i> <i>Dhatri lauha</i>
<i>Mandur</i>	<i>Shatavari mandur</i>
<i>Rasaushadhi</i>	<i>Sutashekhara ras</i> <i>Kamadudha ras</i> <i>Pravala panchamrit</i>
<i>Khanda</i>	<i>Narikelakhanda</i>

The main principle of treatment of *parinamashula* is aimed at the following: (1) alleviation of excited *vata*, (2) controlling or reducing the hyperactivity of *pitta*, and (3) repairing and maintaining the integrity of the *rasavaha srotas* situated at the site of lesion by increasing the *kapha*.

22.7.1 Diet and Regularized Lifestyle (*Pathya ahara* and *Vihara*)

Bajra, wheat, Indian gooseberry (*amalaki*), milk, buttermilk, and all bitter and sweet foods are compatible. Mental and physical rest are also helpful to control the symptoms of the disease. Anger, grief, waking up at night, and excessive exposure to sun, cereals (e.g., blackgram, sesame), excessive sour and salty foods, alcohol, and foods that are heavy for digestion are all considered incompatible for a proper lifestyle and diet.

22.7.2 Conventional Therapies

Conventional drugs are reviewed here to understand their mechanism of action, which can be correlated with Ayurvedic drugs discussed later. This section also highlights their

balance of therapeutic efficiency to adverse effects, drug interactions, and the need for alternative strategies.

22.7.2.1 Antacids

The effect of antacids on the stomach is due to partial neutralization of gastric HCl and inhibition of pepsin. Although adverse effects with antacids are minimal, significant adverse reactions can occur with long-term use.⁴³

22.7.2.2 Histamine H₂-Receptor Antagonists

Histamine H₂-receptor antagonists (H₂RAs) competitively inhibit histamine action at all H₂ receptors. Their main clinical use is as inhibitors of gastric-acid secretion. There are reports suggesting non-acid-related effects contributing to the activity of H₂RAs defying the previously held view as only antisecretory drugs.⁴⁴ Reports suggest that the central nervous system (CNS) may be partially responsible for antiulcerogenic activities of H₂RAs.⁴⁵ This correlates well with the Ayurvedic view of influence of psychological factors and the use of adaptogens (*rasayanas*) for ulcer therapy. Human gastric carcinoid was detected during the development of tolerance and rebound acid secretion during the long-term treatment with H₂RAs.⁴⁶

22.7.2.3 Proton Pump Inhibitors (PPIs)

Proton pump inhibitors (PPIs) act by irreversible interaction with the SH group of the H⁺-K⁺adenosine triphosphatase (ATPase), forming a disulphide bond. Reported side effects include headache, diarrhea, skin rash, and reversible abnormalities in biochemical liver function tests. Long-term inhibition has been reported to produce gastric carcinoma,⁴⁷ achlorhydria, and hypergastrinemia.⁴⁸ Omeprazole has been reported to reduce the secretion, synthesis, and gene expression of pepsinogen, which may create problems in the protein digestion and result in diarrhea.⁴⁹ There are reports of potential CNS side effects after PPI treatment, which need further evaluation.⁵⁰ PPIs have also been reported to have some drug interactions. Recently, lansoprazole has been reported to potentiate vecuronium-induced paralysis in patients.⁵¹

22.8 Preventive Measures

Preventive measures essentially include dietary intervention and lifestyle changes based on the factors as suggested in Ayurveda (see [Sections 22.4](#) and [Section 22.7](#)).

22.9 Scientific Basis of Ayurvedic Therapies

Because the etiological factors for gastroduodenal ulcers in Ayurveda and conventional medicine are similar (e.g., imbalance of offensive [*vata*] and defensive factors [*kapha*]), animal models used to evaluate modern medicine are often used to evaluate Ayurvedic therapies.

The upcoming review of some selected drugs gives an elaborate and clear view of how modern evaluation models are suitably used for screening natural drugs. The success of Ayurvedic drugs in the commercial market depends on stringent evaluation followed closely in line with modern experimentation and clinical evaluation. One of the major hurdles is to clearly define the composition of Ayurvedic drugs and the identification of active ingredients. Without the knowledge of active ingredients, the product cannot be standardized or the amount of active ingredient cannot be quantified and would fall short of drug enforcement standards. Investigations not taken to this level will be of academic interest only and not of any practical value. This review gives major trust for drugs that have been actively quantified or have been standardized for active ingredients.

22.9.1 Animal Studies

22.9.1.1 *Musa sapientum* var. *paradisiaca* (Unripe Plantain Banana)

Plantain banana has been subjected to extensive experimental and clinical studies. Dried powder of banana pulp (DRBP) was effective against experimental ulcers.⁵² The antiulcer activity of DRBP was reported to be due to its predominant effect on mucosal defensive factors rather than on the offensive acid-pepsin secretion.^{53–55}

Ethanol extract of banana was reported to increase the accumulation of eicosonoids like PGE, prostaglandin I₂ (PGI₂), and leukotrienes B₄ and C₄/D₄ (LTB₄, C₄/D₄) in the human gastric and colonic mucosal incubates.⁵⁶ Many active principles such as steryl-aclyglycosides, and sitoinosides I-IV were isolated and characterized from many vegetables and bananas and were reported to have antiulcerogenic activity both in animal and human models.^{57–59} Methanolic extract of banana was reported to have an antioxidant effect but was devoid of anti-*H. pylori* activity *in vitro*.²⁴

22.9.1.2 *Tamrabhasma*

Herbomineral preparations have been widely mentioned in Ayurveda.⁶⁰ *Tamrabhasma* (TMB), a traditional preparation of copper (containing CuO $\geq$ 44.45%–66.13%, Fe₂O₃ < 6.03%, and S < 2.75%) has been advocated for use in *amlapitta*. *Tamrabhasma* showed antiulcer activity against experimental GUs and DUs. The activity of *tamrabhasma* was ascribed due to a decrease in offensive acid-pepsin secretion and an increase in defensive factors; the effect was also free from toxicity.^{61,62}

22.9.1.3 *Mahakasya* Drugs

A water decoction of ginger, which makes up one of the constituents of *mahakasyaya* drugs, along with a water decoction of *Piper longum* and a colloidal solution of *Ferula foetida*, has been reported to protect against cold-restraint stress, aspirin, and pyloric-ligation-induced GUs in rats. Although increase in offensive acid-pepsin secretion was observed in this study, the increase in defensive mucin secretion was sufficient enough to protect against experimental ulcers.⁶³

22.9.1.4 *Bacopa monniera* (Bramhi)

Bacopa monniera Wettst. (syn. *Herpestis monniera* L.; scrophulariaceae) is classified as a *medhya rasayana*, a class of plant drugs used to promote mental health and improve memory and intellect.⁶⁴ It is a classic example where an adaptogenic drug has antiulcer activity. It is

intriguing that *Bacopa monniera* has shown both anxiolytic and antidepressant activity.^{65–67} *Bacopa monniera* contains active chemicals, such as bacoside A, which have been reported to be responsible for facilitation of memory⁶⁸ and the antiulcerogenic effect.⁶⁹

Fresh juice of *Bacopa monniera* and its standardized methanolic extract, containing 35% of bacoside A,²³ has been reported to have an antiulcerogenic effect. *Bacopa monniera* extract (20 mg/kg) showed no effect on offensive factors, but it increased the defensive factors. *Bacopa monniera* showed significant antioxidant effect *per se* in stressed animals. *In vitro* studies with *H. pylori* showed significant inhibition at the concentration of 1000 (g/ml of *Bacopa monniera* extract.⁷⁰ Even though effective antimicrobial therapy is available to eradicate *H. pylori* infection, current therapies require patients to ingest multiple agents several times a day for at least 1 week, leading to noncompliance of treatment schedules. An antiulcer agent having anti-*H. pylori* activity can have better compliance among patients and can also decrease incidences of drug interactions. *Bacopa monniera* extract also increased *in vitro* accumulation of PGs on human colonic mucosa cells.⁷⁰

22.9.2 Clinical Studies

22.9.2.1 DRBP

The clinical effectiveness of DRBP was confirmed by radiological and endoscopic studies in the several clinical trials. DRBP was found to decrease or delay the relapse of peptic ulcer for 6 to 12 months after a 3-month continuous treatment at the dose level of 1 g four times/day. It became commercially available as Musapep[®] after phase IV clinical trials. A double-blind study⁷¹ done at multiple centers has shown that about 40 to 70% of endoscopically proved DUs healed after 12 weeks of treatment with DRBP, as compared with about 16% with placebo. In another clinical study⁷² with Musapep in nonulcer dyspepsia (NUD), there was a reported 75% relief in symptoms after 8 weeks of treatment compared with 20% in the control. The authors advocated that the therapeutic trials with this dose justified the use of DRBP on clinical grounds to be safe and effective in the treatment of NUD.

22.9.2.2 *Satavari mandur*

Effect of *Satavari mandur* was studied in 40 cases of *parinamashula*, including 10 cases of peptic ulcer, 15 cases of erosive mucosal disorder, and 15 cases of NUD. The improvement was assessed by clinical and radiological or endoscopic findings, together with the biochemical study of gastric juice in eight patients. The drug was given in a dose of 3 g/day in two divided doses with ghee for 2 months in NUD cases and 3 months in peptic ulcer cases. A significant improvement was observed in all the groups, but it was more prominent in ulcer dyspepsia than NUD. *Satavari mandur* showed significant improvement in healing ulcers (marked healing of 50%, partial healing of 30% in ulcer group) as well as significant improvement in symptoms. The biochemical study of gastric juice in eight patients showed a significant increase in mucosal defensive factors such as decreased cell shedding, increased individual carbohydrates, and increased total carbohydrate to protein ratio. These findings indicate the drug was promoting the mucosal defensive mechanism rather than affecting the acid and pepsin secretion.⁴¹

22.9.2.3 *Eclipta alba* (*Bhringaraja*)

In a clinical trial (n = 55), *Eclipta alba* showed marked symptomatic relief in NUD and peptic ulcer patients. Total response in NUD was excellent in 62.9%, good in 17.1%, poor

TABLE 22.3

Antiulcerogenic Activity of Some Medicinal Plants

Drugs (Common/Ayurvedic Name)	Experimental Models	Mechanism of Action	Ref.
<i>Abies pindrow</i> Royle, Fam. Pinaceae (<i>Talisapatra</i>) Petroleum ether (PE), acetone (AE), chloroform (CE), and ethanolic (EE) extracts of leaves	CRS- induced GU in rats	Antistress activity	74
<i>Benincasa hispida</i> , Fam. Cucurbitaceae (<i>Petha</i> or <i>Golkaddu</i>) Fresh juice, supernatant and residue fraction of centrifuged juice, PE and EE	Aspirin (ASP) + restraint, swimming stress, indomethacin plus histamine- and serotonin- induced ulcers in rats and mice	—	75
<i>Camellia sinensis</i> , Fam. Ternstroemiaceae (green tea) Hot water extract	Cold + restraint–stress- induced ulcers in rats	—	76
<i>Centella asiatica</i> , Fam. Umbelliferae (<i>Brahma-manduki</i>) Fresh juice of whole plant	CRS-, EtOH-, ASP-, and PL- induced GU in rats	No effect on acid- pepsin secretion, increase in mucin secretion and life span of mucosal cells	77
<i>Convolvulus pluricaulis</i> , Fam. Convolvulaceae (<i>Shankupuspi</i>) Fresh juice of whole plant	CRS-, EtOH-, ASP-, and PL- induced GU in rats	No effect on acid- pepsin secretion, increase in mucin secretion and life span of mucosal cells	78
<i>Emblica officinalis</i> Gaertn. Fam. Euphorbiaceae (<i>Amla</i>) Fresh juice of fruit plup	CRS-, EtOH-, ASP-, and PL- induced GU in rats and acetic-acid-induced healing model	Decreased the acid and pepsin secretion and increased mucus secretion	79
<i>Garcinia cambogia</i> , Fam. Guttiferae (<i>Kukum</i>) Fruit extract	Indomethacin	Decrease in acid-pepsin secretion and promotion of mucosal defensive	80
Honey	Ethanol-induced increased vascular permeability changes in the rat stomach	Free-radical scavenger	81
<i>Pongamia pinnata</i> , Leguminosae (<i>Karanj</i> or <i>Karanja</i>) PE, AE, CE, and EE extracts of seeds PE, AE, CE, and EE extracts of roots	RS-induced GU in mice RS- and PL- induced GU in rats	—	82, 83
<i>Satavari Mandur</i>	PL- and CRS-induced GU in rats	No effect on acid- pepsin secretion and promotes mucosal defensive factors by enhancing mucin secretion and life span of mucosal cells	84
<i>Shilajit</i> <i>per se</i> effect	PL-, IS-, and ASP-induced GU in rats and CYS- and HIST-induced DU in rats and GP, respectively	Tendency to decrease acid-pepsin secretion and significant increase in mucin secretion	85

TABLE 22.3 (continued)
Antilcerogenic Activity of Some Medicinal Plants

Drugs (Common/Ayurvedic Name)	Experimental Models	Mechanism of Action	Ref.
Sitavirya drugs			86
a. <i>Asparagus racemosus</i> , Fam. Liliaceae (<i>Satawari</i>) Fresh juice of roots	PL- and CRS-induced GU in rats	No effect on acid- pepsin secretion and promotes mucosal defensive factors by enhancing mucin secretion and life span of mucosal cells	
b. <i>Glycyrrhiza glabra</i> , Fam. Leguminosae (<i>Mulhatti</i>) Water decoction of roots	PL- and CRS-induced GU in rats		
c. <i>Holarrhene antidysenterica</i> Wall., Fam. Apocynaceae (<i>Kurachi</i>) water decoction of barks	PL- and CRS-induced GU, CYS-induced DU in rats		
d. <i>Ficus religiosa</i> , Fam. Urticaceae (<i>Peepalbanti</i>) Water decoction of barks	PL- and CRS-induced GU and CYS-induced DU in rats		
<i>Tectona grandis</i> L. Fam. Verbenaceae (Sagwan) Ethanollic fraction of trunk bark and wood chips	Pylorus ligation (PL) restraint stress (RS)- and prednisolone-induced GU in rats Histamine (HIST)- induced GU and DU in guinea pigs (GP)	No effect on acid-pepsin secretion but caused an increase in mucin secretion	87
<i>Withania somnifera</i> , Fam. Solanaceae (<i>Aswagandha</i>) (Roots) SG-1 [total Me OH-H ₂ O (1:1)] SG-2 (sitoindosides VII, VIII, and withaferin-A)	RS-induced GU in rats	Antistress activity	88

in 5.7%, and 14.3% patients dropped out of the study. Radiological assessment confirmed the symptomatic improvement. Of 25 DU cases, 48% were under excellent response, 24% were under good response, and 16% showed poor response; 12% did not show any response to the treatment. The was drug given in the form of powder (30 g) in three divided doses/day for 1 month in NUD cases and for 3 months in DU cases. The drug also showed significant reduction in free HCL and total acidity in both the groups (NUD and UD).⁷³

Various other herbal drugs have been reported to have antiulcerogenic activity. These are summarized in [Table 22.3](#). Some of these drugs have been chemically characterized and the entities involved in the activity have been isolated. These are summarized in [Tables 22.3 and 22.4](#).

22.10 Summary

Ayurveda has a clear description of etiology, pathology, and treatment for gastroduodenal ulcers. Scientific evidence was also found in the literature in support of use of the following Ayurvedic herbs: *Abies pindrow*, *Benincasa hispida*, *Camellia sinensis*, *Centella asiatica*, *Convolvulus pluricaulis*, *Embllica officinalis*, *Tambra bhasma*, *unripe plantain banana*, *Piper longum*,

TABLE 22.4

Ulcer Protective Effect of Some Active Constituents Isolated From Herbal Drugs

Plants	Active Constituents	Models	Mode of Action	Ref.
<i>Aegle marmelos</i> Correa, Fam. rutaceae (Bael (seeds)	Luvangetin	-do-	—	89
<i>Asparagus racemosus</i> , Fam. Liliaceae (Satawari)	Standardized extract to active saponins	CRS-, EtOH-, ASP- and PL-induced GU in rats	No effect on acid pepsin secretion, increase in mucin secretion and life span of mucosal cells	90
<i>Azadirachta indica</i> , Fam. Meliaceae (Neem)	Nimbidin	ASP-, prednisolone, indomethacin-, serotonin stress-, and acetic acid-induced GU in rats; HIST-induced DU in GP CYS- induced DU in rats	—	91
<i>Emblica officinalis</i> Gaertn., Fam. Euphorbiaceae (Amla) methanolic extract	Standardized to the tannoids, emblicanin A and B	CRS-, EtOH-, ASP-, and PL-induced GU in rats and acetic- acid-induced healing model	Decreased the acid and pepsin secretion and increased mucus secretion	22
<i>Garcinia indica</i> , Fam. Guttiferae (Kumkum) Fruit rind	Garcinol	Water-immersion stress	Free-radical scavenger	92
<i>Ocimum basilum</i> , Fam. Leguminosae (Tulsi)	Fixed oil	ASP-, indomethacin-, EtOH-, Hist-, reserpine, Serotonin-, PL-, and stress-induced GU in rats	Antisecretory	93
<i>Shilajit</i>	Fulvic acid, 4 - methoxy 6 carbomethoxybi- phenyl	PL-, PL+ASP-, and RS- induced GU and CYS- induced DU in rats	<i>Per se</i> decrease in acid- pepsin secretion and cell shedding, tendency to increase mucin secretion, but reversed the increase in cell shedding and decrease in mucin secretion induced by ASP	94
<i>Tectona grandis</i> L., Fam. Verbenaceae (Sagwan) (Trunk bark and wood chips)	Lapachol	Immobilization stress (IS)- and ASP-induced GU in rats; CYS- and HIST-induced DU in rats and GP, respectively	<i>Per se</i> no significant effect on both offensive and defensive factors but reversed the ASP- induced increase in peptic activity and decrease in sialic acid and mucin secretion	95
<i>Zingiber officinale</i> <i>Roscoe.</i> (Ginger) (Adrak)	6-Gingesulphonic acid, 6-shogaol, and ar-curcumene	—	Gastric motility	76, 77, 96

Ferula foetida, *Bacopa monniera*, *Eclipta alba*, *Garcinia cambogia*, *Pongamia pinnata*, *Tectona grandis*, *Withania somnifera*, and *Satavari mandur*. These therapies are relatively safe and effective.

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Alzheimer's Disease

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23.1 Introduction

Mental diseases and mental temperaments have been extensively described in Ayurveda (e.g., nervous disorders [*vata vyadhi*], epilepsy [*Apasmara*], insanity, psychosis [*unmada*], loss of consciousness, fainting [*murccha*, *moha*, *tamaka*], impairment in functioning of mind [*Pramoha*], amnesia [*Vismriti*], etc.)^{1,2} Alzheimer's disease (AD) is a neurodegenerative disorder associated with the progressively worsening of cognitive functions and behavioral disturbances; it is the most common form of primary dementia and mortality in the geriatric population. Although there is no precise equivalent for AD in Ayurveda, cognitive dysfunction has been described in convulsive disorders^{1,3} and with the aging process, which is known to be associated with physical and mental disability.^{4,5} Thus it can be seen

that in spite of the terminological differences between the Ayurvedic and the modern systems of medicine, there are areas of similarity between the two. It is necessary to bridge the terminology gap between the Eastern and Western concepts and management for a better understanding and control of cognitive disorders. The present review is aimed toward this objective.

23.2 Clinical Description

The major clinical manifestations of AD are memory disturbances, spatial and temporal disorientation, and getting lost in familiar surroundings. As the disease progresses, aphasia, apraxia, acalculia, and sleepwalking may develop.⁶

23.3 Epidemiology

AD is a worldwide problem that affects 5 million people in the U.S.⁷ As the life span is increasing, AD is becoming a more common disorder. Approximately 10% of the population older than 65 years are affected.⁶ In a recent report, AD was listed as the eighth cause of death in the U.S. in 2000.⁸

23.4 Etiology and Pathophysiology

In Ayurveda, AD is considered age related with no specific etiology except that the *vata dosa* is involved. Conventional medicine is using several working hypotheses to find an effective treatment for AD and develop prevention strategies. These hypotheses include defective beta-amyloid (A β) protein metabolism; abnormalities of glutamatergic, adrenergic, serotonergic, dopaminergic, and cholinergic neurotransmission; and the potential involvement of inflammatory, oxidative, and hormonal pathways.^{9–12} Another suggested possible cause is a diet containing excessive intake of sugar, refined carbohydrates (with high glycemic index), and animal products (with high content of saturated fats), and a decrease intake of unrefined seeds (cereals, legumes, and oleaginous seeds), vegetables (with high fibrous content, vitamins, polyphenols, and other antioxidant substances [e.g., phytoestrogens]), and seafood (rich in omega-3 fatty acids).¹³ The other causative factors hypothesized are insulin resistance, decreased endothelial production of nitric oxide, free-radical excess, inflammatory metabolites, homocysteine deficiency, and estrogen deficiency.¹³

In a prospective analysis of risk factors, increasing age, lack of education, and the apolipoprotein E epsilon4 allele were found to be significantly associated with increased risk of AD. On the other hand, use of nonsteroidal anti-inflammatory drugs (NSAIDs), wine, and regular physical activity were found to be significantly associated with reduced risk of AD.¹⁴ Some evidence suggests that oxidative damage of brain tissue may be

involved in pathogenesis, which precedes the occurrence of symptoms and the formation of amyloid-containing plaques and neurofibrillary tangles.¹⁵

Genetic factors may also be involved, but they do not explain in full the etiopathogenesis of AD, implying a role for environmental factors.¹⁶ The discovery of pathogenic mutations in the beta-amyloid peptide (beta-APP) and the presenilin genes provides strong support for the genetic hypothesis because Abeta production and deposition contribute to the etiology of AD.¹⁷

There is some evidence that alterations in the lysosomal system may have a role in the pathogenesis of AD.¹⁸ The lysosomal system contributes to protein deposits associated with different types of age-related neurodegeneration. Lysosomes are highly susceptible to free-radical oxidative stress in the aging brain and leads to the gradual loss of metabolic activity over the life span of an individual.¹⁹

23.5 Clinical Examination and Diagnosis

Ayurvedic examination primarily consists of an eight-point examination to determine the vitiated *dosas*, body constitution, and manifestations of the disease to determine appropriate treatment (Chapter 2). Patients are examined for memory disturbances, difficulty in learning, spatial and temporal disorientation, and decline in daily living activities (e.g., eating and difficulty in grooming, dressing, and undressing). Patients generally begin to feel spatially lost in familiar surroundings. It is important to rule out the other causes of dementia such as metabolic causes (e.g., diabetes, medication-induced psychiatric illness) before treating the patient with cholinesterase inhibitors. Because many causes of dementia are rare and can be excluded by careful historical information, serial examination, and some structural examination, an accurate diagnosis can be made by using clinical criteria.⁶

23.6 Clinical Course and Prognosis

AD is a uniformly progressive disease. It ultimately results in irreversible debilitating cognitive impairment in patients. Although some medications appear to produce transient improvement, there is no medication available that can halt the progression of AD.¹⁹

23.7 Management

Ayurveda uses natural products (drugs of plant, animal, and mineral origin) for the treatment of all physical and mental ailments, including cognitive dysfunction. There are two approaches: one is aimed at general or total well-being, including mental exercises

(memory games, reading, card games) and meditation, whereas the other specifically targets intellect (*medha*) and memory (*smriti*). Ayurvedic *rasayana tantra*, which describes these approaches, was defined by the Ayurvedic physician Susruta (500 B.C.) as the measure that prolongs longevity, develops positive health and mental faculties, and imparts resistance and immunity against disease.⁴ These measures do not appear to differ from the adaptogenic,²⁰ immunomodulatory,²¹ and antioxidant²² therapeutic approaches used in modern medicine. The host-oriented approach of *rasayana* drugs is believed to vitalize the digestion and metabolism and improve microcirculation and tissue perfusion⁵ to elicit the desired therapeutic effects.

23.7.1 Brain Tonics (*Medhya Rasayana*)

Abhang²³ reviewed the past, present, and future of *medhya rasayana* drugs: a special class of *rasayana* agents that rejuvenate mental faculties, intellect, and memory. In ancient India most of the education was imparted by oral instructions. The students retained and recalled this knowledge by repeated recitation. Therefore, intelligence was considered synonymous with memory, and drugs that helped reduce time for learning (memorizing) the lesson were developed from natural sources. Many such products, particularly of plant origin, have been described in Ayurvedic texts. Some of these are particularly reputed for *medhya rasayana* properties since ancient times of great Ayurvedic physicians Charaka (400 B.C.) and Susruta (500 B.C.). The following brain vitalizers have been mentioned in *Charak Samhita*²⁴:

1. Expressed juice of Indian pennywort (*Centella asiatica*)
2. Powder of licorice (*Glycyrrhiza glabra*) with milk
3. Juice of *guduchi* (*Tinospora cordifolia*)
4. Paste made from small leaves of *shankhpushpi* (*Convolvulus pluricaulis*) mixed with its roots and flowers

A large number of Indian medicinal plants are attributed with brain tonic and memory-enhancing effects. These plants are rarely used singly and are mostly incorporated in polyherbal formulations containing 4 to 20 ingredients. The latter may occasionally include mineral-origin drugs in addition to medicinal plants. Polypharmacy appears to be the rule (rather than exception) in Ayurveda as in other traditional systems of medicine practiced in India (e.g., Unani-Tibb and Siddha). It is claimed that such combinations enhance activity and mitigate side effects of individual components.

Some plant ingredients are common to many formulations. A look at the composition of 6 polyherbal formulations, marketed in India for memory enhancing effects, revealed the presence of 32 medicinal plant ingredients (Table 23.1).

In conventional medicine, the only drugs currently approved for the treatment of AD are cholinomimetics, such as tacrine, which have the pharmacologic profile of acetylcholinesterase (AChE) inhibitors.^{12,25,26} Tacrine has been shown to affect some measures of memory performance, but the magnitude of improvement was very modest.²⁷ The side effects of tacrine, such as abdominal cramping, nausea, vomiting, and diarrhea, may be significant and dose limiting. Considerable research is being conducted in other areas, such as using selective anti-inflammatory drugs (selective cyclooxygenase-2 [COX-2] inhib-

TABLE 23.1
Plant Ingredients of Six Polyherbal Formulations
Marketed in India for Memory-Improving Effects

No.	Latin Name (Vernacular Name)
1	<i>Acorus calamus</i> (<i>vacha</i>)
2	<i>Amomum subulatum</i> (<i>badi ilaichi</i>)
3	<i>Asparagus racemosus</i> (<i>shatawar</i>) ^a
4	<i>Bacopa monniera</i> (<i>brahmi</i>) ^a
5	<i>Butea frondosa</i> (<i>dhak, palash</i>)
6	<i>Canscora decussata</i> (<i>sankha holi</i>)
7	<i>Cinnamomum zeylanicum</i> (<i>dalchini</i>)
8	<i>Convolvulus pluricaulis</i> (<i>shankhpushpi</i>) ^a
9	<i>Delphinium denudatum</i> (<i>jadwar shireen</i>)
10	<i>Elettaria cardamomum</i> (<i>choti elaichi</i>)
11	<i>Emblica officinalis</i> (<i>amla</i>)
12	<i>Embelia ribes</i> (<i>vayu vidang</i>)
13	<i>Eugenia caryophyllus</i> (<i>laung</i>)
14	<i>Foeniculum vulgare</i> (<i>bari saunf</i>)
15	<i>Ipomea paniculata</i> (<i>vidari kand</i>)
16	<i>Nardostachys jatamansi</i> (<i>jatamansi</i>)
17	<i>Operculum tarpethum</i> (<i>nishoth</i>)
18	<i>Paenia emodi</i> (<i>uood saleeb</i>)
19	<i>Pandanus odoratissimus</i> (<i>keora</i>)
20	<i>Pimpinella anisum</i> (<i>saunf</i>)
21	<i>Piper aurantiacum</i> (<i>renuka</i>)
22	<i>Piper longum</i> (<i>peepal, pippali</i>)
23	<i>Prunus amygdalis</i> (<i>badam</i>)
24	<i>Rosa damascena</i> (<i>gulab</i>)
25	<i>Sassurea lappa</i> (<i>kuth</i>)
26	<i>Terminalia belerica</i> (<i>baheera</i>)
27	<i>Terminalia chebula</i> (<i>harar, hareetaki</i>)
28	<i>Tinospora card folia</i> (<i>giloe</i>)
29	<i>Valerian walachia</i> (<i>tagar</i>)
30	<i>Vetiverica zizinooides</i> (<i>khas</i>)
31	<i>Withania somnifera</i> (<i>ashwagandha</i>) ^a
32	<i>Zingiber officinalis</i> (<i>adrakh</i>)

^aTop four most frequently occurring plant ingredients in commercial polyherbal formulations used in the management of AD.

itors) to reduce inflammatory activity in the brain and antioxidants.^{7,11,15} Other areas of research are anti-amyloid strategies (e.g., immunization, aggregation inhibitors, secretase inhibitors), transition metal chelators (e.g., clioquinol), lipid-lowering agents, antihypertensives, vitamins, and neurotransmitter receptors.⁹

Studies on herbal materials with AChE inhibitory properties have shown some promise.^{28,29} Ayurveda has many formulations and herbomineral drugs discussed below that are used to improve cognitive functions. The precise biochemical mechanism of action of these herbs is not clear. It is likely that antioxidant and anti-inflammatory properties of these herbs may be responsible for their beneficial effect in the treatment of AD.

23.8 Scientific Basis

23.8.1 Nootropic Plants

This findings of experimental studies on Indian medicinal plants with nootropic actions (activating or stimulating mental activity or causing cerebral or intellectual activity) are given below.

23.8.1.1 Pharmacological Studies on Single Herbs

23.8.1.1.1 *Bacopa monniera* Linn. (*Herpestis monniera* Linn.) (*Brahmi*)^{30–35}

This plant has been extensively investigated at the Central Drug Research Institute, Lucknow, India, for memory-enhancing effects and is a component of Mentat (commercially marketed polyherbal preparation). Ethanolic extract of *B. monniera* was demonstrated to facilitate acquisition, consolidation, and retention of memory in animal models of the following cognitive functions: (1) active conditioned avoidance response, (2) Sidman's continuous avoidance response, and (3) foot shock-motivated brightness discrimination. Bacosides A and B (Saponins) were identified as active principles. Administration of bacosides attenuated the retrograde amnesia produced by immobilization induced stress, electroconvulsive shock, and scopolamine. Protein kinase activity and protein content of the hippocampus were increased. Improvement was observed in learning capacity and correction of epilepsy-associated abnormal behavior to some extent. Maze learning by albino rats was facilitated and brain glutamic acid and gamma-aminobutyric acid (GABA) concentrations were elevated.

The protective action of *B. monniera* was recently demonstrated against phenytoin-induced cognitive deficit. Phenytoin (25 mg/kg orally for 14 days) adversely affected the passive avoidance task in mice. An extract of this plant (Memory Plus, 40 mg/kg orally for 7 days along with phenytoin in the second week of the 2-week regimen) significantly reversed phenytoin-induced impairment without affecting its anticonvulsant efficacy. The observed cognitive effects of phenytoin and the plant extract were found to be independent of motor stimulation.

23.8.1.1.2 *Centella asiatica* (*Mandookparni*, *Brahmi*)³⁶

Aqueous extract of this plant was shown to improve learning and memory in shuttle box, stepdown paradigm, and elevated plus maze in rats. When tested on oxidative stress parameters, it decreased brain levels of malondialdehyde (MDA) with a simultaneous increase in the level of glutathione. The drug also increased catalase levels. As a result, antioxidant mechanism in the cognition-enhancing effect of the drug was suggested.

23.8.1.1.3 *Convolvulus pluricaulis* Choisy (*Shankhpushpi*)³⁷

Ethanolic extract of the drug was made alcohol free and suspended in water to study the effect on brain neurotransmitter content in normal and stressed rats. The drug decreased the acetylcholine (ACh) content of the whole brain homogenate, but markedly increased ACh content in the cortex. The catecholamine and 5-hydroxytryptamine (5-HT) contents were also raised in treated rats. The histamine content was lowered in the whole brain homogenate but raised in the cortex. The changes were more pronounced in rats subjected

to swimming stress. These changes suggest tranquilizing action and improvement of mental function.

23.8.1.1.4 *Eugenia caryophyllus* spl. (Laung)³⁸

The aqueous extract of *E. caryophyllus* reduced hydrolysis of ACh by AChE, indicating the presence of water-soluble substances with anti-AChE activity in the plant.

23.8.1.1.5 *Glycyrrhiza glabra* Linn. (Yastimadhu, Mulethi)³⁷

The alcoholic extract of *G. glabra* reduced ACh content in the whole brain but increased it in the cortex.

23.8.1.1.6 *Hydrocotyl asiatica* Linn. (Syn. *Centella asiatica* Linn. [Mandookparni])³⁷

The effects of this plant are similar to *G. glabra* described above.

23.8.1.1.7 *Lawsonia inermis* Linn. (Mehndi)³⁹

Significant nootropic effect was observed in acetone soluble fraction of petroleum ether extract of leaves on elevated plus maze and passive avoidance paradigms. It affected 5-HT and noradrenaline-mediated behavior in experimental animals.

23.8.1.1.8 *Nardostachys jatamansi* DC (Jatamansi)^{40–42}

This plant has been demonstrated to reduce brain serotonin content, reaction time of trained animals for passage through tunnel in a Columbia obstruction box, and conditioned-avoidance response. It has also impaired biosynthesis and metabolism of 5-HT in rabbit brain. It is a constituent of Mentat, a polyherbal formulation with cognition facilitatory effects.

23.8.1.1.9 *Paeonia emodi* Wall (Uood Saleb)⁴³

This plant is an ingredient of *shimotsu-to*, a traditional Chinese medicine shown to improve spatial-working memory in rats.

23.8.1.1.10 *Pongamia pinnata* (Karanj)⁴⁴

Petroleum ether extract was tested in experimental models of AD (ibotenic-acid induced lesioning of nucleus basalis magnocellularis). It reversed both the cognitive deficits and reduction in cholinergic markers after 2 weeks of treatment. Reversal of perturbed cholinergic function was suggested as a possible mechanism of action.

23.8.1.1.11 *Tinospora cordifolia* F.Vill (Giloe, Guduchi)³⁷

Ethanol extract of this plant reduced ACh content in the whole brain but increased it in the cortex.

23.8.1.1.12 *Withania somnifera* Dunal (Ashwagandha)^{45–47}

Active principles (equimolar amounts of sitoindosides VII-X and withoferin A) of *W. somnifera* were investigated in animal model of AD (ibotenic-acid-induced lesioning of nucleus basalis magnocellularis) in rats. The lesions caused marked cognitive deficit as evidenced by (1) severe reduction in learned task, (2) significant decrease in ACh levels in the frontal cortex and hippocampus, (3) reduction in choline acetyl transferase activity, and (4) muscarinic cholinergic binding. The drug (50 mg/kg) significantly reversed these alterations after 2 weeks of treatment.

W. somnifera induced depletion of ACh and catecholamines and increased serotonin and histamine concentrations in the whole brain tissue. It also elicited differential effect on AChE activity in basal forebrain nuclei by producing a slight enhancement in the lateral septum and globus pallidus and a reduction in vertical diagonal band. The changes were accompanied by enhanced M1-muscarinic cholinergic receptor binding in lateral and medium septum and frontal cortex. GABA and benzodiazepine receptor binding and *n*-methyl-d-aspartate (NMDA) and α -amino 3-hydroxy 5-methyl 4-isoxazole propionic acid (AMPA) glutamate receptor subtypes were not affected after treatment with *W. somnifera* principles, suggesting preferential effect of the drug on events in the cortical and basal forebrain cholinergic signal transduction cascade. The drug induced an increase in cortical muscarinic ACh-receptor capacity. This finding partly explains the nootropic action of the *W. somnifera*.

23.8.1.2 Clinical Studies on Nootropic Plants

The gist of information on clinical studies on Indian medicinal plants for nootropic effects is presented below.

23.8.1.2.1 *Bacopa monniera* Linn. (Syn. *Herpestis monniera* Linn. [*Brahmi*])^{48–49}

A significant increase in intelligent quotient (IQ) scores was observed after treatment of 110 males aged 10 to 13 years for 9 months with a *suksma* (micro) medicine derived from *B. monniera*. Memory (direct) and arithmetic tests were used. The drug was well tolerated in single (20 to 300 mg) and multiple (100 and 200 mg for 4 weeks) doses in a double-blind placebo-controlled and non-crossover phase I clinical trial in human volunteers.

23.8.1.2.2 *Centella asiatica* Linn. (Syn: *Hydrocotyle asiatica* Linn. [*Mandookparni*, *Brahm mandooki*])^{50–52}

Double-blind studies with this herb revealed no change of the height, weight, and intelligence of mentally normal children but showed favorable action in mentally retarded children (free from epilepsy). A significant increase in general ability and behavioral pattern was noted even when the drug was administered only for 12 weeks. Another double-blind study, carried out on 30 mentally retarded children (9 to 13 years old) with an IQ range of 55 to 90, revealed better intelligence scores and psychological and biochemical parameters after treatment with the test drug for 9 months. The Binet test and Senguin Form Board test were used for assessing intelligence and performance of children.

23.8.1.3 Pharmacological and Clinical Studies on Nootropic Plant Formulations

23.8.1.3.1 *Mentat*

The preparation for *Mentat* (*Himalaya*) was developed at the Central Drug Research Institute, Lucknow, India. The formula (50 to 500 mg/kg) was found to improve the acquisition and retention of passive-avoidance task in a stepdown paradigm in mice. It reversed scopolamine-induced amnesia and attenuated amnesia produced by electroconvulsive shock (ECS) immediately after training. Chronic treatment with ECS for 6 successive days at 24-h intervals disrupted memory consolidation on day 7. Daily administration of *Mentat* (50 and 100 mg/kg for 6 days) significantly improved memory consolidation in mice. Administration of drug on day 7 also attenuated ECS-induced disruption of memory. Scopolamine-induced delay in transfer latency was reversed on day 1. Physostigmine enhanced the efficacy of *Mentat* against scopolamine-induced amnesia. When adminis-

tered in combination with GABA and Aniracetam, the test drug improved learning and memory retrieval. The results suggest nootropic action in naive and amnesic mice and involvement of cholinergic and GABA-ergic modulation as possible mechanisms of action.^{41,53}

The drug was shown to augment acquisition and retention of learning in normal rats and in states of cognitive deficits induced by undernutrition, environmental impoverishment, sodium nitrite hypoxia, aluminum, aging, and ECS-induced anterograde and retrograde amnesia. The parameters included active avoidance learning, food motivational behavior (Hebb Willam maze), and avoidance learning during endurance performance (Runimex circular runway).^{54–56}

23.8.1.3.2 *Trasina (Deys Pharmaceuticals)*

Treatment with Trasina for 21 days exhibited significant nootropic effects in two experimental models of AD: (1) intracerebroventricular injection of colchicine (15 µg/rat) and (2) lesioning of nucleus basalis magnocellularis by ibotenic acid (10 µg/rat). The drug improved both memory and cholinergic markers (e.g., ACh concentration, choline acetyl transferase activity) and muscarinic cholinergic receptor binding in frontal cortex and hippocampus of rat brain. Nootropic action was attributed to correction of cholinergic dysfunction.⁵⁷

23.8.1.3.3 *Memorin (Phytopharma)*

Pretreatment with this drug (200 mg/kg/day) attenuated ECS-induced retrograde amnesia in rats. The method used was a passive-avoidance test in a shuttle box.⁵⁷

23.8.1.3.4 *Syrup Shankhpushpi (Baidyanath)*

This drug exhibited nootropic action in various experimental models of cognitive function such as active-avoidance learning in young and old rats, passive-avoidance learning in normal and scopolamine-amnesic mice, transfer latency using elevated plus maze in mice, and food and thirst motivational behavior in rats. It caused significant reduction in brain AChE activity but elicited no significant effects on L-glutamate, L-aspartate, and GABA content of whole brain tissue. The maximum tolerated dose was found to be >40 ml/kg orally, indicating it to be quite safe. Nootropic activity appears to be mediated through cholinergic mechanism.³⁰

23.8.1.3.5 *Saraswatarisht (Baidyanath)*

This formulation (20 ml two times/day for 1 year) showed good antiepileptic activity in 25 patients; there was a significant reduction in the frequency and severity of convulsions and even complete absence of fits in some cases. The drug was found to be free from side effects.^{58,59}

The drug showed cognition facilitatory effects against various experimental models of learning and memory in rats and mice. The effects were duly noted in normal animals, and no activity was discernible against amnesia induced by scopolamine and ECS. The activity appeared to be mediated through cholinergic mechanisms.³⁰

23.8.1.3.6 *Vidyarthi Amrit (Maharishi Ayurved)*

Vidyarthi amrit exhibited nootropic activity in active-avoidance learning in aged rats, passive-avoidance learning in normal mice, and food and thirst motivational behavior in rats. No activity was discernible on transfer latency and against scopolamine- and ECS-amnesic mice. The drug caused a decrease in AChE and L-glutamate and an increase in

L-aspartate and GABA levels of whole brain tissue. The nootropic effects appear to be mediated through cholinergic mechanisms.³⁰

23.8.1.3.7 *Dimagh Pushtak Rasayan (Baidyanath)*

This drug exhibited nootropic effects in active- and passive-avoidance paradigms in both normal and scopolamine-amnesic animals, transfer latency test, and thirst-motivational behavior. No effects were observed against food motivational behavior in the Hebb William maze, possibly because this test involved a single dose schedule which may not be adequate for effective drug concentrations. At least three of its ingredients, *Centella asiatica*, *Convolvulus pluricaulis*, and *siddh makardhwaja*, are reported to have nootropic effects. The memory-enhancing effects of the formulation were attributed to cholinergic mechanisms.³⁰

23.8.1.3.8 *Geriforte (Himalaya)*

Clinical psychobiological studies on apparently normal-aged subjects 45 to 50 years were treated for 3 months with the test formulation. Results showed an increasing feeling of well-being, physical efficiency, and improvement in mental functions. The therapy reduced anxiety and helped in better nitrogen retention. The drug appears to be a rational combination of herbal components for arresting rapid onset of mental and physical disability in aged persons. The effects were attributed to its constituent drugs (e.g., *chyavanprasha* [chief ingredient *Emblica officinalis*]), *Bacopa monniera*, *Withania somnifera*, etc.).⁶⁰

23.8.1.3.9 *Combination of Four Plants: Convolvulus pluricaulis, Bacopa monniera, Withania somnifera, and Acorus calamas*

The combination exhibited encouraging effects in patients of *unmada* (psychosis).⁶¹

23.8.1.4 *Herbs with Allied Activities*

The following is a summary of studies on Indian medicinal plants with properties considered allied to general well-being and nootropic action such as adaptogenic, immunomodulatory, anxiolytic, antioxidant, and antiaging effects.

23.8.1.4.1 *Asparagus racemosus Wild (Shatawari)*

Aqueous extract exhibited adaptogenic activity as evidenced by protection against a variety of biological, physical, and chemical stressors in experimental animals. It also normalized cisplatin-induced intestinal hypermotility.⁶²

23.8.1.4.2 *Bacopa monniera Linn. (Brahmi)*

With this herb, significant anxiolytic effects in experimental animals have been reported.⁶³

23.8.1.4.3 *Boerhavia diffusa Linn. (Punarnava)*

Milk fortified with this plant extract revealed growth-promoting and hematinic effects in children. The plant did not elicit appreciable effect on body weight, total proteins, and hematological parameters of normal adults (45 to 50 years).^{64,65}

23.8.1.4.4 *Centella asiatica Linn. (Mandookparni)*

Aqueous extract of this plant, which is claimed to be a brain tonic, was given at 25 mg/kg i.p. Results showed decreased spontaneous motor activity and delayed pentylenetetrazol-induced convulsions in mice. The activity was comparable with diazepam (4 mg/kg i.p.). The extract potentiated pentobarbitone-induced sleep but did not affect immobil-

ity time in the swimming test. The results suggest an anxiolytic effect but no action on behavioral despair. Another report found antidepressant activity with 50% of ethanolic extract as evidenced by the reduction of immobility time in the forced swimming test. The effect appears to be mediated through D₂ receptor.^{66,67}

Results of a double blind trial in 43 normal adults (45 to 50 years old) after 6 months revealed that the drug caused an increase in erythrocyte count, hemoglobin, blood sugar and serum cholesterol levels, vital capacity, and total proteins. It decreased mean blood urea and serum phosphatase concentrations.⁶⁵

23.8.1.4.5 Syrup Shankhpushpi (Chief Ingredient: *Convolvulus pluricaulis*)

One month treatment with syrup *Shankhpushpi* formulation (10 ml three times/day) elicited beneficial action in 30 cases of anxiety neurosis. The symptoms and levels of plasma cortisol and urinary catecholamines were reduced.⁶⁸

23.8.1.4.6 *Embelia ribes* (Amla)

The protection of *E. ribes* observed against a variety of stressors (biological, chemical, and physical), indicating adaptogenic action. The drug strengthened the defense against free-radical-induced damage during stress.⁶²

23.8.1.4.7 *Piper longum* Linn. (Pippali)

This plant drug exhibited adaptogenic effects similar to *E. ribes* described above.⁶²

23.8.1.4.8 *Terminalia Chebula* Retz (Harar)

This plant drug exhibited adaptogenic effects similar to *E. ribes* described above.⁶²

23.8.1.4.9 *Tinospora cordifolia* F.Vill (Giloe, Guduchi)

Besides adaptogenic activity against different stressors, *T. cordifolia* normalized phagocytic function irrespective of the direction of change. The protection against stress-induced mucosal damage was lost when macrophage activity was blocked.⁶²

23.8.1.4.10 *Withania somnifera* Dunal (Ashwagandha)

This reputed nootropic plant has been extensively investigated by many workers. It showed antistress, anxiolytic, rejuvenating, anticonvulsant, free-radical scavenging, adaptogenic, and antiaging properties in experimental and clinical studies.^{43,46,69–74}

23.8.2 Nootropic Mineral Preparations

Although the majority of the preparations used in Ayurveda as a whole and for memory-enhancing effects are of plant origin, some reports on mineral preparations are also available. The mineral-origin drugs are used mostly in calcined forms. Ayurvedic physicians use even those metals for therapeutic purposes, which are considered toxic and not used for internal administration in modern medicine (e.g., mercury and arsenic). It is believed that the specialized techniques used during preparation of the oxides (*Bhasmas*) (ash) “purify” the metal and make it therapeutically effective and safe for internal use. These techniques involve (1) incorporation of some herbal juices during the calcination process and (2) repeated calcination and trituration to fine powder forms. The mineral preparations, in contrast to herbal drugs, have not received much attention by researchers. The Department of Medical Elementology and Toxicology of Hamdard University, New Delhi, India, ventured into this neglected field and found encouraging results. Some interesting

findings for effects of calcined metal preparations on animal models of learning and memory are summarized below.

23.8.2.1 Siddh Makardhwaja (Mercury)

The preparation helped in total development (physical and mental) of experimental animals. It revealed growth-promoting, memory-improving, and carbohydrate-sparing properties without affecting other central nervous system (CNS) parameters (e.g., pentobarbitone-induced sleeping time, activity index in traction test, activity counts in water wheel, rectal temperature, behavioral despair in forced swimming test, acetic acid-induced writhing episodes, and amphetamine aggregate toxicity) in rats and mice. Facilitation of cognition was shown by marked increase in acquisition, (control = 81.25%, treated = 100%), retention (control = 69.23%, treated = 80%), and learning scores (control = 4.00, treated = 5.26) in the active-avoidance learning test. The drug restored exercise-depleted liver and muscle glycogen concentrations, suggesting antifatigue action. Incorporation of this mineral in the diet (0.01% w/w) of young rat pups (45 to 70 g) for 6 weeks exhibited better growth rate vs. control pups.⁷⁴

23.8.2.2 Swarna Bhasma (Gold)

The preparation (25 mg/kg orally for 7 to 10 days) exhibited a nootropic effect in active- and passive-avoidance learning in rats and mice. It caused significant increase in acquisition, retention, and learning scores in treated rats vs. the control group. The results compared well with panax ginseng tea (350 mg/kg orally for 10 days). Significant reduction in latency to reach shock free-zone and stepdown errors was observed in treated mice. Brain acetyl cholinesterase was significantly decreased in the frontal cortex, hypothalamus, hippocampus, midbrain, cerebellum, brain stem, and corpus striatum after the 10-day treatment in rats. Decreased AChE activity possibly led to increased ACh levels in brain with an improvement in cognitive function. The drug also exhibited glycogen-sparing, adaptogenic, and immunomodulatory activities. The drug showed a wide therapeutic index (effective dose = 25 mg/kg orally, maximum tolerated dose = >2 g/kg orally) and lack of gross behavioral effects, weight loss, or adverse effects on hematological parameters.^{75–79}

23.9 Summary and Discussion

The revival of Ayurvedic medicine with special emphasis for research on medicinal plants has received considerable attention in India during the past 4 to 5 decades.⁸⁰ Research effort on medicinal plants and nootropic drugs that act on CNS gained momentum during the past 5 to 10 years. This has generated substantial data, giving a rational basis to ancient Ayurvedic remedies for deficits in learning and memory. Experimental and clinical data presented earlier provide support for the use of these herbs for the management of AD. However, further research need to be conducted to better utilize these herbs.

In conventional medicine, the drugs currently approved for the treatment of AD are cholinomimetics, such as tacrine, which has the pharmacologic profile of AChE inhibitors.^{12,25,26} Modern drugs used for ameliorating intellectual impairment, confusion, behavioral disorders of senility, mental retardation, and learning problems in children

include CNS stimulants and activators such as pyritinol (Encephabol) and nootropic agents such as Piracetam (Cerecetam, neurocetam, Nootropil, Normobarin, and Pirtam). Some of the adverse effects associated with these drugs include anorexia, epigastric distress, nausea, vomiting, headache, fatigue, excitement, restlessness, sleep disturbances, and skin rash. In addition, special care is required if these drugs are used for patients with hepatic, renal, or cardiac diseases. Ayurvedic herbal preparations have not been shown to produce adverse side effects. They offer a safe alternative in the management of AD.

In a recent study, leisure activities such as reading, playing board games, musical instruments, and dancing are associated with a reduced risk of dementia. This is consistent with the Ayurvedic concept of regular physical and mental exercise, and meditation as a part of disease prevention and treatment strategy.⁸¹

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24.1 Introduction

Epilepsy is one of the most common neurological disorders and afflicts more than 50 million people worldwide.¹ Data from the World Health Organization (WHO) suggest that as many as 1 in 20 people at some point in their lives have an epileptic seizure and that at least 1 in 200 people have epilepsy.² Over four fifths of the 50 million people with epilepsy are thought to be in developing countries, and 90% of these people do not receive appropriate treatment.³ In India, it was estimated that by the end of the year 2001, the number of patients afflicted with epilepsy would be 5.5 million and the number of new cases each year would be close to half a million.⁴ In India, over 70% of the population is rural, whereas 70% of the medical manpower, and nearly 100% of neurologists, are urban based.⁵

Facilities for neurological investigations such as the computerized tomography (CT) scan, magnetic resonance imaging (MRI), and electroencephalogram (EEG), as well as most first-line and second-line antiepileptic drugs, are beyond the reach of a vast majority of patients. Against this backdrop, the Ayurvedic management of epilepsy, which is sometimes the only form of treatment available to the rural population, assumes great importance. Epilepsy was first documented in the ancient Ayurvedic literature as *apasmaara* in

Sanskrit. This review provides a comparative study of modern and Ayurvedic medicine in hopes of providing the scientific basis for the use of Ayurvedic therapies in epilepsy. In addition, our own experiences in the management of epilepsy with the use of therapeutic drug monitoring, and the identification of interactions of allopathic and Ayurvedic antiepileptic drugs are also discussed.

24.2 Definition

Apasmaara, or epilepsy, in Ayurveda has been described as one of the earliest eight diseases known (diagnosed) and can be controlled only with medical therapies that can sometimes be incurable and remain uncontrolled.⁶ The parallels can be found in allopathic medicine where, despite the best available treatment, 10 to 12% of the patients still have uncontrolled seizures.⁷ Epilepsy as per Ayurveda is defined as the transient derangement of memory, intelligence, and the mind, resulting in a temporary blackout of vision, loathsome activities, and unconsciousness. *Vagbhata* defines it as loss or destruction (*apaya*) of memory (*smriti*).⁸ Modern medicine defines a seizure as a transient alteration of behavior due to the disordered, synchronous, and rhythmic firing of populations of neurons. Epilepsy per se is defined as a disorder of brain function characterized by the periodic and unpredictable occurrence of seizures.⁷ Whereas the Ayurvedic definition describes the effect, the modern definition describes the cause of epilepsy.

24.3 Clinical Description

The clinical description as mentioned in Ayurvedic texts is classified as prodromal signs and symptoms (*purvarupa*) and clinical manifestations (*rupa*). The prodromal signs and symptoms include raising of eyebrows, frequent abnormal movements of eyes, hearing of sounds not perceived by others, excessive oozing of saliva and nasal mucus, aversion to food, anorexia, indigestion, distension of abdomen, debility, body ache, transient blackout, giddiness, profuse sweating, increased thirst, fainting, hallucinations, delusions, falling, aura, and insomnia.⁶ This description is very similar to the description in modern medicine of generalized tonic clonic (GTC) seizures. These seizures are characterized by an abrupt onset, sometimes with premonitory symptoms, loud cry, excessive salivation, postictal unresponsiveness, headache, muscle ache, fatigue, and an increase in heart rate and pupillary size.⁹ The loss of memory as described by *Vagbhata* is similar to the description of absence seizures, which are characterized by loss of consciousness for a longer duration and are less abrupt in onset and cessation.

In Ayurveda, epilepsy is divided into four types according to the dominant *dosa* involved in the disease pathogenesis: *vataja*, *pittaja*, *kaphaja*, and *sannipataja*; signs and symptoms manifest accordingly. The *vataja* type involves *vata* as the dominating *dosa* and is characterized by having frequent fits; regaining consciousness in the shortest time interval, bulging eyes, excessive crying, frothing at the mouth, irregularly contracted fingers, reddish, rough, and blackish nails, eyes, face, and skin, hallucinations, trembling, and visions of unstable, coarse, and rough objects. *Pitta* is the leading *dosa* in *pittaja* type with distinctive features of regaining consciousness in shorter periods, scratching the ground, green-

ish-yellowish and coppery nails, eyes, face, and skin, and visions of bloody, agitated, irritated, frightful, and burning objects. The *kaphaja* type denotes *kapha dosa* as the principal *dosa* including features such as delayed fits, delayed recovery, increased frothing at mouth, white nails, eyes, face, and skin, and visions of white, heavy, unctuous, smooth objects. *Sannipataja* means all the three *dosas* are in conjunction with each other. It is their simultaneous vitiation that gives rise to the combination of signs and symptoms.

Vataja and *kaphaja* types match the description of GTC seizures, *pittaja* matches complex partial seizures, and *sannipataja* matches the characteristics of a mixed-seizure profile.

24.4 Etiology (*Vyaadhi hetu*)

As per the classical texts, the basic etiology is threefold for epilepsy. Endogenous factors (*nija*) include genetic, congenital, constitutional (*prakriti*), enzymatic disturbance (*agni vikruti*), and idiopathic. Exogenous factors (*agantuka*) include habitual intake of unwholesome and unhygienic foods and drinks that have mutually contradictory properties (*apathya aahar*), especially those with properties similar to *vata dosa* causing its aggravation and trauma; this aggravation is due to worms (*krimijanya*) and environmental and idiopathic factors. Psychological trigger factors (*manasika*) include excessive worry, grief, fear, passion, anger, anxiety, attachment, and excitement. Modern medicine describes the etiology of epilepsy in a similar manner while giving an agewise classification of the causes. In neonates, genetic disorders, metabolic disturbances, and acute central nervous system (CNS) infection are the common causes. Infants and children have genetic disorders, febrile seizures, traumas, and CNS infections as common causes. Etiology for adults includes trauma, infection, tumor, and metabolic disorders in the case of older adults.⁹ The relationship between the mind and the somatic response is well established.¹⁰

24.5 Pathogenesis

Of the three *dosas*, *vata* is the rudimentary element responsible for all the beneficial and harmful effects in the body in its normal and abnormal states, respectively, as mentioned in the authentic text, *Charak Samhita*. In epilepsy, *vata* is predominately vitiated and is the fundamental cause in the pathology of the disease. Neurological disorders, as understood in Ayurveda, are considered to be due to imbalance of *vata*.¹¹

All the three etiological factors contribute in the accumulation and vitiation of the psychosomatic *dosa*, specifically *vata* affecting consciousness and all the sense organs. After it is aggravated, the *dosa* spreads throughout the body and through the nerves (*dhamanis*), leading to a manifestation of the epileptic episode. The agitated *vata dosa* abruptly proceeds through the nerves of the body, shaking it in a quick succession (shaking jerks) called "*akshepaka*," which means convulsion.¹² Another term for this condition is *apatana* where the patient falls on the ground without spasm at intervals that could be correlated with absence-seizure type of epilepsy.

According to modern medicine, epilepsy results from a focus of hyperexcitable neurons in the cortex. During seizures, the permeability of the cytoplasmic membrane of the neurons changes. This results in increased levels of intracellular calcium and extracellular

potassium, which contributes to overall excitability of the epileptic neuronal aggregate. The neurons become susceptible to hypoglycemia, hyponatremia, hypocalcemia, sleep deprivation, and photic stimulation.^{7,9} An epileptic seizure has two phases: initiation phase and seizure propagation phase. The initiation phase is characterized by two concurrent events: high frequency bursts of action potentials and hypersynchronization. In epileptogenesis, a normal neuronal network transforms into one that is chronically hyperexcitable. Ayurveda also has the concept of seizure propagation where the *vata dosa* spreads rapidly throughout the body.

24.6 Clinical Examination and Diagnosis

Following the principles of an Ayurvedic diagnosis, a comprehensive examination of the patient (*rogi pariksha*) precedes the disease diagnosis (*roga pariksha*). As in the modern approach of history taking, Ayurveda also emphasizes on the detailed history of the patient as well as features of the case definition of the disease. The type of epilepsy is diagnosed as explained under clinical description. Today, even Ayurvedic physicians establish their diagnosis utilizing modern techniques such as the EEG, CT scan, and MRI. As per Brodie,¹³ the choice of treatment in newly diagnosed epilepsy should take into consideration the patient's age, general health, coexisting disabilities, concomitant medication, and lifestyle that is consistent with the Ayurvedic principles of treatment.

24.7 Clinical Course and Prognosis

According to Ayurveda, because epilepsy arises from the aggravation of the *dosas* of the body and mind together and is localized in an important vital organ (*mahamarma*), the brain (*shiromarma*), it is difficult to treat and cure. Sannipatika-type epilepsy, where all three *dosas* are vitiated, is incurable. Chronic epilepsy and the presence of concomitant diseases compound the problem. The case is worsened if the underlying *dosa* and the constitution (*prakriti*) of the patient are similar. For example, if the patient is of *vata pitta* constitution, his or her epilepsy is more difficult to treat than that patient with a different constitution.⁶ Modern medicine determines the prognosis of epilepsy by the epilepsy syndrome type; these are classified into four groups: excellent, good, uncertain, and poor prognosis.¹⁴ Interestingly, the terms are strikingly similar to the Ayurvedic concept of prognosis terms of *sukhsadhya*, *kashtasadhya*, *yapya*, and *pratyakheya*, respectively.

24.8 Management

All the three streams of Ayurvedic therapy are used in treatment of epilepsy: divine therapy (*deva vyapashraya*), rational therapy (*yukti vyapashraya*), and psychotherapy (*satvavajaya*). Divine therapy is often practiced with the principles of astrology, stones, religious scriptures, and auspicious rites. Rational therapy mainly includes purification

and evacuatory (*samshodhana* and *panchakarma*) and curative and pacificatory (*samshamana*) with diet, drug, and lifestyle modification followed by restorative therapy (*rasayana*). Psychotherapy is practiced by incorporating the philosophy of assurance therapy (*aashwasana*) and the substitute of passions from negative to positive. Psychoneuroimmunology is a well-known concept in modern medicine where interaction between the brain and immune system is studied and used to treat patients. Consulting to produce positive emotions is an important aspect of the therapy.¹⁵ The potential for behavioral interventions to affect psychological adaptations and the course of immune-related diseases should be investigated. Physicians should first restore the activities of the heart, mind, and channels (*srotas*) occluded by the *dosa* by using *samshodhana chikitsa*. *Vataja* should be treated with enema therapy (*basti*), *pittaja* with purgation therapy (*virechana*), and *kaphaja* with emesis therapy (*vamana*). This is followed by medicinal recipes. *Samshamana* therapy does not work well if purification is not made a priority.⁶ A study by Nagashayana et al.¹⁶ establishes the necessity of *panchakarma* therapy in Ayurveda medication before palliative therapy.

Primary therapy described in epilepsy is oleation (*snehana*) where *narayan tail* is commonly used followed by fomentation (*swedana*). These are also the preparatory measures for the *panchakarma* described in Chapter 4. Keeping in mind the particular patient, a physician may use emesis therapy in epilepsy where drugs are suitably prepared; a common vehicle is honey and rock salt followed by a postemesis regimen. Nasal insufflation (*nasya*) is also an important treatment wherein herbal powders and medicated oils are administered in the nostrils as per the case; *vacha* powder and oil, *anu tail*, *panchendriya vardhan tail*, and fresh juice of *nirgundi* are commonly used formulations.¹⁷ *Shirodhara* is another part of the treatment (*shiro* = head and *dhara* = slow uniform dripping of a liquid substance). This is basically the pacificatory therapy wherein liquid medicaments like medicated oil, *ghrita* (clarified butter made out of cow's milk), decoctions of various suitable herbs, and buttermilk are allowed to trickle slowly on the forehead of the patient in a supine posture.

Formulations used in the treatment are as follows:

1. *Brahmi ghruta* containing *vacha*, *brahmi*, *kushtha*, *shankhpushpi*, and *ghruta*
2. *Kushmanda ghruta* made by cooking *ghruta* with 18 times its quantity of *kushmanda* juice and paste of *yashtimadhu*
3. *Kalyanaka churna* containing 17 herbs
4. *Sarasvata churna* containing 12 ingredients having *vacha* as the main constituent

The treatment of chronic epilepsy includes garlic with oil, *brahmi* juice with honey or *vacha* powder with honey, and *shatavari* with milk.

Strong emetics cannot be given in an acute attack of epilepsy but could be beneficial after the patient is stable and suitable to receive the therapy following the guidelines of authentic texts. In the current practice, Ayurvedic physicians use conventional medicines in the acute phase and subsequently *panchakarma* therapy followed by *rasayana* therapy to get the optimum benefit for the patient. It is shown that the blood histamine level decreased from 0.9 to 0.37 µg/ml by *vamana*, and the same is emitted in vomitus. This indicates elimination of toxic materials (*sukshma malas*) from the cellular level.¹⁸

The management of elderly patients is also an important issue.¹⁹ The elderly may have many disorders, take multiple concomitant medications, and have different metabolic features; they are also more sensitive to the adverse effects of drugs. The incidence of epilepsy in the elderly is rising; an ideal anticonvulsant for use in an elderly patient in modern medicine is not currently available.²⁰ Furthermore, febrile seizures are the most

common convulsive events in childhood, occurring in 2 to 5% of children. Approximately 20 to 30% of these children may have recurrence during a subsequent febrile infection. These could be nonepileptic. Because the side effects of the antiepileptic drugs outweigh their benefits, their continuous use is no longer recommended.²¹ In both these circumstances, Ayurveda as an individualized therapy could be of great value.

The dos and don'ts (*pathyapathya*) for an epileptic patient have been mentioned in classical texts. The beneficial regime (*pathya*) includes a hot-water shower, consuming an ample amount of clarified butter (*ghruta*) as per the digestive capacity, massaging the soles in the evenings with *narayan tail* followed by hot water, fomentation, and living in a pleasant atmosphere. Furthermore, the patient should never go out alone and never be around a lake or fireplace or ride a horse. Patients should not be told any good or bad news suddenly so that they get emotionally disturbed. Products and regimes causing aggravation of *vata dosa* should be strictly avoided. The parallels in allopathy can be seen in the use of prophylactic therapy, a ketogenic diet, and mono- and polytherapy based on an individual patient's response.

24.9 Cost of Epilepsy Management

The costs of Ayurvedic medicines and conventional therapy for epilepsy are more or less similar in India contrary to the West.²² However, in epilepsy treatment due to milder and fewer adverse effects and toxicities than modern medicine, the ultimate expenses would be certainly less than the conventional therapy. In many countries the vast majority of sufferers remain untreated.²³ The use of Ayurvedic therapies could help solve this problem worldwide. The Ayurvedic formulations used in the treatment of epilepsy available in the Indian market are given in [Table 24.1](#)²⁴ and the conventional medicines are given in [Table 24.2](#).²⁵

24.10 Scientific Basis

Conventional drugs for epilepsy should be further investigated for a safe and effective drug because none is available at this time. Carbamazepine treatment has potential risks that should be weighed against its benefits before initiating the therapy.²⁶ Weight gain in patients on antiepileptic drugs disturbs the general health. It causes cosmetic adverse effects and can have serious psychological effects that may lead to withdrawal of the drug.²⁷ Hepatotoxicity and teratogenicity of antiepileptic drugs like valproate are rare, but severe adverse effects have been reported.²⁸ Hepatoprotective Ayurvedic therapies may be used as an adjuvant to protect from hepatotoxicity of conventional drugs. All conventional antiepileptic drugs may provoke positive or negative psychiatric reactions in individual patients, and these reactions depend on the strength of the drugs and genetic and biographic psychiatric predisposition of the patient.²⁹ These adverse side effects can be prevented using Ayurvedic therapies where the concept of *prakriti* plays a vital role in the treatment.

Although they provide a scientific basis with modern parameters for Ayurvedic antiepileptic drugs, the majority of the herbs are investigated only in animal models. Rats and

TABLE 24.1

Ayurvedic Medicines for Epilepsy Available in the Indian Market

Formulation	Dose	Manufacturers ^a	Ref. ^b
<i>Asvagandhadyarishta</i>	12–24 ml twice/day	2–6, 8–10, 13	BR
<i>Bala tail</i>	For <i>snehana</i> and <i>abhyanga</i> (local application)	4, 9, 10	AH
<i>Brahmi ghruta</i>	12 g/day	3, 6, 7, 9, 10, 12, 13	AH
<i>Chandanadi tail</i>	For <i>snehana</i> and <i>abhyanga</i> (local application)	3, 4, 9–11, 13	YR
<i>Chaturmukha rasa</i>	125 mg twice/day	2, 6, 13	BR
<i>Haratala bhasma</i>	125–250 mg/day	2, 6	AP
<i>Kalyanaka ghruta</i>	12 g/day	4, 9–11	AH
<i>Kumaryasava</i>	12–24 ml twice/day	2, 4–6, 8–10, 13	YR
<i>Mahakalyanaka ghruta</i>	12 g/day	9, 10	AH
<i>Mahamrutyunjaya rasa</i>	125 mg twice/day	2, 3, 6, 9	ASS
<i>Rajata bhasma</i>	125 mg twice/day	2, 7, 8, 11, 13	Rasamrita
<i>Saarasvatarishta</i>	12–24 ml twice/day	2–6, 8–11, 13	BR
<i>Sarpagandha vati</i>	2–3 pills twice/day	2, 6, 11, 13	ASS
<i>Svarna bhasma</i>	15.5–62.5 mg/day	6, 7, 9, 11, 13	Rasamrita
<i>Svarnamakshika bhasma</i>	125–250 mg/day	2, 6–8	RSS
<i>Vaatakulantaka rasa</i>	125–250 mg/day	3, 6, 8, 12	BR
<i>Yogendra rasa</i>	125–250 mg/day	6–8, 11	BR

^a Numbers denote the manufacturer in the Table. (1) Arya Vaidya Sala, Kottakkal, 676 503, Kerala; (2) Dabur Research Foundation, Ghaziabad, 201 010, Uttar Pradesh; (3) IMIS Pharmaceuticals Pvt. Ltd., Dubagunta nivas, Kari Marx Road, Vijayawada, 520 002; (4) Kerala Ayurveda Pharmacy Ltd., Athani Post Office, 683 585, Aluva, Kerala; (5) Sandu Brothers Pvt. Ltd., Sandu Nagar, DK Sandu Marg, Chembur, Mumbai, 400 071; (6) Shree Baidyanath Ayurveda Bhawan Pvt. Ltd., Bangani, Gopalganj, M.P.; (7) Shree Bhuvaneshwari, Aushadhashhram, Gondal, 360 311, Gujarat; (8) Shree Dhootapapeshwar Ltd., 135, Nanubhai Deesai Road, Khetwadi, Mumbai, 400 004; (9) The Arya Vaidya Pharmacy (Coimbatore) Ltd., 326, Perumal Koli Street, Ramanathapuram, Coimbatore, 641 045; (10) Vaidyaratnam Oushadhasala, Ollur, Thaikattussery, Thrissur, 680 322; (11) Venkateswara Ayurveda Nilayam Pvt. Ltd., Chintalru, 533 232 A.P.; (12) Vyas Pharmaceuticals, 98-C, Sanver Road, Sector-E, Indore; (13) Zandu Pharmaceutical Works Ltd., 70 Gokhale Road (South), Dadar, Mumbai, 400 025.

^b BR = *Bhaishajya ratnavali*; AH = *Ashtang hridiya*; YR = *Yoga ratnakar*; RSS = *Rasendra sara sangraha*; AP = *Ayurveda prakash*; ASS = *Ayurveda sara sangraha*.

mice are subjected to kindling, which involves periodic administration of low-intensity electrical stimuli. Systemic administration of convulsant chemical agents, such as pentylenetetrazole and pilocarpine, are also used for animal models to study the antiepileptic activity of the drugs.

In one study, the first ultrastructural evidence of neuroprotective properties in *Semecarpus anacardium* and *Withania somnifera* has been reported.³⁰ Today's molecular pharmacology has made it possible to understand the mechanism of the action of Ayurvedic drugs. *Ashwagandha* (*W. somnifera*) is gamma-aminobutyric acid-A (GABA-A) receptor agonist, and *katuka* (*P. kurroa*) has antioxidant action equal to alfa-tocopherol, which has an effect on glutathion metabolism in the liver and brain.³¹ It is believed to maintain the levels of cellular endogenous antioxidants in liver and brain. Subacute toxicity studies of drugs like *W. somnifera* did not reveal any toxicity.³² A methanolic extract of *W. somnifera* root contains an ingredient that has a GABA-mimetic activity.³³

A clinical trial of rejuvenating drugs for the nervous system (*medhya rasayana*) on epilepsy suggested that these medicines reduce the frequency, duration, and severity of

TABLE 24.2
Allopathic Antiepileptic Drugs

Generic Name	Dose (mg/day)
Phenytoin sodium (100 mg)	100–600
Carbamazepine (200 mg)	200–1200
Phenobarbitone (30 mg)	30–210
Sodium valproate (200 mg)	600–2500
Lamotrigine (25 mg)	25–150
Primidone (250 mg)	125–500
Oxcarbazepine (300 mg)	600–1200
Topiramate (25 mg)	25–800

seizures and have not shown any side effects.³⁴ *Brahmi rasayan*, an Ayurvedic preparation, was studied in mice and rats for its effects on the CNS at oral doses ranging between 1 and 30 g/kg. The study suggests an involvement of the GABA-ergic system in the mediation of the CNS effects of *brahmi rasayan*.³⁵ Piperine, an active alkaloid of *Piper longum* and *Piper nigrum*, exerted a significant protection against tertbutyl hydroperoxide and carbon tetrachloride hepatotoxicity by reducing both *in vitro* and *in vivo* lipid peroxidation, reducing enzymatic leakage of glutamate pyruvate transaminase (GPT) and alkaline phosphatase (AP), and preventing the depletion of reduced glutathione (GSH) and total thiols in the intoxicated mice.³⁶ Piperine and its derivatives are effective anticonvulsant drugs that antagonize convulsions induced by physical and chemical methods. They also have sedative, tranquilizing, and muscle-relaxing effects.³⁷

The effects of piperine on convulsions induced in mice by agonists at different excitatory amino acid receptor subtypes were studied. Piperine significantly blocked convulsions induced by intracerebroventricular injection of threshold doses of kainate.³⁸ Toxicity evaluation of potassium embelate from *Embelia ribes* Burm. did not indicate adverse effects.³⁹ Vohora et al.⁴⁰ screened the ethanol extract of *Acorus calamus* rhizomes for CNS effects and found that it exhibited a large number of actions similar to alpha-asarone (an active principle of *A. calamus*). The effect of acute and subchronic administration of an alcoholic extract of *Nardostachys jatamansi* roots on norepinephrine (NE), dopamine (DA), serotonin (5-HT), and GABA were studied in male albino Wistar rats. A 15-day treatment resulted in a significant increase in the levels of NE, DA, 5-HT, and GABA, which indicates that the extract causes an overall increase in the levels of central monoamines and inhibitory amino acids.⁴¹

The anticonvulsant activity of *Myristica fragrance* was studied against seizures induced by maximum electric shock (MES), pentylenetetrazole (PTZ), lithium sulphate-pilocarpine nitrate (Li-pilo), and picrotoxin. Results showed that *M. fragrans* inhibited seizures induced by MES, PTZ, and Li-pilo, but not by picrotoxin.⁴² An animal study with *Sesbania grandiflora* (*Agastya*) showed a wide spectrum of anticonvulsant profile and anxiolytic activity.⁴³ *Brahmi rasayan* offered a graded protection against audiogenic seizures in mice and antagonized the mescaline-scratch response in mice, indicating its tranquilizing properties.⁴⁴

The therapeutic efficacy of an Ayurvedic formulation is due to actions of various components of the plant and their interactions. Studies on the efficacy of the whole plant cannot be confirmed by fragmented research using the plant extracts or isolated phytochemicals.

24.11 Drug Interactions in Epilepsy Management

In India, Ayurvedic medicines have been used for thousands of years, and adverse drug reactions are also fewer in frequency and severity.^{45,46} These facts lead to the concomitant use of both Ayurvedic and modern medicine by the patient without the physician's knowledge. This can ultimately lead to drug interactions, which could be useful or harmful for the patient. At our therapeutic drug-monitoring clinic, we observed two patients who had well-controlled seizures who presented with sudden loss of seizure control. A detailed history taking revealed that they had started taking *shankhapushpi*, a purported memory enhancer. Animal studies with rats were carried out with single and multiple doses of the two drugs given alone and in combination. It was seen that on chronic administration, both the antiepileptic effect and plasma levels of phenytoin were reduced by *shankhapushpi*. Single-dose administration of the drugs did not lead to any change in phenytoin levels but decreased its antiepileptic effect. The interaction was found to be pharmacokinetic and pharmacodynamic.^{47,48}

Another marketed memory enhancer, Mentat, has been shown to increase phenytoin levels in rabbits.⁴⁹ Likewise, studies on the possible interactions between traditional Chinese medicines and antiepileptic drugs are in progress.⁵⁰ Honey, a substance widely used as a good vehicle for a given Ayurvedic medicine, has shown to decrease the bioavailability of carbamazepine.⁵¹

24.12 The Future

Instead of using single-drug or various formulations mentioned in the authentic texts, the principle of threefold therapy should be validated where yoga can be practiced under divine therapy. Several parameters have been proposed to monitor the level of *tridosha*, where *vata* can be monitored in terms of membrane-bound signal transduction.⁵² With reference to epilepsy, a *vata*-dominant disease, such a hypothesis should be validated. Allopathic and Ayurvedic practitioners and researchers should also plan effective health-care delivery systems for epilepsy to improve the overall quality of life. Traditional systems of medicine like Ayurveda in India are here to stay. It is also inevitable that they will be used concomitantly with the allopathic system of medicine in the years to come. The worldwide herbal medicine market has grown by leaps and bounds. In the light of this, an understanding of the scientific basis of Ayurvedic therapies based on systematic studies of principles and practices and using current scientific principles and technology would certainly help contribute toward better patient care.

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25

Psychiatric Disorders

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25.1 Introduction

The discipline of psychiatry according to Ayurveda has its roots in the four ancient books of knowledge called *Vedas of India*, where science and knowledge of an evil spirit (*bhutavidya*) is mentioned vividly. As a matter of fact, the word *bhutavidya* of *Vedas* refers to Ayurveda as a whole. There is no direct mention of *tridosha* doctrine (*vata*, *pitta*, and *kapha*) of Ayurveda in *Vedas*. As such, *bhutavidya* may be considered the mother of medicine as a whole in ancient India.¹

Psychiatric disorders are very prevalent diseases. They accounted for 10 to 15% of the prescriptions written in 1996 in the U.S.² The diagnostic criteria for psychiatric disorders in the U.S. are described in the American Psychiatric Association's 2000-941 DSM IV TR publication, *Diagnostic and Statistical Manual of Mental Disorders*.³ Since the mid-1950s, research and development efforts in drug therapies for psychiatric disorders have led to the discovery of many useful drugs.² These drugs, however, are known to have significant adverse side effects. Because Ayurvedic dietary herbal supplements are relatively safe, their use in managing various psychiatric disorders is explored here.

25.2 Basic Ayurvedic Concept of Psychiatry

The contemporary Ayurvedic psychiatry in terms of concepts consists of two components, namely:

1. Ayurvedic science of psychiatric disorders (*Ayurvediya manas roga vijñana*) or rational Ayurvedic psychiatry — This deals with clinical conditions where the disease and its treatment are based on fundamental principles of Ayurveda (i.e., theories of five elements [*pancamahabhuta*], *tridosa*, *triguna*, etc.). The diseases such as psychosis (*unmada*) and others are well characterized.
2. Demonology (science and knowledge of evil spirits) (*bhutavidya*) — This deals with psychiatric problems like psychosis (*bhutonmada*, *grahavesa*) in a nonconventional manner. Psychiatry emerged to the present scientific shape from demonology, where the disease and its treatment is not based on classical principles of Ayurveda and science (*vijñana*). Rather, it is based on paranormal factors like the doctrine of deeds (*karma*), planets (*graham*), evil spirits (*bhuta*), and others. The latter appears to be based more on astrology than on psychiatry. Sometimes this aspect of the psychiatric disorders, such as *bhutavidya*, is equated with demonology. It seems that different kinds of psychosis and effects of planets described in ancient texts are nothing but different forms of psychiatric syndromes or sets of behavioral alterations named symbolically after the name of different planets because of the similarity with their mythological descriptions. The element of demonology exists in Ayurveda but only symbolically.

In Ayurveda, the entire concept of life, health, and disease revolves around the classical theory (*loka-purusa samya*) that proclaims that the individual living being is a miniature replica of the universe; the universe (*loka*) and individuals (*purusa*) exist as a continuum of each other. The individual life entity is called *ayu*, which is four dimensions composed of physical, sensorial, mental, and spiritual attributes. The human being is provided not only with sensorial apparatus (*jnanendriyas*), but also with a highly dynamic psyche and mind (*manas*). According to Ayurveda, the mind is highly active but is unconscious (*acetana*). It derives its consciousness from the soul (*atma*) — an extension of the cosmic or divine consciousness.

The mind (*manas*) is further considered to have three dimensions in terms of the three properties or attributes (*gunas*): *sattva*, *rajas*, and *tamas*. The *rajas* represent activity and dynamism, whereas the *tamas* denotes inertia and darkness. *Sattva* is the state of pure mind with absolute balance when both the extreme qualities of mind, namely, *raja* and *tamas*, cease or merge into each other. It is believed that all mental illnesses are because

of the *rajas* and *tamas*. These properties are also called mental (*manasa*) *dosas* in the same way as *vata*, *pitta*, and *kapha* are called body (*sarira*) *dosas* and all kinds of physical diseases are attributed to them.

In consideration of the *trigunas*,⁴ Ayurveda postulates the idea that there can be three broad categories of mental personalities (*prakritis*): *sattvika prakrti*, *rajasa prakrti*, and *tamasa prakrti*.⁵ These categories are further divided on the basis of finer considerations into 16 mental (*manasa*) personalities or mental traits. These 16 personality traits are characterized with unique features that may also predispose specific mental diseases simulating 16 personality factors (16 PF) of modern psychology.

The entire concept of the mind (*manas*) is psychological. Its neurophysiological attributes have not been described vividly in Ayurveda. However, the text *Bhela Samhita* states that the mind (*manas*) is located in the skull (*sirstalvantrgatam manah*). There are also references describing heart (*hrdaya*) as the seat of consciousness (*cetana*). Ayurvedic texts consider mental health a state of sensorial mental, intellectual, and spiritual well-being. Ill mental health is brought about essentially as a result of unwholesome interaction between the individual and his environment. This interaction operates through three fundamental factors: time rhythm (*kala*), intellect (*buddhi*), and sensorial inputs (*indriyartha*).

25.3 Etiology

Ayurveda believes in the theory of rebirth (*punarjanma*) and actions of past life (*karma*). Accordingly, clusters of mental illnesses primarily fall into two etiological areas: the hereditary and the environment.

Etiological factors in the diagnosis of psychiatric disorders are not clearly defined. Categorical relationships between disorders and the symptom constellation provide an important key in understanding how heterogeneous symptoms may descend upon a final common pathway. Consequently, a number of disorders exist as discrete mental disorders but rely on an aggregate of behavior that may be common among other families of mental disorders.

The role of environmental factors has not been adequately ruled out. In fact, socioeconomic, environmental, and dietary factors are currently considered causal, based on epidemiological studies for many diseases previously thought to be related to genetics (e.g., cancer).⁶ The same may be true for psychiatric disorders.

25.4 Psychopathology

The psychopathology in Ayurveda is described by Charaka Samhita, [Chapter 7](#) (*Nidana Sthana*), in a very systematic manner in terms of eight essential psychological factors considered to be centrally affected in all psychiatric disorders in varying degrees⁷:

1. Emotion, mood, affect (*mana*)
2. Thought and decision (*buddhi*)
3. Orientation (*sajna-jnana*)

4. Memory and learning (*Smrti*)
5. Desire (*bhakti*)
6. Habits (*sila*)
7. Psychomotor function (*cesta*)
8. Conduct and behavior (*acara*)

25.5 Clinical Description of Psychiatric Disorders

A wide range of psychiatric conditions and affective behaviors have been described in Ayurveda. Besides their aetiopathogenesis, the texts describe briefly but vividly the signs, symptoms, and behavioral alterations in different psychiatric diseases and their classifications. The psychiatric conditions described in original texts may be classified as follows:

1. Primary psychological conditions caused purely by *manasa dosas* (i.e., *rajas* and *tamas*). An example is the set of *manas dosas vikaras*⁷: lust (*kama*), anger (*krodha*), greed (*lobha*), delusion (*moha*), jealousy (*irsya*), pride (*mana*), euphoria (*mada*), sorrow (*dukha*), grief (*soka*), anxiety (*cinta*), neurosis (*udvega*), fear (*bhaya*), and happiness (*harsa*).
2. Psychiatric conditions, neurological problems, and symptoms caused by the mixed pathology (*samprapti*) including both the body *dosas* (*vata*, *pitta*, *kapha*) and mental *dosas* (*raja*, *tama*). Examples include psychosis (*unmada*), convulsive diseases (*apasmara*), hysteria (*apatrantra*), obsession (*atattvabhiniवेश*), vertigo (*bhrama*), drowsiness (*tandra*), neuresthenia (*klama*), comas (*mada-murccha-sanyas*), alcoholism (*madatyaya*), and hypochondriasis (*gadodvega*).
3. Personality (*prakrti*) disorders: There are 16 mental disorders (*manasa prakriti*) representing 16 types of behavior traits. When such personality behavior becomes overt and overrides the range of normalcy, the patient may have a psychiatric disorder.
4. Mental retardation (*buddhimandya*).
5. Neurological and behavioral problems of the elderly (*jara-janya manasa vikara*).
6. Psychosomatic diseases, where the cause of disease is mental and manifestation is somatic (*manodaihika vyadhis*, [e.g., *sokatisara*]).

The psychiatric syndromes named symbolically after the name of planets (e.g., *bhuton-mada* and *grahavesa*) warrant prayers (*daivavyapasraya cikitsa*). This is because herbal and purification treatments (*yuktivyapasraya cikitsa*) based on fundamental doctrines of Ayurveda are not possible.

In conventional medicine, mental diseases in general are characterized by altered behavior. The clinical diagnosis is done on the basis of the pattern of alteration of the behavior and certain associated signs and symptoms. Common psychiatric diseases described in the conventional medicine⁸ and Ayurveda⁷ are (1) psychosis (*unmada*), (2) depressive illness (*cittavasada*), (3) anxiety disorders (*cittodvega*), and (4) somatoform disorders, alcoholism, and drug abuse (*mada*). These diseases are briefly discussed here.

25.5.1 Psychosis

Psychosis is described in Ayurveda as a major psychiatric disease. There are several types identified in terms of the set of signs, symptoms, and the pattern of behavior alterations referable to the three *dosas* and their combinations.

A separate category of psychosis, described as *agantuja unmada* by Charaka and as *bhutonmada* by Susruta, forms an interesting dimension of Ayurvedic psychiatry. In this context about a dozen psychiatric syndromes are described and named after different planets or divinities (*grahas*). Thus, *bhutonmada* is a specialized aspect of Ayurvedic psychiatry using symbolic terms and need not be pushed into the realm of demonology.

In the conventional medicine, schizophrenia is a major psychotic (presence of hallucinations or delusions) illness. The major feature of schizophrenia is a decline in psychosocial functioning, with a tendency to move down in a person's social circle. Some patients may have acute full-blown symptoms, whereas others may have chronic and less severe symptoms. Characteristic symptoms are delusions, hallucinations, disorganized speech, grossly disorganized behavior, and hallucinations consisting of a voice or two voices that converse with each other and affect the behavior or thought of the patient. Prevalence of the disease is about 1% for lifetime risk. Schizophrenia affects both sexes equally. It affects all ages, although most cases are known to occur between the age of 15 and 35 years.⁸

25.5.2 Depressive Illness (*Cittavasada*)

Major clinical features of depressive disorders include persistent depressed mood, markedly diminished interest or pleasure, insomnia or too much sleep, psychomotor agitation, fatigue, loss of energy, diminished ability to think, and suicidal tendencies. An examination of the patient history does not reveal any organic factor or pathology or any family disaster to explain symptoms, nor does it reveal delusions, hallucinations in the past 2 weeks, or schizophrenia. In some cases, however, the family history may show biological or genetic propensity.

The mechanism responsible for depression is believed to be the decrease of catecholamine. This is based on the observations in 1950s that reserpine, a catecholamine and indolamine depleter, causes depression and drugs that replenish these amines are therapeutic; it was suggested that a reduction in the levels of these amines in the brain are related to depression. A specific cause responsible for the reduction in amine levels is not clear. It is possible that a thinking process in the cortex responding to an adverse physical or psychosocial environment may trigger the reduction in amine levels. Thus, the reduction in amine levels may be a manifestation of the disease and not the cause. Almost all antidepressant drugs are known to increase the availability of catecholamine and indolamine at the synapse in the central nervous system.⁸

25.5.3 Anxiety Disorders (*cittodvega*)

Major features of anxiety disorders include nervousness, sleeplessness, hypochondriasis, and somatic complaints. No age is immune to these disorders. Clinical symptoms may occur as an episode, such as paniclike anxiety, or may show chronic generalized anxiety. A panic anxiety episode may consist of following symptoms: shortness of breath, dizziness, unsteady feeling, fainting, accelerated heart rate, trembling, sweating, choking, nausea, depersonalization, paresthesia, hot flashes, chills, chest pain, fear of dying, and fear

of going crazy. No clear etiology is known, but a cluster of cases in families indicates some genetic component similar to depression. However, physical and sociopsychological environmental factors must be considered. Neurophysiology and neurobiochemistry of anxiety disorders suggests the involvement of the noradrenergic system.⁸

25.5.4 Somatoform Disorders, Alcoholism, and Drug Abuse (*Mada*)

Somatoform disorders basically have one common characteristics: mimicry of medical diseases involving exaggeration of severity or disability accompanying actual medical illness. There is still no credible neurobiological basis for these disorders. Some examples of these diseases are listed below⁸:

1. Somatization disorder — This is a multisystem disorder characterized by complaints of pain, gastrointestinal, sexual problems, and pseudoneurologic problems. It begins early in life.
2. Hypochondriasis — Patients believe that they are going to contract some serious illness.
3. Conversion disorder — Patients mimic symptoms of neurological or medical illness and show dominant psychological factors as the cause.
4. Body dysmorphic disorder — Patients are concerned with an imagined defect in physical appearance.
5. Pain disorder — Patients complain of pain and indicate some dominant psychological factor as the cause.

Examples of somatiformlike disorders include the following:

1. Dissociative disorder — Patients show disrupted consciousness, memory, identity, or perception judged to be attributed to psychological factors.
2. Factitious disorder — Patients intentionally create physical and psychological signs of illness when no cause (financial gain, avoidance of responsibility, etc.) can be identified.
3. Malingering — Patients intentionally create false physical and psychological signs of illness and an identifiable cause (financial gain, avoidance of responsibility, and others).
4. Chronic fatigue syndrome — The syndrome has the basis in reports appeared in 1980s in the U.S. of pathologic fatiguability. Unconscious psychological factors are believed to be involved as important contributors.

25.6 Diagnosis of Psychiatric Disorders

The clinical examination in Ayurveda has two objectives: (1) to examine and assess the primary nature of the patient as an individual as opposed to a disease and his or her health (*rogi pariksha*) and (2) to examine and assess the nature and severity of the disease (*rog-pariksha*). The examination of the patient is done by a 10-point interrogation (*dasavidha pariksa*):

1. Constitution (*prakriti*)
2. Morbidity (*vikrti*)
3. Quality of tissues (*sara*)
4. Body structure (*samhanana*)
5. Anthropometry (*pramana*)
6. Adaptability (*satmya*)
7. Mental stamina (*sattva*)
8. Digestive power (*ahara sakti*)
9. Physical strength (*vyayama sakti*)
10. Age and aging (*vaya*)

The clinical examination includes an interview (*prasna pariksa*) with the patient, a reliable attendant, or relative. The physical examination of the patients (*pancendriya pariksa*) includes the following:

1. A general survey by eight-point examination (*astavidha pariksa*): pulse (*nadi*), urine (*mutra*), stool (*mala*), tongue (*jihva*), voice (*abha*), skin (*sparsa*), eyes (*netra*), facies (*akrti*).
2. Systemic examination (*sadanga pariksa*) of the six parts of the body: head-neck, trunk, and the four limbs along with 13 channels (*srotamsi*) distributed over the entire body (*sadangas*). The examination of channels (*srotas*) with psychiatric patients must include the examination of mental channels (*manovaha srtas*).

The clinical data available through the above methods are critically examined and evaluated in the light of the doctrine of Ayurvedic knowledge of clinical description, signs and symptoms, pathology, and diagnostic criteria (*pramana vijñana* and *nidana pancake*, *nidana*, *purvairiipa*, *rupa*, *samprapti*, *upasaya-anupasaya*). The diagnosis is made not merely by name of a disease or syndrome, but in a descriptive way identifying the constitution, etiology, and abnormal personality (*prakrti*, *hetu*, *vikrti*); this includes *dosa* tissues (*dusya*) and seat of the vitiated *dosa* (*adhithana*). The Ayurvedic physician pays special attention to the patient's environment, hereditary and genetic background, and original personality makeup in terms of *tridosha* and three personality properties (*triguna*).

25.7 Treatment

The Ayurvedic management of a psychiatric patient is carried out through three broad streams of therapy: (1) divine therapy (*Daivavyapasraya cikitsa*), (2) biological therapy (*Yuktivyapasraya cikitsa*), and (3) psychotherapy (*Sattvavajaya*). The divine therapy includes the use of religious activities (*mantra*, *japa*) and wearing of precious stones. Ayurvedic psychotherapy is practiced incorporating the principles of assurance therapy (*asviisana*), replacement of negative emotions with positive emotions, and psychological shock therapy. In addition, it includes encouraging the patient to find or develop a specific goal, purpose, or meaning in life; find a job that gives better job satisfaction; improve relationships with family members; contribute own share of responsibilities to the household;

maintain a healthy lifestyle with respect to daily routine (sleeping schedule and exercise); and join some sports activity, fitness club, and volunteer groups to help the community.

In biological therapy, the patient is subjected to biopurification by *pancakarma* in order to cleanse the channels of the body. This is followed by *salisamana* therapy or palliative treatment with the help of *ausadhi* (medicinal herbs and herbal formulas), *anna* (dietetics), and *vihara* (lifestyle). The drugs used in the treatment are mostly nootropic (*medhya*) herbs or herbomineral formulas or nootropic *rasayanas*, which are believed to act as brain tonics and adaptogens. The nootropic herbs are considered as specific molecular nutrients for the brain, affording a better mental health and leading to alleviation of behavioral alterations.⁹ The *vajikarana* drugs such as *kapikacchu* and others are used in the treatment of depression. The entire Ayurvedic management is more health oriented than disease oriented. As such, there is a big scope of utilizing Ayurvedic approach and therapeutics as an adjunct to the disease-oriented therapy of modern psychiatry to afford a full treatment.

25.7.1 Ayurvedic Herbs and Herbal Formulas

Besides the nondrug approach to prevention and treatment of mental disorders, the Ayurvedic texts describe a large number of herbal and herbomineral formulas for treating mental diseases. Many of these formulas are still in popular use in the hands of a large number of Ayurvedic practitioners. The scientific validity of many of these herbs and herbal formulas drugs has been tested on scientific parameters, and several of these have been found effective.

Ayurveda describes a class of herbs termed as *medhya* herbs and *medhya rasayana*. All *medhya* and *medhya rasayana* drugs are claimed to be nutraceutical agents specific to neuro-nutrition. They are called *medhya* because they are beneficial for *medha* (i.e., intellect). All such herbs have been found to possess nootropic effect besides varying degrees of anxiolytic activity. In addition to other aspects of the *rasayana* effect, they bring about antistress effect and improve the memory and cognitive power. The popularly used *Medhya* herbs and commercially available herbal compound formulas are listed below:

Single Herbs	Herbal Formulas
1. <i>Sankhapuspi</i> (<i>Convolvulus pluricaulis</i>)	1. <i>Sarasvatarista</i>
2. <i>Mandukaparni</i> (<i>Centella asiatica</i>)	2. <i>Aswagandharist</i>
3. <i>Brahmi</i> (<i>Bacopa monnieri</i>)	3. <i>Smriti sagar rasa</i>
4. <i>Vaca</i> (<i>Acorus calamus</i>)	4. <i>Krisna chaturmukha rasa</i>
5. <i>Jatamansi</i> (<i>N. jatamansi</i>)	5. <i>Unmada gajankusa rasa</i>
6. <i>Ashwagandha</i> (<i>Withania somnifera</i>)	6. <i>Unmada gajakeshri rasa</i>
7. <i>Sarpagandha</i> (<i>Rauwolfia serpentina</i>)	
8. <i>Jyotismati</i> (<i>Celestrus panniculatus</i>)	
9. <i>Yasti madhu</i> (<i>Glycyrrhiza glabra</i>)	
10. <i>Guduchi</i> (<i>Tinospora cardifolia</i>)	

All these plants are tropical and grow in India and other tropical countries profusely. The recent clinical and laboratory studies conducted on these herbs have shown mild to moderate nootropic and antistress effects with high safety profile.^{10–17}

In addition to the *medhya* herbs as mentioned above, it is pertinent to mention that the herbs of *vrishya* and *vajikarana* categories are also used in psychiatric care. *Vrishya* and *vajikarana* herbs are essentially aphrodisiacs and are commonly used for promotion of sexual stamina and virility. Some Ayurvedic physicians use this category of herbs for treating depression. The most popular herbal drug of this category is *kapikacchu* (*Mucuna*

pruriens). The seeds of this plant, which are also edible as a vegetable, are used in the treatment of depression and also for treatment of Parkinsonism because of high levadopa (L-DOPA) content.

25.7.2 Nonherbal Therapies

The Ayurvedic classics describe *sattvavajaya* as one of the three major therapeutic streams of Ayurveda. *Sattvavajaya* is a special form of psychotherapy where the therapist attempts to divert the thought process of the patient from unwanted targets (*ahit bhawas*) to beneficial targets (*hita bhawas*). This is done through supportive therapy, assurance, and replacement of negative emotions by positive emotions.

In addition to this classic Ayurvedic psychotherapy, the role of yoga and meditation can not be over emphasized. Scientific evidence has already accumulated to suggest that several yogic practices have great psychorehabilitative and antistress effect.

Similarly, certain traditional practices of *keraliya panchakarma* (i.e., *sirodhara* and *sirobasti*) are used for treatment of chronic mental diseases and have good results. However, their mode of action has not yet been validated scientifically.

In conventional medicine, treating psychiatric disorders essentially consists of identifying and eliminating socioenvironmental factors and administration of antipsychotic drugs (neuroleptic drugs) similar to that in Ayurveda. The drugs used for the treatment of psychosis include phenothiazines, structurally similar thiozanthenes, and heterocyclic dibenzazepines; butyrophenones and diphenylbutylpiperdines; and indolones and other heterocyclic compounds. These drugs act via blocking the mesolimbic dopamine receptors. The adverse side effects of the drugs are as follows:

1. Acute spasm of the tongue, face, neck, or back may mimic seizures, not hysteria
2. Motor restlessness
3. Bradykinesia, rigidity, variable tremor, mask facies, and shuffling gait
4. Catatonia, stupor, fever, unstable blood pressure, and myoglobinemia can be fatal
5. Perioral tremor
6. Oral facial dyskinesia, widespread choreoathetosis, or dystonia
7. Agranulocytosis⁸

The pharmacologic agents available to treat depression disorders exert their antidepressant effect by inhibiting the reuptake of serotonin at presynaptic membrane selectively (fluoxetine) and the reuptake of both serotonin and norepinephrine or (imipramine), directly inhibiting monoamine oxidase enzyme (phenelzine) or causing dopamine blockage (trazodone). These drugs provide symptomatic relief and remissions but do not cure the disease. The side effects of these drugs include dry mouth, constipation, postural hypotension, tachyarrhythmia, nervousness, insomnia, tremor, agitation, headache, weight loss, hypertensive crisis, and priapism.²

The drugs used for the treatment of anxiety disorders are primarily antihistamines and benzodiazepines. The adverse side effects of these drugs include dry mouth, mental confusion, addiction, ataxia, drowsiness, sedation, nervousness, headache, and bradycardia.⁸

No drug treatment should be given to patients suffering from somatoform disorders, alcoholism, or drug abuse (*mada*) unless there are secondary signs that require drug treatments. Attempts should be made to identify and eliminate sociopsychological factors and other environmental factors associated with the diseases. The psychiatrist and family members need to take care of the patient with assurances, spiritual guidance, and prayers.⁸

25.8 Prevention

Ayurveda places great emphasis on preventing diseases; psychiatric disorders are no exception. As a matter of fact, a healthy lifestyle, a healthy diet, balanced physical activities, living up to social and family responsibilities, enjoying work, regularly exercising, practicing a proper daily routine (*dincharya*), and keeping personal living spaces (such as bedrooms and cars) presentable and pleasing may prevent many psychiatric disorders, particularly if children are also raised with this lifestyle. According to Ayurveda, the human body is like a vehicle driven by the soul (spirit, *atma*). When the vehicle takes control of the travel, a disaster is bound to occur. One must not surrender his or her life to the body; the soul must drive the body.

25.9 Scientific Basis of Ayurvedic Therapies

Although the therapies mentioned above have been in traditional use for thousands of years, with the recent demand of evidence-based Ayurveda, modest efforts have been made to generate scientific evidence for the safety and efficacy of the traditional medications. Reports on such studies are accumulating in scientific journals. Some such studies are highlighted below.

25.9.1 Antistress Agents

A large number of Ayurvedic herbs have been described to possess the *medhya* effect. All such herbs are believed to have neuronutraceutical effect with varying degree of anxiolytic and antistress effect. The potentiation of barbiturate hypnosis by *sankhapuspi* (*C. pluricaulis*) in rats and the evidence of anxiolytic and adaptogenic effect of this drug in patients of anxiety and neurosis has been demonstrated.¹⁸ In another study, similar effects for *mandukaparni* (*C. asiatica*) and *ashwagandha* (*W. somnifera*) have been reported in animal models and in clinical settings, respectively.¹⁹ The effect of *brahmi* (*B. monnieri*) in similar clinical and laboratory models was also investigated and was found to lower the level of anxiety and promote memory and adaptation.¹¹ Singh and associates screened a series of herbal therapies in albino rats for antistress effect by measuring swimming performance, changes in adrenal weight, and ascorbic acid and cortisol content of adrenal glands during stress; they also measured the incidence of stress ulcers in stomachs that were under the influence of herbs in stressed rats. The investigators found highly significant antistress effect in *ashwagandha* (*W. somnifera*) and *tulasi* (*O. sanctum*). These two drugs were found to be twice more effective than *P. ginseng* in terms of ED₅₀.^{16,17}

25.9.2 Memory-Enhancing Agents

All *medhya rasayana* of Ayurveda are conceptually nootropic agents. Recent studies conducted on these herbs have shown evidence for this conceptual claim. Singh and Singh reported a memory-enhancing effect in *brahmi* (*B. monnieri*) followed by several studies conducted at the Central Drug Research Institute, Lucknow, India, by Dhawan

and Associates in recent years characterizing active constituents of this plant responsible for the memory enhancing effect.^{10,20}

25.9.3 Treatment of Residual Psychosis

With the advent of strong antipsychotic drugs in modern medicine, it is no longer difficult to control the acute episodes of psychotic disease. It is becoming more and more difficult, however, to take care of the long-term chronic residual phase of these diseases and their negative symptoms. Modern medicines are not suitable for such long-term use because of drug dependence and adverse side effects. In such cases, Ayurvedic medications are the only logical answer. Certain Ayurvedic herbs, particularly the polyherbomineral compounds like *smrti sagar rasa*^{21,22} and *unnada gaja kesari rasa*, have shown encouraging results.²³

25.9.4 Antidepressants and Mood Elevators

A number of Ayurvedic herbs, particularly the *vajikarana* formulas (Ayurvedic aphrodisiacs), have been evaluated for their possible role in the management of depressive illness. *Aswagandha* (*Withania somnifera*), *kapikacchu* (*Mucuna pruriens*), *vyotismati* (*Celeastrus paniculatus*), and *vaca* (*Acorus calamus*) have shown good results in pilot studies. In one study, *aswagandha* and *kapikacchu* used alone in patients of depressive illness exhibited an antidepressant effect when measured on the Hamilton Depression Rating Scale.²⁰ The potentiation of hypothermia effect of the agonist in *aswagandha*-treated animals suggests that *aswagandha* sensitizes serotonin (5-HT_{1A}) autoreceptors. A similar effect was also found for *sankhapuspi* (*C. pluricaulis*).²⁰

25.9.5 Geriatric Mental Health

Aging is generally associated with a range of psychological symptoms. The *rasayana* formulas of Ayurveda are the logical remedy. The antiaging effects of selected *rasayana* formulas like *aswagandha* (*W. somnifera*) and *tulasi* (*O. sanctum*) were evaluated by using a standardized biological age scale and brief psychiatric rating scale, memory span test, and mental fatigue rate.²⁴ It was observed that use of *aswagandha* and *tulasi* singly for a sample test period of 3 to 6 months exhibited notable beneficial effect in elderly individuals. Such an effect is attributed to the drugs' antistress and adaptogenic activities.

25.9.6 Medhya Rasayana and Mental Retardation

A varying degree of mental retardation is a major health hazard in children prevalent throughout the world and belong to divergent etiology. Besides self-recovery and compensation, there is no definite treatment for this condition in conventional Western medicine. But the Ayurvedic texts describe a range of restorative remedies (i.e., *rasayana*), especially for the promotion of mental health and cognitive functions. Such *rasayanas* are called *medhya rasayana*. It is claimed that such *rasayanas*, because of their nutritive impact on the brain, may help mentally retarded children. *Mandukaparni* (*C. asiatica*) is one of the four *medhya rasayanas* described by Charaka Samhita.⁷ This drug is in traditional use and was studied preliminarily for its utility in mental retardation.¹⁵ More recently, a clinical study on *mandukaparni* in 30 mentally retarded children (6 to 18 years old), fulfilling the

selection criteria for mental retardation and excluding any other major organic cause, was conducted.¹⁴ These children received the *mandukaparni* (whole plant granules) at a dose of 5 g twice/day for 6 months. The observations showed significant improvement in performance intelligence quotient, social quotient (measured by Vineland social maturity scale for behavior and social adoptability), immediate memory span, and reaction time. The authors concluded that such an effect of *mandukaparni* could be due to the *rasayana* effect of the drug both in terms of its microneuronutrient effect and its *srotas* (microvascular) effect by promoting the blood supply to the brain.

25.10 Summary and Conclusions

Mental health and ailments of psyche have been important considerations in the field of medicine from the very beginning. Ayurveda has a well-developed branch of psychiatry, and the principles and practices of mental health care which developed thousands of years ago are again drawing attention because of their unique holistic approaches. There is a need to integrate this novel wisdom into the mainstream of the health care delivery system of today.

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Parkinson’s Disease (Kampa Vata)

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26.1 Introduction

According to Ayurveda, most of the diseases of the *vata* system are essentially the conditions of degenerative diseases of the nervous system. As many as 80 kinds of *vata* diseases are described in Charak Samhita.¹ The syndrome of parkinsonism is commonly compared with *kampa vata* (*kampa* literally means tremors). The syndrome was comprehensively described by James Parkinson in 1817, but the review of much early literature would show that syndromes strikingly similar to parkinsonism were already known and were categorized among the *vata* diseases in Ayurveda.

Parkinson’s disease (PD) is prevalent all over the world and has no definitive treatment in conventional medicine except for the palliative prescription of anticholinergics together with levodopa and dopadecarboxylase inhibitor. The Ayurvedic treatment strategy is to combat *vata dosa* and to sustain neuronutrition by *rasayana* remedies. This chapter discusses *vata* diseases (neurological diseases) as a class with emphasis on Ayurvedic therapies of PD and a possible scientific basis for the therapeutic effect.

26.2 Etiology

As discussed in [Chapter 2](#), Ayurveda adopts a holistic approach to the understanding of whole body physiology by explaining the body functions in terms of three biofactors called

tridosa (*vata*, *pitta*, and *kapha*). In all applied considerations, the *vata* system of the *tridosa* doctrine represents neuroscience in Ayurveda. The *vata dosa* is responsible for the entire neurophysiological phenomena operating in the body.¹ There are five major components of *vata* systems: *prana*, *udana*, *vyana*, *samana*, and *apana*. These components are responsible for five different aspects of neural functions at five different levels in the body. When the *vata dosa* loses its equilibrium due to a wide range of aetiological factors, including tissue degeneration and damage (*dhatu ksaya*) and neuro-obstructive diathesis (*margavarana*), it leads to the development of 80 types of *vata* diseases or neurological diseases.

The etiopathogenesis of PD is not precisely known in conventional medicine, but it is understood to be caused by lesions in the basal ganglia and is especially associated with the damage to the interconnecting system between substantia nigra and corpus striatum. Specific etiology of PD is related to aging, and it develops in people over 50 years old. There is no specific genetic component involved in most cases, but mutation in the gene for alpha synuclein has been linked to PD in some cases.² Exposure to environmental chemicals may cause damage to basal ganglia and brain stem resulting in parkinsonism. For example, exposure to 1-methyl 4-phenyl-1, 2, 3, 6-tetrahydropyridine, a contaminant present in a psychoactive meperidine analog, can cause an acute onset of parkinsonism symptoms and the associated destruction of substantia nigra.

26.3 Pathology

In Ayurveda, no specific structural pathology of PD is described other than its identification as a *vata dosa* disease.

In conventional medicine, PD is described as a chronic progressive disease of extra pyramidal system of the brain where voluntary movement is disturbed with the appearance of involuntary movements and altered muscle tone. The pathology of PD essentially involves a loss of pigmented dopaminergic neurons in the substantia nigra pars compacta (SNPC) of the brain with the appearance of intracellular inclusions known as Lewy bodies.^{3,4} SNPC provides dopaminergic innervation to the dopamine content (caudate and putamen). Early studies have shown 80% reduction in the dopamine content of the striatum. This reduction parallels a loss of neurons in the SNPC, indicating that dopamine loss may be the cause of PD symptoms and that replacement of dopamine could be helpful.^{5,6} The major symptoms of PD are related to the deficiency of dopamine. Cholinergic nucleus basalis degeneration is believed to be related to dementia that affects many patients. Noradrenergic locus nucleus degeneration is believed to be associated with the facial expression of freezing and with depression.⁷

26.4 Classification of *Vata* Disorders

The *vata* diseases as described in Ayurvedic classics include a wide range of neurological morbidities, including inflammatory, degenerative, obstructive, and functional. The *vata* diseases may manifest as neural hyperfunctioning (*vata vridhhi*), neural hypofunctioning (*vata ksaya*), or masked functioning (*avarana*). It is not possible to innumerate and classify

all 80 types of *vata* diseases here. However, Ayurvedic texts describe the etiology, symptomatology, and treatment of a variety of paralytic conditions, painful neuropathies, conditions with tremors, and involuntary movements and convulsions in detail. PD is one of those diseases. The descriptions of these conditions in Ayurveda are very similar to those described in conventional medicine. Some of these conditions are listed here as examples.

1. Paralysis and paresis — Facial paralysis (*ardita*), hemiplegias (*pakavadha*), paraplegias (*adharanga vata*), quadriplegia (*sarvauga vata*), and others
2. Painful neuropathies — Sciatica (*gridhrasi*), limb cramp (*khalli*), brachial neuritis (*avavahuka* and *viswachi*), trigeminal neuralgia (*siragrah*), lumbago (*kati-sula/pristhasula*), cervical neuropathy (*manyastambh*), peripheral neuritis (*rasagata vata*), and others
3. Coordination disorders and convulsive diseases: parkinsonism, tremors, choreas (*kampa vata*), convulsive neuropathies (*aksepa*), epilepsies (*apasmara*), pseudo seizure disorders (*apatantarka*), and others
4. Visceral neuropathies — A large number of diseases and syndromes related to the neuropathy of different visceral organs of the digestive system, urogenital system, and sensory organs as described in Ayurveda

26.5 Clinical Features and Diagnosis of Parkinson's Disease

Clinical features of *kampa vata* in Ayurveda are similar to those described for parkinsonism and Huntington chorea in conventional medicine. These are the two most important neurodegenerative diseases of the basal ganglia and brain stem.⁸ These diseases are frequently associated with movement disorders, rigidity, abnormal posturing, and involuntary movements marked by fine disorganized and random movements of the extremities (usually the hands and to a lesser extent proximal limb and muscles). The four main features of parkinsonism are (1) bradykinesia (abnormal slowness of voluntary movements, often associated with diminution of the range of movements), (2) muscular rigidity, (3) resting tremor, and (4) postural imbalance (leading to disturbance of gait and falling).⁹ These features are also seen in other conditions, such as idiopathic PD, progressive supranuclear palsy, and corticobasal degeneration, with common damage to the nigro-striatal dopaminergic system.

Other symptoms of PD are related to mood, intellect, and the autonomic nervous system. These symptoms are hypophonic dysarthria, monotonous speech pattern, difficulty in getting in and out of a chair or a bed, shuffling gait with short steps, decreased automatic movements, and arm swing. The resting tremors are slow, regular oscillations (3 to 6/sec) that usually cease during voluntary movements of the affected part, though occasionally the tremors may be of action type. Tremors are exacerbated by fatigue and cold; they can also affect the lips, lower jaw, and tongue. Tremors may be a dominant feature in some patients, whereas postural instability may be dominant in others.⁷

Other PD-like conditions can be differentiated from PD on the basis of a low to no degree of tremor and poor response to levodopa. The good response to levodopa in PD is because dopamine receptors are preserved in PD, but they are decreased in other PD-like conditions.

26.6 Prognosis

In Ayurveda, the prognosis of PD is not specifically given. Most *vata* diseases are not curable, but the patient may be able to live with the disease with certain degree of inconvenience and difficulties. As the disease progresses, patients develop rigidity and cannot take care of themselves. In extreme cases, death may occur from the patient's inability to breathe, resulting in aspiration pneumonia or pulmonary embolism. Ayurvedic therapies available can make life much easier and increase life expectancy. Treatment with pharmacologic agents or direct electrical stimulation of target areas (thalamus, subthalamic nucleus, or globus pallidus) can provide relief in symptoms, good functional mobility for many years, and a substantial increase in life expectancy.⁹

26.7 Treatment

Ayurveda takes a unique approach to the management of the above-mentioned neuropathies with a special emphasis on eliminating their causes by *panchakarma*, physiotherapy, and medicinal treatment with the help of a wide range of herbal and herbomineral drugs. Ayurveda describes a large number of nootropic drugs (stimulating to mental activity, causing cerebral or intellectual activity) and nervous system tonics (i.e., *rasayanas* in the treatment of *vata* diseases). *Panchakarma* therapy is especially advocated in the treatment of neurological diseases. Using different kinds of oil massage (*snehana-abhyanga*) and medicated heat treatment (hot room) (*swedana*) is very efficacious. Besides special treatments like *sirovasti* and *sirodhara*, *vasti* therapy is indicated in such diseases and administered in the form of a specially planned therapeutic enema. With all this, contemporary Ayurvedic medicine claims good success in the practice of neuromedicine.

Being a neurodegenerative disease, parkinsonism is essentially treated by *rasayana* or a rejuvenative approach in Ayurveda.^{10,11} The *rasayana* drugs of Ayurveda are essentially nutraceutical agents, and the *medhya rasayana* are specific neuronutrients or nervine tonics with nootropic effect. *Aswagandha* (*Withania somnifera*), *brahmi* (*Bocopa monnieri*), *mandukaparni* (*Centella asiatica*), and *bala* (*Cida cordifolia*) are the common classical drugs advocated for this purpose. This is because Ayurveda considers such movement disorders under *vata vyadhi*, and in common practice the term *kampa vata* is used to describe the syndrome. The strategy is to combat *vata dosa* and to sustain neuronutrition by *rasayana* remedies. *Kapikacchu* (*Mucuna pruriens*), an edible legume and popular Ayurvedic aphrodisiac, has also been used in the hands of Ayurvedic practitioners for treating *kampa vata*. With the recent discovery that *Mucuna pruriens* seeds are a highly rich source of levodopa, the use of *kapikacchu* in *kampa vata* has tremendously increased. *Kapikacchu* (*Atmagupta*) has been traditionally used for treating tremor disorders in Ayurveda.¹²

In conventional medicine, the treatment of PD is primarily based on increasing the levels of dopamine in SNPC. Carbidopa and levodopa, dopamine receptor agonists (pergolide and bromocriptine), and monoamino oxidase inhibitor (selegiline) are used to maintain the levels of dopamine. The other less-used drugs are muscarinic receptor antagonists and

amentidine, an antiviral agent. The common side effects of levodopa are psychiatric problems, dyskinesia, and the wearing-off-effect (loss of effectiveness). The pharmacological treatments are only symptomatic; they do not change the progression of the disease. If used skillfully, these drugs may dramatically improve the quality of daily life and functional ability. Another treatment option is surgery. The main surgical approach commonly used is pallidotomy (incision or partial destruction of globus pallidus) and high-frequency deep-brain stimulation with an electrode implanted in one of these target areas: thalamus, subthalamic nucleus, or globus pallidus. These procedures are effective in providing relief from major symptoms and improving the quality of daily life but do not alter the course of the disease.^{7,13}

26.8 Scientific Basis

The levodopa content of *M. pruriens* appears to be the basis for the therapeutic effect. MP endogenously accumulates L-dihydroxyphenylalanine in the range between 0.2 and 2.0% on a dry-weight basis in tissue culture.¹⁴ In a clinical trial, HP-200 made from MP was found to be an effective treatment for patients with PD.¹⁵

A concoction of powdered *M. pruriens* seeds, *Hyoscyamus reticulatus* seeds, *W. somnifera* roots, and *C. cordifolia* roots in cow's milk was clinically tried in 18 diagnosed patients with PD.¹⁶ Of these 18 patients, 13 cases underwent biopurificatory (*panchakarma*) procedures for 28 days and palliative therapy for 56 days, whereas the remaining 5 patients underwent palliative therapy alone for 84 days. The former group showed significant improvement in daily activities. Symptomatically, they showed a better response in tremor, bradykinesia, stiffness, and cramps as compared with the latter group. Salivation increased in both groups. Chemical analysis of the concoction revealed about 200 mg of levodopa per dose. The study shows the importance of cleansing therapy in Ayurveda medication before palliative therapy. It also confirms the usefulness of some Ayurvedic herbal preparations in PD.¹⁷

In an interesting study, PD was treated with the compound Parkino at 20 ml twice/day for 8 weeks.¹⁸ The response was assessed symptomatically in terms of the subjective feelings of the patient, degree of involuntary movements, and facial expressions. All the patients exhibited highly significant improvement in symptoms during the first 2 to 4 weeks of treatment. However, during subsequent intervals, no further improvement was noted.

26.9 Summary and Conclusions

The review clearly indicates that PD patients can benefit from Ayurvedic therapies by using *M. pruriens* and *panchakarma*. Future research needs to focus on integrating Ayurvedic therapies with conventional therapies in treating PD.

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Male Reproductive Dysfunction

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27.1 Introduction

The role of the male in the sexual process and reproduction is twofold: (1) to successfully perform sexual intercourse, deriving pleasure for himself and satisfying the female partner,

and (2) to contribute the sperm for successful fertilization of the ovum toward procreation. The problem of male infertility and impotency is increasing day by day. The global incidence of couple infertility is estimated at 10 to 15%.¹ The contribution of the male factor alone to couple infertility is believed to be about 20%.² Between 20 and 30 million American men suffer from some degree of sexual dysfunction, and a comparative figure generally applies worldwide.³ Further, there are several lines of evidence to suggest that sperm counts are decreasing globally⁴ and the incidence of erectile dysfunction is also on the increase.⁵ The increase in male factor infertility in recent times is believed to be caused by altered lifestyle, stressful living conditions, diverse pollutants, certain prescription and several nonprescription drugs, dietary toxins, and certain nutritional deficiencies.^{6–21} The causative factors produce many types of derangements that directly or indirectly cause various kinds sexual dysfunction.

The remedy for male sexual dysfunction in modern medicine is limited. The fact remains that, other than correction of obstruction, infections, varicocele, and certain endocrine abnormalities, in most cases there is practically no successful therapeutic measure for male infertility or erectile dysfunction except for surgical implants. The allopathic drugs used for erectile dysfunction are believed to produce side effects and affect other physiological processes and, ultimately, general health.²² Ayurveda realized the problem of male sexual dysfunction thousands of years ago and developed a separate speciality, namely, reproductive medicine (*vaaajeekarana*). It realized, among other things, the role of the nervous system, cardiovascular system, and psychological aspects of fertility and male sexual performance; it has also recommended an appropriate use of plant-based remedies, a proper lifestyle, and nutritious diet for improving overall health and treating male infertility and erectile dysfunction. Over 600 plants are described in the original Ayurvedic compendia like *Charak Samhita* and *Sushruta Samhita*. The potential of Ayurvedic and other traditional medicinal plants in the development of invaluable standardized phytomedicines and allopathic type of drugs is great, particularly in the context of male sexual dysfunction.

This chapter is an attempt to review the Ayurvedic concepts of male sexual dysfunction and the potential of Ayurvedic treatments for the same, with plant-based remedies as seen in the classical Ayurvedic treatises as well as in the modern practice. An attempt has also been made to review the pharmacological and biochemical studies on the plants and plant products that are in use in Ayurvedic therapy for problems in male reproduction. The need for more elaborate scientific investigations on the Ayurvedic therapy for male sexual dysfunction is emphasized.

27.2 Male Reproductive Dysfunction

The description of several ailments relating to male reproduction in Ayurveda has relevance to the modern science and clinical practices. *Charak Samhita* is perhaps the earliest medical treatise recognizing that the semen released during orgasm at the time of sexual intercourse is responsible for the development of the embryo and that defective semen is incapable of producing the fetus. Male reproductive disorders, as dealt with in classical Ayurvedic literatures, can be due to two major causes: (1) defective semen and (2) impotence (*viryalpata* or *klaibyam*).

27.2.1 Defective Semen

Ayurveda describes semen as oily, viscous, nonslimy, sweet, and nonburning with the smell of honey. It is white with a glaze similar to honey, dehydrated butter (ghee), or til (*Sesamum orientale*) oil. The total quantity of semen during ejaculation is 2 to 4 oz. Semen can become defective due to different causes. Excessive indulgence in sex or having sex at wrong times, masturbation, excessive exercise, eating an excess of bitter, astringent, salty, sour, or hot foods, old age, anxiety, grief, suspicion, fear, anger, exorcism, emaciation from disease, and suppression of natural urges and wounds can lead to the derangement of *tridosas* of tissues. This derangement can reach the semen-producing organs and carrying channels causing semen defects.

According to Ayurveda there are eight types of defective semen: *vatarethas*, *pittarethas*, *kapharethas*, *kunaparethas*, *grandhirethas*, *poothipooyarethas*, *ksheenarethas*, and *moothrapureethas*. The definitions, etiology, and prognosis of and treatments for each are furnished in [Table 27.1](#).

27.2.2 Causes of Impotence

Main causes of impotence are basically (1) impaired and deficient semen (see Section 27.2.1) and (2) erectile dysfunction.

Erectile dysfunction (*dhvajabhangam*) results from ingesting excess, sour, salty, heavy, incompatible, and unsuitable foods, and drinking excess water. Other causes are irregular meals, excess use of milk and milk products, weakness from illness, defective seminal tract, excessive discharge, coitus with animals, and failing to clean the penis and injured genitals. Diet and lifestyle are emphasized as the causative factors for erectile dysfunction in addition to genetic causes.

There are five types of erectile dysfunction: *vatajam*, *pittajam*, *kaphajam*, *rakthajam*, and *sannipathajam*; detailed information about each is furnished in [Table 27.2](#).

Old age often causes diminished semen and is related to deficiency of the seven body tissues or transformations (*dhathus*).

27.2.3 Eunuchs (*Napumsak*)

Sushruta Samhita characterizes five types of eunuchs: *asyeka*, *saugandhikam*, *kumbheeka*, *eershyakam*, and *shandakam* ([Table 27.3](#)). These eunuchs are to be inferred essentially as male perversion. Perhaps these eunuchs, except *shandakam*, shall have no relevance in the modern context. *Charak Samhita* describes eight types of eunuchs: *dviretha*, *pavanednriya*, *samskaravahi*, *narashanda*, *nareeshanda*, *vakri*, *eershyabhirathi*, and *vatheekashanda* ([Table 27.4](#)). Although most etiology part may not be acceptable in modern science, at least these types are parallel with modern clinical practices.

27.3 Treatment

Considering the gravity of male infertility, one part of octapartite Ayurvedic therapeutic speciality called semen therapy (*beeja poshana*) is devoted to the therapy for male infertility.

TABLE 27.1

Eight Types of Defective Semen as Identified in Ayurveda

Type	Definition	Etiology	Prognosis	Treatment
<i>Vatarethas</i>	Dark-colored frothy semen; painful ejaculation	<i>Vata</i> ; vitiated by wrong lifestyle and food habits	Easily curable	Oliation, fomentation, purgation, catheterization, enema through penis, etc., along with medicines like <i>jeevaneeya ghrutham</i> , <i>chyavan prash</i> , and <i>shilajit</i>
<i>Pittarethas</i>	Yellow-colored very hot semen; burning sensation and pain during ejaculation	<i>Pitta</i> ; vitiated by wrong lifestyle and food habits	Easily curable	Oliation, fomentation, catheterization, enema through penis, etc., along with medicines like <i>jeevaneya ghrutham</i> , <i>chyavan prash</i> , and <i>shilajit</i>
<i>Kapharethas</i>	Whitish, slimy, semen; obstructed and painful ejaculation	<i>Kapha</i> ; vitiated by wrong lifestyle and food habits	Easily curable	Oliation, fomentation, purgation, catheterization, enema through penis, etc., along with medicines like <i>jeevaneya ghrutham</i> , <i>chyavan prash</i> , and <i>shilajit</i>
<i>Kunaparethas</i>	Cadaver-smelling semen; increased in volume	Vitiated by excessive coitus, injury, wound and by wrong lifestyle and food habits, leading to deterioration of quality of blood	Curable	<i>Woodfordia floribunda</i> , <i>Acacia catechu</i> , <i>Punica granatum</i> (fruit cover), and <i>Terminalia arjuna</i> (bark) prepared in butter
<i>Grandirethas</i>	Tissue-like semisolid semen; obstructed and painful ejaculation	<i>Kapha</i> and <i>vata</i> vitiated by wrong lifestyle and food habits	Curable	<i>Kaempferia galanga</i> , or <i>Rotula aquatica</i> (root), or ash of <i>Butea monosperma</i> prepared in ghee
<i>Poothipooyarethas</i>	Puslike and malodorous semen	<i>Pitta</i> and <i>kapha</i> vitiated by wrong lifestyle and food habits	Curable	<i>Parooshakadi ganam</i> or <i>nyagrodhadi ganam</i> prepared in ghee
<i>Ksheenarethas</i>	Decreased quantity and quality of semen; delayed ejaculation; blood stains in semen	<i>Vata</i> and <i>pitta</i> vitiated by excessive coitus, excessive exercise, wrong lifestyle and food habits	Curable	<i>Panchakarma rasayana</i> and <i>vajeekarana</i> treatment
<i>Moothrapure-sharethas</i>	Semen smells like urine and feces	All three <i>dosas</i> , vitiated by wrong lifestyle and food habits	Incurable	—

The basic principle of treatment in Ayurveda is to assist the body to heal itself. The Ayurvedic fundamental principles of repletion (*samtarpana* and *brmhana*) and depletion (*apatarpana* and *langhana*), consisting of radical (*sodhana*) and conservation (*samana*) treatments, apply to therapy for male sexual dysfunction, also. Radical therapy adopts techniques to drain the waste materials of different body compartments through the nearest channel in a system-friendly manner. Radical therapy, followed by repletion and reproductive medicine therapy, can probably improve the male reproductive function by cre-

TABLE 27.2
Classification of Erectile Dysfunction in Ayurveda

Types	Features
<i>Vatajam</i>	Effusion, pain, redness
<i>Pittajam</i>	Carbuncles and pus
<i>Kaphajam</i>	Increased penis size, lesions, discharge with reddish or dark color, ring-like processes near the root of penis
<i>Rakthajam</i>	Fever, burning sensation, vertigo, dizziness and vomiting; blackish red, red, or blue discharge
<i>Sannipathajam</i> (all the <i>dosas</i> acting together)	Pain and burning sensation in urinary bladder, testicles, and seminal vesicles; white and slimy discharge will be present; penis effused; eventually gets infected with pathogens; foul smell; rotted tip of the penis; entire penis and testicles can rot and wither off

TABLE 27.3
Eunuchs According to *Sushruta Samhita*

Type	Feature	Etiology
<i>Asyeka</i>	Erection occurs only when one drinks the semen of another person	Due to quantitative inadequacy of semen and <i>rethas</i>
<i>Saugandikam</i>	Erection occurs only when one smells the vagina or penis	Birth in a malodorous vagina
<i>Kumbheeka</i>	Erection occurs only when another person enters him through the anus	Not mentioned
<i>Eershyakam</i>	Erection occurs only when one sees another person making love	Not mentioned
<i>Shandakam</i>	Person has body structure and behavior of a woman	Due to father making love lying on his back

ating an optimal environment for spermatogenesis and improving intratesticular availability of nutrients.

The concept of radical therapy is highly relevant today in the light of studies emerging with evidence that the real culprits of infertility are the various tissue toxins (gonadotoxins).¹⁵ These chemicals mimic hormones and impair their secretion, resulting in disruption of the normal physiological processes. Ayurveda recognizes the concept of weak toxins and it considers these toxins as one of the causes of subfertility.

As has been explained earlier, the etiology of the various diseases in Ayurvedic perspective is to be traced to the three basic states of the body (the *tridosas*), namely, *vata*, *pitta*, and *kapha*. Any aspect of male sexual dysfunction shall be reflection of any one or more of the *tridosas*. In fact, onset of a disease is due to an upset of the equilibrium of the three *dosas* and the seven tissues or the body transformation processes.

Charak Samhita recognizes the semen-producing and -transmitting parts of the body (*sukravahastrotas*). Depletion of hormones (*ojas*) and disruption in the semen-producing and -transmitting organs can lead to one or more of the manifestations of male sexual dysfunction. Hormones are considered to be the physical expression of consciousness in the body and play a role in immunity and procreation. The Ayurvedic remedy for male

TABLE 27.4

Eunuchs According to *Charak Samhita*

Type	Feature	Etiology
<i>Dviretha</i>	Person having physical characteristics of a man but behaves like a woman	Zygote produced with sperm and ovum having equal dominance
<i>Pavanendriya</i>	Person has erection but no ejaculation	Testicles of fetus destroyed by <i>vata</i>
<i>Samskaravahi</i>	Person has damaged seminal vesicles	Seminal vesicles damaged by <i>vata</i> due to vitiation of <i>dosas</i>), accident, or by birth
<i>Narashanda</i>	Male with weak sexual urges, poor semen volume, poor ejaculation, or has no interest in sexual intercourse	Father should have had these qualities
<i>Nareeshanda</i>	Female with weak sexual urges, poor menstrual flow, or no interest in sexual intercourse	Mother should have had these qualities
<i>Vakri</i>	No erection at all	Either the mother had little interest in sexual intercourse or sperm of father had weak sexual urges
<i>Eershyabhirathi</i>	Erection occurs only when one sees another person making love	Male and female getting angry during coitus
<i>Vatheekashanda</i>	Due to <i>in utero</i> destruction of testicles; azoospermic	Vitiated by <i>agni</i> or <i>vata</i> , testicles destroyed during early pregnancy

sexual dysfunction is aimed not only at the immediate etiological factors of the disease, but also toward improving the hormonal status of the body. For example, according to Charaka, in the case of seminal debility a patient must first be prepared for the therapy through cleansing the body by purgation and, if necessary, by vomiting. Then he must be given liquid diet and gradually transferred to the natural food so as to improve the body's overall health. The specific treatment for increasing the semen and the aphrodisiacs can be practiced either concurrently or in sequence. This is very clearly reflected in the various combinations of herbals and other traditional medicinal ingredients in the various Ayurvedic formulations for male sexual dysfunction.

27.3.1 Polyherbal Formulations

27.3.1.1 Classic Formulas

Among the several polyherbal formulas in the Ayurvedic system of medicine, such as *ghrutham*, *rasayana*, *kashyam*, *lehyam*, etc., each formula has one or more ingredients that can take care of male sexual debility. Information about a few of the most important classical Ayurvedic polyherbal formulations is furnished in [Table 27.5](#), and more detailed information is found in classic texts.

Ashtangahrudayam, in the volume *Utharasthanam*, lists several combinations of herbals and other ingredients for management of male reproductive dysfunction ([Table 27.6](#)). In the modern practice, the formulations have been changed or altered by the individual Ayurvedic practitioners, based on their experiences.

27.3.1.2 Nonclassic Formulas

Although the classical Ayurvedic approach in therapy is still practiced, there has been a rewarding blend of the Ayurvedic approach and the modern allopathic approach through adopting phytochemistry and pharmacology. Further, though the birthplace of

TABLE 27.5
 Classical Polyherbal Ayurvedic Formulations Used in Treating Male Sexual Debility

No.	Name	Textbook Reference
1	<i>Chyavan prash</i>	<i>Ashtangahrudayam</i>
2	<i>Kalyanaka ghrutham</i>	<i>Ashtangahrudayam</i>
3	<i>Phalasarpsis</i>	<i>Ashtangahrudayam</i>
4	<i>Sathavaryadi ghrutham</i>	<i>Sahasrayogam</i>
5	<i>Sukumaram ghrutham</i>	<i>Ashtangahrudayam</i>
6	<i>Kalyanakam kashayam</i>	<i>Sahasrayogam</i>
7	<i>Sukumaram kashayam</i>	<i>Ashtangahrudayam</i>
8	<i>Vidaryadi kashayam</i>	<i>Ashtangahrudayam</i>
9	<i>Sukumaram lehyam</i>	<i>Ashtangahrudayam</i>
10	<i>Vidaryadi lehyam</i>	<i>Ashtangahrudayam</i>
11	<i>Narasimha rasayanam</i>	<i>Ashtangahrudayam</i>
12	<i>Sathavari gulam</i>	<i>Sahasrayogam</i>
13	<i>Ashwagandhadi lehyam</i>	<i>Bhaishajyaratnavali</i>
14	<i>Amruthaprasa rasayanam</i>	<i>Ashtangahrudayam</i>
15	<i>Ashwagandharishtam</i>	<i>Bhaishajyaratnavali</i>
16	<i>Dhasamoolarishtam</i>	<i>Bhaishajyaratnavali</i>
17	<i>Jeevaneya ganam</i>	<i>Ashtangahrudayam</i>
18	<i>Nyagrodhadi ganam</i>	<i>Ashtangahrudayam</i>
19	<i>Parooshkadi ganam</i>	<i>Ashtangahrudayam</i>
20	<i>Bala thailam</i>	<i>Sahasrayogam</i>

Note: Only some of the Ayurvedic manufacturers, who produce generic Ayurvedic preparations, have all these products. Examples include (1) Vaidyarethnam P.S. Varier's Arya vaidhya sala, Kottakal, Malappuram District, Kerala, India; (2) Vaidyarethnam Oushadasala, Thaikattuseri, Ollur, Thiruvananthapuram District, Kerala, India; (3) Nagarjuna Herbal Concentrates Ltd., Kalayanthani, Thodupuzha, Kerala, India.

Ayurveda is India, it has come to be adopted worldwide in which traditional medicine processes of the different regions of the world have been blended. Currently, many patent formulas are the blend of Ayurvedic herbs along with some herbs in other countries. In this process of blending, micronutrients and hormones are also managed. A few such combinations are currently marketed, and their trade names are in [Table 27.7](#). As evident in the following section, very few of these medicinal plants have been subjected to pharmacological testing.

27.4 Scientific Basis

27.4.1 Ayurvedic Herbs

One of the major pitfalls in Ayurvedic therapy in general, and for male sexual dysfunction in particular, is that there is very little research with regard to pharmacological testing and clinical trials. Further, as could be seen from the various chapters in this book, the same herb (e.g., *Withania somnifera*, *Asparagus racemosus*, *Tribulus terrestris*, etc.) is used therapeutically for several ailments. Within this limitation an attempt has been made to review the scientific investigations made on a few of the herbals consistently used as remedies in male sexual dysfunction in the various combinations.

TABLE 27.6

Ayurvedic Formulations for the Treatment of Male Reproductive Dysfunction (Ashtanga Hrudayam, Utharasthanam)

Combination	Ingredients
Combination 1: To increase the power of erection and number of coitus	<i>Taccharum munja</i> (root), <i>Saccharum officinarum</i> (stem), <i>Desmostachya bipinnata</i> (whole plant), <i>Desmostachys</i> spp., <i>Adenia hondala</i> (tuber), <i>Vetiveria zizanioides</i> (root), <i>Solanum surattense</i> , <i>Malaxis acuminata</i> (wet stem), <i>Malaxis muscifera</i> (wet stem), <i>Sida rhombifolia</i> (root and leaves), <i>Polygonatum cirrhifolium</i> , <i>Polygonatum verticillatum</i> (root stock), <i>Fritillaria roylei</i> (stem), <i>Lilium polyphyllum</i> (bulb), <i>Atylosia goensis</i> (seed), <i>Phaseolus trilobus</i> (seed), <i>Asparagus racemosus</i> (tuber), <i>Withania somnifera</i> (tuber), <i>Velloppam</i> , <i>Mucuna pruriens</i> (seed), <i>Boerhaavia diffusa</i> (whole plant), <i>Coccinia grandis</i> (root), <i>Ipomoea paniculata</i> (tuber), <i>Holostemma ada-kodien</i> (tuber), <i>Cassia absus</i> (seed), <i>Alpinia somnifera</i> (rhizome), <i>Tribulus terrestris</i> (fruit), <i>Glycyrrhiza glabra</i> (root), <i>Desmodium gangeticum</i> (root), <i>Phaseolus roxburghii</i> (seed), <i>Emblica officinalis</i> (fruit), cow's milk, <i>Ficus racemosa</i> (fruit), <i>Piper longum</i> (dried spike), <i>Scindapsus officinalis</i> (dried spike), <i>Vitis vinifera</i> (fruit), <i>Phoenix dactylifera</i> , <i>Madhuca longifolia</i> (flower), sugar, <i>Curcuma angustifolia</i> (rhizome), <i>Piper nigrum</i> , <i>Cinnamomum zeylanicum</i> (bark), <i>Cinnamomum tamala</i> (leaves), <i>Elettaria cardamomum</i> (seed), <i>Mesua ferrea</i> (flower) honey
Combination 2: For vitality	<i>Ipomoea paniculata</i> (tuber), <i>Piper longum</i> (dried spike), <i>Oryza</i> sp. (seed), <i>Buchanania lanzan</i> , <i>Hygrophila auriculata</i> (seed), <i>Mucuna pruriens</i> (seed), honey, sugar, ghee
Combination 3: For vitality	<i>Mucuna pruriens</i> (seed), <i>Triticum aestivum</i> (seed), cow's milk, ghee, honey
Combination 4: For vitality	Cow's milk, ghee, honey, <i>Phaseolus mungo</i> (seed)
Combination 5: For vitality	<i>Mucuna pruriens</i> (seed), cow's milk, <i>Sesamum indicum</i> (seed), sugar
Combination 6: For vitality	<i>Ipomoea paniculata</i> (tuber), ghee, honey
Combination 7: For senile impotence	<i>Emblica officinalis</i> (fruit), ghee, sugar, cow's milk, <i>Piper longum</i>
Combination 8: To quicken ejaculation	<i>Glycyrrhiza glabra</i> (root), ghee, cow's milk, honey
Combination 9: For impotence	<i>Pistacia integerrima</i> (galls), cow's milk, ghee, sugar
Combination 10: To increase sperm count	<i>Holostemma ada-kodien</i> (tuber), cow's milk, ghee, honey
Combination 11: To prolong coitus	<i>Curculigo orchoides</i> (tuber), <i>Asparagus racemosus</i> (tuber), sugar, cow's milk
Combination 12. Senile impotence	Watery portion of curd, sugar, <i>Oryza</i> sp.
Combination 13: Senile impotence	<i>Tribulus terrestris</i> (fruit), <i>Hygrophila auriculata</i> (seed), <i>Phaseolus roxburghii</i> (seed), <i>Mucuna pruriens</i> (seed), <i>Asparagus racemosus</i> (tuber), cow's milk

Note: Information in parentheses refers to the part used.

27.4.1.1 *Asparagus racemosus*

Asparagus racemosus is considered as an aphrodisiac to both males and females.²³ In traditional medicine, the plant is used as a refringent, demulcent, diuretic, aphrodisiac, anticeptic, alterative, antidiarrheal, antidysentery, and galactagogue agent. It is also used for nervous and rheumatic complaints. The roots are rejuvenating and carminative; they are also used as an appetizer, stomachic, and tonic.

27.4.1.2 *Cynomorium coccineum*

The aqueous extract of this plant increased the weight of the testes, diameter of the seminiferous tubules, and the number of tubular cell layers in immature rats. It also elicited notable spermatogenesis in the immature rats. However, serum testosterone and follicle

TABLE 27.7

Aphrodisiac Formulations Available in the Market

Product	Ingredients	Contact
Adam's Secrets	Yohimbe, <i>Avena sativa</i> , androstenedione, saw palmetto, guarana, taurine, Siberian ginseng, <i>Tribulus terrestris</i>	E-mail: info@adamssecrets.com
ArginMax for Men	L-Arginine; ginseng; ginkgo; vitamins A, C, E, B-complex; selenium, zinc, niacin	Web: www.arginmax.com
Ascend for Men	Oats, nettle, saw palmetto, damiana, zinc	Web: www.herbalremedies.com
BetterMan	Radix ginseng (ginseng root), <i>Rhizoma dioscoreae</i> (yam rhizomes), <i>Radix paeoniae alba</i> (white peony root), <i>Herba epimedii</i> (aerial parts of epimedium), <i>Cornu cervi pantotrichum</i> (deer antler), <i>Radix astragali</i> (astragalus root), <i>Poria cocos</i> (poria fungus), <i>Radix morindae officinalis</i> (morinda root), <i>Fructus corni</i> (Cornus fruit), cortex eucommiac (wood cotton bark), <i>Radix angelicae sinensis</i> (dong quai root), <i>Fructus lycii</i> (wolfberry fruit), <i>Radix rehmanniae</i> (rehmannia root), rhizoma chuanxiong (Szechuan Lovage root), <i>Fructus schisandrae</i> (Schisandra fruit), <i>Acanthopanax senticosus</i> (Siberian ginseng), <i>Cynomorium songaricum</i> rupr (fleshy stem of cynomorium), cortex cinnamomi (cinnamon bark)	Web: www.worldclassnutrition/ phonoremor.html
Count Plus	<i>Mucuna pruriens</i> , <i>Withania somnifera</i> , <i>Holostemma adakodien</i> , <i>Sida rhombifolia</i> , <i>Curculigo orchoides</i> , <i>Asparagus racemosus</i> , <i>Hygrophila auriculata</i> , <i>Glycyrrhiza glabra</i> , cow's milk, ghee	E-mail: nhcltdpa@md5.vsnl.net.in
Endow Plus	Yohimbe, damiana, guarana, wild oats, sarsaparilla, ginkgo biloba, saw palmetto	Web: www.tiger-one.com
HerbalArouse-M	Zinc, selenium, manganese, <i>Lepidium meyenii</i> (extract), L-Arginine, L-Methionine, ginkgo biloba (extract), <i>Eleutherococcus senticosus</i> (extract), <i>Withania somnifera</i> (extract), <i>Scocoleus forskohlii</i> (extract), <i>Tribulus terrestris</i> (extract), <i>Lirosma ovata</i> (extract), <i>Avena sativa</i> (extract), gelatin, <i>Tilophora indica</i> (extract), bioperine (piperine), bovine orchic substance, magnesium, dihydroepiandrosterone	Web: www.healingedge.net
Horny Goat Weed	<i>Epimedium grandiflorum</i> (horny goat weed), <i>Lepidium meyenii</i> (maca pure), <i>Mucuna pruriens</i> , <i>Polypodium vulgare</i>	Web: www.pinnaclebody.com
Horny Goat Weed Capsules	<i>Epimedium grandifolium</i> (extract), <i>Schisandra chinensis</i> (extract), <i>Psoralea coryflora</i> (extract)	E-mail: cielo@cieloherbals.com
Horny Goat Weed Plus	<i>Epimedium sagittatum</i> (horny goat weed), <i>Tribulus terrestris</i> (puncture vine), <i>Ptychopetalum uncinatum</i> (muira puama), <i>Avena sativa</i> (wild oats), <i>Turnera diffusa</i> (damiana), Korean ginseng, <i>acanthopanax senticosus</i> (Siberian ginseng), ginkgo biloba, nettle, vitamin B6, zinc	Web: www.discountnaturalherbd.com

TABLE 27.7 (continued)

Aphrodisiac Formulations Available in the Market

Product	Ingredients	Contact
Libido-X	Tea tree oil, yohimbe, damiana, saw palmetto, passion flower, Chinese hibiscus	Web: www.male-sexual-enhancement.com
Nirvana	<i>Tribulus terrestris</i> , <i>Mucuna pruriens</i> , <i>Withania somnifera</i> , <i>Matthiola incana</i> , <i>Cherianthus cheiri</i> , <i>Salvia haematodes</i> , <i>Centauria behen</i>	Web: www.herbal-nirvana.com
Teenex Capsules	<i>Withania somnifera</i> (ashwagandh), <i>Mucuna pruriens</i> (cow hage), <i>trivang</i> , <i>Nux vomica</i> , <i>Hygrophila spinosa</i> , <i>shilajit</i> , <i>Myristica fragrans</i> , <i>Ancylus pyrephrum</i> , <i>mikardhwaj</i> , <i>Crocus sativus</i> (saffron)	E-mail: odpl@sancharnet.in
Vibro Plus	<i>Ptychopetalum uncinatum</i> (muira pauma root), catuaba, Jamaican ginger, echinacea, <i>Turnea diffusa</i> (damiana), panax ginseng (ginseng), <i>Avena sativa</i> (wild oat), cayenne, schizandra, yohimbe bark	E-mail: vhssales@viable-herbal.com
Vigor™	Borage oil, organic flaxseed oil, puncture vine, <i>Withania somnifera</i> (extract), ginkgo biloba (extract), zinc sulphate, vitamin E, gingerol, pyridoxine HCl, folic acid	Fax: (713) 683-8796/920-1917 E-mail: Jac53@prodigy.net
Vita Sex	<i>Epimedium sagittatum</i> (horny goats weed), <i>Tribulus terrestris</i> (puncture vine), <i>Ptychopetalum uncinatum</i> (muira puama), <i>Eleutherococcus senticosus</i> (Siberian ginseng), <i>Withania somnifera</i> (ashwagandha), <i>Turnera diffusa</i> (damiana), <i>Lepidium meyenii</i> (maca), <i>Smilax officinalis</i> (sarsaparilla), ginkgo biloba	E-mail: health@feelgoodnatural.com

Source: From various Web sources. Constituents are listed as found in their respective Web sites. The authors do not hold responsibility to the accuracy; the list is not exhaustive but only indicative.

stimulating hormone (FSH) levels were lowered in the treated animals. It was suggested that the plant extract may have direct spermatogenic effect on the seminiferous tubules of the immature rats, presumably by exerting a testosterone-like effect. This plant extract is better than *Withania somnifera* in this respect.²⁴

In 25-day-old rats, extracts of *Cynomorium coccineum* elicited significant changes in gonadotrophin levels coupled with a significant increase in the ovarian weight and profound folliculogenesis. In 7-day-old animals, these changes were not observed.^{25,26}

No citable clinical studies for this plant showing curative effects for male sexual dysfunction were found.

27.4.1.3 *Mucuna pruriens*

Mucuna pruriens seed powder, when administrated at the dose of 75 mg/kg/day, increased the sexual activity of the male albino rats considerably. The different components of copulatory behavior, including mount frequency, mount latency, intromission frequency, and intromission latency, were found to be influenced by the test drug. The plant contains, among other things, levadopa (L-DOPA), which is reported to arouse sexual desire in patients suffering from Parkinson's disease.²⁷

Earlier, the total alkaloids from the seeds of *M. pruriens*, composed of five alkaloidal bases, were found to increase the population of spermatozoa and weight of the testes,

seminal vesicles, and prostate in the albino rat. Isolation of various chemical constituents has been reported. Beta sitosterol has been isolated from unsaponisable part of the petroleum ether extract of the seeds.^{28,29} No citable clinical studies for this plant showing curative effects for male sexual dysfunction were found.

27.4.1.4 *Piper longum*

In Ayurveda and folk medicine, *Piper longum* fruit is an important ingredient of herbal mixtures used as aphrodisiacs and sexual tonics. This is also supported by animal experiments. Acute (2-h) and chronic (90-day) oral toxicity studies of the ethanolic extract of *Piper longum* were carried out in mice. *Piper longum* extract (0.1 g/kg for 90 days) caused a significant increase in the weight of the lungs and spleen in the treated animals. Interestingly, the extract induced a significant increase in the weight of the reproductive organs, sperm motility, and sperm count but failed to elicit any spermatotoxic effect.³⁰ However, it exerted antifertility effect in female rats.³¹ No citable clinical studies for this plant showing curative effects for male sexual dysfunction were found.

27.4.1.5 *Tribulus terrestris*

Tribulus terrestris is used as a diuretic, cardiotonic, aphrodisiac, emollient, appetiser, expectorant, laxative, etc. It is also used to cure cough, asthma, anemia, and inflammation, and in treatment of male urinary problems and impotence.³² *Tribulus terrestris* extract has been found to increase testosterone and spermatogenesis in male animals.³³

Tribulus terrestris extract (2.5 to 10 mg/kg) treatment for 8 weeks daily resulted in an increase in the relaxation of penile tissue isolated from treated animals compared with that in the control. The relaxant response to electrical field stimulation, acetyl choline, and nitroglycerin in noradrenaline-treated precontracted corpus cavernosal tissue from the treated group was found to be higher compared with that in the control. It was concluded that the enhanced relaxant effect observed was probably due to an increase in the release of nitric oxide from the endothelium and nitrinergic nerve endings, which may account for its claim as an aphrodisiac.³⁴

The alkaloids isolated from *T. terrestris*, harmaline and nor-harmaline (54 mg/kg), caused limb paresis which, in some sheep, was body-side based. It was proposed that these alkaloids accumulate in tryptamine-associated neurons of the central nervous system (CNS) during months of tribulus feeding and gradually interact irreversibly with a specific neuronal gene dextroribonucleic acid sequence.³⁵

A recent study has shown that protodioscin, a phytochemical in *T. terrestris*, improves sexual desire and enhances erection via the conversion of protodioscin to dehydroepiandrosterone.^{36–38} No citable clinical study for this plant showing curative effects for male sexual dysfunction was found.

27.4.1.6 *Tricopus zeylanicus*

Recently, it has been shown that the ethanol extract of *Tricopus zeylanicus* has aphrodisiac activity in male mice as evidenced by an increase in the number of mounts and mating performance. Repeated daily administration was more effective than a single dose. The pups fathered by the treated mice were found to be normal with respect to fetal growth, litter size, and sex ratio.³⁹ No citable clinical studies for this plant showing curative effects for male sexual dysfunction were found.

27.4.1.7 *Vanda tessellata*

Screening of orchids for aphrodisiac activity in male mice resulted in the identification of remarkable aphrodisiac activity in *Vanda tessellata*. The flower and, to some extent, the root (but not the leaf) were found to stimulate the mounting behavior of male mice. This activity was found in the alcoholic extract of the flower. The flower extract (50 or 200 mg/kg) increased mating performance in mice. The pups fathered by the extract-treated mice were found to be normal with an increasing trend in the male/female ratio of the pups. Preliminary toxicity studies have shown no conspicuous toxicity in mice.^{40,41} No citable clinical studies for this plant showing curative effects for male sexual dysfunction were found.

27.4.1.8 *Withania somnifera*

Recent studies have shown that the aqueous extract of *Withania somnifera* (*ashwagandha*) increased the testicular weight, the diameter of seminiferous tubules, and the number of seminiferous tubular cell layers in the immature rats. Further, the extract of the plant elicited notable spermatogenesis in the rat. However, serum testosterone and FSH levels were lower, whereas the luteinizing hormone (LH) level was higher in the treated animals. It was concluded that the extract has a direct spermatogenic influence on the seminiferous tubules of immature rats, presumably by exerting a testosterone-like effect.²⁴ An investigation was conducted to test the curative property of *W. somnifera* root powder in the context of derangements in spermatogenesis caused by treatment of the fungicide carbendazim (methyl 2-benzimidazole carbamate) in Wistar rats. It was found that when the root powder was suspended in milk and fed through oral route to the affected rats, spermatogenesis was restored to the full status (Akbarsha and Kadalmani, unpublished observation). No citable clinical studies for this plant showing curative effect for male sexual dysfunction were found.

27.4.1.9 *Shilajit, the Black Asphaltum*

Shilajit, a popular ingredient in several Ayurvedic preparations, is considered an aphrodisiac. It has been suggested that it be taken with *Withania somnifera* in order to combat sexual weakness.²³

27.4.1.10 *Zingiber officinale*

Studies on experimental animals indicate that *Z. officinalis* (ginger) significantly increases sperm counts and motility.⁴²

27.4.2 Non-Ayurvedic Herbs

The following herbs are not cited in Ayurveda but are used in other countries, such as the U.S., to treat male sexual dysfunction.

27.4.2.1 *Ginseng*

Ginseng is one of the plants subjected to a fairly extensive scientific investigation for application in male sexual dysfunction. The studies have concerned three ginseng plants, *Panax ginseng* (Asian ginseng), *P. quinquefolium* (American ginseng), and *Ganoderma japonicum* (Korean red ginseng). Ginsenosides, which are glycosidic compounds, in the ginseng

plant, are the active compounds under consideration. These compounds are, by and large, antioxidants and enhance nitric oxide (NO) synthesis in the endothelium of many organs, including the corpus cavernosum, which is the erectile tissue of the penis. Ginsenosides also enhance acetylcholine-induced and transmural nerve stimulation-activated relaxation associated with increased tissue cyclic guanosinemonophosphate. Both these properties appear to qualify ginsenosides as male aphrodisiacs.⁴³ However, it is important to test the efficacy of ginseng in humans by means of appropriately designed double-blind, placebo-controlled clinical studies in a large scale and under the various causative conditions of infertility and erectile dysfunction.⁴⁴

27.4.2.1.1 Animal Studies

Under the influence of ginseng, male mice began ejaculation earlier and repeated the action more often in a 45-min observation period and deposited more copulation plugs in 10 days.⁴⁵ Male rats that were administered American ginseng were stimulated in copulatory behavior,⁴⁶ unaccompanied by any change in the circulating levels of testosterone and LH. The prolactin level was reduced, suggesting that ginsenosides affect the pituitary gland directly.⁴⁷ In a different approach, it was shown that daily treatment of mice with an extract of Asian ginseng (2.5 to 100 mg/kg intraperitoneally [i.p.]) or ginsenoside Rg1 (2.5 to 10 mg/kg i.p.) produced a dose-related increase in the number of mice displaying copulatory behavior; ginsenosides Rb1, Rb2, and Ro did not have any effect.⁴⁸ *In vivo* and *in vitro* studies on the corpus cavernosal endothelium of rabbit showed that ginsenosides enhanced the release of NO.^{49,50} Ginsenosides also enhance acetylcholine-induced and transmural nerve stimulation-elicited relaxation in the rabbit corpus cavernosum.⁴⁹ These results suggest that the effects of ginseng extract on corpus cavernosum tissue are mediated by the release of NO from the endothelial cells and perivisceral nerves.⁴⁷

There is at least one animal study that does not recommend the use of ginseng in male sexual dysfunction.⁵¹ In this study, feeding male rats with a diet supplemented with the ginseng extract G115 produced little change in the sexual performance.

27.4.2.1.2 Clinical Studies

There are two reports of clinical studies testing the efficacy of Korean red ginseng in treating erectile dysfunction. The first study⁵² consisted of a total of 90 patients in 3 groups of 30 each (ginseng-treated, placebo-controlled, and trazodone [a proerectile drug treated]). It was reported that in the ginseng-treated patients, changes in early detumescence and erectile parameters, such as penile rigidity and girth, libido, and patient satisfaction, were significantly higher than in the other groups, and the overall therapeutic efficiencies on erectile dysfunction were 60% for ginseng group and 30% for placebo- and trazodone-treated groups. The second investigation⁵³ was a double-blind, placebo-controlled, cross-over study in which a total of 45 patients with clinically diagnosed erectile dysfunction were enrolled. The treatment dosage was 900 mg of Korean ginseng or placebo, 3 times/day for first 8 weeks; this was followed by a 2-week wash-out period where the patients received crossover treatment. It was shown that ginseng treatment increased penetration, maintenance, and Rigiscan test outcome.

27.4.2.2 Yohimbe

The bark of *yohimbe* (*Pausinistalia yohimbe*), an African tree, is the source of yohimbine hydrochloride; in short, yohimbine is an indolequinolonic alkaloid. It has been used historically for several years as a treatment for male and female sexual dysfunction.⁵⁴

Yohimbine is an established alpha 2-adrenoceptor antagonist.⁵⁵ Its action at the CNS and the central endocrine mechanisms is believed to increase sexual arousal.^{56,57}

27.4.2.2.1 Animal Studies

In an early study of the male rat, it was shown that yohimbine increased mounting performance, the percentage of male rats ejaculating in their first heterosexual encounter, and induction of copulatory behavior in sexually inactive male rats.⁵⁶ It was later shown that yohimbine treatment caused a decrease in the ejaculatory latency, postejaculatory interval, and intercopulatory interval.⁵⁸ In a more recent study, male golden hamsters were administered i.p. with yohimbine at a single dose of 2 mg/kg body weight 30 min prior to a number of sexual behavior tests. It was found in the treated animals, when compared with the saline-injected controls, that aspects of male sexual behavior, including ejaculatory frequency, were increased.⁵⁹

27.4.2.2.2 Clinical Studies

There are a few clinical reports encouraging the use of yohimbine in erectile dysfunction.^{60–69} A few authors have opined that these studies are poorly designed trials, that yohimbine does not appear to have any modest therapeutic benefit over placebo,^{54,70} and that the drug has not yet been subjected to scientifically rigorous clinical trials.⁷¹ In the most recent clinical study of 18 men between the ages 40 to 80 years, who were not active smokers,⁷² were given doses of 5.4 mg of yohimbine three times/day for 4 weeks; the doses were then increased to 10.8 mg three times/day for another 4 weeks. Results showed that 50% of patients had a positive effect, an improvement over the already reported success histories.^{60,61,64,65} An earlier clinical study⁶⁹ also supports the view that yohimbine could be therapeutic in the case of nonorganic impotence. Another study⁷³ in Brazil examined 22 patients with organic impotence, all serving as placebo control for 1 month, who were later treated with 100 mg of yohimbine once a day for 1 month. Results did not encourage the use of yohimbine in male sexual dysfunction, supporting the view of Danjou et al.⁷⁴

27.4.2.3 *Lepidium meyenii* (Maca)

Lepidium meyenii is a domesticated crop of the Andean Mountains of Central Peru, cultivated at an altitude of 3000 m and above. Its subterranean part (hypocotyle) is edible and highly valued for its nutritive value. For several hundred years it has been used by the Andean Indians as food and folk medicine and used to increase the fertility and sexual performance of men and women. There have been a few attempts of scientific investigation on the application of this plant as a male aphrodisiac and as a curative of sexual dysfunction.

27.4.2.3.1 Animal Studies

Purified lipidic fraction of the alcoholic extract of dried maca roots was produced into two formulae, namely, M-01 and M-02, differing in the concentration of macaene and macamide. Along with a few additional and new compounds, both formulae were administered to male mice for 22 days, and the mice were allowed to mate with virgin female mice. The sexual performance of the treated male mice almost doubled. In the same report, a test on the latent period of erection (LPE) in the rats with erectile dysfunction revealed that the purified maca products at 180 and 1800 mg/kg body weight were improving the erectile function in the testis-removed rats, as demonstrated by the decrease in LPE. The study revealed, for the first time, an aphrodisiac activity of *L. meyenii*.⁷⁵ In a parallel

study,^{76,77} *L. meyenii* pulverized root (standardized to 0.6% macamides and macaenes), prepared in saline, was administered to male rats through a gastric tube at a daily dose of 15 and 75 mg/kg (acute and chronic doses, respectively) for 15 days. It was shown that the treatment significantly improved the sexual performance parameters of the treated rats.

A third study⁷⁸ was concerned with the spermatogenesis in adult male rats. Male rats receiving aqueous extract of maca root (66.7 mg/ml) twice/day for 14 consecutive days showed increased weight of the testes and epididymis and in the length and frequency of stages IX–XIV seminiferous tubules and a decrease in the stages I–VI. Results indicate that *L. meyenii* root invigorates spermatogenesis acting on its initial stages.

27.4.2.3.2 Clinical Studies

In the only clinical study,⁷⁹ a 4-month oral treatment was conducted on 9 men 24 to 44 years old with *L. meyenii* tablets (1500 or 3000 mg of maca). Results showed increased seminal volume, sperm count per ejaculation, motile sperm count, and sperm motility; the response was not related with any alteration in the serum hormones. The study has limitation in respect of sample size and lack of placebo control.

27.5 Future Perspectives

From the foregoing review it is evident that Ayurveda/phytotherapy has the potential to become into effective therapy for male sexual dysfunction. Additional research is required in order to make it acceptable to health-care providers as an alternative to conventional medicine. It is likely that because sex, infertility, and impotence are highly personal and embarrassing issues, several commercial establishments sought to profit from these problems with cheap advertisements to lure the suffering men. The question of heavy metal concentration in some of these preparations and the likely consequence on the general health are also of concern.^{80,81}

The application of the various therapeutic modalities in the abrogation of male sexual dysfunction needs to be related to the various etiological categories of male sexual dysfunction. Most of the scientific studies, among those conducted so far, have been concerned with impotence and aphrodisiacs. There is an urgent need to conduct scientific studies on the phytochemical therapy for male infertility such as the aspects connected with sperm and the mechanism of action at the cellular and molecular levels.

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Raktaja Krimis (Dermatophytes)

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28.1 Introduction

Raktaja krimis have been described as very minute (invisible with naked eyes) on the basis of the description available in Ayurvedic texts. Their correlation can be made with the dermatophytic fungi, causing superficial infections in man and animals. In Ayurveda, *krimis* of a particular group are responsible for hair loss from the head, face, and other parts of the body; the distortion of nails (*kesha*, *shamarshu*, *nakha*, *loma dhwansa*), hair-raising sensation of a person after a horrific situation (*roma harsha*); itching and pain in the wounds (*vranna*, *kandu*, *toda*); a creeping sensation (*sansarpa*); and the eating away of the skin, vessels, ligaments, muscles, and cartilage (*twak*, *shirasnayu*, *mansa*, *tarunasthi bhakshanana*).

The word *krimi* originated from Dhatu *kramu pad vikcep* (*kramyati sarpati*), meaning that which moves with legs. But this derivation of *krimi* does not include all the *krimis* because there are several *krimis* described in *vedas* and Ayurvedic texts that are without legs. Therefore, they cannot be included in the group of *krimi* according to above definition. However, *yaskacharya* has defined *krimi* as *kravyamedati*, which means the organisms that eat and grow on raw flesh.

28.2 Historical Background

Krimi has been classified as external (*bahya*) and internal (*abhyantra*) in most of the Ayurvedic texts, except *Sushruta Samhita* and *Bhela Samhita*. As compared with other texts, a detailed classification of *krimi* is found in *Charak Samhita*. The classification of *sahaja krimi* is given in *Charak Samhita*. Bhela, who was a prudent student of maharishi Attreya, did not give any account of external *krimi*. It seems that external *krimis* are probably a personal contribution of *Agnivesha*, which was universally accepted by later scholars. The classification of *krimi* by *Kashyap Samhita* is found only in the treatment chapter.

In *Charak Samhita*, *raktaja krimis* have been explained up to a certain extent. On the contrary, *Sushruta* did not pay any attention to the health problems. Sharma et al.³ proposed close similarity of *raktaja krimis* with microbes and parasites. Some *raktaja krimis* can be compared with dermatophytes (etymologically the plants of skin) of the fungal kingdom.

This chapter will focus on the scientific basis of *raktaja krimis* as described in Ayurveda.

28.3 Etiology of *Raktaj Krimis*

According to *Sushruta*, the *kapha* and *pitta* of the body are aggravated by several factors. Examples include eating before the digesting the previous meal; excessively using any indigestible, uncongenial, incompatible, or filthy articles of fare sedentary habits; eating cold, heavy, or fatty meals; sleeping during the day; and excessively using *masha* (meat). In modern system of diagnosis, the minute *krimis* are supposed to be the following species: *Epidermophyton* (1 species), *Microsporium* (14 species), and *Trichophyton* (20 species).

28.4 Clinical Description

28.4.1 Symptoms of *Dadrukustha*

The *dadrukustha* (ringworm caused by dermatophytes) have coloration of linseed flowers or copper. They are also serpigerous and full of eruptions.

The general features of this disease are elevated, spherical, itching, and slowly growing patches. Other types of *kustha* may be considered as tuberculoid leprosy, piodermas, psoriasis, melanodermas, and erysepales. These types are incurable or difficult to treat according to Ayurveda.

28.5 Classification of Krami Roga

Acharya Charak and Sushrut mentioned a special chapter on diseases caused by *krimis*, and classified them into seven types of *purisaj*, six types of *kaphaj*, and seven types of *raktaja*. *Raktaja krimis* circulates in the blood and eventually produce disease symptoms in skin. In Ayurveda, Sushrutra has focused more elaborately on *krimi*. According to him, *raktaja krimi* cannot be seen with the naked eye. He has mentioned seven types of *kustha* (skin disease):

- 1. *Arunkustha* (due to *vata*)
- 2. *Udumbarkustha*
- 3. *Krishyajivhakustha*
- 4. *Kapalkustha*
- 5. *Kakanakakustha* (due to *pitta*)
- 6. *Pandarika*
- 7. *Padrukustha* (due to *kapha*)

Out of these seven types, *dadrukustha* is curable with Ayurvedic treatment. Nomenclature given by Sharma et al.³ is given below.

Ayurvedic Nomenclature	Mycological Nomenclature	Term
<i>Keshad</i>	<i>Trichophyton</i> sp.	Dermatophyte
<i>Lomada</i>	<i>Microsporum</i> sp.	Dermatophyte
<i>Nakhada</i>	<i>Epidermophyton</i> sp.	Dermatophyte
	<i>Trichophyton</i> sp.	

Mycological Types and Species

Raktaja Krimi (Dermatophyte)	Selected Species
<i>Epidermophyton</i>	<i>E. stockdale</i> , <i>E. floccosum</i>
<i>Microsporum</i>	<i>M. audouinii</i> , <i>M. canis</i> , <i>M. gypseum</i> , <i>M. fulvum</i> , <i>M. nanum</i>
<i>Trichophyton</i>	<i>T. tonsurans</i> , <i>T. mentagrophytes</i> , <i>T. rubrum</i> , <i>T. schoenleimii</i> , <i>T. soudanes</i>

28.5.1 Dermatophytes

Dermatophytes (*derma* = skin, *phyton* = plant) are a highly specialized group of keratinophilic fungi that utilizes keratin of the skin. They are also known as ring-worm fungi and tinea

infections. More than 40 dermatophyte species are known to exist.^{1,2,4} The three types of dermatophytes (ringworm) — geophilic, zoophilic, and anthropophilic — are shown below:

Geophilic	Zoophilic	Anthropophilic
Soil as a substrate	Parasitizes on animals and may also cause infections in man	Parasitizes on man
<i>Microsporum gypseum</i> , <i>Trichophyton ajelloi</i> , <i>T. terrestre</i>	<i>Microsporum canis</i> , <i>Trichophyton verrucosum</i> , <i>T. mentagrophytes</i>	<i>Microsporum audouinii</i> , <i>E. floccosum</i> , <i>T. rubrum</i> , <i>T. tonsurans</i> , <i>T. violaceum</i>

Dermatophytes have also been isolated from air, public swimming pools, and barber-shops.

28.5.1.1 Clinical Types

The infection caused by dermatophytes is clinically known as tinea infections. These can be categorized as follows:

28.5.1.1.1 Tinea pedis

This type of infection is also known as athlete’s foot or ringworm of the foot. The causal organism involved in *Tinea pedis* are *Trichophyton rubrum* and *T. mentagrophytes*.

28.5.1.1.2 Tinea unguinum

The ringworm of the nails is referred to as *Tinea unguinum* or onychomycosis. Such infections generally occur in nails that are deformed due to the pressure of tight shoes. The species usually responsible for onychomycosis are *T. rubrum* and *T. mentagrophytes*.

28.5.1.1.3 Tinea corporis

The ringworm of the glabrous skin is called *Tinea corporis* or *Tinea circinata*. It is characterized by a circular lesion exhibiting varying degrees of inflammation. The infection is usually caused by *T. rubrum*, *M. canis*, and *T. mentagrophytes*.

28.5.1.1.4 Tinea imbricata

Tinea imbricata (*dadru*) is a common infection in tropical areas. It is characterized by lesions that are originally circinate but which become irregular and coalescent and do not heal at the center. The causal organism is usually *T. concentricum*.

28.5.1.1.5 Tinea barbae

It is also called ringworm of the beard, barber’s itch, or sycosis. *Tinea barbae* is a folliculitis of the beard and rarely infects the face and neck. *T. verrucosum* of *Tinea barbae* is transmitted from cattle, whereas *T. mentagrophytes* is transmitted from horse or dog.

28.5.1.1.6 Tinea favosa

It is commonly known as favus and is a severe type of chronic ringworm. The causal organisms of favus are *T. schoenleinii*, *T. violaceum*, and *M. gypseum*.

28.5.1.1.7 Tinea cruris

This infection is also known as Dhobi’s itch, ringworm of the groin, jock itch, or eczema marginatum. *Tinea cruris* is an acute or chronic infection usually severely pruritic

dermatophytosis of the groin. The lesions are generally caused by *Epidermophyton floccosum* and various species of *Trichophyton* with special reference to *T. rubrum*.

28.5.1.1.8 *Tinea capitis*

This infection is usually known as ringworm of the scalp. *Tinea capitis* includes dry, diffuse, and scaly lesions of the scalp. Generally, it is caused by *T. tonsurans*, *T. violaceum*, *T. schoenleinii*, *T. mentagrophytes*, *T. audouinii*, and *M. canis*.

28.5.1.1.9 *Mycological Types*

There are three main genera or mycological types: *Epidermophyton*, *Microsporum*, and *Trichophyton*. These genera have been classified purely on the basis of mycology.

28.5.1.1.9.1 *Epidermophyton* — The genus *Epidermophyton* includes only one valid species, *E. floccosum*, which is a good example of anthropophilic dermatophyte. Another species named *E. stockdale* is yet to be recognized. It was reported from only one soil sample. It produces club-shaped macroaleuriospores that are $59.5\text{--}2 \times 12.9\text{--}3.0\text{ }\mu\text{m}$ with two to nine cells. *E. floccosum* rarely parasitizes lower animals. It lives in human and is transmitted from man to man by direct and indirect contact. *E. floccosum* is a cosmopolitan fungus in distribution and invades only human nails and skin. It is not capable of invading human hair. The teleomorphic state of *E. floccosum* is unknown.

The genus *Epidermophyton* is characterized by large, fusiform that obovate and smooth, multiseptate asexual spores known as macroaleuriospores. They are produced in bunches of two or three. They measure $20\text{--}40 \times 6\text{--}3\text{ }\mu\text{m}$ with two to four cells. The microaleuriospores are not formed.

28.5.1.1.9.2 *Microsporum* — There are 15 species of *Microsporum*, which can be categorized on the basis of natural habitat into three broad groups. Eight of them live saprophytically in soil (geophilic). *M. boudardi* is reported only from the African republic of Guinea. There is only one report of *M. boudardi*. The majority of the species of dermatophytes produce two types of conidia in culture. The smaller, pyriform to obovate conidia are known as microaleuriospores; the larger conidia are called macroaleuriospores which are multiseptate, echinulate, variable in size and shape (fusiform to obovate), and thin or thick walled. They are born singly or in hyphae. The length of conidia are 5 to 100 μm and width 3 to 8 μm , with two to five cells. The type species of *Microsporum* is *M. audouinii*. The important species include *M. audouinii*, *M. canis*, *M. gypseum*, *M. fulvum*, and *M. nanum*.

28.5.1.1.9.3 *Trichophyton* — Macroaleuriospores are smooth and either thin or thick-walled and range from clavate to fusiform in shape. They are born singly or in clusters. Microaleuriospores are spherical either pyriform or clavate and are borne singly or in chains (in grapelike bunches). The genus *Trichophyton* has 25 valid species. The type species is *T. tonsurans*.

Some important species are *T. equinum* (teleomorph unknown), *T. mentagrophytes* (teleomorph *Arthoderma benhamiae*, *A. vanbreuseghemii*), *T. rubrum* (teleomorph unknown), *T. schoenleinii* (teleomorph unknown), *T. soudanes* (teleomorph unknown), *T. verrucosum* (teleomorph unknown), and *T. violaceum* (teleomorph-unknown). The common names of fungal diseases are listed in [Table 28.1](#).

TABLE 28.1

Common Names of Fungal Diseases and Corresponding Meanings

General Term	Meaning
<i>Tinea infections</i>	Dermatophytes
<i>Tinea pedis</i>	Ringworm of foot
<i>Tinea unguinum</i> /onychomycosis	Ringworm of the nails
<i>Tinea corporis</i>	Ringworm of glabrous skin
<i>Tinea imbricata</i>	Originally circinate but which become irregular and coalescent
<i>Tinea barbae</i>	Ringworm of the beard
<i>Tinea favosa</i>	
<i>Tinea cruris</i>	Severe type of chronic ringworm
<i>Tinea capitis</i>	Ringworm of the groin
	Ringworm of the scalp

28.6 Diagnostic Methods

The method of diagnosis in the ancient period was according to sign and symptoms of the disease, because pathological investigations were not possible. Generally round, bluish, or reddish spots are formed on the skin. Sometimes boils or eruptions may also be produced. The spots generally spread and sometimes coalesce with other spots to form irregular bigger spots.

In conventional medicine, *krami rogas* are diagnosed by the following laboratory test methods: potassium hydroxide (KOH) squash tests, culture techniques, agar blocks technique.

28.7 Treatment

28.7.1 Ayurvedic Therapies

There are various medicines used in Ayurveda for the cure of *raktaja krimis*. Some of them are described below.

28.7.1.1 Plant-Based Natural Oils

1. *Mahamarichadi taila* — This oil is also prepared from natural products. It is used in all kinds of skin infections, such as boils, ringworm, etc. It consists of *kali marich* (*Piper longum*), *hartal*, *Sweta nishoth*, *lal chandan* (*Santalum* sp.), *nagarmotha* (*Cyperus scarious*), *manasnil*, *jatamansi* (*Nardostachys jatamansi*), *daruhaladi* (*Berberis* sp.), *haldi*, *indrayanmula* (*Citrullus colocynthis*), *kanermula* (*Nerium* sp.), *kuth* (*Saussurea lappa*), *arka* (*Calotropis gigantea*), *dugdha*, and *vatsanabha*.
2. *Laghu-vis-garva taila* — This oil is used in cutaneous infections. It is extracted from *Argemone mexicana* Linn. (Papaveraceae), *Nardostachys jatamansi* DC (Valerianaceae),

Terminalia chebula Retz. (Combretaceae), *Ricinus communis* Linn. (Euphorbiaceae), *Plumbago zeylanica* Linn. (Plumbaginaceae), *Curcuma longa* Linn. (Scitamineae), and *Saussurea lappa* Clarke (Asteraceae).

3. *Karanjadi Taila* — The oil is extracted from *Pongamia pinnata* (Linn.) Merr. (Fabaceae). It is useful in various kinds of skin infections, including ringworm (dermatophytes). It consists of seeds of *karanj* (*Pongamia pinnata*), *kuth*, *chitrakmul* (*Plumbago zeylanica*), *chameli puspa* (*Jasminum* sp.), and *Kaner* (*Nerium oleander* L.).
4. *Ring-ring* — This oil is used for ringworm infections caused by fungi. It is also useful for other cutaneous infections.
5. *Dadrughni-Vati* — This tablet is prepared from *chakramarda* (*Cassia tora* Linn.). It is used for all kinds of skin infections, ringworm, and white spots.
6. *V-Gel*: This gel is useful in vaginitis, vaginal candidiasis and other fungal infections of the vagina. Each gram of V-Gel contains: extracts of *triphal* (4.0 mg), *satapat* (*Rosa damascena* Syn. *R. centifolia*; 3.6 mg), *ela* (*Elettaria cardamomum*; 3.6 mg), *punar-nava* (*Boerhaavia diffusa*; 3.6 mg), *shaileyam* (*Parmelia perlata*; 2.0 mg), *nirgundi* (*Vitex negundo*; 1.6 mg), and *Haridra* (*Curcuma longa*; 1.6 mg). The drug is manufactured by Himalayan Herbal Products Co.
7. *Sankhpushpi taila* — This oil includes extract and oil from the plant *Evolvulus alsinoides*. It is used for ringworm, particularly for infections of scalp (*Tinea capitis*) in children. It is also used to cure scabies and other cutaneous infections.
8. *Tuvarakadi taila* — The oil is extracted from *Hydnocarpus laurifolia*, a plant of tropical origin. It is also known as *chalmongra taila*. The oil is used to cure all kinds of cutaneous fungal infections. The oil is sold in the market. Upadhyay¹¹ reported that *Trichophyton mentagrophytes* was very much sensitive to essential oil of *H. laurifolia*.
9. *Chopchinyadi churna* (powder) — This powder is used for all kinds of skin infections.

28.7.1.2 Role of Different Elements

Certain elements like sulfur, mercury, and copper are used in the treatment of mycotic infections. The metals and nonmetals are used in combination with a powdered mixture of *amla* (*Emblica officinalis*), *baheda* (*Terminalia belerica*), and *harada* (*Terminalia chebula*) referred to as *triphal churna* (powder of three fruits). The formulas that contain different compositions of these herbs and minerals are *arogyavardhini*, *gandhak-rasayana*, and *sukhshura triphala*.

28.7.2 Conventional Medicine Approach

Oral administration of griseofulvin is the best treatment for dermatophytosis in general and for *Tinea pedis* in particular. Earlier, the treatment depended only upon topical application of medicines. Sometimes, results are not favorable with griseofulvin and topical therapy is required. At present, a combination of oral administration of griseofulvin and topical antifungal agents are the main therapies.⁶

In onychomycosis, topical therapy is not successful because the causal pathogen does not respond well to griseofulvin. The patient does not feel relief even if the nail is removed surgically because the growing nail will continuously get infected. Griseofulvin used regularly for consecutive months gives promising results because it acts slowly. Topical application is important in such cases. The current most effective and acceptable preparations are imidazole ointments, such as clotrimazole, miconazole, or ketoconazole, which are fre-

quently used. Oral administration of ketoconazole is recommended in some countries but not in Japan; the rate of hepatic disorders is too high there.

Terbinafine is used as an antimycotic agent, which does not cause hepatic disorders. If terbinafine is orally administered once a day, high concentrations in blood can be achieved.⁷ The transfer of this drug from blood to skin is sufficient enough to be effective.⁸ For deep-seated mycosis, fluconazole is administered. Another drug used is amphotericin B, a polyene-based antimycotic. It is used for the cure of fungal infections like aspergillosis, blastomycosis, candidiasis, cryptococcosis, histoplasmosis, and paracoccidioidomycosis. Fluorocytosine is used especially in Candidiasis or in combination therapy. It inhibits DNA and RNA synthesis. Terbinafine and naftifine are both allylamines used against dermatomycoses and mucosal infections. They inhibit ergosterol biosynthesis by blocking squalene epoxidase.

Azoles include ketoconazole, miconazole, itraconazole, and fluconazole. These are known as imidazoles. They are used for the cure of blastomycosis, coccidioidomycosis, histoplasmosis, and paracoccidioidomycosis and inhibit ergosterol biosynthesis by blocking 14-demethylation of lanosterol. Such infections should also be treated with systemic antibacterial antibiotics because secondary bacterial infection may occur. During the recent years, many antifungal agents were introduced only for topical application owing to their toxicity. Today, a mostly broad spectrum of antifungal agents are preferred.

Increasingly, allergic reactions of the skin are observed today. One reason is the high rate of sensitization power of these antimycotics. There are some adverse reactions such as gastrointestinal and hepatic disorders and photo sensitivity reactions particularly by griseofulvin and ketoconazole.⁶ Allergic contact dermatitis due to these new antimycotic agents of imidazole family is rare. Another problem is the use of application.

28.8 Scientific Basis

Dadrughni-vati is a well-known antifungal agent in Ayurveda. The drug is extracted from pawar (*Cassia tora*), a plant of the Fabaceae family. In Sanskrit it is known as *chakramard*, meaning killer of ringworm. An antifungal phytochemical chrysophanic acid-5-anthrone was isolated by Acharya and Chatterjee.³⁵ *Trichophyton mentagrophytes*, a fungus responsible for causing ringworm infections, can be inhibited by oil of *Hydnocarpus laurifolia* at 1:4 dilution, which is equal to 0.125 µg/ml dilution of miconazole nitrate, followed by seed oil of *Derris indica* (*Pongamia pinnata*) at 1:2 dilution (0.25 µg/ml miconazole nitrate). *Epidermophyton floccosum*, another causal fungus of ringworm, was reported to be the most sensitive to oil of *H. laurifolia*.¹¹

Argemone mexicana, *Ricinus communis*, and *Plumbago zeylanica* are included in *Laghu-visgarva taila* (oil). All the three plants are highly antifungal. Roots of *Plumbago zeylanica* are utilized by people of India and China against skin diseases.^{36,37} The plant extract was found to be active against causal organisms of ringworm fungi.¹¹ The *karanjadi taila* is extracted from *karanj* (*Pongamia pinnata*). The oil was reported as high antifungal agent.²⁹ *Bakuchi taila* is extracted from *Psoralea corylifolia*. The plant is reported to contain antimycotic potential. It showed remarkable activity against *Trichophyton mentagrophytes* and *Epidermophyton floccosum*.¹¹ V-Gel has been very effective in ringworm infections, particularly in severe vaginitis.³⁸⁻⁴⁰

Ayurvedic plants are currently being studied for their antidermatophyte activity to find better treatment for skin infections. Plants are extracted either by boiling in water or by

soxhlet extraction procedures. The extracts are evaluated for their minimum inhibitory concentration (MIC). Colony-forming units of a test fungus are determined in a sabouraud dextrose agar medium, which is prepared and sterilized at 15 lb for 15 min. The MIC of the plant extract is then determined in this system. The commonly used methods are a dry-weight method, in which the actual weight of the harvested fungus is weighed at the end of the incubation period, and the disc diffusion method, in which the effect of the extract placed in the center of the plates is measured on the growth of the fungus by the area of the growth.

There have been numerous studies on the plants used for antidematophyte activity. The antifungal activity of nonvolatile constituents of higher plants has been reviewed earlier by many workers.^{9–14} Many plants produce essential oils as secondary metabolites. Their exact role in the life processes of the plant is unknown. A review of literature reveals that a large number of essential oils were reported to possess fungitoxic activity.^{15–34} Most members of the Asteraceae family are known to contain essential oils that usually have antifungal and cytotoxic sesquiterpene lactones.

28.8.1 Why Ayurvedic Drugs for *Raktaja Krimis*?

Ayurvedic therapy for *raktaja krimis* is mostly based on essential oils and plant extracts. These oils are very effective and are aromatic. The patient feels comfortable with this kind of therapy. The essential oil therapy has become more important after the invasion of secondary infections in dreaded diseases like AIDS and cancer. The risk of opportunistic pathogens is increasing day by day in such patients because of the immunocompromising capacity of the hosts. Generally, the secondary infections are caused by yeasts like *Candida albicans* and other filamentous fungi including *Alternaria*, *Curvularia*, *Phoma*, and *Fusarium*. The oil provides a soothing experience after application on skin and also avoids infections caused by opportunistic pathogens. Moreover, the drugs are relatively cheaper, easily available, and do not have side effects; in most antifungal drugs (allopathy), such as imidazoles, there can be side effects. These ointments are chemical based and not natural like the essential oils used in Ayurvedic therapy.

28.9 Conclusions

The drugs of Ayurveda have a tremendous effect on human beings, as these drugs are natural and easily available. In the case of *raktaja krimis*, most drugs are prescribed for topical application because dermatophytes generally infect the dermis of the human beings. There has been a great need to revitalize this system. Most plants used in Ayurveda are becoming rare and some of them are threatened. Oils present in herbs need to be evaluated in order to know their validity for topical application. The sensitivity of *raktaja krimis* to many oils extracted from herbs have already been evaluated in different parts of the world.

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29

Eye Diseases

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29.1 Introduction

One of the subspecialties of Ayurveda is *shalakyatantra*, which includes diseases of the eye, ear, nose, and throat.¹ In this chapter, management of common eye disorders are discussed along with available scientific information that seems to support various Ayurvedic therapies.

29.2 Anatomy of the Eye in Ayurveda

In intrauterine development of the eye, *pitta dosa*, especially *Alochaka pitta*, is a major contributor along with *kapha* for the development of blood vessels and related structures of circulatory system (*Raktavaha srotasa*).^{1,2} Of five basic elements, the fire element (*teja mahabhoot*) is predominant in the formation of the eye. The differentiation of the eye starts during the fourth month of pregnancy.³ The description of eye development in Ayurveda is very much similar to that known in conventional medicine at this time.⁴ The eye is made up of five orbits (*mandal*), six junctions (*sandhi*), and six coats (*patal*).⁵ The orbits in sequence are as follows: (1) orbit of eyelashes, (2) eyelids, (3) sclera and sclerotic circular margin, (4) corneal area, (5) underneath portion of iris, and (6) pupillary zone or circular area of pupil. The junctions include (1) junction of eye lashes with lids; (2) fornices (singular-fornix), where bulbar conjunctiva gets attached to palpebral conjunctiva; (3) sclerocorneal junction or limbus; (4) pupillary margin; (5) inner canthus; and (6) outer canthus.

Six coats (*patal*) are differentiated as two external and four internal. The external two coats are the upper eyelid and lower eyelid. (Although one may not be able to replace all ayurvedic terms with modern medical term, effort has been made to give more appropriate parallel terms.) The first internal coat is sclera (*tejojalashrit patal*). It is the outermost coat and consists of an aqueous humor (*tejojal*). Other important areas in this layer are the whites of eyes (*shwetapatal*) and the cornea (*karnika*). The second internal coat is the choroid (*mamsashrit patal*), a fleshy coat that has been indicated at times as *krishnapatal*. Important structures in this coat are the choroid (*kalaka*), ciliary body (*upataraka*), pupil (*tara*), and iris (*taraka*). Medas *patal* is the third internal coat. Some scholars consider this as the lens (*drishtimani*) and vitreous humor (*sandrajala*). The fourth internal coat is the retina (*asthipatal* or *drishtipatal*). This coat is said to be placed midway between *krishnapatal* and *medas patal*. Some scholars consider this coat as the pupil (*drihti*), whereas others consider it as the entire retina.⁶

29.3 Etiology of Eye Disorders

Etiological factors of eye disorders discussed in Ayurveda^{7,8} are the following:

1. Constant exposure to cold water
2. Exposure to cold immediately after exposure to heat
3. Constant watching of distant objects
4. Alteration in sleep pattern
5. Tensions
6. Excessive exercise
7. Crying
8. Anger
9. Trauma
10. Consuming sour preparations in excess
11. Controlling essential urges
12. Increased amount of sudation
13. Exposure to gases or fumes, e.g., pollutant on roadways or from industries
14. Purification measures (*panchakarma* [e.g., induced emesis in excess])
15. Observing microscopic objects for long periods

In traumatic eye disorders, the traumatic lesion causes vitiation of *dosas* in the eye.

29.4 Pathogenesis of Eye Disorders

Vitiated *dosas* move upward to the eyes, invade various channels, and lead to malfunctioning of various substructures of the eye. The vitiated *dosas*, especially the *pita dosa*, propagate through the channels or vessels (*srotasa*) toward the head region. They enter various parts of the eye such as the lids, junction, sclera, iris, choroids, ciliary body, pupil, and retina and cause a variety of eye diseases.^{9,10}

29.5 Clinical Features of Eye Disorders

The general symptoms of eye disorders¹¹ are the following:

1. Eyes with muddy discharge
2. Hazy cornea or sclera
3. Congestion of eye
4. Sticking of lids due to eye discharge (due to *kapha*)

5. Heaviness of the lids or eyes (due to *kapha*)
6. Burning sensation or hot feeling of eyes (due to *pitta*)
7. Pricking pain in eyes (due to *vata*)
8. Hyperemia of eyes (in *rakta* predominance)
9. Foreign body sensation
10. Pain in lids
11. Visual disturbances
12. Difficulty in eye movement
13. Difficulty in closing or opening of eye
14. Impaired visual perception

29.6 Classification of Eye Disorders

Ayurvedic texts describe 76 to 96 ocular diseases and have several systems of classification, namely, based on vitiated *dosas* (67), therapeutic management (76), or anatomical structure based (76).^{12–15} The structure-based classification closely resembles the classification mentioned in conventional medicine. The diseases are also characterized based on prognosis: curable, difficult to cure, and incurable. An attempt is made to translate Ayurvedic names of eye disorders, but Ayurvedic scholars of eye diseases may disagree on some of the translations.

29.7 Symptoms, Etiology, Pathology, and Management of Eye Disorders

The management of some of the eye disorders in Ayurvedic texts is very similar to contemporary modern ophthalmology. Only six diseases are selected as examples of Ayurvedic management in this chapter: (1) conjunctivitis (*abhishyanda*), (2) glaucoma (*adhi-mantha*), (3) trichiasis (*pakshmakopa*), (4) chalazion (*utsangini* and *lagan*), (5) cataract (*kaphaja linganash*), and (6) miscellaneous diseases (*Drishtigata vyadhi*). Formulations used in the management of eye diseases are given in [Table 29.1](#).

29.7.1 Conjunctivitis (*Abhishyanda*)

According to Ayurvedic texts, conjunctivitis (*abhishyanda* or *syandam*) refers to a condition in which there mainly exists a congestion in general of all channels situated above the level of clavicle (collar bone) and specifically that of the eye.^{15,16} According to modern ophthalmology, inflammation of conjunctiva characterized by redness of the eye and conjunctival discharge is known as conjunctivitis.¹⁷ According to Ayurvedic texts, it is considered to be one of the major causes for eye disorders. If not treated, in time it can lead to other severe ocular disorders similar to those known in conventional medicine.^{18,19}

TABLE 29.1

Formulations Used in the Management of Eye Diseases

Formula Name	Text Reference	Manufacturer ^a
<i>Triphala ghrita</i>	<i>Siddha yoga sangraha</i> <i>Sahasrayogam</i>	BAB, AVS
<i>Mahatriphala ghrita</i>	<i>Siddha yoga sangraha</i> <i>Sahasrayogam</i>	AVP
<i>Shatavaryadi ghrita</i>	<i>Bharat bhaishajya ratnakara</i>	AVP, AVS
<i>Saptamruta yoga</i>	<i>Siddha yoga sangraha</i>	BAB
<i>Punarnavashatak kwatha</i>	<i>Sharangdhara samhita</i>	BAB, ZP, SP
<i>Rasanjana</i>	<i>Sharangdhara samhita</i>	Classic formula manufacturers
<i>Paradanaga rasanjana</i>	<i>Netra roga vidnyana</i>	Classic formula manufacturers
<i>Punarnavanjana</i>	<i>Netra roga vidnyana</i>	Classic formula manufacturers
<i>Netrashani rasa</i>	<i>Netra roga vidnyana</i>	Classic formula manufacturers
<i>Marichyadi yoga</i>	<i>Yoga ratnakara</i>	Classic formula manufacturers
<i>Nibapatranetrabindu</i>	<i>Dravyaguna vidnyana</i>	Classic formula manufacturers
I-tone eyedrops	Patented and proprietary drug	DP
Ophthalmic eyedrops	Patented and proprietary drug	HDC

^aAVP = Arya Vaidya Pharmacy, Coimbtore, India; AVS = Arya Vaidya Sala, Kottakal, Kerala, India; BAB = Baidyanatha Ayurveda Bhavana, Nagpur, India; DP = Dey's Pharmaceuticals, Calcutta, India; HDC = Himalaya Drug Company, Bangalore, India; SP = Sandu Pharmaceuticals, Mumbai, India; ZP = Zandu Pharmaceuticals, Mumbai, India.

29.7.1.1 Symptoms

The common symptoms of conjunctivitis are muddiness of eyes, inflammation of eyes, epiphora or discharge, itching of eyes, heaviness of the eyes, pain in the eyes, burring of the eyes, and congestion of eyes.^{20,21} Ayurvedic scholars defined this disease as an infectious disease (*aupasargika roga*)²² and classified it into four types of conjunctivitis: *vataja abhishyanda*, *pittaja abhishyanda*, *kaphaja abhishyanda*, and *raktaja abhishyanda*. Descriptions of each are listed below.²³

1. Subacute catarrhal conjunctivitis and catarrhal conjunctivitis (*vataja*) — Symptoms include pricking pain in the eye, stiffness of the appendages of eye (e.g., lids), horripilation, foreign body sensation, roughness, headache, dryness in eye, cool lacrimation, and slight oedema of eye. These symptoms are relieved with oleation and heat treatment.²⁴
2. Acute catarrhal conjunctivitis (*pittaja*) — Symptoms include burning sensation of the eye, suppuration or pus formation, feeling of smokiness, blackish lids, chemosis, edematous lids, aggregation of exudates within the eye, and a feeling of extreme heat. These conditions are relieved with cool therapies.²⁵
3. Catarrhal conjunctivitis or purulent conjunctivitis or mucopurulent conjunctivitis (*kaphaja*) — Symptoms include heaviness in and around eyes, swelling of the eyes, itching of eyes, sticking of the lids due to purulent or mucopurulent discharge, feeling drowsy, and loss of taste. These conditions are relieved with hot therapies.²⁶
4. Acute mucopurulent conjunctivitis (*raktaja*) — Symptoms include red lacrimation along with a pus or blood-stained discharge, red eye, or the entire eye appears to be red with red capillaries. Symptomatology is similar to the *pittaja* type.²⁷

29.7.1.2 Treatment

Patients should be protected from external factors like fumes, dust, bright sunlight, etc. Treatment of conjunctivitis is done in two phases; one is specific for a chronic condition (*amavastha*) and the other is specific for acute condition phase (*teevravastha*).^{28,29} In the chronic phase, the therapy is directed toward toxic waste materials in the body, which can continue up to 4 to 5 days with the same preliminary practices before advising *panchakarma* procedures such as fasting (*langhan*), sudation therapy (*swedan*), applying a layer of thin medicament paste on the lid of the affected eye (*pralepa*), and eating bitter foods to cause elimination of toxic materials into the gut. During this treatment, patients are asked to avoid heavy meals, bathing, intake of decoction, and eye ointment and liners.

After treating the chronic phase (*samavastha*), the patient is given treatment in the phase after the toxic materials are eliminated (*niramavastha*). The treatment includes two aspects: (1) local therapy and (2) general therapy. Certain local therapeutic modalities used in *niramavastha* are *tarpana*, *putapaka*, *parisheka*, eyedrops (*ashchyotana*), eye powder (*anjanna*), medicated smoking (*dhoomra*), and nasal drops (*shirobasti*, *nasya*). Drugs used for treatment of conjunctivitis according to vitiated *dosa*, particularly in the form of eyedrops, are described here. In general therapy, one has to undergo *panchakarma* therapy to eliminate the vitiated *dosa* from the gut and body and its further aggregation and bring them to equilibrium. The specific procedures of general therapy are given only if necessary. The following decoctions are cleaned and filtered through a very fine filter and used as eyedrops.

29.7.1.2.1 Vataja Type

1. Brihat panchamool or brihati (*Solanum indicum*), erand (*Ricinus communis*), and shigru (*Moringa olifera*) in decoction form
2. Nimbapatra (*Azadirrhacta indica*) and lodhra (*Symplocos racemosus*)
3. Saindhav (salt), pippali (*Piper longum*), wala (*Andropogon vetiveria*), and yashtimadhu (*Glycerrhyza glabra*) processed with medicated milk (*siddha dugdha*)

29.7.1.2.2 Pittaja Type

1. Kamal (*Nymphaea* spp.), neelakamal (*Nymphaea* spp.), amalki (*Phyllanthus embelicus*), kantakari (*Solanum xanthocarpum*), brihati (*Solanum indicum*), dashmool, shatavari (*Asparagus racemosus*), talispatra (*Abies webbiana*), and manjishtha (*Rubia cordifolia*) in decoction form
2. Chandan (*Santalinum album*), manjishtha (*Rubia cordifolia*), yashtimadhu (*Glycerrhyza glabra*), lodhra (*Symplocos racemosus*), surarnagarik, and honey; used in painful conditions of the eye

29.7.1.2.3 Kaphaja Type

1. Decoction of roots and bark of bilva (*Aegel marmelos*), kantakari (*Solanum xanthocarpum*), and aragvadha (*Cassia fistula*)
2. Decoction of shunthi (*Zinziber officinalis*), triphala (mixture of fruits of three myrobalans, neem (*Azadiracta indica*), vasa (*Adhatoda vasika*), and lodhra (*Symplocos racemosus*)

29.7.1.2.4 Raktaja Type

A decoction for treating this type includes eyedrops made up of *stree stanya* (breast milk), mixture of *musta* (*Cyperus rotandus*) and *jyeshthamadhū* (*Glycyrrhiza glabra*) in rain water, and *lodhra* (*Symplocos racemosus*) in dehydrated butter from cow's milk (ghee). This is all mixed with water along with triphala and sugar.

A great difference does not exist with regard to the current Ayurvedic therapy and that of traditional practice (textual therapy). However certain formulations cannot be prepared today because of the unavailability of certain ingredients or because of difficulties in proper identification of certain ingredients. Many Ayurvedic physicians use Ayurvedic commercial eyedrops in the treatment of conjunctivitis such as Ophthacare Eyedrops (Himalaya Drug Company, Bangalore, India) or Itone Eyedrops (Dey's Pharmaceutical, Calcutta, India). These products are made under sterile conditions instead of having the companies make their own.

In contemporary ophthalmology, there are various approaches to treating different types of conjunctivitis. Basically, usage of antimicrobial preparations and antiallergics have a major role in the treatment of conjunctivitis.

Ayurveda is especially useful when patients have nonspecific conjunctivitis or have resistance to existing antibiotic preparations. With these types of patients, the use of Ayurvedic therapy either alone or as an adjuvant therapy is useful.

29.7.2 Glaucoma (*Adhimantha*)

According to Ayurvedic texts, glaucoma (*adhimantha*) is an ocular disorder characterized by pain in the eye with a particularly churning or throbbing type of pain. The pain is very severe as if the eye is being scooped out, crushed, or churned.^{30,31} According to modern ophthalmology, glaucoma is a symptomatic condition where there is pain in the eye because of increase in the intraocular pressure of the eyeball. Glaucoma is a condition of the eye in which the ocular tension is raised above the normal. The tension becomes high because of the faulty drainage of the aqueous humor. The tension of the eye is variable in health and is usually between 16 to 23 mmHg measured by the Schiotz tonometer.³²

29.7.2.1 Etiology

According to Ayurveda, if a case of conjunctivitis (*abhischyanda*) is not treated in time, it may lead to glaucoma (*adhimantha*).³³

29.7.2.2 Classification of Glaucoma

According to Ayurveda, glaucoma can be classified into four types (*dosa*-based classification): *vataja*, *pittaja*, *kaphaja*, and *raktaja adhimantha*.³⁴ As per modern contemporary ophthalmology, there are various approaches of classification of glaucoma. Examples include primary and secondary glaucoma, acquired and congenital glaucoma, acute and chronic glaucoma, and open-angle and angle-closure glaucoma.³⁵

29.7.2.3 Symptoms

29.7.2.3.1 Vataja Type

With *vataja* type of glaucoma,³⁶ there is a severe feeling of pain at the eyebrows, temple region, and in the half portion of the head (of the side of an affected eye). There is congestion, muddiness of cornea, tinnitus, and vertigo.

29.7.2.3.2 Pittaja Type

Symptoms of the *pittaja* type of glaucoma³⁷ include severe pain in the eyes as if they were being burnt by fire or a caustic chemical (*kshara*). The eyes have red capillary network throughout and are liver-colored. They are watery with edematous lids, and there is a burning sensation at the head region. The patient is often unconscious.

29.7.2.3.3 Kaphaja Type

With the *kaphaja* type of glaucoma,³⁸ the eyes are heavy, with secretion, edema, echimosi, excessive lacrimation, cold and sticky discharge, hypersensitivity, and headache. The patient sees things as smoky, dirty, and imperfect (suggestive of halos). Other symptoms include a sunken cornea and bulged sclera (suggestive of shallow angle of anterior chamber).

29.7.2.3.4 Raktaja Adhimantha Type

Symptoms of the *raktaja adhimantha* type of glaucoma³⁹ include severe pain in the eyes with tenderness, blood-stained discharge, redness, and ciliary congestion. The patient visualizes colors and flames.

29.7.2.4 Prognosis

If not treated in time, there is a possibility of blindness. According to ancient Ayurvedic texts, one may lose sight within 7 days in the *kaphaj* type, 6 days in *vataj*, 5 days in *raktaj adhimantha*, and 3 days or sudden loss (*tatkal*) in the *pittaj* type.⁴⁰ It is also the understanding in modern ophthalmology that if the glaucoma is not treated properly or remains untreated, irreversible loss of visual field may occur, which can lead to blindness.

29.7.2.5 Treatment

Ayurveda has two lines of treatment for glaucoma: general and specific.^{41–43}

29.7.2.5.1 General Line of Treatment

The patient is advised to rest the affected eye and is given sudation, proper diet, and induced medicated purgation at the period of exacerbation of the disease. The patient should avoid intake of liquors and excess sleep. Detoxification methods (*panchakarma*) such as the oleation, sudation, and induced medicated purgation or enema are advised to eliminate vitiated *dosas* that are responsible for the problem. Another unique technique advised traditionally is bloodletting, where leeches are applied around the temporal region. This technique is not currently popular. The patient is advised to take meat soups with ghee, oils, fats, and bone marrows and processed milk medicated with drugs such as *meshashringi* (*Gymnea sylvestris*) and *ajamoda* (*Carum roxburghianum*). The patient is given *triphala* ghee orally after food. During the acute pain phase, the patient is given *kapitha* medicated milk or *brihat panchamool siddha* ghee or milk.

29.7.2.5.2 Specific Line of Treatment

Many physicians prescribe specific formulations such as *punarnavashtaka kwatha*, *chandra-prabha vati*, *laghu malini vasanta rasa*, *mahatriphala ghrita*, *saptamruta loha*, and many other Ayurvedic medicines orally for treatment, and certain medicaments are applied locally as ophthalmic preparation (*anjanas*). The following ophthalmic preparations are applied gently on palpebral conjunctiva: *parad nag rasanjana* and *punarnava anjana*.

29.7.2.5.3 Cauterization (*Agnikarma*)

Cauterization is done only when there is no relief from other recommended medication. Cauterization is a form of heat therapy. Heat is applied by a special hot metal rod placed in-between the eyebrows or at the lateral ends of the eyebrows. This procedure is currently not practiced.

Ghee is applied to these spots, and the wound is then treated with a paste made of 125 mg of common salt and ten leaves of neem (*Azadirrhacta indica*). If a patient comes with a higher degree of intraocular pressure (IOP), the first line of therapy recommended to Ayurvedic eye departments is surgical procedures (antiglaucoma surgeries [AGS]). For fresh cases with mild to moderate increase in the IOP, Ayurvedic physicians provide Ayurvedic therapy.

In conventional ophthalmology, the treatment of glaucoma includes the use of diuretics, meotic agents (e.g., pilocarpine), and the latest antiglaucoma drugs (e.g., Timolol maleate) as a conservative therapy. Treatment can also include various surgical procedures (AGS) such as iridectomy, cyclodiathermy, and trabeculectomy.⁴⁴

29.7.3 Trichiasis and Entropion (*Pakshmakopa* and *Antarvyavartana*)

29.7.3.1 Pathogenesis

Vata, *pitta*, and *kapha* in their aggravated stages cause dryness and pointed eyelashes. The involvement of *vata* causes the eyelashes to turn inside, touching the sclera and cornea and giving rise to friction and irritation of these surfaces.⁴⁵ Trichiasis involves inversion of a varying numbers of eyelashes, so that they rub against the cornea (eyelid margins are in the normal position). The misdirected eyelashes cause mechanical irritation and injury to the cornea, with pain, lacrimation, photophobia, and ulceration.⁴⁶

29.7.3.2 Treatment

Ayurveda recommends removal of the defective eyelashes.⁴⁷ Again, there is a similarity in the Ayurvedic treatment with the modern therapeutic approach of removing the eye lashes with the epilation forcep.⁴⁸ The contraction of the lid may also cause the above problem. In modern ophthalmology, this can be compared with trichiasis and entropion, respectively. Surgical procedures for trichiasis (*pakshmakopa*) and entropion (*paksh-masankocha* or *antarvyavartana*) discussed by various Aayurvedic scholars are similar to those of the modern surgical procedure of tarsorrhaphy.⁴⁹ In practice, Ayurvedic physicians apply certain pastes on eyelashes or use epilation forceps to epilate the hair follicle of eye lashes. In case of entropion or ectropion, modern surgery usually is recommended.

29.7.4 Chalazion (*Utsangini* and *Lagan*) (Internal Hordeolum, Tarsal Cyst, and Meibomian Cyst)

This consists of a swelling of the meibomian gland following an obstruction of its duct accompanied by chronic inflammation in the surrounding tarsus. More than one chalazion may occur at the same time.⁵⁰

29.7.4.1 Pathogenesis

According to Ayurveda, this condition is due to vitiation of all the three *dosas* (*tridohadushti*) and vitiation of the blood (*raktadushti*).⁵¹

29.7.4.2 Symptoms

The signs and symptoms are itching and pain. In addition, when the eye is opened there is discharge similar to egg yolk.⁵²

29.7.4.3 Treatment

The recommended treatment in both conventional medicine and Ayurveda is surgery. The Ayurvedic practitioner uses the paste of herbal ingredients such as *hirakasis*, *trikatu*, *rasanjana*, and *saindhava* mixed with honey to heal the wound.⁵³

29.7.5 Cataract (*Kaphaja Linganasha*)

Cataract is a well-identified and characterized eye disorder in Ayurveda. *Linganash* is a term derived from two words: *linga* (function) and *nash* (destruction). When there is destruction of the function of the eye, the lens becomes opaque and is known as *linganasha*.⁵⁴ Opacification (formation of opacities) in the crystalline lens is known as cataract.⁵⁵ Ayurveda recognizes three stages of cataract: initial blurred vision stage or refractive error (*timir*), immature cataract (*kacha*), and cataract (*kaphaja linganasha*). These stages are not recognized contemporary medicine. Ayurvedic physicians refer these patients to conventional-medicine doctors for the latest procedures, such as intraocular lens implantations, which are far superior and effective to deal with this condition.

29.7.5.1 Treatment

Although the following methods are not practiced today, the descriptions below are given to show how surgery was practiced thousands of years ago.^{57–60}

Ayurveda mentions preoperative procedures (*poorvakarma*), main operative procedures (*pradhana karma*), and post-operative procedures (*pashchat karma*). The procedure describes piercing the globe at a particular position, specifically at the intersection of two thirds of the limbal side and one third of the outer canthal side. This is where a thin rodlike instrument (*shalaka*) is used to prick in such a way that it does not disturb other important appendages of the eye or blood vessels of the eye; it reaches straight to the opaque lens (*drushtimani*) and produces a typical sound when watery secretions of *linganasha dosa* (liquified lenticular matter) come out. This is somewhat similar to the latest approach of surgery, the phacoemulsification procedure. The procedure needs to be done in such a way that the patient is able to see objects like fingers (*anguli*) and threads (*tantu*) clearly. The eye is covered with a dressing of medicated ghee and is properly bandaged.

Ayurveda also describes the complications resulting from a surgery performed by an inexperienced surgeon (e.g., breaking of the lens in several pieces or lens falling back into vitreous chamber) and how to remedy them.

Ayurvedic ophthalmic surgeons currently follow all modern methods of treatment, including conventional modern drugs for anesthesia. Surgeons also undertake procedures like intraocular lens (IOL) implants as well for cataract.

29.7.6 Miscellaneous Eye Disorders (*Drishtigat Vyadhi*)

Ayurveda mentions 12 disorders of *drishtigat vyadhi*.⁵⁶ Six types are *linganasha* and the other six disorders include *nakulandhya* (eye looks like those of a cat under a flashlight). The pupil

gives a bright light suggestive of retinoblastoma of the eye or glioma of the retina. According to Ayurveda and conventional medicine, it is an incurable disease. Enucleation of the eyeball should be undertaken as soon as possible. In descriptions of *patalgat dushti*, equivalents of vitreous degenerations, cobwebs, mosquitoes, refractive errors, diplopias, and even clues towards retinal detachments are found. A certain description given by *Vagbhata-charya* for *timir* specifically points to hypermetropia and myopia. In *tritiya patalgat dushti*, references are indicative of conditions similar to retinal detachments.

29.8 Scientific Basis

An attempt is made here to review the existing scientific literature on Ayurvedic ophthalmology. The references are taken from different sources such as research journals, postgraduate theses, and Internet databases. Some references do not have details, but they are cited only to show that there has been an interest.

29.8.1 Clinical Studies

29.8.1.1 Conjunctivitis

1. Neem eyedrops on viral conjunctivitis (*netrabhishtyanda* by *nimbapatra netrabindu*) — The drops were prepared in the department of the *shalakyatantra*. There were 30 patients of viral conjunctivitis who were treated for 3 weeks with sterilized eyedrops prepared from the extract of leaves of *Azadirrhacta indica*. The improvement was statistically significant.⁶¹
2. Ophthacare eyedrops (a herbal formulation manufactured by Himalaya Drug Company) — An open multicentric clinical trial was conducted in patients suffering from various ophthalmic disorders, namely conjunctivitis, conjunctival xerosis (dry eye), acute dacryocystitis, and degenerative conditions (e.g., pterygium and pinguecula), and in postoperative patients of cataract with Ophthacare. The formula contains chemical ingredients (details not available) from *Carum copticum*, *Terminalia belerica*, *Emblia officinalis*, *Curcuma longa*, *Ocimum sanctum*, *Cinnamomum camphora*, *Rosa damascena*, and *meldesumpumapum*. These herbs reportedly possess anti-infective and anti-inflammatory properties.⁶²
3. Second study on Ophthacare eyedrops — This was a comparative, double-blind, multicentric-randomized, placebo-controlled clinical trial that included 157 patients suffering from different eye ailments: cataract dry eye syndrome, ocular asthenia, refractive errors, and allergic conjunctivitis with a herbal eyedrop preparation and placebo. In both dry eye syndrome and ocular asthenia, the herbal eyedrop preparation was promising. In a few cataract patients, the vision seemed to improve. No analysis of the data was given. In early myopia, it seems to correct the refractive errors whereas in high myopic conditions it controlled the progressive deteriorations. Subjective improvements were also noted with hypermetropia, presbyopia, and astigmatism. Its healing capacity in allergic conjunctivitis was

certainly better than placebo. It had no short-term or long-term side effects and is considered to be a useful drug in all conditions studied.⁶³

29.8.2 Refractive Errors

Although there are no direct references available on refractive errors in ancient Ayurvedic texts, some of the texts are indicative of refractive disorders. Available studies in this area are described below.

1. *Saptamrut loha* and *yashad bhasma* on refractive errors — Although details of the study are not available, the results were reported to be encouraging.⁶⁴
2. *Saptamrut loha*, *Yashtimadhu ghruta*, *Shatavari ghruta*, *Punarnavadi yoga*, and *Pathyadi kwatha* — The author of this chapter has treated many patients with refractive disorders such as high myopia and high hypermetropia by using these formulas with encouraging results at The Center of Ayurveda and Panchakarma Therapy, Eye Care Clinic.

Some studies done at postgraduate institutes of Ayurveda in India by postgraduate students and others on different eye disorders such as trachoma, pterygium, blephritis, corneal ulcer, and retinal disorders are presented here.

29.8.2.1 Trachoma (Pothaki)

As per Ayurvedic texts,⁶⁵ trachoma consists of multiple projections that are reddish on the inner side of the lid (on palpebral conjunctiva). These symptoms are combined with profuse eye discharge, pain, itching, and heaviness in eyes.

According to modern medicine, trachoma (granular conjunctivitis or granular eyelids) is a specific contagious disease of the conjunctiva and cornea. It is chronic and characterized by a subepithelial infiltration of the conjunctiva by a cellular granulation tissue, which is responsible for the presence of the typical follicles or granulations.

29.8.2.2 Therapeutic Efficacy of Shobhanjana Eyedrops in Trachoma (Pothaki)

In one unpublished study,⁶⁶ there were 30 patients with trachoma, where 22 patients in the prefollicular stage of trachoma and 8 patients in the follicular stage of trachoma. All patients were treated with the eyedrops for 8 weeks. There were 25 patients (85%) who showed complete relief, 3 patients (10%) who have moderate relief, and 2 patients (5%) who showed no relief.

29.8.2.3 Pterygium (Arma)

As per Ayurvedic texts,⁶⁷ this is a disease of sclera (*shwetamandala*). It is a triangular growth from either the inner or outer canthus heading toward the limbus. Ayurveda classifies it into five types.

According to modern ophthalmology, pterygium is a threefold of conjunctiva. It occupies the interpalpebral fissure extending from the inner or outer part of the bulbar conjunctiva on the cornea, and the base spreads out and merges with the subconjunctival tissue.

In Ayurvedic texts, there is mention of both surgical and medical treatment of pterygium. Thirty patients of Pterygium were treated with *Marichyadi yoga* (containing marich [*Piper nigrum*] and triturated with the juice of kesharaj [*Eclipta alba*]) for 4 weeks at one of the institutes by a postgraduate student. Results were encouraging in 75% of the cases.⁶⁸

29.8.2.4 Corneal Ulcer (*Savarna Shukra*)⁶⁹

As per Ayurvedic texts, there is an ulceration of the cornea (*krishnapatal*) because of vitiated increased *pitta*; it is circular in shape, with a little depression on the area of ulcer bed, and it is surrounded by neovascularization. There is pain, redness, and overlacrimation of the ulcer-affected eye. According to modern ophthalmology, the break of the continuous lining of the epithelial line of the cornea is known as a corneal ulcer. There are various causes attributed to the formation of the corneal ulcers.

Yashtimadhu ghruta (medicated ghee composed of *amalki* [*Phyllanthus embelicus*], *haritaki* [*Terminalia chebula*], *bhibhitaki* [*Terminalia belerica*], *musta* [*Cyperus rotandus*], and *haridra* [*Curcuma longa*]) was investigated in 30 patients with corneal ulcers by a postgraduate student at one of the institutes. Significant healing activity was noted within 6 to 7 days ($p < 0.05$).⁷⁰

29.8.2.5 Blepharitis (*Klinnavratma*)

As per Ayurvedic texts,⁷¹ Blepharitis (*klinnavartma*) is explained as two types: simple blepharitis (*Pralinna vartma*) and ulcerative blepharitis (*aklinna vartma*). In simple blepharitis there is swelling of the lids, sticky blood-stained discharge from eyes, and pain and itching of the eyelids. This is a *kaphaja* disorder. In ulcerative blepharitis, there are ulcerations at the level of edges of the eyelids, which may lead to entropion if not treated correctly. According to modern ophthalmology, blepharitis is a chronic inflammatory condition of the margin of the eyelids. It occurs in two forms: nonulcerative (squamous blepharitis) and ulcerative.

In one clinical study⁷² done at the institute by postgraduate students, *rasanjanadi rasakriya* (ointment containing *daruharidra* [*Berberis aristata*] processed in goat's milk) was used to treat 30 patients with blepharitis by external application. The ointment (*anjan*) showed healing activity, reduced edema, discharge, itching, and pricking pain within the first week. The results were significant ($p < 0.05$).

29.8.2.6 Retinal Disorders

29.8.2.6.1 Diabetic Retinopathy

Fifteen patients of diabetic retinopathy (a complication of diabetes mellitus) were given *Vasant kusumakar Rasa*; the other 15 patients were given a placebo for a 12-week period. There was a good improvement in, seven cases, no change observed in seven cases, and deterioration in the condition observed in one case. One case of those seven cases improved. This case initially had retinopathy and showed no evidence of the diabetic retinopathy, but a complete cure was seen at the end of the study. Three cases showed marked improvement, whereas two cases showed marginal improvement. There was no significant change noticed in the biochemical parameters at the end of the therapy.⁷³

29.8.2.6.2 Retinitis Pigmentosa

In an open prospective study⁷⁴ on 15 cases of retinitis pigmentosa having less than 3 ft of vision, 2 cases were screened with electroretinography and visual-evoked potential studies in both pre- and posttherapy. The drugs used in the study were *shatavaryadi ghrita*, *netryatarpaka yoga*, and *mahatraiphal ghrita*, along with certain herbal combinations of *punarnava*, *yashtimadhu*, *amalki*, and other herbs. Treatment was given in either form, orally or locally. *Ghrita* were used locally. All these preparations were prepared at the Center of Ayurveda and Panchakarma Therapy, Eye Care Clinic. Nine patients completed the study, of which

four showed a marked improvement in the visual acuity, four showed moderate improvement, and one patient showed no change.

29.8.3 Studies on Special Therapeutic Procedures

One of the Ayurvedic *panchakarma* therapies used in Ayurvedic ophthalmology is *tarpana* (eye is kept in medicated ghee for a given predetermined period while keeping the patient in a supine position). A preliminary study⁷⁵ was conducted by the author of this chapter on 25 healthy volunteers to assess this therapy's safety profile. Certain parameters such as IOP, temperature, pulse, respiration, blood pressure, and other common ocular symptoms and signs were observed for possible adverse effects. No serious adverse effect were noticed and no significant results were obtained.

29.8.4 Biological Studies

The anticonjunctival activity of the *punarnava* (*Borehaavia diffusa*) and neem (*Azadirachta indica*) was assessed by preparing water extracts and testing them on the experimental model of bacterial conjunctivitis in rats. The experimental study revealed that both extracts as well as marketed gentamicin eyedrops showed statistically significant anticonjunctival activity. Herbal extracts showed comparable effect with the established drug, Gentamycin.⁷⁶

29.8.5 Antimicrobial Activities of Herbs Used in the Treatment of Eye Disorders

A study of *in vitro* antibacterial activity of extracts from the plants *T. chebula*, *E. alba*, and *O. sanctum* was carried out by disk diffusion technique. All showed activity against human pathogenic Gram-positive and Gram-negative bacteria. The activity against salmonella organisms was shown only by *T. chebula*; the action against shigella organisms was shown by *T. chebula* and *E. alba*, but not by *O. sanctum*. The widest spectrum of antibacterial activity was shown by *T. chebula*. It was also most potent. The antibacterial spectrum of *E. alba* was between that of *T. chebula* and *O. sanctum*. The narrowest spectrum of antibacterial activity was observed in *O. sanctum*.⁷⁷

Eugenia aromatica (lavanga) (clove) has been studied for its antibacterial activity. Clove oil inhibited tuberculosis at 1:80 dilution. Ethanol extracts of clove inhibited the growth of *C. botulinum*. The extracts were also found to be effective against several Gram-positive and Gram-negative microorganisms tested on agar media at 5000 ppm.⁷⁸

The chloroform extract of *Azadirachta indica* (neem) oil has been found to be fungicidal for common skin pathogens *Trichophyton* spp. The essential oil has shown antibacterial activity against *Staph. aureus*, *E. coli.*, *S. pyogenes*, etc. *A. flavus* and other fungi are also susceptible to neem oil. Neem has known to have antiviral properties.

The volatile oil from the fruit of *Piper nigrum* (*Maricha*) showed antifungal activity.⁷⁹ A crystalline fraction obtained from bark of *Symplocos racemosa* Roxb. (*Lodhra*) is found to inhibit the growth of staphylococci.⁸⁰ Cold water extract of the fruit of *Terminalia belerica* Roxb. (*bibhitaka*) was found to possess antibacterial activity.⁸¹ *Curcuma longa* (*haridra*) exhibits its antimicrobial activity; sodium curcuminates in a dilution of 1:1 million inhibited micrococcus pyogenes.⁸²

29.8.6 Anti-Inflammatory Activity of Herbs Used in the Treatment of Eye Disorders

The anti-inflammatory activity of various extracts of roots of *Boerhaavia diffusa* Linn. were studied in carragenin-induced edema and formaldehyde-induced arthritis in albino rats. Acetone extract showed most the potent anti-inflammatory effect.⁸³

Commiphora mukul (Guggul) has been extensively studied for its anti-inflammatory effects. The steroidal component of Fraction A of the petroleum ether extract has marked antiarthritic effect comparable with that of hydrocortisone and more potent than phenylbutazone.⁸⁴

Water extract of *Curcuma lona* Linn. (*haridra*) showed significant anti-inflammatory activity in acute carrageenin-induced edema. Active principle curcumin showed anti-inflammatory activity similar with cortisone and phenylbutazone in carrageenin-induced edema in rats with an equivalent dose.⁸⁵

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Ischemic Heart Disease

Karunakaran Gauthaman and Lakshmi Chandra Mishra

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30.1 Introduction

Ischemic heart disease (IHD) is a leading cause of morbidity and mortality in the Western world. Cardiovascular disease (CVD), particularly IHD, has become a worldwide health problem affecting all economic groups of society and is responsible for 25% of all deaths in the world. The mortality due to CVD ranges from 16 to 50% in the developing and developed countries. Because 78% of all deaths in the world occur in the developing countries, the absolute number of deaths is very high in these areas. Due to such high incidence, IHD is now considered a modern epidemic.¹

Considering the huge burden of IHD management, there has been a continuous attempt to develop drugs that will delay the development and halt the progress of the disease. In this regard, there is great potential for identifying outstanding Ayurvedic components or its active principles, especially in consideration of the fact that such substances may provide maximum benefit with cost effectiveness, minimum side effects, and enhancement of patient compliance.

The focus of this chapter will be to explore the scientific basis for the Ayurvedic treatment of IHD with herbs, herbal formulae, and dietary interventions by using modern technologies with a view to establishing a scientific rationale for therapeutic acceptance. The modes of administration and the associated lifestyle changes required for the treatment to be effective will be described.

30.2 Definition

In Ayurveda, IHD, or *hridroga*, is clinically characterized by chest pain produced by increased workload on the heart. The pain is felt mostly over the sternum. The pain spreads toward the left side of the chest and radiates to the left arm, neck, and upper part of the abdomen. There are several types depending upon the characteristic features of the pain. If the pain is acute and has a shifting nature, it is known as *vatika* heart disease. If it is associated with a burning sensation, it is called *pitaja* heart disease. If the pain is mild and is associated with heaviness, nausea, and cough, it is called *kaphaja* heart disease.

In modern medicine, IHD is characterized by the atherosclerotic narrowing of the lumen of coronary arteries and reduction in the blood flow to the heart muscle. The symptoms described are the same as described in Ayurveda. There may be a burning sensation in the chest radiating to the arms, back, abdomen, and jaw; there may also be a pressure — a feeling of tightness within the chest or a feeling of heaviness. These symptoms arise under physical or emotional exertion; later on they appear in the resting state.

30.3 Clinical Description

One of the classical Ayurvedic texts, *Rugviniscaya*, lists a variety of symptoms of heart diseases often experienced by patients:

1. Stiffness of the cardiac region (*hrdayayama*)
2. Stabbing pain in the heart region (*hrdaya dirana*)
3. Heaviness of heart (*hrdaya gautaca*)
4. Weakness of heart (*hrdayaksobha* or *hrdaya klams* or *hridaya upahata*)
5. Cutting pain in the heart (*hrdayapatna*)
6. Fluttering of the heart (*hrdayanirmathana*)
7. Pericardial discomfort (*hridaya pidana*)
8. Pain in the heart (*hridayaruja*)
9. Feeling of emptiness of heart (*hridaya sunyata*)
10. Cutting pain in the heart region (*hridaya sphotana*)
11. Pericardial suppression (*hridayastambha*)
12. Heaviness of heart (*hridayastyana* or *hrtstyana*)
13. Pricking pain in the pericardium (*hridayatoda*)
14. Cardiac pain (*hridayavyadha*, *hrdsula*)
15. Burning sensation in the region of the heart (*hrddaha*)
16. Discomfort in the pericardium (*hrdgraha*)
17. Tachycardia and pericardial muscular twitching (*hrtkampa*)
18. Pericardial pain (*hrdruk*)
19. Pricking pain in the pericardial region (*hrdisula*)
20. Tachycardia (*hrdvega*)
21. Pain and discomfort in the region of the heart (*hrtpida*)

There are five clinical descriptions of ischemic heart disease according to Ayurveda based on vitiated *dosas* and other causes: (1) *vataja*, (2) *pitaja*, (3) *kaphaja*, (4) *Tridosaja*, and (5) *kramija* (caused by worms or parasite infections).

30.3.1 Vataja Heart Disease

The features of *vataja* heart diseases are a stretching, pinching, grinding, cracking, and tearing types of pain in the heart. The other symptoms seen are abnormal complexion,

fainting, fever, cough, hiccups, dyspnea, distaste in mouth, thirst, mental confusion, vomiting, excitement of *kapha* and consequent distress, anorexia, palpitation, wasting, tearing, obstructed movement, severe pricking, piercing, bursting, constricting, or splitting pain, dryness, immobility, emptiness, increased heart rate, unfounded helplessness, grief or fear, tremors, body contractions, dislike for noise, obstructed breathing, dark rings around the eyes, hypersensitivity, and attacks of fear, anxiety, worry, and fainting.

30.3.2 *Pittaja* Heart Disease

Pittaja heart disease is characterized by morbid thirst, heating sensation, burning sensation, cardiac fatigue, a feeling of being inside a smoke-filled place, fainting, sweating, and dryness of the mouth. Other symptoms are feeling of being in a dark place, distress, mental confusion, terror, heat, pyrexia, dizziness, acidity, exhaustion, vomiting (sourness), yellowish skin, eyes, and stool, fever, and flushed or bloodshot eyes.

30.3.3 *Kaphaja* Heart Disease

Kaphaja heart disease is characterized by heaviness, excessive salivation, anorexia, stiffness, suppression of the power of digestion, and sweet taste in the mouth. The movements of the heart become obstructed by *kapha*, which gives the feeling of dullness, fever, cough, and drowsiness. The other features are stiffness feeling of the heart, expectorating mucus, excess of sleep, lassitude, fever, cardiac edema, and heaviness of the limbs.

30.3.4 *Tridosaja* Heart Disease

This type of heart disease is characterized by the manifestation of symptoms of all the three varieties of heart diseases related to three *dosas* described above. The main symptoms are constricting pain and discomfort as a result of the vitiation of all three *dosas* simultaneously, spitting of sticky mucus, stabbing pain, colic pain, sensation as if entering into darkness, anorexia, grayish coloration of eyes, and dryness.

30.3.5 *Kramija* Heart Disease

If a patient suffering from this type of heart disease consumes oils, milk, and jaggery, then nodules (*granthis*) appear on and around the area of the heart, and fluids (*rasa*) located there become sticky. Because of this stickiness, germs appear. The patient shows dark discoloration around the eyes, dry irritated skin, and excess mucus expectoration; the patient is prone to fainting.

In modern medicine, the clinical descriptions of IHD are angina, myocardial infarction, chronic postischemic cardiac failure, and sudden ischemic cardiac death.

30.3.6 Angina

The pain that occurs during periods of myocardial ischemia consists of retrosternal constricting discomfort that may radiate to the arms, throat, or jaw; it is associated with shortness of breath. Three types of angina syndromes are clinically recognized: stable angina, unstable angina, and variant angina.²

30.3.6.1 Stable Angina

Stable angina may be provoked by any stimulus that increases myocardial oxygen demand through increments in blood pressure or heart rate. Typically, it is provoked by exertion (e.g., exercise, hurrying, or sexual activity) or emotions (e.g., stress, anger, fright, or frustration) and is relieved within 2 to 10 min by resting. Symptoms are usually worse in the morning shortly after getting up, probably because the blood pressure is at its peak level at this time of the day; symptoms tend to worsen in cold weather and also after a heavy meal, which increases heart rate. Patients seek medical help for troublesome or fighting chest discomfort, usually described as heaviness, pressure, squeezing, smothering, or choking; it is rarely described as frank pain. This symptom is usually crescendo or decrescendo and lasts 1 to 5 min. Angina can radiate to the left shoulder and to both arms, especially to the ulnar surfaces of the forearm and hand. It can also radiate to the back, neck, jaw, teeth, and epigastrium.²

30.3.6.2 Unstable Angina

This is accompanied by objective electrocardiographic evidence of transient myocardial ischemia (chest pain associated with [ST] segment changes in electrocardiogram [ECG] or T-wave inversions). It is mostly associated with critical stenoses in one or more major epicardial coronary arteries. The following three groups of patients may be said to have unstable angina: (1) patients with new onset (<2 months) angina that is severe or frequent (≥ 3 episode/day), (2) patients with accelerating angina (i.e., those with chronic stable angina who developed angina that is distinctly more frequent, severe, prolonged, or precipitated by less exertion), and (3) those with angina at rest. Unstable angina may be primary (i.e., occurs in the absence of an extra cardiac condition that has intensified myocardial ischemia, such as anemia, fever, infection, tachyarrhythmias, emotional stress, or hypoxemia). Unstable angina may also develop shortly after myocardial infarction.²

30.3.6.3 Variant Angina

This unusual angina symptom is characterized by an unprovoked episode of chest pain which may be associated with ST segment depression in the ECG. IHD is present in 70% of cases, but the remainder of the coronary arteries appears normal. An exaggerated increase in coronary arterial tone (spasm) has been demonstrated in these patients during attacks of angina. These patients are at risk of cardiac arrhythmias, and prolonged attacks of spasm may result in myocardial infarction. The spasm is usually of focal distribution, and even in the absence of coronary artery disease, it can restrict flow sufficiently to produce profound myocardial ischemia.

30.3.7 Myocardial Infarction

In myocardial infarction, the pain is deep and visceral. Patients describe it as a heavy, squeezing, and crushing feeling in the chest around the area of the heart. It is similar in character to the discomfort of angina pectoris, but is usually more severe and lasts longer. Typically, the pain involves the central portion of the chest and radiates to the left arm. Less common sites of radiation are the abdomen, back, lower jaw, and neck. The location of the pain beneath the xiphoid and the patients' denial of heart attack are two main causes responsible for the mistaken diagnosis of indigestion. The pain of myocardial infarction may radiate as high as the occipital area but not below the umbilicus. Weakness, sweating,

nausea, vomiting, giddiness, and anxiety often accompany the pain. The discomfort usually commences when the patient is at rest. When the pain begins during a period of exertion, in contrast to angina pectoris, it usually does not subside with cessation of activity. Other less common symptoms, with or without pain, include sudden loss of consciousness, a confused state, a sensation of profound weakness, the appearance of an arrhythmia, evidence of peripheral embolism, or an unexplained drop in arterial pressure.³

30.3.8 Sudden Ischemic Cardiac Death

Sudden ischemic cardiac death occurs due to abrupt rupture of an atheromatous plaque and coronary thrombosis along with left ventricular failure.⁴

30.4 History and Epidemiology

Heart diseases are well described in Ayurvedic texts. *Charaka Samhita*, compiled during 10th century A.D., describes the reasons for cardiac problems and the means for their prevention as follows:

Cardiac diseases of *kapha* type are born by the intake of fatty meals and over eating, and also by excessive indulgence in sleep, sedentary habits.

The person desiring to be protected from the adverse effects upon his heart, coronary blood vessels, and the contents thereof should particularly avoid all that causes mental affliction.

Charaka Sutra, 30/13

Modern medicine has provided excellent tertiary care and means for the treatment in the form of very effective drugs, angioplasty procedures, radio therapeutics, and sophisticated advanced surgical techniques that can beneficially treat CVD. Despite declines in mortality in the past 2 decades, occlusive atherosclerotic coronary artery disease remains a major cause of morbidity and mortality all over the world. The overall mortality from coronary artery disease has declined by almost 50% since reaching its peak in the mid 1960s. Nonetheless, there is still an excess of 500,000 deaths annually in the U.S. attributable to coronary artery disease. The incidence of myocardial infarction, the most dramatic manifestation of coronary artery disease, remains alarmingly high. Up to 1,500,000 persons sustain a myocardial infarction each year in India; unfortunately, many of them die suddenly, and only 50% survive long enough to receive hospital care.¹

30.4.1 Public Health Scenario

Three of trends of IHD are observed¹:

1. In the U.S., where the epidemic started as early as 1920, the death rate due to IHD is declining without change in morbidity since 1968. Similar trends can be seen in Australia, New Zealand, and Canada.

2. In countries where the epidemic started late, either no change is noticed or there is an increase in the incidence and mortality. Most of the European countries fall into this category.
3. The developing countries have started experiencing the epidemic only recently; no clear trend can be described due to the short period. India is in the third category of trend of IHD. Some of the major surveys conducted indicate that the prevalence of IHD ranges from 22.8 to 65.4/1000 in males and 17.3 to 47.8/1000 in females worldwide. With a progressive adaptation of a Westernized lifestyle and increasing life expectancy, India is experiencing both a health and an epidemiological transition with the progressive decline in mortality rate and disability; this is attributable to infectious disease and nutritional deficiencies and is accompanied by the emergences of lifestyle related to other chronic disease epidemics.⁵

It has been projected by the World Bank Health Sectoral Priority Review that in India, CVD alone will account for 33.5% of deaths at all ages by 2015. There is, therefore, sufficient cause of concern that CVD will be the major public health threat by the end of the millennium (Ministry of Health and Family Welfare, Government of India).

30.5 Etiology

In Ayurveda, *kapha* (atherosclerosis) is the major underlying mechanism of heart diseases. The major causes mentioned are unhealthy lifestyle, lack of physical exercise, excessive indulgence in sleep, and sedentary habits.

In conventional medicine, the etiopathogenesis of IHD includes hyperlipidemia, increased waist-hip ratio, hypertension, insulin resistance, stress, smoking, alcoholism, and environmental pollution. Risk factors are causally related to the development of IHD but do not imply that they are causal to all IHD; they merely indicate that they are the determinants of the disease in a sufficiently large number of individuals in the population studied.

30.5.1 Risk Factors for Ischemic Heart Disease

The differences in the incidence and prevalence of coronary artery disease among countries have stimulated the search for factors that might predict the development of the disease. These are now referred to as risk factors. They include not only the physiochemical characteristics of the individuals, but also certain aspects of lifestyle such as diet, smoking, and behavior patterns. It is hoped that the discovery of such risk factors might help in the effort to prevent coronary heart disease. According to the World Health Organization (WHO) (1985), the major modifiable risk factors associated with IHD are serum cholesterol, cigarette smoking, high blood pressure, obesity, lack of physical activity, and diabetes.

It is recognized that the association of these variables with the disease did not establish any etiological relationship. Unfortunately, a high proportion of patients who develop CVD do not have any treatable risk factors. Furthermore, a large proportion of patients

with risk factors do not develop the disease. Most of the reversible risk factors are predictive of coronary artery disease only in relatively young people; this suggests that in older people the disease process is principally related to aging rather than one of the avoidable risk factors.

30.6 Pathogenesis and Pathology

In Ayurvedic texts, the causes of IHD are the intake of excessively hot, heavy, sour, astringent, and bitter food; physical exhaustion; injury; habitually taking food before the previous meal is digested; worry; and suppression of natural urges. When vitiated *dosas* located in the heart afflict blood (*rasa dhatu*) to cause pain in cardiac region, the condition is called heart disease. It is caused by the obstruction to the coronary arteries (*dhamanis*) of the heart muscles. Because the heart muscles consume a huge amount of energy, they need ceaseless nourishment, meaning they demand a good supply of blood. Any impairment of these blood vessels interferes with adequate blood flow to the heart muscles. If its blood flow is significantly diminished, then the heart signals its difficulties by registering pain or discomfort in the chest. The *vata* located in the heart being obstructed by *kapha* and *pitta* interacts with blood nutrients (*rasa*), causing pain, fainting, and (cardiac) obstruction.

In conventional medicine, IHD implies structural and functional abnormalities of the heart as a consequence of an inadequate supply of blood to its tissue. The heart suffers from ischemia when the blood supply to the heart is decreased, either due to increased demand or decreased supply or more commonly due to a combination of both these factors. Ischemia is caused by the insufficiency of oxygen and reduced availability of nutrient substrates and inadequate removal of metabolites.⁴ Once IHD develops, the cells of the cardiac muscles suffer from a lack of O₂ and nutrition, which may be critically low. The condition is manifested as angina pain. This critical reduction of blood supply is corrected either spontaneously, by resting, or by drugs. In conditions like unstable angina and acute myocardial infarction, more aggressive intervention procedures like thrombolytic drugs and surgical procedures are required to restore coronary blood flow.⁶

IHD is composed of four clinical syndromes caused by myocardial ischemia: angina pectoris, acute myocardial infarction, chronic postischemic cardiac failure, and sudden ischemic cardiac death.⁴ The four major pathogenic components of IHD are based primarily on the consideration that the chronic atherosclerotic background determines manifestation and outcomes of IHD. In addition, the components are also based on multiple acute stimuli that cause transient or persistent myocardial ischemia and the heart's response to ischemia. The pathogenic components of IHD, with their individual etiologic and pathogenic components, collectively cause the four major ischemic syndromes and determine their outcomes. The most common cause of IHD is atherosclerotic, the narrowing of the coronary arteries.

30.6.1 Atherosclerosis

This is a pathological condition that underlines several important disorders, including coronary artery disease, cerebrovascular disease, and disease of aorta and peripheral

arterial circulation. It involves lesions of the intimal layer of the wall of epicardial coronary arteries regardless of its nature or pathogenetic and etiologic mechanism. Serum lipids play a significant role in the genesis and progression of atherosclerosis; low-density lipoprotein (LDL) cholesterol is an especially strong atherogen. Recently, it has been documented that oxidation of LDL molecules is needed for its atherogenic action.^{7,8}

The variable severity of coronary atherosclerosis may result from various types of intimal injury, a variable combination of smooth muscle cell proliferation and lipid ground substance deposition, fibrinogen and fibrin deposition, and from thrombus organization. Stimuli may suddenly reduce regional coronary blood flow, transiently or persistently causing coronary constriction, thrombosis, or both. Weaker stimuli may be sufficient to cause ischemia in the presence of a severe atherosclerotic background. Flow limiting stenosis can reduce the coronary flow reserves and cause ischemia when the increase in myocardial demand is excessive. The responses of the heart to ischemic insult in terms of the development of collateral blood flow are angina, ischemia, necrosis, heart failure, and fatal arrhythmias. Of these components of IHD, undoubtedly the most important is the development of ischemic stimuli. The chronic atherosclerotic background and the response of the heart predispose or modulate the effects of ischemic stimuli.

30.6.2 Natural Course of Myocardial Ischemia

Inadequate oxygenation causes disturbances in the cardiac function. When ischemic events are transient, they may cause angina pectoris. If prolonged, they can lead to irreversible injury to the affected part (myocardial infarction). Reestablishment of blood flow or reperfusion, either spontaneous or drug- or surgery-induced, constitutes the only rational therapeutic option for IHD. But it is an established fact that reperfusion of the ischemic myocardium carries with it some detrimental effects that contribute significantly to the delayed or loss of functional recovery of the heart.⁹ The whole phenomenon has given birth to a well-known entity called ischemic reperfusion injury.^{10,11}

Ischemic reperfusion injury is responsible for delayed or loss of recovery of cardiac function depending on the duration and extent of preceding ischemia. If the ischemic period is brief and is not associated with any irreversible myocardial injury, reperfusion can cause myocardial stunning or postischemic myocardial dysfunction of a relatively short duration (weeks to months). On the other hand, reperfusion of irreversibly injured myocardial cells causes permanent loss of cardiac function. It is now well established that ischemic-reperfusion injury plays a major role in the natural course of almost all forms of IHD, such as exercise-induced angina, variant angina, unstable angina, acute myocardial infarction with early reperfusion, coronary angioplasty, coronary graft surgery, and cardiac transplant.⁶

30.7 Diagnosis and Prognosis

Ayurvedic diagnosis of heart diseases is made by observing the clinical features, description, history, location, and characteristics of the pain as described in Section 30.3. In modern medicine, IHD is diagnosed by physical and laboratory examinations, ECG, stress testing (treadmill testing), and coronary arteriography.

30.7.1 Physical Examination

The patient's appearance may reveal signs of risk factors associated with atherosclerosis or diabetic lesions along with signs of anemia, thyroid disease, and cigarette smoking. Palpitation can reveal thickened or absent peripheral arterial pulse, which are signs of cardiac enlargement and abnormal enlargement of cardiac impulse. Examination of the eyes may reveal increased light reflexes and arteriovenous nicking as evidence of hypertension, whereas auscultation can uncover arterial bruits; a third or fourth heart sound; and, if acute ischemia or previous myocardial infarction has impaired papillary muscle function, a late epical systolic murmur. These disorders may cause angina even in the absence of coronary artery disease.

30.7.2 Laboratory Examination

Urine is examined for evidence of diabetes mellitus and renal disease. Blood is examined for measurements of lipids (cholesterol, total HDL, and LDL), glucose, creatinine kinase, and lactate. A chest x-ray is performed to find gross structural changes secondary to IHD (i.e., cardiac enlargement, ventricular aneurysm, or signs of heart failure). Chest fluoroscopy is done to identify the calcification of the coronary arteries.

30.7.3 Electrocardiogram

A 12-lead ECG recorded at rest is normal in about half of the patients with typical angina pectoris, but there may be signs of an old myocardial infarction. Serial tracings are particularly useful to look for past or evolving myocardial infarction. They show repolarization abnormalities (i.e., T-wave and ST segment changes) and intraventricular conduction disturbances at rest, which are suggestive of ischemic heart disease; they are nonspecific, because they can also occur in pericardial, myocardial, and valvular heart disease. Typical ST segment and T-wave changes that accompany episodes of angina pectoris and disappear thereafter are more specific.

30.7.4 Stress Testing

The most widely used test in the diagnosis of IHD involves recording the 12-lead ECG before, during, and after exercise on a treadmill or using a bicycle ergometer. The test consists of a standardized incremental increase in external workload while the patient's ECG, symptoms, and arm blood pressure are continuously monitored. Performance is usually symptom limited and the test is discontinued upon evidence of chest discomfort, severe shortness of breath, dizziness, and fatigue.

30.7.5 Coronary Arteriography

This invasive diagnostic method outlines the coronary anatomy and can be used to detect important evidence of coronary atherosclerosis to assess the severity of obstructive lesions in the major arteries. Coronary arteriography is indicated for the following:

1. Patients with chronic stable or unstable angina pectoris who are severely symptomatic despite medical therapy and being considered for revascularization (i.e.,

percutaneous transluminal coronary angioplasty or coronary artery bypass graft surgery)

2. Patients with troublesome symptoms that present diagnostic difficulties in whom there is need to confirm or rule out the diagnosis of coronary artery disease
3. Patients suspected of having left main stem or three-vessel coronary artery disease based on signs of severe ischemia on noninvasive testing, regardless of the presence or severity of symptoms

30.7.6 Echocardiography

A two-dimensional echocardiography records the image of the left ventricle (LV), which can identify regional wall motion abnormalities due to myocardial infarction or persistent ischemia. This test can also be used as an aid in the diagnosis of IHD.

30.8 Therapy

In Ayurvedic texts, the treatment for all heart diseases is offered to promote biofire (*agni*) and to purify the channels by *panchakarma* in addition to using natural palliative herbs that have hypolipdemic and antistress activity. For heart diseases with a specific vitiated *dosa*, herbs known to mitigate that *dosa* are given in the form of a decoction or medicated ghee. These herbs are listed in [Appendix 2](#).

A few examples of decoctions, medicated ghee, paste, and powders for each type of heart disease as given in *Ashtanghradaya Samhita* text are cited here just to show the historic use of the formulas. Some of these formulas are still used.

30.8.1 Vataja Heart Disease

1. Water decoction of *punarnava* (*Boerhovia diffusa*), *devadaru*, *pancamula*, *rasna* (*Pluchea lanceolata*), barley grains, *bilva* (*Aegle marmelos*), *kulattha* (*Dolichos biflorus*), and *kola*
2. Medicated ghee prepared with the paste of *haritaki* (*Terminalia chebula*), *sunthi*, (*Zinziber officinale*), *puskaramula* (*Iris germanica*), *vayahstha*, *guduci* (*Tinospora cardifolia*), *kayastha*, *amalaki* (*Tinospora cardifolia*), salt, and *asafetida*
3. Powder of *dadima* (*Punica grantum*), black salt, *sunthi*, *asafetida*, and *amlavetasa*
4. Paste of *puskarahva*, *sathi* (*Hedychium spicatum*), *sunthi*, root of *bijapura* and *haritaki* mixed with *ksara* (*yavaksara*), and ghee
5. Decoction of *yavani*, salt, *ksara*, *vaca* (*Acorus calamus*), *ajaji* (*Cuminum cyminum*), *ausadha*, *putidaru*, *bijahva*, *palasa*, *sathi*, and *pauskara*
6. Powder of *pancakola* (*pippali*, *pippalimula*, *cavya*, *citraka*, *nagra*), *sathi* (*Hedychium spicatum*), *pathya*, *guda* (jaggery, molasses), *bijahva*, and *pauskara* made into a paste with *varuni* (a kind of liquor) fried in *yamaka* (mixture of oil and ghee) and added with salt
7. Decoction of *sunthi* or *varuni*, thin liquor of *dadhi* (curds), or fermented water boiled with corn and made into a drink

8. *Bala taila* or *sukumara* or *satapaka yasti taila* or *mahasneha*
9. Decoction or paste of *rasna*, *jinaka*, *jivanti* (*Leptedenia reticulata*), *bala* (*Sida cordifolia*), *vyaghri*, *punarnava*, *bharangi* (*Cleodendron serratum*), *sthira* (*Desmodium gangeticum*), *vaca*, and *vyosa*, one fourth part of *dadhi*, and sour liquids
10. Warm oil mixed with *sauviraka*, curd water, buttermilk, and salt
11. Paste of *puskaramula*, *sunthi*, and *sati* mixed with alkali, water, dehydrated butter, and salt
12. The paste of *haritaki*, *sathi*, *puskaramula*, *pancakola*, and *matulunga* fried in *yamaka* and with jaggery, clear wine, and salt
13. *Pippalyadi curna* is given along with the decoction of *triphala*, *dhanyamla* (a sour drink), decoction of *kulattha*, curd, *madya* (alcoholic drink), *asava* (a type of fermented drink). The constituents are powders of *pippali*, *ela*, *vaca*, *asafetida*, *yavak-sara*, *saindhava*, *sauvarcala*, *sunthi*, and *ajmoda* (*Carium roxburghiana*).

30.8.2 Pittaja Heart Disease

1. The treatment is directed to mitigate *pitta*. Purgation is given with the juice of *draksa* (*Vitis vinifera*), *Iksu* (*Saccharum officinarum*), *sita* (*Santalum album*), *ksaudra*, and *parusaka*. Along with this emetic therapy is given with *sriparni*, *madhuka* (*Glycyrrhiza glabra*), honey, sugar, and jaggery and is mixed with water.
2. Medicated ghee is prepared with the decoction or paste of *sreyasi*, sugar, *draksa*, *jivaka*, *rsabhaka*, *utpala*, *balam kharjura*, *kakoli* and *meda yugma* with milk, and ghee.
3. Medicated ghee is prepared with the decoction or paste of *prapaundarika*, *madhuka*, *bisagranthi*, *kaseruka*, *sunthi*, *saivala*, and milk.
4. Milk is boiled with the decoction of the bark of *T. arjuna* along with the sugar and decoction of *Pancamula*, or *Madhuka*.
5. Powder of the bark of *Terminalia arjuna* along with ghee, milk, or jaggery.

30.8.3 Kaphaja Heart Disease

The patient suffering from *kaphaja hridroga* is given emetic therapy with the help of the decoction of *vaca* and *nimba* and *pippalyadi curna*. A decoction of *T. arjuna*, as well as other treatment given for *kaphaja* heart disease, is given as a palliative treatment.

30.8.4 Tridosaja Heart Disease

The patient is given a fasting therapy followed by a diet that is suitable for all three *dosas*. After ascertaining less aggravated, more aggravated, and moderately aggravated *dosas*, the patient is given the following therapies that will balance all *dosas*:

1. Wheat flour and *T. arjuna* bark powder boiled with oil, ghee, and jaggery and a diet of milk as well as rice
2. A mixture of equal parts of fine powder of *asafetida*, *ugragandha* (*Acorus calamus*), *vida*, *visva*, *krsna*, *kustha* (*Soussera lappa*), *citraka* (*Plumbago zeylanica*), and *yav ksara* mixed with sufficient quantity of *sauvarcala* and *puskaramula* along with barley
3. Decoction of *dasamula* mixed with salt and *ksara* (alkali preparation)

4. A mixture of equal parts of fine powder of *patha*, *vaca*, *yava*, *haritiki*, *amlaki* (*Phyllanthus emblica*), *vetasa*, *duralabha* (*Fagonia arabica*), *citraka*, *tryusana*, *triphala*, *sathi*, *puskaramula*, *tintidi*, *dadima* (*Punica grantum*), and root of *motulunga*

30.8.5 Parasitic (*Kramija*) Heart Disease

The patients are given rice cooked with meat and ghee along with curd and *patala*. After 3 days, purgation therapy is given. *Dhanyamla* mixed with herbs, salt, *ajaji*, sugar, and *vidanga* is also prescribed.

1. *Vallabha ghrta* — Ghee should be cooked with 50 matured fruits of *haritaki* and two *palas* of *sauvarcala* (a kind of salt).
2. *Baladya ghrta* — Ghee is cooked with the decoction of *bala*, *nagabala*, and *T. arjuna* and one fourth in quantity of the paste of *yastimadhu* (*Glycyrrhiza glabra*). (It is good in *rakta-pitta* condition.)
3. *Arjuna Ghrta* — Ghee is cooked with the paste and juice (or decoction) of *T. arjuna*.

Most of the Ayurvedic formulations contain the bark of *T. arjuna*. It is one of the most popular Ayurvedic herbs being used by the Ayurvedic practitioners for the prevention and management of various CVDs. The stem bark of this plant is used for medicinal purposes. This herb has been scientifically evaluated and has sufficient data to support its use in IHD. The other herbs found useful in IHD are *Aloe vera*, *Coleus forskohlii*, *Inula racemosa*, *Andrographis paniculata*, *Centella asiatica*, *Piper longum*, *Picrorhiza kurrora*, and *Commiphora mukul*. These herbs have also been scientifically evaluated and found to be useful in the management of IHD. Patent herbal formulas such as Abna, Hartone, and Lipistat that contain several herbs are also commercially available.

30.9 Scientific Basis

Pharmacological and clinical investigations are reviewed to explore the scientific basis for the use of Ayurvedic therapies. Clinical and biological studies on several botanicals, including *Crataegus oxyacantha*, *T. arjuna*, *Inula racemosa*, and *Astragalus membranaceus*, have been reviewed and found to have therapeutic benefit for the treatment of CVD.¹² The studies are primarily focused on cardioprotective effects against chemical or biological injuries of the heart. The effects studied include anticoagulant, angiogenic, antiatherosclerotic, anti-infarction, blood vessel endothelium protection, and anticholesterol. All these effects are, directly or indirectly, cardioprotective. They can be useful in the management of IHD.

30.9.1 *Terminalia arjuna*

30.9.1.1 Animal Studies

T. arjuna has been a major treatment for IHD in Ayurveda. Because anticoagulants have been found useful in IHD, *T. arjuna* was investigated for a possible anticoagulant activity. An emulsion of *T. arjuna* bark powder (10 g/kg) was given to rabbits orally for 7 days at

the dosage of 10 g/kg body weight. The treatment caused a significant increase in prothrombin time (20 sec vs. 10.01 sec) and decrease in platelet count (44 vs. 473/thousand). In a similar study¹³⁻¹⁵ with the alcoholic extract, there was no change in prothrombin time.

A water-soluble portion of the total alcoholic extract of *T. arjuna* was found to cause an increase in the force of contraction of a frog heart.¹⁶ In later studies,^{17,18} both negative and positive inotropic effects were observed in isolated perfused frog and rabbit hearts and isolated frog and rat atria. It was suggested that the extract consists of a mixture of subsatances capable of exerting both positive and negative inotropic effects.¹⁹ The aqueous extract of *T. arjuna* was also found to produce dose-dependent sustained hypotension and bradycardia in dogs.²⁰ Intracerebroventricular and intravertebral injection of the extract in chloralose anesthetized dogs caused hypotension and bradycardia in doses as small as 1/10th and 1/20th, respectively, of the intravenous dose. Prior bilateral vagotomy blocked the bradycardia, associated hypotension, and the ability of the intravertebral dose of *T. arjuna* to produce these effects in a lower dose. These observations led the authors to propose that the active constituent in the extract acts centrally.²⁰ The hypotensive effect of the alcoholic extract in dogs is abolished by pretreatment with atropine.¹⁷

Subsequently, a study¹⁹ on the isolated rat thoracic aorta showed that the aqueous extract caused contraction of the aorta followed by relaxation. The initial contraction was blocked by propranolol, whereas the vaso relaxant effect was unaffected. Similarly, in another study,²¹ both the aqueous extract as well as the fraction of the extract containing tannin-related compounds (F_2), produced hypotensive effects. The hypotensive effect of F_2 was not affected by pretreatment of rats with propranolol but was attenuated by pretreatment with atropine. The authors suggested that the hypotensive effect of F_2 may be mediated by cholinergic mechanisms. In another study,²² it was observed that aortic prostaglandin E_2 -like (PGE_2) activity was enhanced in ischemic rabbit aorta pretreated with *T. arjuna*. This finding is significant because PGE_2 causes coronary vasodilatation, and this may explain the beneficial effect of *T. arjuna* in patients with coronary artery disease (CAD). In a subsequent study,²³ myocardial ischemia in rabbits was produced by isoproterenol infusion and confirmed by ECG and later by histopathological examination. In this study, the onset of ischemia and its severity were both reduced by *T. arjuna*. *Abana*, an herbal formula containing *T. arjuna*, significantly increased creatinine phosphokinase (CPK), glutamate oxaloacetate transaminase (GOT), glutamate pyruvate transaminase (GPT), and gamma-glutamyltranspeptidase (γ -GT) in serum following myocardial necrosis.²⁴ The study also showed that 90% of the protection against reduction in glycogen levels in ischemic rats was provided by *T. arjuna*. The beneficial effect of *abana* was further evident from the reduction in mitochondrial enzymes, such as γ -GT and sorbitol dehydrogenase (SDH), by 44 and 48%, respectively.

T. arjuna has also been reported to possess significant hypolipidemic effect in rabbits fed *T. arjuna* bark for 3 months.²³ In another study,^{25,26} it was shown that rabbits fed *T. arjuna* along with a high-cholesterol diet showed a lesser increase in total cholesterol and triglycerides and no change in high-density lipoprotein (HDL) cholesterol as compared with control rabbits. In addition, treating hypercholesterolemic rabbits with *T. arjuna* led to a marked reduction in total cholesterol and triglycerides as well as elevation in HDL-cholesterol as compared with hypercholesterolemic control rabbits. The chronic oral administration of *T. arjuna* was found to augment the endogenous antioxidant compounds of rat heart and also prevented oxidative stress associated with myocardial ischemic reperfusion injury.²⁷ In a recent study,²⁸ pre- and posttreatment of rats with arjunolic acid, a new triterpene and a potent active principle from the bark of *T. arjuna*, provided significant cardiac protection in isoproterenol induced myocardial necrosis in the rats.

30.9.1.2 Clinical Studies

The usefulness of *T. arjuna* in IHD has been confirmed in many clinical studies.^{29–31} In one study,³² *T. arjuna* was found to be a mild diuretic without any cardiotoxic action, but in another study it was reported to be beneficial.³³ A decoction prepared from the bark of *T. arjuna* was administered to patients of chronic heart failure (CHF), essential hypertension, and cirrhosis of the liver. The decoction showed clinical improvement in 42, 62, and 40% of patients, respectively.³³ Improvement in CHF substantiated earlier claims of its cardio-tonic property. On the basis of the description in *Chakradutta Samhita* (an Ayurvedic text) that *T. arjuna* bark powder with milk, water, or ghee causes relief in heart pain,^{34,35} a clinical trial in 12 patients with angina pectoris and hemiplegia following cerebral thrombosis was conducted.¹⁷ A total dose of 20 g of crude bark powder of *T. arjuna* was administered to the patients in divided doses for 1 month. At the end of this period, approximately 80% of the patients showed increase in prothrombin time (24.3 ± 2.65 sec vs. 13.4 ± 1.22 sec; $p < 0.001$) along with some degree of functional improvement. *T. arjuna* was also reported to provide marked improvement in a case of Stokes-Adam's attack following chest pain after 3 months of therapy.¹⁷

T. arjuna was further tested in 30 patients with stable angina pectoris.³⁵ These patients were administered 25 mg/kg body weight dose of *T. arjuna* bark powder divided into three doses per day. The mean anginal frequency by the end of 3 months had declined (1.57 ± 1.28 vs. 3.47 ± 1.04 /day). By the end of 1 month, 10% of patients did not require sublingual nitrate. There was concurrent reduction in systolic blood pressure ($p < 0.001$) without much change in the diastolic blood pressure. Electrocardiographic improvement in terms of reduction in the depth of Q- and T-waves and changes in ST segment configuration, decrease in heart rate, and correction of rhythm disturbance (particularly ventricular premature contraction) was also noted. Biochemical parameters, such as plasma catecholamines, plasma cortisol, blood sugar, and serum cholesterol had declined. A significant reduction in body weight was also evident.³⁵

In a double-blind study³⁶ in 30 patients of decompensated rheumatic valvular heart disease, treatment with 200 mg of *T. arjuna* improved LV fraction significantly (54.41 ± 11.62 vs. $41.47 \pm 8.62\%$; $p < 0.001$). The exercise duration, as obtained on bike ergometry, improved from 98.5 ± 44.2 sec to 178.7 ± 6.1 sec. Heart size also decreased significantly. In a subsequent study,³⁷ 500 mg of *T. arjuna* extract was administered twice daily in 25 patients with CAD. After 3 months of therapy, reduction in grade of positivity of a treadmill test response was observed in 6 patients. Also, improvement in exercise tolerance was evident with concurrent reduction in frequency of anginal attacks and use of sublingual nitrates.

In another similar study,³⁸ 500 mg *T. arjuna* was administered twice daily to 20 patients; 15 had stable angina pectoris (Group A) and 5 had unstable angina pectoris (Group B). In both these groups, patients experienced a reduction in anginal frequency and increase in LV ejection fraction (39.7 ± 9.93 vs. $36.2 \pm 10.08\%$). Group A cases also exhibited a decline in mean systolic blood pressure and body mass index. Treadmill testing on 10 patients of stable angina pectoris showed reversion from moderate to mild changes after 3 months of therapy. The target heart rate during exercise could be achieved without significant chest pain.

In a double-blind, crossover design, placebo-controlled study,³⁹ 500 mg of aqueous and alcoholic extract of the bark of *T. arjuna* was administered every 8 h to 12 patients of refractory chronic CHF (New York Heart Association [NYHA] class IV). It was given in addition to maximal tolerable doses of conventional therapy (i.e., digitalis, diuretics, and

vasodilators). In this study, *T. arjuna* therapy as compared with placebo was associated with the following improvements:

1. Symptoms and signs of heart failure
2. Improvement in NYHA classes (class III vs. class IV)
3. Decrease in echo-LV end diastolic (125.28 ± 27.91 vs. 134.56 ± 29.71 ml/m²; $p < 0.005$) and end systolic volume (81.06 ± 24.6 vs. 94.1 ± 24.62 ml/m²; $p < 0.005$) indices
4. Increase in left ventricular stroke volume index (44.21 ± 11.92 vs. 40.45 ± 11.56 ml/m²; $p < 0.005$)
5. Increase in left ventricular ejection fractions (35.33 ± 7.85 vs. $30.24 \pm 7.13\%$; $p < 0.005$)

These patients were followed up with *T. arjuna* therapy for another 20 to 28 months (mean 24 months) during phase II of the study. There was further improvement in symptoms, signs, effort tolerance, and NYHA class.³⁹

The adjuvant *T. arjuna* therapy has been found to reduce LV mass (140.62 ± 55.65 vs. 159.18 ± 51.11 g/m²) and frequency of angina pectoris (1.08 ± 1.08 vs. 3.5 ± 1.98) per day. Besides these findings, it also improved LV ejection fraction (52.67 ± 12.32 vs. $42.25 \pm 9.96\%$).⁴⁰ Similar observations have been reported in a recent open study⁴¹ conducted on 20 patients of hypertension. Oral administration of Abana, a compound formulation containing *T. arjuna* (30 mg/tablet), resulted in significant reduction of the systolic blood pressure, ECG LV internal diameter, posterior wall thickness, and interventricular septum thickness.⁴¹ The reduction in LV mass was seen from 18 weeks onward and lasted through 42 weeks. Compared with these observations, the decrease in LV mass was found at 12 weeks and maintained up to 36 weeks in those patients who were kept on propranolol.

In an open comparative trial⁴² of safety and efficacy of Hartone (herbal product containing TA) in stable angina pectoris patients, 10 patients were given Hartone 2 capsules twice daily for 6 weeks and 1 capsule twice daily for the next 6 weeks. Hematological and biochemical investigations to assess safety were carried out on days 0, 42, and 84. A serum lipid profile was done before and after therapy. Efficacy was based on the reduction in the number of angina episodes and improvement in stress test. These results were compared with 10 patients of stable angina pectoris on isosorbide mononitrate (ISMN) (20 mg twice/day). In this study, Hartone afforded symptomatic relief in 80% of patients and ISMN in 70%. The number of angina attacks was reduced from 79 to 24/week by Hartone and from 26 to 7/week by ISMN. Although patients of both groups showed improvement in several stress test parameters compared with baseline, the difference was not statistically significant. Hartone improved blood pressure response to stress test in two patients and ejection fraction in one. Hartone was better tolerated than ISMN and showed no evidence of hepatic or renal impairment. Its effects on lipid profile were not consistent. The authors of the study suggested that Hartone is a safe and effective antianginal agent comparable with ISMN and is better tolerated.⁴²

In a double-blind, placebo-controlled crossover study⁴³ comparing *T. arjuna* with ISMN in chronic stable angina patients, 58 male patients with chronic stable angina (NYHA class II-III) with evidence of provokable ischemia on the treadmill exercise test received *T. arjuna* (500 mg), isosorbide mononitrate (40 mg/day), or a matching placebo for 1 week each, separated by a wash-out period of at least 3 days. The patients underwent clinical, biochemical, and treadmill exercise evaluation at the end of each therapy; the

scores were compared during the three therapy periods. *T. arjuna* therapy was associated with a significant decrease in the frequency of angina and need for isosorbide dinitrate (5.69 ± 6.91 mg/week in the treated group vs. 18.22 ± 9.29 mg/week in the placebo therapy; $p < 0.005$). The treadmill exercise test parameters improved significantly during therapy with *T. arjuna* compared with those with placebo. The total duration of exercise increased (6.14 ± 2.51 vs. 4.76 ± 2.38 min; $p < 0.005$), maximal ST depression during the longest equivalent stages of submaximal exercise decreased (1.41 ± 0.55 vs. 2.21 ± 0.56 mm; $p < 0.005$), time to recovery decreased (6.49 ± 2.37 vs. 9.27 ± 3.39 min; $p < 0.005$), and higher double products were achieved (25.75 ± 4.81 vs. $23.11 \pm 4.83 \times 10^3$; $p < 0.005$) during the *T. arjuna* therapy. Similar improvements in clinical and treadmill exercise test parameters were observed with ISMN compared with placebo therapy. No significant differences were observed in clinical or treadmill exercise test parameters when *T. arjuna* and ISMN therapies were compared. No significant untoward effects were reported during *T. arjuna* therapy. *T. arjuna* bark extract (500 mg) given every 8 h to patients with stable angina with provokable ischemia on treadmill exercise. The treatment led to an improvement in clinical and treadmill exercise parameters as compared with placebo therapy. These benefits were similar to those observed with ISMN (40 mg/day) therapy and the extract was well tolerated.⁴³

The antioxidant and hypocholesterolaemic effects of *T. arjuna* bark powder was compared with a known antioxidant, vitamin E, in a randomized controlled trial.⁴⁴ One hundred five patients with coronary heart disease were recruited and separated into 3 groups of 35 each using a Latin-square design.⁴⁴ The groups were matched for age, lifestyle and dietary variables, clinical diagnosis, and drug treatment status. None of the patients took lipid-lowering drugs. Supplemental vitamins were stopped for 1 month before the study began, and American Heart Association Step II dietary advice was given to all. At baseline, total cholesterol, triglycerides, HDL and LDL cholesterol, and lipid peroxide estimated as thiobarbituric acid reactive substances were determined. Group I received placebo capsules, Group II received vitamin E capsules (400 units/day), and Group III received fine powder of *T. arjuna* tree bark (500 mg/day) in capsules. Lipids and lipid peroxide levels were determined at 30 days follow-up. The response rate in various groups varied from 86 to 91%. No significant changes in total HDL, LDL cholesterol, and triglycerides levels were seen in Groups I and II (paired t-test $p < 0.05$). In Group III (*T. arjuna*-treated group) there was a significant decrease in total cholesterol ($9.7 \pm 12.7\%$) and LDL cholesterol ($15.8 \pm 25.6\%$) (paired t-test $p < 0.01$). Lipid peroxide levels decreased significantly in both treatment groups ($p < 0.01$). This decrease was more in the vitamin E group ($36.4 \pm 17.7\%$) as compared with the *T. arjuna*-treated group ($29.3 \pm 18.9\%$). In this study, *T. arjuna* exhibited a significant antioxidant effect that was comparable with vitamin E. In addition, it also had a significant hypocholesterolemic effect.⁴⁴

In toxicological studies¹³ conducted in rats and rabbits, no histopathological changes in the heart, liver, and kidney of these animals were evident even after they had been administered 10 g/kg body weight of *T. arjuna* orally for 40 days. The LD₅₀ of *T. arjuna* extract was 2.5 g/kg by intraperitoneally in albino mice.²⁰ It appears that mice are more sensitive to *T. arjuna* than are rats and rabbits.

In various clinical studies,³⁸ *T. arjuna* has been used in the dose of 1 to 2 g/day, and this dose was found to be the optimum in patients with CAD. At this dose, it is well tolerated.³⁷ However, some patients complained of mild gastritis, headache, and constipation.³⁹ No metabolic, renal, and hepatic toxicity has been reported even when patients were administered *T. arjuna* for more than 24 months.³⁹

30.9.2 *Aloe vera*

In an *in vitro* study,⁴⁵ angiogenic activity of *Aloe vera* gel was investigated by using the most active fraction (F₃) from dichloromethane extract of *Aloe vera* gel. The F₃ increased the proliferation of calf pulmonary artery endothelial (CPAE) cells. In addition, it induced CPAE cells to invade type 1 collagen gel and form a capillary-like tube in an *in vitro* angiogenesis assay; it also increased the invasion of CPAE cells into matrigel through an *in vitro* invasion assay. F₃ also increased the effect on the messenger ribonucleic acid expression of proteolytic enzymes, which are the key participants in regulation of extracellular matrix degradation.

In a clinical study,⁴⁶ 5000 patients with atheromatous heart disease presented as angina pectoris were studied over a period of 5 years with diet containing *Aloe vera* and husk of isabgol. The dietary treatment increased HDL and produced a marked reduction in total serum cholesterol, serum triglycerides, fasting and postprandial blood sugar levels in diabetic patients, and total lipids. The clinical profile showed a reduction in the frequency of anginal attacks; gradually, the doses of the drugs, such as verapamil, nifedipine, beta-blockers, and nitrates, were reduced. Diabetic patients benefited the most. Although the exact mechanism of the action of the above two herbs is not known, it appears that they probably act through their high fiber contents. Both these herbs should be further studied. No undesirable side effects were noted.

30.9.3 *Coleus forskohlii*

C. forskohlii (CF) has been used in Ayurvedic medicine for heart diseases, spasmodic pain, painful micturition, and convulsions. Forskolin, an alkaloid isolated from CF, inhibits adenosine di-phosphate (ADP)-induced and collagen-induced platelet aggregation in human and rat platelet-rich plasma.⁴⁷ These studies demonstrated an important role of plasma adenosine, a natural antiplatelet and vasodilatory agent produced by vascular endothelium, in the antiplatelet activity of forskolin. Findings also proved that the effect can be greatly potentiated by the clinically used drugs dipyridamole and dilazep. The anticoagulant effect of *C. forskohlii* may be helpful in IHD.

Coleonol, a diterpene isolated from *C. forskohlii*, has been shown to lower the blood pressure of anesthetized cats and rats and of spontaneously hypertensive rats due to relaxation of the vascular smooth muscle.⁴⁸ In small doses it showed a positive inotropic effect on isolated rabbit hearts as well as on cat hearts *in vivo*. Coleonol also exhibits nonspecific spasmolytic activity on the smooth muscle of the gastrointestinal tract in various species but not on the bronchial musculature of guinea pig. Large doses of coleonol have a depressant action on the central nervous system. The positive inotropic effect of *C. forskohlii* provides the rationale for its use in IHD. In another study,⁴⁹ forskolin activated a membrane bound adenylatecyclase and a cytoplasmic cAMP-dependent protein kinase to a much higher degree than does isoprenaline. The authors postulated that the adenylatecyclase activation may be correlated with the positive inotropic effect via an enhanced calcium uptake by the heart muscle cell.

30.9.4 *Inula recemosa*

The effects of *I. recemosa* on biochemical parameters in rats with myocardial infarction induced by isoprenaline injection was investigated.⁵⁰ The effects on circulating GOT, LDH, CPK, cAMP, cortisol, pyruvate, lactate glucose, and cardiac cAMP adenyl cyclase levels were gradually increased, and serum and cardiac cAMP-PDE levels were gradually

decreased from 1 to 120 h after the first injection of isoprenaline. The rats pretreated with ciplar (beta-blocker) or *pushkarmula* (indigenous drug) showed fewer changes as compared with untreated infarcted rats. Posttreatment with *I. racemosa* also produced similar results. *I. racemosa* was found to be more effective given before infarction induction rather than given after infarction induction.

30.9.5 *Andrographis paniculata* (Kirata)

A. paniculata (AP) (Kirata) was investigated in seven infarction-induced dogs to study the protective effect.⁵¹ One hour after the development of myocardial infarction by formation of thrombus, aqueous extract of AP was injected intravenously. Six infarction-induced dogs served as the control group. As compared with the control group, the treated group showed increased levels of prostacyclin (PG12), inhibition of thromboxane A2 (TXA2), elevated cAMP in platelets, lowered creatine kinase isoenzyme MB (CK-MB) peak, shortened euglobulin lysis time (ELT), decreased release of platelet beta-(1-4)galactosyl transferase (beta-GT) and inhibited platelet maximum aggregation rate, reduced size of the ischemic area recorded by epicardial ECG, and a lowered amplitude of ST segment elevation; a Q-wave appeared in only one dog. Pathologically, the myocardial structure surrounding the initial ischemic area became relatively normal, whereas the degree of myocardial degeneration and necrosis in the central part of the ischemic area was mild. These data suggest that AP may limit the expansion of ischemic focus, exert marked protective effect on reversibly ischemic myocardium, and demonstrate a weak fibrinolytic action. These observations support the use of AP in Ayurvedic therapies of IHD.

The effect of AP and fish oil was further studied on atherosclerotic stenosis and restenosis after coronary angioplasty in dogs.⁵² Preliminary results showed that AP can significantly relieve atherosclerotic iliac artery stenosis induced by both deendothelialization and high cholesterol diet. The authors concluded that AP may play an important role in preventing restenosis after coronary angioplasty, and fish oil may be useful in reducing the extent of restenosis after coronary angioplasty. AP has also been shown to alleviate the ischemia-reperfusion injury in experimental dogs.⁵³ As compared with a control ischemia group, the AP-treated group showed reversal of the changes in the following parameters: superoxide dismutase (SOD), malondialdehyde (MDA), Ca²⁺ in myocardial cells, and ultrastructural changes of myocardial tissues. The observations were confirmed in additional studies.⁵⁴

A component (API0134) of AP was studied in an experimental atherosclerotic rabbit model to determine the effects on nitric oxide, endothelin, cyclic guanosine monophosphate, lipid peroxide, and superoxide dismutase. The study showed that API0134 possesses the effects of antioxidation, preserving endothelial function, and maintaining the balance of NO/ET. *A. paniculata* crude extracts were further fractionated and studied for various cardiovascular activities, mainly the hypotensive and the antithrombotic effects.^{55,56}

30.9.6 Lipistat

Lipistat is made up of equal proportions of extracts of *T. arjuna*, *I. racemosa* Hook, and latex of *C. mukul*. The formula was given to rats at different doses (225, 350, 450 mg/kg) orally daily for 6 days/week for 60 days.⁵⁸ Thereafter, the rats were subjected to isoproterenol-(ISO) induced (85 mg/kg, s.c. for 2 days) myocardial necrosis. Gross and microscopic examinations (histopathology) were done along with estimations of myocardial tissue high energy phosphates stores and lactate content. The study showed protective

effect of the formula. The authors suggested that the formula may be potentially useful in the prevention of IHD.

30.9.7 *Centella asiatica*

The effects of the total triterpenic fraction of *C. asiatica* on serum levels of the uronic acids and lysosomal enzymes involved in mucopolysaccharide metabolism (beta-glycuronidase, beta-N-acetylglucosaminidase, arylsulfatase) in patients with varicose veins were studied.⁵⁸ The results of this trial provide an indirect confirmation of regulatory effects of the extract of *C. asiatica* on metabolism in the connective tissue of the vascular wall and thus may prove useful in treatment of IHD. Total triterpenic fraction of *C. asiatica* has been found effective in improving venous wall alterations in chronic venous hypertension and in protecting the venous endothelium.⁵⁹

30.9.8 *Piper longum*

An amide, dehydropipernonaline, isolated from *P. longum* has been shown to have coronary vasorelaxant activity.⁶⁰

30.9.9 *Picrorhiza kurrora*

The ethanol extract of *P. kurrora* rhizomes and roots was found to exhibit a cardioprotective effect on ISO-induced myocardial infarction in rats as measured by lipid metabolism in serum and heart tissue.⁶¹

30.9.10 *Commiphora mukul*

Hypercholesterolemia is a known risk factor for IHD. In a clinical trial⁶² with 61 hypercholesterolemia patients, 31 patients were given guggulipid, representing 50 mg of guggulosterones, and 30 patients were given placebo capsules twice daily for 24 weeks. The compliance of patients was greater than 96%. In the treated group, total cholesterol level decreased by 11.7%, the LDL by 12.5%, triglycerides by 12.0%, and the total cholesterol/HDL cholesterol ratio by 11.1% as compared with baseline levels; the levels were unchanged in the placebo group. The lipid peroxides, indicating oxidative stress, declined 33.3% in the guggulipid group without any decrease in the placebo group. The combined effect of diet and guggulipid at 36 weeks was as great as the reported lipid-lowering effect of modern drugs. After a wash-out period of another 12 weeks, changes in blood lipoproteins were reversed in the guggulipid group without any changes in the placebo group. Side effects of guggulipid were headache, mild nausea, eructation, and hiccups in a few patients. In another similar study,⁶³ guggulipid also showed similar results.

30.10 Summary and Discussion

In recent years, considerable attention has been paid to the utilization of traditional systems of medicine including Ayurveda in the management of IHD. WHO has recom-

mended utilization of alternative forms of medicine for total health care. *T. arjuna* is one of several popular Ayurvedic herbs already being utilized by the Ayurvedic practitioners for the prevention and management of IHD. Other herbs with great potential to improve quality of life of individuals with IHD and that are commonly used are *Aloe vera*, *Cole forskohlii*, *Inula recemosa*, *Andrographis paniculata*, *Centella asiatica*, *Piper longum*, *Picrorhiza kurrora*, and *Commiphora mukul*. The experimental data available supports their use in IHD.

According to Ayurveda, IHD is the outcome of faulty diet and stressful lifestyle which leads to an *ama* state (i.e., hyperlipidemia) leading further to *dhamani praticaya* (thickening of arteries) and *dhamani kathinya* (hardening of arteries), resulting into angio-obstruction and aggravation of *vata dosa* showing chest pain and angina. The principle treatment is to promote *agni* (biofire) and to purify the channels by *panchakarma* in addition to the use of natural palliative drugs that have cardiogenic, hypotensive, hypolipidemic, and anticoagulant properties. The recent studies show that *T. arjuna* has shown varying degrees of cardioprotective effect as evaluated in terms of being cardio tonic, hypotensive, hypolipidemic, and anticoagulant. Toxicity studies show that *T. arjuna* is a safe and effective herb in the treatment of IHD without any harmful side effects.

In modern medicine, the treatment of IHD involves expensive and chronic drug therapy or equally expensive interventional procedures, such as thrombolytic therapy and surgical recanalization. Reperfusion injuries and undesired side effects of drugs are the major drawbacks of the conventional therapies. There is a vast and untapped source of medicinal plants with cardioprotective effects used in Ayurvedic therapies. The recent research on *T. arjuna* and other Ayurvedic herbs suggests a better cost-effective and socially acceptable therapeutic option for IHD.

The drugs mentioned in this chapter have been used in the treatment of heart disease for thousands of years. The current research data show that they can be utilized effectively in deadly heart diseases, including IHD. The relief provided by *T. arjuna* in angina and its effect on ST segment changes and T-wave depression in IHD are quite impressive. The Ayurvedic herbs discussed in this chapter have the potential of improving quality of life, while avoiding the side effects of conventional treatment. LV function improvement by *T. arjuna* in IHD cannot be overlooked, especially in view of the fact that IHD has an annual mortality rate of 40% under conventional treatment. The widespread use of *T. arjuna* and the other herbs can improve the quality of life in individuals with IHD and potentially save millions of lives.

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31

Urolithiasis (Mutrashmari)

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31.1 Introduction

Formation of stones in the urinary tract is a global phenomenon and is described in ancient Ayurvedic scriptures as *mutraashamari*. It is said to be one of the eight most troublesome diseases (*mahaorgas*).¹ Ayurvedic texts have classified the stones according to *dosic* profiles, namely, *vata*-, *pitta*-, or *kapha*-related and others. Here we have to interpret that the very mention of *dosic* involvement is indicative of the biochemical influences in the formation of stones.

The formation of a stone (calculus) can be at any level in the urinary system. These stones are most frequently formed in the kidney, but they pass farther down the urinary tract toward the bladder. They are intensely painful as they pass along the ureters and out through the urethra.

There are useful management and herbal treatments for urolithiasis that have been currently investigated extensively. The main aim of this chapter is to summarize the management of urolithiasis with herbs, herbal formulations, and dietary and lifestyle inventions; to understand the Ayurvedic concept of urolithiasis; and to explore the scientific basis of Ayurvedic therapies.

31.2 Epidemiology

The overall probability of forming stones varies in different parts of the world. The risk of developing urolithiasis in normal adults appears to be lower in Asia (1 to 5%) than in Europe (5 to 9%) and in North America (12% in Canada, 13% in the U.S.). The highest risk was reported in Saudi Arabia (20.1%).² The compositions of stones and their location in the urinary tract, bladder, or kidneys may also significantly differ in different countries. Moreover, in the same region, the clinical and metabolic patterns of stone disease can change over time. In India, bladder stones accounted for 30% of all urinary stones in 1965, but their prevalence had dropped to 5% in 1985. Concurrently, the chemical composition of stones in the upper urinary tract changed; the prevalence of calcium oxalate stones rose from 26 to 82%, and the prevalence of struvite stones fell from 20 to 5%.³ In Japan, bladder stones decreased from 50 to 5% from 1950 to 1985.⁴ In Portugal, over a 20-year period, the prevalence of calcium stones rose from 64 to 82%, struvite stones decreased from 14 to 3%, and uric acid stones decreased from 19 to 12%.⁵

31.3 Definition

Urolithiasis is defined as a stone or stonelike hard substance formed in the urinary tract. The definition is consistent with current knowledge of urolithiasis — the accretion of hard,

solid, nonmetallic minerals in the urinary tract consisting of a nucleus of organic material around which urinary salts are deposited in concentric layers.

31.4 Clinical Description

A renal calculus is similar to that of the kadamba flower. It is three-layered, resembles a stone, and is either hard or smooth in texture.⁶ The prodromal signs and symptoms described in Ayurvedic texts consist of severe pain around or near the urinary bladder region, suprapubic region, internal urethral orifice, testicles, and in the penis. Common symptoms include distension of urinary bladder, fever, anorexia, dense and turbid urine, dysuria, fatigue, and odor of urine resembling the smell of a sheep.^{15,16}

The major clinical features described are pain in the umbilical and suprapubic regions and in the penis; obstructed urinary flow; split voiding of urine; hematuria; honey colored or yellowish red urine, turbid urine; sandlike particles passing along with urine; pain aggravated by jumping, swimming, riding a horse or camel, climbing in upward direction; and polyurea.^{17–19}

All these clinical features are very much similar to those currently known. In conventional medicine, clinical features of stones vary according to their size, shape, and location of the stone and the nature of underlying condition. The most common complaint is intermittent dull pain in the loin or back increased by movement. Proteins, red cells, or leucocytes may appear in the urine.

31.4.1 Clinical Features According to the Location of the Calculi

31.4.1.1 Renal Calculus

Pain is characterized by a fixed dull ache in the angle between the lower border of the last rib and lateral border of sacro spinalis. Pain is also felt anteriorly in the corresponding hypochondriac region. Pain worsens with movement like running, jumping, and climbing up stairs and eases with rest. Sudden gripping pain is felt in the loin and tends to radiate toward the groin. Patients may experience fitful sleep because of pain. Pain may be associated with hematuria and may be complained of either during or after an attack.

31.4.1.2 Ureteric Calculus

Ureteric colic starts as soon as the stone enters into the pelviureteric junction and recurs at shorter or longer intervals as long as the stone remains in the ureter. Ureteric colic ceases when the stone is ejected into the bladder or impacted in the ureter. When the stone is present in the upper one third of the ureter, pain starts in the loin or near the renal angle and gradually radiates to the groin. Pain is gripping and starts suddenly. The patient experiences fitful sleep because of pain, which is often associated with hematuria and may be complained of either during or after an attack. When the stone is at a lower level, pain commences rather anteriorly just above the iliac crest and is referred along the two branches of the genitofemoral nerve to the testis in males, labia majora females, and anteromedial aspect of the thigh in both sexes. When the stones enter into the intramural part of the ureter in males, pain is referred to the tip of the penis, and the patient complains of strangury.

When the stone is impacted, colic ceases; a dull ache arises according to the site of impaction. Such pain varies in intensity, and it increases with exercise and is relieved by rest.

31.4.1.3 Vesical Calculus

With increased frequency of micturition, pain is often referred to the tip of penis or the labia majora and becomes aggravated by running and jolting. Children may scream and pull the prepuce for pain after micturition and experience hematuria at end of the micturition.

31.5 Etiology

In Ayurveda the causes of urinary calculi are mainly nonadoption of the purificatory measures such as emesis, purgation, and medicated enemas in order to eliminate the vitiated *dosas* (toxic materials) and practice of unhealthy diets and lifestyles. These factors are responsible for the formation of calculi.⁷⁻¹⁰ They are primarily classified into two categories: unhealthy diet or excessive physical activity.

In conventional medicine, there are three primary factors considered responsible for stone formation. They are the supersaturation of stone-forming compounds in urine, the presence of chemical or physical stimuli in urine that promote stone formation, and the inadequate amount of compounds in urine that inhibit stone formation (e.g., magnesium, citrate).

Categories of specific risk factors for stone formation are listed below.¹¹

1. Diet associated with stone formation — Vitamin A deficiency; a high-oxalate diet rich in purine levels; a diet high in protein from animal sources, glucose, or sucrose; etc.
2. Medication associated with stone formation — Calcium supplements, vitamin D supplements, ascorbic acid in megadoses (4 g/day), sulfonamides, triamterene, indinavir, etc.
3. Diseases associated with stone formation — Hyperparathyroidism, renal tubular acidosis (complete or partial), jejunioileal bypass, Crohn's disease, intestinal resection, malabsorptive conditions, sarcoidosis, hyperthyroidism, etc.
4. Anatomical abnormalities associated with stone formation — Tubular ectasia (MSK), pelviureteral junction obstruction, calix diverticulum or calyceal cyst, ureteral stricture, vesicoureteral reflux, horseshoe kidney, etc.

The additional risk factors include habitually low urine volume, high urine excretion of calcium, uric acid and oxalate, low urine pH (uric acid and cystine are less soluble in acid urine), and high urine pH (struvite and calcium phosphate are less soluble in alkaline urine). Some of the biochemical processes not only become relevant here, but also lay the basis for the drug therapy.

The stone is the outcome of accretion of inorganic material around an organic nidus not soluble in its own solution. Urine that is an end-point excreta, in a liquid form, represents the biochemical status of a person's metabolism. For example, in normal urine, nephrocalcin is an acidic glycoprotein, rich in γ -carboxy glutamic acid, which inhibits calcium oxalate crystal growth. The nephrocalcin present in the organic matrix of calcium oxalate

kidney stones resembles the nephrocalcin present in the urine of the patient from whom the stone was removed, but it differs from the nephrocalcin in normal urine. The stone's former nephrocalcin lacks 'Y'-carboxy glutamic acid, which reduces to air-water interfacial films that are less stable than those formed by nephrocalcin from normal urine. It is safe to presume that the alteration of the biochemical quality of urine can help in the prevention of stone formation. The biochemical quality of urine can change with the quality and quantity of fluid inputs, the type of diets, and the constitutional factors. This in turn can play a great role in the formation or nonformation of stones in the urine. Ayurveda suggests a number of herbs and herbal preparations for stone breakdown. It is possible that the therapeutic agents are capable of altering the chemical composition of the urine and its pH.

31.6 Pathology

In Ayurveda, the concept of renal calculi pathogenesis is indicated as when the *kapha dosa* is vitiated because of the etiological factors, *kapha* reaches to the urinary system and, with the help of *vata* and *pitta dosas*, dries up and forms the calculus.¹² There is another similar opinion regarding the pathogenesis of urolithiasis.¹³

Urinary concretions may vary greatly in size. There may be particles like sand anywhere in the urinary tract or large stones in the bladder. Staghorn calculi fill the whole pelvis and branch into the calyces and are usually associated with pyelonephritis. Deposits may also be present throughout the renal parenchyma, giving rise to nephrocalcinosis.¹⁴

31.7 Classification of Renal Calculi

Ayurvedic texts have described four types of urinary calculi: *sleshmaashmari*, *pittaashmari*, *vataashmari*, and *sukraashmari*.²⁰

1. *Sleshmaashmari* — In *sleshmaashmari*, stones are white, unctuous, and as big as a hen's egg. They produce symptoms such as dysuria, cutting, incising, pricking pain, heaviness, and a cold sensation over area the bladder region.
2. *Pittaashmari* — In *pittaashmari*, stones are reddish, yellowish, and blackish and resemble the color of honey. They produce a sucking type of pain, burning sensation, a warm feeling over the bladder region, and *ushnavata*.
3. *Vataashmari* — In *vataashmari*, stones are dusty in color, hard, irregular, rough and nodular or spiny like the kadamba flower. Patients experience severe pain (and may scream); pull out the prepuce; and have difficulty when passing flatus, urine, and stools.
4. *Sukraashmari* — *Sukraashmari* occurs in adults only. It is due to suppression of ejaculation for months or years and frequent coitus or coitus interruption. The semen to be ejaculated will be obstructed, condensed, and brought in-between the scrotum and penis (prostatic part of the urethra) by *vata*. This causes dysuria, scrotal swelling, and lower abdominal pain. The special characteristic of *sukraashmari* is that handling can dissolve it.

In conventional medicine, urinary calculi are classified according to their chemical components. Examples include uric acid and urates, calcium oxalates calcium and ammonio-magnesium phosphate (Struvite), cystine, combinations of the preceding items, and drugs or their metabolites (e.g., phenytoin, triamterene). Descriptions of these components are highlighted below.

1. The uric acid stone is moderately hard and brown. It is usually multiple, typically faceted, occurs in acid urine, and shows concentric rings on the cut surface. The uric acid usually combines with urates and sometimes with oxalates to form opaque stones.
2. The oxalate stone is the most common form of calculus and consists of calcium oxalate. It is extremely hard and the surface is rough and sometimes spiny (mulberry calculus). It causes marked irritation due to rough surface and becomes dark in color due to bloodstains. The outer layers often contain urates, which are usually a single crystal, and give an exceptionally good shadow on radiography.
3. The phosphatic stone consists of calcium phosphate and triple phosphates (ammonio-magnesium phosphate). It is white, smooth, chalky, and is easily broken up; occurs in alkaline urine; and is easily distinguishable from other types. It often fills the renal calyces taking on their shape such as stag horn calculus. In alkaline urine it grows rapidly because it is smooth; a phosphatic calculus gives rise to few symptoms until it attains a large size. Deposits of phosphates are often formed on the surface of uric acid and oxalate stone due to changes in chemical characteristics of the urine as a result of infection.
4. The uric acid and oxalates stones occur away from gross infection in the renal pelvis and sometimes in the bladder as primary stones. The phosphatic stones that form as a result of infection usually in the bladder, but sometimes in the renal pelvis, are known as secondary stones.
5. Cystine calculi are hexagonal, white, translucent, and occur in acid urine. They are soft like bees wax, pink or yellow when first removed, and change color to a greenish hue on exposure to air and light. They are radio-opaque due to the sulfur atom.
6. Xanthene calculi are extremely rare. They are smooth, round, brick red, and show lamella structure.

It should be noted that with the Ayurvedic description of urinary stones, *sleshmaashmari* can be correlated with the phosphatic stone and *pittaashmari* can be correlated with urate stones, *vataashmari* can be correlated with oxalate stones, and *sukraashmari* can be correlated with spermolith or seminal concretions of conventional medicine.

31.8 Diagnosis

After taking a detailed history, diagnosis is made on the basis of laboratory investigations and diagnostic imaging for mere confirmation.

31.8.1 Laboratory Investigations

The routine laboratory investigation includes examination of the urinary sediment and a dipstick test for red cells, white cells, nitrate, pH of the urine, and measurement of serum creatinine. In patients with fever, an analysis of C-reactive protein, a white blood cell count, and a urine culture are carried out. In cases of vomiting, serum sodium and serum potassium should be measured. To exclude metabolic risk factors, it is important to measure the serum calcium and serum urate.

31.8.2 Diagnostic Imaging

Routine examination involves a plain film of kidneys, ureters, and bladder plus an ultrasound examination or an excretory pyelography (urography). The latter examination must not be carried out in patients with an allergy to contrast media, serum creatinine ($>200 \mu\text{mol/l}$), or treatment with metformine and myelomatosis. Other special examinations that can be useful in diagnosis are spiral (helical) unenhanced computerized tomography (CT), retrograde and antegrade pyelography, and scintigraphy. Because of the risk of impaired renal function due to lactic acidosis, the antidiabetic drug metformine should be stopped 2 to 3 days before administration of iodine containing contrast medium. Spiral (helical) CT is a new noninvasive technique that might be considered in case iodine containing contrast medium cannot be administered.²¹

Health conditions frequently associated with stone formation are as follows²²:

1. Obstruction and infection of urinary tract
2. Climate or occupation giving rise to high concentration of constituent materials in the urine and reduced urine volume
3. Hypercalciuria
4. Hyperoxaluria
5. Inherited disorders (e.g., cystinuria, xanthinuria, gout)

31.9 Prognosis

From an Ayurvedic perspective, except for *sukraashmari* or seminal concretions, the other three types are manageable by surgery. If the renal calculus is associated with swelling in the scrotum and umbilical region, retention of urine with severe pain, and passing of urine with fine particles, the condition is said to be incurable.²³ In conventional medicine, prognosis of renal calculi is generally good. Occasionally, death or significant morbidity occurs, but most of the patients recover.

31.10 Treatment

The management of urolithiasis in Ayurveda basically includes herbal formulas, alkaline liquid, and surgical procedures. Newly formed calculi can be cured by herbal formulas,

but chronic calculi has to be treated surgically. Oiling, induced sweating, medicated emesis, purgation and enemas should be given in the prodromal state of the disease only in order to cure it completely.²⁴

31.10.1 Type of Management

31.10.1.1 Shamana Therapy

Palliative treatment includes administration of herbal drugs and herbal formulas orally. The palliative drugs used to treat renal calculi are analgesic, diuretic, and linthnotriptic agent and are able to balance *vata*. Preparations of *varuna*, *gokshura*, *pashana bheda*, *shilajitu*, *ela*, *veerataru*, *brihati*, *kantakari*, *yava kshara*, *kushmanda*, *trapusa*, *hazrul yahud bhasma*, etc. are commonly used in renal calculi.

A list of commonly used formulations is in [Table 31.1](#).

31.10.1.2 Shodhana Therapy

Cleansing treatment includes *prepanchkarma* procedures such as external and internal oleation and induced sweating, and *panchakarma* procedures, such as medicated emesis, purgation, and enemas. Most of the Ayurvedic classics recommend medicated enemas for the treatment of urolithiasis. *Saindhavadi taila niruha vasti*²⁵ and *vrushadi asthapana vasti*²⁶ are generally used in renal calculi. The idea here is that transmucosal fluxes are encouraged away from the kidney for removal of unwanted metabolites, thereby reducing the ionic load on the kidney filtration system. This may be considered as a type of dialysis procedure.

31.10.1.3 Alkali Therapy

Most of the alkaline materials (*kshara*) act as diuretics, lithotriptic, alkalizer, and antispasmodic agents. These pharmacological activities are shown to be effective in the management of different symptoms of urolithiasis. Examples include *palasa kshara*, *yava kshara*, and *mulaka kshara*.

31.10.2 Medical Management²⁷

31.10.2.1 Vataashmari Treatment

A decoction is made for the following drugs: *pashana bheda*, *vasuka*, *ashmantaka*, *satavari*, *gokshura*, *brihati*, *kantakari*, *bhramhi*, *artagala*, *usira*, *kubjaka*, *vrushadani*, *bhalluka*, and *varuna*, and fruits of *saka*, *yava*, *kulutha*, *kola*, and *kataka*. To this *ushakadi* group of drugs, *kalka* (paste) is added and thus *ghrita* is prepared. This preparation immediately destroys *vataashmari*.

Kshara (alkali), gruels, soups, decoctions, milk preparations, and food prepared from this *vata*-allaying group of substances should be administered.

31.10.2.2 Pittaashmari Treatment

A decoction is made for the following drugs: *kusa*, *kasa*, *sara*, *morata*, *pasana bheda*, *satavari*, *pashana bheda*, *vidari*, *varahi*, *shalimula*, *trikantaka*, *bhalluka*, *patala*, *patha*, *kuruntika*, *punarnava*, and *shirisha*. *Ghritha* has to be prepared by using the above drugs' decoctions to which *silajit*, *madhuka*, seeds of *indivara* and *trapusa*, and seeds of *eravaruka* are added.

TABLE 31.1

Some of Important Formulations Mentioned in Ayurvedic Texts

No.	Medicine	Doses, Vehicle, and Duration	Reference ^a	Manufacturer
1	<i>Gokshuradi guggulu</i>	3 g/day with <i>pashana bheda kwatha</i>	AF Sec. 5:3	Amrita, Arkashala, Baidyanath, Indian Medical Practitioner's Co-operative Pharmacy & Stores, Kerala Ayurvedic Pharmacy Ltd., Sandu, Zandu
2	<i>Ashmari hara kashaya churna</i>	48 g/day	AF Sec. 4:3	—
3	<i>Varuna kwatha churna</i>	48 g/day	AF Sec. 4:22	—
4	<i>Trikantaka ghritha</i>	12 g/day with warm water or warm milk	AF Sec. 6:15	Kottakal
5	<i>Vatsyamayantaka ghritha</i>	12 g/day with warm milk	AF Sec. 6:40	Arya Vaidya Pharma, Kottakal
6	<i>Mulaka kshara</i>	1 g/day with water	AF Sec. 10:10	Dindayal, Prabhat
7	<i>Yava kshara</i>	$\frac{1}{2}$ –1 g/day with water		Dindayal, Prabhat
8	<i>Palasha kshara</i>	$\frac{1}{2}$ –1 g/day with water	AF Sec. 10:9	—
9	<i>Chandra prabha vati</i>	250–500 mg two times/day with water, milk, or ginger powder	AF Sec. 12:10	Arya Vaidya Pharma, Dabur, Dindayal, Dhootpapeshwar, Nagarjuna, etc.
10	<i>Sweta Parpati</i>	725 mg–1.25 g two times/day with coconut water or cold water	AF Sec. 12:2	—
11	<i>Hazrul yahud bhsama</i>	500 mg–1 g two times/day with <i>ashmari hara kashaya</i>	AF Sec. 14:4	—
12	<i>Trivikrama ras</i>	250 mg two times/day with <i>beejapuraka nimba mooola twak</i> and water	AF Sec. 16:24	Bhuvaneshwari
13	<i>Sukramatraka vati</i>	500 mg with juice of pomagranate, goat's milk, or water	AF Sec. 12:29	—
14	<i>Vishnu Tailam</i>	Internal and external application	AF Sec. 8:15	—
15	<i>Brihat Saindhavadya Tailam</i>	External application	AF Sec. 8:40	—

^aAll references are from *The Ayurvedic Formulary of India* (AF), a Government of India Publication.

Alkalis, gruels, soups, decoctions, milk, and foods prepared from these *pitta*-allaying groups of substances should be administered.

31.10.2.3 Sleshmashmari Treatment

Drugs of this decoction include *varunadi gana*, *guggulu*, *ela*, *harenuke*, *kushta*, and drugs of *bhdraadi gana*, *maricha*, *chitraka*, and *devadaru*. The decoction, added with paste of the *ushakadi* group of drugs, has to be taken along with dehydrated butter. This preparation provides relief from *sleshmaashmari*.

Alkalies, gruels, soups, decoctions, milk preparations, and food prepared from these *kapha*-balancing groups of substances should be administered. Karpasa flowers, *ankola*, *nirmali*, *saka*, and *indivara* fruits powder have to be taken internally with water and

jaggery; this mixture quickly reduces blood sugar levels. *Gokshura*, *talamuli*, *ajamoda*, roots of *kadamba*, and *adraka* are taken with wine or hot water to remove calculi. *Gokshura* seeds powder with honey should be taken with milk for 1 week to disintegrate the calculi.

31.10.2.4 *Sukraashmari Treatment*

If seminal concretions or fine particles spontaneously come in the urethra, they should be removed by the *badisha* instrument (a hook). After the wound is healed, the patient is advised not to undertake any physical activity, such as horse riding.

31.10.2.5 *Alkali Treatment*²⁸

In Ayurvedic texts, it is advocated that a formulation with medicated *ghrita* and *kshara* is made from the drugs mentioned above.

Kshara (alkali) prepared from pastes of *tila*, *apamarga*, *kadali*, *palasa*, and *yava* administered as a drink quickly relieves the symptoms. *Kshara* prepared from *patala* and *karavera* can also be effective.

Patients suffering from pain should be given milk processed with the above drugs or *triphala* groups or with *punarnava* as a drink. The drugs of the *veerataru* group can be administered by all modes.

31.10.2.6 *Surgical Treatment*

The surgical procedure (*shastra karma*) depicted in the classical texts may be presented as follows.²⁹

First, the patient should be cleansed of the vitiated *dosas*. Then the patient, who is strong enough and is not nervous, should be laid flat with the upper part of his body resting on the lap of another person sitting on a knee-high plank facing east; the patient's waist should be raised by cushions and his knees and ankles flexed and tied together by straps (lithotomy position). After massaging the left side of the well-oiled umbilical region, pressure should be applied first below the navel until the stone comes down. Introduce the lubricated index and middle fingers into the rectum below the perineal raphe. Thereafter, with manipulation and force bring the stone down between the rectum and the penis. Keeping the bladder tense and distended so as to obliterate the folds, the stone is pressed hard by fingers so that it becomes prominent like a tumor.³⁹

An incision of about the size of the stone is then made just a few millimeters away from the perineal raphe on the left side. Some surgeons prefer the incision on the right side for the sake of technical convenience. Precautions should be taken so that the stone does not get broken or crushed. Even a small particle left behind can increase in size. The stone is then removed with an *aggravaktra* instrument (small-tipped forceps, like mosquito forceps).

In females, as the uterus is situated very near the urinary bladder posteriorly, the incision should be directed upward. An incision in the bladder made at one place for the removal of a stone heals well. After removal of the stone, the patient is put in a tub of hot water and sedated to avoid any blood filling the urinary bladder. If the bladder does fill with blood, it is irrigated through a catheter with the decoction of the latex trees.

The decoction of the latex trees administered as an irrigating fluid through a catheter also removes the stones and the blood from the bladder.

31.10.3 Postoperative Management

To purify the urinary tract after an operation, the patient is given sufficient jaggery-based preparations. The patient is taken out of the tub, and honey and ghee are applied to the wound. The patient is given warm gruel processed with urine-purifying substances twice daily for 3 nights. After 3 nights, milk with treacle and small quantities of well-cooked rice is given to eat for 10 nights. After 10 nights, the patient is given sudation therapy either by oils or by liquids. The wound is washed with the decoction of latex trees.

The paste of *lodhra*, *madhuka*, *manjista*, and *prapaundrika* is applied to the wound. An oil or *ghrita* prepared from same substances along with *haridra* should also be anointed over the wound.

In case blood coagulates, it should be managed by bladder washes. If urine does not come out from its natural passage even after 7 nights, the wounds should be treated by heat treatment (cautery) according to the described method. When urine starts flowing along its natural passage, the patient should be treated by bladder washes, enemas of medicated decoctions, and medicated oils and given preparations made of jaggery to eat.

31.10.4 Management of Urolithiasis in Modern Medicine

The immediate treatment of loin pain or renal colic is bedrest and application of warmth to the site of pain. Pain relief can be achieved by administering the following agents: Diclophenac sodium, Indomethacin, Hydromorphone (Hydrochloride+atropin sulfate [Dilaudid-Atropin®]), Methamizol, Penta-zocin, and Tramadol. Treatment is started with an NSAID and changed to an alternative drug if the pain persists. Hydromorphone and other opiates without simultaneous administration of atropine should be avoided. Diclophenac sodium affects glomerular filtration rate in patients with reduced renal function but not in patients with normal renal function.³⁰ When pain relief cannot be achieved by medical means, drainage by stenting or percutaneous nephrostomy or by stone removal is performed.

The medical therapy depends on the type of stone produced. The therapeutic agents that are generally used in renal calculi are alkalinizing agents (sodium citrate and citric acid, sodium citrate, and potassium citrate mixture), diuretics (hydrochlorothiazide), chelating agents (cellulose sodium phosphate), and Xanthine oxidase inhibitors (Allopurinol).

31.11 Prevention

Ayurvedic texts provide detailed information regarding the dietary habits and lifestyles that are to be adopted in renal calculi. It is advised to take whole rice, wheat, barley, horse gram, green gram, matured pumpkin, *varuna*, ginger, *gokshura* and amaranthus, flesh of birds residing on dry soil or barren land, and measures such as medicated enemas, emesis, purgation, fasting or light diet, and sudation. The intake of sour, dry, and heavy foods, food substances that cause indigestion, and unwholesome food items should be avoided.^{31,32}

In conventional medicine, the dietary habits are not much emphasized, but in some conditions some general advice is given. In idiopathic calcium nephrocalcinosis, patients

are advised to maintain adequate fluid intake to produce at least 2 l/day of urine, maintain adequate calcium intake (at least recommended daily allowances) from food sources, cut down intake of animal protein, cut sodium intake 4 g/day (170 meq), cut sucrose intake, increase dietary K intake, avoid grapefruit juice, and avoid unnecessary vitamin C. Coffee, tea, and alcoholic beverages should neither be avoided nor encouraged.³³

31.12 Scientific Basis for the Use of Ayurvedic Drugs in Urolithiasis

31.12.1 Clinical Studies

1. In one study³⁴ with 30 cases of nephrouretero lithiasis, 17 cases were given 1 g of *svetaparpati* with 50 ml of *kuluttha kwatha* three times/day for 1 month. Marked improvement was noticed in different symptoms and pathological findings after the course of the therapy. Radiological investigations revealed expulsions of ureteric calculi were much more than those of renal calculi.
2. In another study,³⁵ 50 patients with urolithiasis (24 patients with renal calculus and 26 patients with ureteric calculus) were given *palasa kshara* in the dose of 1 g three times/day for 30 days. On the basis of radiological findings, the drug is said to be more effective in the expulsion of ureteric calculi as compared with renal calculi.
3. In one clinical study³⁶ consisting 30 urolithiasis patients (14 patients with renal calculi and 16 patients with ureteric calculi), 1 g of *sveta parpati* with 50 ml of *pasanabheda* and *gokshuru kwatha* was given three times/day for 3 weeks. The study indicated significant effect in majority of the cases. The x-ray findings indicated the clearance of the stone. Some of the stones had also passed through urine during the course of treatment.
4. A study³⁷ was conducted with 71 patients suffering from urolithiasis (*ashmari*) who were diagnosed by kidney, ureter, and bladder intravenous pyelography (KUB, IVP). They were treated with juice of the core of the pseudostem of *Musa paradisiaca* Linn. and *Musa sapientum* Linn. A significant segment of them passed out calculi of varying size after consuming the drugs for 2 weeks. The recurrence of stone formation was also prevented by the treatment. The author concludes that the plant material is quite effective in curing urolithiasis, especially of the calcium oxalate variety.
5. Of a total 110 cases, 30 cases were included in group A-1 and treated with *sveta parpati*, *pasanabheda*, and *gokshuru*; 30 cases were included in group A-2 and treated with *sveta parpati* and *kuluttha kwatha*; and 50 cases were included in group B and treated with *palasa-ksara*. On the basis of the radiological evidence and clinical pathological criteria, it was concluded that the drug combinations in groups A-1 and A-2 had very good lithotriptic, diuretic, alkaliser, coagulant, and *vata shamaka* properties. The drug combination in group A-2 showed better lithotriptic action than that in group A-1. The drugs in group B have diuretic, lithnotriptic, and alkalizer properties and also have an antispasmodic effect. The radiological findings indicate that almost all the three therapies proved more effective in the expulsion of ureteric calculi as compared with the nephrotic calculi.³⁸

6. One hundred cases of nephroureterolithiasis were treated with Cystone tablets (Himalaya Drug Company) in four different combinations (plenty of fluids by mouth, forced diuresis, antispasmodics, and antispasmodic-forced diuresis) for 1 year. The therapy with Cystone tablets and fluids given orally gave 76% positive results. The therapy with Cystone tablets combined with forced diuresis revealed 80% positive results, whereas antispasmodics combined with fluids given orally showed 20% positive results. Therapies with antispasmodics in combination with forced diuresis showed 28% positive results. The study indicates that forced diuresis alone is not as effective as the therapy with Cystone tablets alone in the treatment of nephroureterolithiasis. The mechanism action of Cystone could be attributed to any of its pharmacologic agents: diuretic, spasmolytic action, effect on crystalloid and colloid balance, and disintegrating action on the binding mucin. As Cystone contains no toxic ingredients, no side effects were observed even with prolonged therapy for 6 months.³⁹

31.12.2 Pharmacological Studies

31.12.2.1 *Varuna* (*Crataeva nurvala*)

1. The cytoprotective action of lupeol isolated from *Crataeva nurvala* stem bark against free radical toxicity has been investigated in experimental urolithiasis. The increase in lipid per oxidation and super oxide dismutase activity, associated with decreased catalase activity and glutathione level, are the salient features observed in tissues of stone-forming rats. Lupeol administration induced a remarkable decrease in kidney oxalate level and also was effective in counteracting the free-radical toxicity by bringing about a significant decrease in peroxidative levels and an increase in antioxidant status. The antioxidant property of lupeol and its cytoprotective action against free-radical toxicity has also been studied.⁴⁰
2. The antiurolithiatic activity in the crude extract of *C. nurvala* has already been examined. Further fractionation of this extract led to the isolation of the active constituent lupeol (Lup 20(29)-en-3beta-ol). Antiurolithiatic activity of lupeol was assessed in rats by observing the weight of the stone, biochemical analysis of serum and urine, and histopathology of bladder and kidney. Lupeol not only prevented the formation of vesicle calculi, but also reduced the size of the pre-formed stones.⁴¹
3. Male albino rats (100 ~10 g) were fed daily with a 3% glycolic acid solution (1 g/kg) orally for 4 weeks. It caused the deposition of lithogenic constituents, calcium and oxalate in kidneys, in significant amounts and induced hyperoxaluria, hypercalciuria, and hypercrystalluria. Oral administration of ethanolic extract of *C. nurvala* (stem bark) at the dose levels of 25, 50, and 100 mg/kg for 4 weeks showed 12 to 54% protection against the deposition of stone-forming constituents in the kidney and against hyperoxaluria, hypercalciuria, and hypercrystalluria. Similarly, lupeol, the active constituent of *C. nurvala*, was also found to be active. At an oral dose of 10, 25, and 50 mg/kg for 4 weeks, it showed 24 to 63% activity. The increased urinary excretion of the crystalline constituents, found in the stone-forming rats, was nearly normalized by lupeol treatment in a dose-dependent manner.⁴²

31.12.2.2 *Gokshura (Tribulus terrestris)*

An ethanolic extract of the fruits of *T. terrestris* showed significant dose-dependent protection against urolithiasis induced by glass-bead implantation in albino rats. On subsequent fractionation of the ethanol extract, maximum activity was localized in the 10% aqueous methanol fraction. It provided significant protection against the deposition of calculogenic material around the glass bead. It also protected leucocytosis and elevation in serum urea levels. Further fractionation lead to decreased activity. This could be either due to the loss of active compounds during fractionation or the antiurolithiatic activity of *T. terrestris* being a combined effect of several constituents present in the methanolic fraction.⁴³

31.12.2.3 *Pashanabheda (Bergenea ligulata)*

Acetone extract of the root bark of *B. ligulata* has been subjected to preliminary pharmacological investigations. Observations indicate that the extract contains components possessing sedative action, potentiates analgesia induced by a subanalgesic dose of morphine, and anti-inflammatory activity comparable with that of aspirin. The extract was devoid of possessing antilithiatic activity but exhibited a mild diuretic effect when tested on rats and dogs.⁴⁴

31.13 Discussion and Conclusions

The epidemiology shows that there is a wide variation in the occurrence of urolithiasis in various countries. There are also many variations in the chemical composition of stones. Basically, it is a physicochemical phenomena involving the nature of urine and the accretion of minerals. It is interesting to note that the stone former's nephrocalcin lacks γ -carboxy glutamic acid. This chapter provides a number of herbal treatments that provide scope for research into their ability to alter the chemical composition of the urine, making the stone vulnerable for dissolution. This is a very fruitful area for future research for metabolic management of the patients, particularly after surgical removal of the stone.

31.14 Future Research Areas

Biochemical and biophysical characters of the body's metabolism involving hydrokinetics and dispersal of organic waste matter in urolithiasis is a potentially fruitful area. This may give leads for phytotherapy intervention suited to rectify the aberration.

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32.1 Introduction

Women's health is a topic of concern in the medical field, as women are an important factor in the reproduction of healthy progeny. The environmental factors, fast-changing lifestyles, and various addictions (drug abuse) as well as excess use of drugs (like steroids) have endangered their health.¹ More than 4.5 million women (ages 18 to 50) report at least one chronic gynecological condition each year.² It is estimated that 30 to 40% of women have premenstrual syndrome (PMS) with symptoms severe enough to impair their daily activities.³ Infertility affected 6.1 million people in 1997 (about 10% of the reproductive age population), up from 4.6 million in 1988.⁴

Currently available therapies, which advocate greater use of antibiotics, steroids, or surgery, may not be an ultimate answer for a woman's ill health.^{5,6} So, alternative systems of medicine such as Ayurveda could be an option. *Streeroga* and *prasutitantra* (obstetrics and gynecology) is one of the eight branches of Ayurveda.⁷ Women's disorders are described as diseases of female genitalia (*Yonivyapad*).⁸ Some people perceive *yoni* as vagina, but that is just the literal meaning. The real meaning comprises the whole female reproductive system. *Yonivyapad* consists of diseases of *yoni* and *stanaroga* (diseases of breast). The major ailments described in conventional medicine such as amenorrhea, dysmenorrhea, and menorrhagia come under this category and are discussed here along with breast diseases. Uterine and breast cancers (*raktagulma*, *mamsarbuda*, *stanarbuda*) are dealt with under a different category and are not the subject of this chapter. Etiopathogenesis, diagnosis, prognosis, and treatment of all women's diseases including pregnancy, prenatal, postnatal health, and childcare are described in detail in this section.⁸ This chapter deals with the treatment and management of major gynecological disorders based on clinical, laboratory, or pharmacological research.

32.2 Classification

There are 20 types of gynecological disorders (*yonirogas*).⁸

1. Menorrhagia (*asruja*)
2. Amenorrhea (*arajaska*)
3. Vaginitis (*acharana*)
4. Excessive sexual indulgence (*aticharana*)
5. Loss of libido (*prakcharana*)
6. Leukorrhea (*upapluta*)
7. Sexually transmitted diseases (*paripluta*)
8. Dysmenorrhea (*udavartini*)
9. Uterine polyp (*karnini*)
10. Bad obstetric history (*putraghni*)
11. Vaginal atrophy (*antarmukhi*)
12. Atrophy of uterus-congenital anomaly (*soochimukhi*)
13. Vaginal dryness (*shushka*)
14. Small uterus (*vamini*)
15. Sterility (*shandhi*)
16. Prolapse of vaginal walls or uterus (*mahayoni*)
17. *Vataja*
18. *Pittaja*
19. *Kaphaja*
20. *Sannipataja*

These disorders are also considered predisposing factors of infertility. Traditionally, it is believed that women suffering from these disorders cannot receive or hold semen (*shukra*) and hence cannot conceive. Apart from these complaints, there are some separate entities such as abnormal vaginal bleeding (*pradara*) and infertility (*vandhyatva*) that are described in detail. Uterine fibroid, myoma, breast cancer, and endometriosis are categorized under different headings like *gulma*, *arbuda*, etc. in the Ayurvedic classics and are not listed under gynecological diseases.⁸ Neoplasms are not discussed in this chapter. Even women's diseases are a vast topic, which comprises various ailments of reproductive organs and associated organs such as the endocrine system. It is not possible to describe all of them because of the constraint of space and scope of this textbook. The major diseases that affect a majority of the female population are discussed in this chapter.

32.3 Etiopathogenesis

According to Ayurveda, women who violate proper diet and lifestyle are more prone to ill health.⁹ Heavy, slimy, and excessive intake of sweet, acid, salty, sour, and incompatible

foods, lack of exercise, excessive sleep, laziness, daytime naps, and overeating are some of the general causative factors of gynecological disorders.¹⁰ Women's diseases can also occur due to external causes that are beyond one's purview. One has to remember that specific *dosa*-aggravating or -depleting factors (mentioned elsewhere in the textbook) could also be contributory factors to gynecological diseases in addition to those mentioned above. Essentially, all these factors cause an imbalance in the body tissues and organs and make them prone to infections or internal pathology.¹¹ For any pathogenesis there are two essential factors: structural and functional. The disease could originate from structural changes and then bring about functional changes or vice versa. In this context, an increase or deficiency (imbalance) of the three basic entities, *vata*, *pitta*, and *kapha*, lead to the vitiation of the related structure or tissue of the genitourinary system.¹¹

As *dosas* become aggravated they develop pathogenesis in three different manners. When *vata* is aggravated, it dries up the organ or tissues absorbing necessary unctuousness and water in the tissue and makes the surface rough due to its innate rough (*rooksha*) property. In an excited state, the *vata* becomes hypermotile (*chala*) and travels in all directions. For example, this excited state of *vata* can expel the excretory products in excess, like in menorrhagia, or move them upward and stop the normal expulsion that causes pain in the local area (e.g., dysmenorrhea). *Apana vayu* is the type of *vata* that is involved basically in menstrual disorders.

Pitta, in an aggravated state, also absorbs water and creates excess heat, burning sensation, and inflammation in the organ. It also interferes with clotting because of its hot and liquefying properties and may cause excess flow as in menorrhagia. *Kapha*, when out of balance, obstructs the channels because of its slimy and heavy property. It disturbs proper metabolism and stops or delays functions like ovulation and menstruation as in amenorrhea. It can also produce heaviness, coldness, and sluggishness in the body as well as in the local area. Some fungal infections in the vaginal area or vulval region may be attributed to imbalance in *kapha* where itching (*kandu*) is the major symptom.¹¹

An imbalance of a single *dosa* or combination of *dosas* residing in the genitourinary system may cause a gynecological disorder in women.

32.4 Clinical Features

There is no clear-cut description of clinical symptoms for all gynecological disorders in Ayurveda. The symptoms are usually based on the affected *dosas*.⁸ The characteristic properties of each *dosa* reflected during an ailment are utilized by the physician for easy diagnosis and treatment of the gynecological disorder.

32.5 Diagnosis and Prognosis

Diagnosis of gynecological disorders is also based on the same principles of Ayurveda as are any other disease.¹² The history of the disease, its aggravating and relieving factors, its predominant symptoms, and the patient's personal and family history, pulse, tongue, diet, and lifestyle are all taken into consideration to diagnose the type, subtype, and stage of the specific gynecological ailment. Currently, Ayurvedic physicians in India are

also using the modern diagnostic procedures such as blood chemistries, hormonal assays, x-ray, ultrasound, computerized tomography scan, and magnetic resonance imaging for proper diagnosis and staging of the disease.

32.6 Treatment

The Ayurvedic approach of treatment is always holistic and works at both psycho and somatic levels. As most of the women's diseases are psychosomatic, Ayurvedic therapy has been often found to be effective in treating gynecological disorders. The first goal of treatment is to relieve the symptoms and give comfort, which is followed by alleviation or control of disease. The treatment comprises internal and external therapies.^{7,8} Internally, clearing the channels, correcting *dosa* imbalance, and improving internal environment of the system is emphasized. Internal cleansing, if required, is usually done through *panchakarma* therapy. External modalities such as oleation (*snehana*), fomentation (*swedana*), medicated enema (*basti*), and uterovesical douche (*uttara basti*) are also very effective in women's diseases and are usually combined with internal cleansing procedures.¹³ However, they form the main treatment methods in some conditions. They help relieve the obstructions, pacify aggravated *dosas*, and give symptomatic relief reducing pain and tension. Compound formulas used for the treatment of gynecological diseases are listed in Table 32.1.

Usually this purification therapy is followed by the pacification by oral supplementation of Ayurvedic herbs or herbal formulas.¹⁴ Ayurveda also emphasizes preventive measures using behavioral and dietary regimens to all gynecology patients, because both diet and lifestyle play important roles in the etiopathogenesis of the disease. It helps maintain the health and control the frequency and severity of the ailment. Ayurvedic treatment is comprehensive because it encompasses cleansing (detoxifying), pacification, mental health, diet, and behavioral therapy while treating gynecological disorders.

Specific details of seven gynecological disorders are described in detail below.

32.6.1 Amenorrhea

32.6.1.1 Etiopathogenesis

According to Ayurveda, aggravated *vata* and *pitta* affect blood by drying-up menstrual blood (*raja*), which ceases the regular flow of menses (*nashtartava* or *ksheenartava*).⁸ General symptoms that are associated with this condition are emaciation and pallor. It suggests the deficiency of blood or a severely anemic condition and weakness.

Conventional medicine indicates that amenorrhea is due to a wide range of causes that include anatomic or structural and physiological or functional defects. It can be congenital or develop at a certain age because of a particular change. There can be a defect from the ovarian or uterine level to hypothalamus and pituitary level.¹⁵

32.6.1.1.1 Structural

Anatomically there are four major congenital anomalies: absence of uterus or some part of the vagina, failure of the vagina to develop, imperforate hymen, or intrauterine synechiae. All these consequently result in primary amenorrhea.

TABLE 32.1

Compound Formulas for Gynecological Disorders

Formula	Dosage
Gynecological Diseases in General	
<i>Ashokarishtom</i>	15–20 ml two times/day after meals with equal water
<i>Pushyanuga choornam</i>	3–5 g two times/day with rice water
<i>Kumaryasavam/kumari kalpa</i>	15–20 ml two times/day after meals with equal water
<i>Phalakalyanaka ghrita</i>	5–10 two times/day with milk or warm water
Amenorrhea	
<i>Rajahpravartini vati</i>	2 tablets two to three times/day with warm water or <i>kumaryasavam</i>
<i>Nashtapushpantaka ras</i>	Same as above
<i>Jeerakadyarishtom</i>	15–20 ml two times/day after meals with equal water
<i>Chandraprabha vati</i>	2 tablets two or three times/day with warm water or <i>kumaryasavam</i>
<i>Kumaryasavam</i>	15–20 ml two times/day with equal water after meals
Dysmenorrhea	
<i>Rajahpravartini vati</i>	2 tablets two or three times/day with warm water or <i>kumaryasavam</i>
<i>Jeerakadyarishtom</i>	15–20 ml two times/day after meals with equal water
<i>Ashokarishtom</i>	Same as above
<i>Dhanwantharam thailam</i>	For external application like massage
<i>Kumaryasavam</i>	15–20 ml two times/day with equal water
Menorrhagia	
<i>Pradarantaka rasa/loha</i>	2 tablets two or three times/day with warm water or <i>chandanasavam</i>
<i>Chandanasavam</i>	15–20 ml two times/day after meals with equal water
<i>Chandrakala ras</i>	2 tablets two or three times/day with warm water or <i>chandanasavam</i>
<i>Bolabaddha ras</i>	Same as above
<i>Lodhrasavam</i>	15–20 ml two times/day after meals with equal water
Leukorrhea	
<i>Pushyanuga choorna</i>	3–5 g two times/days with rice water
<i>Chandraprabha vati</i>	2 tablets two or three times/day with warm water or <i>chandanasavam</i> or <i>ashokarishtam</i>
<i>Chandrakala ras</i>	1–2 pills two times/day with warm water
<i>Chandanasavam</i>	15–20 ml two times/day after meals with equal water
<i>Praval bhasma</i>	120–240 mg two times/day with warm milk
PMS	
<i>Ashokarishtom</i>	15–20 ml two times/day after meals with equal water
<i>Kumaryasavam</i>	Same as above
<i>Manasamitra vatakam</i>	1 tablet one or two times/day
<i>Ashvagandhadi choornam</i>	3–5 g two times/day with rice water
Infertility	
<i>Phala ghrita</i>	5 ml two times/day with milk or warm water
<i>Satavari kalpa</i>	5–10 g two times/day with warm milk
<i>Kumaryasavam</i>	15–20 ml two times/day after meals with equal water
<i>Ashokarishtom</i>	Same as above
<i>Jeevaniya rasayana</i>	5–10 ml two times/day with milk or warm water
Menopausal Syndrome	
<i>Ashokarishtom</i>	15–20 ml two times/day after meals with equal water
<i>Shatavari kalpa</i>	5–10 g two times/day with warm milk
<i>Panchagavya ghrita</i>	5–10 ml two times/day with milk or warm water
<i>Kalyanaka ghrita</i>	Same as above
<i>Chandanasavam</i>	15–20 ml two times/day after meals with equal water
<i>Chyavanaprasha</i>	5–10 g with warm milk one time/day in the morning

32.6.1.1.2 Functional

Physiologically, after menarche, women have a menstrual flow every month because of ovulation secondary to certain hormonal changes. The leutinizing hormone (LH) -releasing hormone is the main hormone produced in hypothalamus that further releases the LH

and the follicular stimulating hormone (FSH). These hormones are responsible for the process of ovulation and endometrial changes. There are also gonadotropins and estradiol that stimulate ovarian functions. A defect of any of these glands at any level of hormone production and at any age from puberty to menopause can cause amenorrhea.^{1,15}

32.6.1.2 Clinical Features

Absence of menstrual cycle during the reproductive years is the clinical symptom according to both Ayurveda and conventional medicine.

32.6.1.3 Diagnosis and Prognosis

Diagnosis of amenorrhea is very complicated and requires a series of investigations and physical examinations. To reach a perfect diagnosis, a step-by-step evaluation is necessary from vaginal or uterine level to the hypothalamus. To detect the anatomical anomalies, a pelvic ultrasound or examination under anesthesia may be required in adolescent girls. The hormonal levels can be investigated to help determine the exact cause. Prognosis of amenorrhea is good. It is not a life-threatening disease and can be treated successfully in many cases.¹⁵

32.6.1.4 Treatment

According to Ayurvedic treatment, the major objective is to alleviate *vata* in amenorrhea. The therapy is composed of three stages: (1) cleansing therapy, (2) palliative therapy, and (3) advising a dietary regimen.

32.6.1.4.1 Cleansing Therapy (Shodhana)

Vasti (medicated enema), one of the detoxification procedures (*panchakarma*), is the *vata* alleviation treatment. *Vata*-pacifying herbs such as castor, sesame oil, meat soups, and milk can be used for the enema treatment.⁸

32.6.1.4.2 Palliative Therapy (Shamana)

This therapy works on two principles: one is to control *vata* and *pitta* and the other is to improve tissue status. In amenorrhea, blood is deficient, so supplements like iron (*loha*) and *Suvarna makshika* give good results in weak or anemic women. Where there is a hormonal imbalance, herbs like *Aloe vera*, *ashwagandha*, and black kohosh are more useful. *Ultakambal* (*Abroma augusta*), *vamsha* (*Bambusa arundinacea*), and *shana* (*Crotalaria juncea*) are the herbs described for inducing menstruation (*artavajanana*). Black pepper (*Piper nigrum*), long pepper (*Piper longum*), garlic (*Allium sativum*), ginger (*Zingiber officinale*), sesame (*Sesamum indicum*), and papaya (*Carica papaya*) are some of the stimulating herbs proved effective to induce menstruation. *Rajapravartini vati*, *Nashtapushpantaka rasa*, and *Chandraprabha vati* are some of the effective formulas to treat amenorrhea.¹⁴

Conventional treatment includes the induction of ovulation as the main goal of therapy. It is done with the help of the drugs such as bromocriptine, clomiphene citrate, and steroids. Some patients who are hypoestrogenic can be treated with progesterone and estrogen in combination.¹⁵

32.6.1.4.3 Dietary and Behavioral Regimen

Cold and heavy food, oily deep-fried food, and frozen food tend to increase the *vata dosa* in the body in general and might accumulate in the uteri of women suffering with amenorrhea; such foods also block the free flow of menstrual blood.¹⁰ Foods such as

potatoes and peas produce gas and constipation and need to be avoided. A sedentary lifestyle aggravates the disease. Light exercises such as *yogasanas* or walking are recommended.

32.6.2 Dysmenorrhea

Dysmenorrhea is a painful period or menstrual cramps. It is the most common complaint among all gynecological disorders. Because most women have some discomfort during the period, it is considered natural. It is named dysmenorrhea only when the pain is too severe and disturbs a woman's routine, and it is relieved only with medication.¹⁵

32.6.2.1 Etiopathogenesis

Ayurvedic literature attributes dysmenorrhea to the above-mentioned general causes as well as suppression of natural urges (sneezing, coughing, micturition, defecation, etc). These factors provoke *Apana vata* and disturb its natural route of a downward direction and start moving it in reverse or an upward direction. The change in direction of the *vata dosa* alters the menstrual blood flow. This results in painful and scanty menstruation. When the flow is proper without any obstruction, pain is relieved.¹⁰

In conventional medicine, the prostaglandin theory is currently more accepted as the causal explanation for dysmenorrhea, although endometriosis or adhesions can also cause dysmenorrhea. It is believed that women with dysmenorrhea have much higher concentrations of prostaglandin.¹⁵

32.6.2.2 Clinical Features

The symptoms are cramping and spasmodic pain in the lower abdomen which may spread over to the gluteal region, thighs, and lower back. Generalized pelvic tenderness is usually present. Nausea, diarrhea, or constipation and headaches are the other common symptoms. Vomiting or fainting may occur in some cases.^{3,15}

32.6.2.3 Diagnosis and Prognosis

Because the pain and discomfort is subjective, diagnosis relies upon the proper history. It is also important to differentiate the primary diagnosis from the secondary one. For example, an underlying cause like endometriosis may create confusion or be missed. Prognosis of dysmenorrhea is good. It can be treated successfully with Ayurvedic single herbs or formulations in addition to making diet and lifestyle changes.¹⁵

32.6.2.4 Treatment

Ayurvedic treatment emphasizes sweating, cleansing douches, and medicated enemas. Oil therapy, including massage, is used to relieve any obstructions in the passage, relieve any spasms, facilitate free movement of *vata* in the proper direction, and enhance a proper menstrual flow. Gentle massage with sesame oil, *dhanwanthara thailam*, *narayan thailam*, or *kottamchukkadi thailam* is advised over the lower abdomen, pelvic region, and thighs. Sweating methods and fomentation remove the toxins and also relieve muscular spasms and tenderness. A medicated enema with a mixture of oil, milk, herbal paste, and decoction or tea also pacifies *vata* and reduces any flatulence, removes obstruction of fecal matter, and relieves the tension in the pelvic cavity. These three therapies together help relieve pain and discomfort in the pelvic region.^{7,8}

32.6.2.4.1 Palliative Therapy

Pootikaranja (*Caesalpinia bonducella*) is a very effective herb for improving menstrual flow and relieving abdominal pain. It also checks nausea and vomiting along with clove or black pepper. *Bol*, or myrrh (*Commiphora myrrh*), is another effective drug used in this disease.¹⁴

The first aim of the conventional treatment is to relieve pain and discomfort. Analgesics such as codeine or parenteral medication in severe pain will serve the purpose. If the causative factor is prostaglandin as is currently hypothesized, antiprostaglandins (e.g., ibuprofen) are generally very effective. Cyclic administration of oral contraceptives in the lowest dosage is useful in many cases, especially if the patient cannot tolerate antiprostaglandins. In some cases of severe pain, medication does not help. When the underlying cause is endometriosis or adhesions, surgical treatment is recommended.¹⁵

32.6.2.4.2 Dietary and Behavioral Regimen

Hot, slightly oily, and light food containing digestive herbs such as ginger, pepper, clove, and cinnamon is advised. Light exercises, such as like walking or *yogasanas*, are beneficial.

32.6.3 Menorrhagia

Excessively heavy bleeding or prolonged periods are considered as menorrhagia. It is categorized under abnormal uterine bleeding.¹⁵

32.6.3.1 Etiopathogenesis

According to Ayurveda, the aggravated *pitta* vitiates blood (*rakta*) and causes overflow through *yonis*. This is named *asruja yoni*.⁸

In modern medicine, submucous myomas, complications of pregnancy, adenomyosis, endometrial hyperplasias, or malignant tumors are some of the causes. It is very difficult to determine the exact cause, and there may be more than one diagnosis present. Metrorrhagia (bleeding any time in-between the period), polymenorrhea (too frequent menstruation), and menometrorrhagia (frequent and intermediate bleeding) are other types of abnormal excess bleeding. Dysfunctional uterine bleeding is one more type that is without any pathologic cause; it is due to only hormonal imbalance. In this disorder, the endometrium outgrows its blood supply because of estrogen stimulation, breaks down, and sheds in an irregular manner.¹⁵

32.6.3.2 Clinical Features

The symptoms differ according to type. There are four types of *pradara*; three types are caused by a single *dosa* (*vata*, *pitta*, and *kapha*) and the fourth one is known as *sannipataja*.

1. *Vataja pradara* — Menstrual discharge is painful, frothy, thin, rough, dark, and graying or light red. The associated symptoms are intense pain in pelvic region, thighs, back, and sides of the chest.^{7,8}
2. *Pittaja pradara* — This type of *pradara* is characterized by a dark red a with yellowish or bluish tinge; hot, continuous flow that is associated with a burning sensation; and redness in the local area. Excess thirst, vertigo, and fever are the general symptoms.

3. *Kaphaja pradara* — In this condition, menstrual discharge will be slimy, pale, thick, heavy, and unctuous with dull pain. General features are nausea, vomiting, anorexia, cough, and dyspnea.
4. *Sannipatika pradara* — *Sannipatika* refers to all three *dosas*, hence there will be a variety of severe symptoms. Examples include slimy, dark-red blood; hot, yellowish discharge; or discharge containing body fat or marrow. Flow will be continuous, resulting in extreme weakness and anemia. Thirst, a burning sensation, and fever are some of the general symptoms. They indicate that deep tissues in the body are involved. Therefore, one can infer the prognosis is worst.^{3,10}

According to conventional medicine, the main symptom is heavy bleeding with clots and gushing. Many times it is prolonged more than 7 days. Usually it leads to anemia due to excessive blood loss. Severe weakness, pallor, cramps, and palpitations are usually associated with excess bleeding.¹⁵

32.6.3.3 Diagnosis and Prognosis

The number of sanitary napkins soaking per hour with passage of clots or prolonged periods more than 7 days is the salient feature used to make such a diagnosis. Diagnosis requires proper history, physical examinations to detect abdominal masses and so on, endometrial biopsy to rule out malignancy, ultrasonography, and hysteroscopy to confirm the diagnosis and obtain a clear picture.^{15,16}

32.6.3.4 Treatment

The external therapy includes the following:

1. A continuous thin flow of medicated oil (*dhara*), milk, or buttermilk on forehead or lower abdomen
2. Fomentation (*seka*)
3. Simple oil application (*abhyanga*)
4. A local application of drugs with cotton ball as a plug (*pichu*)

These external treatments are done to alleviate pain, stop the excess blood flow, and relax the patient. Internally *Saraca indica*, *Terminalia arjuna*, *Indian sarsaparila*, *Santalum album*, and *A. vera* are commonly used herbs. *Lodhrasavam*, *chandanasavam*, *kumaryasavam*, *ashokarishtom*, *chandrakala vati*, *bolabaddha ras*, and *pradarantak ras* are some of the most commonly used formulas.¹⁴

Takradhara (a continuous flow of buttermilk processed with *Cyperus rotundus* and *Emblica officinalis* is called a *Takradhara*)¹⁷ was found effective in menorrhagia in a 10-day pilot study conducted in Maharashtra, India. Uterovaginal administration (*uttarabasti*) of a formulation named *kashmarya-kutaja ghrita* was found very effective in menorrhagia in a study conducted at Government Ayurveda College, Trivandrum, India. The duration of the study is not available.¹⁸

Pushyanuga churnam was found highly significant in menorrhagia at Government Ayurveda College, Trivandrum, India. In an animal study, the same formulation showed positive effects on endometrium, such as atrophy of hyperplastic endometrium. The investigators' details are not available. In a study conducted at Government Ayurveda College, Trivandrum,¹⁹ *ashokarishtam* and *musalikhadiradi kwatha*, two multiherbal formulas, have been shown to be highly effective in normalizing the menstrual cycle as well as

reducing associated symptoms such as palpitation, weakness, and myalgia. *Kutajashtaka lehyam*, a classical formula, was also found to be very effective in menorrhagia and dysfunctional uterine bleeding in a study with little details.²⁰

A multiherbal formula containing *Saraca indica*, *A. vera*, *Cyperus rotundus*, etc. produced significant results in oligomenorrhea. Approximately 25% of women conceived during the study. This suggests that the formula may be significantly effective in helping infertility disorders.²¹

According to conventional medicine, conservative therapy is advised; when bleeding is not acute and life threatening, a hormonal therapy (e.g., oral contraceptives) can control the disease. If the cause is myoma or adenoma, the new growths are excised to save the uterus in childbearing age. Otherwise, a hysterectomy is the preferred treatment.¹⁵

32.6.3.4.1 Dietary and Behavioral Regimen

Women should avoid foods that are fried, greasy, spicy, and sour. Indian spices such as coriander, turmeric, mint, and anise seeds are recommended. Hot spices, garlic, ginger, and chilies and hot beverages such tea, coffee, and alcohol should be avoided. Green vegetables including zucchini, squash, cucumber, pumpkin, Indian gooseberry, and dates and soups of legumes and vegetables are recommended, as these have a cooling effect. Women should add fresh fruits (except citrus) and dairy products such as milk, fresh cottage cheese, and ghee (butter oil) to the diet. Pulses and grains are also very beneficial. Coconut water, barley water, and buttermilk are also good. Excessive stress and strain need to be avoided. Maintaining proper hygiene of the genital organs is of primary importance. A proper bowel movement everyday is also considered important. Relaxation of the mind is very important for menorrhagia as in any other disease.^{3,8,12}

32.6.4 Leukorrhea

32.6.4.1 Etiopathogenesis

Leukorrhea is also called as *sweta pradara* in Ayurveda. This disease is caused by the imbalance of *vata* and *kapha* *dosa*. The production and transportation of fluids by the urogenital organs becomes disturbed and leads to leukorrhea.^{8,10}

In conventional medicine, an abnormal vaginal discharge that is watery, cloudy, or white; thick or thin in consistency; and malodorous or odorless is a condition known as leukorrhea. When it is thin, clear, mucuslike, odorless, and not subsequently increased in quantity, it is considered normal. It is generally secreted before menstruation, due to sexual excitement, or due to emotional stress. Fungal, bacterial, and viral infections change the environment in the vagina and make it alkaline. An alkaline environment results in production of vaginal discharge. Foreign bodies also cause vaginal infections and discharge. There are several other causes that also cause this type of discharge (e.g., worms or ulcers).¹⁵

32.6.4.2 Clinical Features

According to Ayurveda, the main symptoms are a pale or white discharge from the vagina that is usually painful and thick or thin in consistency.

Western medicine attributes specific symptoms to specific causative agents. The discharge may be malodorous or show a change in color with fever if it is due to bacterial infections. In viral infections, discharge may be profuse and the cervix tender. In fungal infections, vulvar pruritis followed by burning urination and itching are the principal

symptoms. Along with the profuse vaginal discharge, there may be symptoms like weakness, lower backache, leg pain, etc.¹⁵

32.6.4.3 Diagnosis and Prognosis

Diagnosis is dependent upon a proper history and pelvic examination. In addition, a pap smear is done to identify any pathogens and also to rule out any dysplasia. Prognosis is often very good except in the case of a malignancy of the cervix.¹⁵

32.6.4.4 Treatment

Ayurvedic treatment emphasizes the use of vaginal douches (*uttarbasti* or *yonidhavana*). These douches are intended to treat the uterus, fallopian tubes, and the external vagina. They act as local antibiotics, clear the blockages in the channel pathway, and pacify *dosas*.⁸ The herbs used for douches are *Santalum album* (*chandana*), *Azadiracta indica* (*neem*), *Tinospora cordifolia* (*guduchi*), *Saraca indica* (*ashoka*), *Terminalia arjuna* (*Arjuna*), etc. Internally, ginger, cinnamon, fenugreek, gooseberry (*Emblica officinalis*), *Terminalia chebula*, and *Hemidesmus indicus* are the popular herbs. *Pushyanuga choorna*, *Chandraprabha vati*, *Chandrakala ras*, *Chandanasavam*, *loha*, and *praval bhasma* are some of the effective formulas.¹⁴

Conventional medicine treatment includes cleaning the genitalia, maintaining dryness, avoiding tight clothing, and occasionally douching. Infections can be treated with antibiotic drugs. In some cases, surgical intervention is necessary to eradicate infections.¹⁵

32.6.4.4.1 Dietary Regimen

Food should be simple, easily digestible, and taken at regular intervals. Excessive amounts of puddings, garlic, onion, pickles, potatoes, sour foods, and excessive fried and greasy foods are not recommended. Rice water (water collected after washing the raw rice) is a very useful remedy in this disease.¹⁴

32.6.5 Infertility

In Ayurveda, if the breasts, ovaries, and uterus are not well developed and the patient cannot ovulate, then it is considered primary infertility (*shandi yoni*).^{8,10} Women with this condition cannot conceive due to a genetic defect. The monthly cycle may or may not be regular in such women. Decreased fertility (*vandhyatva*), which is also primary infertility, can be treated. Secondary infertility is known as *kakavandhya* according to Ayurveda.^{7,8}

32.6.5.1 Etiopathogenesis

There are four major causes for primary infertility: systemic (*dosadhatu mediated*), congenital (*beejadosa*), vaginal and organic defects (*yonivyapat*), and idiopathic (*daiva*).^{3,10}

Conventionally, infertility is attributed equally to both factors, male and female. In the female factor, congenital and acquired structural abnormalities, ovulatory defects, and a hormonal imbalance should be investigated. The psychological factor is a very important precipitating factor and needs to be considered.¹⁵

32.6.5.2 Clinical Features

No specific symptoms are seen in infertile patients that can differentiate them from other women who have not had problems conceiving. Infertility is defined as a failure to

conceive after a year of regular intercourse without using any contraceptives. The infertility can be primary or secondary. Primary infertility refers to those who have never conceived. Secondary infertility applies to those who have conceived in the past. Although both partners are equally important in the act, there is more evidence of infertility due to abnormalities in the female reproductive system. The abnormality may be structural (e.g., developmental anomalies) or functional (e.g., hormonal imbalance). There are many diseases of various systems as underlying causative factors. Infertility is different than sterility. Sterility implies an intrinsic inability to conceive.¹⁵ It resembles the concept of *shandhi yoni* in Ayurveda.

32.6.5.3 Diagnosis and Prognosis

History, pelvic examination, ultrasonography (USG), hysterosalpingogram, diagnostic laparoscopy, and blood examination are some of the investigations necessary to reach a proper diagnosis. Prognosis is dependent upon the cause. It is usually difficult to treat with total success in every patient.¹⁵

32.6.5.4 Treatment

Ayurvedic treatment differs according to etiopathogenesis. When infertility is due to congenital or idiopathic causes, it cannot be treated. It is clearly mentioned in the classical texts that when a woman has any local defect, she cannot conceive. Those disorders should be treated first and her menstrual cycle should be regular. Cleansing therapies are carried out using herbs and formulas that have a penetrating capacity (e.g., *Triphala/Trikatu kwatha, kshara taila*).⁸ These therapies are advised to unblock the channels and make the tracts patent. Some uterine tonics and ovulation inducing herbs or formulas should be administered as a palliative therapy. The *jeevaniya gana, rasayana*, and *vajikarana* group of formulations not only enhance strength and vigor, but improve the internal environment of the reproductive system. Formulas such as *ashokarishta, phala ghrita*, and *shatavari kalpa* give excellent results. *Amalaki, guduchi, bala*, and *ashwagandha* are some of the uterine and ovarian tonics which promote fertility.^{1,14}

Conventional treatment is offered at various levels. If infertility is due to anovulation, then using clomiphene citrate alone or in combination with some steroids or human gonadotrophins will be the best choice. In case of endometriosis or adhesions, surgical intervention is necessary. If cervical mucus is inadequate, it should be improved or introduced.¹⁵

Diet and lifestyle are not very specific. Eating healthy foods and doing some simple exercises (e.g., yoga) are believed to be beneficial according to the Ayurvedic treatment.

32.6.6 Premenstrual Syndrome (PMS)

32.6.6.1 Etiopathogenesis

In Ayurveda there is no specific information about this disease. The concept and symptoms can be perceived on the basis of *tridosha* theory. The symptoms like heaviness of breasts, change in appetite, nausea, vomiting, depression, and nervousness can be correlated to *kapha* imbalance. Aggravated *pitta* contributes to irritability, anxiety, anger, vomiting, headache, food cravings, and sleep disorders. Bloating of the stomach, pelvic pain, headache, difficulty in concentrating, sleep disorders, mood swings, and anxiety are the features of provoked *vata*.¹¹

In the conventional medicine, though the exact cause of PMS is not yet known, the most popular theory is cyclic change in hormones or estrogen levels. Another cause is chemical changes in brain, especially serotonin levels. Some habits or addictions like eating salty food and consuming alcohol and cold beverages containing caffeine aggravate the problem.^{1,15}

32.6.6.2 Clinical Features

PMS is a psychoneuroendocrinological disorder. It is characterized by physiological and psychological changes in the body before or during each menstrual period and is severe enough to disturb a woman's routine life. Approximately 75% of women suffer from PMS. Among them, 40% have a disturbed routine due to severity. The range of the symptoms is very high. Common physical symptoms include tenderness and heaviness of breasts, change in appetite and food cravings, nausea, vomiting, headache, pelvic pain, and bloating of the stomach. Common psychological symptoms include mood swings, depression, irritability, anger, sleep disorders, nervousness, anxiety, and difficulty concentrating.^{2,15}

32.6.6.3 Diagnosis and Prognosis

PMS is diagnosed by using a proper history. No major investigations are required to diagnose PMS. From an Ayurvedic point of view, the symptoms are differentiated based on the *dosa* involved to choose the right remedy. Prognosis of PMS is good and is quite successful.

32.6.6.4 Treatment

Externally, oleation with sesame oil and fomentation with herbal decoctions give excellent results. In severe cases, medicated oil enema (*anuvasana basti*) is very beneficial (e.g., sesame oil and *dashamoola kwatha*).⁸ Internally, *ashokarishtam* and *A. vera* in combination with some other drugs are used in Ayurvedic practice. For *vata* and *pitta* symptoms, *shatavari* (asparagus), *ashwagandha* (*Withania somnifera*), *amalaki* (Indian gooseberry), and *yashtimadhu* (licorice) work very well. To reduce *kapha* symptoms such as heaviness of breasts and fluid retention, herbs such as *triphala*, *trikatu* (pepper, ginger, and long pepper), and *vacha* (*Acorus calamus*) are very useful. *Jeerakadyarishtom*, *manasamitra vatakam*, *saraswatarishtam*, *shatavari kalpa*, and *kumaryasavam* are some of the effective herbal formulas.^{10,14}

In conventional medicine, a definite treatment has not yet emerged as a definition, and the etiology of the disease is unclear. Symptomatic treatment can be given using sedatives, diuretics, laxatives, and analgesics. Progesterone or its derivatives are recommended to improve the condition.¹⁵

Women suffering from this disorder should specifically avoid *vata* aggravating a week before the menstruation. Examples include dried fish and dry meat, deep fried and very spicy foods, very cold foods, sprouts, beans, and potatoes.

32.6.7 Menopausal Syndrome

32.6.7.1 Etiopathogenesis

In Ayurveda, menopause is called *rajonivrutti*. It means that the menstrual flow stops forever. As this is a natural phenomenon, it is not considered a disorder in Ayurveda. However, a common understanding is that the three *dosas* are imbalanced during menopause. Symptoms of menopause can be correlated to change in *dosa* dynamics. Among

the three *dosas*, *vata* is of prime importance. It is said that there is not a single disorder of the reproductive system without aggravated *vata*. Women generally reach menopause after the age of 45, which is when *vata* is dominant in the body. Pelvic region is a seat of *vata*. It usually becomes vitiated when monthly flow ceases. All pains, roughness of skin, etc. during menopausal age can be attributed to aggravated *vata*. Finally, *vata* can also provoke *kapha* and *pitta*. Symptoms such as heaviness of body, fluid retention, and hot flashes are some of the symptoms attributed to an imbalance in *kapha* and *pitta* during menopause.^{3,10,11}

Modern medicine defines menopausal age as between 47 and 52. Menopause is exactly a last or final menstruation. Postmenopause refers to the phase that comes after menopause. Menopause occurs when oocytes responsive to gonadotropins disappear from the ovary, or the oocytes remaining do not respond to gonadotropins. When this phenomenon occurs at certain age due to hormonal imbalance (i.e., change in levels of concentration of FSH, estradiol, LH, etc.), it is called physiological menopause. If menopause is due to cessation of ovarian function because of surgery (e.g., hysterectomy, oophorectomy), then it is called as artificial menopause.^{22,23}

32.6.7.2 Clinical Features

Symptoms start to show up from the pre- or perimenopausal period. The first and foremost symptom is the alteration of the menstrual cycle. Abrupt cessation is rare. The usual pattern is a gradual decrease in quantity, duration, and frequency of the menstrual flow. In some patients it is heavy or more frequent before menopause. Another common symptom is hot flashes. This is a feeling of excessive heat or burning sensation on the face, neck, chest, and back, sometimes followed by sweating. Many times it is preceded by pressure in the head or a headache. Other symptoms like regression of breasts, inadequate lubrication in vagina, urinary urgency or incontinence, loss of libido, sleep disturbances, or fatigue may be present. Psychological symptoms are also equally important, as they may disturb a woman's life. Mood swings, irritability, depression, anxiety, and dysphoria are common psychological changes during menopause. Additionally, there are some major metabolic changes that manifest in the body during in the postmenopausal period (e.g., osteoporosis and increased risk of heart attacks).^{22,23}

32.6.7.3 Diagnosis and Prognosis

Menopausal syndrome is essentially diagnosed from the patient's age and history. Routine blood work will help differentiate the fluid retention in menopause from anemia and other disorders. Menopause can be treated successfully using Ayurvedic herbs and dietary advice.

32.6.7.4 Treatment

The major goal of the treatment is to reduce symptoms and discomfort as well as to enhance well-being and prevent cardiovascular changes or osteoporosis. No drastic cleansing procedures are advised to menopausal women. *Shirodhara* (continuous flow of medicated oil, decoctions, or buttermilk on the head) and gentle a body massage are advised, as these treatments relieve muscle spasms, pains, burning, and restlessness. Internally, *chandanasava*, *ushirasavam*, and *draksharishtom* are some of the drugs that reduce hot flashes, burning, and urinary incontinence. *Shatavari*, *vidari*, and *kumari* are the herbs that reduce fatigue and vaginal dryness and give a *rasayana* effect. For psychological symptoms, like irritability, anxiety, or depression, there are very effective drugs

like *brahmi*, *vacha*, *jyotishmati*, and *shankhapushpi* or formulas such as *panchagavya ghrita*, *kalyanaka ghrita*, stresscom cap, and *manasmitra vatakam*. On the whole, *Saraca indica* (*ashoka*) is the herb of choice that takes care of most of the problems.^{3,8,14}

The American Association of Clinical Endocrinologists recommends hormone replacement therapy (HRT) as a preventive medicine for menopausal syndrome. HRT relieves menopausal symptoms and prevents osteoporosis. It also plays a role in protecting the heart. A goal of this treatment also is managing weight, stress reduction, vagina lubrication, etc. Steroids, moisturizers, and oral contraceptives may be used to treat menopause.^{15,22}

No specific diet is advised other than avoiding those that have been mentioned above for other conditions.

32.7 Scientific Basis of Ayurvedic Therapies

Our literature search found very few scientific studies and clinical trials on Ayurvedic therapies. These studies are reviewed and summarized here.

32.7.1 Clinical Studies on Compound Formulas

In one study²⁴ on leucorrhea (n = 52), the resin of *Shorea robusta* was given at a dose of 1 g/day in divided doses for 30 days. Externally, a *Terminalia chebula* decoction was administered for vaginal douche. At the end of 30 days of treatment, there was 100% improvement in the symptoms of leucorrhea. Of the 12 cases infected with trichomonas infection at the baseline, 10 cases showed a negative smear test for trichomonas infection at the end of treatment. Similarly, 5 cases of erosion of cervix showed complete healing of erosion at the end of the treatment. The authors of the study reported that there were no adverse reactions observed during the full length of the study.

In the second study (n = 32) on leucorrhea, Masilamani et al.²⁵ reported that the combination of oxidized alum ash (*padigara barpam*) in the dosage of 300 mg three times/day with milk, along with a vaginal douche and a decoction of *T. chebula*, showed promising results. The duration of the study was 30 days. Of the 32 patients, 26 (81%) were clinically cured and 6 (19%) were clinically relieved of their symptoms of leucorrhea. There are no follow-up data or any adverse event information given in the article.

Ashokarishta, a very popular formulation for gynecological disorders, was studied for its efficacy in 22 subjects suffering from menorrhagia (dysfunctional uterine bleeding) for 3 months. A dose of 25 ml three times/day was given until the onset of the next menstruation. It was continued for another 2 months from the fourth day of the menstrual cycle with another break during the next cycle. The trial drug was continued for three consecutive cycles. Results of the study indicate that *ashokarishta* gave relief from excessive bleeding and pain in all the cases. Subjects who had *vata*- and *kapha*-predominant symptoms showed relief within 3 days after starting the formulation. One patient with a follicular cyst and one patient with an ovarian cyst had to undergo a hysterectomy after 3 months of the study. An interesting finding of this study was that the hemoglobin automatically increased as soon as the bleeding was stopped. With the help of this study, we can infer that the properties of *Saraca indica* — oxytocic, hemostatic, and analgesic — have been investigated.²⁶

32.7.2 Pharmacological Studies on Single Herbs and Bhasmas

32.7.2.1 *Saraca indica*

Research has found that *Saraca indica* (saracin) is a good uterine tonic, which is effective in normalizing menstruation and reducing associated symptoms.^{27,28} In another study,²⁹ saracin seems to be an interesting immunomodulator for the mammalian immune system. It has been found to be mitogenic for human lymphocytes, and this mitogenic activity could be inhibited in presence of fetuin. It also revealed that treatment with saracin could induce secretion of interleukin-2 (IL-2) in a culture of resting human peripheral blood mononuclear cells after 48 h. The study found that saracin has a higher affinity for the CD8(+) than the CD4(+) T-cells.

32.7.2.2 *Aloe vera*

In an interesting study,³⁰ the aqueous extract of *Aloe barbadensis* Miller in an *in vitro* culture of *Trichomonas vaginalis* revealed that within 24 h, percentages of inhibition greater than 50% were obtained from concentrations of 20.8 µg/ml. The researchers studied three strains of this parasite. The aqueous extracts were used as an initial concentration of 400 mg/ml of the extract and double serial dilutions were performed; final concentrations based on the dried weight of the extract were 10.4, 20.8, 41, 83, and 160 mg/ml. The inhibition of growth was greater than 50% even when tried with lower concentrations of the extract but it took 48 to 72 h for such an action.

In another study,³¹ CARN 750 (injectable acemannan), a polydispersed beta-(1,4), linked acetylated mannan isolated from the *A. barbadensis* plant, was evaluated for its multiple therapeutic properties. It was found helpful in wound repair and acts as a biological agent for the treatment of neoplasia in animals; it was also noted for its ability to activate macrophages.

It was also found that CARN 750 directly or indirectly has significant hematoaugmenting properties. Subcutaneous administration of CARN 750 significantly increased splenic and peripheral blood cellularity, as well as hematopoietic progenitors in the spleen and bone marrow as determined by the IL-3-responsive colony-forming unit culture assay and the high-proliferative-potential colony-forming cell assay (a measure of primitive hematopoietic precursors) in myelosuppressed (7 Gy) C57BL/6 mice. It was observed that the greatest hematopoietic effect was obtained after sublethal irradiation in mice receiving 1 mg of CARN 750/animal, with less activity observed at higher or lower doses. Results showed activity equal to or greater than the injection of an optimal dose of granulocyte-colony-stimulating factor in myelosuppressed mice when CARN was injected daily.

In another study,³² lyophilized *A. barbadensis* at concentrations of 7.5 and 10% proved to be spermicidal when used along with zinc acetate. The action was attributed to the multiple microelements boron, barium, calcium, chromium, copper, iron, potassium, magnesium, manganese, phosphorus, and zinc; these elements were toxic to the tail and caused instant immobilization. The two compounds did not show any signs of irritation or cause ulceration of rabbit vaginal epithelium. These results suggest the possibility of using zinc acetate and lyophilized *A. barbadensis* as a new, effective, and safe vaginal contraceptive.

The effects of another species of aloe on some physiological and biochemical parameters of reproduction in immature female rats based on the history of their use by the folklore medicine in Cameroon was studied.³³ *Aloe buettneri*, *Justicia insularis*, *Hibiscus macranthus*, and *Dicliptera verticillata* were given in different doses daily to 22-day-old rats for 5, 10, 15, 20, and 25 days by gastric intubation. The weights of ovaries and uteri, levels of uterine and ovarian proteins, ovarian cholesterol, and serum estradiol were evaluated at the end

of each experimental period. The results of the study showed a decrease in growth rate of animals treated with 94 mg/kg/day at the end of the experimental period. Interestingly enough, the ovarian and uterine weights increased in all treated groups, especially within the pubertal period (36 to 41 days old), when compared with the respective controls. During the same period, other parameters that were also observed showed significant differences in treated rats when compared with the controls. Ovarian and uterine protein levels, as well as serum estradiol, also increased in the groups given 49 or 94 mg/kg/day of the plant extracts. The investigators concluded that these results suggest a possible presence of estrogenic compounds in the plant extracts.³³

The study was repeated with the aqueous extract of the leaf mixtures of *A. buettneri*, *Dicliptera verticillata*, *Hibiscus macranthus*, and *Justicia insularis* at doses of 13, 49, and 94 mg/kg/day for 15 days given by oral route to immature female rats. Results showed that the extracts induced a significant increase in ovarian and uteri weight as well as serum and ovarian estradiol.³⁴

Aloe vera was also studied for its anti-inflammatory property. In this study, the effects of aqueous, chloroform, and ethanol extracts of *A. vera* gel on carrageenan-induced edema in the rat paw and neutrophil migration into the peritoneal cavity stimulated by carrageenan were studied. The results showed that the aqueous and chloroform extracts decreased the edema induced in the hind-paw and the number of neutrophils migrating into the peritoneal cavity; the ethanol extract decreased only the number of neutrophils. The mechanism of action of the aqueous extract was due to the inhibited prostaglandin-E2 production from [¹⁴C]arachidonic acid. The investigators conducted an assay of the extracts and found that the aqueous extract contained anthraglycosides, reductor sugars, and cardiogenic glycosides; in the ethanol extract, the chemical tests performed for saponins, carbohydrates, naftoquinones, sterols, triterpenoids, and anthraquinones turned out positive. The chemical tests performed in the chloroform extract showed positive reaction for sterols type delta 5 and anthraquinones. On the basis of these findings, it was concluded that the extracts of *A. vera* gel have anti-inflammatory activity and suggested its inhibitory action on the arachidonic acid pathway via cyclooxygenase.³⁵

32.7.2.3 *Muktashukti Bhasma*

Muktashukti bhasma, a compound formulation consisting of pearl, *A. vera*, and vinegar, inhibited acute and subacute inflammation in albino rats as induced by subplanter injection of carrageenan, histamine, serotonin (5-HT), nystatin, and subcutaneous implant of cotton pellets. The anti-inflammatory response of 1000 mg/kg MSB was comparable with the response observed with 300 mg/kg acetylsalicylic acid (ASA) in all the test procedures. According to the investigators, the anti-inflammatory activity of the compound was attributed to its ability to cause inhibition of prostaglandins, histamine, and 5-HT and stabilization of the lysosomal membranes. The anti-inflammatory activity of MSB seems to be only one third to half as potent as ASA.³⁶

32.7.2.4 *Glycyrrhiza glabra*

Glycyrrhizin, a constituent of *Glycyrrhiza glabra*, inhibits inflammation and prostaglandin synthesis. It blocks estrogen effects binding to estrogen receptors. In other studies for treating menopausal syndrome, licorice was found an effective source of estrogen.

Glycyrrhizae radix is found to suppress estradiol-17 beta (E2)-induced expression of c-fos/jun in uterine corpus and inhibited *N*-methyl-*N*-nitrosourea and E2-induced endometrial carcinogenesis in mice. The probable mode of action could be through

suppression of estrogen-induced c-fos/jun expression and could be promising in preventing agents for endometrial cancers.^{37,38}

Tamir et al.¹⁶ reported that the stimulatory effects of 2.5 to 25 µg/animal glabridin were similar to those of 5 µg/animal estradiol. Glabridin, the major isoflavon in licorice root, was tested for the estrogenic properties, in view of the resemblance of its structure and lipophilicity to those of estradiol. The results indicate that glabridin is a phytoestrogen, binding to the human estrogen receptor and stimulating creatine kinase activity in rat uterus, epiphyseal cartilage, diaphyseal bone, aorta, and left ventricle of the heart. During the course of the study, the researchers found that the position of the hydroxyl groups has a significant role in binding to the human estrogen receptor and in proliferation-inducing activity. There was a biphasic effect of increasing concentrations of glabridin on the growth of breast-tumor cells. It showed an estrogen-receptor-dependent, growth-promoting effect at low concentrations (10 nM to 10 µM) and estrogen-receptor-independent antiproliferative activity at concentrations of >15 µM. The investigators concluded that glabridin and its derivatives exhibited varying degrees of estrogen-receptor agonism in different tests and demonstrated growth-inhibitory actions on breast cancer cells.

32.7.2.5 *Curcuma longa*

It has been proven that this herb has antibacterial, anticancerous, antihemorrhagic, and anti-inflammatory properties and hence can be effective in menstrual difficulties like menorrhagia or abnormal uterine bleeding, vaginitis, and leukorrhea. Laboratory research has shown that it may modulate estrogen and progesteron activity.³⁹

32.7.2.6 *Nardostachys jatamansi* (*Jatamansi*)

The clinical trials showed reduced restlessness, insomnia, and aggressiveness. These properties are very useful in menopause as well as in PMS.

A study was proposed to evaluate the protective effect of *Nardostachys jatamansi* on neurobehavioral activities, thiobarbituric acid reactive substance, reduced glutathione, thiol group, and catalase and sodium-potassium adenosine triphosphatase activities in the middle cerebral artery (MCA) occlusion model of acute cerebral ischemia in rats. Findings showed that the changes induced by ischemia were significantly attenuated by a 15-day pretreatment of *N. jatamansi* (250 mg/kg orally) and correlated well with histopathology by decreasing the neuronal cell death after MCA occlusion and reperfusion. The study provided first evidence of effectiveness of *N. jatamansi* in focal ischemia probably by virtue of its antioxidant property.⁴⁰

32.7.2.7 *Zingiber officinale* (*Ginger*)

Ginger was found effective in the symptoms such as nausea and vomiting in pregnant women. In a randomized, placebo-controlled, double-blind trial (n = 70),⁴¹ women received either oral ginger 1 g/day or an identical placebo for 4 days. At a follow-up visit 7 days later, all participants except three in the placebo group remained in the study. After the therapy, nausea decreased significantly in the ginger group (2.1 ± 1.9) compared with the placebo group (0.9 ± 2.2 , $p = .014$). The number of vomiting episodes also decreased significantly in the ginger group (1.4 ± 1.3) compared with the placebo group (0.3 ± 1.1 , $p < 0.001$). No adverse effect of ginger on pregnancy outcome was detected. It can also be used in PMS symptoms such as nausea and vomiting.

32.7.2.8 *Commiphora mukul* (Myrrh)

In laboratory research, *Commiphora mukul* showed antibacterial and antifungal properties, especially in *Candida albicans*. Therefore, it can be used in vaginitis and related leukorrhea.⁴²

In a study on the species *Commiphora guidotti*, known as scented myrrh, all sesquiterpenes were isolated and characterized. Seven compounds, with cadinane, guaiane, oplopane, and eudesmane skeletons, were obtained. Two of seven are new and two are reported from a natural source for the first time. The present study was planned to compare the effects of the minor sesquiterpenes with those of the previously isolated major component, T-cadinol, which has shown to possess smooth muscle-relaxing properties. The results showed that the minor sesquiterpenes are more efficient in reducing K(+)-induced contractions than those induced by the alpha-adrenoceptor agonist phenylephrine. However, they were all less potent than T-cadinol in their action.⁴³

In another study, Kimura et al.⁴⁴ reported that myrrhanol A, a new triterpene isolated from *Balsamodendron* or *C. mukul* Hook-gum resin, showed a potent anti-inflammatory effect on exudative pouch fluid, angiogenesis, and granuloma weights in the adjuvant-induced air-pouch granuloma of mice. The interesting finding of this study is that its effects were more marked than those of hydrocortisone and the 50% aqueous methanolic extract of the crude drug.

32.7.2.9 *Withania somnifera*

Withania somnifera is found effective in combination with other rejuvenating drugs like licorice in menopausal syndrome in studies conducted by National Institute of Ayurveda, Jaipur, India. It increases levels of estrogen and helps reduce the symptoms like hot flashes, fatigue, and depression because of its antioxidant.^{45,46} It also has immunomodulatory,⁴⁷ antidepressant,⁴⁸ anti-inflammatory, and adaptogenic effects.^{49,50} Information on the studies on these effects are available in other chapters.

32.7.2.10 *Tinospora cordifolia*

The active principles of *Tinospora cordifolia*, cordioside (TC-2), cordiofolioside A (TC-5), and cordiol (TC-7) are found to possess the macrophage activation. Syringin (TC-4) and TC-7 inhibited the *in vitro* immunohaemolysis of antibody-coated sheep erythrocytes by guinea pig serum. The anticomplementary and immunomodulatory activities were found to be caused by inhibition of the C3-convertase of the classical complement pathway. Higher concentrations showed constant inhibitory effects. The authors reported that the compounds also gave rise to significant increases in immunoglobulin-G (IgG) antibodies in serum. Humoral and cell-mediated immunity were also enhanced but were dose dependent.⁵¹

In another study,⁵² *T. cordifolia* extract (100 mg/kg body weight for 15 days) given to CCl₄ intoxicated rats was found to protect the liver, as indicated by enzyme level in serum. A significant reduction in serum levels of serum glutamic oxaloacetic transaminase (SGOT), serum glutamate pyruvate transaminase (SGPT), alkaline phosphatase (ALP), and bilirubin were observed after *T. cordifolia* treatment during CCl₄ intoxication. The extract also deleted the immunosuppressive effect of CCl₄.

32.7.2.11 *Asparagus recemosus*

Asparagus recemosus was tried in menopausal syndrome for relieving symptoms like hot flashes, excessive sweating, tiredness, and sleep disturbances in a study conducted by

Government Ayurveda College, Trivandrum, India. The report reads that the treatment was effective. No further details were available about the author, dosage, and or duration.

In another study,^{50, 53–55} *A. recemosus*, *T. cordifolia*, *W. somnifera*, and *Picrorhiza kurroa* were tested on the functions of macrophages obtained from mice treated with the carcinogen ochratoxin A (OTA). At the end of 17 weeks of treatment, the chemotactic activity of murine macrophages was significantly decreased with OTA compared with controls. Production of interleukin-1 (IL-1) and tumor necrosis factor (TNF) was also markedly reduced. OTA-induced suppression of chemotactic activity and production of IL-1 and TNF-alpha by macrophages was significantly inhibited with the treatment with *A. racemosus*, *T. cordifolia*, *W. somnifera*, and *P. kurroa*. It was also noted that *W. somnifera* treated macrophage chemotaxis and *A. recemosus*-induced excess production of TNF-alpha when compared with controls.

Mandal et al.⁵⁶ studied the antibacterial activity of *A. racemosus* and compared it with chloramphenicol. The methanol extract of the roots of *A. recemosus* were given in concentrations of 50, 100, and 150 µg/ml. Results showed considerable *in vitro* antibacterial efficacy against *Escherichia coli*, *Shigella dysenteriae*, *Shigella sonnei*, *Shigella flexneri*, *Vibrio cholerae*, *Salmonella typhi*, *Salmonella typhimurium*, *Pseudomonas putida*, *Bacillus subtilis*, and *Staphylococcus aureus*.

Antioxytotic action of saponin isolated from *A. recemosus* on uterine muscle was further studied.⁵⁷ The details of this study could not be accessed. Similar studies^{58,59} on the lactogoggue effect of *Asparagus recemosus* showed positive effects, but details were not available.

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Appendix 1:

Ayurvedic Words and Their English Meanings

Abhyanga	oil massage
Adhimamsa	fleshy growth
Adhimamsarma	pterygium
Adhimantha	glaucoma
Adhisthana	site
Adhoga Amlapitta	regurgitation through the rectum
Adhyarbuda	recurrence of tumor
Adhyasana	frequent eating
Agantuka	exogenous factors
Agni	fire, digestive juice, hormones
Agnikarma	heat therapy
Ahara	diet
Ahara rasa	absorbed digested nutrients
Ahara Rasayana	herbal dietary supplement
Ahara sakti	digestive power
Aheerata	eagerness, anxiety
Ahigata	injury
Ahishyanda	conjunctivitis
Ahita	incompatible
Ajakajat	prolapse of the iris
Ajasrik Rasayana	dietary supplement to keep the process of new tissue formation
Ajirna	indigestion
Akrti	general appearance
Akshepaka	spasm/convulsion
Akshipakatyaya	panophthalmitis
Alochaka pitta	pitta located in the retina responsible for eyesight
Ama-Atisara	enterotoxigenic diarrhea
Ama dosa	undigested food and other materials
Amaja anaha	a variety of constipation due to undigested materials
Amasaya	stomach
Amavata	rheumatoid arthritis
Amla	sour
Amladhyashita	subacute glaucoma
Amlapitta	hyperacidity
Amritikarana	the process of heating the product in presence of some herbal drugs
Amruthaprasa rasayanam	a preparation of herbal drugs in concentrated jaggery syrup
Anaha	bloated abdomen due to the accumulation of stool
Anguli	fingers

- Anjan* ophthalmic preparation to be applied or rubbed gently on the conjunctiva
Anna food
Antarvyavartana entropion
Antuja unmada psychosis due to exterior origin
Anubandh binding
Anubandhya principal
Anuvanshaja hereditary
Anuvasan a kind of enema
Anuvasan vasti enema with medicated oils
Anuvsana basti oil enema
Anyasthaniyakshaya body-weight loss
Anyatovata supraorbit neuralgia
Apachi glandular swelling
Apana vata responsible for the excretion of urine and feces
Aparipakvasana eating improperly cooked food
Apasmaara epilepsy
Apatarparna fat reducing
Apathya harmful food
Apathyaja acquired due to unhealthy lifestyle
Apaya loss, going away, destruction of
Arbud carcinomas
Arbuda neoplasm; fibroid
Arista preparation obtained by fermentation
Arjun subconjunctival hemorrhage
Arshas hemorrhoids
Artava ovum/menses
Artavajanana ovulation inducing
Aruchi anorexia
Aruhyakam general well-being
Asadhya incurable
Asadhya vrana incurable ulcers
Asathmya bhojana hypersensitive or incompatible food items
Ashchyotana eyedrop
Ashmari urolithiasis, urinary calculi
Astavidha pariksa eight-point examination
Asthapana basti herbal decoction enema
Asthayi transient lipids
Asthi bones
Asthipatal/drishtipatal retina
Ati sthulya obesity and morbid obesity
Atisara diarrhea
Atma spirit
Aushadhi medicine
Avagalana or utplavana subluxation of lens
Avapeedna instillation of nasal drops
Avara weak
Avipaka acid indigestion
Avrana Shukla corneal opacities
Awaleha sugar-based semisolid preparation
Ayana path/channel
Ayu-kshaya decrease in life span

- Ayurveda** knowledge of life; Vedic health science
- Bala** strength
- Bala thailam** medicine prepared in oil for improving strength
- Baluka sveda** sand fomentation
- Basti** medicated enema
- Beejadosha** genetic disorder
- Beeja poshana** treatise to better the quality and quantity of semen
- Bhaishajya kalpana** formulation of a dosage form
- Bhasma** literally ash; incinerated metal or minerals
- Bhauma** earth
- Bhavana drava** trifala decoction
- Bheda** histopathological differentiation
- Bhedan and lekhan** incise and drainage
- Bhedana** penetrating
- Bhutavidya** science of evil spirit
- Bhutonmada** *grahavesa* (sychosis)
- Brimhan** repletion of the body tissue
- Buddhi** intellect
- Buddhimandya** mentally retarded
- Cetana** consciousness
- Chakrikas** small cakes
- Chikitsa** therapy
- Chinabandha** bandage of eye
- Churna** powder
- Daiva** spiritual
- Daivavyapasraya cikitsa** spiritual treatment
- Darun vicar** serious diseases
- Dashavidha pariksha** ten methods of examination
- Dashmool** roots of ten herbs
- Deergham ayu** long life
- Deva vyapashraya chikitsa** devine therapy
- Dhamani** nerve supply to artieries
- Dhamapana** insufflation
- Dhara** steady flow of liquids
- Dhasamoola Rishtam** hot-water extract of roots of the ten herbal medicines that is fermented
- Dhatu** tissues
- Dhatukshaya** degeneration of essential constituents
- Dhatupak janya vikriti** a disease caused by a defective metabolism leading to derangement in body-tissue (seven dhatus) transformation process
- Dhatus** tissues; there are seven *dhathus*: *rasa* (ingested food), *raktha* (blood), *mamsa* (muscles), *medas* (fat), *asthi* (bone), *majja* (bone marrow), and *shuklam* (semen); *rasa* is converted to *raktha* and so on until it is finally converted into *shuklam*
- Dhatuvahasrotas** place of conversion of tissues
- Dhatu-*vridhhi*** tissue growth
- Dhatwagni** deranged metabolism
- Dhatwagni chikitsa** correction of metabolic defects
- Dhoomra** medicated smoking
- Dhuma nasya** smoke inhalation
- Dhumapana** medicated smoking
- Dhwaja bhangam** erectile dysfunction

- Dipana** materials that stimulate digestive juices
Divavyapashraya spiritual interventions
Divyausadhi divine medicinal plants
Dola yantra, khalva yantra, musha yantra instruments used in preparation of *bhasmas*
Dooshti vitiation
Dooshyas tissues that are the site of a disease
Dosa biomaterials, bioenergy, bodily humor
Dosasushti vitiation of *dosa*
Dravaka liquefacients
Dravya Rasayana therapy with dietary supplements
Drishtimani lens
Dushtavrana infected ulcer
Dushya that which can get vitiated like tissues
Dushyas culpable
Dwirarbuda metastasis of tumor
Ekadesavridhhi enlargement of particular tissue
Gairika red chalk
Gala throat
Galaganda goiter
Garbhini pregnant women
Ghee, Ghrita, or Ghruta dehydrated butter
Ghritika buttermilk
Gomutra cow urine
Grahani irritable colon
Grahan restrained movement
Grahas planets
Grahavesa influence of planets
Granthi minor neoplasm
Gridhra eagle
Gridhrasi sciatica
Gudabhramsa rectal prolapse
Gulma intra-abdominal swellings
Guna attribute or respect
Gunas-sattva, raja, tama three qualities of mind, mental constitution
Halimaka fulminant hepatic failure
Haridra meha yellowish urine (like turmeric)
Hasti meha urine similar to an elephant in rut
Hima cold infusion
Hingula cinnabar
Hridaya heart
Hritkantha Daha burning sensation rising up from the stomach or lower chest toward the neck
Iksu meha urine similar to sugarcane juice and very sweet
Indriyārtha sensorial
Ingalekam cinnabar
Jangama ausadhis prepared from animal products
Jangha thigh
Janu knee
Jara-janya manasa vikara psychiatric problems of the aged
Jatharagni digestive fire
Jeevaneya ganam group of herbs to increase the vigor and vitality

Jeevaneya ghrutham ghee-based medicine used to prolong life
Jihva tongue
Jvara-Atisara infective diarrhea
Kajjali amalgum of sulfur and mercury
Kala time, rhythm
Kalaka choroid
Kala meha black urine
Kalka crushed plant material
Kalyanakam ghrutham a preparation in liquefied butter used for treating oligospermia
Kalyanakam kashayam water extract of certain herbs indicated for oligospermia
Kamala hepatitis
Kamsya bell, metal
Kamya Rasayana rasayana given with certain special aim
Kandaras tendons
Kandu itching
Kanji rice gruel
Kapha biological material, water humor, mucus, *dosa*
Kapha and raktavaha srotasa lymphatic and circulatory system
Kaphadhara Kala mucous lining of the stomach
Kapha dosa mucus and mucoid, interlinking functions and constituents of the body
Kapha Grahani dysentery predominant
Kaphahara/Kaphashamaka measures or drugs that pacify *kapha*
Kaphaja granthi or arbuda adenoma
Kaphaja Linganasha mature cataract
Kaphaj linganash mature cataract
Karma action
Karna ear
Karnika cornea
Karshan the depletion of body tissues
Kasa cough
Kashay rasa astringent
Kashayalkwatha decoction
Kastha ausadhi herbal preparation
Kati low back
Katu pungent
Katu vipak bitter
Kavalika dressing
Kaya chikitsa medicine
Khalva stone mortar
Khanja monoplegia
Khara hard
Klaibyam impotence
Klama fatigue
Kleda waste products of adipose tissues
Kledaka one of the five types of *kapha dosa*
Klinnavartma blephritis
Kosta intestine
Kostashakhasrita kamala hemolytic jaundice
Kotham necrosis
Kricchravitka, alpavitka passing small quantity of stool
Krimi helminthiasis including microbial infection

- Krimijanya* due to worms
Krishnapatal iris
Krura koshtha extremely harsh bowel
Ksara meha the urine is like a solution of alkali in smell, color, and taste
Kshara alkali
Ksharakarma application of caustics
Kshaudrameha excessive urine with sweet taste like honey
Kshayaja due to emaciation
Ksheernartava amenorrhea
Ksudraka svasa asthma
Kukunder oscoccygis bone
Kumba kamala hepatic failure
Kupipakwa vidhi procedure used for preparing mercury *bhasma*
Kushta bhasma-like preparation of *Unani-tibb*
Kutipravesika patient/concerned specialized living in a hut
Kwatha decoction
Laala meha the urine is slimy and contains threads like that of saliva
Lagan chalazion
Laghu lightweight, small
Lakshanika chikitsa symptomatic treatment
Langhana depletion or reducing therapy, no food intake
Lavan salty
Leen lens falling back into vitreous chamber
Lehyam medicinal preparation in jaggary syrup
Lekhan scraping
Lekhana scraping
Lepa ointment
Lepana application of medicated paste
Linga symbol of male organ
Linganasha dosha liquified lenticular material
Lingarsha genital growths, genital wart
Lingavidhikara yoga drugs used for enlargement of penis
Loha iron
Loka-purusa samya balance with universe
Maans muscle
Maans lolup greedy to flesh
Madhumeha diabetes
Madhur sweet
Madhyama middle path or medium
Madhyama koshtha moderate bowel
Mahamarma very important vital organ
Maharasa main metals, also used for mercury
Mahasrotas gastrointestinal tract
Majja bone marrow
Majja meha the urine looks like marrow
Mala excretion, stool
Mamsa muscle, soft tissue
Mamsarbuda neoplasm of soft tissue
Mamsashrit Patal fleshy part or choroid
Manas mind
Manasika psychological factors

Manas Rasayana	rasayana for the improvement of intellect
Manassila	arsenic
Manayah	gems
Mandal	orbits
Mandbudhi	low intelligence
Manjistha meha	foul-smelling urine that is slightly red
Manodaihika vyadhis	psychosomatic diseases
Manovaha srtoas	psychological channels
Mansa dhatwagni	digestive power in the muscle tissues
Marana	the process of burning or calcination
Margavarana	obstruction of passage
Marma	vital part
Marma ghat	vital's trauma
Matruj avayava	maternal organ ovum
Meda	fat
Medadhātu	adipose tissue
Medaja originated from fatty tissue	
Medas patal	choroid
Medhaja granthi or arbuda lipoma	lipid tumor or lipoma
Medhya	intellect
Medhya Rasayana	brain tonics
Medodhatwagni	fat-specific energy
Medomay	osteostasis
Medoposhakansh	nutritive food, fatty food
Medoroga	disease of the fat tissue
Meha	immunity
Mithyaaharavihara	unwholesome diet and regimen
Mool	root
Mrudu koshtha	extremely weak bowel
Mukha	oral cavity
Mula sthana	center of origin
Mulika	root
Murccha, Moha, Tamaka	loss of consciousness, fainting
Mutra	urine
Nadi	pulse
Nadi svedan	steam fomentation by tube
Nadipariksa	pulse examination
Naimittic Rasayana	rasayana is given only for partial period for certain number of days
Naishthiki chikitsa	spiritual therapy
Nakulandhyatva	retinoblastoma
Nanatmaj vadhis	many kinds of vataj diseases
Napumsaka	eunuch
Narasimha rasayanam	a formula to delay the aging process
Nasa	nose
Nash	destruction
Nashtartava	menopause
Nasya	nasal drops
Nasya karma	errhines therapy
Nasya karma	nasal infiltration
Navana nasya	inunction errhine
Netra	eyes

- Nidan** diagnosis
- Nidana pancake** five methods of diagnosis
- Nidana purvarupa** immediate signs, premonitory symptoms
- Nija** endogenous factors
- Nila meha** bluish urine
- Nirama** stool without enterotoxin
- Niruha vasti** medicated enema with decoction
- Niruttha/varttara** end product of *nirutthikarana*
- Nirutthikarana** filtration/separation
- Ojakashaya** immunity
- Ojas** the final product of all seven *dhathus*
- Ojas meha** urine that looks like honey
- Onychomycosis** ringworm of the nails
- Pachaka Pitta** digestive enzymes
- Pachana** digestant
- Pakshma** heat therapy
- Pakshmakopa** trichiasis, blephritis
- Pakvasaya** lower part of gastrointestinal tract
- Palashmakopa** entropion
- Panaki** hepatorenal syndrome
- Pancamahabhuta** five basic elements
- Pancendriya pariksa** examination of the five sense organs
- Panchakarma** five cleansing procedures (*vaman*, *virechan*, *siro-virechan*, *anuvasan*, and *nirooha vastis*)
- Pandu** anemia
- Pangu** paraplegia
- Parada, rasa, suta, maharasa, rasendra** various names for mercury
- Paratantra kamala** obstructive jaundice
- Parinama Sula** ulcerative dyspepsia, duodenal ulcer
- Parooshakadi ganam** group of herbs starting with *Phoenix pusilla*
- Parpams** term for *bhasmas* used in Siddha medicine system
- Parpati, rasayoga** mercury preparations
- Parthive** substances derived from earth
- Pashchat karma** postoperative procedures
- Patal** coats
- Patana yantra** a pot pertaining to an apparatus used for sublimation or distillation for preparing *bhasma*
- Patra pind sveda** poultice
- Phalasarpis** preparation in liquefied butter, mainly used for women to increase fertility
- Phalavarti, snehavarti** anal suppositories
- Phanita** hot infusion
- Pichu** cotton plug
- Pipali** *Piper longum*
- Pishti** triturated drug with specified liquids prepared by exposing to sun or moon
- Pista meha** urine that is white and thick; similar to a solution of corn flour
- Pitta** biological fire, metabolic catabolic enzymes
- Pitta dosa** the functions of tissue interchange and conversion
- Pittaja** originated from *pitta*
- Pittala** brass
- Pittashaya** gall bladder
- Pittaviridhi** serum bilirubin

<i>Poorvakarma</i>	preoperative procedures
<i>Pooyastrava</i>	panophthalmitis, hypopion ulcer
<i>Pothaki</i>	trachoma
<i>Prabhava</i>	unique action
<i>Pradhana dhatu</i>	primary metal
<i>Pradhana karma</i>	main cleansing procedures
<i>Prakopa</i>	transformation of growth into metastatic tumors
<i>Prakriti</i>	constitution
<i>Prakritisthapani chikitsa</i>	restorative treatment
<i>Pralepa</i>	application of thin medicament paste in the affected part
<i>Pramana</i>	build, size
<i>Pramana vijnana</i>	anthropometry
<i>Pramehas</i>	urinary disorders
<i>Pramoha</i>	impairment in functioning of mind
<i>Prana</i>	respiration
<i>Pranavaha srotas</i>	respiration
<i>Pranyama</i>	breathing exercises
<i>Prasara</i>	metastasis
<i>Prasna pariksa</i>	interrogation
<i>Prasrutha</i>	measure/equivalent of one handful
<i>Prastari Arma</i>	pterygium
<i>Prasuti</i>	delivery
<i>Pratimarsha</i>	topical application
<i>Pravahika</i>	dysentery
<i>Pravala</i>	shells
<i>Pravara</i>	type of body constitution
<i>Prayoga</i>	experiment
<i>Purishaja anaha</i>	variety of constipation due to stool accumulation
<i>Purishasanga, purishanaha</i>	accumulation of feces
<i>Purva karma</i>	preparatory procedures
<i>Purvarupa</i>	prodromal signs/symptoms
<i>Pushta</i>	overweight
<i>Putapaka</i>	heating with cow dung cakes
<i>Puttam</i>	cow dung cake or made pits with cow dung cakes
<i>Puyalasak</i>	dacryocystitis
<i>Raja</i>	menstrual blood
<i>Rajas prakriti</i>	authoritarian mental constitution
<i>Rajata</i>	silver
<i>Rajayakshma</i>	tuberculosis
<i>Rajonivritti</i>	menopause
<i>Rakta</i>	blood
<i>Raktaarbudanasna</i>	anticancer drug
<i>Rakta-Atisara</i>	hemorrhagic diarrhea
<i>Rakta meha</i>	blood red
<i>Raktamokshana</i>	bloodletting
<i>Rakta-pradara</i>	menorrhagia
<i>Raktarbuda</i>	tumors that exudate blood
<i>Raktashodhaka</i>	blood purifier
<i>Raktasrava</i>	profuse hemorrhage
<i>Raktavaha srotas</i>	circulatory system
<i>Rakthasthambhaka</i>	styptics

- Ranjaka pitta* type of biliary fluid
Ras blood
Rasa absorbed food nutrients, mercury preparations
Rasa ausadhi metallic preparation
Rasa dhatu plasma tissue
Rasa karpur mercuric subchloride
Rasalinga symbol of power
Rasashala laboratory of an alchemist
Rasashastra Ayurvedic pharmacy of mercury preparations, metallurgy
Rasavadins alchemists
Rasayana Ayurvedic drug preparation to increase the life span
Rasvidya Indian alchemy
Ratna gems
Rihti pupil
Roga disease
Roganashani chikitsa curative therapy
Rogibala ability to defend against diseases
Rog pariksha clinical examination
Rog viakhya clinical description
Rohini sixth layer of skin, similar to epithelium
Ropana healing
Ruk pain
Ruksha dry
Rupa clinical manifestation
Sabda voice
Sadanga pariksa six-point examination
Sadhya curable
Sadhyasadhyata prognosis
Sahaja genetic
Sahaya dhatu secondary metal
Sama stool with enterotoxin
Samana balanced
Samana vata vata that stimulates the digestive enzymes
Saman vikruti decreased hepatic conjugation
Samanya shodhana general purification
Samasanhanan nara proportionate body
Samhanana body structure
Samhita text
Samprapti pathogenesis
Samshodhana purification procedures
Samshodhana chikitsa detoxification therapy
Samskara process
Samudra-sauviram mercuric perchloride
Samyak vedhan proper pricking
Sanair meha urine passes very slowly
Sanchaya localization of growth
Sandhi junctions
Sandhigat vata osteoarthritis and rheumatoid arthritis
Sandrajala vetrious humor
Sandra meha urine kept overnight
Sang biliary stasis

Santarpan restorative treatment
Sara quality of tissues
Sastra chikitsa surgical treatment
Sathavari gulam preparation of herbs made with jaggery used for female infertility
Sathavaryadi ghrutham medicine prepared in liquefied butter for female infertility
Sathya Yuga time period in history when the rule of justice prevailed and all people were happy
Satmya adaptability
Sattva pharmaceutical process, consciousness, intelligence
Sattvavajaya counseling therapy
Sattvika prakrti examination of the psyche
Satva liquid containing active ingredient
Satvavajaya chikitsa psychotherapy
Sauvarcala, saindhava, vida, audbhida types of salt
Savrana Shukla corneal ulcers with perforations
Seka fomentation
Sesa *Sesamum indicum* oil
Shakhasrita kamala intra- and extra-hepatic jaundice
Shalya tantra surgery
Shamana pacifying, palliative
Sharava mud tray
Sharira human body
Shashkakshipaka ophthalmoplegia
Shatadhoutha Ghrita medicated ghee
Sheeta cold
Shefa penis
Shilajit asphaltum
Shiro head
Shiro vasti flow of liquid on the head
Shleshak kapha type of *kapha dosa*
Shleshma another name for kapha or phlegm
Shodana chikitsa therapy by elimination of vitiated pathogens
Shodhana detoxification
Shodhana karma measures of internal purification
Shonitarma pterygium
Shotha localized swelling/edema
Shothahara anti-inflammatory
Shrunga horn of a cow
Shuddha, sishra, pooti loha quality of iron *bhasma*
Shukadosha diseases of external genitalia caused by the use of irritants
Shuklam semen
Shuklarma pterygium
Shukra semen
Shuktika bitot's spots/xerophthalmia
Shwetapatal sclera-white of eye
Shweta-pradara mucous in the urine
Siddha dugdha medicated milk
Siddha ghruta smedicated ghee
Siddh Makardhwaja preparation containing oxide form of mercury
Sikata meha urine contains sandlike particles
Sindoora drug prepared by the process of sublimation

Sira different channels
Siraharsha scleritis
Sirajal acute orbital cellulites
Siravedha bloodletting
Sirobasti treatment of head and neck with oils
Sirotpat episcleritis
Sirstalvantargatam manah skull
Sisa, naga lead
Sita meha urine is sweet and very cold
Smriti memory
Smruti memory
Smrutim Medha good intellect, good recalling ability
Snana bath
Snayu ligament
Snehana oleation
Snehana chikitsa oleation therapy
Snehapan intake of medicated fats
Sodhana chikitsa purification therapy
Sokatisara diarrhea due to sorrow
Somana chikitsa curative therapy
Sopham swelling
Soul spirit
Spandan muscle spasm
Sparsahsnutwam tenderness
Sparsa skin/touch
Sphik gluteous
Srotamsi channels, microscopic pores
Srotas channels, pores
Srotas pariksa examination of channel
Srotorodh channel blockage
Stambha numbness
Sthana samsraya secondary growth
Sthayi stored form
Sthoulyam hugeness
Sthula obese
Streeroga gynecological disorders
Streeshareer anatomy of the female reproductive system
Stree stanya breast milk
Sukra dhatu reproductive tissue, semen
Sukra meha urine mixed with semen
Sukravahasrotas path through which semen passes
Sukshma minute, delicate
Sukumaram ghrutham a preparation of herbals in liquefied butter for the treatment of male and female infertility
Sukumaram kashayam water extract of certain herbs, used for male and female infertility
Sukumaram lehyam a preparation of herbs in concentrated jaggery syrup used to treat male and female infertility
Sura meha urine resembles wine (sura) with a clear top and cloudy at the bottom
Surya-namaskar sun-yoga-asana (body positions)
Suwarna gold
Svasa increased or difficult breathing, dyspnea

<i>Svedana</i>	fomentation
<i>Svedana karma</i>	sudation therapy
<i>Swaras</i>	fresh juices
<i>Swarna bhasma</i>	gold bhasma
<i>Swasthavritta and ritucharya</i>	daily and seasonal health regimens
<i>Swatantra kamala</i>	hepatocellular jaundice
<i>Taila</i>	oil
<i>Takra</i>	buttermilk
<i>Talaka</i>	arsenic trisulphide
<i>Talu</i>	palate
<i>Tamaka svasa</i>	bronchial asthma
<i>Tamas</i>	inertia
<i>Tamra</i>	copper
<i>Tandra</i>	lassitude
<i>Tantriks</i>	practitioners of black magic
<i>Tantu</i>	threads
<i>Tara</i>	pupil
<i>Taraka</i>	iris
<i>Tarunam vayah</i>	ability to remain young
<i>Teekshna</i>	pointed
<i>Teevravastha</i>	acute phase
<i>Tejojalashrit Patal</i>	outermost layer in which there exists aqueous humor
<i>Tikshna guna</i>	severe taste
<i>Tiksna</i>	sharp
<i>Tikta</i>	bitter
<i>Tikta-Amlodgara</i>	bitter and acid regurgitation
<i>Tilapista nibha</i>	clay-colored stool
<i>Timira</i>	refractive errors, premature cataract, or other serious causes of blindness
<i>Toda</i>	paresthesia
<i>Trapa</i>	tin
<i>Tridosa</i>	three <i>dosas</i> (<i>vatta</i> , <i>pitta</i> , and <i>kapha</i>)
<i>Trikatu</i>	three bitter herbs
<i>Triphala</i>	group of three herbs (<i>Embelica officinalis</i> , <i>Terminalia belerica</i> , and <i>Terminalia chebula</i>)
<i>Tritiya patalgat dushti</i>	disorders of the third coat
<i>Trivanga bhasma</i>	bhasma containing lead, tin, and zinc
<i>Twak</i>	bark
<i>Udaka</i>	water
<i>Udaka meha</i>	urine is clear; is in large amounts; and is white, cold, and odorless
<i>Udara</i>	abdomen
<i>Udaram, jalodhar, or asadhya</i>	ascites
<i>Unmada roga</i>	psychosis
<i>Uparasa</i>	there are eight <i>uparasa</i> : <i>gandhaka</i> (sulphur), <i>gairic</i> (ochre), <i>kasis</i> (ferrous sulphate), <i>sphatica</i> (potash alum), <i>hartal</i> (orpiment), <i>manashila</i> (realgar), <i>anjana</i> (lead), <i>kankustha</i> (rhubarb)
<i>Upasaya-anupasay</i>	exacerbating and relieving factors
<i>Upataraka</i>	ciliary body
<i>Urdhvaga Amlapitta</i>	regurgitation through the mouth
<i>Urdhwajatrugat</i>	of the upper level of the clavicle
<i>Uru</i>	thigh
<i>Ushna veerya</i>	dry

- Ushnavata* cystitis/urethritis
Uthara sthanam last volume
Utklesha hypersalivation
Utsangini chalation
Utsangini lagan chalazion
Vadanasthapana analgesics
Vaikrintaka tourmaline
Vajikarana aphrodisiac therapy
Vaman therapeutic emesis
Vamathu vomiting
Vapavahan visceral and omental fat
Vardhanana Rasayana rasayana dose is increased slowly to the highest level, maintained at that level, and then decreased slowly to the base line
Vartka a mixture of five metals
Vartma eyelids
Vasa fatty substance
Vasa meha urine looks like liquid muscle fat and may be passed frequently
Vasti Ayurvedic enema therapy; karma-enema
Vata biological humor related to nervous system
Vatadoshas represents functions of movements or propulsions
Vata grahani constipation predominant
Vataja caused by vitiated *vata*
Vataparyay supraorbital neuralgia
Vata rakta gouty arthritis
Vatatapika rasayana treatment that can be given to outdoor patients
Vata vyadhi nervous disorders
Vati, gutika, or guggulu preparations in pill form
Vaya age and aging
Vayu air, wind; another name of *vata*
Veda knowledge
Vedana severe pain
Veerya potency
Vega Dhara a controlling natural urges
Viadhi heitu etiology
Vibandha constipation
Vidaryadi kashayam water extract of some herbs to enhance body tissues repair
Vidaryadi lehyam preparation of herbs in jaggery syrup for body tissue repair
Vikriti imbalance of *dosas*
Vilekhan scraping
Vipaka postdigestive effect
Virecana karma purgation therapy
Virek purgation
Viruddha Ahara incompatible food
Viruddh asana incompatible exercise
Viryalpata impotence
Visa poison
Vishaghna detoxifier
Vishesha shodhana special purification
Vishtambha obstruction
Vismriti amnesia
Vitiated dosas toxic materials

Vrana ulcer
Vridhipatra shastra scalpal
Vrisya aphrodisiac
Vrukka fat around the body organs
Vyadhi disease
Vyadhi hetu etiology
Vyakti expression of symptoms
Vyayama sakti physical strength
Yakrut liver
Yakruta dhatwagni manya decreased hepatic uptake
Yakruta shotha hepatocellular damage
Yakrutodara hepatomegaly
Yakrutshosh cirrhosis
Yapya difficult to treat
Yavakshara salt of potassium and sodium
Yonidhavana vaginal douche
Yoniroga/yonivyapat vaginal diseases
Yuktivyapashraya nondrug and drug modalities
Yukti vyapashraya chikitsa rational therapy

Appendix 2:

Ayurvedic Herbs and Botanical Names

Botanical Name	Ayurvedic Name	Parts Used
<i>Abroma angusta</i> Linn.	<i>Bharadwaji</i>	Root
<i>Abies pindrow</i> Royale	<i>Talispatra</i>	Leaves
<i>Abrus precatorius</i> Linn.	<i>Gunja</i>	Leaf, root, seed
<i>Abutilon indicum</i> G. Don	<i>Atibala</i>	Entire plant
<i>Acaia arabica</i> Willd.	<i>Babbula</i>	Entire plant, resin
<i>Acacia catechu</i> Willd.	<i>Khadirasara</i>	Sara
<i>Acacia concinna</i> DC.	<i>Phenika</i>	Fruit
<i>Acacia farnesiana</i> Willd.	<i>Irimesa</i>	Bark
<i>Acacia latiosum</i>	<i>Kinkirat</i>	Stem, bark, fruit
<i>Acalypha indica</i> Linn.	<i>Akakiya</i>	Sara
<i>Achyranthes aspera</i>	<i>Apamarga</i>	Entire plant
<i>Aconitum chasmanthum</i>	<i>Sringivisa</i>	Root
<i>Aconitum heterophyllum</i> Wall.	<i>Ativisa</i>	Root
<i>Aconitum napellus</i> Linn.	<i>Vatsanabha</i>	Root
<i>Acorus calamus</i> Linn.	<i>Vaca</i>	Root
<i>Actinopteris dichotoma</i>	<i>Mayurasikha</i>	Entire plant
<i>Adansonia digitata</i> Linn.	<i>Gorakha cinca</i>	Fruit, leaf, bark
<i>Adenanthera pavonia</i> Linn.		<i>Ravan amlika</i>
<i>Adhatoda vasica</i> Nees	<i>Vasa</i>	Complete plant
<i>Adiantum lunulatum</i> Burm.	<i>Hansaraja</i>	Complete plant
<i>Aegle marmelos</i> Corr.	<i>Bilwa</i>	Leaves, fruit, root
<i>Agaricus campestris</i> Linn.	<i>Chatraka</i>	Complete plant
<i>Ailanthus excelsa</i> Roxb.	<i>Aralu</i>	Bark, seed
<i>Alangium salvifolium</i>	<i>Ankola</i>	Root, bark, oil
<i>Albizia lebeck</i> Benth.	<i>Sirisa</i>	Bark, seed
<i>Alhagi pseudalhagi</i>	<i>Yavasakam</i>	Complete plant
<i>Allium cepa</i>	<i>Palandu</i>	Root (<i>kanda</i>)
<i>Allium sativum</i>	<i>Lasuna</i>	Root
<i>Alocasia indica</i>	<i>Manakanda</i>	<i>Kanda</i>
<i>Aloe barbadensis</i>	<i>Kumari</i>	Exudation
<i>Alpinia galanga</i>	<i>Kulinjana</i>	Root
<i>Alstonia scholaris</i>	<i>Saptaparna</i>	Bark
<i>Alternanthera sessilis</i>	<i>Matsyaksi</i>	Complete plant
<i>Amomum sublatum</i> Roxb.	<i>Sthulaila, Badi Ilaichi</i>	Seeds
<i>Amorphophallus campanulatus</i> Blume.	<i>Surana</i>	<i>Kanda</i>
<i>Amorphophallus sylvaticus</i>	<i>Banya surana</i>	<i>Kanda</i>
<i>Amranthus blitum</i>	<i>Tanuliyaka</i>	Leaf, root
<i>Amranthus spinosus</i> Linn.	<i>Tanuliyaka</i>	Complete plant
<i>Anacardium occidentale</i>	<i>Kaju</i>	Seed
<i>Anacyclus pyrethrum</i> DC.	<i>Akarakarabha</i>	Root
<i>Anamirta cocculus</i>	<i>Kakamari</i>	Fruit
<i>Andrographis paniculata</i> Nees.	<i>Yavatikta, kalamegha</i>	Complete plant

<i>Andropogon iwaraneusa</i> Jones	<i>Lamajjaka</i>	Root
<i>Andropogon schoenanthus</i>	<i>Rohisa trina</i>	Root
<i>Anethum sowa</i> Kurz.	<i>Satapuspā</i>	Seed, oil, arka
<i>Annona atemoya</i>	<i>Sitaphala</i>	Fruits
<i>Anogeissus latifolia</i>	<i>Dhava</i>	Bark, gum
<i>Anisomeles malabarica</i>	<i>Sprikka</i>	Whole plant
<i>Anthocephalus cadamba</i> Mig.	<i>Kadamba</i>	Bark, leaf, flower
<i>Anthocephalus chinensis</i>	<i>Kadamba</i>	Fruits, leaves, bark
<i>Aquilaria agallocha</i>	<i>Agar</i>	Wood
<i>Aquilaria agallocha</i> Roxb.	<i>Aguru</i>	Wood, oil
<i>Areca catechu</i> Linn.	<i>Pugaphala</i>	Fruit
<i>Argemone mexicana</i>	<i>Swarnaksiri</i>	Complete plant
<i>Argyrea speciosa</i>	<i>Briddhadaru</i>	Root, leaf, seed
<i>Aristolochia bracteata</i>	<i>Kitamari</i>	Complete plant
<i>Aristolochia indica</i> Linn.	<i>Ishwarimula</i>	Root
<i>Artemisia siversiana</i>	<i>Damanaka</i>	Complete plant
<i>Artemisia vulgaris</i> Linn.	<i>Saraparna</i>	Flower
<i>Artocarpus lakoocha</i> Roxb.	<i>Lakuca</i>	Fruit
<i>Arundo donax</i>	<i>Nala</i>	Whole plant
<i>Asparagus adscendens</i> Roxb.	<i>Krisna Musali</i>	Root
<i>Asparagus racemosus</i> Willd.	<i>Satavari</i>	Root
<i>Asparagus sarmentosa</i> Linn.	<i>Maha satavari</i>	Root
<i>Asteracantha longifolia</i> Nees.	<i>Kokilaksa, Iksuraka</i>	Seeds, complete plant
<i>Averrhoa carambola</i>	<i>Karmaranga</i>	Fruit
<i>Azadirachta indica</i> A. Juss	<i>Nimba</i>	Seeds, oil, resin
<i>Balanites aegyptiaca</i>	<i>Ingudi</i>	Fruit, oil
<i>Balanites arundinacea</i> Willd.	<i>Vamsalocana</i>	Whole plant
<i>Balospermum mantanum</i>	<i>Danti</i>	Seed, oil, root
<i>Balsamodendron myrrha</i>	<i>Bola, gandharasa</i>	Exudate
<i>Bambusa arundinacea</i> Willd.	<i>Vamsalocana</i>	
<i>Barleria prionitis</i>	<i>Saireyaka, sahacara</i>	Complete plant
<i>Barringtonia acutangula</i>	<i>Samudraphala</i>	Fruit
<i>Bassia latifolia</i>	<i>Madhuka</i>	Flower, seed, oil, bark
<i>Bauhinia racemosa</i>	<i>Asmantaka</i>	Leaf, bark
<i>Bauhinia variegata</i> Linn.	<i>Kancanar</i>	Bark, leaf, root
<i>Benincasa hispida</i> Thunb.	<i>Kusmanda</i>	Fruit, seed
<i>Berberis aristata</i>	<i>Rasanjana</i>	Whole plant
<i>Bergenia ligulata</i>	<i>Pashna bheda</i>	
<i>Betula utilis</i> D. Don.	<i>Bhojapatra</i>	Leaf, bark
<i>Bixa orellana</i>	<i>Sinduri</i>	Seeds
<i>Blepharis edulis</i> Pers.	<i>Ucata</i>	Seeds
<i>Boerhaavia diffusa</i>	<i>Punarnava</i>	Complete plant
<i>Boerhaavia verticillata</i>	<i>Vrushiva</i>	Complete plant
<i>Borassus flabellifer</i> Linn.	<i>Tada</i>	Fruit, root, seed
<i>Boswellia serrata</i> Roxb.	<i>Sallaki</i>	Exudate
<i>Brassica juncea</i>	<i>Rajika</i>	Seed, oil
<i>Brassica nigra</i> Koch.	<i>Sarsapa</i>	Seed, leaf, oil
<i>Bryonopsis laciniosa</i>	<i>Sivalingi</i>	Seed
<i>Buchanania lanzan</i>	<i>Badam, almond</i>	Faruit, seed
<i>Buchanania lanzan</i> Spr.	<i>Priyal</i>	Oil
<i>Butea frondosa</i>	<i>Palasha, dhaka</i>	Seed, root, leaf, flower
<i>Butea monosperma</i>	<i>Palasha, dhaka</i>	Seed, root, leaf, flower
<i>C. angustifolia</i> roxb.	<i>Tavaksiri</i>	<i>Kanda curna</i>
<i>C. cetylissineus</i>	<i>Karkatika</i>	Fruit, seed
<i>C. long</i> Linn.	<i>Rajani</i>	Rhizome
<i>C. zedoaria</i> rose	<i>Kachur</i>	Rhizome

<i>Caesalpinia bonducella</i>	<i>Kuberaksa</i>	Complete plant, seed
<i>Caesalpinia digyna</i> Rottler.	<i>Ghritha karanjii</i>	Root
<i>Caesalpinia sappan</i> Linn.	<i>Patrang</i>	Wood
<i>Cajanus indicus</i>	<i>Adhaki</i>	Leaf, seed
<i>Calamus rotang</i>	<i>Vetra</i>	<i>Prakanda</i>
<i>Calophyllum inophyllum</i> Linn.	<i>Punnag</i>	Oil
<i>Calotropis gigantea</i>	<i>Arka</i>	Complete plant, <i>ksira</i>
<i>Calotropis procera</i>	<i>Arka</i>	Complete plant, <i>ksira</i>
<i>Cannabis sativa</i> Linn.	<i>Bhanga</i>	Leaves, <i>manjari</i> , <i>kandastha</i>
<i>Canscora decussata</i>	<i>Sankh holi</i>	Whole plant
<i>Capparis aphylla</i>	<i>Karira</i>	Fruit, flower, root
<i>Capparis horrida</i>	<i>Kinkini</i>	Fruit
<i>Capparis sepiaria</i>	<i>Himsra</i>	Fruits, flowers, root
<i>Capparis zeylanica</i>	<i>Vyaghranakhi</i>	Root
<i>Capsicum annum</i>	<i>Katuvira</i>	Fruit, seeds
<i>Capsicum frutescens</i>	<i>Toksna</i>	Fruits, seed
<i>Careya arborea</i>	<i>Kumbhi</i>	Fruit, bark
<i>Carica papaya</i> Linn.	<i>Madhu karkati</i>	Fruit, seed, leaf
<i>Carissa carandas</i>	<i>Karamarda</i>	Fruit
<i>Carum ajamoda</i>	<i>Ajamoda</i>	Seed, fruit
<i>Carum carvi</i>	<i>Krishn jiraka</i>	Seeds
<i>Carum copticum</i>	<i>Yavani, ajowan</i>	Fruits, leaves, root
<i>Carum opticum</i>	<i>Yavani</i>	Seed, oil
<i>Carum roxburghianum</i>	<i>Yavani</i>	Seeds
<i>Carthamus tinctorius</i>	<i>Kusumbha</i>	Complete plant, oil
<i>Caryophyllus aromaticus</i>	<i>Lavanga</i>	Flower, oil
<i>Casaria asculenta</i>	<i>Satacakra</i>	Flower, oil
<i>Cassia absus</i>	<i>Caksusya</i> , Indian almond tree	Seed
<i>Cassia angustifolia</i>	<i>Markandi</i>	Leaf, fruit
<i>Cassia auriculata</i>	<i>Avartaki</i>	Leaf, bark, fruit
<i>Cassia deodara</i> Roxb.	<i>Devadaru</i>	Stem, oil
<i>Cassia fistula</i>	<i>Aragwadha</i>	Complete plant, pulp of the fruit
<i>Cassia occidentalis</i>	<i>Kasamarda</i>	Seed, root, leaf
<i>Cassia tora</i> Linn.	<i>Cakramarda</i>	Leaf, seed
<i>Cayratia trifolia</i>	<i>Amlata</i>	Leaves, root
<i>Cedrus deodaru</i>	<i>Devdaru</i>	Leaves, bark
<i>Ceiba pentandra</i>	<i>Salmali</i>	Complete plant, fruit, root
<i>Celastrus paniculata</i> Willd.	<i>Jyotismati</i>	Seed, oil
<i>Celosia argentea</i>	<i>Siri balika</i>	Root, leaf, seed
<i>Centella asiatica</i>	<i>Mandukparni</i>	Whole plant, leaves
<i>Celosia cristata</i>	<i>Murva</i>	Seeds, flowers
<i>Centipeda minima</i> Linn.	<i>Chikkika</i>	Complete plant
<i>Chichorium intybus</i>	<i>Kasani</i>	Leaves
<i>Cicer arietinum</i>	<i>Chanakamla</i>	Leaf
<i>Cinchona</i> sps.	<i>Kunayan</i>	Bark, extract
<i>Cinnamomum camphora</i>	<i>Bhimasenikarpura</i>	Extract
<i>Cinnamomum cassia</i>	<i>Twakpatra</i>	Bark
<i>Cinnamomum tamala</i>	<i>Tamala</i>	Leaf
<i>Cinnamomum zeylancium</i>	<i>Twak</i>	Bark, oil
<i>Cissampelos pareira</i> Linn.	<i>Patha</i>	Root, complete plant
<i>Cissus quadrangularis</i> Linn.	<i>Asthirinkhla</i>	Complete plant, <i>kanda</i>
<i>Citrullus lanatus</i>	<i>Indravaruni</i>	Whole plant
<i>Citrus acida</i>	<i>Nimbuka</i>	Fruit, seed, oil, leaves
<i>Citrus limetta</i>	<i>Madhu jambira</i>	Fruit
<i>Citrus limon</i> Linn.	<i>Jambira</i>	Fruit
<i>Citrus</i> sp.	<i>Amla vetas</i>	Root, fruit
<i>Cleome viscosa</i>	<i>Karna sphota</i>	Leaf, seed

<i>Clerodendrum infortunatum</i>	Bhandir	Root
<i>Clerodendrum multiflorum</i>	Agnimantha	Leaves, roots
<i>Clerodendrum phlomidis</i>	Agnimantha	Complete plant, root
<i>Clerodendrum serratum</i> Linn.	Bharangi	Root
<i>Clitoria ternatea</i>	Aparajita	Complete plant, root, seed
<i>Coccinia indica</i> W & A	Bimbi	Complete plant, root
<i>Cocculus hirsutus</i> Linn.	Patala garudi	Complete plant, leaf
<i>Cocculus villosus</i> Linn.	Patala garudi	Complete plant
<i>Cocos nucifera</i>	Sriphala	Fruit, oil
<i>Coleus vettiveroides</i>	Hribera	Leaves
<i>Commiphora mukul</i>	Guggulu	Resin
<i>Convolvulus pluricaulis</i>	Sankhapuspi	Complete plant
<i>Coptis teeta</i> Wall.	Pikta mula	Root
<i>Corallocarpus epigaeus</i>	Nahi kanda	Kanda
<i>Corchorus fascicularis</i>	Bahuphali	Complete plant
<i>Cordia latifolia</i>	Slesmatak	Leaf, bark, fruit
<i>Coriandrum sativum</i>	Dhanyak	Seed, leaf, oil
<i>Coscinium fenestratum</i>	Kaliyaka	Leaves
<i>Cotoneaster racemiflora</i>	Yavasasarkara	Suskasara
<i>Crataeva nurvala</i>	Varuna	Root, bark, leaf
<i>Cressa cretica</i>	Rudanti	Complete plant
<i>Crinum latifolium</i>	Sudarsan	Kanda, leaf
<i>Crocus sativus</i> Linn.	Kumkum	Pollen grains
<i>Crotalaria juncea</i>	Shana	Seed, fruit, leaf
<i>Crotalaria verrucosa</i>	Shanapuspi	Leaf, seed
<i>Croton oblongifolius</i>	Asti danti	Root
<i>Croton tiglium</i> Linn.	Jayapala	Seed
<i>Cryptolepis buchanan</i>	Krisna sariva	Bark
<i>Cucumis melo</i>	Kharbuja	Fruit, seed
<i>Cucumis utilisissimus</i>	Eravaru	
<i>Cuminum cyminum</i>	Jirak	Seed, fruit
<i>Curculigo orchioidea</i>	Bhutala	Leaves, root
<i>Curculigo orchioidea</i> Gaertn.	Swetamusali	Kanda, root
<i>Curcuma amada</i> Roxb.	Amragandhi hrida	Rhizome
<i>Cyamopsis tetragonoloba</i>	Guwar	Fruit, seed, leaf
<i>Cydonia vulgaris</i>	Vrihi	Seed, fruit
<i>Cymbopogon citratus</i>	Jambira truna	Leaves, oil
<i>Cynodon dactylon</i>	Durva	Complete plant
<i>Cynomorium coccineum</i>	Malta fungus	Whole plant
<i>Cyperus rotundus</i>	Musta	Leaves, flower, root
<i>D. stramonium</i>	Krisa dhattura	Complete plant, seeds, leaves, flowers
<i>D. zalil</i>	Trayamana	Complete plant, flower
<i>Dacus carota</i>	Gajar	Seed, root, kanda
<i>Daemia extensa</i>	Yugma kantaka	Leaf
<i>Dalbergia sissoo</i> Roxb.	Shimsapa	Bark, stems
<i>Datura alba</i>	Dhattura	Complete plant
<i>Datura metal</i>	Dhattura	Complete plant
<i>Delphinium denudatum</i>	Nirovisi, jadwar shireen	Root, kanda, seed
<i>Dendrophoe falcata</i>	Vriksadani	Complete plant
<i>Desmodium gangeticum</i>	Salaparni	Complete plant
<i>Desmostachya bipinnata</i>	Darba, kusa	Root
<i>Dichrostachys cinerea</i>	Virataru	Complete plant
<i>Digitalis purpurea</i>	Tilapuspi	Leaf
<i>Dillenia indica</i>	Bhavya	Fruit, leaf
<i>Dioscorea bulbifera</i>	Varahi kanda	Kanda
<i>Dioscorea opposita</i>	Vidari bheda, common yam	Tuber

<i>Diospyros peregrina</i>	<i>Tinduka</i>	Bark, fruit, oil
<i>Digitalis purpurea</i>	<i>Tilapuspī</i>	Leaf
<i>Dolichos biflorus</i>	<i>Kulattha</i>	Seed
<i>Dracaena cinnabari</i>	<i>Rakta niryaś</i>	Resin
<i>E. glandulifera</i>	<i>Dravanti</i>	Seed, oil, fruit
<i>E. nerifolia</i>	<i>Snuhi</i>	Root, stem, milk
<i>Echinops schinatus</i>	<i>Urkantaka</i>	Root, seed
<i>Eclipta alba</i> Hassk.	<i>Bringaraja</i>	Complete plant
<i>Eichhorinia crassipes</i>	<i>Jalakumbhi</i>	Complete plant
<i>Elaeocarpus ganitrus</i> Roxb.	<i>Rudrakṣa</i>	Seed
<i>Elettaria cardamum</i>	<i>Sukṣma eḷa</i>	Seed, bark
<i>Embelia ribes</i> Burm.	<i>Vidanga</i>	Seed
<i>Emblica officinalis</i> Gaertn.	<i>Amalaki</i>	Fruit, complete plant
<i>Enicostemma littorale</i> Blume.	<i>Mamajjak</i>	Complete plant
<i>Ephedra vulgaris</i>	<i>Soma</i>	Complete plant
<i>Erythrina indica</i>	<i>Paribhadra</i>	Leaf, bark
<i>Eugenia caryophyllus</i>	<i>Long</i>	Dried flower buds
<i>Eulophia campestris</i>	<i>Munjataka</i>	Root, <i>kanda</i>
<i>Eulophia nuda</i>	<i>Malakanda</i>	<i>Kanda</i> (root)
<i>Eupatorium ayapana</i>	<i>Vishalyukarni ajaparna</i>	Leaf
<i>Euphorbia antiquorum</i>	<i>Snuhi (tridhara)</i>	Root
<i>Exogonium purga</i>	<i>Revakandam</i>	Root
<i>F. lacor</i> Buch-Ham.	<i>Plakṣa</i>	Bark, fruit
<i>F. religiosa</i> Linn.	<i>Asvattha</i>	Complete plant
<i>F. retusa</i>	<i>Nandi vrikṣa</i>	Complete plant
<i>Fagonia arabica</i>	<i>Duralabha</i>	Complete plant
<i>Feronia limonia</i>	<i>Kapittha</i>	Fruit, leaf
<i>Ferula alliacea</i> (narthex)	<i>Hingu</i>	Resin
<i>Ficus bengalensis</i> Linn.	<i>Vata</i>	Prop roots, complete plant
<i>Ficus glomerata</i> Roxb.	<i>Udumbara</i>	Fruit, root, complete plant, milk
<i>Ficus hispida</i> Linn.	<i>Kakodumbara</i>	Complete plant
<i>Foeniculum vulgare</i>	<i>Misi</i>	Seed, root, flower
<i>Fritillaria roylei</i>	<i>Fritillary</i>	Whole plant
<i>Fumaria parviflora</i>	<i>Kṣetra parpata</i>	Complete plant
<i>G. herbaceum</i> Linn.	<i>Karpasa</i>	Complete plant, oil, leaf
<i>G. morella</i> Desr.	<i>Kankuṣṭha</i>	Resin
<i>G. tiliaefolia</i>	<i>Dhanvana</i>	<i>Antastwak</i>
<i>Garcinia indica</i> Chois.	<i>Vrikṣamla</i>	Fruit, oil
<i>Garcinia pedunculata</i>	<i>Amlavetas</i>	Fruit
<i>Gardenia gummiifera</i>	<i>Nadihingu</i>	Resin
<i>Gloriosa superba</i> Linn.	<i>Langali</i>	Root, <i>kanda</i>
<i>Glycyrrhiza glabra</i>	<i>Madhu yaśti</i>	<i>Prakṇad</i>
<i>Gmelina arborea</i> Linn.	<i>Gambhari</i>	Root, bark, leaf
<i>Gossypium arboreum</i>	<i>Aranya karpasa</i>	Complete plant
<i>Grewia asiatica</i>	<i>Parusaka</i>	Fruit, bark
<i>Grewia populifolia</i>	<i>Nagabala</i>	Complete plant, root, bark
<i>Gymnema sylvestre</i> R. Br.	<i>Madhu nasini</i>	Leaf
<i>Gymnosporia montana</i>	<i>Vikantak</i>	Complete plant, leaf
<i>Gynandropsis pentaphylla</i>	<i>Tilaparni</i>	Leaf, seed
<i>Hedychium spicatum</i>	<i>Sati</i>	<i>Kanda</i>
<i>Helianthus annuus</i>	<i>Suryapuspī</i>	Seed
<i>Helicteria isora</i>	<i>Avarta phala</i>	Fruit, root
<i>Hemidesmus indicus</i> R. Br.	<i>Anantamul</i>	Root
<i>Hibiscus abelmoschus</i>	<i>Latakasturi</i>	Seed, leaf, root
<i>Hibiscus rosa-sinensis</i>	<i>Japa</i>	Flower, root
<i>Hiptage benghalensis</i>	<i>Madhavi</i>	Root
<i>Holarrhena antidysenterica</i>	<i>Indrayava</i>	Seed, bark

<i>Hordeum vulgare</i>	<i>Yava, yavaksara</i>	Seed, <i>ksara</i>
<i>Hydnocarpus laurifolia</i>	<i>Katukapitha</i>	Seed, fruit, oil, pulp
<i>Hydrocotyl asiatica</i>	<i>Brahmamamanduki</i>	Complete plant
<i>Hyocyamus niger</i> Linn.	<i>Parasik yavani</i>	Seed, leaf
<i>Indigofera tinctoria</i>	<i>Nili</i>	Flower, leaf, root, seed
<i>Inula racemosa</i> Hook.	<i>Puskaramula</i>	Root
<i>Ipomoea arvensis</i>	<i>Prasarinee</i>	Complete plant
<i>Ipomoea digitata</i> Linn.	<i>Ksiravidari</i>	<i>Kanda</i>
<i>Ipomoea paniculata</i>	<i>Vidar Kund</i>	Whole plant, root
<i>Ipomoea reniformis</i>	<i>Akhuparni</i>	Complete plant
<i>Iris germanica</i> Linn.	<i>Hainavati</i>	Root
<i>Jasminum grandifolium</i> Linn.	<i>Jati</i>	Flower, leaf, oil
<i>Jasminum officinale</i>	<i>Jati</i>	Fruits
<i>Jasminum pubescens</i>	<i>Kund</i>	Fruit
<i>Jasminum sambac</i> Linn.	<i>Mallika</i>	Flower, leaf, root
<i>Juglans regia</i>	<i>Aksota</i>	Fruit
<i>Juniperus communis</i>	<i>Hapusa</i>	Fruit
<i>Kalanchoe pinnata</i>	<i>Parnabija</i>	Leaf
<i>Lagenaria siceraria</i>	<i>Iksvaku</i>	Complete plant
<i>Lagenaria vulgaris</i>	<i>Iksvaku</i>	Complete plant
<i>Lannea grandis</i>	<i>Jingini</i>	Bark, resin
<i>Lawsonia alba</i>	<i>Madayantika</i>	Flower, leaf, seed
<i>Lawsonia inermis</i>	<i>Mehndi</i>	Leaves, flower, seeds, bark
<i>Leonotis nepetifolia</i>	<i>Dipamala</i>	Flower, leaf
<i>Lepidium sativum</i>	<i>Chandrasura</i>	Seed
<i>Leptadenia reticulata</i>	<i>Jivanti</i>	Whole plant, leaf
<i>Letaria staltica</i>	<i>Kangu</i>	Seed
<i>Leucas cephalotes</i>	<i>Dronapuspi</i>	Complete plant
<i>Linum usitatissimum</i>	<i>Atasi, flaxseed husk</i>	Seed, oil
<i>Lippia nodiflora</i>	<i>Jalapippali</i>	Whole plant
<i>Liquidambar orientalis</i>	<i>Silhaka</i>	Resin
<i>Loranthus longiflorus</i>	<i>Vandaka</i>	Whole plant
<i>Luffa acutangula</i> Linn.	<i>Kosataki</i>	Fruit, root
<i>Madhuca longiflora</i>	<i>South Indian Mahua</i>	Flower, fruits
<i>Maerua arenaria</i>	<i>Marata</i>	Whole plant
<i>Malaxis acuminata</i>	<i>Jeevak</i>	Whole plant
<i>Malaxis muscifera</i>	<i>Rishabak</i>	Whole plant
<i>Mallotus philippinensis</i>	<i>Kampillaka</i>	Fruits
<i>Mammea suriga</i>	<i>Nagkesar</i>	Flowers
<i>Mangifera indica</i>	<i>Amra</i>	Complete plant, resin
<i>Mannaobambusa arundnaciun</i>	<i>Vasmalochana</i>	Whole plant
<i>Marsdenia roylei</i>	<i>Murva</i>	Complete plant
<i>Melia azadirachta</i>	<i>Mahanimba</i>	Fruit, pulp
<i>Mentha piperita</i>	<i>Podina, peppermint</i>	Leaves
<i>Mentha sylvestris</i>	<i>Putiha</i>	Complete plant
<i>Merium odorum</i>	<i>Karavira</i>	Root, flower, leaf
<i>Mesua ferrea</i>	<i>Nagakesara</i>	Stamen
<i>Michelia champaca</i>	<i>Survarna champak</i>	Flower, bark
<i>Mimosa pudica</i>	<i>Lajjalu</i>	Complete plant, seed
<i>Mimusops elengi</i>	<i>Bakula</i>	Fruit, flower, bark
<i>Mimusops hexandra</i>	<i>Ksirini</i>	Fruit, bark
<i>Mirabilis jalapa</i>	<i>Nakata</i>	Flower, root, <i>kanda</i>
<i>Momordica charantia</i> Linn.	<i>Karavellaka</i>	Fruit, leaf
<i>Momordica cymbalaria</i>	<i>Ksudrakaravellaka</i>	Leaf, root, fruit
<i>Momordica dioica</i>	<i>Karkotak</i>	Fruit, <i>kanda</i> (root)
<i>Moringa oleifera</i>	<i>Sigru</i>	Complete plant, fruit
<i>Morus alba</i>	<i>Tooda</i>	Fruit
<i>Mucuna prurita</i> Hook.	<i>Kapikachu</i>	Seed, fruit
<i>Murraya koenigii</i>	<i>Kaidarya</i>	Leaf
<i>Musa paradisiacal</i>	<i>Kadali, plantain banana</i>	Complete plant, root

<i>Myrica esculenta</i>	<i>Katphala</i>	Fruits, bark
<i>Myrica nagi</i>	<i>Katphala</i>	Stembark
<i>Myristica fragrans</i>	<i>Jayaphal</i>	Flower, seed, <i>kosa</i>
<i>Nardostachys jatamansi</i> DC	<i>Jatamansi</i>	Root, complete plant
<i>Nelumbium speciosum</i>	<i>Padmabija</i>	Seed, flower, leaf
<i>Nelumbo nucifera</i>	<i>Indian lotus</i>	Seeds, flowers, leaves, rhizomes
<i>Nicotiana tabacum</i> Linn.	<i>Tamakhu</i>	Leaf
<i>Nigella sativa</i> Linn.	<i>Prithvika</i>	Seed
<i>Nyctanthes arbortristis</i> Linn.	<i>Parijata</i>	Leaf, flower
<i>Nymphaea alba</i>	<i>Swetakamal</i>	Flower, leaf, root, pulp
<i>Nymphaea rubra</i>	<i>Raktakamal</i>	Flower, leaf, root, pulp
<i>Nymphaea stellata</i>	<i>Nilakamal, Indivari</i>	Flower, leaf, seed, pulp
<i>Ochrocarpus longifolius</i>	<i>Surapunnagaa</i>	Stamen, style, stigma
<i>Ocimum basilum</i>	<i>Tulsi</i>	Complete plant, leaves
<i>Ocimum gratissimum</i>	<i>Ajagandha</i>	Complete plant, seed
<i>Ocimum sanctum</i>	<i>Tulsi</i>	Leaf, seed, root
<i>Operculina turpethum</i>	<i>Trivrita, nishoth</i>	Root, bark, leaves
<i>Orchis latifolia</i>	<i>Munjataka</i>	Root, <i>kanda</i>
<i>Origanum majorana</i>	<i>Maruvaka</i>	Seed
<i>Oroxylum indicum</i>	<i>Syonaka</i>	Root, bark, seed
<i>Oryza sativa</i>	<i>Shalimula</i>	Root
<i>Ougeinia dalbergioides</i>	<i>Tinisa</i>	Seed
<i>Oxalis corniculata</i>	<i>Cangeri</i>	Leaf
<i>Paederia foetida</i>	<i>Gandhaprasarini</i>	Complete plant
<i>Paeonia emodi</i>	<i>Ulood saleeb</i>	Seeds, flowers, roots
<i>Pandanus odoratissimus</i>	<i>Keora</i>	Whole plant
<i>Pandanus toctorius</i>	<i>Ketaki</i>	Flower, root, oil, pollen grain
<i>Papaver somniferum</i>	<i>Ahifena</i>	Fruit extract
<i>Parmelia perforata</i>	<i>Saileya</i>	<i>Sarvanga</i>
<i>Paspalum scrobiculatum</i>	<i>Kodrava</i>	Seed
<i>Pavonia odorata</i>	<i>Balaka</i>	Leaves, root
<i>Pedaliium murex</i> Linn.	<i>Brihad goksura</i>	Fruit, complete plant
<i>Pentapetes phoenicea</i>	<i>Bandhuk</i>	Flower
<i>Peristrophe bicaliculata</i>	<i>Kakajangha</i>	Complete plant
<i>Phaseolus mungo</i>	<i>Mung Bean</i>	Seeds, root
<i>Phaseolus trilobus</i>	<i>Mudgaparni</i>	Complete plant, seed
<i>Phoenix dactylifera</i>	<i>Kharjura</i>	Fruit, seed
<i>Phragmites maxima</i>	<i>Nal</i>	Complete plant
<i>Phyllanthus urinaria</i>	<i>Bhumyamalaki</i>	Complete plant
<i>Picrorhiza kurroa</i>	<i>Kutaki</i>	Root
<i>Pimpinella anisum</i>	<i>Saunf</i>	Seeds
<i>Pinus roxburghii</i> Sargent.	<i>Gandhabiroja</i>	<i>Ral</i> , resin
<i>Pinus succinifera</i>	<i>Trinakant</i>	<i>Niryasa</i>
<i>Piper aurantiacum</i>	<i>Renuka</i>	Seeds, root
<i>Piper betel</i> Linn.	<i>Cavika</i>	Complete plant
<i>Piper chaba</i>	<i>Cavya</i>	Leaves
<i>Piper cubeba</i>	<i>Kankola</i>	Seed, oil
<i>Piper longum</i> Linn.	<i>Pippali</i>	<i>Kanda</i> , root, fruit
<i>Piper nigrum</i> Linn.	<i>Marica</i>	Fruit, grain
<i>Pistacia integerrima</i> Stewart	<i>Pista</i>	Fruit
<i>Plantago ovata</i>	<i>Snigdhajirika, psyllium husk</i>	Seeds
<i>Pluchea lanceolata</i> BC Clarke	<i>Rasna</i>	Leaf, complete plant
<i>Plumbago indica</i>	<i>Rakta Citraka</i>	Root
<i>Plumbago zeylanica</i> Linn.	<i>Citraka</i>	Root, leaf
<i>Plumeria acutifolia</i>	<i>Campak</i>	Flower, leaf
<i>Polyalthia longifolia</i>	<i>Devदारु</i>	Bark
<i>Polygonum aviculare</i>	<i>Granthi truna</i>	Complete plant
<i>Pongamia glabra</i>	<i>Karanja</i>	Bark
<i>Pongamia pinnata</i>	<i>Karnaja</i>	Flowers, seed, leaves, bark, root
<i>Portulaca oleracea</i>	<i>Lonika</i>	Leaf, seed

<i>Prosopis spicigera</i>	<i>Sami</i>	Leaf, bark, flower, buds, <i>kanda</i>
<i>Prunus alochia</i>	<i>Alukam</i>	Fruit
<i>Prunus amygdalus</i>	<i>Badam</i>	Fruit, seeds, bark, oil
<i>Prunus cerasus</i>	<i>Ela waluka</i>	Fruit
<i>Prunus domestica</i>	<i>Arkam</i>	Fruit
<i>Prunus mahaleb</i>	<i>Gandhapriyangu</i>	Fruit
<i>Prunus persica</i>	<i>Aluka, peach</i>	<i>Kanda</i> , fruit
<i>Prunus puddum</i>	<i>Padmakh</i>	Stem
<i>Psidium guajava</i>	<i>Amritaphal, Paruka</i>	Fruit, leaf
<i>Psoralea corylifolia</i> Linn.	<i>Bakuchi</i>	Seed, oil
<i>Pterocarpus marsupium</i> Roxb.	<i>Bijak</i>	Bark, extract, stern, resin
<i>Pterocarpus santalinus</i> Linn.	<i>Raktacandan</i>	<i>Kastha</i>
<i>Ptychotis ajowan</i>	<i>Yavani</i>	Seeds
<i>Pueraria tuberosa</i>	<i>Vidarikanda</i>	<i>Kanda</i>
<i>Punica granatum</i> Linn.	<i>Dadima</i>	Complete plant
<i>Putranjiva roxburghii</i> Wall.	<i>Putrajivak</i>	Fruit, root
<i>Quercus infectoria</i>	<i>Mayaphal</i>	Fruit
<i>Randia dumetorum</i>	<i>Madanaphala</i>	Fruit
<i>Randia uliginosa</i>	<i>Pinditaka</i>	Root, fruit
<i>Raphanus sativa</i>	<i>Mulaka, Muli</i>	Complete plant
<i>Rauwolfia serpentina</i>	<i>Sarpagandha</i>	Root
<i>Rheum emodi</i> Wall.	<i>Cuka</i>	Root, leaf, branch
<i>Rhinecanthus communis</i>	<i>Yudhikaparni</i>	Complete plant
<i>Rhus parviflora</i>	<i>Tntidaka</i>	Fruit
<i>Ricinus communis</i> Linn.	<i>Ernada</i>	Leaf, root, seed, oil
<i>Rosa damascena</i>	<i>Satapatri, rose</i>	Flower
<i>Rubia cordifolia</i>	<i>Manjistha</i>	Root, stem
<i>Rumex vesicarius</i>	<i>Cuka</i>	Leaf, seed
<i>Saccharum arundinaceum</i>	<i>Sara</i>	Root, <i>kanda</i>
<i>Saccharum officinarum</i> Linn.	<i>Iksu</i>	Root, stem
<i>Saccharum spontaneum</i> Linn.	<i>Kasa</i>	Rot
<i>Salix alba</i>	<i>Vetasa</i>	Stem
<i>Salix caprea</i>	<i>Vatasa</i>	Stem
<i>Salmalia malabarica</i>	<i>Salmali</i>	<i>Niryasa</i>
<i>Salvadora oleoides</i>	<i>Brihat pilu</i>	Leaf, fruit, oil
<i>Salvadora persica</i> Linn.	<i>Sweta candan</i>	<i>Kastasana</i> , oil
<i>Santalum album</i>	<i>Chandan</i>	Wood, oil
<i>Sapindus trifoliatus</i>	<i>Aristaka</i>	Fruit
<i>Saraca indica</i>	<i>Asoka</i>	Bark
<i>Saussurea lappa</i>	<i>Kustha</i>	Root
<i>Saxifraga ligulata</i>	<i>Pasanabheda</i>	Root
<i>Schleichera oleosa</i>	<i>Kusumbha</i>	Flower, seed, oil
<i>Schrebera swieteniodes</i>	<i>Muskaka</i>	<i>Ksara</i>
<i>Scindapsus officinalis</i>	<i>Gajapippali</i>	Fruit
<i>Scirpus kysoor</i>	<i>Kaseruka</i>	<i>Kanda</i>
<i>Semecarpus anacardium</i>	<i>Bhallatak</i>	Seed, fruit, pulp, bark, oil
<i>Sesamum indicum</i> Linn.	<i>Tila</i>	Seed, oil, root
<i>Sesbania grandiflora</i>	<i>Agastya</i>	Complete plant
<i>Shorea robusta</i> Gaertn.	<i>Rala, sala</i>	<i>Niryasa</i>
<i>Sida cordifolia</i> Linn.	<i>Bala</i>	Root, leaf, seed
<i>Sida humilis</i>	<i>Nagabala</i>	Complete plant, root
<i>Silkworm cocoon</i>	<i>Kosa</i>	<i>Kosa</i>
<i>Sloanum indicum</i> Linn.	<i>Brihati</i>	Complete plant
<i>Smilax china</i>	<i>Dwipantara vaca</i>	Root, rhizome
<i>Smilax zeylanica</i>	<i>Badara valli</i>	Root
<i>Solanum nigrum</i> Linn.	<i>Kakamaci</i>	Complete plant, fruit
<i>Solanum torvum</i>	<i>Kakamaci</i>	Whole plant, fruits, leaves, root
<i>Solanum xanthocarpum</i>	<i>Kantakari</i>	Complete plant
<i>Soymida febrifuga</i>	<i>Rohini</i>	Bark
<i>Spathodea falcate</i>	<i>Mesasringi</i>	Root, bark, leaf
<i>Sphaeranthus indicus</i>	<i>Sravani/mundi, Mundika</i>	Complete plant

<i>Spilanthes acmella</i>	<i>Maharastri</i>	Flower
<i>Spinacia oleracea</i>	<i>Palakya</i>	Leaf, seed
<i>Spirogyra</i> (rhizome)	<i>Saivala</i>	Complete plant
<i>Spondias mangifera</i>	<i>Amratarka</i>	Fruit
<i>Stephania japonica</i>	<i>Patha</i>	Leaves, bark, root
<i>Stereospermum suaveolens</i>	<i>Patola</i>	Root, fruit
<i>Streblus asper</i>	<i>Sakhotak</i>	Bark, <i>ksira</i>
<i>Strychnos nux-vomica</i> Linn.	<i>Visatinduka</i>	Seed
<i>Strychnos potatorum</i> Linn.	<i>Katak</i>	Seed
<i>Styrax benzoin</i>	<i>Lohabana</i>	Resin
<i>Swertia chirata</i>	<i>Kiratatikta</i>	Complete plant
<i>Symplocos racemosa</i>	<i>Lodhra</i>	Bark
<i>Syzygium aromaticum</i>	<i>Longa</i>	Dried flower buds
<i>Syzygium cumini</i>	<i>Jambu</i>	Fruit, seed, leaves, bark
<i>Syzygium jambos</i>	<i>Jambu</i>	Bark
<i>Tamarindus indica</i>	<i>Chinca</i>	Fruit, complete plant
<i>Tamarix aphylla</i>	<i>Phabuka</i>	Bark, <i>niryasa</i>
<i>Taraxacum officinale</i>	<i>Dugdapheni</i>	Root
<i>Taxus baccata</i>	<i>Talisapatra</i>	Leaf
<i>Tecoma undulata</i>	<i>Raktarohitaka</i>	Bark
<i>Tectona grandis</i>	<i>Saka</i>	Flower, seed, root
<i>Tephrosia purpurea</i>	<i>Sarapunkha</i>	Complete plant
<i>Terminalia arjuna</i>	<i>Arjuna</i>	Bark
<i>Terminalia belerica</i>	<i>Bibhitaki</i>	<i>Kashta</i> , bark, fruit
<i>Terminalia chebula</i> Retz.	<i>Haritaki</i>	Fruit, bark
<i>Terminalia tomentosa</i>	<i>Sadada</i>	Bark
<i>Thespesia populnea</i> Willd.	<i>Parishapippali</i>	Leaf, flower, seed
<i>Tinospora cordifolia</i> Willd.	<i>Amrita</i>	Complete plant
<i>Tragia involucrata</i>	<i>Vriscikali</i>	Root, leaf
<i>Trianthema monogyna</i> Linn.	<i>Varsabhu</i>	Complete plant
<i>Tribulus terrestris</i> Linn.	<i>Goksura</i>	Fruit, complete plant
<i>Trachyspermum ammi</i>	<i>Yavani</i>	Fruits, seeds, leaves, root
<i>Trichosanthes dioica</i>	<i>Patola</i>	Fruits, leaves, root
<i>Trichosanthes indica</i>	<i>Patola</i>	Fruits, leaves, root
<i>Trichosanthes palmata</i>	<i>Mahakal</i>	Fruit
<i>Trigonella foenum-graecum</i>	<i>Methika</i>	Seed, leaf
<i>Triticum aestivum</i>	<i>Godhuma</i>	Whole plant
<i>Typha elephantina</i>	<i>Eraka</i>	Flower, root
<i>Uraria picta</i>	<i>Prisniparni</i>	Complete plant
<i>Urginea indica</i>	<i>Vanapalandu</i>	<i>Kanda</i>
<i>Valeriana officinalis</i>	<i>Tagara</i>	Rhizomes
<i>Valeriana wallichii</i>	<i>Tagar</i>	<i>Kasta</i> , root
<i>Vanda tessellata</i>	<i>Ichneumon</i> plant	Whole plant
<i>Vernonia anthelmintica</i>	<i>Krsnajiraka</i>	Seed, fruit
<i>Vernonia cinerea</i>	<i>Sahadevi</i>	Complete plant
<i>Vetiveira zizanioides</i>	<i>Usira</i>	Leaves, root
<i>Vigna mungo</i>	<i>Masha</i> , black gram	Seeds, root
<i>Vitex agnus-castus</i>	<i>Renuka bija</i>	Seed
<i>Vitex negundo</i>	<i>Nirgundi</i>	Seed, fruit
<i>Vitis vinifera</i>	<i>Draksa</i>	Fruit
<i>Withania somnifera</i>	<i>Asvagandha</i>	Root
<i>Woodfordia fruticosa</i>	<i>Dhataki</i>	Flower, root
<i>Xanthoxylum alatum</i>	<i>Tumbaru</i>	Seed, fruit
<i>Zingiber officinale</i>	<i>Sunthi</i>	Dry root
<i>Zingiber officinale</i>	<i>Ardraka</i>	<i>Kanda</i>
<i>Zizyphus jujuba</i>	<i>Badari</i>	Fruit, seed, leaf, root
<i>Zizyphus vulgaris</i>	<i>Sauvira</i>	Fruit pulp

Appendix 3:

List of Ayurvedic Scientific Journals

1. *Journal of Research in Ayurveda and Siddha*

Publisher: Central Council for Research in Ayurveda and Siddha (CCRAS)

Director: CCRAS

Jawahar Lal Nehru Bhartiya Chikitsa Avum Homeopathy Anusandhan Bhavan
No. 61-65

Institutional Area

Opp. 'D' Block

Janakpuri, New Delhi, 110 058, India

Tel.: 91-011-5614970/71/72

Fax: 91-011-5528748

E-mail: ccras@ndf.vsnl.net.in or ccras@del6.vsnl.net.in

2. *Bulletin of Medico-Ethno-Botanical Research*

Publisher: CCRAS

3. *Bulletin of Indian Institute of History of Medicine*

Publisher: CCRAS

4. *Ancient Science of Life*

Publisher: AVR Educational Foundation of Ayurveda

1382 Trichy Rd.

Coimbatore, 641 018, India

5. *Arya Vaidyan*

Publisher: Publication Division, Arya Vaidya Sala

Kottakkal, 676 503, Kerala State, India

6. *Ayu*

Publisher: Gujarat Ayurved University

Jamnagar, Gujarat, India

For combined general and scientific articles:

1. *Heritage Healing*

Publisher: Healthcare Communications Pvt. Ltd.

#8, 5th Cross Extension

Bakthavachalam Nagar, Adyar

Chennai, 600 020, India

Tel.: 91-44-4422780

E-mail: heritagehealing@usa.net

2. *Holistic Healing*

Publisher: Indian Express Newspapers Pvt. Ltd.

Business Publication Division

1st Floor, Manandi Plaza, Express Towers

Nariman Point, Mumbai, 400 021, India

Tel.: 2301020

E-mail: bpdhqbom4.vsnl.net.in

3. *Ayur Medline*

Publisher: Seetharam Prasad

124/2, near Ganesh Temple

7th B- main road iv the Block Jaynagar

Bangalore, 560 011, India

Tel.: 91-080/6349314

E-mail: ayurmedline@mailcity.com

For publishing matters related to any aspect of medicinal plant research, the following journals can be considered:

1. *Indian Journal of Medical Research (ICMR)*
2. *Indian Journal of Experimental Biology*
3. *Indian Journal of Pharmacology*
4. *Indian Journal of Physiology and Pharmacology*
5. *Current Science*
6. *Journal of Phytotaxonomy*
7. *Indian Drugs*
8. *Indian Journal of Pharmaceutical Sciences*
9. *Indian Journal of Medical Sciences*
10. *Journal of Natural Remedies*

Appendix 4:

Ayurvedic Product Manufacturers

Manufacturers of Ayurvedic products were identified by the chapter contributors and are listed in alphabetic order. Sequoia Inc. USA may be contacted for further information.

AIMIL Pharmaceuticals Pvt. Ltd.
2994/4, Street No. 17
New Delhi, 110 008
India

Almasaguda Village
Badangpet P.O. Hyderabad, 500 058
India
Tel.: 040-24652472(O), 24751490 (R)
Fax: 040-4759109
Web site: www.amritayurveda.com

Anglo-French Drugs and Industries Ltd.
41, 3rd cross, 5th Block
Rajaji Nagar, Bangalore, 560 010
India

Anjali Pharma, 48-B.Sector
'C' Mandi-Deep, Pin: 462 046
India
Tel.: 07480-33905, 52050
Fax: 07480-33634

Arya Aushadhi Pharmaceutical Works
Indore, 452 008
India
E-mail: kuki@bom4.vsnl.net.in

Arya Vaidya Pharmacy (Coimbatore) Ltd.
326, Perumal Koli St.
Ramanathapuram, Coimbatore, 641 045
India

Arya Vaidya Sala
Kottakal, 676 503
Kerala, India
Tel.: 742216-19, 744261-64
Fax: 0493-742210/742572
E-mail: kottakal@vsnl.com
Web site: kottakal@md3.vsnl.net.in

Ayucare Pharma
Near Income Tax Office
Ashram Road
Ahmedabad, India
Tel.: 079-7541814

Ayudrugs
Shop No. 4, Fancy Bazar
Guwahati, 781 001
Assam, India

Ayurvedeeya Arkashala Ltd.
1915/4, Mannar's Market, Shivrampet
Mysore, 570 001
India

Bajaj Consumer Care Ltd.
17-1-204-8, Saidabad
Hyderabad, 500 059
India

Ban Labs Ltd.
Ban House, Gondal Rd.
Rajkot, 360 004
India
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THE COMPLETE BOOK OF AYURVEDIC HOME REMEDIES

Vasant D. Lad, B.A.M.S., M.A.Sc.



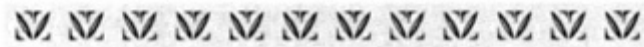
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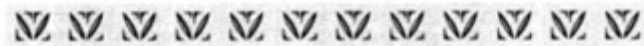
The Complete Book — of Ayurvedic — Home Remedies

Vasant D. Lad, B.A.M.S., M.A.Sc.

Illustrations by Vasant D. Lad



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Although the information contained in this book is based on Ayurvedic principles practiced for thousands of years, it should not be taken or construed as standard medical diagnosis or treatment. For any medical condition, always consult with a qualified physician.

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*This book is dedicated with all my heart to my most loving wife,
Usha, and my children, Aparna and Pranav.*

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Introduction

The Need for Healing

Ayurveda is the art of daily living in harmony with the laws of nature. It is an ancient natural wisdom of health and healing, a science of life. The aims and objectives of this science are to maintain the health of a healthy person and to heal the disease of an unhealthy person. Both prevention (maintenance of good health) and healing are carried out by entirely natural means.

According to Ayurveda, health is a perfect state of balance among the body's three fundamental energies, or *doshas* (*vata*, *pitta*, *kapha*) and an equally vital balance among body, mind, and the soul or consciousness.

Ayurveda is a profound science of living that encompasses the whole of life and relates the life of the individual to the life of the universe. It is a holistic system of healing in the truest sense. Body, mind, and consciousness are in constant interaction and relationship with other people and the environment. In working to create health, Ayurveda takes into consideration these different levels of life and their interconnectedness.

As a science of self-healing, Ayurveda encompasses diet and nutrition, lifestyle, exercise, rest and relaxation, meditation, breathing exercises, and medicinal herbs, along with cleansing and rejuvenation programs for healing body, mind, and spirit. Numerous adjunct therapies such as sound, color, and aromatherapy may also be employed. The purpose of this book is to acquaint you with these natural methods, so you can make the lifestyle choices and learn the self-healing modalities that are right for you in order to create, maintain, or restore health and balance.

Ayurveda is a Sanskrit word that means “the science of life and longevity.” According to this science, every individual is both a creation of cosmic energies and a unique phenomenon, a unique personality. Ayurveda teaches that we all have a constitution, which is our individual psychobiological makeup. From the moment of conception, this individual constitution is created by the universal energies of Space, Air, Fire, Water, and Earth.

These five elements combine into the three fundamental energies, or *doshas*. Ether and air constitute *vata*, which is the energy of movement; fire and water constitute *pitta*, the principle of digestion or metabolism, the transformation of matter into energy; and water and earth make up *kapha*, the energy of structure and lubrication. When the male sperm and the female egg join at the time of fertilization, the *vata*–*pitta*–*kapha* factors from the parents' bodies that are most active and predominant at the moment, due to the season, the time, the emotional state, and the quality of their relationship, form a new individual with a particular constellation of qualities.

In modern terms we speak of this blueprint of the individual as our inherited genetic code; from ancient times Ayurveda has called it our *prakruti* or individual constitution, a constant factor that does not change throughout life. It is our own unique pattern of energy, our combination of physical, mental, and emotional characteristics and predispositions.

Though the underlying structure of our *prakruti* remains a fixed reality, our home base or essential individuality, it is constantly bombarded by numerous forces. Changes in age and in our external environment, alternating heat and cold as the seasons pass, our endlessly shifting thoughts, feelings, and emotions, and the quality and quantity of the food we eat continuously affect us. Unhealthy diet, excess stress, insufficient rest or exercise, and repressed emotions all disturb our *doshic* balance. Depending on the type of changes and the individual's underlying constitution, various ailments may develop:

- Some individuals experience an increase or aggravation of kapha, leading to conditions such as colds, congestion, sneezing, and allergic manifestations, as well as attachment, greed, and possessiveness.
- A pitta individual may become highly critical, angry, or perfectionistic, or may develop physical symptoms such as acid indigestion, heartburn, diarrhea, dysentery, hives, rash, or acne.
- Vata imbalances may manifest as constipation, abdominal distention, sciatica, arthritis, or insomnia, along with psychological symptoms such as fear, anxiety, and insecurity.

All these illnesses and conditions, in addition to the countless others that lead to human suffering, are due to alterations in the body's inner ecology. These upset the individual's balance, creating subtle biochemical changes that ultimately lead to disease. This is why the Ayurvedic system of medicine speaks of the need for healing for every individual in every walk of life.

As the internal and external conditions of our lives change, if we are going to remain healthy we need to constantly adjust in order to maintain equilibrium. Some of this adjusting takes place automatically due to the beautiful wisdom and intelligence with which our bodies have been designed. But much demands conscious choice.

To maintain health and balance, we have to juggle with the three doshas, taking action to increase or decrease vata, pitta, or kapha as conditions demand. This requires moment-to-moment awareness, moment-to-moment consciousness, moment-to-moment healing.

Thus healing—healthy, balanced, conscious living in the fullness of the present moment—is really a way of life. Ayurveda is not a passive form of therapy but rather asks each individual to take responsibility for his or her own daily living. Through our diet, our relationships, our job, our numerous responsibilities, and our daily life as a whole, we can take simple actions for prevention, self-healing, wholeness, and growth

toward fulfillment.

According to Ayurveda, our life has a purpose. Simply stated, that purpose is to know or realize the Creator (Cosmic Consciousness) and to understand our relationship with That, which will entirely influence our daily living. This great purpose is to be achieved by balancing four fundamental aspects of life: *dharma*, which is duty or right action; *artha*, material success or wealth; *kama*, positive desire; and *moksha*, spiritual liberation. These are called the four *purusharthas*, the four great aims or achievements in the life of any individual.

The foundation of all these facets of life is health. To maintain *dharma* and carry out our duties and responsibilities to ourselves and others, we must be healthy. Likewise, in order to create affluence and achieve success in action, good health is indispensable. To have creative, positive desire, we need a healthy mind and consciousness, a healthy body, and healthy perception. (Desire—*kama*—is sometimes translated as sex and refers to progeny and family life, but it is really the positive energy or force of desire that generates and propels any creative work.) And *moksha* or spiritual liberation is nothing but perfect harmony of body, mind, and consciousness or soul. Thus the whole possibility of achievement and fulfillment in life rests on good health.

In the quarter century that I have been practicing medicine, I have worked in surgery, gynecology, obstetrics, and pediatrics, as well as in general medicine, treating thousands of individuals in all stages and walks of life. I have repeatedly observed that lifestyle choices, such as diet, exercise, and daily routine, can be a potent source of healing as well as a cause of disease. Many health problems seem intertwined with the stresses of daily life, family and relationship problems, and worries about job and money. Others are directly connected to eating the wrong kinds of food or getting too much or too little exercise.

I have also grown more and more aware that illness provides us with an invitation for self-transformation, an opportunity to change our way of thinking, feeling, eating, and in general caring for ourselves and our lives. It never ceases to amaze and delight me how quickly and powerfully life can be set on the right track and balance restored simply through a proper diet, herbal medicines, meditation, an appropriate exercise program, and other purely natural means.

The remedies in this book come from my own practical clinical experience, based on principles and practices developed over centuries. The tradition of Ayurveda extends back over more than five thousand years of continuous daily practice, from ancient times to the present day. It is not a recently developed system of “alternative healing” but an enduring science of life that has never lost its integrity and essential nature. You can imagine how much wisdom it contains and how much practical knowledge it has accumulated over a span of five millennia!

About three thousand years ago (around 900 B.C.), the long oral tradition of Ayurveda took new form when three great scholars—Charaka, Sushruta, and Vagbhata—wrote

down the principles of this ancient wisdom. Their textbooks are still used by students, practitioners, and teachers in Ayurvedic medical schools and colleges throughout India.

In a profound sense, Ayurveda is the mother of all healing systems. From its eight principal branches (pediatrics, gynecology and obstetrics, ophthalmology, geriatrics, otolaryngology, toxicology, general medicine, and surgery) have come the main branches of medicine as it is practiced today, as well as many modern healing modalities, including massage, diet and nutritional counseling, herbal remedies, plastic surgery, psychiatry, polarity therapy, kinesiology, shiatsu, acupressure and acupuncture, color and gem therapy, and meditation. All these have roots in Ayurvedic philosophy and practice.

The great sage-physician Charaka, one of the founders of Ayurvedic medicine, said, "A physician, though well versed in the knowledge and treatment of disease, who does not enter into the heart of the patient with the virtue of light and love, will not be able to heal the patient." To the best of my ability, I have followed this advice all my life, and I would urge you to follow it in using this knowledge to help others and to heal yourself.

Love is the essence of our life. I have written this book with love, and I offer it to you, dear reader, with the hope that the suggestions offered here will become a vital part of your self-healing and continued well-being.



Part I



The Science of Life

Chapter 1

Ayurveda: Body, Mind, and Soul

Like other great ancient civilizations, India never separated science from philosophy and religion. Rather, it viewed all knowledge as part of a whole designed to promote human happiness, health, and growth.

Philosophy is the love of truth. *Science* is the discovery of truth through experiment. *Religion* is the experience of truth and application of it in daily living.

Ayurveda, the science of life, is both systematized knowledge and practical wisdom, an art of healthy living that encompasses all phases of life, body, mind, and spirit. Like all sciences, it includes both a practical and a theoretical aspect. In order to make best use of the practical recommendations that come later in this book, it will help if you understand the essentials of Ayurvedic theory. This first chapter may seem a bit abstract, but please be patient and read it carefully, as it forms the basis of all that is to follow.

The Universe and How We Are Connected

According to Ayurveda, the source of all existence is universal Cosmic Consciousness, which manifests as male and female energy. *Purusha*, often associated with the male energy, is choiceless, passive, pure awareness. *Prakruti*, the female energy, is active, choiceful consciousness. Both *Purusha* and *Prakruti* are eternal, timeless, and immeasurable. These two energies are present in all living organisms, including every man and woman, as well as inanimate objects.

Purusha is formless and beyond attributes. Unmanifested pure existence, beyond cause and effect, beyond space and time, *Purusha* takes no active part in creation but remains a silent witness.

Prakruti, which has form, color, and attributes, is the divine creative will that dances the dance of creation. *Prakruti* is the One that becomes many. *Purusha* is the lover, *Prakruti* the beloved. Creation of this universe happens through their love. All of nature is the child born from the womb of *Prakruti*, the Divine Mother.

In the manifestation of nature from *Prakruti*, the first expression is *Mahad* (or *Mahat*), intelligence or cosmic order. (In human beings, it is referred to as *Buddhi*, intellect.) Next is *Ahamkar* or ego, the sense of self-identity, the center in our consciousness from which we think, act, and react. *Ahamkar* expresses itself in three universal qualities:

Sattva is stability, purity, wakefulness, essence, clarity, and light.

Rajas is dynamic movement and causes sensations, feelings, and emotions.

Tamas is the tendency toward inertia, darkness, ignorance, and heaviness. *Tamas* is responsible for deep sleep and periods of confusion. It also leads to the creation of matter.

From the essence of *Sattva* are born the mind, the five sense faculties and their organs (ears to hear, skin to perceive touch, eyes to see, tongue to taste, nose to smell), and the five motor organs or organs of action: the mouth (for speech), the hands, feet, reproductive organs, and organs of excretion.

Rajas is the active force behind the movement of both the sensory and motor organs.

Tamas gives rise to the five elements, which form the basis of material creation: space (ether), air, fire, water, and earth.

Man, a creation of Cosmic Consciousness, is considered to be a microcosm of the macrocosm that is the universe. Whatever is present in the cosmos, the same is present in human beings. Man is a miniature of nature.

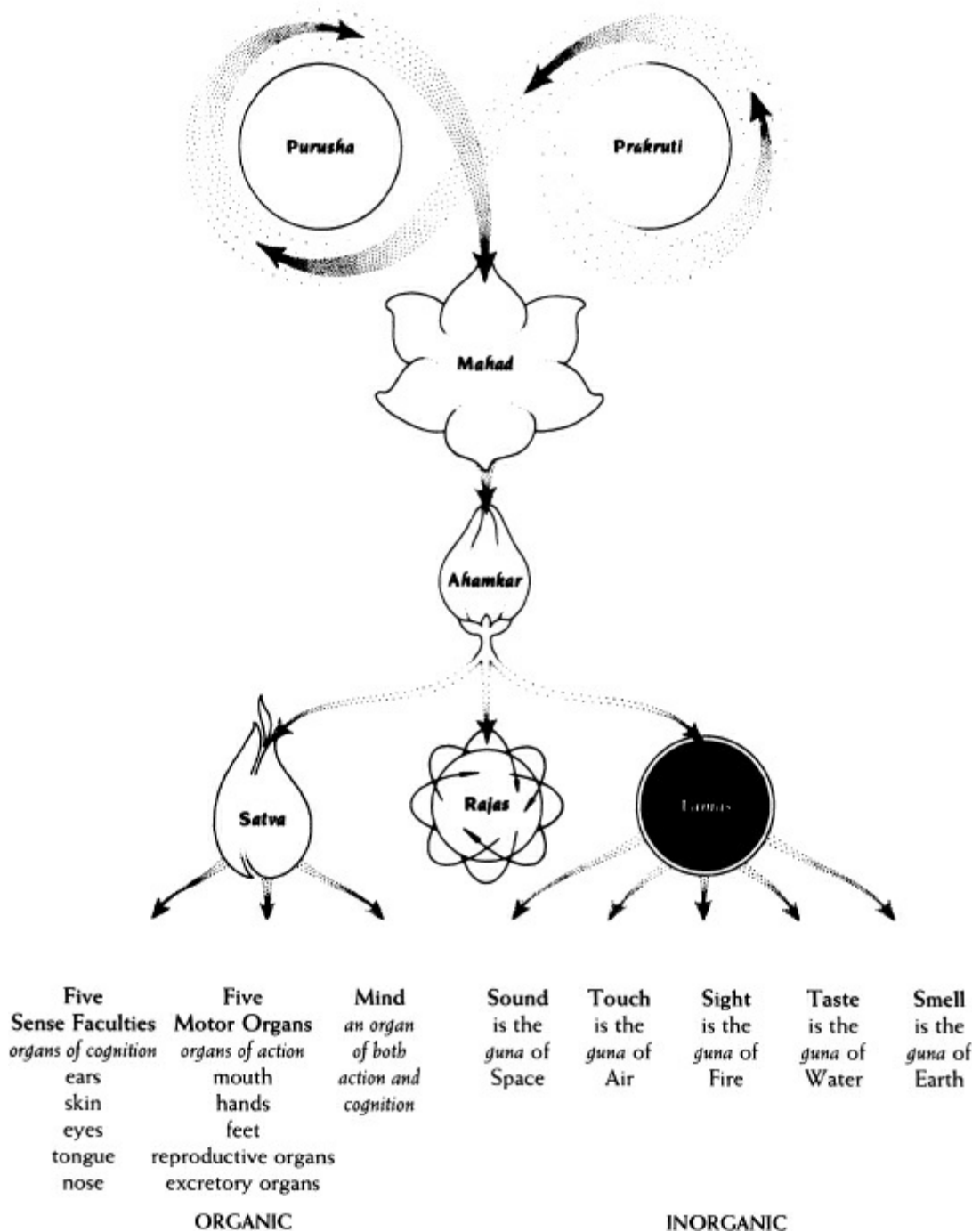
The Five Elements: Building Blocks of Nature

The concept of the five elements is one of the most fundamental in Ayurvedic science. These five elements (space, air, fire, water, and earth) exist in all matter, both organic and inorganic. As man is a microcosm of nature, the five elements also exist within each individual. Our psychological tendencies, as well as our five senses and the various aspects of our body's functioning, are all directly related to the five elements.

According to Ayurveda, the five elements manifest sequentially, beginning with space, from the pure, unified, unmanifested Cosmic Consciousness that is the source of all.

SPACE

Sometimes referred to as "ether," space is empty, light, subtle, all-pervading, omnipresent, and all-enclosing. It is universal, non-moving, and formless. Space is nuclear energy. It appears when the pure unmanifest consciousness begins to vibrate and is associated with sound and the sense of hearing. We need space in order to live, move, grow, and communicate. Spaces in the body include the mouth, nose, gastrointestinal tract, respiratory tract, abdomen, and thorax. Psychologically, space gives freedom, peace, and expansion of consciousness and is responsible for love and compassion as well as feelings of separation, isolation, emptiness, ungroundedness, insecurity, anxiety, and fear.



AIR

Air is dry, light, clear, and mobile. The second manifestation of consciousness, air moves in space. Air is electrical energy—the electron moves because of the air element. It is formless, but it can be perceived by touch, to which it is related. The principle of movement, air expresses itself in the movements of the muscles, the pulsations of the heart, the expansion and contraction of the lungs. Sensory and neural impulses move to and from the brain under the influence of the air principle, which is also responsible for breathing, ingestion, the movement of the intestines, and elimination. The flow of thought, desire, and will are governed by the air principle, which gives us happiness, freshness, joy, and excitement. It is, along with space, also responsible for fear, anxiety,

insecurity, and nervousness.

FIRE

Fire is hot, dry, sharp, penetrating, and luminous. When air begins to move, it produces friction, which generates heat or fire. Fire is radiant energy. On the atomic level, the atom radiates heat and light in the form of a quantum wave. Fire is active and changeable. In our solar system, the sun is the source of fire and light. In the body, our biological “fire” in the solar plexus regulates body temperature and metabolism: digestion, absorption, and assimilation. Fire is associated with light and with vision. Fire is intelligence. It is necessary for transformation, attention, comprehension, appreciation, recognition, and understanding. Fire is also responsible for anger, hatred, envy, criticism, ambition, and competitiveness.

WATER

The next manifestation of consciousness, water is fluid, heavy, soft, viscous, cold, dense, and cohesive. It brings molecules together. Water is chemical energy (it is the universal chemical solvent). Water is associated with the sense of taste; without moisture the tongue cannot taste anything. Water exists in the body as plasma, cytoplasm, serum, saliva, nasal secretion, cerebrospinal fluid, urine, and sweat. It is necessary for nutrition and to maintain life; without it, our cells could not survive. Water is contentment, love, and compassion. It creates thirst, edema, and obesity.

EARTH

Earth is heavy, hard, rough, firm, dense, slow-moving, and bulky—the most solid of the five elements. It is neither hot nor cold. Earth is mechanical or physical energy. According to Ayurveda, it is nothing but crystallized or solidified consciousness. It gives strength, structure, and stamina to the body. All the body’s solid structures (bones, cartilage, nails, teeth, hair, skin) are derived from the earth element. Earth is associated with the sense of smell. It promotes forgiveness, support, groundedness, and growth. It also creates attachment, greed, and depression, and its absence produces feelings of ungroundedness.

In our body, the electrical energy of the neuron becomes the physical energy of the movement of muscles, mediated through the neurotransmitter, which is chemical. Indeed, all the five elements are present on every level of our physiology, starting with a single cell. Within the cell, the cell membrane is earth, cellular vacuoles are space, cytoplasm is water, nucleic acid and other chemical components of the cell are fire, and movement of the cell is due to the air principle. Every single cell also has mind, intelligence, and consciousness, through which it manifests selectivity and choice. From

all the possible nutrients in its environment, every cell chooses its own food—that choice is intelligence at work.

Both in our outer environment and within us, the proportion and balance of these elements is forever shifting, changing with the seasons, the weather, the time of day, the stage of one's life. For health, and often for sheer survival, we have to continuously accommodate ourselves to these changes, through what we eat, what we wear, where we live, and so on. This is a balancing act, playing elements against each other. We use solid earth to build homes, to protect ourselves against changes in air, heat (fire), and water. We use fire to prepare food (made of water and earth).

The Three Doshas: Vata, Pitta, and Kapha

These five great elements combine into three basic energies or functional principles, which are present, in varying degrees, in everything and everybody. Space (ether) and air constitute vata. Fire and water combine to make up pitta. Water and earth constitute kapha.

In our bodies, these three doshas or humors govern our psychobiological functioning. vata–pitta–kapha are present in every cell, tissue, and organ. When in balance, they create health. When out of balance, they are the cause of disease.

These three doshas are responsible for the huge variety of individual differences and preferences, and they influence all we are and all we do, from our choices of food to our modes of relating to others. They govern the biological and psychological processes of our body, mind, and consciousness. They regulate the creation, maintenance, and destruction of bodily tissue, and the elimination of waste products. They also govern our emotions. When in balance, they generate noble qualities such as understanding, compassion, and love. When their balance is disturbed by stress, improper diet, environmental conditions or other factors, they can give rise to negative emotions such as anger, fear, and greed.

In Ayurveda, vata is the bodily air principle. It is the energy of movement. Pitta is the principle of fire, the energy of digestion and metabolism. And kapha is the principle of water, the energy of lubrication and structure.

All people have all of these three doshas, but one of them is usually primary, one secondary, and the third least prominent. Thus, each person has a particular pattern of energy, an individual combination of physical, mental, and emotional characteristics that make up his or her constitution (*prakruti*). Just as everyone has an individual fingerprint that can be identified by a trained practitioner, so everyone has an energy print—a balance or proportion of vata, pitta, and kapha—that is uniquely his or her own.

Health depends on maintaining this proportion in balance. Balance is the natural

order of things; imbalance provokes and reflects disorder. Within our bodies there is a constant interplay between order and disorder, which determines our state of health.

Health is order; disease is disorder. The internal environment of the body is ceaselessly reacting to the external environment. Disorder occurs when these two are out of harmony with each other. But since order is inherent within disorder, the wise person learns to be aware of the presence of disorder and sets about to reestablish order.

In [chapter 2](#) we will see how the three fundamental doshas combine to create the seven constitutional types of Ayurveda, and you will learn your own body type, the key to making lifestyle choices for self-healing and maximum well-being. For the moment, let us look a little more deeply into the characteristics of these three basic energies of life.

VATA

Vata is the energy of movement. Although it is the air *principle*, it is not considered the same as actual air in the external environment, but rather as the subtle energy that governs biological movement.

Vata is intimately related to our vital life essence, known as *prana*. *Prana* is the pure essence of vata. It is the life-force, the play of intelligence. That flow of intelligence is necessary for communication between two cells, and it maintains the life function of both. On a cosmic level, *prana* is said to be the attraction between *Purusha* and *Prakruti*.

As the principle of mobility, vata regulates all activity in the body, both mental and physiological. It is responsible for breathing, the blinking of our eyes, the beating of our hearts, and all movement in the cytoplasm and cell membranes. All the impulses in the vast networks of our nervous system are governed by vata.

When vata is in balance, it promotes creativity and flexibility and evokes feelings of freshness, lightness, happiness, and joy. Out of balance, vata produces fear, nervousness, anxiety, even tremors and spasms.

Vata is dry, light, cold, subtle, clear, mobile, and dispersing. We shall soon see how these qualities are expressed in a person with a vata constitution.

PITTA

Pitta is translated as fire, but this is not meant literally. Rather, it is the *principle* of fire, the energy of heating or metabolism. Pitta governs all the biochemical changes that take place within our bodies, regulating digestion, absorption, assimilation, and body temperature. From the standpoint of modern biology, pitta comprises the enzymes and amino acids that play a major role in metabolism.

Pitta regulates body temperature through the chemical transformation of food. It

promotes appetite and vitality.

Not only food is metabolized by us. Every impression coming in from the outside is also processed or “digested” and made a part of us. Thus pitta (when in balance) promotes intelligence and understanding and is crucial in learning. Out-of-balance pitta may arouse fiery emotions such as frustration, anger, hatred, criticism, and jealousy.

Pitta is hot, sharp, light, oily, liquid, pungent, sour, and spreading. These qualities occur in various ways in people of pitta constitution.

KAPHA

Kapha combines water and earth. It is the energy that forms the body’s structure, the glue that holds the cells together. Kapha also supplies the liquid needed for the life of our cells and bodily systems. It lubricates our joints, moisturizes the skin, helps to heal wounds, and maintains immunity. Kapha provides strength, vigor, and stability.

Psychologically, excess kapha is responsible for the emotions of attachment, greed, lust, and envy. When kapha is in balance it expresses itself in tendencies toward love, calmness, and forgiveness.

The qualities of kapha include heavy, slow, cool, oily, damp, smooth, soft, static, viscous, and sweet. Kapha individuals display these qualities in various ways.

Together, these three doshas govern all the body’s metabolic activities. Kapha promotes anabolism, the process of building up the body, the growth and creation of new cells as well as cell repair. Pitta regulates metabolism, which is digestion and absorption. Vata triggers catabolism, the necessary deterioration process in which larger molecules are broken down into smaller ones.

Vata, the principle of movement, moves both pitta and kapha, which are immobile. Thus when vata is out of balance, it influences and disturbs the other doshas. The majority of illnesses have aggravated vata at their source.

The whole of life’s journey is divided into three major milestones. From birth to age 16 is the kapha age. From 16 to 50 is the age of pitta, and from 50 to 100 the age of vata.

In childhood, kapha and the process of anabolism are predominant, as this is the time of greatest physical growth and the structuring of the body. Kapha disorders, such as lung congestion, cough, colds, and mucus secretions, are common at this time. In adulthood, a time of activity and vitality, pitta is most apparent. Vata and the catabolic processes of deterioration take over in old age, bringing vata disorders such as tremors, emaciation, breathlessness, arthritis, and loss of memory.

The Twenty Qualities: An Important Key to Healing

Now we come to another important aspect of Ayurvedic theory, which will help you to make intelligent choices for self-healing. Ayurveda delineates twenty fundamental qualities, which appear in ten pairs:

The Twenty Basic Attributes or Qualities

Heavy—Light
Oily—Dry
Stable—Mobile
Slimy—Rough
Gross—Subtle
Cold—Hot
Slow—Sharp
Soft—Hard
Dense—Liquid
Cloudy—Clear

These qualities are found both in the world around us and in our bodies. Today's weather may feel light or heavy, and it may be liquid or dry, mobile (windy) or stable, hot or cold, cloudy or clear. Food we eat can partake of any of these qualities. Ice cream, for example, is heavy, oily, cold, soft, and liquid. Our skin may be oily or dry, rough or smooth. Our moods, too, can be heavy or light, cloudy or clear; our thinking may be slow or sharp, our mind quiet and stable or mobile and hyperactive, clear or cloudy.

We are constantly affected by changes in these qualities. Cold, windy, clear, dry weather aggravates vata dosha and may lead to colds and any number of vata ailments such as insomnia, constipation, or arthritis. Hot, humid weather aggravates pitta and may lead to outbreaks of irritation and anger as well as physical complaints like acne, eczema, or skin rashes. Cloudy, gray, humid or rainy weather can aggravate kapha, leading to colds and coughs, depression, lethargy, overeating and oversleeping, and weight gain.

Each of these paired qualities represents the extreme on a continuum. The two qualities in each pair influence or affect one another according to two fundamental principles of Ayurveda:

1. Like increases like.
2. Opposites decrease each other.

These principles are a key to healing with Ayurveda. When an imbalance has manifested, *successful treatment requires increasing opposite qualities*. For example, if there is too much heat (excess pitta), a cool drink, a swim, or some herbs with cooling

properties will greatly help pacify pitta and reduce the heat. A person suffering from too much heat will not be helped by playing tennis in the sun, eating spicy foods, or taking a sauna. Similarly, if you are cold and shivering from exposure to cold windy weather, have a bowl of warm soup, wrap up in a blanket, or take a hot bath. These simple remedies immediately make sense when we hear them because they are so natural.

Ayurvedic physicians have carefully observed nature and located these qualities within all things, both organic and inorganic. Ayurvedic treatment consists to a great extent of identifying a person's disorder in terms of these qualities, and setting right any imbalances.

How is this done? Speaking in very general terms, excessive dryness in the body—constipation, dry skin, emaciation, and so on—is frequently associated with aggravated vata; excessive heat—burning urine, irritated eyes, fever, inflammation, anger, or a critical attitude—with aggravated pitta; and undue heaviness—lethargy, overweight, congestion, and excess mucus—with unbalanced kapha. Whatever the symptoms may be, for self-treatment you need to understand them and then adjust your lifestyle—diet, exercise, and so on—to restore a state of balance and health.

The hundreds of remedies in [Part III](#) of this book will help you to do this, but essentially it is your own moment-to-moment awareness and self-observation, your sensitivity to your own constitution and your own unique requirements for health, and perhaps most importantly, your willingness to act on your knowledge, that will make all the difference between poor health and a vital, happy, healthy, long life.

Discover Your Mental and Physiological Type

This chapter will take you further on your journey toward radiant health as we begin to apply the principles from [chapter 1](#) to discover and understand your own unique constitution.

According to Ayurveda, there are seven main body types:

- Vata
 - Pitta
 - Kapha
- } Mono types
- Vata–Pitta
 - Pitta–Kapha
 - Kapha–Vata
- } Dual types
- Vata–Pitta–Kapha
- } Triple type

All three doshas are present in each individual at all times, but their proportion varies from person to person. Thus, ten vata individuals, or ten kapha-pitta individuals, will have ten different temperaments, ten unique sets of qualities and characteristics. Maintaining our individual qualitative and quantitative proportion of the doshas is our challenge if we are to remain healthy. When we maintain this proportion our health is good, but when the balance is upset, disease may result.

Prakruti and Vikruti

At the time of conception, each person's combination and proportion of vata, pitta, and kapha is determined according to the genetics, diet, lifestyle, and current emotions of the parents. As mentioned in the Introduction, the doshas predominant in the parents combine to form the constitution of the new life they are creating.

If, for example, the father is pitta predominant and the mother vata predominant, and the pitta factor is stronger than the vata, and the union is taking place on a hot summer night after eating a spicy meal, then the baby that is born will have a constitution that is pitta predominant. Or if both parents are kapha, and they have a kaphagenic diet and are making love in kapha season, then their child will have a predominance of kapha dosha.

A few fortunate individuals are born with a constitution in which all three doshas are equally present, which gives them the likelihood of exceptionally good health and a long life span, but most of us have one or two doshas predominant.

NOTE: When Ayurveda says health comes from balancing your doshas, this does not mean you should try to have equal amounts of vata, pitta, and kapha. Rather, it means *maintaining the balance with which you were conceived*.

The unique and specific combination of the three doshas at conception is called your *prakruti*, which means “nature.” It is your psychobiological temperament. *Prakruti* does not change during a person’s lifetime. Your *prakruti* may, for example, be predominantly pitta, with vata secondary and a little kapha. (This can be written as $V_2P_3K_1$.) For you, maintaining balance means keeping this proportion. If your vata or kapha should increase, moving toward an equal proportion of the doshas, it would not be healthy for you.

As conditions change—due to weather, dietary choices, fatigue, stress, emotional state, exercise or lack of it—the balance of the doshas in our mind-body system also changes. This altered state of the doshas, reflecting the current state of our health, is called our *vikruti*. If your health is excellent, your current doshic status will be the same as your *prakruti*. But more likely there will be a discrepancy, and it is this difference between the two that gives a direction for healing. Your aim will be to reestablish the balance indicated by your *prakruti*.

I know you are eager to find out what your constitution is, so let’s move on to that. Afterward, we will look more deeply into the characteristics of each doshic type.

How to Determine Your Constitutional Type

The chart on [this page](#) to [this page](#) is a self-assessment that will enable you to determine your unique constitution according to Ayurveda.

Please remember that this can provide only a rough guideline. The subtleties of each person’s mental, emotional, and physical makeup are manifold and can be accurately assessed only by a physician thoroughly trained and experienced in Ayurvedic diagnosis. So please do not draw any absolute conclusions about yourself based on the self-assessment or the descriptions of the doshas; rather, use this information to help you grow in self-understanding, and as a guide to plan your diet, exercise regimen, and other aspects of your lifestyle for maximum health.

It is best to fill out the self-evaluation twice. (You might want to photocopy it, to have it available for others or for future use.) First, base your choices on what is most consistently true about your life as a whole, over many years. This indicates your *prakruti*. Then fill it out a second time, considering how you have been feeling recently, in the last month or two. This is your *vikruti* or present condition.

It often helps to have a spouse or good friend verify your answers, as they may have good insights and some objectivity to offer as you make your responses.

After filling out the chart, add up the number of marks under vata, pitta, and kapha

to discover your own balance of the doshas in your *prakruti* and *vikruti*. Most people will have one dosha predominant, a few will have two doshas approximately equal, and even fewer will have all three doshas in equal proportion.

After adding up the numbers, make them into a ratio, with 3 as the highest number. For example, suppose you come out with $V = 10$, $P = 6$, $K = 3$. This would translate into $V_3P_2K_1$.

Once you have determined your predominant doshas, studying the following characteristics of vata, pitta, and kapha will help you gain a deeper and more comprehensive understanding.

Characteristics of the Vata Individual

Vata individuals have light, flexible bodies. Their frame is on the small side, with light muscles and little fat, so that they tend to be slim or even underweight. They often appear to be “too tall” or “too short,” or they may appear physically underdeveloped, with flat chests and less strength and stamina than the other types. Their veins and muscles are often quite prominent.

Vatas generally have dry skin tending toward roughness. Their circulation is poor, with the result that their hands and feet are often cold. Because vata dosha is cold, dry, light, and mobile and people with a vata constitution tend to lack insulating material (the fatty tissue under the skin), they are uncomfortable in cold weather, especially if it is dry and windy, and they much prefer spring and summer.

These individuals have a variable appetite and thirst and variable digestive strength. They are often attracted to astringent food such as salads and vegetables, but their bodies actually need sweet, sour, and salty tastes. (We will discuss the effect of tastes in [chapter 8](#).) Raw vegetables increase, rather than balance, vata. Vata individuals often experience digestive difficulties and problems with absorption of nutrients. They tend to produce scanty urine, and their feces are hard, dry, and small in size and quantity. Constipation is one of their most common ailments.

Vatas are the most likely of the body types to fast or to eat very little, but this actually increases vata and tends toward imbalance.

Other physical characteristics typical of vata types include small, recessed eyes, which are often quite lusterless; dry, thin hair, often curly or kinky; dry, rough skin and nails; cracking, popping joints; and teeth that may be irregular, broken, or protruding.

Vatas walk quickly and are always in a rush. Due to the mobile quality of vata, they do not like sitting idle but prefer constant activity. They also like to do a lot of traveling. Not doing anything is a punishment for them. They are attracted to jogging, jumping, and vigorous physical activity, but because they tend to have less stamina, they can easily get strained or overtired.

OBSERVATIONS	V	P	K	VATA	PITTA	KAPHA
Body size	☐	☐	☐	Slim	Medium	Large
Body weight	☐	☐	☐	Low	Medium	Overweight
Skin	☐	☐	☐	Thin, Dry, Cold, Rough, Dark	Smooth, Oily, Warm, Rosy	Thick, Oily, Cool, White, Pale
Hair	☐	☐	☐	Dry Brown, Black, Knotted, Brittle, Thin	Straight, Oily, Blond, Gray, Red, Bald	Thick, Curly, Oily, Wavy, Luxuriant, All colors
Teeth	☐	☐	☐	Protruding, Big, Roomy, Thin gums	Medium, Soft, Tender gums	Healthy, White, Strong gums
Nose	☐	☐	☐	Uneven shape, Deviated septum	Long pointed, Red nose-tip	Short Rounded, Button nose
Eyes	☐	☐	☐	Small, Sunken, Dry, Active, Black, Brown, Nervous	Sharp, Bright, Gray, Green, Yellow/red, Sensitive to light	Big, Beautiful, Blue, Calm, Loving
Nails	☐	☐	☐	Dry, Rough, Brittle, Break easily	Sharp, Flexible, Pink, Lustrous	Thick, Oily, Smooth, Polished,
Lips	☐	☐	☐	Dry, Cracked, Black/brown tinged	Red, Inflamed, Yellowish	Smooth, Oily, Pale, Whitish
Chin	☐	☐	☐	Thin, Angular	Tapering	Rounded, Double
Cheeks	☐	☐	☐	Wrinkled, Sunken	Smooth Flat	Rounded, Plump
Neck	☐	☐	☐	Thin, Tall	Medium	Big, Folded
Chest	☐	☐	☐	Flat, Sunken	Moderate	Expanded, Round
Belly	☐	☐	☐	Thin, Flat, Sunken	Moderate	Big, Potbellied
Belly button	☐	☐	☐	Small, Irregular, Herniated	Oval, Superficial	Big, Deep, Round, Stretched
Hips	☐	☐	☐	Slender, Thin	Moderate	Heavy, Big
Joints	☐	☐	☐	Cold, Cracking	Moderate	Large, Lubricated
Appetite	☐	☐	☐	Irregular, Scanty	Strong, Unbearable	Slow but steady

Digestion	☐	☐	☐	Irregular, forms gas	Quick, Causes burning	Prolonged, Forms mucus
Taste, healthy preference	☐	☐	☐	Sweet, Sour, Salty	Sweet, Bitter, Astringent	Bitter, Pungent, Astringent
Thirst	☐	☐	☐	Changeable	Surplus	Sparse
Elimination	☐	☐	☐	Constipation	Loose	Thick, Oily, Sluggish
Physical activity	☐	☐	☐	Hyperactive	Moderate	Sedentary
Mental activity	☐	☐	☐	Always Active	Moderate	Dull, Slow
Emotions	☐	☐	☐	Anxiety, Fear, Uncertainty, Flexible	Anger, Hate, Jealousy, Determined	Calm, Greedy, Attachment
Faith	☐	☐	☐	Variable, Changeable	Intense, Extremist	Consistent, Deep, Mellow
Intellect	☐	☐	☐	Quick but faulty response	Accurate response	Slow, Exact
Recollection	☐	☐	☐	Recent good, remote poor	Distinct	Slow and sustained
Dreams	☐	☐	☐	Quick, Active, Many, Fearful	Fiery, War, Violence	Lakes, Snow, Romantic
Sleep	☐	☐	☐	Scanty, Broken up, Sleeplessness	Little but sound	Deep, Prolonged
Speech	☐	☐	☐	Rapid, Unclear	Sharp, Penetrating	Slow, monotonous
Financial	☐	☐	☐	Poor, spends on trifles	Spends money on luxuries	Rich, Good money preserver
TOTAL						

Vatas are drawn toward a lot of sexual activity. But excess sex is one of the causes of aggravated vata. Vatas generally have a difficult time prolonging sex, and vata men may experience premature ejaculation.

Vata individuals sleep less than the other body types and have a tendency toward interrupted sleep or insomnia, especially when vata is aggravated. Nevertheless they generally wake up feeling alert and fresh and ready to go.

Psychologically, vatas are blessed with quick minds, mental flexibility, and creativity. They have excellent imaginations and excel at coming up with new ideas. When in balance, they are joyful and happy. Vatas tend to talk quickly and to talk a lot. They are easily excited, alert, and quick to act—but may not think things through before acting, so they may give a wrong answer or make a wrong decision with great confidence!

Vatas are quite loving people, but may love someone out of fear and loneliness. In fact, fear is one of the symptoms of unbalanced vata. These individuals may experience fear of loneliness, darkness, heights, and closed spaces. Anxiety, insecurity, and nervousness are also common among them. They are worriers.

One of the main psychological qualities of vata individuals is readiness to change or, to put it the other way, difficulty with stability and commitment. They often change furniture, housing, jobs, or towns, and get easily bored. They don't like to stay in any place more than a year! Their faith is also quite variable. Vatas may be low on willpower and often feel unstable or ungrounded.

Clarity is one of the attributes of vata, and vata individuals generally are clear-minded and even clairvoyant. Along with their lively minds and fertile imaginations, they are usually highly alert and grasp new ideas quickly. However, they are also quick to forget. They think and speak quickly, but are restless and easily fatigued. They generally have less tolerance, confidence, and boldness.

Vata types tend to make money quickly, but they spend it quickly, too, often impulsively or on trifles. A vata may go to the flea market and come home with a lot of junk! They are not good savers. Nor are they good planners, and as a consequence they may suffer economic hardship.

The word *vata* is derived from a root that means "to move," and this gives an important clue to the character of vata individuals. As the principle of mobility, vata provides the motive power for all our mental and bodily processes. It regulates all activity in the body, from the number of thoughts we have to how quickly and efficiently food moves through our digestive tract.

The behavior that vatas are drawn to—travel, erratic hours, continual stimulation, frequent change—can easily upset their balance and lead to vata disorders such as constipation, gaseous distention, weakness, arthritis, pneumonia, excessively dry skin, dry lips, dry hair, dry, cracked nipples, and cracked heels. Nerve disorders, twitches and tics, mental confusion, palpitations, and breathlessness, as well as muscle tightness, low backache, and sciatica, are also due to aggravated vata. Excess vata makes the mind restless and hyperactive. Loud noises, drugs, sugar, caffeine, and alcohol also derange vata dosha, as does exposure to cold weather and cold foods.

The Attributes of Vata Individuals

Following are the main attributes of vata dosha and how they are expressed in the physical, mental, and behavioral characteristics of a vata individual.

ATTRIBUTES

MANIFESTATIONS IN THE BODY

Dry	Dry skin, hair, lips, tongue; dry colon, tending toward constipation; hoarse voice
Light	Light muscles, bones, thin body frame, light scanty sleep; underweight
Cold	Cold hands, feet, poor circulation; hates cold and loves hot; stiffness

	of muscles
Rough	Rough, cracked skin, nails, hair, teeth, hands and feet; cracking joints
Subtle	Subtle fear, anxiety, insecurity; fine goose pimples; minute muscle twitching, fine tremors; delicate body
Mobile	Fast walking, talking, doing many things at a time; restless eyes, eyebrows, hands, feet; unstable joints; many dreams; loves traveling but does not stay at one place; swinging moods and shaky faith
Clear	Clairvoyant; understands and forgets immediately; clear, open mind, experiences void and loneliness
Astringent	Dry choking sensation in the throat; gets hiccoughs, burping; loves oily, mushy soups; craving for sweet, sour, and salty tastes

Excess vata is a major factor in PMS (premenstrual syndrome). When, as her period approaches, a woman experiences bloating, low back ache, pain in the lower abdomen, cramps, pain in the calf muscles, and insomnia, and emotionally she feels anxiety, fear, and insecurity, this is due to aggravation of vata dosha.

Like the wind, vata types have a hard time settling down and staying grounded. When their vata becomes aggravated, it is difficult to calm them down. Sticking to a routine is difficult for them, but it is vital if they are to remain healthy.

The dry, cold, windy seasons of autumn and winter tend to increase and aggravate vata dosha, so at these times vata individuals need to be particularly careful to stay in balance. They need to dress warmly and eat warm, heavier foods. Warm, moist, slightly oily foods are beneficial, as are most warming spices. Steam baths, humidifiers, and moisture in general are helpful.

General Guidelines for Balancing Vata

Keep warm

Keep calm

Avoid raw foods

Avoid cold foods

Avoid extreme cold temperatures

Eat warm foods and spices

Keep a regular routine

Characteristics of the Pitta Individual

The pitta body type is one of medium height and build, though some individuals are slender with a delicate frame. They seldom gain or lose much weight. Their muscle development is moderate, and they are generally stronger physically than vata types. Pitta eyes are bright and may be gray, green, or copper-brown, and their eyeballs are of medium prominence. These individuals tend to have ruddy or coppery skin and may have reddish hair, which tends to be silky. They often experience early graying or hair loss, so that a pitta man frequently has a receding hairline or a big, beautiful bald head!

Moles and freckles are common on pitta skin, which tends to be oily, warm, and less wrinkled than vata skin. Pittas have sharp, slightly yellowish teeth and frequently have bleeding gums.

The normal body temperature of people with a pitta constitution is a little higher, and their hands and feet are usually warm and may be sweaty. Pittas may feel quite warm when both vatas and kaphas are cold. They perspire quite a lot, even when it's 50 degrees, while a vata person will not perspire even at a much higher temperature. Their body perspiration often has a strong, sulfury smell; their feet also perspire and may have a strong smell.

This heat is the main characteristic of pitta types, which is not surprising as the word pitta is derived from the Sanskrit word *tapa*, which means to heat. (The word can also be translated as “austerity,” and pitta individuals can be quite austere.) They have a low tolerance for hot weather, sunshine, or hard physical work. Although they are a fiery type, their sex drive may not be very strong. Pittas may use sex to release anger.

Pittas have a strong appetite, strong metabolism, and strong digestion. They consume large quantities of food and drink, and they also produce large quantities of urine and feces, which tend to be yellowish and soft. When out of balance, they crave hot spicy dishes, which are not good for them. They should eat food with the sweet, bitter, and astringent tastes. When hungry, a pitta person needs to eat soon, otherwise he or she will become irritable and hypoglycemic.

Pitta sleep is of medium duration, but it is uninterrupted and sound. These individuals like to read before they go to sleep and often fall asleep with a book on their chest.

Girls with a pitta constitution begin menstruating and reach puberty early. They can start menstruation as early as the age of ten.

Pitta physical ailments tend to be related to heat and the fire principle. They are prone to fevers, inflammatory diseases, acid indigestion, excessive hunger, jaundice, profuse perspiration, hives and rashes, burning sensations, ulceration, burning eyes, colitis, and sore throats. All “itises” are inflammatory disorders and are due to excess pitta. These individuals are susceptible to sunburn and do not like bright light.

Pitta-type PMS symptoms include tenderness in the breasts, hot flashes, hives, urethritis, and sometimes a burning sensation when passing urine.

Pitta individuals are alert and intelligent and have good powers of comprehension and concentration. Their intellects are penetrating and keen, and their memories are sharp. They have good, logical, investigating minds. They love to go deeply into problems and find solutions. Their minds are always at work, and they like to solve problems and puzzles of all kinds. They also tend to be good speakers. They are lovers of knowledge and have a great capacity for organization and leadership.

Pittas are night people. They become alert around midnight and love to read late at night.

Orderliness is important to them. A pitta person's home or room is always clean and neat. Clothes are kept in a designated place, shoes are in orderly rows, and books are arranged according to height or another definite system.

Pittas love noble professions. They are doctors, engineers, lawyers, judges—very bright, brainy people. They have good administrative abilities and like to be in a leadership role. They are good planners and are ambitious and disciplined. Aggressive by nature, they easily take charge of situations. They may become political figures. They have a lot of charisma. People are attracted to them.

Pittas are often wise, brilliant people, but they can also have a controlling, dominating personality. They have a tendency toward comparison, competition, and aggressiveness, and they are meticulous and perfectionistic. Everything has to be done on time, and correctly! Pitta individuals never yield an inch from their principles, which sometimes leads to fanaticism. They tend to be critical, especially when pitta dosha is aggravated; if there is no one to criticize, pitta people will criticize and judge themselves.

A pitta's life span is only moderately long. These individuals burn their life energy through too much mental activity, perfectionism, aggressiveness, and the constant search for success. They have a deep-seated fear of failure. They don't like the words *no* or *fail* and therefore may be highly stressful. They are the typical workaholics.

Pittas generally seek material prosperity and tend to be moderately well off, though they spend rather than save their money. They like to live in luxurious homes and drive fancy cars; they love perfumes, gems, jewelry, and other costly items; and they enjoy exhibiting their wealth and possessions.

A number of factors can increase pitta to the point of aggravation. One is simply eating too much spicy food, including black pepper, cayenne pepper, curry peppers, and jalapeño peppers. Pitta can also be increased by sour and citrus fruits, such as grapefruits and sour oranges. Eating rancid yogurt, smoking cigarettes, and drinking sour wine can also be harmful. Working near fire or lying in the sun are causes of increased pitta.

The Attributes of Pitta Individuals

Following are the main attributes of pitta dosha and how they are expressed in the physical, mental, and behavioral characteristics of a pitta individual.

ATTRIBUTES	MANIFESTATIONS IN THE BODY
Hot	Good digestive fire; strong appetite; body temperature tends to be higher than normal; hates heat; gray hair with receding hairline or baldness; soft brown hair.
Sharp	Sharp teeth, distinct eyes, pointed nose, tapering chin, heart-shaped face; good absorption and digestion; sharp memory and understanding; irritable
Light	Light/medium body frame; does not tolerate bright light; fair shiny skin, bright eyes
Oily	Soft oily skin, hair, feces; does not like deep-fried food (which may cause headache)
Liquid	Loose liquid stools; soft delicate muscles; excess urine, sweat and thirst
Spreading	Pitta spreads as rash, acne, inflammation all over the body or on affected areas; pitta subjects want to spread their name and fame all over the country
Sour	Sour acid stomach, acidic pH; sensitive teeth; excess salivation
Bitter	Bitter taste in the mouth, nausea, vomiting; repulsion toward bitter taste; cynical
Pungent	Heartburn, burning sensations in general; strong feelings of anger and hate
Fleshy smell	Fetid smell under armpits, mouth, soles of feet; socks smell
Red	Red flushed skin, eyes, cheeks and nose; red color aggravates pitta
Yellow	Yellow eyes, skin, urine and feces; may lead to jaundice, overproduction of bile; yellow color increases pitta

Eating fatty fried food, or oily food such as peanut butter, can create nausea or headaches for a pitta.

Summer is the most difficult time for pitta individuals. In hot humid weather pitta dosha can easily become aggravated. Heat builds up in the system, and pitta individuals become more susceptible to the heat-related ailments mentioned above. They may become quite irritable and are easily agitated and angered. Tempers flare. Their sharp minds become hypercritical and judgmental. Jealousy and envy may blaze up. They need to cool down!

General Guidelines for Balancing Pitta

Avoid excessive heat

Avoid excessive oil

Avoid excessive steam

Limit salt intake

Eat cooling, nonspicy foods

Drink cool (but not iced) drinks

Exercise during the cooler part of the day

Characteristics of the Kapha Individual

Kapha people are blessed with a strong, healthy, well-developed body. Their chests are broad and expanded, and they have strong muscles and large, heavy bones. With their larger frames and constitutions dominated by the water and earth elements, kaphas tend to gain weight and have difficulty taking it off. To complicate matters, kaphas generally have a slow digestion and metabolism. As a result, they tend to carry excess weight and to be on the chubby side. A kapha person may even do a water fast and gain weight!

In addition to their large frames, kapha individuals have strong vital capacity and stamina and tend to be healthy. Their skin is soft, smooth, lustrous and thick, and it tends to be oily. Their eyes are large, dark, and attractive, with long, thick lashes and brows. The whites of their eyes are very white. They have large, strong, white teeth. Their hair tends to be thick, dark, soft, wavy, and plentiful. They have hair everywhere!

Individuals with a kapha body type have a steady appetite and thirst, though digestion is slow. They can comfortably skip a meal or work without food, while it is difficult for a pitta person to concentrate without eating.

Because of their slow metabolic rate, kaphas who maintain health and balance generally enjoy a long span of life, longer than the other two doshic types, who tend to “burn out” more quickly. However, if kapha dosha is allowed to become aggravated, the

person is likely to become obese, which is one of the main causes of diabetes, hypertension, and heart attack. Such a person cannot live a very long life.

Kaphas have quite a sweet tooth and love candy, cookies, and chocolate. They are generally attracted to sweet, salty, and oily foods, but these contribute to water retention and weight gain; their bodies need lighter fare and do better with the bitter, astringent, and pungent tastes.

Because of the cloudy and heavy qualities of kapha, these individuals often feel heavy and foggy in the morning and may find it hard to get going without a cup of coffee or tea. Morning is not their time. They prefer midday, yet might feel like taking a nap after lunch; they often feel lethargic after a full meal. Unfortunately, daytime sleep increases kapha and is not good for them.

Kaphas evacuate slowly and their stools tend to be soft and pale in color. Their perspiration is moderate, more than vatas but less than pitta types. Their sleep is deep and prolonged.

Despite their strong bodies and great stamina, kaphas shun exercise. Vigorous exercise is good for them, but they prefer to sit, eat, and do nothing! Rather than jog, they prefer to walk—slowly! Kaphas do tend to like swimming, but it is not particularly good for them, as their bodies will absorb some water. When they do exercise, they become hungry afterward and will want to eat. After a workout at the gym they will go to a restaurant for a snack.

Kapha dosha is slow and steady in every way. These individuals move slowly and talk slowly (their speech pattern may become monotonous). They eat slowly and are slow to decide and slow to act. They move slowly and gracefully.

Kapha individuals are blessed with a sweet, loving disposition. By nature they are peaceful, patient, tolerant, caring, compassionate, and forgiving. They love to hug people. Kaphas are stable, solid, and faithful. Their spiritual or religious faith is deep and abiding, and their minds calm and steady.

One of the dominant qualities of kapha is softness, which manifests as soft skin, soft hair, soft gentle speech, a soft nature, and a soft, gentle, loving look. A pitta person's look is sharp and penetrating. A vata's look is spacy! But a kapha person looks calm, quiet, grounded, stable. He or she is *here*, right now!

Kaphas can be slow to comprehend, but once they know something, that knowledge is permanently retained. They have excellent long-term memories.

Although a kapha individual is forgiving, if you insult them or hurt their feelings, they will forgive you, but they will never forget! A kapha person will tell you, "On the twenty-fourth of January, 1972, at three-thirty in the afternoon, when we were having tea, you said such and such to me—but I have forgiven you!"

Their tendencies toward groundedness and stability help them to earn and hold on to money, and they are good at saving. Their extravagances are minor, mostly in the area

of spending a little on cheese, candy, and cakes.

A kapha person has a steady sex drive, and he or she can enjoy sex for hours at a time, without dissipation of energy, without orgasm or ejaculation of semen. It may take them some time to become interested, but once they are stimulated, they tend to stay that way.

Kapha dosha is aggravated by kapha-producing food, such as watermelon, sweet fruits, candy, cookies, yogurt, and other dairy products. Cold and frozen food and chilled water, sleeping in the daytime, and sitting and doing nothing all increase kapha. Sedentary work, especially when combined with steady munching at the desk, produces too much kapha in the body. Excess kapha slows digestion and metabolism and lowers the digestive fire, and the person may become chubby or even obese.

The Attributes of Kapha Individuals

Following are the main attributes of kapha dosha and how they are expressed in the physical, mental, and behavioral characteristics of a kapha individual.

ATTRIBUTES	MANIFESTATIONS IN THE BODY
Heavy	Heavy bones, muscles, large body frame; tends to be overweight; grounded; deep heavy voice
Slow	Slow walk, talk; slow digestion, metabolism; sluggish gestures
Cool	Cold clammy skin; steady appetite and thirst with slow metabolism and digestion; repeated cold, congestion and cough; desire for sweets
Oily	Oily skin, hair and feces; lubricated, unctuous joints and other organs
Damp	Congestion in the chest, sinuses, throat and head
Smooth	Smooth skin; gentle calm nature; smoothness of organs
Dense	Dense pad of fat; thick skin, hair, nail and feces; plump rounded organs
Soft	Soft pleasing look; love, care, compassion and kindness
Static	Loves sitting, sleeping and doing nothing
Viscous	Viscous, sticky, cohesive quality causes compactness, firmness of joints and organs; loves to hug; is deeply attached in love and relationships

Cloudy	In early morning mind is cloudy and foggy; often needs coffee as a stimulant to start the day
Sweet	The anabolic action of sweet taste stimulates sperm formation, increasing quantity of semen; strong desire for sex and procreation; abnormal function may cause craving for sweets
Salty	Helps digestion and growth, gives energy; maintains osmotic condition; abnormal function may create craving for salt, water retention

The difficult time of year for kapha individuals is winter and early spring, when the weather is heavy, wet, cloudy, and cold. Then kapha accumulates in the system and leads to physical, emotional, and mental imbalances of the kapha variety. Physical problems will tend to be related to the water principle, such as colds, flu, sinus congestion and other diseases involving mucus, such as bronchial congestion. Sluggishness, excess weight, diabetes, water retention, and sinus headaches are also common.

Emotionally, when kapha becomes unbalanced, these individuals may suffer from greed, attachment, envy, possessiveness, lust, and laziness, leading to kapha-type depression.

Interestingly, kapha can become aggravated as the moon gets full because, as biologists have discovered, there is a tendency toward water retention in the body at that time.

Kapha women may suffer from PMS symptoms such as excessive emotionality, water retention, white vaginal discharge, and overurination. They may feel attachment, greed, and lethargy at that time and will probably have a tendency toward excess sleep.

General Guidelines for Balancing Kapha

Get plenty of exercise

Avoid heavy foods

Keep active

Vary your routine

Avoid dairy foods

Avoid iced food and drinks

Avoid fatty or oily foods

Eat light, dry food

How to Use This Knowledge

Knowing your Ayurvedic constitution (*prakruti*) has many benefits for your life and health:

- Self-understanding, which is the foundation of life, is greatly increased. Ayurveda says that every person is a unique and divine book. To read that book is a great art. The knowledge of *prakruti* can help you to read your own book, which is your life. By understanding your constitution, you can better understand your psychological tendencies, your strengths, and your weaknesses, as well as your physiological strong and weak areas.

- You may see that your habits and tendencies, such as erratic lifestyle and schedule (*vata*), irascibility (*pitta*), or laziness (*kapha*), or physical problems such as overweight (*kapha*), ulcers (*pitta*), or constipation (*vata*), are directly related to your constitution. The tendency toward such imbalances is inherent in the way your mind-body system is designed.

- When you can anticipate the kinds of illnesses and imbalances you are likely to have, you can take precautions to prevent them from arising. You can adjust your lifestyle—daily routine, diet, amount or type of exercise, and so on—to keep your doshas in balance and your health at its best.

- You can also use the knowledge of constitutional types to understand others with whom you are related, whether in your personal life or at work. To succeed in relationships, which is such a confused and problematic area today, it is helpful to know the constitution of your husband or wife, boyfriend or girlfriend. Understanding one another brings clarity; clarity brings compassion, and compassion is love. Such a relationship brings happiness, joy, and longevity.

Using this knowledge of *prakruti*, if your spouse is upset and angry you can say to him or her, “Honey, it’s not you, it’s your pitta!” That will open a new dimension of understanding of emotional reactions in the relationship.

Use the knowledge of your *prakruti* as a baseline, to see where you should be. Then look at your *vikruti*, your current imbalances, as a clue to help you restore balance, using the food charts, yoga postures, herbs, recommended exercises, and the like, presented throughout this book.

For instance, if your *vikruti* shows more pitta than your *prakruti*, you will want to follow the guidelines for pacifying pitta. If you have a disorder that is caused by excess vata, pitta, or kapha, follow the guidelines for pacifying that dosha. For sinus congestion, for example, follow a kapha-reducing diet until the condition subsides.

If your *prakruti* and *vikruti* seem about the same, then choose the diet and lifestyle guidelines for your strongest dosha.

Finally, remember that “balance” does not mean equal amounts of vata, pitta, and

kapha; rather, it means maintaining *your proportion* of the three doshas, according to your constitution. It is not a static state, but a dynamic equilibrium that needs constant renewal.

Chapter 3

Why We Get Sick

What is health? What is disease? Are sickness and health just a matter of luck or of which bacteria you happen to encounter in your daily life? What can we do to maintain a positive state of health and avoid getting sick?

These are questions that the five-thousand-year-old tradition of Ayurvedic medicine has considered in depth. The answers, drawn from deep insight and generations of practical experience, can help us prevent illness from developing and heal it if it arises.

Let's begin by examining the Ayurvedic understanding of health. Then we will look at ten potential causes of illness and how you can counteract them. Once you are aware of the factors that can either maintain health or disturb your body's equilibrium and set the disease process in motion, you can organize your life for health and balance. Finally, we will consider the Ayurvedic understanding of how illness develops, from its earliest, invisible stages until it is fully grown.

The Definition of Health

According to Ayurveda, health is not simply the absence of disease. It is rather a state of balance among body, mind, and consciousness.

Health consists of a balanced state of the three humors (doshas), the seven tissues (*dhatus*), the three wastes (*malas*), and the gastric fire (agni), together with the clarity and balance of the senses, mind, and spirit.

Although you will not need to master all these terms and considerations in order to effectively use the remedies in [Part III](#), an acquaintance with them will give you a bigger picture of the depth and practicality of this science.

You are already familiar with the three doshas, the biological humors or principles that govern all activity in the body: vata, the energy or principle of movement; pitta, the energy of digestion and metabolism; and kapha, the principle of lubrication and structure. Balance of the three doshas maintains health; imbalance leads to disease.

The *dhatus* are the basic bodily tissues. They are responsible for the entire structure of the body and the functioning of the different organs and systems. Crucial to the development and nourishment of the body, the *dhatus* unfold successively as follows, starting with the nourishment derived from the product of digestion:

1. *Rasa* (plasma or cytoplasm) contains nutrients from digested food and subsequently

nourishes all tissues, organs, and systems.

2. *Rakta* (blood) governs oxygenation in all tissues and vital organs and thus maintains life-function.
3. *Mamsa* (muscle) covers the delicate vital organs, performs the movements of the joints, and maintains the physical strength of the body.
4. *Meda* (fat) maintains the lubrication of the tissues and serves as insulating material to protect the body's heat.
5. *Asthi* (bone and cartilage) gives support to the body's structure.
6. *Majja* (bone marrow and nerves) fills up the bony spaces, carries motor and sensory impulses, and facilitates communication among the body's cells and organs.
7. *Shukra* and *artava* (male and female reproductive tissues) contain the pure essence of all bodily tissues and can create a new life.

Each *dhatu* is dependent on the previous one. If the raw materials of digestion are inadequate, or if there is a problem in any stage, each successive *dhatu* will not receive the nourishment it needs and the respective tissues or organ systems will suffer. So for good health, all seven *dhatu*s must develop and function properly.

The three waste products (*malas*) are feces, urine, and sweat. The body must be able to produce these in appropriate amounts, and to eliminate them through their respective channels.

Agni is the biological fire or heat energy that governs metabolism. It can be equated with the digestive enzymes and metabolic processes involved in breaking down, digesting, absorbing, and assimilating our food. Agni maintains the nutrition of the tissues and the strength of the immune system. It destroys microorganisms, foreign bacteria, and toxins in the stomach and intestines. It is an extremely vital factor in maintaining good health.

Agni sustains life and vitality. An individual endowed with adequate agni lives long and has excellent health. But when agni becomes impaired because of an imbalance in the doshas, metabolism is adversely affected. The body's resistance and immunity are impaired, and the person begins to feel unwell. When this vital fire is extinguished, death soon follows.

In addition to these bodily factors, the senses, mind, and spirit also play a vital role in maintaining good health, as we will discuss in the next section. When all these factors are balanced, it produces a state called *swastha*, which means "totally happy within oneself."

Agni

There is a saying in Ayurveda that a person is as old as his or her agni. According to the *Charaka Samhita*, one of the great classics of Ayurvedic medicine:

“The span of life, health, immunity, energy, metabolism, complexion, strength, enthusiasm, luster, and the vital breath are all dependent on agni (bodily fire). One lives a long healthy life if it is functioning properly, becomes sick if it is deranged, or dies if this fire is extinguished. Proper nourishment of the body, *dhatu*s, *ojas*, etc., depends upon the proper functioning of agni in digestion.

“The five types of agni, corresponding to ether, air, fire, water, and earth, digest the respective components of the food.... In this way, balanced agni cooks the appropriately chosen and timely consumed food, and leads to promotion of health.

...

“Agni is necessary for the normal process of digestion, and the subtle energy of *agni* transforms the lifeless molecules of food, water, and air into the consciousness of the cell.”

This state of happiness and balance can be created and sustained by maintaining a healthy lifestyle in accordance with nature and the requirements of your own constitution. Proper nutrition, proper exercise, healthy relationships, positive emotions, and a regulated daily routine all contribute to a healthy life. On the other hand, wrong diet, inadequate exercise, troubled relationships, negative or repressed emotions, and an erratic schedule are at the root of disease. These causative factors upset the balance of the doshas, weaken agni and the *dhatu*s, and lead to poor health.

Ten Factors in Health and Illness

Illness does not suddenly appear. There is a direct causal link between the factors that influence us and the effects they produce. The *cause is the concealed effect, and the effect is the revealed cause*. The cause is like a seed, in which the as-yet-unmanifested tree is concealed. The tree is the expressed value of the seed. Health is the effect of a healthy lifestyle and healthy habits; disease is the “tree” sprouted from unhealthy habits.

According to the *Charaka Samhita*,

Both the patient and the patient’s environment need to be examined in order to arrive at an understanding of the disease and the causes of disease. It is important to know where the patient was born and raised, and the time of onset of the imbalance. It is also important to know the climate, customs, common local diseases, diet, habits, likes and dislikes, strength, mental condition, etc.

This enumeration opens the door to the wide variety of factors constantly influencing our health. Let us consider some of them.

LIKE INCREASES LIKE

The first important principle in considering the potential causes of disease is “Like increases like.” A dosha is increased by experiences and influences (such as food, weather, and seasons) with qualities similar to it. Dry foods, dry fruit, running, jogging, jumping, always being in a rush, and working too hard are all factors that aggravate vata in the system. Pittagenic factors, such as hot spicy food, citrus fruit, fermented food, and hot humid weather, provoke excess pitta. Cold, cloudy, damp weather, eating dairy products, wheat, and meat, and sitting and doing nothing increase kapha.

The antidote to “like increases like” is “opposite qualities decrease or balance.” This is the key to healing.

NOTE: In general, one’s *prakruti* indicates one’s disease proneness. Individuals of pitta constitution, for example, tend to have pitta diseases. But this is not inevitable. A person of vata constitution who eats a lot of hot spicy food, drinks alcohol, lies in the sun, smokes cigarettes, and represses anger will definitely get a pitta disease. If he or she eats candy, cookies, ice cream, and other dairy products and is exposed to cold weather, there will be a susceptibility to congestive kapha disorders.

FOOD AND DIET

We have already touched on the effects of food on the doshas, and [chapter 8](#) will discuss this important topic in depth, so we won’t go into it at length here. The principle is simply that eating the right kinds of food for your *prakruti* maintains vitality and balance, while eating the wrong kinds creates imbalance in the doshas, the first step in the genesis of disease.

Eating spicy food or sour or citrus fruit and drinking alcohol all increase heat and acidity in the body, something a pitta person cannot afford. For a vata individual, dried fruits, beans (including garbanzo, pinto, and aduki) are hard to digest and will provoke vata. Raw salads, which are cold and astringent, will likewise increase vata. For a kapha individual, dairy products, cold drinks, and fatty fried foods definitely add to kapha. So a vata person eating a vatagenic diet, a pitta person eating a pitta-provoking diet, and a kapha person eating kapha-aggravating food are definitely creating imbalance and sowing the seeds for ill health.

Wrong food combinations (see table [this page](#)), stale food, food with chemical additives, and wrong eating habits, such as eating too much late at night or eating in a rush, also contribute to imbalance and lead to poor digestion and poor health. Diet is thus one of the main potential causes of ill health—but by understanding these principles and eating according to the guidelines for our constitutional type, it is also one of the major ways we can take control of our lives and maintain healthy balance.

SEASONS

Ayurveda classifies the seasons according to their predominant dosha. The windy, cool, dry weather of autumn is largely vata, followed by the dark, heavy, damp, cloudy kapha qualities of winter. Early spring is still primarily kapha, but as late spring arrives, the increased warmth, light, and brightness express pitta qualities, which blossom in their full intensity in the summer.

Each of these seasons brings its own challenges to health. The predominant dosha of the season will tend to build up at that time and can cause aggravation especially in someone of the same *prakruti*. If we act intelligently, we can avoid this accumulation and aggravation.

For example, because autumn and early winter tend to increase vata, individuals with a vata-predominant constitution need to eat warm foods, dress warmly, avoid cold food and drinks, and stay out of nasty weather. Otherwise they will fall prey to vata illnesses and discomforts, such as constipation, insomnia, and lower back pain. If pitta individuals want to remain free of anger, as well as hives, rash, and diarrhea, they need to keep cool in the summer, avoiding spicy foods, overexertion, and overexposure to the hot sun. Kaphas need care in the heart of winter and early spring if they are going to avoid colds, coughs, allergies, and sinus congestion in the damp, cool, heavy weather.

We will look further at the seasons, their effects, and how we can best live in harmony with their rhythms and changes in [chapter 5](#), where we discuss the ideal Ayurvedic lifestyle, including daily and seasonal routines.

EXERCISE

Exercise is another factor that can profoundly influence your health for better or worse. Regular exercise improves circulation and increases strength, stamina, and immunity. It helps one to relax and to sleep peacefully. It benefits the heart and lungs, is vital for effective digestion and elimination, and helps the body purify itself of toxins through sweating and deep breathing. Exercise increases the rate of combustion of calories, so it is good for maintaining body weight and for weight loss. It also makes the mind alert and sharp and develops keen perception.

On the other hand, insufficient exercise, overexertion, or exercise that is inappropriate for one's constitution can lead to ill health.

Lack of exercise eventually brings a loss of flexibility and strength and puts one at greater risk for many diseases, such as diabetes, hypertension, osteoporosis, and heart disease.

Some amount of sweating helps to eliminate toxins, reduces fat, and makes you feel good. But overexertion may cause dehydration, breathlessness, chest pain, or muscle aches, ultimately leading to arthritis, sciatica, or a heart condition.

Yoga stretching and some aerobic exercise are valuable for all body types, but the

amount and intensity of your exercise should be based upon your constitution. Kaphas can do the most strenuous exercise, pittas can handle a moderate amount, and vatas require the gentlest exercise. Even though fast-moving vatas are attracted to active sports, quieter exercises such as walking and yoga stretching are better for them. They should leave jogging, fast bicycling, aerobic dancing, and fast walking to pitta and kapha types. Kaphas are the most reluctant exercisers, preferring to do little or nothing, but it is important for them, or they will tend to put on weight and feel emotionally heavy and dull.

So here again, self-knowledge—knowledge of your constitution—plus a few pieces of vital information give you the opportunity and the challenge to maintain good health or fall into imbalance and illness.

You will find additional information about exercise in [Part II](#), where we discuss the Ayurvedic daily routine.

AGE

As briefly mentioned in [chapter 1](#), Ayurveda divides the human life span into three stages. At each stage, certain diseases and types of disease are more common. Childhood is the age of kapha. Children's bodies are growing and building up their structure, so kapha dosha is more predominant. Their bodies are soft and gentle (qualities of kapha), they require more sleep than adults, and they are susceptible to kapha illnesses such as colds and congestion.

Adulthood exhibits more characteristics of pitta. Adults are more competitive, aggressive, and ambitious than children; they work hard, they require less sleep, and they fall prey to pitta-type disorders such as gastritis, colitis, and peptic ulcers.

Old age is the age of vata. Elderly people sleep quite a bit less, and their sleep is broken. They tend to get constipation, cracking and popping of joints, degenerative diseases such as rheumatoid arthritis and Alzheimer's disease, and suffer from forgetfulness, all characteristic of vata dosha.

This shows that our age and stage of life are factors that have to be considered in the choices we make to keep our doshas in balance and remain healthy. Elderly people, for example, should not engage in strenuous exercise, and if possible they should minimize travel, among many factors that increase vata. They should favor a vata-balancing diet, with more warm, moist foods, more oil, and less salad and dried fruit.

MENTAL AND EMOTIONAL FACTORS

Our life is a whole, consisting of body, mind, and pure consciousness. Both health and disease have psychological as well as physical origins. Illness may begin in the mind and emotions and then affect the body; mental imbalance creates physical imbalance.

Equally, physical disorders and imbalances can generate mental disorders. Because of this, mind and body are never considered separately in Ayurveda.

Every perception, thought, feeling, and emotion, whether positive or negative, is a biochemical event that influences the doshas and affects the cells, tissues, and organs of the body. Fear, anger, grief, hatred, envy, possessiveness, and other negative emotions disturb our doshic balance; likewise, when the doshas are already out of balance, they may give rise to these same negative emotions.

- Increased vata is associated with anxiety, insecurity, fear, nervousness, restlessness, confusion, grief, and sadness.

- *Increased pitta* is associated with anger, envy, hate, ambition, competitiveness, criticism, judgmental attitude, sharp speech, perfectionism, and the need to be in control.

- *Increased kapha* is associated with greed, attachment, possessiveness, boredom, laziness, and lethargy.

Emotions have an affinity with certain organs: grief and sadness with the lungs, anger with the liver, and hatred with the gall bladder. The kidneys may become the seat of fear, and the heart (as well as the lungs) the abode of grief and sadness. Nervousness is associated with the colon, while the stomach is the home of agitation and temptation, and the spleen may be related to attachment.

As we have discussed, emotions have a physical as well as a psychological aspect. Emotions are reactions to situations. If we do not understand and maintain clear awareness of the total movement of an emotion, from its arising to its dissolution, it will tend to adversely affect a particular organ, causing stress and weakness and creating what is known as a “defective space” (*khavaigunya*), where a future disease may manifest. (See [this page](#), “How Disease Develops.”)

STRESS

Modern medicine often views stress as the result of a particular lifestyle, or of overwork, emotional trauma, and so on. Ayurveda considers stress less as a result or condition than as a causal factor in disease. A regular daily routine, nourishing diet, positive emotions, and loving relationships result in strength and health. But keeping late hours, eating food that is aggravating to one’s constitution, traveling a lot, using the mind or stimulating the senses too much, repressing negative emotions such as anger or fear, and maintaining problematic relationships all put stress on the body and mind. In addition, toxins in food, water, air pollution, excessive noise, and many other environmental factors are also stressful.

Stress is a major factor in many diseases. It may trigger allergies, asthma, and herpes, and it may even lead to heart conditions.

Stress disturbs the doshas and can create disequilibrium of vata, pitta, or kapha, depending on the individual's constitution. Vata individuals may develop vata conditions such as anxiety or fearfulness. Pitta individuals may react to stress in the form of anger, or they may suffer from hypertension, peptic ulcer, ulcerative colitis, and other pitta disorders. Kapha individuals under stress tend to eat and eat and eat.

In [Part III](#) you will find many suggestions to minimize the impact of stress on your life, and to relieve symptoms caused by stress if they develop.

OVERUSE, UNDERUSE, AND WRONG USE OF THE SENSES

Our senses give us great pleasure as well as vital information. Through ordinary experience our senses of taste, touch, smell, sight, and hearing can nourish us, and we can also find healing through sense therapies such as aromatherapy, color therapy, mantras and other healing sounds, massage, and the tastes in herbs and foods.

But because all our perceptions, as well as our thoughts and feelings, are biochemical events as well as experiences in consciousness, improper use of the senses can create imbalance or damage in the body and result in illness.

Overuse of the senses strains and stresses our nervous system. To use a simple example, repeated exposure to bright light hurts the retina and strains the optic nerve, which triggers pitta, and sooner or later a person's eyesight will be affected or neuritis-like symptoms will arise. If we listen to loud music or hear loud sounds, the eardrum and the rest of our hearing apparatus are hurt and weakened; if it occurs often, the person can become deaf. Loud sounds also affect systemic vata dosha, giving rise to vata symptoms such as arthritis or degenerative changes in the bones. Lying in the sun strains the sense of touch, aggravates pitta, and may lead to skin cancer.

Misuse of the senses means using them in a wrong way, such as trying to read very small letters, or looking through a microscope or telescope (which creates a strain on the eyes), or reading while lying down (which changes the angle of focus and builds up stress on the muscles of the eyeball), which will eventually result in pitta or vata disorders. Eating a large quantity of wrong food, such as hot, spicy, stimulating food containing cayenne pepper, is a misuse of the taste organ. Listening to loud sounds over the telephone, and long phone conversations, both aggravate vata. Exposing the senses to wrong inputs, such as watching violent movies on television, is also a misuse of the senses.

Underuse of the senses means not perceiving with total attention, ignoring what we perceive, or not making full use of our wonderful sensory equipment. This can lead, for example, to accidents. SAD (seasonal affective disorder) is a form of depression that affects people who don't get enough sunlight during the winter—a kind of underuse of the sense of sight. "Cabin fever," the discomfort and restlessness born of staying indoors for a long time, is at least partly the result of sensory deprivation. Prolonged fasting—

underuse of the sense of taste—aggravates vata.

“KNOWING BETTER”

Very often we get sick because we disregard our own knowledge or wisdom. Understanding our *prakruti*, our psychobiological constitution, is self-knowledge; understanding how certain foods, for instance, can disturb the balance of our mind-body system and lead to illness, while other food is balancing and strengthening for us, is knowledge we can use to remain healthy. And yet often we follow the impulses of the moment and choose foods that will cause us problems.

If a person who knows that her constitution is largely pitta decides to eat hot spicy food for lunch and then spends the rest of the summer afternoon working in the garden, she is disregarding her intelligence and understanding and asking for trouble.

As individuals, we are all part of the Cosmic Consciousness, the universal intelligence that so beautifully organizes all of nature. That intelligence is within us, and by following the time-tested principles of Ayurveda and paying attention to our own intuition and inner wisdom about what is right for us, we can regulate our lives in harmony with it.

RELATIONSHIPS

Our life is relationship. We are related to the earth, the moon, the sun, the air we breathe, the water we drink, the food we eat. You are related to your friends, your parents and children, your spouse, and your co-workers, as well as to your own body, your thoughts and feelings, your job, and your bank account. In our daily life, relationships are most important.

Often we use our personal relationships as a sort of power game, to control others. Then relationships become a battlefield rather than a field of love. When a negative emotion comes up in a relationship, such as resentment of a past hurt or insult, anger, fear, anxiety, or criticism, pay attention to the feeling. Don't judge the other person or yourself. When your spouse says something and you feel hurt or angry, look inside to see what your thoughts and feelings are saying to you. Be honest. Out of honesty, clarity comes.

When clarity is lacking, feelings are repressed, or communication is absent at times of crisis in our relationships, stress builds up, and this is one of the causes of illness. Stress disrupts our inner biochemistry, the doshas are thrown out of balance, and the seeds of disease are sown.

Husband and wife, brother and sister, parent and child—all our relationships must be absolutely clear. Clarity in relationships develops compassion, and compassion is love. Therefore love is clarity. And as we all know, love is the key to successful relationships.

If you look back over the ten factors presented in this section, you will see that you have a great deal of choice and control over whether they will create a potentially disease-producing imbalance in the doshas. This is true even of such apparently uncontrollable factors as the seasons and the weather: if it is cold, you can dress warmly; if it is hot, you can take it easy and stay out of the sun.

How Disease Develops

According to Ayurveda, illness is the end result of a long process that can be detected and addressed at any stage. This process has been thoroughly studied and its phases delineated in great detail.

The disease process begins with disturbances in the balance of the doshas. Temporary imbalances are common and quite normal; problems arise if the aggravated condition is not corrected. In the normal course of events, vata, pitta, and kapha go through cycles of change in three stages: accumulation, provocation or aggravation, and pacification. Pitta, for example, begins to build up and accumulate in the late spring. It is provoked or aggravated in the hot summer months, and it naturally becomes pacified when the weather cools down in the autumn.

If the increased dosha isn't pacified naturally through a change in seasons, it undergoes further changes and disease may result. If a person with a predominantly vata constitution experiences some degree of increased vata in the fall due to the cool, dry, windy weather, but it returns to normal soon after, disease will not develop. The person can aid the process of restoring balance, for example, by eating moist, warming foods and dressing warmly in the windy weather.

How to Transform Negative Feelings

Negative feelings can cause hurt both to ourselves and to others. If we express anger or criticism, for example, we inflict pain on someone else. On the other hand, repressing such feelings creates problems for ourselves, as the stressful biochemistry affects the internal organs and systems down to the cellular level.

If both expressing and holding back negative feelings can be harmful, what shall we do when these emotions boil up in us? Ayurveda offers a way to learn from such situations and resolve them in a positive manner.

At the moment the feeling comes up, look into it. Let's suppose it is a feeling of anger. Take a long, deep breath, let yourself feel the anger, and exhale it out. Give the feeling total freedom to express itself *within you*, so that you look at it honestly and feel it. Breathe into it, surrender to it, and be with it. Breathe into it, and breathe out. Soon it will dissolve by itself.

You have to be aware not only of the external thing—what your spouse or friend is saying—but at the same time you have to bring awareness to your inner self. When awareness goes both ways, outer and inner, understanding is total. This approach doesn't put a scar on the mind.

Look at the feeling—any feeling or emotion—without labeling it or naming it. Then the observer and the observed become one. Observe with total awareness, with no division between subject and object, no separation between yourself and the feeling. Give freedom to the feeling; let it flower; let it fade away.

If the condition of aggravated vata continues, vata will move into the general circulation and into the deep connective tissue, where it will generate pathological changes. Disease will develop. Imbalance is disorder, and disorder is disease.

Disease is like a child. It has its own creation within the womb of the body, according to a process known as *samprapti* or pathogenesis, literally “the birth of pain.” In brief, this is how it happens:

1. ACCUMULATION

Due to various causes, such as diet, weather, seasons, emotions, and others we have discussed, the doshas begin to accumulate in their respective sites: vata in the colon, pitta in the intestines, and kapha in the stomach. This is the easiest stage at which to treat any incipient health problem. A trained Ayurvedic physician can feel the imbalance in your pulse even at this stage, and you may be able to detect it yourself.

Vata accumulation may be experienced as constipation, abdominal distension, or gases in the colon. Pitta buildup may be felt as heat around the belly button area and can be observed as a slightly yellowish discoloration in the whites of the eyes, or dark yellow-colored urine. The person will be very hungry and will crave candy and sugar. Accumulated kapha leads to feelings of heaviness, lethargy, and loss of appetite.

At this stage the individual is still quite healthy, and when a dosha starts to build up, the body's intelligence creates an aversion to the causal factor and a craving for opposite qualities, which can restore balance. For example, if you've eaten ice cream three days in a row and kapha is building up, the thought of more ice cream will not be appealing; rather, your body will crave cayenne pepper or other spicy food to burn up the kapha and counteract it. One should listen to this wisdom and not continue increasing the cause.

2. AGGRAVATION

The accumulated dosha continues to build up in its own site. The stomach gets brimmed up with kapha, the intestines fill with pitta, or the colon overflows with vata.

These accumulated doshas then try to move from their sites. Kapha tries to go up into the lungs, pitta tries to move into the stomach and gallbladder, and vata tries to move into the flanks.

You can feel this stage, too. For example, if you eat too much kapha food on Saturday night, you might feel full when you wake up on Sunday and think to yourself, “Maybe I should fast or eat very lightly today.” But then someone invites you out for Sunday brunch, and you eat heavily again. The next day you might get a cough or a feeling of congestion in the lungs as the kapha starts to move upward. Excess pitta in the second stage may cause heartburn or acid indigestion, even nausea. Vata rising up may cause pain in the flanks or midback, or even breathlessness.

According to Ayurvedic therapeutics, the disease process can be addressed at any stage, but specific treatments are needed for specific stages. In these first two stages, one can reverse the process by oneself, using common sense to apply the principle of opposite qualities, and taking some home remedies. Once the disease process has gone beyond the gastrointestinal tract and entered the third phase, it is no longer under one’s own control, and trained medical help is needed. (See [sidebar](#) on ama.)

3. SPREAD

The dosha begins to spread from its place of origin, overflowing into the bloodstream and the general circulation of the body, “looking” for a place to enter. Here the disease process has progressed to the point where eliminating the causal factor will not be enough. A *panchakarma* purification program (or a similar cleansing regimen) is needed in order to return the doshas to their respective sites in the gastrointestinal tract so they can be excreted from the body.

Ama, Agni, and the Disease Process

The body’s biological fire, which governs the transformation of matter into energy, is of thirteen major types. The central fire, called *jatharagni*, governs the digestion and assimilation of food. The other agnis (the fire component in the cells, tissues, and organs) perform the local process of digestion and nutrition. When agni is robust and healthy, then whatever a person eats, the system digests, assimilates, and absorbs it, then eliminates the impurities. But when the doshas are aggravated because of poor diet, an unhealthy lifestyle, or negative emotions, they first affect agni, which becomes unbalanced. When agni becomes weakened or disturbed, food is not properly digested.

The undigested, unabsorbed food particles accumulate in the gastrointestinal tract and other subtle sites in the body and turn into a toxic, sticky, foul-smelling substance called *ama*. (Ama may also be formed by bacterial invasion and cellular metabolic waste.) In the third (“spread”) stage of the disease process, ama

overflows from its site of origin to other bodily channels such as the blood vessels, capillaries, and lymphatics, and clogs the channels and the cell membranes.

When these molecules of ama clog the channels, the cellular intelligence (*prana*) which is constantly flowing between the cells gets blocked, and some cells become isolated. An isolated cell is a lonely cell, and a lonely cell is a confused cell. Pathological changes begin to occur. But the root cause of cytopathological changes is the movement of these molecules of ama. So the ama has to be eliminated from the body by *panchakarma* or other means. (See “[Techniques for Cleansing and Purification](#)” in [chapter 4](#).)

4. DEPOSITION OR INFILTRATION

The aggravated dosha enters an organ, tissue, or system that is weak or defective, due to previous trauma, genetic predisposition, accumulated emotional stress, repressed emotions, or other factors. These weak areas in the body can be described as negative locations, like potholes in the road. Smoking cigarettes, for example, creates weakness in the lungs; eating too much sugar creates weakness in the pancreas and blood tissue, and so on.

The newly arrived, aggravated humor (dosha) creates confusion within the cellular intelligence of the weaker tissue and overwhelms it, changing its normal qualities and functions. The quality of the aggravated dosha suppresses the normal qualities of the tissue and combines with it, creating an altered state, changed in structure and function. In this way, the “seeds” of disease begin to sprout.

Up to this point, the disease has not appeared on the surface, but it can be detected by a skilled physician or recognized by imbalances in the doshas such as those mentioned above. An alert person can feel subtle changes in the body. If the condition is not interrupted at this stage, it will erupt as a full-blown disease.

5. MANIFESTATION

In this stage, qualitative changes become apparent. The signs and symptoms of an actual disease appear on the surface; the person becomes sick. Whether in the lungs, kidneys, liver, joints, heart, brain, or wherever, the seeds of disease now sprout and begin to manifest in the area of the defective tissue.

6. CELLULAR DEFORMITY LEADING TO STRUCTURAL DISTORTION

Now the pathological process is fully developed and the disease completely manifested. Structural changes appear, and complications of other organs, tissues, or

systems become evident. This is also the stage at which the disease, now fully developed, is therefore most difficult to treat.

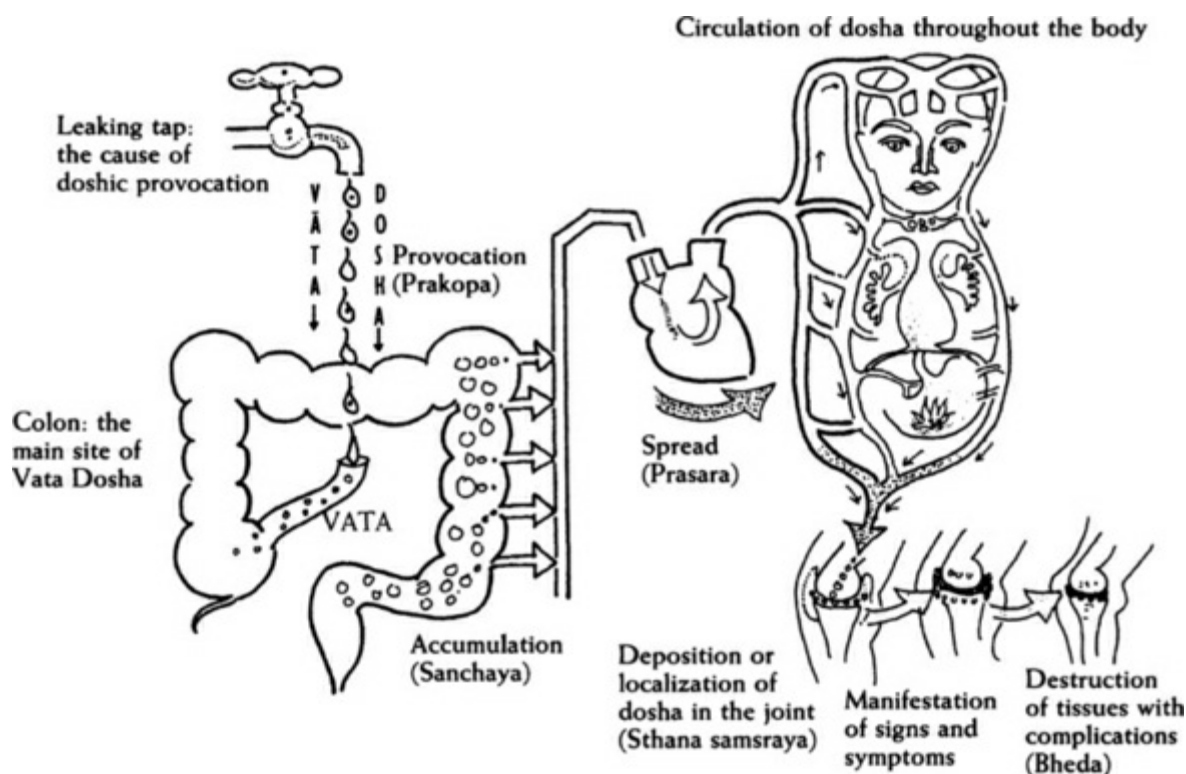
In the fifth stage, for example, when aggravated pitta dosha is invading the wall of the stomach, it may manifest as an ulcer. But in the sixth stage, the pitta will perforate the ulcer and cause hemorrhaging, or it may provoke a tumor. Function begins to be disturbed in the fifth stage, but here the structure of the tissue is affected, as well as the surrounding tissues and systems.

Obviously, treatment—restoration of balance and normal functioning—is far easier at earlier stages. That is why prevention is emphasized so strongly in Ayurveda. It is much more effective to treat the illness in its seed stage, before it sprouts and grows.

Both health and disease are processes. Disease is a process of abnormal movement of the doshas, while health is a process of their normal functioning. The wise person understands that the normal rhythm and quality of the process can be reestablished by changing diet and lifestyle and avoiding the etiological factors that cause disease.

The key is awareness. The more you are alert to how your mind, body, and emotions are reacting to changing circumstances; the more you are aware of your constitution and the moment-to-moment choices you can make to maintain health, the less opportunity you create for becoming sick.

SAMPRAPTI (PATHOGENESIS): THE SIX STAGES OF THE DISEASE PROCESS





Part II



Putting Ayurveda to Work

Chapter 4

How We Can Stay Healthy

The goal of Ayurveda is to maintain the health of a healthy person and heal the illness of a sick person. [Part III](#) of this book contains hundreds of suggestions to help you if you have fallen ill. But staying well is far easier than curing an illness, especially once an imbalance has progressed through the later stages of the disease process. That is why prevention is so strongly emphasized in Ayurvedic medicine. In this chapter we will consider some of the fundamental principles and approaches recommended by Ayurveda for remaining healthy.

Awareness

The master key to remaining healthy is *awareness*. If you know your constitution, and you can remain alert to how your mind, body, and emotions respond to the changing conditions in your environment and the numerous facets of your daily life, such as the food you eat, you can make informed choices to maintain good health.

As we saw in [chapter 3](#), the cause is the concealed effect and the effect is the revealed cause, as the seed contains the potential tree and the tree reveals the potency of the seed. To treat the cause is to treat the effect, to prevent it from coming to fruition. If a kapha person always has kapha problems in the spring season, such as hay fever, colds, congestion, sinus headaches, and weight gain, such a person should watch his diet and eliminate kapha-producing food like wheat, watermelon, cucumber, yogurt, cheese, candy, ice cream, and cold drinks. (Ice is not good for a kapha person; it will produce congestive disorders.)

The knowledge of the causes of disease, and the understanding that “like increases like” and “opposites balance,” give us all the information we need to maintain or restore our health, simply through conscious attention, moment-to-moment awareness of our behavior.

If I am living consciously, I may observe that after I ate yogurt two weeks ago, I felt congested and a cold developed. Then it cleared up and I was okay for a few days. When yogurt comes my way again, the memory will come up and my body will say, “Hey, last time you ate yogurt, you got sick!” If I bring lively awareness and listen to my body, it will tell me, “I don’t want yogurt.” To listen to the body’s wisdom, the body’s intelligence, is to be aware, and this is one of the most effective ways to prevent disease.

Developing an awareness of the potential causes of imbalance, and of one’s moment-to-moment state of well-being, is the necessary first step to maintaining health. The

second step is taking action.

Taking Action to Modify the Cause

You can't control the weather, but you can dress properly, so that cold winds, or rain, or summer's heat will not aggravate the doshas. Changes in the weather are a potential cause of doshic imbalance. Windy, cold, dry weather will aggravate vata dosha; hot, sticky weather is sure to provoke pitta; cold, cloudy, wet weather will increase kapha dosha. Once we have knowledge and understanding, it is time to take action. Put on a hat, a scarf, a warm coat; stay out of direct sunlight. Modify the cause.

Potential causes of illness and imbalance are constantly arising, both within us and on the outside. The weather is changing, our surroundings are changing, our thoughts and feelings are changing, and stressful situations are coming and going. In response to these changes, we have to act skillfully. As the Bhagavad Gita says, "Skill in action is called *yoga*."

I have to be smart enough to know my previous history and to learn from it. When I eat garbanzos, I get a stomachache, so *this* time I should not eat them. Or if there is nothing to eat except garbanzos, then I can add cumin powder, ghee, and a little mustard seed, and it will be suitable for me to eat. The garbanzos' dry, light vatagenic effect will be modified by the moist, oily ghee and the warming spices.

A substantial part of the Ayurvedic pharmacy is the Ayurvedic art of cooking. Adding specific seasonings changes the property of food and can cause a "forbidden" food, one that might have provoked imbalance, to become acceptable. Some people, for example, are sensitive to potatoes. Potatoes give them gas and little aches and pains in the muscles and around the joints. But if they peel off the skin and sauté the potato with ghee and a little turmeric, mustard seed, cumin powder, and cilantro, it mitigates the vata-provoking property of the potato and the body can then handle it. One can take action to modify the cause; the body's response will be different, and that particular causative factor will not have an adverse effect.

This principle applies equally well to psychological factors. You may know that watching violent movies upsets you and gives you nightmares. The violent imagery disturbs your doshic balance, provoking anxiety and fear. You have observed this happening to you; the next time you are confronted with the "opportunity" to subject yourself to a violent movie, you can just say no.

It keeps coming down to the same central issue: consciousness, awareness, finding out, "What is my role in this situation? What do I know? What can I do?"

Restoring Balance

The first step in staying healthy is developing awareness of the potential causes of disease so you can avoid them or deal with them intelligently. The second step is taking action to modify causes you can't avoid or control (such as the weather). The next step is to restore balance once it begins to be lost. The main method for doing this is to apply the opposite quality or qualities.

If you're cold, have some hot soup or take something warm to drink. If you're agitated or upset (perhaps you watched that violent movie against your better judgment), sit down and do some meditation to calm your mind and emotions. If your pitta has been provoked and you're feeling hot under the collar, take a swim or have some sweet cooling fruit.

This principle seems so simple and makes such good sense that it is easy to overlook it in practical daily life. But it is extremely powerful and effective. If you apply it, you will find that you can quickly and effortlessly restore balance to your mind and body.

Techniques for Cleansing and Purification

Now we have to consider still another level of self-healing. What if you haven't taken the opportunity to develop awareness, to modify the cause, or to apply opposite qualities to restore balance, and you have begun to get sick? What to do now?

The principle of opposites is almost universally valid and helpful at any stage of disease. But once disease has begun to develop, it will not be sufficient. At this stage it becomes necessary to use techniques for cleansing and purifying your body of excess doshas and accumulated toxins.

As we have seen, when the doshas are aggravated because of poor diet, unhealthy lifestyle, negative emotions, or other factors, they first affect agni (the body's biological fire, which governs digestion and assimilation). When agni becomes weakened or disturbed, food is not properly digested. The undigested, unabsorbed food particles accumulate in the gastrointestinal tract and turn into the toxic, sticky substance called ama. In the third ("spread") stage of the disease process, ama clogs the intestines, overflows through other bodily channels such as the blood vessels, and infiltrates the bodily tissues, causing disease.

Ama is thus the root cause of disease. The presence of ama in the system can be felt as fatigue, or a feeling of heaviness. It may induce constipation, indigestion, gas, and diarrhea, or it may generate bad breath, a bad taste in the mouth, stiffness in the body, or mental confusion. Ama can most easily be detected as a thick coating on the tongue.

According to Ayurveda, disease is actually a crisis of ama, in which the body seeks to eliminate the accumulated toxicity. Thus the key to prevention of disease—once ama has begun to build up—is to help the body eliminate the toxins.

To remove ama from the system, Ayurveda employs many internal cleansing

programs. One of these, most widely known in the West, is a five-procedure program known as *panchakarma* (“five actions”). The *panchakarma* programs used at Ayurvedic treatment centers include prepurification methods to prepare the body to let go of the toxins, followed by the purification methods themselves.

The first preparatory step is internal oleation. The patient is asked to drink a specific, small quantity of ghee (clarified butter) every day for several days. The ghee creates a thin film in the body’s channels that lubricates them, allowing the ama lodged in the deep connective tissues to move freely, without sticking to the channels, to the gastrointestinal tract for elimination. Internal oleation is done for three to five days or even longer, depending on the individual circumstances.

This is followed by external oleation in the form of oil massage (*snehana*) and sweating (*swedana*). Oil is applied to the entire body with a particular kind of massage that helps the toxins move toward the gastrointestinal tract. The massage also softens both the superficial and deep tissues, helping to relieve stress and to nourish the nervous system. Then the individual is given a steam bath, which further loosens the toxins and increases their movement toward the gastrointestinal tract.

After three to seven days of these procedures, the doshas will have become well “ripened.” At this point the physician will determine that the patient is ready to eliminate the aggravated doshas and accumulated ama. One of the five *karmas* or actions is selected as the most expedient route to eliminate the excess doshas. These procedures may include:

- therapeutic vomiting (*vamana*) to remove toxins and excess kapha from the stomach;
- purgation or laxative therapy (*virechana*) to help remove ama and excess pitta from the small intestines, colon, kidneys, stomach, liver, and spleen;
- medicated enema therapy (*basti*) to help remove excess vata from the colon. Aggravated vata is one of the main etiological factors in the manifestation of diseases. If we can control vata through the use of *bastis*, we have gone a long way toward eliminating the cause of the vast majority of diseases.
- *nasya* or nasal administration of medication, in which dry herbal powders or oils such as ghee are inserted into the nose to help remove accumulated doshas in the head, sinus, and throat areas, and to clear up breathing.
- *rakta moksha*, purification of the blood, which is traditionally done in one of two ways. Bloodletting, in which a small amount of blood is extracted from a vein, is one method, though it is illegal in the United States and is therefore not available here. The second way is to cleanse the blood using blood-purifying herbs such as burdock.

Panchakarma is not the only method used by Ayurveda to remove ama from the body. Depending on the individual’s strength and the seriousness of the disease, one of two main approaches will be employed. If the person is weak and debilitated and the disease

is strong, the preferred method is palliation and pacification (*shamanam*), which neutralizes ama through gentler methods of purification, including herbs. If the patient has more strength and energy and the illness is not so complicated or serious, then *panchakarma* is appropriate.

IMPORTANT NOTE: *Panchakarma* is a special, powerful procedure requiring guidance from a properly trained medical staff, not just someone with a modest amount of Ayurvedic training. It is performed individually for each person, with his or her specific constitution and medical condition in mind, and it requires close observation and supervision at every stage, including post-*panchakarma* support.

Should You Use Ghee?

The use of ghee for internal oleation is recommended for most people. However, *individuals with high blood levels of cholesterol, triglycerides, and sugar should not use it. So before you begin your home treatment, see a doctor and have your blood tested for these factors.*

If they are within the normal range, there is no problem. If they are high, then instead of ghee use flaxseed oil, which provides effective oleation and also contains fatty acids, which help to reduce cholesterol levels.

Take 2 tablespoons of the flaxseed oil three times a day for three days, fifteen minutes before eating.

A Simple Home Purification

Both for periodic prevention (to reverse any buildup of ama) and to deal with a specific health problem, *panchakarma* is a highly recommended art of cleansing and detoxification. If you are not near a center where *panchakarma* is available under the supervision of a trained Ayurvedic physician, you can do an effective purification program at home.

Begin your home detoxification program with internal oleation. For three days in a row, take about 2 ounces of warmed, liquefied ghee early in the morning. (See [appendix 2](#) for instructions on making ghee.) For a vata person, take the ghee with a pinch of rock salt. For a pitta individual, take the 2 ounces of ghee plain. The kapha individual should add to the ghee just a pinch of *trikatu* (a mixture of equal amounts of ginger, black pepper, and *pippali*, or Indian long pepper).

The ghee provides internal oleation and lubrication, which is necessary so that the ama or toxins begin to come back from the deep tissue to the gastrointestinal tract for elimination.

After your three days of internal oleation, it is time for external oleation. For the next five to seven days, apply 7 to 8 ounces of warmed (not hot!) oil to your body from head to toe, rubbing it in well. The best oil for vata types is sesame, which is heavy and warming; pittas should use sunflower oil, which is less heating; kaphas do best with corn oil. You can do this oil massage for fifteen to twenty minutes.

After the oil is well rubbed in and absorbed, take a hot bath or shower. Then wash with some Ayurvedic herbal soap, such as *neem*. Let some of the oil remain on your skin.

The ancient Ayurvedic textbooks recommend rubbing some chickpea flour over the skin to absorb and help remove the oil. This works very well to remove the oil, but it is more suited to a culture in which individuals bathe outdoors. Today, if you use chickpea flour, be aware that oil, flour, and hot water combine into a formidable mass that can easily clog your plumbing. Flushing the drain with extra hot water immediately following your bath can help.

During your home purification, every night at least one hour after supper take ½ to 1 teaspoon of *triphala*. (For information on *triphala*, see [appendix 2](#).) Add about half a cup of boiling water to the *triphala* powder, and let it steep ten minutes or until it has cooled down, then drink it. Along with its many healing and nourishing properties, *triphala* is a mild but effective laxative. It will provide the benefits of a more potent *virechana* or purgative treatment, but more gently and over a longer span of time. *Triphala* is safe and can be effectively used for months at a time.

To complete your home *panchakarma* treatment, on the last three days perform an Ayurvedic medicated enema, or *basti*, after your hot bath or shower. Use *dashamoola* tea for the enema. Boil 1 tablespoon of the herbal compound *dashamoola* in 1 pint of water for five minutes to make a tea. Cool it, strain it, and use the liquid as an enema. (See instructions for *basti* in [appendix 3](#).) Retain the liquid as long as you comfortably can. And don't worry if little or no liquid comes out. For certain individuals, particularly vata types, the colon may be so dry and dehydrated that the liquid may all be absorbed. This is not harmful in any way.

This *snchana* (oleation both internal and external with ghee and oil), *swedana* (sweating using a hot shower or hot bath), and *virechana* (purgation) using *triphala*, followed by *basti* using *dashamoola* tea, constitute an effective *panchakarma* that you can easily do on your own at home.

During this entire time it is important to get plenty of rest, and to observe a light diet. From day four to day eight, eat only *kitchari* (equal amounts of basmati rice and mung dal cooked with cumin, mustard seed, and coriander, with about 2 teaspoons of ghee added to it). Kitchari is a wholesome, nourishing, balanced food that is an excellent protein combination. It is easy to digest and good for all three doshas, and it is also cleansing.

Be your own healer. Do this simple home purification, preferably at the junction between seasons. Take responsibility for your own healing. You will start to experience

a great change in your thinking and in your feelings, and you will really fall in love with your life!

Schedule for Home Panchakarma

Here is an ideal schedule for your home *panchakarma* purification treatment:

DAY	INTERNAL OIL	EXTERNAL OIL & BATH	BASTI	DIET
1	X			Your doshic diet
2	X			"
3	X			"
4		X		Monodiet—kitchari with coriander/cumin/fennel tea
5		X		"
6		X	X	"
7		X	X	"
8		X	X	"
9				Kitchari with steamed vegetables

Rejuvenation and Rebuilding

The purpose of *panchakarma* is not just to get well but to purify the body and strengthen it so that future diseases will not occur, and you can enjoy a long life in good health. In this regard the *panchakarma* purification can be seen as a preliminary to rejuvenation. If you want to dye your shirt, don't color it while it's dirty. Wash it first, then dye it. The washing is the *panchakarma* detoxification program, and the dyeing is the rejuvenation and revitalization.

Ayurvedic rejuvenatives (*rasayanas*) bring renewal and longevity to the cells, and when the cells live longer, the person lives longer. *Rasayanas* give strength, vitality, and longevity, strengthen tone, increase energy, and build immunity. The body's various agnis become more robust, so health becomes more robust.

For a vata individual, an excellent rejuvenative tonic is the herb *ashwagandha*. Take 1 teaspoon of *ashwagandha* in a cup of hot milk twice a day, morning and evening.

An excellent rejuvenative herb for pittas is *shatavari*. Take 1 teaspoon twice a day in a cup of warm milk. Kaphas can use *punarnava*, 1 teaspoon twice a day, but in a cup of warm water.

You can also use various herbal mixtures designed to tonify the system, such as the traditional recipe *chyavanprash*.

Three Cautions About Home Panchakarma

1. *Panchakarma*, even in this gentle home program, has a powerful effect and

should be done only by individuals of sufficient strength. If you are anemic, or feel weak and debilitated, even this home procedure is not for you.

2. Do not do *panchakarma* in a clinic, or even this home purification, if you are pregnant.

3. One result of *panchakarma*, even in this mild home version, is that the deep connective tissue may start releasing unresolved past emotions, such as grief, sadness, fear, or anger along with the built-up ama and excess doshas. If this happens, make yourself some Tranquillity Tea (see recipe on [this page](#)), and meditate, using whatever method you have learned or the Empty Bowl meditation described in [chapter 7](#). The releasing of emotions may happen weeks or even several months after you finish your home *panchakarma*.

To make your rejuvenation more effective, after completing your *panchakarma* purification program, set some time aside to build up your strength. Whether you take a weekend, a week, a month, or even more, use the time as a purposeful period of rest, relaxation, and rebuilding of body, mind, and spirit. Here are a few suggestions:

- Get plenty of rest.
- Observe celibacy so that you don't waste your vital energy.
- Eat carefully, according to the guidelines for your constitution.
- Meditate and do yoga postures regularly.

Quite a few more suggestions for rejuvenative herbs, foods, and tonics for all constitutional types are offered in [Part III](#). See, for example, the recommendations under "[Low Libido](#)" and "[Fatigue](#)."

Self-Esteem

Self-esteem is at the core of healing. Because of the connectedness of mind and body, our sense of self-esteem is our cells' sense of self-esteem. This is because, according to Ayurveda, every cell is a center of intelligence and awareness. Every cell carries the sense of self for its own survival. It is the sense of self in the cell that maintains the size and shape of the cell. Self-esteem, self-confidence, and self-respect promote cellular intelligence, which is necessary for proper cell function and immunity.

Modern science is just now acknowledging the importance of the mind-body connection, but knowledge of it has been part of Ayurveda for five thousand years. Our sense of self, our attitudes and understandings, our feelings, are all psychobiological events. Self-esteem is one such event, one that is strengthening to our cells and to all aspects of our bodies. A lack of self-confidence and self-love is detrimental.

Cancer is an example of this lack. Cancer cells have lost their intelligence and grow

separate from the body. They are irregular and robust and have an isolated, selfish sense of self which is in conflict with the life of normal, healthy cells. When cancer occurs, it's as if a war is going on between the cancerous cells and the healthy cells. If the healthy cells are strong enough in self-esteem, they can conquer and kill the cancer cells. But if we do not have enough self-esteem and self-respect, then the cancer cells will win and will conquer the healthy cells.

Thus self-esteem is important for maintaining immunity. If you love yourself as you are, you will develop confidence, and that will heal disease. That is why cellular immunity, or natural resistance, depends upon self-esteem.

Ayurvedic Lifestyle: The Ultimate Preventive Medicine

How you live your daily life is the key factor in determining your health and your quality of experience. It is also the factor over which you have the most control. You can't control the weather or your genetic makeup, but what you do every day either builds up your health, vitality, and resistance to disease, or wears you down. Your moment-to-moment choices—what to eat, how much to eat, how to respond to others, whether to exercise or not, how late to stay up at night, and so on—play a major role in your mental and physical health.

But how do you create your lifestyle, the rhythms of your daily living? Is it just pure habit, based on how your parents lived and how you grew up? Should the time you wake up be dictated by when you need to get to work, and should what you eat be determined by what's available at the fast-food shops? If you decide to take control of your lifestyle and structure new, healthier habits, what principles will guide you?

According to Ayurveda, you couldn't do better than to strive to live your life in harmony with Mother Nature.

In Tune with Nature

Ayurveda flourished in a civilization vastly different from life today, a world in which human life was intimately intertwined with the life of nature. The great rhythms and forces of nature—the alternation of day and night, the rhythmic cycle of seasons—all affect us, as do the inevitable seasons and cycles of human life, birth and growth, aging and death. Through the plants we eat for food, the water we drink, and the air we breathe in common with all beings, we are inextricably one with nature.

The sages of settled mind who unfolded the wisdom of Ayurveda saw this, and they saw that the master key to good health is to get ourselves into harmony with nature. Thus the ideal Ayurvedic daily routine that follows is, as you will see, based on patterns of nature.

Being in tune with nature also means being in tune with *your* nature, your constitution or *prakruti* (which means nature). It means being true to your own nature, to how you are built, mentally and emotionally as well as physically. It means that your food and exercise requirements, how much you need to sleep, how much sexual activity is healthy for you, what kind of climate is beneficial, all revolve around your doshic makeup, your individual nature.

Living in accordance with nature and natural law means continually balancing our

inner ecology by adjusting to our ever-changing environment.

Ayurvedic Daily Routine

A daily routine is essential for maintaining good health and for transforming our body, mind, and consciousness to a higher level of functioning. A regulated daily routine puts us in harmony with nature's rhythms. It establishes balance in our constitution and helps to regularize our biological clock. It indirectly aids in digestion, absorption, and assimilation of food, and it generates self-esteem, discipline, peace, happiness, and long life.

Waking up too early or too late, undisciplined eating, staying up too late, job stress, and untimely bowel movements are a few habits that can unsettle us. Regularity in sleeping, waking, eating, and eliminating, indeed following a regular daily routine, brings discipline to life and helps maintain the integrity of the doshas.

Our body is a clock. Or rather, it is several clocks at once. According to Ayurveda, every organ has a definite time of maximum functioning. Morning time is the lung time. Midday is stomach time, when we feel hungry. Afternoon is liver time, and late afternoon is when the colon and kidneys operate at their peak.

This biological clock works in conjunction with the doshic clock. Morning and evening (dawn and dusk) are the times when the influence of vata is greatest. In the early morning, from about 2 A.M. to sunrise, vata creates movement and people awaken and tend to excrete waste. Again in the late afternoon, from about 2 P.M. until sunset, the influence of vata makes one feel light and active.

Early morning and evening are kapha times. From sunrise until about 10 A.M., kapha makes one feel fresh but a little heavy. Then again in the evening, from about 6 P.M. until around 10, kapha ushers in a period of cooling air, inertia, and declining energy.

Midday and midnight are pitta times. At midmorning, kapha slowly merges into pitta, and by noon one feels hungry and ready for lunch. Again from 10 P.M. until around 2 A.M., pitta is at its peak, and food is digested.

Thus there is a daily cycle of vata–pitta–kapha:

6 A.M.–10 A.M. = kapha

10 A.M.–2 P.M. = pitta

2 P.M.–6 P.M. = vata

6 P.M.–10 P.M. = kapha

10 P.M.–2 A.M. = pitta

2 A.M.–6 A.M. = vata

So there is a doshic clock (when a particular dosha is operating at its peak) and a biological clock (when a particular organ is operating at its peak). Based on these clocks, the Ayurvedic sages developed the *dinacharya*, or daily routine. This daily routine is the art of bringing harmony between the biological and doshic clocks and chronological time. Here are its most salient features:

WAKE UP EARLY

It is beneficial to wake up before the sun rises. At this time of the morning, pure qualities are lively in nature, which can bring freshness to the doors of perception and peace of mind.

Ideally, vata people should get up at about 6 A.M., pitta people by 5:30, and kapha people by 4:30. This is the ideal: do the best you can. If you can wake up at 5:30, it will be very good.

Right after awakening, look at your hands for a few moments, then gently move them over your face, neck, and chest down to your waist. This will bring more alertness.

SAY A PRAYER

It is good to start the day by remembering the Divine Reality that is our life. You may do this in your own way, as your religion or personal experience dictates. Or you may use this simple prayer:

*Dear God, you are inside of me
Within my very breath
Within each bird, each mighty mountain.
Your sweet touch reaches everything
and I am well protected.
Thank you God
for this beautiful day before me.
May joy, love, peace, and compassion
be part of my life
and all those around me on this day.
I am healing and I am healed.*

WASH YOUR FACE, MOUTH, AND EYES

Splash your face with cold water a couple of times. Swish and rinse out your mouth. Then wash your eyes with cool water, and massage the eyelids by gently rubbing them.

Blink your eyes seven times, and then rotate your eyes in all directions: side to side, up and down, diagonally, clockwise, and counterclockwise. All this will help you feel alert and fresh. (See “[Eyes—Ayurvedic Care](#)” in [Part III](#) for more on eye exercises and eye washes.)

DRINK A GLASS OF WATER

Drink a glass of room-temperature water, preferably from a pure copper cup or tumbler. (Fill the cup the night before and let it sit overnight.) If the water is too cold, it may provoke kapha disorders such as colds, coughs, and sore throat. For kapha and vata individuals, it is actually better to drink hot water, but for a pitta person, lukewarm is best.

This water will not be absorbed but will wash the gastrointestinal tract and flush the kidneys. It also stimulates peristalsis in the intestines, stimulates the descending colon and ileocecal valve, and helps with having a good bowel movement.

It is not a good idea to start the day with coffee or black tea. These drain kidney energy, overstimulate the adrenals, and promote constipation. They are also habit-forming.

EVACUATION

Sit (or better, squat) on the toilet, and have a bowel movement. Even if you don’t have the urge, sit for a few minutes, without forcing. If you do this every day, following your glass of warm water, the habit will develop. (See “[Constipation](#)” for suggestions on promoting healthy bowel movements.)

After evacuation, wash the anal orifice with warm water, then wash your hands with a gentle soap.

CLEAN YOUR TEETH AND TONGUE

Use a soft toothbrush for your teeth, and an herbal powder made of astringent, pungent, and bitter herbs. (See “[Teeth and Gums—Ayurvedic Care](#)” for further suggestions.)

Scrape your tongue every morning. This is an important part of daily hygiene, from which you can learn a lot about your health and habits. Note how coated your tongue is, and how your breath smells. If you get the smell of last night’s pizza, that means the food is not yet thoroughly digested. If there is a lot of coating on the tongue, that means there is much ama or toxicity in the system. Perhaps you ate too late, or your dinner was hard to digest.

If there is ama on the tongue and a bad smell on the breath, don't eat breakfast. Eating breakfast is not good if you have not digested last night's dinner.

You can see that this daily regimen brings more awareness. By following this routine, you come in contact with your body and observe the functioning of your system. You know exactly what is happening. This knowledge gives you the power to create better health by altering your behavior.

To scrape your tongue, use a stainless-steel tongue scraper. You can also use a spoon. Gently scrape from the back or base of the tongue forward, until you have scraped the whole surface (seven to fourteen strokes). In addition to removing bacteria from the tongue, scraping sends an indirect message to all the internal organs and stimulates gastric fire and digestive enzymes.

GARGLE

To strengthen the teeth, gums, and jaw, to improve the voice and remove wrinkles from the cheeks, gargle twice a day with warm sesame oil. Also, hold the oil in your mouth and swish it around vigorously. Then spit it out and gently massage the gums with your index finger.

NOSE DROPS (*NASYA*)

Now put 3 to 5 drops of warm ghee, *brahmi* ghee, or sesame oil into each nostril. This helps to clean the sinuses and also improves voice, vision, and mental clarity. In dry climates, and during cold winters when the house is heated with dry air, nose drops help to keep the nostrils lubricated. (For more on *nasya*, see [appendix 3](#).)

The nose is the doorway to the brain. Use of nose drops nourishes *prana* and enlivens consciousness and intelligence.

OIL MASSAGE

Take 4 or 5 ounces of warm (not hot) oil, and rub it all over your head and body. Gently massaging the scalp with oil can bring happiness into your day, as well as help prevent headaches and slow balding and graying of your hair. If you oil your body again before going to bed, it will help induce sound sleep.

Oil massage improves circulation, calms the mind, and reduces excess vata. The skin of the entire body becomes soft, smooth, and brightened.

Best Oils by Body Type

For Ayurvedic oil massage, use one of the following oils, according to your

constitutional type:

Vata = sesame oil

Pitta = sunflower oil

Kapha = corn oil

BATHING

Following your oil massage, take a bath or shower. Bathing is cleansing and refreshing. It removes fatigue, brings energy and alertness, and promotes long life. Bathing every day brings holiness into your life.

EXERCISE

Everyone should do some exercise every day. A walk in the fresh early-morning air and some yoga stretching are good enough for many people; some additional aerobic exercise may also be beneficial, depending on your *prakruti*.

Kapha individuals, with their stronger, heavier physiques, can do the most strenuous exercise, and they benefit from it. Jogging, bicycling, tennis, aerobics, hiking, and mountain climbing are great for kaphas (though they don't like such vigorous exercise!). Pittas do well with a moderate amount (swimming is especially helpful for cooling pitta), while vata individuals do best with quieter exercises like walking, easy swimming, or yoga *asanas*.

As a general rule, Ayurveda recommends exercising up to one half of one's capacity. A good gauge is to exercise until sweat forms on the forehead, under the arms, and along the spinal column. Straining is absolutely not recommended.

Yoga stretching is recommended for all body types. Postures particularly beneficial for vata individuals include the Sun Salutation (twelve cycles, done slowly). The most important seat of vata in the body is in the pelvic cavity, and any exercise that stretches the pelvic muscles helps to calm vata. These include the Forward Bend, Backward Bend, Spinal Twist, Shoulder Stand, Plow, Camel, Cobra, Locust, Cat, and Cow poses, and Leg Lifts. The Headstand, Half Wheel, and Yoga *Mudra* are also beneficial. (For illustrations of yoga postures see [appendix 4](#).)

The major seat of pitta is the solar plexus, so exercises that stretch the muscles around the solar plexus are especially beneficial for individuals with a pitta *prakruti* and will help to pacify pitta. These include the Fish, Boat, Camel, Locust, and Bow poses. Pittas should also do the Moon Salutation (sixteen cycles, moderately fast). Avoid the Headstand, Shoulder Stand, Plow, and other inverted poses.

The important seat of kapha is in the chest. Exercises that stretch the pulmonary

cavity and increase circulation in the chest are effective for kaphas and will help relieve and prevent bronchial congestion, cough, and other kapha illnesses. Beneficial postures include the Sun Salutation (twelve cycles, done rapidly) and the Shoulder Stand, Plow, Locust, Bridge, Peacock, Palm Tree, and Lion postures. (Illustrations of yoga postures are found in [appendix 4](#).)

PRANAYAMA

After finishing your exercises, sit quietly and do some deep breathing: twelve Alternate Nostril breaths for vata; sixteen Cooling (*shitali*) breaths for pitta; one hundred Breath of Fire (*bhastrika*) breaths for kapha. (Instructions for these breathing exercises appear in [chapter 6](#).)

MEDITATION

End your *pranayama* by going right into your meditation. Whatever system or technique of meditation you do, do it now. If you don't presently do any meditation practice, try the Empty Bowl meditation explained in [chapter 7](#). You will find that meditation brings peace and balance into your life.

BREAKFAST

Now it is time for you to enjoy your breakfast! Your meal should be fairly light in the hot months, and more substantial in cold weather. Vata and pitta persons should eat some breakfast; kaphas are usually better off if they don't eat, since eating during kapha time will increase kapha in the body. Follow the dietary guidelines for the three doshas which appear in [chapter 8](#).

Mealtimes for Each Dosha			
	VATA	PITTA	KAPHA
Breakfast	8 A.M.	7:30 A.M.	7 A.M.
Lunch	11 A.M.–Noon	Noon	Noon–1 P.M.
Supper	6 P.M.	6–7 P.M.	7–8 P.M. No Snacks!

OFF TO WORK

After breakfast go to work or to your studies if you are a student. While walking to work (or to and from your car, the train, or the bus), be aware of every step. Carry your meditative mind with you. When you look at your boss or colleague, at the same time look inside. Then your work will become a meditation. You will find yourself looking at

others with compassion and greater awareness.

It is better not to drink tea or coffee at work. If you are thirsty, have some warm water or some fruit juice if you prefer.

LUNCHTIME

By around noon you will become quite hungry. Have a bowl of soup and some salad, or some rice and vegetables, following the guidelines for your constitution. And don't drink too much during your meal. Take a cup of water (preferably warm but definitely not iced), and just take a sip between two mouthfuls of food. Drinking a little water improves digestion.

One can drink a cup of water an hour before lunch or an hour after lunch, but not immediately afterward, as that slows down digestion and creates ama.

SIT STRAIGHT, WALK STRAIGHT

Maintain your vertebral column straight. When you keep the backbone straight, energy flows upward and you maintain your awareness. It is difficult to be aware when the spine is crunched.

TAKE A WALK

When you've finished your job for the day, go home and take a walk, alone, silently, in the woods, in the park, or on the bank of the river. Listen to the water, the birds, the rustle of leaves, the barking of a dog. In that listening, the meditative mind is regained.

In this way, every day becomes heavenly. Every day becomes a celebration, something new. That's why the routine is most important. The discipline of the routine leaves room for awareness, openness, and freshness.

SUPPER TIME

At around six o'clock (see "[Mealtimes for Each Dosha](#)" box) have your supper. If you like to cook, you can cook according to the *Ayurvedic Cookbook for Self-Healing* that I have written with my wife, Usha Lad (see the [Reading List](#)). Don't watch television while eating. Pay attention to the food. Eating food with attention becomes meditation. And when you are eating with awareness, you will not eat too much; you'll eat just a sufficient amount.

It is better to eat when the sun is up. Eating late at night will change the body chemistry, sleep will be disturbed, and you will not feel rested in the morning. If you eat

supper around 6, by 9 the stomach will be empty and you will sleep soundly.

AFTER DINNER

Sing songs while you wash the dishes. Be happy. Keep smiling.

About an hour after dinner, if you are taking *triphala* (an herbal compound that is both strengthening and purifying), take ½ teaspoon with some warm water.

Then if you like, you can watch TV, perhaps some news. You should know what's happening in this world of ours. Or you can read a magazine or a book.

BEFORE GOING TO BED

Before you go to bed, some spiritual reading is important, even if only for a few minutes.

And don't forget to drink a cup of hot milk, with a little ginger, cardamom, and turmeric. Drinking milk at bedtime helps to induce sound sleep. According to Ayurveda, that milk also nourishes *shukra dhatu*, the body's highly refined reproductive tissue.

Rubbing a little oil on the soles of your feet and on the scalp is also soothing and promotes restful sleep.

Finally, before you go to bed, do a few minutes of meditation. Sit quietly and watch your breath. In the pauses between breaths, you'll meet with nothingness, and nothingness is energy and intelligence. Allow that intelligence to deal with your problems. In this way, you'll begin and end your day with meditation, and meditation will stay with you even during deep sleep.

BEDTIME

It is recommended that vatas go to bed by 10 P.M. and sleep on their left side. Pittas should sleep on the right side, retiring between 10 and 11 P.M. The best bedtime for kapha individuals is between 11 and midnight; they should sleep on their left side.

Kapha individuals generally like to sleep about nine hours, and they feel it is good for them. But this is an illusion. Sleeping this long will slow down their metabolism, and they will put on weight and become chubby. The best schedule for them is to stay up until about 11 P.M. or midnight, then to wake up early, around 4:30 or 5:00 A.M. and go out for a walk. That shorter sleep will help to induce a light quality in their body, and they will start losing weight.

SEX

Ayurveda has some definite suggestions about the proper role of sex in our lives. Sex is a tremendous creative force, and through sex people share their love and compassion and can derive great pleasure.

Ideal Wake-up and Bed Times			
	VATA	PITTA	KAPHA
Wake Up	6:00	5:30	4:30
Go to Bed	10:00	10:00–11:00	11:00–midnight

Sex is also correlated with constitutional type. The recommended frequency of sexual activity is quite different for the different types. Kaphas, with their strong constitutions, can make love two to three times a week, whereas the suggestion for vatas is once or at most twice a month. Pitta individuals are in the middle; every two weeks is recommended for them.

Too-frequent lovemaking reduces *ojas*, the body's vital energy, and leaves the person weak and open to diseases. It also aggravates vata dosha.

To restore strength and replenish *ojas*, after each time you make love a massage is helpful, as are nourishing drinks such as almond milk. (See recipe for almond milk on [this page](#).) The best time for lovemaking is between 10 and 11 P.M. Sex in the morning or in the daytime is not recommended.

This entire daily routine is very important.

I set more store by a good regimen that maintains my humors in balance and procures me a sound sleep. Drink hot when it freezes, drink cool in the dog days; in everything, neither too much nor too little; digest, sleep, have pleasure, and snap your fingers at the rest of it.

—Voltaire

Seasonal Routines

The seasons, like the times of day, are characterized by cycles of vata, pitta, and kapha. Maintaining good health during all four seasons requires living in harmony with these natural cycles, continually adjusting to the changes in the outer environment through the food we choose to eat, the type and amount of exercise we do, the clothes we wear, and so on. The suggestions in this section will help you be at your best all year round.

Please remember that you cannot determine the seasons just by dates on the calendar. Ayurveda is a system of natural medicine, which means that you have to see what is happening in nature! In different geographic areas the seasons come at different times and have varied characteristics. In addition, in just one day there may be four

seasons: sunshine and singing birds creating a springlike air in the morning; warm summery breezes at midday; gusts of cool, dry autumnal wind in the afternoon; cold, cloudy, wintry weather after dark. So look at nature as it is, and apply the appropriate principles and practices.

GUIDELINES FOR SUMMER

Summer is hot, bright, and sharp, the season of pitta. Thus the main recommendation for everyone, especially for individuals whose *prakruti* is primarily pitta, is to keep cool and not allow pitta dosha to become aggravated.

- In the morning, as part of your daily routine, rub 5 to 6 ounces of coconut oil or sunflower oil on your body before bathing. Coconut oil is calming, cooling, and soothing to the skin.

- Wear cotton or silk clothing; it is cooling, light, and allows the skin to breathe. Loose-fitting clothes are best; they permit the air to pass through and cool the body.

- The best colors to wear in hot weather are white, gray, blue, purple, and green. Avoid red, orange, dark yellow, and black, which absorb and retain heat and will aggravate pitta.

- Follow the pitta-pacifying diet from the food guidelines in [chapter 8](#). Good fruits for summer include apples, pears, melons, plums, and prunes. Watermelon and lime juice are also good in summer. Try steamed asparagus, broccoli, brussels sprouts, cucumber raita, and basmati rice. Kitchari made of basmati rice and mung dal, with a little ghee and grated coconut, makes a delicious light meal. Avoid sour fruits, citrus fruits, and even beets and carrots, which are all heating. Garlic, onion, chili, tomato, sour cream, and salted cheeses are also not recommended. You can eat more salads in summer than at any other time, as they are cooling, but they are best eaten for lunch. If you eat meat, you can have some light meat—chicken, turkey, or shrimp—once a week. Avoid dark meats, which are heating.

- Don't drink hot water or hot drinks in the summer. Room-temperature or cool drinks are best. Ice and iced drinks, however, inhibit digestion and create toxins (ama) in the body; it is best never to drink them.

- A refreshing drink is cool lassi. Mix 1 part yogurt with 4 parts water, and blend 2 or 3 minutes until creamy. You can add ¼ teaspoon roasted cumin seed before blending, or for a sweet-flavored drink, add 2 tablespoons Sucanat or other sweetener and 1 drop of rose water. The juice of ¼ lime in a cup of cool water with a pinch of cumin powder is also refreshing.

- Working in a hot kitchen provokes pitta. If you cook, cook in the early morning or in the evening. If someone cooks three days in a row, on the fourth day you should treat that person to dinner in a restaurant. This will avoid conflicts in the relationship.

- If you customarily drink alcoholic beverages, avoid whiskey, brandy, rum, and red wine, which are heating. Some cool beer during hot days will be all right.
- This is a season of generalized low energy. Thus it is all right to take a short nap in the daytime.
- If you have to work outside, wear a wide-brimmed hat.
- Wear sunglasses outdoors during the brightest part of the day. Lenses should be smoky gray or green, not red or yellow and especially not blue or purple, which will damage the eyes.
- If you can, work indoors. Have some air conditioning in your car and in your room or office.
- Never lie in the sun in summer. If the weather is very hot, don't wear shorts or short sleeves, but wear loose-fitting clothing to protect your skin. No person having multiple moles should lie in the sun; it may provoke extreme pitta aggravation and lead to skin cancer.
- If you feel really hot, take a swim in a cool lake or pool, then drink a little lime juice in water.
- Avoid strenuous exercise. If you are accustomed to running or other vigorous aerobic exercise, do it early in the morning at the coolest part of the day.
- Do some mild yoga exercises and quiet meditation twice a day. Good postures for summer include the Fish, Camel, Boat, Cobra, Cow, and Palm Tree poses. Pitta individuals should not do inverted poses such as Headstand and Shoulder Stand, which can be pitta-provoking. Also, do the Moon Salutation. (See illustrations in [appendix 4](#).)
- Perform *shitali pranayama*, a cooling breathing exercise described in [chapter 6](#).
- Certain jewelry and gems will help cool pitta. These include a necklace of sandalwood beads, a jade or pearl necklace, amethyst crystals, moonstone, malachite crystals, and any silver jewelry.
- In the evening, after dinner, go for a walk in the moonlight. Dress in white clothes, with white flowers in your hair or a garland of white flowers around your neck.
- You can go to bed a little later on summer nights, around 11 P.M. or midnight. Rub some coconut oil on your scalp and the soles of your feet for a cooling effect before going to sleep. Sleep on your right side.
- Sandalwood, jasmine, and khus oils are cooling and are good fragrances to wear in the summer. Also, place a few drops of sandalwood oil on your pillow, and you will be sleeping with sandalwood perfume all night.
- Sex should be minimized in the summer, as it is heating and will provoke pitta. If you want to have sex, do it between 9 and 10 P.M., when it is cooler but not yet pitta time.

During the summer, the sun evaporates the moisture of the earth and therefore induces hot, dry and sharp qualities in the atmosphere, resulting in pitta aggravation. In summer sweet, cold, liquid, and fatty food and drinks are beneficial. One should avoid or minimize excessive exercise and sex, alcohol, and diets which are salty, sour, pungent, or hot. In summer time one should enjoy forests, gardens, flowers, and cool water. During the night one should sleep on the open airy roof of the house, which is cooled by the rays of the moon.

—*Charaka Samhita*

GUIDELINES FOR FALL

Autumn is dry, light, cold, windy, rough, and empty (trees drop their leaves). All these qualities provoke vata dosha. So naturally the guidelines for autumn revolve around pacifying vata.

- If you can, wake up early, around 5 A.M., when the air is calm and the birds are not yet out of bed. There is an extraordinary silence and peace at this time of day.
- Good yoga *asanas* for the autumn season include the Lotus pose, Forward Bend, Backward Bend, Vajrasana (Sitting on the Heels), Spinal Twist, Camel, Cobra, Cow, and Cat. Shoulder Stand and Headstand are all right in moderation. Also do the Sun Salutation a minimum of twelve cycles. As a maximum, you can do as many Sun Salutations as your age, but you have to build up to this through regular daily practice. Finish your yoga session with *savasana*, the relaxation pose.
- Gentle Alternate Nostril *pranayama* is good following yoga postures. Then meditate for at least ten to fifteen minutes.
- Every morning before your bath or shower, rub 6 to 9 ounces of warm sesame oil all over your body, from head to toe. Sesame oil is warming and heavy and will help to balance vata. Then take a nice warm shower. Leave a little of the oil on your skin.
- Good fall colors for pacifying vata are red, yellow, and orange. White is also helpful.
- After your yoga, meditation, and bath, have some breakfast. Try oatmeal, cream of rice, cream of wheat, tapioca, or any grain that will help to settle vata. (See the food recommendations for vata in [chapter 8](#).) For lunch and supper, tortillas, chapatis, basmati rice, mung dal kitchari, and steamed vegetables are all good fall foods to balance vata. Salads are not recommended. Mushy, soft soups and stews are good, and be sure to use some ghee.
- Don't drink black tea or coffee after dinner. Try some herbal tea, such as cumin-coriander-fennel tea (equal proportions), or ginger-cinnamon-clove tea.
- Fasting is not good during the autumn season. It generates too much lightness and

emptiness, which provoke vata.

- Be sure to keep warm. Dress warmly enough both indoors and out. On a windy, gusty day, cover your head and ears.
- Very active, vigorous exercise should be avoided, especially by individuals with a vata constitution.
- A short afternoon nap is acceptable for vatas.
- Try to be in bed by 10 P.M.
- Drinking a cup of warm milk at bedtime is good in the autumn season. It induces sound, natural sleep. Heat the milk until it begins to boil and rise up, then let it cool enough to drink comfortably. You might add a pinch each of ginger and cardamom and a small pinch of nutmeg. These herbs are warming and soothing and will help both with digesting the milk and with relaxation.
- At the junction between summer and fall, a *panchakarma* treatment will help remove excess vata from the system. (See [chapter 4](#).) If you can't go to an Ayurvedic clinic, try the home purification treatment outlined in [chapter 4](#). A crucial component of this treatment should be the *basti* or medicated enema, as follows:
 1. Boil 2 tablespoons of *dashamoola* powder in 1 pint of water for 5 minutes.
 2. Strain out the herbs, and to the liquid add ½ cup warm sesame oil.
 3. When this mixture has cooled to a comfortable temperature, use it for the enema. Try to retain it for 30 minutes.
 4. After half an hour or after a good bowel movement, add another ½ cup warm sesame oil to the rectum. Try to retain this oil for at least 10 minutes.
- This procedure will lubricate the colon, calm vata, and remove stress from the lower back and neck areas. You can do this *basti* once a week during the autumn season to keep vata in check.
- During this season, take particular care to avoid loud noise, loud music (such as rock), fast driving, and too much sexual activity. Avoid cold drafts and cold winds. These all aggravate vata.
- Excellent herbs for pacifying vata dosha in the autumn are *dashamoola* (actually a formula consisting of ten herbs), *ashwagandha*, *bala*, and *vidari*.

GUIDELINES FOR WINTER

In winter, the sky is cloudy, the weather is cold, damp, and heavy, and life in the cities moves slowly; it is generally a season of kapha. A kapha-pacifying regimen should be adopted, especially by kapha individuals. However, certain vata-provoking qualities, such as dry, cold, windy, and clear, are sometimes prominent on winter days, so vata

individuals need to keep this in mind.

- In winter there is no need to get up early. The 5:00 rising time suggested for summer and autumn is not necessary now. Unless you have to get up earlier to go to work, you can get up around 7 A.M.

- After brushing your teeth and scraping your tongue (see the Daily Routine, [this page](#)), do some yoga *asanas*, including the Sun Salutation. Beneficial postures for winter season include the Fish, Locust, Boat, Bow, Lion, and Camel poses, Shoulder Stand and Headstand. These postures help to open the chest, stretch the throat, drain the sinuses, and relieve congestion of the chest.

- Follow your yoga postures with some breathing exercises. *Bhastrika* (Breath of Fire) will cleanse kapha dosha. Follow this with a few minutes of Right Nostril breathing, which promotes circulation and heat. (See instructions in [chapter 6](#).)

- Winter is a season of kapha. So, like slow and steady kapha, don't be in a rush. Be sure to follow your breathing exercises with some quiet meditation.

- After meditation apply some warm sesame oil to your entire body, then take a hot shower. Sesame oil, which is warming, is beneficial for all constitutional types in the winter.

- For a good winter breakfast, have some oatmeal, cornmeal, barley soup, tapioca, kitchari, or poha (cooked rice flakes). About an hour later, drink tea made of these herbs:

dry ginger ½ teaspoon

cinnamon ½ teaspoon

clove a pinch

Boil these herbs in a cup of hot water for five minutes, and drink the tea. It will increase heat and pitta, improve circulation, and eliminate mucus from the system. However, if you have an ulcer, don't drink this tea; it will be too heating.

- Wear bright warming colors such as red and orange.

- Always wear a hat outdoors in winter. More than 60 percent of the body's heat is lost through the head. Also cover your neck and ears.

- For lunch, eat kapha-soothing food but not food that is vata-aggravating. Whole-wheat bread, steamed vegetables, and hot mushy soup with much ghee and some crunchy croutons would be just right.

- If you like to eat meat, Ayurveda says that winter is the time to eat it, because agni (digestive fire) is strong. Chicken and turkey are good choices.

- Although a nap may be acceptable in summer and autumn (especially for pitta and vata individuals, respectively), sleeping in the daytime is not recommended during

winter because it will increase kapha, slow down metabolism, and reduce the gastric fire.

- Ayurveda recommends drinking a little dry red wine—a few ounces at most—in the winter to improve digestion and circulation. *Draksha* (Ayurvedic herbal wine) is a good choice. Take 4 teaspoons of draksha with an equal amount of water before or after dinner.

- The winter season, when the sky is covered by clouds and it is gray outdoors, is conducive to loneliness and depression. Following the kapha-pacifying routine will definitely help. If possible, don't be away from your wife, husband, boyfriend, or girlfriend in the winter. When it is cold outside and inside there is no one to sleep with, you will definitely feel lonely. When you have your companion in the winter, you feel great!

- At the end of the day, rub a small amount of sesame oil on your scalp and on the soles of your feet.

- According to Ayurvedic tradition, winter is the season in which you can have sex more often.

- The best herbs for winter are *pippali*, licorice, ginger, *punarnava*, black pepper, and *kutki*. You can also use the herbal tonic *chyavanprash*.

- Some light fasting, for a day or two, is all right if your digestive fire is strong. You can drink some apple juice or pomegranate juice during your fast if you like.

- At the junction between autumn and winter, individuals who tend to get kapha problems in winter (colds, coughs, flu, sinus congestion, and the like) should receive *panchakarma* at an Ayurvedic clinic, under the care of an Ayurvedic physician, to remove excess kapha dosha. This will help give you a problem-free winter.

GUIDELINES FOR SPRING

Spring is the king of the seasons. In the Bhagavad Gita Lord Krishna reveals his predominant attributes in the eleventh chapter: “I am the Soul in the body, the Mind in the senses, the Eagle among birds, the Lion among animals. Among all the trees I am the sacred *Bodhi* tree, and of the seasons, I am Spring.”

In spring Mother Earth wakes up and causes sprouting; energy moves up; everything is blooming and flowering, full of colors and greenery. People feel energetic and love to go outdoors. It is the season of celebration.

The qualities of spring are warm, moist, gentle, and unctuous. Due to the warmth, the accumulated snow and ice of winter begin to melt. Similarly, accumulated kapha in the body starts liquefying and running. That is why so many people get spring colds. In addition, as flowers shed their pollen, fragrance, and perfume, making vata and pitta

people happy, many kapha individuals get hay fever and allergies.

As early winter carries some of the qualities of fall, so early spring is much like winter, and many of the recommendations are the same. For example, *panchakarma* is highly recommended, to clear the system of accumulated kapha dosha and help prevent allergies, hay fever, colds, and sinus congestion.

- Good herbs for spring include ginger, black pepper, *pippali*, and a tea made of cumin, coriander, and fennel in equal proportions. *Sitopaladi*, *punarnava*, and *sudarshan* are also beneficial.

- Strictly avoid heavy, oily food. Also, it is better not to eat sour, sweet, and salty foods, as they provoke kapha. Stay away from dairy products, especially in the morning. Avoid ice cream and cold drinks, which are especially kaphagenic.

- Favor bitter, pungent, and astringent foods. All legumes, such as yellow split peas, red lentils, and garbanzo and pinto beans, are recommended. Radishes, spinach, okra, onions, and garlic can be used, along with hot spices such as ginger, black pepper, cayenne pepper, and chili pepper. (But don't overdo these hot spices if your constitution is predominantly vata or especially if it is pitta.) After each meal, drink some tea made from ginger, black pepper, and cinnamon.

- Use less ghee and fewer dairy products, and use more honey, which is heating. A cup of hot water with a teaspoon of honey helps balance kapha during the spring season. (But never cook honey; it clogs the subtle channels and acts as a toxin in the system.) You can end your meal with a cup of freshly made lassi (see [this page](#) for recipe).

- For those who eat meat, chicken, turkey, rabbit, and venison are permissible; seafood, crab, lobster, and duck are not recommended during spring season.

- This is a good season to observe a juice fast of apple, pomegranate, or berry juice.

- Wake up early, and go for a morning walk. Also, do the Sun Salutation and kapha-reducing yoga postures, such as the Fish, Boat, Bow, Locust, Lion, and Camel poses, and the Headstand and/or Shoulder Stand. *Bhastrika* and Right Nostril breathing are also helpful (see [chapter 6](#)).

- Sleeping in the daytime aggravates kapha; hence it is not recommended during this season.

As spring advances and the weather heats up, you will want to change from a kapha-pacifying regimen to the pitta-pacifying guidelines suggested for summer. In fact, as the weather alternates between cold and hot, you will need to be alert day-to-day and use your common sense to remain in balance.

Breathing Techniques

P*rana* is the bridge between body, mind, and consciousness. It is the constant movement of awareness. *Prana* carries awareness to the object of our perception; that movement of awareness through *prana* is called attention. The internal movement of *prana* is the movement of sensation, thought, feeling, and emotion. Thus *prana* and mind are deeply connected.

The physical manifestation of *prana* is breath. Breathing and mind are closely connected. Ayurveda says that breathing is the physical part of thinking and thinking is the psychological part of breathing. Every thought changes the rhythm of breath, and every breath changes the rhythm of thinking. When one is happy, blissful, and silent, breathing is rhythmic. If one is disturbed with anxiety, fear, or nervousness, the breathing is irregular and interrupted.

Ancient Vedic seers (*rishis*) discovered this intimate relationship between breathing and mental activity and uncovered the art of *pranayama*. *Pranayama* equals *prana* plus *ayam*. *Ayam* means “to control,” *prana* is “breath.” By controlling the breath, we can control mental activity.

The Secret of Pranayama

The *rishis* also found an intimate relationship between the right and the left breath cycle. You may have noticed that sometimes you breathe more easily through the left nostril, and sometimes your breathing shifts more to the right nostril. This shift happens about every 45 to 90 minutes. Just as the right side of our body is governed by the left side of the brain and vice versa, breathing better through the left nostril activates the right hemisphere of the brain, and breathing better through the right nostril activates the left brain.

The left brain is associated with male energy, the right with female energy. The left brain is for logical thinking, investigation, inquiry, aggressiveness, competition, and judgment. Whenever we are judging, investigating, and so on, our right breath cycle is dominant and our left brain hemisphere is operating. Exactly opposite, when the right hemisphere is acting and the left breath cycle is working, we have female energy, which is associated with love, compassion, intuition, art, poetry, and religion. So when an artist is drawing a picture or a poet writing a poem, he or she is using some part of the right brain. And when the scientist is working in the lab, investigating and solving a problem, at that time he or she is using some part of the left brain.

The secret of *pranayama* is the secret of handling the male and female energy operating in our nervous system. In Alternate Nostril *pranayama*, when we inhale through the left nostril, we charge the right brain. When we inhale through the right nostril, we charge our left brain. When yogis do Alternate Nostril breathing, their male and female energies become equally balanced. When these energies are balanced, the neutral energy is awakened and one experiences pure awareness, which is called *brahman*.

When we do *pranayama*, the *nadis* (subtle channels of the nervous system) become purified, the mind becomes controlled, and we can go beyond male and female energies to pure, choiceless, passive awareness.

This is the basis for *pranayama*. Then there are numerous types of *pranayama*: practices to heat or cool the body; Right Nostril breathing to awaken more male energy; Left Nostril breathing to awaken more female energy; and so on.

Six Breathing Techniques

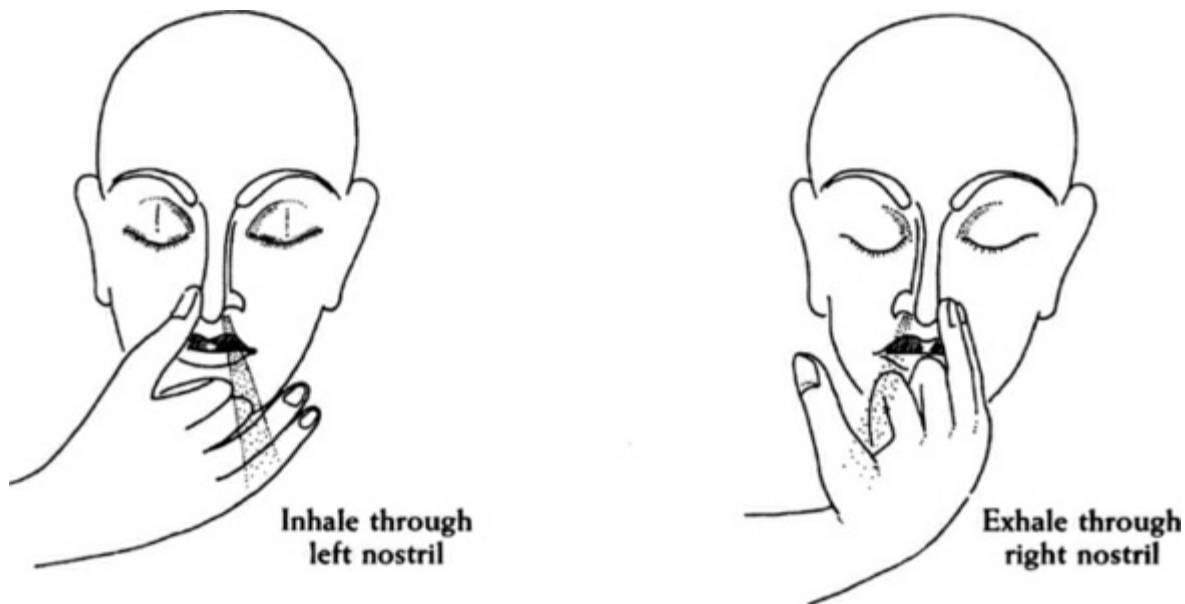
ALTERNATE NOSTRIL *PRANAYAMA*

One of the simplest breathing practices, and one that is very effective, as we have discussed, is Alternate Nostril *pranayama*.

1. Sit comfortably on the floor in a crosslegged posture, keeping the spine straight. If you are not comfortable in this position, sit upright on the front edge of a chair with your feet flat on the floor.
2. Close the right nostril with your right thumb, and inhale through the left nostril. Inhale into the belly, not into the chest.
3. After inhaling, hold your breath for just a moment.
4. Exhale through your right nostril while closing the left with the ring and little finger of your right hand.
5. Repeat steps 1 to 3, but this time start by inhaling through the *right* nostril (while you close the left nostril with your ring and little finger).

You can do this breathing exercise for five to ten minutes.

NOTE: This *pranayama*, as well as the others in this book, is best learned under the guidance of an accomplished teacher.



ALTERNATE NOSTRIL PRANAYAMA

SHITALI PRANAYAMA (COOLING BREATH)

Curl your tongue into a tube. Inhale slowly through the curled tongue, swallow, and then exhale normally through the nose, keeping the mouth closed. You will feel the incoming air cool your saliva, your tongue, and the oral mucous membranes.

This form of breathing is helpful for pacifying high pitta. It lowers the oral temperature, makes the saliva cool, helps to quench thirst, and improves digestion, absorption, and assimilation. *Shitali* is effective for high blood pressure, burning throat or tongue, and a burning sensation in the eyes. It cools the entire body.

If you can't curl your tongue into a tube, an alternative way to perform *shitali* is with your teeth lightly clenched together and your tongue pressed up against the teeth. The air is then inhaled through the teeth. Some people feel pain when cool air is drawn through the teeth; keeping your tongue against your teeth will provide warmth and prevent this discomfort.

BHASTRIKA PRANAYAMA (BREATH OF FIRE)

This breathing exercise increases the vital capacity of the lungs, relieves allergies and asthma, and helps make the lungs strong and healthy. It also heats the body.

Inhale passively (through the nose), but exhale actively and with *a little* force. Start slowly and increase the speed. Imagine a steam locomotive moving slowly and picking up speed. Do one round of 30 strokes or exhalations, then rest for one minute. You can do up to five such rounds of *bhastrika* in the morning and five in the evening.



SHITALI PRANAYAMA (COOLING BREATH)



BHASTRIKA PRANAYAMA (BREATH OF FIRE)

BHRAMARI PRANAYAMA (HUMMING BREATH)

On inhalation, constrict the epiglottis so as to create a humming sound. On exhalation, the sound is long and low. The inhalation, which is more high-pitched, is traditionally said to be like a female bee; the exhalation, which has a deeper sound, like a male bee.

If you find it difficult to make the humming sound on the inhalation, just inhale naturally, take a deep breath into the belly, and then do the humming on the exhalation.

When doing *bhramari*, touch the tip of your tongue lightly to the edge of the soft palate near the back of the roof of your mouth. Be sure the teeth are not clenched.

Bhramari improves the melodiousness of the voice. The humming vibrates the nervous system and is a form of sound therapy for the brain. It is also good for the thyroid, thymus, and parathyroid glands. Do ten cycles.

UJJAYI PRANAYAMA (BREATH OF VICTORY)

Sit in either Vajrasana or the Lotus posture, with your hands resting on your knees, palms up. Keep your head, neck, and chest in a straight line. Lower your head into a slight chin lock by moving your head in and down, toward your chest. Bring your awareness to the throat area.

Now comes the slightly tricky part. Without actually swallowing, start the action of swallowing, to raise the trachea upward. At the same time, while constricting your epiglottis, as in silently “saying” the letter *e*, slowly and deeply inhale into the belly. The inhaled air will create a soft, gentle whispering sound of rushing air as it brushes the throat, trachea, heart, and diaphragm.

After inhaling, swallow and hold your breath at the belly for a moment, then slowly exhale the air by again constricting the epiglottis—as if humming, but without producing any humming sound.

Ujjayi pranayama brings great joy. It calms the mind, relaxes the intercostal muscles, and really brings a sense of victory. *Ujjayi* is good for all three doshas and helps to reestablish constitutional balance. It promotes longevity. Do twelve cycles (repetitions) at a time.

SURYA BHEDI PRANAYAMA (RIGHT NOSTRIL BREATHING)

Put a cotton “plug” in your left nostril so that you will breathe through the right nostril, or block the left nostril by gently pressing with the ring and little finger of the right hand. Sit comfortably. Breathe in and out through the right nostril only. Repeat ten times.

Meditation and Mental Discipline

Meditation is an art of bringing harmony to body, mind, and consciousness. Life with meditation is a flowering of bliss and beauty. Life without meditation is stress, confusion, and illusion.

During ancient times meditation was often considered a way of life. Truly, meditation is not separate from daily living, but as a discipline, we have to practice certain techniques, methods, and systems. Once we have practiced a form of meditation and mastered it, that discipline stays with us in every aspect of our lives. So whatever technique you do, whatever system you follow, according to the instruction given by your teacher, please do that.

But what is meditation, and what is not?

Meditation is not concentration. In concentration we narrow the mind, and a narrow mind is a limited mind. We need that limited, pointed, concentrated mind to probe into any subject, to solve problems, to learn a language, to fly an airplane. We need it. But not in meditation.

In concentration we build a wall of resistance, and in the effort to control the mind, we lose energy. Some people meditate that way for an hour, and when they're finished, they feel tired, because for that hour they were fighting and fighting, negating everything, saying no to all thoughts and perceptions, trying to focus the mind.

Concentration is all-exclusive, but meditation is all-inclusive. Meditation is open, choiceless awareness. Everything is welcome. Meditation says yes to everything, while concentration says no to everything.

Concentration is effort. Wherever there is effort, there is a maker of the effort. The maker of the effort is the ego. Concentration nourishes the ego, the maker of the effort. The more the concentration, the more ego.

In meditation, there is no effort and no effort-maker. Therefore there is freedom. You are just sitting quietly and listening to everything, whether it is the call of a bird, the cry of a child, the rustle of the leaves. Every sound is welcome. Whatever the sound, allow it to come to you. When you listen to the sound, you become the center and the sound is peripheral, rushing toward you, to meet with you.

In listening to any kind of sound without judgment, without criticism, without liking or disliking, you become the center and all sounds rush toward you, to dissolve into you. Follow the sound. Allow it to pass through you. Don't resist. Then a magical phenomenon happens. You become empty. You become silent, pure existence.

When a breeze comes to you, allow that breeze to pass through you. No effort, no resistance. Remember, peace is not the opposite of sound. Every sound dissolves into peace. You are that peace, and sound comes to meet with you and dissolve into you.

Look at any object, a tree, a flower, even the wall. There is no choice in the looking, no judgment, just choiceless observation.

Awareness is the act of listening, the act of looking. No effort is required, no concentration. In awareness, in meditation, concentration happens naturally. It is given to you as a gift. But in concentration, in choosing, you miss meditation.

In expanded, empty consciousness, thinking stops, breathing becomes quiet, and one simply exists as pure awareness. In that state there is great joy, beauty, and love. Individual consciousness merges with Cosmic Consciousness, and one goes beyond time and thought.

In that state, whether the eyes are open or closed does not matter. It comes like a breeze without invitation, because this state is your true nature—love, bliss, beauty, and awareness. There is no fear, no depression, no anxiety, no worry, no stress. One becomes the witness of anxieties, worries, and stress. In that state, healing occurs.

This is what is called discipline. Discipline means learning, and one who is learning is called a disciple. So we have to learn the art of discipline. Discipline means putting everything in its right place. Thought has a right place, desire has a right place, job has a right place, duty has a right place. Discipline brings harmony in our life. Therefore discipline and meditation go together. There is no meditation without discipline, and no discipline without meditation. They are one. Mind in meditation is mind in discipline.

The so-called concentrated mind is a controlling mind. A confused mind controls. But a mind that is free, alert, and aware is blissful. That mind is a disciplined mind. And discipline is the perfume of life. Without that perfume, life never becomes celebration.

When you meditate, sit with your back straight. If you can, sit in the Lotus pose (or Half Lotus if that is more comfortable for you). If that is not comfortable, you can sit on a chair but keep your vertebral column erect.

With persistent practice, you can increase the time you spend in the Lotus to one, two, or even three hours. If one sits properly in a Lotus pose for three hours a day, enlightenment will soon come.

Sitting in the Lotus pose helps to open the heart. Breathing becomes quiet, and thinking automatically slows and stops. To go beyond thought is to go beyond suffering, because it is thought that creates suffering.

Empty Bowl Meditation

Sit comfortably and quietly with your palms up and open and placed on your knees,

like empty bowls. Open your mouth slightly, and touch the tongue to the roof of the mouth, behind the front teeth.

Begin by paying attention to your breath. Let your lungs breathe with no effort on your part. Simply watch the movement of your breath. Inhale. Exhale.

During inhalation, air touches the inside of the nostrils. Be aware of that breath. During exhalation, again air touches the nostrils. The ingoing air feels cool, the outgoing air is warm. For a fraction of a second, enter into your nose! Sit in the nostril and watch your breath: ingoing, outgoing, ingoing, outgoing. Let your lungs do their job. You are simply sitting and watching.

After five minutes, follow the breath. When the lungs inhale, go with the air into the nose, to the back of the throat, the trachea, lungs, heart, diaphragm. Go deep down behind the belly button, where you will experience a natural stop. For a fraction of a second, the breath stops. Stay in that stop, then when the lungs exhale, again follow the breath as it reverses its course. Come up from the belly button to the diaphragm, heart, lungs, trachea, and throat, back to the nose, then out of the body.

During exhalation the air goes out of the body to about nine inches in front of the nose, where there is a second stop. Again, stay in that stop for a moment.

These two stops are very important. The first stop is behind the belly button, the second outside the body in space. As your awareness rests in these two stops, time stops, because time is the movement of breath. When breath stops, mind stops, because mind is the movement of breath. When the mind becomes quiet, you simply exist, without body, without mind, without breath.

In that stop, you become like an empty bowl, and when you become an empty bowl, divine lips will touch you. God will come to you, to pour out his love. You do not need to seek God, for God is seeking you. From ages past, God is seeking an empty bowl, to fill with his love. But all bowls are full of desire, ambition, business, competition, success and failure.

Just sit quietly and stay in the stop. That stop is a door. Simply enter the door and jump into the inner abyss. You will feel an extraordinary tranquillity and peace surrounding you.

Practice this meditation for 15 minutes in the morning and in the evening. Over the days, weeks, and months, you will find your time in the stops naturally increasing until, eventually, inner and outer will merge and everything will happen within you.

NOTE: If you are more comfortable, you may practice this meditation in a prone position.



So-Hum Meditation

In *So-Hum* meditation we sit quietly and watch our breath, as in the Empty Bowl meditation, but we add the sound *So* on inhalation, *Hum* on exhalation. (Only silently; we don't speak the sounds aloud.)

When sound, breath, and awareness come together, it becomes light. We have seen that every atom radiates light and heat energy, which is a quantum wave. The moment we pay attention to our breath and start feeling *So-Hum*, *So-Hum* along with the breath, our breath becomes a quantum wave and radiates light. You may see that light of life at the third eye.

Inspiration (breathing in) is living; expiration (breathing out) is dying. When a child is born, with its first breath life expresses itself with inspiration. When someone dies, we say he has expired. The breath has gone out.

Hum means "I" or "individual ego"; *So* means "He, the Divine." So in the natural course of *So-Hum* meditation, when *So* goes in, life energy goes in and *Hum*, ego, our limited individuality, goes out. That is the significance of *So-Hum* meditation. When you inhale *So*, you are inhaling life. When you exhale *Hum*, you are exhaling ego and limitation.

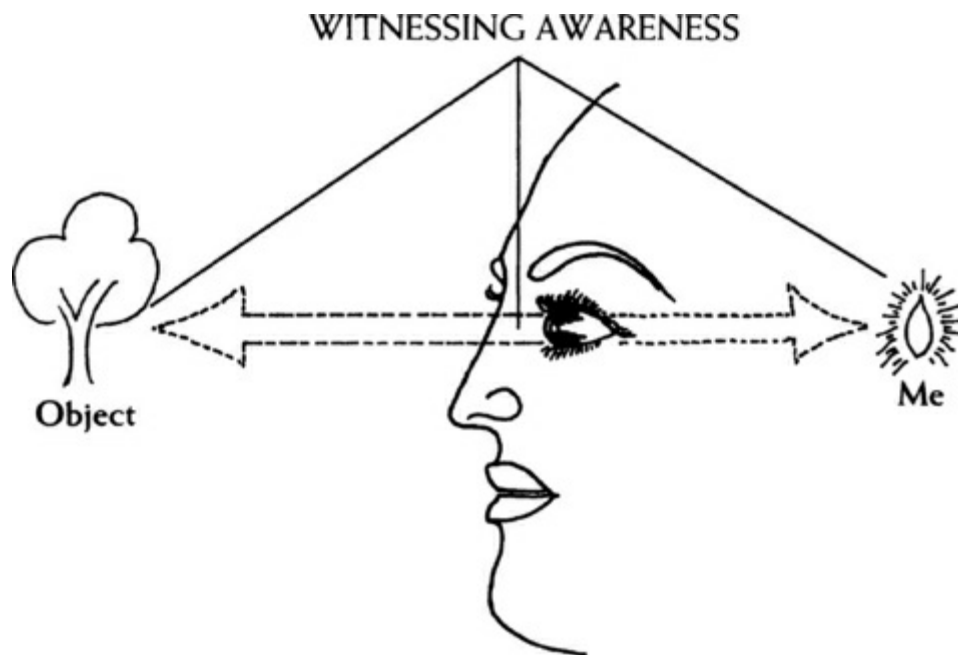
So-Hum meditation properly practiced leads to the union of the individual with the universal Cosmic Consciousness. You will go beyond thought, beyond time and space, beyond cause and effect. Limitations will vanish. Your consciousness will empty itself, and in that emptying it will expand and peace and joy will descend as a benediction.

Double-Arrowed Attention (Witnessing)

In Vedic science, witnessing is called *samyag darshan*. This is a process of looking outward and inward simultaneously.

When we look at a tree, a star, a mountain, or a flower, something goes out from our eyes, touches the object, and comes back to us. That which goes out of our eyes in order to touch the object of perception, we call attention. Ayurveda says that attention happens when *prana* goes out and carries the vibration of awareness toward the object. Thus, attention is awareness plus *prana*, movement.

One arrow goes out and touches the object. At the same time, a second arrow of attention should go inside, toward the center of our heart, to observe the observer. At the time of looking, when you look at the object outside, at the same time, look at the looker; watch the watcher; observe the observer. What happens when the watcher is watched is that the watcher disappears. This simple watching, without a watcher, is called witnessing. In that witnessing, you develop intimacy, relationship, with the object of perception.



Ayurvedic Dietary Guidelines

The purpose of this chapter is to help you choose a suitable diet for balance, harmony, and health in your life, based on Ayurvedic principles. Health-conscious people today are interested in the role good nourishment can play in their healing and in their health. Many have come to realize that proper food and diet can make a vital contribution to good health, while inappropriate eating is often responsible for poor health, lack of vitality, and susceptibility to disease.

The Ayurvedic tradition offers much insight into what food will suit and balance each individual, how to prepare and cook this food properly, how to avoid food combinations that will create toxins in the body, and what eating habits to cultivate—and which to avoid—in order to receive the most nourishment from what you eat. All these topics, except specific guidelines on how to prepare and cook the food, will be discussed in this chapter. (Interested readers may consult *Ayurvedic Cooking for Self-Healing* by Usha Lad and Dr. Vasant Lad, a complete reference guide to Ayurvedic cooking, including spices and herbs, healing qualities of common foods, menu planning, and dozens of delicious recipes.)

Food Guidelines for the Constitutional Types

What you eat should be suited to your individual constitution. Ideally, in deciding what to eat, you would know your constitution and understand its relationship to the qualities of various kinds of food, including whether each food would be helpful or aggravating to your unique doshic balance. You would have to take into account the taste of the food (we will discuss that issue later in this chapter), and whether its qualities are heavy or light, oily or dry, liquid or solid. You would also have to know whether the food is cooling or heating (*virya*), and its postdigestive effect (*vipaka*).

If you are interested, you can go more deeply into Ayurvedic theory in order to fully comprehend these factors (see the [Reading List](#)). Otherwise, the following charts take these factors into consideration in recommending what foods to eat or avoid.

The charts categorize foods according to their suitability for each doshic type. Here are a few points to remember:

- Foods marked “no” tend to aggravate that particular dosha, while foods marked “yes” pacify or balance that dosha. In planning your diet, choose foods that create balance, and avoid those that might provoke your predominant doshas or the dosha that is currently aggravated or increased.

- The recommendations are not meant to be absolute, but are guidelines. If a food is on your “no” list, that means you should avoid it most of the time, and if you eat it, eat a modest amount or do something to modify its effects. Apples, for example, are quite vata-provoking if eaten raw. But if you cook them and eat them warm, with a little ghee and warming spices such as cardamom or cinnamon, they are fine for vata individuals in modest amounts.

- Keep the seasons in mind. Summer, for example, is pitta season, and it is not good—especially for people with a predominantly pitta constitution—to eat too many hot, spicy foods, or pitta dosha will become aggravated. Similarly, during autumn, when the air is dry and cool and more vata is present in the atmosphere, everyone—but especially individuals with a vata constitution—should avoid dry fruit, salads, cold foods, and other vata-provoking items. In winter and early spring, the heavy, cold, moist season of kapha, one should make an extra effort to avoid cold food and drinks, ice cream, cheese, yogurt, melons, and other kaphaincreasing foods.

- For individuals with a dual constitution (two doshas approximately equal), a little extra care is needed, but you can figure it out. For example, a vata-pitta individual needs to avoid vata-increasing foods in the fall and winter (but without increasing pitta too much) and minimize pitta-provoking foods in the summer (but without aggravating vata). Stated in positive terms, favor vata-balancing foods in the fall, pitta-pacifying foods in the summer.

Here are some general dietary guidelines for balancing the doshas:

Vata

- 50 percent whole grains—whole-grain cooked cereals, some breads and crackers
- 20 percent protein—eggs, high-quality dairy products, poultry, fish, seafood, beef, tofu, black and red lentils
- 20–30 percent fresh vegetables—with an optional 10 percent for fresh fruit

Pitta

- 50 percent whole grains—whole-wheat breads, cereals, cooked grains
- 20 percent protein—beans (except lentils), tofu, tempeh, cottage cheese, ricotta cheese, raw milk, egg white, chicken and turkey (white meat only), shrimp, rabbit, venison
- 20–30 percent vegetables—with an optional 10 percent for fresh fruit

Kapha

- 30–40 percent whole grains—rye crackers, dry cereals, cooked grains
- 20 percent protein—chicken, turkey, boiled and poached eggs, small amount of goat’s milk, and most beans (including garbanzos, adzukis, pintos, black beans, red lentils, navy and white beans, split peas, and black-eyed peas)

- 40–50 percent fresh vegetables—with an optional 10 percent for fresh or dried fruit. A daily salad is good.

FOOD GUIDELINES FOR THE BASIC CONSTITUTIONAL TYPES

Note: Guidelines provided in this table are general. Specific adjustments for individual requirements may need to be made, e.g., food allergies, strength of agni, season of the year, and degree of dosha predominance or aggravation. *okay in moderation; **okay rarely.

	Vata		Pitta		Kapha	
	NO	YES	NO	YES	NO	YES
FRUITS	Generally most dried fruit	Generally most sweet fruit	Generally most sour fruit	Generally most sweet fruit	Generally most sweet & sour fruit	Generally most astringent fruit
	Apples (raw)	Apples (cooked)	Apples (sour)	Apples (sweet)	Avocado	Apples
	Cranberries	Applesauce	Apricots (sour)	Applesauce	Bananas	Applesauce
	Dates (dry)	Apricots	Bananas	Apricots (sweet)	Coconut	Apricots
	Figs (dry)	Avocado	Berries (sour)	Avocado	Dates	Berries
	Pears	Bananas	Cherries (sour)	Berries (sweet)	Figs (fresh)	Cherries
	Persimmons	Berries	Cranberries	Cherries (sweet)	Grapefruit	Cranberries
	Pomegranates	Cherries	Grapefruit	Coconut	Kiwi	Figs (dry)*
	Prunes (dry)	Coconut	Grapes (green)	Dates	Mangoes**	Grapes*
	Raisins (dry)	Dates (fresh)	Kiwi**	Figs	Melons	Lemons*
	Watermelon	Figs (fresh)	Lemons	Grapes (red & purple)	Oranges	Limes*
		Grapefruit	Mangoes (green)	Limes*	Papaya	Peaches
		Grapes	Peaches (sour)	Mangoes (ripe)	Pineapple	Pears
		Kiwi	Persimmons	Melons	Plums	Persimmons
		Lemons	Pineapple (sour)	Oranges (sweet)	Rhubarb	Pomegranates
		Limes	Plums (sour)	Papaya*	Tamarind	Prunes
		Mangoes	Rhubarb	Pears	Watermelon	Raisins
		Melons	Strawberries	Pineapple (sweet)		Strawberries*
		Oranges	Tamarind	Plums (sweet)		
		Papaya		Pomegranates		
		Peaches		Prunes		
		Pineapple				
		Plums				
		Prunes (soaked)		Raisins		
		Raisins (soaked)		Watermelon		
		Rhubarb				
		Strawberries				
		Tamarind				
VEGETABLES	Generally frozen, raw, or dried vegetables	In general, vegetables should be cooked	In general, pungent vegetables	In general, sweet & bitter vegetables	In general, sweet & juicy vegetables	In general, most pungent & bitter vegetables
	Artichoke	Asparagus	Beet greens	Artichoke	Cucumber	Artichoke
	Beet greens**	Beets	Beets (raw)	Asparagus	Olives, black or green	Asparagus
	Bitter melon	Cabbage* (cooked)	Burdock root	Beets (cooked)	Parsnips**	Beet greens
	Broccoli	Carrots	Corn (fresh)**	Bitter melon	Potatoes, sweet	Beets
	Brussels sprouts	Cauliflower*	Daikon radish	Broccoli	Pumpkin	Bitter melon
	Burdock root	Cilantro	Eggplant**	Brussels sprouts	Squash, winter	Broccoli
	Cabbage (raw)	Cucumber	Garlic	Cabbage	Taro root	Brussels sprouts
	Cauliflower (raw)	Daikon radish*	Green chilies	Carrots (cooked)	Tomatoes (raw)	Burdock root
	Celery	Fennel (anise)	Horseradish	Carrots (raw)*	Zucchini	Cabbage
	Corn (fresh)**	Garlic	Kohlrabi**	Cauliflower		Carrots
	Dandelion greens	Green beans	Leeks (raw)	Celery		Cauliflower
	Eggplant	Green chilies	Mustard greens	Cilantro		Celery
	Horseradish**	Jerusalem artichoke*	Olives, green	Cucumber		Cilantro
	Kale	Leafy greens*	Onions (raw)	Dandelion greens		Corn
	Kohlrabi	Leeks	Peppers (hot)	Fennel (anise)		Daikon radish
	Mushrooms	Lettuce*	Prickly pear (fruit)	Green beans		Dandelion greens
			Radishes (raw)	Jerusalem artichoke		Eggplant

Olives, green	Mustard greens*	Spinach	Kale	Fennel (anise)
Onions (raw)	Okra	(cooked)**	Leafy greens	Garlic
Peas (raw)	Olives, black	Spinach (raw)	Leeks (cooked)	Green beans
Peppers, sweet & hot	Onions (cooked)*	Tomatoes	Lettuce	Green chilies
Potatoes, white	Parsley*	Turnip greens	Mushrooms	Horseradish
Prickly pear (fruit & leaves)	Parsnip	Turnips	Okra	Jerusalem artichoke
Radish (raw)	Peas (cooked)		Olives, black	Kale
Tomatoes (cooked)**	Potatoes, sweet		Onions (cooked)	Kohlrabi
Tomatoes (raw)	Pumpkin		Parsley	Leafy greens
Turnips	Radishes (cooked)*		Parsnips	Leeks
Wheatgrass sprouts	Rutabaga		Peas	Lettuce
	Spaghetti squash*		Peppers, sweet	Mushrooms
	Spinach (cooked)*		Potatoes, sweet & white	Mustard greens
	Spinach (raw)*		Prickly pear (leaves)	Okra
	Sprouts*		Pumpkin	Onions
	Squash, summer & winter		Radishes (cooked)	Parsley
	Taro root		Rutabaga	Peas
	Turnip greens*		Spaghetti squash	Peppers, sweet & hot
	Watercress		Sprouts (not spicy)	Potatoes, white
	Zucchini		Squash, winter & summer	Prickly pear (fruit & leaves)
			Taro root	Radishes
			Watercress*	Rutabaga
			Wheatgrass sprouts	Spaghetti squash*
			Zucchini	Spinach
				Sprouts
				Squash (summer)
				Tomatoes (cooked)

	Vata		Pitta		Kapha	
	NO	YES	NO	YES	NO	YES
VEGETABLES						Turnip greens Turnips Watercress Wheatgrass sprouts
GRAINS						
Always use suitable grains when "generic" things are listed						
Barley		Amaranth*	Bread (with yeast)	Amaranth	Bread (with yeast)	Amaranth*
Bread (with yeast)		Durham flour	Buckwheat	Barley	Oats (cooked)	Barley
Buckwheat		Oats (cooked)	Corn	Cereal, dry	Pancakes	Buckwheat
Cereals (cold, dry, or puffed)		Pancakes	Millet	Couscous	Pasta**	Cereal (cold, dry, or puffed)
Corn		Quinoa	Muesli**	Crackers	Rice (brown, white)	Corn
Couscous		Rice (all kinds)	Oats (dry)	Durham flour	Rice cakes**	Couscous
Crackers		Seitan (wheat meat)	Polenta**	Granola	Wheat	Crackers
Granola		Sprouted wheat bread (Essene)	Quinoa	Oat bran		Durham flour*
Millet		Wheat	Rice (brown)**	Oats (cooked)		Granola
Muesli			Rye	Pancakes		Millet
Oat bran				Pasta		Muesli
Oats (dry)				Rice (basmati, white, wild)		Oat bran
Pasta**				Rice cakes		Oats (dry)
Polenta**				Sago		Polenta
Rice cakes**				Seitan (wheat meat)		Quinoa*
Rye				Spelt		Rice (basmati, wild)*
Sago				Sprouted wheat bread (Essene)		Rye
						Sago

	Spelt			Tapioca		Seitan (wheat meat)
	Tapioca			Wheat		Sprouted wheat bread (Essene)
	Wheat bran			Wheat bran		Tapioca
						Wheat bran
LEGUMES	Adzuki beans	Lentils (red)*	Miso	Adzuki beans	Kidney beans	Adzuki beans
	Black beans	Mung beans	Soy sauce	Black beans	Soybeans	Black beans
	Black-eyed peas	Mung dal	Soy sausages	Black-eyed peas	Soy cheese	Black-eyed peas
	Chickpeas	Soy cheese*	Tur dal	Chickpeas	Soy flour	Chickpeas
	(garbanzo beans)	Soy milk*	Urad dal	(garbanzo beans)	Soy powder	(garbanzo beans)
	Kidney beans	Soy sauce*		Kidney beans	Soy sauce	Lentils (red & brown)
	Lentils (brown)	Soy sausages*		Lentils, brown & red	Tofu (cold)	Lima beans
	Lima beans	Tofu*		Lima beans	Urad dal	Mung beans*
	Miso**	Tur dal		Mung beans	Miso	Mung dal*
	Navy beans	Urad dal		Mung dal		Navy beans
	Peas (dried)			Navy beans		Peas (dried)
	Pinto beans			Peas (dried)		Pinto beans
	Soybeans			Pinto beans		Soy milk
	Soy flour			Soybeans		Soy sausages
	Soy powder			Soy cheese		Split peas
	Split peas			Soy flour*		Tempeh
	Tempeh			Soy milk		Tofu (hot)*
	White beans			Soy powder*		Tur dal
				Split peas		White beans
				Tempeh		
				Tofu		
				White beans		

	Vata		Pitta		Kapha	
	NO	YES	NO	YES	NO	YES
DAIRY	Cow's milk (powdered)	<i>Most dairy is good!</i>	Butter (salted)	Butter (unsalted)	Butter (salted)	Buttermilk*
	Goat's milk (powdered)	Butter	Buttermilk	Cheese (soft, not aged, unsalted)	Butter (unsalted)**	Cottage cheese (from skimmed goat's milk)
	Yogurt (plain, frozen, or w/fruit)	Buttermilk	Cheese (hard)	Cottage cheese	Cheese (soft & hard)	Ghee*
		Cheese (hard)*	Sour cream	Cow's milk	Cow's milk	Goat's cheese (unsalted & not aged)*
		Cheese (soft)	Yogurt (plain, or frozen, w/fruit)	Ghee	Ice cream	Goat's milk, skim
		Cottage cheese		Goat's milk	Sour cream	Yogurt (diluted)
		Cow's milk		Goat's cheese (soft, unsalted)	Yogurt (plain, frozen, or w/fruit)	
		Ghee		Ice cream		
		Goat's cheese		Yogurt (freshly made & diluted)*		
		Goat's milk				
ANIMAL FOODS	Lamb	Beef	Beef	Buffalo	Beef	Chicken (white)
	Pork	Buffalo	Chicken (dark)	Chicken (white)	Buffalo	Eggs
	Rabbit	Chicken (dark)	Duck	Eggs (albumen or white only)	Chicken (dark)	Fish (freshwater)
	Venison	Chicken (white)*	Eggs (yolk)	Fish (freshwater)	Duck	Rabbit
	Turkey (white)	Duck	Fish (sea)	Rabbit	Fish (sea)	Shrimp
		Eggs	Lamb	Shrimp*	Lamb	Turkey (white)
		Fish (freshwater or sea)	Pork	Turkey (white)	Pork	Venison
		Salmon	Salmon	Venison	Salmon	
		Sardines	Sardines		Sardines	
		Seafood	Seafood		Seafood	
		Shrimp	Tuna fish		Tuna fish	
			Turkey (dark)		Turkey (dark)	

		Tuna fish				
		Turkey (dark)				
CONDIMENTS	Chocolate Horseradish	Black pepper*	Chili pepper	Black pepper*	Chocolate	Black pepper
		Chutney, mango	Chocolate	Chutney, mango	Chutney,	Chili peppers
		(sweet or spicy)	Chutney, mango	(sweet)	mango	Chutney, mango
		Chili peppers*	(spicy)	Coriander leaves	(sweet)	(spicy)
		Coriander leaves*	Gomasio	Dulse*	Gomasio	Coriander leaves
		Dulse	Horseradish	Hijiki*	Kelp	Dulse*
		Gomasio	Kelp	Kombu*	Ketchup**	Hijiki*
		Hijiki	Ketchup	Lime*	Lime	Horseradish
		Kelp	Mustard	Sprouts	Lime pickle	Lemon*
		Ketchup	Lemon	Tamari*	Mango pickle	Mustard
		Kombu	Lime pickle		Mayonnaise	(without
		Lemon	Mango pickle		Pickles	vinegar)
		Lime	Mayonnaise		Salt	Scallions
		Lime pickle	Pickles		Soy sauce	Seaweed*
		Mango pickle	Salt (in excess)		Tamari	Sprouts
		Mayonnaise	Scallions		Vinegar	
		Mustard	Seaweed			
		Pickles	Soy sauce			
		Salt	Vinegar			
		Scallions				
		Seaweed				
		Soy sauce				
		Sprouts*				
		Tamari				
		Vinegar				
NUTS	None	<i>In moderation:</i>				
		Almonds	Almonds (with	Almonds (soaked	Almonds	Charole
		Black walnuts	skin)	and peeled)	(soaked &	
			Black walnuts	Charole	peeled)**	
		Vata	Pitta		Kapha	
		NO	YES	NO	YES	NO
NUTS						
		Brazil nuts	Brazil nuts	Coconut	Black walnuts	
		Cashews	Cashews		Brazil nuts	
		Charole	Filberts		Cashews	
		Coconut	Hazelnuts		Coconut	
		Filberts	Macadamia nuts		Filberts	
		Hazelnuts	Peanuts		Hazelnuts	
		Macadamia nuts	Pecans		Macadamia nuts	
		Peanuts	Pine nuts		Peanuts	
		Pecans	Pistachios		Pecans	
		Pine nuts	Walnuts		Pine nuts	
		Pistachios			Pistachios	
		Walnuts			Walnuts	
SEEDS	Popcorn Psyllium**	Chia	Chia	Flax	Halva	Chia
		Flax	Sesame	Halva	Psyllium**	Flax*
		Halva	Tahini	Popcorn (no salt,	Sesame	Popcorn (no
		Pumpkin		buttered)	Tahini	salt, no butter)
		Sesame		Psyllium		Pumpkin*
		Sunflower		Pumpkin*		Sunflower*
		Tahini		Sunflower		
OILS	Flaxseed	<i>For internal &</i>	Almond	<i>For internal &</i>	Avocado	<i>For internal &</i>
		<i>external use</i>	Apricot	<i>external use</i>	Apricot	<i>external use in</i>
		<i>(most suitable at</i>	Corn	<i>(most suitable at</i>	Coconut	<i>small amounts</i>
		<i>top of list):</i>	Safflower	<i>top of list):</i>	Flaxseed**	<i>(most suitable at</i>
		Sesame	Sesame	Sunflower	Olive	<i>top of list):</i>

Ghee
Olive
Most other oils
External use only:
Coconut
Avocado

Ghee
Canola
Olive
Soy
Flaxseed
Primrose
Walnut
External use only:
Avocado
Coconut

Primrose
Safflower
Sesame
(internal)
Soy
Walnut
Corn
Canola
Sesame
(external)
Sunflower
Ghee
Almond

BEVERAGES	Apple juice	Alcohol (beer or wine)*	Alcohol (hard or wine)	Alcohol, beer*	Alcohol (beer, hard, sweet wine)	Alcohol (dry wine, red or white)
	Black tea	Almond milk	Apple cider	Almond milk	Almond milk	Aloe vera juice
	Caffeinated beverages	Aloe vera juice	Berry juice (sour)	Apple juice	Caffeinated beverages**	Apple cider
	Carbonated drinks	Apricot juice	Caffeinated beverages	Apricot juice	Carbonated drinks	Apple juice*
	Chocolate milk	Berry juice (except for cranberry)	Carbonated drinks	Black tea	Cherry juice (sour)	Apricot juice
	Coffee	Carob*	Carrot juice	Carob	Cherry juice (sour)	Berry juice
	Cold dairy drinks	Carrot juice	Chai (hot, spiced milk)*	Chai (hot, spiced milk)*	Chocolate milk	Black tea (spiced)
	Cranberry juice	Chai (hot, spiced milk)	Cherry juice (sour)	Cherry juice (sweet)	Coffee	Carob
	Iced tea	Cherry juice	Chocolate milk	Cool dairy drinks	Cold dairy drinks	Chai (hot, spiced milk)*
	Icy cold drinks	Grain "coffee"	Coffee	Grain "coffee"	Grapefruit juice	Cherry juice (sweet)
	Mixed veg. juice	Grape juice	Cranberry juice	Grape juice	Iced tea	Cranberry juice
	Pear juice	Grapefruit juice	Grapefruit juice	Mango juice	Icy cold drinks	Grain "coffee"
	Pomegranate juice	Lemonade	Iced tea	Miso broth*	Lemonade	Grape juice
	Prune juice**	Mango juice	Icy cold drinks	Mixed veg. juice	Miso broth	Mango juice
		Miso broth	Lemonade	Orange juice*	Orange juice	Peach nectar
		Orange juice	Papaya juice	Peach nectar		
		Papaya juice	Pineapple juice	Pear juice		

Vata

Pitta

Kapha

	NO	YES	NO	YES	NO	YES
BEVERAGES	Soy milk (cold)	Peach nectar	Sour juices	Pomegranate juice	Papaya juice	Pear juice
	Tomato juice**	Pineapple juice	Tomato juice	Prune juice	Rice milk	Pineapple juice*
	V-8 juice	Rice milk	V-8 juice	Rice milk	Sour juices	Pomegranate juice
	Vegetable bouillon	Sour juices		Soy milk	Soy milk (cold)	Prune juice
		Soy milk (hot & well-spiced)*		Vegetable bouillon	Tomato juice	Soy milk (hot & well-spiced)
	<i>Herb Teas:</i>	<i>Herb Teas:</i>	<i>Herb Teas:</i>	<i>Herb Teas:</i>	<i>Herb Teas:</i>	<i>Herb Teas:</i>
	Alfalfa**	Ajwan	Ajwan	Alfalfa	Marshmallow	Alfalfa
	Barley**	Bancha	Basil**	Bancha	Red Zinger	Bancha
	Basil**	Catnip*	Cinnamon*	Barley	Rosehip**	Barley
	Blackberry	Chamomile	Clove	Blackberry		Blackberry
	Borage**	Chicory	Eucalyptus	Borage		Burdock
	Burdock	Chrysanthemum*	Fenugreek	Burdock		Chamomile
	Cinnamon**	Clove	Ginger (dry)	Catnip		Chicory
	Cornsilk	Comfrey	Ginseng	Chamomile		Cinnamon
	Dandelion	Elder Flower	Hawthorn	Chicory		Clove
	Ginseng	Eucalyptus	Hyssop	Comfrey		Comfrey*
	Hibiscus	Fennel	Juniper berry	Dandelion		Dandelion
	Hops**	Fenugreek	Mormon tea	Fennel		Fenugreek
	Jasmine**	Ginger (fresh)	Pennyroyal	Ginger (fresh)		Ginger
	Lemon balm**	Hawthorn	Red Zinger	Hibiscus		Ginseng*
	Mormon tea	Juniper berry	Rosehip**	Hops		Hibiscus
	Nettle**	Kukicha*	Sage	Jasmine		Hyssop
	Passion flower**	Lavender	Sassafras	Kukicha		Jasmine
	Red clover**	Lemongrass	Yerba maté	Lavender		Juniper berry
		Licorice		Lemon balm		Kukicha

Red Zinger**	Marshmallow	Lemongrass	Lavender
Violet**	Oat straw	Licorice	Licorice*
Yarrow	Orange peel	Marshmallow	Lemon balm
Yerba maté**	Pennyroyal	Nettle	Lemongrass
	Peppermint	Oat Straw	Mormon tea
	Raspberry*	Passion flower	Nettle
	Rosehips	Peppermint	Passion flower
	Saffron	Raspberry	Peppermint
	Sage	Red clover	Raspberry
	Sarsaparilla	Sarsaparilla	Red clover
	Sassafras	Spearmint	Sarsaparilla*
	Spearmint	Strawberry	Sassafras
	Strawberry*	Violet	Spearmint
	Wintergreen*	Wintergreen	Strawberry
		Yarrow	Wintergreen
			Yarrow
			Yerba maté

SPICES	Caraway	<i>All other spices are good!</i>	Ajwan	Basil (fresh)	Salt	<i>All spices are good!</i>
		Ajwan	Allspice	Black pepper*		Ajwan
		Allspice	Almond extract	Caraway*		Allspice
		Almond extract	Anise	Cardamom*		Almond extract
		Anise	Asafoetida (hing)	Cinnamon		Anise
		Asafoetida (hing)	Basil (dry)	Cumin		Asafoetida (hing)
		Basil	Bay leaf	Curry leaves		Basil
		Bay leaf	Cayenne	Dill		Bay leaf
		Black pepper	Cloves	Fennel		Black pepper
		Cardamom	Fenugreek	Ginger (fresh)		Cardamom
		Cayenne*	Garlic	Mint		Cayenne
		Cinnamon	Ginger (dry)	Neem leaves*		Cinnamon
		Cloves	Mace	Orange peel*		Cloves

SPICES	V a t a		P i t t a		K a p h a	
	NO	YES	NO	YES	NO	YES
		Coriander	Marjoram	Parsley*		Cloves
		Cumin	Mustard seeds	Peppermint		Cinnamon
		Curry leaves	Nutmeg	Saffron		Coriander
		Dill	Oregano	Spearmint		Cumin
		Fennel	Paprika	Tarragon*		Curry leaves
		Fenugreek*	Pippali	Turmeric		Dill
		Garlic	Poppy seeds	Vanilla*		Fennel*
		Ginger	Rosemary	Wintergreen		Fenugreek
		Mace	Sage			Garlic
		Marjoram	Salt			Ginger
		Mint	Savory			Mace
		Mustard seeds	Star anise			Marjoram
		Nutmeg	Thyme			Mint
		Orange peel				Mustard seeds
		Oregano				Neem leaves
		Paprika				Nutmeg
		Parsley				Orange peel
		Peppermint				Oregano
		Pippali				Paprika
		Poppy seeds				Parsley
		Rosemary				Peppermint
		Saffron				Pippali
		Salt				Poppy seeds
		Savory				Rosemary
		Spearmint				Saffron
		Star anise				Sage
		Tarragon				Savory
		Thyme				Spearmint

		Turmeric Vanilla Wintergreen				Star anise Tarragon Thyme Turmeric Vanilla* Wintergreen
SWEETENERS	Maple syrup** White sugar	Barley malt Fructose Fruit juice concentrates Honey (raw & not processed) Jaggary Molasses Rice syrup Sucanat Turbinado	Honey** (raw & not processed) White sugar** jaggary Molasses	Barley malt Fructose Fruit juice concentrates Maple syrup Rice syrup Sucanat Turbinado	Barley malt Fructose Jaggary Maple syrup Molasses Rice syrup Sucanat Turbinado White sugar	Fruit juice concentrates Honey (raw & not processed)
FOOD SUPPLEMENTS	Barley green Brewer's yeast	Aloe vera juice* Bee pollen Amino acids <i>Minerals:</i> calcium, copper, iron, magnesium, zinc Royal jelly Spirulina Blue-green algae Vitamins A, B complex, B ₁₂ , C, D, & E	Amino acids Bee pollen** Royal jelly** <i>Minerals:</i> copper, iron Vitamins A, B complex, B ₁₂ & C	Aloe vera juice Barley green Brewer's yeast <i>Minerals:</i> calcium, magnesium, zinc Spirulina Blue-green algae Vitamins D & E	<i>Minerals:</i> potassium	Aloe vera juice Amino acids Barley green Bee pollen Brewer's yeast <i>Minerals:</i> copper, calcium, iron, magnesium, zinc Royal jelly Spirulina Blue-green algae Vitamins A, B complex, B ₁₂ , C, D, & E

The Six Tastes

Taste is important and has a direct effect on bodily doshas. According to Ayurveda, each food substance (and also each medicinal herb) has a specific taste. When the tastes are used in the proper amounts, individually and collectively, they bring about balance of our bodily systems.

The taste buds on our tongue are organized in six groups, corresponding to the six tastes recognized by Ayurveda: sweet, sour, salty, bitter, pungent, and astringent. These six basic tastes are derived from the five elements:

Earth + Water = Sweet

Earth + Fire = Sour

Water + Fire = Salty

Fire + Air = Pungent (Spicy)

Air + Space = Bitter

Air + Earth = Astringent

Different groups of taste buds on the tongue perceive taste and send a signal to the brain; from there, messages go out which not only directly influence digestion but also affect the doshas and all the body's cells, tissues, organs, and systems.

SWEET

The sweet taste is present in foods such as rice, sugar, milk, wheat, dates, and maple syrup. The qualities of sweet foods are usually oily, cooling, and heavy. The sweet taste increases the vital essence of life. When used moderately, it is wholesome to the body and promotes growth of all seven *dhatus* (plasma, blood, muscles, fat, bones, marrow and nerve tissue, and reproductive fluids). Proper use gives strength and longevity. It encourages the senses, improves complexion, and promotes healthy skin, hair, and a good voice. Sweet taste can relieve thirst and burning sensations and can be invigorating. It promotes stability.

Despite all these good qualities, excessive use of the sweet taste can produce many disorders. Sweet foods aggravate kapha and cause colds, cough, congestion, heaviness, loss of appetite, laziness, and obesity. They may also provoke lymphatic congestion, tumors, edema, diabetes, and fibrocystic changes in the breast.

SOUR

The sour taste is found in foods such as citrus fruits, sour cream, yogurt, vinegar, cheese, lemon, green grapes, and fermented food. Sour substances are liquid, light, heating, and oily in nature. When used in moderation, they are refreshing and delicious, stimulate appetite and salivation, improve digestion, energize the body, nourish the heart, and enlighten the mind.

If one uses the sour taste in excess, it can cause excessive thirst, hyperacidity, heartburn, acid indigestion, ulcers, and sensitive teeth. As it has a fermenting action, it may be toxic to the blood and can cause skin conditions such as dermatitis, acne, eczema, boils, and psoriasis. The hot quality may lead to an acid pH in the body and may cause burning in the throat, chest, heart, bladder, and urethra.

SALTY

Sea salt, rock salt, and kelp are examples of the salty taste. Salt is heating, heavy, and oily. Used moderately, it relieves vata and increases pitta and kapha. Due to its water element, it is laxative, and due to its fire element, it lessens spasm and pain of the colon. In moderation it promotes growth and maintains water electrolyte balance. It stimulates salivation, improves the flavor of food, and aids in digestion, absorption, and the elimination of wastes.

Too much salt in the diet may cause aggravation of pitta and kapha. It makes the blood thick and viscous, causes hypertension, and worsens skin conditions. Feeling hot, fainting, skin wrinkling, and baldness may be due to excessive use of the salty taste. Salt may also induce water retention and edema. Patchy hair loss, ulcers, bleeding disorders, skin eruptions, and hyperacidity may all result from overuse of the salty taste.

PUNGENT

The pungent taste is present in various hot peppers (cayenne, chili, black), as well as in onions, radishes, garlic, mustard, and ginger. It is light, drying, and heating in nature. Used in moderation, it improves digestion and absorption and cleans the mouth. It clears the sinuses by stimulating nasal secretions and tearing of the eyes. The pungent taste aids circulation, breaks up clots, helps in the elimination of waste products, and kills germs and parasites. It brings clarity of perception.

On the other hand, overuse of the pungent taste in the daily diet may cause negative reactions. It can kill sperm and ova, causing sexual debility in both sexes. It may induce burning, choking, fainting, and fatigue with feelings of heat and thirst. By aggravating pitta, it can cause diarrhea, heartburn, and nausea. Pungency can also aggravate vata (it is derived from both the fire and air elements), resulting in giddiness, tremors, insomnia, or pain in the leg muscles. Peptic ulcers, asthma, colitis, and skin conditions may result from excessive use.

BITTER

This taste is found in coffee, bitter melon, aloe vera, rhubarb, and the herbs yellow dock, fenugreek, turmeric root, dandelion root, and sandalwood. Bitter is the taste most lacking in the North American diet. It is cool, light, and dry in nature, increases vata, and decreases pitta and kapha. Though bitter is not delicious in itself, it promotes the flavor of other tastes. It is antitoxic and kills germs. It helps to relieve burning sensations, itching, fainting, and obstinate skin disorders. It reduces fever and stimulates firmness of the skin and muscles. In a small dose it can relieve intestinal gas and works as a digestive tonic. It is drying to the system and causes a reduction in fat, bone marrow, urine, and feces.

Overuse of the bitter taste may deplete plasma, blood, muscles, fat, bone marrow, and semen and may result in sexual debility. Extreme dryness and roughness, emaciation, and weariness may be the result of excessive eating of the bitter taste. It may at times induce dizziness and unconsciousness.

ASTRINGENT

The astringent taste is present in unripe bananas, pomegranates, chickpeas, green beans, yellow split peas, okra, alfalfa sprouts, and the herbs goldenseal, turmeric, lotus seed, arjuna, and alum. It is cooling, drying, and heavy in nature and produces a dry, choking sensation in the throat. Taken in moderation, the astringent taste calms pitta and kapha but excites vata. It helps in the healing of ulcers and stops bleeding by promoting clotting.

Excess use may cause dryness in the mouth, difficulty in speech, and constipation, as

well as abdominal distention, heart spasms, and stagnation of circulation. It may affect the sex drive and lead to depletion of sperm. It can give rise to emaciation, convulsions, Bell's palsy, stroke paralysis, and other neuromuscular vata disorders.

EFFECTS OF TASTES ON THE DOSHAS

The tastes have the following effects upon the doshas:

VATA. People of vata constitution should avoid bitter, pungent, and astringent substances in excess, because they increase air and have a tendency to cause gas. Foods and herbs containing sweet, sour, and salty tastes are good for individuals of vata constitution.

How the Tastes Affect the Doshas

TASTE	VATA	PITTA	KAPHA
Sweet	↓	↓	↑
Sour	↓	↑	↑
Salty	↓	↑	↑
Pungent	↑	↑	↓
Bitter	↑	↓	↓
Astringent	↑	↓	↓

Herbs and the Six Tastes

The taste of an herb is not incidental but is directly related—indeed, directly responsible—for much of its therapeutic value. That is why Ayurvedic herbs are generally taken in a form that requires tasting them, rather than concealing the taste in a capsule.

There is no problem in taking an herb that has a sweet, pungent, or otherwise tempting taste. But most people, particularly in Western culture, don't like the bitter or astringent tastes, and if they have to take an herb with either of these tastes, they want to put the herb into a capsule and swallow it without tasting it. Since the stomach has no taste buds, when the herb is taken this way, the effects and benefits derived from the taste are lessened, because they are not perceived. When we eat food, we don't lose the effect of the tastes because we have to chew; when we use capsules, we miss the taste of the herb.

One of the reasons the Ayurvedic physician prescribes an herb is to balance whatever taste is lacking in the body. The herb transmits that taste and its effects into *rasa dhatu* (plasma). *Triphala*, for example, provides all the tastes except salty, but it tends to yield the predominant taste that is lacking in the body, which for

most Westerners is the bitter taste. That's why for many people *triphala* tastes bitter for some time. Later, after regular use, the bitter taste will have been received into the *rasa dhatu*, and *triphala* may taste sour or sweet.

In Ayurvedic medicine, most herbs are classified according to their predominant taste, secondary aftertaste, and "potential" taste. The main taste acts on *rasa dhatu*, the aftertaste acts on the nervous system, and the third taste has either a heating or a cooling effect.

This explains why it is important to have the effect of taste on the tongue when taking Ayurvedic medications.

PITTA. Pitta individuals should avoid sour, salty, and pungent substances, which aggravate bodily fire. However, sweet, bitter, and astringent tastes are beneficial for pittas.

KAPHA. Kapha individuals should avoid foods containing the sweet, sour, and salty tastes, for they increase bodily water. Better for them are foods with pungent, bitter, and astringent tastes.

Healthy and Unhealthy Eating Habits

How you eat is as important as what you eat. Here are some suggestions for healthy eating, followed by a list of habits to avoid.

EATING HABITS TO CULTIVATE

- Choose foods according to your constitution. They will nourish you and not aggravate your doshas.
- Choose foods according to the season.
- Eat fresh, *sattvic* food of the best quality you can afford.
- Do not eat unless you feel hungry.
- Do not drink unless you feel thirsty. If you are hungry and you drink instead of eating, the liquid will dissolve the digestive enzymes and reduce your gastric fire.
- Sit, don't stand, to eat.
- When eating, eat. That is, don't read, watch TV, or be distracted by too much conversation. Focus on the food.
- Chew well, at least 32 times per mouthful. This enables the digestive enzymes in the mouth to do their work properly.
- Eat at a moderate speed. Don't gobble your food.
- Fill one-third of your stomach with food, one-third with water, and leave one-third empty.

- Don't eat more at a meal than the amount of food you can hold in two cupped hands. Overeating expands the stomach so that you will feel the need for additional food. Overeating also creates toxins in the digestive tract.
- During meals, don't drink iced drinks or fruit juice, sip a little warm water between mouthfuls of food.
- Honey should never be cooked. If it is cooked, the molecules become like a glue that adheres to mucous membranes and clogs the subtle channels, producing toxins.

UNHEALTHY EATING HABITS

- Overeating
- Eating too soon after a full meal
- Drinking too much water, or no water, during a meal
- Drinking very chilled water during a meal, or indeed at any time
- Eating when constipated
- Eating at the wrong time of day, either too early or too late (see "[Ayurvedic Daily Routine](#)," starting [this page](#))
- Eating too much heavy food or too little light food
- Eating fruit or drinking fruit juice with a meal
- Eating without real hunger
- Emotional eating
- Eating incompatible food combinations (see chart on [this page](#))
- Munching between meals

Incompatible Food Combinations

The shelves of pharmacies and health food stores these days are lined with digestive aids and pills for indigestion and gas. It is likely that most of these gastrointestinal problems begin with poor food combining.

According to Ayurveda, certain food combinations disturb the normal functioning of the gastric fire and upset the balance of the doshas. Combining foods improperly can produce indigestion, fermentation, putrefaction, and gas formation. If such a situation in your stomach and intestines is frequent or prolonged, it can lead to disease. As just one example, eating bananas with milk can diminish agni (gastric fire) and change the intestinal flora, resulting in toxins and causing sinus congestion, cold, cough, allergies, hives, and rash. Such disturbances generate ama, the toxic substance that is the root cause of most ailments.

The following table lists some (but far from all) of the incompatible food combinations worth avoiding. You can alleviate some of the ill effects of these combinations by using spices and herbs in your cooking. A strong digestive fire can be

the most powerful means of dealing with these combinations. Chew a bit of fresh ginger (sprinkled with salt and lime juice if you like) before meals to stimulate digestion.

NAME OF FOOD	INCOMPATIBLE WITH
MILK	BANANAS Fish, Melons, Yogurt, Sour Fruits, Kitchari (mung dal and basmati rice), Bread made with yeast
YOGURT	MILK Sour fruits, Melons, Hot drinks—including coffee and tea— Fish, Mango (thus mango lassi is not a good idea), Starches, Cheese, Banana
MELONS	EVERYTHING, especially: Grains, Starches, Fried foods, Cheese
“Eat them alone or leave them alone”	
EGGS	MILK Yogurt, Melons, Cheese, Fruits, Potatoes
STARCHES	BANANAS Eggs, Milk, Dates
HONEY	GHEE in equal proportions (by weight)
(never cook honey)	Grains
CORN	Dates, Raisins, Bananas
LEMONS	Yogurt, Milk, Cucumber, Tomato
NIGHTSHADES (Potato, tomato, eggplant)	Yogurt, Milk, Melon, Cucumber

Particularly to be avoided are such concoctions as banana milkshakes and “fruit smoothies” made with milk. Mixed fruit salads are also incompatible. Some blended fruit drinks made with all fruit may be all right, but check this chart first.

Recommendations Regarding Milk and Milk Products

In Ayurveda, milk and dairy products such as ghee and freshly made yogurt are considered highly important to the diet. However, the process of pasteurization, which kills bacteria and other potentially harmful microorganisms, may also destroy the enzymes necessary for proper digestion. If the milk is heated for a fairly

long period of time, such as fifteen or twenty minutes, the enzymes will definitely be destroyed, and calcium and other nutrients may not be absorbed.

When milk is heated just until it reaches the boiling point, its enzymes are not destroyed, and it becomes less kaphagenic. So if you can obtain organic, unpasteurized milk from certified dairies and heat it just to the boiling point, that would be best.

Nevertheless, pasteurized milk from the supermarket, and dairy products made from that milk, are still better than no dairy products at all.

Note that for each food in capital letters on the left, the food in capitals on the right is the most incompatible; foods in small letters are less incompatible.

Food and the Three Gunas

The Ayurvedic tradition teaches that food is not only for nutrition, to nourish the body, but also affects the mind and consciousness. As we have a physical constitution (vata–pitta–kapha), we also have a mental constitution characterized by the three *gunas*: *sattva*, *rajas*, and *tamas*.

According to the Sankhya philosophy of creation, *sattva*, *rajas*, and *tamas* are universal qualities necessary for the creation of the universe (see [this page](#)). They are equally necessary for maintaining our psychobiological functions.

Because of *sattva*, we remain conscious and reawaken every morning. Because of *rajas*, our thoughts, feelings, and emotions move in a creative way. Because of *tamas*, we become tired, exhausted, and heavy; without *tamas* there is no sleep. Another way to look at it is that *sattva* brings clarity, *rajas* brings perception, and *tamas* gives solid, concrete experience.

These three qualities are also necessary for the functioning of every cell. *Satva* is potential energy, *rajas* is kinetic energy, and *tamas* is inertia. The potential energy in the cell is awareness; it becomes active due to the kinetic energy of *rajas*; then the cell becomes inert because of the *tamasic* quality. Thus these three qualities are absolutely necessary for the psychobiological activities of the human body.

Psychological Constitutions

Indian philosophy classifies human temperaments into three basic types: *sattvic*, *rajasic*, and *tamasic*. These types all differ in psychological and moral disposition, as well as in their reactions to social, cultural, and physical conditions, as is described in the classical texts of Ayurveda.

Sattvic qualities imply essence, reality, consciousness, purity, and clarity of

perception. People in whom *sattvic* qualities predominate are loving, compassionate, religious, and pure-minded, following truth and righteousness. They tend to have good manners and positive behavior, and they do not easily become upset or angry. Although they work hard mentally, they do not get mental fatigue, so they need only four to five hours of sleep at night. They look fresh, alert, aware, and full of luster and are recognized for their wisdom, happiness, and joy. They are creative, humble, and respectful of their teachers. Worshiping God and humanity, they love all. They care for people, birds, animals, and trees and are respectful of every life and existence.

Rajasic individuals are loving, calm, and patient—so long as their own interests are served! All their activities are self-centered and egotistical. They are kind, friendly, and faithful only to those who are helpful to them.

All movement and activity is due to *rajas*, which leads to the life of sensual enjoyment, pleasure and pain, effort and restlessness. People in whom *rajasic* qualities predominate tend to be egoistic, ambitious, aggressive, proud, and competitive and have a tendency to control others. They like power, prestige, and position and are perfectionists. They are hard-working people but may be lacking in proper planning and direction. Emotionally they tend to be angry, jealous, and ambitious and to have few moments of joy. They suffer from a fear of failure, are subject to stress, and are quickly drained of mental energy. They require eight hours of sleep.

Tamas is darkness, inertia, heaviness, and a tendency toward materialism. Individuals dominated by *tamas* are often less intelligent. They tend toward depression, laziness, and excess sleep, even during the day. A little mental work tires them easily. They like jobs with less responsibility, and they love to eat, drink, sleep, and have sex. They tend to be greedy, possessive, attached, irritable, and uncaring toward others. They are willing to harm others for their own self-interest.

There is a constant interplay of these three *gunas* in everyone's consciousness, but the relative predominance of either *sattva*, *rajas*, or *tamas* is responsible for an individual's psychological constitution.

In the Ayurvedic literature, food is classified as *sattvic*, *rajasic*, or *tamasic* according to the mental qualities it promotes. In brief, *sattvic* food is light, healthy food that increases clarity of mind, *rajasic* food is tempting food that increases activity and agitation, and *tamasic* food is heavy, dulling food that creates depression and heaviness and leads to many disorders.

Sattvic food is light and easy to digest. It brings clarity of perception, unfolds love and compassion, and promotes the qualities of forgiveness and austerity. *Sattvic* foods

include fruit, steamed vegetables, and fresh vegetable juice. Milk and ghee are *sattvic* foods that build up *ojas* and give vitality to *prana*.

Rajasic foods are hot, spicy, and salty. They are irritants and stimulants, and they are tempting foods (once your hand goes into the bag, you cannot stop eating them), such as salty crackers and potato chips. *Rajasic* foods also include certain heavily spiced foods, such as hot pickles and chutneys, which stimulate the senses. These foods make the mind more agitated and susceptible to temptation. Gradually, from eating these foods, the mind becomes more *rajasic*, which means it tends toward anger, hate, and manipulation.

Tamasic food is heavy, dull, and depressing and induces deep sleep. Under that category comes any dark meat, lamb, pork, and beef, as well as thick cheese. Old and stale food is also *tamasic*.

However, the heavy, dulling effect of *tamasic* food occurs only when it is eaten in excess. In moderation, *tamasic* food is grounding and promotes stability. If, for example, an individual has an excess of the *rajasic* quality—the mind is hyper and ungrounded and there is insomnia—some *tamasic* food eaten in moderation will help the person become more grounded and get some sleep.

We can classify food into categories of *sattvic*, *rajasic*, or *tamasic* according to the table on the facing page.

RELATIONSHIP OF THE *GUNAS* AND THE DOSHAS

Students of Ayurveda frequently ask whether there is a relationship between the three *gunas* and the three doshas. There is not a direct correspondence, but there is a relationship.

Sattva is present in the doshas in this order:

1. in pitta as knowledge and understanding
2. in vata as clarity and lightness
3. in kapha as forgiveness and love

Tamas is present in the doshas in this order:

1. It is heavy, dull, and sleepy in kapha.
2. In pitta, it expresses as aggressiveness and competitiveness.
3. There is very little of it in vata, but it is represented as confusion.

Rajas, active and hyper, is present in vata and in pitta but is virtually absent from

kapha.

Vata is approximately 75 percent *rajas*, 20 percent *sattva*, and 5 percent *tamas*. Pitta is 50 percent or more *sattva*, 45 percent *rajas*, and up to 5 percent *tamas*. Kapha is maybe 75 percent *tamas* and 15 to 20 percent *sattva*, with very little *rajas*. Here is another way we can see these relationships:

Food Categories of Tamasic, Rajasic and Sattvic			
	TAMASIC	RAJASIC	SATTVIC
FRUIT	Avocado Watermelon Plums Apricots	Sour fruits Apples Bananas Guava	Mango Pomegranate Coconut Figs Peaches Pears
GRAINS	Wheat Brown Rice	Millet Corn Buckwheat	Rice Tapioca Blue corn
VEGETABLES	Mushrooms Garlic Onion Pumpkin	Potato Nightshades Cauliflower Broccoli Spinach Tamarind Pickles Winter squash	Sweet potato Lettuce Parsley Sprouts Yellow squash
BEANS	Urad dal Black Pinto Pink	Red lentils Toor dal Adzuki	Mung Yellow lentils Kidney Lima
DAIRY	Cheese (hard, aged)	Old, sour milk Sour cream	Milk Fresh homemade yogurt or cheese
MEAT	Beef Lamb Pork	Fish Shrimp Chicken	None
	SATTVA	RAJAS	TAMAS
VATA	Clarity Creativity Lightness	Hyperactivity Nervousness Fear Anxiety Ungroundedness	Confusion Lack of direction Indecisiveness Sadness Grief
PITTA	Knowledge Understanding Comprehension Recognition	Aggressiveness Competitiveness Power Prestige	Anger Hatred Envy Jealousy
KAPHA	Love Compassion Forgiveness	Attachment Greed Possessiveness	Deep confusion Unconsciousness Coma Depression

Now you have more than enough background information to benefit from the remedies and recommendations made in [Part III](#). I hope you have enjoyed this introduction to Ayurveda and that you will incorporate its principles and practices into your life. If you do, I know your health will improve and your life will blossom physically, mentally, emotionally, and spiritually.



Part III



Secrets of Ayurvedic Self-healing: An Encyclopedia of Illnesses and Remedies

How to Use the Encyclopedia

The ancient Ayurvedic art of healing deals with every individual's life as a whole. To the Ayurvedic *vaidya* (physician), every individual is an indivisible, complete, unique being. At the same time, Vedic philosophy teaches that there is a concurrent and inherent relationship between the macrocosm (the universe) and the microcosm (the individual). The individual is constantly exposed to environmental changes, seasonal changes, changes in diet, lifestyle, emotions, job, financial status, and relationships. These changes are constantly bombarding every human being. To remain healthy or to regain health, all these factors have to be taken into consideration.

As you have learned, when the body's doshic balance of vata–pitta–kapha is disturbed, illness may result. The purpose of Ayurvedic healing is not just to relieve the symptoms of a specific illness but to bring the out-of-balance factors back into harmony.

Thus, the purpose of Ayurvedic remedies, whether they be herbal medications, dietary and nutritional changes, yoga postures, cleansing procedures, or breathing exercises, is to eradicate the underlying causes of the disease, not merely to remove the symptoms. Of course one must deal directly with emergency or life-threatening symptoms, such as wheezing in asthma, heart pain in heart conditions, or high fever in infections. But if the fundamental causes of the illness are not dealt with, the problem will manifest again in the same or another form.

Components of Ayurvedic Healing

The Ayurvedic approach to restoring health is known as *chikitsa* (disease management) and traditionally consists of eight components. If you glance at the accompanying box, you will note that *chikitsa* is a complete healing program that begins with identifying and eliminating the underlying cause of the illness, proceeds by purifying the body and reestablishing balance, and ends with strengthening and revitalizing the affected organs, tissues, and systems so that the disease will not recur.

If you become sick, your illness is not likely to become completely healed unless you change the behaviors that gave rise to it. Dietary indiscretions, a stressful lifestyle, unresolved emotions, insufficient exercise—these are some of the factors that are at the root of most illness. Identifying and eliminating these causal factors are essential components of the healing process. If you take some of the recommended herbal formulas but allow unhealthy lifestyle habits to continue unabated, it's not likely that you will experience lasting or significant improvement.

That is why, for each condition, I have suggested not only medications but numerous other measures you can use for healing, such as specific yoga postures and breathing exercises, foods to favor or avoid, massage oils, teas, healing pastes, and many other ways to promote healing. In addition, [Part II](#) of the book provides suggestions for your

diet and daily routine that can get you established in a healthy way of living in harmony with nature. Please make use of these recommendations as part of your own personal holistic program for creating and supporting health.

Your body's innate healing mechanisms are always at work, striving to maintain or restore total health and balance. The recommendations offered here will support the body in its natural process of healing.

Ayurvedic Disease Management (Chikitsa)

The traditional Ayurvedic program of healing has eight essential components:

1. Find out the person's *prakruti* (constitution).
2. Find out the *vikruti* (the present altered state of the doshas in the body).
3. Find out the cause or causes of the illness, such as diet, lifestyle, emotional patterns, quality of relationships, genetic predisposition, and so on.
4. As the first line of treatment, remove the cause.
5. Provide the proper regimen (diet, exercise, *pranayama*, etc.), according to the person's *prakruti*, *vikruti*, seasons, climate, age, and so on.
6. Provide a detoxification procedure: either palliation (*shamana*) or elimination (*shodana*, such as *panchakarma*).
7. Provide rejuvenation (*rasayana*) for the body in general, to increase immunity and to strengthen specific organs and tissues.
8. Provide therapies that are (a) antagonistic to the provoked dosha and (b) antagonistic to the disease, based on the principle that opposite qualities balance.

The following points will help you to make best use of this encyclopedia of Ayurvedic healing.

Diagnosis and Treatment

Ayurvedic first-aid treatments are effective if you first do a careful, discriminating diagnosis to determine whether the condition is vata, pitta, or kapha in nature. A specific treatment, for a specific type of condition, will be beneficial.

On the other hand, if you do not take the time to diagnose carefully, the treatment may not be appropriate, and consequently you will not get the desired result. So before deciding on an Ayurvedic line of treatment, look carefully at the physical signs and symptoms that will help you determine whether it is a vata, pitta, or kapha illness or condition. Then choose the recommended treatment.

On the few occasions when it is not mentioned whether a condition is vata, pitta, or

kapha, you can feel comfortable using any general treatment suggestions made for that condition.

A NOTE ABOUT HERBAL MEDICATIONS. The pharmacopoeia of Ayurveda is vast, including literally thousands of medicines, many of them herbal preparations. For each condition listed here, I have tried to include several very simple remedies, utilizing common kitchen herbs or household strategies such as taking a warm bath. I also recommend some common Ayurvedic herbs, most of which are easily obtained from various sources (see [Resources](#)). For guidelines on how to prepare your own herbal formulas, how to take herbs, how to make ghee, and other important information, please see [appendix 2](#).

SEE WHAT WORKS. For each condition, several suggestions are made. No one expects you to use all of them. Try what sounds good to you, and observe how it works. If it solves the problem, you don't need to try any others. If it doesn't seem to work, then try a different approach. Each person is different and will react differently to these remedies.

DON'T BE DISCOURAGED IF A REMEDY DOESN'T WORK. In self-treatment using Ayurvedic principles and remedies, keep in mind a fundamental therapeutic guideline. If you diagnose your problem to be, for example, a vata disorder and embark on a treatment to reduce vata, then find that the treatment is not working or appears to be exacerbating the problem, this indicates that your diagnosis was not entirely correct.

This does not mean you should give up on Ayurvedic remedies! Simply try using the treatment recommended for either pitta or kapha, whichever makes more sense to you considering your symptoms. If you are untrained in Ayurvedic diagnosis, you may simply have made a mistake. So use your best judgment and try again.

Sometimes the effect of the first remedy can give you a clue for a more beneficial treatment. For example, perhaps you thought your condition was due to excess kapha, and you decided to use a heating herb such as ginger, pepper, or *trikatu* to treat it. If the condition worsens and you develop symptoms of aggravated pitta, such as irritability or a skin rash, you can infer that perhaps the problem was actually related to excess pitta. Reassess the situation, and take the next step.

BE SURE TO TREAT YOUR ACTUAL CONDITION. Most imbalances and illnesses are related to constitution. As it is often said, “*Prakruti* indicates disease proneness.” Vata individuals are likely to have vata-related problems, pitta individuals will have primarily pitta conditions, and kaphas will have kapha illnesses.

However, in a small percentage of cases, your illness or condition may not coincide with your own constitutional type. For example, you may be predominantly a pitta type, but you may have a vata condition such as insomnia or constipation. It may be the result of any combination of vata-aggravating factors in your life, such as a vata diet, a cold dry climate, or irregular hours. So when remedies are suggested, be careful to choose options that fit the symptoms of your *condition*, not necessarily your body type.

HOW LONG TO TAKE A REMEDY. In general, remedies should be used until your symptoms

disappear. This may be a few days, a few weeks, or a few months, depending on several important factors, such as: How severe is the disease or condition? How long have you had it? How motivated are you to get well?

First, assess your desire to get well. If you have a strong desire and are motivated to get results, the first thing you need is diligence in following the prescribed Ayurvedic regimen. Because of the law of *karma* (cause and effect), to a great extent you will derive benefits according to your own actions. So in addition to taking your herbs two or three times a day as recommended (no skipping, no forgetting them at home ...), you also need to address the underlying causes of your condition by rethinking your diet, daily routine, exercise program, etc. Chances are that the herbal remedies alone, without some changes in your lifestyle, will not be powerful enough to override the very behavior patterns that caused the illness in the first place.

If your condition is serious, and you are diligent with your remedies and make appropriate lifestyle changes and still the symptoms persist, you need to see your doctor for help. On the other hand, if the condition is chronic, it is unrealistic to expect that something that has persisted for years is going to disappear in a week or a month.

So in deciding how long to use the recommended remedies, use your common sense. And do your best right from the start: If you don't, and you end up not getting the results you would like, it's human nature that you will question whether you could have done better.

Every disease process in the body, and every process of healing, has its own momentum, speed, and duration. You cannot make the mango ripen quickly; it ripens of its own accord. So in order to eradicate the disease completely, allow sufficient time, and have patience. Ayurveda is not a quick fix. Anything that is quickly fixed does not resolve the problem totally.

Cautions

CONSULT YOUR DOCTOR. Some of the conditions in this book are serious medical conditions which require the supervision of a medical doctor. Mild or preclinical cases may sometimes be completely healed using these Ayurvedic recommendations, but these remedies are not a substitute for consulting with a physician.

If you are already under a doctor's care for a specific illness or condition, the Ayurvedic remedies suggested here may be used in conjunction with the regimen outlined by your physician. But it is only fair and proper to do so with his or her knowledge and supervision.

Ask your doctor to carefully monitor your progress. As time goes by, you may be able to minimize or eliminate your dependence on strong medications if your body's balance can be brought to a point at which diet, exercise, herbs, and other Ayurvedic methods

are sufficient to control or eliminate the condition.

DETERMINE THE SEVERITY OF THE DISEASE. The vast majority of minor injuries and illnesses can be treated at home, with such natural means as lifestyle changes, diet, herbs, and simple yoga stretching exercises. But sometimes it is absolutely crucial to get medical attention from a qualified physician or even from a trained hospital staff. This is extremely important and must always be remembered.

For example, diarrhea that lasts for a day or even a couple of days can be effectively treated at home (see “[Diarrhea](#)”). But suppose a person has profuse diarrhea that goes on for some time, so that he or she becomes severely dehydrated. This is a serious condition that requires hospitalization. It cannot be treated at home or even in the outpatient department; it needs admission to a hospital so that intravenous feeding can begin immediately.

Or suppose someone has a high fever, perhaps 104 or 105 degrees Fahrenheit, and is babbling incoherently, appears delirious, and is losing consciousness. This is a serious situation and one has to act promptly to get help.

So always be alert to the severity, intensity, persistence, or recurrence of any disease condition, and be sure to treat it properly. And remember: If you are not a trained medical professional, you may not always know. When in doubt, see a doctor.

A WORD ABOUT LICORICE TEA. On several occasions I recommend the use of licorice tea. Individuals with hypertension (high blood pressure) may use this tea for emergencies (such as to relieve an asthma attack—see “[Asthma](#)”), but they should not use it on a regular basis; it promotes sodium retention, which may increase blood pressure.

Illnesses and Remedies, A-Z

▮▮ Abdominal Cramps ▮▮

See “[Muscle Cramps and Spasms](#)”

▮▮ Acne ▮▮

See also “[Skin—Ayurvedic Care](#)”

Acne is the result of high pitta moving under the skin and breaking out in pimples. Ayurveda recommends several natural approaches that, taken together, can effectively control acne.

The possible pitta-provoking causes are numerous. They include emotional stress, premenstrual hormonal changes, and exposure to chemicals or too much sunlight. The problem might also be a bacterial infection. It is important to find out the cause, so it can be properly treated or, in the case of exposure to chemicals or sunlight, simply avoided.

FOLLOW THE PITTA-PACIFYING DIET. Because acne is a pitta condition, the first step is to follow the pitta-pacifying diet detailed in [chapter 8](#). Avoid spicy and fermented foods, salt, fried foods, and citrus fruits. Favor blander foods such as rice, oatmeal, and applesauce.

USE THESE HERBS TO BALANCE PITTA. The following formula is excellent for pacifying the excess pitta that causes acne:

kutki

guduchi

shatavari

Make a mixture of equal proportions of these three herbs (you might start with 1 teaspoon of each), and take ¼ teaspoon of your mixture 2 or 3 times a day. After meals, place the powder on your tongue and wash it down with warm water.

- A helpful tea, made from common household herbs, is cumin-coriander-fennel tea. After each meal, steep ⅓ teaspoon of each of these three seeds in hot water for 10 minutes, strain, and drink. Use this tea 3 times a day.

DRINK BLUE WATER. Fill a clear glass bottle or jar with water, and cover it with some translucent blue paper (such as blue cellophane, available at art supply and some grocery stores). Put the bottle in the sun for about 2 hours. Drink 1 to 3 cups of the water each day. This will have a cooling, soothing effect. Believe it or not, it works!

Healing Pastes for Your Skin

Here are three pastes you can make at home and apply to your skin. They can be effective in reducing acne.

1. Mix 1 teaspoon of chickpea flour (available at Indian grocery stores and natural food stores) with enough water to make a paste, and wash your face with this mixture. Rinse off, then apply either of the following:
2. Mix almond powder with a little water, and apply the paste to your face. Let it dry and remain on your skin for up to half an hour, then rinse it off. (You can easily make the almond powder yourself in a coffee or nut grinder.)
3. A paste of sandalwood and turmeric powders mixed in goat's milk is healing for the skin. Take $\frac{1}{4}$ teaspoon turmeric and $\frac{1}{2}$ teaspoon sandalwood powder, mix them together, and add sufficient goat's milk to make a paste. Apply this mixture to your face. *Note:* Your face will look yellow for some time—up to 5 days—but this formula is quite effective in alleviating acne.

DRINK ALOE VERA JUICE. You might try drinking half a cup of pure aloe vera juice twice a day.

KEEP YOUR COLON CLEAN. Keeping your colon clean is important, in order to remove toxins from the body. You can easily accomplish this by taking the herb *amalaki*, $\frac{1}{2}$ to 1 teaspoon daily, as a powder on the tongue. Take it before bedtime, and wash it down with warm water.

APPLY MELON. Rub some melon on the skin at bedtime, and leave it on overnight. Its cooling, anti-pitta quality will help heal acne. It also makes the skin soft.

YOGA POSTURES. Recommended yoga *asanas* for acne are the Lion Pose and the sequence of postures called the Moon Salutation. (See [appendix 4](#) for illustrations.)

BREATHING EXERCISE. Breathing through the left nostril only, for 5 to 10 minutes, will help to reduce pitta. (This is called the Moon Breath and is said to be cooling; breathing through the right nostril is called the Sun Breath and is heating.) Simply cover the right nostril with your thumb, and breathe normally through the left side. If the nostril is blocked, don't force it; try again later.

RELAX YOUR FACE. Rub your hands together vigorously to create a bit of warmth, and place both hands over your face for a couple of minutes. This will relax the facial muscles and increase the blood supply.

You can also rub your hands until the palms become warm, then gently touch just the heart of the palm to the eyelashes. Our eyelashes are intensely electrically charged; they take that warmth and relieve the pitta under the skin that is coming out in the form of

acne.

VISUALIZATION. The root cause of acne is emotional stress. One effective way to relieve that stress is visualization. Close your eyes and visualize that the acne is clearing up and going away—as if you are communicating with the tissues in your skin that are bursting out in acne. This works.

A final suggestion: Avoid frequent looking in the mirror and feeling bad about the acne.

❧ ❧ Addictions ❧ ❧

See also “[Smoking](#)”

Why are people addicted? In the majority of cases (excluding tragic situations such as babies who are born addicted because of their mother’s addiction), people who are addicted start out simply seeking more pleasure and joy in their life. Their life is difficult and unhappy, their relationships may be painful and unfulfilling, they may be dissatisfied and stressed at work, and they simply don’t know how to deal with the situation. So they escape from the reality of their circumstances into drugs or alcohol.

Whether the addictive substance is tobacco, marijuana, alcohol, or something else, it soon goes beyond being a psychological escape and becomes a chemical dependency. Then, unless a certain level of the addictive substance is present in the blood, the person’s brain doesn’t function properly.

Treatment depends on how serious and long-standing the addiction is. For milder addictions, such as a recent smoking habit, the person may simply be able to stop. But if a chronic alcoholic suddenly stops drinking, it creates alcohol withdrawal syndrome that is difficult to deal with.

CLEANSING. To effectively handle the problem of addiction, it is important to do *panchakarma*, an effective Ayurvedic cleansing and detox program. See [chapter 4](#) for a description of *panchakarma* treatments you can receive at an Ayurvedic clinic, and a home *panchakarma* program you can do for yourself.

DOSE REDUCTION. Along with this cleansing program, slowly decrease the dose of the addictive substance. According to Ayurveda, unless some strong medications are available to deal with withdrawal, it is not good to completely stop using the addictive substance all at once, or a stressful withdrawal syndrome will probably occur.

- With nicotine toxicity (which affects the lungs and cardiovascular system) and alcohol toxicity (which affects the liver) we have to strengthen the affected organs. For alcohol toxicity, use this formula:

chitrak 3 parts

kutki 3 parts

Take ½ teaspoon of these herbs, with 2 tablespoons of aloe vera juice, 3 times a day.

- Ayurveda suggests a bitter wine made of aloe vera juice. (It is called *kumari asava*.) In place of hard liquor or other alcoholic drinks, the person who is addicted to alcohol can take small amounts of this light, dry wine. Try 4 teaspoons diluted with an equal amount of water. Then gradually reduce the amount of herbal wine in the dose, while at the same time using the above herbal formula to strengthen the damaged liver.

- The same applies to tobacco. For a person who is addicted to nicotine, remove one-third to one-half of the tobacco from each cigarette (at the end you light), and fill the paper with a mixture of rose petals, *brahmi*, and *jatamansi* (equal proportions). Smoke it until the tobacco starts burning. The moment the tobacco starts, put out the cigarette and discard it.

NASAL MEDICINE. Doing *nasya* with *brahmi* ghee will also reduce the toxicity of nicotine (see [appendix 3](#)).

EXERCISE. Most of the time, whenever there is the desire to drink or to smoke, the person should go out for a walk in the fresh

TIME TO SEE THE DOCTOR

If there is strong alcohol addiction and the person experiences a headache, tremors, drowsiness, depression, or other alcohol withdrawal symptoms when he or she stops drinking, see a doctor right away: Medical help is needed.

air, or do some other exercise, or go for a swim.

APPETITE STIMULATION. Some people drink because they have a low appetite. Unless they have a drink, they never feel hungry. In such cases, instead of alcohol, they can have some ginger tea to stimulate their appetite (see also “[Appetite, Low](#)”). Or try this recipe for a tea to stimulate agni, the digestive fire:

Agni Tea

1 quart water

⅛ pinch cayenne pepper

½ handful minced ginger root

2 tablespoons Sucanat or other sweetener

⅛ to ½ teaspoon rock salt

Put all the above ingredients in a pot and boil for 20 minutes.

Take the pot off the burner, cool for a few minutes, then add the juice of half a lime. Do not boil the lime juice.

YOGA ASANAS. Some yoga exercises will also be beneficial. Sun Salutations Should be included, and some Alternate Nostril breathing. *So-Hum* Meditation will also be helpful. (See [chapters 6](#) and [7](#).)

☸ ☸ Allergies ☸ ☸

See also “[Food Allergies](#)”

According to Ayurvedic *samprapti* (pathogenesis), allergies are a doshic reaction to a specific allergen, such as pollen, dust, chemicals on a rug, ragweed, or any strong chemical smell. These allergic reactions are classified as vata type, pitta type, and kapha type.

- *Vata-type allergies* are characterized by bloating of the stomach, gastric discomfort, or even intestinal colic. A vata allergy may lead to sudden wheezing, sneezing, headache, ringing in the ears, or insomnia. For example, some individuals, when exposed to dust or pollen, suddenly start wheezing. The wheezing is due to narrowing of the bronchial tree due to vata dosha. That person may also experience insomnia and other vata-type symptoms.

- *In a pitta type of allergy*, pitta dosha is already present under the skin. If the person comes in contact with an allergen, such as chemicals, ragweed, or certain synthetic fibers, then the pitta penetrates through the capillaries due to its hot and sharp qualities and creates a rash, itching, hives, urticaria, allergic dermatitis, or eczema—all pitta-type allergic reactions.

- *Kapha allergies* are often experienced during spring season, when plants and trees shed their pollens into the atmosphere. When the pollens, such as juniper or any other flower pollen, are inhaled, they enter the nasal-respiratory passage, and in some people they irritate the delicate mucous membrane, leading to hay fever, colds, congestion, a cough, sinus infection, and even asthma.

In order to treat allergies effectively, first we have to find out whether it is vata, pitta, or kapha type. Then we can determine the specific line of treatment.

In most cases, perhaps 80 percent, your *prakruti* (constitution) predicts your allergy proneness. That is, there is usually a correspondence between a person's constitution and the type of allergic reaction. A person of pitta *prakruti* is more likely to have a pitta allergic reaction, especially when the *vikruti* or current status of the system shows a pitta imbalance. But it may also happen that due to diet, environmental conditions, emotional factors, or other causes, a kapha person may have a vata imbalance, and so forth.

TREATMENT FOR VATA-TYPE ALLERGIES

BASTI. One of the most effective remedies for vata-type allergies is a *dashamoola* tea *basti* (enema). Boil 1 tablespoon of the herbal compound *dashamoola* in 1 pint of water for 5 minutes to make a tea. Cool it, strain it, and use the liquid as an enema. (See [appendix 3](#) for complete directions.) Vata symptoms, such as wheezing, sneezing, dryness of the throat, dryness of the colon leading to distension, constipation, and abdominal discomfort, can be immediately corrected by this *dashamoola* tea *basti*.

HERBAL REMEDIES. Use this herbal formula:

ashwagandha 1 part

bala 1 part

vidari 1 part

Mix these herbs in equal proportion, and take $\frac{1}{4}$ teaspoon of the powder 3 times a day, washed down with warm water, to relieve vata allergies.

- To soothe an extreme wheezing condition, make one cup of either ginger or licorice tea, boiling 1 teaspoon of the herb for about 3 minutes in 1 cup of water. Then add 5 to 10 drops of *mahanarayan* oil, mix thoroughly, and take 1 sip every 10 to 15 minutes. (If you do not have *mahanarayan* oil, you can substitute $\frac{1}{2}$ teaspoon of plain ghee.)

TREATMENT FOR PITTA-TYPE ALLERGIES

HERBAL REMEDIES. This herbal formula is effective to pacify pitta:

shatavari 8 parts

kama dudha $\frac{1}{2}$ part

guduchi 1 part

shanka bhasma $\frac{1}{4}$ part

Take $\frac{1}{2}$ teaspoon of this mixture 2 or 3 times a day after meals, with a little warm water.

- For hives, rash, urticaria, dermatitis, or eczema, apply *neem* oil or *tikta ghrita* (bitter ghee) on the skin.

BLOOD PURIFICATION. Traditionally, Ayurveda has suggested that individuals with high pitta, who are prone to developing pitta-type problems such as sunburn in the summer season, do *rakta moksha*, or bloodletting, before the onset of the summer. Although this practice is currently not very well respected in the West, it is still used widely in India, as it has proven to be an effective preventive and healing measure. To make use of it today, you might consider donating about $\frac{1}{2}$ pint or 100 cc. of blood to a blood bank. That will help to defuse pitta conditions such as allergic dermatitis and allergic eczema.

- To produce a similar effect, you can use a blood-cleansing herbal combination. For

example, mix the herbs *manjistha* and *neem* in equal amounts.

manjistha 1 part

neem 1 part

Take ½ teaspoon of this mixture 3 times a day with warm water after meals. It will cleanse the blood and help to heal pitta-type allergies.

- The common Western herb burdock is also an effective blood purifier; you can make a tea from ½ teaspoon burdock per cup of boiling water and drink it 2 or 3 times a day.

TREATMENT FOR KAPHA-TYPE ALLERGIES

HERBAL REMEDIES. Kapha allergies generally take the form of respiratory-pulmonary congestion, cough, cold, asthma, or hay fever. For relief from these conditions, use the following herbal formula:

sitopaladi 4 parts

yashti madhu 4 parts

abrak bhasma ⅓ part

Take about ¼ teaspoon of this mixture 3 times a day with honey.

PURGATION THERAPY. Kapha-type allergies occur when excess kapha collects in the stomach and lungs. One way to relieve this congestion is purgation therapy (*virechana*). Use flaxseed oil (available in most natural food stores), and take 1 teaspoon 2 or 3 times a day for 2 or 3 days. This will be quite effective. Or you can use *triphala* (see below).

VOMITING THERAPY. The Ayurvedic therapy that is particularly effective for removing excess kapha from the stomach and respiratory tract is *vamana*, or vomiting therapy. I have noticed, however, that people in the West have a strong cultural bias against vomiting, and many seem particularly uncomfortable with this procedure. It not only seems physically repugnant but may be emotionally difficult as well, as some emotional purification may arise as a result of the physical purification. So if you tend to have strong emotions or have trouble dealing with them, it might be better for you not to try *vamana*.

If you want to try it—and I want to emphasize that it is very effective for eliminating excess kapha—the procedure is to drink a stomach full of licorice tea and salt water and then to regurgitate it, emptying the stomach. Start by drinking several cups of licorice tea, followed by a pint of water with about 1 teaspoon of salt mixed in. Drink enough to fill your stomach, then rub the back of the tongue and vomit it out.

IMPORTANT CAUTION: If you have high blood pressure, low blood pressure, hiatal hernia, or a history of heart problems, do not do *vaman* therapy.

HEALING GUIDELINES FOR ALL TYPES OF ALLERGIES

USE TRIPHALA. For all three types of allergies, one can take ½ to 1 teaspoon of *triphala* at night. (See [appendix 2](#) for instructions for preparing *triphala*.) *Triphala* acts as both a laxative and a purgative. It consists of three herbs: *amalaki*, *bibhitaki*, and *haritaki*. *Haritaki* works on vata dosha, *amalaki* on pitta dosha, and *bibhitaki* on kapha dosha.

DIETARY CHANGES. For vata allergy, follow a vata-soothing diet; for pitta type of allergy, the pitta-pacifying diet; and for kapha allergy, the kapha-reducing diet. (Diet guidelines may be found in [chapter 8](#).)

WATCH YOUR FOOD COMBINATIONS. It is important for individuals with allergies not to eat incompatible food combinations, such as milk and yogurt, meat and dairy, poultry and dairy, melon and grains, or fruits and grains. Avoid such things as banana milk shakes and “fruit smoothies” made with milk. For a more complete list of food incompatibilities, please turn to [this page](#).

AVOID THE CAUSE. For most allergies, one should try to avoid the immediate cause: the allergen. People who are allergic to cats, dogs, hair, pollen, mold, and so on should simply try to avoid them. Also try to stay away from synthetic fibers such as polyester and rayon, which can cause pitta-type skin allergies. It is best to wear cotton clothing. Because of the large quantity of pesticides routinely sprayed on cotton, you might consider using only organic cotton products, though they tend to be more expensive.

BLOCK THE ALLERGENS. Generally, the respiratory passage is open to dust and other allergens. One way to minimize the effect of allergens that you can’t avoid is to lubricate the nasal mucous membrane with ghee. This prevents direct contact of the allergen with the mucous membrane.

USE NEEM OIL. Another way to reduce or avoid the effect of environmental allergens is to apply *neem* oil to the exposed part of the body. The presence of the oil on the skin, as well as the disinfectant properties of *neem*, will minimize contact with the allergen.

NOTE: Use *neem* herbalized oil—that is, *neem* leaves cooked in a base of sesame or another oil. Pure *neem* extract will be too strong. If you find that even this herbalized *neem* oil is too strong and creates an itching or burning sensation, mix it half and half with coconut oil.

MEDITATE FOR STRESS REDUCTION. Most allergies are stress related. Because of stress, imbalance is created in mind and body. The practice known as Empty Bowl meditation helps to restore balance and thus helps take care of stress-related allergies. (See [chapter 7](#) for guidelines.)

YOGA POSTURES. The most helpful yoga *asana* for kapha and vata allergies is the Sun Salutation. For pitta allergies, do the Moon Salutation. For help with yoga *asanas*, see [appendix 4](#).

BREATHING EXERCISES. Alternate Nostril breathing is effective for respiratory allergies such as hay fever, wheezing, and sneezing. *Bhastrika* (Breath of Fire) is good for kapha-type congestive allergies (see [chapter 6](#)). Also, *ujjayi pranayama* helps to improve immunity and is beneficial for all types of allergies.

❧ ❧ Anemia ❧ ❧

Modern medicine has delineated several types of anemia: iron-deficiency anemia, pernicious anemia, sickle-cell anemia, and hypoproteinemic anemia (lack of protein in the blood), as well as certain vitamin-deficiency anemias, such as B-12 and folic acid-deficiency anemia. Certain bleeding disorders, such as profuse menstrual bleeding, bleeding hemorrhoids, or bleeding gums, can lead to anemia because of blood loss. Whenever modern medicine deals with the problem of anemia, it considers all these etiological factors.

Ayurveda looks at anemia quite differently. Ayurvedic classification of anemia falls under three basic doshic conditions: vata type, pitta type, and kapha type. It doesn't matter whether a person has iron deficiency or folic acid deficiency; what is important is how the anemia is expressing itself through a particular individual. As it turns out (see box [this page](#)), there is also a correspondence between the Ayurvedic interpretation and the understanding according to modern medicine.

TYPES OF ANEMIA

- In vata-type anemia, the person looks thin, with dry, rough, scaly skin, and has cracking of the joints. He or she looks emaciated and pale, may suffer from breathlessness and constipation, and may pass tarry black stool.
- In pitta-type anemia, the eyes are slightly yellowish, the person may get high-colored urine, and the stools are dark brown or have a slightly yellowish tinge. There may be nausea or pain in the liver and/or spleen area. Dizziness or vertigo may be experienced, and the person may become easily irritated by light.
- In kapha-type anemia, there is often swelling (edema), and the person's skin feels cold and clammy and looks shiny. Because of the edema, the skin gets stretched so much that you can often see the reflection of the window on it.

By careful observation one can tell whether the anemia is vata, pitta, or kapha type. By treating that dosha, one treats the root cause and can alleviate the anemia.

TREATMENT

FOR VATA-TYPE ANEMIA. For vata anemia, Ayurveda suggests taking *tikta ghrita*, which is

bitter ghee (see [appendix 2](#)). One teaspoon of *tikta ghrita* 5 to 10 minutes before breakfast, lunch, and dinner will help to improve the blood volume.

- One can also use a mixture of

kaishore guggulu 2 parts

abrak bhasma 1/8 part

ashwagandha 5 parts

dashamoola 5 parts

Take 1/2 teaspoon of this mixture 3 times a day with warm milk to help correct vata-type anemia.

- A specific formula is given in Vedic literature for a cleansing, detoxifying herbal remedy for vata-type anemia. It is called *gandharva haritaki*, which is *haritaki* powder roasted in castor oil in an iron pan. Place 1 tablespoon castor oil in an iron pan and heat on the stove; when the oil is warm enough, sauté 1 ounce of the herb *haritaki*. The *haritaki* will become thick and will turn slightly brown. (You have to stir it.) Take 1/2 teaspoon of *gandharva haritaki* at bedtime, washed down with warm water. Take it for 2 months or until the blood returns to normal.

NOTE: This mixture may cause loose stools; if this happens, reduce your dosage until comfortable.

FOR PITTA-TYPE ANEMIA. For pitta anemia, Ayurveda suggests *shatavari* ghee. (The *shatavari* and the ghee are cooked together. Directions for preparing herbalized and medicated ghees and oils appear in [appendix 2](#).) Take 1 teaspoon of *shatavari* ghee 3 times a day, before breakfast, lunch, and dinner.

- One can also use this herbal formula:

shatavari 5 parts

brahmi 3 parts

neem 2 parts

loha bhasma 1/8 part

Mix these herbs, and take 1/2 teaspoon of the mixture 3 times a day with 2 tablespoons of aloe vera gel. This combination will be quite effective in treating pitta-type anemia.

FOR KAPHA-TYPE ANEMIA. Where there is swelling, use:

punarnava 5 parts

gokshura 3 parts

kutki 2 parts

Take ½ teaspoon of this herbal mixture twice a day with a few sips of warm water. Or you can mix it with a little honey and then wash it down with water.

FOR ALL TYPES OF ANEMIA

EAT IRON-RICH FOODS. Iron is a good blood builder, so foods rich in iron, such as beets, carrots, grapes, raisins, and currants, are used in the Ayurvedic treatment of most anemias. Figs, dates, and date sugar are also good sources of iron. Pomegranate juice and cranberry juice can be used as blood builders, as can a combination of beet and carrot juice. (Add a pinch of cumin to your carrot/beet juice for maximum effect.) Chlorophyll is also a good source of iron, and many times Ayurveda does suggest using chlorophyll, generally in the form of spinach, chard, and other fresh green vegetables.

YOGURT AND TURMERIC. Eat a cup of plain yogurt with up to 1 teaspoon turmeric on an empty stomach, morning and afternoon. Do not eat this after sunset. If kapha is unbalanced, eat this at noon only.

BLUE-GREEN ALGAE. Blue-green algae can also be effectively used, but primarily for pitta anemia. Because it is a rich source of *prana*, it is not good for vata individuals, as it will make them quite hyper. Kapha types may also find it beneficial.

COPPER WATER. Vata and kapha individuals may find copper water effective. Fill a genuine copper glass or cup with water and let it stand overnight, then drink it in the morning (see [appendix 1](#)).

DIET AND LIFESTYLE CHOICES. These choices should follow the general guidelines (diet, exercise, and so on) for each doshic constitutional type. For vata problems, follow the anti-vata guidelines; for pitta anemia follow the pitta-soothing diet and other guidelines; and for kapha-type anemia, follow the kapha-reducing guidelines.

Correlation of Ayurvedic and Western Types of Anemia

Clinical observation has shown that the various types of anemia classified by modern medicine can be correlated with the types of anemia delineated by Ayurveda. For example, pitta-type anemia has been associated with mononucleosis and hepatitis and may lead to problems with the liver. Cobalamine (vitamin B-12)-deficiency anemia is also associated with pitta. Kapha-type anemia may lead to hypo-proteinemic anemia and swelling, while vata-type anemia may be associated with iron deficiency and folic acid-deficiency anemia. By treating the vata-pitta-kapha types of anemia, Ayurveda can at the same time treat the types of anemia categorized by modern medicine.

YOGA ASANAS. Yoga postures good for anemia of both the vata and kapha varieties

include the Locust pose, Lotus pose, and inverted poses (Shoulder Stand, Plow pose, Headstand), which bring the blood supply to the vital organs such as the thyroid, thymus, and brain. Headstand will not be good for pitta-type anemia, but the Sun Salutation, as well as the Boat, Bow, and Bridge poses, will be effective.

BREATHING EXERCISE. For all anemic conditions, *surya pranayama* (Right Nostril breathing) is recommended. Block your left nostril with your right ring finger and breathe only through the right nostril. Right Nostril breathing stimulates the liver, which plays an important role in building the blood.

⌘⌘ Anger and Hostility ⌘⌘

Anger and hostility are signs of aggravated pitta in the nervous system. Pitta is necessary for right understanding and judgment, but when it gets disturbed or out of balance, it creates misunderstanding and wrong judgment, leading to anger and hostility. The aim is to bring the pitta back to its normal constitutional function.

Here are several simple home remedies to cool down that hot pitta and keep tempers under control.

DIET. Perhaps most important, a person who becomes angry easily or often should follow the pitta-pacifying diet (see [chapter 8](#)), especially avoiding hot, spicy and fermented foods, citrus fruit, and sour fruit. Favor simple, bland foods and cool drinks, and avoid alcohol and drinks with caffeine.

KEEP COOL. It's also not recommended for people with a pitta body type to take saunas or steam baths, to get overheated from exercise or sports, or to be in too much direct sun. In other words: keep cool.

OIL MASSAGE. Rub some *bhringaraj* oil or coconut oil on your scalp and on the soles of the feet. That will help to bring down the excess pitta. (See instructions for preparing herbalized ghees and oils in [appendix 2](#).) You can do this every night before getting in bed to regularly moderate pitta. Be sure to wear some old socks and a hat, or put a towel on your pillow, to prevent the oil from staining.

USE SANDALWOOD OIL. Another simple and effective way to help balance your emotions is to place a drop of sandalwood essential oil on the “third eye” area between your eyebrows, as well as on the throat, breastbone, navel, temples, and wrists. Just a small amount of oil is sufficient.

HERBAL TEAS. Drink chamomile-tulsi-rose tea.

chamomile 1 part

tulsi (holy basil) 1 part

rose petal powder 2 parts

Steep ½ teaspoon of this mixture in 1 cup of hot water, cool it, and then drink. You can drink it 3 times a day, after each meal.

- You can use an even simpler formula. Take ½ teaspoon of chamomile and 1 teaspoon of fresh, finely chopped cilantro leaves, and steep them in 1 cup hot water for about 10 minutes. Allow this tea to cool before you drink it.

HAVE A PITTA-PACIFYING DRINK. Into 1 cup of grape juice, add ½ teaspoon cumin, ½ teaspoon fennel, and ½ teaspoon sandalwood powder. This cooling, pitta-pacifying drink will help to settle angry feelings and other pitta symptoms such as burning in the stomach.

GHEE NASYA. Dip your little finger into a jar of *brahmi* ghee (or plain ghee if you haven't made *brahmi* ghee), and lubricate the inside of your nostrils with a small amount. (Make sure your nails are trimmed so you don't scratch yourself.) Then gently inhale the ghee upward. This sends a calming message to the brain. You will become quite tranquil; anger and hostility will dissolve like a cloud in the sky.

DO BREATHING EXERCISES. A cooling *pranayama* to help dissipate anger is *shitali pranayama*. Make a tube of your tongue; breathe deeply through your mouth down into your belly; hold the breath for a few seconds; exhale through your nose. Do about 12 repetitions. (See the illustration in [chapter 6](#).)

YOGA POSTURES. Good yoga *asanas* for pitta include the Camel, Cobra, Cow, Boat, Goat, and Bridge poses. (See [appendix 4](#) for illustrations of yoga postures.) Avoid the Headstand or other inverted poses such as the Plow and Shoulder Stand. Do not perform the Sun Salutation; do the Moon Salutation instead.

MEDITATE. There is an ancient method of meditation that involves watching your every emotion come and go, without either naming it or trying to tame it. As the feelings arise, breathe deeply, and exhale the emotions out.

❧❧ Angina ❧❧

Angina, or to give it its full name, angina pectoris (chest pain), is a condition created by kapha dosha. Accumulated kapha blocks the flow of *prana* vata into the coronary artery, so that the heart muscles do not receive sufficient blood and oxygen supply. It is a kind of local anemia, resulting in pain that can be severe and frightening. Typically, the pain starts from the breastbone in the center of the chest, goes to the left shoulder, and passes along the inner side of the upper arm to the tip of the little finger.

HERBAL REMEDIES. To heal angina, the following herbal remedy is effective:

shringa bhasma ⅓ part

musta 3 parts

arjuna 3 parts

Take ½ teaspoon of this mixture 2 or 3 times a day with warm water.

Another effective remedy is to boil ½ cup each of milk and water, add ½ teaspoon of *arjuna* and 2 pinches of saffron, and take twice or 3 times a day. You may find this remedy beneficial for both chest pain and heart palpitations.

SPECIAL HERBS. In India, angina is often directly dealt with by taking certain powerful herbs sublingually (under the tongue) to produce immediate relief, as nitroglycerin tablets are often used in modern medicine. Ask your Ayurvedic physician about these herbs.

A HEALING PASTE. Topically, one can apply a paste to the chest. Make the paste of ginger powder (1 teaspoon) and *shringa bhasma* (just a pinch). Add sufficient warm water, and apply.

DEER HORN. Try to get hold of some deer horn. (Indian groceries and Chinese herb stores are possible sources.) Using a mortar and pestle or another kind of grinding stone, rub the deer horn on the rough stone until a little paste is formed. Applying a little of that paste on the chest can instantaneously relieve angina pain.

TIME TO SEE THE DOCTOR

The heart is a precious, vital organ. Any symptom related to the heart could be a sign of heart disease or a signal that heart disease is developing. So if you have any chest pain that could be due to your heart, please consult your doctor. Especially—but not exclusively—if you get chest pain from less exertion than usual, or if your chest pain lasts longer than a few minutes, treat it as a medical emergency.

GOLD WATER. Taking 1 teaspoon of gold water 2 or 3 times a day before food is also effective. (See [appendix 1](#) for instructions on making gold water.)

YOGA POSTURES. If there is no acute angina pain, you can improve coronary circulation by doing some gentle yoga stretching. Beneficial postures include the Camel pose, Boat pose, Locust pose, gentle Spinal Twist, and Cobra pose. These postures stretch the coronary arteries and increase blood supply to the heart.

❧ ❧ Anxiety ❧ ❧

Anxiety, which often is associated with insomnia and feelings of fear, is due primarily to aggravation of vata dosha in the nervous system. So to heal anxiety, we have to balance vata.

Here are several effective Ayurvedic remedies to pacify vata, heal anxiety and fear, and improve your sleep.

CALMING TEA. Make a tea of the following herbs:

tagar or valerian 1 part

musta 1 part

Steep ½ teaspoon of these herbs in 1 cup hot water for 5 to 10 minutes, and drink.

TIME TO SEE THE DOCTOR

Everyone experiences worry and anxiety from time to time. But if severe anxiety continues for a long time or becomes overwhelming and interferes with your social or occupational functioning, medical attention is required. Here are three reasons to see a doctor:

1. You chronically experience severe symptoms such as shortness of breath, chest pressure or pain, and dizziness, along with extreme worry and tension.
2. You have panic attacks—short, unexplained periods of intense fear.
3. You avoid people, places, or situations in order to avoid feeling anxious.

This formula effectively pacifies vata and reduces anxiety. You can drink this tea twice a day.

RELAXING BATH. A warm bath of ginger and baking soda will help you pacify anxiety. Use ⅓ cup ginger and ⅓ cup baking soda in a tubful of water. Soak for 10 to 15 minutes.

ALMOND MILK. Almond milk helps to eliminate anxiety. Soak about 10 raw (not toasted) almonds overnight in water. Peel off the skins, and put the almonds in the blender. Add 1 cup warm milk. While blending, add a pinch of ginger and a small pinch of nutmeg and saffron.

ORANGE JUICE. For anxiety accompanied by a fast heart rate, a cup of orange juice with 1 teaspoon of honey and a pinch of nutmeg powder can be effective.

CALMING PRESSURE POINT. Make a fist with your left hand, so that the fingers rest in the middle of the palm. Locate the point where the middle finger ends, in the “heart” of the palm. Then, with the thumb of your right hand, press firmly on this point in the center of your left hand. Press for 1 minute. This will calm down the agitation of *prana*, which causes anxiety.

Ayurvedic Oil Massage

Giving yourself a full-body oil massage will greatly help to reduce anxiety. Vatas should use sesame oil; pittas, sunflower or coconut oil; kaphas, corn oil. Use 6 or 7 ounces of warmed-up (*not hot*) oil, and rub it over your whole body from head to toes. Ordinarily this massage is done before the morning bath, but if you have high anxiety or insomnia, you can also do it before going to bed.

A minimassage is also effective: Using the appropriate oil for your constitution,

rub some on your scalp and spend a few minutes rubbing the bottom of your feet.

RELAXATION POSE. Lie down on your back in the yoga posture known as *savasana*, the “Corpse” or Relaxation pose, arms by your sides.

MEDITATE TO RELAX. Sitting quietly, focus your attention on the top of your head while doing the *So-Hum* meditation (described in [chapter 7](#)).

❧ ❧ Appetite, Low ❧ ❧

Poor appetite is a condition of low *jatharagni* (digestive fire). Low agni may be due to slow metabolism, or slow metabolism caused by low agni; each affects the other. Low agni creates not only poor appetite but also indigestion, bloating, ama (toxins) in the gastrointestinal tract, coating on the tongue, and bad breath. Lack of energy is also common.

The most effective treatment for this situation may surprise you: Don’t eat. A short fast will help to kindle the digestive fire. Skip breakfast, and don’t nibble anything. By noon your appetite will probably return, and you will be hungry and ready to eat.

Lack of appetite is often due to continual munching, combined with drinking cold drinks, which depress agni. For the sake of good digestion and long-term health, as well as reviving your appetite, both of these bad habits have to be stopped.

If you are not hungry even by lunchtime, take some fresh ginger, chop a little into small pieces, add some lime juice and a pinch of rock salt, and chew it up. That will kindle agni and stimulate the appetite.

Low appetite may also be due to emotional factors. If that is the case, make a tea of ginger, *brahmi*, and chamomile in equal proportions. Use 1 teaspoon of this mixture per cup of water, steep for 5 to 10 minutes, and drink.

Also, take ½ teaspoon of *triphala* in a cup of warm water every night before going to bed. Pour boiling water into the *triphala*, and let it cool until it’s comfortable to drink.

These few simple measures should be enough to help you regain a healthy appetite. If you try them and still don’t feel like eating, see your doctor, as lack of appetite can be a symptom of a more serious illness.

❧ ❧ Arthritis ❧ ❧

Ayurveda distinguishes three categories of arthritis, corresponding to vata, pitta, and kapha. To treat this condition properly, it is vital to carefully diagnose which type you have.

TYPES OF ARTHRITIS

- If *arthritis is due to vata*, your joints will crack and pop. They become dry and are not swollen as they may be if excess vata is not the cause. The joints may also feel cold to the touch. They are painful mostly upon movement, and there is usually one particular tender spot. Jogging, jumping, trampolining, or any strenuous exercise tends to aggravate the pain.

- *Pitta-type arthritis* is characterized by inflammation; the joint becomes swollen and is painful even without movement. It often looks red and feels hot to the touch.

- In *kapha-type arthritis*, the joint also becomes stiff and swollen, but it feels cold and clammy rather than hot. A little movement, rather than aggravating the pain, tends to relieve it. The pain is greater in the morning, and as the person starts moving around, the pain diminishes.

FOR ALL TYPES OF ARTHRITIS

Treatment for each type of arthritis is unique, as we shall see in a moment. But for all cases, it is important to know that arthritis begins in the colon.

Depending on the person's lifestyle, diet, and emotional pattern, either vata, pitta, or kapha goes out of balance. Then that particular dosha slows down agni (digestive fire), resulting in the toxic, sticky by-product of inadequate digestion known as ama.

Vata, the main active dosha, brings the ama into the colon, and from there it travels through the system and lodges in the *asthi dhatu* (bone tissue) and in the joints, giving rise to the stiffness and pain characteristic of arthritis.

So our aim in treating arthritis is to remove the ama from the joint and bring it back to the colon, and then to eliminate it.

This is why, to relieve arthritis, it is important to keep the colon clean. To accomplish this, if you do not know positively whether it is vata, pitta, or kapha arthritis, taking *triphala* at night (1 teaspoon) with some warm water (½ to 1 cup) will be effective for all types. Alternatively, if you know positively which type it is, you can use *haritaki* for vata-type arthritis, *amalaki* for pitta-type, and *bibhitaki* for kapha-type arthritis (½ to 1 teaspoon with warm water in all cases).

Now for the complete treatments.

FOR VATA-TYPE ARTHRITIS

- Follow the vata-pacifying diet ([chapter 8](#)). Favor warm, easy-to-digest foods, and avoid cold foods and drinks, including salads. Avoid beans, drying grains such as barley and corn, and the nightshades: tomato, potato, and eggplant.

- Take *yogaraj guggulu*, 1 tablet 3 times per day.

- Apply *mahanarayan* oil on the affected joint, followed by application of local moist heat. For example, say you have an arthritic ankle. Apply the *mahanarayan* oil, rub it into the skin, and then soak your foot in warm to bearably hot water. Add a “teabag” of brown mustard seeds to the water. You can make the bag by wrapping 2 tablespoons of mustard seeds in a handkerchief or cheesecloth.

- Helpful yoga postures include Forward Bend (don’t strain), the Chest-Knee pose, Maha Mudra, and the Half Bridge pose. (For illustrations of yoga postures, see [appendix 4](#).)

FOR PITTA-TYPE ARTHRITIS

Pitta arthritis often has more pain and inflammation associated with it than other types.

- Follow the pitta-pacifying diet ([chapter 8](#)). Particularly avoid hot, spicy foods, pickles, spinach, and tomatoes.

- These Ayurvedic herbal formulas will help you. Take 1 tablet of *kaishore guggulu* (350 mg.) 3 times a day, and ½ teaspoon *sudarshan* twice a day, washed down with warm water.

- Externally, apply cool castor oil or coconut oil to the painful area.

- Application of a cooling substance, such as sandalwood powder paste, is soothing. Make the paste by taking 1 teaspoon of sandalwood powder and adding sufficient water to make a paste. Rub it gently onto the joint.

- If the joint is hot and inflamed, you can put an icebag on it. This will help to ease the pain and inflammation.

- Helpful yoga postures include the Boat, Bow, Camel, Cow, and Locust poses, as well as the series of postures known as the Moon Salutation. (See [appendix 4](#) for illustrations of yoga postures.)

- You can also perform *shitali pranayama*, as follows: Make a tube of your tongue. Breathe deeply through your mouth down into your belly. Hold the breath for a few seconds, then exhale through your nose. Do about 12 repetitions. (See illustration in [chapter 6](#).)

FOR KAPHA-TYPE ARTHRITIS

Arthritis is classified as kapha when the joint is painful, swollen, stiff, and feels cold and clammy to the touch.

- Follow the kapha-reducing diet ([chapter 8](#)). Especially, take no dairy products and

no cold drinks.

- A potent herbal formula to help with this type of arthritis is *punarnava guggulu* tablets (250 mg.). Take 1 tablet 3 times a day.
- Externally, you can apply a paste of *vacha* (calamus root) powder. Add sufficient warm water to 1 teaspoon of powder to make a paste, and apply to the joint.
- When there is an effusion (when the joint fills with fluid), you can make an effective paste of equal amounts of *punarnava* powder and ginger powder. Mix 1 teaspoon of each with enough warm water to form a paste, and apply.
- Yoga postures that are helpful for kapha-type arthritis include the Tree, Triangle, Forward Bend, and Spinal Twist. (Illustrations of yoga postures appear in [appendix 4](#).)

RHEUMATOID AND OSTEOARTHRITIS

In addition to the vata–pitta–kapha method of classifying arthritis, this condition can also be classified as either rheumatoid arthritis or osteoarthritis. If you are sure which condition you have, the following guidelines will add more specificity to your treatment.

FOR RHEUMATOID ARTHRITIS. Take 1 *simhanada guggulu* tablet (350 mg.) 3 times a day, and 1 *chitrak-adhivati* (200 mg.) tablet twice a day.

- A quarter teaspoon of *yogaraj guggulu* washed down with a little warm water 3 times a day is also recommended.
- You can also drink 1 cup of ginger tea with 2 teaspoons of castor oil added. Castor oil contains natural precursors of steroids, which help to heal the inflammatory condition of rheumatoid arthritis. Take this tea before going to bed. Expect some laxative effect from the castor oil.

FOR OSTEOARTHRITIS. Take 1 tablet of *yogaraj guggulu* twice a day. At night, take ½ teaspoon *gandharva haritaki* (*haritaki* sautéed in castor oil) with warm water. If you don't have *gandharva haritaki*, use ginger tea with castor oil, as described above.

☞ ☞ Asthma and Wheezing ☞ ☞

Bronchial asthma is characterized by sudden attacks of short, gasping breaths accompanied by wheezing. If the attack is not stopped, the person may have increasing difficulty breathing.

The underlying cause of all asthmatic conditions is increased kapha dosha in the stomach. From there it moves into the lungs, trachea, and bronchi. The increased kapha blocks the natural flow of air, creating spasm in the bronchial tree and resulting in asthma and wheezing.

Ayurvedic treatment for asthma aims to bring the kapha lodged in the lungs and bronchi back to the stomach, from which it can be eliminated.

Asthma may be brought on by allergies, a cold, congestion, cough, or hay fever. It can be instigated by pollen, dust, animal hair, or various foods, or an increase of kapha internally. Regardless of the cause, during an asthmatic attack it is important to immediately relieve the difficult breathing and asthmatic wheezing.

To Immediately Stop Wheezing

Boil 1 teaspoon of licorice root (*yashti madhu*) in a cup of water for a couple of minutes to make a licorice tea. Just before drinking the tea, add 5 to 10 drops of *mahanarayan* oil if you have some, or use ½ teaspoon of plain ghee. Take one sip of this tea every 5 to 10 minutes.

In some instances, the licorice tea may induce vomiting. This is beneficial: It eliminates kapha and relieves the spasm of the bronchial tubes, and the person usually feels better immediately.

You can use this licorice tea not only for emergencies but, if you are prone to develop asthma, every day as a preventive. The only exception is that individuals with hypertension should not use much licorice tea, as it makes the body retain sodium. They may take it as an emergency measure to avert an asthma attack, but they should not drink it regularly.

NOTE: Prepare the tea as soon as you begin to feel an attack coming on, when there is tightness in the chest, some difficulty breathing, or whatever warning bell symptoms you recognize from past experiences. Don't wait until you are already having serious problems breathing.

Once you prepare the tea, it can last for 72 hours and not lose its effectiveness.

HERBAL REMEDIES

The following herbal remedies can be taken on a regular basis for long-term prevention of asthma.

- Mix 1 teaspoon cinnamon and ¼ teaspoon *trikatu* into a cup of boiling water. Let it steep for 10 minutes, and add 1 teaspoon of honey before drinking. You can take this tea twice a day.

- A tea made of half licorice and half ginger is also beneficial for asthma prevention. Use half a teaspoon of the combined herbs per cupful of water.

- Also try ½ teaspoon bay leaf and ¼ teaspoon *pippali* mixed into 1 teaspoon honey, taken 2 or 3 times a day.

- Another remedy that can relieve congestion and alleviate breathlessness is ¼ cup onion juice with 1 teaspoon honey and ⅛ teaspoon black pepper. This remedy will also be effective for immediate relief of asthma.

- This herbal formula is helpful both for prevention and immediate relief:

sitopaladi ½ teaspoon

punarnava ½ teaspoon

pippali pinch

abrak bhasma pinch

For immediate relief, take this entire mixture with honey, a little bit at a time. For long-term usage, take it once a day.

- You may also find ⅓ cup spinach juice with a pinch of *pippali* effective. Drink this twice a day.

THREE MUSTARD SEED REMEDIES

Mustard seeds are effective in healing the bronchial system. Here are three ways to take advantage of their heating and healing power:

1. Rubbing a little brown mustard oil onto your chest will give some relief.
2. Make a tea by mixing ground mustard seeds and *pippali* (or black pepper if you don't have *pippali*). Steep ¼ teaspoon of each for 10 minutes in 1 cup hot water, then add 1 to 2 teaspoons honey. Drink 2 or 3 times a day, or for better results sip it every 15 minutes throughout the day.
3. Mix 1 teaspoon of brown mustard oil with 1 teaspoon natural organic sugar. Take 2 or 3 times a day on an empty stomach.

OTHER APPROACHES

IF YOU HAVE AN INFECTION. In some people, the underlying cause of asthmatic wheezing may be an infection descending from the nose and sinuses. If this is the case, putting 5 to 10 drops of warm ghee in each nostril will help.

TRY TO AVOID ALLERGENS. If your asthma and wheezing are due to a food allergy, then avoid the problematic food. Similarly, avoid any object that may provoke your condition, such as dusty books, moldy basements, and some chemicals.

FOODS TO AVOID. Avoid most dairy products, including all cheeses. Avoid fermented foods and all hydrophilic food substances such as salty items, cucumber, and tuna fish. Some people need to avoid mushrooms, peanuts, walnuts and other nuts, and yeast. The reaction may be immediate, for people extremely sensitive to these substances, or it may take several hours to develop.

FOR CHRONIC BRONCHIAL ASTHMA. If you have chronic bronchial asthma, try this remedy. Insert about 7 cloves into a peeled banana, and keep it overnight. Next morning eat the banana and the cloves. Don't eat anything for an hour, then drink 1 cup of hot

TIME TO SEE THE DOCTOR

Asthma can ordinarily be controlled using these Ayurvedic remedies. However, if you find that medicines that ordinarily help you breathe more comfortably no longer seem to be working, or if, along with your difficulty breathing, you have chest pain, swollen feet, and profuse sweating, and you have a history of heart problems, you need to seek immediate medical attention.

water with 1 teaspoon of honey. This will energize the lungs and should reduce asthmatic wheezing.

YOGA FOR ASTHMA. Effective yoga *asanas* to help relieve asthma are the Bow and Cobra poses, sitting in the Vajrasana, and the inverted poses including Shoulder Stand and Plow. (Illustrations of yoga postures appear in [appendix 4](#).)

❧ ❧ Athlete's Foot ❧ ❧

People with kapha–pitta constitutions, who sweat a lot, are most prone to get athlete's foot. This is an itchy, inflammatory condition between the toes, often accompanied by sweating on the soles of the feet.

Athlete's foot can be effectively treated with Ayurvedic remedies. Begin by cleaning the problem area with some tea tree oil on a cotton swab. This natural antiseptic oil is widely available in natural food stores and elsewhere.

Then apply a mixture of aloe vera gel and turmeric. Mix 1 teaspoon of aloe vera gel with ½ teaspoon turmeric, and apply some of the mixture to the affected areas. But be a little careful: This mixture will turn your skin and socks yellow! If you use it at night, it will stain your sheets, so you might wear a pair of old socks to prevent the discoloration. Continue with this treatment twice a day for at least 2 weeks

An alternative treatment is to wash your feet with *neem* soap. Then dry thoroughly with a hair drier or soft towel, and apply some *neem* oil (about ¼ teaspoon) mixed with about 10 drops of tea tree oil. Apply that mixture topically to the affected area with a cotton swab.

If you have athlete's foot or are prone to get it, avoid fermented food and sugar.

❧ ❧ Backache ❧ ❧

Backache is uncommonly common these days. It has become an occupational hazard in many different kinds of work. People may pull or strain their back muscles while lifting a heavy load or while sitting at a desk in front of a computer. Emotional factors can also lead to back pain, as can injuries, such as from a car accident. Some people may even have a slipped disk, which can lead to severe back pain. Whatever the cause of your backache, the following natural Ayurvedic home remedies will be helpful.

HERBAL REMEDIES. Take *yogaraj guggulu*, 1 tablet 3 times a day, or 1 tablet of *kaishore guggulu* 2 or 3 times a day. Both of these special Ayurvedic formulas are available from most sources of Ayurvedic herbs (see [Resources](#)).

- Backache can also be relieved by the use of the herb *musta*, which is a muscle painkiller. Take ¼ to ½ teaspoon 2 or 3 times a day with warm water.

- The herbs *tagara* and valerian are muscle relaxants. Taking ½ teaspoon of either with some warm water will relax the muscles that may be causing back pain. It will also help induce restful sleep.

Most back pain can be effectively treated with these herbs, but a ruptured or slipped disk often requires intensive medical care.

RUB IN SOME OIL. Rubbing the painful area of the back with *mahanarayan* oil is also effective for relieving the pain. Vata and pitta types should just rub the oil on the surface, while kapha types should give a deeper massage to the area for some time.

Alternatively, try this procedure: Apply a paste made of ginger powder mixed with sufficient water to the affected area. Leave it on for 10 to 15 minutes, wash it off, and then rub the back with some eucalyptus oil.

(Unless your back pain is in the neck or shoulder area, these back rubs will have to be done by a friend!)

TAKE A HOT HERBALIZED BATH. For extra healing and muscle relaxation, apply the *mahanarayan* oil on your back and then follow with a hot bath in which you put some ginger powder and baking soda (⅓ cup of each). Soak in the tub for 10 to 15 minutes. You may repeat this bath 2 or 3 times a week, perhaps on Tuesday, Thursday, and Saturday.

AN ENEMA CAN HELP. Individuals who suffer from backache often get constipated, and it may be difficult to tell which is the effect and which is the cause. The backache may be due to chronic constipation, or the spasming muscles and anxiety caused by the back pain may induce constipation. In either case, a simple enema of *dashamoola* tea will

help.

Boil 1 tablespoon of *dashamoola* powder in a pint of water for about 5 minutes, cool it down, and add ½ cup sesame oil. When it is cool enough, use it as an enema, retaining the liquid for 5 to 10 minutes if you can. A *dashamoola*-sesame oil enema is soothing to vata and will help relieve both the constipation and the pain. (See [appendix 3](#) for enema guidelines.)

GENTLE STRETCHES. Some gentle yoga exercises can help with back pain.

IMPORTANT: All yoga postures should be learned with a trained yoga teacher, but especially when you have a backache, you should not do any yoga postures without expert guidance. This is especially true if your pain comes from a slipped disk.

In general, the following postures may be helpful:

Camel pose Lotus pose

Cow pose Forward Bend

Spinal Twist Palm Tree pose

Locust pose gentle, modified Fish pose

All these postures may be used both as a preventive measure and to help remedy back pain. But again, be sure to get advice from a trained teacher. (Illustrations of yoga postures appear in [appendix 4](#).)

MORE TIPS TO HELP HEAL BACKACHE

- Backache is often due to excess vata, so it is helpful to reduce your consumption of vata-increasing foods. Avoid most beans, including black beans, pinto beans, adzuki beans, and garbanzo beans. Avoid raw, cold salads. (See [chapter 8](#) for more on the vata-pacifying diet.)

- Avoid exposure to cold weather or cold winds.

- Sit quietly and meditate, or observe your breathing. This will help relax tense muscles. (For help with meditation, see [chapter 7](#).)

- Don't walk in high-heeled shoes.

- Don't try to do jogging, jumping, or other strenuous exercise, rather, do some gentle yoga stretching as mentioned above.

- Sexual activity should be minimized.

Following these guidelines will help you heal your aching back as well as avoid back

pain in the future.

❧ ❧ Bad Breath ❧ ❧

Bad breath is often a sign of systemic toxicity, either in the colon, intestine, or mouth. It can also be due to chronic indigestion or malabsorption. When digestion is weak or sluggish, the food you eat undergoes fermentation and putrefaction in the gastrointestinal tract, leading to the formation of ama, which has a foul smell.

Stand in front of a mirror, and stick out your tongue. If the back portion of the tongue is coated, that is the sign of ama, which is responsible for the bad breath.

The primary Ayurvedic aim in treating bad breath is to kindle gastric fire (agni), which in turn burns ama and alleviates the root cause of the condition. Here are several effective home remedies to prevent and treat bad breath.

DIETARY AND HERBAL REMEDIES

- First monitor your diet. It's important not to eat heavy meals, and to stay away from cold drinks, ice cream, cheese, and yogurt, all of which reduce the gastric fire and slow down digestion, with a likely increase of ama.
- Then, after each meal (generally after lunch and dinner), chew about 1 teaspoon of roasted fennel and cumin seeds (mixed half and half). This will improve digestion, which indirectly helps to detoxify the colon. The licorice-flavored fennel seeds alone would be delicious and helpful, but this mixture will have a better effect.
- Drink ½ cup of aloe vera juice twice a day until freshness is restored to the breath.
- Slowly chewing one or two cardamom seeds also helps to minimize bad breath. Cardamom aids digestion and helps to reduce ama.
- After each meal, drink a cup of cumin-coriander-fennel tea (equal proportions) as a digestive aid. Steep about ¼ to ½ teaspoon of each herb per cup of hot water.

OTHER REMEDIES

TAKE CARE OF YOUR TEETH AND GUMS. Another cause of bad breath is poor oral hygiene. It's important to clean the teeth after each meal. Use an Ayurvedic toothpaste containing *neem* or an herbal formula. Also use dental floss every day. Applying some tea tree oil mixed half and half with *neem* oil to the gums and gently massaging will help prevent receding gums. Be sure to spit out the residue rather than swallowing. (See [“Teeth and Gums—Ayurvedic Care.”](#))

YOGA POSTURES. The yoga posture known as Yoga Mudra, the Lion pose, and sitting in the

Lotus posture with a Forward Bend are good *asanas* for combating bad breath (see [appendix 4](#)).

BREATHING EXERCISE. You can also do the *pranayama* known as *shitali* (see instructions in [chapter 6](#)).

If you follow these guidelines, you can say good-bye to bad breath.

❧ ❧ Baldness ❧ ❧

See also “[Hair Care Secrets](#)”

Hair loss is a subtle metabolic disorder. It may be related to disease—I have seen hair loss in persons with diabetes, for example, or following typhoid infection. It may be due to some fungal infection on the scalp, or to hormonal imbalance. A deficiency of calcium, magnesium, and zinc may affect the nourishment of the hair enough that the hair starts to fall out. And there is a definite hereditary factor, in which genes appear to trigger hair loss at a certain age.

According to Ayurveda, early hair loss is often related to body type and the balance of the doshas. Pitta individuals, and those who have excess pitta in their system, are more likely than the other body types to lose their hair early in life, or to find that their hair becomes prematurely thin or gray. Excess pitta in the sebaceous gland, at the root of the hair, or folliculitis (inflammation of the hair follicles) can make the person start losing hair.

ALOE VERA. To reduce pitta and preserve the health of your hair, drink aloe vera juice ($\frac{1}{3}$ cup) or take some aloe vera gel (1 tablespoon with a pinch of cumin) 3 times a day for about 3 months.

OIL MASSAGE. Another effective way to pacify pitta is to rub some coconut oil on your scalp and on the soles of your feet at bedtime. Wear some old socks, and to preserve the health of your pillow, wear a loose woolen cap or cover the pillow with a towel so it won't get damaged by the oil.

- Massaging the scalp with *brahmi* oil or *bhringaraj* oil at bedtime can help to prevent hair loss. In addition to the qualities of the oil, the massage itself will improve circulation at the root of the hair, thus bringing more nutrients to support hair growth.

- Massaging the hair with vitamin E oil may also be effective in preventing or slowing hair loss.

FOOD FOR YOUR HAIR. Healthy hair depends upon a nourishing diet. Dairy products such as cheese, milk, and yogurt are beneficial for the hair (assuming you digest them well), as are white radish and daikon. Coconut, cooked apples, and cabbage are also useful.

- Eat a handful of white sesame seeds every morning. One handful of these small seeds contains about 1,200 mg. of calcium and magnesium and is a good source of nourishment for your hair.

HERBS FOR YOUR HAIR. Certain herbs are useful for nourishing the hair. Try this beneficial formula:

dashamoola 5 parts

bhringaraj 4 parts

jatamamsi 3 parts

At bedtime, add ½ teaspoon of this mixture to 1 cup of goat's milk, heat to boiling, and drink. This will build up the bones and nourish the hair.

MINERAL SUPPLEMENTS. You may also be able to improve the condition of your hair by taking mineral supplements to be sure you have enough calcium, magnesium, and zinc. Take a supplement containing approximately the following daily dose of minerals:

calcium 1,200 mg.

magnesium 600 mg.

zinc 60 mg.

Take these supplements at bedtime.

MASSAGE. Stress, stiffness in the neck, and whiplash from a car accident can also contribute to hair loss. To soften the neck muscles, relieve pain, and reduce stress, massage your neck and shoulder muscles before showering.

NECK EXERCISES. You can also do some simple neck exercises, such as turning your head to the left 3 times, to the right 3 times, lifting it up 3 times, moving it down toward your chest 3 times, and then rolling it gently in a circle 3 times in each direction.

ANTISTRESS TEA. To help you deal with stress, make a tea of equal proportions of *jatamamsi* and *brahmi*. Steep 1 teaspoon of the mix in 1 cup of hot water, and drink 2 or 3 times a day.

YOGA POSTURES. Yoga postures can help to relieve tension in your neck and indirectly aid in keeping hair healthy. Recommended postures include the Shoulder Stand, Camel pose, Cobra pose, and Cow pose. (Illustrations appear in [appendix 4](#).)

MEDITATE FOR RELAXATION. You will also find meditation an effective means for reducing stress and tension. Try sitting quietly and observing your breathing. Or try the Empty Bowl meditation (described in [chapter 7](#)).

❧ ❧ Bites and Stings ❧ ❧

Any bite or sting of an insect can trigger a local irritation of pitta under the skin. As long as the venom of the insect remains there, it may keep on creating periodic allergic reactions, or may even create sting-bite nephritis, a serious condition involving generalized edema (swelling) and breathlessness, and the person can choke. So stings and bites, though usually quite innocuous, may occasionally be very serious; one has to be watchful.

CILANTRO. AS SOON as possible after receiving a sting or bite, take some cilantro juice. Place a handful of cilantro in a blender with about $\frac{1}{3}$ cup water, blend thoroughly, and strain it. Drink the juice (use 2 teaspoons 3 times a day), and apply the pulp locally to the skin at the affected area. It will instantaneously pacify the itching, burning, and hives or rash created by the sting or bite.

DRINK COCONUT WATER. One can drink $\frac{1}{3}$ cup of coconut water (the “juice” inside the coconut) with about $\frac{1}{8}$ teaspoon *kama dudha* added. Drinking this mixture 2 or 3 times will help heal the reaction to the sting bite.

COCONUT ASH. Here is another simple and fascinating remedy. Take a piece of dried coconut, and set fire to it. It will catch fire like wax. Let it burn for about $\frac{1}{2}$ inch and then blow it out. There will be a little smoke, and when the smoke disappears, a tarry black residue will remain. Apply that residue directly to the bite for instant relief.

Why does this work? Because coconut is a good source both for antihistamines and for natural steroids.

You may do the same thing with the ash from burning some of the outer coconut shell.

APPLY NEEM OIL OR NEEM PASTE. At the site of the bite, you can also apply *neem* oil or a *neem* paste. *Neem* is an antidote to most poisonous insect venoms. To make a paste, take a little *neem* powder and mix it with a little water. Apply it to the skin and leave it on for 10 to 20 minutes, then rinse it off. Do not use pure *neem* extract; instead use an herbalized oil made by boiling *neem* leaves in a sesame oil base. This is generally available in natural food stores or Indian groceries.

A HEALING PASTE. You will also find a paste made from $\frac{1}{2}$ teaspoon of sandalwood powder plus $\frac{1}{2}$ teaspoon turmeric soothing and healing. Mix the two herbs together with sufficient water to make a paste, and apply topically to the site of the bite.

PREVENTION. *Neem* oil is a much-used insect repellent in India and around the world. It contains a natural chemical compound that repels insects. Rub a little onto exposed skin before going outdoors.

❧❧ Bladder Problems ❧❧

See also “[Urinary Incontinence](#)”

Problems with the bladder and urination may indicate cystitis, an inflammation of the bladder that causes a burning sensation when passing urine. Other bladder problems include frequent urination, or its opposite, stagnation or retention of urine in the bladder, leading to bladder distension. There may be pain in the bladder area while urinating, a condition called strangury. Let's look at each of these.

RETENTION OF URINE IN THE BLADDER

In this condition the bladder is distended but the person doesn't pass urine. It may be due to constriction of the urethra, enlargement of the prostate gland, or perhaps a stone in the urethra. The causes may be many, but the cure is simple:

- Take two towels or sponges, one dipped in hot water, the other dipped in cool water. About every minute, alternate placing them in the bladder area. This alternation of hot and cold stimulates the bladder, and the person easily passes urine.
- If alternating hot and cold compresses isn't completely successful, apply *punarnava* paste (*punarnava* powder with sufficient water to make a paste) to the skin directly above the distended bladder. Leave it on for about half an hour.
- If the retention is due to stricture (narrowing) in the urethra or enlargement of the prostate gland, then use this formula:

punarnava guggulu 4 parts

shilajit 1 part

Taking ½ teaspoon of this mixture 3 times a day with warm water will help to dilate the stricture or relax the prostate gland and help to restore the easy flow of the urine.

CYSTITIS

Cystitis causes a burning sensation while urinating. To relieve this condition, drink coriander tea, cumin tea, or fennel tea. Or make a tea of equal proportions of these three herbs. Cumin-coriander-fennel tea is widely used in Ayurveda to relieve irritation of the bladder while passing urine.

You may also find this mixture effective for cystitis:

punarnava 5 parts

gokshura 4 parts

musta 3 parts

Take ½ teaspoon of this herbal mixture 2 or 3 times a day with warm water.

BLADDER ATONIA

This is a condition in which the bladder's sphincter loses its tone or strength, and the bladder leaks urine. It is more common among women. A woman may sneeze or cough and inadvertently pass some urine. For this condition, take one handful of white sesame seeds along with 1 teaspoon of jaggery or natural, unrefined brown sugar, and chew it well, followed by half a cup of water to wash it down. This is a very simple remedy to bring tone back to the bladder. Take it once or twice a day until your condition is better.

For further discussion and suggestions, see “[Urinary Incontinence](#)” and “[Prostate Problems](#).”

❧❧ Bleeding, External ❧❧

Generally, within 5 to 6 minutes after getting a cut, the bleeding will stop by itself. The blood will clot, the bleeding will stop, and the cut will be sealed. In such cases—the vast majority—there is nothing much to do unless the cut is severe and bleeding is excessive.

Some people, however, bleed for a longer time, because the blood does not cooperate and clot quickly enough. When a person continues to bleed, it basically means that the blood is too thin. Although the problem usually has a relatively simple and benign cause, failure of the blood to clot—or gums, cuts, or wounds that start to bleed—may also be an early sign of blood cancer, leukemia, or hemophilia purpura, a pitta disorder that causes profuse bleeding under the skin.

From the Ayurvedic perspective, blood that fails to clot in a timely manner is due to a pitta imbalance. Excess pitta in the blood makes the blood hot, sharp, and penetrating and doesn't allow natural coagulation and clotting to occur. The basic prescription, then, is to follow a pitta-soothing diet, use pitta-soothing herbs, and take specifically hemostatic herbs, herbs that directly help to stop bleeding.

APPLY COLD. To stop external bleeding, start by applying cold. Use some ice (plain or wrapped in a cloth), which helps to constrict the blood vessels and stop bleeding. A bag of frozen vegetables from the freezer will do the job in an emergency.

APPLY PRESSURE. Other simple ways to stop bleeding, well known to most people, are

1. tying a tourniquet

2. applying pressure directly over the bleeding area
3. if the bleeding is in an extremity, raising the arm or leg higher than the rest of the body.

ALOE. Another effective approach is to apply some aloe. A pinch of aloe powder mixed into a paste with a pinch of turmeric powder will immediately stop most bleeding. Aloe vera gel will also be effective.

ASTRINGENT HERBS. Other astringent herbs are also effective. The Ayurvedic herbs *lodhra*, *kushtha*, and *bilva* are effective to stop bleeding, either alone or mixed in equal proportions into a paste that can be applied directly. They are also effective taken internally for continuing problems with bleeding, use ½ teaspoon 2 or 3 times a day.

COTTON ASH. For external bleeding, you will find this ancient, simple remedy effective. Take a small ball of sterilized cotton, and burn it. (Be sure it is real cotton, not the synthetic material often found these days, which will be totally ineffective.) When the cotton turns to black ash, wait for it to cool down, and then apply it to the bleeding wound and press. The ash will stick to the bleeding point and stop the bleeding instantaneously. Within a couple of days a scab will form, and the wound will completely heal.

NOTE: Don't remove the ash; leave it on the wound to form a scab. Otherwise you will open the cut again.

DRINK COLD WATER. Many times just drinking some cold water will stop the bleeding, as coolness constricts blood vessels.

❧ ❧ Bleeding, Internal ❧ ❧

See also "[Rectal Bleeding](#)"

Cases of internal bleeding include peptic ulcer, hematoma (a blood-filled swelling), and bleeding through the urethra. To help put a stop to the bleeding, make an herbal compound out of these Ayurvedic herbs:

lodhra

kushtha

bilva

Mix these herbs in equal amounts, and take ½ teaspoon 2 or 3 times a day. Each of these herbs helps to stop bleeding, so you can use just one if that's all you can locate, though the mixture will be most effective.

Pitta people bruise more easily; their blood vessels are thin and are more easily

ruptured. People with pitta constitutions can help prevent internal bleeding by taking the same three Ayurvedic herbs, *lodhra*, *kushtha*, and *bilva*. Mix the three herbs in equal amounts, and take ½ teaspoon of the mixture 2 or 3 times a day with warm water, for as long as the condition lasts.

TIME TO SEE THE DOCTOR

The appearance of blood in the urine or stools can be a sign of serious illness such as kidney problems or cancer, and it should be investigated more deeply in consultation with your physician.

TURMERIC PASTE. When a person has been injured, the blood vessels sometimes rupture, causing a large bruise and a blood-filled swelling known as a hematoma. To stop the internal bleeding and pacify the hematomalike swelling, apply a paste made of 1 teaspoon turmeric powder, 1 teaspoon sandalwood powder, and a pinch of alum powder. (Mix the powders together with a little water to make a paste.) When the paste is on the skin, apply some pressure on the hematoma.

SAFFRON MILK. Another aid to stopping internal bleeding is to drink a cup of warm milk, to which ½ teaspoon of turmeric powder and a pinch of saffron have been added.

JUICES. Drinking cranberry or pomegranate juice is also helpful to stop internal bleeding.

❧ ❧ Boils ❧ ❧

Boils—painful, pus-filled inflammations of the skin and subcutaneous tissue—have many causes. They may be due to chronic constipation, or to high pitta in the blood. A toxic liver can also create boils. Repeated boils may be a sign of diabetes, so if you get boils repeatedly, check on your blood sugar.

NEEM POWDER PASTE. At the site of the boil, apply a paste of *neem* powder (preferably) or some *neem* oil. To make the paste, simply mix a little *neem* powder with warm water.

TRIPHALA WASH. Wash the affected area with *triphala* tea. Boil 1 teaspoon *triphala* in 1 cup of water. Cool, and wash your face or other affected area with the tea. Let it dry on the skin. (For information about *triphala*, see [appendix 2](#).)

FOR DIABETES. If there is a history of diabetes in the family and you get repeated boils, use this formula:

neem 1 part

turmeric 1 part

kutki ½ part

Taking ½ teaspoon of this mixture 2 or 3 times a day with warm water will

help take care of the root cause of the boil. Continue taking it until the boil disappears.

FOR CHRONIC CONSTIPATION. If the boil appears to be due to chronic constipation, do a *basti* (enema) using *dashamoola* tea. Boil 1 tablespoon of the herb *dashamoola* in a pint of water for 5 minutes. Let the liquid cool, strain it, and use it for an enema.

In addition, take ½ teaspoon of *amalaki* or the herbal compound *triphala* at night. Steep the *triphala* in a cup of hot water for 5 to 10 minutes and then drink. This purgation will help to remove excess pitta from the hematopoietic (blood-building) system, which is the cause of the boil. You may continue taking the *triphala* or *amalaki* indefinitely, even after the boil is healed, as a preventive and general health tonic.

COOLING, HEALING PASTE. Apply a paste of red sandalwood and turmeric powder locally. Use ½ teaspoon of each powder, and mix them together in warm water to make a paste.

BRING THE BOIL TO A HEAD. Apply cooked onions as a poultice, or apply a paste of ginger powder and turmeric (½ teaspoon of each) directly to the boil, to bring it to a head.

LIVER CLEANSERS. A boil may develop into an abscess, if the boil is due to an infection of the sebaceous glands (a pitta condition). It becomes inflamed, raised, and red. If you use a formula to help cleanse the liver, the condition will be improved.

A simple and effective liver cleanser is aloe vera gel. Take 2 tablespoons 3 times a day.

Or you may try this Ayurvedic formula:

shanka pushpi 3 parts

kutki 2 parts

gulwel sattva ⅓ part

Take ½ teaspoon of this mixture 3 times a day with warm water.

❧❧ Breastfeeding Problems ❧❧

There are several types of breastfeeding problems. Let's take them one at a time.

THE CHILD HAS NO APPETITE

In this case, the mother is producing a lot of milk, but the child has no appetite. Here are a number of effective, natural ways to help.

DRINK FENNEL TEA. Make a tea by steeping 1 teaspoon of fennel seeds in a cupful of boiled water. When the tea cools, give the baby 1 teaspoonful every 10 to 15 minutes.

Childhood is the kapha stage of life, when the body is building. It is also the time when many kapha disorders occur (such as colds and runny noses) and when kapha may stagnate in the stomach, slowing down the appetite. Giving this fennel tea will help to wash out the kapha and in a gentle way stimulate the secretion of digestive enzymes.

MAKE GHRITA MADHU. Mix a pinch of *pippali* in ½ teaspoon of honey combined with ½ teaspoon ghee. Give it to your baby to lick. The more he or she licks the mixture, the more the appetite will come back.

MONITOR YOUR DIET. One possible reason for a baby's apparent lack of appetite is that your milk may not taste good to the child. If you tend to have excess pitta in your constitution, and especially if your diet is hot and spicy or includes sour foods and fruits, your milk may become bitter, and the baby won't like it. This unpleasant taste could be the cause of the child's apparent lack of appetite. So it is important to determine the mother's *prakruti* (constitutional type) and to be sure the diet she is eating is appropriate.

EMPTY YOUR BREASTS. If your child has a diminished appetite and you are secreting more milk than gets used, it is important to empty your breasts. This will avoid congestion of the mastic and lymphatic tissue. Be sure the breasts are emptied of milk at least 2 or 3 times a day.

IF YOU DECIDE NOT TO BREASTFEED. A related situation occurs when a woman chooses not to breastfeed her child. Then the milk is stagnant, which may be one of the causes of fibrocystic changes in the breast. So it is important to empty the breasts whenever milk is present.

THERE IS INSUFFICIENT MILK

The child has a strong appetite, but lactation is scanty. This problem is the reverse of the first. Here are several suggestions to increase the quality and quantity of milk.

SHATAVARI KALPA. To increase lactation, Ayurveda recommends a delicious concoction called *shatavari kalpa*: the herb *shatavari* roasted in a pan with ghee and natural brown sugar. Take one teaspoon of that sweet, roasted *shatavari* in warm milk 2 or 3 times a day.

PLAIN SHATAVARI. You can also use plain *shatavari* with ghee and sugar. Mix together equal amounts of *shatavari* and natural sugar, and take 1 teaspoon of the mixture with 1 teaspoon of ghee along with a cup of hot milk.

ALMOND MILK. Another formula to increase breast milk is almond milk. Soak 10 almonds overnight in water. In the morning, peel them and blend them in the blender with a cup of hot water or hot milk. Pour the mixture into a cup or glass, and add 1 teaspoon honey or date sugar, and a pinch each of ginger, cardamom, and saffron. Drink twice a day.

HERBAL FORMULA. To maintain healthy lactation, use this herbal formula:

kutki 2 parts

shilajit 2 parts

shatavari 3 parts

Take $\frac{1}{4}$ teaspoon of this mixture 2 or 3 times a day with a spoonful of honey. If you wish, you may use this formula the entire time you are breastfeeding.

INFECTED NIPPLE

A third problem is that during the time you are nursing your child, the nipple is not properly cleaned and a fungal infection develops. So take care to wash carefully to prevent this from happening.

PREVENTION

To prevent breast abscess, mastitis, congestion, and stagnation of milk in the breast, gently massage your breasts with warm castor oil. Take 1 teaspoon of the oil, and gently massage the breast from inside outward—that is, from the sternum (breastbone) back toward the armpit, both underneath and around the nipple, and to the side.

Do not apply the castor oil to the areola and nipple. If you apply castor oil to the nipple and your baby sucks the oil, he or she may get diarrhea. So either avoid the nipple or wash off the oil before nursing the child.

☞ ☞ Breasts, Sore ☞ ☞

Sore breasts are generally symptomatic of hormonal imbalance, lymphatic congestion, or premenstrual syndrome. Or the physical discomfort may be associated with some emotional factor, such as grief or sadness. Here are several suggestions for effective self-treatment:

GENTLE MASSAGE. Take 1 teaspoon of warm castor oil, and gently massage the breast from inside outward, that is, from the sternum (breastbone) back toward the armpit, both underneath and around the nipple, and to the side. This kind of gentle massage before the morning bath, as well as at bedtime, will help to relieve the soreness.

HERBS FOR WATER RETENTION. Water retention may also be responsible for sore breasts. The breasts become tender, swollen, and enlarged, and the bra becomes tight. To reduce the swelling, make the following herbal mixture:

For a Constipated Baby

In India, if a nursing baby becomes constipated, the mother intentionally applies a few drops of castor oil to her nipple. Then, when the baby sucks the milk, those few drops of castor oil are taken in along with the milk, and the constipation is relieved in a gentle way.

punarnava 1 part

shatavari 1 part

musta 1 part

Use ½ teaspoon of this mix in 1 cup hot water to make a tea, and drink it. Then massage the breasts as above, and you will see remarkable improvement. You may drink the tea twice a day until the soreness goes away.

ANOTHER EFFECTIVE HERBAL REMEDY. Sore breasts can also be treated with another herbal combination:

jatamamsi 2 parts

shatavari 3 parts

tagar 3 parts

Mix the herbs together in the above proportions, and use ½ teaspoon in a cup of water to make a tea. Steep for 5 to 10 minutes, and drink. You can use this tea twice a day until the soreness is gone.

NOTE: Never use an overly tight bra for long periods of time. The pressure obstructs circulation, and the breast tissue stops breathing. It is better to use a cotton bra that allows proper breathing of the mastic tissue.

❧ ❧ Brittle Nails ❧ ❧

The nails of our fingers and toes are considered in Ayurveda to be a by-product of bone formation (*asthi dhatu*). Proper nutrition for building strong bones is thus essential if you want to have healthy nails. If there is insufficient intake of calcium and magnesium, or malabsorption of these minerals, the nails become rough, brittle, cracked, and split, and ridges and creases may appear. If you have these symptoms, you can be sure they are signs of malnutrition of the bone tissue.

To strengthen the bones and nails, take over-the-counter supplements of calcium, magnesium, and zinc. Your formula should include a daily dose of approximately

calcium 1,200 mg.

magnesium 600 mg.

zinc 60 mg.

Take the supplements at bedtime for best results. A general mineral supplement might also be helpful.

You may be getting enough minerals, but toxins in the colon may be preventing their complete absorption. If you are taking calcium, magnesium, and zinc as a dietary supplement and your nails are still brittle, that indicates that you are not absorbing these minerals. The culprit is overaccumulation of toxins (ama) in the colon.

A safe and simple way to cleanse the colon of the ama is to regularly take the herbal compound *triphala* (see [appendix 2](#)). Steep about ½ teaspoon in ½ to 1 full cup of warm water, strain, and drink. You can take this at night before going to bed, or steep it overnight in cold water and drink it first thing in the morning. It will gradually eliminate the ama.

Another way to promote the growth and strength of your nails is to eat a handful of white sesame seeds every day. A handful of sesame seeds contains about 1,200 mg. of calcium and magnesium.

Either one of the Ayurvedic herbs *ashwagandha* or *shatavari* (½ teaspoon) taken in a cup of hot milk twice a day may also help to prevent brittle nails.

Aerobic exercise such as swimming, jogging, or aerobic dance, or the sequence of yoga postures known as the Sun Salutation (see illustration in [appendix 4](#)), should also be helpful. Exercise improves the circulation and helps carry the minerals to the tissue at the root of the nails.

Because the nails are connected with the *asthi dhatu* (bone tissue), strengthening *asthi* will be helpful. Taking *triphala guggulu* tablets (200 mg.) twice a day, after lunch and dinner, will help to strengthen your nails.

In some individuals, brittle nails and bone loss go together. Specifically, in women of menopausal age, brittle nails may suggest weakness in the *asthi dhatu*. So if you have brittle nails, it would be wise to investigate the possibility that osteoporosis may be developing.

☞ ☞ Burns ☞ ☞

Many people believe the best thing you can do for a burn is to put something fatty or greasy on it, such as butter. But this is not true. Any burn is pitta, the dosha of fire. The hot, sharp, burning quality of pitta immediately creates searing pain and inflammation. A fatty substance is in fact contraindicated, as it would serve to aggravate the pitta.

TIME TO SEE THE DOCTOR

We are not discussing serious burns in this section but common household burns, such as from inadvertently touching an iron, a skillet, or a cigarette. For serious burns, particularly over a larger area of the body, a person needs hospitalization, plasma, and intensive care.

The most effective way to treat a burn is with cold. Immediate application of something cold, such as ice or cold water, is the best remedy. If you have no ice cubes, use a bag of frozen vegetables from the freezer.

After applying cold, make a paste of sandalwood and turmeric powders (equal amounts) mixed *not* in water but in aloe vera gel. Use about 1 tablespoon aloe vera and ¼ teaspoon each of the sandalwood and turmeric. Mix them together, and apply the paste topically. This will be soothing and healing.

Or, after the ice, when the burning sensation stops, apply bitter ghee (*tikta ghrita*).

Cilantro is beneficial for burns. Make fresh cilantro juice by placing a handful of cilantro in the blender with about ⅓ cup of water. Strain. Take the juice internally (2 teaspoons 3 times a day), and put some of the pulp directly on the skin.

It is important not to apply a bandage to cover the burn. Keep it open to the air. If you apply a bandage, the body's heat may affect the burn adversely. So apply the herbal paste or bitter ghee, and leave it alone. If you have to cover it to keep the medication from rubbing off, use some light gauze.

❧ ❧ Bursitis ❧ ❧

See also “[Arthritis](#)”

Bursitis is an inflammation of the bursae, the little fluid sacs around the shoulder, knee, and other parts of the body. Bursitis is a pitta condition that is similar to arthritis. Effective treatment for it is similar to the treatment for pitta arthritis.

Taking *kaishore guggulu* (1 tablet 3 times a day) should help. This herbal remedy is available by mail from various sources of Ayurvedic herbs (see [Resources](#)).

Application of sandalwood paste will be soothing. Make the paste by taking 1 teaspoon of sandalwood powder and adding sufficient water to form a paste. Rub it gently onto the painful area.

Nasya, or the application of warm ghee nose drops to the nostrils (5 drops in each nostril), will help to relieve the pain. *Nasya* opens up the flow of *prana* and helps the *prana* to flow freely through the connective tissue of the joint, which will alleviate pain. (For instructions, see [appendix 3](#).)

Topically, try gently rubbing some sesame oil, eucalyptus oil, tea tree oil, *mahanarayan* oil, or *neem* oil into the swollen joint or other painful area.

As with arthritis, the pathological process of bursitis begins with toxicity accumulated in the colon. These toxins get absorbed in the bloodstream, go into the general circulation, and lodge in the bursae, leading to the symptoms of bursitis. So it is important to keep the colon clean by using the following strategies:

- Before going to bed at night, take 1 cup of hot milk with 2 teaspoons of castor oil added. The laxative effect will eliminate pitta-type toxicity in the colon. If 2 teaspoons do not work, use 3 teaspoons the following night, adjusting your own dose. You should get a couple of good bowel movements in the morning, which will help to cleanse the impurities of the colon. You may continue using this remedy until your symptoms clear up.

- As a second choice, you can take *triphala* or *amalaki* (1 teaspoon) at night in 1 cup of warm water.

Your diet should be pitta-soothing but not vata-provoking. Strictly avoid hot, spicy foods and fermented foods such as pickles. Also avoid raw vegetables and salad. Strictly avoid ice water and other ice-cold drinks. Don't eat beans (pinto beans, adzuki beans, black beans, or garbanzos).

As long as you have bursitis, you should not do strenuous exercise. Gentle yoga stretching is beneficial. Try the Camel, Cobra, Cow, and Cat poses, the Spinal Twist, and Forward Bend—under the guidance of a yoga teacher.

❧ ❧ Canker Sores ❧ ❧

Canker sores are generally traumatic, that is, they are due to a cut or bruise in the mouth. When people eat some sharp, dried, or hard foods, such as popcorn, corn chips, crackers, or dried bread, or chew fennel seeds after a meal, the hard, sharp food can hurt the oral mucous membrane, and within a couple of days it will manifest as a canker sore.

People who use a rough, hard-bristled toothbrush, or who press unduly hard while brushing their teeth, may irritate the mucous membranes and create a canker sore. Some individuals inadvertently bite the delicate tissues in their cheeks or lips during sleep, or even while chewing or talking. This is more likely if they have TMJ and their bite is uneven, which can easily lead to lacerations and canker sores. If they have high pitta in their saliva, their teeth may be sharp because the crown of the tooth becomes eroded, a combination that may result in repeated sores.

TOPICAL REMEDIES

- The simplest Ayurvedic solution to canker sores is local application of turmeric and honey. Mix together 1 teaspoon honey with ¼ teaspoon turmeric, and rub it on the sore.

It will burn a little at first, but the sore area will heal quickly.

- Rinse your mouth several times a day with a little aloe vera juice.
- Aloe vera gel, 2 tablespoons 3 times a day, can also help heal canker sores.
- Mixing aloe vera gel with *neem* powder is also helpful. Use 1 teaspoon of gel with a pinch of *neem* powder, mix together, and apply directly to the canker sore.
- Put 10 drops of tea tree oil in $\frac{1}{3}$ cup of water, and swish the liquid in your mouth. The mild solution will act as an antiseptic to help prevent secondary infection, and it will also help to heal the sore.
- Another topical Ayurvedic remedy is the herb *kama dudha*. Take $\frac{1}{4}$ teaspoon and 1 teaspoon fresh cream. Mix together and rub it on the sore.

INTERNAL REMEDIES

- Generally, people with high pitta are more likely to get canker sores. So follow the pitta-soothing diet, avoiding hot, spicy foods and fermented food ([chapter 8](#)). Also keep away from strong alcoholic drinks, which will aggravate the sores.
- Between meals, drink $\frac{1}{2}$ cup cranberry juice. That will help to heal the sore and relieve the burning sensation and irritation.
- Eating a mixture of rock candy powder ($\frac{1}{2}$ teaspoon) and cumin powder ($\frac{1}{2}$ teaspoon) will help stop the pain, as well as reduce inflammation and irritation.
- Sometimes canker sores are accompanied by either diarrhea or constipation. Taking $\frac{1}{2}$ teaspoon arrowroot with 1 cup warm milk helps relieve constipation; taking $\frac{1}{4}$ teaspoon arrowroot with 1 teaspoon ghee corrects diarrhea.

❧❧ Cataracts ❧❧

Cataracts are a kapha disorder. Molecules of kapha accumulate in the lens of the eye and affect its translucence and transparency, making it increasingly opaque. As the cataract grows, it creates increasingly smoky, blurred vision. Generally persons with diabetes are prone to cataracts, as are young people with juvenile diabetes, though cataracts are primarily associated with elderly people.

EFFECTIVE HERBAL EYEWASH. If your eye doctor has detected an early sign of cataracts developing, this *triphala* tea eyewash will be effective in dissolving the molecules of kapha that cause the cataract.

Boil 1 teaspoon *triphala* in a cup of water for 2 to 3 minutes. Let the tea cool, and strain it with a double or triple layer of cheesecloth so that not a single particle of *triphala* is left in the strained tea. Then, with an eye cup, wash your eyes with the *triphala*

tea. You may wish to repeat the wash 2 or 3 times, depending on how it feels and how much tea actually got into the eye.

To keep the lens clear and prevent the further growth of cataracts, wash your eyes with *triphala* tea in the morning and at bedtime. Do it regularly for one month, and if it proves effective, you can continue indefinitely. It will help to arrest the process of cataract formation.

CASTOR OIL EYE DROPS. One drop of pure castor oil (without preservatives) placed in the eye at bedtime lubricates the cornea and the conjunctiva and helps to remove molecules of kapha from the lens. In this way one can prevent cataracts from developing.

HERBAL REMEDY. A third approach is to take this herbal mixture internally:

punarnava 5 parts

shatavari 3 parts

brahmi 3 parts

Take ½ teaspoon of this herbal mixture twice a day with warm water as a preventative measure against cataracts. As with the triphala eyewash, you may continue using this formula as long as you wish.

❧❧ Cellulite ❧❧

Cellulite is more a sociological problem than a health problem! Subcutaneous fat that accumulates under the skin and causes little dimples on the skin surface, cellulite is certainly not a disease. People with high cholesterol, who eat fried, fatty food and whose diet is excessively kapha-provoking, appear to be more likely to have it, as do individuals who use a lot of olive oil in their cooking. As cellulite may gradually lead to obesity, this condition can be seen as an early phase of being overweight.

In Ayurvedic terms, *meda* agni (the agni or fiery quality responsible for metabolizing fat) under the skin becomes low. Unprocessed fatty molecules lodge there and create cellulite. The Ayurvedic aim in treatment is to kindle or enliven the *meda* agni.

EXERCISE. Regular exercise is the first approach. Walking, swimming, or other aerobic exercise is important, and at least some of the exercise should be “local” to the area where the cellulite is forming. In other words, if cellulite is developing on the thighs, don’t restrict your exercise to weight lifting and upper-body development!

WATCH OUT FOR KAPHA. Watch your diet and be sure it isn’t kapha-increasing (see [chapter 8](#)). Minimize dairy products, sweets, cold food and drinks, and fatty fried food. Strictly avoid olive oil in cooking.

VITAMIN K. Rub vitamin K cream onto the skin. This cream will remove spider veins and

minimize cellulite.

LOCALIZED MASSAGE. Massage the affected area with sesame oil and mustard oil, mixed half and half. After the massage, dust with a powder of the herb *vacha* (calamus), and rub the skin. This will help remove the cellulite.

❧❧ Cholesterol ❧❧

High cholesterol means increased lipids (fats) in the blood. It is essentially a metabolic disorder. People having low liver function or diminished thyroid activity, who have taken steroids in the past, or whose diet is very kaphagenic, seem most prone to develop high cholesterol.

Your cholesterol level should be below 200. About 160 to 190 is normal, but a cholesterol reading of 200 or above is worrisome, as a high cholesterol level in the blood tends to create plaque on the artery walls, resulting in atherosclerotic changes, cardiovascular disorders, hypertension, stroke, and heart problems.

There are two kinds of cholesterol, HDL (high-density lipoprotein), which is the “good” cholesterol, and LDL (low-density lipoprotein), which is the “bad” cholesterol. Researchers these days are saying that what is more important than the total level of cholesterol, as a predictive factor for cardiovascular and other health problems, is the ratio of total cholesterol to HDL.

To reduce high cholesterol levels, and to prevent cholesterol from building up any higher, follow these guidelines.

WATCH YOUR DIET. Stick to a kapha-pacifying diet ([chapter 8](#)). No fatty fried food. No cheese. No high-fat milk or yogurt. Minimize sweets and cold food and drinks. Use garlic and onion in cooking.

GET REGULAR EXERCISE. Each day from Monday to Friday, walk for at least half an hour. Go swimming or participate in some other aerobic exercise at least three times each week.

Just by regulating diet and exercise, you can control cholesterol. But there is much more that you can do.

HERBS TO COMBAT CHOLESTEROL. Garlic is effective against high cholesterol. Mix together 1 clove of fresh garlic chopped fine, ½ teaspoon grated ginger root, and ½ teaspoon lime juice; eat before each meal.

- Drink a tea made of 1 teaspoon cinnamon and ¼ teaspoon of the herbal mixture *trikatu*. Steep it for 10 minutes in a cupful of water, add 1 teaspoon honey, and drink. Take it twice a day.

- Taking ½ teaspoon *trikatu* with 1 teaspoon honey 2 or 3 times a day is good for burning ama and excess kapha and helps to regulate cholesterol.

- This herbal mixture can help control high cholesterol levels:

kutki 3 parts

chitrak 3 parts

shilajit ¼ part

Take ½ teaspoon twice a day, with honey and hot water.

- Take one 200-mg. tablet of *triphala guggulu* 3 times a day.

• Another herb that has been remarkably effective in bringing down cholesterol levels is *chitrak-adhivati*. Taking 1 200-mg. tablet twice a day, after lunch and dinner, will help bring cholesterol back to normal.

HOT WATER AND HONEY. Early in the morning, drink 1 cup of hot water into which you add 1 teaspoon of honey. This will help to “scrape” fat from your system and reduce cholesterol. Adding 1 teaspoon of lime juice or 10 drops of apple cider vinegar will make the drink more effective.

FOODS THAT REDUCE CHOLESTEROL. In addition to avoiding high-fat foods, you can eat certain foods that in themselves help to reduce cholesterol. These include blue corn, quinoa, millet, and oatmeal. Some research suggests that apples, grapefruit, and almonds can also help reduce cholesterol.

YOGA POSTURES. Yoga postures that are good for controlling cholesterol include the Sun Salutation, Shoulder Stand, Peacock, Cobra, Spinal Twist, Locust, and Lotus.

BREATHING EXERCISE. The breathing exercise known as Breath of Fire (*bhastrika*) is also helpful. (See [chapter 6](#) for instructions.)

❧ ❧ Chronic Fatigue ❧ ❧

See “[Fatigue and Chronic Fatigue](#)”

❧ ❧ Colds and Flu ❧ ❧

Individuals often get colds and flu during the winter and spring seasons. The symptoms are all too familiar: runny nose, cough, congestion, headache, an achy body, and sometimes fever.

Ayurvedically speaking, colds are a kapha-vata disorder. The body builds up an excess of cool and moist kapha qualities, resulting in congestion and a runny nose, and at the same time it may suffer from excess vata, which reduces agni (gastric fire), leading to chills, loss of appetite, and/or poor digestion.

GINGER REMEDIES

The best remedy for colds is ginger. Here are several simple home remedies using ginger that will greatly relieve your cold symptoms and hasten full recovery.

- Combine the following herbs together:

ginger 1 part

cinnamon 1 part

lemongrass 2 parts

Steep 1 teaspoon of this formula for about 10 minutes in 1 cup of hot water; strain it, and add honey for sweetness if you like. If you drink this delicious tea several times a day, it will help to take care of cold, congestion, and flu.

- Another excellent remedy is ginger-cardamom-cinnamon tea. Here is the formula:

ginger 2 parts

cinnamon 3 parts

cardamom just a pinch

Steep 1 teaspoon in a cup of hot water for 10 to 15 minutes. When the tea has cooled down somewhat, you can add about ½ to 1 teaspoon of honey for taste.

- Boil 1 teaspoon ginger, or a few eucalyptus leaves, in a pint of water. Turn off the stove, put a towel over your head, and inhale the steam. This will relieve congestion and help you feel much better. Just steam alone, with no herbs added at all, will also be beneficial.

ADDITIONAL HERBAL REMEDIES

- Try ½ teaspoon fennel seed powder mixed with 1 teaspoon natural sugar, 2 or 3 times a day.

- For a cold with cough and congestion, mix ½ teaspoon cinnamon and 1 teaspoon honey. Eat this mixture 2 or 3 times a day. (For help with your cough, please see [“Cough.”](#))

- For flu, make a tea from 1 teaspoon *tulsi* (holy basil) in 1 cup water. Boil for just 1 minute and then drink.

- An ancient Ayurvedic herbal formula that is effective for colds is the following:

sitopaladi 1 part

maha sudarshan churna 1 part

Combine these herbs in equal proportions and take $\frac{1}{4}$ teaspoon with 1 teaspoon honey 2 or 3 times a day after eating.

- Western herbology has some helpful herbs for colds. Try this mixture:

echinacea 1 part

goldenseal 1 part

cinnamon 2 parts

Take $\frac{1}{4}$ teaspoon of this mixture with honey twice a day.

CAUTION: Don't Combine Ginger and Aspirin. Ginger and aspirin are both blood thinners and should not be taken together. Therefore it is wise to drink ginger tea—or use any other ginger remedy—either two hours before or two hours after you take any aspirin.

OTHER REMEDIES AND RECOMMENDATIONS

VITAMIN C. Taking some Vitamin C will be beneficial.

NATURAL NOSE DROPS. Put some liquefied ghee (3 to 5 drops) in each nostril in the morning and evening. This will lubricate the nasal passages and relieve the irritation and sneezing of a cold.

HOT WATER. Drinking hot water several times a day is an effective way to remove toxins from the system and hasten recovery from a cold.

NO DAIRY PRODUCTS. Strictly avoid dairy products such as yogurt, cottage cheese, and milk, and all cold drinks.

BE SURE TO REST. Rest is very important for healing. As much as possible, rest, read, and relax.

ONLY MILD EXERCISE. Ayurveda says that when you have a cold, it is best not to do vigorous exercise, which can set up the conditions for the cold to move into the chest. Just do some gentle yoga *asanas*. *Surya namaskar* (Sun Salutation) is beneficial. The inverted postures, including Shoulder Stand and Headstand (hold for only about 1 minute), as well as Forward Bend, help to prevent postnasal drip and help to drain the mucus through the nose.

BREATHING EXERCISE. Use the Breath of Fire breathing exercise to help burn up your cold. Inhale normally and passively, but exhale forcefully, and repeat rapidly several times. This exercise will help to eliminate mucus from the respiratory tract. For more detailed instructions, see [chapter 6](#).

- Another effective breathing exercise is deep Alternate Nostril breathing, without

retention of breath. This will also help to relieve congestion (see [chapter 6](#)).

PREVENTION. As a preventive measure, take the herb *amalaki*. It is a *rasayana* (rejuvenative tonic) and a good source of vitamin C and iron. Taking 1 teaspoon of *amalaki* daily, with warm water at night, will help to prevent the common cold.

If you are taking *triphala* at night, you are already taking *amalaki*, it is one of the three herbs that constitute *triphala* (along with *haritaki* and *bibhitaki*). Taking extra *amalaki* is not recommended, as it would create diarrhea.

❧❧ Colitis ❧❧

Colitis is caused when vata pushes pitta into the colon and inflammation occurs. The basic line of treatment is to pacify pitta.

HERBAL REMEDIES

- An excellent herbal remedy for colitis is:

shatavari 4 parts

shanka bhasma 1/8 part

kama dudha 1/8 part

sanjivani 2 parts

Take 1/4 teaspoon of this mixture with warm water 2 or 3 times a day for 1 or 2 months.

- You can also take aloe vera gel, 1 tablespoon twice a day. Aloe vera is cooling and good for reducing pitta.

ENEMAS

HEALING ENEMA FOR ULCERATIVE COLITIS. Ulcerative colitis is characterized by diarrhea, mucus, and blood from the rectum. For that condition, Ayurveda suggests a *basti* (enema) using (instead of plain water) a tea made from an astringent herb such as *bilva*, *ashoka*, sandalwood, or licorice root. This is how to prepare the enema:

Boil 1 tablespoon of the herb (such as licorice powder) in 1 pint of water for 5 minutes. Strain, and add about 2 tablespoons of ghee while the tea is still warm. Let the liquid cool to room temperature, and use it for an enema. Retain the liquid inside for 5 minutes if you can. Do this procedure once or twice a week. (More complete instructions for *basti* are found in [appendix 3](#).)

Licorice contains food precursors of natural steroids, which will help heal the ulcer. This is a safe, simple way to correct colitis or ulcerative colitis.

SOOTHING OIL ENEMA. The colon is the seat of vata. This vata dosha is pulling or pushing pitta into the colon, which then causes the colitis. To combat the excess pitta and at the same time pacify vata, Ayurveda suggests injecting a cooling oil such as coconut oil into the rectum. Use about 1 cup of slightly warmed oil as a *basti*, and try to retain it for 5 minutes, but don't worry if it comes out more quickly.

DIETARY REMEDIES

TWO APPLE REMEDIES.

- A simple and beneficial remedy for colitis is to eat cooked apples with a pinch of nutmeg. Peel a couple of apples, remove the seeds, and cook. Make them into a pulp (you can use a potato masher), and add 1 teaspoon ghee and a pinch of nutmeg. This will help to pacify the irritation of colitis and ulcerative colitis.

- Apple juice will also help relieve the burning sensation.

DIETARY PRECAUTIONS. A person with ulcerative colitis should never eat hot spicy food, drink alcohol, or use tobacco, all of which provoke pitta and will significantly irritate the colon.

EXERCISES TO STRENGTHEN THE COLON

LEG LIFT. When a person has ulcerative colitis, the colon is very weak. To strengthen the wall of the colon, lie flat on your back and gradually lift both your legs, keeping the knees as straight as you can, until the legs are at a 45-degree angle to the floor. This exercise is called a Leg Lift. If you find it difficult to lift both legs together, you can lift one leg and then the other. Hold your legs in the raised position for just a few seconds at first, up to 1 minute after several weeks of practice.

YOGA POSTURES. Slowly move into the Chest-Knee pose, then the Plow pose, Locust pose, and if you can, the Elevated Lotus. Also, exercise your abdomen by drawing it gradually in and out. This will strengthen the wall of the colon. Illustrations of yoga postures are in [appendix 4](#).

❧ ❧ Conjunctivitis ❧ ❧

This is a pitta condition involving inflammation of the conjunctiva of the eyes, which makes the eyes red and photophobic (highly sensitive to light), with discharge and a burning sensation. Here are some effective remedies:

CILANTRO LEAVES. Apply the pulp of fresh cilantro leaves on closed eyelids. Blend a handful of cilantro leaves with $\frac{1}{4}$ to $\frac{1}{3}$ cup water; strain out the juice, and apply the pulp to your closed eyelids. Drinking the liquid will also be beneficial.

CORIANDER EYEWASH. Make an eyewash by steeping 1 teaspoon of coriander seeds in 1 cup boiling water for at least 15 minutes. Strain thoroughly, and cool before using the water on your closed eyes. (Don't worry if a little goes into the eyes.) Careful: Don't use it either too hot or too cold.

GOATS MILK COMPRESS. Dip a sterilized cotton ball in goat's milk, and put it over your eyes. This will cool down pitta and allow conjunctivitis to be healed.

HERBAL REMEDY. Internally, you can take a mixture of equal parts of *kama dudha* and *gulwel sattva*. Take $\frac{1}{4}$ teaspoon of this mixture with warm water twice a day for 1 week.

HOW TO HEAL CONJUNCTIVITIS IN CHILDREN

The best way to heal conjunctivitis in young children is to place a drop of mother's milk in the eye. Mother's milk is healing for her child. If the mother is still lactating, and her child gets conjunctivitis, with irritation and discharge in the eye, just one drop of her milk may heal it.

NATURAL HERBAL ANTIBIOTICS

- Make a turmeric solution by stirring some turmeric into a few ounces of pure water. Immerse a clean handkerchief into the solution, and let it dry. (It will be quite yellow.) Then use it to mop the affected eyes.

TIME TO SEE THE DOCTOR

If your conjunctivitis is not cleared up within 3 to 4 days, immediately see your doctor.

The natural antiseptic, antibiotic properties of the turmeric will help take care of the bacteria and facilitate healing.

- Herbal antibiotics are also helpful taken internally. You can make a mixture of these three:

turmeric 1 part

neem 1 part

manjistha 1 part

Take $\frac{1}{2}$ teaspoon twice a day with warm water, after meals.

❧❧ Constipation ❧❧

Constipation is a vata condition expressing such vata qualities as dryness and hardness. It is caused by insufficient fiber in the diet, insufficient water intake, lack of exercise, heavy meat eating, and numerous other factors. Constipation may create distension and discomfort, flatulence and pain, headache and bad breath, and may lead to absorption of toxins from the colon. Thus it is best to prevent constipation by keeping vata in balance.

FOLLOW THE VATA-PACIFYING DIET. One of the best ways to prevent constipation, especially if you have a predominantly vata constitution, is to follow the vata-balancing diet (see [chapter 8](#)). Stay away from cold foods and drinks, dried fruit, salads, and most beans; favor warm foods, warm drinks, and well-cooked vegetables. Some oil in the diet is helpful.

TRIPHALA. Probably the best Ayurvedic remedy for constipation is *triphala*, a combination of three herbs beneficial for all doshic types (see [appendix 2](#)). Most problems with constipation can be corrected by taking ½ to 1 teaspoon of *triphala* at night. Steep the herbs in a cup of hot water for 5 to 10 minutes, and drink.

Some people find that taking *triphala* at night creates a diuretic action and they have to get up several times to urinate. If that happens to you, you can steep the *triphala* in a cup of warm water overnight and drink it first thing in the morning. Actually, the best time to take *triphala* is in the very early morning, around 4 or 5 A.M., but do your best within your daily schedule.

Here are some further recommendations to help relieve constipation:

SNACK ON FRUIT. Many varieties of fruit can help remedy constipation. So in between meals, eat some fruit. A banana, for example, is a mild laxative. Two ripe yellow bananas, taken between meals, will help relieve constipation. (But don't eat the bananas with meals. They don't combine well with other foods. See [chapter 8](#) for tips on healthy food combining.)

NOTE: Bananas should be eaten when ripe. You can tell a ripe banana by its bright yellow skin. The inside will be studded with tiny black dots. Green bananas are constipating and should be avoided. Also avoid eating bananas once the skin turns black. They are overripe.

AN APPLE A DAY KEEPS THE DOCTOR AWAY. There is a lot of truth in this old folk saying. Apples are effective both to help regularize the bowels and to clean the tongue and teeth. To combat constipation, peel and thoroughly chew a raw apple about an hour after a meal.

Also effective:

- Pineapple juice

- Raisins—a handful every day, at least an hour after meals
- Prunes
- Peaches—one or two about an hour after meals

EAT MORE FIBER. Fiber in the diet, such as wheat bran, oatmeal, or oat bran, will help keep the bowels regular. Don't forget that fresh fruit and vegetables as well as whole grains are high in fiber, too.

MILK AND GHEE—MILD AND EFFECTIVE. Taking 1 or 2 teaspoonfuls of ghee in a cup of hot milk at bedtime is an effective but gentle means of relieving constipation. This is especially good for vata and pitta constitutions, but it may be too kapha-increasing for kapha types to use regularly.

CASTOR OIL. Castor oil can also be used, but only when the constipation is more obstinate. At bedtime, make yourself a cup of ginger tea (either by boiling some fresh, sliced ginger in water, or adding some powdered ginger to a cup of hot water), and then add 2 teaspoons of castor oil to the tea and drink it.

If 2 teaspoons don't bring the desired result, try the procedure again the next night, increasing the dose to 3 teaspoons, and then go to 4 teaspoons if necessary. Adjust your dose according to what works.

SUGGESTION: Use this castor oil laxative treatment on weekends, when you can stay home!

Castor oil tends to create dependency, in the sense that once you use it, other purgatives rarely work. That is why it is recommended to use castor oil for constipation only in severe cases or in emergencies, as a last resort, but not on a regular basis.

However, there is one way to avoid this dependency. Take 2 teaspoons of castor oil with 1 cup warm milk. This will give a good bowel movement without creating dependency.

FLAXSEED. At night, boil 1 tablespoon of flaxseed in a cup of water for at least 2 to 3 minutes. Then drink the whole cupful, including the tea and the seeds.

FOR CONSTIPATION IN CHILDREN. Give the child 3 figs soaked in warm water.

FOR SEVERE CONSTIPATION. If there is absolute constipation for three days, don't use purgatives. The use of drastic purgatives when constipation is severe can create an intestinal obstruction or even perforation, which can be dangerous. A better approach is to do an enema and then take steps to regulate the system.

Do the enema with plain warm water, or you can use *triphala* tea or *dashamoola* tea instead of plain water. That will relieve the immediate problem. Then, to regulate bowel movements, follow the suggestions in the following section.

PREVENTION. Here are several suggestions for avoiding constipation in the future.

- Follow the vata-pacifying diet ([chapter 8](#)).
- Use a lot of fiber in your diet.
- Drink 4 to 5 glasses of water a day, in addition to whatever juices or teas you may drink.
- Get regular exercise. Half an hour of walking, light jogging, swimming, or other aerobic exercise (suitable to your constitution, age, and level of fitness) from Monday to Friday will be very beneficial.
- Yoga *asanas* will also help prevent constipation. In particular, practice the Sun Salutation (12 cycles a day), Chest-Knee pose, and Leg Lift (see [appendix 4](#)).
- The yoga exercise known as *navli* will also be beneficial (see [appendix 3](#)).

❧❧ Constipation During Pregnancy ❧❧

See also “[Constipation](#)”

The best remedy for constipation during pregnancy is the herb *sat isabgol* (psyllium husks). Take 1 teaspoon with a glass of warm milk.

A cup of hot milk with 1 teaspoon of ghee added is also effective. Ghee and milk mixed together is a wonderful combination for gentle, mild laxative action during pregnancy.

One of the most effective herbal remedies for constipation, *triphala*, should *not* be used during pregnancy. It will irritate the child, and the baby will become hyperactive.

Also, do not take castor oil or any other drastic purgative during pregnancy; like *triphala*, it will make the baby hyperactive.

❧❧ Cough ❧❧

A tickling sensation in the throat, dryness, irritation, or inflammation of the trachea or bronchus can all cause a cough.

From an Ayurvedic perspective, most coughs are caused by excess pitta or kapha in the bronchial tree, causing congestion and irritation of the bronchial mucous membrane. The basic strategy for managing this condition is to reduce the unwanted pitta or kapha that is creating the congestion.

To treat a cough most effectively, you need to determine whether it is a dry cough (vata) or a productive cough with mucus coming up (kapha), or whether pitta has also become involved.

DRY COUGH

For a dry cough or a cough without much mucus, eat a ripe banana with 1 teaspoon of honey and 2 pinches of ground black pepper. You can eat this 2 or 3 times a day.

Rx for a Stubborn Cough

Chop a clove of garlic, and boil it in a cup of milk. Then add $\frac{1}{4}$ teaspoon turmeric. This creates a golden yellow milk that tastes like garlic soup. This garlic-turmeric milk is effective for soothing and healing most types of cough.

- Try chewing $\frac{1}{4}$ teaspoon of *ajwan* mixed with 1 teaspoon natural organic sugar.
- Make *talisadi* tea, made of
talisadi powder $\frac{1}{2}$ teaspoon
licorice powder $\frac{1}{2}$ teaspoon
with a little honey added. This tea is quite effective.
- Dry cough or irritation in the throat may be due to slight congestion in the tonsils, or a congestive condition in the throat such as pharyngitis or laryngitis. To relieve this condition, boil 1 cup milk with $\frac{1}{2}$ teaspoon turmeric and $\frac{1}{4}$ teaspoon ginger, making a golden yellow milk. If you drink this at night, it will relieve the irritation in the throat and take care of a dry cough.

PRODUCTIVE COUGH

For a productive or kapha cough, the simplest home remedy is black pepper. Mix $\frac{1}{4}$ teaspoon of the powder with 1 teaspoon of honey, and eat it on a full stomach. (If your voice is hoarse, use 1 teaspoon of ghee instead of the honey.) The heating quality of black pepper helps relieve congestion and drives out the cough. Take 2 or 3 times a day for 3 to 5 days.

- A tea made from $\frac{1}{2}$ teaspoon ginger powder, plus a pinch of clove and a pinch of cinnamon powder in a cupful of boiled water, can offer relief from your cough.
- If your cough persists, try this formula:
ground mustard $\frac{1}{2}$ teaspoon
ginger powder $\frac{1}{2}$ teaspoon

Mix together into 1 teaspoon of honey, and eat slowly. (Ginger relieves congestion, and mustard has a heating action.) You can use this mixture 2 to 3 times a day for as long as the cough persists.

- Another helpful natural remedy for a productive cough is the following:

bay leaf ½ teaspoon

pippali ¼ teaspoon

Take this mixture in 1 teaspoon honey 2 or 3 times a day.

- You can also try 1 teaspoon honey mixed with a pinch of clove powder, 2 or 3 times a day.

OTHER COUGHS

PRODUCTIVE COUGH WITH GREENISH-YELLOWISH MUCUS. In such a cough there is some secondary infection due to the involvement of pitta dosha. For this condition, you can use a tea made of equal proportions of

sitopaladi

maha sudarshan

Use ¼ teaspoon of the mixture to make a tea, and drink it 3 times a day with honey.

TIME TO SEE THE DOCTOR

If your cough persists for more than a week, you should see a doctor.

It will encourage elimination of the mucus and help heal the cough quickly.

FOR A CHILD'S COUGH. Try giving the child a drink of ½ cup pomegranate juice with a pinch of ginger powder and a pinch of *pippali* powder.

• Mix ¼ teaspoon of *sitopaladi* with 1 teaspoon of honey as an effective cough remedy for children. However, if the mother is sensitive or allergic to pollen, the child may be allergic to honey. In that case, instead of honey, use maple syrup.

FOR A CHRONIC COUGH. Make a mixture of 4 parts garlic powder to 1 part *trikatu*. Add a little honey. Take this twice a day.

☞ ☞ Cramps, Abdominal ☞ ☞

See “[Muscle Cramps and Spasms](#)”

☞ ☞ Cramps, Menstrual ☞ ☞

See “[Menstrual Difficulties](#)”

❧ ❧ Dandruff ❧ ❧

Although dandruff can sometimes be caused by a fungal infection or other skin disease, in most cases it is produced when the scalp doesn't receive a sufficient supply of blood. Consequently there is a lack of protein in the skin, which becomes dry and flaky.

Dandruff may also be due to a deficiency of vitamin B-6, or to an excess of vata dosha, which also makes the skin dry.

The treatment is simple. To improve circulation to the scalp, massage for a few minutes daily with *neem* oil (in a sesame oil base). If a fungal infection of the skin is causing the dandruff, the *neem* oil, which has disinfectant properties, will help heal this also.

A second option is to use some eggwhite mixed with lime juice. Put two eggwhites in a small jar or container with the fresh juice of one lime, mix together, and apply to your hair. Let it stay in your hair for a half hour, then wash the hair with *neem* soap. The eggwhite will provide the scalp with the lacking protein, and the dandruff will soon disappear.

❧ ❧ Depression ❧ ❧

PLEASE NOTE: Depression is a serious medical condition that requires the supervision of a medical doctor. Mild or preclinical cases may sometimes be completely healed using these Ayurvedic recommendations, but they are not a substitute for consulting with a physician.

If you are already under a doctor's care for depression, the Ayurvedic remedies suggested here can be used in conjunction with the regimen outlined by your physician. But it is only fair and proper to do so with his or her approval and supervision.

Ask your doctor to carefully monitor your progress. As time goes by, you may be able to minimize or eliminate your dependence on strong medications if your body's balance can be brought to a point where diet, exercise, and other Ayurvedic programs are sufficient to control or eliminate the depression.

Clinical depression is more than just a low or heavy mood. The symptoms of depression include a loss of interest in friends and usual activities; sleep disturbances, such as insomnia, early morning awakening, or oversleeping; anxiety, irritability, or restlessness; low energy and fatigue; poor appetite and weight loss, or sometimes the reverse, overeating and weight gain; difficulty concentrating and making decisions; decreased sex drive; feelings of worthlessness and guilt; feelings of hopelessness and helplessness; frequent crying spells; and suicidal thoughts.

The genesis of depression, from an Ayurvedic perspective, is too complex to lay out

here. In brief we can say that because of specific etiological factors, vata from the colon, pitta from the intestine, or kapha from the stomach enters the general circulation and lodges in the nervous system, interferes with normal functioning of the mind and nervous system, and causes depression.

The resulting depression may be vata, pitta, or kapha. Each of these three types of depression is treated in different ways, though the first step for each is to bring the diet in line with the guidelines for vata-, pitta-, and kapha-pacifying diets (see [chapter 8](#)). This is important—please don't overlook it!

VATA DEPRESSION

Vata-type depression is generally associated with fear, anxiety, nervousness, and insomnia. The following home remedies will help dissolve a mild vata depression:

- Drink *dashamoola* tea. Steep 1 teaspoon of the herb *dashamoola* in 1 cup of hot water, and drink. Take twice a day.
- Make a tea from equal proportions of *ashwagandha* and *brahmi* (about $\frac{1}{3}$ to $\frac{1}{2}$ teaspoon each), steep in a cup of water for about 10 minutes, and drink 2 or 3 times a day.
- Another domestic remedy for depression is a tea made of holy basil (*tulsi*) and sage. Use $\frac{1}{4}$ teaspoon *tulsi* and $\frac{1}{2}$ teaspoon of sage per cup of hot water; drink twice a day.
- Nose drops of warm sesame oil (3 to 5 drops in each nostril) are effective for relieving depression (see [appendix 3](#)). Do this *nasya* procedure morning and evening on an empty stomach.
- Rubbing the top of the head and the soles of the feet with sesame oil is quite soothing to vata and healing for vata depression.
- Psychologically, one factor that sustains vata depression is loneliness. Try to spend more time relating to people; it will help lift the depression.

PITTA DEPRESSION

Pitta-type depression is generally associated with anger, or with fear of failure, of losing control, or of making mistakes; it often involves thoughts of committing suicide. It is serious; you should consult your doctor.

Of course it is possible to have a mild pitta depression, caused by failing an examination, not getting a promotion at work, or some such thing. A pitta person can be quite addicted to success and, when he or she doesn't succeed, can easily become upset and depressed. This type of depression may not last long or be too severe.

Pittas are most vulnerable to SAD, seasonal affective disorder, a fairly mild form of

depression that usually occurs in the winter.

For all types of pitta depression, use the following simple but effective remedies:

- Rub some coconut oil or sunflower oil onto your scalp and the soles of your feet at bedtime.

- Drink gotu kola or *brahmi* tea or ginkgo tea 2 or 3 times a day. Use ½ teaspoon of herbs in a cup of hot water.

- Mix equal amounts of these three herbs:

brahmi

jatamamsi

shatavari

Take ½ to 1 teaspoon of this mixture 2 or 3 times a day with warm water, as a tea.

- Use *brahmi* ghee nose drops, 3 to 5 drops in each nostril, twice a day on an empty stomach.

- Meditate. A few minutes of meditation will be helpful in healing pitta depression. See [chapter 7](#) for some hints on meditation.

KAPHA DEPRESSION

Kapha depression creates a sense of mental heaviness and is associated with excess sleep, weight gain, drowsiness, and lousiness! The following natural remedies may bring great relief from kapha depression:

- Fast for 3 to 4 days on apple juice. This will work wonders to lighten the heaviness of a kapha depression.

- Increase the amount of exercise you do.

- Drink ginger tea (½ to 1 teaspoon of ginger powder steeped in hot water) twice a day.

TIME TO SEE THE DOCTOR

If your depression doesn't soon begin to lift as a result of using these Ayurvedic home remedies, please consult a doctor.

- Make the following herbal compound:

sarasvati 2 parts

punarnava 3 parts

chitrak 3 parts

Take this mixture 3 times a day. Put ½ teaspoon of the powder on your tongue, and wash it down with warm water.

- Put 5 drops of *punarnava* ghee in each nostril twice a day. (See [appendix 2](#) for instructions on how to prepare your own medicated ghees and oils.)
- The Sun Salutation (12 repetitions a day), Shoulder Stand, and Plow pose are recommended yoga *asanas* for depression. Also do the Maha Mudra. Other recommended postures are the Bow pose and the Vajrasana (Sitting on the Heels). (Illustrations of yoga *asanas* appear in [appendix 4](#).)
- The breathing exercise known as *ujjayi pranayama* is also beneficial for healing kapha depression. (See instructions in [chapter 6](#).)

Diabetes

Diabetes is a metabolic kapha type of disorder in which diminished functioning of agni (digestive fire) leads to a tendency toward high blood sugar. To control high blood sugar, Ayurveda uses the following herbal mixture:

guduchi 1 part

shardunika 1 part

kutki 1 part

punarnava 2 parts

Take ½ teaspoon 2 or 3 times a day with warm water.

• Another simple and effective herbal way to control blood sugar is to use turmeric. Fill some 00-size capsules (available at a pharmacy or natural food store) with turmeric, and take 2 capsules 3 times a day, a few minutes before meals. You can continue this program for up to a month, and then reevaluate your condition. Clinical observation suggests that a person who is insulin dependent will experience a markedly diminished requirement for insulin; the diabetes can often be brought under control.

• To help regulate your blood sugar level, you can try taking ½ teaspoon of ground bay leaf and ½ teaspoon turmeric, mixed in 1 tablespoon aloe vera gel. Take the mixture twice a day before lunch and dinner.

DIET. To reduce kapha, you should follow the kapha-pacifying diet (see [chapter 8](#)), especially avoiding excess intake of sweets, carbohydrates, and dairy products. Take more fresh vegetables and bitter herbs.

COPPER WATER. Put one cup of water into a copper vessel at night, and drink the water in the morning.

YOGA POSTURES. Beneficial yoga postures for diabetic conditions include the Sun

Salutation and the Peacock pose, Locust pose, Leg Lift, and Chest-Knee pose. Alternate Nostril breathing is also helpful. (Please see illustrations for yoga postures in [appendix 4](#), and the *pranayama* instructions in [chapter 6](#).)

❧ ❧ Diarrhea ❧ ❧

See also “[Diarrhea—Babies](#)”

Generally, diarrhea occurs when agni (the digestive fire) becomes weak. As a result, absorption and assimilation become minimal, and the undigested foodstuff gets eliminated as a liquid, watery stool. To relieve diarrhea, Ayurveda aims to strengthen agni and to pacify whatever dosha—generally pitta—is aggravated.

Indigestion, nervousness, or eating some wrong food or food combination can also create diarrhea.

PACIFY PITTA

The first line of defense against diarrhea is to immediately reduce pitta in your diet. Follow the guidelines for the pitta-pacifying diet (see [chapter 8](#)), especially keeping away from spicy and fermented foods.

TIME TO SEE THE DOCTOR

If there is no serious illness, diarrhea is usually quite easy to control. However, this condition can also be a symptom of a more serious illness, so if these home remedies don't work in 2 or 3 days, consult a doctor.

FOUR FOOD CURES

- Cook one or two apples until they are mushy, then add 1 teaspoon ghee, a pinch of cardamom, and a pinch of nutmeg. Eat slowly. This mixture is not only delicious, it will also help stop diarrhea right away.
- If you don't have apples, you can use bananas, only you don't have to cook them. Chop 1 or 2 ripe bananas into pieces, and as with the apples, add 1 teaspoon of warm ghee and a pinch each of cardamom and nutmeg. Bananas are high in potassium, which helps to bind the stool.
- Another effective remedy for diarrhea is cooked rice with yogurt. Take about a cupful of cooked basmati rice, add 1 tablespoon of ghee and 3 or 4 tablespoons of plain fresh yogurt, stir together, and eat.
- Another remedy using yogurt is to blend together equal parts of yogurt and water (about ½ cup each), add about ⅛ teaspoon of fresh grated ginger, and drink.

HERBAL REMEDIES

- Try ginger powder (about $\frac{1}{2}$ teaspoon) with 1 teaspoon raw natural sugar. Mix together, and chew the mixture with some warm water. Take 2 or 3 times a day for 2 to 3 days.

- Another simple remedy is the following:

ghee 1 teaspoon

nutmeg $\frac{1}{4}$ teaspoon

ginger powder $\frac{1}{4}$ teaspoon

natural sugar 1 teaspoon

Mix together and eat. Like the ginger-sugar mixture above, take this formula 2 or 3 times a day for 2 to 3 days.

- For acute diarrhea, mix $\frac{1}{2}$ teaspoon fennel powder with $\frac{1}{2}$ teaspoon ginger powder, and chew this mixture 2 or 3 times a day.

- Try drinking a cup of hot black coffee with a little lime juice (about 10 drops) and a pinch of cardamom or nutmeg.

- If your diarrhea appears to be the result of high pitta, use this herbal formula:

shatavari $\frac{1}{2}$ teaspoon

arrowroot $\frac{1}{2}$ teaspoon

Mix together, and take with $\frac{1}{2}$ cup warm water 2 or 3 times a day.

- You can also use *sat isabgol* (psyllium husks) to combat pitta diarrhea. This may sound strange, as psyllium is often used as a laxative. However, in pitta diarrhea, an excess of pitta accumulates in the gastrointestinal tract and irritates the wall of the colon, creating the diarrhea. So at bedtime, take 1 teaspoon *sat isabgol* mixed into a cup of fresh yogurt. That will absorb pitta and bind the stool, helping to correct the diarrhea. Be sure the yogurt is freshly made, not old.

AVOID DEHYDRATION

Diarrhea can sometimes cause dehydration. To prevent this from happening, add 1 teaspoon of natural sugar, 1 teaspoon of lime juice, and a pinch of salt to a pint of room-temperature water, and sip throughout the day.

NOTE: If the diarrhea continues beyond three days, it would be wise to consult a doctor.

Diarrhea—Babies

Diarrhea in the babies of nursing mothers may be due to the mother's diet. For example, if the mother eats stale and leftover food, or food that is heavy to digest, then her milk will become harder for the child to digest. As a general rule, Ayurveda suggests that when babies six months old or younger get diarrhea, the mother should follow a pitta-pacifying diet.

AN APPLE A DAY ... A baby's diarrhea can usually be corrected simply by feeding the child some cooked apple. Remove the skin and seeds, cook the apple, and add ½ teaspoon ghee, a pinch of cardamom, and a small pinch of ginger. Stir it up well, and cool to room temperature.

SUGGESTION: To feed this applesauce to your child, use a standard baby bottle. With a clean scissors, cut off the tip of the nipple, making an opening large enough for the applesauce to flow through.

MAKE WHEY. Whey is a natural source of lactobacillus. Since a baby's diarrhea is often due to altered flora in the gastrointestinal tract, the whey should help restore the normal flora. It is also a good source of potassium and of calcium, which binds the stool.

Bring 1 cup of milk to a boil; when it just begins to boil, squeeze a little lime juice (about ½ teaspoon) into the pot. The milk will curdle. Then strain the thick part out by pouring the mixture through cheesecloth or a sieve.

The remaining watery part is the whey. Giving 4 to 5 teaspoons of the whey to the child every 10 to 15 minutes should stop the diarrhea.

(The thick, solid part is a delicious fresh cheese called *paneer*, much used in Indian cooking. Try it!)

A SIMPLE HERBAL REMEDY. Another simple domestic remedy is to mix 1 teaspoon honey, ½ teaspoon ghee, a pinch of nutmeg, and a pinch of saffron.

POPPYSEED PORRIDGE. You can also make your child a porridge of poppy seeds. Bring ½ cup milk and ½ cup water to a boil, then add about 1 teaspoon poppy seeds. The seeds will swell and become soft; they will make a nice, easily digested porridge. It is a good food that helps to stop diarrhea. As a bonus, your child will also sleep soundly.

TIME TO SEE THE DOCTOR

If your infant has diarrhea, watch him or her carefully. If bowel movements are becoming less frequent and are firmer and denser, your child is getting better. But if the stools continue to be watery and occur several times in a day, and if your baby's eyes appear sunken, the lips are dry, and she or he looks drowsy, these are all warning signs of dehydration. This is a serious condition, and you need to consult a doctor.

CALCIUM SUPPLEMENT. Babies may also get diarrhea when they are teething. Teeth are a by-product of *asthi dhatu*, or bone formation. When babies start growing teeth, especially the incisors or canine teeth, vata in the *asthi dhatu* gets provoked, their digestive fire becomes weak, and they may get diarrhea. At that time, they need more calcium. You can give the child a simple calcium supplement in the correct dose for children.

❧ ❧ Dizziness ❧ ❧

See “[Fainting and Dizziness](#)”

❧ ❧ Dry Skin ❧ ❧

See also “[Skin—Ayurvedic Care](#)”

Dry skin can have several causes. It may be due to a lack of sebaceous (oily) secretions; insufficient sweating; an excess of hot, sharp pitta; or too much vata. External causes of dry skin include sun, wind, hot dry air, excess washing, and excess use of soap or dishwashing soap.

MOISTURIZING CREAM MAY NOT BE THE ANSWER. Many people use moisturizing creams to counteract dry skin. But dryness of the skin usually comes from within rather than from external causes. For this reason, mere application of a moisturizer doesn’t truly solve the problem.

In general, moisturizing creams work only temporarily. They stimulate secretions of the sebaceous glands, and for a while the skin looks soft and oily. But then the glands become tired and exhausted, resulting in more dryness. You will have much greater success if you treat the dryness both externally, with natural oils, and internally, by lubrication of the colon with oil enemas.

Here are several effective Ayurvedic home remedies to keep your skin smooth and lustrous.

APPLY SOME OIL. In some cases, applying some oil to your skin may be all you need to eliminate dry skin. If your constitution is predominantly vata, apply sesame oil; if you are pitta, apply sunflower or coconut oil. If you are kapha, apply corn oil.

However, application of oil to the skin will probably not be sufficient. To treat the internal cause of the dryness, you’ll have to lubricate the colon with a gentle oil enema.

OIL ENEMA. Here is the procedure:

- *Step 1:* Begin with a cleansing enema. Either in the morning or evening, give yourself a regular water enema. After a good bowel movement, wait one hour before proceeding to step 2.

• **Step 2:** For the oil enema, use 1 cup of sesame, sunflower, or corn oil, according to the above vata–pitta–kapha recommendations. Inject the oil into the rectum (using a hot water bottle or a syringe), and try to retain it for 5 to 10 minutes. If it goes out, don't worry, let it out. (More complete guidelines on how to do an enema are in [appendix 3](#).)

This simple oil enema will make your skin soft, delicate, and gorgeous. The colon is an important part of the absorption process for food nutrients. Similarly, the oil in the colon from the enema easily gets absorbed into the system and helps to lubricate the skin from within.

For best results, perform the oil enema according to this schedule:

Week 1 Every day

Week 2 Every 2 days

Week 3 Every 3 days

Week 4 Once

CHERRY MASK. You may also find relief from dry skin by applying a pulp of fresh cherries to your face at night, before going to bed. Leave it on for 15 minutes, and then rinse off. This will give you a beautiful complexion.

❧ ❧ Earache ❧ ❧

If the ear passage gets dried and crusty due to excess vata, it may start aching. The eardrum may even become tight and painful.

However, before treating earache, it is important to first rule out several possibilities, such as infection (*otitis externa* or *otitis interna*), perforated eardrum, or excess wax causing pressure on the ear (see "[Earwax](#)"). Having ruled these out, one can then treat the vata problem.

TEA TREE OIL. Begin by pulling down on the lobe of your painful ear. If this hurts, it means there is *otitis externa*, external ear infection. To heal the infection, take a cotton swab and dip it into tea tree oil, a wonderful natural disinfectant widely available at natural food stores and some pharmacies. Then apply the oil to the ear with the swab.

NOTE: Using plain tea tree oil may create a burning sensation on sensitive skin, so it is usually best to dilute it, using 10 to 20 drops of tea tree oil mixed in 1 ounce of sesame oil.

TEA TREE OIL WITH NEEM. For an even more effective treatment, combine the tea tree oil with *neem* oil. Here again, don't use pure *neem* extract. Mix 10 to 20 drops of *neem* in sesame oil, then add it to the tea tree oil. Gently apply a few drops of the oil mixture to the ear passage.

ANTIBIOTIC HERBS. At the same time you are treating the infection on the outside with tea tree oil, internally you can take turmeric-echinacea-goldenseal tea:

turmeric 1 part

goldenseal 1 part

echinacea 1 part

Stir ½ teaspoon of this mixture into hot water, steep for a few minutes, and drink. Alternatively, simply swallow ½ teaspoon of the powder mixed in 1 teaspoon of honey. Take 3 times a day after food, for 1 week. This powerful antiseptic, antibiotic formula will help control the ear infection.

ASAFETIDA. You can also take a small amount of cotton, put a pinch of asafetida into it, and roll it into a capsulelike shape. Place that ball of cotton into the outer ear. The fumes of the asafetida will quickly relieve ear pain.

ONION JUICE. Combine 1 teaspoon fresh onion juice with ½ teaspoon honey. Mix well; introduce 5 to 10 drops into the affected ear. The mixture should be room temperature or a little warmer before you put it in your ear.

HEAT. Earache can also be moderated by heat. Take a handkerchief and put it on a warm (not hot) pan, fold it, and place it on the ear to give a little soothing external heat.

❧ ❧ Ears, Ringing ❧ ❧

See “*Tinnitus*”

❧ ❧ Earwax ❧ ❧

Earwax is one of the natural secretions of the body. It has the function of protecting the eardrum from dust, and it also keeps the ear canal lubricated. But because it is constantly exposed to the air, the wax may accumulate dust and dirt and become tarry black or brown-black and thick, packed tightly into the ear canal. It may obstruct hearing or create uncomfortable pressure, and so it has to be periodically removed.

WASH IT OUT. Gentle irrigation of the ear canal with warm water will usually take care of it. You can pick up an ear syringe in a pharmacy quite inexpensively. Prepare a pint of warm water (body temperature), add about ½ teaspoon of baking soda, and use that water to clean the ear. Follow the directions that come with the syringe, which essentially are to hold the syringe at the edge of the ear canal (do not insert it; the water has to be free to come out again), and squirt the water *gently* into the ear. You will want to tilt your head toward the shoulder of the ear you are cleaning and hold your head

over a basin or over the sink. Do a final rinse with plain warm water.

After irrigating the ear, it is important to dry the ear thoroughly. A good way is to dip a cotton swab into some rubbing alcohol, and wipe the ear with the swab.

SOFTEN IT WITH OIL. Sometimes the wax is stubborn and doesn't come out easily. In that case, for a day or two before doing the irrigation, soften the wax in the ear canal by lubricating it with warm garlic oil. Take about 1 tablespoon of sesame oil, add ½ clove of freshly chopped garlic, and boil it until the garlic pieces turn brown. It will have a pleasant smell. Then press the oil from the garlic, and strain it into a jar or other container. (Plain sesame oil will also work, but the garlic oil will be more effective.)

Then, 2 or 3 times a day, put a few drops of that oil—when it is warm, not hot—into the ear. That will lubricate the ear passage and soften the wax for easy removal.

DISSOLVE IT WITH HYDROGEN PEROXIDE. You can also use a mild hydrogen peroxide solution (about a 3 percent solution) to dissolve the wax. You can buy this solution in most pharmacies. Put a few drops in the ear. It will oxygenate the ear, and the wax will simply dissolve. When you're finished, clean the ear with some warm sesame oil.

MESSAGE TO LOOSEN STUBBORN WAX. For stubborn earwax, massage the outside of the ear. Put a little sesame oil or castor oil on the mastoid bone (the bone behind the ear), and gently massage. Pull the ear lobe, and at the same time open your mouth. This will help to loosen the ear wax, and it will come out quite easily when you irrigate with water.

HERBAL REMEDY. The root cause of excessive earwax production is increased vata in the *mamsa dhatu* (muscle tissue). To deal with this, take *triphala guggulu* tablets, 200 mg. twice a day, for one month. That will definitely break down the body's habit of forming excess earwax.

PREVENTION. Here is another excellent way to prevent the buildup of earwax. Once a month, sleep on your left side. Fill your right ear with warm sesame oil, and go to bed. (Put an old towel on the pillow to catch the excess oil.) The whole night, the ear passage will be soaked in the sesame oil. The wax will rise up toward the surface of the ear canal, and you can clean it out in the morning with a dry cotton swab. (Don't stay awake all night trying not to turn over; just sleep comfortably. Even a couple of hours with the oil filling the canal will do the trick.)

The next night, sleep on the other side and treat the other ear in the same way. This way you can avoid the tendency for excess earwax to form.

❧❧ Eating Disorders ❧❧

See also "[Overeating](#)" and "[Obesity](#)"

NOTE: The Ayurvedic recommendations in this and related sections can help

you deal with eating disorders, but to fully heal them you may also require psychiatric counseling.

KINDS OF EATING DISORDERS

BULIMIA AND BINGE EATING. Almost everyone does some extra munching now and then. But if you often continue eating even though you are full, you might be a victim of binge eating disorder or bulimia.

Individuals with bulimia eat to excess and then induce vomiting to avoid gaining weight, while those with binge eating disorder overeat but don't induce vomiting. Binge eating may lead to obesity, and bulimia may lead to metabolic disorders.

OVEREATING. Overeating of all kinds is most often the result of psychological and emotional factors, such as low self-esteem, anxiety, grief, and sadness. To compensate for those emotions, people go on eating.

ANOREXIA. Anorexia nervosa is a serious problem, usually among young women who have a fear of being fat and purposefully starve themselves. The root cause of anorexia is often depression.

SUGGESTED REMEDIES

Here are some brief suggestions to help you deal effectively with these eating disorders. For more complete treatments, please see "[Overeating](#)" and "[Obesity](#)."

FOLLOW A LOW-FAT DIET. Avoid fatty fried food, cheese, yogurt, excess carbohydrates, ice cream. These kapha-inducing substances lead to weight gain and possibly to obesity.

EAT HOT SPICY FOOD. When cooking, add garam masala, chili pepper, cayenne pepper, curry pepper, turmeric, cumin, and the like to your food. These spices will burn the ama that causes overeating.

EAT TWO OR THREE MEALS A DAY. That's all. Eat breakfast, lunch, and dinner, but skip between-meal snacking. While eating, play some soft, gentle music such as classical Indian music; choose a gentle, loving raga. Don't listen to jazz or rock; that loud music will overstimulate the system and cause you to eat more.

LICORICE. Whenever you desire to munch, eat licorice candy, which is a mild diuretic that acts to reduce kapha. Or eat a handful of raisins.

HERBAL TEA. To help heal the emotional factor in all eating disorders, drink herbal tea. Excellent choices are chamomile, comfrey, *brahmi*, or *jatamamsi*. These are good individually, but a tea made from all of these mixed together in equal amounts is especially effective to reduce stress and balance the emotions. Use ½ to 1 teaspoon of

herbs steeped in boiling water for 10 minutes.

NOTE: If there is depression, *jatamamsi* or *brahmi* tea will help, but it may also be necessary to see a psychiatrist or other mental health professional.

OIL MASSAGE. Rub some *bhringaraj* oil on the soles of the feet and on the scalp at night. That will help you relax and sleep. Lightly massaging with a little of the oil in the morning will also help you deal with the stress.

YOGA POSTURES. Good yoga *asanas* for eating disorders include the Bow, Boat, Peacock, and Rooster poses. The Lion pose is also quite effective for reducing stress (see [appendix 4](#)).

BREATHING AND MEDITATION. *Ujjayi pranayama* and *So-Hum* meditation will also be helpful (see [chapters 6](#) and [7](#)).

❧ ❧ Edema ❧ ❧

See also “[Swelling During Pregnancy](#)”

Suddenly one morning a person may wake up with swollen eyes, a swollen foot, a swollen nose, or a swollen toe—any part of the body can swell. There may be associated symptoms such as pain or itching. Sometimes the edema may be related to an injury, such as a bump, or it may be due to torn ligaments. Or fluid may seep from the blood vessels due to prolonged standing or walking. Because of poor circulation, a person’s feet or ankles may swell. Edema may also be an allergic reaction, or the result of an insect bite.

Because there are so many possible reasons for the swelling, finding out the cause is essential for maximally effective treatment. Nevertheless, the following recommendations should prove helpful.

APPLY A HEALING PASTE. At the site of the swelling, apply a paste made of turmeric and red sandalwood. Mix an equal amount of the herbal powders together, add sufficient water to make a paste, and apply.

NOTE: Don’t get this mixture of turmeric and sandalwood in your eyes; it may irritate the eye and create conjunctivitis.

GIVE YOUR FEET A RAISE. If the edema is in your feet, raise the feet. Sit in a comfortable chair with your legs resting on a footstool, or use a small table and place a couple of pillows under your feet. When you go to bed, place pillows under your feet. This will gradually drain the excess water and relieve the swelling.

FOR INSECT BITES. For swelling due to insect bites, use *neem* and tea tree oil topically. The venom of the insect irritates the skin, and *neem* and tea tree oil mixed together in equal

proportion will neutralize the venom toxicity and minimize the edema. *Neem* oil is also effective by itself. (See “[Bites and Stings](#)” for further suggestions.)

NATURAL ANTIHISTAMINE FOR ALLERGIES. If the swelling is due to allergy, take antihistamine in the form of fresh cilantro juice. Chop some fresh cilantro and place in the blender with $\frac{1}{3}$ cup of water. Blend, strain, and drink immediately. Also apply the pulp locally to the swollen area. (See “[Allergies](#)” for many more suggestions.)

FINGER RING SWELLING. Swelling on a finger may be due to a too-tight ring. Simply remove the ring. If it will not easily slip off, raise your hand high above your head or soak your hand in ice water for a few minutes, then use some soap or oil to make your finger slippery. If this does not work, the ring may have to be cut. That will release the blood flow and relieve the edema.

HERBS TO IMPROVE CIRCULATION. Poor circulation can lead to edema. To increase circulation, mix:

punarnava 5 parts

manjistha 3 parts

gokshura 3 parts

Take $\frac{1}{2}$ teaspoon twice a day with warm water, after meals.

LOCAL EXERCISE TO IMPROVE CIRCULATION. In addition to the above herbal formula, giving some exercise to the part of the body that is swollen should improve circulation and reduce the swelling. Fill a pan or bucket with hot water, and steep a “teabag” of mustard seeds in it. Make the bag by wrapping 2 tablespoons of mustard seeds in a handkerchief or some cheesecloth. Put it into the bucket you’re using to soak your ankle, finger, or whatever part is swollen. While you soak, flex the swollen part and do some underwater exercises to increase circulation.

REMEDIES FOR LOCALIZED SWELLING

- *For swelling of the nose*, do *nasya* using *brahmi* oil or plain ghee (see [appendix 3](#)).
- *For swelling of the eyes*, apply a few drops of pure rose water. You can generally buy a 3 percent solution of rose water; to make it at home, use organic roses and soak both petals and the hip in distilled water for several hours, then strain and use the water.
- *For swollen toes*, apply a paste of turmeric and red sandalwood.
- *For a torn ligament*, apply some *mahanarayan* oil topically. Internally, take *kaishore guggulu*, 200 mg. twice a day.
- *For facial swelling*. Some parasites, such as amoebas, giardia, and pinworms, can create swelling of the face. If you have determined that this is the cause of your facial swelling, the following formula will be effective:

vidanga 3 parts

neem 3 parts

shardunika 3 parts

Take ½ teaspoon twice a day after lunch and dinner.

☞ ☞ Eye Irritation ☞ ☞

See also “[Eyes—Ayurvedic Care](#)”

Around the age of 40, *alochaka* pitta (a subtype of pitta associated with vision) tends to become low, and people frequently find that their eyes become irritated. Individuals who work in front of a computer for a long time, watch a lot of TV or movies, or do a lot of driving or close-up work, or who live in a city with high air pollution, may find that their eyes become irritated and build up quite a lot of strain.

The irritation may be due to dryness of the conjunctiva (which results from excess vata), or it may be related to excess hyperacidity or pitta in the stomach. Or, because the liver and eyes are closely related, high or stagnant pitta in the liver may be a causal factor. It is important to pin down the cause in order to treat it with maximum effectiveness. But most of the following remedies will be helpful in all situations.

TAKE A BREAK. If you are using your eyes a lot, perhaps in front of a computer or driving, stop every hour or two for a few minutes to give your eyes a rest.

USE A PROTECTIVE SCREEN. Use a protective screen on your video terminal, to cut radiation and glare.

CHANGE YOUR FOCUS. If you are doing a lot of reading, stop for a minute or two and refocus on a distant object. Every half hour, close the book and look at something across the room or out the window. Doing something different with the eyes is restful and will help prevent eye irritation.

TAKE A TEA BREAK. In the midst of a lot of close-up work or sitting at the computer, stop for a few minutes and have a cup of tea: chamomile, comfrey, mint, or even *chai* (which has some caffeine and will not be as soothing as the other three). This will be relaxing and will help to relieve the irritation and strain.

ROSE WATER EYE DROPS. You can also prepare a solution of rose water. Take 1 ounce of distilled or purified water, add 5 drops of pure rose water, and use the solution to rinse your eyes. Use a dropper or an eye cup, and be sure the water is neither too cold nor too hot. This solution will immediately soothe any eye irritation.

SPLASH YOUR EYES WITH WATER. Simply washing the eyes with clean water will also help. As if rinsing your face after washing, splash some water onto your eyes, opening the eyes for

a moment to allow some water to go in. Or you can use an eye cup and rinse the eyes.

THE SIMPLEST AND BEST REMEDY FOR IRRITATED EYES. Put a single drop of pure, genuine castor oil (without preservatives) into your eyes at bedtime. Rubbing a little castor oil on the soles of your feet at bedtime also helps; remember to wear some old socks to prevent the oil from staining your sheets.

FOR BLOODSHOT EYES. If your eyes are irritated and bloodshot, drink a cup of fresh orange juice with ½ teaspoon of natural sugar and a pinch of cumin.

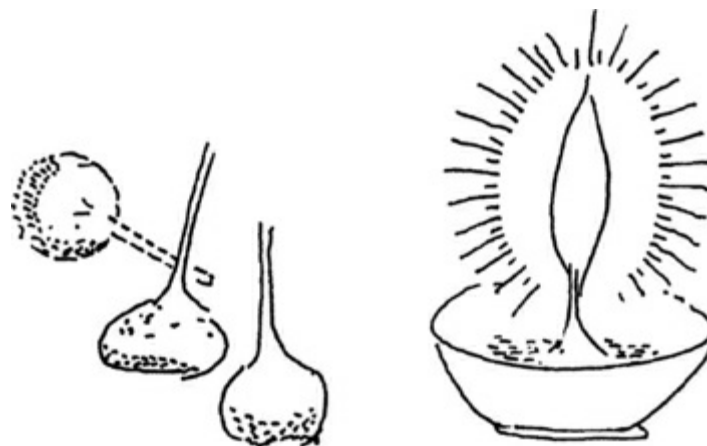
TO PACIFY ACIDITY. If excess acid and high pitta in your system seems to be the cause of the eye irritation, take some *shatavari* (½ teaspoon), add just a pinch of *shanka bhasma*, and take it twice a day for 2 weeks, with a little warm water.

- Aloe vera gel (2 tablespoons 3 times a day) also pacifies acidity.
- If you determine that high pitta might be the cause of the eye irritation, make a mixture of the herbs *shanka pushpi* and *jatamamsi* in equal proportions. Take ½ teaspoon twice a day for 2 weeks.

❧ ❧ Eyes-Ayurvedic Care ❧ ❧

Ayurveda has a number of excellent suggestions for maintaining your eyes in strength and health.

COOL WATER WASH. Early in the morning, when you wash your face, fill your mouth with cool water and hold it there; then splash cool water over your open eyes. Traditionally, you're not supposed to swallow or spit out the water, but keep it in your mouth as you sprinkle the cool water on your eyes. This has a double cooling effect, both from outside and from the oral cavity, that will make your eyes feel fresh, happy, and cheerful. (If you prefer, you can use an eye cup instead.)



Cotton Wick

Ghee Lamp

GHEE LAMP. A traditional Ayurvedic way to strengthen and soothe the eyes is to gaze at the flame of a ghee lamp. Prepare a ghee lamp by taking a small bowl, placing a cotton

wick in it, and adding ghee. Be sure the wick is made of genuine cotton; synthetic cotton will burn up in an instant. Also be sure the wick is not too thick. Apply a little ghee to the tip of the wick and then light it.

Ayurvedic Eye Exercises

The following exercises will strengthen your eyes.

- First, blink the eyes rapidly several times. Then, with eyes open, move your eyes in this pattern:

up and down

side to side

diagonally from top left to bottom right

diagonally from top right to bottom left

clockwise in a circle

counterclockwise in a circle

- Hold your arm outstretched in front of you. Look at the tip of your index finger, and gradually bring the finger close to you, following it with your eyes, until it touches your “third eye,” between the eyebrows.

- Look at the tip of your nose, and then up toward your “third eye.”

- Finally, close your eyes tightly, then release. These exercises are beneficial for improving the circulation of the eccentric muscles of the eyeball.

After finishing the exercises, or anytime your eyes feel tired or strained, rub your palms vigorously together for a few seconds to generate some warmth, then place them *lightly* over your eyes. Feel how soothing the warmth is! Keep them there for a minute or two to strengthen and nourish your eyes.

Set the ghee lamp at a distance of 2 or 3 feet from you, remove your glasses, and gaze at the flame for 2 or 3 minutes *without blinking*. This procedure will improve the *tejas* or lustrous quality of the eyes.

SALUTE THE SUN. Another effective way to keep eyes strong is to do the Sun Salutation exercise. Performing 12 sets of Sun Salutations is one of the best exercises for the entire body, including the eyes (see [appendix 4](#)).

FOR PITTA TYPES

The following six remedies are especially helpful for pitta types, who are more likely

to find their eyes burning.

TRIPHALA EYEWASH. Boil 1 teaspoon of *triphalā* in a cup of water for 3 minutes. Cool the tea, and strain it with double- or triple-folded cheesecloth or a paper coffee filter so that no particle of *triphalā* is left in the tea. Then wash your eyes with the tea.

CASTOR OIL REMEDY. At bedtime, put 1 drop of pure castor oil, with no preservatives, into each eye. Also rub 1 teaspoon of the oil into the soles of your feet. Next morning your eyes will feel really happy and fresh!

TO SOOTHE BURNING EYES. For a burning sensation in the eyes, put 1 drop of lukewarm liquid ghee in each eye at bedtime. That will lubricate the eyelid and eyelashes and will also soothe and strengthen your eyes.

ROSE WATER. You can also put 3 drops of pure rose water into each affected eye. Rose water is cooling.

A HEALING SALVE. Collyrium is effective for soothing your eyes. (It is available in most Indian grocery stores. Ask for *kajal*.) It is made of castor oil and natural camphor. It is black and people use it cosmetically as an eye liner, but it is actually medicinal and good for the eyes. Scoop out a small amount of salve with your little finger. Make sure the nail is well trimmed. With the other hand, pull the lower eyelid down while looking in the mirror and with your little finger apply the salve sparingly to the inner edge of the eyelid.

STRESS REDUCER. For eyestrain, Ayurveda also suggests taking a sterilized cotton ball or a piece of gauze, dipping it into cool goat's milk, and putting it over your closed eyes. This will ease stress and make your eyes feel better. (If goat's milk is not available, you may use cow's milk, but goat's milk is preferable.)

ADDITIONAL SUGGESTIONS

READ SITTING UP. While reading, keep your vertebral column straight. Avoid reading while lying down.

USE A NONGLARE COMPUTER SCREEN. If you are using a computer, be sure it has a nonglare screen, or use a protecting filter to reduce glare so it won't hurt your eyes.

❧❧ Eyestrain ❧❧

See “[Eyes—Ayurvedic Care](#)” and “[Eye Irritation](#)”

❧❧ Fainting and Dizziness ❧❧

Increased vata dosha, along with excess pitta moving in the nervous system, cause fainting and dizziness. There are two types of dizziness:

In the first kind, one feels that external objects in the environment are moving: “I am still, but the room is moving around me.” That is called *objective dizziness*, and it is primarily due to aggravated vata.

The second type is called *subjective dizziness*. Here, one has the feeling of swinging or disequilibrium, of being subjectively in motion. This type of dizziness is caused primarily by excess pitta. The difference between the two is subtle and not easy to pinpoint, especially since vata and pitta are both always involved. But it is helpful to distinguish between the two, as some aspects of treatment will be different.

TIME TO SEE THE DOCTOR

The remedies in this section are simple, effective means of relieving dizziness. But if dizziness or fainting persists after using them, it would be wise to see a neurologist or another medical expert, because the dizziness might be due to some serious pathology.

Vertigo, a spinning, merry-go-round kind of feeling, may be related to infection in the inner ear, head injury, or viral infection. It may be due to Ménière’s disease, a condition in which increased pressure in the middle ear generates a feeling of vertigo. Ménière’s disease will result in progressive hearing loss if it’s not treated. All these conditions have to be suspected and ruled out before undertaking home treatment for dizziness. However, the domestic remedies below will help most cases of dizziness.

Dizziness or feeling faint may also occur when there is insufficient oxygen in the room.

If you feel dizzy and you’re not sure whether you’re spinning or the room is moving, focus on a fixed point, such as a window frame. When the eyeball becomes steady by focusing on a stable point, it sends a message to the brain that calms down the *rajasic* quality of vata dosha. This works well for objective dizziness.

The kind of dizziness known as *motion sickness* is often associated with nausea and vomiting. To help with motion sickness, before starting the trip, take one 00-size capsule filled with ginger powder. Bring some more along on the trip. You might also bring some candied ginger and chew on a piece of it from time to time. If you are in a car and you feel that everything seems to be in motion, try to look at a fixed point, such as the horizon or a distant unmoving object. This should help to stop the dizziness.

Sometimes people feel dizzy when they stand up quickly. There are many possible causes for this, including low adrenal energy, low blood pressure, or the use of some antihypertensive drugs, especially beta blockers, which may weaken the adrenals. If you find yourself getting dizzy upon standing up, simply get up *slowly*. This will help.

Pitta individuals with hypoglycemia may experience sweating and dizziness and may

even verge on fainting if they go too long without eating. This is a form of subjective dizziness. Hypoglycemics have to eat on time, or they may even become unconscious. Drinking some apple juice or any sweet fruit juice will be effective.

QUICK REMEDY FOR FAINTING. If a person faints, collapses, and becomes unconscious, sprinkle some cold water on the face.

DIZZINESS FROM EXERCISING. Individuals who exercise vigorously may become dehydrated from sweating a lot, which may bring on dizziness. The cure for this is about as simple as it gets: Drink some water. Even just one glass of cool water will help to reduce pitta as well as correct the dehydration, and the dizziness will subside.

DEEP BREATHING. Another simple remedy for dizziness is breathing deeply, as in *ujjayi pranayama* (see [chapter 6](#)) and holding the breath behind the belly button.

EAR PULL. Here's another simple remedy: Insert your index finger into your ear and gently pull the ear upward, forward, and downward. This will regulate intracranial pressure, which will greatly if not entirely relieve the feeling of faintness or dizziness.

NATURAL SMELLING SALTS. For dizziness or when feeling faint, slice or chop up an onion and inhale the smell forcibly until tears come to your eyes. Onions contain a lot of ammonia; inhaling them produces vasodilation, which brings more blood supply to the brain, and dizziness is automatically corrected.

HERBAL FORMULAS. If you determine that the dizziness is primarily due to pitta, a wonderful Ayurvedic herbal formula may prove helpful. Mix these herbs:

brahmi 1 part

jatamamsi 1 part

saraswati ¼ part

Take ½ teaspoon of the mixture 2 or 3 times a day with water after meals.

The following formula is also helpful when dizziness is due to excess pitta:

shatavari 1 part

jatamamsi 1 part

kama dudha 2 pinches

If you take ½ teaspoon of this mixture a couple of times a day with warm water, it will help take care of your dizziness.

SANDALWOOD. Pitta-type dizziness can also be helped by the smell of sandalwood oil or incense.

GHEE NOSE DROPS. Using ghee nose drops will also help. Warm up a little bit of ghee until it is liquefied. When cool enough not to injure delicate tissue, put 3 to 5 drops in each

nostril, and inhale. This will improve blood supply to the brain and will relieve the dizziness.

COLON CLEANSING. Sometimes dizzy spells are due to chronic constipation and gases. So keep the colon clean with *triphala*, ½ teaspoon taken at bedtime in a cupful of warm water.

❧ ❧ Fatigue and Chronic Fatigue ❧ ❧

Fatigue is physical and mental stress. However, it is not always due to overwork. In fact, sometimes people feel tired because they're not doing enough, not working hard enough. For such people, fatigue can be due to boredom or lack of motivation.

In such cases, I've had to ask patients to walk or to do some physical work in order to get rid of fatigue and increase their energy level. So the first thing to determine is whether the tiredness is due to too much physical work or too much idleness!

Fatigue may be due to low gastric fire, weakness of the liver, low adrenal energy, or anemia. It may be caused by Epstein-Barr virus, a form of chronic fatigue syndrome related to high stagnant pitta in the liver. People having a history of infectious mononucleosis can feel very tired.

Here are some treatment recommendations for fatigue of various causes:

FOR FATIGUE AFTER EXERCISE OR HARD PHYSICAL WORK. Drink a cup of fresh orange juice with a pinch of rock salt. Adding 10 drops of fresh lime juice will help pacify pitta.

- Drinking 1 cup of ginseng or *ashwagandha* tea once or twice a day will help.

FOR ANEMIA. If the fatigue is due to anemia, treat the anemia with blood builders such as pomegranate juice, grapes and/or grape juice, beets or beet/carrot juice, or the herbs *abrak bhasma* and *loha bhasma*. (See "[Anemia](#)" for many more recommendations.)

FOR EPSTEIN-BARR VIRUS. If the fatigue is due to Epstein-Barr virus, treat it as a pitta disorder.

- Follow the pitta-pacifying diet (see [chapter 8](#)).
- Use the following herbal formula:

shatavari 5 parts

bala 4 parts

vidari 3 parts

kama dudha ¼ part

Take ½ teaspoon of this mixture 2 or 3 times a day with 1 teaspoon of *shatavari* ghee. This will strengthen the liver and help to remedy chronic fatigue

syndrome (see the *shatavari* ghee recipe in [appendix 2](#)).

TO STRENGTHEN GASTRIC FIRE

When your agni (gastric fire) is low and your digestion slow, nourishment from the food you eat will be poorly absorbed and assimilated. If the nourishment you derive from eating is insufficient, naturally your energy will be low.

- One of the best ways to kindle agni is to use some ginger. Before each meal, chop or grate a little fresh ginger, add a few drops of lime juice and a pinch of salt, and chew it up. Or just cut a thin slice of fresh ginger, put on a pinch of salt, and chew that.
- Avoid cold and iced drinks, especially during or after meals. They counteract agni and impede effective digestion. Take small sips of warm water during the meal.
- Taking *chitrak-ativati*, one 200-mg. tablet twice a day, after lunch and dinner, will help kindle the fire and the fatigue will go.

TIME TO SEE THE DOCTOR

Unusual exertion, stress, lack of sleep, and a number of other factors can make you understandably tired. But if you use the Ayurvedic remedies recommended for fatigue for a few weeks and you still have unexplainable feelings of exhaustion or continue to feel unusually listless, lethargic, and drained of energy, your tiredness may be due to a more serious illness. Fatigue is a symptom of many illnesses, including anemia, lung disease, diabetes, hepatitis, mononucleosis, thyroid disease, and cancer. You may also have developed chronic fatigue syndrome. So please be wise and consult a physician.

- You will find many additional tips for strengthening the digestive fire in “Indigestion.”

YOGA POSTURES AND PRANAYAMA. Alternate Nostril Breathing and some gentle yoga stretching can be beneficial to kindle the gastric fire. See [appendix 4](#) and [chapter 6](#) for help with yoga postures and *pranayama*. In general, unless fatigue is due to idleness, Ayurveda does not recommend much exercise for cases of fatigue. Exercise will burn *ojas* and may increase rather than decrease the feelings of fatigue.

TO BUILD STRENGTH AND ENERGY

Use the following food remedies for added nourishment and strength.

DATES. Soak 10 fresh dates in a quart jar of ghee. Add 1 teaspoon ginger, $\frac{1}{8}$ teaspoon cardamom, and a pinch of saffron. Cover and keep in a warm place for at least 2 weeks. Then eat 1 date daily, in the early morning. Believe it or not, it tastes delicious, and it works to remedy anemia, sexual debility, and chronic fatigue.

A simpler tonic using dates is this date drink: Soak 5 fresh dates in a glass of water overnight. Next morning, liquefy in a blender and drink. It will give you energy and vitality. (Be sure to remove the pits before blending!)

MANGOES. Eating one ripe mango daily, and an hour or so later drinking 1 cup of warm milk with 1 teaspoon ghee added, is also good for increasing vitality.

A variation is to drink 1 cup of fresh mango juice, followed an hour or so later by ½ cup of warm milk with a pinch of cardamom, a pinch of nutmeg, and 1 teaspoon of ghee.

❧ ❧ Fever ❧ ❧

Fever is a sign of ama (toxins) moving in the circulatory system. Contrary to what some people believe, fever is not often a sign of infection. In some cases there is infection, but most of the time the fever is due to toxicity in the *rasa dhatu*, the body's basic vital tissue (see [this page](#)). When the ama has been eliminated, the fever will subside.

DON'T EAT. There is an old saying, “Feed a cold and starve a fever,” and truly the first treatment recommended by Ayurveda for a fever is to observe a fast. For acute fever, a total fast is recommended if the person is strong enough. If the person is debilitated or weak, it is better to drink water, some kind of fruit juice, or one of the herbal teas suggested below, such as holy basil (*tulsi*) or lemongrass. *Don't drink milk*; it will worsen the fever and create diarrhea.

In addition to fasting, the following recommendations will be helpful.

HERBAL REMEDIES

The simplest herbal remedy for fever is cilantro juice. Put a handful of cilantro leaves in a blender with about ⅓ cup water, and blend thoroughly. Strain out the pulp. Take 2 teaspoons of the remaining liquid 3 times a day to help lower the fever.

- You can make an effective herbal tea for fever consisting of

lemongrass tulsi (holy basil) fennel

in equal proportions. For each cup, use 1 teaspoon of the mixture and steep in boiling water for 10 minutes; strain and drink. This is an excellent diaphoretic; that is, it makes you sweat, which brings down the temperature.

TIME TO SEE THE DOCTOR

Fever is a sign that your body is fighting toxins and/or infection and is healing and purifying itself. It is usually self-limiting—that is, it will disappear

when the needed healing is done. But there are definitely times when fever requires the attention of a medical professional:

- any fever in a baby under 4 months old
- fever above 104 degrees Fahrenheit in an adult
- fever above 101 degrees in a person over age 60
- fever that lasts longer than 3 days
- fever accompanied by a severe headache and a stiff neck
- any fever in a person who has a chronic illness such as heart disease, diabetes, or respiratory disease

If any of these conditions are present, call a doctor or seek immediate attention.

- Another excellent herbal remedy for reducing fever is a tea made of

coriander 2 parts

cinnamon 2 parts

ginger 1 part

Steep 1 teaspoon of this mixture in a cup of hot water for 10 minutes before drinking. You can drink this every few hours until the fever breaks.

- Another simple three-ingredient tea made of household herbs is this one:

cumin seeds

coriander seeds

fennel seeds

Mix seeds in equal proportions. Use 1 teaspoon of your mixture in a cup of boiling water; steep for 10 minutes, strain, and drink.

OTHER REMEDIES AND RECOMMENDATIONS

GRAPE JUICE WITH HERBS. Grapes are cooling. Into a cup of grape juice add ½ teaspoon cumin, ½ teaspoon fennel, and ½ teaspoon sandalwood powder, and drink. This will help to relieve fever.

AVOID COLD DRINKS. When you have a fever, it's best not to drink anything cold. Use the lemongrass-tulsi-fennel tea mentioned above, or any of the other herbal suggestions. Any hot tea will help to kindle the body's digestive fire (agni) and burn the ama (toxins). Again, fever is a sign of ama in the system; once the ama is burned out, the body's temperature will come back to normal.

If the Fever Is High

If the temperature is high, prepare a bowl of cool water into which you have added 1 teaspoon of salt. Fold two pieces of clean cloth (such as handkerchiefs), dip them into the water, and put one on the forehead and one over the belly button. Repeat as needed. This will bring down the temperature quickly.

If you can get *maha sudarshan churna* from an Ayurvedic pharmacy, take ½ teaspoon twice a day with warm water. This formula will bring down any kind of fever.

If the person with high fever has a pitta constitution, there may be a danger of febrile convulsions. To help relieve this condition, make a slight modification of the procedure described above. Grate an onion and wrap half in one of the damp handkerchiefs, the other half in the second handkerchief. Place on the forehead and belly button, as above.

The belly is the seat of pitta, and the onion will help absorb the pitta. Tears will come to the eyes, the convulsions will stop, and the temperature will come down. If this procedure doesn't bring the fever down, see a doctor.

STAY PUT. It is not a good idea to exercise or to travel when you have a fever. If you do get a fever when on a journey, follow any of the above recommendations that you can; if possible, take *maha sudarshan churna* (½ teaspoon with a little warm water).

FOR CHRONIC FEVER. Make a tea of 1 teaspoon holy basil (*tulsi*) steeped in 1 cup hot water. Add ¼ teaspoon black pepper and 1 teaspoon honey. Take this 2 or 3 times a day.

- Another excellent formula to bring down chronic fever is ½ teaspoon *maha sudarshan churna* mixed with 1 teaspoon *tikta ghrita* (bitter ghee). Take this 3 times a day on an empty stomach.

❧❧ Fibrocystic Breast Disease ❧❧

According to Ayurvedic principles, fibrocystic breast disease is a kapha disorder. Excess kapha builds up, leading to congestion, enlargement of the breasts, tenderness, and development of fibrocystic tissue.

GENTLE MASSAGE. To help reduce the accumulation of kapha, apply 1 teaspoon of warm castor oil to the breasts, and gently massage from inside to outside, that is, from the breastbone toward the armpits. Do this gentle massage before taking a warm shower. Or you can do a soap massage of the breasts during your shower, again massaging from the center of your chest outward.

Massaging the breasts in this way will improve the circulation of the mastic tissue and encourage lymphatic drainage into the armpits; in this way, fibrocystic changes can

be minimized.

This massage will also take care of tenderness in the breasts. (See “[Breasts, Sore](#)” for more suggestions.)

EFFECTIVE HERBAL REMEDY. To help prevent fibrocystic breasts, use the following herbal formula:

kutki 2 parts
chitrak 2 parts
punarnava 5 parts

This herbal combination (½ teaspoon twice a day) will help to prevent the accumulation of kapha in the breast that leads to the development of fibrocystic tissue.

KAPHA-REDUCING DIET. As fibrocystic breasts are due to excess kapha, you will find it helpful to follow the kapha-reducing diet. Avoid dairy products, cold food and drink, heavy meats, wheat, and all sweets except honey. (See dietary guidelines in [chapter 8](#).)

YOGA POSTURES. Doing some yoga postures daily will be helpful. Include the Locust, Bow, Boat, Spinal Twist, and Shoulder Stand in your “routine” of postures. (See [appendix 4](#) for illustrations.)

❧ ❧ Food Allergies ❧ ❧

See also “[Allergies](#)”

To deal effectively with food allergies and minimize their negative impact on your life, begin by making a list of foods you appear to be allergic to. You will usually find, according to Ayurvedic dietetics, that you have reactions to foods with the same doshic quality as your constitution, particularly if you have an excess of that quality at the present time.

DETERMINING YOUR FOOD ALLERGY TYPE

The following descriptions should help you determine the type of food allergy you have.

- Kapha-type individuals, with excess kapha in their system, will be allergic to kapha foods, including dairy products such as milk, yogurt, and cheese; wheat; cucumber; and watermelon. Their food allergies may manifest as heaviness in the stomach, slow digestion, sleeping disorders, colds, congestion, cough, or water retention. In more serious conditions kapha food allergies may lead to bronchial congestion and kapha-type asthma.

- Pitta individuals, whose systemic pitta is unduly high, will show allergic reactions to such high-pitta foods as hot, spicy dishes; citrus fruits; sour fruits; tomatoes; potatoes; eggplant; and fermented foods. Their symptoms are likely to include heartburn, acid indigestion, stomach upset, nausea, and even vomiting. The person may suddenly get hot flashes, and may have bloodshot eyes.

- Vata individuals, especially when vata is in excess, are prone to get allergies to raw foods; beans (black beans, adzuki beans, pinto beans, and so on) and certain animal proteins. Vata food allergies are likely to manifest as bloating of the stomach, burping, gas, gurgling of the stomach, and vague abdominal discomfort and pain. It may also lead to insomnia and nightmares, pain in the joint, sciatica, muscle twitching, and muscle spasms.

So food allergies need to be studied according to an individual's *prakruti* (constitution) and *vikruti* (current status of the doshas).

TREATMENT

The best approach is simply to avoid the problematic food items and to follow the diet appropriate for your body type. For example, a vata person having vata-type food allergies should avoid a vata-provoking diet and should eat vata-pacifying foods. The same is true for pitta and kapha. (See the diet recommendations in [chapter 8](#).)

Here are some additional suggestions for each of the three main types of food allergy.

FOR VATA-TYPE FOOD ALLERGIES. An effective way to help bring vata food allergies under control is to take a *dashamoola basti* (enema) twice a week, such as on Sunday and Thursday. Boil up 1 pint of water with 1 tablespoon of *dashamoola*, and use the resulting tea (after it cools down) for the enema. Try to retain the liquid at least 10 minutes. (For more on enemas, see [appendix 3](#).) This herbal enema will help to pacify vata and remove toxicity from the colon, and vata-type food allergies can be minimized. Continue taking the *dashamoola* enema twice a week for one month.

Drinking licorice tea made from ½ teaspoon licorice root powder, ½ teaspoon honey, and 1 teaspoon ghee can also be helpful. Sip a little of the tea every half hour to an hour throughout the day until symptoms are relieved. Don't forget to add the honey only after the tea starts to cool; honey should never be cooked. *Note:* People with high blood pressure should not drink licorice tea. Substitute ½ teaspoon cinnamon and 2 to 3 cloves for the licorice, and make the tea as above.

FOR PITTA FOOD ALLERGIES. To control pitta food allergies, it is helpful to do *virechana chikitsa* (purgation). Take 1 teaspoon of *amalaki* or *sat isabgol* (psyllium husks) at night. Add the herbs to 1 cup of hot water, let it stand for 10 to 15 minutes, and drink. Stagnation of pitta in the small intestine is the root cause of a pitta-type food allergy. Purgation

therapy clears away the pitta so that the allergy can be brought under control. Continue with *virechana* daily for a month, or until the allergy subsides.

You will also find it helpful to eat cooked apples. Peel and core a couple of apples, cook them a few minutes to soften them, and mash them up with a pinch of cumin and 1 teaspoon ghee. Eat about ½ cup once or twice daily, at least an hour before or after meals.

FOR KAPHA FOOD ALLERGIES. The Ayurvedic purification known as *vamana* (vomiting) will be helpful. Every Sunday (or at least for a couple of weeks) in the early morning, drink 2 pints of salt water. Add 2 teaspoons of salt to 2 pints of water, drink it down, and then try to vomit it out. (Rubbing the back part of your tongue until you get a “gag” reflex will help.) This purification process will remove a lot of excess mucus from the stomach and help to clear up food allergies.

If vomiting doesn’t take place, don’t worry about it; the remaining salty water won’t hurt you. It will simply go through your system and will actually have some beneficial cleansing effect.

NOTE: Many people feel a lot of fear or discomfort with the idea or the actuality of vomiting. If you feel this way, do not force yourself to do this purification. Do *vamana* only if it is comfortable for you to do so.

Kaphas should also fast on Sunday. The fasting is important; it will help cleanse the system.

Make a tea out of ½ teaspoon licorice root powder, ¼ teaspoon cinnamon, and ½ teaspoon coriander. Steep in 1 cup of water for 10 minutes, strain, and sip a little of the tea throughout the day, about an ounce every half hour to an hour. Again, people with high blood pressure should not drink licorice tea. Substitute ½ teaspoon cinnamon and 2 to 3 cloves for the licorice.

FOR ALL FOOD ALLERGIES. If you don’t know whether the allergy is vata, pitta, or kapha, then try this simple remedy that is effective for all types: Roast some cumin, fennel, and white sesame seeds, and eat a handful after meals.

To prepare the mixture, take one ounce of each kind of seed, and dry-roast it *separately* (one type at a time) on a heavy cast-iron pan. You will have to stir the seeds constantly to keep them from burning. The fennel takes a few minutes, the others only about a minute or two before they become fragrant and slightly brown. Put the seeds together, add about ½ teaspoon rock salt (don’t use sea salt), and mix it all up. Store in a glass jar.

Chewing a little of this mixture after eating any food will aid digestion and help prevent any kind of food allergy.

⌘ ⌘ Frequent Urination ⌘ ⌘

See “[Bladder Problems](#),” “[Prostate Problems](#),” and “[Urinary Incontinence](#)”

⌘ ⌘ Frequent Urination During Pregnancy ⌘ ⌘

Frequent urination during pregnancy is an entirely natural phenomenon. When the uterus becomes enlarged due to the growth of the fetus in the womb, it creates pressure on the bladder. The bladder hasn't got sufficient space to accumulate urine and expand, so just a little accumulation stimulates the desire for relief.

If frequent urination disturbs the woman's sleep, this is not good, as she needs her rest. Ayurveda suggests this simple home remedy: Eat a handful of white sesame seeds with about ½ teaspoon of jaggery or natural brown sugar. This mixture will pacify vata dosha; by soothing vata, it prevents the excess stimulation that contracts the bladder. In this way, frequent urination during pregnancy can be corrected.

Additional suggestions:

- Don't drink anything for at least 2 hours before going to bed.
- Don't drink anything with caffeine, such as coffee, tea, or cola, especially in the evening. Caffeine is a diuretic (it promotes urination) and is exactly what you don't need if you want to counteract frequent urination.
- For further recommendations, see “[Urinary Incontinence](#).”

⌘ ⌘ Frigidity ⌘ ⌘

See “[Low Libido](#)”

☞ ☞ Gallstones ☞ ☞

Gallstones are a kapha disorder associated with underactive thyroid and slow metabolism. Gallstones begin with stagnation of bile in the gallbladder. The bile becomes thick, accumulates, coagulates, and slowly leads to stone formation.

Prevention of gallstones, and what to do when gallstones are present, are two different lines of treatment. First let's consider what to do when gallstones have already developed.

TO ALLEVIATE THE PAIN. In general, gallstones are not painful. They may remain in the gallbladder for a long time without causing any pain; indeed, you will not know they are there. Pain occurs when the gallbladder tries to push the stones out of the gallbladder through the bile duct.

To reduce pain, put a warm castor oil pack on your abdomen. Castor oil produces a slow, sustained heat that is soothing and healing. Warm up about 3 tablespoons of castor oil and pour it onto a handkerchief or other soft cloth, spreading it equally on the cloth. Place this compress on the abdomen above the gallbladder (on the right side of your abdomen, above the line of your navel and below the ribs). If you have a hot water bottle, you may place it on top of the pack to keep it warm. (Electric heating pads are not recommended.)

FOR ACUTE ATTACK. During an acute gallstone attack, use this herbal formula:

musta 4 parts

trikatu 3 parts

guduchi 6 parts

Take ¼ teaspoon of this mixture 2 or 3 times a day with honey. It will reduce the pain and ease the gallbladder attack.

LIVER FLUSH TO ELIMINATE THE GALLSTONE. When the pain is gone, you can do a liver flush to wash out the gallstone. This liver flush should not be done if the stone is large, *so before doing the treatment it is vital to get an ultrasound report on the exact size of the gallstone and consult with your physician about using the treatment.* If it is a small and recently formed stone, this treatment will help. If it is quite large, such as 3 to 4 mm. in diameter, then the flushing treatment will not be good.

TIME TO SEE THE DOCTOR

IMPORTANT: This liver-flushing treatment should *not* be done without the approval and guidance of your physician or the direct supervision of an Ayurvedic physician. Otherwise, you may damage the gallbladder and experience dangerous complications.

Combine 8 ounces olive oil, 4 ounces lemon juice, a chopped fresh garlic clove, and

¼ teaspoon cayenne pepper. Drink this entire mixture in the early morning (at about six o'clock) on an empty stomach. Don't eat anything until at least noon. If you feel thirsty, drink hot water or a little lime juice.

The treatment is a shock to the gallbladder, which contracts and squeezes the gallstone(s) out into the duodenum.

That night, take ½ teaspoon of *triphala* with warm water. The next day you will pass some green stuff in the stools. This is thick, coagulated bile containing the gallstone crystals.

PREVENTION OF GALLSTONES. In order to prevent gallstone formation, one has to improve both thyroid function and metabolism. In general, the following formula is effective:

punarnava 5 parts

shatavari 4 parts

kutki 2 parts

chitrak 2 parts

musta 3 parts

shilajit ¼ part

This mixture (¼ teaspoon 3 times a day with honey), taken regularly for 2 to 3 months, will help to prevent gallstone formation.

YOGA POSTURES. Several yoga *asanas* are beneficial for prevention of gallstones. The Bow pose, Peacock pose, Spinal Twist, and Narayan pose (lying on the left side) will help to empty the gallbladder. (For illustrations of yoga postures, see [appendix 4](#).) These postures will improve circulation in the gallbladder, helping to prevent the crystallization process.

DIET. Stay away from deep-fried food, dairy products such as yogurt and cheese, and all fatty foods, especially animal fat and any saturated fat. These accelerate coagulation of bile into gallstones.

❧❧ Gas and Flatulence ❧❧

No one under the sun is exempt from flatulence. Every person at some time or another gets gases and disturbances in the colon.

We are all vulnerable to this condition for several reasons. First, the colon is the main seat of vata dosha, the dosha that is derived from ether and air. If vata increases in the colon, due to eating vata-aggravating foods, cold weather, anxiety, insomnia, and other factors, gases may build up. Also, whenever we eat anything, we swallow a small amount of air, which increases vata. And any food we eat undergoes slight

fermentation, which produces gases. These gases, in the segmented colon, create flatulence, distension, and discomfort.

Here are some effective ways to control flatulence.

GINGER REMEDY. Grate some fresh ginger root until you have about 1 teaspoon of pulp, and add 1 teaspoon of lime juice. Take this mixture immediately after eating.

LEMON JUICE REMEDY. Another simple method to reduce excess gas is to stir 1 teaspoon lemon juice and ½ teaspoon baking soda into 1 cup cool water. Drink it down quickly, right after meals, for best results! (It forms carbon dioxide, which facilitates digestion.)

CUMIN-FENNEL-CELERY SEED MIXTURE. Prepare a mixture of roasted cumin, fennel, and *ajwan* seeds (Indian celery seed) in equal proportions. (See [this page](#) for suggested method of preparation of roasted seeds.) After each meal, take about ½ to 1 teaspoon of this mixture, chew well, and swallow with about ⅓ cup warm water.

CHARCOAL TABLETS. Another simple remedy is charcoal tablets, which you can buy in most health food stores. Swallow two tablets after lunch and dinner. The charcoal absorbs gases and helps to prevent flatulence.

HERBAL TABLETS. Ayurveda also suggests the herbs *shankavati* and *lasunadivati*. These herbs are useful for an aching stomach and are helpful to reduce flatulence. Take 1 tablet (available from most sources of Ayurvedic herbs) at night for 5 days.

DIETARY GUIDELINES. Flatulence is mostly a vata condition, so following a vata-pacifying diet can help prevent it. Avoid raw foods, cold food and drinks, and most beans. (See [chapter 8](#) for guidelines.) Fermented foods also increase gases in the colon, so it's best to avoid them.

IRIPHALA. Finally, it is helpful to take the herbal compound *triphala*. At night before going to bed, steep ½ to 1 teaspoon *triphala* in a cup of boiling hot water for 5 to 10 minutes, then drink it.

❧ ❧ Glaucoma ❧ ❧

Increasing intraocular pressure, due to accumulation of kapha dosha in the vitreous humor (the viscous fluid inside the eyeball), is called glaucoma. When glaucoma is occurring in an eye, palpation will show a tenseness in that eye. If pressure in the eye becomes high, headaches may occur. Glaucoma may become a serious condition and can cause blindness, so one has to be very watchful.

In persons who lift heavy weights (either at work or for exercise), who strain in exercise, who have high cholesterol or high triglycerides, or who have diabetes or nicotine toxicity due to smoking, intraocular pressure has a tendency to increase and may lead to glaucoma.

If upon examination by an eye doctor it is determined that you have higher than normal intraocular pressure, these remedies may be helpful.

AN HERBAL REMEDY. In earlier stages of glaucoma, Ayurveda treats this problem with the following formula, which will help to relieve the tension in the eye:

punarnava 5 parts

jatamamsi 3 parts

shanka pushpi 3 parts

Boil 1 teaspoon of this mixture in a cup of water for a few minutes to make a tea. Drink twice a day.

TRIPHALA TEA EYEWASH. To relieve the tension in the eye, wash the eyeball with *triphala* tea, which helps to regulate pressure in the eye. Boil ½ teaspoon *triphala* in 1 cup water for 2 minutes, strain it thoroughly (through cheesecloth double-folded, or through a coffee filter) so that no particles of *triphala* remain in the tea, cool it down, and wash the eye. (For more on *triphala*, see [appendix 2](#).)

TREAT THE SOURCE OF THE PROBLEM. In addition, one has to determine and then treat the root cause of the glaucoma. If the problem is diabetes, follow the instructions in the section “Diabetes.” If high blood pressure is the problem, then try to regulate the blood pressure (see “[Hypertension](#)”). If you have high triglycerides and high cholesterol, then you have to control that (see “[Cholesterol](#)”).

REDUCE KAPHA. Follow a kapha-pacifying diet. Especially avoid coffee, white sugar, and dairy products.

BE CAREFUL OF EXERCISE. Strictly avoid heavy weightlifting and similar straining. When you do yoga postures, avoid inverted poses such as Headstand and Shoulder Stand.

☞ ☞ Gum Disease ☞ ☞

See also “[Teeth and Gums—Ayurvedic Care](#)”

Gum disease includes receding gums, bleeding gums, gingivitis, and swollen gums. From an Ayurvedic perspective, excess vata dosha leads to receding gums, while pitta dosha is responsible for bleeding gums, gingivitis, and swollen gums.

FOR GENERAL CARE. For general cleaning of the teeth and caring for the gums, Ayurvedic dentistry recommends the use of certain bitter and astringent herbs, particularly *neem*, which is bitter, and *lodhra*, *kushtha*, and *bilva*, which are all astringent. You can make an excellent cleanser for brushing your teeth by mixing the powdered form of these herbs. You can use *neem* plus any one of the other three, mixed in equal proportions.

Rinsing your mouth with a tea made from these herbs is also beneficial.

You can also buy any one of several commercial toothpastes with *neem* and other Ayurvedic herbs in natural food stores or by mail order (see [Resources](#)).

FOR RECEDING GUMS. Receding gums expose the roots of the teeth, and then both the gums and teeth become sensitive to cold and more susceptible to infection. To help with this problem, take a mouthful of warm sesame oil, and swish it around in your mouth for about 3 minutes before going to bed. Then massage your gums with your index finger. It is better not to rinse with water afterward; leave the oil residue in your mouth.

FOR BLEEDING GUMS AND GINGIVITIS. *Triphala* tea is effective for gingivitis and bleeding gums. *Triphala* has astringent qualities, and it is a hemostatic—that is, it stops bleeding. Gargling and swishing the mouth with *triphal*a tea are helpful for both gingivitis and bleeding gums.

- One cup orange juice with ½ teaspoon natural sugar and a pinch of cumin will help bleeding gums.
- Drink the juice of ½ lemon squeezed into a cupful of water.
- Raw apples are also beneficial. Eating a raw apple about an hour after meals helps to clean the teeth and heal the gums. Pears are also effective.
- Eat some melon, chewing it slowly. (Again, at least an hour after meals.)
- Try eating about 10 to 20 raspberries 2 or 3 times a day on an empty stomach. (Don't combine them with any dairy products.)
- Massaging the gums with coconut oil can also help heal gingivitis and bleeding gums.

FOR INFECTED GUMS. Tea tree oil is effective for painful infected gums, as is clove oil. Both help to reduce pain and heal the infection. All you need is one drop of tea tree or clove oil directly at the site of the pain. A small piece of natural, edible camphor will also take care of pain in the gums. (Do not use synthetic camphor; it is poisonous.)

- Dental floss with tea tree oil will take care of infected pockets in the gums below the gumline. Some dental floss treated with tea tree oil is available commercially; otherwise, just dip the floss in the oil prior to flossing.

❧❧ Hair Care Secrets ❧❧

In Ayurveda, hair (along with nails) is considered to be a by-product of bone formation. The *dhatu* or tissue responsible for building bones (*asthi*) also gives rise to the hair. (For an explanation of the *dhatu*s, please see [this page](#).)

Thus, proper nutrition of the bones is necessary for healthy growth of your hair. If you don't completely absorb calcium and magnesium, for example, not only will your bones be adversely affected, but your hair may become brittle, develop split ends, break

easily, and even begin to fall out.

It is important to note that if your hair is unhealthy, it may be an indication that you are not absorbing these minerals effectively; this suggests that the health of your bones will be—or may already be—adversely affected. So the health of your hair (as well as your nails) is a good indicator of the health of your bones.

The following Ayurvedic secrets of healthy hair will help you keep your hair's natural strength and luster.

PROPER DIET. Healthy hair depends first upon eating nutritious food. Dairy products such as cheese, milk, and freshly prepared yogurt are good for bones and hair, as are white radish and daikon. Coconut, cooked apples, and cabbage are also beneficial.

TAKE A MINERAL SUPPLEMENT. You can improve the condition of your hair (and also strengthen your bones) by taking supplements of calcium, magnesium, zinc, and other minerals. An effective formula will contain a daily dose of about

calcium 1,200 mg.

magnesium 600 mg.

zinc 60 mg.

Oil Massage for Hair Health and Beauty

Rubbing a little oil on your scalp is beneficial for your hair. *Amla* oil, *bhringaraj* oil, and *brahmi* oil are all cooling, are quite favorable for healthy growth of your hair, and help maintain the hair's natural luster. These oils are fine for all body types. (Please see instructions on how to prepare these oils in [appendix 2](#).)

Before going to bed, rub 2 teaspoons of the oil onto your hair. Note that the object here is to apply the oil to the *scalp*, not the hair. Gently massaging the scalp improves circulation at the root of the hair and thus increases the supply of nourishing minerals that support the roots.

First pour the oil into a small dish. (You may wish to warm it up a little.) Dip your fingertips into the oil, and then run your fingers through your hair, with the intention of getting the oil to the scalp, not oiling the hair. *Gently* massage from the center of your scalp (the crown of your head) down toward your ears. A rough or rapid application may disturb the roots of your hair.

SESAME SEEDS. Every morning, eat a handful of white sesame seeds. One handful of sesame seeds contains about 1,200 mg. of calcium and magnesium and promotes healthy growth of your hair.

COCONUT WATER FOR CALCIUM. Drinking coconut water (the “juice” inside the fresh coconut) also helps to provide essential calcium for hair growth. You can have ½ cup a day. Drinking coconut milk (made from coconut “meat” blended in water) is also helpful, but is a second choice.

NOTE: If you have high cholesterol, you may not want to use so much coconut, as coconut is high in saturated fat, which increases cholesterol in the blood.

STIMULATE YOUR HAIR BY COMBING. Gently combing your hair with a comb, a little in the opposite direction of its natural tendencies, helps to improve the circulation at the root of the hair and will make your hair healthier.

Vigorous brushing of the hair is not recommended.

YOGA FOR YOUR HAIR. Several yoga postures are effective for relieving pressure and tension in the neck, which will increase circulation to the scalp. These include the Shoulder Stand; the Camel, Cobra, and Cow poses; and the Spinal Twist.

NOTE: For strategies to combat hair loss, see “[Baldness](#).”

☞ ☞ Hangover ☞ ☞

The effects of overconsumption of alcohol—headache, dullness, inability to focus the mind clearly, nausea, dizziness, and so on—are all symptoms of excess pitta. Drinking too much alcohol may become toxic to the stomach and liver, which triggers pitta and can eventually lead to serious illness.

The following recommendations will help you get over the effects of drinking too much alcohol the night before and will help restore normal functioning.

- Drink a glass of water with about 1 teaspoon lime juice, ½ teaspoon sugar, and a pinch of salt added. Just before drinking, add ½ teaspoon baking soda. This will immediately take care of pitta aggravation, and you will feel much better.

- A glass of fresh orange juice with 1 teaspoon lime juice and a pinch of cumin powder can help with both alcohol and drug-induced hangover.

- If you are feeling drowsy and dull, with an aching head, a burning stomach and no appetite, try a cup of cool lassi. Blend 1 tablespoon of fresh yogurt with 1 cup of water and a pinch of cumin powder. Drink this 3 or 4 times in the day. It will be effective to prevent dehydration, relieve nausea, and soothe the burning in your stomach.

- You can also use this herbal formula:

shatavari 5 parts

shanka bhasma ⅛ part

kama dudha ⅛ part

jatamamsi 3 parts

Take ½ teaspoon of this mixture 2 or 3 times in a day. Put it on your tongue, and wash down with warm water.

- Most of the time, drinking coconut water (the natural liquid inside the coconut) is beneficial for hangover.
- Doing *nasya* with *bhringaraj* oil or *brahmi* ghee may also be effective. (Instructions for preparing medicated oils and ghee are in [appendix 2](#). To buy them already prepared, see [Resources](#). Guidelines for *nasya* appear in [appendix 3](#).)
- The herbal compound *tikta* is an effective antidote for alcohol toxicity. If you take ½ teaspoon of the powder 3 times during the day that you need it, washed down with warm water, it should take care of the hangover. If you cannot locate *tikta*, you may substitute aloe vera, myrrh, or *sudharshan*.

▮▮ Headaches ▮▮

Headaches are a very complex phenomenon. Ayurveda speaks a great deal about the etiological factors behind headaches and the many ways headaches manifest.

Generally, headaches are classified as vata type, pitta type, and kapha type. In vata individuals, fear, anxiety, stress, nervousness, constipation, and physical overactivity can aggravate systemic vata, which can go into the skeletal, muscular, or nervous system and cause headaches. Vata-caused headaches will tend to be in the occipital area (the back of the head) or on the left side.

In pitta individuals, acid indigestion, hyperacidity, acidic pH of the saliva and stomach, excess pitta in the intestine and colon, and getting overheated, as well as a diet high in pitta-provoking food, can create a headache. That headache will be more in the temple or temporal area.

Because of a kapha-producing diet, systemic kapha in the stomach increases, enters into the general circulation, and can lodge in the sinuses and create kapha-type sinus headaches. Kapha headaches tend to be more in the frontal and nasal areas of the head.

Headaches can also be due to ear problems, eye problems, insomnia, food allergies, exposure to cold temperatures, tension in the neck, or working too long (for instance, in front of a computer) in a wrong position. Even using two pillows below the head for sleeping can cause headaches.

Clearly the causes are extremely varied. Remember that in Ayurveda, treatment is determined by the specifics of each situation. Thus in order to successfully treat your headache you need to know as much as possible about its cause.

FOR VATA-TYPE HEADACHES

These headaches are in the back (occipital) portion of the head. They are characterized by throbbing, pulsating, migrating pain that radiates from the back of the head and may go to the front. A vata headache may be associated with tension in the neck and shoulder muscles, back stiffness, constipation, and sciatica. This kind of headache is aggravated by high altitude. It gets worse when you move your body and subsides when you rest.

WARM WATER ENEMA. Vata headaches are often due to toxins accumulated in the colon. Ayurveda recommends using a warm water enema to relieve any constipation and taking *triphala* (½ teaspoon at night with ½ to 1 cup warm water) over several weeks to systematically detoxify the colon.

OIL ENEMA. Probably the best way to pacify vata is by oil *basti* (enema). Half a cup of warm sesame oil injected into the rectum and retained for at least 5 to 10 minutes helps to calm vata. (Detailed instructions for *basti* are in [appendix 3](#).)

OIL MASSAGE. For tension in the neck and shoulders, massage the tight muscles there with sesame oil. Then take a hot shower.

GHEE NASYA. Putting 3 to 5 drops of warm ghee in each nostril will help reduce vata and will be effective for soothing your headache (see [appendix 3](#)).

NIGHTTIME FOOT AND SCALP MASSAGE. Before you go to sleep at night, rub some sesame oil gently on the top of your head and on the soles of your feet. This is one of the most effective ways to keep vata under control.

IN CASE OF DEHYDRATION. Vata headache is frequently associated with dehydration, especially if you have just gone to a higher altitude. If dehydration has occurred, make some homemade dextrosaline: Mix 1 tablespoon sugar, ¼ teaspoon salt, and about 10 drops of lime juice in a pint of water and drink it. The moment the dehydration is corrected, a vata headache will disappear or at least be greatly reduced.

A SOOTHING PASTE. If, after this treatment, the headache does not subside, then take ¼ teaspoon nutmeg powder in your palm, and add sufficient water to make a paste by rubbing your hands together. Apply the paste to your forehead. Leave it on for about half an hour and then wash it off. This should help to soothe a vata-type headache.

FOLLOW THE VATA-BALANCING DIET. Remember that if you are prone to vata headaches and other vata-related problems such as constipation and insomnia, following a vata-pacifying diet will help a lot (see [chapter 8](#)).

FOR PITTA-TYPE HEADACHES. Pitta headaches start in the temple area and go to the central part of the head. A pitta headache is characterized by shooting, burning, piercing, or penetrating pain, and it is worsened by bright light, hot sun, or high temperatures, or by eating sour fruits, pickles, or highly spicy food. It may be associated with nausea and/or

burning of the eyes. The person may also become quite irritable. A pitta headache is often felt behind the eyes and may be associated with dizziness.

These headaches are related to the stomach and intestines.

ALOE VERA. If you have a pitta-type headache, take 2 tablespoons of aloe vera gel, up to 3 times in a day.

A COOLING TEA. Drinking cumin-coriander tea (equal amounts of each, about 1 teaspoon of the mixture per cup) can help to relieve a pitta-type headache. Let the tea cool to room temperature before you drink it.

A COOLING PASTE. This cooling paste may help to quickly relieve a pitta headache. Mix 1 teaspoon sandalwood powder with sufficient water to make a paste, and apply it to your forehead and/or temples. Leave it on your skin for about half an hour, then wash it off.

SOOTHING GHEE NASYA. A few drops of warm ghee placed in the nostrils will be beneficial for soothing a pitta-type headache.

EAT SOMETHING SWEET. Sometimes a pitta headache responds quickly if you just have something sweet to eat. Try a piece of sweet fruit, or some ice cream.

NIGHTTIME MINIMASSAGE. At night, rub a little *bhringaraj* oil or *brahmi* oil on the soles of your feet and on your scalp. Take precautions not to get oil on your pillow and sheets.

COVER YOUR HEAD. If you have a pitta-type headache—or are prone to getting them—don't walk or work in the sun without wearing a hat. A hat on your head protects against aggravating pitta and helps to prevent the headache.

FOR KAPHA-TYPE HEADACHES

If your headache occurs in winter or spring, strikes in the morning or evening, and gets worse when you bend down, it is a kapha headache. It is often associated with sinus congestion and clogging of the nose, and it often accompanies a cold or a cough. It may go along with hay fever and other allergies. The pain of a kapha headache is usually dull and deep-seated. It starts in the upper frontal area of the skull, moves down to the forehead, and sometimes comes down to the sinuses.

EUCALYPTUS STEAM. To immediately relieve a kapha headache, put 10 drops of eucalyptus oil in boiled water, cover your head with a towel, and inhale the steam. This helps to relieve the congestion and often completely takes care of the headache. Ginger steam—boiling fresh ginger or dried ginger powder and then inhaling the steam—is also effective.

USE THIS WARMING PASTE. A warming ginger paste can be quite helpful. Take 1 teaspoon ginger powder, mix together with sufficient water to form a paste, and apply it to your

forehead. You can also put some of the paste across the bridge of your nose and over your cheekbones. A paste of *vacha* powder (calamus) is also helpful and is preferable for pitta individuals, as the ginger powder may burn their skin. Leave the paste on for about half an hour, then wash it off. *Be careful when washing off ginger paste:* Avoid getting any in your eyes.

NOTE: Ginger paste can sometimes create a burning sensation on the skin, especially in pitta individuals. It is not dangerous, but if you begin to feel an uncomfortable burning feeling, wash the ginger off with warm water.

A PASTE FOR SINUS HEADACHES. For sinus headaches (usually related to kapha), make a paste out of ½ teaspoon cinnamon and sufficient water, and apply it locally.

SALTWATER REMEDY. For some individuals, a kapha-type headache can be quickly relieved by this simple remedy: Mix 1 teaspoon warm water and at least ⅛ teaspoon salt to make a thick, concentrated solution. Put 3 to 5 drops of this salt water in each nostril. This greatly helps to drain and unclog the sinuses and relieve the headache.

TIME TO SEE THE DOCTOR

Generally headaches can be relieved using Ayurvedic remedies. However, if you have a headache that persists for more than a couple of days; if your headache is accompanied by a fever or stiff neck; if you also experience neurological symptoms such as blurred vision, difficulty with coordination or speech, memory loss, numbness, or weakness in your arms or legs; if you wake up at midnight with a headache; or if you are having headaches often that seem to be becoming more severe, then please see a doctor.

YOGA POSTURES FOR HEADACHES

Generally, a person having headaches should do the Moon Salutation. Certain yoga postures are also helpful, such as the Boat pose, Hidden Lotus, Bow pose, Spinal Twist, Palm Tree pose, and Standing on the Toes. (For illustrations of yoga poses see [appendix 4](#).) Inverted poses such as Headstand, Shoulder Stand, and the Plow pose are not recommended.

❧❧ Hearing Loss ❧❧

Hearing is governed by an aspect of vata known as *prana* vata. In older people, *prana* vata tends to get debilitated, leading to what is known as conductive or nerve deafness: The person doesn't hear properly because the nerves become weak.

To improve hearing, try the following natural remedies:

TAKE YOGARAJ GUGGULU. This special compound of Ayurvedic herbs pacifies vata dosha and

strengthens weakened nerves. Take 200 mg. of this compound 2 or 3 times a day with warm water, after meals. *Yogaraj guggulu* can be ordered in capsules from many sellers of Ayurvedic herbs. (see [Resources](#) for addresses.)

DIET. Avoid vatagenic foods such as popcorn, corn chips, beans, raw vegetables, and cold drinks. (Guidelines for the vata-balancing diet maybe found in [chapter 8](#).)

KEEP OUT OF COLD DRAFTS.

TRY GARLIC OIL. Pour about 1 tablespoon of sesame oil into a small pan, and into it place 1 clove of garlic, well chopped. Cook till the garlic turns brown, then let it cool. While cooking, press the garlic into the oil. This helps the healing properties of the garlic ooze out and permeate the oil. The resulting mixture, which has an excellent fragrance, is called garlic oil. Place 5 to 10 drops (of body-temperature oil) into the ear. This will improve the hearing capacity of the auditory nerve.

NOTE: Oil should be used in the ear only when there is no infection.

YOGA POSTURES. The following yoga postures may improve hearing: Lion, Camel, Cobra, and Cow (see [appendix 4](#)). Alternate Nostril breathing may also be effective (see [chapter 6](#)).

TIME TO SEE THE DOCTOR

Ayurveda offers several recommendations to restore hearing or retard hearing loss. However, if these treatment suggestions don't result in any improvement after a couple of months, or if your hearing loss seems to be increasing rather than diminishing, seek a doctor's advice.

❧❧ Heart—Ayurvedic Care ❧❧

According to Ayurveda, the heart is the seat of *prana*, *ojas*, and mind. It is a most vital organ. In fact, a person is as old as his or her heart. So we have to take good care of this precious organ.

If a person has high blood pressure, high cholesterol, and high triglycerides, and in addition lives a stressful life, that person runs a great risk of heart problems. So to keep the heart healthy, it is important to control these factors as much as possible.

High blood pressure is discussed at length in a separate section ("[Hypertension](#)"), as is high cholesterol ("[Cholesterol](#)"). Please see those sections for more complete recommendations. A few suggestions follow:

DIET

In order to control cholesterol and triglycerides, it's important not to eat food with a

high fat content, such as fatty fried food, ice cream, heavy meats, and cheese. Yogurt is also not helpful.

EXERCISE

To keep your heart healthy, you need some daily exercise, though the quantity of exercise and the degree of strenuousness depend on your age, level of fitness, and constitutional type. Kaphas need the most vigorous exercise, vatas the least, with pittas in the middle. (See [this page](#) for further guidelines.)

For most people, walking at least two miles a day is very beneficial. Some more vigorous aerobic cardiovascular exercise may also be beneficial, such as fast walking, gentle jogging, or jogging in place on a trampoline. (You don't need to use the large gymnastic trampoline; you can use the small variety, usually about 3 feet in diameter.)

Cholesterol-Reducing Foods

Some foods may actually help to *reduce* cholesterol. These include:

- oatmeal
- corn
- apples
- fresh fruit juice, such as orange or grapefruit
- millet
- most fresh vegetables

Be sure to include some of these foods every day in your diet if you have high cholesterol.

STRESS MANAGEMENT

To keep stress levels low, here are two important recommendations.

DO SOME QUIET MEDITATION. Meditation is one of the best ways to relax, dissolve stress, and allow the body to heal. By meditating for 10 to 20 minutes twice a day, a person's heart may be healed. Please see [chapter 7](#) for guidelines on how to meditate.

DO DAILY PRACTICE OF SAVASANA. *Savasana* is the yogic rest pose. While lying quietly, flat on your back with your arms by your sides, watch the flow of your breath. Inhale and exhale, inhale and exhale.... You'll notice that after exhalation (and before inhalation), there is a brief, natural stop. Similarly there is a natural stop after inhalation and before exhalation. In that stop, stay naturally quiet, silent, for just a few seconds. This practice brings tranquillity and rest, which are healing for the heart. Remain in *savasana*

practicing this quiet breathing for 10 to 15 minutes.

OTHER REMEDIES

In addition to these recommendations for diet, exercise, and stress management, several other simple home remedies can help keep your heart healthy.

GOLD WATER. Gold is healing for the heart. It is good for the coronary artery and is said to gradually reduce cholesterol. For instructions on making gold water, see [appendix 1](#).

HERBS FOR YOUR HEART. Certain Ayurvedic herbs are strengthening and healing to the heart.

- First is the herb *arjuna*. Take ½ teaspoon 3 times a day with honey and warm water. *Arjuna* does much the same thing as gold: It is a coronary vasodilator, it protects the heart, it strengthens circulation, and it helps to maintain the tone and health of the heart muscle.

- Ginger is also important for a healthy heart. Make fresh ginger tea by boiling a little grated or sliced ginger in a cup or two of water. Or grate a little ginger and add it to your rice and/or your soup. Eating a little bit of ginger every day will help to prevent heart attack.

- The following formula of four Ayurvedic herbs is good for the heart:

punarnava 4 parts

kutki 3 parts

gulwel sattva ¼ part

shilajit ¼ part

Steep ½ teaspoon of this herb mixture in a cup of hot water to make a tea. Drinking that tea twice a day after lunch and dinner will help your heart.

- Another simple domestic remedy to protect the heart and keep it healthy is to include a little garlic in your diet. Garlic reduces cholesterol, strengthens the circulation, and acts as a decongestant.

RUDRAKSHA. *Rudraksha*, the “tears of Shiva,” are the dried seeds from the fruit of the *rudraksha* tree. An ancient story says that when Lord Shiva came out of deep meditation, a few tears dropped from his eyes and fell to earth, from which sprang up the *rudraksha* tree. The seeds are good for the heart both physically and spiritually; they are said to be good for meditation and for “opening the heart chakra.”

You can wear a necklace of the beads externally, in front of the heart. Or soak a *rudraksha* bead overnight in water and drink the water in the morning. Drinking *rudraksha* water can reduce blood pressure and strengthen the heart.

YOGA POSTURES. *Unless there are acute heart problems*, Ayurveda suggests daily

performance of the set of yoga postures known as *surya namaskar*, the Sun Salutation. Do at least 6 to 12 cycles a day. That will help to strengthen the health of the heart and help to prevent heart attack.

Ancient Advice for a Healthy Heart

Charaka, one of the ancient sage-physicians who wrote down the principles and practices of Ayurveda thousands of years ago, gave the following advice for care of the heart:

“One who wishes to protect the heart, circulatory system and vital essence should avoid, above all else, those causes leading to mental stress and instability. One should regularly adopt measures that support the heart and vital essence, cleanse the blood vessels, increase knowledge and calm the mind.

“The practice of nonviolence is the best among life-promoting practices, conservation of vital energy among the strength-promoting, and acquisition of knowledge among the nourishing practices. Control of the sensory organs is best to achieve happiness, and knowledge of reality for pleasure. Among all of these, celibacy is regarded as the best.

“From the heart, as the root, ten great vessels carry *ojas* throughout the body. In its importance, the heart is to be regarded as the central supporting member of a house.”

TIME TO SEE THE DOCTOR

IMPORTANT: If you have any heart problems, or you are over 40, it is wise to consult your physician before beginning any new exercise program.

If there are heart problems, the Sun Salutation may be too strenuous. In that case, substitute the following postures: Locust, Lotus, Bridge, Cow, Camel, Bow, and Cobra poses, Forward Bend, and Standing on One Leg—the Palm Tree pose. (Illustrations of yoga postures appear in [appendix 4](#).)

BREATHING EXERCISE. A deep breathing exercise such as *ujjayi pranayama* will also be helpful (see [chapter 6](#)).

⌘ ⌘ Heartburn and Acid Stomach ⌘ ⌘

Although there are several rather strong medications on the market for heartburn and acid stomach, these conditions are usually quite easy to control with the following natural Ayurvedic home remedies:

ALOE VERA GEL. Take 2 tablespoons of aloe vera gel with a pinch of baking soda. This will have an immediate soothing effect.

INSTANT EFFERVESCENCE. You might also try this formula. Into 1 cup of water add

lime juice 10 drops
organic sugar ½ teaspoon
baking soda ¼ teaspoon

Put the baking soda into the cup last. When you add it, an effervescent reaction will occur. Immediately drink the mixture to neutralize heartburn and acid stomach.

PAPAYA JUICE. For hyperacidity and indigestion, try drinking 1 cup papaya juice with 1 teaspoon organic sugar and 2 pinches cardamom.

NOTE: Pregnant women should not eat papayas, which contain natural estrogen and may create the danger of miscarriage.

FOLLOW THE PITTA-PACIFYING DIET. Generally, acid stomach can be controlled by a pitta-pacifying diet. Strictly avoid all hot spicy food. No pickles, and no fermented food. Minimize or cut out citrus fruit and sour fruit. And avoid overeating.

TIME TO SEE THE DOCTOR

If your heartburn is not helped by the remedies recommended in this section or if it occurs often for no apparent reason, you should consult a doctor. If you feel you have heartburn and it is accompanied by vomiting, dizziness, chest pain radiating into your neck or shoulder, or shortness of breath, you need help *immediately*: You may be having a heart attack.

A SOOTHING BREATHING EXERCISE. The breathing technique known as *shitali pranayama* is also helpful. It is not only cooling (to combat the excess pitta) but also stimulates digestion.

☞ ☞ Hemorrhoids ☞ ☞

According to Ayurveda, there are two basic kinds of hemorrhoids, associated with vata and pitta imbalances.

- Vata hemorrhoids are small, dry, and irregular in shape and may be accompanied by fissures or cracking of the anus. They are rough and hard to the touch and look like raisins. Vata hemorrhoids may become active when the person takes antibiotics or does a lot of cycling or physical exercise.

- Pitta-type hemorrhoids tend to get red, irritated, and inflamed and to bleed. They may look like purple grapes and are painful—sometimes very painful—to the touch. When they burst, they bleed extensively.

There are also kapha hemorrhoids, which look like green grapes. They don't bleed, and people generally live with them without problems, so we won't consider them in this section.

We can also distinguish between “internal” and “external” hemorrhoids. The internal kind are usually of the kapha variety and are more like polyps. They are not painful and are generally not problematic.

In most cases, hemorrhoids can be completely healed by Ayurvedic treatment, but one must first understand the distinction between the two basic problematic types in order to treat them properly.

EFFECTIVE TREATMENT FOR VATA HEMORRHOIDS

- A person with vata hemorrhoids should follow the vata-pacifying diet. Especially, it is vital to keep away from the nightshade vegetables—potatoes, tomatoes, and eggplant—all of which aggravate hemorrhoids.
- Take *triphala guggulu*, 1 tablet 3 times a day.
- Another helpful herbal formula for vata-type hemorrhoids is a combination of these herbs:

hingwastak 1/8 part

dashamoola 2 parts

Take 1/2 teaspoon 2 or 3 times a day with warm water.

- It is important to keep the stools loose and soft, as hard stools irritate the hemorrhoids. Taking 1 teaspoon of *sat isabgol* (psyllium husks) with a glass of warm milk at night is beneficial for vata hemorrhoids.
- Another way to help keep the stools soft is to take 1/2 to 1 teaspoon of *triphala* powder at bedtime with warm water.
- Prepare a warm castor oil or sesame oil pack, and sit on it for a while. Both oils are warming; castor oil especially produces a slow, sustained heat that is soothing and healing. Warm up about 3 tablespoons of oil, and pour it onto a handkerchief or other soft cloth, spreading it equally on the cloth (or dip the cloth in the oil).

EFFECTIVE TREATMENT FOR PITTA HEMORRHOIDS

- For pitta-type hemorrhoids, the first step is to follow the pitta-pacifying diet, especially avoiding spicy and fermented foods (see [chapter 8](#)).
- For hemorrhoids which become active and inflamed and start bleeding, one should prepare an herbal mixture of:

guduchi 1 part

neem 2 parts

kama dudha 1/8 part

Take 1/2 teaspoon of this mixture twice a day with warm water.

- If you have rectal bleeding from the hemorrhoid, drink a few ounces of cranberry juice and pomegranate juice (mixed half and half) between meals. That juice will act as a hemostatic, to stop the bleeding.
- Locally, you can apply coconut oil, which helps to control inflammation and irritation as well as bleeding.
- As in the case of vata hemorrhoids, it is important to keep the stool soft to avoid aggravating the condition. To accomplish this, take 1/2 to 1 teaspoon of *amalaki* at night with cool water. Or you can take 1 teaspoon of *sat isabgol* (psyllium husks) with a glass of warm milk at night.
- Steep *triphala* (1 teaspoon) overnight in a glass of water, and then the next day, early in the morning, after brushing your teeth, drink that tea.

GENERAL RECOMMENDATIONS

- All types of hemorrhoids respond well to aloe vera juice. Drink 1/2 cupful of the pure juice 3 times a day.
- Or you can add a pinch of ginger to 1 tablespoon aloe vera gel, and take it twice a day.
- Drink 1 cup carrot juice mixed with 2 teaspoons cilantro juice twice a day on an empty stomach for relief of hemorrhoids.
- Externally, you can also apply a mixture of 1/2 teaspoon turmeric and 1 teaspoon ghee directly to the hemorrhoid at bedtime. (But remember that the yellow color of the turmeric will stain whatever cloth it touches.)
- After each bowel movement, instead of using dry toilet paper, wash the anal orifice with warm water, and then apply some castor oil to the hemorrhoid. The dry paper may irritate the delicate mucous membrane and aggravate the hemorrhoid. It can also spread some fecal matter to the site of the hemorrhoid and lead to complications.

⌘ ⌘ Herpes ⌘ ⌘

ORAL HERPES

Herpes labialis comes through the oral contact of kissing, or drinking from the same cup or glass. It is essentially a pitta disorder that comes from *rakta dhatu* (blood) and breaks out on the skin, particularly in the corner of the mouth or on the upper lip.

Many people confuse herpes and canker sores, which are somewhat similar in appearance. With canker sores, however, only one ulcer appears; herpes produces a rash consisting of numerous small blisters, possibly clustered around one central, larger bump. Also, canker sores are usually *in* the mouth, while herpes sores are on the outside.

- Externally, one can apply *tikta ghrita* (bitter ghee) directly onto the affected area. Aloe vera gel or ointment is also quite effective.
- You can also use $\frac{1}{4}$ teaspoon *kama dudha*, mixed into 1 teaspoon dairy cream. Apply to the sore in the morning and at bedtime.
- Internally, use this Ayurvedic herbal mixture:

neem 3 parts
kama dudha $\frac{1}{8}$ part
maha sudarshan 3 parts

Take $\frac{1}{2}$ teaspoon of this mixture 3 times a day, either with 1 teaspoon *tikta ghrita*, if you have it, or with warm water.

- At night, take $\frac{1}{2}$ teaspoon *triphala* in warm water. This will pacify systemic pitta and will help relieve the herpes.

GENITAL HERPES

Genital herpes is similar to oral herpes. But they are transmitted differently; oral herpes through kissing or drinking glasses, as mentioned above, and genital herpes through sexual contact.

- For internal treatment, use the same herbs and herbal formulas recommended above.
- Externally, *tikta ghrita* can also be helpful. Also, dry genital intercourse can sometimes aggravate the herpes, so apply some *tikta ghrita* to the glans penis and/or labia minora before intercourse.

TO NEUTRALIZE STRESS

One of the primary causes of herpes eruptions appears to be psychological stress. The virus lies locked in the neuromuscular cleft and comes out during times of stress. To minimize stress:

TAKE HERBAL TEA. Make some tea of chamomile, *jatamamsi*, and brahmi. Mix these herbs in

equal proportions and use ½ teaspoon to make a tea; take it twice a day.

MEDITATE. Meditation is also effective to soothe the system and reduce stress. (See [chapter 7](#) for meditation suggestions.)

NOTE TO MEN: If you know you are susceptible to herpes outbreaks, be careful shaving around your lips and the corners of your mouth. The lips form a delicate and sensitive muco-cutaneous junction that is easily injured. Any small cut or scratch can activate the herpes virus. To help prevent this from happening, apply some *neem* oil after shaving.

❧ ❧ Hiccups ❧ ❧

Hiccups are caused by ischemia, or lack of blood supply to the diaphragm. As a result, the diaphragm undergoes the spasmodic periodic movements that we call hiccups.

HOLD YOUR BREATH. The simplest remedy for hiccups is to hold your breath. Take a deep breath; hold the breath behind the belly button; then gradually exhale.

BROWN BAG REMEDY. If you find it difficult to do this, take a paper bag, open it, hold the edges near your nose, and breathe into the bag (both exhaling and then inhaling). This will force you to inhale your own carbon dioxide, which naturally relaxes the muscles of the diaphragm. Continue for 1 or 2 minutes. Your hiccups should quickly be relieved.

HONEY AND CASTOR OIL. If the above procedure doesn't stop the hiccupping, mix 1 teaspoon honey with 1 teaspoon castor oil. Every 2 or 3 minutes dip your index finger into the mixture and then lick your finger.

DEEP BREATHING EXERCISE. Another simple 'remedy is Alternate Nostril *pranayama*, slightly modified as follows:

1. Inhale through the left nostril while closing the right nostril with your thumb.
2. After inhaling, hold your breath for just a moment, then
3. *Swallow*.
4. Exhale through your right nostril while closing the left nostril with your ring and little finger.
5. Repeat steps 1 to 4, this time starting by inhaling through the right nostril.

You can do this breathing exercise for 5 minutes.

TIME TO SEE THE DOCTOR

There are certain serious pathological hiccups, such as cardiac hiccups, which occur in persons having congestive heart failure; uremic hiccups, which occur due to renal failure (when the kidneys do not perform their function) and the resulting toxicity of uric acid in the blood; and cerebral hiccups, due to cerebral

pathology. These serious hiccups require medical attention and treatment.

If the remedies in this section aren't helpful and your hiccups are continuing longer than two days, see your doctor.

CONSTIPATION MAY BE THE CAUSE. Hiccups can also occur due to chronic constipation, bloating, and gases in the colon. If this appears to be the situation, do *basti* (enema), using 1 cup warm sesame oil for the enema. Try to retain the oil for at least 2 to 3 minutes, longer if possible. The warm oil in the rectum will relax the diaphragm and internal muscles and help alleviate the hiccups.

- If your hiccups are still continuing after another half hour, do a regular warm water enema.

❧ ❧ Hypertension ❧ ❧

A healthy heart pumps the blood through the veins and arteries with a certain optimum amount of pressure. But sometimes, due to various causes, the pressure increases, and when it does, the person is at greater risk for heart disease and possibly paralyzing stroke.

Blood pressure will increase when there is increased viscosity of the blood, increased velocity, or constriction due to decreased diameter of the blood vessel.

TYPES OF HYPERTENSION

From the Ayurvedic point of view, high blood pressure falls under three categories, primarily vata and pitta but also kapha.

Vata is responsible for constriction of the blood vessels. This frequently happens in old age. Somewhere around 65, the walls of the blood vessels often become thicker and the passage becomes narrower, with the result that many individuals develop a type of hypertension known as essential hypertension. It is a vata condition and is different from the narrowing of the arteries that occurs as a result of fatty deposits on the artery walls, which is a kapha condition.

Pitta is responsible for the rushing of the blood with more force. Kapha is related to increased blood viscosity.

Physical and emotional stress, including anger and anxiety, constrict the blood vessels and can increase blood pressure for a time. Heavy responsibilities, or stressful situations such as public speaking, may send your blood pressure soaring. There is even a phenomenon known as “white coat syndrome”; when a person goes to the doctor, anxiety and tension may increase, with the result that blood pressure goes up. This is all quite common and quite physiological, and fortunately it is also usually temporary. But

if blood pressure *stays* high, it can become dangerous.

TIME TO SEE THE DOCTOR

If high blood pressure is temporary and related to a stressful situation, some rest and relaxation will generally take care of it. And even in the long term, just because pressure is high does not necessarily mean it has to be treated with medications. Entirely natural means, such as diet, exercise, herbs, meditation, and yoga, which have no negative side effects, may be quite sufficient to deal with hypertension.

However, high blood pressure does require a doctor's supervision. Hypertension is a potentially life-threatening condition. I do not recommend that you use the following treatments in place of standard medical care.

Rather, try these Ayurvedic remedies—preferably with your doctor's knowledge and approval—as an adjunct to your medical care, and ask your doctor to monitor your progress. If the remedies are successful, he or she will find less and less need to supervise your condition and keep you on medications. At the least you should be able to gradually reduce the dosage of your medicines.

FOODS FOR HYPERTENSION

Several foods can help you control hypertension:

- Drink 1 cup mango juice, followed about an hour later by $\frac{1}{2}$ cup warm milk, with a pinch of cardamom, a pinch of nutmeg, and 1 teaspoon ghee. (*Note:* If your cholesterol level is high, skip the ghee.)
- Mix orange juice and coconut water (the natural juice inside a fresh coconut) in a ratio of 2 parts orange to 1 part coconut. Drink $\frac{1}{2}$ to 1 cup 2 or 3 times a day.
- Add 1 teaspoon coriander and 1 pinch cardamom to 1 cup freshly squeezed (not canned) peach juice. Drink this as many as 2 or 3 times a day to help with high blood pressure.
- Eat some watermelon with a pinch of cardamom and a pinch of coriander added. This will act as a mild diuretic and will help to regulate blood pressure.
- Try some cucumber raita with your meal. Cucumber is a good diuretic. Raita is a yogurt-based condiment often used in Indian cooking. (See the recipe in the accompanying sidebar.)
- Mung dal soup, made of mung dal with cilantro, cumin, and a pinch of turmeric, is good for persons with hypertension.
- Honey water can also help. Add a teaspoon of honey and 5 to 10 drops of apple cider vinegar to a cup of hot water, and drink it in the early morning. This drink helps

to reduce cholesterol, maintains vasodilation, and helps to regulate blood pressure.

Cucumber Raita

2 cucumbers

3 tablespoons ghee

½ teaspoon black mustard seeds

½ teaspoon cumin seeds

1 pinch hing

4 curry leaves

1 pinch cayenne or ½ small chili, chopped

1 small handful fresh cilantro leaves, chopped

½ cup plain fresh yogurt

Skin and grate the cucumbers. Pour off and discard any excess juice.

Heat the ghee in saucepan over medium heat, and add the mustard, cumin, hing, and curry leaves. Cook a moment until seeds pop.

Add the cayenne or chili and the cilantro, shake the pan, and remove from heat.

Stir the yogurt and grated cucumber together in a bowl.

Add the cooled spices to the yogurt mixture, mix well, and serve.

Serves 4–6 as side dish (1–2 spoonfuls per person).

NOTE: This recipe is from Vasant Lad and Usha Lad, *Ayurvedic Cooking for Self-healing* (Albuquerque: Ayurvedic Press, 1994), [this page](#).

AVOID PITTA-PROVOKING FOOD. Persons with hypertension should not eat salt, fatty fried foods, or hot spicy food.

HERBS FOR HYPERTENSION

Ayurveda suggests the following mixture of herbs for hypertension:

punarnava 1 part

passion flower 1 part

hawthorn berry 2 parts

Steep ½ teaspoon of this mixture in a cup of hot water for 5 to 10 minutes,

and drink the tea after lunch and dinner.

Another formula of Ayurvedic herbs is also effective for regulating blood pressure:

jatamamsi 2 parts

musta 2 parts

tagar 1 part

Use as above: ½ teaspoon steeped in a cup of hot water for 5 to 10 minutes as a tea, after lunch and dinner.

OTHER REMEDIES AND RECOMMENDATIONS

KEEP COOL. Working hard under the hot sun should be avoided by anyone with high blood pressure, as that may promote hemorrhage in the brain. Be cautious about this.

MAGNETIC WATER. You may be able to keep your blood pressure under control by drinking magnetic water. Put a cupful of water (preferably in a glass container) next to the north pole of a magnet. Let it sit for 2 hours. Drink a cupful of magnetic water twice a day.

Charging the water in this way increases its diuretic properties and thus helps bring down high blood pressure, in the same way as several widely used current hypertension medications, which are diuretics.

RUDRAKSHA. Soak 1 or 2 *rudraksha* beads in a cup of water overnight, and the next day drink that water. This is beneficial for regulating blood pressure.

Deep Relaxation to Help High Blood Pressure

Tension and stress increase high blood pressure. A wonderful and effective way to relax is the yogic rest pose, *savasana*.

Lie quietly, flat on your back with arms by your sides. Observe the flow of your breath. You will notice that after exhalation there is a brief, natural stop, and another natural pause after inhalation and before the next exhalation. In that stop, stay naturally quiet for just a few seconds. This practice will bring you deep relaxation, a natural antidote for hypertension. Remain in *savasana* practicing this quiet breathing for 10 to 15 minutes.

MEDITATION. Meditation is excellent for regulating blood pressure. (See [chapter 7](#) for guidelines to help you meditate.) Several research studies, including a study funded by the National Institutes of Health, have shown that meditation can be as effective as medications in controlling hypertension, and it produces none of the negative side effects often associated with hypertensive drugs.

BREATHING EXERCISE. Some gentle *shitali pranayama* can help control blood pressure. Make a tube of your tongue, and inhale through that tube into the abdomen. Hold the breath for a few seconds and then breathe it out through your nose.

YOGA POSTURES. Effective yoga *asanas* for helping control hypertension include Yoga Mudra and the Moon Salutation. (See the illustrations in [appendix 4](#).)

EXERCISE AND WORK WITH CARE. No one with elevated blood pressure should practice the Headstand or engage in weightlifting or vigorous exercise without a physician's guidance.

IMPORTANT NOTE: As mentioned above, hypertension may lead to serious complications. Although these Ayurvedic recommendations are time-tested, safe, and effective, it would be wise to use them in conjunction with the advice and care of your physician.

❧ ❧ Hypoglycemia ❧ ❧

Hypoglycemia (low blood sugar) is a very common condition. If you are late to eat at your regular mealtime and you feel light-headed or dizzy while standing up, or experience palpitations, tremor, nausea, drowsiness, sweating, nervousness, or mental confusion, all these indicate hypoglycemia. In some serious cases of hypoglycemia, a person can even have convulsions and fall into coma.

The brain uses blood sugar (glucose) as its sole food, and it depends on it for the necessary energy for its activity. If the brain doesn't receive sufficient blood sugar, it goes into crisis; it will create tremors, headache, sweating, nausea, drowsiness, and the other symptoms mentioned because it is starving for blood sugar.

According to Ayurveda, hypoglycemia is common in persons with pitta *prakruti* (a pitta constitution) or pitta *vikruti* (a present pitta imbalance). Increased pitta stimulates the secretion of insulin, which lowers the blood sugar level and creates hypoglycemia. Hypoglycemia in turn induces the secretion of adrenaline, which causes rapid heartbeat and tremors.

An unduly large dose of insulin taken by a diabetic person may lead to hypoglycemia. This condition is also common among alcoholics.

Hypoglycemia is a disorder that needs careful attention. To maintain their blood sugar level, pitta individuals should eat regular meals—breakfast, lunch, and dinner—and also have some fruit or another snack between meals or whenever they feel hypoglycemic symptoms. Emphasize proteins and complex carbohydrates for main meals; they digest more slowly, so the rises and falls in blood sugar will be less rapid.

TIME TO SEE THE DOCTOR

Reactive hypoglycemia may indicate a pancreatic tumor. When a middle-aged person craves sugar about 90 minutes after a full meal, it may mean that he or she has, or is going to develop, a pancreatic tumor. This is a serious health concern and requires medical attention.

Hypoglycemia can be classified into two types: fasting and reactive. Fasting hypoglycemia simply results from not eating, which is why people with a tendency toward low blood sugar need to eat regular meals. In addition, some individuals choose to fast, whether for religious reasons or for cleansing purposes. If they do too much prolonged fasting, hypoglycemia may result.

Reactive hypoglycemia occurs when the pancreas, often in response to a large intake of sugar, secretes too much insulin, which then reduces the blood sugar level. This is also called postprandial (after-eating) hypoglycemia. About 90 minutes after eating, the blood sugar level drops, and the person craves something sweet.

To deal with reactive hypoglycemia, one should treat pitta dosha.

DIET. Follow the pitta-pacifying diet (outlined in [chapter 8](#)). Stay away from hot spicy foods, fermented food, sour and citrus fruits, and alcoholic drinks. Smoking cigarettes should be curtailed or eliminated, as cigarettes also aggravate pitta.

LICORICE TEA. Drink licorice tea. When you are feeling lightheaded or faint or have some other hypoglycemic symptoms, make a cup of licorice (*yashti madhu*) tea, using 1 teaspoon of licorice root per cup of water. This tea will safely increase your blood sugar level. (However, individuals with hypertension should use licorice tea only sparingly; it increases water retention and may raise blood pressure.)

HERBAL REMEDIES.

- The best herbs to take are *brahmi*, *jatamamsi*, *shanka pushpi*, and licorice. These herbs are brain tonics; they nourish the brain and allow it to function on limited sugar. Mix these herbs together in equal proportions and use the mixture to make a tea. Steep ½ teaspoon of the herbs in 1 cup hot water, and drink after lunch and dinner.

- If the problem is postprandial hypoglycemia due to gastrojejunostomy, Ayurveda suggests using

guduchi 5 parts

shanka bhasma 2 parts

kama dudha 2 parts

Take ½ teaspoon twice a day (after lunch and dinner) with some water.

- If there is a suspicion that reactive hypoglycemia may lead to pancreatic tumor (see [“Time to See the Doctor”](#)), see your doctor. In addition, use *shilajit*, a nervine tonic that is rejuvenating for the pancreas. A good formula for using *shilajit* is the following:

shilajit 1 part

shardunika 1 part

Take ½ teaspoon of this mixture after lunch and dinner with some water. This formula will also help to prevent adult-onset diabetes (diabetes mellitus).

YOGA POSTURES. Some yoga *asanas* to strengthen the pancreas and help prevent hypoglycemia include the Peacock, Camel, and Locust poses, and the Elevated Lotus, as well as *navli* (an abdominal exercise explained in [appendix 3](#)). Individuals susceptible to hypoglycemia should do these yoga exercises and Alternate Nostril breathing.

NASYA. *Brahmi* ghee *nasya*, 5 drops in each nostril, will quickly relieve nausea, sweating, and mental confusion (see [appendix 3](#)).

FOR DROWSINESS. Drowsiness can be corrected by taking some sweet juice, such as pomegranate or sweet orange juice.

❧❧ Impotence ❧❧

See also “[Low Libido](#)”, “[Premature Ejaculation](#)”

Impotence, which is a man’s inability to have or to maintain an erection, may have several causes. Many people think it is always an emotional or psychological issue, but it can also be due to stress or to certain physical problems.

For example, when a person has high cholesterol, the fat and plaque may get deposited on the coronary arteries, blocking the flow of blood to the heart and triggering a heart attack due to the insufficient blood supply. Similarly, the plaque may get deposited in the blood vessels leading to the penis, resulting in a “penis attack”—in which the blood supply to the penis is insufficient to create or sustain an erection, resulting in impotence.

MASSAGE TO IMPROVE CIRCULATION. A simple and effective remedy is to massage the pubic area (the lower part of your abdomen) and the root of the penis with a few drops of *mahanarayan* oil. This massage will improve the circulation and may be sufficient to eliminate the problem.

APPLY SOME HERBALIZED OIL. You can also apply some *bala* oil or *ashwagandha* oil directly to the penis.

MASSAGE THE PROSTATE AREA. Massaging above the prostate gland (midway between the scrotum and the anus) with any of the three oils mentioned above will also be beneficial. If you do not have these oils, you can just use a little plain ghee. First rub in a circular motion, then finish with strokes from the anus toward the base of the penis. Use a light pressure. Like the massage of the pubic area, this will help improve circulation.

STRENGTHENING HERBAL FORMULA. For internal use, make a mixture of equal proportions of the following herbs:

ashwagandha

bala

vidari

and take 1 teaspoon of this mixture twice a day in warm milk, continuing for 3 months. While the milk is heating, add a few pieces of chopped fresh garlic. The garlic improves the blood supply and increases the dilation of the blood vessels. The combination of the garlic with these strengthening herbs will help alleviate the problem.

MILD APHRODISIAC DRINK. Every night, drink a cup of warm milk with a pinch of saffron added. Saffron is an aphrodisiac and also increases sperm count.

RELAX WITH TRANQUILLITY TEA. If the cause of impotence is psychological, such as fear or anxiety, you may be able to correct the problem by drinking some Tranquillity Tea, an herbal compound that contains *jatamansi*, *brahmi*, and *shanka pushpi* in equal proportions. Make a tea from ½ teaspoon of this mixture, and drink it about an hour before going to bed. It will definitely help with the emotional and psychological stress that may be at the root of the problem.

YOGA POSTURES. Certain yoga exercises may also help. The Rooster pose, which involves sitting in such a way that pressure is applied on the prostate gland, is particularly beneficial. Also try the Elevated Lotus, Vajrasana, Chakra Asana, and the Bow pose. (For illustrations, see [appendix 4](#).)

❧❧ Incontinence, Urinary ❧❧

See “[Urinary Incontinence](#)”

❧❧ Indigestion ❧❧

As discussed in [chapter 3](#), the effectiveness of your digestion depends on the strength of your digestive fire (agni). If your food intake is large in quantity and heavy, very liquid, or quite dense in quality, these properties are antagonistic to the properties of gastric fire and can inhibit the normal function of the agni, leading to indigestion.

Emotional eating—eating for emotional reasons when the system is not in need of food or is given too much to comfortably digest—is another potential cause of indigestion. A third major causative factor is wrong food combining. Eating bananas and milk, melon and grains, and other incompatible food combinations adversely affects

the digestive fire, leading to indigestion. (See [chapter 8](#) for a chart of incompatible food combinations.)

These various factors promote excess secretion of acid, leading to acid indigestion, heartburn, nausea, or even diarrhea; fermentation of the food in the stomach or intestines may also occur, leading to gas, bloating, and a possible stomachache, depending on the severity of the cause.

So one has to deal with indigestion first by avoiding these causative factors, and second by using the herbal treatments recommended below.

FOUR WAYS TO INCREASE YOUR DIGESTIVE FIRE

The first key to prevention of indigestion is to enhance the digestive fire. Here are several suggestions:

GINGER. One of the best herbs to kindle agni is ginger. Before each meal, chop or grate a little fresh ginger, add a few drops of lime juice and a pinch of salt, and chew it up. Alternatively, you can simply cut a thin slice of ginger, put on a pinch of salt, and chew that.

GARLIC AND TRIKATU. Here is another before-eating stimulant for your digestion. Make a mixture of $\frac{1}{4}$ teaspoon garlic powder, $\frac{1}{2}$ teaspoon *trikatu*, and a pinch of rock salt. (*Trikatu* is composed of ginger, black pepper, and Indian long pepper in equal proportions.) Take it before lunch and dinner.

HERBAL MIX. A similar mixture is 1 clove of fresh garlic, chopped up with $\frac{1}{4}$ teaspoon cumin powder, a pinch of rock salt, a pinch of *trikatu*, and 1 teaspoon lime juice. Take before meals.

BAY LEAF. Enliven your gastric fire with the common spice bay leaf. Steep $\frac{1}{2}$ teaspoon crushed or ground-up bay leaf in a cup of hot water for about 10 minutes to make a tea. Add a pinch of cardamom, and drink after eating.

FASTING

Fasting can be beneficial to dispel indigestion. Fasting not only kindles the digestive fire, it also gives the digestive system a rest. When you have indigestion, you can either observe a complete fast, or try this: Drink 1 cup of sweet fresh pineapple juice with a pinch of ginger, a pinch of black pepper, and $\frac{1}{2}$ teaspoon organic sugar. Take this 3 times a day.

WHEN YOU HAVE INDIGESTION

To relieve a case of indigestion:

ONION JUICE. Take $\frac{1}{4}$ cup of fresh onion juice with $\frac{1}{2}$ teaspoon honey and $\frac{1}{2}$ teaspoon black pepper.

GARLIC. Or chop up a clove of fresh garlic, add a pinch of salt and a pinch of baking soda, and eat it.

LIME JUICE. For acute indigestion, squeeze the juice from $\frac{1}{4}$ lime in 1 cup warm water. Just before drinking, add $\frac{1}{2}$ teaspoon baking soda, and then drink it down quickly.

FOR CHRONIC INDIGESTION

For people with chronic poor digestion (that is, weak agni), prepare this herbal mixture:

trikatu 1 part

chitrak 2 parts

kutki 1 part

Take $\frac{1}{4}$ teaspoon of this effective formula before meals, with a little honey and fresh ginger juice. If you don't have fresh ginger, just use honey. This mixture will strengthen the digestive fire.

TEN TIPS TO PREVENT INDIGESTION

- Don't eat unless you are really hungry.
- Don't eat emotionally. Emotional eating can affect the digestive fire adversely.
- Eat only two or at most three meals a day: breakfast, lunch, and dinner. Avoid snacking between meals.
- Avoid cold and iced drinks, especially during or after meals. They cool the digestive fire. For best digestion, take small sips of warm water during a meal.
- Fill your stomach to one-third of its capacity with food and one-third with liquid, leaving one-third empty. This aids in proper digestion and promotes mental clarity.
- Chew your food well to make sure the saliva is well mixed in. Saliva plays a major role in digestion.
- You can finish your meal by drinking a cup of lassi. This is made by blending 4 teaspoons of yogurt with 2 pinches each of ginger and cumin powder in 1 cup of water.
- Yoga postures that can be helpful for improving digestion include Leg Lifts and the Peacock pose. (See illustrations in [appendix 4](#).)

- You might also want to try the Ayurvedic procedure known as *navli* (see [appendix 3](#)).
- The *pranayama* (breathing exercise) called Breath of Fire will help to stoke your digestive fires (see [chapter 6](#)).

❧ ❧ Ingrown Toenail ❧ ❧

See “[Toenail Infection](#)”

❧ ❧ Insomnia ❧ ❧

Insomnia is an uncommonly common disorder in the modern world, caused primarily by an increase of vata dosha in the mind or nervous system. Insomnia is a cause or a complicating factor in many other problems. It may be related to constipation. It may be a result of stress or being overtired, or it may create fatigue and lead to greater stress. It may be a symptom of depression, or it may intensify depression. So we have to deal with it effectively.

DIETARY REMEDIES

WARM MILK. It’s definitely true that drinking a cup of warm milk before going to bed helps bring on a peaceful sleep. You can have it plain if you prefer, but the following suggestions will make it more delicious as well as more effective:

- Add a pinch (up to ⅛ teaspoon) of nutmeg.
- Add some crushed almonds (blanched is better), a pinch of nutmeg, and a pinch of cardamom. You can prepare the almonds in a nut grinder or coffee grinder.
- Try garlic milk. Mix together 1 cup milk, ¼ cup water, and 1 clove of fresh, chopped garlic. Boil gently until 1 cup of liquid remains.

TRY CHERRIES. Cherries are good for mental fatigue and stress, both of which can contribute to insomnia. Eating 10 to 20 cherries daily may help relieve these conditions and help you sleep.

TOMATO JUICE. Here’s a use for tomato juice you probably never thought of. Drink 1 cup with 2 teaspoons natural sugar and 2 pinches of nutmeg. Drink the juice between four and five in the afternoon; have dinner between six and seven. That evening you should get a sound sleep.

HERBAL REMEDIES

HERBAL FORMULA. An effective herbal formula to help you sleep is:

tagar 1 part

valerian root powder 1 part

chamomile 1 part

Take ¼ teaspoon of this powdered mixture with a little warm water just before going to bed.

CHAMOMILE TEA. Justly famous around the world, a cup of chamomile tea at bedtime is truly beneficial for inducing sleep.

ADDITIONAL REMEDIES AND RECOMMENDATIONS

WARM OIL MASSAGE. One of the simplest and most effective ways to induce sleep is to rub some oil on the scalp and the soles of the feet before going to bed. Use sesame oil, *brahmi* oil, or *jatamamsi* oil, and massage gently for a few minutes. Slightly warming the oil before applying is helpful.

NUTMEG. Using this common spice can help induce sleep. A fine paste made of nutmeg powder mixed with an equal amount of ghee can be applied around your eyes and on your forehead before bed to help you fall asleep.

TAKE A HOT BATH. A hot bath or shower at bedtime helps to soothe vata and promote sound sleep.

TRY YOGA MEDITATION. Sleep disturbances are often due to worries and anxieties that keep the mind agitated at night. To help dissolve those anxieties, meditate for a while before going to sleep. Sit comfortably on your bed and put your attention on the “third eye” (the area on your forehead between your eyebrows). Follow your breath in and out, or do the traditional *So-Hum* meditation: Inhale while thinking the syllable “So” and exhale with “Hum.” (Further guidelines are in [chapter 7](#).)

Then lie on your back. Watch your breath, and continue the *So-Hum* meditation, focusing your mind gently on the “third eye.” You will sleep like a child.

❧❧ Irritable Bowel Syndrome ❧❧

According to Ayurveda, irritable bowel syndrome is due to vata pushing pitta into the colon. To help correct the situation, combine the following herbs:

shatavari 1 part

kama dudha ½ part

shanka bhasma ⅛ part

arrowroot 2 parts

Take ½ teaspoon of this mixture a couple of times a day with a little warm water, just after eating.

- You can also take 1 teaspoon of *sat isabgol* (psyllium husks) with ½ cup of fresh yogurt 1 hour after dinner.

- To create another simple remedy, boil 1 teaspoon of flaxseed in a cupful of water to make a tea, and drink it at bedtime.

- In certain chronic cases of irritable bowel syndrome, Ayurveda recommends introducing ½ to 1 cup of warm sesame oil into the rectum. If you use this enema treatment, try to retain the oil for 5 minutes. Once the colon is well lubricated with sesame oil, irritable bowel syndrome will be controlled. You can do this oil enema (*basti*) once or twice a week, as needed. (For more about *basti*, see [appendix 3](#).)

⌘ ⌘ Jet Lag ⌘ ⌘

Jet lag is essentially a condition of excess vata in the body. Traveling on a jet at a tremendous speed induces a light, mobile, and spacy quality in the system, which aggravates vata. To prevent jet lag, Ayurveda suggests the following three-part strategy:

1. *An hour before flying*, swallow 2 capsules (00 size) of ginger, with a cup of water.
2. *While on the plane*, drink at least 2 to 3 cups of water, at intervals of 1 to 2 hours. Flying creates a slight dehydration of the body, which can be corrected by drinking sufficient liquid. Dehydration increases vata. Don't drink coffee or any other caffeinated beverage caffeine also provokes vata.
3. *When you reach your destination*, rub a little warm sesame oil on your scalp and on the soles of your feet. Also, drink 1 cup of hot milk with a pinch each of nutmeg and ginger. These two simple actions will help pacify vata.

If you reach your destination before nightfall, you can drink a tea made of equal proportions of chamomile, mint, and *jatamamsi* (⅓ teaspoon each), steeped in a cup of hot water for 10 minutes.

If you don't expect to have herbs available where you are going, you can mix them at home and take them with you in a small plastic sandwich bag or other convenient container.

⌘ ⌘ Kidney Problems ⌘ ⌘

See also “[Kidney Stones](#)”

Congestion, high pitta, or crystal urea in the kidney, as well as kidney stones, can all create tightness and pain in the lower back area. For these problems, Ayurveda suggests the following simple remedies.

First, prepare a mixture of these herbs:

punarnava 1 part

gokshura 1 part

fennel seed 1 part

Take 1 full teaspoon of this mixture 2 or 3 times a day after meals, washing it down with warm water.

Another simple, natural treatment for kidneys is cumin-coriander-fennel tea. Prepare the tea by mixing equal amounts of these herbs, boil in water, and drink 2 or 3 times a day. (Use about ¼ teaspoon of each herb per cup of water.)

For another effective remedy, combine about ½ teaspoon of the Ayurvedic herb *musta* with ½ teaspoon fennel, steep in a cup of hot water for 10 minutes, strain, and drink. This drink will strengthen the kidneys.

☒ ☒ Kidney Stones ☒ ☒

There are different types of kidney stones, corresponding to vata, pitta, and kapha dosha.

- Calcium stones are kapha stones. They are generally soft and painless, though they may become painful when they start leaving the pelvis of the kidney and enter the ureter. An individual with an underactive thyroid or parathyroid gland can develop calcium stones.

- Phosphate stones are rough; they irritate the bladder and cause pain because of their rough quality. They are primarily caused by increased phosphates in the system, due to excess eating of nightshade plants (potatoes, eggplant, tomatoes).

- Oxalate stones are sharp. They are pitta stones. They irritate, burn, and create bleeding, and they may create a great deal of pain from loin to groin. A diet high in oxalic acid promotes formation of pitta stones. Thus individuals who eat spinach, potato, tomato, and rhubarb, which are rich in oxalic acid, have a tendency to form stones. These food items should be strictly avoided by anyone concerned about preventing the formation of kidney stones.

HERBAL REMEDIES. If you already have a kidney stone, to decrystallize the pelvis of the kidney Ayurveda suggests *punarnava guggulu* and *gokshuradi guggulu*. Take one tablet of each twice a day after lunch and dinner.

If you have begun to pass a kidney stone and are having a great deal of pain, prepare the following herbal formula:

punarnava 1 part

mutral 1 part

coriander 1 part

Take 1 teaspoon of this mixture twice a day, with a can of beer. Beer, which is essentially fermented barley soup, is a diuretic. It accelerates the passing of the kidney stone and, in combination with the herbs, effects a decrystallization of the kidney. You will pass the stone more easily, without much pain. NOTE: You can use nonalcoholic beer if you prefer, or you can heat up regular beer in a saucepan and the alcohol will quickly evaporate.

Another effective herbal remedy is this formula:

punarnava 5 parts

gokshura 3 parts

mutral 2 parts

shilajit 1/8 part

Taking 1/2 teaspoon of this mixture twice a day with a can of beer can help the kidney stone pass with less discomfort.

Again, if you don't like to drink beer, you can substitute nonalcoholic beer, or drink barley tea or barley soup instead. All are diuretic and will accomplish the same action.

HOT AND COLD APPLICATIONS. Another way to ease the pain of passing a kidney stone is to alternate hot and cold applications to the kidney area. Use a hot water bottle or hot compress, and an ice bag or a bag of frozen vegetables. Alternate them every 30 seconds to 1 minute, until the pain subsides.

WATERMELON JUICE. Try drinking a cup of watermelon juice with 1/4 teaspoon coriander power. Watermelon is a diuretic (as is coriander), so this mixture will give the kidneys a good flushing and help to remove small stones and crystals. Use this 2 to 3 times a day.

❧❧ Low Libido ❧❧

Libido is the desire for pleasure and satisfaction. According to Ayurveda, this desire derives from *shukra dhatu*, the male reproductive tissue, and *artava dhatu*, the female reproductive tissue. When there is weakness or debility in the male or female reproductive tissue, libido is low.

Low libido is a symptom that many people have in middle age or even earlier. In

addition to weakness in the reproductive tissue, emotional factors and high stress are the primary causes of a reduced sex drive. It can be effectively treated with Ayurvedic programs and remedies that reduce stress and strengthen the reproductive system.

But as always in Ayurveda, the question arises, What is the total context of the situation? Low libido can be a problem that affects one's marriage or partnership. On the other hand, reduced sexual desire may sometimes be a healthy response by the body, shutting off excess loss of the health-giving vital fluid. From this perspective, low libido may be seen as an expression of the body's intelligence.

Intentional celibacy is quite different from low libido. In celibacy there is incredible sexual power, but the person consciously controls that sexual energy and transforms it into supreme bliss or supreme intelligence.

In cases of low libido, that energy is simply lacking. In this section we will explore several ways to increase it.

FOR MEN

A simple but effective technique is to gently press the glans penis (the head or tip of the penis) with the top of your index finger. Specifically, press on the groove about one inch behind the tip of the penis. A *marmā* point is located at the center of that groove. (See [glossary](#) for a brief explanation of *marmā* points.) Gently press for a minute or two, then release. This technique will also help remedy premature ejaculation. (See further suggestions under “[Premature Ejaculation](#).”)

- You can also gently massage the glans penis with *bala* oil or *shatavari* ghee. Or apply a little castor oil or *brahmi* ghee. (See [appendix 2](#) for guidance in preparing herbalized oils and ghees.)

- Internally, taking the herb *ashwagandha* is effective. Add 1 teaspoon *ashwagandha* and ½ teaspoon *vidari* to a cup of warm milk, and drink it at night. This formula is strengthening for men with low libido. NOTE: It is better to cook the herbs in the milk for a few minutes than to just mix the powder in the cup.

FOR WOMEN

A similar herbal formula is helpful for women, but use *shatavari* instead of *ashwagandha*. Mix 1 teaspoon *shatavari* with ½ teaspoon *vidari*, and take it with a cup of warm milk at night before going to bed.

- You can also gently massage the pubic bone with *bala* oil, *shatavari* ghee, castor oil, or *brahmi* ghee.

FOOD REMEDIES FOR BOTH MEN AND WOMEN

ALMONDS. Eat 10 raw (unroasted) almonds for breakfast. Soak them overnight in water, and the next morning peel off the skins before eating.

- Make this strengthening almond drink: As above, soak 10 raw almonds in water overnight, and peel off the skins the next morning. But then put the almonds in a blender, and add:

warm milk 1 cup

ghee 1 teaspoon

natural sugar 1 teaspoon

nutmeg a pinch

saffron a pinch

Blend thoroughly. It is yummy, and excellent for combating low libido!

DATES. Soak 10 fresh dates in a quart jar of ghee. Add:

ginger 1 teaspoon

cardamom $\frac{1}{8}$ teaspoon

saffron 1 pinch

Cover, and keep in a warm place for at least 2 weeks. Then eat 1 date daily, in the early morning. These dates taste delicious and help to remedy low libido and sexual debility as well as chronic fatigue.

APPLE DESSERT. Another delicious and strengthening concoction is the following apple dessert. Remove the skins and core from 5 raw apples. Blend or mash them to make a pulp, and mix thoroughly, adding some honey according to your taste. Then add:

powdered cardamom $\frac{1}{8}$ teaspoon

saffron 1 pinch

nutmeg 1 pinch

rose water 10 drops

Enjoy $\frac{1}{2}$ cup of this dessert at least an hour after meals.

NOTE: Avoid milk, yogurt, and fish for at least 4 hours before and after eating this apple dessert.

FIGS AND HONEY. After breakfast, eat 3 figs with 1 teaspoon honey. An hour later, take a glass of lassi. (See [this page](#) for lassi recipe.) This will help restore sexual energy.

GARLIC AND ONIONS. Adding some more garlic and onions to the diet is helpful. However, these foods are not considered good for meditation as they are somewhat dulling to the mind, so if that is a concern of yours, you can skip this and the following two

recommendations.

- Garlic milk is said to have aphrodisiac qualities. Mix together 1 cup milk, $\frac{1}{4}$ cup water, and 1 clove chopped garlic. Boil gently until only 1 cup of liquid remains, and drink at bedtime.
- Take 1 tablespoon onion juice mixed with 1 teaspoon fresh ginger juice twice a day.

OTHER REMEDIES FOR MEN AND WOMEN

HERBAL REMEDY. Low libido can be effectively treated with this herbal formula:

shatavari 1 part

vidari 1 part

nutmeg $\frac{1}{8}$ part

tagar $\frac{1}{2}$ part

Take 1 teaspoon of this mixture in the morning and another teaspoon in the evening, with warm milk. Continue taking it twice a day for 1 month.

LOW LIBIDO AND CONSTIPATION. These two conditions are often found together. To easily overcome constipation, take *triphala* daily, $\frac{1}{2}$ teaspoon at night with warm water. Brew the *triphala* powder in about a cup of hot water, let it steep for 10 minutes, strain, and drink.

FOR PSYCHOLOGICAL AND EMOTIONAL ISSUES. If stress and psychological problems such as anxiety or hostility are contributing to your state of low libido, it will be helpful to meditate regularly, practice yoga *asanas*, and do breathing exercises. Particularly helpful *asanas* include *Vajra*, Rooster, Camel, and *Nataraj*. (Guidelines for yoga postures are found in [appendix 4](#).)

Memory Problems

Everyone has occasional episodes of forgetfulness. Forgetting a name, date, or other information that seems to be on the tip of the tongue can make a person frustrated. One may forget where one has parked the car or left one's keys. Such experiences are common to everyone. On the other extreme, a severe form of forgetfulness happens in Alzheimer's disease, in which one may fail to recognize one's wife, husband, children, friends, or family members.

Memory problems are often the result of an inadequate supply of nutrients to the brain. Also, as age advances, memory frequently becomes poorer. Alcohol destroys brain cells, so memory loss among alcoholics is quite common. Memory problems may also be due to the use of drugs such as LSD, marijuana, and cocaine, as well as to alcohol abuse.

All these can damage the brain tissue and memory will be affected.

According to Ayurvedic principles, memory is recorded on the sensitive film of the nerve cells within the brain, which are of a kapha nature; memory is enlivened and brought back at a proper time by means of vata. Most memory problems are due either to stagnation of kapha or to aggravation of vata dosha, with its light, airy, even spacy qualities. Thus, in order to improve memory, we need to control vata and kapha. Pitta, on the other hand, is sharp and penetrating and supports good memory.

The following recommendations will help you improve your memory and prevent memory loss.

MEMORY FOODS

CARROTS. Carrots, which contain carotene, are good for memory. Carrots also enhance pitta, which brings sharpness of recall. Drink carrot or beet juice, both of which are blood builders and will help improve memory.

KITCHARI FAST. Kitchari is a simple, nourishing dish made essentially of 50 percent basmati rice and 50 percent split yellow mung dal, with some spices usually added for flavor. A five-day kitchari fast, using plain kitchari with just some chopped cilantro leaves added, will cleanse the system and help to strengthen memory. (For more delicious kitchari recipes, use your imagination or consult a cookbook such as *Ayurvedic Cooking for Self-Healing* by Usha Lad and Vasant Lad; see [Bibliography](#).)

Simple Kitchari Recipe

1 cup basmati rice

1 cup yellow split mung dal

1 small handful cilantro leaves, chopped

6 cups water

Wash the rice and mung dal twice, using plenty of water. If you have time, let the mung dal soak for a few hours before cooking, to help with digestibility.

Add rice, dal, and cilantro to the water.

Bring to a boil, and boil 5 minutes uncovered, stirring occasionally.

Turn down heat to low, and cover, leaving the lid slightly ajar. Cook until tender, about 25 to 30 minutes.

FRUIT FAST. A 3- to 5-day fruit fast, with *triphala* at night ($\frac{1}{2}$ teaspoon steeped for 5 to 10 minutes in a cup of hot water), will sharpen the nervous system and make the brain

more capable of investigating deeply into the memory. *Note:* A fruit fast should not be attempted by anyone who tends to get hypoglycemia.

The following fruits are recommended for the fast:

For vata	Papayas, prunes, mangoes
For pitta	Grapes, pomegranates, apples
For kapha	Apples, cranberries, pomegranates

OTHER FOODS. Additional foods that are helpful for improving memory include sweet potatoes, tapioca, okra (frequently used as a brain tonic), and spinach, which promotes pitta and stimulates memory. *Sattvic* foods in general are good for memory. Foods traditionally considered most *sattvic* include fresh fruits and vegetables, almonds, oranges, ghee, and milk. One food that is particularly bad for memory is heavy meat. This should be strictly avoided by anyone with memory problems.

HERBS FOR MEMORY

- Ginkgo and gotu kola have recently been touted as good for improving memory, and indeed they are truly beneficial. Both of these herbs serve to dilate the cerebral blood vessels, increasing circulation to the brain, and are good memory tonics.

- In Ayurveda there are specific herbs known as *medhya* herbs. *Medhya* means “that which improves memory.” The first and foremost is *brahmi* (which is similar to gotu kola). Also, *jatamamsi*, *bhringaraj*, and *shanka pushpi* are valuable for the brain and memory. You can use these herbs separately, or mixed together in equal proportions to make a tea.

brahmi

jatamamsi

bhringaraj

shanka pushpi

Steep 1 teaspoon herbs in 1 cup hot water for 10 minutes, and drink it on an empty stomach in the morning and evening. Continue taking this tea for 1 month to help improve circulation to the brain and eliminate memory problems. If it seems to be beneficial, you can continue using it indefinitely.

- Drink *brahmi* milk— $\frac{1}{2}$ teaspoon *brahmi* boiled for a couple of minutes in a cup of milk—at bedtime. It will improve your memory remarkably. Add a pinch of saffron for increased benefit. You can drink *brahmi* milk every day for a month, or continue indefinitely.

- A teaspoon of *brahmi* ghee, taken 5 to 10 minutes before breakfast and before dinner, also helps.
- Aloe vera gel (1 tablespoon) with a pinch of black pepper and ¼ teaspoon *bhringaraj* powder, taken 2 or 3 times a day, is also beneficial.

PRACTICAL TIPS FOR GOOD MEMORY

WRITE IT DOWN. Write down important information in order to remember it. Then you can go over it, and over it. Also, make a list of things you want to do or buy. Then you won't forget the milk or bananas!

DEVELOP A POETIC ATTITUDE. Think in rhymes and rhythms. In the oral traditions of India and other countries, students memorize vast amounts of knowledge that is usually set in rhyme and rhythm.

ASSOCIATE. Use association to aid memory. Suppose somebody tells you his name. Try to associate that name with something familiar. You might even imagine a picture to go along with the sound.

MEDITATION ON FORGETFULNESS. There is an ancient Vedic technique to recapture a lost memory: If you forget something, just sit quietly and stay in the forgetfulness. Breathe into the forgetfulness, and try to dig out the memory. Suddenly it will come back!

OTHER REMEDIES

EXERCISE. Daily walking, especially fast walking if you are fit for it, improves circulation and helps strengthen memory. Walk for half an hour, 5 times a week, Monday to Friday.

YOGA POSTURES. Yoga postures are helpful, especially the inverted poses (Shoulder Stand, Headstand, Plow pose, and Camel pose), which help to bring more blood to the brain. The Bow and Cobra poses are also helpful, as is Savasana, the yogic rest pose. Also do the Sun Salutation, 12 cycles a day.

OIL MASSAGE. *Brahmi* oil rubbed on the soles of the feet and on the scalp stimulates cerebral neural receptors under the skin, which send messages to the brain cells and can activate memory.

NASYA. The nose is the doorway to the brain and memory. Nose drops of warmed-up *brahmi* ghee, 5 drops in each nostril, can help to improve memory. (See [appendix 2](#) for instructions on how to make medicated oils and ghees.)

PRANAYAMA. Alternate Nostril Breathing helps to improve cerebral circulation (see [chapter 6](#)).

MEDITATE FOR BETTER MEMORY. Memory problems may be caused by stress, anxiety, and worries. Regular meditation is beneficial for relieving stress. Try the Empty Bowl or So-Hum meditation explained in [chapter 7](#).

AVOID TOXIC SUBSTANCES. A person who has memory problems should avoid alcohol and marijuana. Also, strictly avoid drugs that will directly affect the brain, such as LSD. Smoking cigarettes can also adversely affect memory, as nicotine toxicity constricts the cerebral blood vessels, which will damage the brain cells. Some research indicates that certain commonly prescribed medications, such as diazepam (Valium), may also seriously impair memory.

These do's and don'ts will be effective in protecting and improving your memory.

⌘⌘ Menopause Problems ⌘⌘

Menopause is a natural phenomenon. The body stops producing female hormones, and menstruation ceases. In addition to their reproductive and other functions, female hormones are necessary for regulation of bone metabolism. That is why some women may end up with osteoporosis. This is particularly true of vata and pitta individuals. (See "[Osteoporosis](#)" for suggestions on preventing this condition.)

Some women may experience menopausal syndrome, characterized by hot flashes, retention of water, and mood swings. Some may have insomnia. The following recommendations will help you move through this natural stage of your life gracefully and comfortably.

DIETARY GUIDELINES. The main dietary recommendation for menopause is to follow the vata-pacifying diet (see [chapter 8](#)). This is true for all constitutions but is especially crucial for vata types.

ALOE VERA. Taking fresh aloe vera gel (1 teaspoon 3 times a day) will help prevent and relieve uncomfortable symptoms.

MINERAL SUPPLEMENTS. It is important to take some mineral supplements. Specifically, take a formula of calcium, magnesium, and zinc that provides a daily amount approximately as follows:

calcium	1,200 mg.
magnesium	600 mg.
zinc	60 mg.

Natural Hormone Sources

It has become commonplace for modern medicine to prescribe hormone replacement therapy for menopausal women. Ayurveda has long recognized the value of female rejuvenating herbs at this stage of life, to prevent and/or alleviate menopausal syndrome. However, these herbs, rather than being a synthetic formula, provide your body with natural food precursors of estrogen and progesterone.

The herbs *shatavari* and wild yam (which is similar to the Ayurvedic herb *vidari*) are most effective. A mixture of the two will be strengthening and healing to your system.

vidari or wild yam ½ teaspoon

shatavari ½ teaspoon

Take this formula twice a day after lunch and dinner during the entire menopausal stage, with a few sips of warm water or ½ cup of aloe vera juice.

Take these supplements at night. They should help with menopausal symptoms such as hot flashes, and they can also help prevent osteoporosis.

FOR HOT FLASHES. Try drinking 1 cup of pomegranate juice with 1 teaspoon rock candy powder or organic sugar and 5 to 10 drops of lime juice. You can drink this 2 or 3 times per day, as needed to relieve hot flashes.

FOR VAGINAL DRYNESS. To alleviate vaginal dryness, you can pour some sesame oil on a piece of sterile cotton, mold it into a tamponlike shape, and insert it into the vagina at night. Use a sanitary pad to catch any leakage of the oil. (Consider tying a clean thread or string to the cotton to help you remove it in the morning.)

HELPFUL YOGA POSTURES. Certain yoga *asanas* are beneficial. Do the Sun Salutation at least 12 cycles per day, as well as postures that will strengthen the lower abdominal area, such as the Lotus, Locust, Bow, Boat, and Spinal Twist. Leg Lifts and the Chest-Knee pose are also helpful. (See [appendix 4](#).)

☞ ☞ Menstrual Difficulties ☞ ☞

See also “[PMS](#)”

TYPES OF DYSMENORRHEA

According to Ayurvedic theory, painful or difficult menstruation (dysmenorrhea) can be divided into three types: vata, pitta, and kapha. It is important to know as specifically as possible what the problem is (vata, pitta, or kapha) in order to treat it

effectively. Please note that these difficulties are independent of one's *own* constitutional type. That is, somebody with a pitta constitution could be having vata-type menstrual problems. So read the following descriptions carefully and compare them to your own experience.

- *Vata*. More pain before the onset of menstruation, bloating, lower abdominal pain, lower back ache, constipation, cramps, and insomnia are associated with vata-type dysmenorrhea. Menstrual flow tends to be scanty.

- *Pitta*. Congestive dysmenorrhea is a pitta problem. Pitta creates congestion, inflammation, and irritation. The breasts become tender, the bladder becomes sensitive, and there can be a burning sensation while passing urine. There may be hot flashes and irritability. It may lead to profuse menstrual flow.

- *Kapha*. In kapha-type menstrual difficulties, there is also congestion, and pain occurs more in the later part of the menstrual period, associated with white discharge, bloating, water retention, feelings of heaviness, lethargy, and drowsiness. The woman feels like sleeping during the daytime.

TREATMENTS FOR DYSMENORRHEA

Now for some effective treatments for each type of disorder.

VATA. For vata-type menstrual discomfort, make an herbal compound of

ashwagandha

vidari

tagar

Mix equal proportions of these herbs together, and take 1 teaspoon of the mixture with warm water after lunch and dinner.

Also effective for relieving vata-type cramps and discomfort is applying castor oil on the lower abdomen.

Take 1 tablespoon of aloe vera gel with 2 pinches of black pepper 3 times a day until the cramps subside.

PITTA. Women with pitta-type symptoms should use

shatavari 2 parts

kama dudha 1/8 part

musta 1 part

This formula is effective for pitta-type menstrual pain. Take 1/2 teaspoon of this mixture twice a day with warm water, after meals.

You will also find application of some coconut oil on the lower abdomen quite soothing.

KAPHA. Kapha-type menstrual problems can be effectively treated by using a mixture of

punarnava ½ part

manjistha ½ part

trikatu ⅓ part

Take about ½ teaspoon of this mixture twice a day, after lunch and dinner, washed down with warm water.

Application of mustard oil and castor oil (half and half) to the lower abdomen will also help with kapha-type menstrual discomfort.

NOTE: This may be a good time to remind you of an important Ayurvedic principle. If you are finding that your self-treatment is not working or even appears to be making the condition worse, you may have made a mistake in your diagnosis of the problem.

Don't give up! Remember, you are not a trained Ayurvedic diagnostician. Simply reassess the situation and try a different line of treatment.

FOR ALL BODY TYPES

HERBAL REMEDIES. Ayurvedic medicine includes a series of powerful herbal compounds based on the herb *guggulu*. In addition to other healing qualities, these compounds are especially good for regulating menstruation. For vata-type pain, use *triphala guggulu* or *yogaraj guggulu*. Pittas may use *kaishore guggulu*. Kaphas will do best with *punarnava guggulu*. In each case, take one tablet twice a day. These tablets are generally available from sources of Ayurvedic herbs (see [Resources](#)).

- Another universal remedy that should bring some relief from menstrual pain is to roast some cumin seeds in an ungreased pan until they smell pungent. (It will just take a few minutes.) When they are cool, chew about 1 teaspoon slowly, and follow with one tablespoon aloe vera juice.

FOR EXCESSIVE MENSTRUAL FLOW. A tea made of raspberry leaves and hibiscus flowers (equal amounts, 1 to 2 teaspoons of herbs per cup of water) is often quite effective.

- You might also try drinking a cup of coconut water (the natural juice inside a fresh coconut) with ½ teaspoon rock candy powder or natural sugar added.
- Eating about 10 to 20 fresh raspberries on an empty stomach, up to 2 or 3 times a day, may be helpful.

STRATEGY TO PREVENT MENSTRUAL PROBLEMS

Perhaps more important than all these pain remedies is a strategy to *prevent* menstrual problems. You can follow this strategy quite easily, effectively, safely, and inexpensively.

ALOE VERA GEL. For the entire week before you expect your period to start, take 1 tablespoon of aloe vera gel, 3 times a day. This will help prevent all types of menstrual pain and discomfort.

DIETARY GUIDELINES. Throughout the month, follow the dietary guidelines for your constitutional type (see [chapter 8](#)).

YOGA POSITIONS. Throughout the month, spend a few minutes every day on the yoga postures recommended for your body type (see [this page](#)). Yoga *asanas* are not recommended during the menstrual period. Just rest, read, and relax as much as possible.

If you follow these recommendations, your menstrual problems may soon be nothing but a memory.

☼ ☼ Migraine ☼ ☼

See also “[Headaches](#)”

Although migraine headaches can result from a vata, pitta, or kapha imbalance, they most frequently occur when systemic pitta moves into the cardiovascular system, circulates, and affects the blood vessels around the brain. The hot, sharp quality of pitta dilates the blood vessels and creates pressure on the nerves, causing this painful condition.

FOLLOW THE PITTA-PACIFYING DIET. To treat migraine headaches, it is vital first to take care of pitta with a proper pitta-soothing diet. (See the dietary guidelines in [chapter 8](#).) Especially avoid hot, spicy foods, fermented foods, and sour or citrus fruits. Carefully following a pitta-soothing diet is effective both for migraine relief and as a preventive measure.

PREVENTIVE BREAKFAST. Certain individuals get migraines at midday, which then subside later in the evening. For such individuals, try this preventive method. It may sound too simple, but it is effective. First thing in the morning, take 1 ripe banana. Peel it, chop it into pieces, and add 1 teaspoon warm ghee, 1 teaspoon date sugar, and a pinch of cardamom on top. This is delicious, and it will help to reduce pitta and prevent a headache from arising.

HERBAL REMEDY. The following herbal compound will be beneficial:

shatavari 5 parts

brahmi 4 parts

jatamamsi 3 parts

musta 3 parts

Prepare this mixture, and take ½ teaspoon twice a day, morning and evening, after breakfast and dinner, with a little lukewarm water. This formula is designed to pacify the aggravated pitta and help relieve migraine headaches.

AVOID DIRECT SUN. Because migraine headaches are predominantly a pitta disorder, they are affected by the hot sun. When the sun rises, its hot, sharp, penetrating rays increase pitta in the cardiovascular system and cause the dilation of the blood vessels in the brain, which results in the painful headaches. So it is important to avoid direct exposure to the sun, and if you do go out in the sun, wear a hat.

SOOTHING NOSE DROPS. Once a headache has developed, putting about 5 drops of warm *brahmi* ghee in each nostril will help relieve the pain.

RECOMMENDED YOGA POSTURES. Generally, a person having migraines should do the Moon Salutation (see [appendix 4](#)). Helpful yoga postures include the Hidden Lotus, Boat pose, Bow pose, Spinal Twist, Palm Tree pose, and Standing on the Toes.

A COOLING PRANAYAMA. You will also find a cooling breathing exercise such as *shitali* helpful (see [chapter 8](#)).

A HEALING YAWN. When you have a migraine, gently squeeze your earlobes, pulling the ear down, and do the act of yawning. That will relieve the pressure on the blood vessels and help to pacify the headache.

See “[Headaches](#)” for a detailed analysis of vata, pitta, and kapha headaches and their proper treatment.

☞ ☞ Morning Sickness ☞ ☞

Early in pregnancy, morning sickness—nausea and vomiting in the early morning, immediately or soon after waking up—is quite common. This condition is due to aggravation of pitta and is especially common among pitta individuals. It usually occurs from approximately the sixth to the tenth week of pregnancy.

Some medical researchers have said that the level of estrogen in the blood rises during pregnancy. Estrogen is pittagenic. The higher estrogen level in the blood triggers pitta in the stomach and increases acid secretion, so that the stomach becomes more acidic. In the early morning, when the stomach is empty and acidic, morning sickness can occur.

Also, certain odors can trigger nausea and vomiting at any time of the day or night, but sensitivity to smell is greatest in the morning. During pregnancy, a woman's sense of smell becomes unusually sensitive. The reason is interesting. According to Ayurveda, the earth element is particularly prominent during pregnancy, because the fetus is building and growing and the earth element is responsible for solidity and structure. The earth element is also associated with the sense of smell. (In Ayurveda, the senses are related to the elements like this: Space = touch; Air = hearing, Fire = sight; Water = taste, Earth = smell.)

Ayurvedic literature talks poetically about morning sickness, saying that it is common among women whose babies will develop a copious head of hair after birth.

START THE DAY WITH A LITTLE FOOD. Believe it or not, the first thing to do upon waking up in the morning is to put something in your stomach. Have a little light food. You might try some crackers with a slightly salty taste. Salt is generally pitta-provoking, but a small quantity of salt stimulates salivary secretion and helps to reduce pitta. Fresh lime juice is also helpful, with a little salt and sugar.

EAT OFTEN. If you are troubled by morning sickness, eat small frequent meals, as many as five or six in a day. An empty stomach has more acid secretion, and irritation and nausea easily follow.

COCONUT WATER. Add 1 teaspoon of lemon juice to a cup of coconut water (the natural juice inside a fresh coconut), and take a sip every 15 minutes to settle your stomach.

EFFECTIVE HERBAL REMEDY. The following herbal formula is effective for settling morning sickness:

shatavari 5 parts

shanka bhasma 1/8 part

kama dudha 1/8 part

Take 1/2 teaspoon of this mixture with lime juice or lemon juice in the early morning and at bedtime to reduce acid secretion in the stomach and relieve nausea.

TRY ALMONDS. Soak 10 raw (unroasted) almonds overnight, and next morning peel off the skin and eat them. In addition to providing a high-quality protein, almonds are a good source of calcium. Pregnant mothers need both. And they settle the stomach.

WALK IT OFF. Sometimes walking in the fresh morning air helps with morning sickness, because fresh cool air reduces pitta. It also helps to relieve stress. It sometimes happens that a pitta woman, working for a demanding boss and coming home to a criticizing husband, develops some unresolved anger. Stress builds up in the solar plexus and manifests as morning sickness. It is important for her to reduce her stress level by doing some early morning walking or other appropriate exercise.

MINIMASSAGE. In the morning, before taking a shower, warm up 3 to 5 ounces of sesame oil (for vata constitutions), coconut oil (for pittas) or sunflower oil (for kaphas), and rub it over your body for 5 to 10 minutes. Be sure to get some on your scalp and feet. Then take a nice warm shower. This soothing oil massage (called *abhyanga*) will minimize stress and help with morning sickness.

MILK WITH ROSE WATER. Buy some rose essence or rose water. When you are feeling nauseated, put 1 drop in a cupful of milk, boil the milk, and then drink it warm. (You can just as well use 5 fresh rose petals from your garden to boil in the milk.) This will help to eliminate the nausea. As a preventive measure, taking a cup of this rose milk with a teaspoon of ghee at bedtime will pacify pitta and help to control morning sickness.

DRINK A LOT. Vomiting can lead to dehydration, so you need to drink extra liquid to compensate. Better still, make some homemade dextrosaline by adding 2 teaspoons sugar, the juice of ½ lime, and a pinch of salt to 1 pint of water. Drink 1 cup every 2 hours to eliminate dehydration and help the nausea and vomiting subside.

PEACOCK FEATHER ASH. This remedy might sound strange, but it is quite effective. Ayurvedic literature says that morning sickness can be remedied by using peacock feather ash. Burn a peacock feather in such a way that you can collect the ash. The smell of the burning feather is quite awful, so someone else should burn the feather, not the pregnant woman. Take just a pinch of the powdery ash with a teaspoon of honey. This can immediately stop morning sickness.

EFFECTIVE HERBAL FORMULA. No matter what your constitutional type is, this formula will be beneficial for you:

shatavari 5 parts

kama dudha ⅛ part

shanka bhasma ⅛ part

moti bhasma ⅛ part

Take ½ teaspoon 2 or 3 times a day with ghee. Or, if you don't want to use ghee, use warm water. Sometimes during pregnancy women may not feel like taking ghee.

❧ ❧ Muscle Cramps and Spasms ❧ ❧

While running, walking, riding a bicycle, standing up for a long time, doing yoga stretches incorrectly, or even sleeping, a person can get muscle cramps. Any muscle—upper arm, forearm, leg, calf, or even a toe or the baby finger—can undergo sudden spasm.

Muscle cramps and spasms have numerous causes. They may be due to insufficient blood supply to the muscle or, in some cases, to excess blood supply, as in the case of “writer’s cramp.” Muscle spasms can be due to lack of calcium in the diet or malabsorption of calcium, since calcium plays a significant role in relaxing the muscle. A related problem is hypoparathyroidism (underactive parathyroid), in which the person loses calcium; this may also cause the muscles to undergo spasm. Exposure to cold and poor circulation may also be responsible.

The Ayurvedic understanding is, in simplest terms, that muscle spasms are caused by vata dosha. Vata, increased by mobile, cold, or rough qualities, makes a muscle stiff and hard, and it undergoes spasm.

FOR IMMEDIATE RELIEF. Whenever a muscle is under spasm

- Grab hold of it.

- At the same time, press your index finger deeply into the “belly” of the muscle (the bulging central part of the muscle) for 15 to 20 seconds. A *marma* point (like an acupressure point) is located in the center of the muscle; pressing on it helps the muscle relax.

- Take a few deep breaths.

This procedure will increase circulation, and the muscle will relax.

ANOTHER PRESSURE POINT REMEDY. A *marma* point is located at the midpoint of both the lips. If there is a spasm in the upper body, such as in the arms or fingers, grab hold of your upper lip, in the middle, between your thumb and index finger. If the spasm is in the legs or the lower part of the body, grab the lower lip.

These energy points on the lips send messages to the brain, which in turn relays a signal to the motor system to relax the muscles. Just grabbing the lips for 30 seconds should help relieve the spasm. Press fairly strongly, but not hard enough to be painful.

CONTRACT AND RELAX THE MUSCLE. Repeatedly alternate contraction and relaxation of the muscle. This will improve circulation and help to release lactic acid, which will relax the muscle.

MASSAGE. Another approach is to apply a little oil to the muscle belly and gently massage. *Mahanarayan* oil is best if you have it; otherwise, some sesame oil or other oil will do. Gentle massage of the painful cramped muscle will relax the muscle fibers, improve the circulation, and pacify the vata. That will help relieve the spasm.

After rubbing in the *mahanarayan* oil, apply a little heat. For best results, use a hot water bottle (not an electric heating pad).

Winter, with its dry, cold weather, is vata season. During winter season, when vata individuals get exposed to the cold, their muscles may cramp. Rubbing the muscle with *mahanarayan* oil (or sesame oil) and then applying heat will be soothing and healing.

SOAK IT. For a cramp in your foot, soak the foot in a bucket of warm water that contains salt (1 or 2 tablespoons).

A hot ginger powder-baking soda bath (2/3 cup baking soda and ¼ cup ginger per tubful of water) is also effective for muscle relaxation.

HERBAL RELAXANTS. Make yourself a cup of chamomile tea, *jatamamsi* tea, or comfrey tea. Or, even better, you can make a tea using equal amounts of all three of these herbs (⅓ teaspoon of each per cup). It will help to relax your muscles.

FOR PREVENTION. As a preventive measure against future cramping, drink some *dashamoola* tea—½ teaspoon of *dashamoola* powder steeped for a few minutes in 1 cup hot water. Have a cup or two of *dashamoola* tea every Saturday.

MINERAL SUPPLEMENTS. Getting repeated muscle cramps suggests that you either have a calcium deficiency or are not absorbing enough of the calcium in your food. First, take some supplements of calcium, magnesium, and zinc. Your formula should contain approximately 1,200 mg. calcium, 600 mg. magnesium, and 60 mg. zinc. Take these supplements at bedtime.

Second, to improve absorption, take *triphala* every night or in the very early morning. Use about ½ teaspoon *triphala* powder per cup of boiling water.

YOGARAJ GUGGULU. *Yogaraj guggulu* tablets (200 mg. per tablet), taken 2 or 3 times a day for 1 month, are effective for pacifying vata in the muscles, which is the root cause of muscle cramps.

EFFECTIVE HOME TREATMENT FOR ABDOMINAL CRAMPS. Muscle cramps and cramps in the stomach may be related. Muscle pain can happen anywhere in the body, in the skeletal muscles as well as in the smooth muscles of the abdomen. As with spasm in the arm or leg, abdominal cramps may have many causes, such as eating too big a meal, or lifting too heavy a weight, which strains the abdominal muscles. Gases in the stomach, constipation, or acid indigestion can also create cramps in the stomach or abdomen.

- For painful cramps in the abdominal muscles, take the herbal compound *shankavati* (one 200-mg. tablet after dinner).

- This antacid formula may also be effective:

shatavari ½ teaspoon

guduchi ¼ teaspoon

shanka bhasma pinch

Take this entire amount once or twice a day, after meals.

- Warm milk is helpful for abdominal muscle cramps. Its alkaline property helps pacify acidity, and it is also a good source of calcium, which helps muscles to relax. One cup of warm milk taken at bedtime will help to dispel acid irritation and spasm of the

stomach.

- *Lasunadivati* (garlic compound) is effectively used for smooth muscle spasm as well as skeletal muscle spasm. Take 1 tablet after dinner for 5 days. You can also buy odorless garlic tablets; take them as directed on the package. Garlic relaxes the muscles, calms down vata dosha, and helps to take care of muscle cramps.

- *Hingwastak churna*, ¼ teaspoon taken twice a day after lunch and dinner, will also help relieve abdominal muscle pain. It may also create some gas, however.

- *Triphala* is very effective in relieving gases, promoting proper elimination, and facilitating absorption of calcium and other key minerals. Take ½ teaspoon of *triphala* daily at bedtime with warm water. It will minimize your chances of getting muscle cramps in the stomach.

❧ ❧ Nasal Crust ❧ ❧

Some individuals, when their systemic kapha dosha becomes high (perhaps due to eating too many kapha-increasing foods), become sensitive to pollen, dust, ragweed, cat hair, dog hair, and other allergens, as well as to cold temperatures. As a result, they may develop rhinitis, with nasal congestion and nasal discharge. Even in the absence of infection, dryness in the atmosphere may dry up the mucous membranes and nasal passage; to compensate for that, the body will produce more mucus. Then, due to continuing dry heat in the environment, the nasal discharge becomes thick, dry, and crusty. This is known as nasal crust.

People with deviated nasal septum can also accumulate nasal discharge, and because of dryness in the air it may form a crust. Nasal crust can create stuffy nose, sinus headache, and difficulty breathing. It can be one of the causes of snoring and of sleep apnea. Nosebleeds may also be due to nasal crust.

Ayurvedic medicine offers a number of effective remedies:

STEAM IT. The simplest remedy is to inhale steam. You can use plain water, water with some ginger boiled in it, or a tea of the following ingredients:

ginger

ajwan (Indian celery seed)

turmeric

Put equal amounts of each of these herbs in a pint of water, and boil it up. Then turn off the fire, put a towel over your head, and inhale the steam. This will ease out the discharge. The crust will come out, and you will breathe freely. Though simple, this is an effective remedy.

MENTHOL AND EUCALYPTUS. Rubbing menthol on the forehead and on the sinus area will help.

Placing a few drops of *mild* eucalyptus oil in the nose will also help.

NOTE: Do not use pure eucalyptus oil. Dilute a few drops of eucalyptus oil with sesame oil or some other mild oil so that it doesn't burn your skin or the sensitive tissue in the nose.

USE AN ONION. Chop up an onion, and sniff its fragrance. Onions contain ammonia, which is a powerful decongestant. It brings tears to the eyes and promotes sneezing. The tears from the eyes will pass through the tear ducts and into the nasal passage, which will lubricate and loosen the crust; then the sneezing will help the crust to be eliminated.

LUBRICATE THE NOSTRILS. Putting a few drops of *brahmi* ghee or saline solution into the nose will also lubricate the nasal passage and facilitate removal of the crust. You can make an effective saline solution by adding $\frac{1}{8}$ teaspoon salt to $\frac{1}{2}$ cup water.

BURN IT OUT. A meal of spicy food will also help. For example, a hot soup or vegetables spiced with cayenne pepper, curry powder, or chili pepper (within your limits of comfort!) will increase circulation and help to eliminate stuffy nose and nasal crust.

USE A HUMIDIFIER. At night, run a humidifier so that the room will be warm and moist. If possible, don't use an ultrasound humidifier. A hot-water type is best.

VITAMINS AND HERBS. Finally, take some or all of the following:

- Vitamin C—1,000 mg. (1 gram) twice a day
- *Amalaki* (a good source of vitamin C)—1 teaspoon at bedtime in warm water. (Don't use it if you are already taking *triphala* at night; *amalaki* is one of the ingredients of *triphala*.)
- Zinc—60 mg.
- *Sitopaladi churna*— $\frac{1}{2}$ to 1 teaspoon, with 1 teaspoon honey and 1 teaspoon ghee.

❧ ❧ Nausea and Vomiting ❧ ❧

See also "[Morning Sickness](#)"

Nausea and vomiting have numerous possible causes, including excess acid secretion, toxins in the liver, pregnancy, worms in the colon, food poisoning, and flu. (For suggestions on reducing nausea and vomiting during pregnancy, see "[Morning Sickness](#).”)

In the event of food poisoning or excess acid secretion in the stomach, vomiting occurs as a protective response of the body to get rid of toxins. With flu also, excess bile may build up in the stomach, and vomiting occurs to cast it out. In such cases vomiting is a sign of health, of the body taking care of itself.

But when vomiting becomes persistent, it may lead to dehydration or other problems, and it has to be stopped. Morning sickness, for example, may adversely affect the flow of nutrition going to the fetus.

Ayurveda recommends quite a few effective ways to relieve nausea and put a stop to vomiting.

PACIFY PITTA. Nausea and vomiting indicate high pitta in the stomach, with increased acid secretion irritating the gastric mucous membrane. Therefore, it is good to follow a pitta-soothing diet, especially abstaining from hot, spicy food or fermented food.

TRY FASTING. Fasting gives a healing rest to the digestive system. Don't eat for a day—and drink 1 cup of sweet fresh pineapple juice with a pinch of ginger, a pinch of black pepper, and ½ teaspoon organic sugar. Take this 3 times during the day.

- Alternatively, you can drink cranberry juice or pomegranate juice on your fast.

EIGHT WAYS TO SETTLE NAUSEA AND VOMITING. Here are eight simple and effective suggestions to help you soothe nausea and vomiting:

- To 1 cup of water, add 10 drops of lime juice and ½ teaspoon sugar. Last, add ¼ teaspoon baking soda. Stir and drink. This can immediately stop nausea and vomiting.
- An effective remedy is to chew 1 or 2 cardamom seeds.
- A mixture of 1 teaspoon ginger juice (or freshly grated ginger pulp) and 1 teaspoon onion juice will help to settle nausea and vomiting.
- Make a mixture of equal parts lemon juice and honey. Dip your index finger into the mixture and lick it, consuming the mixture slowly.
- Try stirring ½ teaspoon honey and 2 pinches of cardamom into half a cup of plain yogurt.
- A tea made from 1 teaspoon cumin seeds and a pinch of nutmeg steeped in a cup of hot water will be quite soothing.
- Drinking sugar cane juice can also be helpful, as is cranberry juice with a little lime juice added.

• Ayurveda also recommends the following herbal formula to quickly stop nausea and vomiting:

rose petal powder ½ teaspoon

sandalwood powder ¼ teaspoon

rock candy powder ½ teaspoon

lime juice 10 drops

Take this entire mixture in room temperature water.

Vomit for Healing

Most people find vomiting a rather unpleasant experience, but there are times when one might want to induce vomiting. When someone has a flu or a bad cold, excess kapha may build up and the person suffers from persistent headache, congestion, and coughing. Mother Nature may bring on vomiting to remove the kapha, but if that doesn't happen, Ayurveda suggests taking matters into your own hands.

Drink a glass of water with $\frac{1}{4}$ teaspoon salt dissolved in it. The salty water itself is emetic (provokes vomiting), but you can also rub the back of the tongue to stimulate the "gag" reflex and vomit out the water. The moment vomiting occurs, fever will generally come down, the headache will disappear, congestion in the chest will be greatly relieved, and you will feel much better.

FOR NAUSEA IN CHILDREN. Try giving the child some coconut water. Add 1 teaspoon lemon juice to a cup of coconut water (the natural juice inside a fresh coconut), and have the child take a sip every 15 minutes or so to settle the stomach.

FOR WORMS. Nausea and vomiting may also be a sign of worms. If a person has a history of passing worms in the stool and gets repeated attacks of nausea and vomiting, use the following strategy:

- Take the herb *vidanga*, about $\frac{1}{2}$ teaspoon twice a day with a little warm water.
- Keep the colon clean by taking $\frac{1}{2}$ teaspoon *triphala* at night for several weeks. Mix the *triphala* into $\frac{1}{2}$ cup of warm water, let it steep for 10 minutes, then strain and drink.

TO DETOXYFY THE LIVER. Nausea and vomiting may be a sign that there are excess toxins in the liver. To detoxify the liver, the following formula is effective:

kutki $\frac{1}{4}$ teaspoon

shatavari $\frac{1}{2}$ teaspoon

shanka bhasma pinch

kama dudha pinch

Take this mixture 2 or 3 times a day with water to relieve nausea and vomiting.

⚡ ⚡ Nightmares ⚡ ⚡

Nightmares are quite common in children up to the age of 12 but are more rarely found in adults. The main causes of nightmares in adults are (1) fears, anxieties,

worries, and other psychological stresses, and (2) eating too much food too late at night. There may be other physical causes, such as problems with adenoids, or sleep apnea, or nasal crust that doesn't allow adequate breathing. Whenever there is cerebral hypoxia—lack of oxygen and *prana* to the brain—the person can get nightmares. This can occur even due to insufficient fresh air in the room.

TREATING NIGHTMARES IN CHILDREN

The main cause of nightmares in children is psychological—fear and anxiety based on frightening images they have seen or scary stories they have heard. So it is important not to feed the child's imagination with disturbing images. Keep children busy with creative play; don't let them watch violent or horrifying television programs or read frightening stories.

The child's room should be pleasant and filled with sweetness, with nice music and perhaps some tinkling bells. Bells have a happy sound that children like. You can tell the child, "When there is a bell, monsters won't come." And they'll sleep peacefully.

Rather than watching frightening or violent movies, tell children positive, uplifting stories, such as from the *Ramayana*, about baby Krishna, or any other beautiful, happy story.

Give the child a mini-oil massage before bed. Rub some oil, especially *brahmi* oil or *bhringaraj* oil, on the soles of the feet and on the scalp. This will help relax the child. (See [appendix 2](#) for instructions on making herbalized oils.)

Sometimes nightmares come as a result of bedwetting. To help prevent this, see that the child doesn't drink much for at least two hours before going to sleep. Some cumin-coriander-fennel tea (again, not just before bedtime) can help prevent bedwetting.

REMEDIES FOR ADULTS AND CHILDREN

TRANQUILLITY TEA. Make a tea from equal proportions of

jatamamsi

brahmi

ginkgo

yashti madhu (licorice root)

Drinking a cup of this tea (made from 1 teaspoon of the herbal mix steeped in 1 cup hot water) before going to bed will help create a more peaceful mind and body. This tea is good for children as well as adults.

- You can also make a similar tea of equal amounts of *jatamamsi* and *shanka pushpi*.

HERBS FOR ALLERGIES. If allergies are the cause of the nightmares, you can help remedy them with *sitopaladi* and *yashti madhu*. Mix them in equal proportions, and take ½ teaspoon of each with honey twice a day. For children, use ¼ teaspoon of each.

JATAMAMSI. If you sew an ounce or two of the herb *jatamamsi* inside a small silk bag and place it under the pillow, its fragrance will help create a tranquil night.

NASYA. Doing *nasya* will help. Put 2 to 3 drops of warm ghee or any Ayurvedic nose drops (such as *brahmi* ghee) in each nostril, and inhale (see [appendix 3](#)). *Nasya* is equally effective for children.

Are Your Dreams Vata, Pitta, or Kapha?

To eliminate nightmares, it can be helpful to know if they are the result of a doshic imbalance of kapha, pitta, or vata, so that the imbalance can be corrected. By analyzing the nature and content of the dreams, it is usually possible to find out.

- **Vata dreams.** Vata dreams are active and hyper. They are plentiful, and the dreamer may well forget them in the morning. Horror, fear, running, jumping, flying high in the sky, falling deep down into a valley, being attacked or pursued, being locked up—these are vata dreams.

- **Pitta dreams.** These dreams can be rather violent. In addition to such themes as teaching, studying, trying to solve a problem, or failing at an examination, pitta dreams may involve fire, war, nuclear weapons, fighting, killing, and murder.

- **Kapha dreams.** Kapha dreams are generally mild and romantic. Water figures prominently, such as swimming in the ocean. Seeing gardens, lotus flowers, swans, and elephants and eating candy are all characteristic of kapha dreams. Drowning, or seeing oneself as dead, are “negative” aspects of the kapha dream repertory.

EAT EARLY. Eat dinner before seven o'clock. Eating too late at night may create nightmares.

CUT DOWN ON STRESS. Yoga *asanas*, regular exercise, Alternate Nostril *pranayama*, and meditation morning and evening help to relax the nervous system and reduce stress. (You will find instructions for *pranayama* in [chapter 6](#) and guidelines for meditation in [chapter 7](#).)

CRYSTALS. Wearing crystals, or putting some amethyst crystals on the four corners of the bed, may also help. You can tell your child, “Look, I’ve put these crystals around your bed; it will keep the ghosts and monsters away.” They will feel comfortable and will sleep well.

The root cause of nightmares is wrong thinking: negative imagination, loneliness, isolation, fear, disturbed relationships. That is why prayer, positive thinking, positive

affirmations, and positive imagination are the best medications for nightmares.

❧ ❧ Nosebleed ❧ ❧

Nosebleed has many possible origins. Trauma to the nose, extreme dryness of the nasal passage so that the nasal mucous membranes crack and bleed, allergies, rhinitis, a nasal polyp, or high blood pressure may all cause nosebleed. Going to a high altitude or consuming excess alcohol are also possible causes.

Usually we don't have any time to investigate the cause; we need to treat it immediately. Here are several effective remedies:

DRINK COOL WATER. This alone will stop many nosebleeds.

USE A COLD COMPRESS. Dip a handkerchief or any clean soft cloth into cold water, and place it on the forehead and the nose. Then *gently* blow the nose so that the clot, if there is one, comes out. (If there is a dry nasal crust, it will irritate the nasal passage and cause bleeding.)

SNIFF COLD WATER. Take a little cool water in your palm, inhale it up into the nose, and gently blow the nose.

SQUEEZE YOUR NOSE. If the cold water doesn't stop the bleeding, then pinch the nose with your thumb and index finger, as if you were about to dip under water. Hold for 2 or 3 minutes, breathing normally through the mouth. That should stop the bleeding.

GHEE OINTMENT. If the bleeding still doesn't stop, another simple remedy is to put a couple of drops of lukewarm ghee in each nostril. Dip a cotton swab into a jar of ghee, and apply it to the nose. Ghee is hemostatic—that is, it stops bleeding.

STAND OR SIT UPRIGHT. Don't lie down, which will encourage bleeding. Also don't do any inverted yoga postures such as Headstand, Shoulder Stand or Plow. Remaining upright will minimize bleeding.

HUMIDIFY YOUR SURROUNDINGS. Many nosebleeds are due to a dry nose, caused by hot, dry air. So as a preventive measure in dry climates or in winter when heating the house can create a lot of dry air, be sure to humidify your bedroom, workroom, or the entire living space. It is better not to use an ultrasound humidifier; a hot water unit is best.

HERBAL REMEDY. Orally, take a mixture of:

manjistha ⅓ teaspoon

kama dudha ⅓ teaspoon

Take this mixture with a little warm water twice a day.

JUICE CURE. To stop or prevent nosebleeds, you can drink cranberry juice, pomegranate

juice, or a half-and-half mixture of the two.

POMEGRANATE NOSE DROPS. When making the pomegranate juice, if you pick up a few drops of the fresh juice in an eyedropper and place them in your nostrils, it should instantaneously stop the bleeding.

TIME TO SEE THE DOCTOR

If, after trying these remedies, your nose is still bleeding, or if you have repeated nosebleeds over a couple of weeks, see your doctor. You may have a serious health problem. The bleeding may be due to high blood pressure, which needs to be cared for. Or it could possibly be caused by leukemia, a type of blood cancer.

FOLLOW THE PITTA-PACIFYING DIET. According to Ayurvedic principles, even though nosebleed has some vata symptoms, such as dryness and cracking of the nasal passages, it is essentially a pitta disorder, in which pitta becomes hot and sharp and causes the bleeding. So when you have a nosebleed, don't eat hot and spicy foods, abstain from alcohol and cigarettes, and don't work under the hot sun. These are all pitta-provoking.

❧ ❧ Obesity ❧ ❧

Obesity is a condition in which an individual is significantly overweight, and an excessive amount of body fat has accumulated under the chin and on the breasts, belly, buttocks, and/or thighs. Though it is not a serious disease in itself, it may shorten the span of life, as well as create diminished efficiency and a predisposition to diabetes, hypertension, low libido, and arthritis. Ultimately, obesity reduces happiness.

To a great extent, obesity is due to the socioeconomic problems of an affluent society. A prosperous life, a sedentary job, and lack of exercise are the major contributors to obesity. From an Ayurvedic point of view, the main causes of this condition are eating too much, sitting too much, and doing too little.

Obesity is a kapha disorder. In obese individuals the gastric fire is strong, but the cellular fire in the tissue is relatively low. Whatever excess food or calories a person consumes are not burned and instead turn into adipose tissue, leading to overweight and obesity.

There are numerous other potential causes. Certain hereditary factors of the endocrine system, such as excess production of growth hormone, may contribute to the condition. When women are pregnant, they may eat too much and be unable to lose the weight afterward. Stress may induce repeated emotional eating, leading to significant weight gain. Frequent munching between meals is also detrimental to maintaining a healthy weight. Certain drugs, including steroids and oral contraceptives, can change the metabolism and produce weight gain, as can insulin. Addictions, including alcohol

and cigarettes, are often associated with obesity. But the main factor is usually eating too much, along with insufficient exercise.

Habitually drinking cold drinks and eating fatty fried foods, dairy products such as cheese, yogurt, and ice cream, and consuming excess sugar and carbohydrates are all causative factors.

WATCH YOUR DIET. In treating obesity, the first step is to control what you eat. Follow the kapha-pacifying diet (see guidelines in [chapter 8](#)). Avoid habitually drinking cold drinks and eating fatty fried foods. Minimize dairy products such as cheese, yogurt, and ice cream. Be sure to include salad (without creamy dressings) and beans in your meals. Drink hot water instead of ice-cold drinks. Obese people generally hate hot water, but they should drink it, either plain or in herbal teas such as ginger, mint, or cinnamon.

If you like to eat meat, you can have some fish or chicken once a month, but no beef, lamb, or pork.

GET ADEQUATE EXERCISE. Do some regular exercise. A daily walk of at least half an hour is essential. And do some aerobic exercise, such as gentle jogging. Obese people hate jogging, but they should at least walk fast, carrying 2.5-pound hand-held weights. Swimming is also good exercise.

Lift weights to reduce body weight. Do some gentle weight lifting, using 5-pound weights to get started. This will help to burn adipose tissue. Also, muscle tissue burns calories more quickly than fat.

If you want to lose weight, you must understand some simple arithmetic. When you take in more calories than you burn off, you will gain weight. In order to lose weight, you *must* burn off more calories than you take in. In practical terms, this means two things: reducing your caloric intake, and increasing your output in the form of exercise. Follow the kapha-reducing diet, and increase the amount of exercise you do every day.

YOUR POSTEXERCISE PRESCRIPTION. Immediately after exercise, kapha individuals will feel hungry and thirsty and will want to rush to the restaurant to have a cold drink and a bite to eat. But the cold drink will slow down the metabolism, defeating the gain from the exercise. And we know that the eating will be counterproductive. So after exercise heavy and obese people should skip snacks and stay away from cold water and other cold drinks, and choose hot drinks such as herbal teas.

EAT YOUR BIGGEST MEAL EARLY IN THE DAY. According to Ayurveda, the best policy regarding meals, if you are overweight, is to skip breakfast entirely (maybe have some hot herbal tea), then take your biggest meal at noon. Supper should be light. Don't snack between the two meals.

If you can't seem to skip breakfast, take your main meal then, early in the day. Have a light lunch and a lighter supper; skip supper entirely if you can.

DROP ALCOHOL AND TOBACCO. Quit drinking alcohol and smoking. These emotional habits

unduly stimulate *jatar agni* (gastric fire) and make a person hungry.

LISTEN TO MUSIC. While eating your meals, listen to soft music and chew your food more, so that moderate eating gradually becomes habitual.

LEARN TO LOVE YOURSELF. Most obese people do not love themselves. This is significant because there is a deep relationship between food and love. Food is the food of the body; love is the food of the soul. When an individual is missing love in a relationship, he or she may try to find love through eating, and food will become a substitute for love. When a woman misses her husband or a man misses his wife, they frequently begin to eat too much.

TIME TO SEE THE DOCTOR

If you are very heavy and have not exercised for a long time, and especially if you are over 40, you *must* see your doctor before you start an exercise program more strenuous than walking.

Obese people also hate how they look, and they hate looking in the mirror because they don't like what they see. When people hate their body, they become anxious and worried; then suddenly they become hungry and need to eat. This is not real hunger; it is false, emotional hunger. (See "[Eating Disorders](#)" and "[Overeating](#)" for further discussion of emotional eating.)

To help develop more love for yourself just as you are, try this technique. Go into your bathroom and remove all your clothes, or wear some shorts. Then take a good look at the person you see in the full-length mirror. Look at that image, starting at the head, the eyes, cheeks, lips, and neck; look at your chest, your belly, and so on.

As you look at the image in your mirror, at the same time look within. Ask yourself, do you like those eyes of yours? Do you love your nose? Don't you have some affection toward your own lips, your own chest?

In this way, gradually, by looking outside, into the mirror, and at the same time inside at your own inner observer, two things will happen. First, you will begin to feel that you are beyond your body, that you are something higher, nobler, greater, and more beautiful, that you are pure existence. Second, this process will also bring greater acceptance, and you will start feeling more love for yourself. So look at yourself in the mirror, and love that person that you see.

A second important factor in loving yourself is to stop judging, comparing, and criticizing yourself. What you are is unique, and it is divine. Stopping judgment, criticism, and comparison is the beginning of self-love.

These two exercises will really help.

DRINK HOT WATER AND HONEY. Whenever you become hungry, drink a cup of hot water with 1 teaspoon honey and 10 drops of lime juice added. This will be a good substitute for

eating and will help to melt the fat.

HERBAL HELPERS. Here is an herbal formula that will help you lose weight.

kutki 3 parts

chitrak 3 parts

shilajit 2 parts

punarnava 5 parts

Take ½ teaspoon of this mixture twice a day with 1 teaspoon honey, before meals.

- In addition, take 1 tablet of the following herbs all together, with warm water, 3 times a day after meals: *triphala guggulu*; *chitrak-adhivati*; *punarnava guggulu*.

- It will also be helpful for you to take *triphala* every night. At least 1 hour after dinner, pour 1 cup of boiling water over ½ to 1 teaspoon *triphala*; let it steep for 10 minutes, and drink.

SNACKS YOU CAN AFFORD TO EAT. In between meals, if you like to munch, eat raisins, which are a mild laxative. Don't eat corn chips, which are salty and fatty and difficult to stop eating; don't eat popcorn either. You might also try celery sticks or carrot sticks.

USE HOT SPICES. Use spices in cooking. The kapha-pacifying diet uses many spices, such as cumin, coriander, fennel, cardamom, ginger, cinnamon, and garam masala: These spices are good for kindling the gastric fire. (See [chapter 8](#) for details of the kapha diet.)

HAVE COMPANY WHEN YOU EAT. It is better to eat with friends, particularly with skinny people and people with good eating habits! You'll be happy to be surrounded by them. But don't compare, thinking, "All my friends are thin, and I'm chubby." Be in the company of slim people, and be active with them. That will help to reduce your weight.

HELPFUL YOGA POSTURES. Certain gentle yoga *asanas* are helpful, including the Palm Tree pose and the Triangle pose. Also, while sitting on the floor, bend forward as far as you comfortably can, with the goal of eventually touching your head to your knees. (You'll have to work up to this one gradually.) The Fish, Camel, Cobra, and Cow poses are simple, helpful postures you can easily perform. Remember, don't try to do them perfectly right away. (See the illustrations of yoga postures in [appendix 4](#).)

BREATHE AWAY FAT. *Bhastrika pranayama* (Breath of Fire) will increase the rate at which your body burns off fat. Right Nostril breathing (*surya bhedi*) will also be helpful (see [chapter 6](#)).

DON'T NAP IN THE DAYTIME. Obese people frequently like to take siestas, but this is not a helpful practice. Don't sleep during the day. Daytime sleep slows down agni (metabolism) and increases kapha dosha. Rather, do some hard physical work, and watch less TV. Generally I find that obese people get fixed to the couch, watch TV, and

drink soda pop.

If you follow these suggestions, you can definitely take control of your weight problem. Don't try to lose a lot of weight all at once. That almost never brings long-term improvement. Kapha individuals are noted for their ability to make steady, determined, consistent progress. So be persistent, and over time you will be successful.

❧ ❧ Osteoporosis ❧ ❧

Osteoporosis is a thinning and increasing porosity of the bone due to increased vata. Bones are normally porous, but because of increasing vata—a normal occurrence as a person grows older—the porosity increases. Sometimes the individual loses so much bone that weak spots develop in the skeletal structure. Then the hips, forearms, or even the spine can fracture quite easily. The bones may crack under the body's own weight, or a minor injury may be enough to create a fracture.

Osteoporosis is more common in women than men. Women lose bone rapidly after menopause. This is because the postmenopausal body produces little or no estrogen, which is necessary for maintaining bone metabolism utilizing calcium, magnesium, zinc, and other materials for building the bone. So it is during the postmenopausal years that women may have a tendency toward osteoporosis.

Men also need estrogen to maintain strong bones, but testosterone and prostatic secretions also play an important role. However, men may lose bone mass due to heavy drinking, heavy smoking, chewing tobacco, and taking steroids.

Lack of exercise can also reduce bone mass. To some extent, people need some stress to the body, in the form of exercise. Research has shown that if a person is confined to bed for several weeks, the bones become significantly weaker. Once the ill effects of lack of exercise became clear, exercise programs were designed even for astronauts in space. Exercise is a food for the bones.

For women, the combination of increased vata simply from growing older (see [chapter 2](#), where we talk about the stages of life), plus the menopausal cessation of estrogen, may have a powerfully deleterious effect on bone mass.

EXERCISE. Gentle, daily exercise for 30 minutes a day, 5 days a week, can help to treat osteoporosis. Walking is excellent and quite sufficient, but you may do swimming, gentle jogging, or whatever suits your constitution and level of fitness, and the condition of your bones.

Some people recommend weight-bearing exercise, even including weight lifting, for building bone. While this is good in general, for a person with osteoporosis it can be quite dangerous; as mentioned above, even a slight injury may crack the fragile bones of a person with osteoporosis. Therefore a good, safe way to begin an exercise program is to do some underwater exercise. As the bones become stronger, weight-bearing exercise,

even including some gentle weight lifting, may be acceptable and effective.

CALCIUM. It is important to get a plentiful supply of calcium from natural food sources, such as sesame seeds, soybeans, soy milk, cow's milk, cheese, carrots, and coconut. Calcium supplements, such as from oyster shells, may also be helpful. Your daily dose should include about 1,200 mg. calcium, along with 600 mg. magnesium and 60 mg. zinc for maximum absorption and effectiveness.

Almond milk also contains a significant amount of calcium. Soak ten almonds overnight in water. In the morning, peel them and blend in the blender with a cup of warm milk. (You can use goat's milk or soy milk if you prefer them to cow's milk.) Pour into a cup or glass, and add a pinch each of ginger, cardamom, and saffron. Drink twice a day, before breakfast and before bedtime.

Daily chewing a handful of white sesame seeds in the morning provides at least 1,200 mg. of natural calcium. These seeds won't create clogged arteries, as dependence upon calcium from dairy products may. This is an effective way to help prevent osteoporosis in menopausal women.

However, merely taking calcium may not be sufficient. Along with it you need to bring some physical stress to the system through exercise.

BE CAREFUL WITH YOGA POSTURES. If osteoporosis has begun to develop, yoga exercises should be done gently, with great care, as there is a real danger of breaking a bone.

HERBAL HELPERS. Certain herbs can help to make up for estrogen in the metabolic cycle. Try the following formula:

shatavari 5 parts

vidari 3 parts

wild yam 2 parts

These herbs are food precursors of estrogen and progesterone. To this formula you can add $\frac{1}{8}$ part each of *shanka bhasma* (conch shell ash) and *kama dudha* (coral shell ash). They contain a natural source of calcium bicarbonate, which can help to prevent osteoporosis.

Take $\frac{1}{4}$ teaspoon of this mixture twice a day with warm milk, whether cow's, goat's, or soy milk. Treat this as a daily maintenance dose, and take it indefinitely to help prevent osteoporosis.

❧❧ Overeating ❧❧

See also "[Obesity](#)" and "[Eating Disorders](#)"

Because of hard work, especially physical labor, some people need to eat a large

amount of food to replenish the body. In such individuals, overeating may occasionally occur. But most overeating takes place because of emotional factors, and that is what we will be considering here.

Food nourishes the body; love nourishes the soul. When you are with a loving friend or close family members, you can become so happy that you forget to eat. You feel no need for food at that time because you receive a higher food—love.

But when a person doesn't receive that love or feel that happiness, he or she becomes lonely or has a feeling of rejection or of not being loved. Food may become the substitute for love. So eating, in order to suppress feelings of loneliness, grief, sadness, or depression, is the emotional, psychological cause of overeating. Statistically, overeating happens more in women than in men.

From an Ayurvedic perspective, due to emotional factors and stress, *prana* vata stimulates *jatharagni*, the gastric fire, and this stimulation activates the stomach. The stimulation is translated or experienced as hunger, and that is the reason people may eat more food.

There are many ways to deal with the problem of overeating. With a little care, it can be overcome.

EXPRESS YOUR FEELINGS. First, you have to let go of your feelings of loneliness and not being loved. Write about your feelings. Express them. That way, the energy that is blocked in the solar plexus starts releasing, and emotional hunger will begin to subside.

MEDITATE AND BREATHE. Whenever you feel emotionally hungry, sit quietly and pay attention to your breath. Or do 10 to 15 minutes of *So-Hum* meditation (see [chapter 7](#)).

Shitali pranayama (making a tube of the tongue and inhaling through that tube into the belly) will also be quite helpful. (See [chapter 6](#).)

- Or take 12 deep breaths, then drink a cup of warm water. This will help dissolve the emotional hunger, and overeating can be avoided.

YOGA. Yoga exercises such as the Moon Salutation, Camel pose, Cobra pose, and Spinal Twist will help you control overeating due to emotional factors (see [appendix 4](#)).

WALK OFF THE CRAVINGS. Whenever you have emotional cravings for food, take a brisk walk for 20 minutes in the fresh air. That will help reduce the cravings.

EAT LIGHTLY. If you feel you must eat, then eat some light food. Try some light crackers, or some cereal or grains such as millet or rye. Or drink some fruit juice. Follow a low-fat diet. (See the guidelines for the kapha diet in [chapter 8](#).) That way you won't have to deny yourself when you feel like eating, but the light food won't put on weight or fat.

TRY BANANAS. Eating 1 ripe banana, chopped up with 1 teaspoon ghee and a pinch of cardamom, is effective for pacifying emotional, obsessive eating habits.

TEST YOUR HUNGER. When you are hungry, here is one way to find out whether it is an emotional craving or a real biological need. Drink some licorice tea, chamomile tea, or mint tea. If it was emotional hunger, the warm soothing tea will take care of it, and you will feel better. If you are really hungry and need some food, the tea won't diminish your appetite.

FOR HYPERTHYROIDISM. If your overeating is due to hyperthyroidism, which is a metabolic disorder, Ayurveda suggests using *kaishore guggulu*. This herbal compound helps to regulate metabolism and pacifies overactive thyroid.

DRINK BRAHMI MILK. When you feel hungry, drinking 1 cup of warm milk boiled with ½ teaspoon of *brahmi* will help you control overeating.

IF YOU HAVE ALREADY EATEN TOO MUCH. Roast 1 teaspoon coriander seeds and 1 teaspoon fennel seeds on a heavy iron pan, without any oil. (Stir constantly to avoid burning.) Add a pinch of salt, let the mixture cool, and eat it. It will help with indigestion.

Another aid to indigestion from overeating is to drink a cupful of water into which you have added the juice of half a lemon; just before drinking, add a pinch of baking soda, stir, and drink quickly.

⌘ ⌘ Overweight ⌘ ⌘

See “[Obesity](#)”

⌘ ⌘ PMS ⌘ ⌘

See also “[Menstrual Difficulties](#)”

TYPES OF PMS

Premenstrual syndrome or PMS is classified in Ayurveda in three types: vata, pitta, and kapha.

- Vata type is characterized by low backache, lower abdominal pain, distention, anxiety, fear, insomnia, and mood swings.
- Pitta-type PMS symptoms include tenderness in the breasts, urethritis, hives, hot flashes, irritability, and sometimes a burning sensation when passing urine.
- Kapha-type PMS involves water retention (breasts become enlarged and tender) and drowsiness, so that the woman loves to drink coffee!

PMS can be cared for successfully using the following Ayurvedic remedies and

preventive measures.

IMPORTANT: In all cases, start the preventive program one week before the anticipated start of your period.

FOR VATA-TYPE PMS

- Drink *dashamoola* tea ($\frac{1}{2}$ teaspoon *dashamoola* steeped in a cup of hot water for 10 minutes); you can add a little honey for taste. Take twice a day.
- Eat about 10 cherries daily on an empty stomach for a week before the onset of menstruation.
- Use *kaishore guggulu* or *yogaraj guggulu*, 1 tablet twice a day.
- You can also take 1 tablespoon aloe vera gel with a pinch of black pepper, 3 times a day before food.

FOR PITTA-TYPE PMS

- Take the following herbal mixture:

shatavari 2 parts

brahmi 1 part

musta 1 part

Take $\frac{1}{2}$ teaspoon of this mixture twice a day with warm water.

- Aloe vera gel (1 tablespoon) taken with a pinch of cumin powder is also effective.

FOR KAPHA-TYPE PMS

- Make this herbal mixture;

punarnava 2 parts

kutki 1 part

musta 2 parts

Take $\frac{1}{2}$ teaspoon of this mixture twice a day with a little warm water.

- Eat about 10 cherries daily on an empty stomach for a week before the expected onset of your period.
- You can also take 1 tablespoon aloe vera gel along with a pinch of *trikatu* (a traditional Ayurvedic formula consisting of equal amounts of black pepper, *pippali*, and ginger).

FOR ALL BODY TYPES

- Warm ghee nose drops (5 drops in each nostril) stimulate natural hormones and help regulate balance of the system.
- When there is abdominal bloating and cramps, all constitutional types can put a warm castor oil pack on the lower abdomen. One of the qualities of castor oil is that it produces a slow, sustained heat that is soothing and healing. Warm up about 3 tablespoons of castor oil, and pour it onto a handkerchief or other soft cloth, spreading it equally on the cloth. Place this compress on the lower abdomen. If you have a hot water bottle; you may place it on top of the pack to keep it warm. An electric heating pad is not recommended.

NOTE: A warm castor oil pack will also help relieve the congestion and discomfort of endometriosis.

FOR PREVENTION

- Be sure to get regular exercise during the month, including half an hour of walking or other aerobic exercise at least five days a week. Yoga stretching is also helpful. However, Ayurveda recommends no exercise or yoga during the actual menstrual period. As much as possible, rest, read, and relax!
- To maintain health and balance, follow the dietary guidelines for your constitutional type (see [chapter 8](#)).

❧ ❧ Premature ❧ ❧ Ejaculation

See also “[Impotence](#)”

For the man who repeatedly ejaculates prematurely, sex can become a nightmare. He may run away from his partner out of fear that his sexual performance is inadequate, causing serious difficulties in the relationship.

Premature ejaculation is primarily caused by aggravated vata. Vata, with its qualities of quickness and heightened sensitivity to the sense of touch (as pitta brings greater sensitivity to light), gives a predisposition toward faster ejaculation; in general, persons with a vata constitution cannot sustain sex for very long. When vata is unduly increased in a vata individual, premature ejaculation is common.

A psychological, emotional factor of nervousness, fear, or anxiety may also be involved, but this too is due largely to the aggravated vata dosha. Thus the main avenue of treatment is to balance vata dosha.

Another possible cause is high cholesterol (and the related situation, high triglycerides). In men who have this problem, cholesterol deposition takes place in the blood vessels of the penis and the entire reproductive system. These blood vessels become thick and narrow (ischemia), so that the blood supply to the muscles of the penis and prostate is insufficient, causing lack of sphincter control and leading to premature ejaculation.

Premature ejaculation can be brought under control. Here are some effective ways to deal with it.

MASSAGE THE PENIS. Do a gentle massage of the penis. To 1 ounce of sesame oil, add 5 to 10 drops of mustard oil. The diluted mustard oil has a heating effect, which will dilate the blood vessels and improve the circulation of blood to the penis.

IMPORTANT: Do this massage about 1 hour before intercourse, and be sure to wash the penis thoroughly beforehand, or your partner's delicate skin may feel burned by the hot mustard oil. You can also use castor oil, which will improve the tone of the sphincter muscles.

- Certain medicated ghees, such as *brahmi* ghee, *shatavari* ghee, or *ashwagandha* ghee, can also be used for massaging the pubic bone and the glans penis before making love.

PRACTICE. Practice stimulating the organ up to the point of ejaculation, but don't ejaculate. Then sit up straight (in a Lotus posture, if you can), and raise the energy by doing Ashwini Mudra, pulling or sucking the anus inward in a series of tense-release movements. Do about 10 repetitions of the Ashwini Mudra. This way you train your organ to be ready but not to ejaculate. To succeed at this, you have to practice with great care.

Do this exercise about 1 to 2 hours before making love.

IMPORTANT: This is not masturbation, nor an endorsement of masturbation.

- You can also learn to control ejaculation by contracting the muscles of the buttocks.
- Locate the *marma* point (an energy center like an acupuncture point) at the center of the glans penis, on the underside. Gently press that point, and at the same time suck the anus inward and hold for five seconds. Release. Do this 10 times. This will improve the blood supply to the penis and will help to maintain the erection. Do this about 1 hour before sex.

YOGA POSTURES. Helpful yoga *asanas* include the Bow, Fish, and Camel poses, Elevated Lotus, and Vajrasana.

DIET SUGGESTIONS. As mentioned, premature ejaculation is a vata disorder, so a man with this problem should adhere to a vata-soothing diet (see guidelines in [chapter 8](#)). If you have high cholesterol, follow a low-cholesterol diet. And if you have diabetes, keep your sugar intake low. Most diabetic men suffer from premature ejaculation.

EFFECTIVE HERBS. Make an herbal formula, mixing equal amounts of the following herbs:

ashwagandha

bala

vidari

Take ½ teaspoon of this mixture twice a day with warm goat's milk. Cow's milk is acceptable if you can't find goat's milk, but goat's milk is more effective.

- Ginseng is also helpful. It has properties similar to the Ayurvedic herb *ashwagandha*. Take ½ teaspoon after lunch and dinner with warm goat's milk. Again, cow's milk is acceptable if you can't locate fresh goat's milk.

ALMOND MILK. Soak 10 almonds overnight in water. In the morning peel off the skin, put the almonds in a blender, and add 1 cup hot cow's milk and a pinch each of ginger, cardamom, and saffron. (Saffron is a mild aphrodisiac.) Drink almond milk every morning. You can also make a second cupful in the evening.

HEAL THE PROSTATE. Premature ejaculation often accompanies prostate problems such as prostatitis, or it may be a warning of developing problems. To help with this, massage the area of the prostate gland with castor oil. Apply a small amount of the oil (sesame oil is also beneficial) to the perineum, the area in between the anus and testicles. First rub in a circular motion, then finish with strokes from the anus toward the base of the penis. Don't press hard; use light strokes.

⌘ ⌘ Prostate Problems ⌘ ⌘

In middle-aged men, it is uncommonly common for the prostate gland to become enlarged, resulting in a number of uncomfortable symptoms. For example, the man may wake up several times during the night to pass urine. He may have difficulty urinating, so that it may take some moments for the flow to start, or the flow may be slow, or it may "dribble" at the end. He may find that the need to urinate occurs rather suddenly, or that he feels a frequent need to go. This happens when the bladder does not empty completely during urination; even though there may not be much urine left in the bladder, the residual fluid creates the sensation of needing to go.

Another problem that sometimes develops is prostatitis (inflammation of the prostate), which has many of the same symptoms described above but also is characterized by a burning sensation while passing urine.

HERBAL REMEDIES

A number of herbal remedies are effective in treating prostate problems.

- Make this formula of Ayurvedic herbs:

punarnava 2 parts

shilajit $\frac{1}{8}$ part

gokshura 2 parts

Take $\frac{1}{4}$ teaspoon twice a day, after meals, with a little warm water. Continue to take it until your symptoms resolve.

• You can also use ginseng, hibiscus, or horsetail tea, available in packages or as bulk herbs in most natural food stores. These teas can be drunk several times daily, as desired. Follow the directions on the package, or if you use bulk herbs, add about 1 teaspoon of mixture to a cup of boiling hot water, steep 5 minutes, cool, and drink.

TIME TO SEE THE DOCTOR

Since prostate cancer is a possible diagnosis based on the above symptoms, you should see a medical professional to determine the source of your problem. If the cause is benign, the following remedies may be effectively used for self-treatment.

- Cumin-coriander-fennel tea will help relieve the burning sensation and other symptoms as well. Mix the herbs in equal amounts, and steep 1 teaspoon of the mixture in a cup of hot water for 5 to 10 minutes. Drink 2 or 3 times a day.
- Another Ayurvedic herbal compound that is helpful for enlarged prostate is *punarnava guggulu*. Take 1 tablet twice a day.
- Make a mixture of equal amounts of *vidari* and *ashwagandha*, and take $\frac{1}{2}$ teaspoon 2 or 3 times a day, washed down with warm water.

OTHER HELPFUL REMEDIES

OIL MASSAGE. A gentle massage of the prostate area is also helpful. Apply a small amount of castor oil or sesame oil to the perineum (the area in the middle between the anus and testicles). First rub in a circular motion, then finish with strokes from the anus toward the base of the penis. Don't press hard; use light strokes.

FROM THE YOGA TRADITION. It will be helpful if you perform what is known in yoga as the Ashwini Mudra: pulling or sucking the anus inward in a series of tense-release movements. Do about 10 repetitions of the Ashwini Mudra morning and evening. They should be done while you are sitting.

The Kukutasana, or Elevated Lotus pose, is also helpful for prostate problems. (See illustration in [appendix 4](#).)

❧ ❧ Rashes and Hives ❧ ❧

Skin rashes and hives indicate excess pitta or heat in the body; the Ayurvedic approach is to provide a cooling effect from both the inside and outside.

FOR IMMEDIATE RELIEF. Whatever the cause of the rash, whether allergies, an insect bite, or something else, cilantro juice will be immediately effective. Wash fresh cilantro, chop it into pieces, put it into the blender, add $\frac{1}{3}$ cup of water, and blend. Drink the juice, and apply the pulp directly onto the skin.

TOPICAL SOLUTIONS. The following remedies applied directly to the skin will help soothe and heal rashes and hives:

- If you have a fresh coconut, break it open and apply the coconut water to the rash.
- Melon can also be soothing to rashes and hives. Eat some melon, and rub the remaining rind (not the outer tough skin) over your skin. With watermelon, eat the red part and rub the white part of the rind onto your skin.
- For rashes, hives, and other high-pitta conditions such as nausea, steep 1 teaspoon coriander, $\frac{1}{2}$ teaspoon cumin, and 1 teaspoon natural sugar in 1 cup hot milk. Drink once or twice a day.
- You can also use a paste made of sandalwood and turmeric powders mixed in goat's milk. This paste is healing for the skin. The formula is simple:

turmeric 1 part

sandalwood powder 2 parts

Mix up about a teaspoon of powder in this proportion, add sufficient goat's milk to make a paste, and apply it to the affected area. Cow's milk is acceptable, but goat's milk is more effective. PLEASE NOTE: Your skin will look yellow for some time—up to 3 or 4 days—after you use the paste.

FOR INTERNAL HEALING. Here is an effective formula to help restore health to your skin from the inside:

coriander 2 parts

cumin 1 part

raw natural sugar 2 parts

Steep $\frac{1}{2}$ teaspoon of this mixture in a cup of hot milk, and drink it once or twice a day until the condition is healed.

❧ ❧ Rectal Bleeding ❧ ❧

See also “[Hemorrhoids](#)”, “[Bleeding, Internal](#)”

Numerous factors can give rise to rectal bleeding. These include hemorrhoids; inflammation of the rectal area due to aggravated pitta; anal fissures or polyps; passing dry, hard stools, which hurt the mucous membrane of the rectum; straining due to constipation; excessive pressure on the rectal veins due to liver cirrhosis; and congestive heart failure. In the last stage of pregnancy, prolonged pushing of the child during labor exerts pressure on the rectal blood vessels and leads to bleeding. Rectal bleeding is also common among people who eat refined foods containing little fiber. Eating hot spicy food may tend to produce constipation and rectal bleeding.

Whatever the cause, the following recommendations will be helpful.

WASH WITH COOL WATER. After each bowel movement, wash the anal orifice with cool water. This will help stop the bleeding, and if there is irritation, itching, and cracking, it will help to minimize it.

APPLY GHEE OR CASTOR OIL. This will help to soothe the irritation of the blood vessels.

DRINK CRANBERRY JUICE OR POMEGRANATE JUICE. These are both hemostatics (they stop bleeding) and will be helpful. Drink 1 cup twice a day.

TIME TO SEE THE DOCTOR

If these remedies do not stop your rectal bleeding in a week to 10 days, it would be wise to see your doctor. If the bleeding is profuse, go sooner. The bleeding could be a symptom of a serious illness.

AVOID HOT SPICY FOOD. Follow the pitta-pacifying diet (see [chapter 8](#)). Also avoid fermented food, sour fruit, citrus fruit, and alcohol.

KEEP THE STOOLS SOFT. This will help to minimize irritation and bleeding. Here are three ways to keep stools soft:

- Drink a cup of warm milk with 1 teaspoon plain ghee at bedtime. (This is a very mild laxative.)
- Take 1 teaspoon of *amalaki* or ½ teaspoon *triphala* at bedtime in warm water. Steep the herb in hot water for 5 to 10 minutes, then when it has cooled down, drink it.
- Take 1 teaspoon psyllium husks (*sat isabgol*) at night in 1 cup of warm water. This bulk laxative will keep the stools soft and avoid pressure on the rectal blood vessels.

USE VITAMIN K. If the bleeding is serious (that is, profuse or repeated), take vitamin K supplements according to the dosage recommended on the package. Also, you can buy vitamin K cream at most health food stores. Apply this cream to the anal orifice to stop the bleeding.

PACIFY PITTA. To reduce pitta, take vitamin E supplements, and drink cranberry juice and/or parsley juice.

PREVENTIVE MEASURES. To avoid rectal bleeding in the future, follow these recommendations:

- Strictly follow the pitta-pacifying diet: no sour fruit, citrus fruit, fermented food, or hot spicy food.
- Avoid hard physical labor if possible.
- Do abdominal yoga exercises such as *navli* (see [appendix 3](#) for instructions).
- The following yoga postures should improve elimination and help to prevent rectal bleeding: Camel, Cobra, Cow, and Spinal Twist.
- Take this herbal formula:

shatavari 5 parts

kama dudha 1/8 part

gulwel sattva 1/8 part

This herbal mixture (1/2 teaspoon with warm water), taken twice a day for 1 month, will help to prevent future rectal bleeding.

Rectal Itching

Rectal itching has several possible causes. One is worms and parasites, such as roundworms and pinworms. Itching may also be caused by hemorrhoids, inflamed ulcers, a yeast infection, or a fungal infection. Excess toxicity—ama in the colon—can also create itching of the anus.

FOR WORMS. If worms are the problem, you may be able to eliminate them entirely with this herbal formula:

vidanga 5 parts

shardunika 2 parts

trikatu 1/8 part

Use 1/4 teaspoon of this mixture twice a day washed down with warm water after meals.

- Also, take 1/2 teaspoon *triphala* every night in a cup of warm water. (Steep 5 to 10 minutes before drinking.)

FOR YEAST INFECTION. If there is a yeast infection, apply a little yogurt to the anal orifice, then wash the anus with water.

FOR HEMORRHOIDS. If the problem is hemorrhoids, take a baking soda bath, soaking the hemorrhoids in warm water with 1/3 cup baking soda added. Then apply *neem* oil to the

hemorrhoids. (See also “[Hemorrhoids](#).”)

FOR FUNGAL INFECTION. Mix a little tea tree oil and *neem* oil together, and apply it directly to the anal area to heal the rectal itch.

PREVENTIVE DIET. Avoid hot spicy foods, fermented foods, and yeast-containing bread. Stay away from alcohol entirely.

❧ ❧ Sinus Problems ❧ ❧

The sinuses are air-filled cavities located on either side of the nose. There are ten sinus cavities, five on each side, all connected to the nose. They are covered with a superfine mucous membrane. The sinuses drain constantly into the nose; their chief function is to keep the nasal cavity moist. They also serve to amplify the voice when we speak.

Due to allergies, colds, or bacterial infections, the sinuses in both adults and children sometimes get clogged or infected. This is a condition of excess kapha that can be aggravated by many factors, including cold drinks, dairy products, and smoking.

Sinus problems can create complications of various kinds, from sinus headaches, snoring, and difficulty breathing to bad breath, ear infections, and sleep apnea. In serious cases, sinus infections can lead to brain infection, meningitis, or osteomyelitis.

HERBAL DECONGESTANT. When the sinuses become clogged and congested, it is important to drain them. A mixture of fresh ginger juice (or freshly grated ginger pulp) with 1 teaspoon honey taken 2 or 3 times a day will be quite helpful.

You can prepare this safe, effective decongestant formula:

sitopaladi 5 parts

trikatu 1/8 part

mahasudarhan churna 2 parts

Take 1/4 teaspoon of this mixture 2 or 3 times a day with warm water after meals.

HERBAL ANTIBIOTICS. To help prevent secondary infection, mix equal amounts of the following antibacterial herbs:

goldenseal

osha

turmeric

neem

Fill some 00-size capsules with the herbal mixture, and swallow 2 capsules twice a day.

NOSE DROPS AND NASAL WASH TO DRAIN THE SINUSES. Make a mild saline solution by dissolving $\frac{1}{2}$ teaspoon salt into $\frac{1}{2}$ cup lukewarm water. Instill 5 drops of the solution into each nostril with a dropper, or take a little in the palm of your hand and snuff it into your nostrils. Repeat as often as needed to keep sinuses clear. (Several times a day is fine.)

Use Medicated Steam to Clear up Your Sinuses

You will be amazed at how effective this simple method is to help clear up painful, congested sinuses.

Heat up about a cupful of water, and add 3 to 5 drops of eucalyptus oil. Turn off the flame, cover your head with a towel, lean over the pot, and inhale the steam.

You can also use ginger in the same way. Take 1 inch of fresh ginger, chop it into pieces, and boil it in about a cup of water. Then cover your head and inhale the ginger steam. It will be effective for draining the sinuses. (You can use powdered ginger as a backup if you don't have fresh.)

Here is a more powerful remedy that you can use for severe sinus congestion and pain. You may not enjoy doing this, but it works. Use a garlic press to squeeze out some fresh garlic juice. With an eye dropper, pick up some juice and insert just a few drops into each nostril. Keep your head tilted back for about five minutes to let the juice work, then sit up and let it drain out onto a tissue. You will be surprised at how clear your sinuses feel. Do this once a day as needed; for a severe sinus attack, you may do it up to three times, in the morning, afternoon, and evening.

FOR A SINUS HEADACHE. Try mixing $\frac{1}{2}$ teaspoon cinnamon with enough water to make a paste, and apply locally.

STRATEGIES FOR PREVENTION. Here are several ways you can help save yourself from future sinus problems.

- Avoid dairy products, especially cheese, yogurt, and ice cream.
- Avoid cold drinks.
- Avoid exposure to cold weather.
- Do not smoke cigarettes.
- Once a day, instill a little warm ghee in each nostril and sniff. You may use an eye-dropper or use a clean finger dipped in ghee.
- Take this herbal formula on a regular basis:

sitopaladi 5 parts

mahasudarshan churna 3 parts

abrak bhasma 1/8 part

Take 1/4 teaspoon of this mixture 3 times a day, after meals. Wash down with warm water. Take it for 3 months.

❧ ❧ Skin—Ayurvedic Care ❧ ❧

Ayurveda has many wonderful suggestions for maintaining the health and beauty of your skin. Some you will find here. Others you will find, along with remedies for various skin problems, in the sections on “Acne,” “Dandruff,” “Dry Skin,” “Rashes and Hives,” and others.

The following suggestions will help you keep your skin healthy, glowing, and beautiful.

OIL MASSAGE. Daily oil massage over your whole body is very effective for keeping your skin healthy and beautiful. If you are vata or have a vata imbalance, use sesame oil. If you are pitta or have a pitta imbalance, use sunflower oil. If you are kapha or have a kapha imbalance, use corn oil or canola oil. Gentle oil massage maintains the beauty and texture of the skin.

TAKE TURMERIC. To have beautiful skin, take a capsule of turmeric daily. Ayurvedic tradition states that if a pregnant woman takes turmeric regularly, her child will have gorgeous skin!

GET ENOUGH IRON IN YOUR DIET. If your skin looks pale, it may be a sign of anemia. Drink some carrot juice, and eat cooked beets. This will give you natural iron and should improve skin color. (See further suggestions under “[Anemia](#).”)

SOAK UP SOME SUN. Apply the appropriate doshic oil to your skin (sesame for vata constitutions, coconut or sunflower for pitta, corn for kapha), and lie in the sun for a little while—10 to 15 minutes, or at the most half an hour. This will improve the circulation and strengthen skin tone.

NOTE: Sunbathing should be done either before noon or in the late afternoon (after three o'clock), when the rays are not as direct. Also, the danger of sunburn is greater at high altitudes, so take care to limit your exposure when you're in the mountains.

USE HERBALIZED OILS. For a wonderful face lotion, apply some *neem* oil or *brahmi* oil to your face. These oils are acceptable for all constitutions.

SHIKAKAI SHAMPOO. For washing your hair, use a shampoo that includes the herb *shikakai*.

NEEM SOAP. Use a *neem* or sandalwood soap for bathing.

BE SPARING WITH SOAP. AS a general rule, it is beneficial to use soap on your skin only once or twice a week, not on a regular everyday basis. In a tropical climate, where people sweat a lot and their sweat contains salts and minerals that lodge in the skin, they have to take a soap bath every day. But in colder countries, where there is not much sweating, so much soap bathing is usually not necessary.

Of course, it depends on a person's job. Someone who does hard physical work and sweats a lot needs some kind of soap. But to maintain softness and luster, it's important not to wash off the sebaceous secretions that maintain the oiliness of the skin. If we apply soap daily, the oil will be washed off and the skin will become dry.

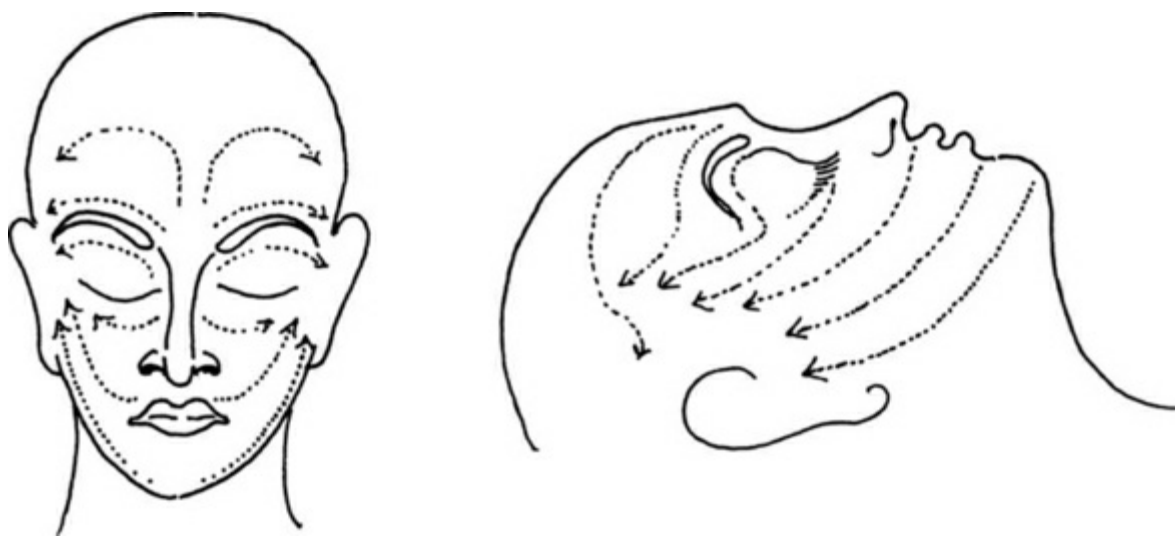
Pitta constitutional types may need soap a little more often, perhaps three times a week, both because they tend to perspire more and because pitta skin tends to be more oily.

MATCH WATER TEMPERATURE TO YOUR CONSTITUTION. Generally, washing in cool water is recommended for pitta, warm water for kapha, and hot water for vata. Vata individuals often have poor circulation, and hot water will improve circulation and help to keep skin healthy and beautiful.

GENTLE TOWEL MASSAGE. After bathing, gently rub your towel over the skin of your face and body. This improves cutaneous circulation, removes dead skin, and helps your skin look young.

Home Face-lift

You can do your own home facial massage and face-lift by applying gentle pressure from the chin to the forehead.



Face-lift Massage

Put your two index fingers together, and your two thumbs. Place the two index fingers between your lower lip and your chin, and the two thumbs just under the chin. Then, pressing lightly with your index fingers, sweep along the jawbone, with

your thumbs underneath. When you reach the ears, go up in front of them, letting the thumb trail. Keep going up until your thumbs are just behind the temple area, above the ear. A *marma* point is located there. When you reach this point press lightly, with a lifting motion, for about 30 seconds. This procedure will stimulate the nerves that control the facial muscles, and it will improve the tone of those muscles and help remove wrinkles from the facial skin. Repeat seven times once a day, preferably in the morning.

FRESH CHERRY MASK. Apply a pulp of fresh cherries to your face at night as a mask, before going to bed. Leave it on for 15 minutes. This will relieve dry skin and give you a beautiful complexion.

YOUR SMILE HELPS YOUR SKIN. Keeping a smile on your face will help maintain the tone of your facial muscles and skin; people may say you look ten years younger than you are!

❧ ❧ Sleep Apnea ❧ ❧

Apnea is a temporary suspension of breath. Sleep apnea is a brief interruption in breathing during deep sleep—sometimes numerous interruptions in a single night—that is quite common in young children but also occurs in some adults. It may also happen at high altitudes. Sleep apnea is often accompanied by loud snoring and an abnormal breathing pattern.

Apnea in older children and adults is less life-threatening. But because the person's sleep is briefly interrupted each time the breathing stops, sometimes dozens of times in a night, it can be physically exhausting. It strains the cardiovascular system and respiratory system and may create excess carbon dioxide in the blood. It may lead to drowsiness and irritability during the wakeful state and an inability to concentrate, due to insufficient sleep.

In Ayurvedic terms, sleep apnea occurs because *tarpaka* kapha is blocking *prana* vata. So treatment centers on controlling excess kapha.

TO CONTROL KAPHA. The first remedy is *pippali*. Take ¼ teaspoon with 1 teaspoon honey and 1 teaspoon ghee on an empty stomach in the morning and evening.

- Instead of *pippali*, you can substitute *trikatu churna* (which consists of equal amounts of *pippali*, black pepper, and ginger).

- For an effective decongestant, take ½ teaspoon *sitopaladi* along with ¼ teaspoon *yashti madhu* in 1 teaspoon honey, twice a day. This will help both as a congestion remedy and as a preventive measure.

Sleep Apnea in Premature Babies

Premature infants sometimes have sleep apnea. Because the respiratory center in the hypothalamus has not fully matured in these babies, from time to time the child may completely stop breathing and can turn blue or purple.

If this happens, don't panic, but act quickly: *If you tickle the soles of the feet, or sprinkle cool water on the belly at the diaphragm, the baby will start breathing again.*

This condition may cause SIDS, sudden infant death syndrome. It is a serious condition that needs prompt medical care. But when the breathing control center becomes mature, the baby's sleep apnea should go away.

NASYA. Instill 5 drops of warm *brahmi* ghee or plain ghee in each nostril in the morning and before going to bed.

FOR OBESITY. One of the primary causes of sleep apnea is being significantly overweight. If that is the problem, you can treat obesity with this herbal formula:

kutki 1 part

chitrak 1 part

shilajit 1/8 part

Take 1/4 teaspoon of this mixture 2 or 3 times a day before breakfast, lunch, and dinner with warm water.

- Regular walking or other exercise will also help the obese person to breathe better. Be careful not to undertake exercise that might be too much for you if you are very heavy and have not exercised in some time. Stick to walking, or consult with your doctor if you want to do more.

CHANGE YOUR SLEEPING POSITION. Anyone with sleep apnea would do well to experiment with different sleeping positions to see if the condition eases. In particular, obese people, who are often in the habit of sleeping on their chest or on their back with their hands on their chest, might try sleeping on their left side. Just this one shift could remedy or at least reduce the problem.

KEEP THE HOUSE HUMIDIFIED. Sometimes dry, hot air creates a choking sensation in the nose and can be the cause of sleep apnea. A warm, comfortably humid atmosphere will be best. Preferably, use a hot water humidifier; ultrasound humidifiers are not recommended.

PANCHAKARMA. Under the supervision of an Ayurvedic physician, it would be helpful for a person with sleep apnea to undergo *panchakarma* purification therapy (see [chapter 4](#)). This includes *abhyanga* (oil massage), *virechana* (purgation therapy), and *nasya* (nasal administration of medications) among others. This procedure is cleansing and rejuvenating.

See also [“Addictions”](#)

Addiction to smoking has two main causes: nicotine toxicity and stress. Once a person becomes a smoker, he or she has to achieve just the “right” amount of nicotine toxicity to maintain proper functioning of the brain as well as normal digestion and elimination. Also, at times of pressure or emotional disturbance, smokers habitually reach for a cigarette. So two parallel strategies are needed: detoxifying the nicotine toxicity and dealing with the stress.

HERBAL CIGARETTE. To gradually wean yourself from smoking, Ayurveda suggests that you prepare the following herbal mixture. It will help both with reducing your stress level and with detoxifying your body. Mix together equal amounts of:

brahmi

jatamamsi

rose petal powder

Remove $\frac{1}{3}$ of the tobacco from a few of your cigarettes (at the end you light) and replace it with this mixture. When you feel like smoking, light and smoke the herbal mixture. When you reach the tobacco, stop. Soon you will see that the desire to smoke begins to diminish.

OVERWHELM YOURSELF. Here’s another helpful procedure: Place a few drops of *brahmi* ghee directly on the cigarette, then light up. The smoke produced will be so strong, it will eliminate your desire to smoke.

EFFECTIVE HERBAL TEA. At times of stress, instead of smoking a cigarette, prepare and drink a tea made of equal proportions of *jatamamsi*, chamomile, and *brahmi*.

jatamamsi 1 part

chamomile 1 part

brahmi 1 part

Steep 1 teaspoon of this mixture in a cup of hot water, and drink. Take it slowly, sip by sip, to help relieve the desire to smoke.

CHEW INSTEAD OF SMOKING. Whenever the desire for a cigarette arises, chew one or two small pieces of dried pineapple instead, mixed with $\frac{1}{2}$ teaspoon honey.

REDUCE STRESS BY MEDITATING. For Stress management, spend some time every day meditating. If you know a practice, use it! Or try the Empty Bowl meditation explained in [chapter 7](#). Most people find that when stress is reduced, their desire to smoke spontaneously decreases.

YOGA POSTURES. Yoga postures are helpful. Try the Moon Salutation sequence, as well as

the Locust pose, Bow, Shoulder Stand, Plow, and Palm Tree. (Illustrations of yoga *asanas* are found in [appendix 4](#).)

BREATHING EXERCISES. The *pranayama* (breathing exercise) known as Breath of Fire will also help you in your effort to quit smoking. It is explained in [chapter 6](#).

❧ ❧ Sore Throat ❧ ❧

A sore throat is caused by irritation and inflammation of the throat. This condition is generally quite easy to remedy using Ayurvedic methods.

GARGLE. Perhaps the simplest remedy, and one that is quite effective, is gargling. Ayurveda recommends using 1 cup of hot water (not so hot that it might burn your throat) with ½ teaspoon turmeric and ½ teaspoon salt mixed in. Gargle with this mixture morning and evening.

TURMERIC MILK. Drinking 1 cup of hot milk boiled with ½ teaspoon turmeric is also helpful for sore throat.

SOOTHING HERBAL TEA. Another remedy that works well is ginger-cinnamon-licorice tea in the following proportions:

ginger 2 parts

cinnamon 2 parts

licorice 3 parts

Steep 1 teaspoon of the herbal mixture in water for 5 to 10 minutes, and drink up to 3 times a day.

TIME TO SEE THE DOCTOR

If your sore throat persists for more than a few days even though you are trying these remedies, see a medical professional.

FOODS TO AVOID. When you have a sore throat, it is important not to eat dairy products such as cheese, yogurt, and ice cream. Also avoid fermented foods.

NOTE: A cup of warm turmeric milk, as recommended above, is an exception. Do stay away from *chilled* dairy products, which are mucus-forming and will exacerbate your sore throat.

YOGA POSTURES. The yoga posture known as the Lion pose is traditionally recommended for sore throats. The Yoga Mudra is also effective. (See yoga illustrations in [appendix 4](#).)

BREATHING EXERCISE. You can also practice *bhramari* (Humming Breath), which is described in [chapter 6](#).

❧ ❧ Spastic Colon ❧ ❧

For help with spastic colon, you can use the remedies suggested under “Irritable Bowel Syndrome.” In addition, try the following Ayurvedic herbal formula:

shatavari 4 parts

hingwastak churna 1/8 part

ajwan 1/4 part

chitrak 1 part

Take about 1/4 teaspoon of this mixture 3 times a day with warm water, after meals.

❧ ❧ Sprains and Strains ❧ ❧

Stepping off the curb in a wrong way, losing your balance when walking (especially easy to do when you wear high-heeled shoes), falling on an outstretched hand—all of these can lead to a sudden twisting of an ankle, wrist, or hip joint, resulting in a sprain or strain.

Although sprains and strains are similar, there is a difference. A *sprain* affects ligaments and is a pitta condition. It is caused by excessive stretching or even tearing of the ligament and involves pain, swelling, and blackish-bluish discoloration. A *strain* affects muscle, is due to vata, and does not produce bruise discoloration, although there is pain.

WRAP IT UP. Whenever there is a strain or sprain, as soon as possible wrap the area in an elastic bandage.

DRINK SOME JUICE. Then drink some pineapple or pomegranate juice. These juices contain an enzyme that acts as an anti-irritant and anti-inflammatory. This helps to pacify pitta and accelerates the healing process.

SOAK IT. For ankle sprain, soak your foot in a pot of hot water with a homemade mustard seed tea bag. Make the bag by tying up 2 teaspoons of brown mustard seeds in a handkerchief, some cheesecloth, or other light cloth. Immerse the bag in the hot water.

To relieve the swelling, soak the foot for 15 minutes in hot water with 2 tablespoons of salt per gallon of water.

APPLY A HEALING PASTE. Make a paste out of 1/2 teaspoon turmeric and 1/2 teaspoon salt, with enough water to form the paste. Use cool water. Applying it to the injury will help reduce swelling.

If it is a strain rather than a sprain, use a *hot* paste of turmeric and salt to help pacify

the vata.

HERBAL REMEDY. If you can't tell whether it is a strain or a sprain, take capsules of *kaishore guggulu* (200 mg. twice a day); this herbal formula will help heal both conditions.

SUGGESTION: If you want to avoid sprained and strained ankles, don't wear high-heeled shoes.

☒ ☒ Stomachache ☒ ☒

See also "[Indigestion](#)"

Stomachache is a very ambiguous symptom. Many causes are possible: acid stomach, acid indigestion, constipation, eating the wrong kind of food, even toxicity in the liver. All these, and other factors, can create abdominal aches and pains.

To treat a stomachache intelligently and effectively, first we have to rule out the serious causes, such as appendicitis, enteritis, gastritis, and colic. You may need a physician's help to determine the cause. But for a common tummy ache, Ayurveda offers numerous simple, natural, and effective home remedies.

HERBAL REMEDIES

- Mix together $\frac{1}{3}$ teaspoon cumin powder, a pinch of hing (asafetida), and a pinch of rock salt. Chew well; wash down with warm water.
- For stomachache associated with diarrhea, rub a little fresh ginger juice on your stomach, around the belly button.
- Try the Ayurvedic herb *shankavati*. It ordinarily comes in tablets; just take 1 pill twice a day, morning and evening. It will take care of an aching stomach.
- If you don't find *shankavati*, take *lasunadivati*, 1 tablet twice a day after food.
- Another helpful herb is *ajwan* (Indian celery seed), which you can usually buy from an Indian grocery shop. Mix $\frac{1}{2}$ teaspoon with $\frac{1}{4}$ teaspoon baking soda, chew the mixture, and drink it down with a little warm water.
- If this doesn't relieve your stomachache, mix together some roasted fennel, roasted cumin, and roasted coriander seed, and chew on about $\frac{1}{2}$ teaspoon of the mixture. (Roast these seeds individually—that is, one type at a time—in a heavy cast iron pan, stirring constantly so they don't burn. Then mix them together.)
- To relieve indigestion, take $\frac{1}{4}$ cup of fresh onion juice with $\frac{1}{2}$ teaspoon honey and $\frac{1}{2}$ teaspoon black pepper.

TWO HERBAL TEAS

- Make some cumin-coriander-fennel tea. Mix the three herbs in equal proportions, and use ½ teaspoon per cup of water. Drink 2 to 3 times a day to help soothe an aching stomach.
- Another simple tea can be made of equal amounts of the common Western herbs angelica, chamomile, and comfrey. Mix them together, and steep about ½ teaspoon of the mix in hot water.

THE MASTER KEY TO PREVENT STOMACHACHE

Most of the time, a stomachache is due to indigestion and low agni (digestive fire). Here are four ways you can kindle agni:

- One of the best herbs to strengthen digestive fire is ginger. Before each meal, chop or grate a little fresh ginger, add a few drops of lime juice and a pinch of salt, and eat the mixture. Or just cut a thin slice of ginger, put on a pinch of salt, and chew that.
- Ginger tea will also increase the gastric fire and reduce the stomachache. Boil a little fresh ginger (grated, chopped, or sliced), or use powdered ginger to make a tea, and drink it 2 or 3 times a day.
- *Draksha* (Ayurvedic herbal wine) before meals will help to kindle the digestive fire. Use 2 to 4 tablespoons mixed with an equal amount of water. Or take a few sips of port or another sweet wine.
- Another simple way to enliven your gastric fire is to use the common spice bay leaf. Steep ½ teaspoon of crushed or ground bay leaf in a cup of hot water for about 10 minutes to make a tea. Add a pinch of cardamom, and drink after eating.

ADDITIONAL REMEDIES AND RECOMMENDATIONS

EAT LIGHTLY. If your stomach hurts, avoid eating any heavy meals, beans, meat, and heavier grains like wheat. It is better to be on a diet of kitchari, a combination of rice and dal that is easy to digest. (See [this page](#) for a basic kitchari recipe). Between meals, take some fruit juice.

FOR CHRONIC INDIGESTION. For chronic poor digestion and frequent stomachaches, prepare this herbal mixture:

trikatu 1 part

chitrak 2 parts

kutki 1 part

Take ¼ teaspoon before meals, with a little honey and fresh ginger juice. If you don't have fresh ginger, just use honey. This mixture will help strengthen the digestive fire.

TAKE A BREAK FROM EXERCISE. When you have an upset stomach, Ayurveda recommends that you just rest, read, and relax as much as you can until the condition clears. Exercise, including *yoga asanas*, is not recommended.

☞ ☞ Stress ☞ ☞

Stress is a psychosomatic disorder that has many causes in our day-to-day life. We may get stressed when traffic is jammed, when we have to stand in a long line, or when we deal with a difficult work situation. An all-too-common cause of stress is the feeling that we have too much to do and too little time to do it. Unemployment is another potential source, as are pollution and crime. Even watching crime stories day after day on television builds up stress. An unhappy relationship, a dominating spouse, faulty plumbing, school exams, burn-out at work—the list of causes is virtually endless.

Stress, in turn, may trigger allergies, asthma, herpes, high cholesterol, and hypertension. It may even lead to a heart condition. It can induce an imbalance of vata, pitta, or kapha, depending on the individual's *prakruti* (constitution).

Generally, vata individuals are likely to develop vata-aggravated stress reactions, such as anxiety or fearfulness, even phobias or anxiety neurosis. Pitta individuals increase in pitta during stressful situations and typically react to stress in the form of anger. They may also suffer from hypertension, peptic ulcer, ulcerative colitis, and other pitta disorders. Kapha individuals under stress can develop underactive thyroid function, slow metabolism, and even increased blood sugar, leading to a prediabetic condition. They tend to eat and eat and eat and become chubby.

TAKE A RELAXATION BREAK. To prevent the buildup of stress, the first line of defense is to stay calm and cool during potentially stressful circumstances. Take long, deep breaths, and breathe out your stress. Relax. Have a massage, or give yourself an oil massage by rubbing a few ounces of warm oil on your body, from head to toe. Vata individuals should use sesame oil, pitta individuals sunflower oil, and kapha individuals corn oil. After your massage, take a hot shower or a hot tub bath. Mental techniques such as positive imagery, prayer, singing and chanting, and meditation, as well as regular yoga exercises, are all effective to minimize and remedy stress.

ANALYZE YOUR STRESS. Separate the things in your life that you find stressful into two categories: things you can do something about, and things you can't. If you can do something about it, then do it! If there's nothing you can do, then surrender to it. Accept it. When there's nothing I can do about a situation, I have to surrender to it, and in accepting it, there is peace.

MONITOR YOUR NEGATIVE THINKING. Stress is often the result of fear that is based largely on imagination. Look at your negative thinking, and replace it with positive thinking. Just changing your thinking or your attitude can alleviate much stress.

EXAMINE YOUR ROLE AND YOUR GOAL. Find the right match between your job and your personality. Job stress is a terrible burden on many people when work and personality are not appropriately matched. If you love what you are doing, there is no stress. If you don't love what you are doing and you still have to do it, that is very stressful. So you have to discover your true role and your goal.

A SOOTHING BATH. A ginger-baking soda bath is quite soothing. Add $\frac{1}{3}$ cup ginger and $\frac{1}{3}$ cup baking soda to a hot bath for greater relaxation and healing.

OILS FOR RELAXATION. Rub a little *brahmi* oil on the soles of your feet and on your scalp at bedtime.

- Put one drop of pure castor oil (with no preservatives) into each of your eyes, and rub a little on the soles of your feet for a calming, soothing effect.

USE MEDICATED NOSE DROPS. Do *nasya* with *brahmi* ghee or plain ghee. Put 5 drops of the ghee into each nostril. (Refer to the *nasya* instructions in [appendix 3](#).)

STRESS-REDUCING TEA. A tea made from equal proportions of the herbs chamomile, comfrey, and angelica will be relaxing. So will *brahmi* tea, which you can make by adding a cup of boiling water to $\frac{1}{2}$ teaspoon *brahmi*. Or try a tea made from equal amounts of these herbs:

brahmi

bhringaraj

jatamamsi

shanka pushpi

Steep $\frac{1}{2}$ teaspoon in 1 cup of hot water for 10 minutes. You can drink this tea 2 or 3 times a day for stress management.

YOGA STRETCHING. Certain specific *asanas* are effectively used for management of stress, especially the Shoulder Stand, Plow pose, Spinal Twist, and Locust pose. The Lion pose is also effective for relieving stress. (*Asana* illustrations appear in [appendix 4](#).)

MEDITATION. Sit in the Lotus pose or the Easy pose (legs comfortably crossed) facing east, and meditate. Just observe the inflow and outflow of your breath, or do the *So-Hum* meditation (described in [chapter 7](#)).

BREATHE AWAY STRESS. *Ujjayi pranayama* is deeply calming and helpful for relieving stress. You can do it sitting up, or try lying on your back in Savasana (the “corpse” or rest pose) and doing this breathing exercise. (See instructions in [chapter 6](#).)

LET YOURSELF CRY. If you have a lot of grief and sadness, it will help to cry out your stressful feelings. Crying is an excellent release for emotions.

LAUGHTER IS GOOD MEDICINE. Laughter is another good way to relieve stress. Try it, even if it is forced at first. Just start laughing! Soon real laughter will come and with it the release of tension and stress.

START YOUR WEEK WITH MEDITATION. Many people experience extra stress on Monday. They have to travel a long distance to work and start another week at a job they don't enjoy. It's a fact that heart attacks are more common on Mondays. To help you reduce stress at this crucial time, keep in mind that Monday is the day of the moon, and the moon represents the mind. So begin Monday morning—and your week—with 15 to 20 minutes of meditation before going to work. Regular daily meditation, morning and evening, is one of the best things you can do to keep your stress level low.

☼ ☼ Sunburn ☼ ☼

Sunburn is an acute inflammatory condition of the skin cells due to overexposure to ultraviolet radiation from the sun (or a solar lamp). The inflammation may be mild or serious, depending on the degree of exposure.

Something more may also be involved, however. Many people use large amounts of chemical products on their skin, as well as internally—chemical deodorants, chemical soaps, chemical perfumes, cough medicines, and numerous other medications. These substances all weaken the skin. When a person who uses a large number of these products lies in the sun, he or she is more prone to burn.

Also, according to Ayurveda, people of pitta constitution, who are generally more fair-skinned, are more susceptible to sunburn.

When sunburn is severe, the person may have many symptoms, including dizziness, nausea, blisters, undue sensitivity to light, and peeling of the skin. Repeated sunburn may create premature aging and wrinkling of the skin, and the person looks old, like a roasted potato!

Ayurveda has numerous recommendations both to prevent sunburn and to treat it effectively if it happens.

TOPICAL REMEDIES TO SOOTHE SUNBURN

- Apply aloe vera cream to the site of the burn. You can also use some pure aloe vera gel (with no preservatives added), or if you have access to an aloe plant, lightly rub a piece of the plant on the sunburned area.
- Coconut oil is also effective for soothing sunburned skin.

- Take a gauze pad, dip it into cool milk (either cow's milk or goat's milk), and apply it directly to the sunburn. If you have no milk available, use a cloth dipped in cool water, but milk is better.
- Apply *tikta ghrita* (bitter ghee) topically.
- Pulverize some lettuce and apply the pulp directly to the sunburn.
- Place an icebag or a bag of frozen food (like corn, peas, or beans) on the affected area to cool the skin. But don't leave ice touching your skin for more than a minute or two without a break.
- Apply a little fresh cream (from milk) directly on the sunburned skin.
- Make a paste of sandalwood and turmeric by mixing equal amounts of these herbs with a little cool water. Apply gently to the sunburned area. It will have a cooling effect. *Note:* This paste will stain your skin yellow for at least a couple of days and will also stain any clothing it touches.

TO PREVENT SUNBURN

- Avoid or minimize exposure to sunlight from 10 A.M. to 3 P.M. Those are the peak hours of the sun's strength.
- Limit your time in the direct sun to no more than half an hour. At high altitudes, even this may be too long.
- Before going into the sun, apply *neem* oil to the exposed parts of the body. *Neem* is a good sun blocker that will help protect your skin.

NOTE: As always in this book, "*neem* oil" does not mean pure *neem* extract, but a few drops of *neem* mixed with sesame oil or another mild oil. This is the way *neem* oil is commercially sold.

- Drink coconut water or coconut milk.
- Follow a pitta-soothing program. Don't take saunas or get overheated. Eat pitta-pacifying foods, especially avoiding spicy and fermented foods (see [chapter 8](#)).
- Before and after showering, apply *neem* oil to your skin. Coconut oil is also helpful.

☞ ☞ Swelling (Edema) During Pregnancy ☞ ☞

See also "[Edema](#)"

During pregnancy, the enlarged uterus exerts pressure on the pelvic blood vessels and constricts them, which leads to swelling of the feet. Swelling may also be due to lack of protein, lack of iron, or lack of exercise. Or it may be due to high blood pressure or

poor circulation. Whatever the cause, it is important to treat it.

EFFECTIVE HERBAL REMEDIES

DRINK CUMIN-CORIANDER-FENNEL TEA. Mix equal proportions of these three herbs, and use $\frac{1}{4}$ to $\frac{1}{2}$ teaspoon of each herb per cup of hot water. Drink 2 or 3 times a day.

USE A DIURETIC TEA. If the swelling is severe, certain herbal teas made of diuretic herbs can help to stimulate the kidneys. Make a tea of *punarnava* or *gokshura*, or combine the two in equal proportions, using 1 teaspoon of the mixture steeped in a cup of hot water. Drink 2 or 3 times a day. This tea will act as a mild diuretic and will remove the swelling.

TIME TO SEE THE DOCTOR

In certain extreme cases, during the last stage of pregnancy the placenta releases toxins and creates a condition characterized by increased blood pressure and swelling of the extremities; proteinurea, convulsions, and coma can develop. This dangerous condition is called eclampsia or pregnancy toxemia. It cannot be treated by home remedies; it requires prompt, expert medical supervision and care.

However, the following herbal formula will help, as an adjunct to treatment, to relieve the convulsions:

brahmi

jatamamsi

shanka pushpi

Mix these herbs in equal amounts. Use $\frac{1}{2}$ teaspoon of the mixture to make a tea, and drink it twice a day.

NOTE: If there is ever a generalized swelling of the entire body (including, in men, a swelling of the penis and scrotum), this is a very serious, life-threatening condition that requires immediate treatment.

FOR HIGH BLOOD PRESSURE. Use these herbs:

hawthorn berry

passion flower

arjuna

Mix in equal proportion, and use 1 teaspoon to make a tea. Take twice a day after meals until the situation is normalized. This simple tea will effectively regulate blood pressure. (For more suggestions, see “[Hypertension](#).”)

IF THE PROBLEM IS ANEMIA. The swelling may be due to anemia. Folic acid anemia is common during pregnancy, and iron deficiency anemia not uncommon. You can have tests to determine whether you have these deficiencies. If you do, providing the missing ingredients will help reduce or eliminate the swelling. See “[Anemia](#)” for suggestions.

PROTEIN DEFICIENCY. Lack of protein may also be the culprit. If you determine that this is the case, eat more protein foods, such as soy products.

ADDITIONAL RECOMMENDATIONS TO RELIEVE EDEMA

EXERCISE. Walking every day for 20 to 30 minutes will improve circulation and help reduce swelling.

AVOID SALT. Salt makes the body retain water and promotes swelling.

RAISE YOUR FEET. When lying down, put a pillow under your feet. This will drain the water and help to gradually decrease the swelling.

☞ ☞ Teeth and Gums—Ayurvedic Care ☞ ☞

According to Ayurveda, teeth are a by-product of bone. Cavities in the teeth, and receding gums, are signs of aggravation of vata in the skeletal system.

FOR MINERAL DEFICIENCY. Problems with teeth are often related to a deficiency of calcium, magnesium, and zinc. To prevent future problems or to alleviate a deficiency:

- Chew a handful of calcium-rich white sesame seeds every morning. (Then brush your teeth without putting any toothpaste on the brush, so the residue of the sesame seeds can rub against your teeth, polishing and cleaning them.)
- You can also use a mineral supplement containing a daily dose of approximately these amounts: calcium (1,200 mg.), magnesium (600 mg.), and zinc (60 mg.) per tablet.

MASSAGE YOUR GUMS. You can make your teeth healthier and more beautiful by massaging your gums daily with sesame oil. Take a mouthful of warm sesame oil and swish it from side to side for 2 to 3 minutes. Don't swallow it. Spit out the oil, then gently massage your gums with your index finger. This is an excellent preventive measure for receding gums, tooth infection, and cavities.

HERBAL TOOTH CLEANSER. Ayurvedic dentistry recommends the use of bitter and astringent herbs for cleaning the teeth. The main herbs used are *neem*, which is bitter, and *lohdra*, *kushta*, and *bilva*, which are all astringent. You can make an excellent cleanser for brushing your teeth by mixing the powdered form of *neem* and an equal amount of any of the astringent herbs and use the mixture to brush your teeth as usual. You can also buy commercial toothpastes and powders containing these herbs in natural food stores

and from most suppliers of Ayurvedic herbs.

- You may also be able to find powders for the teeth made out of finely ground, roasted almond shells. These are highly beneficial for maintaining the good health of the teeth.

USE TEA TREE OIL FOR RECEDING GUMS AND SENSITIVE TEETH. Receding gums, and sensitivity of teeth to cold and hot temperatures, indicate a bacterial infection in the roots of the teeth.

- To treat the infection, wet a clean toothbrush, and put a few drops of tea tree oil directly onto the brush. Brush your teeth. Then use a cotton swab to apply some tea tree oil to the exposed part of the gums. This treatment will help arrest further infection of the teeth and will take care of pain and sensitivity to hot and cold.

- You can also apply some tea tree oil to your dental floss. This will help you reach some possible deep pockets of infection that the brush doesn't reach. (Commercial floss treated with tea tree oil is also available.)

CHEW WELL. Ayurveda emphasizes the importance of chewing food well. This not only helps the digestive process, but also stimulates the gums.

TRY FIGS. For strengthening teeth and gums, try eating 4 figs and chewing them well, once a day.

TAP YOUR TEETH. *Gently* tap your teeth together 5 or 6 times, almost clenching them—but gently, so that you don't break the crowns! This is said to stimulate the energy meridians related to the teeth.

⌘ ⌘ Tendinitis ⌘ ⌘

In strenuous sports and athletic activities such as running or jumping, a person can pull a muscle and cause inflammation of the tendon, leading to tendinitis or bursitis. This condition can also result from a nonstrenuous activity repeated often enough to become stressful. An example is carpal tunnel syndrome, the result of working for hours a day at a repetitive task such as typing at a computer keyboard.

FOR MILD CONDITIONS

COLD COMPRESS. At the site of tendinitis, apply a cold compress such as an ice pack or a bag of frozen vegetables. That will minimize the inflammation.

APPLY A COOL PASTE. Make a paste of sandalwood and turmeric powder in equal amounts. Mix the powders in enough cold water to form a paste, and apply it on the painful area.

- A paste of salt and turmeric will also be effective.

GENTLE STRETCHING. Careful, gentle stretching of the affected muscles will gradually help to improve circulation and heal the pain and inflammation.

TAKE ANTI-INFLAMMATORY HERBS. Taking some anti-inflammatory herbs will accelerate healing. Make this formula:

kaishore guggulu 2 parts

manjistha 2 parts

musta 2 parts

guduchi 3 parts

Take ¼ teaspoon of this mixture 2 or 3 times a day with warm water after meals.

FOR ACUTE OR CHRONIC CONDITIONS

APPLY HEAT. For both acute tendinitis, where there is much pain, and chronic tendinitis, apply soothing heat, such as a liniment (a penetrating, heating ointment) or a warm water compress. Wet or damp heat is effective to minimize the pain or inflammation.

OIL MASSAGE. Gently massage the affected area with *mahanarayan* oil, and then soak it in warm water.

❧❧ Tinnitus ❧❧ (Ringing in the Ears)

NOTE: The same line of treatment used to improve hearing is absolutely good to silence tinnitus, or ringing in the ear, so please consult that section (“Hearing Loss”) in addition to the following recommendations.

HERBAL REMEDIES. According to Ayurveda, ringing in the ear is a vata disorder. To alleviate this root cause—aggravation of vata in the nervous system—prepare a tea made from equal amounts of comfrey, cinnamon, and chamomile. Steep up to 1 teaspoon of this mixture per cup, and drink 2 or 3 times a day.

- Also, you can take *yogaraj guggulu* (200 mg. 2 or 3 times a day) with warm water, after food.

- Gently rubbing the mastoid bone (behind your ear) with warm sesame oil may be helpful. Try it twice day, morning and evening, for a week, and see if it helps.

- Garlic oil is often effective. Place 3 drops into your ear at night before going to bed. (See [appendix 2](#) for how to prepare oils like garlic oil.)

❧ ❧ Toenail Infection ❧ ❧

When persons who have thick, tough toenails cut their nails in a convex curve, as the nails grow the edges may start to penetrate the skin and cut the soft tissue. This may lead to inflammation, swelling, pain, and possible infection. Generally, the nails of vata and kapha individuals grow stronger, so they are more prone to toenail infection. Shoes that are too tight can create pressure on the nails, also potentially leading to toenail infection.

The long-term solution is simply to cut the nail straight across, or with a slightly half-moon shape (a concave curve). That allows the nail to grow straight forward instead of into the skin at the sides.

If the nails have already begun to grow under the skin, soak your foot in warm water to soften the nails. When the nails are soft, clean them and apply some tea tree oil mixed half and half with *neem* oil under the nail. Then cut the nail straight across.

To prevent recurrences, wear soft shoes that are not tight, and cut the nails straight across or in a shallow half-moon shape.

❧ ❧ Toothache ❧ ❧

See also [“Teeth and Gums—Ayurvedic Care”](#), [“Gum Disease”](#)

Toothache may be caused by receding gums, cavities, sensitivity due to hyperacidity, and/or infection.

If high acidity is the problem, the person will tend to get heartburn and acid indigestion in addition to toothache. You can control the acidity by following a pitta-soothing diet, particularly avoiding spicy foods, pickles, citrus fruits, and fermented foods (see [chapter 8](#)).

When the teeth have cavities or the gums are receding, the roots of the teeth may become exposed. These nerve-filled areas become sensitive to cold or heat. (Sensitivity to cold is a sign of receding gums, while sensitivity to heat indicates infection.)

For any toothache, use a cotton swab to apply a little tea tree oil or clove oil at the site of the pain. Or a small piece of natural, edible camphor (not the synthetic kind, which is poisonous) can be placed next to the painful tooth. The saliva will mix with the camphor and will relieve the toothache. (See [appendix 2](#) for directions on making medicated oils.)

Don't just settle for getting rid of the pain, however. If you have a cavity, be sure to get it taken care of. And to avoid future problems, follow the recommendations for healthy teeth and gums in the section entitled, [“Teeth and Gums—Ayurvedic Care.”](#)

❧❧ Ulcers ❧❧

Ulcers in the gastrointestinal tract are a pitta disorder. Pitta is hot, sharp, and penetrating, and when aggravated, it can erode the surface of internal organs or soft tissues. The mucous membrane lining the esophagus, stomach, duodenum, or colon can develop ulceration because of excess pitta.

Individuals with a pitta constitution or pitta disorder, or whose blood type is O and Rh-positive, are more prone to developing ulcers. A person with ulcers may experience pain, heartburn, nausea, vomiting, midback ache, and shoulder pain.

THE MAIN PRINCIPLE FOR TREATMENT: PACIFY PITTA. Ulcers can be effectively treated by treating the high pitta. If you have ulcers, strictly follow the pitta-pacifying diet: avoid hot spicy food, fermented or sour food, sour fruit, and citrus fruit. (See [chapter 8](#) for complete guidelines.) Stay away from alcohol, tobacco, and coffee. And do not take aspirins or steroids, which can accelerate erosion of the gastric mucosal membrane.

TWO HERBAL FORMULAS. An effective herbal treatment for ulcers is the following formula:

shatavari 5 parts

yashti madhu 3 parts

shanka bhasma 1/8 part

kama dudha 1/8 part

Take 1/2 teaspoon of this mixture twice a day, after lunch and dinner, with a little warm water.

- Taking 1 teaspoon of *sat isabgol* (psyllium husks) at bedtime with a cup of warm milk will also be beneficial.

REDUCE STRESS. Stress is often the cause of or a complicating factor in ulcers. To help dissolve stress, make yourself some Tranquillity Tea from these herbs:

jatamamsi

brahmi

shanka pushpi

Combine the herbs in equal proportions, and use about 1/2 teaspoon of the mixture steeped in 1 cup of boiling water to make a tea. Drink it about an hour before going to bed.

YOGA POSTURES. Yoga *asanas* can help you relax and relieve stress, which is beneficial for ulcers. The following positions are particularly recommended for helping with ulcers: Leg Lift, Camel, Cobra, Bow, Bridge, and Spinal Twist. (See illustrations of yoga postures in [appendix 4](#).)

BREATHE AWAY STRESS. *Shitali pranayama*, a breathing exercise that has a cooling, soothing effect, will be helpful. Moon breathing (breathing through the left nostril only) for 5 to 10 minutes has a cooling effect that may also be helpful. Breathing exercises are described in [chapter 6](#).

EAT EARLY IN THE EVENING, AND EAT SMALL MEALS. Persons with ulcers should not eat late at night. They also do best if they eat small, frequent meals. Don't keep the stomach empty for long. At least have breakfast, lunch, and dinner, so the acid secreted in the stomach can be utilized for digestion.

AN EFFECTIVE HERBAL ACID BLOCKER. Today many people use acid-blocking medications for ulcers. The best acid blocker in Ayurveda is a combination of the herbs

brahmi

jatamamsi

yashti madhu

in equal proportions. A tea made from ½ teaspoon of these herbs per cup of water, taken after lunch and dinner, can help to inhibit acid secretions and prevent ulceration.

❧ ❧ Urinary Incontinence ❧ ❧

Urinary incontinence is primarily a vata disorder, caused by weakness of the muscles of the bladder, especially the bladder sphincter. If that becomes weak and uncontrolled, a person may lose voluntary control of urination to a greater or lesser extent.

Another cause of incontinence, especially in the West, seems to be the frequent use of rest rooms. When some people see a rest room, they just use it, whether the need is urgent or not. They apparently think (perhaps subconsciously), "Maybe I won't have another chance for a while." But the bladder muscles become weak and lose the capacity to retain a large amount of urine and pass it out only when really necessary.

Urinary incontinence is more common among women than men, because their urethra is quite short. When women cough, sneeze, or strain, or even when laughing, their bladder may leak a little urine.

This disorder can be controlled by certain herbs, a periodic oil enema, and certain yoga exercises. Let's begin with the latter.

YOGA

ASHWINI MUDRA. This exercise involves using the perineal muscles to suck the anus inward, tightening the gluteal muscles. It will also strengthen the bladder sphincter. Do

10 to 12 repetitions, 2 or 3 times a day (while sitting down).

OTHER POSTURES. Sitting in the Vajrasana pose (see illustration in [appendix 4](#)) will help, as will the Forward Bend, Shoulder Stand, and Plow pose.

HERBS

Prepare the following mixture:

ashwagandha 5 parts

bala 3 parts

vidari 2 parts

Take ½ teaspoon with warm water twice a day, after meals.

- Every day eat a handful of white sesame seeds with some jaggery or natural brown sugar. This will strengthen the bladder and help to correct the incontinence.

OTHER REMEDIES AND RECOMMENDATIONS

SESAME OIL ENEMA. Once a week, do a *basti* (enema) using warm sesame oil (see [appendix 2](#)). Insert about 1 cup of oil into the rectum, and try to retain it for at least 5 to 10 minutes. If the oil runs out, don't worry. Sesame oil enema is one of the most effective ways to balance vata.

AVOID ALCOHOL AND CAFFEINE. Caffeine is a diuretic and intensifies the need to urinate. Therefore, incontinence and urgency of urination are quite common among coffee drinkers or users of other caffeinated beverages, such as tea or cola. Alcoholic beverages also promote incontinence. So it is better to avoid caffeine and alcohol if you have any problem with urinary control.

VATA-PACIFYING DIET. Since urinary incontinence is largely due to excess vata, following a vata-balancing diet will help. (See the dietary guidelines in [chapter 8](#).)

❧❧ Yeast Infections ❧❧

Yeast infections are a result of a mixed pitta and kapha disorder. To heal them, first one has to follow a proper diet, which must not increase either pitta or kapha. Strictly avoid sugar, fermented food, and yeast-containing bread. Ideally, it is best to follow a diet that is pitta-soothing, but not kapha-provoking. The best way to manage this balance is to look in the food charts for each doshic type (see [chapter 8](#)) and favor foods in the “yes” column for both kapha and pitta.

EFFECTIVE HERBAL REMEDY. A potent herbal formula that will help to heal a yeast infection is:

turmeric $\frac{1}{4}$ teaspoon

licorice $\frac{1}{2}$ teaspoon

shardunika $\frac{1}{4}$ teaspoon

Take this complete mixture (1 teaspoon of herbs) twice a day with warm water until your symptoms are gone.

DOUCHE FOR VAGINAL YEAST INFECTION. If the yeast infection is vaginal, you will find it helpful to douche with licorice tea. Boil 1 tablespoon of licorice powder in 1 pint of water for 5 minutes. Cool, strain, and use that tea for the douche. You will notice the healing effect quickly.

NOTE: If you have a thick discharge from the genitals, along with an itching and burning sensation, it will be better to do the douche with *triphala* tea instead of licorice. Use the same procedure: take 1 tablespoon of *triphala*, boil it in a pint of water, cool, strain, and use the tea as a douche.

Conclusion

Taking Responsibility for Your Own Health: How to Integrate Ayurveda into Your Life

Ayurveda is a comprehensive approach to health that encompasses all facets of our life and living. Body, mind, and spirit; work and relationships; diet and the external environment; season of the year and daily routine; physical exercise and spiritual practices—all these and many additional factors are treated in the classical Ayurvedic texts.

From our deepest spiritual concerns (Who am I? Where do I come from? What is the purpose of my life?) to the most practical and mundane (How can I heal a sore throat? How much exercise is best for me? What foods should I eat?), the five-thousand-year-old living tradition of Ayurveda has answers that are practical and meaningful.

As a system of natural medicine, Ayurveda is not about curing symptoms, although it certainly accomplishes that. Rather, it is about building a way of life that creates health and healing. To make the best use of Ayurveda—and of this book—you have to put its principles into action. Just to run to the remedies section ([Part III](#)) when you have a health problem is truly to miss the richness and beauty of Ayurveda, which is a complete science of life that can enable every individual to be healthy and happy.

I would not be telling the truth if I said that incorporating Ayurveda into your life is totally simple. It is not. You will probably have to learn some new principles, as well as understanding the nature of your constitution, your mind-body type. It is likely that from what you learn, you will want to make changes in your daily routine, such as waking up earlier or changing the amount and type of exercise that you do. You may decide it would be wise to modify your diet, perhaps dropping some favorite foods that may be inappropriate for you. Such changes in our habitual way of living do not happen overnight.

On the other hand, I would also not be telling the truth if I didn't say that every small step you take toward an Ayurvedic lifestyle will have an immediate and positive effect on your body, mind, and consciousness.

To incorporate Ayurveda into your life, you have to start somewhere. Many people find it easiest to begin by following some of the dietary guidelines for their body type. Then gradually, little by little, you can adopt some of the recommendations for the daily routine, adjusting your daily schedule to be more in tune with nature, or using some of the suggested breathing exercises or meditation practices.

The underlying assumption of Ayurveda is that each individual has the power to heal himself or herself. We each have the ability and the freedom to recover our health if we

become ill, or to maintain vitality and joy of living. We can do this by understanding our body and its needs, and by attending to those needs as they change in response to the ever-changing outer environment and our inner world of feelings. For this, consciousness is the key: moment-to-moment awareness of what is happening.

In this book, I discuss more than a hundred health conditions, from acne to yeast infections, from headaches to athlete's foot. Within those major categories of symptoms, I describe literally hundreds of smaller signs and symptoms. These signs and symptoms are nothing but the body's language. By presenting us with a headache, diarrhea, fever, toothache, joint pain, insomnia, emotional anxiety, fear, or insecurity, the body is talking to us, letting us know that something is wrong, something is out of balance and needs our attention. It is the language of *tridosha*, of vata–pitta–kapha.

Ayurveda says that whatever symptom we experience is an expression of doshic imbalance. To restore good health, we have to reestablish balance by juggling the three doshas, favoring this one or pacifying that, in order to attain harmony between *vikruti*, the present doshic state, and *prakruti*, the original state of our constitution.

Ayurveda teaches us how to read this language of signs and symptoms. When, for example, there are signs of excess heat in the body—skin rash, heartburn, a flaring temper—we know that pitta dosha is in excess and needs to be pacified. Similarly, out-of-balance vata may speak to us in terms of insomnia, constipation, or anxiety, while kapha imbalance may communicate its presence as lethargy, overeating, or congestion in the lungs, sinuses, and chest.

We should never disregard this language of *tridosha*. It is the foundation of health and happiness, as well as of ill health.

In this book I have tried to share with you how to take care of those symptoms—not symptomatically but radically, fundamentally, basically. I have tried to show you how to use the symptoms as a catalyst to reestablish balance within the *tridosha*, as well as between body, mind, and consciousness.

By any method, creating balance within body, mind, and consciousness, and within vata–pitta–kapha, is called healing. The purpose of this book is to help you in your healing efforts, so that you can attain total health in your life.

In health and healing, there is an equal sharing of responsibility between the physician and the patient. In this book, really, every reader is taking on both sides of that responsibility. By using your own insight, perception, observation, and judgment, you are becoming your own physician and healer. By following the recommendations given in this book in order to bring healing, you are taking the role of the patient. You are the healer, and you are the person who receives healing.

Use this knowledge well to bring health, harmony, and happiness into your life.

Appendix 1

How to Use the Healing Properties of Metals, Gemstones, Colors, and Aromas

Ayurveda teaches that everything in existence is imbued with the energy and intelligence of Universal Consciousness. That is because all forms of matter, organic and inorganic, are simply the outer manifestations of this most subtle creative energy. Matter is the trapped light of consciousness. The vital force of life flows from the universal source, the essence of all matter, and manifests in the myriad forms and phenomena of nature.

The classic texts of Ayurveda make it clear that all substances in nature contain this cosmic creative intelligence, and thus have a healing value when used in the proper manner. That is why, in its quest to create and maintain perfect health, Ayurvedic medicine makes use of almost everything in nature and in daily life, including food, breathing, exercise, meditation, relationships, yoga, and massage, as well as regulated daily and seasonal routines. It also uses thousands of herbs and herbal formulas.

In addition, Ayurveda utilizes the healing properties of metals, gemstones, colors, and aromas. These contain special, potent forms of energy which may be drawn upon for healing purposes. Most of these methods, clearly described in the ancient textbooks, have been used safely and successfully for healing for thousands of years, though they have been scarcely known and little appreciated in the West until very recently. This appendix will provide a brief introduction to these healing modalities.

Metals

For medicinal purposes, metals are traditionally processed to be taken internally in small doses, after undergoing rigorous and extensive purification to negate any toxic effects upon the body's vital organs. The following recommendations are safe, as they do not involve ingestion of the actual metal.

COPPER

Copper alleviates excess kapha and reduces fat. It is a good tonic for the liver, spleen, and lymphatic system and helps in curing anemia. To treat obesity as well as liver and spleen disorders, thoroughly wash some copper pennies and boil them in a quart of water (or boil a quart of water in a copper vessel) until half the water remains. Take 2 teaspoons of this copper water 3 times a day for a month. It is also helpful to buy

a copper drinking glass, fill it every night with pure water, and drink the water in the morning.

GOLD

Gold is strengthening to the nervous system and the heart, improves memory and intelligence, and increases stamina. It is also good for weak lungs. Gold can be helpful for students' pre-exam tension, for arthritis, and for heart arrhythmia.

The energy of gold may be harnessed by preparing gold medicated water. Use pure gold (24 karat is best), such as a gold band. Place the gold into 2 cups of water, and boil until 1 cup evaporates. Take 1 teaspoon of this gold water 2 or 3 times a day to energize the heart, strengthen mental faculties, and awaken pure awareness. (This process won't hurt your gold.)

You can also make golden rice. While cooking rice, place a piece of gold in the rice pot, and cook as usual. When the rice is finished, remove the gold before serving.

NOTE: Gold has heating properties and should be used sparingly by individuals with a pitta constitution.

SILVER

Silver has cooling properties and is beneficial for treating excess pitta. Silver increases strength and stamina and is thus helpful for balancing vata. Emaciation, chronic fever and weakness after fever, heartburn, inflammatory conditions of the intestines, and profuse menstrual bleeding may all be helped by silver. Silver is antiseptic, antibacterial, and disinfectant. Make silver water following the above directions for gold water, and take 1 teaspoon 2 to 3 times a day. Drink warm milk heated in a silver vessel to build up strength and stamina.

IRON

This metal is beneficial for bone marrow, bone tissue, liver, and spleen. It increases the production of red blood cells and helps to cure anemia. Iron also strengthens muscle and nerve tissues and is rejuvenating. For extra iron, try cooking in cast iron pots and pans. However, excess iron in the system is harmful, so be careful about using it. Although women during their menstruating years may be iron-deficient and may benefit from additional iron, very few men in Western society need extra iron. An exception might be longtime strict vegetarians.

Gems and Stones

Gems and precious stones contain healing energies that can be activated by wearing them as ornaments, such as rings or necklaces, or by placing them in water overnight and drinking the water the following day. Gems enliven the vital energy centers in the body (the chakras) and have a direct influence on vata, pitta, and kapha. They may be used to pacify or activate specific organs of the body, or to enhance or neutralize the effects of particular planets in the person's astrological birth chart.

Before we go into the effects of specific gems and stones, here are a few important general points.

- Gems tend to absorb the qualities and energy vibrations of their owners. It is beneficial to purify any stone before using it. Soaking it for two days in saltwater or milk should be sufficient. This will not harm the stone.
- When you wear a gemstone, it should touch the skin through a small window in the setting, so that the subtle energies of the stone can interact directly with the energies of the body.
- Where you wear the stone is important. Here are some recommendations:
 - diamond—ring finger
 - pearl—little finger
 - red coral—ring finger
 - emerald—little finger
 - opal—ring finger
 - yellow sapphire—index finger
 - blue sapphire—middle finger

Ayurveda generally recommends that rings be worn on the right hand, though in the West, if someone wants to wear their wedding ring on the left hand to conform with tradition, that is all right.

- Processed or chemically treated stones may not have the same healing energy. It is best to get authentic, unprocessed, clean stones without a flaw or crack. When you are considering buying a stone, be sure to use a magnifying glass to examine it for cracks or imperfections.
- Stones should be 3 to 5 karats if possible, but a 1-karat diamond is large enough. A stone that is too small will not produce much of an effect.
- Unless you are knowledgeable both in stones and in Vedic astrology (*jyotish*), it is wise to consult an expert before investing in a stone. The wrong gem for you, or one worn on the wrong part of the body, can have a negative influence.

Here are some of the characteristics of the main gems and stones.

RUBY

Astrologically, the ruby represents the sun. It is a life-protecting stone that promotes longevity, especially for vata and kapha individuals, and brings prosperity. This gem strengthens concentration and bestows mental power. It also strengthens the heart. Rubies pacify vata and kapha but may elevate pitta. Garnets have the same vibration as rubies; they are the poor man's ruby. Wear both rubies and garnets either in a ring on the ring finger, or in a necklace.

PEARL

As rubies represent the sun, pearls symbolize the moon. They have a cooling effect and a calming, healing vibration. Pearls are balancing to all the doshas, though their cooling action is particularly good for pitta. Pearls confer mental peace and tranquillity. Pearl ash is used internally to effectively treat many ailments. You can gain many of the strengthening effects of pearls by making pearl water. Place 4 or 5 pearls in a glass of water; let it stand overnight, and drink the water in the morning.

YELLOW SAPPHIRE

This precious stone, which represents Jupiter, brings groundedness, stability, and wisdom. It helps to calm both vata and pitta and may slightly increase kapha qualities. It strengthens the heart and also builds lung and kidney energy. Yellow sapphire should always be worn on the index finger, the finger of Jupiter. Yellow topaz, the poor man's sapphire, has many of the same qualities and produces similar benefits.

BLUE SAPPHIRE

This beautiful precious stone represents Saturn and brings the benefits of that very spiritual planet. Saturn, a deity of earth and iron, confers enlightenment. Blue sapphire calms vata and kapha and may stimulate pitta. It builds up muscles and the skeletal system and helps to heal arthritis. Wear blue sapphire on the right middle finger, preferably in a silver setting. Do not wear it with diamonds; this will create disharmony.

LAPIS LAZULI

This stone, which has Saturn-like energy, is heavenly and sacred. It gives strength to the body, mind, and consciousness, and it sensitizes the wearer to higher spiritual vibrations. It strengthens the eyes, calms vata and pitta, and is helpful for anxiety, fear, and weakness of the heart. It is also good for the liver and for skin diseases. Lapis should be set in gold and worn on the little finger, or worn as a necklace.

EMERALD

This powerful precious stone brings prosperity and spiritual awakening. It calms vata and pitta, settles the nervous system, and relieves nervousness. Symbolic of the planet Mercury, emeralds improve writing skills, enhance the power of speech, and promote intelligence. They are best set in gold and worn on the little finger.

DIAMOND

This very powerful gemstone combats premature aging, enhances the span of life, and strengthens immunity. Its energy brings subtle energy vibrations to the heart, brain, and deeper bodily tissues. It is the best stone for rejuvenation. It brings prosperity and is spiritually uplifting.

The doshic effects of diamonds vary according to their color. Red diamonds have a fiery energy that stimulates pitta; blue diamonds are cooling and calm pitta while increasing kapha. Clear, colorless diamonds calm pitta but increase both vata and kapha.

Symbolic of the planet Venus, diamonds actually do help to create a close bond in relationships and are rightfully associated with marriage. These stones stimulate *shukra*, the body's reproductive tissue. Art, music, romance, and sex all go together with this stone. Wear your diamond set in gold, either as a necklace or as a ring on the ring finger. But note: Diamonds of low quality may have negative effects upon the body.

RED CORAL

This gemstone from the sea represents the planet Mars. It calms pitta and helps one to control anger, hatred, and jealousy. Coral gives energy to the liver, spleen, and pericardium. Wear your red coral as a necklace or as a ring set in copper (preferably), silver, or white gold and worn on the ring finger. Red coral is strength-giving and imparts gracefulness.

OPAL

This semiprecious stone represents the planet Neptune. It strengthens *majja dhatu* (bone marrow and nerves) as well as *shukra dhatu* (reproductive tissue). It improves vision, relieves fever, calms pitta, and is good for migraine headaches. Opals enhance spiritual feelings, increase devotion, and help to unfold intuition. This gem is particularly beneficial for individuals with Neptune in their third, fourth, sixth, tenth, or twelfth astrological house. It should be set in gold or silver and worn on the ring finger.

CAT'S-EYE

This stone is good for allergies, repeated colds and congestion, and allergic asthma. It pacifies kapha and vata while slightly increasing pitta. It aids in healing kidney dysfunction. Cat's-eye enhances awareness and helps a person not get caught up in emotions. People working in psychological healing should wear this stone in a gold setting on their ring or little finger; it will help protect them from negative influences.

QUARTZ CRYSTALS

These stones have a Venus-like vibratory energy that is somewhat like that of diamonds. They calm vata, improve the quality of perception, strengthen communication, and enhance intuition. You can wear quartz crystals as a necklace, or set in either silver or gold and worn on the ring finger.

ONYX

This stone is excellent for vata disorders. It is good for old age, debilitating disorders, and neurological dysfunction, and it helps with epilepsy, Parkinson's disease, and even schizophrenia. It induces quiet, deep sleep yet combats lethargy. It is good for memory and promotes positive thinking. Onyx makes life peaceful and happy and enhances love in relationships. It has energy vibrations that are Sun-like and Jupiter-like. This stone should be set in silver and worn on the ring finger. (If your Sun sign is in Sagittarius or Gemini, it is better not to wear this stone.)

JADE

Jade is beneficial for longevity. It strengthens kidney energy and is reputed to bestow success upon its wearer. This stone is also good for the power of speech. It helps to prevent cataracts and is strengthening for the prostate. Wear a jade ring set in silver, on your little finger.

AMETHYST

Amethyst is a stone for the crown chakra and is good for mental clarity. To bring prosperity, it should be set in gold. You can also wear it around the neck on a gold necklace. A person with neuromuscular weakness can be helped by wearing amethysts and by putting them at the four corners of the bed. Some amethysts have a darker color, which gives them a Saturn-like energy similar to blue sapphire. Amethysts bestow dignity, love, compassion, and hope. This gem helps the individual to control emotions and is good for vata and pitta imbalance.

AQUAMARINE

A substitute for emerald, which symbolizes Mercury, aquamarine reduces dullness of mind, promotes happiness and intelligence, enhances the power of speech, and improves memory. This stone also has Venus-like qualities; it is good for married couples to wear aquamarine to enhance love in their relationship. Aquamarine should be set in silver and worn on the little finger.

Remember that in all these cases, simply wearing the correct stone is not enough to take care of a doshic imbalance; you need to watch your diet, meditate, do appropriate exercise and yoga postures, and consciously and conscientiously look after your day-to-day and moment-to-moment health.

Four Inexpensive Stones to Help Balance the Doshas

Though some of the gemstones discussed in this appendix can be obtained for a moderate amount of money, many may be prohibitively priced for you at this time. If so, here are four inexpensive stones you can use to help create balance in your mind and body.

When vata dosha is excessive, you can use *rose quartz* to promote balance. The warming color and energy of rose quartz can bring relief to vata ailments such as nervousness, dry skin, constipation, intestinal gas, and lower back pain.

For aggravated pitta, use *red coral* or *pearls*. Their cooling energy will help with pitta disorders such as angry emotions, various inflammatory conditions and “-itises” such as colitis and conjunctivitis, as well as hyperacidity.

Kapha dosha can be balanced by wearing *garnets*. The deep red color of this stone enlivens the energy in the body and reduces the effects of excess kapha, such as water retention, lethargy, depression, and overweight.

More information about these and other gemstones, and how to wear them (settings, correct fingers, and so on) is provided in [chapter 8](#) of my book *Secrets of the Pulse* (see the [Reading List](#)).

Colors

Ayurvedic treatments also make use of the healing properties inherent in colors. Because the basic colors of the rainbow are correlated with the bodily tissues (*dhatus*) and the doshas, the vibratory energy of the colors may be used to help establish balance in mind and body.

Color is nothing but light, and light is radiant energy imparted from every atom. The

source of light and color is the sun. In our solar system, whatever colors we perceive come from sun rays. Every color has a different wavelength, frequency, and vibration. When we put a prism in the sunlight, the seven colors of the rainbow can be separated, but the equal presence of all seven gives white light. The absence of any color is black, darkness. So black is a negative color, and white a positive color.

You can influence your health and happiness by choosing appropriate colors for your clothing and surroundings at home and at work. Also, if you place colored, translucent paper or plastic wrap around a jar or glass of water and place it in sunlight for four hours, the water will become infused with the vibrations of that color. Drinking the water will then bring beneficial results.

RED

Red is warming and stimulating. It relieves aggravated vata and reduces excess kapha. However, because of its heating effect, overexposure to this color may aggravate pitta and result in inflammatory ailments such as conjunctivitis. Red is related to our blood. It stimulates the formation of red blood cells and improves circulation. It also helps to maintain color in the skin and gives energy to nerve tissue and bone marrow. *Pink* has a gentler effect, promoting love and calmness, but it may be conducive to lethargy in kapha individuals.

ORANGE

Like red, orange is warming and has a healing energy. It is a sexually stimulating color that gives energy and strength to the sex organs. Paradoxically, in spiritual seekers who have chosen to be celibate, orange helps with renunciation and transforming sexual energy into Supreme Consciousness. Orange is balancing to both vata and kapha but may be aggravating to pitta. It has antibacterial and bacteriostatic properties; it hinders the growth of bacteria.

YELLOW

Yellow relieves excess vata and kapha. It promotes understanding and intelligence and helps energy rise to the crown chakra for spiritual realization. Yellow is a decongestant that helps to relieve kapha congestion. It also acts as an antibacterial. Overexposure to yellow causes excess bile to accumulate and increases pitta dosha.

GREEN

This color has a calming effect upon mind and body and creates a feeling of

freshness. It is soothing to the emotions and brings energy to the heart chakra and feelings of happiness to the heart. Green is calming and pacifying to excess pitta and may aggravate vata and kapha. Green helps to heal ulcers and promotes the growth of granulation tissue.

BLUE

Blue is a cooling color that relieves aggravated pitta. It has a calming effect on the body and mind and helps to correct liver disorders. When a baby has jaundice, placing the baby under a blue light will help it heal faster. Blue is the color of Pure Consciousness. Overexposure to blue may cause aggravation of vata and kapha and may provoke congestion.

PURPLE

This is the color of Cosmic Consciousness and brings an awakening of awareness. It creates lightness in the body and helps to open the doors of perception. Purple relieves excess pitta and kapha but may aggravate vata.

GOLD AND SILVER

Gold, the color of the sun, is a warming color that is beneficial for vata and kapha.

Beneficial Colors for Constitutional Types

For each constitutional type, certain colors are soothing and balancing while others are aggravating. Here is a summary of healthful colors:

- *Vata*: Vata types should minimize the use of dark and cooling colors such as blues, browns, and black. On the other hand, very hot, vivid colors may be overstimulating to vata, which has a tendency toward being hyperactive. So your best bets are warm pastels, sunny yellows, and green, with some warming red and orange.

- *Pitta*: Cool, soft colors are the best for your health and balance of body and mind. Blues and purples/violets are excellent, along with silver (including silver jewelry) and blue-greens. Watch out for reds and oranges, which can inflame pitta dosha, and minimize yellow and gold. Avoid black.

- *Kapha*: Bright, lively, bold colors are good to balance kapha, with its tendency toward lethargy and mental and physical heaviness. Red, yellow, orange, and gold are all good. Even if you feel you look good in green, dark blue, or white, these are

not the best for you from a health standpoint.

Silver, associated with the moon, is cooling and soothes pitta.

Aromas

Every human being has five senses, which are associated with the five elements. Sound and hearing are related to the element of space, color and sight to fire, taste to water, smell and aroma to earth, and touch to air. These five senses are the gateways of perception for the human being, and they can be used for healing purposes.

Aromatherapy uses incense and essential oils made from flowers, plants, trees, and grasses to relay fragrances through the olfactory sense to the brain, in order to bring healing energy to mind and body. Ayurveda teaches that smells are directly related to doshic balance and imbalance, and that certain aromas are heating, cooling, or neutral.

Deer musk, for example, and hina, are heating: they calm vata and kapha but may provoke pitta. Camphor is cooling and fragrant, but its aftereffect is heating; it too calms and pacifies vata and kapha but may stimulate pitta. The aroma of sandalwood is anti-inflammatory and cooling; it is calming and soothing for pitta but may increase kapha or vata.

Khus (the essence of khus grass) is grounding, pleasant, and cooling. It has a sweet smell and pacifies pitta, but it may provoke kapha and vata. Jasmine too is cooling and sweet and good for pitta, but it may build up kapha.

The effect of rose depends somewhat on the color of the flower. Dark red roses are warming, while white- and yellow-colored roses are relatively cooling. In general, the aroma of rose flowers is anti-inflammatory and soothing and has an aphrodisiac quality. Rose aroma can be used for cooling pitta but may provoke vata and kapha.

AROMAS AND THE DOSHAS

- Vata can be balanced by using sweet, warming, grounding aromas such as musk, hina, and camphor. Other good fragrances for vata include orange, clove, cardamom, lavender, pine, angelica, and frankincense.

- Pitta is soothed by the use of cooling, calming, sweet aromas such as sandalwood, khus, jasmine, and rose. Rose geranium, lemongrass, fennel, peppermint, gardenia, and mint may also be beneficial.

- Kapha is pacified and balanced by using aromas with a warming, somewhat stimulating effect. Musk, hina, and camphor are helpful. Some more pungent aromas are also helpful for kapha. Some of these are eucalyptus, cinnamon, myrrh, thyme, basil, rosemary, and sage.

Appendix 2

How to Prepare and Use Herbs, Ghees, and Oils

This appendix briefly explains the preparation of herbs and special remedies such as medicated oils and ghees and offers suggestions on their use. For a complete description of herbs and their uses, as well as thorough directions for the preparation of herbal remedies, please refer to *The Yoga of Herbs* by Dr. Vasant Lad and Dr. David Frawley (see [Reading List](#)).

Herbal Mixtures

USE WHOLE HERBS AND FOODS

Ayurveda believes strongly in the use of whole herbs, foods, and plants. Isolated active ingredients or chemically produced analogues are not equivalent to natural whole food sources. For sources of herbs, see [Resources](#).

HOW MUCH TO PREPARE

If you are going to be taking an herbal mixture for several days, weeks, or even months, to save time you will probably want to prepare a fairly large quantity. Then each time you use it, take $\frac{1}{4}$ to $\frac{1}{2}$ teaspoon unless otherwise directed.

In this book formulas for herbal mixtures are given in “parts” rather than in milligrams or teaspoons. For example, a formula may call for

herb 1 2 parts

herb 2 3 parts

herb 3 $\frac{1}{4}$ part

Choose your own measure, according to how much you want to make. If, for example, you are measuring in teaspoons, you will use

2 teaspoons of herb 1

3 teaspoons of herb 2

$\frac{1}{4}$ teaspoon of herb 3

If you are making a large batch for long-term use, you can measure in tablespoons. A 1- to 2-month supply will require 2.5 to 5.0 ounces of herbs, a palm-size quantity. Again,

when you take the herb each time, use only $\frac{1}{4}$ or $\frac{1}{2}$ teaspoon, as recommended.

WHERE TO OBTAIN HERBS

Most of the herbs mentioned in this book can be obtained in bulk or capsules at a good natural foods store. Always try to get organic, nonirradiated herbs. Special Ayurvedic herbs and herbal formulas are available from The Ayurvedic Institute, the suppliers listed in the Resources, and some Indian groceries.

HOW TO TAKE HERBS

Herbs are almost always taken with a vehicle or medium (*anupana*) to facilitate absorption by the body and to carry the quality of the herbs to the specific tissue or site of the disease. The most common vehicles used in Ayurveda are water, milk, honey, aloe vera, and ghee. Raw sugar may also be used at times. The vehicle varies according to the herb, the illness or condition being treated, your constitution, and other factors. Typically, you might take your herbal dose with a spoonful of ghee or honey; mixed in a cup of warm milk; or placed dry on your tongue and then washed down with some warm or room-temperature water.

Most of the recommendations in this book suggest which vehicle to use. If none is suggested, use warm water. I do not recommend using capsules because taste is important (see [this page](#)), but if you must use them, you can buy some empty 00-size vege-capsules (available at most natural food stores) and fill them with the herbal mixture. This is better than not taking the herbs at all, and may be more convenient when you are traveling or at work.

USING TRIPHALA

Triphala (“the three fruits”) is a wonderful remedy consisting of three of the most important Ayurvedic herbs: *amalaki*, *bibhitaki*, and *haritaki*. *Amalaki* works on pitta dosha, *bibhitaki* on kapha dosha, and *haritaki* on vata dosha. This compound is rejuvenative and strengthening for all three doshas and all seven *dhatu*s; it balances *ojas*, *tejas*, and *prana* and is also an excellent, mild laxative.

Take *triphal*a in the evening, at least 1 hour after your evening meal. Add $\frac{1}{2}$ to 1 teaspoon *triphal*a powder to about 1 cup boiling water. Steep for 10 minutes or until the water is cool enough to drink. You may strain out the herbs before drinking or just leave them at the bottom of your cup.

You may not like the taste of *triphal*a at first—it may taste quite bitter. But if you use it regularly, your health is bound to improve, and you will eventually find that the taste becomes less unpleasant.

Triphala can also be taken in these ways:

1. Place ½ to 1 teaspoon of dry *triphala* powder on your tongue, and wash it down with warm water.
2. For some people, *triphala* acts as a mild diuretic and may disturb sleep if taken at night. If you are one of these people, drink the tea in the morning and it will act in about one hour.
3. If you really hate the taste, you can mix the *triphala* powder with honey and take it that way.

Ghee

HOW TO MAKE GHEE

Two pounds of butter will make one quart of ghee. Put the butter (sweet and unsalted, organic if available) in a heavy, medium-size pot, turn the heat to medium, and heat until the butter melts, taking care not to burn the butter.

Then turn down the heat, cook until the butter just boils, and continue to cook at this temperature. Do not cover the pot, as it is important to boil the water out and separate the solids. The butter will foam and sputter for a while and then begin to quiet down. Stir it occasionally with a stainless steel spatula, scraping the bottom of the pan.

In 12 to 15 minutes your ghee will begin to smell like popcorn and will turn a lovely golden color. Whitish curds will form and separate from the clear ghee. When these whitish curds turn a light tan color and the boiling quiets down, the ghee is ready. Take it off the heat immediately, for it is most likely to burn at this stage. The cooking time should not be longer than 15 to 20 minutes, depending on the kind of pan and the heat source.

Let the ghee cool until it is just warm. The solid curds will have settled to the bottom of the pot. Decant the clear ghee into a container, and discard the curds left on the bottom.

STORING GHEE

Ghee can be kept on the kitchen shelf. It does not need refrigeration. Its medicinal properties are said to improve with age. Don't ladle out the ghee with a wet spoon or allow any water to get into the container, as this will create the conditions for bacteria to grow and spoil the ghee.

EFFECTS OF GHEE

Ghee increases digestive fire and improves absorption and assimilation. It nourishes *ojas*, the subtle essence of all the body's tissues, strengthens the brain and nervous system, and improves memory. It lubricates the connective tissue and makes the body more flexible. Ghee carries the medicinal properties of herbs to all seven *dhatu*s. It pacifies pitta and vata and is okay for kapha in moderation.

NOTE: People who have high cholesterol or who suffer from obesity should be cautious about using ghee.

HOW TO MAKE MEDICATED GHEES

Ghee is a highly effective *anupana* (vehicle) for carrying herbs to the deeper tissues of the body. Hence many remedies are made by cooking herbs into ghee. Examples include *shatavari* ghee, *brahmi* ghee, *tikta ghrita* (bitter ghee), *triphala* ghee, and many others. The process is quite lengthy, and you will probably prefer to purchase these medicated ghees (and medicated oils as well). But if you wish to make it yourself, proceed as follows:

First, make the ghee, as described above.

Next, make a decoction of the desired herbs by cooking 1 part dry herbs to 16 parts of water, or about ½ ounce of herbs per cup (8 ounces) of water. Boil the herbs *slowly* over a low flame until the water is reduced to one-quarter its original amount. For example, 4 cups would be reduced to 1, or 1 cup to ¼ cup. Then strain out the herbs. This process takes several hours. The liquid can be used as a prepared medicinal decoction, but in this case, you are going to use it for your medicated ghee.

Finally, mix equal parts of ghee and the herbal decoction, and cook it over a low flame until all the water evaporates.

Medicated Oils

Medicated oils (*amla* oil, *brahmi* oil, *bhringaraj* oil, garlic oil, clove oil, and others) are made the same way, except that you use oil instead of *ghee* in the final stage.

NEEM OIL

Neem oil is an herbalized oil made from cooking *neem* leaves in a base of sesame (usually) or some other oil. It is not pure *neem* extract, which would be too strong. *Neem* oil is generally available from suppliers of Ayurvedic products.

How Long to Take a Remedy

The general rule is to use remedies until your symptoms disappear. This may take from a few days to a few months, depending on the severity of the disease or condition, how long you have had it, how motivated you are to get well, and other factors.

In addition to taking your remedies, please also look into the underlying causes of your condition. You may need to rethink your diet, daily routine, exercise program, and so on. Simply taking herbal remedies, without making some changes in your lifestyle, will probably not be sufficient to override the behavior patterns that caused your illness in the first place.

So use your common sense. If the condition is chronic, it is unrealistic to expect that something that has persisted for years is going to disappear in a week or a month. On the other hand, if your condition is serious, and you are diligent with your remedies and make appropriate lifestyle changes, and still the symptoms persist, you need to see your doctor for help.

Appendix 3

Special Ayurvedic Procedures

This brief appendix offers guidelines on some of the Ayurvedic procedures recommended throughout [Part III](#).

Basti (Ayurvedic Enema)

Ayurvedic enema treatment (*basti*) introduces into the rectum medicinals such as sesame oil, or herbal decoctions such as *dashamoola*, in a liquid medium. Medicated enemas pacify vata and alleviate many vata disorders, such as constipation, abdominal distension, insomnia, backache, neck pain, arthritis, sciatica, anxiety, and various nervous disorders. It is said that there are at least eighty vata-related disorders, and that *basti* is a complete treatment for 80 percent of them. *Basti* also is effective for treating chronic fever, sexual disorders, kidney stones, hyperacidity, and numerous other conditions.

NOTE: Medicated enemas are not to be used by anyone suffering from diarrhea, bleeding from the rectum, indigestion, cough, breathlessness, ascites, profuse edema, or active hemorrhoids. Individuals with diabetes or anemia, the aged, and children below the age of seven should also not receive medicated enemas. Oil enemas should not be used if there is acute fever, diarrhea, cold, paralysis, heart pain, severe abdominal pain, or emaciation.

The best times for *basti* are in the morning or evening. The stomach should be empty, so wait at least three hours after eating. Make sure the environment is clean, warm, and comfortable: An area where you can lie down near the toilet is best. You will need an enema bag or syringe, a measuring cup, a hot plate or stove (this doesn't have to be in the bathroom!), the oil and/or herbal substance, and towels.

The usual procedure for *basti* is first to introduce 5 ounces of warm (not hot) sesame oil into the rectum and retain it for 10 minutes. Then, without expelling the oil, introduce a mixture of oil and herbal tea and retain it for at least 30 minutes. The mixture should consist of another 5 ounces of sesame oil, mixed with 16 ounces (1 pint) of tea made from herbs steeped in hot water, then strained and cooled to about body temperature. In this book, the most frequently suggested herbal formula is *dashamoola*, which is particularly effective for balancing vata.

To introduce the fluid into the rectum, first fill the enema bag, which should be suspended about 3 feet above your position. Allow all air to be expelled from the tube, and close the clip on the hose. Then lie on the floor on your left side, your left (bottom)

leg extended and your right leg flexed at the knee. (For comfort, prepare a rug or a couple of towels to lie on rather than lying on a bare floor.) Lubricate the tip of the syringe with oil or ghee. Make sure the anal area is clean and lubricated. Carefully and slowly insert the tip of the syringe into the rectum, then release the clip and allow all the fluid to enter. When the enema bag is empty, remove the tip.

Again, try to hold the oil for 10 minutes, and then hold the oil/herbal mixture for another 30 minutes if you can. While holding the fluid inside, assume a hands-and-knees position for a while and elevate the buttocks; this relaxes the colon. Periodically massage the colon area lightly with a counterclockwise motion (as seen when you look down toward your navel). Massage up the left side to the rib cage, across to the right side, and down. (This is opposite to the way food journeys through and serves to propel the enema fluid up into the higher areas of the colon.)

When the fluid has been retained for the suggested time (or if you find you just can't hold it), sit on the toilet and allow the passage of the fluid and fecal matter.

You may wish to wear a sanitary pad of some kind in your underwear for a few hours following *basti*, as there is likely to be some seepage of residual oil.

You may note that for some conditions discussed in this book, only an oil enema, or only a *dashamoola* enema—and occasionally only a warm water enema—is recommended. Follow the recommendations for that condition.

For some individuals, the fluid does not come back out. That simply means that the colon was very dry and that all the liquid has been absorbed. This is entirely natural and is nothing to worry about.

Nasya

Nasya is the nasal administration of herbal oils, ghee, or fine powders. If you were to receive *nasya* in a *panchakarma* treatment at an Ayurvedic clinic, you would lie face up on a table with your head tilted back, nostrils “facing the sky.” A small amount of an appropriate powder might be placed into your nostrils, or 3 to 5 drops of a medicated oil or ghee.

When you do *nasya* yourself, simply dip your little finger (clean and with the nail closely trimmed) into ghee or whatever herbalized oil might be recommended, and lightly massage the inside of your nostril with your little finger. Then gently sniff to draw the oil upward.

Nauli

Nauli is a simple method of massaging the internal organs, particularly the colon,

intestines, liver, and spleen. It also maintains abdominal fire and helps to keep the colon clean.

Stand with your feet about shoulder-width apart, and slightly flex your knees. Bend forward as shown in the diagram, and put each hand on its respective knee. Breathe in, a long deep breath, and gradually exhale it. After a complete exhalation, hold your breath out.

Contract your abdominal muscles so that you form a ropelike structure at the abdominal wall. Then, by alternating the pressure on your right and left hand, you can move the abdominal muscles from right to left and left to right. Do this seven times.



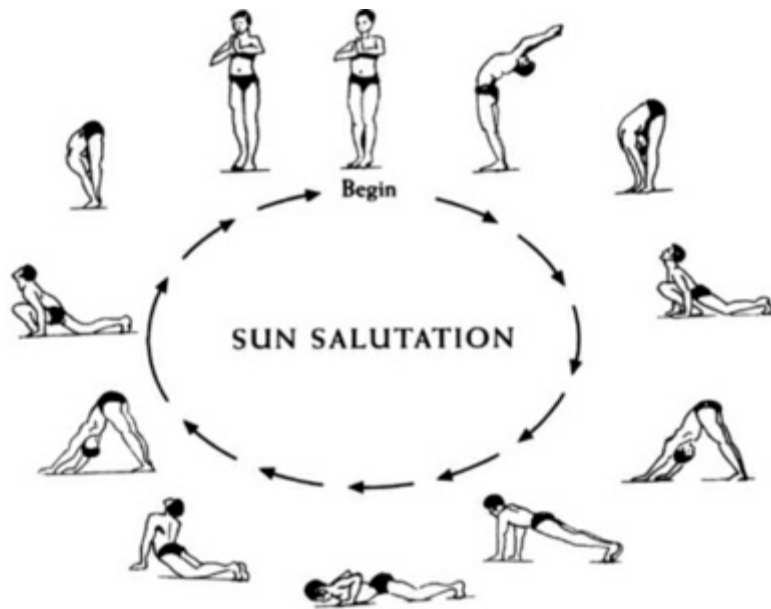
Nauli

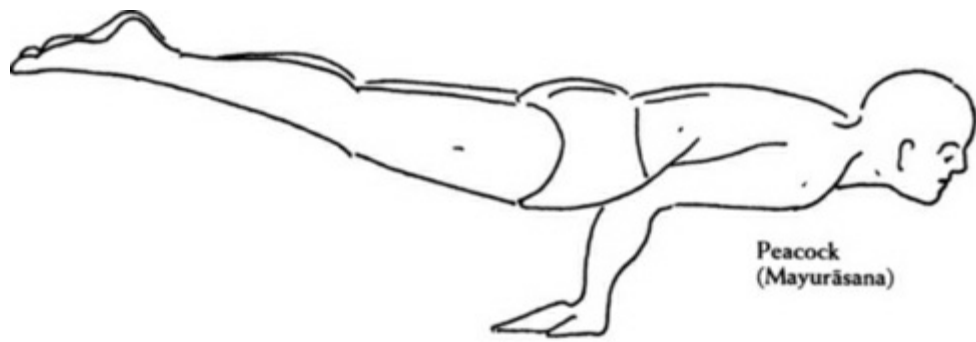
Appendix 4

Yoga Asanas

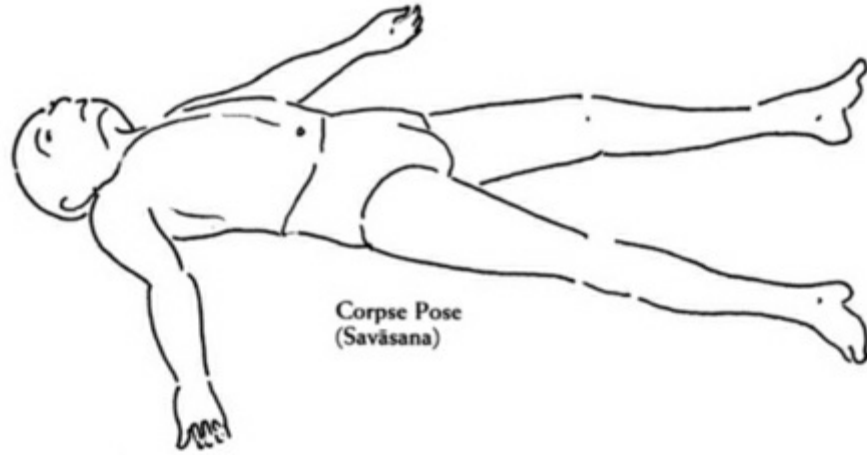
Throughout this book, I recommend specific yoga *asanas* for vata–pitta–kapha and for various ailments. Although simple illustrations of the recommended postures are provided here, instruction in yoga is not intended and is beyond the scope of this book. You cannot really learn how to do yoga *asanas* properly from written instructions and some illustrations.

If you have already learned how to do yoga *asanas*, these illustrations will serve to refresh your memory. To learn the postures for the first time, please see a qualified yoga instructor.





Peacock
(Mayurāsana)



Corpse Pose
(Savāsana)



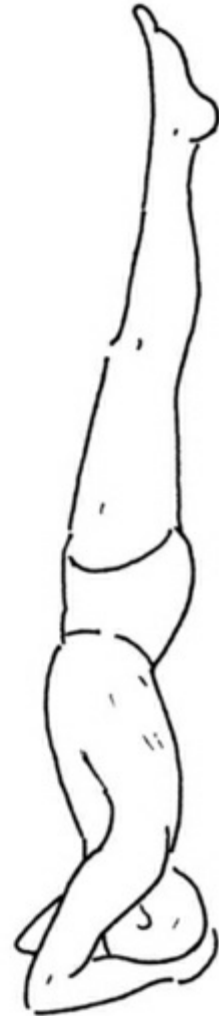
Plow
(Halāsana)



Locust
(Shalabhāsana)



Unilateral Head-to-Knee
Pose (Maha Mudra)



Head Stand
(Shirshāsana)



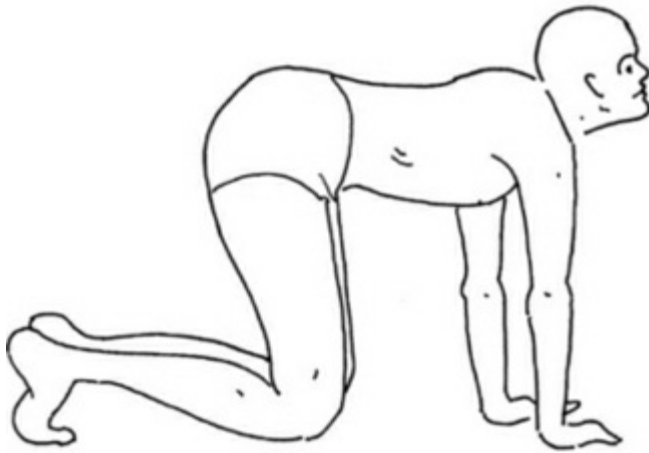
Cow-Head Pose
(Go Mukhāsana)



Cow
(side view)



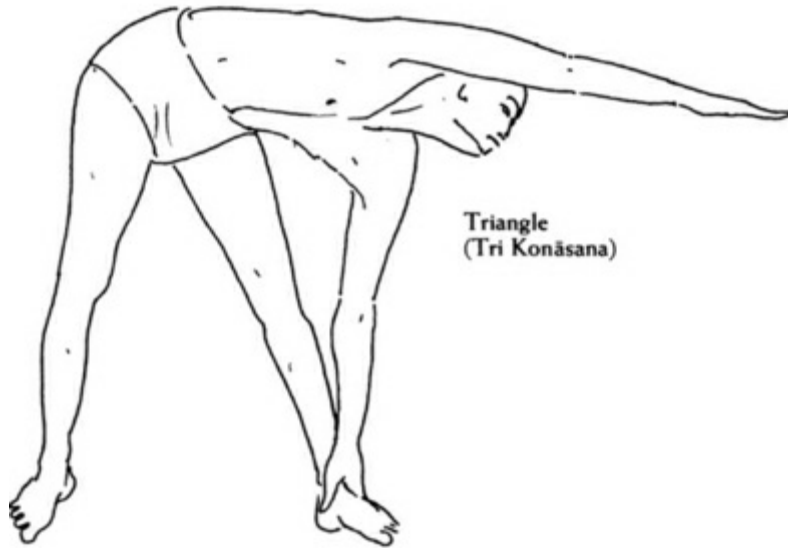
Palm Tree
(Tādāsana)



Cat
(Marjārasana)



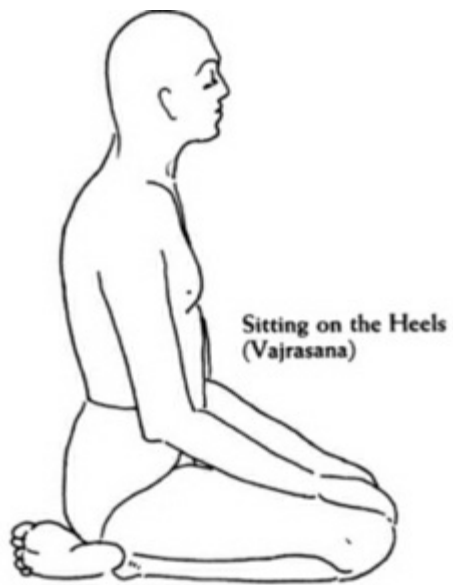
Lotus Pose with
Forward Bend
(Yoga Mudra)



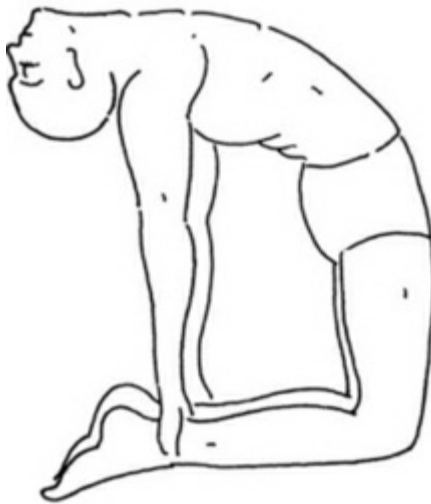
Triangle
(Tri Konāsana)



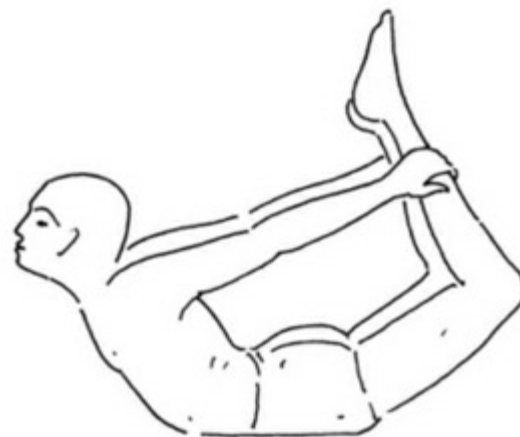
Left Lateral Relaxed Pose with Head Resting on Hand
(Narayana)



Elevated Lotus
(Kukutasana)



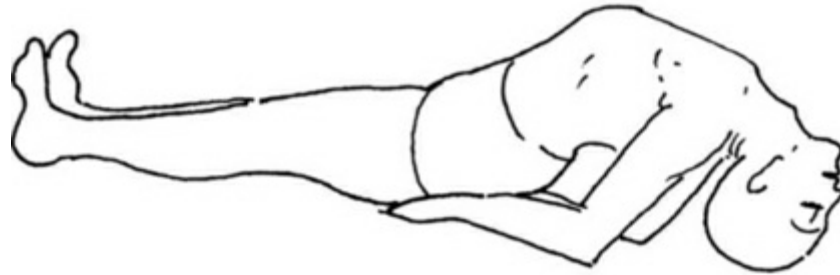
Camel
(Ushtrasana)



Bow
(Dhanurasana)



The Cobra Pose
(Bhujangāsana)



The Fish Pose
(Matsyāsana)



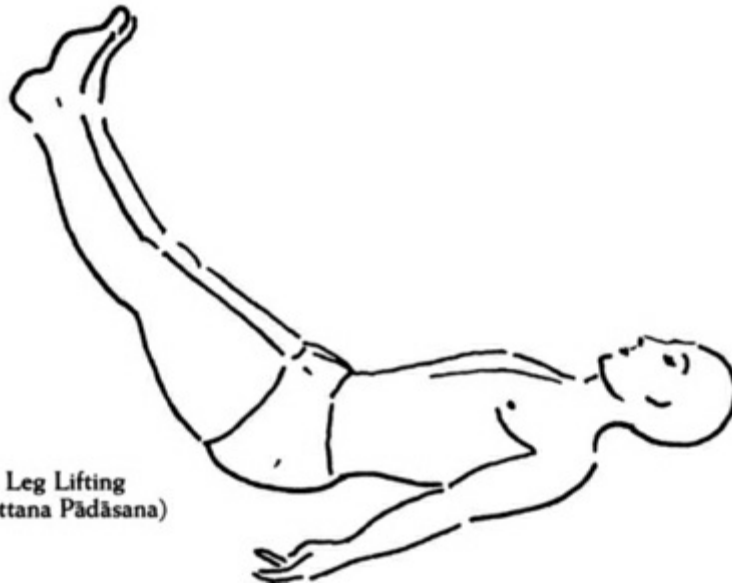
The Bridge Variation
(Setu Bandhāsana)



The Forward Bend
(Purvottanāsana)



The Knee Chest Pose
(Pavana Muktasana)



Leg Lifting
(Zittana Pādāsana)



The Lion Pose
(Simhāsana)



The Lotus Pose
(Padmāsana)



The Kneeling Wheel Pose
(Ardha Chakra Asana)



The Spinal Twist
(Matsyendrāsana)

Glossary

AGNI. The biological fire that provides energy for the body to function. Agni regulates body heat and aids digestion, absorption, and assimilation of food. It transforms food into energy or consciousness.

AHAMKARA. Literally, the “I-former”; the ego; sense of separate self; the feeling of “I am.”

AMA. A toxic, morbid substance (both systemic and cellular) produced by undigested food which is the root cause of many diseases.

ANUPANA. Substance (such as milk, water, ghee, etc.) that serves as a medium for taking herbs.

ARTAVA DHATU. The female reproductive tissue, one of the seven *dhatu*s or bodily tissues.

ASTHI DHATU. One of the seven *dhatu*s or bodily tissues; specifically, the bone tissue that supports the body, giving protection, shape, and longevity.

AYURVEDA. The science of life; derived from the Sanskrit words *ayur* meaning life, and *veda*, knowledge or science. The Vedas are the authentic, ancient, spiritual scriptures of India.

BASMATI RICE. A long-grained scented rice originating in the foothills of the Himalayas in India. Easily digestible and nutritious.

BASTI. One of the five important cleansing measures of *panchakarma*, it eliminates excess vata dosha from the system via medicated herbal tea or oil enemas. Helps greatly to heal all vata disorders. The word *basti* literally means bladder. In ancient times, the apparatus used for the procedure was made out of leather.

BHASMA. A specialized Ayurvedic compound prepared and purified by being burned into ash; *bhasmas* have a high potency and release *prana* into the system.

BHASTRIKA. A breathing practice (*pranayama*) in which air is passively drawn in and forcibly pushed out, as in a bellows. Increases heat and improves circulation.

BHRAMARI. A type of breathing practice (*pranayama*) in which a soft humming sound, like a bee, is made during exhalation and/or inhalation. Calms the mind and cools pitta.

CARDAMOM. Pungent spice from a tropical plant.

CHAI. General word for tea; often refers to a spiced black tea made with milk and sugar.

CHAKRAS. The energy centers in the body, related to nerve plexus centers, which govern bodily functions. Each chakra is a reservoir of consciousness.

CHICKPEA FLOUR. A finely ground yellow flour. Also called gram.

CILANTRO. Fresh coriander leaf. This herb is used extensively in Indian cooking and valued for its zesty and cooling taste. Balances spicy dishes.

COCONUT MILK. Made from grating the white flesh of the coconut and mixing with a cup of water.

COCONUT WATER. The natural juice inside the coconut.

DAL. Any type of dried bean, pea, or lentil is called dal. Most dal is husked and split for quick cooking and greater ease of digestion.

DHATU. The structural, building, elemental tissue of the body. There are seven *dhatu*s defined in Ayurveda: *rasa* (plasma); *rakta* (blood tissue); *mamsa* (muscle tissue); *meda* (adipose tissue); *asthi* (bone marrow); *majja* (bone and nerves); *shukra* and *artava* (male and female reproductive tissue).

DOSHA. The three main psycho-physiological functional principles of the body (vata, pitta, and kapha). They determine each individual's constitution and maintain the integrity of the human body. The *doshas* govern the individual's response to changes. When disturbed, they can initiate the disease process.

GHEE. Clarified butter; made from unsalted butter that has been gently cooked and the milk solids removed.

GUGGULU. Main ingredient in a number of herbal preparations (*yogaraj guggulu*, *kaishore guggulu*, etc.). A resin from a small tree, it has many useful medical actions, including benefits for the nervous system, tonification, and anti-inflammatory action on muscle tissues. Helps increase white blood count (good for the immune system) and is a nervine, rejuvenating tonic.

GUNAS. Three qualities influencing all creation: *sattva*, *rajas*, and *tamas*. *Sattvic* qualities imply essence, reality, consciousness, purity, and clarity of perception. All movement and activity are due to *rajas*. *Tamas* brings darkness, inertia, heaviness, and materialistic attitudes. There is a constant interplay among these three *gunas* in all creation. Also refers to the qualities (hard/soft, hot/cold, etc.) of the three doshas, seven *dhatu*s, and three *malas*.

JAGGERY. An unrefined sugar made from the juice of crushed sugarcane stalks.

KAPHA. One of the three doshas, combining the water and earth elements. Kapha is the energy that forms the body's structure—bones, muscles, tendons—and provides the “glue” that holds the cells together. It supplies the water for all bodily parts and systems, lubricates joints, moisturizes the skin, and maintains immunity. In balance, kapha is expressed as love, calmness, and forgiveness. Out of balance, it leads to attachment, greed, and envy.

KHAIVAIGUNYA. A weak or defective space within an organ or tissue of the body where a pathological condition is likely to begin.

KITCHARI. A cooked mixture of rice and dal and spices that is easy to digest and high in protein. Often used as a nourishing food for a mono-fast.

LASSI. A refreshing drink made from yogurt, water, and spices and often served at the end of a meal as a digestive. Can be sweet or salty.

MAHAT (OR MAHAD). The “great principle,” intelligence, the cosmic aspect of intellect; also contains the individual intellect, called *Buddhi*.

MAJJA DHATU. One of the seven *dhatus* or bodily tissues; the bone marrow and nerve tissue. It is unctuous and soft. Its main function is to oleate the body, to fill up the bone, and to nourish the *shukra dhatu*. It plays an important role in communication.

MAMSA DHATU. One of the seven *dhatus* or bodily tissues; the muscle tissue. Produced by *rasa* and *rakta*, its main functions are to provide physical strength, coordination, movement, covering, form, and protection.

MANTRA. A sacred word or phrase of spiritual significance and power that transcends the mind and yields bliss.

MARMA. An energy point on the skin that has a door receptor and is connected to the inner pathways of healing.

MUNG DAL. A small bean that has been husked and split. Usually a medium yellow color. Easy to digest.

NASYA. Method of administering medication through the nose; one of the five measures of *panchakarma*.

NIGHTSHADE. Common name for a family of plants including tomatoes, potatoes, eggplant, tobacco, petunias, and belladonna, which have strong medicinal properties. Frequent use may disturb the doshic equilibrium.

OJAS. The pure essence of all the bodily tissues (*dhatus*); the superfine essence of kapha; maintains immunity, strength, and vitality. *Ojas* creates bliss and awareness in the mental faculties and governs the body’s immune function. If it is depleted, it can lead to death.

PANCHAKARMA. Five measures for elimination of excess dosha and/or ama from the body. Used for the purpose of internal purification. They are: vomiting (*vamana*); purgation (*virechana*); medicated oil or decoction enema (*basti*); bloodletting (*rakta moksha*); and nasal administration of specific medication (*nasya*).

PIPPALI. *Piper longum*; a close relative of black pepper, which has many medicinal applications, especially for digestion and respiration. A rejuvenative tonic (*rasayana*) for

the lungs and liver.

PITTA. One of the three doshas; it corresponds to the elements of fire and water. Sometimes referred to as the fire or bile principle, pitta governs digestion, absorption, assimilation, metabolism, and body temperature. In balance, pitta promotes understanding and intelligence; out of balance pitta arouses anger, hatred, jealousy.

PRAKRUTI. *Prakruti* (spelled with a capital *P*) is the Cosmic Creativity, the primordial matter.

PRAKRUTI. The inherent nature or psychosomatic, biological constitution of the individual, *prakruti* is the fixed constitution of a person, which reflects the proportion of the three doshas (vata, pitta, and kapha) established at conception.

PRANA. The vital life energy. Without it, life cannot exist. The flow of cellular intelligence from one cell to another. Equivalent to the Oriental *Ch'i* or *Ki*.

PRANAYAMA. The control of life energy by various techniques which regulate and restrain breath, through which one can control the mind and improve one's quality of awareness and perception. Helpful with all types of meditation.

PURUSHA. Choiceless, passive awareness; the pure Cosmic Being.

RAJAS. One of the three universal qualities (*gunas*) of *Prakruti*, Cosmic Creativity. *Rajas* is active, mobile, dynamic.

RAKTA DHATU. The second of the seven tissues (*dhatus*), *rakta* mainly contains red blood cells, which carry life energy (*prana*) to all bodily tissues. This oxygenates, or provides the life function, for all the tissues.

RASA DHATU. The first of the seven *dhatus*, *rasa* (plasma) is nourished from digested food, and after absorption, it circulates in the entire body via specific channels. Its main function is to provide nutrition to each and every cell of the body.

RASAYANA. Rejuvenation therapy which brings about renewal, regeneration, and restoration of bodily cells, tissues, and organs, giving longevity to the cells and enhancing immunity and stamina.

RISHI. A seer, a Vedic sage. The ancient *rishis* perceived and/or recorded the Vedic hymns. These enlightened sages shared their knowledge, medicine, philosophy, and spiritual teachings.

RUDRAKSHA. The "tears of Shiva"; the dried seeds from the fruit of the rudraksha tree. Said to be good for the heart both physically and spiritually, helpful for meditation and for "opening the heart chakra."

SAFFRON. A golden yellow spice that comes from the stigma of a particular crocus. The best quality saffron is grown in Spain and Kashmir.

SAMPRAPTI. The pathogenesis of disease; the entire disease process from its cause through its various stages to the complete manifestation of the disease.

SANKHYA. One of the schools of Indian philosophy, Sankhya denotes both “discriminative knowledge” and “enumeration.” It gives a systematic account of cosmic evolution from *Purusha* (Cosmic Spirit) and *Prakruti* (Primordial Matter) through the stages of creation: *Mahad* (Cosmic Intelligence); *Ahamkara* (individuating principle); *Mana* (mind); *Indriyas* (the inner doors of perception); *Tanmatras* (the objects of perception); and *Mahat Bhutas* (five great elements). *Sat* means truth and *khyā* means to realize; thus Sankhya means to realize the theory of the creation of the universe in order to realize the ultimate truth of human life. Sankhya reveals the journey of consciousness into matter.

SATTVA. One of the three *gunas* of *Prakruti*, *sattva* denotes light, clarity, purity of perception; it is the essence of pure awareness.

SHITALI. A practice of *pranayama* (breath control) that cools the system. Inhalation is through the curled tongue; exhalation is slow, steady, and complete.

SHUKRA DHATU. The seventh tissue (*dhatu*); the male reproductive tissue.

SROTAS. Bodily channels.

SUCANAT. A granulated natural sugar made from pure sugarcane juice.

SURYA NAMASKAR. The Sun Salutation, a series of yoga postures done in a flowing sequence with coordinated breathing.

TAMAS. One of the three *gunas* of *Prakruti* or Nature; its characteristics are darkness, inertia, and ignorance; it is responsible for sleep, drowsiness, dullness, unconsciousness.

TEJAS. The pure essence of the fire element; the superfine essence of pitta dosha, which governs the transformation of matter into energy and of food, water, and air into consciousness.

TIKTA GHRITA. “Bitter ghee,” a specific Ayurvedic compound made of clarified butter with various bitter herbs; used for medicinal purposes.

TRIDOSHA. The three organizations or codes of intelligence within the body, mind, and consciousness; the three bodily humors: air (*vata*), fire/bile (*pitta*), and water (*kapha*).

TRIKATU. An Ayurvedic compound of ginger, black pepper, and *pippali* (*piper longum*) that burns ama, detoxifies the body, and improves digestion, absorption, and assimilation.

TRIPHALA. An important Ayurvedic compound consisting of three herbs: *amalaki*, *bibhitaki*, and *haritaki*. It is the best laxative and bowel tonic and a balanced *rasayana* that is good for *vata*, *pitta*, and *kapha*.

TULSI. Indian holy basil. The sacred plant of Krishna, this herb is said to open the heart and mind, bestowing the energy of love and devotion.

TURBINADO. A granulated sugar made from pure sugarcane.

TURMERIC ROOT. An underground rhizome from a perennial plant native to southern India and Asia. Comes in a red and yellow form, but only the yellow is eaten. One of the most important herbs for both internal and external use, it is also essential in most Indian cooking.

VATA. One of the three doshas, combining the space and air elements; it is the subtle energy associated with bodily movement and governs breathing, blinking, muscle and tissue movement, pulsation of the heart, and all movements in the cytoplasm and cell membranes. In balance, vata promotes creativity and flexibility; out of balance, vata produces fear and anxiety.

VIKRUTI. The current state of the individual, as opposed to the original constitution (*prakruti*) at conception. It may also denote disorder.

YOGA. In its deeper sense, Yoga is union of the lower self with the higher self, of the inner with the outer, mortality with immortality. Yoga postures (*asanas*) promote health, flexibility, and purity toward achieving the state of Yoga.

Resources

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Silver Lake, WI 53170

414-889-8569

The Ayurvedic Institute

P.O. Box 23445

Albuquerque, NM 87192-1445

505-291-9698

Fax: 505-294-7572

Ayush Herbs, Inc.

10025 N.E. 4th Street

Bellevue, WA 98004

800-925-1371

Bazaar of India Imports, Inc.

1810 University Avenue

Berkeley, CA 94703

510-548-4110

Bioveda

P.O. Box 420

Congers, NY 10920

Dr. Singha's Mustard Bath and More

Attn.: Anna Searles

Natural Therapeutic Centre

2500 Side Cove

Austin, TX 78704

800-856-2862

Herbalvedic Products

P.O. Box 6390

Santa Fe, NM 87502

Kanak

P.O. Box 13653

Albuquerque, NM 87192-3653

Internatural (retail)

33719 116th Street, Dept. VL

Twin Lakes, WI 53181

800-643-4221

www.internatural.com

Lotus Herbs

1505 42nd Ave. Suite 19

Capitola, CA 95010

408-479-1667

Lotus Light (wholesale)

P.O. Box 1008, Dept. VL

Silver Lake, WI 53170

414-889-8501

Yoga of Life Center

2726 Tramway N.E.

Albuquerque, NM 87122

505-275-6141

Vinayak Ayurveda Center

2509 Virginia NE, Suite D

Albuquerque, NM 87110

505-296-6522

505-298-2932 (fax)

Maharishi Ayur-Veda Products International Inc.

417 Bolton Road

P.O. Box 54

Lancaster, Mass 01523

800-All-Veda

508-368-7475 (fax)

New Moon Extracts

P.O. Box 1947

Brattleboro, VT 05302-1947

NATURAL INGREDIENTS

Aloe Farms

Box 125

Los Fresnos, TX 78566

800-262-6771

(For aloe vera juice, gel, powder, and capsules)

Arya Laya Skin Care Center

Rolling Hills Estates, CA 90274

(For carrot oil)

Aubrey Organics

4419 North Manhattan Ave.

Tampa, FL 33614

(For rosa mosquita oil and a large variety of natural shampoos and cosmetics)

Body Shop

45 Hosehill Road

Cedar Knolls, NJ 07927-2014

800-541-2535

(Aloe vera, nut and seed oils, cosmetics, makeup, brushes, loofas, and much more)

Culpepper Ltd.

21 Bruton Street

London W1 × 7DA

England

(Variety of natural seed, nut, and kernel oils, essential oils, herbs, books, and cosmetics)

Desert Whale Jojoba Co.

P.O. Box 41594

Tucson, AZ 85717

602-882-4195

(For jojoba products and many other natural oils, including rice bran, pecan, macadamia nut, and apricot kernel)

Everybody Ltd.

1738 Pearl Street

Boulder, CO 80302

800-748-5675

(Large variety of oils, oil blends, and cosmetics)

Flora Inc.

P.O. Box 950

805 East Badger Road

Lynden, WA 98264

800-446-2110

(For flaxseed oil, herbal supplements for skin, hair, nails, and cosmetics)

Green Earth Farm

P.O. Box 672

65½ North 8th Street

Saguache, CO 81449

(For calendula, oil, cream, and herbal bath)

The Heritage Store Inc.

P.O. Box 444

Virginia Beach, Virginia 23458

804-428-0100

(Castor oil, organic ghee, cocoa butter, massage oils, flower waters, essential oils,

cosmetics, and natural home remedies)

Janca's Jojoba Oil and Seed company

456 E. Juanita No. 7

Mesa, AZ 85204

602-497-9494

(Jojoba oil, butter, wax, and seeds. Also a large variety of naturally pressed nut and grapeseed. Also have clay, aloe products, essential oils, and their own line of cosmetics.)

Reading List

CLASSICAL AYURVEDIC TEXTS

Caraka Samhita. 4 volumes. Priyavrat V. Sharma, editor-translator. Chowkhamba Sanskrit Series Office: Varanasi, India, 1981–1994.

Caraka Samhita. 3rd edition, 3 volumes. Ram Karan Sharma and Vaidya Bhagwan Dash, editors-translators. Chowkhamba Sanskrit Series Office: Varanasi, India, 1992.

Sushruta Samhita. 4th edition, 2 volumes. Kaviraj Kunjalal Bhishagratna, editor-translator. Chowkhamba Sanskrit Series Office: Varanasi, India, 1991.

Vagbhata, *Ashtanga Hridayam*. 2 volumes. K. R. Srikantha Murthy, translator. Krishnadas Academy: Varanasi, India, 1991–1992.

GENERAL READING

Frawley, David. *Ayurvedic Healing*. Morson Publishing: Salt Lake City, 1989.

Frawley, David, and Vasant Lad. *The Yoga of Herbs*. Lotus Press: Santa Fe, 1986.

Lad, Vasant. *Ayurveda: The Science of Self-Healing*. Lotus Press: Santa Fe, 1984.

Morrison, Judith. *The Book of Ayurveda: A Holistic Approach to Health and Longevity*. New York: Simon and Schuster, 1995.

Svoboda, Robert E. *Ayurveda: Life, Health, and Longevity*. Penguin: London, 1992.

—. *The Hidden Secret of Ayurveda*. Pune, India, 1980. Reprint, The Ayurvedic Press: Albuquerque, 1994.

—. *Prakruti: Your Ayurvedic Constitution*. Geocom Limited: Albuquerque, 1989.

SPECIFIC TOPICS

Lad, Usha, and Dr. Vasant Lad. *Ayurvedic Cooking for Self-Healing*. 2nd edition. The Ayurvedic Press: Albuquerque, 1997.

Lad, Dr. Vasant Dattatray. *Secrets of the Pulse: The Ancient Art of Ayurvedic Pulse Diagnosis*. The Ayurvedic Press: Albuquerque, 1996.

Morningstar, Amadea. *The Ayurvedic Cookbook*. Wilmot: Lotus Press, 1990.

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The Ayurvedic Institute was established to promote the traditional knowledge of Ayurveda. In support of this, the Institute also offers programs in the sister disciplines of Ayurveda Sanskrit, Yoga, and Jyotisha (Vedic astrology). The Ayurvedic programs reflect the style of sitting with a traditional Indian teacher. The Vedic educational model is quite different from the Western experience with which most of us are familiar. The knowledge is taught with the body, mind, and spiritual components intact, along with practical examples, ceremonies, and stories.

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